SEG TINTINALLI’S EMERGENCY MEDICINE MANUAL

Educational book production draft

Source edition: Eighth Edition
Content ID: SEG-TINTINALLI-EMM8-FULL
Version: v1.0

DRAFT — MEDICAL REVIEW REQUIRED BEFORE RELEASE

Title Page

Tintinalli’s Emergency Medicine Manual
SEG Educational Transformation

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition
Authors: Cline, Ma, Cydulka, Wang, Joing, Fitch
Publisher metadata: McGraw Hill / Medical

Table of Contents

Chapter 1: Advanced Airway Support — source pages 22–29

Chapter 2: Management of Cardiac Rhythm Disturbances — source pages 30–49

Chapter 3: Resuscitation of Children and Neonates — source pages 50–57

Chapter 4: Fluids, Electrolytes, and Acid-Base Disorders — source pages 58–79

Chapter 5: Therapeutic Approach to The Hypotensive Patient — source pages 80–83

Chapter 6: Anaphylaxis, Acute Allergic Reactions, and Angioedema — source pages 84–87

Chapter 7: Acute Pain Management and Procedural Sedation — source pages 88–97

Chapter 8: Chronic Pain — source pages 98–101

Chapter 9: Evaluating and Preparing Wounds — source pages 102–105

Chapter 10: Methods for Wound Closure — source pages 106–115

Chapter 11: Lacerations to The Face and Scalp — source pages 116–121

Chapter 12: Injuries to the Arm, Hand, Fingertip, and Nail — source pages 122–127

Chapter 13: Lacerations to the Leg and Foot — source pages 128–131

Chapter 14: Soft Tissue Foreign Bodies — source pages 132–135

Chapter 15: Puncture Wounds and Bites — source pages 136–141

Chapter 16: Postrepair Wound Care — source pages 142–145

Chapter 17: Chest Pain: Cardiac or Not — source pages 146–151

Chapter 18: Acute Coronary Syndromes — source pages 152–157

Chapter 19: Cardiogenic Shock — source pages 158–161

Chapter 20: Low-Probability Acute Coronary Syndrome — source pages 162–165

Chapter 21: Syncope — source pages 166–169

Chapter 22: Acute Heart Failure — source pages 170–171

Chapter 23: Valvular Emergencies — source pages 172–179

Chapter 24: The Cardiomyopathies, Myocarditis, and Pericardial Disease — source pages 180–187

Chapter 25: Venous Thromboembolism — source pages 188–195

Chapter 26: Systemic and Pulmonary Hypertension — source pages 196–201

Chapter 27: Aortic Aneurysms and Aortic Dissection — source pages 202–207

Chapter 28: Arterial Occlusion — source pages 208–211

Chapter 29: Respiratory Distress — source pages 212–219

Chapter 30: Bronchitis, Pneumonia, and Novel Respiratory Infections — source pages 220–223

Chapter 31: Tuberculosis — source pages 224–227

Chapter 32: Spontaneous and Iatrogenic Pneumothorax — source pages 228–233

Chapter 34: Asthma and Chronic Obstructive Pulmonary Disease — source pages 234–237

Chapter 35: Acute Abdominal Pain — source pages 238–241

Chapter 36: Nausea and Vomiting — source pages 242–245

Chapter 37: Disorders Presenting Primarily with Diarrhea — source pages 246–253

Chapter 38: Acute and Chronic Constipation — source pages 254–259

Chapter 40: Esophageal Emergencies — source pages 260–263

Chapter 41: Peptic Ulcer Disease and Gastritis — source pages 264–267

Chapter 42: Pancreatitis and Cholecystitis — source pages 268–271

Chapter 43: Acute Appendicitis — source pages 272–275

Chapter 44: Diverticulitis — source pages 276–279

Chapter 45: Intestinal Obstruction and Volvulus — source pages 280–281

Chapter 46: Hernia in Adults and Children — source pages 282–285

Chapter 47: Anorectal Disorders — source pages 286–293

Chapter 48: Jaundice, Hepatic Disorders, and Hepatic Failure — source pages 294–301

Chapter 49: Complications of General Surgical Procedures — source pages 302–307

Chapter 50: Acute Kidney Injury — source pages 308–311

Chapter 51: Rhabdomyolysis — source pages 312–315

Chapter 52: Emergencies in Renal Failure and Dialysis Patients — source pages 316–319

Chapter 53: Urinary Tract Infections and Hematuria — source pages 320–323

Chapter 54: Acute Urinary Retention — source pages 324–325

Chapter 55: Male Genital Problems — source pages 326–331

Chapter 56: Urologic Stone Disease — source pages 332–335

Chapter 57: Complications of Urologic Procedures and Devices — source pages 336–339

Chapter 58: Vaginal Bleeding and Pelvic Pain in the Nonpregnant Patient — source pages 340–343

Chapter 59: Ectopic Pregnancy and Emergencies — source pages 344–347

Chapter 60: Comorbid Diseases in Pregnancy — source pages 348–355

Chapter 61: Emergencies After 20 Weeks — source pages 356–359

Chapter 62: Emergency Delivery — source pages 360–363

Chapter 63: Vulvovaginitis — source pages 364–367

Chapter 64: Pelvic Inflammatory Disease — source pages 368–369

Chapter 65: Complications of Gynecologic Procedures — source pages 370–373

Chapter 66: Fever and Serious Bacterial Illness in Children — source pages 374–381

Chapter 67: Common Neonatal Problems — source pages 382–385

Chapter 68: Common Infections of the Ears, Nose, Neck, and Throat — source pages 386–391

Chapter 69: Upper Respiratory Emergencies—Stridor and Drooling — source pages 392–399

Chapter 70: Wheezing in Infants and Children — source pages 400–405

Chapter 71: Pneumonia in Infants and Children — source pages 406–407

Chapter 72: Pediatric Heart Disease — source pages 408–417

Chapter 73: Vomiting and Diarrhea in Infants and Children — source pages 418–419

Chapter 74: Pediatric Abdominal Emergencies — source pages 420–429

Chapter 76: Seizures and Status Epilepticus in Children — source pages 430–431

Chapter 77: Altered Mental Status and Headache in Children — source pages 432–437

Chapter 78: Syncope and Sudden Death in Children and Adolescents — source pages 438–439

Chapter 79: Hypoglycemia and Metabolic Emergencies in Infants and Children — source pages 440–445

Chapter 80: Diabetes in Children — source pages 446–447

Chapter 81: Fluid and Electrolyte Therapy in Infants and Children — source pages 448–451

Chapter 82: Musculoskeletal Disorders in Children — source pages 452–463

Chapter 83: Rashes in Children — source pages 464–475

Chapter 84: Sickle Cell Anemia in Children — source pages 476–481

Chapter 85: Hematologic-Oncologic Emergencies in Children — source pages 482–487

Chapter 86: Renal Emergencies in Infants and Children — source pages 488–493

Chapter 87: Sexually Transmitted Infections — source pages 494–499

Chapter 88: Toxic Shock Syndromes — source pages 500–503

Chapter 89: Sepsis — source pages 504–511

Chapter 90: Soft Tissue Infections — source pages 512–517

Chapter 91: Serious Viral Infections — source pages 518–525

Chapter 92: HIV Infection and AIDS — source pages 526–533

Chapter 93: Infective Endocarditis — source pages 534–537

Chapter 94: Tetanus and Rabies — source pages 538–543

Chapter 95: Malaria — source pages 544–547

Chapter 96: Foodborne and Waterborne Diseases — source pages 548–551

Chapter 97: Zoonotic Infections — source pages 552–559

Chapter 98: World Travelers — source pages 560–565

Chapter 99: The Transplant Patient — source pages 566–575

Chapter 100: General Management of the Poisoned Patient — source pages 576–583

Chapter 101: Anticholinergic Toxicity — source pages 584–585

Chapter 102: Psychopharmacologic Agents — source pages 586–593

Chapter 103: Sedatives and Hypnotics — source pages 594–599

Chapter 104: Alcohols — source pages 600–603

Chapter 105: Drugs of Abuse — source pages 604–611

Chapter 106: Analgesics — source pages 612–617

Chapter 107: Xanthines and Nicotine — source pages 618–621

Chapter 108: Cardiac Medications — source pages 622–633

Chapter 109: Anticonvulsants — source pages 634–637

Chapter 110: Iron — source pages 638–641

Chapter 111: Hydrocarbons and Volatile Substances — source pages 642–643

Chapter 112: Caustics — source pages 644–645

Chapter 113: Pesticides — source pages 646–651

Chapter 114: Metals and Metalloids — source pages 652–657

Chapter 115: Industrial Toxins — source pages 658–663

Chapter 116: Vitamins and Herbals — source pages 664–665

Chapter 117: Dyshemoglobinemias — source pages 666–669

Chapter 118: Cold Injuries — source pages 670–675

Chapter 119: Heat Emergencies — source pages 676–677

Chapter 120: Bites and Stings — source pages 678–685

Chapter 121: Trauma and Envenomation from Marine Fauna — source pages 686–689

Chapter 122: High-Altitude Disorders — source pages 690–695

Chapter 124: Near Drowning — source pages 696–697

Chapter 125: Thermal and Chemical Burns — source pages 698–705

Chapter 126: Electrical and Lightning Injuries — source pages 706–711

Chapter 127: Carbon Monoxide — source pages 712–715

Chapter 128: Mushroom and Plant Poisoning — source pages 716–721

Chapter 129: Diabetic Emergencies — source pages 722–731

Chapter 131: Thyroid Disease Emergencies — source pages 732–735

Chapter 132: Adrenal Insufficiency — source pages 736–743

Chapter 133: Evaluation of Anemia and the Bleeding Patient — source pages 744–745

Chapter 134: Acquired Bleeding Disorders — source pages 746–749

Chapter 135: Hemophilias and von Willebrand Disease — source pages 750–753

Chapter 136: Sickle Cell Disease and Other Hereditary Hemolytic Anemias — source pages 754–759

Chapter 137: Transfusion Therapy — source pages 760–765

Chapter 138: Anticoagulants, Antiplatelet Agents, and Fibrinolytics — source pages 766–771

Chapter 139: Emergency Complications of Malignancy — source pages 772–779

Chapter 140: Headache — source pages 780–785

Chapter 141: Stroke Syndromes and Spontaneous Subarachnoid Hemorrhage — source pages 786–795

Chapter 142: Altered Mental Status and Coma — source pages 796–801

Chapter 143: Ataxia and Gait Disturbances — source pages 802–803

Chapter 144: Acute Vertigo — source pages 804–811

Chapter 145: Seizures and Status Epilepticus in Adults — source pages 812–815

Chapter 146: Acute Peripheral Neurologic Lesions — source pages 816–819

Chapter 147: Chronic Neurologic Disorders — source pages 820–825

Chapter 148: Central Nervous System and Spinal Infections — source pages 826–833

Chapter 149: Eye Emergencies — source pages 834–845

Chapter 150: Face and Jaw Emergencies — source pages 846–851

Chapter 151: Ear, Nose, and Sinus Emergencies — source pages 852–857

Chapter 152: Oral and Dental Emergencies — source pages 858–865

Chapter 153: Neck and Upper Airway Disorders — source pages 866–871

Chapter 154: Dermatologic Emergencies — source pages 872–875

Chapter 155: Other Dermatologic Disorders — source pages 876–885

Chapter 156: Trauma in Adults — source pages 886–889

Chapter 157: Trauma in Children — source pages 890–893

Chapter 158: Trauma in the Elderly — source pages 894–897

Chapter 159: Trauma in Pregnancy — source pages 898–899

Chapter 160: Head Trauma — source pages 900–905

Chapter 161: Spine Trauma — source pages 906–911

Chapter 162: Facial Injuries — source pages 912–917

Chapter 163: Neck Injuries — source pages 918–923

Chapter 164: Cardiothoracic Injuries — source pages 924–929

Chapter 165: Abdominal Injuries — source pages 930–933

Chapter 166: Penetrating Trauma to The Flank and Buttocks — source pages 934–935

Chapter 167: Genitourinary Injuries — source pages 936–939

Chapter 168: Trauma to The Extremities — source pages 940–943

Chapter 169: Initial Evaluation and Management of Orthopedic Injuries — source pages 944–949

Chapter 170: Hand and Wrist Injuries — source pages 950–953

Chapter 171: Forearm and Elbow Injuries — source pages 954–961

Chapter 172: Shoulder and Humerus Injuries — source pages 962–969

Chapter 173: Pelvis, Hip, and Femur Injuries — source pages 970–975

Chapter 174: Knee and Leg Injuries — source pages 976–981

Chapter 175: Ankle and Foot Injuries — source pages 982–985

Chapter 176: Compartment Syndrome — source pages 986–989

Chapter 177: Neck and Back Pain — source pages 990–995

Chapter 178: Shoulder Pain — source pages 996–999

Chapter 179: Hip and Knee Pain — source pages 1000–1003

Chapter 180: Acute Disorders of The Joints and Bursae — source pages 1004–1007

Chapter 181: Emergencies in Systemic Rheumatic Diseases — source pages 1008–1015

Chapter 182: Nontraumatic Disorders of The Hand — source pages 1016–1017

Chapter 183: Soft Tissue Problems of The Foot — source pages 1018–1023

Chapter 184: Clinical Features of Behavioral Disorders — source pages 1024–1027

Chapter 185: Emergency Assessment and Stabilization of Behavioral Disorders — source pages 1028–1029

Chapter 186: Panic and Conversion Disorders — source pages 1030–1033

Chapter 187: Child and Elderly Abuse — source pages 1034–1035

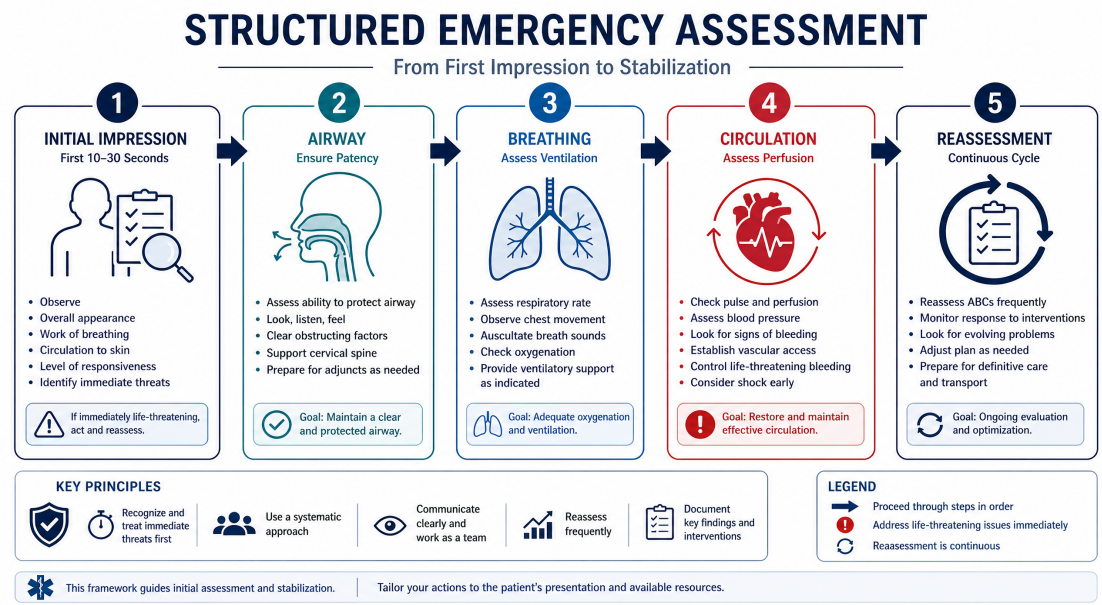
Chapter 188: Sexual Assault and Intimate Partner Violence and Abuse — source pages 1036–1041

Chapter 189: Palliative Care — source pages 1042–1089

Visual Educational Assets

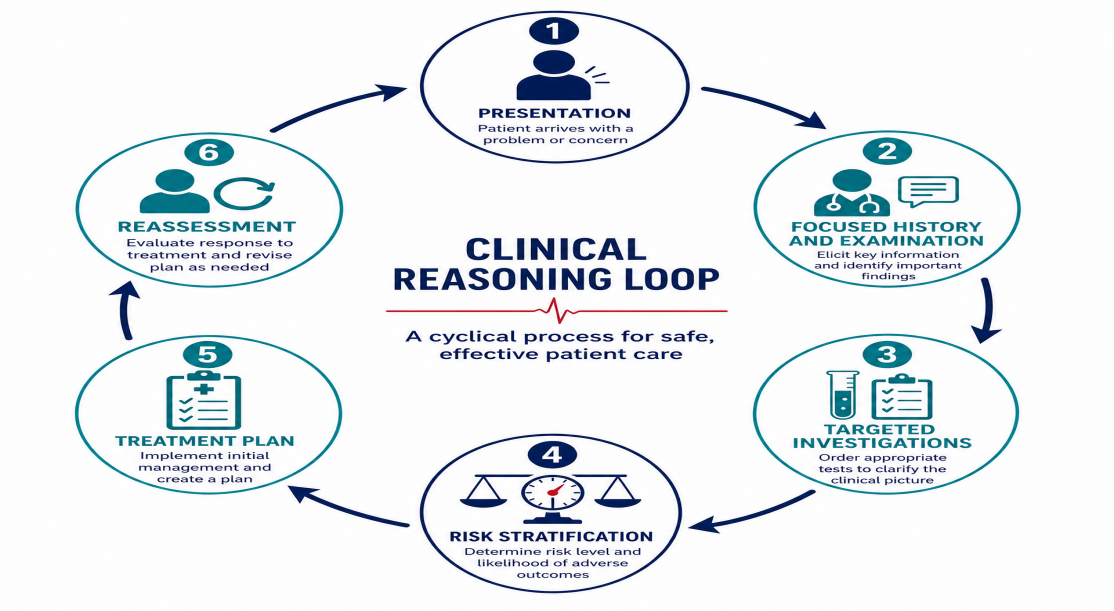
Original SEG assets are included as draft educational supplements and require medical review before release.

SEG VA 001 initial assessment



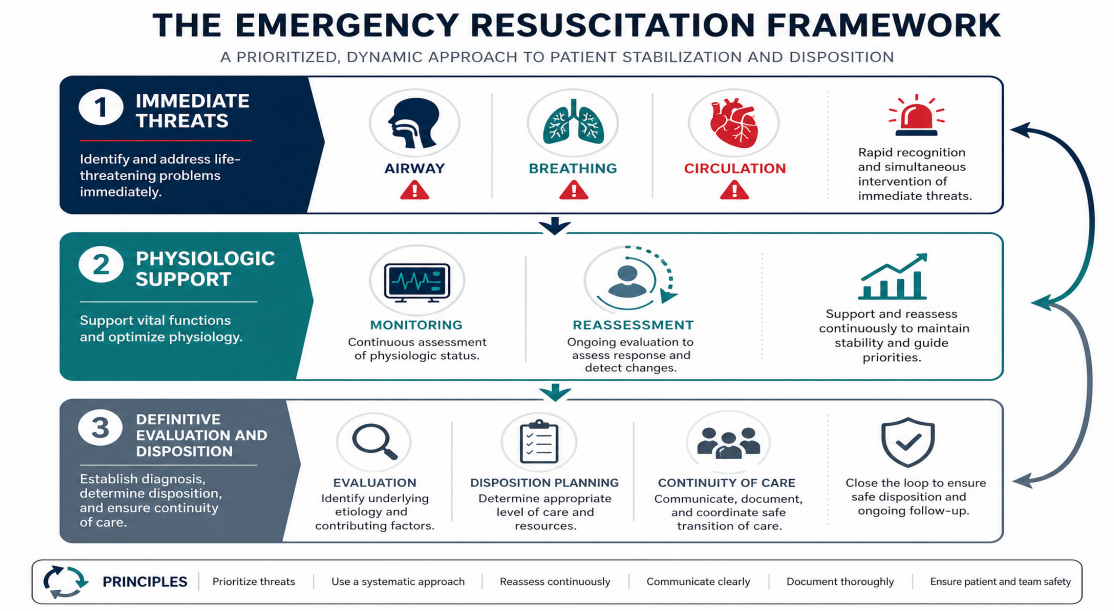
Asset ID: SEG-VA | Topic: Initial emergency assessment | Review: MEDICAL_REVIEW_REQUIRED

SEG VA 002 clinical reasoning



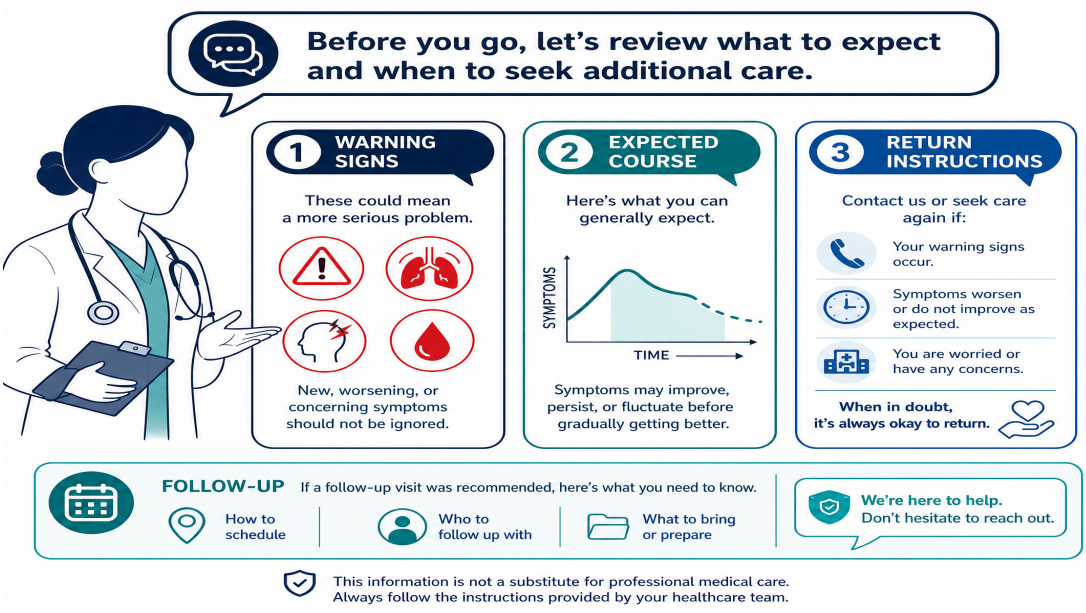
Asset ID: SEG-VA | Topic: Clinical reasoning | Review: MEDICAL_REVIEW_REQUIRED

SEG VA 003 resuscitation framework



Asset ID: SEG-VA | Topic: Resuscitation framework | Review: MEDICAL_REVIEW_REQUIRED

SEG VA 004 safety netting



Asset ID: SEG-VA | Topic: Safety-netting and follow-up | Review: MEDICAL_REVIEW_REQUIRED

Chapter 1: Advanced Airway Support

Source pages 22–29 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition \n**Source pages:** 22–29 \n**Content ID:** SEG-TINTINALLI-EMM8-FULL \n**Review status:** MACHINE-AUTHORED; MEDICAL REVIEW REQUIRED

Chapter scaffold — Advanced Airway Support (source: pages 22–29)

Learning objectives

- Describe indications for intubation and basic decision points for airway approach (RSI, awake technique, surgical airway). (pp. 23–24, 26–27)

- List preparatory steps and equipment considerations for orotracheal intubation. (pp. 23–25)

- Summarize the stepwise orotracheal intubation procedure and immediate confirmation steps. (pp. 23–26)

- Recognize indications, contraindications, and the bougie-aided surgical (cricothyroid) airway technique. (pp. 26–28)

- Identify key high-yield clinical pearls and common pitfalls explicitly noted in the source. (pp. 22–28)

Core concepts (statements drawn exactly from the excerpt)

- Always have an appropriately sized extraglottic device (EGD) available during airway management; do not rely on an EGD to the exclusion of surgical airway when critical hypoxemia is encountered. (p. 22)

- Intubate patients in cardiac arrest after other critical resuscitation steps have been assured. (p. 23)

- Intubation is indicated for unconscious, nonarrested patients unless a rapidly correctable situation is suspected (examples given: opioid overdose, simple postictal state). Consider intubation for conscious patients with refractory hypoxemia or a deteriorating clinical course. (p. 23)

- Rapid sequence intubation (RSI) should be used unless the patient’s condition makes it unnecessary (e.g., cardiac arrest) or RSI is contraindicated because of an anticipated difficult airway. RSI consists of simultaneous administration of an induction agent and a neuromuscular blocking agent to facilitate orotracheal intubation in the nonarrested/peri-arrested patient. (p. 23)

- Anticipated difficulty in mask ventilation, intubation, rescue with an extraglottic device, and surgical airway placement are relative contraindications to RSI; awake techniques should be considered in these circumstances. (p. 23)

- Multiple intubation attempts are associated with adverse events; efforts should be made to set up success on the first attempt. (p. 23)

Evaluation / decision logic (explicitly supported)

- If preparing for intubation: assess airway difficulty and anticipate required airway rescue; assemble suction, bag-valve-mask, and rescue devices within easy reach; ensure sufficient personnel and assign tasks in advance; use of a checklist is strongly encouraged. (p. 23)

- Preoxygenation: use non-rebreather at maximal flow, NIPPV, or mask ventilation if inadequate ventilation; place nasal cannula with up to 15 L/min under the mask for apneic oxygenation. Failure to achieve SpO2 > 93% with these maneuvers places the patient at risk for critical desaturation after apnea is induced — be prepared to perform controlled positive pressure mask ventilation. (p. 23)

- Optimize physiology before intubation (e.g., IV fluids, inotropes/vasopressors) where possible to reduce peri-intubation complications. (p. 23)

- Device selection logic: video laryngoscopy (VL) is a good first choice if the operator is familiar with it; direct laryngoscopy (DL) is reasonable if the operator has more experience with it. (p. 23)

- If oxygen saturation is dropping during an attempt: abort the attempt early and begin immediate mask ventilation. If unable to maintain saturations with mask ventilation, insert an EGD while preparing for a possible surgical airway. If saturations are maintained but intubation is unsuccessful within three attempts (or deemed unlikely to be successful at any point), place an EGD. (p. 26)

Management principles / stepwise procedure (explicitly supported)

- Preparation: prepare equipment, personnel, and drugs; test laryngoscope and blade; select and test endotracheal tube (commonly 7.5 mm for women and 8 mm for men); use a stylet shaped “straight-to-cuff” for DL/non-hyperangulated VL blades; hyperangulated VL blades often have proprietary stylets. (pp. 23–24)

- Positioning: place patient in sniffing or ramped position to align external ear canal and sternal notch; if C-spine injury suspected, maintain neutral head/neck with inline stabilization and jaw thrust. (p. 24)

- Pretreatment: evidence is mixed; routine pretreatment is no longer recommended. Fentanyl 3 μg/kg may be considered in normotensive patients with possible raised intracranial pressure, cardiac ischemia, or aortic dissection. (p. 24)

- Induction agents: etomidate 0.3 mg/kg is presented as an excellent choice in most circumstances; ketamine 1–2 mg/kg is an alternative (generally safe though hypotension and hypertension have been reported); propofol 0.5–1.5 mg/kg is an option in patients not at risk for hypotension. (p. 25)

- Paralysis: immediately flush induction agent with a paralytic. Succinylcholine 1–2 mg/kg total body weight commonly used unless risk of serious hyperkalemia (examples given: renal failure, neuromuscular disorders, subacute spinal cord injury, crush injury, burns). Rocuronium 1–1.5 mg/kg ideal body weight is an increasingly common alternative. (p. 25)

- Timing: wait for paralysis to occur to diminish risk of vomiting/aspiration. Succinylcholine onset usually 30–45 seconds; rocuronium ~60 seconds. Continue oxygenation (non-rebreather or gentle mask ventilation) during this interval. (p. 25)

- Laryngoscope technique: insert hyperangulated VL blades midline; insert traditional curved blades (direct or video) on the right side of the mouth and sweep the tongue left; advance blades into the vallecula to engage the hyoepiglottic ligament. Do not over-insert hyperangulated blades; keep the blade shallow and airway visualized in top half of screen. For straight blades, use paraglossal approach without sweeping the tongue; advance until epiglottis is seen, then lift with blade tip. Lift blades along the axis of the laryngoscope handle to avoid levering on teeth. (p. 25)

- If only epiglottis visible: use an intubating stylet (bougie) and/or external laryngeal manipulation (operator's right hand on assistant's hand applying laryngeal pressure) to improve view. (p. 25–26; fig. 1-4)

- Tube passage and depth: once vocal cords/posterior cartilages visualized, pass tube until balloon disappears, then remove stylet. With hyperangulated blade stylet, withdraw stylet 2–3 cm once tip of tube enters airway before advancing. Advance tube to ~21 cm at mouth corner in adult females and ~23 cm in adult males, then remove stylet. (p. 26)

- Confirmation: confirm tracheal tube placement immediately with end-tidal CO2 (ETCO2). Confirm appropriate depth by auscultation of bilateral lung sounds and secure tube. Obtain portable chest x-ray to further evaluate tube depth and lung pathology; chest x-ray should never be used to assess tracheal versus esophageal positioning. (p. 26)

Surgical airway (bougie-aided cricothyroidotomy) — indications, contraindication, and steps (explicitly supported)

- Indication: perform surgical airway when oral or nasal intubation is not a reasonable clinical option or when intubation has failed and critical oxygen saturation cannot be maintained via other means. (p. 26)

- Contraindication: surgical airway is contraindicated in children younger than 10 years of age; transtracheal jet ventilation is preferred subglottic technique in that group. (p. 26)

- Bougie-aided technique steps (as described):

1. Use sterile technique if possible. (p. 27)

2. Palpate cricothyroid membrane and stabilize larynx. (p. 27)

3. With scalpel, make a vertical 3–4 cm incision starting at superior border of thyroid cartilage, incise caudally toward suprasternal notch. (p. 27; fig. 1-5)

4. Identify cricothyroid membrane (use blunt dissection if necessary) and make a 2-cm horizontal incision; immediately withdraw and secure blade while inserting a gloved finger into incision. (p. 27)

5. Place adult bougie into incision with coude tip directed distally; bougie should pass easily until hold-up in smaller airways, generally confirming tracheal placement. (p. 27)

6. Pass a 6.0-cuffed endotracheal tube or #4 cuffed tracheostomy tube over bougie into airway; if using an ETT, stop advancing as soon as cuff is completely within the airway; inflate cuff. (p. 27; fig. 1-6)

7. Confirm with capnography and easy chest rise with bilateral breath sounds. (p. 27)

8. Secure the tube. (p. 27)

High-yield / red flags / exam points / common mistakes (only those explicitly supported)

- Multiple intubation attempts are associated with adverse events — aim for first-pass success. (p. 23)

- Cricoid pressure is no longer recommended; it has limited evidence of benefit and clearly worsens laryngoscopic view. (p. 25)

- If preoxygenation maneuvers cannot achieve SpO2 > 93%, patient is at risk for critical desaturation after apnea — be prepared to perform controlled positive pressure mask ventilation. (p. 23)

- Confirm placement immediately with ETCO2; do not rely on chest x-ray to distinguish tracheal from esophageal tube placement. (p. 26)

- Inability to maintain oxygenation with mask ventilation during failed intubation requires insertion of an EGD and preparation for surgical airway. (p. 26)

- Surgical airway is contraindicated in children <10 years — prefer transtracheal jet ventilation in that population. (p. 26)

Figures and maneuvers (as shown/described in excerpt)

- Mask ventilation grips: traditional “E-C” and modified “T-E” hand grips (figure referenced). (p. 22, fig. 1-1)

- Stylet shapes: hyperangulated VL proprietary stylets and straight-to-cuff stylet for DL/non-hyperangulated VL (figure referenced). (p. 24, fig. 1-2)

- Sniffing position/ramped position image referenced for positioning when cervical precautions not indicated. (p. 24, fig. 1-3)

- External laryngeal manipulation technique: intubator’s right hand on top of assistant’s hand holding laryngeal cartilage; assistant provides inline stabilization and jaw thrust (figure referenced). (p. 26, fig. 1-4)

- Cricothyroid incision and bougie-tube passage images referenced. (pp. 27–28, figs. 1-5, 1-6)

Unclear or incomplete items (explicitly labeled)

- The excerpt shows figures for mask grip and stylet shapes but does not provide full textual descriptions of technique variations or indications beyond figure captions. [UNCLEAR — figure details, p. 22 and p. 24]

- The exact recommended blade selection algorithm for all difficult-airway scenarios (beyond “VL if familiar; DL if more experienced”) is not fully specified in the excerpt. [UNCLEAR — p. 23]

- Any local institutional preferences, detailed complication rates, or comparative effectiveness data for specific drugs/devices are not provided in the excerpt. [UNCLEAR — pp. 23–26]

Evidence ledger — mapping substantive statements to source page(s)

- Always have an appropriately sized EGD available; do not rely on EGD to exclusion of surgical airway when critical hypoxemia encountered. (p. 22)

- Mask ventilation grips (E-C and T-E) illustrated. (p. 22)

- Intubate patients in cardiac arrest after other critical resuscitation steps assured. (p. 23)

- Indications for intubation: unconscious nonarrested unless rapidly correctable (opioid overdose, postictal); consider for conscious with refractory hypoxemia or deteriorating course. (p. 23)

- RSI definition and use: simultaneous induction and neuromuscular blocking agent for nonarrested/peri-arrested; do not use when unnecessary (cardiac arrest) or contraindicated with anticipated difficult airway. (p. 23)

- Anticipated difficulty in mask ventilation, intubation, rescue with EGD, and surgical airway placement are relative contraindications to RSI; consider awake techniques. (p. 23)

- Multiple intubation attempts associated with adverse events; aim for first-pass success. (p. 23)

- Preparation steps (equipment, personnel, suction, BVM, rescue devices, assign tasks, checklist encouraged). (p. 23)

- Preoxygenation methods (NRB at max flow, NIPPV, mask ventilation); nasal cannula up to 15 L/min under mask for apneic oxygenation. (p. 23)

- SpO2 > 93% threshold and implications (risk for critical desaturation and need for controlled positive-pressure mask ventilation). (p. 23)

- Optimize physiology prior to intubation (IV fluids, inotropes, vasopressors). (p. 23)

- VL preferred if operator familiar; DL reasonable if more experienced. (p. 23)

- ETT sizes commonly 7.5 mm women, 8 mm men. (p. 23)

- Stylet recommendations: straight-to-cuff for DL/non-hyperangulated VL; hyperangulated blades often have proprietary stylets. (p. 23–24)

- Positioning: sniffing/ramped to align ear canal and sternal notch; neutral position and inline stabilization/jaw thrust if C-spine injury suspected. (p. 24)

- Pretreatment: evidence mixed; not routinely recommended; fentanyl 3 μg/kg may be considered in normotensive patients with possible raised ICP, cardiac ischemia, or aortic dissection. (p. 24)

- Induction agents and doses (etomidate 0.3 mg/kg; ketamine 1–2 mg/kg; propofol 0.5–1.5 mg/kg) and general remarks. (p. 25)

- Paralytic agents and doses (succinylcholine 1–2 mg/kg total body weight; rocuronium 1–1.5 mg/kg ideal body weight) and contraindications for succinylcholine (risk of serious hyperkalemia: renal failure, neuromuscular disorders, subacute spinal cord injury, crush injury, burns). (p. 25)

- Cricoid pressure no longer recommended; limited benefit and worsens laryngoscopic view. (p. 25)

- Wait for paralysis timing (succinylcholine 30–45s; rocuronium ~60s); continue oxygenation during interval. (p. 25)

- Laryngoscope insertion details for hyperangulated, curved, and straight blades; vallecula/hyoepiglottic ligament engagement and technique to avoid dental trauma. (p. 25)

- Use of bougie and external laryngeal manipulation when only epiglottis visible (and figure). (pp. 25–26)

- Tube passage technique including stylet withdrawal with hyperangulated blades; recommended depths (~21 cm female, ~23 cm male). (p. 26)

- Immediate confirmation with ETCO2; auscultation for bilateral breath sounds; secure tube; portable chest x-ray to evaluate depth/lung pathology but not to assess tracheal vs esophageal positioning. (p. 26)

- Abort attempt early if SpO2 dropping and begin immediate mask ventilation; options thereafter (repeat with modifications, insert EGD, prepare for surgical airway depending on ability to maintain oxygenation and number of attempts). (p. 26)

- Indications for surgical airway (failed/unreasonable oral/nasal intubation and inability to maintain critical SpO2 with other means). (p. 26)

- Contraindication to surgical airway in children <10 years; recommend transtracheal jet ventilation in that group. (p. 26)

- Bougie-aided cricothyroidotomy steps (sterile technique; palpate cricothyroid membrane; vertical 3–4 cm incision from superior border of thyroid cartilage caudally toward suprasternal notch; identify membrane and make 2-cm horizontal incision; finger into incision; place adult bougie with coude tip distally until hold-up; pass 6.0 cuffed ETT or #4 cuffed tracheostomy tube over bougie; stop advancing when cuff in airway; inflate cuff; confirm with capnography and bilateral breath sounds; secure tube). (pp. 27–28; figs. 1-5, 1-6)

End of scaffold.

(Note: this scaffold contains only statements and relationships explicitly present in the supplied excerpt. Items identified as [UNCLEAR] indicate where the excerpt is incomplete or lacks detail.)

Chapter 2: Management of Cardiac Rhythm Disturbances

Source pages 30–49 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 30–49

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Management of Cardiac Rhythm Disturbances**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 2: Management of Cardiac Rhythm Disturbances 11 Figure 2-2. [Source pages 30–49]

- absence of significant heart disease, but can precipitate sustained atrial tachycardia, flutter, or fibrillation under certain circumstances. [Source pages 30–49]

- Discontinue precipitating drugs (alcohol, tobacco, or coffee) or toxins. [Source pages 30–49]

- Treat underlying disorders (stress or fatigue). [Source pages 30–49]

- Premature Ventricular Contractions Clinical Features Premature ventricular contractions (PVCs) are due to impulses originating from single or multiple areas in the ventricles. [Source pages 30–49]

Clinical reasoning structure

- **Problem representation:** Management of Cardiac Rhythm Disturbances; source pages 30–49.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Management of Cardiac Rhythm Disturbances**.

- Source-page range: **30–49**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 30

CHAPTER 2: Management of Cardiac Rhythm Disturbances 11

Figure 2-2. Premature atrial contractions (PACs). A. Ectopic P′ waves (arrows).

B. Atrial bigeminy.

absence of significant heart disease, but can precipitate sustained atrial tachycardia, flutter, or fibrillation under certain circumstances.

Emergency Department Care and Disposition

1. Discontinue precipitating drugs (alcohol, tobacco, or coffee) or toxins.

2. Treat underlying disorders (stress or fatigue).

Premature Ventricular Contractions

Clinical Features

Premature ventricular contractions (PVCs) are due to impulses originating from single or multiple areas in the ventricles. The ECG characteristics of PVCs are as follows: (a) a premature and wide QRS complex; (b) no preceding P wave; (c) the ST segment and T wave of the PVC are directed opposite the preceding major QRS deflection; (d) most PVCs do not affect the sinus node, so there is usually a fully compensatory postectopic pause, or the PVC may be interpolated between two sinus beats; (e) many PVCs have a fixed coupling interval (within 40 milliseconds) from the preceding sinus beat; and (f) many PVCs are conducted into the atria, thus producing a retrograde P wave (Fig. 2-3). If three or more PVCs occur in a row, patients are considered to have nonsustained ventricular tachycardia. PVCs are very common, occurring in most patients with ischemic heart disease and acute myocardial infarction (MI). Other common causes of PVCs include digoxin toxicity, congestive heart failure (CHF), hypokalemia, alkalosis, hypoxia, and sympathomimetic drugs. Pooled data and meta-analyses have found no reduction in mortality from suppressive or prophylactic treatment of PVCs. Ventricular parasystole occurs when the ectopic ventricular focus fires frequently enough to compete with the SA node and is associated with cardiac ischemia, electrolyte imbalance, and

hypertensive or ischemic heart disease.

02_Cydulka_Ch02.indd 11 20/04/17 2:34 pm

Source page 31

12 SECTION 1: Resuscitation Techniques

Figure 2-3. Premature ventricular contractions (PVCs). A. Unifocal PVC. B. Interpolated PVC. C. Multifocal PVCs.

Emergency Department Care and Disposition

1. Stable patients require no treatment.

2. Patients with three or more sequential PVCs should be managed as non-sustained VT.

3. Potential causes such as hypoxia, drug effect, or electrolyte disturbances should be treated.

■■ BRADYDYSRHYTHMIAS

Sinus Bradycardia

Clinical Features

Sinus bradycardia occurs when the SA node rate becomes slower than 60 beats/min. The ECG characteristics of sinus bradycardia are (a) normal sinus P waves and PR intervals, (b) 1:1 AV conduction, and (c) atrial rate slower than 60 beats/min. Sinus bradycardia represents a suppression of the sinus node discharge rate, usually in response to three categories of stimuli: (a) physiologic (vagal tone), (b) pharmacologic (calcium channel blockers, β-blockers, or digoxin), and (c) pathologic (acute inferior myocardial infarction (MI), increased intracranial pressure, carotid sinus hypersensitivity, hypothyroidism, or sick sinus syndrome).

02_Cydulka_Ch02.indd 12 20/04/17 2:34 pm

Source page 32

CHAPTER 2: Management of Cardiac Rhythm Disturbances 13

Emergency Department Care and Disposition

Sinus bradycardia usually does not require specific treatment unless the heart rate is slower than 50 beats/min and there is evidence of hypoperfusion.

1. Transcutaneous cardiac pacing is the only Class I treatment for unstable patients.

a. Attach the patient to the monitor leads of the external pacing device.

b. When placing transcutaneous pacing pads, place the anterior pad over the left lateral precordium and the posterior pad at the level of the heart in the right infrascapular area. Do not use multifunction pacing defibrillation pads unless the patient is unconscious as the pads cause a lot of discomfort.

c. Slowly increase the pacing output from 0 mA to the lowest point

where capture is observed, usually at 50 to 100 mA, but may be up to 200 mA. A widened QRS after each pacing spike denotes electrical capture.

d. If needed, administer a sedative, such as lorazepam, 1 to 2 mg IV, or an opiate, such as morphine, 2 to 4 mg IV, for pain control.

2. Atropine is a Class IIa treatment for symptomatic bradycardia. The dose is 0.5 mg IV push, repeated every 3 to 5 minutes as needed up to a total of 3 mg IV. If given via endotracheal tube, increase the dose by 2 to 2.5 times over the IV dose. Slow administration or lower doses may cause paradoxical bradycardia. Atropine may not be effective in cardiac transplant patients since the heart is denervated and has no vagal stimulation.

3. Epinephrine, 2 to 10 μg/min IV, or dopamine, 3 to 10 μg/kg/min IV, may be used if external pacing is not available.

4. Permanent pacemaker placement is indicated in the patient with symptomatic recurrent or persistent sinus bradycardia due to sick sinus syndrome.

5. Glucagon 3 to 10 mg IV over 1 to 2 minutes, followed by an infusion of 1 to 5 mg/h may be used in β-blocker or calcium channel blocker toxicity.

Junctional Rhythms

Clinical Features

In patients with sinus bradycardia, SA node exit block, or AV block, junctional escape beats may occur, usually at a rate between 40 and 60 beats/min, depending on the level of the rescue pacemaker within the conduction system. Junctional escape beats may conduct retrogradely into the atria, but the QRS complex usually will mask any retrograde P wave (Fig. 2-4). When alternating rhythmically with the SA node, junctional escape beats

Figure 2-4. Junctional escape rhythm, rate 42.

02_Cydulka_Ch02.indd 13 20/04/17 2:34 pm

Source page 33

may cause bigeminal or trigeminal rhythms. Sustained junctional escape rhythms may be seen with CHF, myocarditis, acute MI (especially inferior MI), hyperkalemia, or digoxin toxicity (“regularized Afib”). If the ventricular rate is too slow, myocardial or cerebral ischemia may develop. In cases of enhanced junctional automaticity, junctional rhythms may be accelerated (60 to 100 beats/min) or tachycardic (≥100 beats/min), thus overriding the SA node rate.

Emergency Department Care and Disposition

1. Isolated, infrequent junctional escape beats usually do not require spe-

cific treatment.

2. If sustained junctional escape rhythms are producing symptoms, treat the underlying cause.

3. In unstable patients, give atropine 0.5 mg IV every 5 minutes to a total of 3 mg. This will accelerate the SA node discharge rate and enhance AV nodal conduction.

4. Use transcutaneous or transvenous pacing in unstable patients not responsive to atropine.

5. Manage patients with digoxin toxicity as discussed for SVT.

Idioventricular Rhythm

Clinical Features

The ECG characteristics of idioventricular rhythm (IVR) are (a) wide and regular QRS complexes; (b) a rate between 40 and 100 beats/min, often close to the preceding sinus rate; (c) mostly runs of short duration (3 to 30 beats/min); and (d) an AIVR often beginning with a fusion beat (Fig. 2-5). This condition is found most commonly with an acute MI or in the setting of reperfusion after successful thrombolysis.

Emergency Department Care and Disposition

Treatment is not necessary unless the patient is unstable or pulseless. On occasion, especially after reperfusion therapy, the IVR may be the only functioning pacemaker, and suppression with lidocaine can lead to cardiac asystole.

If the patient is hypotensive or in arrest, treatment includes identifying contributing mechanical factors (e.g., aggressive volume resuscitation) and α-adrenergic agents.

Figure 2-5. Accelerated idioventricular rhythms (AIVRs).

02_Cydulka_Ch02.indd 14 20/04/17 2:34 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 30–49.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 3: Resuscitation of Children and Neonates

Source pages 50–57 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 50–57

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Resuscitation of Children and Neonates**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 3: Resuscitation of Children and Neonates 31 Table 3-2 Common Rapid-Sequence Intubation Medications in Children* Medication Dose† Comments Induction agents Etomidate 0.3 mg/kg Preserves hemodynamic stability; may suppress adrenal axis even in a single dose; short acting, requires anxiolysis or analgesia after intubation Ketamine 1–2 mg/kg Bronchodilator, preserves respiratory drive, cardiovascular stimulant; drug of choice for intubation for asthma and sepsis Propofol 1–2 mg/kg Rapid push, higher dose in infants, may cause hypotension; short acting, requires ongoing anxiolysis or analgesia after intubation Paralytics Rocuronium 1 mg/kg Nondepolarizing agent; longer duration than succinylcholine Succinylcholine <10 kg: 1.5–2.0 mg/kg Shorter duration than rocuronium; better >10 kg: 1.0–1.5 mg/kg intubating conditions at 60 s; may cause bradycardia in children and hyperkalemic cardiac arrest in children with undiag- nosed neuromuscular disease Sedatives Midazolam 0.1 mg/kg Short-acting sedative Lorazepam 0.1 mg/kg Longer-acting sedative Analgesics Fentanyl 1–2 μg/kg Short-acting analgesic; preserves hemo- dynamic stability Morphine 0.1–0.2 mg/kg Longer-acting analgesic; may cause histamine release * Premedication is no longer routinely recommended in children due to a lack of supporting evidence. [Source pages 50–57]

- † Rapid-sequence intubation medications can be given IO when IV access cannot be obtained. [Source pages 50–57]

- Cricoid pressure may not be needed because it has been associated with difficulty with intubation and bag-mask ventilation, and in children, cricoid pressure can occlude the pliable trachea. [Source pages 50–57]

- If cricoid pressure is applied, release pressure if laryngoscopy and intubation are difficult. [Source pages 50–57]

- Visualization may or may not be enhanced by the Backward-Upward- Rightward-Pressure on the thyroid cartilage (the “BURP” maneuver), moving the cords into better view. [Source pages 50–57]

Clinical reasoning structure

- **Problem representation:** Resuscitation of Children and Neonates; source pages 50–57.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Resuscitation of Children and Neonates**.

- Source-page range: **50–57**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 50

CHAPTER 3: Resuscitation of Children and Neonates 31

Table 3-2 Common Rapid-Sequence Intubation Medications in Children*

Medication Dose† Comments

Induction agents

Etomidate 0.3 mg/kg Preserves hemodynamic stability; may

suppress adrenal axis even in a single

dose; short acting, requires anxiolysis or

analgesia after intubation

Ketamine 1–2 mg/kg Bronchodilator, preserves respiratory drive,

cardiovascular stimulant; drug of choice

for intubation for asthma and sepsis

Propofol 1–2 mg/kg Rapid push, higher dose in infants, may

cause hypotension; short acting, requires

ongoing anxiolysis or analgesia after

intubation

Paralytics

Rocuronium 1 mg/kg Nondepolarizing agent; longer duration

than succinylcholine

Succinylcholine <10 kg: 1.5–2.0 mg/kg Shorter duration than rocuronium; better

>10 kg: 1.0–1.5 mg/kg intubating conditions at 60 s; may cause

bradycardia in children and hyperkalemic

cardiac arrest in children with undiag-

nosed neuromuscular disease

Sedatives

Midazolam 0.1 mg/kg Short-acting sedative

Lorazepam 0.1 mg/kg Longer-acting sedative

Analgesics

Fentanyl 1–2 μg/kg Short-acting analgesic; preserves hemo-

dynamic stability

Morphine 0.1–0.2 mg/kg Longer-acting analgesic; may cause

histamine release

*

Premedication is no longer routinely recommended in children due to a lack of supporting evidence.

†

Rapid-sequence intubation medications can be given IO when IV access cannot be obtained.

2. Preoxygenate with 100% oxygen.

3. Cricoid pressure may not be needed because it has been associated with

difficulty with intubation and bag-mask ventilation, and in children,

cricoid pressure can occlude the pliable trachea. If cricoid pressure is

applied, release pressure if laryngoscopy and intubation are difficult.

Visualization may or may not be enhanced by the Backward-Upward-

Rightward-Pressure on the thyroid cartilage (the “BURP” maneuver),

moving the cords into better view.

4. Atropine 0.02 mg/kg (minimum dose 0.1 mg; maximum dose 1 mg)

may be used if symptomatic reflex bradycardia occurs.

5. The trachea should be intubated, proper placement confirmed, and the

tube secured.

03_Cydulka_Ch03.indd 31 20/04/17 2:35 pm

Source page 51

Vascular Access

Vascular access is obtained in the quickest, least invasive manner possible;

peripheral veins (antecubital, hand, foot, or scalp) are tried first. Intraosse-

ous access is also a quick, safe, and reliable route for resuscitation medica-

tions, especially in the critically ill infant or child suffering from

hypovolemia or shock. Percutaneous access of the femoral vein or saphe-

nous vein cutdown may be attempted, but is more time consuming. Airway

management is paramount in pediatric arrest, and should not be delayed

while obtaining vascular access.

The technique for insertion of the intraosseous line is as follows: the

bone most commonly used is the proximal tibia. The anterior tibial

tuberosity is palpated with the index finger. The cannulation site is 1 to

3 cm below this tuberosity and in the middle of the anteromedial surface

of the tibia. Manual and powered devices are commonly used. If a bone marrow needle is not available, an 18-G spinal needle can be used but is prone to bending. With sterile technique, the needle is inserted in a slightly caudal direction until the needle punctures the cortex. The stylet is removed, and marrow is aspirated to confirm placement. If no marrow is aspirated but the needle is thought to be in place, flushing the needle may be attempted. Fluids or drugs (including glucose, epinephrine, dopamine, anticonvulsants, and antibiotics) may then be administered as with a standard IV.

Fluids

In hypotension or frank shock, IV isotonic fluid (i.e., normal saline solution) boluses of 20 mL/kg should be given as rapidly as possible and potentially repeated, depending on the clinical response. If hypovolemia has been corrected and shock or hypotension persist, a pressor agent should be considered.

Drugs

The indications for resuscitation drugs are the same for children as for adults; the exception is epinephrine, which is considered first-line therapy in children with bradycardia (Table 3-3).

Proper drug dosages in children require knowledge of the patient's weight. The use of a length-based system for estimating the weight of a child in an emergency situation reduces dosage errors when an exact weight is unavailable and cannot be safely obtained due to the patient's medical condition. The rule of 6s may be used to quickly calculate continuous drug infusions of drugs such as dopamine and dobutamine. The calculation is 6 mg times weight in kilograms: fill to 100 mL with 5% dextrose in water. The infusion rate in milliliters per hour equals the microgram per kilogram per minute rate (i.e., an infusion running at 1 mL/h = 1 μg/kg/min, or 5 mL/h = 5 μg/kg/min).

Epinephrine is indicated in pulseless arrest and in hypoxia-induced bradycardia unresponsive to oxygenation and ventilation. The initial dose is 0.01 mg/kg (0.1 mL/kg of 1:10,000 solution) IV/IO or 0.1 mg/kg (0.1 mL/kg) of 1:1,000 solution by endotracheal route. Subsequent doses of epinephrine are at the same dose and concentration and may be administered every 3–5 minutes.

03_Cydulka_Ch03.indd 32 20/04/17 2:35 pm

Source page 52

Table 3-3 Drugs for Pediatric Resuscitation

Drug Pediatric Dosage Remarks

Adenosine IV/IO: 0.1 mg/kg, followed by Maximum single dose: 6 mg first
2–5 mL NS bolus dose, 12 mg second dose.
Double dose and repeat once, if
needed
Amiodarone IV/IO: 5 mg/kg over 20–60 min; Maximum bolus repetition to
then 5–15 μg/kg/min infusion 15 mg/kg/d. Use lowest effective
dose.
Bolus may be given more rapidly in
shock states.
Atropine IV/IO: 0.02 mg/kg, repeat in 5 min Maximum single dose: 0.5 mg
(minimum single dose is 0.1 mg) (child) and 1 mg (adolescent).
Endotracheal: 0.04–0.06 mg/kg Maximum cumulative dose: 1 mg
diluted with NS to 3–5 mL (child) and 2 mg (adolescent).
Calcium IV/IO: 20 mg/kg Not routinely recommended. Use in
chloride (10%) (maximum dose 2 g) documented hypocalcemia, calcium
channel blocker overdose, hyper-
magnesemia, or hyperkalemia.
Administer slowly.
Epinephrine Bradycardia: Maximum dose: 1 mg IV/IO;
IV/IO: 0.01 mg/kg (0.1 mL/kg of 2.5 mg ETT.
1:10,000) Unlike other agents, epinephrine
Endotracheal: 0.1 mg/kg (0.1 mL/ per endotracheal tube is 10 × the
kg of 1:1000) IV dose. Follow endotracheal dose
Pulseless arrest: with several positive pressure
IV/IO: 0.01 mg/kg (0.1 mL/kg of ventilations.
1:10,000) Maximum dose: 1 mg IV/IO;
Endotracheal: 0.1 mg/kg (0.1 mL/ 2.5 mg ETT.
kg of 1:1000) No evidence for high-dose par-
enteral epinephrine (may worsen
outcomes).
Glucose IV/IO:
Newborn: 2 mL/kg D10W
Infants and children: 2 mL/kg D25W
Adolescents: 1 mL/kg D50W
Lidocaine IV/IO: 1 mg/kg bolus
Endotracheal: double IV dose and
dilute with NS to 3–5 mL
Naloxone IV/IO: If <5 years or ≤20 kg: 0.1 Titrate to desired effect.
mg/kg
If >5 years and >20 kg: 2.0 mg

Sodium IV/IO: 1 mEq/kg (1 mEq/mL) Not routinely recommended.
bicarbonate Infuse slowly and use only
if ventilation is adequate for
tricyclic antidepressant overdose
and hyperkalemia.
Abbreviation: D10W, 10% dextrose in water; D25W, 25% dextrose in water; D50W, 50% dextrose in water; ETT, endotracheal tube; NS, normal saline.
03_Cydulka_Ch03.indd 33 20/04/17 2:35 pm

Source page 53

34 SECTION 1: Resuscitation Techniques

Sodium bicarbonate is no longer recommended as a first-line resuscitation drug. It is recommended only with persistent acidosis after the establishment of effective ventilation, administration of epinephrine, and performance of chest compressions to ensure adequate circulation.

Calcium also is not recommended in routine resuscitation but may be useful in hyperkalemia, hypocalcemia, and calcium-channel blocker overdose. Calcium may be given as calcium chloride, 20 mg/kg (0.2 mL/kg of 10% solution) or calcium gluconate, 60–100 mg/kg (0.6–1 mL/kg of 10% solution) via IV or IO route.

Dysrhythmias

Dysrhythmias in infants and children are most often the result of respiratory insufficiency or hypoxia, not of primary cardiac causes, as in adults. Careful attention to oxygenation and ventilation therefore are cornerstones of dysrhythmia management in children.

The most common rhythm seen in pediatric arrest situations is bradycardia leading to asystole. Oxygenation and ventilation are often sufficient therapy in this situation. Epinephrine may be useful if the child is unresponsive to this respiratory intervention.

The next most common dysrhythmia in children is supraventricular tachycardia (SVT), which presents with a narrow-complex tachycardia with rates between 250 and 350 beats/min. On EKG, p waves are either absent or abnormal. Adenosine 0.1 mg/kg given via rapid IV push followed by a normal saline flush through a well-functioning IV line as close to the central circulation as possible is the recommended treatment for stable SVT in children. This dose may be doubled if the first dose is unsuccessful.

Treatment of unstable SVT is synchronized cardioversion at 0.5–1 J/kg. If not effective, increase to 2 J/kg. Sedation prior to cardioversion is recommended.

It may be difficult to distinguish between a fast sinus tachycardia and SVT. Young infants may have sinus tachycardia with rates as fast as 200–

220 beats/min. The presence of normal p waves is strongly suggestive of sinus tachycardia rather than SVT. Patients with sinus tachycardia may have a history of fever, dehydration, or shock, while SVT is usually associated with a vague, nonspecific history.

Defibrillation and Cardioversion

Ventricular fibrillation and ventricular tachycardia are rare in children. When present, immediate defibrillation at 2 J/kg is recommended. Immediately after defibrillation, give 2 minutes of high-quality uninterrupted CPR (five cycles of 15:2 compressions and ventilations) to restore coronary perfusion and improve oxygen delivery to the myocardium before additional attempts at defibrillation. If the first defibrillation attempt is unsuccessful, the energy is doubled to 4 J/kg for each subsequent attempt. Epinephrine may also be given and oxygenation and acid–base status should be reassessed. Synchronized cardioversion is used to treat unstable tachydysrhythmias at a dose of 0.5 J/kg. Double the energy level to 1 J/kg if the first attempt is unsuccessful.

The largest paddles that still allow contact of the entire paddle with the chest wall are used. Electrode cream or paste is used to prevent burns. One

03_Cydulka_Ch03.indd 34 20/04/17 2:35 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 50–57.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 4: Fluids, Electrolytes, and Acid-Base Disorders

Source pages 58–79 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition \n**Source pages:** 58–79 \n**Content ID:** SEG-TINTINALLI-EMM8-FULL \n**Review status:** MACHINE-AUTHORED; MEDICAL REVIEW REQUIRED

Chapter scaffold — Chapter 4 excerpt (pages 58–79)

Note: All content below is drawn only from the supplied excerpt. Statements that are not fully supported by the excerpt are labeled [UNCLEAR] with page reference.

Learning objectives

- Classify hyponatremia and hypernatremia by volume status and identify common causes (pp. 58, 60–61).

- Recognize diagnostic criteria for SIADH and important hyponatremic syndromes (p. 58).

- Apply ED decision logic for acute symptomatic hyponatremia, including indications and steps for 3% hypertonic saline (pp. 58–59).

- Calculate expected change in serum sodium and free-water deficit using formulas supplied (p. 59, 61), and understand TBW estimates used in those calculations (p. 59).

- Summarize ED management principles for hypernatremia, hypokalemia, and hyperkalemia as presented (pp. 60–63).

- Appreciate limits and risks of rapid sodium correction (osmotic demyelination) and key supportive measures for respiratory acid–base disturbances and shock as presented (pp. 59, 63, 79).

Core concepts (with source)

- Hyponatremia classification by volume status (hypervolemic, normovolemic, renal and extrarenal hypovolemic) with examples and physical findings such as orthostatic hypotension and edema (Table 4-2, p. 58).

- SIADH: diagnostic criteria include hypotonic hyponatremia (Posm <275 mOsm/kg), inappropriately elevated urine osmolality (usually >200 mOsm/kg), elevated urinary Na+ (typically >20 mEq/L), clinical euvolemia, and normal adrenal, renal, cardiac, hepatic, and thyroid function (Table 4-3, p. 58).

- Two important hyponatremic disorders named: SIADH and cerebral salt-wasting syndrome; both are diagnoses of exclusion (p. 58).

- MDMA (Ecstasy) can induce inappropriate ADH secretion and profound hyponatremia (p. 58).

- Indications for 3% hypertonic saline in ED: [Na+] <120 mEq/L, severe neurologic symptoms (seizures or coma), and acute hyponatremia (Table 4-4 and text, pp. 58–59).

- Hypertonic (3%) saline contains 513 mEq/L Na+; isotonic (0.9%) saline contains 154 mEq/L Na+ (p. 59).

- Expected change in serum Na+ formula: (Infusate Na+ – Serum Na+) / (TBW + 1) (p. 59).

- TBW estimates by age/sex: adult males <65 y = 60% body weight; adult females <65 y and elderly males = 50%; elderly females = 45%; (children and other groups noted) (p. 59).

- Limits for correction of chronic hyponatremia: ≤6 mEq/24 h in high-risk patients; ≤12 mEq/24 h in low-risk patients. Rapid correction (>12 mEq/L/24 h) risks osmotic demyelination syndrome (p. 59).

- Hypernatremia classification by volume status (hypervolemic, normovolemic, hypovolemic) with clinical features (Table 4-5, pp. 60–61).

- Free water deficit (FWD) formula: FWD = TBW × (Posm – 285) / Posm; Posm = 2 × [Na+] + glucose/18 (p. 61).

- Correction rates: acute hypernatremia (<48 h) — correct at 1 mEq/L/h; chronic hypernatremia — correct no more than 0.5 mEq/L/h or 10–12 mEq/24 h (p. 61).

- ED resuscitation for shock and its classification into four categories: hypovolemic, cardiogenic, distributive, and obstructive (p. 79).

- Respiratory acidosis: improve alveolar ventilation; consider ventilatory assistance (intubation or noninvasive support) if inadequate response, particularly if pH <7.25; treat underlying disorder (p. 63).

- Respiratory alkalosis: identify/treat underlying cause; rule out life-threatening causes before diagnosing anxiety; avoid paper-bag rebreathing (p. 63).

Evaluation / decision logic (explicitly supported)

- Hyponatremia by volume status: use clinical volume assessment (orthostatic hypotension, edema) and clinical conditions in Table 4-2 to narrow differential (p. 58).

- SIADH diagnosis requires hypotonic hyponatremia with Posm <275 mOsm/kg, inappropriately concentrated urine (>200 mOsm/kg), urinary Na+ typically >20 mEq/L, clinical euvolemia, and normal endocrine and organ function (Table 4-3, p. 58).

- Indication algorithm for 3% hypertonic saline in ED (from Table 4-4, pp. 58–59):

- Step 1: Confirm severe symptoms (seizures/coma) with suspected impending herniation in acute or chronic hyponatremia.

- Step 2: Infuse 100 mL 3% hypertonic saline IV over 10–15 min.

- Step 3: Measure serum Na after each infusion.

- Step 4: Stop infusion when symptoms improve or target increase of ~5 mEq/L (range 4–6 mEq/L) achieved.

- Step 5: May repeat 100 mL up to 3 total doses (total 300 mL).

- Step 6: Keep IV open with minimal 0.9% saline and limit overall increase to ≤8 mEq/L during first 24 h (pp. 58–59).

- For chronic hyponatremia, apply the specified 24-h correction caps (6 vs 12 mEq/24 h) to reduce risk of osmotic demyelination (p. 59).

- For hypernatremia: treat shock/hypoperfusion with isotonic crystalloids; treat underlying causes; calculate and replace FWD using the formula and route (oral two-thirds if safe, one-third IV as D5W) (p. 61).

Management principles (explicitly supported)

- Hyponatremia:

- Use 3% hypertonic saline for severe neurologic symptoms with the stepwise dosing and monitoring described (pp. 58–59).

- After clinical improvement, stop hypertonic infusion and transition to slower/appropriate fluids (e.g., 0.9% saline) (p. 58).

- Use the expected-change formula and TBW estimates to plan therapy (p. 59).

- Limit correction in chronic hyponatremia as specified (p. 59).

- Hypernatremia:

- Resuscitate shock with isotonic (0.9%) saline or Lactated Ringer’s (p. 60).

- Correct FWD using formula; in alert patients able to drink give two-thirds orally and one-third IV as D5W (p. 61).

- Acute hypernatremia: correct at 1 mEq/L/h; chronic: ≤0.5 mEq/L/h or 10–12 mEq/24 h (p. 61).

- Hemodialysis may be used in life-threatening acute salt ingestion to replace FWD (p. 61).

- Hypokalemia:

- For stable patients with mild hypokalemia (>3.0 mEq/L) who can take PO, give oral K+ (food, salt substitutes, supplements) (p. 62).

- IV KCl: maximum recommended concentration in 500 mL saline is 40 mEq, infused over 4–6 h via peripheral line; a second peripheral line may be used for an identical solution if more aggressive correction needed (p. 62).

- Central line allows higher concentrations but infusion rates should never exceed 20 mEq/h (p. 62).

- ECG monitoring is recommended (p. 62).

- Replace Mg2+ (20–60 mEq/24 h) when hypomagnesemia coexists to optimize potassium repletion and reduce proarrhythmic risk (p. 62).

- Hyperkalemia:

- Recognize common causes listed (p. 63); specific ED therapeutic steps are not detailed in the supplied pages (see evidence ledger).

Red flags / high-yield / exam points / clinical pearls / common mistakes (supported)

- Osmotic demyelination syndrome is a risk of rapid correction of hyponatremia; avoid raising serum Na+ >12 mEq/L/24 h and use stricter limits in high-risk patients (≤6 mEq/24 h) (p. 59).

- In acute severe hyponatremia with seizures/coma, brief boluses of 3% hypertonic saline (100 mL q10–15 min up to 3 doses) can rapidly improve symptoms; measure Na+ after each bolus (Table 4-4, pp. 58–59).

- Know Na+ concentration in common fluids: 3% hypertonic saline 513 mEq/L, 0.9% saline 154 mEq/L — important when using the expected-change formula (p. 59).

- MDMA intoxication is an important cause of SIADH-related profound hyponatremia (p. 58).

- For hypernatremia, give two-thirds of free water orally if patient can safely drink; otherwise provide IV D5W for part of deficit (p. 61).

- ECG monitoring is recommended during IV potassium repletion (p. 62).

- Avoid paper-bag rebreathing for respiratory alkalosis because it may cause hypoxia; favor treating underlying cause and calm reassurance (p. 63).

- Be aware that blood pressure may be normal early in shock and is an insensitive detector of global tissue hypoperfusion (p. 79).

Management steps (condensed, per excerpt)

- Severe symptomatic hyponatremia with seizures/coma:

1. Indication confirmed (Table 4-4) (pp. 58–59).

2. Give 100 mL 3% hypertonic saline IV over 10–15 min.

3. Measure Na+ after each bolus; stop when symptoms improve or ~5 mEq/L rise achieved; may repeat up to 3 total doses (pp. 58–59).

4. Transition to slower fluids; keep IV open with minimal 0.9% saline; adhere to correction limits (pp. 58–59).

- Hypokalemia:

- PO replacement for mild (>3.0) if tolerating (p. 62).

- IV KCl: 40 mEq in 500 mL over 4–6 h via peripheral line; central line allows higher concentrations but infusion rate limit 20 mEq/h; ECG monitor and correct Mg2+ if low (p. 62).

- Hypernatremia:

- Treat shock with isotonic crystalloids; calculate FWD and replace (oral two-thirds if safe; IV D5W one-third) and observe rate limits depending on acuity (pp. 60–61).

- Respiratory acidosis:

- Improve alveolar ventilation; consider intubation/ventilatory assistance especially if pH <7.25; treat underlying disorder (p. 63).

- Respiratory alkalosis:

- Evaluate for CNS, hypoxic, toxic, metabolic or iatrogenic causes; treat underlying cause; avoid rebreathing; rule out life-threatening causes prior to labeling anxiety (p. 63).

Items or areas that are incomplete or ambiguous in the excerpt

- The paragraph beginning with “ventilation most frequently results from head trauma, chest trauma, lung disease, or excess sedation…” and surrounding discussion appears truncated in the supplied excerpt; details on causes and management of hypoventilation and the full context are incomplete. Marked here as [UNCLEAR] (page 63).

- [UNCLEAR] page 63: full scope of causes, diagnostic approach, and management details for hypoventilation/ventilation failure are not fully present in the excerpt.

Evidence ledger — mapping statements to specific page(s)

- Table 4-2 classification and clinical conditions/orthostatic hypotension/edema columns — p. 58 (Table 4-2).

- SIADH diagnostic criteria (Posm <275, urine osmolality >200, urinary Na+ >20, clinical euvolemia, normal organ functions) — p. 58 (Table 4-3).

- SIADH and cerebral salt-wasting are diagnoses of exclusion — p. 58 (text).

- MDMA induces inappropriate ADH secretion and profound hyponatremia — p. 58 (text).

- ED care: infuse 3% hypertonic saline when [Na+] <120 mEq/L with severe neuro symptoms and acute hyponatremia — p. 58 (Emergency Department Care and Disposition).

- After improvement stop 3% and reduce rate or switch to isotonic saline — p. 58 (Emergency Department Care and Disposition).

- Table 4-4 steps 1–6 (100 mL 3% over 10–15 min; measure Na after each infusion; stop when symptoms improve or target 5 mEq/L increase; may repeat up to three doses or total 300 mL; keep IV open with minimal 0.9% saline; limit increase ≤8 mEq/L in first 24 h) — pp. 58–59 (Table 4-4).

- Footnotes re: guideline variations and alternative 150 mL over 20 min regimens (European guidelines) — p. 59 (Table 4-4 footnotes).

- Expected change in serum Na+ formula — p. 59 (text).

- TBW estimates by age/sex and Na+ content of 3% and 0.9% saline — p. 59 (text).

- Correction limits for chronic hyponatremia (≤6 mEq/24 h in high-risk; ≤12 mEq/24 h in low-risk) and osmotic demyelination risk with >12 mEq/L/24 h — p. 59 (text).

- Hypernatremia definition (>145 mEq/L, Posm >295 mOsm/L) and clinical features — p. 60 (text).

- Hypernatremia classification by volume status and Table 4-5 entries — pp. 60–61 (Table 4-5).

- ED care for hypernatremia: treat shock with isotonic fluids; treat underlying cause; correct FWD using formula presented — p. 60–61 (Emergency Department Care and Disposition and FWD formula).

- Free water deficit formula and Posm calculation (Posm = 2 × [Na+] + glucose/18; 285 as normal plasma osmolality) and oral/IV replacement proportions (two-thirds oral, one-third IV as D5W) — p. 61 (text).

- Acute hypernatremia correction rate 1 mEq/L/h and chronic rate ≤0.5 mEq/L/h or 10–12 mEq/24 h; hemodyalysis as an alternative in life-threatening acute salt ingestion — p. 61 (text).

- Hypokalemia onset of symptoms around 2.5 mEq/L and symptoms table (Table 4-6) — p. 62 (Table 4-6 and text).

- Causes of hypokalemia (Table 4-7) — p. 62 (Table 4-7).

- ED care for hypokalemia: oral replacement for mild (>3.0) if tolerating; IV KCl recommendations (40 mEq in 500 mL over 4–6 h peripheral; second peripheral line allowable; central line for higher concentrations but rate ≤20 mEq/h); ECG monitoring recommended; Mg2+ replacement 20–60 mEq/24 h may be added — p. 62 (Emergency Department Care and Disposition and text).

- Hyperkalemia clinical features and causes table (Table 4-8) — p. 63 (Table 4-8).

- Respiratory acidosis: improve alveolar ventilation; consider intubation/noninvasive support if inadequate response, particularly if pH <7.25; treat underlying disorder — p. 63 (Emergency Department Care and Disposition).

- Respiratory alkalosis: causes (CNS, pregnancy, hypoxia, toxins, anxiety, pain, iatrogenic overventilation); management — identify/treat underlying cause, rule out life-threatening causes before diagnosing anxiety, avoid paper-bag rebreathing — p. 63 (text).

- Shock definition and classification into four categories; clinical features and limitations of vital signs in shock (BP may be normal early; hypotension not always present) — p. 79 (Chapter 5 opening paragraph and Clinical Features).

If you want, I can convert this scaffold into a slide outline, a teaching case keyed to these learning objectives, or a one-page high-yield summary for quick exam review — using only the same page-referenced material.

Chapter 5: Therapeutic Approach to The Hypotensive Patient

Source pages 80–83 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition \n**Source pages:** 80–83 \n**Content ID:** SEG-TINTINALLI-EMM8-FULL \n**Review status:** MACHINE-AUTHORED; MEDICAL REVIEW REQUIRED

Chapter scaffold — Therapeutic Approach to the Hypotensive Patient (based ONLY on source pp. 80–83)

Note: This scaffold uses only the supplied excerpt (Chapter 5, pp. 80–83). It is a concise educational outline (not a clinical guideline). Every substantive statement below is traceable to the source pages in the Evidence Ledger at the end. If the excerpt lacked necessary detail for a given decision, that item is marked [UNCLEAR] with the page cited.

Learning objectives

- Describe early physiologic changes and physical findings that may precede overt hypotension in shock. [p80]

- List common metabolic and end-organ manifestations of progressing shock. [p80]

- Identify initial diagnostic studies and monitoring modalities commonly used in shock evaluation. [p80–81]

- Apply the stepwise resuscitation priorities (Airway, Breathing, Circulation) and key management principles explicitly described in the excerpt. [p81–82]

- Recognize specific fluid and blood product guidance and when vasopressors are used according to the excerpt. [p81–82]

- Recall goal-directed resuscitation targets cited for ED resuscitation of septic shock. [p82]

Core concepts (explicit in source)

- Pulse pressure can be a sensitive early indicator: it often rises early in shock and then decreases before systolic BP falls. [p80]

- Physical exam may show neck vein distention or flattening, cardiac dysrhythmias, and S3 in high-output states. Decreased coronary perfusion can cause myocardial ischemia, decreased ventricular compliance, raised LV diastolic pressures, and pulmonary edema. [p80]

- Cerebral hypoperfusion produces progressive mental-status changes (weakness → confusion → delirium → syncope → coma); patients with chronic hypertension may show these changes without severe hypotension. [p80]

- Cutaneous signs include pallor/dusky skin, diaphoresis, bruising, petechiae, cyanosis (may be masked if Hb <5 g/dL), temperature changes, and delayed capillary refill. [p80]

- GI, renal, and endocrine/metabolic consequences of low-flow states include ileus, GI bleeding, pancreatitis, acalculous cholecystitis, mesenteric ischemia; oliguria from aldosterone/ADH-mediated renal effects; paradoxical polyuria in sepsis; early respiratory alkalosis progressing to lactic metabolic acidosis and other metabolic abnormalities (hyper/hypoglycemia, hyperkalemia). [p80]

- No single laboratory value is sensitive/specific for shock; diagnostic testing and monitoring must be individualized to clinical presentation. Common labs and imaging used are listed in the excerpt. [p80]

Evaluation / decision logic (as stated)

- Continuous monitoring of vital signs is required for all patients; advanced modalities (pulse oximetry, end-tidal CO2, CVP, ScvO2, cardiac output, SVR, systemic oxygen delivery) may be indicated. [p81]

- A focused search for the etiology of shock must be undertaken; lack of response to stabilization should prompt reassessment for occult causes and confirmation that basic resuscitation steps were performed appropriately. [p81]

- Before assuming refractory shock, ensure: adequate volume resuscitation (early vasopressors can elevate CVP and mask hypovolemia), equipment functioning, full exposure and exam for occult wounds, and consideration of less common causes (cardiac tamponade, tension pneumothorax, adrenal insufficiency, toxic/allergic reactions, occult bleeding such as ruptured ectopic or intraabdominal/pelvic hemorrhage). [p81]

Management principles (explicit)

- Overall goal: restore adequate tissue perfusion while identifying and treating underlying etiology. [p81]

Airway

- Aggressive airway control (endotracheal intubation) is indicated; note that sedative medications can worsen hypotension and positive-pressure ventilation may reduce preload and cardiac output, contributing to collapse. [p81]

Breathing

- All patients should receive supplemental high-flow oxygen. With mechanical ventilation, neuromuscular blockade should be used to decrease lactic acidosis from muscle fatigue/increased O2 consumption. Target arterial oxygen saturation >93% and PaCO2 35–40 mm Hg. [p81]

Circulation — vascular access and positioning

- Begin with large-bore peripheral IV access; central venous access is helpful for assessing preload and monitoring ScvO2 and is preferred for long-term vasopressor administration. Ultrasound guidance is useful for these procedures. [p81]

- Trendelenburg position does not improve cardiopulmonary performance and may worsen gas exchange and aspiration risk; passive leg raising above the heart in the supine patient can be effective. [p81]

Circulation — fluids

- The type, amount, and rate of fluid resuscitation are controversial. There is no difference in survival comparing crystalloid with colloid; isotonic crystalloids (0.9% NaCl, Ringer lactate) are most commonly used. [p81–82]

- Standard initial fluid therapy for the hemodynamically unstable patient: 20–40 mL/kg given rapidly (over 10–20 minutes). [p82]

- Because ~30% of infused isotonic crystalloids remain intravascular, approximately three times the estimated blood loss is recommended in acute hemorrhagic shock. [p82]

- Benefits of early/aggressive fluid replacement in trauma and benefits of permissive hypotension remain unproven per the excerpt. [p82]

Circulation — blood products

- Blood is the ideal resuscitative fluid; use fully cross-matched PRBCs when possible. If rapid transfusion needed, type-specific or type O may be used; use Rh-negative O blood for females of childbearing potential. [p82]

- Platelets: generally given if ongoing hemorrhage with platelet count ≤50,000/mm3; administer 6 units initially. [p82]

- FFP: indicated if prothrombin time is prolonged beyond 1.5 seconds; administer 2 units initially. [p82]

- Trauma patients requiring multiple RBC units should receive FFP and platelets early in ratios approaching 1:1:1 to address trauma-induced coagulopathy; fresh whole blood has been advocated and may be very effective. [p82]

Circulation — vasopressors

- Vasopressors are used after appropriate volume resuscitation if hypotension persists. Possible choices and dosing ranges in the excerpt: dobutamine 2.0–20.0 μg/kg/min; dopamine 5.0–20.0 μg/kg/min; norepinephrine 0.5–30.0 μg/min; epinephrine 2.0–10.0 μg/min. [p82]

Goal-directed resuscitation targets (excerpt-cited, for ED resuscitation of septic shock)

- Urine output >0.5 mL/kg/h; CVP 8–12 mm Hg; MAP 65–90 mm Hg; ScvO2 >70% — a goal-directed approach using these targets during ED resuscitation of septic shock significantly decreases mortality (as stated). [p82]

Acidosis

- Treat acidosis with adequate ventilation and fluid resuscitation. Use of sodium bicarbonate 1 mEq/kg is controversial; use only for severe acidosis refractory to ventilation/fluids and correct only to arterial pH 7.25. [p82]

Other

- Early surgical or medical consultation for admission/transfer is indicated. [p82]

Red flags / high-yield points / common pitfalls (explicit)

- Pulse pressure may change before systolic BP falls — early cue to evolving shock. [p80]

- Early use of vasopressors can raise CVP and mask persistent hypovolemia — reassess volume status before assuming vasopressor-only management. [p81]

- Trendelenburg positioning is not beneficial and may be harmful (worse gas exchange, aspiration risk). [p81]

- In sepsis, paradoxical polyuria can be mistaken for adequate hydration — monitor other perfusion indicators. [p80]

- No single lab confirms shock; use serial clinical assessment and appropriate targeted testing. [p80]

- When transfusion is needed urgently in women of childbearing potential, Rh-negative O blood is recommended if type-specific not available. [p82]

Management steps (concise, as presented in source)

1. Continuous vital-sign monitoring and establish oxygenation/airway (intubate if indicated). [p81]

2. High-flow oxygen; if ventilated, use neuromuscular blockade as indicated and target SpO2 >93%, PaCO2 35–40 mm Hg. [p81]

3. Obtain large-bore peripheral IVs; consider central venous access for monitoring and vasopressors (use US guidance). [p81]

4. Rapid crystalloid bolus (20–40 mL/kg over 10–20 min) as initial resuscitation; remember about 30% remains intravascular and adjust for hemorrhage (≈3× estimated blood loss). [p82]

5. If ongoing hemorrhage or coagulopathy: transfuse PRBCs (preferably cross-matched), consider platelets (if ≤50,000/mm3) and FFP (if PT >1.5× control) and early balanced transfusion ratios in massive hemorrhage (approach 1:1:1). [p82]

6. If hypotension persists after adequate volume resuscitation, initiate vasopressors (doses listed in excerpt). [p82]

7. Apply goal-directed targets (urine output, CVP, MAP, ScvO2) during ED septic shock resuscitation. [p82]

8. Treat severe acidosis refractory to ventilation/fluids with sodium bicarbonate cautiously (1 mEq/kg, correct only to pH 7.25). [p82]

9. Reassess for occult causes and consult surgery/medicine early for admission/transfer. [p81–82]

Items the excerpt does NOT provide in sufficient detail [UNCLEAR]

- Specific algorithms for selection among vasopressors for different shock etiologies (e.g., septic vs cardiogenic) — regimen choice rationale beyond listed agents/doses is not detailed in the excerpt. [p82] — [UNCLEAR: p82]

- Exact triggers and stepwise increments for vasopressor titration or weaning protocols are not provided. [p82] — [UNCLEAR: p82]

- Detailed transfusion thresholds for red cells (hemoglobin targets) are not stated. [p82] — [UNCLEAR: p82]

- Management algorithms for specific shock subtypes (e.g., cardiogenic, obstructive) beyond general causes to consider are deferred to other chapters. [p81] — [UNCLEAR: p81]

Evidence ledger — mapping of substantive statements to source pages

(Note: each ledger item corresponds to the location in the excerpt where that statement is made.)

- Pulse pressure behavior as early indicator (rises early then decreases before systolic BP change): p80.

- Neck vein distention/flattening and cardiac dysrhythmias as exam findings in shock: p80.

- Presence of third heart sound (S3) in high-output states: p80.

- Decreased coronary perfusion → myocardial ischemia, decreased ventricular compliance, increased LV diastolic pressures, pulmonary edema: p80.

- Cerebral hypoperfusion produces weakness, restlessness, confusion, disorientation, delirium, syncope, coma; patients with long-standing HTN may show these without severe hypotension: p80.

- Cutaneous manifestations (pallor, dusky skin, sweating, bruising, petechiae, cyanosis may be absent if Hb <5 g/dL, altered temperature, delayed capillary refill): p80.

- GI manifestations (ileus, GI bleeding, pancreatitis, acalculous cholecystitis, mesenteric ischemia): p80.

- Aldosterone/ADH increase → reduced GFR, redistribution of flow renal cortex→medulla, oliguria; paradoxical polyuria in sepsis may be mistaken for adequate hydration: p80.

- Early respiratory alkalosis progressing to lactic metabolic acidosis; other metabolic abnormalities (hyperglycemia, hypoglycemia, hyperkalemia): p80.

- Common laboratory studies to perform (CBC, platelet count, electrolytes, BUN, creatinine, PT/PTT, urinalysis) and additional tests (ABG, lactic acid, fibrinogen, fibrin split products, d-dimer, cortisol, hepatic panel, CSF, cultures, pregnancy test): p80.

- No single laboratory value is sensitive or specific for shock: p80.

- Common diagnostic tests include radiographs (chest, abdomen), ECG, CT scans (chest, head, abdomen, pelvis), echocardiograms; bedside US may help with listed views (subcostal cardiac, parasternal long-axis, apical four-chamber cardiac, IVC, RUQ abdominal, pelvic, abdominal aorta): p80.

- Continuous monitoring of vital signs should be instituted in all patients; modalities including pulse oximetry, end-tidal CO2, CVP, ScvO2, cardiac output, SVR, systemic oxygen delivery may be indicated: p81.

- Search for etiology must be undertaken; lack of response should prompt evaluation for occult causes and ensure basic resuscitation steps completed: p81.

- Consider adequacy of volume resuscitation; early vasopressors may elevate CVP and mask ongoing hypovolemia: p81.

- Ensure equipment is connected/functioning; fully expose and examine for occult wounds: p81.

- Consider less common diagnoses in nonresponding patients: cardiac tamponade, tension pneumothorax, adrenal insufficiency, toxic/allergic reactions, occult bleeding (ruptured ectopic pregnancy, intraabdominal/pelvic bleeding): p81.

- Overall goal: restore adequate tissue perfusion while identifying/treating underlying etiology: p81.

- Airway: aggressive airway control (endotracheal intubation) indicated; sedatives and positive-pressure ventilation may reduce preload/cardiac output and worsen hypotension: p81.

- Breathing: supplemental high-flow oxygen for all; if mechanically ventilated, use neuromuscular blockade to decrease lactic acidosis from muscle fatigue/increased O2 consumption; target SpO2 >93% and PaCO2 35–40 mm Hg: p81.

- Circulation: begin with large-bore peripheral IVs; central venous access helps assess volume status and ScvO2; US guidance helpful; central access preferred for prolonged vasopressor therapy: p81.

- Trendelenburg position does not improve cardiopulmonary performance and may worsen pulmonary gas exchange and aspiration risk; passive leg raising above the heart with patient supine can be effective: p81.

- Early surgical consultation indicated for internal bleeding; most external hemorrhage controlled by direct compression; vessel clamping/tying rarely needed: p81.

- Type/amount/rate of fluid replacement controversial; no survival difference crystalloid vs colloid; crystalloids recommended because of colloid cost: p81–82.

- Most practitioners use isotonic crystalloid fluids (0.9% NaCl, Ringer lactate) initially: p82.

- Routine use of colloids (5% albumin, purified protein fraction, FFP, hydroxyethyl starch, dextran 70) is questionable due to cost, lack of benefit, potential disease transmission (with FFP): p82.

- Standard therapy in hemodynamically unstable patient: 20–40 mL/kg given rapidly (over 10–20 minutes): p82.

- ~30% of infused isotonic crystalloids remain intravascular; recommend infuse ~3× estimated blood loss in acute hemorrhagic shock: p82.

- Benefits of early/aggressive fluid replacement in trauma and benefits of permissive hypotension remain unproven: p82.

- Blood is ideal resuscitative fluid; use fully cross-matched PRBCs when possible; if rapid needed, type-specific or type O (Rh negative for females of childbearing years) may be used: p82.

- Platelets: generally given if ongoing hemorrhage and platelet count ≤50,000/mm3; administer 6 units initially: p82.

- FFP: indicated if prothrombin time prolonged beyond 1.5 seconds; administer 2 units initially: p82.

- Trauma patients requiring multiple RBC units should receive FFP and platelets early in ratios approaching 1:1:1 to address trauma-induced coagulopathy; fresh whole blood has been advocated and may be most effective: p82.

- Consider potential need for FFP/platelet transfusion early and reassess frequently to detect/limit trauma-induced coagulopathy adverse effects: p82.

- Vasopressors used after appropriate volume resuscitation if persistent hypotension; listed agents/doses: dobutamine 2.0–20.0 µg/kg/min, dopamine 5.0–20.0 µg/kg/min, norepinephrine 0.5–30.0 µg/min, epinephrine 2.0–10.0 µg/min: p82.

- Goal-directed resuscitation (urine >0.5 mL/kg/h, CVP 8–12 mm Hg, MAP 65–90 mm Hg, ScvO2 >70%) during ED resuscitation of septic shock significantly decreases mortality: p82.

- Acidosis: treat with ventilation and fluid resuscitation; sodium bicarbonate 1 mEq/kg is controversial and should be used only in severe refractory acidosis and correct only to arterial pH 7.25: p82.

- Early surgical/medical consultation for admission or transfer is indicated: p82.

End scaffold.

If you want, I can convert this scaffold into a slide set, learner checklist, or an expanded stepwise flowchart — but I will use only the same excerpt for any further expansion.

Chapter 6: Anaphylaxis, Acute Allergic Reactions, and Angioedema

Source pages 84–87 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition \n**Source pages:** 84–87 \n**Content ID:** SEG-TINTINALLI-EMM8-FULL \n**Review status:** MACHINE-AUTHORED; MEDICAL REVIEW REQUIRED

Chapter scaffold — Chapter 6: Anaphylaxis, Acute Allergic Reactions, and Angioedema

(Use only source: pages 84–87 of supplied excerpt; all substantive statements below are traceable to the cited page(s).)

Learning objectives

- Recognize immediate evaluation triggers and indications for airway intervention in anaphylaxis/angioedema. (p84)

- Describe first-line pharmacologic therapy and routes/doses of epinephrine for anaphylaxis, including pediatric dosing and when to escalate to IV therapy. (p84)

- Outline adjunctive therapies (fluids, steroids, antihistamines, bronchodilators, glucagon for β■blocked patients) and their roles/limitations. (p84)

- Differentiate management considerations for ACE inhibitor–induced and hereditary angioedema, including specific targeted therapies and replacement options. (p84)

- Summarize disposition and observation recommendations after allergic reactions and anaphylaxis, and discharge counselling/medications. (p85)

Core concepts (concise)

- Airway: give oxygen as indicated by oximetry; angioedema or respiratory distress should prompt early consideration for intubation. (p84)

- Remove/limit allergen exposure (stop offending IV drug, remove stinger); first-aid/ice/elevation for local symptoms; do not perform gastric decontamination for ingested allergens. (p84)

- Epinephrine is first■line for anaphylaxis: IM 0.3–0.5 mg (0.3–0.5 mL of 1:1000) in thigh (pediatric 0.01 mg/kg); repeat q5 min as needed. (p84)

- IV epinephrine for refractory cases or cardiovascular compromise: bolus 100 μg (prepared as 0.1 mL of 1:1000 in 10 mL NS) over 5–10 min; if needed proceed to infusion (mix described; start at 1 μg/min; pediatric infusion 0.1–0.3 μg/kg/min). Stop infusion for chest pain or arrhythmias. (p84)

- No evidence that any specific vasopressor is superior for patients who cannot tolerate IV epinephrine. (p84)

- Hypotension: aggressive fluid resuscitation with normal saline 1–2 L (pediatric 10–20 mL/kg). (p84)

- Steroids: generally recommended for persistent or delayed reactions though evidence is sparse; severe steroid dosing examples given (methylprednisolone 125 mg IV; hydrocortisone 500 mg IV; pediatric doses listed). (p84)

- Antihistamines: diphenhydramine 50 mg IV/IM (pediatric 1 mg/kg) and consider H2 blocker (eg, ranitidine 50 mg IV; pediatric 0.5 mg/kg) in severe cases. (p84)

- Bronchospasm: nebulized β■agonist (albuterol 2.5 mg); if refractory add nebulized ipratropium 250 μg and IV magnesium 2 g (25–50 mg/kg in children) over 20 minutes. (p84)

- β■blocker patients with hypotension refractory to epinephrine/fluids: give glucagon 1 mg IV q5 min (pediatric 50 μg/kg); start infusion 5–15 μg/min once BP improves. (p84)

- Angioedema: supportive care, early intubation if airway compromised; mild swelling without airway involvement may be observed 12–24 h and discharged if improving. (p84)

- ACE inhibitor–induced angioedema: discontinue ACEI; icatibant 30 mg SC can be used; C1 esterase inhibitor (human) 1000 U IV has case■series evidence. (p84)

- Hereditary angioedema: treat with C1 esterase inhibitor replacement (human 20 U/kg; recombinant 50 U/kg to max 4200 U); FFP 2–3 U IV has been used if replacement unavailable; icatibant 30 mg SC and ecallantide 30 mg SC also effective in shortening attacks. (p84)

- Disposition: unstable/refractory → ICU. Moderate–severe symptoms → admit for observation. Mild reactions → ED observation then discharge if stable/improving. (p85)

- Observation interval after epinephrine: generally safe for discharge after 4 hours without symptoms; consider longer observation for prior severe reactions, β■blocker use, or social factors. (p85)

- Discharge medication/counselling: 3■day course of antihistamines and prednisone (longer for idiopathic anaphylaxis); counsel to return for recurrent symptoms; instruct avoidance of known allergens; severe reactions/anaphylaxis → provide and teach epinephrine autoinjector use; consider Medic■Alert and allergist referral. (p85)

Evaluation / decision logic (explicitly supported statements only)

- If oximetry indicates hypoxia or if angioedema/respiratory distress present → consider early intubation. (p84)

- If allergen exposure ongoing (eg, IV drug, stinger) → limit further exposure (stop drug/remove stinger). (p84)

- If anaphylaxis diagnosed and no cardiovascular collapse → give IM epinephrine 0.3–0.5 mg into thigh; repeat q5 min as needed. (p84)

- If refractory to IM epinephrine or cardiovascular compromise present → use IV epinephrine bolus then infusion as specified. (p84)

- If hypotension present → initiate aggressive NS fluid resuscitation (1–2 L adult; 10–20 mL/kg pediatric). (p84)

- If bronchospasm persists after nebulized β■agonist → add ipratropium and IV magnesium as specified. (p84)

- If hypotension refractory to epinephrine/fluids in a patient on β■blockers → administer glucagon, then start infusion when BP improves. (p84)

- For ACEI■induced angioedema → discontinue ACEI; consider icatibant or C1 esterase inhibitor (human). (p84)

- Disposition decision informed by stability: unstable/refractory → ICU; moderate–severe → admission; mild → ED observation and possible discharge. (p85)

- After epinephrine administration, consider 4■hour symptom■free observation before discharge for stable patients; extend observation for higher risk patients. (p85)

Management principles (explicitly supported)

- Prioritize airway and oxygenation as guided by oximetry and clinical signs. (p84)

- Remove or stop exposure to allergen; do not use gastric decontamination for ingested allergens. (p84)

- Use epinephrine promptly as first■line therapy; select route based on hemodynamics and response. (p84)

- Escalate to IV epinephrine bolus and infusion for refractory or cardiovascularly compromised patients, with monitoring for cardiac ischemia/arrhythmias and stopping infusion if chest pain/arrhythmias occur. (p84)

- Provide aggressive IV fluids for hypotension. (p84)

- Use steroids and antihistamines as adjuncts (steroids for persistent/delayed reactions despite sparse evidence). (p84)

- Treat bronchospasm with inhaled β■agonists and escalate to anticholinergic and IV magnesium if needed. (p84)

- For special scenarios (β■blocker patients, ACEI■induced angioedema, hereditary angioedema), employ the specific therapies listed. (p84)

- Observe and admit based on severity and risk factors; ensure patients with severe reactions receive an epinephrine autoinjector and counseling. (p85)

Red flags / high■yield points / common exam pearls (supported)

- Early consideration for intubation if angioedema or respiratory distress. (p84)

- Epinephrine is first■line — IM into the thigh is preferred for most patients; repeat q5 min as needed. (p84)

- IV epinephrine requires careful preparation and monitoring; stop infusion with chest pain or arrhythmias. (p84)

- For patients on β■blockers with refractory hypotension, glucagon is the recommended next pharmacologic step. (p84)

- ACE inhibitors must be discontinued in ACEI■induced angioedema. (p84)

- Patients who received epinephrine are commonly observed for ~4 hours symptom■free before discharge if stable. (p85)

Common mistakes to avoid (supported)

- Do not rely on gastric decontamination for ingested allergens — it is not recommended. (p84)

- Do not delay epinephrine when indicated — epinephrine is first■line therapy. (p84)

- When using IV epinephrine infusion, monitor for ischemia/arrhythmias and stop infusion if chest pain/arrhythmias occur. (p84)

Items or areas not specified or ambiguous in the excerpt

- Diagnostic criteria and formal clinical definition/thresholds for diagnosing anaphylaxis are not provided in the excerpt. [UNCLEAR — p84]

- Specific prednisone dosing for the discharge course is not given (excerpt only states “prednisone” for discharge). [UNCLEAR — p85]

- Detailed airway management techniques (eg, when and how to perform cricothyrotomy, video laryngoscopy vs direct laryngoscopy) are not provided. [UNCLEAR — p84]

- Detailed mixing/infusion concentration steps could be interpreted from text but some providers may find the brief description insufficient for bedside preparation (if more granular pharmacy instructions needed, not provided). [UNCLEAR — p84]

Evidence ledger — mapping of every substantive statement above to source page(s)

(Each ledger item mirrors the core statements; page citations are exact as in supplied excerpt.)

1. "Administer oxygen as indicated by oximetry." — p84

2. "Angioedema or respiratory distress should prompt early consideration for intubation." — p84

3. "Limit further exposure (stop IV drug, remove stinger)." — p84

4. "First■aid measures, ice, and elevation may be helpful for local symptoms." — p84

5. "Gastric decontamination is not recommended for ingested allergens." — p84

6. "First■line therapy for anaphylaxis is epinephrine." — p84

7. "IM epinephrine 0.3–0.5 mg (0.3–0.5 mL of 1:1000); pediatric 0.01 mg/kg; administer in thigh; may repeat every 5 minutes." — p84

8. "Patients refractory to IM dosing or with cardiovascular compromise should receive IV epinephrine." — p84

9. "IV bolus of 100 μg (0.1 mL of 1:1000 in 10 mL NS) can be infused over 5–10 minutes." — p84

10. "If needed, proceed to an infusion (start at 1 μg/min of mix of 1 mL 1:1000 added to 500 mL NS infused at 0.5 mL/min, titrate as needed); pediatric infusion dose 0.1–0.3 μg/kg/min." — p84

11. "Stop the infusion for chest pain or arrhythmias." — p84

12. "For patients with refractory shock who do not tolerate IV epinephrine, another vasopressor should be chosen; there is no evidence for superiority of any specific agent." — p84

13. "Hypotensive patients require aggressive fluid resuscitation with normal saline 1–2 L (pediatric 10–20 mL/kg)." — p84

14. "Steroids are generally recommended to control persistent or delayed reactions, although supportive evidence is sparse." — p84

15. "Severe steroid dosing examples: methylprednisolone 125 mg IV (pediatric 2 mg/kg) or hydrocortisone 500 mg IV (pediatric 10 mg/kg)." — p84

16. "Antihistamines have theoretical benefit; diphenhydramine 50 mg IV/IM (pediatric 1 mg/kg); H2 blocker (eg, ranitidine 50 mg IV, pediatric 0.5 mg/kg) recommended in severe cases." — p84

17. "Bronchospasm should be treated with nebulized β■agonists such as albuterol 2.5 mg." — p84

18. "If bronchospasm refractory, add nebulized ipratropium bromide 250 μg and IV magnesium 2 g (25–50 mg/kg in children) infused over 20 minutes." — p84

19. "For patients on β■blockers with hypotension refractory to epinephrine and fluids, use glucagon 1 mg IV every 5 minutes (pediatric 50 μg/kg); start infusion 5–15 μg/min once BP improves." — p84

20. "Angioedema is treated with supportive care and consideration for early intubation if the airway is compromised." — p84

21. "Episodes of angioedema may last hours to days." — p84

22. "Patients with mild swelling without airway involvement can be observed for 12–24 hours and discharged if symptoms are improving." — p84

23. "ACEI■induced angioedema can be treated with icatibant (30 mg SC)." — p84

24. "C1 esterase inhibitor (human) (1000 U IV) has been shown to be effective in case series." — p84

25. "The ACEI must be discontinued in ACEI■induced angioedema." — p84

26. "Hereditary angioedema can be treated with IV C1 esterase inhibitor replacement: human 20 U/kg; recombinant 50 U/kg to a maximum of 4200 units." — p84

27. "Fresh frozen plasma (2–3 U IV) has been used as an alternative when C1 esterase inhibitor replacement is not available." — p84

28. "Icatibant (30 mg SC) and ecallantide (30 mg SC) are also effective in shortening attacks of hereditary angioedema." — p84

29. "Unstable or refractory patients merit admission to the intensive care unit." — p85

30. "Patients with moderate to severe symptoms should be admitted for observation." — p85

31. "Patients with mild allergic reactions should be observed in the ED and may be sent home if symptoms are stable or improving." — p85

32. "Stable patients who received epinephrine are generally felt to be safe for discharge after 4 hours without symptoms." — p85

33. "Consider longer observation for patients with a history of severe reactions, patients on β■blockers, and patients who live alone or might have difficulty returning." — p85

34. "Discharge patients on a 3■day course of antihistamines and prednisone; longer courses may be necessary for idiopathic anaphylaxis." — p85

35. "Counsel all patients about the need to return to the ED for recurrence of symptoms and instruct in avoiding future exposures to the allergen, if known." — p85

36. "All patients who have experienced a severe allergic reaction or anaphylaxis should have and know how to use an epinephrine autoinjector." — p85

37. "Consider Medic■Alert bracelets and referral to an allergist in these patients." — p85

[UNCLEAR items — exact pages]

- "Formal diagnostic criteria/definition for anaphylaxis" — not provided in excerpt. [UNCLEAR — p84]

- "Specific prednisone dosing for the discharge course" — not provided in excerpt. [UNCLEAR — p85]

- "Detailed airway management techniques (eg, cricothyrotomy indications/techniques)" — not provided. [UNCLEAR — p84]

- "Detailed pharmacy■level infusion mixing/step■by■step instructions beyond the brief mixing description" — may be insufficient for bedside prep without local protocol. [UNCLEAR — p84]

Notes on scope

- This scaffold preserves statements, dosing, and decision points only as they appear in the supplied excerpt (pages 84–85). Pages 86–87 of the supplied excerpt pertain to a different chapter (acute pain/procedural sedation) and were not used to support Chapter 6 content. Do not use this scaffold as an exhaustive clinical guideline; it is an educational framework strictly reflecting the supplied text.

Chapter 7: Acute Pain Management and Procedural Sedation

Source pages 88–97 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 88–97

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Acute Pain Management and Procedural Sedation**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 7: Acute Pain Management and Procedural Sedation 69 those with severe systemic disease, at the extremes of age, and those with predicted difficult airway. [Source pages 88–97]

- When PSA is performed, necessary equipment includes a continuous cardiac monitor with pulse oximetry, oxygen, suction, and immediate availability of appropriately sized resuscitation equipment. [Source pages 88–97]

- The patient should be under constant observation by a provider trained in airway management. [Source pages 88–97]

- Blood pressure, heart rate, respiratory rate, and level of consciousness should be monitored. [Source pages 88–97]

- Some advocate routine use of capnography to monitor ventilation in sedated patients, as it has been shown to detect changes in ventilation before clinical observation allowing for earlier recognition of adverse events. [Source pages 88–97]

Clinical reasoning structure

- **Problem representation:** Acute Pain Management and Procedural Sedation; source pages 88–97.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Acute Pain Management and Procedural Sedation**.

- Source-page range: **88–97**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 88

CHAPTER 7: Acute Pain Management and Procedural Sedation 69

those with severe systemic disease, at the extremes of age, and those with predicted difficult airway.

When PSA is performed, necessary equipment includes a continuous cardiac monitor with pulse oximetry, oxygen, suction, and immediate availability of appropriately sized resuscitation equipment. Informed consent should be obtained. The patient should be under constant observation by a provider trained in airway management. Blood pressure, heart rate, respiratory rate, and level of consciousness should be monitored. Some advocate routine use of capnography to monitor ventilation in sedated patients, as it has been shown to detect changes in ventilation before clinical observation allowing for earlier recognition of adverse events. The application of supplemental oxygen reduces hypoxemia and has no adverse clinical effects.

The analgesic or sedative agents chosen should be individualized to the patient and the planned procedure. The agents used for PSA often have a narrow therapeutic index. Therefore, these medications should be administered in small, incremental intravenous doses, with adequate time between doses to determine peak effect. All patients undergoing PSA should be reassessed continuously. Patients experiencing transient respiratory depression can usually be managed by bag-mask ventilation.

Sedation Management

Table 7-2 describes selected sedation agents for procedural sedation. Nitrous Oxide is used for minimal sedation and is supplied in a 50:50 mixture with oxygen. It requires a special setup of a demand delivery system and a scavenging system to prevent exam room accumulation of nitrous oxide. It is contraindicated in pregnant women and those with pneumothorax, pneumocephalus, and vascular air embolus.

Fentanyl has a rapid onset of action and is easily titratable. Chest wall rigidity unresponsive to naloxone may occur at higher doses (5 to 15 mcg/kg) or when rapidly administered potentially necessitating neuromuscular blockade and mechanical ventilation. Slow administration (e.g., over 3 to 5 minutes) and careful flushing of the IV can prevent rigid chest syndrome.

Midazolam is commonly used as a sole agent for minimal sedation. Respiratory depression and hypotension may develop. Flumazenil quickly reverses sedation and respiratory depression due to benzodiazepines. Routine use to reverse sedation is not recommended.

Ketamine is a dissociative analgesic with sedative, analgesic, and amnestic properties that causes minimal respiratory depression and has a minimal effect on blood pressure. Ketamine may be administered IV or IM. In lower doses, it can produce analgesia. In higher doses (approximately 1mg/kg and above IV) a critical dosing threshold is exceeded and produces a dissociative state. Ketamine may cause increased intraocular pressure, hypersalivation, laryngospasm, and a hallucinatory emergence reaction in older children and adults. Ketamine can cause vomiting, especially in adolescents. Pretreatment with ondansetron can reduce the incidence of vomiting. There is conflicting data on whether ketamine causes increased intracranial pressure, thus there is no concluding evidence to restrict its use in head injury patients. Emergence reaction is best treated with midazolam (0.1 mg/kg IM or IV). Ketamine is contraindicated in children less than 3 months as well as those with increased intraocular pressure or obstructive

07_Cydulka_Ch07.indd 69 20/04/17 2:40 pm

Source page 89

07_Cydulka_Ch07.indd 70

Table 7-2 Sedation Agents for Adult Procedural Sedation and Analgesia

Medication	Recommended Dosage	Route of Administration	Approximate Peak Effect	Duration	Use
Nitrous oxide	50:50 mixture with oxygen	Inhalational	1–2 min	3–5 min	Minimal sedation
Midazolam	0.05–0.1 mg/kg. May repeat every 2 min until adequately sedated	IV	2–3 min	20–30 min	Minimal or moderate sedation
	0.1 mg/kg	IM	15–30 min	1–2 h	Minimal sedation
Fentanyl	1–3 µg/kg, can be titrated up to 5 µg/kg	IV	2–3 min	30–60 min	Minimal sedation
Fentanyl and midazolam	1–2 µg/kg fentanyl plus 0.1 mg/kg midazolam	IV	2–3 min	1h	Moderate and deep sedation
Methohexital	1 mg/kg	IV	1 min	3–5 min	Moderate or deep sedation
Pentobarbital	2.5 mg/kg followed by 1.25 mg/kg, as IV rate should be	IV	3–5 min	15 min	Minimal and moderate sedation.

needed, up to two times <50 mg/min Frequently used for radiologic procedures.

Ketamine 1 mg/kg IV 1–3 min 15–30 min Dissociative sedation

2–5 mg/kg IM 5–20 min 30–60 min Dissociative sedation

Ketamine and midazolam Ketamine as above plus IV 1–3 min 30–60 min Dissociative sedation

midazolam, 0.03–0.05 mg/kg

Etomidate 0.15 mg/kg, followed by IV 15–30 s 3–8 min Moderate and deep sedation

0.1 mg/kg every 2 min, if needed

Propofol 0.5–1.0 mg/kg, followed by 0.5 mg/kg IV 30–60 s 5–6 min Moderate and deep sedation

every 3 min, if needed

Ketamine and propofol Ketamine 0.125–0.5 mg/kg and Propofol IV 30–60 s 15 min Moderate and deep sedation

20/04/17 2:40 pm

“ketofol” 0.5 mg/kg

Source page 90

CHAPTER 7: Acute Pain Management and Procedural Sedation 71

hydrocephalus and should be used in caution in patients with significant upper respiratory tract infectious symptoms or those undergoing a procedure involving major stimulation of the oropharynx (e.g., endoscopy).

Etomidate is a rapid onset sedative with short duration of effect. It causes temporary adrenal insufficiency which is associated with increased mortality in critically-ill patients; however, this effect has not been shown to cause adverse events when used in stable patients. Up to 20% of patients can have myoclonic jerking.

Propofol is an anesthetic agent administered by intravenous infusion that has a short duration of effect, which results in shorter recovery time and ED length of stay. The most common side effect is respiratory depression and apnea. Side effects include hypotension, therefore hypovolemia should be corrected before propofol administration. Adjunct analgesic is mandatory for painful procedures. Pain at the IV site is common during administration and can be blunted with adjunct use of lidocaine (e.g., 0.05 mg/kg IV). Propofol use is contraindicated in patients who are allergic to eggs or soy products.

Ketamine and Propofol can be used in combination and may have synergy with each other. Ketamine may blunt the hypotension and respiratory depression seen with propofol. Propofol may prevent the emergence reaction and vomiting seen with ketamine. Ketamine use precludes the use of an analgesic with propofol alone. The greatest advantage of using this combination is likely that sedation can be achieved with lower doses of either drug used alone.

Elderly

Procedural sedation in the elderly is associated with increased adverse events. The risk of respiratory depression is increased with all agents. Etomidate is a good choice owing to its minimal cardiovascular effects.

When using propofol in this age group, the initial and subsequent doses should be 50% of those recommended for younger adults.

Children

Children of all ages feel pain, even neonates. Anxiety issues, pain control, and need for sedation must be addressed. Anxiety may be a significant barrier to a successful procedure performance, especially when patient's cooperation is needed. Parents can provide significant anxiety relief and should be allowed to stay with children. Age appropriate distraction techniques should also be employed. Benzodiazepines, such as midazolam, provide effective pharmacologic anxiety relief when needed. Oral sucrose is an effective analgesic in infants undergoing painful procedures such as IV catheter placement. Procedural sedation and analgesia should be used when performing painful procedures or when procedures require the patient to be still. Common medications used for pediatric procedural sedation are listed in Table 7-3.

Disposition

Patients are eligible for discharge only when fully recovered. When discharged, the patient must be accompanied by an adult and should not drive or operate machinery for 24 hours. Because many of the agents used for PSA produce anterograde amnesia, discharge instructions must be given to responsible accompanying adults.

07_Cydulka_Ch07.indd 71 20/04/17 2:40 pm

Source page 91

07_Cydulka_Ch07.indd 72

Table 7-3 Medications for Procedural Sedation

Class	Drug	Route	Dose	Onset	Duration	Advantages	Disadvantages	Examples	Comments
Anxiolytic	Midazolam	PO, PR, IV, IM, IN	PO/PR 0.5 mg/kg IV/IM 0.05–0.1 mg/kg IM 0.2–0.4 mg/kg IN 0.2–0.4 mg/kg	PO/PR 20–30 min IV 3–5 min IM 10–20 min IN 5–10 min	1–4 h using local anesthetic onset, less predictable	Flexible route of administration	No analgesia, paradoxical reaction	Premedication for IV start, laceration repair	Acidic, nasal administration stings, may cause increased secretions; oral/rectal slow
Hypnotic/ sedative	Propofol	IV	1–2 mg/kg, repeat doses as needed	Seconds	Minutes	Rapid onset and short duration, respiratory depressant	No analgesia, short duration myoclonus, procedures	CT scan, LP with motionlessness, cardiovascular reduction of painful dislocation	Non-analgesic, topical analgesic, increased requirement for younger patients, muscle relaxant injection
	Etomidate	IV	0.2–0.3 mg/kg	Seconds	Minutes	Rapid onset, short duration myoclonus, procedures	No analgesia, increased tone (e.g.,	CT scan,	Avoid in patients with

respiratory requiring CP) due to myoclonic

depressant motionlessness jerks, painful injection

Pentobarbital IV 1–2 mg/kg, <1 min 15–45 min Well studied, No analgesia, CT scan, no Variable dosing, long

repeated every motionlessness, respiratory and reversal agent recovery times

3–5 min as neuroprotective cardiovascular

needed depressant

Methohexital IV 0.5–1 mg/kg Seconds 10–60 min Rapid onset No analgesia, CT scan, no —

respiratory and reversal agent

cardiovascular

depressant

20/04/17 2:40 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.

- Content ID: SEG-TINTINALLI-EMM8-FULL.

- Source pages: 88–97.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 8: Chronic Pain

Source pages 98–101 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 98–101

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Chronic Pain**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 8: Chronic Pain 79 Gabapentin is started with an initial dose of 300 mg daily and is increased up to a maximum of 1200 mg three times daily according to response. [Source pages 98–101]

- Referral to the appropriate specialist is one of the most productive means to aid in the care of chronic pain patients who present to the ED. [Source pages 98–101]

- Possible outcomes of referral to pain specialists include optimization of medical therapy, trigger point injections, dedicated exercise pro- grams, physical therapy, epidural steroid injections, or nerve blocks as indicated. [Source pages 98–101]

- ■■ MANAGEMENT OF PATIENTS WITH ABERRANT DRUG-RELATED BEHAVIOR The spectrum of patients with aberrant drug-related behavior includes those who have chronic pain and are now dependent on narcotics to “hustlers” who are obtaining prescription drugs to sell on the street. [Source pages 98–101]

- Clinical Features Because of the wide spectrum of patients with aberrant drug-related behav- ior, the history given may be factual or fraudulent. [Source pages 98–101]

Clinical reasoning structure

- **Problem representation:** Chronic Pain; source pages 98–101.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Chronic Pain**.

- Source-page range: **98–101**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 98

CHAPTER 8: Chronic Pain 79

Gabapentin is started with an initial dose of 300 mg daily and is increased up to a maximum of 1200 mg three times daily according to response. Pregabalin—start 50 mg three times daily. Duloxetine—start 30 mg PO daily. Lamotrigine—begin with 25 mg daily.

3. Referral to the appropriate specialist is one of the most productive means to aid in the care of chronic pain patients who present to the ED. Possible outcomes of referral to pain specialists include optimization of medical therapy, trigger point injections, dedicated exercise programs, physical therapy, epidural steroid injections, or nerve blocks as indicated.

■■ MANAGEMENT OF PATIENTS WITH ABERRANT DRUG-RELATED BEHAVIOR

The spectrum of patients with aberrant drug-related behavior includes those who have chronic pain and are now dependent on narcotics to “hustlers” who are obtaining prescription drugs to sell on the street.

Clinical Features

Because of the wide spectrum of patients with aberrant drug-related behavior, the history given may be factual or fraudulent. Patients with aberrant drug-related behavior may be demanding, intimidating, or flattering. In one ED study, the most common complaints of patients seeking drugs were (in decreasing order) back pain, headache, extremity pain, and dental pain. Many fraudulent techniques are used, including “lost” prescriptions, or doctor’s note explaining need for opioids, “impending” surgery, fictitious hematuria with a complaint of kidney stones, self-mutilation, and fictitious injury.

Diagnosis and Differential

The diagnosis of aberrant drug-related behaviors may not be possible in the

ED. The medical record can provide a wealth of information regarding the patient, including documentation proving that the patient is supplying false information. Most states now have controlled drug databases that can be queried. Aberrant drug-related behaviors are listed in Table 8-3. The predictive behaviors are illegal in many states and form a solid basis to refuse narcotics to the patient.

Table 8-3 Aberrant Drug-Related Behaviors

Forges/alters prescriptions*

Sells controlled/illicit drugs*

Uses aliases to receive opioids*

Current illicit drug use*

Fictitious illness, requests opioids

Conceals multiple ED visits for opioids

Conceals multiple physicians prescribing opioids

Abusive when refused

*

Unlawful in many states.

008_Cydulka_Ch008.indd 79 10/03/17 4:28 PM

Source page 99

80 SECTION 2: Analgesia, Anesthesia, and Sedation

Emergency Department Care and Disposition

The treatment of aberrant drug-related behavior is to refuse the controlled substance, consider the need for alternative medication or treatment, and consider referral for drug counseling.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 38 "Chronic Pain," by David M. Cline.

008_Cydulka_Ch008.indd 80 10/03/17 4:28 PM

Source page 100

SECTION

Emergency Wound

Management 3

C h a p ter Evaluating and Preparing Wounds

9 Timothy J. Reeder

■■ CLINICAL FEATURES

Traumatic wounds are regularly encountered problems in the emergency department. It is important to document important historical information such as the mechanism, timing, location of injury, and the degree of contamination. Associated symptoms of pain, swelling, paresthesias, and loss

of function should be identified. Determine factors that affect wound healing, such as patient age, location of injury, medications, chronic medical conditions, and previous keloid or scar formation. Adults with the sensation of a foreign body are much more likely to have retained a foreign body that should be removed. Review allergies, particularly to latex, and determine whether tetanus immunization is required (see Chapter 94). When caring for wounds, the ultimate goal is to restore the physical integrity and function of the injured tissue without infection.

When treating a wound, consider factors that impact risks for infection, such as time since injury, mechanism, and location. Shear forces are produced by sharp objects with relatively low energy, resulting in a wound with a straight edge and little contamination that can be expected to heal with a good result. Wounds caused by compression forces crush the skin against the underlying bone and often produce stellate lacerations. Tension forces may produce flap-type lacerations with surrounding devitalized tissue and may be more susceptible to infection. Other predictive factors for infection include location, depth, characteristics, contamination, and patient age. The risk of infection also relates to the interaction of bacterial contamination and blood supply. The density of bacteria is quite low over the trunk and proximal arms and legs, and thus these areas have lower risks for infection. Moist areas such as the axilla, perineum, and exposed hands and feet have a higher degree of colonization and may be at higher risk of infection. Wounds located on the face or scalp, both highly vascularized areas, are at lower risk for infection.

Wounds of the oral cavity are heavily contaminated with bacteria, although evidence supporting routine antibiotic use in simple intra-oral laceration is inconclusive. Wounds sustained from contaminated objects or environments and animal and human bites have an increased infection risk.

81

09_Cydulka_Ch09.indd 81 15/03/17 10:49 AM

Source page 101

Wounds contaminated with feces have a high risk of infection even when treated with antibiotic therapy. Although there is no clearly defined relationship between time to closure and infection rate, consider time since injury when making decisions about wound repair. Delayed primary closure after 4 days of open wound management is recommended for wounds with a high risk for infection during the first care encounter.

■■ DIAGNOSIS AND DIFFERENTIAL

Wound examination is greatly facilitated by a cooperative patient, good

positioning, optimal lighting, and little or no bleeding. A thorough examination will minimize the risk of missed foreign bodies, tendon injuries, and nerve injuries. While performing wound assessment, repositioning the joint or extremity in the position assumed during injury may help identify underlying damaged structures.

Assess wound location, size, shape, margins, and depth. Pay particular attention to sensory and motor functions, tendon injury, vascular compromise, and injuries to specialized structures. Carefully palpate and inspect the wound and surrounding area to identify a retained foreign body or bone injury, and consider radiographic imaging when these complicating features are suspected. For wounds overlying joints, consider additional assessment to determine whether a violation of the joint space may have occurred.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

Proper wound preparation is an important first step to restoring the integrity and function of the injured tissue, preventing infection, and maximizing cosmetic results.

Anesthesia

1. Control pain with local or regional anesthesia to enable better preparation and evaluation of the wound.
2. Perform a neurovascular examination of the involved and distal areas before anesthesia. Two-point discrimination (normal <6 mm) will help identify digital nerve injury.

Irrigation

1. Wound irrigation reduces the risk of infection.
2. Use low-pressure irrigation with a slow, gentle wash for simple uncontaminated wounds in highly vascular areas such as the face and scalp.
3. Use high-pressure irrigation with an appropriate 18-gauge catheter, syringe, and splash guard for contaminated wounds.
4. Consider anesthesia before irrigation.
5. Wound soaking is not effective in cleaning contaminated wounds and may increase wound bacterial counts.
6. Although sterile normal saline solution is commonly used, tap water is safe and effective. There is no added benefit of adding povidone iodine or hydrogen peroxide to irrigation fluid. Polyhexanide irrigation may have benefit in high-risk wounds.
7. Warmed fluids may be more comfortable for wound irrigation.

09_Cydulka_Ch09.indd 82 15/03/17 10:49 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.

- Content ID: SEG-TINTINALLI-EMM8-FULL.

- Source pages: 98–101.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 9: Evaluating and Preparing Wounds

Source pages 102–105 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 102–105

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Evaluating and Preparing Wounds**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 9: Evaluating and Preparing Wounds 83 Skin Disinfection and Sterile Technique 1. [Source pages 102–105]

- Sterilizing the surrounding skin with a povidone iodine- or chlorhexi- dine-containing agent is common practice. [Source pages 102–105]

- Avoid contact with nonintact skin when these agents are used. [Source pages 102–105]

- Full sterile technique has not been shown to reduce infection after repair of wounds in the acute setting. [Source pages 102–105]

- Clean, nonsterile gloves and attention to cleanliness may be used to improve efficiency and cost savings. [Source pages 102–105]

Clinical reasoning structure

- **Problem representation:** Evaluating and Preparing Wounds; source pages 102–105.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Evaluating and Preparing Wounds**.

- Source-page range: **102–105**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 102

CHAPTER 9: Evaluating and Preparing Wounds 83

Skin Disinfection and Sterile Technique

1. Sterilizing the surrounding skin with a povidone iodine- or chlorhexidine-containing agent is common practice. Avoid contact with nonintact skin when these agents are used.

2. Full sterile technique has not been shown to reduce infection after repair of wounds in the acute setting. Clean, nonsterile gloves and attention to cleanliness may be used to improve efficiency and cost savings.

Hemostasis

1. Control of bleeding is necessary for proper wound evaluation and treatment.

2. Direct pressure is the preferred method and is usually effective.

3. Epinephrine-containing local anesthetics can be helpful in controlling bleeding associated with wounds and can be used for digital nerve blocks, nose, and ears except in patients with underlying small vessel disease.

4. Other means of hemostasis can be used, such as applying pressure with gelatin, cellulose, or collagen sponges placed directly into the wound.

5. Ligation of minor vessels may be necessary and can be achieved by ligating with an absorbable suture material after isolating and clamping the involved vessel.

6. Cautery, with a bipolar device for vessels <2 mm and battery-powered device for capillaries, can be used to control extensive bleeding.

7. Finger tourniquets or a blood pressure cuff placed proximal to the injury and inflated above the systolic blood pressure will control bleeding. The duration of tourniquet use should be minimized and not left in place for more than 20 to 30 min at a time.

Foreign Body and Hair Removal

1. Visually inspect the full depth and course of all wounds for foreign bodies. Carefully remove foreign debris with forceps.

2. Many opaque foreign bodies can be detected by routine radiographs. Foreign bodies with densities similar to those of soft tissue may require

the use of computed tomography, magnetic resonance imaging, or ultrasound to visualize.

3. Routine hair removal does not appear to have an effect on infections, and shaving may increase the rate of postoperative skin and soft tissue infections.

4. When hair removal is required to adequately assess or repair a wound, clip with scissors 1 to 2 mm above the skin surface.

Debridement

1. Devitalized tissue may increase the risk of infection and delay healing. Debridement removes foreign matter, bacteria, and devitalized tissue and creates a clean wound edge.

2. Elliptical excision around the wound edges with a standard surgical blade is the easiest technique for excisional debridement.

3. Wounds with an extensive amount of nonviable tissue may require a large amount of tissue removal and will need more delayed wound closure or grafting. In general, a surgical specialist should be consulted to manage these wounds.

09_Cydulka_Ch09.indd 83 15/03/17 10:49 AM

Source page 103

Prophylactic Antibiotics

Although there is no clear evidence that antibiotic prophylaxis prevents wound infection in most emergency department patients, it may have a role in selected higher-risk wounds and populations.

1. When used, antibiotic prophylaxis should be initiated before significant tissue manipulation, performed with agents that are effective against predicted pathogens, and administered through routes that rapidly achieve desired blood levels. Oral antibiotics may be as effective as intravenous if the agent has sufficient spectrum of coverage and rapid absorption.

2. Most non-bite infections are caused by staphylococci or streptococci, and coverage with a β-lactam antibiotics is typically adequate.

3. Prophylactic antibiotics are recommended for human bites, mammalian bites to the hand, and other high-risk bite wounds (see Chapter 15, Puncture Wounds and Bites).

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 37, "Wound Evaluation," by Adam J. Singer and Judd E. Hollander; and Chapter 40, "Wound Preparation," by Aleksandr M. Tichter, Wallace A. Carter, and Susan C. Stone.

09_Cydulka_Ch09.indd 84 15/03/17 10:49 AM

Source page 104

CHAPTER Methods for Wound Closure

10 Corey R. Heitz

Wound closure in the emergency department can reduce the risk of infection, provide for improved cosmetic outcome, and maintain the skin's protective functions. Methods for wound repair include primary closure (immediately after injury), secondary closure (allowing a wound to heal on its own, which may be useful for contaminated or infected wounds), and delayed closure (packing the wound and performing closure at a later date once infection is ruled out).

Clinicians should consider several factors when choosing a wound closure method: time since injury, wound tension, whether tension is static or dynamic, and risk of scar formation. The cosmetic outcome after wound closure is primarily a function of technique and not necessarily reflective of the specific closure method or device that is selected. Options for closure include sutures, staples, adhesive tape, tissue adhesives, and hair apposition.

■■ SUTURES

Sutures are strong, reliable, and adaptable and allow good approximation of wounds that are under tension. Absorbable suture material loses its tensile strength in <60 days and thus does not require removal after healing. Nonabsorbable sutures maintain their tensile strength and will require removal after a suitable time period for tissue healing. For most emergency department applications, either choice is acceptable as both have similar rates of infection, cosmetic outcomes, and wound dehiscence risks. Absorbable sutures are suitable for deep tissue structures such as dermis and fascia or when avoiding the need for suture removal is desired. Often, monofilament nonabsorbable sutures are used due to ease of use, good tensile strength, and low rates of infection. Suture material is available in a wide variety of sizes, with higher gauge sizes (such as 5-0 and 6-0) being very thin with less tensile strength. Smaller gauge sizes (such as 3-0 and 4-0) are larger in diameter with greater strength. Large gauge (e.g., 6-0) is most often chosen for areas of low tension where minimizing scars is desired, such as on the face. Smaller gauge suture (e.g., 3-0) is often used for the scalp, extremities, or other areas where increased tension requires suture material with greater tensile strength. When placing sutures of any size, be sure to evert the skin edges and avoid applying excessive crushing force on the tissue that can further damage the area and impair healing. Direct pressure, topical vasoconstrictors, or local anesthetic with vasoconstrictor are all useful for achieving hemostasis when repairing a wound.

■■ SUTURING TECHNIQUES

Simple Interrupted Percutaneous Sutures

Placing simple interrupted sutures is the easiest and most commonly used technique for wound repair. The needle is introduced on one side of the wound, through the deep tissues, and exits the other side of the wound

85

10_Cydulka_Ch10.indd 85 15/03/17 11:20 AM

Source page 105

86 SECTION 3: Emergency Wound Management

(Fig. 10-1). Be sure to keep the needle tip at 90 degrees to the skin and evert the wound edges for best results, and take care not to pucker the skin when tying the knot. Square knots are generally tied with the number of knots corresponding to the suture gauge (e.g.,four knots when using 4-0 suture material). Position the knots so that they all remain on the same side of the wound.

Continuous (Running) Percutaneous Sutures

A rapid method for closure of high-tension wounds, this method can be challenging to get cosmetically optimal results in irregularly shaped wounds and

A B

C D

E F

G

Figure 10-1. Placement of simple interrupted sutures. (A) Enter at 90 degrees. (B–D) Exit the same distance as the entrance. (E–G) Be consistent and tie knots on one side of wound.

Reproduced with permission from E. F. Reichman: Emergency Medicine Procedures. 2nd ed. New York: McGraw-Hill Education; 2013.

10_Cydulka_Ch10.indd 86 15/03/17 11:20 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 102–105.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 10: Methods for Wound Closure

Source pages 106–115 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 106–115

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Methods for Wound Closure**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 10: Methods for Wound Closure 87 thus is most appropriate for long linear lacerations. [Source pages 106–115]

- Place a suture at one end of the wound, tie, but do not cut the suture material. [Source pages 106–115]

- Then sew back and forth down the wound at a 65 degree angle until the opposite end is reached, at which point the suture material is tied off and cut (Fig. [Source pages 106–115]

- 10_Cydulka_Ch10.indd 87 15/03/17 11:20 AM 88 SECTION 3: Emergency Wound Management Buried Dermal Sutures These sutures are placed to reduce tension on the wound and close dead spaces by approximating underlying structures. [Source pages 106–115]

- Place these sutures by entering the deep tissues first and exiting in the subcutaneous layer, then entering the other side of the wound in the subcutaneous layer and exiting in the deep tissue (Fig. [Source pages 106–115]

Clinical reasoning structure

- **Problem representation:** Methods for Wound Closure; source pages 106–115.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Methods for Wound Closure**.

- Source-page range: **106–115**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 106

CHAPTER 10: Methods for Wound Closure 87

thus is most appropriate for long linear lacerations. Place a suture at one end of the wound, tie, but do not cut the suture material. Then sew back and forth down the wound at a 65 degree angle until the opposite end is reached, at which point the suture material is tied off and cut (Fig. 10-2).

A B

C D

E F

H

G

I

Figure 10-2. Continuous running. (A–C) Begin like a simple interrupted. (D–F) Be consistent, use a 65 degree angle. (G–I) Tie off at the end.

Reproduced with permission from E. F. Reichman: Emergency Medicine Procedures. 2nd ed. New York: McGraw-Hill Education; 2013.

10_Cydulka_Ch10.indd 87 15/03/17 11:20 AM

Source page 107

88 SECTION 3: Emergency Wound Management

Buried Dermal Sutures

These sutures are placed to reduce tension on the wound and close dead spaces by approximating underlying structures. Place these sutures by entering the deep tissues first and exiting in the subcutaneous layer, then entering the other side of the wound in the subcutaneous layer and exiting in the deep tissue (Fig. 10-3). This allows the knot to be buried deep within the tissue and thus should be completed with absorbable suture material. Closing adipose tissue with deep sutures is unnecessary as it does not

A B

C D

E

Figure 10-3. Buried (deep) dermal suture. Reverse of a simple interrupted, to bury

the knot. (A) Insert the needle into one side of the base of the wound, and drive the needle from deep to superficial and exiting at the dermal–epidermal junction. (B) Insert the needle through the dermal–epidermal junction on the opposite side of the wound and drive it through the base of the wound. The suture should exit the base of the wound across from and level with the entrance site of the first throw. (C) Pull both free ends of the suture up and out through the laceration. (D) Tie a knot in the suture. (E) Pull both free ends of the suture to lower the knot to the base of the wound and oppose the tissue. Tie two additional knots to secure the suture. Cut off any excess suture.

Reproduced with permission from E. F. Reichman: Emergency Medicine Procedures. 2nd ed. New York: McGraw-Hill Education; 2013.

10_Cydulka_Ch10.indd 88 15/03/17 11:20 AM

Source page 108

enhance cosmetic outcome or wound healing. Infection risks can be decreased when clinicians minimize the number of buried sutures used for wound repair.

Continuous Subcuticular Sutures

A more complex method, subcuticular sutures are performed with absorbable sutures and do not require removal. Begin by anchoring the suture at one end, and then weave horizontally in and out of the dermal–epidermal junction, subcutaneously (Fig. 10-4). This technique allows for excellent approximation of wound edges.

Vertical Mattress Sutures

Vertical mattress sutures allow for closure of deep and epidermal layers with a single suture placement, and allow for excellent wound edge eversion. Enter the wound on one side a few millimeters from the edge, take a

A B C

D E F

Figure 10-4. Continuous subcuticular suture. (A–C) Tie the anchor knot. (D, E) Use horizontal subcutaneous stitching. (F) Bury the knot at the end.

Reproduced with permission from E. F. Reichman: Emergency Medicine Procedures. 2nd ed. New York: McGraw-Hill Education; 2013.

10_Cydulka_Ch10.indd 89 15/03/17 11:20 AM

Source page 109

A B

C D

E

Figure 10-5. Vertical mattress suture. (A, B) Far-far. (C, D) Near-near. (E) Finished

product, knots on one side.

Reproduced with permission from E. F. Reichman: Emergency Medicine Procedures. 2nd ed. New York: McGraw-Hill Education; 2013.

deep bite of tissue, and exit on the other side, similar to a simple interrupted suture technique. Then before tying the knot, reenter the same side of the wound that was just exited at a point closer to the wound edge and then exit in a similar position on the other side in a “far-far/near-near” fashion for each suture placement (Fig. 10-5).

Horizontal Mattress Sutures

For horizontal mattress sutures, begin similarly to a simple interrupted suture. Instead of tying the knot after the first suture throw, go several millimeters parallel to the wound and perform another throw, tying the knot on the original side of the wound (Fig 10-6). This technique is useful for wounds with poor circulation and other delicate areas as it distributes wound tension and prevents further skin disruption close to the wound edge.

Horizontal Half-Buried Mattress Sutures

This technique is useful for skin flaps and stellate lacerations because the suture passes through the dermis of the skin tip and not the epidermal layer.

10_Cydulka_Ch10.indd 90 15/03/17 11:20 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 106–115.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 11: Lacerations to The Face and Scalp

Source pages 116–121 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 116–121

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Lacerations to The Face and Scalp**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 11: Lacerations to The Face and Scalp 97 Consider a pressure dressing over deep scalp lacerations for the first 24 hours to reduce the chance of hematoma formation. [Source pages 116–121]

- Superficial forehead lacerations are not associated with injury to the frontalis muscle, while deep lacerations do have damage to the frontalis muscle. [Source pages 116–121]

- Close superficial forehead lacerations with 6-0 nonabsorbable interrupted suture, rapidly absorbable suture, or tissue adhesive. [Source pages 116–121]

- Close the muscular layer of deep forehead lacerations with buried 5-0 absorb- able suture, and then close the epidermal layer with 6-0 nonabsorbable suture, skin closure strips, or tissue adhesive. [Source pages 116–121]

- Eyebrows are important landmarks to assist with reapproximation of wound edges. [Source pages 116–121]

Clinical reasoning structure

- **Problem representation:** Lacerations to The Face and Scalp; source pages 116–121.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Lacerations to The Face and Scalp**.

- Source-page range: **116–121**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 116

CHAPTER 11: Lacerations to The Face and Scalp 97

Consider a pressure dressing over deep scalp lacerations for the first 24 hours to reduce the chance of hematoma formation.

Superficial forehead lacerations are not associated with injury to the frontalis muscle, while deep lacerations do have damage to the frontalis muscle. Close superficial forehead lacerations with 6-0 nonabsorbable interrupted suture, rapidly absorbable suture, or tissue adhesive. Close the muscular layer of deep forehead lacerations with buried 5-0 absorbable suture, and then close the epidermal layer with 6-0 nonabsorbable suture, skin closure strips, or tissue adhesive. Eyebrows are important landmarks to assist with reapproximation of wound edges. When wounds involve the eyebrows, minimize skin debridement and leave suture tails long to facilitate removal. Remove scalp staples or sutures in 10 to 14 days, while forehead nonabsorbable sutures can be removed in 5 to 7 days.

■■ EYELIDS

The eyelids are thin and offer limited protection from injuries to the globe and surrounding structures. Examine lid injuries for involvement of the canthi, the lacrimal system, or penetration through the tarsal plate or lid margin (Fig. 11-2). Eyelid injuries within 6 to 8 mm of the medial canthus are at risk for canalicular laceration, particularly when associated with medial wall blow-out fractures. Consider consultation with an ophthalmologist when ptosis is present or for complex eyelid wounds such as those involving the inner surface of the lid, lid margins, lacrimal duct, and tarsal plate. Close uncomplicated eyelid lacerations with nonabsorbable 6-0 or 7-0 simple interrupted percutaneous sutures. Avoid the use of tissue adhesive near the eye. Remove nonabsorbable sutures in 3 to 5 days.

Figure 11-2. Periorbital anatomy.

11_Cydulka_Ch11.indd 97 15/03/17 11:26 AM

Source page 117

98 SECTION 3: Emergency Wound Management

■■ NOSE

Examine nasal lacerations to determine depth and involvement of deep tissue layers. Exposed cartilage or deep involvement of the tissue layers increases the risk of infection. Topical lidocaine into the nasal cavity may provide sufficient anesthesia for some local wound repair, and injected intradermal lidocaine with or without epinephrine is appropriate when needed to facilitate wound evaluation, cleansing, and repair.

Close superficial nasal lacerations with 6-0 nonabsorbable monofilament simple interrupted sutures. When the laceration extends through all tissue layers, begin closure with a 5-0 nonabsorbable, monofilament suture that aligns the skin surrounding the entrances of the nasal canals to prevent malposition and notching of the alar rim. Gentle traction on the long, untied ends of this suture facilitates alignment of the mucosa and cartilage layers during subsequent suture placement. Close the mucosal layer with 5-0 rapidly absorbable interrupted sutures, and then irrigate the area gently from the outside before completing the repair. Avoid suture placement directly into the cartilage. Complete the repair with 6-0 nonabsorbable monofilament material close to the wound edges. Percutaneous nonabsorbable sutures should be removed in 3 to 5 days.

Drain small unilateral septal hematomas with an 18G needle, or incise the hematoma if it is larger. After the hematoma has been evacuated, place an anterior nasal packing to prevent reaccumulation. Consider a course of oral antibiotics to prevent infection while nasal packing is in place.

■■ EAR

Close superficial lacerations of the ear with 6-0 nonabsorbable monofilament interrupted sutures, taking care to cover any exposed cartilage and minimize skin debridement. In most through-and-through lacerations of the ear, the skin can be approximated and the underlying cartilage will be supported adequately (Fig. 11-3). After laceration repair is complete,

Figure 11-3. Repair of auricular laceration. A. Laceration through auricle. B. Interrupted 6-0 nonabsorbable sutures approximate the skin edges.

11_Cydulka_Ch11.indd 98 15/03/17 11:26 AM

Source page 118

place a small piece of nonadherent gauze over the laceration and apply a

pressure dressing to prevent development of a perichondral hematoma. To create a pressure dressing, place gauze squares behind the ear to apply pressure, and wrap the head circumferentially with gauze. Discontinue the pressure dressing after 24 hours when reevaluation demonstrates that no hematoma has accumulated. Consider consultation with an otolaryngologist or plastic surgeon for complex lacerations, ear avulsions, or auricular hematomas. Remove nonabsorbable percutaneous sutures in 5 to 7 days.

■■ LIPS

The external surface of the lips has three distinct regions: the skin, the vermilion border (junction of the skin and red portion of the lips), and the oral mucosa. Isolated intraoral lesions of the lip often do not need to be sutured, especially if they are <1 cm in length. Repair large or gaping intraoral lip wounds with a rapidly absorbable 5-0 suture. Through-and-through lacerations of the lip that do not include the vermilion border can be closed in layers. Begin repair with 5-0 rapidly absorbable suture for the mucosal surface, then gently irrigate the wound from the outside before closing the orbicularis oris muscle with 4-0 or 5-0 absorbable suture. Finally, close the skin layer with 6-0 nonabsorbable monofilament sutures in a simple interrupted fashion.

Begin closure of a complicated lip laceration at the junction between the vermilion and the skin with a 6-0 nonabsorbable, monofilament suture to precisely align the edges of the vermilion border (Fig. 11-4). After placing that first stitch, continue to repair the vermilion and skin with the same 6-0 nonabsorbable monofilament suture and then use 5-0 rapidly absorbable suture for any needed mucosal surface repair. Consider oral antibiotics for contaminated wounds or through-and-through lacerations. Remove nonabsorbable percutaneous sutures in 5 to 7 days.

A B C

Orbicularis oris muscle

Figure 11-4. Lip laceration repair. A. The first suture is placed to align the vermilion skin junction. B. The orbicularis oris muscle is then repaired with 5-0 absorbable sutures. C. The irregular edges of the skin are then approximated with 6-0 nonabsorbable sutures.

11_Cydulka_Ch11.indd 99 15/03/17 11:27 AM

Source page 119

Temporal nerve

Zygomatic nerve

Buccal nerve

Facial nerve

Parotid duct Parotid

salivary gland

Marginal mandibular

nerve

Cervical nerve

Figure 11-5. Cheek anatomy. The course of the parotid duct is within 1.5 cm of the midportion of a line drawn from the lower border of the tragus to the cheilion. The five branches of the facial nerve are the temporal, zygomatic, buccal, marginal mandibular, and cervical.

■■ CHEEKS AND FACE

Evaluate cheek and other facial lacerations to determine wound depth and to assess for damage to the parotid gland, parotid duct, and facial nerve (Fig. 11-5), which would necessitate specialty consultation and operative repair. Repair superficial facial lacerations without underlying complications with 6-0 nonabsorbable monofilament, simple interrupted sutures. Tissue adhesive may also be used for wounds of appropriate size and location. Full-thickness cheek lacerations can be repaired in layers beginning with the intraoral mucosa using absorbable sutures. After irrigation of the external wound, close the subcutaneous and skin layers using 5-0 or 6-0 suture material. Remove nonabsorbable percutaneous sutures in 5 to 7 days.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 42, "Face and Scalp Lacerations," by Wendy C. Coates.

11_Cydulka_Ch11.indd 100 15/03/17 11:27 AM

References/provenance

- Source: *Tintinalli's Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 116–121.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 12: Injuries to the Arm, Hand, Fingertip, and Nail

Source pages 122–127 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 122–127

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Injuries to the Arm, Hand, Fingertip, and Nail**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 12: Injuries to the Arm, Hand, Fingertip, and Nail 103 Figure 12-1. [Source pages 122–127]

- Extensor tendon laceration repair with a figure-of-eight stitch. [Source pages 122–127]

- Lacerations to the extensor tendons over the distal interphalangeal (DIP) joint may produce a mallet deformity, and if not repaired may result in a swan neck deformity. [Source pages 122–127]

- Lacerations over the proximal interphalangeal joint may produce a boutonniere deformity. [Source pages 122–127]

- Open DIP tendon lacera- tions require operative repair by a hand surgeon; closed tendon injuries should be splinted in extension for up to 6 weeks or until operative re- pair. [Source pages 122–127]

Clinical reasoning structure

- **Problem representation:** Injuries to the Arm, Hand, Fingertip, and Nail; source pages 122–127.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Injuries to the Arm, Hand, Fingertip, and Nail**.

- Source-page range: **122–127**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 122

CHAPTER 12: Injuries to the Arm, Hand, Fingertip, and Nail 103

Figure 12-1. Extensor tendon laceration repair with a figure-of-eight stitch.

3. Lacerations to the extensor tendons over the distal interphalangeal (DIP) joint may produce a mallet deformity, and if not repaired may result in a swan neck deformity. Lacerations over the proximal interphalangeal joint may produce a boutonniere deformity. Open DIP tendon lacerations require operative repair by a hand surgeon; closed tendon injuries should be splinted in extension for up to 6 weeks or until operative repair. Refer these cases to a hand specialist.

12_Cydulka_Ch12.indd 103 20/04/17 4:30 pm

Source page 123

104 SECTION 3: Emergency Wound Management

Volar Forearm, Wrist, and Hand Lacerations

1. Linear lacerations over the volar aspect of the wrist or forearm may raise the possibility of self-inflicted injuries. In appropriate circumstances, question the patient about how the injury occurred and assess risks for possible self-injurious behavior.

2. Most simple lacerations to the volar surface of the forearm and wrist can be repaired with 4-0 or 5-0 nonabsorbable monofilament sutures.

3. Injuries that involve more than one parallel laceration, which may be seen with self-inflicted injuries, can sometimes be repaired with horizontal mattress sutures that cross adjacent lacerations. This can help to prevent compromising the vascular supply of any skin islands located between incisions (Fig. 12-2).

4. Examine the function of tendons and distal nerves individually to assess for possible damage. (See Tables 12-3 and 12-4.)

Palm Lacerations

1. Injuries to the palm can damage deep structures, even those with a benign superficial appearance. Regional anesthesia with a median or ulnar

nerve block may be useful for exploring such wounds.

Figure 12-2. Horizontal mattress sutures for multiple parallel lacerations.

12_Cydulka_Ch12.indd 104 20/04/17 4:30 pm

Source page 124

CHAPTER 12: Injuries to the Arm, Hand, Fingertip, and Nail 105

Table 12-3 Extensor Compartments in the Forearm Compartment

Compartment	Muscle	Function
First compartment	Abductor pollicis longus	Abducts and extends thumb
	Extensor pollicis brevis	Extends thumb at MCP joint
Second compartment	Extensor carpi radialis longus	Extends and radially deviates wrist
	Extensor carpi radialis brevis	Extends and radially deviates wrist
Third compartment	Extensor pollicis longus	Extends thumb at interphalangeal joint
Fourth compartment	Extensor digitorum communis	Splits into four tendons at level of the wrist; extends index, long, ring, and little digits
	Extensor indicis proprius	Extends index finger
Fifth compartment	Extensor digiti minimi	Extends little finger at MCP joint
Sixth compartment	Extensor carpi ulnaris	Extends and radially deviates wrist

Abbreviation: MCP, metacarpophalangeal.

2. If no deep injury is identified, close the wound with sutures to re-oppose the skin creases accurately. Interrupted horizontal mattress sutures with 5-0 monofilament suture can be helpful to promote appropriate skin eversion.

Flexor Tendon Lacerations

1. Refer all flexor tendon injuries to a hand specialist. Many surgeons prefer to repair these injuries within 24 hours after injury.

2. Timely repair by a hand surgeon is important, as post-injury scarring and retraction make flexor tendon repairs more difficult after 10 to 14 days.

Table 12-4 Flexor Tendons in the Forearm

Flexor Tendon	Function
Flexor carpi radialis	Flexes and radially deviates wrist
Flexor carpi ulnaris	Flexes and ulnarly deviates wrist
Palmaris longus	Flexes wrist
Flexor pollicis longus	Flexes thumb at MCP and interphalangeal joints

Flexor digitorum superficialis Flexes index, long, ring, and little digits at MCP and PIP joints

Flexor digitorum profundus Flexes index, long, ring, and little digits at MCP, PIP, and DIP joints

Abbreviations: DIP, distal interphalangeal; MCP, metacarpophalangeal; PIP, proximal interphalangeal.

12_Cydulka_Ch12.indd 105 20/04/17 4:30 pm

Source page 125

106 SECTION 3: Emergency Wound Management

3. If operative repair is delayed, clean the wound and suture the overlying skin. Splint with the wrist and metacarpophalangeal joint flexed and the proximal interphalangeal and DIP joints in extension.

Finger Injuries

1. Many finger lacerations are straightforward and can be repaired by using 5-0 nonabsorbable suture materials.

2. Assess for underlying injuries, such as sensory nerve damage or extensor and flexor tendon injuries.

3. Partial or complete digit amputations should involve consultation with a hand surgeon to discuss whether implantation is a feasible option.

4. Digital nerve injuries are suspected when static two-point discrimination is >10 mm or distinctly greater on one side of the volar pad than the other.

Finger Tip Injuries

1. Successful repair of fingertip injuries requires knowledge of anatomy (Fig. 12-3) and an understanding of techniques of reconstruction.

2. Distal fingertip amputations with skin or pulp loss only are usually managed conservatively, with serial dressing changes.

3. For some distal fingertip amputations without exposed bone, a full-thickness skin graft using the severed tip itself may be considered. For larger injuries, skin can also be harvested from a distant site by a hand specialist with skin graft expertise.

4. Injuries with exposed bone are not amenable to skin grafting, and most of these injuries require specialist advice and management.

Injuries Involving the Nail and Nail Bed

1. Subungual hematomas of >50% of the nail bed are often decompressed by simple trephination of the nail plate. This produces good to excellent

Nail well

Nail fold

Eponychium Ventral floor

Lunula Dorsal roof

Nail bed Insertion of extensor tendon

Distal

interphalangeal

joint

Hyponychium

Periosteum

Insertion of deep

flexor tendon

Figure 12-3. Anatomy of the perionychium.

12_Cydulka_Ch12.indd 106 20/04/17 4:30 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.

- Content ID: SEG-TINTINALLI-EMM8-FULL.

- Source pages: 122–127.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 13: Lacerations to the Leg and Foot

Source pages 128–131 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 128–131

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Lacerations to the Leg and Foot**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 13: Lacerations to the Leg and Foot 109 4. [Source pages 128–131]

- Determine each patient’s tetanus immunization status and update as necessary (see Chapter 94). [Source pages 128–131]

- Cyanoacrylate glue is not used as often for wound repair on the lower extremities because of the greater wound tension associated with these injuries. [Source pages 128–131]

- Once repaired, lacerations involving the joint or tendons should be splinted in a position of function. [Source pages 128–131]

- Evidence does not support routine antibiotic prophylaxis in uncompli- cated lacerations of the lower extremity. [Source pages 128–131]

Clinical reasoning structure

- **Problem representation:** Lacerations to the Leg and Foot; source pages 128–131.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Lacerations to the Leg and Foot**.

- Source-page range: **128–131**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 128

CHAPTER 13: Lacerations to the Leg and Foot 109

4. Determine each patient’s tetanus immunization status and update as necessary (see Chapter 94).

5. Cyanoacrylate glue is not used as often for wound repair on the lower extremities because of the greater wound tension associated with these injuries.

6. Once repaired, lacerations involving the joint or tendons should be splinted in a position of function.

7. Evidence does not support routine antibiotic prophylaxis in uncomplicated lacerations of the lower extremity. Antibiotics are recommended for open fractures, tendon or joint involvement, heavily contaminated wounds, bite wounds, patients with higher risks for infection, and wounds that exhibit early signs of infection.

Knee Injuries

1. Examine knee wounds throughout the range of movement of the joint. Assess patellar and quadriceps tendon function.

2. Evaluate these wounds to determine whether penetration of the joint capsule may have occurred. An x-ray may show air in the joint. Another approach to diagnose joint penetration is to inject 65 to 95 mL of sterile saline into the joint at a site separate from the laceration. Leakage of the solution from the wound indicates joint capsule injury.

3. Assess for potential injury to the popliteal artery, the popliteal nerve, and the tibial nerve.

Ankle Injuries

1. Move the ankle through its full range of motion and directly inspect the wound to ensure there are no tendon injuries. The Achilles tendon, the tibialis anterior, and the extensor hallucis longus in particular are at risk for injury and should be repaired if an injury is identified.

2. The Achilles tendon can rupture without a penetrating injury when a

tensed gastrocnemius is suddenly contracted. The Thompson test can be used to assess the Achilles tendon. Have the patient kneel on a chair or lie supine with feet extending beyond the stretcher (see Fig. 13-1). Squeeze the patient's calf gently at the midpoint. Absent plantar flexion of the foot indicates complete Achilles tendon laceration.

Foot Injures

1. Explore lacerations on the sole of the foot to assess for tendon injury or retained foreign bodies. Place the patient in a prone position with the foot supported on a pillow or overhanging the bed.
2. Regional anesthesia can be helpful to facilitate exploration and repair of lacerations in this area.
3. Repair wounds as soon as possible after injury, ideally within 6 hours to minimize infection risks. Wounds presenting between 6 and 12 hours from the time of injury can often be repaired primarily. Evaluate each wound individually for infection risks.
4. When repairing foot wounds, a large needle is often required to penetrate the thick dermis of the sole. Use nonabsorbable 3-0 or 4-0 suture material to repair wounds located on the sole and 4-0 or 5-0 nonabsorbable sutures to repair the dorsal foot.

13_Cydulka_Ch13.indd 109 15/03/17 11:55 AM

Source page 129

Figure 13-1. Thompson test. "With the patient's feet extending beyond the stretcher, squeeze both calfs at the midpoint. Normal response is shown on the patient's left, absent plantar flexion (abnormal) shown on patient's right."

5. Lacerations between the toes can be difficult to repair and may be facilitated by an assistant to hold the toes apart during repair. An interrupted mattress suture may be helpful to optimize adequate skin apposition.
6. Consider prescribing crutches and/or a walking boot after repair of a foot laceration.
7. Lawn mowers and bicycle spokes may cause extensive soft tissue injury along with underlying fractures and tendon lacerations. Severe injuries may require consultation with an orthopedic specialist.

Hair-Thread Tourniquet Syndrome

1. Hair-thread tourniquet syndrome is an unusual type of injury seen in infants. A long strand of hair or thread wraps around one of the toes and can lead to vascular compromise.
2. Remove the hair or thread completely to relieve neurovascular compromise by unwinding it, if possible. Otherwise, consider making a midline longitudinal incision along the extensor surface of the toe to cut the hair or thread.

■■ DISPOSITION

1. Instruct patients to keep wounds clean and dry.

2. Recommend suture removal in 10 to 14 days for wounds of the lower limb and in 14 days for lacerations over joints.

13_Cydulka_Ch13.indd 110 15/03/17 11:55 AM

Source page 130

CHAPTER 13: Lacerations to the Leg and Foot 111

3. Provide wound care instructions. Elevation of the affected limb can help to reduce edema and may aid healing.

4. Consider recommending a wound recheck after 48 hours for a heavily contaminated wound, when a complex repair is required, or if the patient is at higher risk for wound infections.

5. Crutches may be recommended for 7 to 10 days after a wound repair, as needed, to prevent additional tension on the wound during the initial healing process.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 44, "Leg and Foot Lacerations," by Annabella Salvador-Kelly and Nancy Kwon.

13_Cydulka_Ch13.indd 111 15/03/17 11:55 AM

Source page 131

C h a p ter Soft Tissue Foreign Bodies

14 Michael T. Fitch

Emergency providers are often called upon to evaluate acute wounds that are at risk for embedded foreign bodies. Assessment of older wounds may also be complicated by retained foreign material that was initially unrecognized and can impact the healing process and risk of subsequent infection. Careful evaluation of wounds, radiographic imaging when indicated, and local exploration allow identification of most foreign bodies. Once identified, many foreign bodies can be removed in the emergency department, although some may be left in place or referred to appropriate specialists for delayed removal.

■■ CLINICAL FEATURES

While only a small percentage of lacerations and puncture wounds contain foreign bodies, carefully assess all injuries to evaluate for potentially unrecognized retained material. The mechanism of injury, composition and shape of the wounding object, and the shape and location of the wound are all factors to consider. Objects that have broken, shattered, or splintered at the time of injury may increase the risk of an embedded foreign body. Brittle

materials such as thorns, spikes, or branches may penetrate deeply into tissue before breaking. Wood splinters often fragment when pulled from a puncture wound.

Adult patients who complain of a foreign body sensation in an acute wound more than doubles the likelihood of one being present. Wounds that have healed but continue to be the source of sharp pain with movement or pressure over the site may represent a retained foreign object. Other potential signs of an unrecognized foreign body may include poor wound healing, recurrent infections, or the development of soft tissue masses.

Most foreign bodies can be identified through deliberate and careful exploration of wounds determined to be at risk. Use adequate lighting, appropriate anesthesia, and hemostasis techniques to optimize wound evaluation. Visually inspect all recesses of a wound, when possible, recognizing that wounds deeper than 5 mm or where the full depth cannot be visualized have a higher association with foreign bodies. Consider extending the wound margins with a scalpel to permit a more complete exploration when appropriate. Blind and gentle probing with a closed hemostat can be an effective method to identify some foreign bodies, such as glass fragments, when direct visualization is not possible.

■■ DIAGNOSIS AND DIFFERENTIAL

When a foreign body is suspected but not identified and removed during visual inspection, consider radiographic imaging for further evaluation. Most foreign bodies can be seen on plain radiographs, although CT scan, ultrasound, or MRI may be indicated in some circumstances (Table 14-1). Use an underpenetrated soft tissue plain radiography technique or adjust

14_Cydulka_Ch14.indd 112 15/03/17 12:06 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 128–131.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 14: Soft Tissue Foreign Bodies

Source pages 132–135 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 132–135

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Soft Tissue Foreign Bodies**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 14: Soft Tissue Foreign Bodies 113 Table 14-1 Imaging Modalities for Detection of Soft Tissue Foreign Bodies Material Plain Radiographs High-Resolution US CT MRI Wood Poor Good Moderate Moderate to good Metal Excellent Good Excellent Poor Glass Excellent Good Excellent Good Organic (e.g., Poor Good Good Good plant thorns and cactus spines) Plastic Moderate Moderate to good Good Good Palm thorn Poor Moderate Good Good the contrast and brightness when using a digital system to increase the likelihood of identifying a foreign body. [Source pages 132–135]

- CT scan is capable of detecting more types of materials than plain film radiography, and may be useful for thorns, spines, wood splinters, or plastic foreign bodies. [Source pages 132–135]

- Ultrasound can be useful at the bedside for directing exploration and removal of foreign bod- ies. [Source pages 132–135]

- It is >90% sensitive for detecting foreign bodies larger than 4 to 5 mm taking into account the composition of the foreign body, proximity to echo- genic structures, and operator experience. [Source pages 132–135]

- MRI is more accurate than the other modalities for identifying wood, plastic, spines, and thorns, but is often less available for emergency use. [Source pages 132–135]

Clinical reasoning structure

- **Problem representation:** Soft Tissue Foreign Bodies; source pages 132–135.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Soft Tissue Foreign Bodies**.

- Source-page range: **132–135**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 132

CHAPTER 14: Soft Tissue Foreign Bodies 113

Table 14-1 Imaging Modalities for Detection of Soft Tissue Foreign Bodies

Material Plain Radiographs High-Resolution US CT MRI

Wood Poor Good Moderate Moderate

to good

Metal Excellent Good Excellent Poor

Glass Excellent Good Excellent Good

Organic (e.g., Poor Good Good Good

plant thorns and

cactus spines)

Plastic Moderate Moderate to good Good Good

Palm thorn Poor Moderate Good Good

the contrast and brightness when using a digital system to increase the likelihood of identifying a foreign body. CT scan is capable of detecting more types of materials than plain film radiography, and may be useful for thorns, spines, wood splinters, or plastic foreign bodies. Ultrasound can be useful at the bedside for directing exploration and removal of foreign bodies. It is >90% sensitive for detecting foreign bodies larger than 4 to 5 mm taking into account the composition of the foreign body, proximity to echogenic structures, and operator experience. MRI is more accurate than the other modalities for identifying wood, plastic, spines, and thorns, but is often less available for emergency use.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. Once a foreign body has been identified, consider the risks and benefits of immediate removal, delayed removal, or leaving in place. Consider the potential for infection, toxicity, functional problems, or persistent pain, recognizing that not all foreign bodies require removal. Objects that are small, inert, deeply embedded, and causing no symptoms can

often be left in place. Thorns, wood splinters, spines, vegetative material, or contaminated objects are often best removed promptly to prevent subsequent inflammation, pain, or infection.

2. Techniques for soft tissue foreign body removal are based on clinical experience, as there are no systematic studies comparing different techniques. Many objects can be removed by the emergency provider when facilitated by adequate lighting, hemostasis, and anesthesia. Consider referral to a specialist for complicated or delayed foreign body removal. There is no proven benefit for prophylactic antibiotics for wounds that are not infected.

3. Provide tetanus immunization and appropriate wound care after foreign objects are removed. Primary closure of wounds can be considered after foreign contaminants are removed and the risks of infection are low. Consider delayed primary closure for wounds with higher risks of infection.

4. Superficially embedded metallic needles can be removed with a hemostat after a small incision over one end. Remove more deeply embedded

14_Cydulka_Ch14.indd 113 15/03/17 12:06 PM

Source page 133

needles by making an incision at the midpoint of the needle, grasp it with a hemostat, and push it back out through the entrance wound. For needles perpendicular to the skin, extend an incision and use pressure on the wound edges to expose the needle and facilitate removal.

5. Wood splinters and organic spines can be removed using forceps when they are small and superficially located. Removal for deeper objects can be facilitated by an incision along the long axis of the foreign body when it is parallel to the skin surface. If the object is small and localization is difficult, a small elliptical block of tissue may be removed and the remaining wound closed primarily. Remove subungual splinters using traction or by excising a portion of nail over the splinter. Cactus spines embedded in the dermis may be removed individually or using an adhesive such as facial gel, rubber cement, or household glue applied to the skin and removed when dry.

6. Fishing hooks can be removed using one of several described techniques after providing adequate anesthesia to the skin around the site of entry. Superficial fishing hooks can be removed using a retrograde technique of gentle downward pressure on the shank while the hook is pulled back out the path of entry. The string pull method uses one hand to depress the shank of the hook to disengage the barb while the other hand gives

a quick tug on a string that has been wrapped around the bend in the hook. When the advance-and-cut technique is used, the point and the barb of the fishhook are pushed through the skin and clipped with wire cutters, and the remaining part of the hook is threaded back through the original wound. The incision technique is nearly always successful, and involves enlarging the entrance wound to expose the bend in the hook and the barb until disengaged from the subcutaneous tissue. This technique allows for easier wound exploration and cleaning.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 45, "Soft Tissue Foreign Bodies," by Richard L. Lammers.

14_Cydulka_Ch14.indd 114 15/03/17 12:06 PM

Source page 134

C h a pter Puncture Wounds and Bites

15 Michael T. Fitch

■■ PUNCTURE WOUNDS

Puncture wounds can be challenging to manage due to difficulty visualizing and cleaning the full depth of an injury. These injuries commonly occur to the extremities, such as on the plantar surface of the foot. Puncture wounds also include injuries caused by high-pressure injection equipment, health care associated needle-stick injuries, and some bite wounds from animals. Infections are reported in 6% to 11% of puncture wounds, with Staphylococcus aureus predominating (including methicillin-resistant S. aureus—MRSA). Pseudomonas aeruginosa is the most common pathogen in post-puncture wound osteomyelitis, particularly when penetration occurs through the sole of an athletic shoe. Post-puncture wound infections despite treatment with antibiotics suggest the possibility of a retained foreign body.

Clinical Features

Puncture wounds treated more than 6 hours after they occur have a greater risk of infection. Wood, glass, or plastic materials may break or splinter when an injury occurs, increasing the chance for retained fragments within the puncture track. Patient perception of a foreign body may be useful for predicting the presence of such a contaminant or fragment.

On physical examination, assess the wound and the likelihood of injury to structures beneath the skin by evaluating distal function of tendons, nerves, and blood vessels. Inspect the wound location, condition of surrounding skin, and the potential presence of foreign matter, debris, or devitalized tissue. Signs of infection include significant pain, swelling, erythema, warmth, fluctuance, decreased range of motion, or drainage from the site.

Diagnosis and Differential

Plain film radiographs should be obtained for any wound suspected of having a potential radiopaque retained foreign body (Table 15-1).

Table 15-1 Indications for Imaging in Puncture Wounds

Plain radiographs

Suspicion of fracture

Infected wound

Wound caused by materials prone to fragment (wood, glass, etc.)

Foreign body sensation reported by patient

CT or MRI

Suspected deep-space infection

Persistent pain after injury

Failure to respond to treatment

115

15_Cydulka_Ch15.indd 115 15/03/17 12:12 PM

Source page 135

116 SECTION 3: Emergency Wound Management

Organic substances, such as wood or plant matter, are not reliably detected by plan radiographs, but >90% of radiopaque foreign bodies that are >1 mm in diameter can be found this way. CT scan is the imaging modality to use when a retained foreign body continues to be suspected after negative plain film radiography.

Emergency Department Care and Disposition

Treatment recommendations for puncture wounds are based almost entirely on anecdotal evidence and uncontrolled case series.

1. Uncomplicated, clean puncture wounds less than 6 hours after injury and without foreign body should have superficial wound cleansing and tetanus prophylaxis as indicated. Soaking has no proven benefit. Debridement or coring of the wound tract does not reduce rates of infection. There is no proven benefit of prophylactic antibiotics for any type of puncture wound, and therefore routine use of antibiotics for healthy patients with clean puncture wounds is not recommended.

2. High-risk patients with plantar puncture wounds are recommended to receive prophylactic antibiotics along with superficial wound cleansing and tetanus prophylaxis. Consider antibiotics for diabetic patients or patients with other sources of impaired host defenses, forefoot puncture wound injuries, or punctures through athletic shoes. A first-generation cephalosporin, anti-staphylococcal penicillin, or macrolide are suitable antibiotics for most puncture wounds where antibiotics are prescribed. Consider an oral fluoroquinolone with anti-pseudomonal activity, such as ciprofloxacin 500

mg twice daily, for plantar puncture wounds through athletic shoes.

3. Cellulitis is a complication of puncture wounds that is most often localized without significant drainage and develops within the first 4 days after injury. This can usually be treated with a 7- to10-day course of a first-generation cephalosporin, anti-staphylococcal penicillin, trimethoprim-sulfamethoxazole, or clindamycin.

4. A local abscess may develop at the puncture site, especially if a foreign body remains. Treatment includes incision, drainage, and careful exploration for a retained foreign body. Antibiotics are typically only required if there is surrounding cellulitis.

5. Osteomyelitis is an uncommon but serious complication of puncture wounds and may have a greater prevalence in forefoot punctures through athletic shoes. Symptoms typically present more than 7 days after injury and diagnosis can be challenging as radiographs are normal in early stages. Elevations of erythrocyte sedimentation rate or C-reactive protein may support the diagnosis. MRI is the most commonly used imaging modality from the emergency department to help identify this complication. Surgical consultation with discussion of timing of IV antibiotics and admission for surgical treatment is important when osteomyelitis is diagnosed.

■■ NEEDLE-STICK INJURIES

Needle-stick injuries in health care professionals create the most concerns about infection risk from potential exposure to hepatitis viruses or human immunodeficiency virus (HIV). The risk of infection after an inadvertent

15_Cydulka_Ch15.indd 116 15/03/17 12:12 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 132–135.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 15: Puncture Wounds and Bites

Source pages 136–141 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 136–141

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Puncture Wounds and Bites**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 15: Puncture Wounds and Bites 117 needle stick contaminated from an infectious source has been estimated to be negligible for hepatitis A, 6% for hepatitis B, 2% for hepatitis C, and 0.3% for HIV. [Source pages 136–141]

- Because recommendations in this area are complex and evolving, each hospital should have a predesigned protocol developed by infectious disease specialists for the evaluation, testing, and treatment of needle-stick injuries, including hepatitis B and HIV prophylaxis. [Source pages 136–141]

- ■■ HIGH-PRESSURE-INJECTION INJURIES High-pressure-injection injuries may present as puncture wounds, usually to the hand or foot. [Source pages 136–141]

- These are caused by industrial equipment that forces grease, paint, or other liquids through a small nozzle under high pressure. [Source pages 136–141]

- Patients initially may have pain with minimal swelling, but these injuries can still be severe as the injected liquid spreads along fascial planes and causes a delayed inflammatory response. [Source pages 136–141]

Clinical reasoning structure

- **Problem representation:** Puncture Wounds and Bites; source pages 136–141.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Puncture Wounds and Bites**.

- Source-page range: **136–141**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 136

CHAPTER 15: Puncture Wounds and Bites 117

needle stick contaminated from an infectious source has been estimated to be negligible for hepatitis A, 6% for hepatitis B, 2% for hepatitis C, and 0.3% for HIV. Because recommendations in this area are complex and evolving, each hospital should have a predesigned protocol developed by infectious disease specialists for the evaluation, testing, and treatment of needle-stick injuries, including hepatitis B and HIV prophylaxis.

■■ HIGH-PRESSURE-INJECTION INJURIES

High-pressure-injection injuries may present as puncture wounds, usually to the hand or foot. These are caused by industrial equipment that forces grease, paint, or other liquids through a small nozzle under high pressure. Patients initially may have pain with minimal swelling, but these injuries can still be severe as the injected liquid spreads along fascial planes and causes a delayed inflammatory response. Pain can be controlled with parenteral analgesics while avoiding digital blocks with local anesthesia as these can further increase tissue pressure. An appropriate hand specialist should be consulted for consideration of early surgical debridement of these injuries.

■■ EPINEPHRINE AUTOINJECTOR INJURIES

These injuries typically occur when a patient inadvertently injects epinephrine into the hand or finger while attempting to treat an allergic reaction. Patients present with pain due to the needle stick, paresthesias, and epinephrine-induced vasospasm to the injected area. In the case of finger injections, the entire digit may be blanched and cold. There is no clear evidence that active treatment is better than observation alone, so many patients can be effectively treated with supportive care. Subcutaneous phentolamine injected into the original puncture site is the one treatment consistently described that rapidly reverses digital ischemia from accidental epinephrine injection. A mixture of 0.5 mL of standard phentolamine

solution (5 mg/mL concentration) and 0.5 mL of 1% lidocaine solution will produce a 1 mL total volume containing 2.5 mg of phentolamine that can be subcutaneously injected directly through the site of an autoinjector puncture. Patients can be discharged home once ischemia is resolved, as relapse appears unlikely.

■■ MAMMALIAN BITES

Clinical Features

Complications from bite wounds include tissue injury from the bite itself, local infections, or systemic illness. All wounds should be examined and cleaned well with irrigation and debridement as necessary. Evaluate for underlying injury to joints or tendons. Patient evaluation should also include assessment for the potential need for tetanus and/or rabies prophylaxis (see Chapter 94).

Diagnosis and Differential

Bite wounds with low risks for infection can often be repaired primarily. Face or scalp wounds without devitalized tissue that have no underlying fractures can be candidates for primary closure in patients with normal

15_Cydulka_Ch15.indd 117 15/03/17 12:12 PM

Source page 137

immune systems and wound healing properties. Delayed primary closure can be considered for wounds at higher risk of infection.

Emergency Department Care and Disposition

1. Consider primary closure for low-risk wounds after careful assessment, cleaning, and irrigation.
2. Avoid primary closure in patients with immunodeficiency or comorbidities that may impact successful wound healing.
3. High-risk wounds should be cleaned, debrided, and dressed to maintain a moist environment for healing. Recommend wound reevaluation for signs of infection in 24 to 48 hours.
4. Prophylactic antibiotics are recommended for human bites and for higher risk wounds, particularly if primary wound closure is required (Table 15-1).

■■ INFECTIONS FROM CAT AND DOG BITES

Clinical Features

Most dog bite wounds are superficial and have a relatively low risk (5%) of infection. Cat bites tend to be deeper puncture wounds and are reported to become infected in up to 80% of patients who present for emergency care. However, the true rate of infection is likely much lower than this because most patients with cat bites do not present for care unless the bite appears

severe or already has signs of infection.

Diagnosis and Differential

Despite a lack of evidence that antibiotics are effective for reducing the rates of infection after dog or cat bites, prophylactic antibiotics are often suggested for treatment of higher risk uninfected wounds. Many clinicians recommend antibiotics for cat bites, bites in immunocompromised patients, deep dog bite puncture wounds, hand wounds, or wounds needing debridement.

Emergency Department Care and Disposition

1. Common first-line antibiotics can be used when prophylactic antibiotics are desired (Table 15-2). Amoxicillin-clavulanate, 500 to 800 mg by mouth twice daily, is the antibiotic most commonly recommended for treatment of local infections after dog or cat bites.
2. Penicillin V 500 mg four times daily, or ampicillin 500 mg four times daily, may be adequate to cover Pasteurella multocida infections from cat bites and may be a lower cost alternative for prophylaxis of cat bites.
3. Cephalexin, dicloxacillin, erythromycin, or clindamycin alone are likely not sufficient for dog or cat bites, as they do not reliably provide coverage for Pasteurella species.
4. Serious systemic infections after dog or cat bites are rare, but can develop days after the injury. Capnocytophaga canimorsus produces a rare but fulminant bacteremic illness after a dog bite and is more common in alcoholics, post-splenectomy, or other immunosuppressed patients.

15_Cydulka_Ch15.indd 118 15/03/17 12:12 PM

Source page 138

Table 15-2 Common Bites and First-Line Treatment

Animal	Organism	First-Line Antibiotic
Cat	Pasteurella multocida	Amoxicillin-clavulanate
	Bartonella henselae (cat-scratch fever)	Azithromycin
Dog	Pasteurella, strepto-cocci, staphylococci, Capnocytophaga canimorsus	Amoxicillin-clavulanate
Human	Eikenella, staphylococci, streptococci	Amoxicillin-clavulanate
	Herpes simplex (herpetic whitlow)	Acyclovir or valacyclovir
Rats, mice, squirrels, gerbils	Streptobacillus moniliformis	Amoxicillin-clavulanate

(North America) or Spirillum
minus/minor (Asia)
Livestock, large game Multiple organisms Amoxicillin-clavulanate or
animals Brucella, Leptospira, specific agent for disease
Francisella tularensis
Bats, monkeys, dogs, Rabies Rabies immune globulin,
skunks, raccoons, foxes (all rabies vaccine
carnivores and omnivores)
Monkeys Herpes B virus Acyclovir or valacyclovir
(Cercopithecine herpesvirus)
Freshwater fish Aeromonas, staphylococci, Fluoroquinolone or trime-
streptococci thoprim-sulfamethoxazole
Saltwater fish Vibrio, staphylococci, Fluoroquinolone
streptococci

Broad-spectrum antibiotic coverage and appropriate resuscitative efforts are indicated with any such systemic infection.

5. Cat-scratch disease is characterized by regional lymphadenopathy 7 to 12 days after a cat bite or scratch, and is caused by Bartonella henselae. Antibiotics are not usually indicated as patients can be managed with pain relief and symptomatic care. Patients with severe painful lymph-adenopathy may benefit from a 5-day course of azithromycin. Immu-nosuppressed patients can be treated with a 7- to 10-day course of trimethoprim-sulfamethoxazole, ciprofloxacin, or rifampin.

■■ HUMAN BITES

Human bite wounds are at higher risk for polymicrobial infection with staphylococcal and streptococcal species or Eikenella corrodens. Treatment with prophylactic amoxicillin-clavulanate is recommended and typically suture repair is not advised with the possible exception of facial wounds. Infections requiring parenteral antibiotics can be treated with ampicillin-sulbactam, cefoxitin, or piperacillin-tazobactam. Patients with closed fist injuries, or “fight bite” wounds, to the knuckle caused by impact with

15_Cydulka_Ch15.indd 119 15/03/17 12:12 PM

Source page 139

another individual’s teeth may require surgical treatment (see Chapter 12, Injuries to the Arm, Hand, Fingertip, and Nail).

■■ RODENTS, LIVESTOCK, AND EXOTIC AND WILD ANIMALS

Rodent bites are typically trivial, rodents are not known to carry rabies, and these bites have a low risk for infection. Most patients can be treated con-

servatively with standard wound care. Livestock and large game animals can cause serious injury with a significant risk of infection and systemic illness caused by brucellosis, leptospirosis, or tularemia (see Chapters 97 and 98). Aggressive wound care and broad-spectrum prophylactic antibiotics are recommended.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 46, "Puncture Wounds and Bites," by James Quinn

15_Cydulka_Ch15.indd 120 15/03/17 12:12 PM

References/provenance

- Source: *Tintinalli's Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 136–141.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 16: Postrepair Wound Care

Source pages 142–145 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 142–145

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Postrepair Wound Care**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 16: Postrepair Wound Care 123 ■■ WOUND CLEANSING Wounds that have been repaired with sutures or staples may be cleansed as early as 8 hours after closure without increasing the risk of wound infec- tion. [Source pages 142–145]

- Clean the wound area with soap and water and examine for signs of infection daily. [Source pages 142–145]

- Application of topical antibiotics for the first 3 to 5 days may decrease scab formation and help to prevent wound edge separation. [Source pages 142–145]

- Patients with wounds closed with tissue adhesives may shower, but should not completely immerse the wound or apply topical antibiotics, as this will loosen the adhesive bond and cause earlier sloughing of the adhesive. [Source pages 142–145]

- ■■ WOUND DRAINS Drains or wound packing may be placed in healing wounds to remove interstitial fluid or blood, keep an open tract for drainage of infectious material, or to prevent an abscess from forming by allowing drainage from a contaminated area. [Source pages 142–145]

Clinical reasoning structure

- **Problem representation:** Postrepair Wound Care; source pages 142–145.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Postrepair Wound Care**.

- Source-page range: **142–145**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 142

CHAPTER 16: Postrepair Wound Care 123

■■ WOUND CLEANSING

Wounds that have been repaired with sutures or staples may be cleansed as early as 8 hours after closure without increasing the risk of wound infection. Clean the wound area with soap and water and examine for signs of infection daily. Application of topical antibiotics for the first 3 to 5 days may decrease scab formation and help to prevent wound edge separation. Patients with wounds closed with tissue adhesives may shower, but should not completely immerse the wound or apply topical antibiotics, as this will loosen the adhesive bond and cause earlier sloughing of the adhesive.

■■ WOUND DRAINS

Drains or wound packing may be placed in healing wounds to remove interstitial fluid or blood, keep an open tract for drainage of infectious material, or to prevent an abscess from forming by allowing drainage from a contaminated area. Ribbon gauze packing is often used to pack an abscess cavity after incision and drainage, and can be changed regularly as long as the wound continues to drain. Closed drainage systems have largely replaced open wound drains placed after surgical procedures, as closed systems prevent secondary bacterial contamination into the wound. Provide appropriate follow-up instructions for patients with drains or packing in place.

■■ PAIN CONTROL

Educate patients about the expected degree of pain associated with a wound and the measures they can take to reduce that pain. Splints may help to reduce pain and swelling in significant extremity lacerations. Analgesics may be needed for a short time period, although narcotic analgesia is rarely necessary after the first 48 hours of healing for most wounds.

■■ FOLLOW-UP

Provide instructions for regular wound examination during the healing

process and when to return for suture or staple removal. Recommend reevaluation in 24 to 48 hours for patients with high-risk wounds or comorbidities that may impair successful wound healing. Facial sutures are typically removed in 3 to 5 days, while sutures in most other areas of the body can be removed after 7 to 10 days of healing. Sutures in some locations such as the hands, feet, buttocks, or over joints should remain in place for 10 to 14 days to reduce the chance of wound dehiscence after suture removal.

When removing sutures or adhesive tapes, take care to avoid tension perpendicular to the wound that could cause dehiscence. When needed, gently debride scab or crusting over each suture with hydrogen peroxide–soaked gauze. Grasp the suture knot with forceps and use scissors or an appropriate scalpel blade to cut the suture. Remove skin staples with an appropriate device that deforms the center and extracts the legs from the skin. For wounds closed with tissue adhesive, advise patients to avoid rubbing, picking, scrubbing, or exposing the area to water for more than brief periods of time during the healing process. Tissue adhesive will slough off

16_Cydulka_Ch16.indd 123 15/03/17 12:19 PM

Source page 143

itself over time, typically within 5 to 10 days of application, as wound healing progresses.

■■ PATIENT EDUCATION ABOUT LONG-TERM COSMETIC OUTCOME

Inform patients that all traumatic lacerations result in some degree of tissue scarring and that the short-term cosmetic appearance may not be predictive of the ultimate cosmetic outcome. Instruct patients to avoid sun exposure while wounds are healing, as wounds exposed to the sun can develop permanent hyperpigmentation. Patients should wear sunblock for at least 6 to 12 months after injury to optimize the cosmetic outcome.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 47, "Postrepair Wound Care," by Adam J. Singer and Judd E. Hollander.

16_Cydulka_Ch16.indd 124 15/03/17 12:19 PM

Source page 144

SECTION

Cardiovascular Diseases 4

C h a p ter Chest Pain: Cardiac or Not

17 Andrew Nyce

Millions of patients present to emergency departments (EDs) each year with acute nontraumatic chest pain. Varied clinical presentations coupled with a wide differential diagnosis make patients with chest pain some of the most challenging cared for by emergency care providers. An organized approach will assist clinicians when differentiating acute coronary syndrome (ACS) from other causes of chest pain.

■■ CLINICAL FEATURES

Classically described cardiac chest pain is retrosternal in the left anterior chest with crushing, tightness, squeezing, or pressure that is often brought on by exertion and relieved with rest. Patients may also complain of dyspnea, diaphoresis, and nausea with pain radiating to the left shoulder, jaw, arm, or hand. Some patients, such as premenopausal and early menopausal women, racial minorities, diabetics, the elderly, and patients with psychiatric disease or altered mental status, may have nonclassic presentations of ACS that may or may not be associated with chest pain. Patients with acute myocardial infarction (AMI) who present without chest pain have diagnostic and treatment delays and a higher mortality rate compared to AMI patients who do have chest pain. The onset of symptoms attributed to cardiac disease may be sudden or gradual, and traditionally angina pain lasts 2 to 10 minutes, unstable angina lasts 10 to 30 minutes, and AMI pain often lasts longer than 30 minutes. Dyspnea at rest or with exertion, nausea, light-headedness, generalized weakness, acute changes in mental status, diaphoresis, or shoulder, arm, or jaw discomfort may be the only presenting symptoms for ACS for some patients.

Cardiac risk factors are useful in predicting coronary artery disease in patient populations but may be less useful when applied to an individual patient. Cocaine abuse and HIV infection can accelerate atherosclerosis. Classic symptoms such as radiation of pain to the arms, an exertional component, associated diaphoresis, nausea, and vomiting increase the likelihood that a patient is suffering from an AMI while other symptoms such as pain that is pleuritic in nature, positional, sharp and reproducible with

17_Cydulka_Ch17.indd 125 01/05/17 10:23 am

Source page 145

palpation/positioning decrease the likelihood of disease. Unfortunately, there is no identifiable symptom complex that definitively rules in or out the disease without objective testing.

■■ DIAGNOSIS AND DIFFERENTIAL

Patients with ACS often have a normal physical exam but may present with abnormal vital signs. Tachycardia may result from increased sympathetic tone and decreased left ventricular stroke volume. Bradycardia may result from ischemia to the conduction system. The degree of hemodynamic instability is dependent on the amount of myocardium at risk, associated dysrhythmias, or preexisting valvular or myocardial dysfunction. Patients with acute ischemia may have a third or fourth heart sound from changes in ventricular compliance, a new murmur from ruptured cordae tendineae or an aortic root dissection, or crackles on lung auscultation from congestive heart failure. Chest wall tenderness has been demonstrated in up to 15% of patients with AMI, making this physical examination finding unlikely to be useful by itself to completely exclude the possibility of ACS. Response to a particular treatment such as nitroglycerin or a “gastrointestinal (GI) cocktail” poorly discriminates between cardiac and noncardiac chest pain in isolation.

A detailed history and physical exam allows an emergency provider to risk stratify serious pathology in the differential diagnosis of a chest pain patient. Focused diagnostic testing should be performed based on the likelihood of serious pathology. When history and exam make ACS a likely potential cause, initial testing commonly includes an electrocardiogram (ECG), chest x-ray, and cardiac biomarkers.

Electrocardiography

Guidelines recommend ECG screening within 10 minutes of ED arrival for patients with chest pain of potential cardiac etiology. A normal ECG lacks the sensitivity to exclude ACS in isolation; notably patients with unstable angina or non-ST-segment elevation MI may have nonspecific or normal ECG tracings. ECGs demonstrating new ST-segment elevations ≥1 mm in two contiguous leads represent acute MI that will benefit from rapid reperfusion interventions. New ST-segment elevations, Q waves, left bundle branch block, and T-wave inversions or normalizations in symptomatic patients are suggestive of ischemia and should be further investigated. Misinterpretation of ECGs occurs in up to 40% of missed AMI cases, and the diagnostic value of an ECG can be improved by comparing it to a prior ECG and/or repeating it during the patient evaluation process.

Imaging

Chest radiography is commonly performed for patients with chest pain. Patients with ACS will most often have a normal chest x-ray, but images are useful to evaluate for other diagnostic possibilities such as thoracic aortic

aneurysm, aortic dissection, pneumonia or pneumothorax. Chest CT may be useful to diagnose other possible conditions such as pulmonary embolism or aortic dissection.

17_Cydulka_Ch17.indd 126 01/05/17 10:23 am

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 142–145.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 17: Chest Pain: Cardiac or Not

Source pages 146–151 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 146–151

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Chest Pain: Cardiac or Not**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 17: Chest Pain: Cardiac or Not 127 Serum Markers Due to its high sensitivity and nearly complete cardiac specificity, cardiac troponin (cTn) is the biomarker of choice for the detection of myocardial injury. [Source pages 146–151]

- Serum cTn should be obtained in patients with suspected ACS. [Source pages 146–151]

- While other clinical conditions such as aortic dissection, pulmonary embo- lism, acute CHF, aortic valve disease, and cardiac procedures can also be associated with an elevated troponin level, acute myocardial ischemia can be differentiated from nonischemic troponin elevations based on the pattern of elevation and the clinical context. [Source pages 146–151]

- With contemporary assays, cTn is detected as early as 2 hours after symptom onset in AMI patients but may not be reliably elevated until 6 hours after symptoms. [Source pages 146–151]

- Elevations peak at 48 hours from symptom onset and may remain elevated for 10 days. [Source pages 146–151]

Clinical reasoning structure

- **Problem representation:** Chest Pain: Cardiac or Not; source pages 146–151.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Chest Pain: Cardiac or Not**.

- Source-page range: **146–151**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 146

CHAPTER 17: Chest Pain: Cardiac or Not 127

Serum Markers

Due to its high sensitivity and nearly complete cardiac specificity, cardiac troponin (cTn) is the biomarker of choice for the detection of myocardial injury. Serum cTn should be obtained in patients with suspected ACS. While other clinical conditions such as aortic dissection, pulmonary embolism, acute CHF, aortic valve disease, and cardiac procedures can also be associated with an elevated troponin level, acute myocardial ischemia can be differentiated from nonischemic troponin elevations based on the pattern of elevation and the clinical context. With contemporary assays, cTn is detected as early as 2 hours after symptom onset in AMI patients but may not be reliably elevated until 6 hours after symptoms. Elevations peak at 48 hours from symptom onset and may remain elevated for 10 days. Patients with early presentation (within 6 hours of symptom onset) or those with intermittent symptoms should have serial measurements over time. A single cTn may be sufficient to exclude AMI for constant symptoms >8 to 12 hours. The measurement of cTn at short time intervals (delta cTn) is more sensitive for AMI than a single cTn. Patients with renal failure often have elevation of cTn and comparing baseline measures can assist the provider in risk stratifying the patient. Although almost obsolete, creatine kinase-MB testing may be useful in a subset of patients in whom the timing of infarction remains unclear.

Differential Diagnosis

Common causes of chest pain are listed in Table 17-1.

The classic symptoms of life-threatening causes of acute chest pain are listed in Table 17-2.

Pulmonary Embolism

Patients with pulmonary embolism often complain of sudden onset pleuritic chest pain associated with dyspnea, tachypnea, tachycardia, or hypoxemia. Risk factors include prolonged immobilization, active cancer, recent surgery

or trauma, procoagulant syndromes, exogenous estrogen, or previous thromboembolic disease. Clinical decision aids such as the Wells and Revised Geneva Scores and the Pulmonary Embolism Rule-Out Criteria (PERC) can help risk stratify patients for diagnostic evaluation. Serum D-dimer may be a useful test in low-risk patients to help determine which patients need further

Table 17-1 Common Causes of Acute Chest Pain

Visceral Pain	Pleuritic Pain	Chest Wall Pain
Typical angina	Pulmonary embolism	Costosternal syndrome
Unstable angina	Pneumonia	Costochondritis (Tietze's syndrome)
Acute myocardial infarction	Spontaneous pneumothorax	Precordial catch syndrome
Aortic dissection	Pericarditis	Xiphodynia
Esophageal rupture	Pleurisy	Radicular syndromes
Esophageal reflux or spasm		Intercostal nerve syndromes
Mitral valve prolapse		Fibromyalgia

17_Cydulka_Ch17.indd 127 01/05/17 10:23 am

Source page 147

Table 17-2 Classic Symptoms of Potentially Life-Threatening Causes of Chest Pain*

Disorder	Pain Location	Pain Character	Radiation and Associated Signs and Symptoms
Acute coronary syndrome	Retrosternal, L chest, or epigastric	Crushing, tight- ness, squeezing, pressure	R or L shoulder, R or L arm/ hand, jaw	Dyspnea, diaphoresis, nausea
Pulmonary embolism	Focal chest	Pleuritic	None	Tachycardia, tachypnea, hypoxia, may have hemoptysis
Aortic dissection	Midline, substernal	Ripping, tearing	Intrascapular area of back	Secondary arterial branch occlusion
Pneumonia	Focal chest	Sharp, pleuritic	None	Fever, hypoxia, may see signs of sepsis
Esophageal rupture after forceful vomiting	Substernal	Sudden, sharp, resis, may see signs of sepsis	Back	Dyspnea, diaphoresis,
Pneumothorax	One side of chest	Sudden, sharp, lancinating, pleuritic	Shoulder, back	Dyspnea

Pericarditis Substernal Sharp, constant Back, neck, Fever, pericardial
or pleuritic shoulder friction rub
Perforated Epigastric Severe, sharp Back, up into Acute distress,
peptic ulcer chest diaphoresis
A typical presentations are common.
*
Abbreviations: L, left; R, right.
testing with CT pulmonary angiography to identify pulmonary embolism. For more details, see Chapter 25, “Thromboembolism.”

Aortic Dissection

Patients with aortic dissection classically describe a sudden onset of severe, tearing pain radiating to the intrascapular area of the back. Secondary symptoms result from arterial branch occlusions and may include ischemic stroke, AMI, or limb ischemia. Risk factors include male sex, age >50, uncontrolled hypertension, connective tissue disorders, cocaine use, bicuspid valve or aortic valve replacement, and pregnancy. Physical exam findings may include unilateral pulse deficits or focal neurologic deficits. If aortic dissection is suspected, obtain a CT aortogram or transesophageal echocardiogram. A normal chest x-ray and negative D-dimer lowers the probability but does not completely exclude the diagnosis of an aortic dissection. Additionally, this diagnosis can be associated with nonspecific ST segment or T-wave changes on ECG testing. For more details see Chapter 27, “Aortic Dissection and Aneurysms.”

Esophageal Rupture (Boerhaave’s Syndrome)

Patients with esophageal rupture often present with sudden-onset, sharp substernal chest pain that follows an episode of forceful vomiting. Patients are usually ill appearing with tachycardia, fever, dyspnea, and diaphoresis.

17_Cydulka_Ch17.indd 128 01/05/17 10:23 am

Source page 148

Physical examination may reveal crepitus in the neck or chest from subcutaneous emphysema, while audible crepitus on cardiac auscultation (Hamman’s crunch) is a rare finding. A chest x-ray may be normal or demonstrate a pleural effusion (left more common), pneumothorax, pneumomediastinum, pneumoperitoneum, or subcutaneous air. Diagnosis is made via CT of chest with oral water-soluble contrast. For more details see Chapter 40, “Esophageal Emergencies.”

Spontaneous Pneumothorax

A spontaneous pneumothorax causes sudden-onset, sharp, pleuritic chest pain with dyspnea. These classically occur in tall, slender male patients and risk factors include smoking, COPD, and asthma. Physical exam findings

are inconsistent but auscultation may reveal decreased breath sounds on the affected side. The diagnosis is made by chest x-ray with only a small percentage of patients progressing to develop a tension pneumothorax. See more details in Chapter 32, “Spontaneous and Iatrogenic Pneumothorax.”

Acute Pericarditis

Pain from acute pericarditis is typically sharp, severe, constant, and retrosternal that radiates to the back, neck, or jaw. Pain is classically worsened by lying supine and is relieved by sitting forward. The presence of a pericardial friction rub supports the diagnosis. ECG may show PR-segment depressions, diffuse ST-segment elevations, or T-wave inversions that are typically diffuse. See more details in Chapter 24, “The Cardiomyopathies, Myocarditis, and Pericardial Disease.”

Musculoskeletal Causes

Chest pain due to irritation or inflammation of structures in the chest wall is commonly seen in the ED. Possible causes include costochondritis, xiphodynia (inflammation of the xiphoid process), precordial catch syndrome, intercostal strain due to coughing, and pectoralis muscle strain in the setting of recent physical exertion. Patients often complain of sharp pain that is worsened with movement of the chest wall and palpation. While chest wall tenderness is also present in some patients with ACS or other significant disease processes, a clear musculoskeletal etiology with completely reproducible pain in a patient without other symptoms or risk factors support this diagnosis.

Gastrointestinal Causes

Gastrointestinal disorders such as esophageal reflux, dyspepsia syndromes, and esophageal motility disorders often cannot be reliably differentiated from ACS by history and physical exam alone. Symptoms can range from a gnawing or burning pain in the lower chest with gastritis to postprandial dull, boring pain in the epigastric region with peptic ulcer disease. Esophageal spasm is often associated with reflux disease and is characterized by a sudden onset of dull or tight substernal chest pain. The pain is typically precipitated by drinking cold liquids and can be relieved by nitroglycerin. Clinicians should determine whether the symptoms are due to a GI disorder based on the clinical presentation and the absence of findings and/or risk factors suggesting an ischemic cause. Diagnostic decisions should not be made solely on the basis of a response to a therapeutic trial of antacids,

17_Cydulka_Ch17.indd 129 01/05/17 10:23 am

Source page 149

GI cocktails, or nitroglycerin. See more details in Chapter 40, "Esophageal Emergencies” and Chapter 41, “Peptic Ulcer Disease and Gastritis.”

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. Patients with abnormal vital signs, concerning ECG findings, or clinical history concerning for an acute cardiac event should be promptly evaluated and treated according to immediate airway, breathing, or circulation needs.
2. Place patients with suspicious histories for serious pathology on cardiac monitors, establish IV access, and provide supplemental oxygen as necessary. Vital signs and pulse oximetry should be monitored at regular intervals.
3. Perform an ECG within 10 minutes in patients for whom there is a reasonable suspicion of myocardial ischemia.
4. Obtain a focused history to elicit potential features of life-threatening causes of chest pain such as ACS, aortic dissection, pulmonary embolism, severe pneumonia, and esophageal rupture. Ask about onset, timing, severity, radiation and character of the chest pain; alleviating and exacerbating factors; and presence of associated symptoms such as vomiting, diaphoresis, and dyspnea.
5. Perform physical examination of the thorax for prior surgical incisions, chest wall deformities, and symmetric rise and fall of the chest. Palpate for tenderness, masses, or crepitus. Auscultate to identify chest consolidation or pneumothorax, murmurs, gallops, or friction rubs.
6. Administer aspirin for patients at risk for AMI (see Chapter 18, “Acute Coronary Syndromes: Management of Myocardial Infarction and Unstable Angina,” for details and additional medications).
7. Consider serum laboratory studies including cardiac troponin levels when concern for ACS remains after history and physical examination.
8. Consider chest radiography for patients with acute chest pain where the etiology remains uncertain after history and physical examination. Evaluate the radiograph for findings of pneumonia, pneumothorax, aortic dissection, pneumomediastinum, or other diagnoses.
9. ED treatment and disposition is contingent on the suspected etiology of the patient’s chest pain. Patients in whom ACS cannot be reliably excluded after ED evaluation and diagnostic workup may require further diagnostic evaluation in the hospital.
10. The Thrombosis in Myocardial Infarction (TIMI) risk score or Global Registry of Acute Coronary Events score can aid clinicians when risk stratifying patients for ACS. However, a low-risk score is not sensitive enough to reliably exclude ACS or identify patients for early discharge without further evaluation.
11. If the etiology of chest pain remains unclear after the initial ED

workup, clinicians should consider further testing and observation or

admission as guided by clinical suspicion and findings.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study

Guide, 8th ed., see Chapter 48, “Chest Pain” by Simon A. Mahler.

17_Cydulka_Ch17.indd 130 01/05/17 10:23 am

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.

- Content ID: SEG-TINTINALLI-EMM8-FULL.

- Source pages: 146–151.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 18: Acute Coronary Syndromes

Source pages 152–157 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 152–157

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Acute Coronary Syndromes**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 18: Acute Coronary Syndromes 133 Unstable angina is a clinical diagnosis based on history, physical examination findings, and diagnostic testing that does not reveal a STEMI or NSTEMI. [Source pages 152–157]

- This may precede STEMI or NSTEMI or may be the present- ing diagnosis that leads to a new diagnosis of CAD. [Source pages 152–157]

- Unstable angina characteristically presents with chest pain (or atypical ACS symptoms) associated with evidence of obstructive coronary artery disease and has one of the following three characteristics: (1) began within the past 2 months; (2) has increasing frequency, intensity, or duration of existing angina symptoms; or (3) existing angina begins to occur at rest. [Source pages 152–157]

- Additional diagnostic testing that may be helpful when unstable angina is considered includes coronary CT angiogram, exercise treadmill testing, cardiac nuclear perfusion imaging, or cardiac MRI. [Source pages 152–157]

- ■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION 1. [Source pages 152–157]

Clinical reasoning structure

- **Problem representation:** Acute Coronary Syndromes; source pages 152–157.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Acute Coronary Syndromes**.

- Source-page range: **152–157**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 152

CHAPTER 18: Acute Coronary Syndromes 133

Unstable angina is a clinical diagnosis based on history, physical examination findings, and diagnostic testing that does not reveal a STEMI or NSTEMI. This may precede STEMI or NSTEMI or may be the presenting diagnosis that leads to a new diagnosis of CAD. Unstable angina characteristically presents with chest pain (or atypical ACS symptoms) associated with evidence of obstructive coronary artery disease and has one of the following three characteristics: (1) began within the past 2 months; (2) has increasing frequency, intensity, or duration of existing angina symptoms; or (3) existing angina begins to occur at rest. Additional diagnostic testing that may be helpful when unstable angina is considered includes coronary CT angiogram, exercise treadmill testing, cardiac nuclear perfusion imaging, or cardiac MRI.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. STEMI treatment begins immediately upon recognition of diagnostic ECG findings. Goals include reperfusion by reducing thrombus, limiting thrombus extension, and relieving obstructive CAD. The effectiveness of interventions is time sensitive. Options include administering systemic thrombolytic therapy within 30 minutes of arrival or initiating percutaneous coronary intervention (PCI) within 90 minutes. PCI is the preferred therapy, when possible, based on greater benefits and fewer risks in patients without contraindications to thrombolysis (Table 18-3) who can achieve PCI within 120 minutes. See Table 18-3 for medications used in the treatment of STEMI.

2. For patients with a suspected ACS, begin cardiac monitoring, place an intravenous line, and provide supplemental oxygen if oxygen saturation is less than 95%. Administer aspirin 160 to 325 mg orally chewed. Consider oral, transdermal, or intravenous nitroglycerin to treat any ongoing angina. Morphine sulfate may be used as an adjunct if pain

continues despite treatment with nitroglycerin.

3. Clopidogrel is recommended for use along with aspirin for patients with moderate to high-risk NSTEMI and STEMI, and in patients in whom PCI is planned. Use as an alternative to aspirin in patients allergic to aspirin. Clopidogrel increases risk of bleeding, and should be held at least 5 days before coronary artery bypass grafting (CABG).

4. Begin anticoagulation for patients with unstable angina or NSTEMI using unfractionated heparin or low-molecular-weight heparins (LMWH). These are also options for patients undergoing PCI revascularization, in consultation with your cardiology consultant. Unfractionated heparin is preferred for patients in whom CABG is planned.

5. Factor Xa inhibitors such as fondaparinux have similar efficacy to unfractionated heparin in patients with unstable angina or NSTEMI, and current guidelines consider it an option as an antithrombin. In STEMI patients lacking renal impairment, fondaparinux may be considered for those patients treated with thrombolytics that are not fibrin specific such as streptokinase.

6. Direct thrombin inhibitors, such as bivalirudin, bind directly to thrombin in clot and are resistant to agents that degrade heparin. Comparison of bivalirudin with unfractionated heparin found no outcomes benefit

18_Cydulka_Ch18.indd 133 15/03/17 12:31 PM

Source page 153

Table 18-3 Drugs Used in the Emergency Treatment of STEMI

Antiplatelet Agents	
Aspirin	162–325 mg
Clopidogrel	Loading dose of 600 mg PO followed by 75 mg/d. No loading dose is administered in patients > 75 years old receiving fibrinolytics
Prasugrel	Loading dose of 60 mg promptly and no more than 1 h after PCI once coronary anatomy is defined and a decision is made to proceed with PCI
Ticagrelor	Loading dose is 180 mg PO followed by 90 mg twice a day
Antithrombins	
Unfractionated heparin	Bolus of 60 U/kg (maximum, 4000 U) followed by infusion of 12 U/kg/h (maximum, 1000 U/h) titrated to a partial thromboplastin time 1.5-2.5 × control
Enoxaparin	30 mg IV bolus followed by 1 mg/kg SC every 12 h
Fondaparinux	2.5 mg SC*

Fibrinolytic Agents

Streptokinase 1.5 MU over 60 min

Anistreplase 30 U IV over 2–5 min

Alteplase Body weight > 67 kg: 15 mg initial IV bolus; 50 mg infused over next 30 min; 35 mg infused over next 60 min

Body weight < 67 kg: 15 mg initial IV bolus; 0.75 mg/kg infused over next 30 min; 0.5 mg/kg infused over next 60 min

Reteplase 10 U IV over 2 min followed by 10 U IV bolus 30 min later

Tenecteplase Weight Dose (total dose not to exceed 50 mg)

<60 kg 30 mg

≥60 but <70 kg 35 mg

≥70 but <80 kg 40 mg

≥80 but <90 45 mg

≥90 50 mg

Glycoprotein IIb/IIIa Inhibitors†

Abciximab 0.25 mg/kg bolus followed by infusion of 0.125 µg/kg/min (maximum, 10 µg/min) for 12–24 h

Eptifibatide 180 µg/kg bolus followed by infusion of 2.0 µg/kg/min for 72–96 h

Tirofiban 0.4 µg/kg/min for 30 min followed by infusion of 0.1 µg/kg/min for 48–96 h

Other Anti-Ischemic Therapies

Nitroglycerin Sublingual: 0.4 mg every 5 min × 3 PRN pain

IV: Start at 10 µg/min, titrate to 10% reduction in MAP if normotensive, 30% reduction in MAP if hypertensive

Morphine 2–5 mg IV every 5–15 min PRN pain

(Continued)

18_Cydulka_Ch18.indd 134 15/03/17 12:31 PM

Source page 154

Table 18-3 Drugs Used in the Emergency Treatment of STEMI (Continued)

Metoprolol 50 mg PO every 12 h on first day, unless significant hypertension, may consider 5 mg IV over 2 min every 5 min up to 15 mg; withhold β-blockers initially if the patient is at risk for cardiogenic

shock/adverse effects‡

Atenolol 25–50 mg PO, unless significant hypertension, may consider 5 mg IV over 5 min, repeat once 10 min later; withhold β-blockers initially if the patient is at risk for cardiogenic shock/adverse effects‡

*Fondaparinux should not be used as monotherapy for PCI; if used, addition of unfractionated heparin or bivalirudin is recommended before PCI.

†American College of Cardiology/American Heart Association 2009 focused update for STEMI patients recommended glycoprotein IIB/IIa inhibitors be given at the time of PCI; benefit prior to arrival in the cardiac catheterization laboratory is uncertain.

‡Risk factors for cardiogenic shock/adverse effects: (1) signs of heart failure; (2) evidence of a low cardiac output state; (3) increased risk for cardiogenic shock (cumulatively: age >70 years old, systolic blood pressure <120 mm Hg, sinus tachycardia >110 beats/min or bradycardia <60 beats/min, and longer duration of STEMI symptoms before diagnosis and treatment); or (4) standard relative contraindications to β-blockade (PR interval >0.24 s, second- or third-degree heart block, active asthma, or reactive airway disease).

Abbreviations: MAP, mean arterial pressure; PCI, percutaneous coronary intervention; PRN, as needed; STEMI, ST-segment elevation myocardial infarction.

in NSTEMI patients, but less bleeding occurred and no dosage adjustment is required in renal impairment. For patients with STEMI, bivalirudin may be considered as an alternative to unfractionated heparin and GP IIb/IIIa inhibitors.

7. In treatment settings without timely access to PCI, fibrinolytics are indicated for patients with STEMI if time to treatment is <6 to 12 hours from symptom onset, and the ECG has at least 1-mm ST-segment elevation in two or more contiguous leads. The dosages of individual fibrinolytic agents are listed in Table 18-3.

8. STEMI patients who have received fibrinolytics should receive full-dose anticoagulation started in the ED and maintained for a minimum of 48 hours. Similar efficacy and safety profiles have been demonstrated for tPA, rtPA, and TNK. Contraindications for fibrinolytics are listed in Table 18-4. Before administering thrombolytics, obtain informed consent. Arterial puncture should be avoided, as should venipuncture or central line placement in areas which are not readily compressible.

9. Tissue plasminogen activator (tPA) is a naturally occurring human protein and is not antigenic. tPA is fibrin-specific and has a half-life of 5 minutes. When compared with traditional dosing, front-loaded tPA has been shown to have superior 90-minute patency rates and reocclusion rates, with no increase in bleeding risk.

10. Reteplase (rPA) is a non-fibrin-specific deletion mutant of tPA with a prolonged half-life of 18 minutes and its use may have a faster time to perfusion. The main advantage of reteplase is that it is given as a double bolus rather than continuous infusion.

18_Cydulka_Ch18.indd 135 15/03/17 12:31 PM

Source page 155

136 SECTION 4: Cardiovascular Diseases

Table 18-4 Contraindications to Fibrinolytic Therapy in ST-Segment Elevation Myocardial Infarction

Absolute contraindications

Any prior intracranial hemorrhage

Known structural cerebral vascular lesion (e.g., arteriovenous malformation)

Known intracranial neoplasm

Ischemic stroke within 3 months

Active internal bleeding (excluding menses)

Suspected aortic dissection or pericarditis

Relative contraindications

Severe uncontrolled blood pressure (>180/100 mm Hg)

History of chronic, severe, poorly controlled hypertension

History of prior ischemic stroke >3 months or known intracranial pathology not covered in contraindications

Current use of anticoagulants with known INR >2–3

Known bleeding diathesis

Recent trauma (past 2 weeks)

Prolonged CPR (>10 min)

Major surgery (<3 weeks)

Noncompressible vascular punctures (including subclavian and internal jugular central lines)

Recent internal bleeding (within 2–4 weeks)

Patients treated previously with streptokinase should not receive streptokinase a second time

Pregnancy

Active peptic ulcer disease

Other medical conditions likely to increase risk of bleeding (e.g., diabetic retinopathy)

11. Tenecteplase (TNK) is a fibrin-specific substitution mutant of tPA that is given as a single weight-based bolus.

12. Streptokinase (SK) activates circulating plasminogen, is not fibrin-specific, and is capable of generating an allergic reaction (minor: 5% to 5.7%, anaphylaxis: <0.2% to 0.7%). Hypotension occurs in up to 15% of patients and is usually responsive to fluids and slowing of SK

infusion. Contraindications include hypotension, prior SK administration within 6 months, and streptococcal infection within a year. SK's half-life is 23 minutes, but systemic fibrinolysis persists for 24 hours. Administer heparin within 4 hours of starting SK.

13. The most significant complication of systemic thrombolytics is bleeding, particularly intracranial hemorrhage. If significant bleeding occurs, discontinue thrombolytics, heparin, and aspirin. Crystalloid and red blood cell infusion may be necessary. Cryoprecipitate and fresh frozen plasma (FFP) may be used to reverse fibrinolysis by replenishing thrombotic factors. Begin with 10 units of cryoprecipitate and obtain fibrinogen levels. If the fibrinogen level is <1 g/L, administer a second 10 U of cryoprecipitate. If bleeding continues despite a fibrinogen >1 g/L, or if the fibrinogen level is <1 g/L after 20 U of cryoprecipitate, administer 2 U of FFP. If hemorrhage continues, administer platelets or antifibrinolytic agents such as aminocaproic acid or tranexamic acid.

18_Cydulka_Ch18.indd 136 15/03/17 12:31 PM

References/provenance

- Source: *Tintinalli's Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 152–157.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 19: Cardiogenic Shock

Source pages 158–161 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 158–161

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Cardiogenic Shock**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 19: Cardiogenic Shock 139 arrhythmias, electrolyte abnormalities, or signs of drug toxicity with car- diac manifestations. [Source pages 158–161]

- ST-segment depression in the lateral leads of the ECG may be suggestive of a right ventricular infarction, which is associated with increased mortality and may not manifest itself with ST-segment elevation in a standard twelve lead ECG. [Source pages 158–161]

- Look for pulmonary edema, widened medi- astinum, enlarged cardiac silhouette, or other alternative diagnoses such as pneumonia or pneumothorax on chest radiograph. [Source pages 158–161]

- Bedside echocardiography, while not a substitute for emergent formal transthoracic echocardiography when clinically indicated, can rapidly exclude other causes of hypotension, identify some mechanical precipitants of shock, and help guide therapy. [Source pages 158–161]

- Visualization of the inferior cava can help to determine volume status when considering hypovolemic shock. [Source pages 158–161]

Clinical reasoning structure

- **Problem representation:** Cardiogenic Shock; source pages 158–161.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Cardiogenic Shock**.

- Source-page range: **158–161**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 158

CHAPTER 19: Cardiogenic Shock 139

arrhythmias, electrolyte abnormalities, or signs of drug toxicity with cardiac manifestations. ST-segment depression in the lateral leads of the ECG may be suggestive of a right ventricular infarction, which is associated with increased mortality and may not manifest itself with ST-segment elevation in a standard twelve lead ECG. Look for pulmonary edema, widened mediastinum, enlarged cardiac silhouette, or other alternative diagnoses such as pneumonia or pneumothorax on chest radiograph.

Bedside echocardiography, while not a substitute for emergent formal transthoracic echocardiography when clinically indicated, can rapidly exclude other causes of hypotension, identify some mechanical precipitants of shock, and help guide therapy. Visualization of the inferior cava can help to determine volume status when considering hypovolemic shock. A pericardial effusion can be diagnosed using limited bedside echocardiography and along with findings such as diastolic collapse of the right atrium and ventricle can be diagnostic for cardiac tamponade. Left ventricular function and cardiac contractility can also be evaluated with bedside testing and may support the clinician’s impression of cardiogenic shock when decreased from normal cardiac function.

There is no single laboratory test that is diagnostic for cardiogenic shock. Obtain a complete blood count and chemistries (including liver function tests) to evaluate for anemia, potential signs of infection, and some components of end organ dysfunction. In the absence of ST-segment elevation that is diagnostic for an acute MI, elevations of cardiac markers such as troponin can establish the diagnosis of non-ST-segment elevation MI (NSTEMI). In addition, these markers may add prognostic value in patients with other causes of shock such as acute heart failure and sepsis. Given their high negative predictive value, natriuretic peptides, such as cardiac BNP or n-terminal pro-BNP, suggest a noncardiac etiol-

ogy of patient condition if normal. In the setting of shock, serum lactate levels indicate the degree of hypoperfusion present and may be useful to follow during resuscitative efforts. Blood gas measurements will provide insight into acid-base status and CO2 retention. Specialized laboratory testing, such as toxicology studies, should be guided by the specific clinical situation.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

Airway management when necessary, circulatory stabilization, and arrangements for definitive cardiac care often occur simultaneously for patients in cardiogenic shock. Cardiology and/or cardiothoracic surgery should be consulted early when cardiogenic shock is diagnosed, as emergent coronary revascularization is an important intervention when ischemia is the underlying cause. Transfer to a facility with such specialized care should be arranged when indicated.

1. ■Supplemental oxygen should be provided. Noninvasive positive pressure ventilation may provide temporary support, but many patients in cardiogenic shock will require definitive airway management and thus endotracheal intubation should be considered as needed for respiratory support.

19_Cydulka_Ch19.indd 139 15/03/17 1:53 PM

Source page 159

2. ■Intravenous access should be obtained. Cardiac rhythm and pulse oximetry monitoring should be initiated. Rhythm disturbances, hypoxia, hypovolemia, and electrolyte abnormalities should be identified and treated.

3. ■Early revascularization is required for cardiogenic shock due to acute cardiac ischemia. Percutaneous coronary intervention (PCI) is superior to fibrinolysis in the setting of cardiogenic shock. However, when PCI is not available (or prolonged transfer times are anticipated) fibrinolysis is superior to supportive measures alone.

4. ■In AMI, aspirin and heparin should be given unless there is an absolute contraindication present. Typical anti-anginal therapies may precipitate cardiovascular collapse, especially in patients with right ventricular ischemia. To treat acute chest pain in the setting of cardiogenic shock, consider cautious use of titrated intravenous nitroglycerin 5 to 100 μg/min or morphine sulfate given in 2 mg increments with close monitoring of blood pressure. Avoid the use of β-blockers in patients with cardiogenic shock as this may worsen cardiac function.

5. ■A small intravenous fluid challenge (250 to 500 mL bolus) with close monitoring for effectiveness may be considered for patients with mild hypotension and no pulmonary edema. Hypotension in the setting of right ventricular ischemia will warrant a more robust fluid resuscitation.
6. ■Norepinephrine may be considered for severe hypotension as a vasopressor and positive inotrope. An infusion should begin at 2 µg/min and titrated to the desired effect.
7. ■Dobutamine 2 to 20.0 µg/kg/min can be used to support blood pressure for hypotension due to cardiogenic shock in the absence of hypovolemia. Dobutamine may cause peripheral vasodilatation, and concomitant use of dopamine 2 to 20.0 µg/kg/min can be added and titrated to the desired effect with the lowest dose possible.
8. ■Milrinone may be considered as a positive inotrope and can be started with a loading dose of 50 µg/kg IV over 10 minutes followed by an infusion of 0.5 µg/kg/min.
9. ■In the setting of acute mitral regurgitation, afterload reduction via intravenous sodium nitroprusside 0.5 to 10.0 µg/kg/min should be combined with inotropic support via dobutamine 2 to 20.0 µg/kg/min.
10. ■Consider using intraaortic balloon pump counterpulsation (if available) as a temporizing measure to decrease afterload and to augment coronary perfusion. This is contraindicated in severe aortic regurgitation.
11. ■Consider more advanced mechanical circulatory support, such as percutaneous left ventricular assist devices or extracorporeal membrane oxygenation, in selected refractory cases.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 50, "Cardiogenic Shock," by Casey Glass and David Manthey.

19_Cydulka_Ch19.indd 140 15/03/17 1:53 PM

Source page 160

C h a p ter Low-Probability Acute Coronary
20 Syndrome

David A. Wald

Patients presenting to the emergency department (ED) with chest pain or other symptoms suggesting possible coronary ischemia should be risk-stratified based on the probability of having an acute coronary syndrome (ACS). Patients with a low-probability ACS have no objective evidence of acute coronary ischemia or infarction. These patients do not have charac-

teristic ST-segment elevation or depression on an electrocardiogram (ECG), and initial cardiac biomarkers are not elevated.

■■ CLINICAL FEATURES

After performing an initial history and physical examination, an emergency provider needs to determine how much of a diagnostic evaluation to undertake for a patient presenting with chest pain. Approximately 3% to 6% of patients with an initial diagnosis of noncardiac chest pain or another alternative diagnosis may later develop a short-term adverse cardiac event, making risk stratification an important aspect of clinical decision making. Among patients who are assessed as low-probability for ACS, aspects that have been shown to be associated with a low-risk profile include chest pain that is described as pleuritic, positional, sharp, stabbing, or that is found to be reproducible. High-risk historical features include chest pain that radiates to the arm or shoulders, is exertional, described as pressure, is accompanied by nausea or diaphoresis, or is similar to prior cardiac pain. However, even patients who present with atypical features may have some risk, and the absence of high-risk features alone cannot completely exclude the possibility of ACS. Significant coronary artery disease is rare in patients <30 years old, although age alone does not completely eliminate ACS as a cause of acute chest pain. In addition, treatment responsiveness to nitrates, antacids, or nonsteroidal anti-inflammatory medications cannot reliably confirm or exclude ACS. Focus the initial evaluation to identify potential alternative diagnoses and detect findings that may be consistent with heart failure or other underlying conditions.

The results and timing of previous cardiac testing, such as ECG, stress test, and cardiac catheterization, can be helpful to consider when determining the appropriate evaluation for possible ACS. For example, new ECG changes consistent with cardiac ischemia offer strong evidence of underlying cardiac disease. Conversely, a recent negative cardiac catheterization with no coronary luminal irregularities is associated with a very low incidence of myocardial infarction or ACS within a 2-year period. Previous stress test results can add evidence to the clinician's diagnostic decision making, but cannot confirm the presence or absence of disease.

■■ DIAGNOSIS AND DIFFERENTIAL

The evaluation of patients presenting to the ED with possible ACS can be conceptualized into primary and secondary assessments (see Fig. 20-1). The goal of the primary evaluation is to identify patients with definite ACS

20_Cydulka_Ch20.indd 141 22/04/17 4:36 pm

Source page 161

Primary evaluation Secondary evaluation
Emergency department Observation unit or
History Possible ACS inpatient
Physical exam Serial ECGs
ECG Serial cardiac markers
First set of cardiac (Excludes AMI)
biomarkers (biomarkers Definite ACS or Objective cardiac testing
may not be needed for High-risk features (Excludes unstable
noncardiac chest pain) angina)
Chest pain not Admission to Chest pain not
from ACS coronary care unit or from ACS
revascularization
(see chapter 49,
Treat alternative Acute Coronary Treat alternative
diagnosis Syndromes) diagnosis

Figure 20-1. Evaluation process for patients with possible acute coronary syndrome (ACS). AMI, acute myocardial infarction.

and differentiate them from those with probable or possible ACS. Identifying patients with alternative causes of chest pain that are unlikely to be ACS should also be considered (see Chapter 17).

The primary evaluation for patients with suspected ACS includes a detailed history, physical examination, ECG, chest radiography as indicated, and cardiac biomarkers. Serial ECGs can be helpful in further evaluating patients with ongoing symptoms or changes in condition. Data obtained from this primary survey should be used to further risk-stratify patients presenting to the ED with possible ACS. Some experts have determined that when the pretest probability of ACS is ≤2% further testing is not indicated, whereas others have suggested a threshold of <1% to determine when additional testing is unnecessary.

Upon completion of the primary evaluation, classify patients into one of the following categories of the prognosis-based classification system: acute myocardial infarction (AMI), probable acute ischemia, possible acute ischemia, or definitely not ischemia (see Table 20-1). Patients with probable or possible acute ischemia can be further stratified based on their risk for adverse events using scoring systems such as the TIMI risk score or the HEART pathway based on clinical features, ECG findings, and initial diagnostic testing results. These scoring systems can assist clinicians in determining the extent of additional testing that is recommended.

The secondary assessment can be conducted in an ED, a hospital-based observation unit, or in the inpatient setting. The specific components of this

secondary evaluation is informed by appropriate risk stratification. The goal of this assessment should be to exclude both components of ACS: myocardial infarction and unstable angina. In the absence of ST-segment elevation on the ECG, a myocardial infarction is excluded through the use of serial troponin measurements to detect myocardial necrosis. Serum troponin levels can take as long as 8 hours from the time of infarction to

20_Cydulka_Ch20.indd 142 22/04/17 4:36 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 158–161.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 20: Low-Probability Acute Coronary Syndrome

Source pages 162–165 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 162–165

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Low-Probability Acute Coronary Syndrome**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 20: Low-Probability Acute Coronary Syndrome 143 Table 20-1 Prognosis-Based Classification System for ED Chest Pain Patients* I. [Source pages 162–165]

- Acute myocardial infarction: immediate revascularization candidate II. [Source pages 162–165]

- Probable acute ischemia: high risk for adverse events (any of the following): Evidence of clinical instability (i.e., pulmonary edema, hypotension, arrhythmia, transient mitral regurgitation murmur, diaphoresis) Ongoing pain thought to be ischemic (consider chest pain or discomfort as chief symptom, reproducing documented angina, or pain in setting of known coronary artery disease, including myocardial infarction) Pain at rest associated with ischemic ECG changes (consider new, or presumably new, transient, ST-segment deviation, 1 mm or greater, or T-wave inversion in multiple precordial leads) One or more positive myocardial marker measurements Positive perfusion imaging study III. [Source pages 162–165]

- ■Possible acute ischemia: intermediate risk for adverse events. [Source pages 162–165]

- History suggestive of ischemia with absence of high-risk features, and any of the following: Rest pain, now resolved New onset of pain Crescendo pattern of pain Ischemic pattern on ECG not associated with pain (may include ST-segment depres- sion <1 mm or T-wave inversion >1 mm) IV. [Source pages 162–165]

Clinical reasoning structure

- **Problem representation:** Low-Probability Acute Coronary Syndrome; source pages 162–165.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Low-Probability Acute Coronary Syndrome**.

- Source-page range: **162–165**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 162

CHAPTER 20: Low-Probability Acute Coronary Syndrome 143

Table 20-1 Prognosis-Based Classification System for ED Chest Pain Patients*

I. Acute myocardial infarction: immediate revascularization candidate

II. Probable acute ischemia: high risk for adverse events (any of the following):

Evidence of clinical instability (i.e., pulmonary edema, hypotension, arrhythmia, transient mitral regurgitation murmur, diaphoresis)

Ongoing pain thought to be ischemic (consider chest pain or discomfort as chief symptom, reproducing documented angina, or pain in setting of known coronary artery disease, including myocardial infarction)

Pain at rest associated with ischemic ECG changes (consider new, or presumably new, transient, ST-segment deviation, 1 mm or greater, or T-wave inversion in multiple precordial leads)

One or more positive myocardial marker measurements

Positive perfusion imaging study

III. ■Possible acute ischemia: intermediate risk for adverse events. History suggestive of ischemia with absence of high-risk features, and any of the following:

Rest pain, now resolved

New onset of pain

Crescendo pattern of pain

Ischemic pattern on ECG not associated with pain (may include ST-segment depression <1 mm or T-wave inversion >1 mm)

IV. ■Possible acute ischemia: low risk for adverse events. History not strongly suggestive of ischemia and all of the following:

ECG normal, unchanged from previous, or nonspecific changes

Negative myocardial marker measurement

or (all of the following)

>2 weeks of unchanged symptom pattern or long-standing symptoms with only mild change in exertional pain threshold

ECG normal, unchanged from previous, or nonspecific changes

Negative initial myocardial marker measurement

V. ■Definitely not ischemia: very low risk for adverse events. All of the following:

Clear evidence of nonischemic symptom etiology

ECG normal, unchanged from previous, or nonspecific changes

Negative initial myocardial marker measurement†

or

Unstructured clinician estimate of acute coronary syndrome ≤2%

*Authors' analyses from multiple sources.

†Literature not conclusive.

become elevated. Therefore, using cardiac biomarkers to diagnose a myocardial infarction in the absence of dynamic ECG changes should take into account the time from symptom onset and generally include multiple measurements. A modified approach in appropriate low-risk patients is to obtain two troponin measurements, at least 2 hours apart with one measurement at least 6 hours after the onset of pain.

Normal serial cardiac biomarker measurements reduce the likelihood of AMI but do not exclude unstable angina. Patients with a continued suspicion for ACS despite negative biomarkers often undergo some form of advanced cardiac testing. Common modalities of advanced cardiac testing include exercise or pharmacologically induced ECG stress test, stress echocardiography, nuclear imaging, cardiac magnetic resonance imaging

20_Cydulka_Ch20.indd 143 22/04/17 4:36 pm

Source page 163

(MRI), computed tomography coronary angiography (CTCA), or cardiac catheterization. Based on risk stratification and institutional resources, some patients will undergo this type of testing during the initial hospitalization. However, additional testing as an outpatient is an option for low-risk patients in whom AMI has been excluded.

The use of advanced cardiac testing should be guided by general principles discussed below along with available resources and institutional practice patterns:

1. Cardiac imaging with CTCA may play a role in the evaluation of low-risk patients presenting with chest pain, as this approach can result in a more rapid and cost-effective diagnosis than stress myocardial perfusion

imaging.

2. Noninvasive stress testing is often recommended for patients risk-stratified as low to intermediate risk. Treadmill exercise testing may be useful in patients who are able to exercise. In patients with physical limitations, pharmacologic stress testing may be necessary.

3. When stress testing is ordered, select a method of cardiac assessment such as electrocardiography, nuclear imaging, echocardiography, or magnetic resonance imaging. Selection from these options is often based on local practice, institutional expertise, and resource availability.

a. ECG-based exercise treadmill testing is the least costly and most widely available but has the lowest sensitivity (68%) of the imaging options which may limit its use for high-risk patients. In addition, ECG-based exercise treadmill testing should not be used in patients with abnormal baseline ECGs due to difficulties in interpretation.

b. Stress echocardiography has the advantages of no radiation exposure, improved sensitivity (80%), and wide availability.

c. Nuclear imaging is also widely available, allows assessment of myocardial perfusion, and has high accuracy. However, there is associated radiation exposure and radioisotope-related delays.

d. Cardiac MRI is also highly accurate and does not expose patients to radiation, but is less widely available.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

The evaluation and treatment of patients with suspected ACS is typically initiated prior to either confirmation or exclusion of this condition. If an alternative diagnosis is confirmed and ACS is excluded, treatment should begin for the alternative diagnosis. Treat patients with suspected ACS as follows:

1. Provide supplemental oxygen via nasal cannula or as needed to maintain appropriate pulse oxygenation.

2. Administer aspirin 160 to 325 mg orally.

3. Provide nitroglycerin 0.4 mg spray or sublingual to treat suspected ischemic chest pain. Intravenous nitroglycerin may be used for patients with persistent ischemic chest pain unrelieved by other methods. Nitroglycerin paste applied to the skin is an alternative to intravenous nitroglycerin in patients without ongoing or recurrent ischemia.

4. Consider morphine 1 to 5 mg IV in patients with persistent ischemic chest pain.

20_Cydulka_Ch20.indd 144 22/04/17 4:36 pm

Source page 164

5. Prescribe metoprolol 25 to 50 mg PO in the first 24 hours when ACS is suspected.
6. Antiplatelet and/or anticoagulant therapy may be appropriate for patients with definitive or likely ACS. The decision to administer these medications and specific doses may be institution-specific based on protocols or guidelines and are balanced with the patient's bleeding risk. These decisions may be determined through multidisciplinary discussions. These options include (a) dual antiplatelet therapy: a common regimen is clopidogrel 300 to 600 mg PO in addition to aspirin, (b) antithrombin therapy: common regimens are heparin 60 U/kg IV bolus (maximum bolus 4000 units) with 12 U/kg/h IV infusion (maximum infusion 1000 U/h) or enoxaparin 1 mg/kg SC every 12 hours.
7. Selected patients can be managed primarily in the ED or within a dedicated ED Observation Unit. Appropriately risk-stratified patients with negative serial cardiac markers, without diagnostic ECG changes, and who have normal advanced cardiac testing (when indicated) are unlikely to have ACS as a cause of their symptoms. Evaluate for alternative diagnoses and recommend appropriate close follow-up resources.
8. Patients with positive cardiac markers, diagnostic ECG changes, or positive diagnostic testing suggesting ACS should receive appropriate consultation and admission to the hospital for inpatient management.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 51, "Low-Probability Acute Coronary Syndromes," by Kathleen A. Hosmer and Chadwick D. Miller.

20_Cydulka_Ch20.indd 145 22/04/17 4:36 pm

Source page 165

C h a p ter Syncope

21 Jo Anna Leuck

Syncope accounts for up to 2% of all emergency department (ED) visits and 6% of hospital admissions. Syncope is defined as a transient loss of consciousness accompanied by loss of postural tone, followed by complete resolution without intervention. Although syncope often is a benign vasovagal event, it may represent a life-threatening dysrhythmia or other condition, particularly in the elderly. Near-syncope, or feeling an impending loss of consciousness without syncope, may carry the same risk as syncope. In up to half of syncope cases presenting to the ED, there is no definite etiology established for the syncopal episode.

■■ CLINICAL FEATURES

Cardiac-related syncope can be due to a structural cardiac lesion that limits

the heart’s ability to appropriately increase cardiac output. Examples of structural cardiac disease that can cause syncope include hypertrophic cardiomyopathy, aortic stenosis, pulmonary embolism, and myocardial infarction. Tachydysrhythmias such as ventricular tachycardia, torsades des pointes, and supraventricular tachycardia are common causes of syncope, but bradycardic syndromes may cause it as well. Syncope from a dysrhythmia is typically sudden and without prodrome. In young people, familial dysrhythmias such as Brugada or QT syndromes are uncommon but potentially serious causes of syncope.

Syncope is most commonly caused by vasovagal reflexes. Inappropriate vagal or sympathetic tone may lead to bradycardia, hypotension, or both. The hallmark of vasovagal syncope is a slow progressive prodrome of dizziness, nausea, pallor, diaphoresis, and diminished vision. The history should include a search for possible stimuli that are known to be associated with vasovagal syncope, such as phlebotomy, prolonged standing in a warm place, or fear. In situational syncope, the autonomic reflexive response results from a specific physical stimulus such as micturition, defecation, or extreme coughing. Carotid sinus hypersensitivity is another type of reflex-mediated syncope that is suggested by a history of presyncope when shaving, head-turning, or wearing of a constricting collar. This should be considered as a potential cause in elderly patients with recurrent syncope despite a negative cardiac workup.

Orthostatic syncope occurs when a sudden change in posture after recumbence is associated with inadequate compensatory increases in heart rate and peripheral vascular resistance. Orthostatic syncope can be due to decreased intravascular volume or poor vascular tone, which has a myriad of potential causes such as peripheral neuropathy, spinal cord injury, or medication side-effects. Since orthostatic changes can also be associated with other serious illnesses, alternative causes of syncope should still be considered even in the presence of orthostatic changes in blood pressure.

21_Cydulka_Ch21.indd 146 15/03/17 2:31 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 162–165.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 21: Syncope

Source pages 166–169 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 166–169

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Syncope**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 21: Syncope 147 Neurologic syncope is rarely the primary cause of syncopal episodes, as patients with loss of consciousness with persistent neurological deficits or altered mental status do not meet the usual diagnostic criteria for syn- cope. [Source pages 166–169]

- When brainstem ischemia or vertebrobasilar insufficiency is the cause of syncope, patients will often report other posterior circulation deficits such as diplopia, vertigo, focal neurological deficits, or nausea associated with the syncopal episode. [Source pages 166–169]

- If patients report that upper extrem- ity exercise preceded the event, there may be intermittent obstruction of the brachiocephalic or subclavian artery such as subclavian steal syn- drome. [Source pages 166–169]

- Subarachnoid hemorrhage may also present with syncope, but it is usually accompanied by persistent symptoms such as headache. [Source pages 166–169]

- A trau- matic subarachnoid hemorrhage can also occur as a result of injuries associated with syncope from another cause. [Source pages 166–169]

Clinical reasoning structure

- **Problem representation:** Syncope; source pages 166–169.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Syncope**.

- Source-page range: **166–169**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 166

CHAPTER 21: Syncope 147

Neurologic syncope is rarely the primary cause of syncopal episodes, as patients with loss of consciousness with persistent neurological deficits or altered mental status do not meet the usual diagnostic criteria for syncope. When brainstem ischemia or vertebrobasilar insufficiency is the cause of syncope, patients will often report other posterior circulation deficits such as diplopia, vertigo, focal neurological deficits, or nausea associated with the syncopal episode. If patients report that upper extremity exercise preceded the event, there may be intermittent obstruction of the brachiocephalic or subclavian artery such as subclavian steal syndrome. Subarachnoid hemorrhage may also present with syncope, but it is usually accompanied by persistent symptoms such as headache. A traumatic subarachnoid hemorrhage can also occur as a result of injuries associated with syncope from another cause. Seizure is the most common disorder mistaken for syncope. Brief tonic-clonic movements are often seen with syncope, but the presence of a postictal state, tongue biting, incontinence, or an epileptic aura can point toward seizure as a more likely alternative diagnosis.

Medication-induced syncope can be a common complication of pharmacotherapy. Because of poor autonomic responses and multiple medications, the elderly are particularly prone to syncope as a side effect of medication use. Cardiovascular responses to orthostatic or vasodilatory challenges may be blunted by antihypertensive agents such as β-blockers and calcium channel antagonists. Cardiovascular medications may also cause conduction abnormalities or life-threatening dysrhythmias. Diuretics also contribute to the risk of orthostatic hypotension due to their volume-depleting effect.

■■ DIAGNOSIS AND DIFFERENTIAL

Although an etiology for a syncopal episode may be difficult to establish, the

most important components of the syncope workup are a comprehensive history, physical examination, and ECG. From there, risk stratification is a practical approach. The history should be directed to high-risk factors, including age, medications, and prodromal associations. Sudden events that occur without warning suggest dysrhythmias, while preceding exertion may imply a structural cardiopulmonary lesion. Associated symptoms such as palpitations or chest pains may suggest an underlying cardiac etiology, while vertigo or focal weakness can point toward a neurologic cause. Back or abdominal pain may suggest a leaking abdominal aortic aneurysm or ruptured ectopic pregnancy in the appropriate patient populations. Single-vehicle crashes or trauma in the absence of defensive injuries may prompt a consideration of syncope as a precipitating event. The medical history is useful in revealing likely cardiac or psychiatric causes for syncope. When present, a family history of cardiac disease or sudden death may also be informative.
Physical examination may occasionally reveal the cause of syncope.
The cardiac examination may uncover a ventricular flow obstruction or a cardiac murmur suggestive of aortic stenosis or hypertrophic cardiomyopathy. Consider obtaining orthostatic vital sign measurements, as orthostatic hypotension with a systolic blood pressure drop of at least 20 mm Hg upon standing may support a diagnosis of vasovagal syncope. A significant blood pressure differential between upper extremities (>20 mm Hg) suggests
21_Cydulka_Ch21.indd 147 15/03/17 2:31 PM

Source page 167

subclavian steal syndrome as a possible etiology. In appropriate circumstances, a complete neurologic assessment or stool guaiac testing may yield information about secondary causes for syncope.
An ECG should be obtained for patients with syncope to assess for signs of ischemia or dysrhythmia. Suggestive ECG findings include prolonged QT interval, which may indicate a propensity for torsades des pointes, or PR interval shortening with a delta wave diagnostic for Wolf–Parkinson–White syndrome. Prolonged cardiac monitoring may show a transient recurrent dysrhythmia. Laboratory testing should be selective based on history and physical examination. An abnormal hemoglobin level in a patient with symptoms suggestive of anemia may explain orthostatic syncope and direct further workup. Women of childbearing age warrant a pregnancy test. Although not recommended routinely, serum electrolytes may be helpful in limited cases. If a patient is asymptomatic or does not have trauma, neuroimaging is not typically indicated in the routine workup of syncope.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

By definition, syncope results in spontaneous recovery of consciousness. Therefore, the main goal of ED care is to identify those patients at risk for further medical problems.

1. Patients with an established medical condition leading to syncope, such as cardiac dysrhythmia, pulmonary embolism, or GI bleeding, can be appropriately managed by appropriately treating the underlying diagnosis. Patients for whom a life-threatening etiology is identified, including those with cardiac or neurologic causes, warrant admission.
2. Patients with unclear diagnoses who are at high-risk for morbidity and mortality are those for whom there is concern about sudden cardiac death or ventricular dysrhythmia. Per the San Francisco Syncope Rule and other studies, features that suggest a risk for adverse events include abnormal ECG, complaint of shortness of breath, systolic blood pressure of <90 mm Hg on arrival, hematocrit <30%, age older than 45 years, or a history of ventricular dysrhythmia or congestive heart failure. Consider admission for patients determined to be at higher risk to monitor for cardiac arrhythmias and to facilitate an expedited workup.
3. Patients with unclear diagnoses who are at low-risk are unlikely to have a cardiac etiology for their syncope. These patients lack the high-risk criteria noted above, tend to be young with few comorbidities, and have a normal physical examination and ECG. Low-risk patients can be safely discharged home with instructions for outpatient follow-up and to return for any recurrence of syncope or nearsyncopal symptoms.
4. Additional factors to consider when making inpatient versus outpatient decisions may include syncope while supine or during exercise, absence of prodromal symptoms, palpitations, or elderly patients.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 52, "Syncope," by James Quinn.

21_Cydulka_Ch21.indd 148 15/03/17 2:31 PM

Source page 168

C h a p ter Acute Heart Failure

22 Lori J. Whelan

Acute heart failure covers a wide spectrum of illness, with symptoms ranging from a gradual increase in leg swelling, shortness of breath, or decreased exercise tolerance to the abrupt onset of pulmonary edema and respiratory distress. While the term congestive heart failure was historically used to describe volume overload, current terminology describes patients as having acute heart failure when they present with an acute exacerbation of

chronic heart failure or when new-onset heart failure is diagnosed. Heart failure has a poor prognosis with an approximately 50% mortality rate within 5 years of initial diagnosis. The most common precipitating factors of acute heart failure are atrial fibrillation, acute myocardial infarction or ischemia, discontinuation of medications (diuretics), increased sodium load, drugs that impair myocardial function, and physical overexertion.

■■ CLINICAL FEATURES

No single historical or physical finding is sensitive and specific enough to accurately diagnose acute heart failure in all patients. On physical examination, patients with acute heart failure may present with dyspnea, frothy pink sputum, or respiratory distress. Patients are frequently tachycardic and hypertensive, and a third heart sound (S3) may be identified on auscultation. Abdominojugular reflux and jugular venous distension may also be seen. Acute heart failure can be further classified as follows:

1. ■Hypertensive acute heart failure is characterized by signs and symptoms of acute heart failure with relatively preserved left ventricular function, systolic blood pressure >140 mmHg, a chest radiograph compatible with pulmonary edema, and symptom onset less than 48 hours.
2. ■Pulmonary edema presents with respiratory distress, rales on chest auscultation, reduced oxygen saturation from baseline, and characteristic chest radiograph findings.
3. ■Cardiogenic shock is characterized by evidence of tissue hypoperfusion and systolic blood pressure <90 mmHg.
4. ■Acute-on-chronic heart failure has signs and symptoms of acute heart failure that are mild to moderate and do not meet criteria for hypertensive heart failure, pulmonary edema, or cardiogenic shock. Systolic blood pressure is typically <140 mmHg and >90 mmHg, associated with increased peripheral edema, and has symptom onset over several days.
5. ■High-output failure presents with high cardiac output, tachycardia, warm extremities, and pulmonary congestion.
6. ■Right heart failure is a low-output syndrome with jugular venous distention, hepatomegaly, and variable hypotension.

■■ DIAGNOSIS AND DIFFERENTIAL

Commonly, patients with acute heart failure present with dyspnea and the differential diagnosis may include other conditions such as COPD, asthma, pneumonia, pneumothorax, pleural effusion, pulmonary embolus, and

22_Cydulka_Ch22.indd 149 21/04/17 3:38 pm

Source page 169

acute coronary syndrome. There is no single diagnostic test for heart failure; it is a clinical diagnosis based on the history, physical examination, and diagnostic testing. The most useful historical parameter is a history of acute heart failure. The symptom with the highest sensitivity for diagnosis is dyspnea on exertion (84%) and the most specific symptoms are paroxysmal nocturnal dyspnea, orthopnea, and edema (77% to 84%).

Chest radiographs showing pulmonary venous congestion, cardiomegaly, and interstitial edema are the most specific for a diagnosis of acute heart failure, although up to 20% of patients may have an initially negative CXR. ECG is not useful for diagnosis, although it may reveal an underlying cause or precipitant. BNP (B-type natriuretic peptide/N-terminal B-type natriuretic peptide) testing is useful to supplement provider assessment when diagnostic uncertainty exists. Bedside cardiac ultrasound may be useful to determine other causes for acute dyspnea, such as cardiac tamponade, and may also be useful to determine left ventricle function and volume status or identify signs of pulmonary congestion. These ancillary tests combined with an appropriate history and physical can help determine the likelihood of acute heart failure.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. ■Provide supplemental oxygen to keep saturations above 95%.
2. ■Consider noninvasive ventilation with continuous positive airway pressure (CPAP) or bilevel positive airway pressure (BiPAP). Ensure adequate facemask seal, hemodynamic stability, and close monitoring for adequate tidal volumes, patient cooperation, and effectiveness.
3. ■While many patients with hypertensive acute heart failure present with severe hypertension, pulmonary edema can occur with systolic blood pressures as low as 150 mmHg. Prompt recognition of this condition and afterload reduction with vasodilators can improve symptoms and avoid the need for emergent intubation.
4. ■Administer nitroglycerin 0.4 mg sublingual up to one dose per minute or nitroglycerin 0.5 to 0.7 μg/kg/min IV up to 200 μg/min. Titrate dosages based on patient symptoms blood pressure levels.
5. ■Nitroprusside 0.3 μg/kg/min titrated upward every 5 to 10 minutes as needed to maximum 10 μg/kg/min may be initiated if elevated blood pressure is unresponsive to nitroglycerin.
6. ■When volume overload is suspected, consider loop diuretics such as furosemide 40 mg IV, bumetanide 1 to 3 mg IV, or torsemide 10 to 20 mg IV.
7. ■Shortness of breath, orthopnea, jugular venous distention, rales and possibly an S3 may be evident in patients with normotensive heart

failure, even in the presence of normal vital signs, oxygen and ventilation. For these patients, treat with loop diuretics initially and then base additional treatment on responses to therapy.

8. ■Transient hypotension may develop after nitroglycerin is started. This may be due to marked clinical improvement, and should improve with decreasing the dose or discontinuing nitroglycerin. If hypotension persists, initiate an IV fluid bolus 250 to 1000 cc and consider causes such as right ventricular infarction, valvular pathology such as severe aortic stenosis, hypovolemia, or recent use of medications for erectile dysfunction

22_Cydulka_Ch22.indd 150 15/03/17 2:49 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 166–169.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 22: Acute Heart Failure

Source pages 170–171 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 170–171

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Acute Heart Failure**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 22: Acute Heart Failure 151 Comprehensive AHF Evaluation for Disposition ACUTE HEART FAILURE Provider Risk Determination Patient-Centric Self-Care Evaluation Facilitators or Barriers Clinical Physiologic Risk to Gestalt Profile Ideal Self-Care Caregiver support Vital signs Hospitalization history May include Examination Symptom monitoring consideration of some Laboratory testing Level education Medication/healthcare Components physiologic or self- Radiography care measures Past cardiac measures provider access Geospatial environment Disease knowledge Medication adherence Highly variable Risk models Interviews Severity scores Past medical history Established measures of activation, Quantification education, disease knowledge, literacy, numeracy & adherence GIS measures Level of care needs Ideal Morbidity Disposition Mortality Plan Readmission risk Shared Decision Making Figure 22-1. [Source pages 170–171]

- Factors impacting disposition decisions in ED patients with acute heart failure. [Source pages 170–171]

- ■Morphine 2 to 5 mg IV can relieve congestion and anxiety but is asso- ciated with adverse events such as need for mechanical ventilation, prolonged hospitalization, ICU admission, and mortality. [Source pages 170–171]

- For hypotensive patients or patients in need of additional inotropic sup- port, see Chapter 19, “Cardiogenic Shock,” for management recom- mendations. [Source pages 170–171]

- Identify high-risk patient characteristics for severe illness that may require intensive care unit admission such as altered mental status, per- sistent hypoxia, hypotension, troponin elevation, ischemic ECG changes, BUN >43, creatinine >2.75, tachypnea, or inadequate urine output. [Source pages 170–171]

Clinical reasoning structure

- **Problem representation:** Acute Heart Failure; source pages 170–171.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Acute Heart Failure**.

- Source-page range: **170–171**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 170

CHAPTER 22: Acute Heart Failure 151

Comprehensive AHF Evaluation for Disposition

ACUTE HEART FAILURE

Provider Risk Determination Patient-Centric Self-Care Evaluation

Facilitators or Barriers

Clinical Physiologic Risk

to

Gestalt Profile

Ideal Self-Care

Caregiver support

Vital signs Hospitalization history

May include Examination Symptom monitoring

consideration of some Laboratory testing Level education

Medication/healthcare Components

physiologic or self- Radiography

care measures Past cardiac measures provider access

Geospatial environment

Disease knowledge

Medication adherence

Highly variable Risk models Interviews

Severity scores Past medical history

Established measures of activation, Quantification

education, disease knowledge,

literacy, numeracy & adherence

GIS measures

Level of care needs Ideal

Morbidity Disposition

Mortality Plan

Readmission risk Shared Decision

Making

Figure 22-1. Factors impacting disposition decisions in ED patients with acute heart failure.

9. ■Morphine 2 to 5 mg IV can relieve congestion and anxiety but is associated with adverse events such as need for mechanical ventilation, prolonged hospitalization, ICU admission, and mortality.

10. For hypotensive patients or patients in need of additional inotropic support, see Chapter 19, “Cardiogenic Shock,” for management recommendations.

11. Identify high-risk patient characteristics for severe illness that may require intensive care unit admission such as altered mental status, persistent hypoxia, hypotension, troponin elevation, ischemic ECG changes, BUN >43, creatinine >2.75, tachypnea, or inadequate urine output.

12. See Fig. 22-1 for some of the many factors to consider when making disposition decisions for outpatient management, observation care, inpatient admission, or need for intensive care for patients with acute heart failure.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 53, “Acute Heart Failure,” by Sean P. Collins and Alan B.Storrow.

22_Cydulka_Ch22.indd 151 15/03/17 2:49 PM

Source page 171

C ha p ter Valvular Emergencies

23 Boyd Burns

Many patients with chronic cardiac valve abnormalities who present for emergency care have previously established disease. Acute valvular dysfunction can present with significant symptoms, and emergency physicians must be alert to the presenting signs and symptoms to identify the undiagnosed patient. With bedside echocardiography becoming more commonly available, the initial diagnosis of valvular disease may become more common in the ED.

■■ THE NEWLY DISCOVERED MURMUR

When an emergency provider identifies a new murmur on exam, the first step in the ED is to determine the potential clinical significance of this finding. Benign murmurs are not likely to cause symptoms and may be found incidentally. These are typically soft systolic ejection murmurs and occur

after S1 and end before S2 and are not associated with specific symptoms. In contrast, a new diastolic murmur or a new systolic murmur with symptoms at rest warrants emergent echocardiographic imaging and further evaluation. In addition, risks for possible infections such as endocarditis should be considered in patients with a newly discovered murmur and the presence of fever (Fig. 23-1).

■■ MITRAL STENOSIS

Clinical Features

Mitral stenosis prevents normal diastolic filling of the left ventricle, and dyspnea with exertion is the most common presenting symptom. In the past, hemoptysis was the second most common presenting symptom, but this is less common now that patients are typically diagnosed and treated earlier in the disease course. As the obstructive process progresses, left atrial pressures rise and lead to left atrial enlargement and symptoms of heart failure. Systemic emboli may develop, especially when accompanied by atrial fibrillation, and can result in myocardial, kidney, central nervous system, or peripheral infarction. Most patients with mitral stenosis eventually develop atrial fibrillation due to progressive dilatation of the atria.

The classic murmur of mitral stenosis and associated signs are listed in Table 23-1.

Diagnosis and Differential

In the presence of mitral stenosis, the electrocardiogram (ECG) may demonstrate notched or diphasic P waves and right axis deviation. On the chest radiograph, straightening of the left heart border is a typical early radiographic finding suggestive of left atrial enlargement. Later in the illness, findings of pulmonary congestion develop, such as redistribution of flow to the upper lung fields, Kerley B lines, and an increase in vascular markings.

23_Cydulka_Ch23.indd 152 22/04/17 4:37 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 170–171.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 23: Valvular Emergencies

Source pages 172–179 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 172–179

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Valvular Emergencies**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 23: Valvular Emergencies 153 New cardiac murmur Systolic murmur Diastolic murmur Mid-systolic grade 2 ● Mid-systolic grade 3 ● Early or late systolic ● Holosystolic ● Asymptomatic ● Symptomatic ● No signs of cardiovascular ● Signs of cardiovascular disease disease ● Normal ECG and CXR ● Abnormal ECG or CXR ● Murmur does not increase ● Murmur increases in in intensity with Valsalva intensity with Valsalva or or standing standing No further workup Echocardiography Figure 23-1. [Source pages 172–179]

- Algorithm for evaluation of newly discovered systolic murmur. [Source pages 172–179]

- The diagnosis of mitral stenosis is confirmed with echocardiography. [Source pages 172–179]

- The urgency for an accurate diagnosis and appropriate referral depends on the severity of symptoms and initial management will typically focus on symptom control. [Source pages 172–179]

- Table 23-1 Comparison of Heart Murmurs, Sounds, and Signs Mitral Mid-diastolic rumble, crescendos Loud snapping S1, small apical stenosis into S2 impulse, tapping due to under- filled ventricle Mitral Acute: harsh apical systolic murmur S3 and S4 may be heard regurgitation starts with S1 and may end before S2 Chronic: high-pitched apical holosys- tolic murmur radiating into S2 Mitral valve Click may be followed by a late sys- Mid-systolic click; S2 may be dimin- prolapse tolic murmur that crescendos into S2 ished by the late systolic murmur Aortic Harsh systolic ejection murmur Paradoxical splitting of S2, S3, stenosis and S4 may be present; pulse of small amplitude with a slow rise and sustained peak Aortic High-pitched blowing diastolic S3 may be present; wide pulse regurgitation murmur immediately after S2 pressure 23_Cydulka_Ch23.indd 153 22/04/17 4:37 pm 154 SECTION 4: Cardiovascular Diseases Emergency Department Care and Disposition 1. [Source pages 172–179]

Clinical reasoning structure

- **Problem representation:** Valvular Emergencies; source pages 172–179.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Valvular Emergencies**.

- Source-page range: **172–179**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 172

CHAPTER 23: Valvular Emergencies 153

New cardiac murmur

Systolic murmur Diastolic murmur

Mid-systolic grade 2

● Mid-systolic grade 3

● Early or late systolic

● Holosystolic

● Asymptomatic ● Symptomatic

● No signs of cardiovascular ● Signs of cardiovascular

disease disease

● Normal ECG and CXR ● Abnormal ECG or CXR

● Murmur does not increase ● Murmur increases in

in intensity with Valsalva intensity with Valsalva or

or standing standing

No further workup Echocardiography

Figure 23-1. Algorithm for evaluation of newly discovered systolic murmur. CXR,

chest x-ray.

The diagnosis of mitral stenosis is confirmed with echocardiography. The

urgency for an accurate diagnosis and appropriate referral depends on the

severity of symptoms and initial management will typically focus on

symptom control.

Table 23-1 Comparison of Heart Murmurs, Sounds, and Signs

Mitral Mid-diastolic rumble, crescendos Loud snapping S1, small apical

stenosis into S2 impulse, tapping due to under-

filled ventricle

Mitral Acute: harsh apical systolic murmur S3 and S4 may be heard
regurgitation starts with S1 and may end before S2
Chronic: high-pitched apical holosys-
tolic murmur radiating into S2
Mitral valve Click may be followed by a late sys- Mid-systolic click; S2 may be dimin-
prolapse tolic murmur that crescendos into S2 ished by the late systolic murmur
Aortic Harsh systolic ejection murmur Paradoxical splitting of S2, S3,
stenosis and S4 may be present; pulse of
small amplitude with a slow rise
and sustained peak
Aortic High-pitched blowing diastolic S3 may be present; wide pulse
regurgitation murmur immediately after S2 pressure
23_Cydulka_Ch23.indd 153 22/04/17 4:37 pm

Source page 173

154 SECTION 4: Cardiovascular Diseases
Emergency Department Care and Disposition
1. ■Heart rate control is typically necessary for patients with persistent tachy-
cardia and dyspnea who have atrial fibrillation with rapid ventricular
response.
2. ■Anticoagulation is recommended for patients who have an atrial diam-
eter greater than 55 mm, atrial fibrillation, left atrial thrombus, or a
history of embolic disease.
3. ■Patients with asymptomatic mitral stenosis should be referred for outpa-
tient evaluation, as mechanical intervention, balloon valvotomy, and
valve repair or replacements are recommended before the disease
progresses to pulmonary hypertension.

■■ MITRAL REGURGITATION

Clinical Features
Acute mitral regurgitation secondary to rupture of the chordae tendineae or
papillary muscles (or rarely due to blunt thoracic trauma) presents with
dyspnea, tachycardia, and pulmonary edema as the acute valvular dysfunc-
tion allows retrograde flow of blood to occur from the left ventricle to the
left atrium. These patients may quickly develop cardiogenic shock or
cardiac arrest. Intermittent mitral regurgitation usually presents with acute
episodes of respiratory distress due to pulmonary edema and may be
asymptomatic between attacks. In contrast, chronic mitral regurgitation
may be tolerated by patients for many years without specific treatment. In
these cases, the first symptom is usually dyspnea with exertion that is some-
times precipitated by new-onset atrial fibrillation. If patients are not appro-
priately anticoagulated, systemic emboli may occur in up to 20% of

patients and may initially be asymptomatic. The classic murmur and signs of mitral regurgitation are listed in Table 23-1.

Diagnosis and Differential

Consider mitral regurgitation in a patient without significant cardiomegaly who presents with new onset and marked pulmonary edema. In an acute valvular rupture, the ECG may show evidence of acute inferior wall infarction or ischemia. On chest radiography, acute mitral regurgitation from papillary muscle rupture may result in a minimally enlarged left atrium and pulmonary edema. In chronic disease, the ECG may demonstrate findings of left atrial and left ventricular hypertrophy associated with corresponding enlargement visible on chest radiography.

Echocardiography can confirm the diagnosis, and bedside testing may be needed in an acutely ill patient. While transthoracic echocardiography can help to make the diagnosis, this technique may underestimate lesion severity. Therefore, transesophageal echo should be used for definitive diagnostic purposes when a patient is stable and able to have this testing completed.

Emergency Department Care and Disposition

1. ■For acute mitral regurgitation due to papillary muscle rupture, emergent consultation for surgical repair is indicated.

23_Cydulka_Ch23.indd 154 22/04/17 4:37 pm

Source page 174

2. ■Nitrates and noninvasive ventilation can be used to provide preload and afterload reduction and to decrease the work of breathing. Dobutamine may be used to provide inotropic support when necessary.

3. ■In severe cases, aortic balloon counterpulsation can be utilized to increase forward flow and mean arterial pressure while decreasing the severity of the regurgitation. This approach may be necessary to stabilize patients prior to emergent surgical repair.

■■ MITRAL VALVE PROLAPSE

Clinical Features

Most patients with mitral valve prolapse are asymptomatic. When symptoms do occur, patients may report atypical chest pain, palpitations, fatigue, and dyspnea. The abnormal heart sounds are listed in Table 23-1. In patients with mitral valve prolapse who do not have mitral regurgitation at rest, exercise can provoke mitral regurgitation in about one-third of patients and is associated with a higher risk for morbid events.

Diagnosis and Differential

This diagnosis is unlikely to be made in the ED as the ECG and chest x-ray

are usually normal. Echocardiography is recommended nonemergently to confirm the clinical diagnosis of mitral valve prolapse and to identify any associated mitral regurgitation. Echocardiography or consultation with a cardiologist can be performed on an outpatient basis and specific ED treatment for this conditionis rarely required.

Emergency Department Care and Disposition

1. Acute treatment for mitral valve prolapse is rarely required.
2. ■Patients with mitral valve prolapse who also have concomitant mitral regurgitation should receive recommendations for endocarditis prophylaxis.
3. ■Patients with palpitations attributed to mitral valve prolapse may be prescribed β-blocker therapy to provide symptomatic relief.

■■ AORTIC STENOSIS

Clinical Features

Aortic stenosis presents an obstruction to left ventricular outflow. In the United Sates, the most common cause in adults is degenerative calcification associated with increasing age, hypertension, smoking, elevated cholesterol, and diabetes. The classic triad of symptoms is dyspnea, chest pain, and syncope. Dyspnea is usually the first symptom to appear, followed by paroxysmal nocturnal dyspnea, syncope on exertion, angina, and myocardial infarction. The classic murmur and associated signs of aortic stenosis are listed in Table 23-1. Blood pressure is typically normal or low, with a narrow pulse pressure.

Brachioradial delay is often present in aortic stenosis. To evaluate for this finding, the examiner palpates simultaneously the right brachial artery

23_Cydulka_Ch23.indd 155 22/04/17 4:37 pm

Source page 175

and the right radial artery of the patient, and any palpable delay is considered abnormal. Because patients with aortic stenosis are very preload dependent, the presence of atrial fibrillation may have significant consequences, as the absence of effective atrial contraction leads to a drop in cardiac output. This can lead to hypotension, especially when patients are treated with nitroglycerin for chest pain or dyspnea.

Diagnosis and Differential

Electrocardiogram and chest radiographic findings lack sensitivity and specificity for aortic stenosis. The ECG often demonstrates criteria for LVH and may have a left or right bundle branch block in up to 10% of patients. The chest radiograph is normal early, but eventually LVH and findings of congestive heart failure will develop if the patient does not have valve

replacement. Transthoracic echocardiography can confirm the suspected diagnosis of aortic stenosis and determine disease severity.

Emergency Department Care and Disposition

1. ■Positive pressure ventilation and supplemental oxygen can be helpful in the treatment of pulmonary edema.
2. ■Negative inotropic drugs and vasodilators are often poorly tolerated and should be used with caution.
3. ■New-onset atrial fibrillation may require cardioversion to maintain cardiac output.
4. ■Patients discharged from the ED should avoid vigorous activity and be promptly seen in follow-up by a cardiologist.

■■ AORTIC REGURGITATION

Clinical Features

Aortic regurgitation occurs when valve leaflets do not fully close and result in retrograde flow of blood from the aorta into the left ventricle during diastole. In acute disease, dyspnea from pulmonary edema is the most common presenting symptom. Patients may also complain of fever and chills when endocarditis is the underlying cause. Dissection of the ascending aorta that impacts the function of the aortic valve typically produces a "tearing" chest pain that may radiate between the shoulder blades. The classic murmur and signs of aortic regurgitation are listed in Table 23-1. Tachycardia develops in an attempt to maintain cardiac output in the setting of acute aortic regurgitation, although this may be inadequate and cardiogenic shock or arrest may occur.

Chronic aortic regurgitation typically presents with gradually worsening fatigue or dyspnea with exertion. Signs include a wide pulse pressure with a prominent ventricular impulse, and may be associated with head bobbing. A "water hammer pulse" characterized by a quick rise in peripheral pulse upstroke followed by a rapid collapse may also be present. Other classic findings may include an accentuated precordial apical thrust, pulsus bisferiens, a "to-and-fro" femoral murmur (Duroziez sign), and capillary pulsations visible at the proximal nailbed while pressure is applied at the tip (Quincke sign).

23_Cydulka_Ch23.indd 156 22/04/17 4:37 pm

References/provenance

- Source: *Tintinalli's Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 172–179.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 24: The Cardiomyopathies, Myocarditis, and Pericardial Disease

Source pages 180–187 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 180–187

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **The Cardiomyopathies, Myocarditis, and Pericardial Disease**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 24: The Cardiomyopathies, Myocarditis, and Pericardial Disease 161 Emergency Department Care and Disposition Patients with symptomatic or newly diagnosed dilated cardiomyopathy typically require admission to a monitored bed or intensive care unit depending on severity of presentation. [Source pages 180–187]

- Patients with known dilated cardio myopathy benefit from identification of an underlying cause for an exacer bation, including myocardial ischemia, anemia, infection, new-onset atrial fibrillation, bradydysrhythmia, valvular insufficiency, renal dysfunction, pulmonary embolism, or thyroid dysfunction. [Source pages 180–187]

- Establish intravenous access, provide supplemental oxygen, and initiate cardiac monitoring. [Source pages 180–187]

- Standard therapies for the treatment of acute heart failure should be employed. [Source pages 180–187]

- Guidelines from the American College of Cardiology that include issues important for the management of heart failure are available online at http://www.cardiosource.org/Science-And-Quality/ Practice-Guidelines-and-Quality-Standards.aspx. [Source pages 180–187]

Clinical reasoning structure

- **Problem representation:** The Cardiomyopathies, Myocarditis, and Pericardial Disease; source pages 180–187.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **The Cardiomyopathies, Myocarditis, and Pericardial Disease**.

- Source-page range: **180–187**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 180

CHAPTER 24: The Cardiomyopathies, Myocarditis, and Pericardial Disease 161

Emergency Department Care and Disposition

Patients with symptomatic or newly diagnosed dilated cardiomyopathy typically require admission to a monitored bed or intensive care unit depending on severity of presentation. Patients with known dilated cardio myopathy benefit from identification of an underlying cause for an exacer bation, including myocardial ischemia, anemia, infection, new-onset atrial fibrillation, bradydysrhythmia, valvular insufficiency, renal dysfunction, pulmonary embolism, or thyroid dysfunction.

1. Establish intravenous access, provide supplemental oxygen, and initiate cardiac monitoring.
2. Standard therapies for the treatment of acute heart failure should be employed. Guidelines from the American College of Cardiology that include issues important for the management of heart failure are available online at http://www.cardiosource.org/Science-And-Quality/Practice-Guidelines-and-Quality-Standards.aspx.
3. Complex ventricular ectopy can be treated with amiodarone 150 mg IV over 10 min and then 1 mg/min for 6 hours.
4. Chronic therapy includes diuretics and digoxin, which improve symp toms but have not been shown to impact survival. Angiotensin-converting enzyme (ACE) inhibitors and ß-blockers, such as carvedilol, are often used as they have been shown to improve patient survival.
5. A growing subset of advanced dilated cardiomyopathy patients are treated with Left Ventricular Assist Devices (LVADs) while awaiting heart transplant.

Myocarditis

Myocarditis is a common cause of dilated cardiomyopathy. Myocarditis is

an inflammation of the myocardium and may be the result of a systemic disorder or an infectious agent including viral or bacterial etiologies. Peri carditis frequently accompanies myocarditis.

Clinical Features

Systemic signs and symptoms of myocarditis are nonspecific and can include myalgias, headache, rigors, and fever and tachycardia. Chest pain with a coexisting pericarditis is frequently present and may be accompanied by a pericardial friction rub. In severe cases, there may be symptoms of progressive heart failure with dyspnea on exertion, pulmonary rales, pedal edema, and/or cardiogenic shock.

Diagnosis and Differential

ECG may be normal or may demonstrate nonspecific changes such as atrioventricular block, prolonged QRS duration, or ST-segment elevation and PR depression if there is accompanying pericarditis. Chest radiogra phy is typically normal, but may demonstrate pulmonary congestion if symptoms are severe. Cardiac enzymes may be elevated. Differential diagnosis includes cardiac ischemia or infarction, valvular disease, and sepsis.

24_Cydulka_Ch24.indd 161 15/03/17 3:17 PM

Source page 181

Emergency Department Care and Disposition

1. Admission with supportive care is the mainstay of treatment for myo carditis.
2. Initiate appropriate antibiotics for patients with suspected bacterial myocarditis.
3. Patients with progressive heart failure symptoms associated with myo carditis may require admission to an ICU setting.

Left Ventricular Assist Devices

In patients with severe cardiomyopathy, LVADs may be used to augment left ventricular output. These devices consist of an implanted pump that transfers blood from the apex of the left ventricle to the proximal aorta. The pump receives its power from an external source (a battery or bedside power unit outside of the body) that is connected to a controller outside of the body. The controller drives the pump via a drive line that connects the pump to the outside battery and controller through a small abdominal wall incision.

Patients and their families usually have extensive training on the devices and should be involved in the evaluation of the LVAD patient. In addition, most patients will carry a card with their LVAD coordinator’s contact infor

mation in the bag with their spare battery and controller. It is imperative to contact these coordinators as soon as possible to discuss diagnostic and management options.

Clinical Features

Most LVADs now employ pumps that create continuous blood flow to maintain a normal mean arterial pressure. However, because flow is continuous, many LVAD patients may not have a normal palpable pulse. Blood pressure readings can be obtained via Doppler or mechanical cuff. Auscultation reveals heart sounds associated with a whirr from the LVAD pump. The ECG should have discernible QRS complexes. The chest radiograph should show the LVAD components in place.

Emergency Department Care and Disposition

1. Many experts recommend to never perform CPR on an LVAD patient with hemodynamic instability, as chest compressions may dislodge the LVAD causing LV rupture with intractable hemorrhage.
2. Listen to the heart. If the whirr of the pump is heard, go on to step 3. If the whirr is not heard, enlist the patient and family to help search for a cause of mechanical failure and to change the battery and controller, but do not disconnect any of the equipment.
3. If the whirr of the pump is heard, obtain a blood pressure via Doppler or mechanical cuff, place the patient on the monitor, obtain IV access, and administer a bolus of normal saline.
4. Evaluate for the common complications (infection, anemia, bleeding, and thromboembolism). If identified, treat according to the following:

a. ■Infection at the abdominal wall outlet for the drive site is treated with antibiotics.

24_Cydulka_Ch24.indd 162 15/03/17 3:17 PM

Source page 182

b. Anemia secondary to red cell destruction from the pump or hemorrhage from anticoagulation is treated with blood transfusions as needed.
c. Thromboembolic events (such as pulmonary embolism, mesenteric ischemia, or stroke) are treated with heparin once bleeding has been ruled out.
5. If hypotension persists and/or if right ventricular failure is present, administer pressors to support appropriate hemodynamics.
6. Communication with the LVAD coordinator can be of considerable assistance to augment diagnostic and management strategies for any LVAD patient.

■■ CARDIOMYOPATHIES WITH DIASTOLIC DYSFUNCTION

Hypertrophic Cardiomyopathy

Hypertrophic cardiomyopathy is characterized by asymmetric left ven tricular and/or right ventricular hypertrophy primarily involving the intra ventricular septum. The result is decreased compliance of the left ventricle leading to impaired diastolic relaxation and diastolic filling. Cardiac out put and ejection fraction are usually normal. This condition is often hereditary.

Clinical Features

The type and severity of symptoms depend on age and tend to be more severe as patients get older. Dyspnea on exertion is the most common symptom, followed by chest pain, palpitations, and syncope. Patients may be aware of forceful ventricular contractions as well. Physical exam usually reveals a fourth heart sound and a systolic ejection murmur heard best at the lower left sternal boarder or apex that does not typically radiate to the neck. The examiner may enhance this murmur by having the patient Val salva or stand, and a decrease in the murmur may be heard with maneuvers that increase LV filling such as squatting and passive leg elevation.

Diagnosis and Differential

The ECG in hypertrophic cardiomyopathy is nonspecific, but often demon strates left ventricular hypertrophy and left atrial enlargement. Deep S waves may be seen with large septal Q waves (>0.3 mV) and upright T waves. In contrast, inverted T waves can suggest ischemia. Chest radio graph is usually normal. Echocardiography is the diagnostic study of choice and will demonstrate disproportionate septal hypertrophy.

Emergency Department Care and Disposition

Treatment is primarily supportive. Patients with suspected hypertrophic cardiomyopathy should be referred for echocardiographic evaluation.

1. Patients with suspected hypertrophic cardiomyopathy who have syncope should be hospitalized for cardiac monitoring and a thorough evaluation. This presentation can be a precursor to sudden cardiac death.

2. Following definitive diagnosis, ß-blockers, such as atenolol 25 to 50 mg orally every day, are the mainstay of treatment for patients with hyper trophic cardiomyopathy and chest pain.

24_Cydulka_Ch24.indd 163 15/03/17 3:17 PM

Source page 183

Restrictive Cardiomyopathy

This type of cardiomyopathy is relatively uncommon and can be idiopathic or caused by a number of different etiologies such as sarcoidosis, sclero

derma, and amyloidosis. In this disorder, ventricular filling is restricted leading to normal or diminished diastolic volume while systolic function is usually normal.

Clinical Features

The predominant symptoms are dyspnea, orthopnea, and pedal edema that occur without associated cardiomegaly or systolic dysfunction. Chest pain is uncommon. Physical examination is consistent with the degree of dis ease, and may show a third or fourth heart sound, cardiac gallop, pulmonary rales, jugular venous distension, inspiratory jugular venous distention (Kussmaul's sign), hepatomegaly, pedal edema, or ascites.

Diagnosis and Differential

The ECG is usually nonspecific, but may show conduction disturbances or low voltage in the setting of sarcoidosis or amyloidosis. The chest radio graph may show signs of acute heart failure without cardiomegaly. The differential diagnosis includes constrictive pericarditis and diastolic left ventricular dysfunction. Differentiating between restrictive cardiomyopa thy and constrictive pericarditis is important as constrictive pericarditis can be treated surgically.

Emergency Department Care and Disposition

1. Most patients should be admitted for further diagnostic and therapeutic interventions.
2. Direct treatment towards symptom control by using diuretics and ACE inhibitors.
3. Some causes of restrictive cardiomyopathy may have specific treatment therapies, such as corticosteroids for sarcoidosis or chelation for the treatment of hemochromatosis.

Acute Pericarditis

Pericarditis involves inflammation of the layers covering the heart and may be idiopathic or due to infections such as a virus, bacteria, or fungus. Other etiologies include malignancy, drugs, radiation, connective tissue disease, uremia, myxedema, or postmyocardial infarction (Dressler's syndrome).

Clinical Features

The most common symptom of pericarditis is sharp or stabbing precordial or retrosternal chest pain that may radiate to the back, neck, shoulder, or arm. The classic description is pain that is made worse by lying supine, and lessened by sitting up and leaning slightly forward. Movement, swallowing, or inspiration may also aggravate the pain. Radiation to the left trapezial ridge is distinctive. Associated symptoms include recent or current infec tion, low-grade intermittent fever, dyspnea, and dysphagia. The physical exam is usually normal, but an intermittent friction rub heard best at the

lower left sternal border or apex may be present.

24_Cydulka_Ch24.indd 164 15/03/17 3:17 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 180–187.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 25: Venous Thromboembolism

Source pages 188–195 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 188–195

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Venous Thromboembolism**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 25: Venous Thromboembolism 169 Table 25-2 Wells’ Score for Pulmonary Embolism Factors Points* Suspected deep venous thrombosis 3 Alternative diagnosis less likely than PE 3 Heart rate >100 beats/min 1.5 Prior venous thromboembolism 1.5 Immobilization within prior 4 weeks 1.5 Active malignancy 1 Hemoptysis 1 * Risk score interpretation (probability of PE): >6 points = high risk (78.4%); 2–6 points = moderate risk (27.8%); and <2 points = low risk (3.4%). [Source pages 188–195]

- Source: Adapted with permission from Wells PS, Anderson DR, Rodger M, et al. [Source pages 188–195]

- Derivation of a simple clinical model to categorize patients probability of pulmonary embolism: increasing the models utility with the SimpliRED d-dimer. [Source pages 188–195]

- Diagnostic Testing For patients who need additional testing and have low or intermediate pretest probability, serum d-dimer testing is the recommended initial test. [Source pages 188–195]

- The diag- nostic sensitivity of automated quantitative d-dimer assays ranges from 94% to 98% and the specificity from 50% to 60% for PE and DVT. [Source pages 188–195]

Clinical reasoning structure

- **Problem representation:** Venous Thromboembolism; source pages 188–195.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Venous Thromboembolism**.

- Source-page range: **188–195**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 188

CHAPTER 25: Venous Thromboembolism 169

Table 25-2 Wells' Score for Pulmonary Embolism

Factors Points*

Suspected deep venous thrombosis

3

Alternative diagnosis less likely than PE

3

Heart rate >100 beats/min 1.5

Prior venous thromboembolism 1.5

Immobilization within prior 4 weeks 1.5

Active malignancy

1

Hemoptysis

1

*

Risk score interpretation (probability of PE): >6 points = high risk (78.4%); 2–6 points = moderate risk (27.8%); and <2 points = low risk (3.4%).

Source: Adapted with permission from Wells PS, Anderson DR, Rodger M, et al. Derivation of a simple clinical model to categorize patients probability of pulmonary embolism: increasing the models utility with the SimpliRED d-dimer. Thromb Haemost. 2000;83(3):416–420.

Diagnostic Testing

For patients who need additional testing and have low or intermediate pretest probability, serum d-dimer testing is the recommended initial test. The diagnostic sensitivity of automated quantitative d-dimer assays ranges from 94% to 98% and the specificity from 50% to 60% for PE and DVT. The d-dimer has a half-life of approximately 8 hours and can be elevated for at least 3 days after symptomatic VTE. Advanced age, pregnancy, active malignancy, recent surgery, liver disease, rheumatologic disease, infection,

trauma, and sickle cell disease can all elevate d-dimer levels in the absence of VTE. d-dimer cutoffs for use in screening for VTE can be adjusted for age to maintain adequate exclusionary ability. The most commonly used formula

Table 25-3 Wells' Score for Deep Vein Thrombosis

Clinical Features	Points*
Active cancer (treatment within 6 months, or palliation)	1
Paralysis, paresis, or immobilization of lower extremity	1
Bedridden for >3 days because of surgery (within 12 weeks)	1
Localized tenderness along distribution of deep veins	1
Entire leg swollen	1
Unilateral calf swelling of >3 cm (below tibial tuberosity)	1
Unilateral pitting edema	1
Collateral superficial veins	1
Alternative diagnosis as likely as or more likely than deep venous thrombosis	–2
Prior history of DVT or PE†	1

Risk score interpretation (probability of deep venous thrombosis) in original Wells' DVT model:

*

≥3 points: high risk (75%); 1 or 2 points: moderate risk (17%); <1 point: low risk (3%).

†

Only awarde

Source page 189

Determine pretest probability of DVT using Wells' criteria

High or moderate probability score

Low probability score (<1 point)

(>0 points)

D-dimer testing US

– + + –

DVT ruled + Treat D-dimer

US

out DVT testing

– +

–

DVT ruled

out DVT ruled Repeat

out US in 1 week

Figure 25-1. Diagnostic algorithm for deep vein thrombosis, applied in patients with leg symptoms compatible with DVT. +, positive test result; –, negative test result.

is age × 10 ng/mL to determine the upper limit of normal. For patients with high pretest probability of VTE, those with a positive d-dimer, or those in whom d-dimer testing is very likely to be positive, imaging is indicated. Figure 25-1 illustrates a recommended diagnostic pathway for suspected DVT. Patients with a low pretest probability of DVT can be adequately evaluated with a serum d-dimer. Patients with a moderate or high pretest probability for DVT or those who have a positive d-dimer should undergo venous ultrasound (US). Venous US is nonionizing and has high sensitivity (90% to 95%) and specificity (95%) for lower extremity DVT. Sensitivity of venous US is lower for pelvic and isolated calf DVT and in obese patients. To rule out DVT in patients with high clinical probability, some diagnostic algorithms recommend both a negative initial ultrasound and a concurrent negative d-dimer or a follow-up ultrasound performed 1 week after the initial negative testing.

Figure 25-2 shows one recommended diagnostic algorithm for PE, though no singular diagnostic test or approach perfectly diagnoses or excludes PE. For patients who need radiographic imaging to evaluate for PE, a computed tomography (CT) pulmonary angiography of the chest is the test of choice (see Fig. 25-3) where a blood clot appears as a filling defect in a contrast-enhanced pulmonary artery. The sensitivity of a technically adequate CT for PE is about 90%, and specificity is 95%. Unfortunately, about 10% of scans are technically inadequate and should be interpreted with this limitation in mind. The negative likelihood ratio of a CT scan is 0.12, which is similar to that of a negative d-dimer. Therefore,

25_Cydulka_Ch25.indd 170 15/03/17 3:37 PM

Source page 190

Assess gestalt

pretest probability*

<15% 15–40% >40%

LMWH if no

PERC rule contraindication

– +

Order

imaging

Quant

No PE – +

D -dimer

– +

Serum Cr – Serum Cr –

GFR <60 mL/min GFR >59 mL/min

V/Q CTPA

+ –

Nl High Nondx

PE (+)

No PE

Figure 25-2. Pulmonary Embolism Rule-Out Criteria (PERC) rule—diagnostic algorithm for pulmonary embolism (PE). +, positive for PE; –, negative for PE; Cr, creatinine; High, high probability scan findings; LMWH, low-molecular-weight heparin; Nl, normal; Nondx, nondiagnostic (any reading other than normal or high probability); quant, quantitative.* Some physicians prefer to start with a clinical decision rule such as the Wells' score (where <2, 2 to 6, and >6 are used instead of <15%, 15% to 40%, and >40%, respectively). Note: Determine renal function by clinical picture (healthy, no risk factors for reduced glomerular filtration rate (GFR)) or calculated GFR. Nondiagnostic ventilation-perfusion (V■/Q■) scan findings require confirmation from results of another test, such as CT pulmonary angiography (CTPA), if benefits outweigh risks.

as with d-dimer testing, CT scan results should be interpreted in the appropriate context of pretest probability. Compared to other imaging modalities, CT scanning has the advantage of demonstrating important alternative diagnoses in 18% to 22% of patients.

Ventilation-perfusion (V■/Q■) scanning is another testing option to assess for possible PE. This is performed by comparing emission of radioisotope that has been injected into the pulmonary arteries to emission of radioisotope

25_Cydulka_Ch25.indd 171 15/03/17 3:37 PM

Source page 191

Figure 25-3. Axial image from a chest CT angiogram demonstrating a filling defect consistent with acute pulmonary embolism. Two white arrowheads outline

a circular filling defect in the right middle lobar pulmonary artery. The long white arrow projecting in the left lung points to a filling defect in a segmental artery in the posterior medial segmental artery.

that has been inhaled into the alveoli. A V■/Q■ scan that demonstrates homogeneous scintillation throughout the lung in the perfusion portion rules out PE with 100% sensitivity. A V■/Q■ scan with two or more unmatched filling defects indicates 80% probability of PE. However, only about one-third of V■/Q■ scans demonstrate findings sufficient to diagnose or rule out PE with certainty.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

The treatment of VTE consists of initial stabilization, anticoagulation, and thrombolytic or surgical therapy in severe cases. All patients suspected of having PE should have their cardiac rhythm, blood pressure, and pulse oxygenation measured.

1. Utilize supplemental oxygen when necessary to maintain adequate pulse oximetry readings.

2. Administer intravenous crystalloid fluids as needed to augment preload and correct hypotension.

25_Cydulka_Ch25.indd 172 15/03/17 3:37 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 188–195.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 26: Systemic and Pulmonary Hypertension

Source pages 196–201 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 196–201

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Systemic and Pulmonary Hypertension**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 26: Systemic and Pulmonary Hypertension 177 Table 26-2 Hypertensive Emergencies Diagnostic Category Evidence of Acute End-Organ Damage Acute aortic dissection Abnormal CT angiogram of chest and abdomen/pel- vis or transesophageal echocardiogram of the aorta Acute pulmonary edema Interstitial edema on chest radiograph Acute myocardial infarction Changes on ECG or elevated levels of cardiac biomarkers Acute coronary syndrome Clinical diagnosis, changes on ECG, or elevated levels of cardiac biomarkers Acute renal failure Elevated serum creatinine level, proteinuria Severe preeclampsia, HELLP Proteinuria, hemolysis, elevated liver enzyme levels, syndrome, eclampsia low platelet counts Hypertensive retinopathy Retinal hemorrhages and cotton-wool spots, hard exudates, and sausage-shaped veins Hypertensive encephalopathy May see papilledema or arteriolar hemorrhage or exudates on funduscopic examination, may note cerebral edema with a predilection for the posterior white matter of the brain on MRI Subarachnoid hemorrhage Abnormal CT of the brain Intracranial hemorrhage Abnormal CT of the brain; RBC on lumbar puncture Acute ischemic stroke Abnormal MRI or CT of the brain Acute perioperative hypertension Clinical diagnosis; manifestations of other hyperten- sive emergencies Sympathetic crisis Clinical diagnosis in the setting of sympathomimetic drug use (i.e., cocaine or amphetamines) or pheochro- mocytoma (24-h urine assay for catecholamines and metanephrine or plasma fractionated metanephrines) Abbreviation: HELLP, hemolysis, elevated liver enzymes, low platelets. [Source pages 196–201]

- Formulate a problem-specific plan of care with a primary therapeutic goal of safely reducing arterial pressure while avoiding hypoperfusion. [Source pages 196–201]

- Initiate treatments to reduce heart rate to 60 beats per minute or less and systolic blood pressure between 100 and 140 mm Hg, with an ideal target of less than 120 mm Hg. [Source pages 196–201]

- Initiate beta- adrenergic antagonists before starting vasodilator agents. [Source pages 196–201]

- Initial options include esmolol 250 to 500 µg/kg IV over 1 to 3 minutes followed by continuous infusion of 50 µg/kg/min, or labetalol 10 to 20 mg IV over 2 minutes followed by subsequent doses of 40 to 80 mg every 10 minutes as needed to a maximum dose of 300 mg. [Source pages 196–201]

Clinical reasoning structure

- **Problem representation:** Systemic and Pulmonary Hypertension; source pages 196–201.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Systemic and Pulmonary Hypertension**.
- Source-page range: **196–201**.
- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 196

CHAPTER 26: Systemic and Pulmonary Hypertension 177

Table 26-2 Hypertensive Emergencies

Diagnostic Category Evidence of Acute End-Organ Damage

Acute aortic dissection Abnormal CT angiogram of chest and abdomen/pel-

vis or transesophageal echocardiogram of the aorta

Acute pulmonary edema Interstitial edema on chest radiograph

Acute myocardial infarction Changes on ECG or elevated levels of cardiac

biomarkers

Acute coronary syndrome Clinical diagnosis, changes on ECG, or elevated

levels of cardiac biomarkers

Acute renal failure Elevated serum creatinine level, proteinuria

Severe preeclampsia, HELLP Proteinuria, hemolysis, elevated liver enzyme levels,

syndrome, eclampsia low platelet counts

Hypertensive retinopathy Retinal hemorrhages and cotton-wool spots, hard

exudates, and sausage-shaped veins

Hypertensive encephalopathy May see papilledema or arteriolar hemorrhage or

exudates on funduscopic examination, may note

cerebral edema with a predilection for the posterior

white matter of the brain on MRI

Subarachnoid hemorrhage Abnormal CT of the brain

Intracranial hemorrhage Abnormal CT of the brain; RBC on lumbar puncture

Acute ischemic stroke Abnormal MRI or CT of the brain

Acute perioperative hypertension Clinical diagnosis; manifestations of other hypertensive emergencies

Sympathetic crisis Clinical diagnosis in the setting of sympathomimetic drug use (i.e., cocaine or amphetamines) or pheochromocytoma (24-h urine assay for catecholamines and metanephrine or plasma fractionated metanephrines)

Abbreviation: HELLP, hemolysis, elevated liver enzymes, low platelets.

Formulate a problem-specific plan of care with a primary therapeutic goal of safely reducing arterial pressure while avoiding hypoperfusion.

1. Aortic dissection. Initiate treatments to reduce heart rate to 60 beats per minute or less and systolic blood pressure between 100 and 140 mm Hg, with an ideal target of less than 120 mm Hg. Initiate beta-adrenergic antagonists before starting vasodilator agents. Initial options include esmolol 250 to 500 µg/kg IV over 1 to 3 minutes followed by continuous infusion of 50 µg/kg/min, or labetalol 10 to 20 mg IV over 2 minutes followed by subsequent doses of 40 to 80 mg every 10 minutes as needed to a maximum dose of 300 mg. If beta-adrenergic antagonists are contraindicated, diltiazem 0.25 mg/kg over 2 minutes, alpha-adrenergic antagonists, benzodiazepines, or vasodilators may be necessary. When additional blood pressure control is needed, consider nicardipine 5 mg/h IV with incremental increases by 2.5 mg/h every 5 to 15 minutes to a maximum infusion rate of 15 mg/h. An alternative blood pressure control agent is sodium nitroprusside 0.3 to 0.5 µg/kg IV with incremental increases by 0.5 µg/kg/min.

2. Acute hypertensive pulmonary edema. The primary therapeutic goal is to reduce systemic afterload by 20% to 30%. There are no clear

26_Cydulka_Ch26.indd 177 15/03/17 3:44 PM

Source page 197

evidence-based guidelines for the single best treatment for hypertensive pulmonary edema, although diuresis and vasodilation are commonly used. Nitroglycerin 0.4 mg sublingual up to three doses, topical 1 to 2 inches transdermal, or IV infusion starting at 5 µg/min with increases by 5 µg/min every 3 to 5 minutes up to 20 µg/min. Refractory cases can continue nitroglycerin IV infusion with incremental increases by 10 µg/min every 3 to 5 minutes to a maximum of 200 µg/min or add nicardipine (see dose above, #1). Consider sodium nitroprusside (see dose above, #1) when other agents are unsuccessful,

keeping in mind its association with cyanide and thiocyanate toxicity in patients with prolonged therapy or renal failure. Intravenous enalaprilat 0.625 to 1.25 mg IV over 5 minutes every 4 to 6 hours, incrementally increased by 1.25 mg to a maximum of 5 mg every 6 hours is an additional adjunct, but may induce significant “first-dose” hypotension, and is pregnancy category D.

3. Acute coronary syndromes. The primary therapeutic goals are to limit ischemia and reduce the mean arterial pressure by no more than 25%. Sublingual or intravenous nitroglycerin (see dose above, #2) is preferred. Routine use of intravenous beta-adrenergic antagonists has fallen out of favor, and these agents are currently recommended only when patients have severe hypertension and should be avoided for those with risk for progression to cardiogenic shock. Oral beta-adrenergic antagonists may not provide sufficiently rapid blood pressure control in a hypertensive emergency.

4. Acute sympathetic crisis. The primary therapeutic goals are to decrease adrenergic stimulation and to relieve symptoms. Benzodiazepines are the recommended first intervention. Nitroglycerin (see dose above, #2), phentolamine 5 to 15 mg IV followed by continuous infusion of 0.2 to 0.5 mg/min, or nicardipine (see dose above, #1) are secondary adjuncts to consider.

5. Acute renal failure. The primary therapeutic goals are to reduce systemic vascular resistance with acute blood pressure reductions of less than 20% to help preserve renal blood flow. Recommended agents include fenoldopam 0.1 µg/kg/min IV titrated to desired effect every 15 minutes to a maximum of 1.6 µg/kg/min, clevidipine 1 to 2 mg/h IV, or nicardipine (see dose above, #1).

6. Preeclampsia. The primary therapeutic goals are to reduce significant elevations of blood pressure and prevent progression to eclampsia. Labetalol (see dose above, #1) and hydralazine 5 to 10 mg IV are commonly prescribed agents for blood pressure levels greater than 160/100 mm Hg.

7. Hypertensive encephalopathy. The primary therapeutic goal is to rapidly reduce significantly elevated mean arterial pressures by 20% to 25% once other neurologic emergencies are excluded. Recommended agents include nicardipine (see dose above, #1), labetalol (see dose above, #1), fenoldopam (see dose above, #5), or clevidipine (see dose above, #5).

8. Subarachnoid hemorrhage. The primary therapeutic goals for blood pressure control in subarachnoid hemorrhage are unclear, although

reduction of systolic blood pressure to 160 mm Hg or lower is generally recommended. Recommended agents include nicardipine (see dose

26_Cydulka_Ch26.indd 178 15/03/17 3:44 PM

Source page 198

above, #1), labetalol (see dose above, #1), esmolol (see dose above, #1), or clevidipine (see dose above, #5). Oral nimodipine may reduce vasospasm and subsequent cerebral infarction rate in patients with moderately increased blood pressure not requiring an intravenous agent.

9. Intracranial hemorrhage. The primary therapeutic goals for blood pressure control in intracranial hemorrhage are to reduce hemorrhage expansion and to lower systolic blood pressures to 120 to 160 mm Hg. Recommended agents include nicardipine (see dose above, #1), labetalol (see dose above, #1), or esmolol (see dose above, #1).

10. Acute ischemic stroke: The primary therapeutic goals for blood pressure control in acute ischemic stroke depend on the likelihood of fibrinolytic therapy, and should be balanced to avoid worsening ischemia from excessive blood pressure reduction. If fibrinolytic therapy is planned, blood pressure should be below 185/110 mm Hg, with a systolic blood pressure goal of 141 to 150 mm Hg. Recommended agents include labetalol (see dose above, #1), nicardipine (see dose above, #1), or sodium nitroprusside (see dose above, #1). If no fibrinolytic therapy is planned and significantly elevated blood pressures remain (e.g., 220/120 mm Hg), blood pressure should be lowered by no more than 10% to 15% in the first 24 hours.

11. For hypertensive urgency, useful agents include labetalol 200 to 400 mg orally, captopril 12.5 to 25 mg orally every 4 to 6 hours, losartan 50 mg orally per day, or clonidine 0.1 to 0.2 mg orally.

12. Consider starting an oral agent on an outpatient basis for asymptomatic patients with severe hypertension, such as those with systolic blood pressures above 180 to 200 mm Hg or diastolic blood pressures above 110 to 120 mm Hg. A diuretic agent such as hydrochlorothiazide 25 mg orally each day is an appropriate medication to consider starting in patients with uncomplicated asymptomatic severe hypertension. For patients with angina, postmyocardial infarction, migraines, or supraventricular arrhythmias, consider a beta-adrenergic antagonists such as metoprolol 50 mg orally once or twice daily. An angiotensin-converting enzyme inhibitor such as lisinopril 10 mg orally each day can be used in patients with a history of heart failure, renal disease, recurrent strokes, or diabetes mellitus.

■■ CHILDHOOD HYPERTENSIVE EMERGENCIES

Hypertensive emergencies in children, as for adults, refer to profoundly elevated blood pressures associated with target organ damage. Neonates often present with nonspecific signs such as apnea, cyanosis, or poor feeding. Older children may present with symptoms and physical examination findings more typical of adults such as throbbing frontal headaches or blurred vision.

Renovascular lesions and pheochromocytoma are the most commonly identified etiologies for pediatric hypertensive emergencies. The decision to treat a hypertensive emergency in a child is based on the blood pressure and associated symptoms indicating target organ damage. Consider urgent treatment if the hypertensive blood pressure exceeds prior measurements by 30%, or if diastolic or systolic blood pressures exceed the 90th to 95th percentiles on a standardized table of blood pressure by age. The primary

26_Cydulka_Ch26.indd 179 15/03/17 3:44 PM

Source page 199

therapeutic goal for blood pressure control in an acutely symptomatic child with a hypertensive emergency is to lower the mean arterial pressure by 25% within 1 hour. Preferred medications include labetalol 0.2 to 1 mg/kg IV to a maximum of 40 mg per dose or a continuous infusion of 0.25 to 3 mg/kg/h or nicardipine 0.5 to 3 µg/kg/min IV.

The second-line agents include hydralazine 0.2 to 0.6 mg/kg IV to a maximum of 20 mg per dose, esmolol 100 to 500 µg/kg followed by an infusion of 50 to 150 µg/kg/min, or clevidipine 0.5 to 3.5 µg/kg/min IV. Refractory cases may be treated with nitroprusside 0.5 to 10 µg/kg/min IV. Phentolamine 0.1 mg/kg IV to a maximum of 5 mg is the drug of choice for significantly elevated blood pressure in the setting of a pheochromocytoma, while awaiting definitive surgical treatment.

■■ PULMONARY HYPERTENSION

Pulmonary hypertension is characterized by an elevated pressure within the pulmonary vascular system compromising right ventricular function. Pulmonary hypertension is defined as a mean pulmonary artery pressure exceeding 25 mm Hg at rest or 30 mm Hg with exertion. Pulmonary hypertension can manifest with isolated pulmonary arterial pressure elevations or combined pulmonary arterial and venous pressure elevations. Definitive diagnosis requires right heart catheterization.

The most common symptom of pulmonary hypertension is dyspnea, either at rest or with exertion. Other symptoms include fatigue, syncope or near-syncope, and chest pain. Typically, this disorder will be seen in asso-

ciation with other cardiovascular or pulmonary disorders such as chronic obstructive pulmonary disease, left ventricular dysfunction, or disorders associated with hypoxemia.

When treating patients with suspected or known pulmonary hypertension:

● ■Provide supplemental oxygen to maintain saturation levels greater than 90% in patients presenting with hypoxia. Intubation in patients with severe pulmonary hypertension can cause rapid cardiovascular collapse due to increased intrathoracic pressure from positive pressure ventilation.

● ■Optimize intravascular volume to treat or prevent hypotension.

● ■Augment right ventricular function when necessary by using dobutamine 2 to 10 µg/kg/min IV or milrinone 0.375 µg/kg/min IV.

● ■Maintain coronary artery perfusion when necessary by using norepinephrine 0.05 to 0.75 µg/kg/min IV.

● ■Reduce right ventricular afterload with prostanoids (e.g., epoprostenol) and phosphodiesterase-5 inhibitors (e.g., sildenafil). Although these agents are rarely initiated in the emergency department, patients with previously diagnosed pulmonary hypertension may present already receiving continuous infusions of one of these agents. When assessing these patients in the acute setting, confirm catheter and pump function as one possible cause of clinical decompensation.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 57, "Systemic Hypertension," by Brigitte M. Baumann; Chapter 58, "Pulmonary Hypertension," by Michael E. Winters.

26_Cydulka_Ch26.indd 180 15/03/17 3:44 PM

References/provenance

- Source: *Tintinalli's Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 196–201.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 27: Aortic Aneurysms and Aortic Dissection

Source pages 202–207 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 202–207

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Aortic Aneurysms and Aortic Dissection**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 27: Aortic Aneurysms and Aortic Dissection 183 Table 27-1 Nonaortic Large-Artery Aneurysms Artery Risk Factors Clinical Presentation Management Popliteal (>2 cm Advanced age, male Most common Thrombolysis, liga- or >150% of gender, trauma, peripheral aneurysm; tion, arterial bypass, normal caliber) congenital disorders discomfort behind knee endovascular repair with swelling with or without deep venous thrombosis Subclavian Arteriosclerosis, Pulsatile mass above Surgical repair thoracic outlet or below clavicle, dys- obstruction phagia, stridor, chest pain, hoarseness, upper extremity fatigue or numbness and tingling, limb ischemic symptoms Femoral Advanced age, male Pulsatile mass with Thrombolysis, liga- gender, trauma, or without pain, limb tion, arterial bypass, congenital disorders ischemic symptoms, endovascular repair peripheral embolic symptoms Femoral Prior femoral artery Pulsatile mass with or Surgical repair pseudoaneurysm catheterization, without pain trauma, infection Iliac Pain in groin, scro- Surgical repair tum, or lower abdo- men; sciatica; vulvar or groin hematoma with rupture Renal Age 40–60 years, Flank pain, hematuria, Surgical repair, no gender pref- collecting system nephrectomy erence, HTN, obstruction, shock if fibrodysplasia, ruptured arteriosclerosis Splenic Advanced age, Rapid symptom onset; Surgical repair, female gender, HTN, epigastric or left upper splenectomy, congenital, arterio- quadrant pain first, embolization if sclerosis, liver dis- then diffuse abdominal unruptured ease, multiparous, pain with rupture, rupture increased in shock pregnancy Hepatic Infection, Obstructive jaundice, Surgical ligation, arteriosclerosis, hemobilia from rupture embolization trauma, vasculitis into common bile duct, right upper quadrant pa

- 27_Cydulka_Ch27.indd 183 01/05/17 10:25 am 184 SECTION 4: Cardiovascular Diseases ■■ AORTIC DISSECTION Clinical Features Aortic dissection occurs when blood dissects between the intimal and adventitial layers of the aorta, and classically presents with acute chest pain that is most severe at onset and radiates to the back. [Source pages 202–207]

- The location of the pain may indicate the area of the aorta that is involved. [Source pages 202–207]

- Sixty percent of patients with dissection of the ascending aorta have anterior chest pain, and involve- ment of the descending aorta may cause abdominal or back pain. [Source pages 202–207]

- The pain pattern may change as the dissection progresses from one anatomic area to another. [Source pages 202–207]

Clinical reasoning structure

- **Problem representation:** Aortic Aneurysms and Aortic Dissection; source pages 202–207.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Aortic Aneurysms and Aortic Dissection**.

- Source-page range: **202–207**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 202

CHAPTER 27: Aortic Aneurysms and Aortic Dissection 183

Table 27-1 Nonaortic Large-Artery Aneurysms

Artery Risk Factors Clinical Presentation Management

Popliteal (>2 cm Advanced age, male Most common Thrombolysis, liga-

or >150% of gender, trauma, peripheral aneurysm; tion, arterial bypass,

normal caliber) congenital disorders discomfort behind knee endovascular repair

with swelling with or

without deep venous

thrombosis

Subclavian Arteriosclerosis, Pulsatile mass above Surgical repair

thoracic outlet or below clavicle, dys-

obstruction phagia, stridor, chest

pain, hoarseness,

upper extremity fatigue

or numbness and

tingling, limb ischemic

symptoms

Femoral Advanced age, male Pulsatile mass with Thrombolysis, liga-

gender, trauma, or without pain, limb tion, arterial bypass,

congenital disorders ischemic symptoms, endovascular repair

peripheral embolic
symptoms
Femoral Prior femoral artery Pulsatile mass with or Surgical repair
pseudoaneurysm catheterization, without pain
trauma, infection
Iliac Pain in groin, scro- Surgical repair
tum, or lower abdo-
men; sciatica; vulvar
or groin hematoma
with rupture
Renal Age 40–60 years, Flank pain, hematuria, Surgical repair,
no gender pref- collecting system nephrectomy
erence, HTN, obstruction, shock if
fibrodysplasia, ruptured
arteriosclerosis
Splenic Advanced age, Rapid symptom onset; Surgical repair,
female gender, HTN, epigastric or left upper splenectomy,
congenital, arterio- quadrant pain first, embolization if
sclerosis, liver dis- then diffuse abdominal unruptured
ease, multiparous, pain with rupture,
rupture increased in shock
pregnancy
Hepatic Infection, Obstructive jaundice, Surgical ligation,
arteriosclerosis, hemobilia from rupture embolization
trauma, vasculitis into common bile duct,
right upper quadrant
pain, peritonitis, upper
GI bleed
Abbreviation: HTN, hypertension.
27_Cydulka_Ch27.indd 183 01/05/17 10:25 am

Source page 203

184 SECTION 4: Cardiovascular Diseases

■■ AORTIC DISSECTION

Clinical Features

Aortic dissection occurs when blood dissects between the intimal and adventitial layers of the aorta, and classically presents with acute chest pain that is most severe at onset and radiates to the back. The location of the pain may indicate the area of the aorta that is involved. Sixty percent of patients with dissection of the ascending aorta have anterior chest pain, and involvement of the descending aorta may cause abdominal or back pain. The pain

pattern may change as the dissection progresses from one anatomic area to another. The pain is most commonly described as sharp, ripping, or tearing pain. Syncope can occur in 10% of patients.

Most patients with aortic dissection are male (66%), older than 50 years, and have a history of hypertension. Chronic cocaine use and prior cardiac surgery are additional risk factors. Younger patients with identifiable risk factors such as connective tissue disorders, congenital heart disease, and pregnancy are also at risk. Up to 30% of patients with Marfan's syndrome will develop a dissection. Iatrogenic aortic dissection may occur after aortic catheterization or cardiac surgery.

The Stanford classification divides dissections into those that involve the ascending aorta (type A) and those that are restricted to the descending aorta (type B). The DeBakey classification divides dissections into three groups: involvement of the ascending and descending aortas (type I), involvement of only the ascending aorta (type II), or involvement of only the descending aorta (type III).

As an aortic dissection progresses, seemingly unrelated symptom complexes may present themselves. Presentations include aortic valve insufficiency, coronary artery occlusion with myocardial infarction, carotid involvement with stroke symptoms, occlusion of vertebral blood supply with paraplegia, cardiac tamponade with shock and jugular venous distention, compression of the recurrent laryngeal nerve with hoarseness of the voice, and compression of the superior cervical sympathetic ganglion with Horner's syndrome.

Findings on physical examination will depend on the location and progression of the dissection. A diastolic murmur of aortic insufficiency may be heard. Hypertension and tachycardia are common, but hypotension also may be present. Fifty percent of patients have decreased pulsation in the radial, femoral, or carotid arteries, although no specific threshold values have been defined for blood pressure differences in extremities. Forty percent of patients have neurologic sequelae.

Diagnosis and Differential

The differential diagnosis to be considered depends on the location and progression of the dissection, but may include myocardial infarction, pericardial disease, stroke, spinal cord disorders, and primary conditions causing abdominal, back, or chest pain.

d-dimer testing is one of several biomarkers that have been investigated for potential utility in identifying or excluding dissection. No current guideline endorses the use of d-dimer as the sole means for excluding aortic dissection, partially due to a false-negative rate as high as 18%.

27_Cydulka_Ch27.indd 184 01/05/17 10:25 am

Source page 204

Figure 27-2. CT image of a type A aortic dissection. True and false lumens are present in the ascending aorta and descending aorta (descending false lumen at arrow) on noncontrast (left) and contrast (right) images. AF, ascending false lumen; AT, ascending true lumen; DT, descending true lumen.

The diagnosis of aortic dissection depends on radiographic confirmation. The most commonly found abnormalities on chest x-ray are an abnormal aortic contour and widening of the mediastinum, while other findings may include deviation of the trachea, mainstem bronchi, or esophagus, apical capping, pleural effusion, or displacement of aortic intimal calcifications. The chest x-ray is normal in 12% to 37% of patients with aortic dissection. CT scan with IV contrast is the imaging modality of choice and can reliably detect a false lumen as well as help to identify the extent of the dissection including extension into other vessels (see Fig. 27-2). Transesophageal echocardiograms are 97% to 100% sensitive and 97% to 99% specific. The preferential use of these studies may be institutionally dependent, and clinicians should coordinate diagnostic testing with the consulting vascular or thoracic surgeon.

Emergency Department Care and Disposition

1. Consult with a vascular or thoracic surgeon for patients with confirmed or strongly suspected aortic dissection to determine if operative intervention is indicated.
2. Stabilize hemodynamics with large-bore IV access, IV fluids for hypotension, and blood transfusion when required.
3. Manage hypertension with β-blockers such as esmolol 0.1 to 0.5 mg/kg IV bolus followed by a 0.025 to 0.2 mg/kg/min infusion, or labetolol 20 mg IV followed by subsequent doses of 20 to 40 mg IV every 10 minutes. Reduce the heart rate to a goal of between 60 and 70 beats per minute and the systolic blood pressure to a goal of 100 to 120 mm Hg.

27_Cydulka_Ch27.indd 185 01/05/17 10:25 am

Source page 205

4. Additional blood pressure reduction with agents such as nitroprusside or nicardipine should be used when needed after adequate inotropic blockade has been achieved and systolic blood pressures remain above 120 mm Hg.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 59, “Aortic Dissection and Related Aortic Syndromes,” by Gary A. Johnson and Louise A. Prince; and Chapter 60, “Aneurysmal Disease” by Louise A. Prince and Gary A. Johnson.

27_Cydulka_Ch27.indd 186 01/05/17 10:25 am

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 202–207.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 28: Arterial Occlusion

Source pages 208–211 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 208–211

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Arterial Occlusion**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 28: Arterial Occlusion 189 Table 28-1 Rutherford Criteria for Acute Limb Ischemia Findings Doppler Signals Description/ Muscle Category Prognosis Sensory Loss Weakness Arterial Venous I. [Source pages 208–211]

- Viable Not immediately None None Audible Audible threatened II. [Source pages 208–211]

- Marginally Salvageable if Minimal (toes) None Inaudible Audible promptly treated or none b. [Source pages 208–211]

- Immediately Salvageable More than toes, Mild, Inaudible Audible with immediate associated with moderate revascularization rest pain III. [Source pages 208–211]

- Irreversible Major tissue loss Profound, Profound, Inaudible Inaudible or permanent anesthetic paralysis nerve damage (rigor) inevitable Reproduced with permission from Rutherford RB, Baker JD, Ernst C, et al. [Source pages 208–211]

Clinical reasoning structure

- **Problem representation:** Arterial Occlusion; source pages 208–211.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Arterial Occlusion**.

- Source-page range: **208–211**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 208

CHAPTER 28: Arterial Occlusion 189

Table 28-1 Rutherford Criteria for Acute Limb Ischemia

Findings Doppler Signals

Description/ Muscle

Category Prognosis Sensory Loss Weakness Arterial Venous

I. Viable Not immediately None None Audible Audible

threatened

II. Threatened

a. Marginally Salvageable if Minimal (toes) None Inaudible Audible

promptly treated or none

b. Immediately Salvageable More than toes, Mild, Inaudible Audible

with immediate associated with moderate

revascularization rest pain

III. Irreversible Major tissue loss Profound, Profound, Inaudible Inaudible

or permanent anesthetic paralysis

nerve damage (rigor)

inevitable

Reproduced with permission from Rutherford RB, Baker JD, Ernst C, et al. Recommended standards for

reports dealing with lower extremity ischemia: revised version. J Vasc Surg. September 1997;26(3): 517–538.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study

Guide, 8th ed., see Chapter 61, “Arterial Occlusion,” by Anil Chopra and

David Carr.

28_Cydulka_Ch28.indd 189 15/03/17 4:27 PM

Source page 209

This page intentionally left blank

28_Cydulka_Ch28.indd 190 15/03/17 4:27 PM

Source page 210

SECTION

Pulmonary Emergencies 5

CHAPTER Respiratory Distress

29 Baruch S. Fertel

The term respiratory distress includes both symptoms of dyspnea and signs indicating difficulty breathing. The cause of respiratory distress is often multifactorial and may include the findings of hypoxia, hypercapnia, and cyanosis.

■■ DYSPNEA

Dyspnea is the subjective feeling of difficult, labored, or uncomfortable breathing. There is no single pathophysiologic mechanism that causes dyspnea and its etiology may be secondary to pulmonary, cardiac, or neurological dysfunction.

Clinical Features

The initial assessment of any patient with dyspnea should be directed toward identifying respiratory failure, which requires more immediate action. Dyspnea alone is a subjective complaint often difficult to quantify; therefore, objective findings are often needed to aid in diagnosis. Assess for signs of impending respiratory failure which include tachycardia, tachypnea, stridor, the use of accessory respiratory muscles (intercostals, sternocleidomastoid), stridor, lethargy, agitation, altered mental status, and inability to speak due to breathlessness. If present, oxygen should be administered immediately. Early use of noninvasive ventilation is often helpful in reversing the downward trajectory, although the need for aggressive airway management and mechanical ventilation should be anticipated. Lesser degrees of distress allow for a more detailed approach.

Diagnosis and Differential

The history and physical examination, together with ancillary testing, will help identify the etiology of dyspnea (Table 29-1).

It can be challenging to differentiate cardiac (CHF) and pulmonary causes. Signs such as an S3 gallop, edema, jugular venous distention, a history of orthopnea, an elevated brain natriuretic peptide (BNP), or troponin and chest radiograph findings of edema and cardiomegaly may point to cardiac causes. Pulse oximetry is a rapid but insensitive screen for

29_Cydulka_Ch29.indd 191 15/03/17 4:27 PM

Source page 211

Table 29-1 Common Causes of Dyspnea in the ED

Most Immediately Life-Threatening

Most Common Causes Causes

Obstructive airway disease: asthma, chronic Upper airway obstruction: foreign body,
obstructive pulmonary disease angioedema, hemorrhage
Decompensated heart failure/cardiogenic Tension pneumothorax
pulmonary edema
Ischemic heart disease: unstable angina and Pulmonary embolism
myocardial infarction
Pneumonia Neuromuscular weakness: myasthenia gra-
vis, Guillain-Barré syndrome, botulism
Psychogenic Fat embolism

disorders of gas exchange as it may be falsely elevated as in methemoglobinemia. Arterial blood gas (ABG) analysis is more sensitive, detects both hypoxia and hypercarbia, and is useful for identifying metabolic causes, such as acidosis with compensatory tachypnea, but does not take into account work of breathing and fatigue. A peak expiratory flow rate may indicate reactive airway disease and a negative inspiratory force (NIF) may identify neurological causes of dyspnea. Additional ancillary tests that may prove helpful include a complete blood count to look for anemia, d-dimer assay when used with clinical decision rules to look for pulmonary embolus (PE), chest radiograph, electrocardiogram, and computed tomography of the chest. Bedside cardiopulmonary ultrasound is increasingly being used to look for pneumothorax, tamponade, pulmonary consolidation or effusion, right heart strain suggesting PE, and congestive heart failure.

Emergency Department Care and Disposition

1. ■The goal of oxygen therapy is to maintain the PaO2 above 60 mm Hg or the oxygen saturation above 90%. Lower PaO2 or saturation may be appropriate in those with longstanding lung disease such as chronic obstructive pulmonary disease (COPD).
2. ■After oxygenation has been ensured with an appropriate delivery device (nasal cannula, facemask, non-rebreather, BVM, NIV, or intubation) and the patient is stabilized, disorder-specific treatment and evaluation can be pursued.
3. ■The disposition of patients with dyspnea often depends on its etiology and acuity. Patients with hypoxia and an unclear cause of dyspnea will require hospital admission.

■■ HYPOXIA AND HYPOXEMIA

Hypoxia is the insufficient delivery of oxygen to the tissues. Oxygen delivery is a function of cardiac output, hemoglobin concentration, and oxygen saturation. Hypoxemia is defined as a PaO2 below 60 mm Hg. While hypoxia is often the result of hypoxemia, these two terms are not interchangeable and one can occur without the other. Hypoxemia results from a

29_Cydulka_Ch29.indd 192 15/03/17 4:27 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 208–211.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 29: Respiratory Distress

Source pages 212–219 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 212–219

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Respiratory Distress**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 29: Respiratory Distress 193 combination of five distinct mechanisms: (a) hypoventilation in which lack of ventilation increases PaCO2, thereby displacing oxygen from the alveo- lus and lowering the amount delivered to the alveolar capillaries; (b) right-to-left shunt in which blood bypasses the lungs, thereby increasing the amount of unoxygenated blood entering the systemic circulation; (c) venti- lation/perfusion mismatch in which areas of the lung are perfused but not ventilated; (d) diffusion impairment in which alveolar-blood barrier abnor- mality causes impairment of oxygenation; and (e) low inspired oxygen, such as that occurs at high altitude. [Source pages 212–219]

- Clinical Features Signs and symptoms of hypoxemia are nonspecific. [Source pages 212–219]

- Acute physiologic responses to hypoxemia include pulmonary arterial vasoconstriction and increases in minute ventilation and sympathetic tone manifesting as tachy- pnea, tachycardia, and an initial hyperdynamic cardiac state. [Source pages 212–219]

- The predomi- nant features are often neurologic, and may include headache, somnolence, lethargy, anxiety, agitation, coma, or seizures. [Source pages 212–219]

- Chronic hypoxemia may result in polycythemia, digital clubbing, cor pulmonale, and changes in body habi- tus (e.g., pulmonary cachexia or barreled chest of COPD). [Source pages 212–219]

Clinical reasoning structure

- **Problem representation:** Respiratory Distress; source pages 212–219.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Respiratory Distress**.

- Source-page range: **212–219**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 212

CHAPTER 29: Respiratory Distress 193

combination of five distinct mechanisms: (a) hypoventilation in which lack of ventilation increases PaCO2, thereby displacing oxygen from the alveolus and lowering the amount delivered to the alveolar capillaries; (b) right-to-left shunt in which blood bypasses the lungs, thereby increasing the amount of unoxygenated blood entering the systemic circulation; (c) ventilation/perfusion mismatch in which areas of the lung are perfused but not ventilated; (d) diffusion impairment in which alveolar-blood barrier abnormality causes impairment of oxygenation; and (e) low inspired oxygen, such as that occurs at high altitude.

Clinical Features

Signs and symptoms of hypoxemia are nonspecific. Acute physiologic responses to hypoxemia include pulmonary arterial vasoconstriction and increases in minute ventilation and sympathetic tone manifesting as tachypnea, tachycardia, and an initial hyperdynamic cardiac state. The predominant features are often neurologic, and may include headache, somnolence, lethargy, anxiety, agitation, coma, or seizures. Chronic hypoxemia may result in polycythemia, digital clubbing, cor pulmonale, and changes in body habitus (e.g., pulmonary cachexia or barreled chest of COPD). Cyanosis may be present but is not a sensitive or specific indicator of hypoxemia.

Diagnosis and Differential

A formal diagnosis of hypoxemia requires ABG analysis; however, pulse oximetry may be useful for gross abnormalities or trends. Hypoxemia may be quantified, and clues to its etiology may be obtained, by calculation of the alveolar-arterial oxygen gradient (“A-a gradient,” where the capital “A” represents alveolar oxygen tension and “a” indicates arterial oxygen level). The formula for calculating A-a gradient while breathing room air at sea

level is

$P(A - a)O_2 = 147 - (PaCO_2 \times 1.25) - PaCO_2$

The A-a gradient is increased in cases of right-to-left shunts, ventilation-perfusion mismatch, and diffusion impairment. The normal value for a 20-year-old seated upright and healthy nonsmoker is 5 to 10; the upper limit of normal increases by 1 for each decade of life. Patients with shunt will often not respond to supplemental oxygen while those with VQ mismatch, diffusion impairment, low inspired oxygen will improve their Pa02.

Emergency Department Care and Disposition

Regardless of the specific cause of hypoxemia, the initial approach remains the same. The following are general treatment guidelines for hypoxia:

1. ■Supplemental oxygen is administered to achieve an O_2 saturation greater than 90%.
2. ■The airway is managed aggressively if there are signs of respiratory failure (see Chapter 1).
3. Cause-specific treatment and evaluation should be pursued.
4. ■All patients with new hypoxemia should be admitted and monitored until their condition is stabilized.

29_Cydulka_Ch29.indd 193 15/03/17 4:27 PM

Source page 213

■■ HYPERCAPNIA

Hypercapnia occurs exclusively due to alveolar hypoventilation (releasing of CO_2) and is defined as a $PaCO_2$ above 45 mm Hg. Factors that affect alveolar ventilation include respiratory rate, tidal volume, and dead space volume, all of which are controlled by the body to maintain $PaCO_2$ in a narrow range.

Clinical Features

The signs and symptoms of hypercapnia depend on the absolute value of the $PaCO_2$ and the rate of change. Acute elevations result in increased intracranial pressure, prompting patient complaints of headache, confusion, and lethargy. Coma, encephalopathy, and seizures may occur when the $PaCO_2$ acutely rises above 80 mm Hg; similar $PaCO_2$ levels may be well tolerated if elevations are chronic.

Diagnosis and Differential

The diagnosis of hypercapnia requires clinical suspicion and ABG analysis as pulse oximetry may be completely normal. In acute cases, the ABG will demonstrate an elevation in $PaCO_2$ with a respiratory acidosis and minimal metabolic compensation. Common causes of hypercapnia include COPD,

respiratory center depression from drugs (e.g., opiates, sedatives, and anesthetics), neuromuscular impairment from disease (e.g., Guillain–Barré syndrome) or toxin (e.g., botulism), and finally thoracic cage disorders (e.g., morbid obesity, kyphoscoliosis).

Emergency Department Care and Disposition

Treatment of acute hypercapnia requires aggressive measures to increase minute ventilation.

1. Airway maintenance is crucial.
2. ■A trial of biphasic positive airway pressure or continuous positive airway pressure may prove helpful and improve minute ventilation; however, vigilance should be maintained in patients with depressed mental status as mechanical ventilation may be needed. Ensure that the respiratory rate is set to facilitate the removal of CO2.
3. ■Where indicated, treatment should include condition-specific therapies, such as bronchodilators for COPD, or reversal agents for opiate overdose.
4. ■Disposition depends on acuity, but many patients with hypercapnia require hospital admission and monitoring.

■■ WHEEZING

Wheezes are “musical” adventitious lung sounds produced by airflow through the central and distal airways. It is usually more prominent on exhalation, in contrast to upper airway stridor which is more prominent during inspiration, but it may be difficult to tell the two apart.

Wheezing is usually associated with lower airway disease such as asthma or other obstructive pulmonary diseases; however, the differential

29_Cydulka_Ch29.indd 194 15/03/17 4:27 PM

Source page 214

is broad and may also include pulmonary edema (commonly referred to as “cardiac asthma”), foreign body, bronchiolitis, and other pathologies.

■■ COUGH

Cough is a protective reflex for clearing secretions and foreign debris from the tracheobronchial tree. Coughing is initiated by stimulation of irritant receptors located throughout the respiratory tract. Such irritants may include inflammation as in asthma, irritants such as mold or dust or pulmonary secretions.

Clinical Features

Acute cough (<3 weeks) is most often caused by infection of the respiratory tract, or allergic reactions. Common upper respiratory infections are associated with a combination of rhinorrhea, sinusitis, pharyngitis, and

laryngitis, with the cough a result of drainage from the nasopharynx onto cough receptors in the pharynx and larynx. A productive cough is the hallmark of acute bronchitis. Pertussis may last 1 to 6 weeks. A subacute cough may last for 3 to 8 weeks and is often post-infectious and related to airway inflammation and secretions.

The most common cause of a chronic cough lasting more than 6 weeks is smoking. Features of smoking-related cough include peak severity in the morning and sputum production. Other causes of chronic cough include upper airway cough syndrome (formerly postnasal discharge) which is associated with mucus drainage from the nose, a history of “allergies or sinus problems,” and frequent clearing of the throat or swallowing of mucus. Asthma-related chronic cough is often worse at night, exacerbated by irritants, and associated with episodic wheezing and dyspnea. Gastroesophageal reflux-related cough often has a history of heartburn, is worse when lying down, and improves with anti-acid therapy (e.g., antacids, H2 blockers, or proton pump blockers). Angiotensin-converting enzyme (ACE) inhibitor or angiotensin II receptor blocker (ARB) therapy cough is highly variable in onset, severity, and variation during the day and can begin as early as 1 week to as late as 1 year after starting treatment. The cough is dry and typically resolves in 1 to 4 weeks after ACE or ARB therapy is stopped but may linger for up to 3 months.

Diagnosis and Differential

Obtain a chest radiograph in patients with purulent sputum and/or fever. Use spirometry to document the presence of airflow obstruction in patients with asthma. In both children and adults, acute coughing illnesses can last up to 3 weeks and appropriate guidance should be given.

In addition to disease-specific therapy, patients with acute cough may benefit from symptomatic treatment with antitussives, which block the cough reflex at various locations, or demulcents, which soothe the pharynx and somewhat suppress the cough reflex. For intractable coughing paroxysms in the ED, nebulized lidocaine may provide relief.

Figure 29-1 outlines a sequential approach for evaluation of subacute and chronic cough.

29_Cydulka_Ch29.indd 195 15/03/17 4:27 PM

Source page 215

196 SECTION 5: Pulmonary Emergencies

Subacute or chronic cough

History and physical exam,

chest x-ray

3–8 weeks' duration

Postinfectious Not Greater than

postinfectious 8 weeks' duration

Pneumonia New onset or Etiology suggested?

Pertussis exacerbation of:

Bronchitis UACS: 1st generation A/D

Acute Asthma: bronchodilators,

Chronic Yes No

ICS, LTRA

See relevant

NAEB: ICS

chapters for Stop smoking UACS: 1st generation A/D

treatment GERD: PPI, diet/lifestyle

Stop ACEI Asthma: bronchodilators,

changes

Treat ICS, LTRA

presumed NAEB: ICS

etiology GERD: PPI, diet/lifestyle

changes

Figure 29-1. Evaluation of subacute and chronic cough for patients 15 years of age and older. ACEI, angiotensin-converting enzyme inhibitor; A/D, antihistamine/decongestant; GERD, gastroesophageal reflux disease; ICS, inhaled corticosteroids; LTRA, leukotriene receptor antagonist; NAEB, nonasthmatic eosinophilic bronchitis; PPI, proton pump inhibitor; UACS, upper airway cough syndrome.

■■ HICCUPS

Hiccups are the involuntary spastic contraction of the inspiratory muscles.

Clinical Features

Benign hiccups are generally initiated by gastric distention from food or drink, especially carbonated beverages or alcohol. Persistent hiccups are usually the result of injury or irritation to a branch of the vagus or phrenic nerves.

Diagnosis and Differential

Determine whether a specific triggering event exists. Ask about the relationship with sleep; if there is resolution during sleep it suggests a psychogenic cause. Consider obtaining a chest radiograph to evaluate for intrathoracic pathology.

Emergency Department Care and Disposition

There are various physical maneuvers and drugs that can be used to terminate an acute episode.

29_Cydulka_Ch29.indd 196 15/03/17 4:27 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 212–219.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 30: Bronchitis, Pneumonia, and Novel Respiratory Infections

Source pages 220–223 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 220–223

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Bronchitis, Pneumonia, and Novel Respiratory Infections**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 30: Bronchitis, Pneumonia, and Novel Respiratory Infections 201 distress that may progress to pulmonary failure, most cases of aspiration pneumonia progress insidiously. [Source pages 220–223]

- While aspiration pneumonias may occur anywhere in the lung, aspirated material has a predilection for the right lower lobe due to gravity and tracheobronchial tree anatomy. [Source pages 220–223]

- Untreated or partially treated aspiration pneumonia may progress to empyema, defined as pus in the pleural space, or a lung abscess. [Source pages 220–223]

- Diagnosis and Differential Uncomplicated presentations in otherwise healthy patients may not require use of radiology, laboratory, or pathology resources; however, chest radiography is most commonly used for diagnosis (see Fig. [Source pages 220–223]

- Depending upon the anticipated etiology and disposition course, assess- ing white blood count with differential analysis, serum electrolytes, blood urea nitrogen, creatinine, glucose, blood gases, sputum Gram staining, and cultures of sputum and blood provide benefit, particularly in patients requiring intensive care unit (ICU) admission. [Source pages 220–223]

Clinical reasoning structure

- **Problem representation:** Bronchitis, Pneumonia, and Novel Respiratory Infections; source pages 220–223.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Bronchitis, Pneumonia, and Novel Respiratory Infections**.

- Source-page range: **220–223**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 220

CHAPTER 30: Bronchitis, Pneumonia, and Novel Respiratory Infections 201

distress that may progress to pulmonary failure, most cases of aspiration pneumonia progress insidiously. While aspiration pneumonias may occur anywhere in the lung, aspirated material has a predilection for the right lower lobe due to gravity and tracheobronchial tree anatomy. Untreated or partially treated aspiration pneumonia may progress to empyema, defined as pus in the pleural space, or a lung abscess.

Diagnosis and Differential

Uncomplicated presentations in otherwise healthy patients may not require use of radiology, laboratory, or pathology resources; however, chest radiography is most commonly used for diagnosis (see Fig. 30-1). Depending upon the anticipated etiology and disposition course, assessing white blood count with differential analysis, serum electrolytes, blood urea nitrogen, creatinine, glucose, blood gases, sputum Gram staining, and cultures of sputum and blood provide benefit, particularly in patients requiring intensive care unit (ICU) admission. Most patients do not require identification of a specific organism to make a diagnosis and begin treatment.

The differential diagnosis of nontrauma patients with respiratory complaints and radiographic abnormality is lengthy and partially includes noninfectious atelectasis; chronic pulmonary fibrosis; pleural effusion; chemical pneumonitis; inflammatory disorders, such as sarcoidosis; neoplasm; postsurgical changes; tuberculosis; bronchiolitis obliterans; pulmonary embolus;

Figure 30-1. Lobar pneumonia.

30_Cydulka_Ch30.indd 201 15/03/17 4:29 PM

Source page 221

202 SECTION 5: Pulmonary Emergencies

congestive heart failure; and pulmonary vasculitides, such as Goodpasture disease or Wegener granulomatosis.

Emergency Department Care and Disposition

1. Support vital respiratory function (oxygenation, ventilation) as indicated, with rapidly impending or unresponsive respiratory failure managed via intubation and mechanical ventilation. Noninvasive positive pressure ventilation may prevent the need for intubation.
2. In general, progressive degrees of abnormal vital signs, comorbidities, and advancing age confer increased need for inpatient management. The complexity of pneumonia severity scoring as a means to determine discharge or admission prevents inclusion in this manual.
3. Begin antibiotic treatment in all cases of suspected bacterial pneumonia, with the specific choice(s) made considering the patient's recent environment, differentiating community acquired from HCAPs, comorbidities, drug allergies, drug–drug interactions, and local resistance patterns.
4. Specialty society guidelines and infectious disease consultation advice change with the advent of antimicrobials and resistance patterns. The antimicrobials listed here represent a summary of current and generally accepted antibiotic regimens for adults with the indicated clinical situations. Dosages may require adjustment for renal insufficiency.
5. Outpatient management of uncomplicated CAP in otherwise healthy patients: azithromycin, day 1 with 500 mg orally, followed by days 2 to 5 with 250 mg orally or doxycycline 100 mg orally twice daily for 10 days (this is a low-cost alternative). The Centers for Disease Control and Prevention (CDC) recommends reserving oral fluoroquinolones for those failing macrolide or tetracycline class therapy to minimize resistance.
6. Outpatient management of CAP in patients with significant comorbi dities (and without health care–associated pneumonia suspected): levo floxacin 750 mg orally daily for 5 days OR amoxicillin-clavulanate 875/125 mg orally twice daily for 10 days PLUS azithromycin, day 1 with 500 mg orally, followed by days 2 to 5 with 250 mg orally.
7. Inpatient management of CAP in patients not requiring ICU admission: start antibiotic therapy with levofloxacin 750 mg IV OR ceftriaxone 1 g IV PLUS azithromycin 500 mg IV. Utilize antibiotics early in the course of any pneumonia requiring admission.
8. Inpatient management of CAP in patients requiring ICU admission: start antibiotic therapy with ceftriaxone 1 g IV PLUS levofloxacin 750 mg IV. If MRSA suspected, add vancomycin 10 to 15 mg/kg IV.

9. Inpatient management of suspected HCAP: start double coverage against Pseudomonas with (1) levofloxacin 750 mg IV PLUS (2) cefepime 1 to 2 g IV OR piperacillin/tazobactam 4.5 g IV. Also, cover suspected MRSA with vancomycin 10 to 15 mg/kg IV OR line zolid 600 mg IV.

10. Aspiration pneumonitis: prophylactic antibiotics are not recommended and their indiscriminate use may contribute to organism resistance. For witnessed aspirations, immediate tracheal suction followed by bronchoscopy, if needed, to remove large particles is indicated. In pneumonitis

30_Cydulka_Ch30.indd 202 15/03/17 4:29 PM

Source page 222

that has already progressed to pneumonia prior to or shortly after emergency department presentation, start antibiotic therapy with levofloxacin 750 mg IV PLUS clindamycin 600 mg IV.

11. Empyema: piperacillin/tazobactam 4.5 g IV. If MRSA suspected, add vancomycin 10 to 15 mg/kg IV. Admit the patient and consult with a pulmonologist or thoracic surgeon early for further consideration of definitive diagnostic measures and treatment options to promote drainage.

12. Lung abscess: clindamycin 600 mg IV for anaerobic coverage plus ceftriaxone 1 g IV. A significant majority of lung abscesses are successfully treated with inpatient medical management and surgical consultation is required in only a minority of cases.

13. For patients with uncomplicated pneumonia, discharge instructions should, at a minimum, include timely follow-up with a primary care physician, smoking cessation (when applicable), and delineation of symptoms that should prompt a return visit to the emergency department.

■■ NOVEL RESPIRATORY INFECTIONS

Severe acute respiratory syndrome (SARS) came to worldwide attention in the winter of 2003. Numerous deaths were reported in Asia, North America, and Europe. The etiologic agent is a coronavirus, SARS-CoV, spread by “droplet infection.” In the event of SARS outbreak, up-to-date information can be found at the CDC website (http://www.cdc.gov/sars/who/clinicians.html) as well as at the World Health Organization (WHO) website (http://www.who.int/topics/sars/en/).

Middle East respiratory syndrome (MERS) was named in latter 2012 and became a more prominent global concern in the spring of 2014. A fatality rate approaching 50% was reported in the initial 100 diagnosed cases. Since

these initial cases, mortality has decreased. The etiologic agent is a coronavirus. Respiratory symptoms predominate, though GI symptoms of abdominal pain and diarrhea may be present. Health care workers exposed via airborne transmission have become more common in case reports. Current, basic steps to reduce MERS coronavirus transmission to health care workers include wearing gloves, gown, eye protection and using a fit-tested N95 respirator facemask. Up-to-date information can be found at the CDC website (http://www.cdc.gov/coronavirus/mers/hcp.html) as well as at the WHO website (http://www.who.int/csr/disease/coronavirus_infections/en/).

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 64, "Acute Bronchitis and Upper Respiratory Tract Infections," by Cedric W. Lefebvre; Chapter 65, "Pneumonia and Pulmonary Infiltrates," by Gerald Maloney, Eric Anderson, and Donald M. Yealy, and Chapter 66, "Lung Empyema and Abscess," by Eric Anderson and Sharon E. Mace.

30_Cydulka_Ch30.indd 203 15/03/17 4:29 PM

Source page 223

C h a p ter Tuberculosis

31 Amy J. Behrman

Tuberculosis (TB) is the second most common cause of infectious disease deaths globally, with one-third of the world's population infected. Although active TB infection rates continue to decline in the United States, TB remains an important public health problem, particularly among immigrants, whose active TB case rate is 12 times higher than the US-born population. Other risk factors include HIV infection; living or working in prison, shelters, and long-term care facilities; and alcohol/drug abuse. Transmission occurs by inhalation of droplet nuclei and may lead to active primary infection or latent disease (which may reactivate later). Identifying and treating high-risk patients for active and latent TB is key to ongoing TB control.

■■ CLINICAL FEATURES

Primary TB

Initial TB infection is usually asymptomatic in immune-competent adults, generally presenting with only a new positive reaction to TB skin testing (TST) or a new positive interferon gamma release assay (IGRA). These patients have normal chest radiographs (CXRs) and are presumed to have latent infections. When active primary TB does develop, symptoms often include fever, cough, weight loss, malaise, and chest pain. Some patients may present with active pneumonitis (which may be mistaken for

community-acquired pneumonia) or extra-pulmonary disease.

Children are more likely to present with active early disease, although the presenting symptoms may be subtle even when CXRs are abnormal. Presenting symptoms may include fever, cough, wheezing, poor feeding, and fatigue. TB meningitis and miliary TB are more common in children than adults.

Immunocompromised patients are much more likely to develop rapidly progressive primary infections. All patients with active TB should be evaluated for immune-compromising conditions. Symptoms may be pulmonary (fever, cough, dyspnea, hemoptysis) or extrapulmonary, reflecting early hematogenous spread to the liver, bones, central nervous system, or other sites.

Reactivation TB

Latent TB infections will progress to active disease (i.e., reactivation TB) in 5% of cases within 2 years of primary infection; an additional 5% will reactivate over their lifetimes. Reactivation rates are higher in children, the elderly, persons with recent primary infection, those with immune compromise (in particular HIV), and those with chronic diseases such as diabetes and renal failure.

Most patients with reactivation TB present subacutely with fever, malaise, weight loss, fatigue, and night sweats. Most patients with active

31_Cydulka_Ch31.indd 204 15/03/17 5:49 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 220–223.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 31: Tuberculosis

Source pages 224–227 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 224–227

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Tuberculosis**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 31: Tuberculosis 205 TB will have pulmonary involvement and will develop productive coughs. [Source pages 224–227]

- Hemoptysis, pleuritic chest pain, and dyspnea may occur. [Source pages 224–227]

- Rales and rhonchi may be found, but the physical examination is not usually diagnostic. [Source pages 224–227]

- Extrapulmonary TB develops in up to 20% of active TB cases. [Source pages 224–227]

- Lymph- adenitis, with painless enlargement and possible draining sinuses, is a com- mon presentation. [Source pages 224–227]

Clinical reasoning structure

- **Problem representation:** Tuberculosis; source pages 224–227.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Tuberculosis**.

- Source-page range: **224–227**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 224

CHAPTER 31: Tuberculosis 205

TB will have pulmonary involvement and will develop productive coughs. Hemoptysis, pleuritic chest pain, and dyspnea may occur. Rales and rhonchi may be found, but the physical examination is not usually diagnostic.

Extrapulmonary TB develops in up to 20% of active TB cases. Lymphadenitis, with painless enlargement and possible draining sinuses, is a common presentation. Patients may also present with symptomatic pleural effusion, pericarditis, peritonitis, or meningitis. Additional sites of reactivation TB after hematogenous spread include bones, joints, adrenals, GI tract, and GU tract. Extrapulmonary reactivation TB is more common and often more severe in young children and immunocompromised patients as noted for primary TB infection above.

Miliary TB is a multisystem disease caused by massive hematogenous dissemination. It is also more common in immunocompromised patients and children. Symptoms are systemic with fever, weight loss, adenopathy, and malaise. Patients may present with multiorgan failure or Adult Respiratory Distress Syndrome (ARDS).

■■ DIAGNOSIS AND DIFFERENTIAL

Variable presentations and limited testing options make TB diagnosis particularly challenging in the ED. Differential diagnoses may include other infectious causes of pulmonary and extrapulmonary lesions as well as malignancy. TB should always be considered in patients with prolonged febrile cough illnesses, particularly in known risk groups. TB should be considered in any HIV patient with respiratory symptoms, even if chest radiographs are normal (see Chapter 92 HIV Infections and AIDS).

Imaging

CXR remains the most useful ED diagnostic tool for TB. Active primary pulmonary TB usually presents with parenchymal infiltrates in any lung area. Hilar and/or mediastinal adenopathy may occur with or without infiltrates. Effusions, usually unilateral, may be seen with or without infiltrates. Reactivation TB classically presents with lesions in the upper lobes or

superior segments of the lower lobes (Fig. 31-1). Cavitation, calcification, scarring, atelectasis, hilar adenopathy, and effusions may be seen. Cavitation is associated with increased infectivity. Miliary TB may cause diffuse, small (1 to 3 mm) nodular infiltrates. Atypical CXRs are progressively more common with worsening immune compromise. Patients co-infected with HIV and TB are particularly likely to present with atypical imaging. Stable scarring, volume loss, and calcified or noncalcified nodules may be found (often as incidental findings) in patients with asymptomatic latent TB infection; these patients are not infectious and do not require urgent treatment or isolation. Comparison with prior films is useful in determining the likelihood of active TB infection.

Laboratory Tests

Acid-fast staining of sputum can detect mycobacteria in many patients with pulmonary TB, although the yield is lower in HIV patients. Results may be available within several hours, with potential ED utility, but there

31_Cydulka_Ch31.indd 205 15/03/17 5:49 PM

Source page 225

Figure 31-1. Cavitary tuberculosis of the right upper lobe.

are serious limitations. Many patients will have false negatives on a single sputum sample. Microscopy of nonsputum samples (e.g., pleural fluid, cerebrospinal fluid) is even less sensitive. Microscopy cannot differentiate between TB and nontuberculous mycobacteria. Culture of sputum (or other specimens) is the gold standard for diagnosing active TB. Unfortunately, definitive culture results take weeks. When available, newer technologies such as TB-specific nucleic acid amplification tests (NAATs) can produce results within 24 hours, a time frame potentially useful in ED management. These tests have poor negative predictive value, but a patient with positive microscopy and a positive NAAT should be treated for active TB unless subsequent cultures rule it out.

Tuberculin skin tests (TSTs) identify most patients with latent, prior, or active TB, but results are read 48 to 72 hours after placement, limiting the ED utility of this approach. Patients with disseminated TB, early active TB, or major immune compromise (especially HIV) may have false-negative TSTs. Immigrants who received BCG vaccine in childhood may have false-positive TSTs. Interferon gamma release assays (IGRA) of whole blood may become more useful than TST for ED evaluation of suspected TB, since these tests may be resulted within hours and may have better sensitivity and specificity in some situations. For instance, BCG

31_Cydulka_Ch31.indd 206 15/03/17 5:49 PM

Source page 226

alone should not cause a positive IGRA, making it useful for evaluating immigrants from high-prevalence countries.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. Train ED and pre-hospital staff to identify patients at risk for active TB as early as possible during their care so that airborne isolation and respiratory protection for staff can be implemented as soon as possible to protect staff, visitors, and other patients. Consider the diagnosis of active TB in any patient with respiratory or systemic complaints, particularly if they are in higher risk groups, to facilitate treatment and reduce exposure risks.

2. Initial therapy for active TB should include at least four drugs until susceptibility profiles are available. The regimen usually includes isoniazid 5 mg/kg up to 300 mg PO daily in adults, rifampin 10 mg/kg up to 600 mg PO daily in adults, pyrazinamide 15 to 30 mg/kg up to 2 g PO daily in adults, and ethambutol 15 to 20 mg/kg up to 1.6 g PO daily in adults. Give pyridoxine 50 mg/day with INH to prevent neuropathy. Caution all patients to avoid alcohol and hepatotoxins during treatment. Directly observed therapy (DOT) may improve outpatient compliance with these complex regimens. (See http://www.cdc.gov/tb for DOT recommendations and long-term treatment options.)

3. Consider the possibility of multi-drug resistant (MDR) TB in patients with a prior history of TB, suboptimal prior treatment of TB, known or likely exposure to MDR TB, or persistently positive smears and symptoms after several months of treatment, and when there is known local drug resistance. Treatment of known or suspected MDR TB begins with four to six drugs judged likely to be effective by Infectious Disease consultation.

4. Admission is indicated for patients with clinical instability, hypoxia, dyspnea, diagnostic uncertainty, unreliable outpatient follow-up or compliance, and suspected or known MDR TB. ED physicians should know local laws and public health resources for case reporting, involuntary hospitalization, and treatment (including DOT). Patients with suspected TB should wear masks during all transport, and should be admitted to airborne-isolation single rooms.

5. Patients discharged from the ED with known or likely active TB infection must have urgent documented referral to a physician or local public health department for long-term treatment and contact

tracing.

6. Persons with latent TB infection (defined as a positive TST and/or IGRA without active TB findings on imaging) should be referred to primary care or public health clinics for prophylaxis against reactivation TB.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 67, "Tuberculosis," by Vu D. Phan and Janet M. Poponick.

31_Cydulka_Ch31.indd 207 15/03/17 5:49 PM

Source page 227

C h a p ter Spontaneous and Iatrogenic

32 Pneumothorax

Mike Cadogan

Pneumothorax occurs when air or gas enters the pleural cavity, the potential space between the visceral and parietal pleura, leading to partial lung collapse. Smoking is the most common risk factor for spontaneous pneumothorax.

Primary pneumothorax occurs in patients without known lung disease and secondary pneumothorax occurs in the presence of known lung disease, such as chronic obstructive pulmonary disease, asthma, cystic fibrosis, interstitial lung disease, infection, connective tissue disease, and cancer. Latrogenic pneumothorax occurs secondary to invasive procedures such as needle biopsy of the lung, placement of a subclavian line, nasogastric tube placement or positive pressure ventilation, Tension pneumothorax results from positive pressure in the pleural space leading to decreased venous return, hypotension, and hypoxia. Hemopneumothorax occurs in 2% to 7% of patients with spontaneous pneumothorax. Treating the underlying disease may help decrease the risk of pneumothorax.

■■ CLINICAL FEATURES

Sudden-onset dyspnea and ipsilateral, pleuritic chest pain are the most common presenting symptoms. The physical examination findings are often subtle. Sinus tachycardia is the most common physical finding. In spontaneous pneumothorax the classic examination findings of ipsilateral decreased breath sounds, reduced chest expansion, and hyperresonance to percussion are often absent. However, in traumatic pneumothorax, the positive predictive value of ipsilateral decreased breath sounds is 86% to 97%. Clinical hallmarks of tension pneumothorax include severe progressive dyspnea, tachycardia (>140 bpm), hypoxia, and ipsilateral decreased breath sounds. Tracheal deviations away from the affected side, distended neck veins, and cardiac apical displacement are late and infrequent signs of progressive tension.

■■ DIAGNOSIS AND DIFFERENTIAL

Pneumothorax is an important differential consideration in patients with pleuritic chest pain, especially in those with underlying lung disease. Patients with pulmonary embolism, pneumonia, pericarditis, pleural effusions, or shingles can present with pain similar to those with pneumothorax. As clinical signs and symptoms are often subtle and variable, the definitive diagnosis is usually established with appropriate imaging. In stable patients with suspected pneumothorax, an erect posteroanterior (PA) chest radiograph is usually the primary investigation. Characteristically this demonstrates a displaced pleural line with absent lung markings extending from the visceral pleura (lung edge) to the chest wall (parietal pleura). Routine expiratory radiographs do not significantly enhance diagnostic yield. The presence of cardiophrenic recess hyperlucency and costophrenic angle enlargement (deep sulcus sign) on a supine anteroposterior radiograph is suggestive of pneumothorax.

208

32_Cydulka_Ch32.indd 208 16/03/17 9:35 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 224–227.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 32: Spontaneous and Iatrogenic Pneumothorax

Source pages 228–233 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 228–233

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Spontaneous and Iatrogenic Pneumothorax**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 32: Spontaneous and Iatrogenic Pneumothorax 209 Large emphysematous bullae may mimic pneumothorax. [Source pages 228–233]

- To prevent the potentially disastrous consequence of inserting a chest drain into a lung bulla mistaken for a pneumothorax, thoracic computed tomography (CT) should be used to clarify the diagnosis. [Source pages 228–233]

- Bedside ultrasound is a rapid, non- invasive method to diagnose pneumothorax in young, healthy patients with no underlying lung disease. [Source pages 228–233]

- In normal lung, the visceral and parietal pleura are in direct contact and ultrasound can be used to demonstrate the lung moving back and forth beneath the ribs during respiration (lung sliding), and vertical reverberation artefacts (comet-tails). [Source pages 228–233]

- Features suggesting pneumothorax include absence of lung sliding, comet tail artefacts and a lung pulse in the presence of a distinct A lines and visualized lung point. [Source pages 228–233]

Clinical reasoning structure

- **Problem representation:** Spontaneous and Iatrogenic Pneumothorax; source pages 228–233.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Spontaneous and Iatrogenic Pneumothorax**.

- Source-page range: **228–233**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 228

CHAPTER 32: Spontaneous and Iatrogenic Pneumothorax 209

Large emphysematous bullae may mimic pneumothorax. To prevent the potentially disastrous consequence of inserting a chest drain into a lung bulla mistaken for a pneumothorax, thoracic computed tomography (CT) should be used to clarify the diagnosis. Bedside ultrasound is a rapid, non-invasive method to diagnose pneumothorax in young, healthy patients with no underlying lung disease.

In normal lung, the visceral and parietal pleura are in direct contact and ultrasound can be used to demonstrate the lung moving back and forth beneath the ribs during respiration (lung sliding), and vertical reverberation artefacts (comet-tails). Features suggesting pneumothorax include absence of lung sliding, comet tail artefacts and a lung pulse in the presence of a distinct A lines and visualized lung point. Chest CT is effective at detecting pneumothorax, determining their size and volume, and detecting other lung pathology, and can differentiate large bullae from intrapleural air.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

The ED treatment goal is the elimination of intrapleural air.

1. ■Tension pneumothorax should be diagnosed clinically—before a radiograph—and immediately treated by needle decompression followed by tube thoracostomy.

2. ■Administer oxygen >28% or 2 to 4 L by nasal cannula to increase pleural air resorption. Monitor for hypercapnia in patients with chronic obstructive pulmonary disease.

3. ■Observe patients with a small primary pneumothorax for at least 4 hours on supplemental oxygen, and then repeat the chest radiograph. If symptoms and chest radiograph have improved, the patient should return in 24 hours for repeat examination and then weekly until resolution. First-

time spontaneous pneumothorax of <20% lung volume in a stable, healthy adult may be treated initially with oxygen therapy and observation.

4. ■Aspiration or tube thoracostomy is performed in order to rapidly reduce the volume of a closed pneumothorax and significantly shorten the time to pneumothorax resolution.

a. ■For aspiration of a small primary or secondary spontaneous pneumothorax, insert a 14-G needle in adults (18-G needle in children) that is at least 2 in. (5 cm) long, followed by syringe aspiration and needle removal. Alternatively, a small catheter or pigtail catheter may be used and attached to a Heimlich valve or suction. Either of two locations may be used: anteriorly in the second intercostal space (ICS) at the midclavicular line (MCL) and laterally in the fourth or fifth ICS at the anterior axillary line (AAL). The fifth ICS AAL is the preferred location for needle thoracostomy decompression.

b. ■Treat patients with a failed aspiration, large pneumothorax, recurrent pneumothorax, bilateral pneumothoraces, hemothorax, abnormal vital signs, or severe dyspnea using chest tube thoracostomy. Use a small 10- to 14-French chest tube for nontrauma, reserving larger 14- to 22-French chest tubes if a large air leak is probable, such as from mechanical ventilation or with underlying pulmonary disease.

5. ■Patients who require chest tube thoracostomy require admission to the hospital.

32_Cydulka_Ch32.indd 209 16/03/17 9:35 AM

Source page 229

■■ TREATMENT COMPLICATIONS

Intervention complications include intercostal vessel hemorrhage, lung parenchymal injury, empyema, and tube malfunction (development of an air leak or tension pneumothorax), and re-expansion lung injury. Re-expansion lung injury is uncommon and seen more often when there is collapse of the lung for greater than 72 hours, large pneumothoraces, rapid re-expansion, or negative pleural pressure suction of greater than 20 cm. Most patients with re-expansion injury require only observation and supplemental oxygen and rarely suffer adverse outcomes.

Iatrogenic Pneumothorax

Iatrogenic pneumothorax is a subset of traumatic pneumothorax and occurs more often than spontaneous pneumothorax. Transthoracic needle procedures (needle biopsy and thoracentesis) account for 50% of iatrogenic pneumothoraces with subclavian vein catheterization accounting for a further 25%. US guidance for central venous catheter insertion for thoracente-

sis reduces the pneumothorax complication rate. Factors associated with the increasing frequency of iatrogenic pneumothorax include underlying disease, body habitus, and experience of the operator. Although it is routine to obtain a chest radiograph after central line placement or transthoracic needle procedures, chest radiograph may not identify a pneumothorax if the patient is supine or if there is inadequate time for the pneumothorax to develop, with up to one-third detected later. Treatment for iatrogenic pneumothorax is generally the same as for spontaneous pneumothorax.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 68, "Pneumothorax," by Bret A. Nicks and David Manthey.

32_Cydulka_Ch32.indd 210 16/03/17 9:35 AM

Source page 230

C h apter Hemoptysis

33 Nilesh Patel

Hemoptysis is the expectoration of blood from the lungs or tracheobronchial tree. Massive hemoptysis is defined as 100 mL of blood per 24 hours up to >1000 mL per 24 hours. Minor hemoptysis is defined as the production of smaller quantities of blood in a patient with no comorbid lung disease and stable vital signs. Hemoptysis results from the disruption of blood vessels within the bronchial circulation which supplies oxygenated blood to the bronchi, bronchioles, and lung parenchyma.

■■ CLINICAL FEATURES

Hemoptysis may be the presenting symptom for many different diseases. Massive hemoptysis can often be life threatening. A careful history and physical can raise suspicion for the underlying diagnosis and guide the appropriate workup. The acute onset of fever, cough, and bloody sputum may indicate pneumonia or bronchitis. An indolent productive cough can indicate bronchitis or bronchiectasis. Dyspnea and pleuritic chest pain are potential indicators of pulmonary embolism, particularly in the presence of venous thromboembolic risk factors. Tuberculosis should be considered in the setting of fever, night sweats, and risk factors such as travel from endemic regions. Bronchogenic carcinoma may present with tobacco use, chronic weight loss, and a change in cough. Chronic dyspnea and minor hemoptysis may indicate mitral stenosis or alveolar hemorrhage syndromes (most commonly seen in patients with renal disease). Consider Goodpasture's syndrome in patients with hemoptysis, hematuria, and renal insufficiency.

The physical examination should begin with an assessment of airway, breathing, and circulation, with a focus on the vital signs. Common abnor-

mal vital signs include fever and tachypnea. Tachypnea may be a sign of respiratory compromise with hypoxemia. Hypotension is an ominous sign, usually seen only in massive hemoptysis. The cardiac examination may reveal signs of valvular heart disease (e.g., the diastolic murmur of mitral stenosis). The nasal and oral cavities should be inspected carefully to help rule out an extrapulmonary source of bleeding (pseudohemoptysis).

■■ DIAGNOSIS AND DIFFERENTIAL

A careful history and physical examination may suggest a diagnosis, although the cause of hemoptysis is undetermined in up to 30% of cases. Pulse oximetry and a chest x-ray (PA and lateral, if the patient's condition allows) are always indicated. Other tests that may be helpful include arterial blood gas, hemoglobin and hematocrit levels, platelet count, coagulation studies, urinalysis, and electrocardiogram. Chest CT should be considered if there is hemoptysis with an abnormal chest radiograph or if considering pulmonary embolism or carcinoma on the differential diagnosis. The long differential diagnosis list includes infectious, neoplastic, and cardiac etiologies. Infectious etiologies include bronchitis, bronchiectasis, bacterial pneumonia, tuberculosis, fungal pneumonia, and lung abscess.

33_Cydulka_Ch33.indd 211 16/03/17 9:40 AM

Source page 231

Neoplastic etiologies include bronchogenic carcinoma and bronchial adenoma. Cardiogenic etiologies include mitral stenosis and left ventricular failure. Trauma, foreign body aspiration, pulmonary embolism (hemoptysis is one of the Wells criteria), primary pulmonary hypertension, pulmonary vasculitis, and bleeding diathesis, and coagulopathies secondary to medications are other potential causes.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. Assess airway, breathing, and circulation in all patients who present with hemoptysis.

2. Administer supplemental oxygen as needed to maintain adequate oxygenation.

3. Administer normal saline or lactated Ringer's solution initially for resuscitation.

4. Type and cross-match blood if transfusion is necessary. Transfuse packed red blood cells as needed.

5. Administer fresh frozen plasma or prothrombin complex concentrates to patients with coagulopathies, including those taking warfarin; give platelets for thrombocytopenia (see Chapter 137, "Transfusion

Therapy”). Other blood products can be considered as indicated.

6. Patients with ongoing massive hemoptysis may benefit from being placed in the decubitus position, with the bleeding lung in dependent position. This maneuver can also be performed after intubation in patients on mechanical ventilation.

7. Perform endotracheal intubation with a large diameter (8.0 mm) tube, which facilitates bronchoscopy, if there is respiratory failure or if the patient cannot clear blood or secretions from the airway. If bleeding is uncontrolled, consider preferentially intubating the mainstem bronchus of the unaffected lung (and using a tamponade device to stop bleeding in the alternate bronchus).

8. Admit any patient with moderate to severe hemoptysis to the hospital, and strongly consider placement in the intensive care unit. Patients with mild hemoptysis who have conditions that predispose them to severe bleeding also should be considered for admission. The advice of a pulmonologist or thoracic surgeon is required for decisions as to whether bronchoscopy, computed tomography, or angiography for bronchial artery embolization might be needed. For massive hemoptysis, emergent bronchoscopy may be indicated in the ED. If the appropriate specialists are not available, the patient should be stabilized and then transferred to another facility.

9. Treat patients who are discharged home for several days with cough suppressants, inhaled β-agonist bronchodilators as needed, and antibiotics if bacterial infection is thought to be the cause. Close follow-up is essential, particularly in those at high risk for neoplasm.

10. Most cases of mild hemoptysis are self-limited and patients can be discharged home with appropriate follow-up.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 63, “Hemoptysis,” by Troy Sims.

33_Cydulka_Ch33.indd 212 16/03/17 9:40 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 228–233.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 34: Asthma and Chronic Obstructive Pulmonary Disease

Source pages 234–237 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 234–237

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Asthma and Chronic Obstructive Pulmonary Disease**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 34: Asthma and Chronic Obstructive Pulmonary Disease 215 nebulization (10 to 15 mg/h), titrating treatment to clinical response and signs of toxicity (tachycardia, hypertension, and palpitations). [Source pages 234–237]

- Terbutaline 0.25 mg or epinephrine 1:1000 0.3 to 0.5 mg) every 20 minutes for three doses SC may be administered to patients not tolerat- ing aerosolized therapy. [Source pages 234–237]

- Epinephrine should be used with caution in the presence of underlying cardiovascular disease. [Source pages 234–237]

- Anticholinergics are useful adjuvants for effect on the large, central airways. [Source pages 234–237]

- Nebulized ipratropium (0.5 mg = 2.5 mL) added to SABAs every 20 minutes for three doses, then as needed, may improve bron- chodilation and decrease the need for hospitalization. [Source pages 234–237]

Clinical reasoning structure

- **Problem representation:** Asthma and Chronic Obstructive Pulmonary Disease; source pages 234–237.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Asthma and Chronic Obstructive Pulmonary Disease**.

- Source-page range: **234–237**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 234

CHAPTER 34: Asthma and Chronic Obstructive Pulmonary Disease 215

nebulization (10 to 15 mg/h), titrating treatment to clinical response and signs of toxicity (tachycardia, hypertension, and palpitations). Terbutaline 0.25 mg or epinephrine 1:1000 0.3 to 0.5 mg) every 20 minutes for three doses SC may be administered to patients not tolerating aerosolized therapy. Epinephrine should be used with caution in the presence of underlying cardiovascular disease.

3. Anticholinergics are useful adjuvants for effect on the large, central airways. Nebulized ipratropium (0.5 mg = 2.5 mL) added to SABAs every 20 minutes for three doses, then as needed, may improve bronchodilation and decrease the need for hospitalization.

4. Give corticosteroids in the ED to patients with exacerbations of asthma and COPD. The initial dose is the equivalent of 40 to 60 mg of prednisone. Neither the choice of steroid nor the route of administration is critical. If the patient is unable to take oral medication, intravenous methylprednisolone 1 mg/kg may be used. Additional doses may be given every 4 to 6 hours. Inhaled steroids are not indicated for the treatment of acute symptoms.

5. Give antibiotics directed at Streptococcus pneumoniae, Haemophilus influenzae, and Moraxella catarrhalis to patients with COPD exacerbation who have a change in sputum volume or purulence. Commonly used agents include doxycycline, the macrolides (azithromycin), or amoxicillin with or without clavulanic acid. Antibiotic use in asthma should be reserved for concurrent bacterial infections such as pneumonia.

6. Administer intravenous magnesium sulfate (1 to 2 g over 30 minutes) to patients with severe asthma exacerbations (FEV1 <25% predicted). Magnesium sulfate is not recommended for mild or moderate asthma

exacerbations and should not be substituted for standard regimens. Current evidence does not support its use in COPD exacerbations.

7. Mechanical ventilation is necessary in patients with respiratory muscle fatigue, respiratory acidosis, altered mental status, or hypoxia refractory to standard therapies. Noninvasive partial pressure ventilation (NPPV) has become a useful alternative to intubation and invasive ventilation. NPPV lowers intubation rates, short-term mortality, and length of hospitalization in COPD. NPPV can be given by continuous positive airway pressure (CPAP) or bilevel positive airway pressure (BiPAP). BiPAP has the advantage of reducing the work of breathing. CPAP is titrated up to 15 cm H2O, while BiPAP settings are between 8 and 20 cm H2O for inspiration and 4 and 15 cm H2O for expiration. NPPV is likely to fail when there is poor mask fit, or when the patient is unable to cooperate with (or tolerate) NPPV, is obtunded, cannot clear airway secretions, or is hemodynamically unstable.

8. When NPPV is not a viable option, oral intubation is indicated. Rapid inspiratory flow rates at a reduced respiratory frequency (12 to 14 breaths/min) plus adequate expiratory phase may help reduce air-trapping and subsequent barotrauma. Therapy is guided by pulse oximetry, capnography, and ABG results. Continue sedation and therapy for bronchospasm after the patient has been placed on mechanical ventilation as the purpose of mechanical ventilation is to eliminate the work of breathing and enable rest. Mechanical ventilation itself does not relieve the airflow obstruction.

34_Cydulka_Ch34.indd 215 16/03/17 9:44 AM

Source page 235

9. Ketamine has been reported as therapy for refractory asthma, but controlled studies are lacking. An IV bolus dose of 0.2 mg/kg followed by an infusion of 0.5 mg/kg/h is sometimes used; ketamine is also a good agent if intubation is needed.

10. Admission criteria for patients with asthma include failure of outpatient treatment, persistent or worsening dyspnea, PEFR or FEV1 <40% of predicted, hypoxia, hypercarbia, and altered mental status; presence of comorbidities increases likelihood of need for admission. As compared to asthmatics, patients with acute COPD exacerbations are more likely to require admission. Indications for COPD admission include therapeutic failure, frequent exacerbations or relapse after ED treatment, severe dyspnea, significant comorbidities, arrhythmias, older age, insufficient home support, worsening hypoxia and hypercapnia

with acidosis, and impaired mental status.

11. Agents of uncertain or no benefit: In the absence of intubation, sedatives, hypnotics, and other medications that depress respiratory drive are generally contraindicated. The data supporting methylxanthines in acute asthma or COPD are conflicting; these medications have significant side effects, and they should be a third-line option at best.

12. Beta-blockers may exacerbate bronchospasm. Antihistamines and decongestants should be avoided as they diminish clearance of respiratory secretions. Mucolytics should also be withheld; they may exacerbate bronchospasm. Mast cell and leukotriene modifiers have no role in the treatment of acute exacerbations of asthma or COPD.

13. An 80%:20% mixture of helium and oxygen (Heliox) can lower airway resistance and aid in drug delivery in the patient with very severe asthma exacerbation; however, it does not reliably avert intubation, change admission rates, or decrease mortality. Care must be taken in the oxygen-dependent patient since Heliox administration may entail an FiO2 less than 21%.

14. Administer a 5- to 10-day course of oral steroids (prednisone 40 to 60 mg/dose) to patients who improve enough to be discharged home. Consider adding inhaled corticosteroids for asthmatic patients with chronic symptoms. Close follow-up care must be arranged for discharged patients to ensure resolution of the exacerbation and review the management plan. Despite appropriate therapy, these patients have high relapse rates. Education of asthma and COPD patients before discharge (i.e., review of medications, inhaler techniques, use of PEFR measurements, avoidance of noxious stimuli, and need to follow-up) should be an integral part of ED care.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 69, "Acute Asthma," by Rita K. Cydulka; Chapter 70, "Chronic Obstructive Pulmonary Disease," by Rita K. Cydulka and Craig G. Bates.

34_Cydulka_Ch34.indd 216 16/03/17 9:44 AM

Source page 236

SECTION

Gastrointestinal Emergencies 6

C h a p ter Acute Abdominal Pain

35 Bryan E. Baskin

Acute abdominal pain may be due to numerous etiologies including gastrointestinal, genitourinary, cardiovascular, pulmonary, musculoskeletal,

dermatologic, neurogenic, and other sources creating visceral, parietal, or referred pain to the abdomen.

■■ CLINICAL FEATURES

Consider immediate life threats that might require emergency intervention. Elicit time of pain onset; character, severity, location of pain and its referral (Fig. 35-1); aggravating and alleviating factors; and similar prior episodes. Cardiorespiratory symptoms, such as chest pain, dyspnea, and cough; genitourinary symptoms, such as urgency, dysuria, and vaginal discharge; and any history of trauma should be elicited. In older patients it is also important to obtain a history of myocardial infarction, dysrhythmias, coagulopathies, and vasculopathies. Past medical and surgical histories should be elicited, and a list of medications, particularly steroids, antibiotics, or nonsteroidal anti-inflammatory drugs (NSAIDs), should be noted. A thorough gynecologic history is indicated in female patients.

The physical examination should include the patient's general appearance. Patients with peritonitis tend to lie still. The skin should be evaluated for pallor, jaundice, or rash. The vital signs should be reviewed for signs of hypovolemia due to blood loss or volume depletion. Due to medications and/or the physiology of aging, elderly patients may not exhibit tachycardia in the face of hypovolemia or illness. A core temperature should be obtained; however, absence of fever does not rule out infection, particularly in the elderly. The abdomen should be inspected for contour, scars, peristalsis, masses, distention, and pulsation. The presence of hyperactive or high-pitched or tinkling bowel sounds increases the likelihood of small bowel obstruction.

Palpation is the most important aspect of the physical examination. The abdomen and genitals should be assessed for tenderness, guarding, masses,

35_Cydulka_Ch35.indd 217 16/03/17 9:49 AM

Source page 237

Diffuse Pain

Aortic aneurysm (leaking, ruptured) Mesenteric ischemia

Aortic dissection Metabolic disorder

Appendicitis (early) (Addisonian crisis, AKA,

Bowel obstruction DKA, porphyria, uremia)

Diabetic gastric paresis Narcotic withdrawal

Familial Mediterranean fever Pancreatitis

Gastroenteritis Perforated bowel

Heavy metal poisoning Peritonitis (of any cause)

Hereditary angioedema Sickle cell crisis

Malaria Volvulus

Right Upper Quadrant Pain Left Upper Quadrant Pain

Appendicitis (retrocecal) Gastric ulcer

Biliary colic Gastritis

Cholangitis Herpes zoster

Cholecystitis Myocardial ischemia

Fitz-Hugh-Curtis syndrome Pancreatitis

Hepatitis Pneumonia (LLL)

Hepatic abscess Pulmonary embolism

Hepatic congestion Splenic rupture/distention

Herpes zoster

Myocardial ischemia

Perforated duodenal ulcer

Pneumonia (RLL)

Pulmonary embolism

Right Lower Quadrant Pain Left Lower Quadrant Pain

Aortic aneurysm (leaking, ruptured) Aortic aneurysm (leaking,

Appendicitis ruptured)

Crohn’s disease (terminal ileitis) Diverticulitis (sigmoid)

Diverticulitis (cecal) Ectopic pregnancy

Ectopic pregnancy Endometriosis

Endometriosis Epiploic appendagitis

Epiploic appendagitis Herpes zoster

Herpes zoster Inguinal hernia

Inguinal hernia (incarcerated, strangulated)

(incarcerated, strangulated) Ischemic colitis

Ischemic colitis Mittelschmerz

Meckel’s diverticulum Ovarian cyst (ruptured)

Mittelschmerz Ovarian torsion

Ovarian cyst (ruptured) Pelvic inflammatory disease

Ovarian torsion Psoas abscess

Pelvic inflammatory disease Regional enteritis

Psoas abscess Testicular torsion

Regional enteritis Ureteral calculi

Testicular torsion

Ureteral calculi

Figure 35-1. Differential diagnosis of acute abdominal pain by location. AKA,

alcoholic ketoacidosis; DKA, diabetic ketoacidoisis; LLL, lower left lobe; RLL, right

lower lobe.

organomegaly, and hernias. Rebound tenderness, often regarded as the clinical criterion standard of peritonitis, has several important limitations. In patients with peritonitis, the combination of rigidity, referred tenderness, and, especially, cough pain usually provides sufficient diagnostic confirmation; false-positive rebound tenderness occurs in about one patient out of four without peritonitis. This has led some investigators to conclude that rebound tenderness, in contrast to cough pain, is of no predictive value. A useful and underused test to diagnose abdominal wall pain is the sit-up test, also known as the Carnett sign. After identification of the site of maximum

35_Cydulka_Ch35.i

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 234–237.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 35: Acute Abdominal Pain

Source pages 238–241 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 238–241

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Acute Abdominal Pain**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 35: Acute Abdominal Pain 219 abdominal tenderness, the patient is asked to fold his or her arms across the chest and sit up halfway. [Source pages 238–241]

- The examiner maintains a finger on the tender area, and if palpation in the semisitting position produces the same or increased tenderness, the test is said to be positive for an abdominal wall syndrome. [Source pages 238–241]

- Perform a pelvic examination in all postpubertal females. [Source pages 238–241]

- During the rectal examination, the lower pelvis should be assessed for tenderness, bleeding, and masses. [Source pages 238–241]

- Elderly patients often fail to manifest the same signs and symptoms as younger patients, with decreased pain perception and decreased febrile or muscular response to infection or inflammation. [Source pages 238–241]

Clinical reasoning structure

- **Problem representation:** Acute Abdominal Pain; source pages 238–241.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Acute Abdominal Pain**.

- Source-page range: **238–241**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 238

CHAPTER 35: Acute Abdominal Pain 219

abdominal tenderness, the patient is asked to fold his or her arms across the chest and sit up halfway. The examiner maintains a finger on the tender area, and if palpation in the semisitting position produces the same or increased tenderness, the test is said to be positive for an abdominal wall syndrome.

Perform a pelvic examination in all postpubertal females. During the rectal examination, the lower pelvis should be assessed for tenderness, bleeding, and masses.

Elderly patients often fail to manifest the same signs and symptoms as younger patients, with decreased pain perception and decreased febrile or muscular response to infection or inflammation. This is similarly true of diabetics and immunocompromised patients. Due to this, morbidity and mortality of elderly patients presenting to EDs with abdominal pain are high. Biliary disease, bowel obstruction, diverticulitis, cancer, and hernia are more common causes of abdominal pain in patients over 50 years old. Hypotension from volume contraction, hemorrhage, or sepsis can be missed if a normally hypertensive patient appears normotensive. Conditions, somewhat less frequent but proportionately higher in occurrence, among the elderly include sigmoid volvulus, diverticulitis, acute mesenteric ischemia, and abdominal aortic aneurysm. Mesenteric ischemia should be considered in any patient older than 50 years with abdominal pain out of proportion to physical findings.

■■ DIAGNOSIS AND DIFFERENTIAL

Suggested laboratory studies for goal-directed clinical testing are listed in Table 35-1. All women of child-bearing age with abdominal pain and/or abnormal vaginal bleeding should receive a qualitative pregnancy test. A complete blood count is neither sensitive nor specific to identify abdominal pathology; however, it remains the most commonly ordered test for ED

patients with abdominal pain.

Plain abdominal radiographs can be helpful in patients with suspected obstruction, perforation (looking for free air), or to follow previously identified stones in renal colic patients; however, in general have low sensitivity. Ultrasonography is useful for the diagnosis of cholelithiasis, choledocholithiasis, cholecystitis, biliary duct dilatation, pancreatic masses, hydroureter or hydronephrosis, intrauterine or ectopic pregnancies, ovarian and tubal pathologies, free intraperitoneal fluid, suspected appendicitis (institution specific), and abdominal aortic aneurysm. Since ultrasound machines are portable, a bedside ultrasound can be performed quickly by an emergency provider. Computed tomography (CT) is currently the preferred imaging method for mesenteric ischemia, pancreatitis, aortic aneurysm, appendicitis, and urolithiasis and is superior for identifying virtually any abnormality that can be seen on plain films. Intravenous contrast is essential to identify vascular lesions, is helpful to identify inflammatory conditions (i.e., appendicitis), but is not needed for urolithiasis. Oral contrast aids in the diagnosis of bowel obstruction, but otherwise is less useful.

35_Cydulka_Ch35.indd 219 16/03/17 9:49 AM

Source page 239

Table 35-1 Suggested Laboratory Studies for Goal-Directed Clinical Testing in Acute Abdominal Pain

Laboratory Test	Clinical Suspicion
Amylase	Pancreatitis (if lipase is not available)
Lipase	Pancreatitis
β-Human chorionic gonadotrophin	Pregnancy
	Ectopic or molar pregnancy
Coagulation studies (prothrombin time/ partial thromboplastin time)	GI bleeding End-stage liver disease
	Coagulopathy
Electrolytes	Dehydration
	Endocrine or metabolic disorder
Glucose	Diabetic ketoacidosis
	Pancreatitis
Gonococcal/chlamydia testing	Cervicitis/urethritis
	Pelvic inflammatory disease
Hemoglobin	GI bleeding
Lactate	Mesenteric ischemia
Liver function tests	Cholecystitis

Cholelithiasis

Hepatitis

Platelets GI bleeding

Renal function tests Dehydration

Renal insufficiency

Acute renal failure

Urinalysis Urinary tract infection

Pyelonephritis

Nephrolithiasis

ECG Myocardial ischemia or infarction

Source: Reproduced with permission from Fitch M. Utility and limitations of laboratory studies, in Cline DM, Stead LG (eds), Abdominal Emergencies. New York: McGraw-Hill Medical; 2008, p. 19.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

Unstable patients should be resuscitated immediately, then diagnosed clinically with emergent surgical consultation.

1. The most common resuscitation need for abdominal pain patients is intravenous fluids with normal saline or lactated Ringer's (LR) solution. During the initial evaluation, the patient should have nothing by mouth.

2. The judicious use of analgesics is appropriate and may facilitate the ability to obtain a better history and more accurate physical examination. Consider morphine 0.1 mg/kg IV or hydromorphone 0.2 to 1 mg IV, which can be reversed by naloxone 0.4 to 2 mg SC/IV if necessary. NSAIDs such as ketorolac, 30 mg IV, or ibuprofen, (200 to 400 mg PO) are useful in patients with renal colic, but their use in other conditions is controversial and they can mask peritoneal inflammation.

35_Cydulka_Ch35.indd 220 16/03/17 9:49 AM

Source page 240

3. Antiemetics, such as ondansetron 4 mg IM/IV, or metoclopramide 10 mg IM or slow IV, also increase the patient's comfort and facilitate assessment of the patient's signs and symptoms.

4. When appropriate, antibiotic treatment (i.e., gentamicin 1.5 mg/kg IV plus metronidazole 1 g IV; or piperacillin-tazobactam, 3.375 g IV) should be initiated, depending on the suspected source of infection. See specific chapters that follow in this section for additional antibiotics and guidelines.

5. Surgical or obstetric and gynecologic consultation should be obtained for patients with suspected acute abdominal or pelvic pathology requiring immediate intervention, including, but not limited to, abdominal

aortic aneurysm, intraabdominal hemorrhage, perforated viscus, intestinal obstruction or infarction, ectopic pregnancy, or gynecologic emergencies. Historically, the “acute abdomen” or “surgical abdomen” has been identified by the presence of pain, guarding, and rebound as indicating a likely need for emergent surgery.

6. Indications for admission include toxic appearance, unclear diagnosis in elderly or immunocompromised patients, inability to reasonably exclude serious etiologies, intractable pain or vomiting, altered mental status, and inability to follow discharge or follow-up instructions. Continued observation with serial examinations is an alternative. Many patients with nonspecific abdominal pain can be discharged safely with 12 to 24 hours of follow-up and instructions to return immediately for increased pain, vomiting, fever, or failure of symptoms to resolve.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 71, “Acute Abdominal Pain,” by Mary Claire O’Brien.

35_Cydulka_Ch35.indd 221 16/03/17 9:49 AM

Source page 241

C h a p ter Nausea and Vomiting

36 Jonathan A. Maisel

Although nausea and vomiting are typically caused by gastrointestinal disorders, the clinician must consider systemic causes as well. Neurologic, infectious, cardiac, endocrine, renal, obstetric, pharmacologic, toxicologic, and psychiatric disorders may all be the cause of nausea and vomiting. A comprehensive history and physical examination, as well as the use of various diagnostic modalities, are needed to determine the cause and its complications.

■■ CLINICAL FEATURES

History is essential in determining the cause of vomiting. Important features to elicit include the onset and duration of symptoms, the frequency and timing of episodes, the content of the vomitus (e.g., undigested food, bile-tinged, feculent), associated symptoms (e.g., fever, abdominal pain, diarrhea), exposure to foodborne pathogens, and the presence of sick contacts. A thorough past medical and surgical history (e.g., prior abdominal surgery) will also be valuable. Review the patient’s medication list, as a variety of agents in therapeutic or toxic doses can trigger nausea and vomiting. The physical examination should initially focus on determining the presence or absence of a critical, life-threatening condition. Hypotension, tachycardia, lethargy, poor skin turgor, dry mucous membranes, and delayed capillary refill suggest significant dehydration. A careful abdominal examination will help clarify the presence or absence of a primary GI

etiology. The extent to which the balance of the physical examination will be of value will be dictated by the history. In the event that a reliable history is not available (e.g., drug overdose, cognitive impairment), a comprehensive physical examination is warranted.

■■ DIAGNOSIS AND DIFFERENTIAL

Vomiting with blood could represent gastritis, peptic ulcer disease, or carcinoma. However, aggressive nonbloody vomiting followed by hematemesis is more consistent with a Mallory-Weiss tear. The presence of bile rules out gastric outlet obstruction, such as from pyloric stenosis or strictures. The presence of abdominal distension, surgical scars, or an incarcerated hernia suggests a small bowel obstruction. The presence of fever would suggest an infectious (e.g., gastroenteritis, appendicitis, cholecystitis) or inflammatory cause. Vomiting with chest pain suggests myocardial infarction. Post-tussive vomiting suggests pneumonia. Vomiting with back or flank pain can be seen with aortic aneurysm or dissection, pancreatitis, pyelonephritis or renal colic. Headache with vomiting suggests increased intracranial pressure, such as with subarachnoid hemorrhage, tumor, or head injury. The presence of vertigo and nystagmus suggests either vestibular or CNS pathology. Vomiting in a pregnant patient is consistent with hyperemesis gravidarum in the first trimester; but in the third trimester, could represent preeclampsia if accompanied by hypertension. Associated

36_Cydulka_Ch36.indd 222 16/03/17 10:11 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 238–241.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 36: Nausea and Vomiting

Source pages 242–245 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 242–245

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Nausea and Vomiting**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 36: Nausea and Vomiting 223 medical conditions are also useful in discerning the cause of vomiting: diabetes mellitus suggests ketoacidosis, peripheral vascular disease sug- gests mesenteric ischemia, and medication use or overdose (e.g., lithium or digoxin) suggests toxicity. [Source pages 242–245]

- All women of childbearing age warrant a pregnancy test. [Source pages 242–245]

- In vomiting associated with abdominal pain, liver function tests, urinalysis, and lipase or amylase determinations may be useful. [Source pages 242–245]

- Electrolyte determinations and renal function tests are usually of benefit only in patients with severe dehydration or prolonged vomiting. [Source pages 242–245]

- In addition, they may confirm the presence of adrenal insufficiency, with hyperkalemia and hyponatremia. [Source pages 242–245]

Clinical reasoning structure

- **Problem representation:** Nausea and Vomiting; source pages 242–245.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Nausea and Vomiting**.

- Source-page range: **242–245**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 242

CHAPTER 36: Nausea and Vomiting 223

medical conditions are also useful in discerning the cause of vomiting: diabetes mellitus suggests ketoacidosis, peripheral vascular disease suggests mesenteric ischemia, and medication use or overdose (e.g., lithium or digoxin) suggests toxicity.

All women of childbearing age warrant a pregnancy test. In vomiting associated with abdominal pain, liver function tests, urinalysis, and lipase or amylase determinations may be useful. Electrolyte determinations and renal function tests are usually of benefit only in patients with severe dehydration or prolonged vomiting. In addition, they may confirm the presence of adrenal insufficiency, with hyperkalemia and hyponatremia. Obtain specific drug levels for acetaminophen, salicylates, and digoxin when toxicity is suspected, and urine and/or serum toxicology screens when ethanol or illicit drug use is suspected. Urinalysis revealing ketones may suggest dehydration or diabetic ketoacidosis. The presence of nitrates, leukocyte esterase, bacteria and white blood cells is diagnostic of a urinary tract infection. The presence of red blood cells may suggest nephrolithiasis. The electrocardiogram and chest radiograph can be reserved for patients with suspected cardiac ischemia or pulmonary infection. Abdominal x-rays can be used to confirm the presence of intestinal obstruction. If plain x-rays are unrevealing, CT scan of the abdomen and pelvis with IV and PO contrast is not only helpful for revealing the location of a mechanical obstruction, but may also clarify alternative explanations for the patient's symptoms. CT scan of the brain will be helpful if a CNS lesion is suspected. Measuring intraocular pressure with a Tono-Pen® (Reichert, Inc., Depew, NY) is useful if glaucoma is suspected.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

The treatment of nausea and vomiting consists of correcting fluid and electrolyte problems. In addition, one must initiate specific therapy for any life-threatening cause identified in the initial workup.

1. Resuscitation of seriously ill patients requires intravenous boluses of

normal saline 20 mL/kg. Boluses may be repeated as necessary, targeting euvolemia. Caution should be used in the elderly, and those with compromised left ventricular function. Mildly dehydrated patients may tolerate an oral rehydration solution containing sodium as well as glucose to enhance fluid absorption. Many commercial products (e.g., Pedialyte®) are available.

2. Nutritional supplementation should be started as soon as nausea and vomiting subside. Patients can quickly advance from clear liquids to solids, such as rice and bread. Patients may benefit from avoiding raw fruit, caffeine, and lactose and sorbitol-containing products

3. Antiemetic agents are useful in actively vomiting patients with dehydration. Ondansetron 4 to 8 mg IV or ODT (children 0.15 mg/kg) is very effective and well tolerated, and can be administered to pregnant women (category B). Promethazine 25 mg (0.25 to 1 mg/kg in children over 2 years) IM or PR every 4 to 6 hours can be effective. Prochlorperazine 5 to 10 mg IM every 6 hours, or 25 mg PR every 12 hours is effective. Metoclopramide 10 mg (children 0.1 mg/kg) IV/IM every 6 to 8 hours is useful and can be administered to pregnant women (category B). Meclizine 25 mg PO every 6 hours is effective for vomiting associated with vertigo.

36_Cydulka_Ch36.indd 223 16/03/17 10:11 AM

Source page 243

Patients with a life-threatening cause of vomiting require admission. In addition, toxic or severely dehydrated patients, particularly infants and the elderly, or those still intolerant of oral fluids after hydration, warrant admission. Patients with an unclear diagnosis, but favorable examination findings after hydration, can be discharged home safely with antiemetics. Work excuses are indicated for patients in the food, day care, and health care industries.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 72, "Nausea and Vomiting," by Bophal Sarha Hang, Susan Bork, and Jeffrey Ditkoff.

36_Cydulka_Ch36.indd 224 16/03/17 10:11 AM

Source page 244

C ha p t er Disorders Presenting Primarily 37 with Diarrhea

Jonathan A. Maisel

Diarrhea is defined as three or more watery stools per day. There are four

basic mechanisms: increased intestinal secretion (e.g., cholera), decreased intestinal absorption (e.g., enterotoxins, inflammation, or ischemia), increased osmotic load (e.g., laxatives, lactose intolerance), and abnormal intestinal motility (e.g., irritable bowel syndrome). Most cases are infectious in etiology.

■■ DIARRHEA

Clinical Features

Determine if the diarrhea is acute (<3 weeks duration) or chronic (>3 weeks duration). Acute diarrhea is more likely to represent a serious problem, such as infection, ischemia, intoxication, or inflammation. Inquire about associated symptoms. Features such as fever, pain, presence of blood, or type of food ingested may help in the diagnosis of infectious gastroenteritis, food poisoning, diverticulitis, or inflammatory bowel disease. Neurological symptoms can be seen in certain diarrheal illnesses, such as seizure with shigellosis or hyponatremia, or paresthesias and reverse temperature sensation with ciguatoxin.

Details about the host can also better define the diagnosis. Malabsorption from pancreatic insufficiency or HIV-related bowel disorders need not be considered in a healthy host. Dietary practices, including frequent restaurant meals, exposure to day care centers, consumption of street vendor food or raw seafood, overseas travel, and camping with the ingestion of lake or stream water, may isolate the vector and narrow the differential diagnosis for infectious diarrhea (e.g., lakes or streams—Giardia; oysters suggest Vibrio; rice suggests Bacillus cereus; eggs suggest Salmonella; and meat suggests Campylobacter, Staphylococcus, Yersinia, Escherichia coli, or Clostridium). Certain medications, particularly antibiotics, colchicine, lithium, and laxatives, can all contribute to diarrhea. Travel may predispose the patient to enterotoxigenic E. coli or Giardia. Social history, such as sexual preference, drug use, and occupation, may suggest diagnosis such as HIV-related illness or organophosphate poisoning.

The physical examination begins with assessment of hydration status. Abdominal examination can narrow the differential diagnosis and reveal the need for surgical intervention. Even appendicitis can present with diarrhea in up to 20% of cases. Rectal examination can rule out impaction or presence of blood, the latter suggesting inflammation, infection, or mesenteric ischemia.

Diagnosis and Differential

The most specific tests in diarrheal illness all involve examination of the stool in the laboratory. Stool culture testing should be limited to severely

37_Cydulka_Ch37.indd 225 01/05/17 10:26 am

Source page 245

dehydrated or toxic patients, those with blood or pus in their stool, immunocompromised patients, and those with diarrhea lasting longer than 3 days. Consider testing for Salmonella, Shigella, Campylobacter, Shiga toxin-producing E. coli, or amoebic infection. Make the laboratory aware of which pathogens you suspect. In patients with diarrhea >7 days, those who have traveled abroad, or consumed untreated water, an examination for ova and parasites may be useful to rule out Giardia or Cryptosporidium. Multiple samples may be required. Assay for Clostridium difficile toxin may be useful in ill patients with antibiotic-associated diarrhea or recent hospitalization.

Because most diarrheal illnesses are viral or self-limited, laboratory testing in routine cases is not indicated. However, in extremely dehydrated or toxic patients, electrolyte determinations and renal function tests may be useful. (Hemolytic-uremic syndrome, characterized by acute renal failure, thrombocytopenia, and hemolytic anemia, may complicate E. coli 0157:H7 infections in children and the elderly). If toxicity is suspected, tests for levels for theophylline, lithium, or heavy metals will aid in the diagnosis. Radiographs are reserved for ruling out intestinal obstruction or pneumonia, particularly Legionella. In addition, CT scanning or angiography may be indicated in acute mesenteric ischemia.

Emergency Department Care and Disposition

The treatment of diarrhea consists of correcting fluid and electrolyte problems. (Antibiotic-associated diarrhea often responds to withdrawal of the offending drug.) Initiate specific therapy for any life-threatening cause identified in the initial workup.

1. Replacement of fluids can be intravenous (boluses of 500 mL IV in adults, 20 mL/kg in children), with normal saline solution in seriously ill patients. Mildly dehydrated patients who are not vomiting may tolerate an oral rehydrating solution containing sodium (e.g., Pedialyte®, Gatorade®) and glucose to enhance fluid absorption (glucose transport unaffected by enterotoxins). The goal is 30 to 100 mL/kg over the first 4 hours.
2. Once patients tolerate oral fluids, introduce a “BRAT” diet (bananas, rice, apples, toast). Patients should avoid raw fruit, caffeine, lactose, and sorbitol-containing products.

Patients with an unclear diagnosis but favorable examination findings

after hydration can be discharged home safely. Toxic or severely dehydrated patients, particularly infants and the elderly, warrant admission.

■■ ACUTE INFECTIOUS AND TRAVELER'S DIARRHEA

Norovirus causes 50% to 80% of all infectious diarrheas in the United States, followed much less frequently by non-Shiga toxin-producing E. coli, C. difficile, invasive bacteria (Campylobacter, Shigella, Salmonella), Shiga toxin-producing E. coli, and protozoa. A history of foreign travel, with consumption of contaminated food or drink, is associated with an 80% probability of bacterial diarrhea, primarily toxin and nontoxin-producing strains of E. coli.

37_Cydulka_Ch37.indd 226 01/05/17 10:26 am

References/provenance

- Source: *Tintinalli's Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 242–245.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 37: Disorders Presenting Primarily with Diarrhea

Source pages 246–253 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 246–253

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Disorders Presenting Primarily with Diarrhea**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 37: Disorders Presenting Primarily with Diarrhea 227 Diagnosis and Differential Patients with severe abdominal pain, fever, and bloody stool should undergo stool studies for specific pathogens, including culture for Salmo- nella, Shigella, Campylobacter, and E. [Source pages 246–253]

- coli O157:H7; assay for Shiga toxin; and microscopy or antigen assay for Entamoeba histolytica. [Source pages 246–253]

- Expo- sure of a traveler or hiker to untreated water, and illness that persists for more than 7 days, should prompt an evaluation for a protozoal pathogen. [Source pages 246–253]

- Stool should be tested by enzyme immunoassay for E. [Source pages 246–253]

- histolytica antigen, Giardia intestinalis antigen, and Cryptosporidium parvum antigen. [Source pages 246–253]

Clinical reasoning structure

- **Problem representation:** Disorders Presenting Primarily with Diarrhea; source pages 246–253.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Disorders Presenting Primarily with Diarrhea**.

- Source-page range: **246–253**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 246

CHAPTER 37: Disorders Presenting Primarily with Diarrhea 227

Diagnosis and Differential

Patients with severe abdominal pain, fever, and bloody stool should undergo stool studies for specific pathogens, including culture for Salmonella, Shigella, Campylobacter, and E. coli O157:H7; assay for Shiga toxin; and microscopy or antigen assay for Entamoeba histolytica. Exposure of a traveler or hiker to untreated water, and illness that persists for more than 7 days, should prompt an evaluation for a protozoal pathogen. Stool should be tested by enzyme immunoassay for E. histolytica antigen, Giardia intestinalis antigen, and Cryptosporidium parvum antigen.

Emergency Department Care and Disposition

Treatment of infectious diarrhea (including viral causes) includes antibiotics (see Table 37-1), antimotility agents, fluid resuscitation (oral or parenteral), and dietary modification. Probiotics are safe and may be beneficial.

Antimotility agents, such as loperamide 4 mg initially, then 2 mg following each unformed stool (16 mg/day maximum), will shorten the

Table 37-1 Antibiotic Recommendations for Infectious Diarrhea

Organism	Primary Treatment	Alternative Treatment
Empiric Tx – Traveler's Diarrhea	Ciprofloxacin 500 mg single dose or 500 mg BID × 3 days	Azithromycin 1000 mg single dose or trimethoprim/sulfamethoxazole 160 mg/800 mg single dose or BID × 3 days; or rifaximin 200 mg TID × 3 days
Empiric Tx – Infectious Diarrhea	Ciprofloxacin 500 mg BID × 5 days	Trimethoprim/sulfamethoxazole 160 mg/800 mg BID × 5 days

Clostridium difficile Metronidazole 500 mg Vancomycin 125 mg

TID × 14 days QID × 14 days

Escherichia coli 0157:H7 No antibiotics No antibiotics

Salmonella non-typhi Ciprofloxacin 750 mg Azithromycin 500 mg

BID × 5 days daily × 7 days

Shigella Ciprofloxacin 750 mg Azithromycin 500 mg

BID × 3 days daily × 3 days

Cyclospora Trimethoprim/sulfamethoxa-

zole 160 mg/800 mg

BID × 10 days

Giardia Tinidazole 2 g single dose Nitazoxanide 500 mg

BID × 3 days

Vibrio cholerae Doxycycline 500 mg single Trimethoprim/sulfamethoxa-

dose or azithromycin 1 g zole 160 mg/800 mg

single dose BID × 3 days

Entamoeba histolytica Metronidazole 750 mg TID × Metronidazole AND iodoquinol

10 days AND paromomycin 650 mg TID × 20 days

10 mg/kg TID × 7 days

Listeria monocytogenes No antibiotics No antibiotics

37_Cydulka_Ch37.indd 227 01/05/17 10:26 am

Source page 247

228 SECTION 6: Gastrointestinal Emergencies

duration of symptoms when combined with an antibiotic. Alternative agents include bismuth subsalicylate 30 mL or 2 tablets every 30 minutes for 8 doses, or diphenoxylate and atropine 4 mg four times daily. Avoid antimotility agents in the subset of patients with bloody or suspected inflammatory diarrhea because of the potential for prolonged fever, toxic megacolon in C. difficile patients, and hemolytic uremic syndrome in children infected with Shiga toxin-producing E. coli.

Most patients can be discharged home. Educate patients regarding the need for frequent hand washing to minimize transmission. Accordingly, patients employed in the food, day care, and health care industries should not return to work until diarrhea has resolved. Any toxic-appearing patient should be admitted. Consider admission for those at extremes of age as well. Individuals should be counseled about the proper selection of food and beverages consumed when traveling abroad, as well as the use of bottled or boiled water for drinking, brushing teeth, and the preparation of food and infant formula.

■■ CLOSTRIDIUM DIFFICILE-ASSOCIATED DIARRHEA AND COLITIS

Clostridium difficile is an anaerobic bacillus, which secretes two toxins that interact in a complex manner to cause illness ranging from diarrhea to pseudomembranous colitis. Pseudomembranous colitis is an inflammatory bowel disorder in which membrane-like yellowish plaques of exudate overlay and replace necrotic intestinal mucosa. Broad-spectrum antibiotics, most notably clindamycin, cephalosporins, ampicillin, amoxicillin, and fluoroquinolones, alter gut flora in such a way that C. difficile can flourish within the colon, causing enteropathy. Transmission of the organism can occur from contact with humans and fomites. C. difficile is the most common cause of infectious diarrhea in hospitalized patients and is now reported to affect healthy adults who were not exposed to a hospital setting.

Clinical Features

Onset is typically 7 to 10 days after initiating antibiotic treatment, but may occur up to several weeks following treatment. Clinical manifestations can vary from frequent, watery, mucoid stools to a toxic illness, characterized by profuse diarrhea, crampy abdominal pain, fever, leukocytosis, and dehydration.

Diagnosis and Differential

The diagnosis is confirmed by the demonstration of C. difficile toxin in stool. Colonoscopy is not routinely needed to confirm the diagnosis.

Emergency Department Care and Disposition

1. Ensure contact isolation, use of personal protective equipment, and appropriate hand hygiene with soap and water (alcohol-based rubs ineffective).

2. Mild C. difficile infection in an otherwise healthy patient can be treated with discontinuing the offending antibiotic, confirmation of infection, and clinical monitoring.

37_Cydulka_Ch37.indd 228 01/05/17 10:26 am

Source page 248

3. Oral metronidazole 500 mg three times daily for 14 days is the treatment of choice in patients with mild to moderate disease who do not respond to conservative measures.

4. Patients with severe diarrhea, those with a systemic response (e.g., fever, leukocytosis, or severe abdominal pain), and those whose symptoms persist despite appropriate outpatient management, must be hospitalized and should receive vancomycin 125 mg orally four times daily for 14 days.

5. Patients with pseudomembranous colitis complicated by toxic megacolon should receive metronidazole 500 mg IV every 6 hours AND vancomycin 500 mg orally every 6 hours. Toxic megacolon or intestinal

perforation require immediate surgical consultation. Rarely, emergency colectomy may be required for fulminant colitis.

Use of antidiarrheal agents is controversial. Probiotic treatment is not a useful adjunct. Relapses occur in 20% to 30% of patients. A first relapse is treated with metronidazole 500 mg orally three times daily for 14 days. A second relapse is treated with vancomycin 125 mg orally four times daily for 14 days, followed by a four-week taper.

■■ INFLAMMATORY BOWEL DISEASE/CROHN'S DISEASE

Crohn's disease is a chronic, idiopathic, granulomatous inflammatory disease, characterized by segmental ulceration of the GI tract anywhere from the mouth to the anus.

Clinical Features

The clinical course is variable and unpredictable, with multiple remissions and exacerbations. Patients commonly report a history of recurring fever, abdominal pain, and diarrhea over several years before a definitive diagnosis is made. Abdominal pain, anorexia, diarrhea, and weight loss occur in most patients. Patients may also present with complications of the disease, such as intestinal obstruction, intraabdominal abscess, or a variety of extraintestinal manifestations. Perianal fissures or fistulas, abscesses, and rectal prolapse can be seen in those with colonic involvement. Fistulas occur between the ileum and sigmoid colon; the cecum, another ileal segment, or the skin; or between the colon and the vagina. Abscesses can be intraperitoneal, retroperitoneal, interloop, or intramesenteric. Obstruction, hemorrhage, and toxic megacolon also occur. Toxic megacolon can be associated with massive GI bleeding.

Up to 40% of patients develop extraintestinal manifestations, including arthritis, uveitis, nephrolithiasis, and skin disease (e.g., erythema nodosum, pyoderma gangrenosum). Hepatobiliary disease, including gallstones, pericholangitis, and chronic active hepatitis, is commonly seen, as is pancreatitis. Some patients develop thromboembolic disease as a result of a hypercoaguable state. Malabsorption, malnutrition, and chronic anemia develop in longstanding disease. The recurrence rate for those with Crohn's disease is 25% to 50% when treated medically; higher for patients treated surgically.

Diagnosis and Differential

The definitive diagnosis of Crohn's disease is usually established months or years after the onset of symptoms. A careful and detailed history for

37_Cydulka_Ch37.indd 229 01/05/17 10:26 am

Source page 249

previous bowel symptoms that preceded the acute presentation may pro-

vide clues to the correct diagnosis. Abdominal CT scanning is the most useful diagnostic test, potentially revealing bowel wall thickening, segmental narrowing, destruction of the normal mucosal pattern, mesenteric edema, abscess formation, and fistulas, as well as extraintestinal complications (e.g., gallstones, renal stones, sacroiliitis). Colonoscopy can detect early mucosal lesions, define the extent of colonic involvement, and identify colon cancer.

The differential diagnosis of Crohn's disease includes lymphoma, ileocecal amebiasis, sarcoidosis, chronic mycotic infections, tuberculosis, Kaposi sarcoma, Campylobacter enteritis, and Yersinia ileocolitis. Most of these conditions are uncommon, and the latter two can be differentiated by stool cultures. When confined to the colon, ischemic colitis, infectious colitis, pseudomembranous enterocolitis, irritable bowel syndrome, and ulcerative colitis should be considered.

Emergency Department Care and Disposition

Initial evaluation should determine the severity of the attack and identify significant complications. Laboratory evaluation includes complete blood count, chemistries, and type and cross match when indicated. Plain abdominal x-rays may identify obstruction, perforation, and toxic megacolon, which may appear as a long, continuous segment of air-filled colon greater than 6 cm in diameter. CT of the abdomen is the most useful test to confirm the diagnosis, and identify both intraintestinal and extraintestinal manifestations. Initial ED management includes intravenous fluid replacement, parenteral analgesia, bowel rest, correction of electrolyte abnormalities, and nasogastric suction if obstruction, ileus, or toxic megacolon is present. Additional treatment may include the following:

1. Sulfasalazine 3 to 5 g/day is effective for mild to moderate Crohn's disease, but has multiple toxic side effects, including GI and hypersensitivity reactions. Mesalamine, up to 4 g/day, is equally effective, with fewer side effects.
2. Intravenous hydrocortisone 300 mg/day is recommended for severe disease. Oral glucocorticoids (e.g., prednisone 40 to 60 mg/day) are often used as induction therapy.
3. Immunosuppressive drugs, 6-mercaptopurine up to 1.5 mg/kg/day or azathioprine up to 2.5 mg/kg/day, are used as steroid-sparing agents, in healing fistulas, and in patients with serious surgical contraindications.
4. Oral antibiotics are first-line agents for perianal disease and can help induce remission. Ciprofloxacin 1 to 1.5 mg/kg/day, metronidazole 10

to 20 mg/kg/day, and rifaximin 800 mg twice daily are effective. Fulminant colitis should be treated intravenously with broad spectrum coverage, such as piperacillin-tazobactam 4.5 g four times daily.

5. Patients with medically resistant, moderate to severe Crohn’s disease, may benefit from the antitumor necrosis factor antibody infliximab 5 mg/kg intravenously, or adalimumab 160 mg subcutaneously.

6. Diarrhea can be controlled by loperamide 4 to 16 mg/day, diphenoxylate 5 to 20 mg/day, or cholestyramine 4 g one to six times daily.

37_Cydulka_Ch37.indd 230 01/05/17 10:26 am

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 246–253.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 38: Acute and Chronic Constipation

Source pages 254–259 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 254–259

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Acute and Chronic Constipation**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 38: Acute and Chronic Constipation 235 suppositories, or mineral oil (age 5 to 11 years: 5 to 15 mL PO daily; age >12 years: 15 to 45 mL PO daily) have been advocated. [Source pages 254–259]

- ● ■ Management of constipation in palliative care, chronic opioid use, or abuse has no definitive regimen. [Source pages 254–259]

- When the above measures are unsuccessful, sup- port is increasing for methylnatrexone 0.15 mg/kg subcutaneous a maxi- mum dose of 12 mg. [Source pages 254–259]

- ● ■ Many constipated patients can be safely discharged from the ED with established fecal flow or key aspects addressed (Tables 38-2 and 38-3). [Source pages 254–259]

- Early follow-up is indicated in patients with recent severe constipation; Table 38-2 Suggested Laboratory Studies for Goal-Directed Clinical Testing in Acute Abdominal Pain Laboratory Test Clinical Suspicion Amylase Pancreatitis (if lipase is not available) Lipase Pancreatitis β-Human chorionic gonadotrophin Pregnancy Ectopic or molar pregnancy Coagulation studies (prothrombin time/ GI bleeding partial thromboplastin time) End-stage liver disease Coagulopathy Electrolytes Dehydration Endocrine or metabolic disorder Glucose Diabetic ketoacidosis Pancreatitis Gonococcal/chlamydia testing Cervicitis/urethritis Pelvic inflammatory disease Hemoglobin GI bleeding Lactate Mesenteric ischemia Liver function tests Cholecystitis Cholelithiasis Hepatitis Platelets GI bleeding Renal function tests Dehydration Renal insufficiency Acute renal failure Urinalysis Urinary tract infection Pyelonephritis Nephrolithiasis ECG Myocardial ischemia or infarction Source: Reproduced with permission from Cline DM, Stead LG. [Source pages 254–259]

Clinical reasoning structure

- **Problem representation:** Acute and Chronic Constipation; source pages 254–259.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Acute and Chronic Constipation**.

- Source-page range: **254–259**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 254

CHAPTER 38: Acute and Chronic Constipation 235

suppositories, or mineral oil (age 5 to 11 years: 5 to 15 mL PO daily; age

>12 years: 15 to 45 mL PO daily) have been advocated.

● ■

Management of constipation in palliative care, chronic opioid use, or abuse

has no definitive regimen. When the above measures are unsuccessful, sup-

port is increasing for methylnatrexone 0.15 mg/kg subcutaneous a maxi-

mum dose of 12 mg.

● ■

Many constipated patients can be safely discharged from the ED with

established fecal flow or key aspects addressed (Tables 38-2 and 38-3).

Early follow-up is indicated in patients with recent severe constipation;

Table 38-2 Suggested Laboratory Studies for Goal-Directed Clinical Testing in

Acute Abdominal Pain

Laboratory Test Clinical Suspicion

Amylase Pancreatitis (if lipase is not available)

Lipase Pancreatitis

β-Human chorionic gonadotrophin Pregnancy

Ectopic or molar pregnancy

Coagulation studies (prothrombin time/ GI bleeding

partial thromboplastin time) End-stage liver disease

Coagulopathy

Electrolytes Dehydration

Endocrine or metabolic disorder

Glucose Diabetic ketoacidosis

Pancreatitis

Gonococcal/chlamydia testing Cervicitis/urethritis

Pelvic inflammatory disease

Hemoglobin GI bleeding

Lactate Mesenteric ischemia

Liver function tests Cholecystitis

Cholelithiasis

Hepatitis

Platelets GI bleeding

Renal function tests Dehydration

Renal insufficiency

Acute renal failure

Urinalysis Urinary tract infection

Pyelonephritis

Nephrolithiasis

ECG Myocardial ischemia or infarction

Source: Reproduced with permission from Cline DM, Stead LG. Abdominal Emergencies. New York: McGraw-Hill Medical; 2008.

Table 38-3 Key Aspects to Address Before Discharging a Constipated Patient

Possible obstructing lesion

Systemic illness

Medication interaction/effect

Electrolyte imbalance

Potential for intestinal perforation with self-administered enemas

38_Cydulka_Ch38.indd 235 21/04/17 10:45 am

Source page 255

chronic constipation associated with systemic symptoms, such as weight loss, anemia, or change in stool caliber; refractory constipation; and con stipation requiring chronic laxative use. Patients with organic constipation secondary to obstruction require hospitalization and surgical evaluation.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 71, “Acute Abdominal Pain,” by Mary Claire O’Brien.

38_Cydulka_Ch38.indd 236 21/04/17 10:45 am

Source page 256

C h a p ter Gastrointestinal Bleeding

39 Mitchell C. Sokolosky

Gastrointestinal (GI) bleeding is a common problem in emergency medicine and should be considered life-threatening until proven otherwise. Acute upper GI bleeding is more common than lower GI bleeding. Upper GI bleeding is defined as that originating proximal to the ligament of Treitz. Upper GI bleeds can result from peptic ulcer disease, erosive gastritis and esophagitis, esophageal and gastric varices, and Mallory–Weiss syndrome. Lower GI bleeds most commonly result from diverticular disease, followed by colitis, adenomatous polyps, and malignancies. What may initially appear to be lower GI bleeding may be upper GI bleeding in disguise.

■■ CLINICAL FEATURES

Most patients complain of hematemesis, coffee-ground emesis, hematochezia, or melena. Others will present with hypotension, tachycardia, angina, syncope, weakness, and confusion. Hematemesis or coffee-ground emesis suggests an upper GI source. Melena suggests a source proximal to the right colon. Hematochezia (bright red or maroon-colored) indicates a more distal colorectal lesion; however, approximately 10% of hematochezia may be associated with upper GI bleeding. Weight loss and changes in bowel habits are classic symptoms of malignancy. Vomiting and retching, followed by hematemesis, is suggestive of a Mallory–Weiss tear. A history of medication or alcohol use should be sought. This history may suggest peptic ulcer disease, gastritis, or esophageal varices. Spider angiomata, palmar erythema, jaundice, and gynecomastia suggest underlying liver disease. Ingestion of iron or bismuth can simulate melena, and certain foods, such as beets, can simulate hematochezia. However, stool heme (guaiac) testing will be negative.

■■ DIAGNOSIS AND DIFFERENTIAL

The diagnosis may be obvious with the finding of hematemesis, coffee ground emesis, hematochezia, or melena. A careful ear, nose, and throat (ENT) examination can exclude swallowed blood as a source. Nasogastric (NG) tube placement and aspiration may detect occult upper GI bleeding. A negative NG aspirate does not conclusively exclude an upper GI source. Guaiac testing of NG aspirate can yield both false-negative and false-positive results. Most reliable is gross inspection of the aspirate for a bloody, maroon, or coffee-ground appearance. A rectal examination can detect the presence of blood, its appearance (bright red, maroon, or melanotic), and the presence of masses. All patients with significant GI bleeding require type and crossmatch for blood. Other important tests include a complete blood count, electrolytes, blood urea nitrogen, creatinine, glucose, coagulation studies, and liver function tests. The initial hematocrit level may not reflect the actual amount of blood loss. Upper GI bleeding

may elevate the blood urea nitrogen level. Routine plain radiographs are of limited value. The initial diagnostic procedure of choice for lower GI

39_Cydulka_Ch39.indd 237 16/03/17 10:50 AM

Source page 257

bleeds—angiography, scintigraphy, or endoscopy—depends on resource ability and consultant preference. In one study, a cause for lower GI bleeding was found in <50% of cases.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. Emergency stabilization (airway, breathing, and circulation) takes priority. Administer oxygen, insert large-bore intravenous catheters, and institute continuous monitoring.
2. Replace volume loss immediately with isotonic crystalloids (e.g., normal saline or Ringer lactate). The decision to transfuse blood is based on clinical factors (continued active bleeding and no improvement in perfusion after administration of 2 L of crystalloids) rather than initial hematocrit values. The threshold for blood transfusion should be lower in the elderly.
3. Replace coagulation factors, as needed.
4. A nasogastric tube is recommended in most patients with significant GI bleeding, regardless of the presumed source. If bright red blood or clots are found, perform gentle gastric lavage.
5. Consider early therapeutic endoscopy for significant upper GI bleeding. Timing of endoscopy for diagnosis and treatment of some lower GI bleeding sources can vary. It is estimated that 80% of lower GI bleeding will resolve spontaneously.
6. Proton pump inhibitors (e.g., pantoprazole 80 mg bolus followed by an infusion of 8 mg/h) are recommended for patients with nonvariceal bleeding from peptic ulcer disease.
7. Consider octreotide 25 to 50 μg bolus followed by 25 to 50 μg/h intravenously for patients with upper GI bleeding.
8. Balloon tamponade with the Sengstaken–Blakemore tube or its variants should only be considered an adjunctive or temporizing measure and is rarely used due to the high complication rate.
9. A surgical and gastroenterology consult should be obtained in patients with uncontrolled bleeding. Patients who do not respond to both pharmacologic and endoscopic treatments may require emergent surgery.
10. Most patients with GI bleeding will require hospital admission and early referral to an endoscopist.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 75, “Upper Gastrointestinal Bleeding,” by Christopher M. Ziebell, Andy Kitlowski, Janna M. Welch and Phillip A. Friesen; Chapter 76, “Lower Gastrointestinal Bleeding,” by Bruce M. Lo.

39_Cydulka_Ch39.indd 238 16/03/17 10:50 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 254–259.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 40: Esophageal Emergencies

Source pages 260–263 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 260–263

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Esophageal Emergencies**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 40: Esophageal Emergencies 241 Esophagitis Esophagitis can cause prolonged periods of chest pain and odynophagia. [Source pages 260–263]

- Esophagitis may be inflammatory (e.g., GERD, NSAIDs, potassium chlo- ride, doxycycline, tetracycline, and clindamycin) or infectious if immuno- suppressed (e.g., Candida most common, herpes simplex, cytomegalovirus (CMV), and aphthous ulceration). [Source pages 260–263]

- Withdrawal of offending agent is gener- ally curative with medication-induced esophagitis. [Source pages 260–263]

- Patients with reflux- induced esophagitis require acid-suppressive medications. [Source pages 260–263]

- Infectious causes are established by endoscopy often with biopsy and specimen cultures. [Source pages 260–263]

Clinical reasoning structure

- **Problem representation:** Esophageal Emergencies; source pages 260–263.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Esophageal Emergencies**.

- Source-page range: **260–263**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 260

CHAPTER 40: Esophageal Emergencies 241

Esophagitis

Esophagitis can cause prolonged periods of chest pain and odynophagia. Esophagitis may be inflammatory (e.g., GERD, NSAIDs, potassium chloride, doxycycline, tetracycline, and clindamycin) or infectious if immunosuppressed (e.g., Candida most common, herpes simplex, cytomegalovirus (CMV), and aphthous ulceration). Withdrawal of offending agent is generally curative with medication-induced esophagitis. Patients with reflux-induced esophagitis require acid-suppressive medications. Infectious causes are established by endoscopy often with biopsy and specimen cultures.

Esophageal Perforation

Iatrogentic perforation is most common. Other causes include transient increase in intraesophageal pressure (Boerhaave syndrome), trauma, foreign body, infection, tumor, and aortic pathology. Perforation of the esophagus is associated with a high mortality rate.

Clinical Features

Pain is classically described as acute, severe, unrelenting, and diffuse and is reported in the chest, neck, and abdomen. Pain can radiate to the back and shoulders, or back pain may be the predominant symptom. Swallowing often exacerbates pain. Physical examination varies with the severity of the rupture and the elapsed time between the rupture and presentation. Abdominal rigidity with hypotension and fever often occur early. Tachycardia and tachypnea are common. Mediastinal emphysema takes time to develop. It is less commonly detected by examination or radiography in lower esophageal perforation and its absence does not rule out perforation. A Hammon crunch can sometimes be auscultated. Pleural effusions develop in 50% of patients with intrathoracic perforations and are uncommon in cervical perforations.

Diagnosis and Differential

Chest radiography can suggest the diagnosis. Given the limitations of CT, an esophagram or emergency endoscopy is most often used to confirm the diag-

nosis. Selection of the procedure depends upon the clinical setting and the resources available. Mistaking perforation for acute myocardial infarction, pulmonary embolism, or an acute abdomen can lead to delays in therapy.

Emergency Department Care and Disposition

1. Rapid, aggressive management is essential.

2. In the ED, initiate fluid resuscitation (see Chapter 5) and give broad-spectrum parental antibiotics to cover aerobic and anaerobic organisms. Examples include single drug coverage such as piperacillin/tazobactam 3.375 g intravenously (IV) or double drug coverage with cefotaxime 2 g IV or ceftriaxone 2 g IV plus clindamycin 600 mg IV or metronidazole 15 mg/kg IV once, then 7.5 mg/kg q6h (maximum 1 g/dose).

3. Obtain emergent surgical consultation.

4. All of these patients require hospitalization.

Swallowed Foreign Bodies

Children (18 to 48 months old) and those with mental illness account for most cases. Coins, toys, and crayons typically lodge in the anatomically

40_Cydulka_Ch40.indd 241 16/03/17 11:14 AM

Source page 261

narrow proximal esophagus. Adult candidates are those with esophageal disease, prisoners, and psychiatric patients. In adults, most impactions are distal. Complications include airway obstruction, stricture, and perforation. Once an object transverses the pylorus, it usually continues through the GI tract. Objects that become lodged distal to the pylorus are usually irregular, have sharp edges, and are wide (>2.5 cm) or long (>6 cm)

Clinical Features

Objects lodged in the esophagus can produce retrosternal pain, dysphagia, coughing, choking, vomiting, and aspiration, and the patient may be unable to swallow secretions. Adults with an esophageal foreign body generally provide unequivocal history. In the pediatric patient it may be necessary to rely on clues such as refusal or inability to eat, vomiting, gagging, choking, stridor, neck or throat pain, dysphagia, and drooling.

Diagnosis and Differential

Physical examination starts with an assessment of the airway. The nasopharynx, oropharynx, neck, and chest should also be examined. Occasionally, a foreign body can be directly visualized in the oropharynx. Plain films are used to screen for radiopaque objects. CT scanning has replaced the barium swallow test to evaluate for nonradiopaque objects.

Emergency Department Care and Disposition

1. Patients in extremis or with pending airway compromise are resuscitated

in standard fashion and may require active airway management.

2. Emergent endoscopy is indicated for complete distal obstruction of the esophagus with pooling of secretions (often distal esophageal food impaction).

3. Hospital admission is generally not needed if the foreign body is easily removed by endoscopy without complications.

4. In stable patients, indirect or fiberoptic laryngoscopy may allow removal of very proximal objects.

5. Consult surgery for worrisome foreign bodies that are in the more distal GI tract.

Food Impaction

Meat is the most common cause of food impaction.

1. Complete esophageal obstruction requires emergency endoscopy.

2. Uncomplicated food impaction may be treated expectantly but should not be allowed to remain impacted for >12 to 24 hours.

3. The use of proteolytic enzymes (e.g., Adolph Meat Tenderizer, which contains papain) to dissolve a meat bolus is contraindicated.

4. Glucagon 1 to 2 mg for adults may be attempted but success rates are poor.

Coin Ingestion

1. Obtain radiographs on all children suspected of swallowing coins to determine the presence and location of the object. Coins in the esophagus present their circular face on anteroposterior films, as opposed to coins in the trachea, which show that face on lateral films.

40_Cydulka_Ch40.indd 242 16/03/17 11:14 AM

Source page 262

2. Coins should be removed by endoscopy if lodged in the esophagus.

3. Removal of a coin with a Foley balloon catheter should be done under fluoroscopy by experienced hands. Complications include aspiration, airway compromise, and mucosal laceration.

4. Once in the stomach, coins almost always pass spontaneously.

Button Battery Ingestion

A button battery lodged in the esophagus is a true emergency requiring prompt removal because the battery may quickly induce mucosal injury and necrosis. Perforation may occur within 6 hours of ingestion.

1. Resuscitate the patient as needed.

2. Obtain radiographs to locate position of the battery.

3. Emergency endoscopy is indicated if battery is lodged in the esophagus. Foley balloon catheter technique may be considered if reliable history of ingestion ≤2 hours is obtained.

4. Batteries that have passed the esophagus can be managed expectantly with 24-hour follow-up examination. Repeat x-rays at 48 hour to ensure passage through pylorus. Most batteries pass through the body in 48 to 72 hours but may take longer.
5. Consult surgery if the patient develops symptoms or signs of GI tract injury.
6. The National Button Battery Ingestion Hotline at 202-625-3333 is a 24-hours, 7 days-a-week resource for help with management decisions.

Ingestion of Sharp Objects

1. Sharp objects in the esophagus, stomach, or duodenum require immediate removal by endoscopy in order to prevent complications such as perforation.
2. If the object is distal to the duodenum at presentation and the patient is asymptomatic, obtain daily plain films to document passage.
3. Consider surgical removal if 3 days elapse without passage.
4. Consult surgery immediately if the patient develops symptoms or signs of intestinal injury (e.g., pain, emesis, fever, and GI bleeding).

Narcotic Ingestion

1. The packets (condoms containing up to 5 g of narcotic) ingested by a narcotic courier (body packer) are often visible on plain x-ray.
2. Endoscopy is contraindicated because of the risk of iatrogenic packet rupture, which may be fatal.
3. Observation until the packet reaches the rectum is the favored treatment if the packets appear to be passing intact through the GI tract.
4. Whole-bowel irrigation may aid in the process of packet removal.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 77, “Esophageal Emergencies” by Moss Mendelson.

40_Cydulka_Ch40.indd 243 16/03/17 11:14 AM

Source page 263

C h a p ter Peptic Ulcer Disease
41 and Gastritis

Teresa Bowen-Spinelli

Peptic ulcer disease (PUD) is a chronic illness manifested by recurrent ulcerations in the stomach and duodenum. Acid and pepsin are crucial for ulcer development, but the great majority of ulcers are directly related to infection with Helicobacter pylori or nonsteroidal anti-inflammatory drugs (NSAIDs) use. Gastritis is an acute or chronic gastric mucosal inflammation and has various causes. Dyspepsia is upper abdominal discomfort with or without other symptoms that can have various causes or be functional.

■■ CLINICAL FEATURES

Peptic ulcer disease typically presents with burning epigastric pain, though it may be described as sharp, dull, and an ache, or an “empty” or “hungry” feeling. It may be relieved by the ingestion of food, milk, or antacids, presumably due to an acid buffering or a dilutional effect. The pain recurs as the gastric contents empty and the recurrent pain classically awakens the patient at night. Atypical presentations are common in the elderly and may include no pain, pain that is not relieved by food, nausea, vomiting, anorexia, weight loss, and/or bleeding.

A change in the character of the pain may herald the onset of a complication. Abrupt onset of severe pain is typical of perforation with spillage of gastric or duodenal contents into the peritoneal cavity. Back pain may represent pancreatitis from a posterior perforation. Nausea, vomiting, early satiety, and weight loss may occur with gastric outlet obstruction or cancer. Vomiting blood or passing melanotic stools with or without hemodynamic instability represents a bleeding complication.

■■ DIAGNOSIS AND DIFFERENTIAL

Peptic ulcer disease cannot be definitively diagnosed on clinical grounds, but it can be strongly suspected in the presence of a “classic” history (as above) accompanied by “benign” physical examination findings and normal vital signs with or without mild epigastric tenderness. Examination findings that may be indicative of PUD complications include a rigid abdomen consistent with peritonitis in perforation, abdominal distension and succussion splash consistent with gastric outlet obstruction, occult or gross rectal blood, or blood in nasogastric aspirate consistent with bleeding.

The differential diagnosis of epigastric pain is extensive. Pain, radiating into the chest, and belching may point to gastroesophageal reflux disease; more severe pain in the right upper quadrant (RUQ) radiating around the right side of the abdomen with tenderness suggests cholelithiasis or biliary colic; pain radiating into the back is common with pancreatitis and/or a concomitant mass may represent a pseudocyst or if the mass is pulsatile may represent an abdominal aortic aneurysm. Chronic pain, anorexia, and weight loss with or without a mass may represent cancer. Myocardial

41_Cydulka_Ch41.indd 244 16/03/17 11:24 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 260–263.
- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 41: Peptic Ulcer Disease and Gastritis

Source pages 264–267 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 264–267

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Peptic Ulcer Disease and Gastritis**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 41: Peptic Ulcer Disease and Gastritis 245 ischemia may present as epigastric pain and should be strongly considered in the appropriate clinical setting, especially in the population of diabetics and the elderly. [Source pages 264–267]

- Some ancillary tests may be helpful to exclude PUD complications and to narrow the differential. [Source pages 264–267]

- A normal CBC rules out chronic (but not acute) gastrointestinal (GI) bleeding. [Source pages 264–267]

- Elevated liver enzymes may indicate hepati- tis. [Source pages 264–267]

- An upright xray may show free air in the setting a perforation and an abdominal US examination may show cholecystitis, cholelithiasis, or an abdominal arotic aneurysm. [Source pages 264–267]

Clinical reasoning structure

- **Problem representation:** Peptic Ulcer Disease and Gastritis; source pages 264–267.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Peptic Ulcer Disease and Gastritis**.

- Source-page range: **264–267**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 264

CHAPTER 41: Peptic Ulcer Disease and Gastritis 245

ischemia may present as epigastric pain and should be strongly considered in the appropriate clinical setting, especially in the population of diabetics and the elderly.

Some ancillary tests may be helpful to exclude PUD complications and to narrow the differential. A normal CBC rules out chronic (but not acute) gastrointestinal (GI) bleeding. Elevated liver enzymes may indicate hepatitis. Elevated lipase may indicate pancreatitis. An upright xray may show free air in the setting a perforation and an abdominal US examination may show cholecystitis, cholelithiasis, or an abdominal arotic aneurysm. An ECG and troponin are indicated if myocardial ischemia is suspected.

The gold standard for diagnosis of PUD is visualization of an ulcer by upper GI endoscopy. Endoscopy is indicated in most patients with upper GI bleeding and in any patient with certain “alarm” features consistent with cancer: age >55 year, unexplained weight loss, early satiety or anorexia, persistent vomiting, dysphagia, anemia, abdominal mass, or jaundice.

Because of the strong association of H. pylori infection with PUD, testing for the presence of H. pylori with PUD is usually indicated, but this is generally more appropriate at the time of follow-up with gastroenterologist.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

After PUD is diagnosed, the goal of treatment is to heal the ulcer while relieving pain and preventing complications and avoiding recurrence. If the patient is infected with H. pylori then it must be eradicated in order to prevent ulcer recurrence. NSAIDs should be stopped whenever possible.

1. Proton pump inhibitors (PPIs) act to decrease acid production by blocking H+ ion secretion. They serve to heal ulcers faster than other therapies and also have an inhibitory effect on H. pylori. PPIs should be taken about 30 to 60 minutes prior to a meal. Include omeprazole 20 to 40 mg daily, esomeprazole 20 to 40 mg daily, 20 to 40 mg daily, lansoprazole

15 to 30 mg daily, pantoprazole 20 to 40 mg daily, or rabeprazole 20 mg daily.

2. Histamine-2 receptor antagonists (H2RAs) inhibit acid secretion and are available over the counter. H2RAs include cimetidine 200 to 400 mg twice a day, famotidine 10 to 20 mg twice a day; nizatidine 75 to 150 mg twice a day; and ranitidine 75 to 150 mg twice a day.

3. Liquid antacids relieve pain and heal ulcers by buffering gastric acid. Due to the minimal side effects of PPIs and H2RAs, liquid antacids are generally used on an as needed basis for pain relief. Typical dosing is 15 mL, 1 hour after meals and at bedtime.

4. If present, H. pylori infection should be treated, though this would rarely be initiated in the ED. There are multiple therapies, but the most common is “triple therapy” formulation taken for 14 days which consists of a PPI, clarithromycin, and either amoxicillin or metronidazole.

5. Patients with complications always require consultation and most require admission for continued treatment. For the treatment of bleeding (see Chapter 39). For perforation provide resuscitation as needed, start broad-spectrum antibiotics, and obtain immediate surgical consultation. For gastric outlet obstruction provide resuscitation as needed, place a nasogastric tube, and admit for continued treatment.

41_Cydulka_Ch41.indd 245 16/03/17 11:24 AM

Source page 265

6. When uncomplicated PUD is suspected in a stable patient, the great majority can be discharged home on a PPI or an H2RA with a liquid antacid for breakthrough pain and with recommendations to follow-up with their primary care provider for further evaluation as indicated. Patients with “alarm” features who are stable enough for discharge should be referred for endoscopy.

7. Patients should be told that PUD is a presumptive diagnosis and that for definitive diagnosis gastroenterologists should perform an endoscopy. They should return for further evaluation or treatment if any of the following occur: worsening pain, increased vomiting, hematemesis or melena, weakness or syncope, fever, or chest pain.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 7th ed., see Chapter 78, “Peptic Ulcer Disease and Gastritis,” by Matthew C. Gratton and Angela Bogle.

41_Cydulka_Ch41.indd 246 16/03/17 11:24 AM

Source page 266

C h apter Pancreatitis and Cholecystitis

42 Rita K. Cydulka

Acute pancreatitis (AP) is an inflammation of the pancreas. Disease severity ranges from mild local inflammation to multisystem organ failure secondary to a systemic inflammatory response. Cholelithiasis and alcohol abuse are the most common causes, but there are many potential etiologies. Patients without risk factors often develop pancreatitis secondary to medications or severe hyperlipidemia. Commonly used medications associated with pancreatitis include acetaminophen, carbamazepine, enalapril, estrogens, erythromycin, furosemide, hydrochlorothiazide, opiates, steroids, tetracycline, and trimethoprim-sulfamethoxazole.

■■ PANCREATITIS

Clinical Features

The most common symptom is a midepigastric, constant, boring pain radiating to the back that is worse when the patient is supine and is often associated with nausea, vomiting, and abdominal distention. Low-grade fever, tachycardia, and hypotension may be present. Epigastric tenderness is common, whereas peritonitis is a late finding.

Physical findings are dependent on the severity of disease. Physical examination findings include epigastric tenderness, but tenderness may localize more to the right or left upper quadrant of the abdomen. Bowel sounds may be diminished and abdominal distention may be present secondary to ileus. Refractory hypotensive shock, renal failure, fever, altered mental status, and respiratory failure may accompany the most severe disease.

Diagnosis and Differential

The diagnosis should be suspected by the history and physical examination. The presence of two of the three following features makes the diagnosis more likely: (1) history and examination findings consistent with AP, (2) lipase or amylase levels at least two to three times the upper limit of normal, or (3) imaging findings consistent with pancreatic inflammation. Lipase is the preferred diagnostic test as it is more accurate. There are many sources of extrapancreatic amylase, making it relatively nonspecific. Normal serum amylase does not exclude the diagnosis of AP. There is no benefit to ordering both tests.

A CBC will identify leukocytosis or anemia. Liver studies can demonstrate associated biliary involvement. An elevated alkaline phosphatase level suggests biliary disease and gallstone pancreatitis. Persistent hypocalcemia (<7 mg/100 mL), hypoxia, increasing serum urea nitrogen, and

metabolic acidosis are associated with a potentially complicated course. Imaging can help confirm the diagnosis of pancreatitis, evaluate biliary involvement, and exclude causes of abdominal pain. Abdominal computed tomography (CT) scan is preferred over ultrasound as the latter is often limited by bowel gas overlying the pancreas. In the face of a typical clinical picture and laboratory results, emergency imaging may not be needed.

247
42_Cydulka_Ch42.indd 247 16/03/17 11:34 AM

Source page 267

The differential diagnosis includes referred chest pain secondary to ischemic heart disease, pulmonary pathology such as pneumonia or empyema, hepatitis, cholecystitis or biliary colic, ascending cholangitis, renal colic, small bowel obstruction, peptic ulcer disease or gastritis, and acute aortic pathology such as aneurysm or dissection.

Emergency Department Care and Disposition

Care for the patient with pancreatitis includes fluid resuscitation; management of nausea, vomiting, and pain; and diligent monitoring of vital signs and pulse oximetry.

1. Initiate aggressive fluid resuscitation with crystalloid intravenous fluid. Pressors are indicated for hypotension not responsive to adequate fluid resuscitation.
2. All patients with nausea and vomiting should initially be NPO.
3. Administer antiemetics, such as ondansetron 4 mg or prochlorperazine 5 to 10 mg to reduce vomiting.
4. Administer parenteral analgesia for patient comfort. Intravenous opioids such as morphine 0.1 mg/kg are often required.
5. Administer oxygen to maintain a pulse oximetry reading of 95% oxygen saturation. Treat respiratory failure aggressively.
6. Treat patients with infected pseudocyst, abscess, or infected peripancreatic fluid with imipenem-cilastatin 500 mg IV, meropenem 1 g IV, or ciprofloxacin 400 mg IV and metronidazole 500 mg IV.
7. Patients with severe systemic disease will require intubation, intensive monitoring, bladder catheterization, and transfusion of blood and blood products as needed. Symptomatic hypocalcemia should be corrected. Laparotomy may be indicated for hemorrhage or abscess drainage.
8. Consult gastroenterology for patients with gallstone pancreatitis for endoscopic retrograde cholangiopancreatography (ERCP) and sphincterotomy.
9. Most patients will require hospitalization. Patients who demonstrate

poor prognostic signs (dropping hemoglobin, poor urine output, persistent hypotension, hypoxia, acidosis, or hypocalcemia) despite aggressive early treatment should be admitted to the intensive care unit with surgical consultation.

10. Patients with mild disease, no biliary tract disease, and no evidence of systemic complications may be managed as outpatients with close follow-up if they tolerate clear liquids and oral analgesics in the ED. Instruct patients to increase their diet as tolerated once nausea is controlled.

■■ CHOLECYSTITIS

Biliary tract emergencies most often result from obstruction of the gallbladder or biliary duct by gallstones. The four most common biliary tract emergencies caused by gallstones are biliary colic, cholecystitis, gallstone pancreatitis, and ascending cholangitis. Biliary disease affects all age groups, especially diabetics and the elderly. Gallstones, although common in the general population, remain asymptomatic in most patients. Common

42_Cydulka_Ch42.indd 248 16/03/17 11:34 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 264–267.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 42: Pancreatitis and Cholecystitis

Source pages 268–271 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 268–271

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Pancreatitis and Cholecystitis**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 42: Pancreatitis and Cholecystitis 249 risk factors for gallstones and cholecystitis include advanced age, female sex and parity, obesity, rapid weight loss or prolonged fasting, familial tendency, use of some medications, Asian ancestry, chronic liver disease, and hemolytic disorders (e.g., sickle cell disease). [Source pages 268–271]

- Clinical Features Patients with biliary disease present with a wide range of symptoms. [Source pages 268–271]

- Biliary colic may present with epigastric or right upper quadrant pain, may range from mild to severe, and, although classically described as intermit- tent or colicky, is often constant. [Source pages 268–271]

- Pain may be referred to the right shoulder or left upper back. [Source pages 268–271]

- It may begin after eating but often bears no association to meals. [Source pages 268–271]

Clinical reasoning structure

- **Problem representation:** Pancreatitis and Cholecystitis; source pages 268–271.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Pancreatitis and Cholecystitis**.

- Source-page range: **268–271**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 268

CHAPTER 42: Pancreatitis and Cholecystitis 249

risk factors for gallstones and cholecystitis include advanced age, female sex and parity, obesity, rapid weight loss or prolonged fasting, familial tendency, use of some medications, Asian ancestry, chronic liver disease, and hemolytic disorders (e.g., sickle cell disease).

Clinical Features

Patients with biliary disease present with a wide range of symptoms. Biliary colic may present with epigastric or right upper quadrant pain, may range from mild to severe, and, although classically described as intermittent or colicky, is often constant. Nausea and vomiting are usually present. Pain may be referred to the right shoulder or left upper back. It may begin after eating but often bears no association to meals. Acute episodes of biliary colic typically last for 1 to 5 hours, followed by a gradual or sudden resolution of symptoms. Recurrent episodes are usually infrequent, generally at intervals longer than 1 week. Biliary colic seems to follow a circadian pattern, with highest incidence of symptoms between 9 pm and 4 am. Physical examination commonly demonstrates right upper quadrant or epigastric tenderness without findings of peritonitis.

Acute cholecystitis presents with pain similar to that of biliary colic that persists for longer than the typical 5 hours. Fever, chills, nausea, emesis, and anorexia are common. Past history of similar attacks or known gallstones may be reported. As the gallbladder becomes progressively inflamed, the initial poorly localized upper abdominal pain often becomes sharp and localized to the right upper quadrant. The patient may have moderate to severe distress and may appear toxic. Choledocholithiasis often presents with midline pain that radiates to the middle of the back.

Examination findings include tenderness in the right upper quadrant or epigastrum, and Murphy’s sign (increased pain or inspiratory arrest during

deep subcostal palpation of the right upper quadrant during deep inspiration). Murphy's sign is the most sensitive physical examination finding for the diagnosis of cholecystitis. Generalized abdominal rigidity suggests perforation and diffuse peritonitis. Volume depletion is common, but jaundice is unusual. Acalculous cholecystitis occurs in 5% to 10% of patients with cholecystitis, has a more rapid, aggressive clinical course, and occurs more frequently in patients with diabetes, the elderly, trauma or burn victims, after prolonged labor or major surgery, or with systemic vasculitides.

Ascending cholangitis, a life-threatening condition with high mortality, results from complete biliary obstruction (often a common bile duct stone; less commonly a tumor) with bacterial superinfection. Patients often present in extremis with jaundice, fever, confusion, and shock. Examination findings can be subtle. Patients commonly have focal right upper quadrant pain and nausea. Jaundice may or may not be present. The Charcot triad of fever, jaundice, and right upper quadrant pain is suggestive but all three components are usually not present at once.

Diagnosis and Differential

Suspicion of gallbladder or biliary tract disease must be maintained in any patient who presents with upper abdominal pain. The differential diagnosis is similar to that of AP (see Diagnosis and Differential section under the section Pancreatitis, earlier).

42_Cydulka_Ch42.indd 249 16/03/17 11:34 AM

Source page 269

Patients with uncomplicated biliary colic usually have normal laboratory findings. The diagnosis is usually made based on the patient presentation, response to therapy, and examining the test results in aggregate.

Laboratory studies that may aid in diagnosis include a white blood cell count; leukocytosis with left shift suggests acute cholecystitis, pancreatitis, or cholangitis, but a normal white blood cell count does not exclude them. Serum bilirubin and alkaline phosphatase levels may be normal or mildly elevated in patients suffering from biliary colic or cholecystitis. Serum bilirubin and alkaline phosphatase levels are usually elevated in cases of choledocholithiasis and ascending cholangitis. Serum lipase or amylase levels should be checked to help exclude associated pancreatitis.

Ultrasound of the hepatobiliary tract is the initial diagnostic study of choice for patients with suspected biliary colic or cholecystitis (see Fig. 42-1). It can detect stones as small as 2 mm and signs of cholecystitis which include a thickened gallbladder wall (>3 to 5 mm), gallbladder distention (>4 cm in short-axis view), and pericholecystic fluid. A

positive sonographic Murphy's sign is very sensitive for diagnosis of cholecystitis when it is elicited during the scan. Ultrasound has a strong positive predictive value (92%) when both a sonographic Murphy's sign and gallstones are present. Choledocholithiasis is suggested when the common bile duct diameter is greater than 5 to 7 mm.

Computed tomography of the abdomen is most useful when additional intraabdominal processes are suspected. Radionuclide cholescintigraphy (technetium-iminodiacetic acid [HIDA]) or diisopropyl iminodiacetic acid ([DISIDA] scans) offers a sensitivity of 97% and a specificity of 90% for cholecystitis. A reasonable emergency department (ED) approach to suspected cholecystitis would be to obtain an ultrasound scan and then a radionuclide scan if ultrasound fails to establish the diagnosis.

Figure 42-1. Abdominal US demonstrating acute cholecystitis with a gallstone (arrow-head), gallbladder sludge (asterisk), and pericholecystic fluid (arrow). Used with permission from Bart Besinger, MD, FAAEM.

42_Cydulka_Ch42.indd 250 16/03/17 11:34 AM

Source page 270

Emergency Department Care and Disposition

Care for the patient with biliary disease includes fluid resuscitation and management of nausea, vomiting, and pain. Uncomplicated biliary colic can be managed without the aid of consultants. ED treatment includes the following measures:

1. ■Initiate aggressive fluid resuscitation with crystalloid intravenous fluid. Pressors are indicated for hypotension not responsive to adequate fluid resuscitation.
2. ■Patients should be made NPO.
3. ■Administer antiemetics, such as ondansetron 4 mg or prochlorperazine 5 to 10 mg to reduce vomiting.
4. ■Administer parenteral analgesia for patient comfort. Intravenous opioids such as morphine 0.1 mg/kg are often required. The intravenous nonsteroidal anti-inflammatory drug (NSAID) ketorolac 30 mg IV may also be helpful.
5. ■A nasogastric tube to low suction should be considered if the patient is distended or actively vomiting, or if vomiting is intractable to antiemetics.
6. ■Patients with acute biliary obstruction may require urgent decompression via endoscopic sphincterotomy of the ampulla of Vater.
7. ■Initiate early antibiotic therapy in any patient with suspected cholecystitis or cholangitis. Adequate therapy for uncomplicated cases of cholecystitis includes a parenteral third-generation cephalosporin

(cefotaxime or ceftriaxone 1 g IV q12 to q24h) plus metronidazole 500 mg IV. Patients with ascending cholangitis, sepsis, or obvious peritonitis are best managed with triple coverage by using ampicillin (0.5 to 1.0 g IV q6h), gentamicin (1 to 2 mg/kg IV q8h), and clindamycin (600 mg IV q6h), or the equivalent substitutes (e.g., metronidazole for clindamycin, third-generation cephalosporins or piperacillin/tazobactam, or a fluoroquinolone for ampicillin).

8. ■Patients diagnosed with acute cholecystitis, gallstone pancreatitis, or ascending cholangitis require immediate surgical consultation with hospital admission. Patients with choledocholithiasis, gallstone pancreatitis, or ascending cholangitis may also require urgent gastroenterology consultation to facilitate ERCP and sphincterotomy. Signs of systemic toxicity or sepsis warrant admission to the intensive care unit pending surgical treatment.

9. ■Patients with uncomplicated biliary colic whose symptoms abate with supportive therapy within 4 to 6 hours of onset can be discharged home if they are able to maintain oral hydration. Oral opioid analgesics may be prescribed for the next 24 to 48 hours for the common residual abdominal aching. Arrange timely outpatient follow-up with a surgical consultant or the patient's primary care physician. Instruct patients to return to the ED if fever develops, abdominal pain worsens, for intractable vomiting, or if another significant attack occurs before follow-up.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 79, "Pancreatitis and Cholecystitis," by Bart Besinger and Christine R. Stehman.

42_Cydulka_Ch42.indd 251 16/03/17 11:34 AM

Source page 271

C h a p ter Acute Appendicitis

43 Charles E. Stewart

Appendicitis is one of the most common surgical emergencies. Despite advances in laboratory testing and imaging, accurate diagnosis of appendicitis remains a challenge. Complications from misdiagnosis of appendicitis include intraabdominal abscess, wound infection, adhesion formation, bowel obstruction, and infertility.

■■ CLINICAL FEATURES

The early signs and symptoms of appendicitis are quite nonspecific and progress with time. The most reliable symptom in appendicitis is abdominal pain. Pain commonly begins in the periumbilical or epigastric region. As peritoneal irritation occurs, the pain often localizes to the right lower quad-

rant. The final location of the pain depends on the location of the appendix. Other symptoms associated with appendicitis include anorexia, nausea, and vomiting, but these symptoms are neither sensitive nor specific. As the pain increases, irritation of the bladder and/or colon may cause dysuria, tenesmus, or other symptoms. Many patients have the “bump” sign, where the patient notes an increase in the abdominal pain associated with bumps in the ride to the hospital. Other physicians will have the patient jump up and down in the examining room to evoke the pain. (Such maneuvers illustrate peritoneal irritation, but are nonspecific for appendicitis.) If the pain suddenly decreases, the examiner should consider appendiceal perforation. The classic point of maximal tenderness is in the right lower quadrant just below the middle of a line connecting the umbilicus and the right anterior superior iliac spine (McBurney’s point). Patients may also have pain referred to the right lower quadrant when palpating the left lower quadrant (Rovsing’s sign), pain elicited by extending the right leg to the hip while lying in the left lateral decubitus position (psoas sign), or pain elicited by passively flexing the right hip and knee and internally rotating the hip (obturator sign). Patients with a pelvic appendix may be quite tender on rectal examination, and patients with a retrocecal appendix may have more prominent flank pain than abdominal pain. No individual physical finding is sensitive or specific enough to rule in or rule out the diagnosis, and all physical findings and maneuvers depend on irritation of the peritoneum. Fever is a relatively late finding in appendicitis and rarely exceeds 39°C (102.2°F), unless rupture or other complications occur. Meta-analysis of 42 studies found that fever was the single most useful sign, followed by rebound tenderness and migration of the pain to the right lower quadrant.

■■ DIAGNOSIS AND DIFFERENTIAL

Even with that caveat, the diagnosis of acute appendicitis is primarily clinical. Symptoms with high sensitivity for appendicitis include fever, right lower quadrant pain, pain that occurs before vomiting, and absence of prior similar pain. Migration of the pain is thought to be a strong predictor

43_Cydulka_Ch43.indd 252 16/03/17 11:41 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 268–271.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 43: Acute Appendicitis

Source pages 272–275 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 272–275

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Acute Appendicitis**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 43: Acute Appendicitis 253 for appendicitis. [Source pages 272–275]

- Physical signs with high specificity include right lower abdominal rigidity and positive psoas sign. [Source pages 272–275]

- In both children and adults, no single historical or physical examination finding is sufficient to make an unequivocal diagnosis of appendicitis. [Source pages 272–275]

- Consider appendicitis in any patient with atraumatic right-sided abdominal, periumbilical, or flank pain who has not previously undergone appendectomy. [Source pages 272–275]

- Additional studies, such as complete blood count, C-reactive protein, urinalysis, and imaging studies, may be performed if the diagnosis is unclear, but lack the sensitivity to rule in or rule out the diagnosis. [Source pages 272–275]

Clinical reasoning structure

- **Problem representation:** Acute Appendicitis; source pages 272–275.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Acute Appendicitis**.

- Source-page range: **272–275**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 272

CHAPTER 43: Acute Appendicitis 253

for appendicitis. Physical signs with high specificity include right lower abdominal rigidity and positive psoas sign. In both children and adults, no single historical or physical examination finding is sufficient to make an unequivocal diagnosis of appendicitis. Consider appendicitis in any patient with atraumatic right-sided abdominal, periumbilical, or flank pain who has not previously undergone appendectomy.

Additional studies, such as complete blood count, C-reactive protein, urinalysis, and imaging studies, may be performed if the diagnosis is unclear, but lack the sensitivity to rule in or rule out the diagnosis. The combination of a normal WBC and C-reactive protein may have some utility as a negative screening test, as the likelihood of both being negative despite a pathologic diagnosis of appendicitis in a patient with a low pre-test probability is quite low. Unfortunately, these tests are often elevated in multiple conditions, so the reverse has almost no value. A pregnancy test must be performed in all females of reproductive age. A normal WBC does not rule out appendicitis. Urinalysis is useful to help rule out other diagnoses, but pyuria and hematuria can occur when an inflamed appendix irritates the ureter.

The differential diagnosis of right lower quadrant pain is wide and includes other gastrointestinal processes (e.g., inflammatory bowel disease, hernia, abscess, volvulus, diverticulitis), gynecologic or urological processes (e.g., ectopic pregnancy, ovarian torsion, renal colic, genitourinary (GU) infection or abscess), or musculoskeletal processes (e.g., muscular hematoma or abscess).

Scoring systems such as the Alvarado and Samuel scoring systems have been developed to aid in diagnosis. These scoring systems have uniformly had less sensitivity for diagnosis of appendicitis than the clinical judgment of an experienced examiner and should not replace this experienced-based decision.

The goal of imaging is to establish the diagnosis of appendicitis, avoid

a negative appendectomy, and identify other potential causes of abdominal pain. Increased concerns about accumulated radiation exposure for children, potentially childbearing females, and pregnant patients have led to growing interest in alternatives to the CT scan. Unfortunately, imaging can also escalate the cost of the medical care.

Plain radiographs of the abdomen are not helpful.

Computed tomography (CT) is probably the imaging study of choice for most patients with an overall sensitivity of 96% and PPV of 96%. CT findings suggesting acute appendicitis include pericecal inflammation, abscess, and periappendiceal phlegmon or fluid collections (Fig. 43-1). CT has been shown to change management in women, decreasing unnecessary tests. CT may be conducted with or without contrast administered orally, intravenously, or rectally, depending on institutional experience and preference. Unenhanced CT scanning has 92% sensitivity and 96% specificity. Noncontrast CT should be considered an acceptable imaging modality in the workup of acute appendicitis. Clearly, no contrast should be used for patients with allergy to contrast or for patients with renal disease. CT findings of appendicitis may be obscured in the thin patient, as intraperitoneal fat serves as an intrinsic contrast medium in unenhanced CT.

Ultrasonography has a high sensitivity but is limited both by operator skill and in evaluating a ruptured appendix or an abnormally located (e.g., retrocecal) appendix. Reports of the effectiveness of ultrasound (US)

43_Cydulka_Ch43.indd 253 16/03/17 11:41 AM

Source page 273

Figure 43-1. Acute appendicitis on contrast CT scan as evidenced by dilated and inflamed appendix.

diagnosis of appendicitis in pregnancy are conflicting, with some reporting US as useful and others reporting it as ineffective for diagnosis.

Graded compression ultrasonography is the initial modality of choice in both children and the pregnant patient to decrease radiation exposure. Overall sensitivity of US is 86% with positive predictive value (PPV) of 95%. Given the highly operator-dependent nature of ultrasonography, hospitals that frequently perform US for diagnosis of appendicitis may have greater reproducibility of high-quality studies. The diagnostic accuracy of abdominal US in children is better at ruling in acute appendicitis than excluding it.

Magnetic resonance imaging (MRI) for the diagnosis of appendicitis is accepted in many hospitals as a reliable technology that avoids completely the ionizing radiation risks. IV gadolinium contrast with MRI should be avoided in the pregnant patient as it crosses the placenta, given

the teratogenic effects seen in animal studies. Gadolinium is not given to patients with renal disease as it may exacerbate the renal insufficiency.

The elderly patient is more likely to have preexisting comorbidities and may be taking medications that alter presentation, management, and outcomes. Elderly patients may have decreased perceptions of symptoms.

Patients with communication difficulties and those with poor access to medical care may also have vague complaints, including diffuse pain, fever,

43_Cydulka_Ch43.indd 254 16/03/17 11:41 AM

Source page 274

or alteration in mental status. Such individuals commonly present later in the course of the disease and are more likely to have worse outcomes. In these patients, an atypical presentation is quite common and alternative or comorbid diagnoses are often found on imaging.

The pediatric patient who cannot verbalize symptoms also presents a significant challenge. Patients younger than 6 years have a high misdiagnosis rate due to poor communication skills and the association of many nonspecific symptoms. Careful history from the parent or guardian and equally careful physical examination are essential to accurate diagnosis. Coordinate imaging studies in these patients with your surgical consultant to ensure most accurate diagnosis with least radiation exposure.

Pregnant patients are at risk for misdiagnosis. Acute appendicitis is the most common surgical emergency in pregnancy, and delay in diagnosis is the greatest cause of increased morbidity in the pregnant woman with an acute abdomen. Ovarian torsion and ectopic or heterotopic pregnancy are additional considerations in the fertile female. If an abdominal US is nondiagnostic or not available, consider pelvic US, CT, or MRI.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. Obtain surgical consultation before imaging when the diagnosis is thought to be clear. Patient should have nothing by mouth and should have intravenous (IV) access and fluids.
2. The treatment for acute appendicitis is appendectomy. If the local surgical services are inadequate or unavailable, transfer the patient to an appropriate facility.
3. Control pain with opioid analgesics, such as fentanyl 1 to 2 μg/kg IV every 1 to 4 hours or morphine 0.1 mg/kg.
4. Antibiotics given before surgery decrease the incidence of postoperative wound infection or, in cases of perforation, postoperative abscess formation. Several antibiotic regimens to cover anaerobes, enterococci, and gram-negative intestinal flora have been recommended, including

piperacillin/tazobactam 3.375 g IV or ampicillin/sulbactam 3 g IV.
Consult with the surgeon regarding the antibiotic regimen and timing.
5. In patients for whom the diagnosis is not clear, admit for observation, serial examinations, and surgical consultation. This is a safe option for high-risk patients (pediatric, geriatric, pregnant, or immunocompromised).
6. Stable, nontoxic-appearing patients with adequate pain control who can tolerate oral hydration have no significant comorbidities, and are able to return for reevaluation in 12 hours may be considered for discharge and 12-hour follow-up. These patients should be instructed to avoid strong analgesics, and should return if they develop increased pain, localization of the pain, fever, nausea, or other signs or symptoms of illness that are worsening or not resolving.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 81, "Acute Appendicitis," by E. Paul DeKoning.

43_Cydulka_Ch43.indd 255 16/03/17 11:41 AM

Source page 275

C h a p ter Diverticulitis

44 James O'Neill

Diverticulitis is a common GI disorder that occurs when small herniations through the wall of the colon, or diverticula, become inflamed or infected.

■■ CLINICAL FEATURES

Classically, diverticulitis presents with left lower abdominal pain, fever, and leukocytosis. The most common symptom is a steady, deep discomfort in the left lower quadrant of the abdomen. Pain may be constant or intermittent, with associated symptoms of change in bowel habits (constipation or diarrhea), nausea, vomiting, and anorexia. Urinary tract symptoms are less common. Patients with a redundant sigmoid colon, of Asian descent, or with right-sided disease may complain of pain in the suprapubic area or right lower quadrant. The presentation can mimic other diseases, such as appendicitis.

Patients have a low-grade fever, but the temperature may be higher in patients with generalized peritonitis and in those with an abscess. Physical findings range from mild abdominal tenderness to severe pain, obstruction, and peritonitis. Occult blood may be present in the stool.

■■ DIAGNOSIS AND DIFFERENTIAL

The differential diagnosis includes acute appendicitis, colitis (ischemic or infectious), inflammatory bowel disease (Crohn's disease or ulcerative

colitis), colon cancer, irritable bowel syndrome, pseudomembranous colitis, epiploic appendagitis, gallbladder disease, incarcerated hernia, mesenteric infarction, complicated ulcer disease, peritonitis, obstruction, ovarian torsion, ectopic pregnancy, ovarian cyst or mass, pelvic inflammatory disease, sarcoidosis, collagen vascular disease, cystitis, kidney stone, renal pathology, and pancreatic disease.

Diverticulitis can be diagnosed by clinical history and examination alone. In stable patients with past similar acute presentations, no further diagnostic evaluation is necessary unless the patient fails to improve with conservative medical treatment. If a patient does not have a prior diagnosis or the current episode is different from past episodes, diagnostic imaging should be performed to rule out other intraabdominal pathology and evaluate for complications. CT scan is the preferred imaging modality for its ability to evaluate the severity of disease and the presence of complications. CT with IV and oral contrast has documented sensitivities of 97% and specificities approaching 100%. Compression ultrasound is operator dependent and has been shown to have sensitivity and specificity greater than 80% with experienced operators. Laboratory tests, such as a CBC, liver function tests, and urinalysis, are not diagnostic but may help exclude other diagnoses.

256

44_Cydulka_Ch44.indd 256 16/03/17 11:51 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 272–275.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 44: Diverticulitis

Source pages 276–279 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 276–279

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Diverticulitis**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 44: Diverticulitis 257 ■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION ED care begins with fluid and electrolyte replacement, pain, and nausea control. [Source pages 276–279]

- ■Ill-appearing patients, those with uncontrolled pain, vomiting, peritoneal signs, signs of systemic infection, comorbidities, or immunosuppression, and those with complicated diverticulitis (e.g., phlegmon, abscess, obstruc- tion, fistula, or perforation), require admission and surgical consultation. [Source pages 276–279]

- ■Uncomplicated diverticulitis is managed with oral antibiotics and a liq- uid diet, although recent data suggests that antibiotics may not be required in uncomplicated diverticulitis. [Source pages 276–279]

- ■Outpatients should follow up with a gastroenterology specialist for an outpatient colonoscopy in 6 weeks if they show improvement. [Source pages 276–279]

- Patients with worsening of their condition during outpatient treatment will re- quire admission to the hospital. [Source pages 276–279]

Clinical reasoning structure

- **Problem representation:** Diverticulitis; source pages 276–279.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Diverticulitis**.

- Source-page range: **276–279**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 276

CHAPTER 44: Diverticulitis 257

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

ED care begins with fluid and electrolyte replacement, pain, and nausea control.

1. ■Ill-appearing patients, those with uncontrolled pain, vomiting, peritoneal signs, signs of systemic infection, comorbidities, or immunosuppression, and those with complicated diverticulitis (e.g., phlegmon, abscess, obstruction, fistula, or perforation), require admission and surgical consultation.
2. ■Uncomplicated diverticulitis is managed with oral antibiotics and a liquid diet, although recent data suggests that antibiotics may not be required in uncomplicated diverticulitis.
3. ■Outpatients should follow up with a gastroenterology specialist for an outpatient colonoscopy in 6 weeks if they show improvement. Patients with worsening of their condition during outpatient treatment will require admission to the hospital.

Please see Table 44-1 for antibiotic recommendations.

Table 44-1 Antibiotics for Diverticulitis

Outpatient First line Metronidazole 500 mg PO q6h

4–7 days PLUS

Ciprofloxacin 750 mg PO BID

OR

Levofloxacin 750 mg PO q24h

OR

Trimethoprim-sulfamethoxazole (Bactrim) 1 tab PO BID

Alternate Amoxicillin-clavulanate extended release 1000/62.5 mg

2 tabs PO BID

Moxifloxacin 400 mg PO q24h

Inpatient

Moderate First line Metronidazole 500 mg IV TID

disease PLUS

Ciprofloxacin 400 mg IV BID

OR

Levofloxacin 750 mg IV q24h

OR

Aztreonam 2 g IV TID

OR

Ceftriaxone 1 g IV q24h

Alternative Ertapenem 1 g IV q24h

Piperacillin-tazobactam 4.5 g IV q8h

Moxifloxacin 400 mg IV q24h

Severe, life- First line Imipenem 500 mg IV q6h

threatening Meropenem 1 g IV q8h

Piperacillin-tazobactam 4.5 mg IV q8h

Alternative Ampicillin 2 g IV q6h

PLUS

Metronidazole 500 mg IV q6h

PLUS

Ciprofloxacin 400 mg IV q12h

OR

Amikacin, gentamicin, or tobramycin (Penicillin allergy:

Aztreonam 2 g IV q6h PLUS metronidazole 500 mg IV q6h)

44_Cydulka_Ch44.indd 257 16/03/17 11:51 AM

Source page 277

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 82, "Diverticulitis," by Graham Autumn.

44_Cydulka_Ch44.indd 258 16/03/17 11:51 AM

Source page 278

C h a p ter Intestinal Obstruction

45 and Volvulus

Olumayowa U. Kolade

Intestinal obstruction results from mechanical blockage or the loss of normal peristalsis. Adynamic or paralytic ileus is more common and usually self-limiting. Common causes of mechanical small bowel obstruction (SBO) are adhesions due to previous surgery, incarcerated hernias, or inflammatory diseases. Other causes to consider are inflammatory bowel diseases, congenital anomalies, and foreign bodies. The most frequent causes of large bowel obstructions are cancer, diverticulitis with stricture, sigmoid volvulus,

and fecal impaction. Consider intussusception in children. Sigmoid volvulus is more common in the elderly taking anticholinergic medications while cecal volvulus is more common in gravid patients. Intestinal pseudoobstruction (Ogilvie syndrome) may mimic large bowel obstruction. The elderly and bedridden and patients taking anticholinergic medications or tricyclic antidepressants are at increased risk for pseudoobstruction.

■■ CLINICAL FEATURES

Crampy, intermittent, progressive abdominal pain and inability to have a bowel movement or to pass flatus are common presenting complaints. Vomiting, bilious in proximal obstructions and feculent in distal obstruction, is usually present. Patients with partial SBO can still pass flatus. Physical signs vary from abdominal distention, localized or general tenderness, to obvious signs of peritonitis. Localization of pain and the presence of abdominal surgical scars, hernia, or masses may provide clues to the site of obstruction. The abdomen may be tympanitic to percussion. Active, high-pitched bowel sounds can be heard in mechanical SBO. Bowel sounds may be diminished or absent if the obstruction has been present for many hours. Rectal examination may demonstrate fecal impaction, rectal carcinoma, or occult blood. Key features of ileus and mechanical bowel obstruction are described in Table 45-1. The presence of stool in the rectum does not exclude obstruction. Consider a pelvic examination in women. Systemic symptoms and signs depend on the extent of dehydration and the presence of bowel necrosis or infection.

Table 45-1 Key Features of Ileus and Mechanical Bowel Obstruction

	Ileus	Bowel Obstruction
Pain	Mild to moderate	Moderate to severe
Location	Diffuse	May localize
Physical examination	Mild distention, ± tenderness, decreased bowel sounds	Mild distention, tenderness, high-pitched bowel sounds
Laboratory	Possible dehydration	Leukocytosis
Imaging	May be normal	Abnormal
Treatment	Observation, hydration	Nasogastric tube, surgery

45_Cydulka_Ch45.indd 259 01/05/17 10:27 am

Source page 279

■■ DIAGNOSIS AND DIFFERENTIAL

Suspect intestinal obstruction in any patient with abdominal pain, distention, and vomiting, especially in patients with previous abdominal surgery, abdominal/pelvic radiotherapy, or groin hernias.

Flat and upright abdominal radiographs and an upright chest x-ray can screen for obstruction (see Fig. 45-1), confirm severe constipation, or diagnose hollow viscous perforation with free air. The diagnostic procedure of choice in the ED is CT scanning using IV and oral contrast when possible. CT scanning can delineate partial versus complete bowel obstruction, partial SBO versus ileus, and strangulated versus simple SBO.

Laboratory tests may include a complete blood count, electrolytes, blood urea nitrogen, creatinine, lactate levels, coagulation profile, and type and cross-match. Suspect abscess, gangrene, or peritonitis if leukocytosis > 20,000 or left shift is noted. An elevated hematocrit is consistent with dehydration.

Figure 45-1. Sigmoid volvulus. Note that the open portion of the “C” formed by the twisted large bowel points toward the left side in the case of sigmoid volvulus. Reproduced with permission from Wikiradiography.com.

45_Cydulka_Ch45.indd 260 01/05/17 10:27 am

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 276–279.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 45: Intestinal Obstruction and Volvulus

Source pages 280–281 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 280–281

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Intestinal Obstruction and Volvulus**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 45: Intestinal Obstruction and Volvulus 261 ■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION ED care is directed at vigorous fluid resuscitation with crystalloids, careful monitoring of response, and prompt surgical consultation. [Source pages 280–281]

- Surgical inter- vention is usually necessary to treat a mechanical obstruction. [Source pages 280–281]

- Decompress the bowel with a nasogastric tube especially if vomiting or distension is present. [Source pages 280–281]

- Administer preoperative broad-spectrum intravenous antibiotics cov- erage such as piperacillin/tazobactam 3.375 g IV every 6 hours, tircarcillin-clavulanate 3.1 g IV every 6 hours or ampicillin/sulbac- tam 3.0 g or double drug coverage with cefotaxime 2 g or ceftriax- one 2 g plus clindamycin 600 mg or metronidazole 1 g or a carbapenem, such as meropenem 1 g IV every 8 hours. [Source pages 280–281]

- When the diagnosis is uncertain or if adynamic ileus is suspected, con- servative measures, such as intravenous fluids and observation without surgical intervention, may be appropriate. [Source pages 280–281]

Clinical reasoning structure

- **Problem representation:** Intestinal Obstruction and Volvulus; source pages 280–281.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Intestinal Obstruction and Volvulus**.

- Source-page range: **280–281**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 280

CHAPTER 45: Intestinal Obstruction and Volvulus 261

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

ED care is directed at vigorous fluid resuscitation with crystalloids, careful monitoring of response, and prompt surgical consultation. Surgical intervention is usually necessary to treat a mechanical obstruction.

1. Decompress the bowel with a nasogastric tube especially if vomiting or distension is present.

2. Administer preoperative broad-spectrum intravenous antibiotics coverage such as piperacillin/tazobactam 3.375 g IV every 6 hours, tircarcillin-clavulanate 3.1 g IV every 6 hours or ampicillin/sulbactam 3.0 g or double drug coverage with cefotaxime 2 g or ceftriaxone 2 g plus clindamycin 600 mg or metronidazole 1 g or a carbapenem, such as meropenem 1 g IV every 8 hours.

3. When the diagnosis is uncertain or if adynamic ileus is suspected, conservative measures, such as intravenous fluids and observation without surgical intervention, may be appropriate.

4. In patients with pseudoobstruction, colonoscopy is both diagnostic and therapeutic. Surgery is not indicated.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 83, “Bowel Obstruction,” by Timothy G. Price and Raymond J. Orthober.

45_Cydulka_Ch45.indd 261 01/05/17 10:27 am

Source page 281

C h a p ter Hernia in Adults and Children

46 Louise Finnel

A hernia is a protrusion of any viscus from its normal cavity, for example,

bowel bulging through the abdominal wall. Hernias are classified by anatomic location, hernia contents, and the status of those contents (e.g., reducible, incarcerated, or strangulated). The most common abdominal hernias are inguinal, ventral, and femoral hernias (Fig. 46-1).

Predisposing factors include family history, lack of developmental maturity, undescended testes, genitourinary abnormalities, conditions that increase intraabdominal pressure (e.g., ascites or pregnancy), chronic obstructive pulmonary disease, and surgical incision sites.

■■ CLINICAL FEATURES

Most hernias are detected on routine physical examination or inadvertently by the patient. When the contents of a hernia can be easily returned to their original cavity by manipulation, the hernia is defined as reducible. A hernia becomes incarcerated when its contents are not reducible. Incarcerated hernias may lead to bowel obstruction and strangulation. Strangulation refers to vascular compromise of the incarcerated contents and is an acute surgical emergency. When not relieved, strangulation may lead to gangrene, perforation, peritonitis, and septic shock.

Symptoms other than an obvious protruding mass from the abdominal wall include localized pain, nausea, and vomiting. Signs of strangulation include severe pain and tenderness, induration, and erythema over the site. Children may exhibit irritability and poor feeding. Careful evaluation for obstruction is essential.

Inguinal ligament

Indirect inguinal hernia

Femoral hernia Direct inguinal hernia

Figure 46-1. Groin hernias.

46_Cydulka_Ch46.indd 262 16/03/17 12:03 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 280–281.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 46: Hernia in Adults and Children

Source pages 282–285 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 282–285

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Hernia in Adults and Children**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 46: Hernia in Adults and Children 263 ■■ DIAGNOSIS AND DIFFERENTIAL Physical examination is the predominant means of diagnosis. [Source pages 282–285]

- Ultrasonographic detection of hernias is opera- tor and body habitus dependent but can be helpful in pediatric and pregnant patients where radiation exposure is a concern (Fig. [Source pages 282–285]

- Computed tomography remains the best radiographic test for the evaluation of hernias and can more easily identify the less common hernia types such as Spige- lian or obturator. [Source pages 282–285]

- The differential diagnosis of a groin mass includes direct or indirect hernia, testicular torsion, tumor, groin abscess, hydrocele, varicocele, and hidradenitis. [Source pages 282–285]

- In children, retracted or undescended testes may be mistaken for inguinal hernias. [Source pages 282–285]

Clinical reasoning structure

- **Problem representation:** Hernia in Adults and Children; source pages 282–285.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Hernia in Adults and Children**.

- Source-page range: **282–285**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 282

CHAPTER 46: Hernia in Adults and Children 263

■■ DIAGNOSIS AND DIFFERENTIAL

Physical examination is the predominant means of diagnosis. Laboratory testing is of minimal value. Ultrasonographic detection of hernias is operator and body habitus dependent but can be helpful in pediatric and pregnant patients where radiation exposure is a concern (Fig. 46-2). Computed tomography remains the best radiographic test for the evaluation of hernias and can more easily identify the less common hernia types such as Spigelian or obturator.

The differential diagnosis of a groin mass includes direct or indirect hernia, testicular torsion, tumor, groin abscess, hydrocele, varicocele, and hidradenitis. In children, retracted or undescended testes may be mistaken for inguinal hernias.

A

B

C D

Figure 46-2. Ultrasonographic detection of incarcerated hernia. A. An incarcerated femoral hernia is demonstrated as a small - bowel segment herniated through the femoral canal. B. In an incarcerated incisional hernia, a small-bowel segment (arrow) is demonstrated as herniated through a small orifice in the abdominal wall. Dilated small-bowel loops are evident proximal to the incarceration. C. In an umbilical hernia, a herniated small - bowel segment is demonstrated within the fluid space in the hernia sac. The segment was softly strangulated at the femoral orifice (arrow) formed by a defect of the fascia and was easily reduced by manipulation in this case. D. An incarcerated obturator hernia is demonstrated deep in the femoral region. It locates posterior to the pectineus muscle (arrows) and medial to the femoral artery (A) and vein (V).Reproduced with permission from Ma OJ, Mateer JR, Reardon RF, et al: Ma and Mateer’s Emergency Ultrasound, 3rd ed. New York:

The McGraw-Hill Companies; 2014

46_Cydulka_Ch46.indd 263 16/03/17 12:04 PM

Source page 283

264 SECTION 6: Gastrointestinal Emergencies

Table 46-1 Steps for Hernia Reduction

1. NPO—in case reduction is unsuccessful
2. Adequate analgesia; children will require procedural sedation
3. Place patient in Trendelenberg position
4. Externally rotate and flex ipsilateral leg
5. Place a padded icepack to reduce swelling and blood flow to the area
6. Grasp and elongate the hernia neck with one hand
7. ■With the other hand apply firm steady pressure to proximal hernia at the site of the fascial defect

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

Do not attempt reduction if signs of strangulation exist, as dead bowel may then be introduced into the abdomen (Table 46-1).

Definitive Management

1. Adults with easily reducible hernias can be referred for outpatient surgical evaluation and repair. Patients should avoid heavy lifting and return to the ED if herniation recurs and cannot be reduced promptly. Discuss signs of obstruction with them prior to discharge.
2. Incarcerated hernias that can’t be reduced with one or two attempts and strangulated hernias require emergent surgical consultation and intervention. Give nothing by mouth, initiate intravenous fluid resuscitation and administer intravenous opioid analgesia. Broad-spectrum antibiotics, such as cefoxitin 2 g IV or piperacillin/tazobactam 3.375 g IV, should be given if there is evidence of perforation or strangulation.
3. Infants with successfully reduced inguinal hernias should have surgical repair within 24 to 72 hours because one-third will redevelop incarceration.
4. Children with uncomplicated umbilical hernias may be discharged and followed longitudinally by their primary care providers. Refer children older than 4 years or those with hernias greater than 2 cm in diameter for surgical evaluation.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 84, “Hernias in Adults,” by Donald Byars and Turan Kayagil

46_Cydulka_Ch46.indd 264 16/03/17 12:04 PM

Source page 284

C ha p ter Anorectal Disorders

47 Chad E. Branecki

Anorectal disorders range from simple local disease processes to underlying serious systemic disorders. Most anorectal diseases originate in the anal crypts, glands, internal hemorrhoidal plexus, and external hemorrhoid veins. More serious life-threatening infections tend to lie in the deeper tissues such as the ischiorectal and pelvic-rectal spaces.

After a detailed history, a digital examination of the rectum should be performed, followed by anoscopy in the left lateral decubitus position. The supine or lithotomy position can be used for debilitated patients.

■■ ANAL TAGS

Skin tags are usually asymptomatic minor projections of the skin at the anal verge, which may be from residual prior hemorrhoids. Most are asymptomatic but inflammation may cause itching or pain. Inflammatory bowel disease may be associated with sentinel tags; therefore, surgical referral is warranted.

■■ HEMORRHOIDS

Engorgement, prolapse, or thrombosis of the internal or external hemorrhoidal vein(s) is termed hemorrhoids.

Clinical Features

Patients report painless, bright red rectal blood on the surface of the stool, toilet tissue, or dripping into the toilet bowl after defecation. Thrombosed hemorrhoids are usually painful and may appear as a bluish-purple mass protruding from the rectum. Large hemorrhoids may result in prolapse that may spontaneously reduce or require periodic manual reduction by patients or clinicians. They may become incarcerated and gangrenous, and require surgical intervention. Prolapse may cause mucous discharge and pruritus. If not reduced, severe bleeding, thrombosis, infarction, incarceration, urinary retention, or sepsis may occur.

Diagnosis and Differential

Internal hemorrhoids are not readily palpable and are best visualized through an anoscope. They are found at 2, 5, and 9 o’clock positions when patients are prone. Constipation, pregnancy, ascites, ovarian tumors, radiation fibrosis, and increased portal venous pressure are some of the common causes of hemorrhoids. Rectal and sigmoid colon tumors should be considered in patients older than 40 years.

Emergency Department Care and Disposition

Unless a complication is present, management is usually nonsurgical.

1. Hot sitz baths for at least 15 minutes, three times per day, and after each bowel movement will reduce pain and swelling. After the sitz baths, the anus should be gently but thoroughly dried.

47_Cydulka_Ch47.indd 265 16/03/17 12:14 PM

Source page 285

2. Topical steroids and analgesics may provide temporary relief. Bulk laxatives, such as psyllium seed compounds or stool softeners, should be used after the acute phase has subsided. Laxatives causing liquid stool are contraindicated as they may result in cryptitis and sepsis.

3. Surgical treatment is indicated for severe, intractable pain, continued bleeding, incarceration, or strangulation.

4. Acute and recently thrombosed painful hemorrhoids (< 48 hours) can be treated with clot excision. After analgesia, with a long acting local anesthetic such as 0.5% bupivacaine with epinephrine, an elliptical skin incision is made over the hemorrhoids and the thrombosed clot is evacuated (Fig. 47-1). Hemostasis is achieved by packing and pressure dressing. The pressure dressing may be removed after about 6 hours, when the patient takes the first sitz bath. Refer to a surgeon for definitive hemorrhoidectomy.

■■ CRYPTITIS

Sphincter spasm, and superficial trauma from diarrhea, or repeated passage of large hard stools cause breakdown of the mucosa over the crypts. Infecting organisms enter the crypts and cause inflammation of anal glands, abscesses formation, fissures, and fistulae. Common symptoms include anal pain during bowel movements and itching, with or without rectal bleeding. Diagnosis is made by palpation of tender, swollen crypts with associated hypertrophied papillae. Anoscopy allows visualization of the inflamed crypts in posterior midline of the anal ring. Bulk laxatives, additional roughage, hot sitz baths, and warm rectal irrigations enhance healing. Surgical treatment may be needed in refractory cases.

■■ ANAL FISSURES

Anal fissures are superficial linear tears of the anal canal usually caused by local trauma (e.g., passage of hard stool) and are the most common cause of painful rectal bleeding.

Clinical Features

Patients complain of sharp cutting pain with defecation that subsides between bowel movements. Bleeding is bright and in small quantities. Rectal examination is very painful and often not possible without application of topical anesthetic agents. Most fissures are located in the posterior midline. A sentinel pile may be noted in patients with chronic fissures. A non-posterior midline fissure should alert the physician to consider serious

causes, such as Crohn’s disease, ulcerative colitis, carcinomas, AIDS, tuberculosis, and sexually transmitted diseases.

Treatment is aimed at relieving sphincter spasm and pain, and preventing stricture formation. Hot sitz baths and the addition of bran (fiber) to the diet are helpful. Use of topical analgesics or steroids may be temporarily helpful. Surgical excision of the fissure may be required if the area does not heal in 6 weeks after adequate treatment.

47_Cydulka_Ch47.indd 266 16/03/17 12:14 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 282–285.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 47: Anorectal Disorders

Source pages 286–293 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 286–293

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Anorectal Disorders**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 47: Anorectal Disorders 267 A B C Figure 47-1. [Source pages 286–293]

- Elliptical incision of thrombosed external hemorrhoid. [Source pages 286–293]

- 47_Cydulka_Ch47.indd 267 16/03/17 12:14 PM 268 SECTION 6: Gastrointestinal Emergencies ■■ FISTULA IN ANO An anal fistula is an abnormal inflammatory tract, originating from an infected anal gland. [Source pages 286–293]

- Fistulae commonly result from perianal or ischiorectal abscess. [Source pages 286–293]

- Crohn’s disease, ulcerative colitis, tuberculosis, gonococcal proc- titis, and carcinomas should also be considered in the etiology. [Source pages 286–293]

Clinical reasoning structure

- **Problem representation:** Anorectal Disorders; source pages 286–293.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Anorectal Disorders**.

- Source-page range: **286–293**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 286

CHAPTER 47: Anorectal Disorders 267

A

B

C

Figure 47-1. Elliptical incision of thrombosed external hemorrhoid. A. Elliptical incision. B. Unroofing of thrombosed external hemorrhoid. C. Evacuation of clot.

47_Cydulka_Ch47.indd 267 16/03/17 12:14 PM

Source page 287

268 SECTION 6: Gastrointestinal Emergencies

■■ FISTULA IN ANO

An anal fistula is an abnormal inflammatory tract, originating from an infected anal gland. Fistulae commonly result from perianal or ischiorectal abscess. Crohn’s disease, ulcerative colitis, tuberculosis, gonococcal proctitis, and carcinomas should also be considered in the etiology. Persistent bloody, malodorous discharge occurs as long as the fistula remains open. Blockage of the tract causes recurrent bouts of inflammation and abscess formation.

Ultrasonography with a 7 MHz endoprobe and enhanced with 3% hydrogen peroxide may aid in the diagnosis. Non-ill appearing patients can be treated with analgesics, antipyretics, and oral antibiotics such as ciprofloxacin 750 mg twice daily and metronidazole 500 mg four times daily for 7 days. Sitz baths and local cleaning will temporize the condition before surgery. Surgical excision is the definitive treatment and should not be delayed in ill-appearing patients.

■■ ANORECTAL ABSCESSES

Abscesses start from the anal crypts and spread to involve the perianal, intersphincteric, ischiorectal, or deep perianal space. Perianal abscess is the most common and found at the anal verge (Fig. 47-2).

Clinical Features

Persistent dull, aching, throbbing pain that increases prior to defecation is typical. As the abscess progresses, pain and tenderness interfere with walk-

ing or sitting. Fever, leukocytosis, and a painful tender mass may be present upon digital rectal examination.

Emergency Department Care and Disposition

Simple isolated perianal abscess without systemic illness may be safely incised in the ED. Other perirectal abscesses, such as supralevator or ischiorectal abscesses, should be drained in the operating room. After adequate local and systemic analgesia, a cruciate incision is made over the abscess, and the “dog ears” are excised. Packing usually is not required. Sitz baths should be started the next day. Antibiotics usually are not necessary unless systemic infection or toxicity is present.

■■ PROCTITIS

Proctitis is inflammation of the rectal mucosa. Common causes include prior radiation treatments, autoimmune disorders, vasculitis, ischemia, sexually transmitted infections (e.g., syphilis, gonorrhea, chlamydia, lymphogranuloma venereum, herpes simplex, chancroid, human papillomavirus), and other infectious diseases.

Clinical Features

Symptoms include anorectal pain, itching, discharge, diarrhea, bleeding, and lower abdominal cramping. Mucosal inflammation, erythema, bleeding, ulcers, or discharge may be noted during anoscopy.

47_Cydulka_Ch47.indd 268 16/03/17 12:14 PM

Source page 288

CHAPTER 47: Anorectal Disorders 269

Supralevator Peritoneum

space

Retrorectal

space

Supralevator

space

Deep postanal

space

Superficial

postanal space

A B

Ischiorectal Perianal Submucosal Intersphincter

space space space space

Higher

intermuscular

Pelvirectal

Submucosal

Ischiorectal Intersphincteric

Perianal

C

D E

Figure 47-2. A, B. Anatomic classification of anorectal spaces. C. Anorectal abscesses. D. Perianal abscess. E. Complicated perirectal abscess.

A, B: Reproduced with permission from Reichman EF: Emergency Medicine Procedures. 2nd ed. New York: McGraw-Hill Education; 2013. D: Reproduced with permission from Knoop K, Stack L, Storrow A, et al: Atlas of Emergency Medicine, 3rd ed. New York: McGraw-Hill Education; 2010. Photo contributor: The American Society of Colon and Rectal Surgeons. E: Reproduced with permission from Knoop K, Stack L, Storrow A, et al: Atlas of Emergency Medicine, 3rd ed. New York: McGraw-Hill Education; 2010. Figure 9-35. Photo contributor: Lawrence B. Stack, MD.

Emergency Department Care and Disposition

Obtain cultures if an infectious cause is suspected. Stool softeners, sitz baths, good anal hygiene, and analgesics will provide some relief. Patient with enteric pathogens and sexually transmitted infections will require

47_Cydulka_Ch47.indd 269 16/03/17 12:14 PM

Source page 289

antibiotics directed at the suspected underlying pathogen. Arrange outpatient follow-up.

■■ RECTAL PROLAPSE

Prolapse (procidentia) may involve the mucosa alone or all layers of the rectum. In addition, intussusception of the rectum may present as a prolapse.

Clinical Features

Most patients complain of protruding mass, mucous discharge, associated bleeding, and pruritus. Partial prolapse involves only the rectal mucosa and tends to protrude only a few centimeters from the dentate line. Complete prolapse involves all layers of the rectum, and appears like a red, ball-like mass and may extend up to 15 cm. Often, pain is not a significant feature.

Emergency Department Care and Disposition

In children, the prolapse can be gently reduced under proper analgesia and sedation. The child should then be referred to a specialist to ensure the prolapse is not due to an underlying condition. Every effort should be made to prevent the child from being constipated. In adults, reduction can be more difficult if the rectal walls have become edematous. Generous amounts of granulated sugar applied 15 minutes prior to the reduction may aid in the process with direct, continuous pressure. Lubricated gauze should

be taped in place over the anal verge for a few hours after reduction. If the prolapse cannot be easily reduced, or there is evidence of ischemia, emergency surgical consultation and hospitalization are warranted.

■■ ANORECTAL TUMORS

Factors such as smoking, anal intercourse, HIV, and genital warts are associated with anorectal cancer. Neoplasms that occur in this group include adenocarcinoma, malignant melanoma, and Kaposi's sarcoma. Patients present with nonspecific symptoms including sensation of a mass, pruritus, pain, and blood on the stool. Constipation, anorexia, weight loss, narrowing of the stool caliber, and tenesmus eventually develop. Anal margin neoplasms frequently present as an ulcer that fails to heal in a timely manner. Virtually all anorectal tumors can be detected by careful visual examination of the perianal area, digital palpation of the distal rectum and anal canal, and procto- or sigmoidoscopic examination. Complications of anorectal tumors include rectal prolapse, prolonged blood loss, perirectal abscesses, or fistulae. Refer all patients for proctoscopic or sigmoidoscopic examination and biopsy if the history or physical examination is suspicious for neoplasms.

■■ RECTAL FOREIGN BODIES

Not all patients are forthcoming with accurate history of rectal foreign body insertion. Patients may instead complain of abdominal pain, anorectal bleeding, or discharge.

47_Cydulka_Ch47.indd 270 16/03/17 12:14 PM

References/provenance

- Source: *Tintinalli's Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 286–293.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 48: Jaundice, Hepatic Disorders, and Hepatic Failure

Source pages 294–301 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 294–301

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Jaundice, Hepatic Disorders, and Hepatic Failure**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 48: Jaundice, Hepatic Disorders, and Hepatic Failure 275 ■■ HEPATIC DISORDERS Specific entities addressed in this chapter include acute hepatitis, chronic liver disease, and complications of cirrhosis including ascites, spontaneous bacterial peritonitis (SBP), and hepatic encephalopathy. [Source pages 294–301]

- ■■ ACUTE HEPATITIS Hepatitis is an inflammation of the liver owing to infectious, toxic, or metabolic injury to hepatocytes. [Source pages 294–301]

- Patients can present to the ED anywhere along the spectrum of the disease from asymptomatic infection to an acute, or fulminant, liver failure to chronic cirrhosis. [Source pages 294–301]

- The most common causes are viral infection and toxic ingestion. [Source pages 294–301]

- Clinical Features Acute hepatitis typically presents with nausea, vomiting, and right upper quadrant abdominal pain. [Source pages 294–301]

Clinical reasoning structure

- **Problem representation:** Jaundice, Hepatic Disorders, and Hepatic Failure; source pages 294–301.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Jaundice, Hepatic Disorders, and Hepatic Failure**.

- Source-page range: **294–301**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 294

CHAPTER 48: Jaundice, Hepatic Disorders, and Hepatic Failure 275

■■ HEPATIC DISORDERS

Specific entities addressed in this chapter include acute hepatitis, chronic liver disease, and complications of cirrhosis including ascites, spontaneous bacterial peritonitis (SBP), and hepatic encephalopathy.

■■ ACUTE HEPATITIS

Hepatitis is an inflammation of the liver owing to infectious, toxic, or metabolic injury to hepatocytes. Patients can present to the ED anywhere along the spectrum of the disease from asymptomatic infection to an acute, or fulminant, liver failure to chronic cirrhosis. The most common causes are viral infection and toxic ingestion.

Clinical Features

Acute hepatitis typically presents with nausea, vomiting, and right upper quadrant abdominal pain. The patient with acute hepatitis can also have fever, diarrhea, jaundice, bilirubinuria, and an enlarged, tender liver. The presence of altered mental status, abnormal bleeding (bruising, bleeding gums, epistaxis, blood in the stool), ascites, and lower body edema suggests chronic disease or a fulminant liver failure.

Paying attention to the historical clues and risk factors is crucial in determining the etiology of hepatitis. Ingestion of acetaminophen (in one-time overdose or chronically high doses), mushrooms, raw oysters, and herbal remedies should be assessed in the history. Risk factors in the past medical history include chronic hepatitis, transfusion of blood products, positive human immunodeficiency virus status, frequent use of pain medications, or depression. A social history positive for injection drug use, chronic alcohol abuse, sexual promiscuity, or travel to countries with endemic parasitic liver diseases represents increased risk for liver disease.

Hepatitis A virus is transmitted by fecal-oral contamination that is associated with improper food handling. The most common transmission occurs from asymptomatic children to adults.

Hepatitis B virus (HBV) is transmitted sexually, by blood transfusion, by contaminated needles, and by perinatal transmission.

Hepatitis C virus (HCV) transmission occurs primarily through exposure to contaminated blood or blood products. HBV and HCV can lead to chronic infection, cirrhosis, and hepatocellular carcinoma.

Other hepatotropic viruses such as Hepatitis D and E viruses, cytomegalovirus, herpes simplex virus, coxsackievirus, and Epstein-Barr virus can cause acute hepatitis.

Toxic insults to the liver can cause acute hepatitis and fulminant liver failure. In addition to the most common toxic insult of acetaminophen, alcohol, a variety of prescription medications (e.g., certain antibiotics, statins, isoniazid), herbal remedies (e.g., black cohosh, chaparral, Echinacea, kava), and dietary supplements are associated with acute hepatitis and liver failure.

The clinical presentation of infectious hepatitis varies with the individual as well as with the specific causative virus. Asymptomatic incubation period is followed by a prodrome of nausea, vomiting, and malaise. Later on, patients may note dark urine (bilirubinuria) and clay-colored stools and jaundice.

48_Cydulka_Ch48.indd 275 16/03/17 12:25 PM

Source page 295

Alcoholic liver disease can range from asymptomatic, reversible fatty liver to acute alcoholic hepatitis, cirrhosis, or a combination of acute and cirrhotic features. Acute alcoholic hepatitis develops in patients with asymptomatic liver disease (fatty liver can be seen on imaging) if they continue drinking. These patients may complain of gradual onset of anorexia, nausea, fever, dark urine, jaundice, weight loss, abdominal pain, and generalized weakness.

Fulminant hepatic failure is generally used to describe the development of encephalopathy within 8 weeks of the onset of symptoms in a patient with a previously healthy liver. Signs and symptoms of acute failure may include encephalopathy, cerebral edema, jaundice, ascites, right upper quadrant tenderness, and coagulopathy. Clinical features of hepatitis are listed in Table 48-1.

Table 48-1 Clinical Features of Hepatitis

	Acute Hepatitis	Chronic Disease/ Cirrhosis	Acute Liver Failure
Symptoms			
Nausea/vomiting/diarrhea	+	±	+

Fever + – –

Pain + ± ±

Altered mental status – ± +

Bruising/bleeding – ± +

Physical examination

Jaundice + + +

Hepatomegaly + – ±

Ascites – + +

Edema – + –

Skin findings (bruising, vascular malformations) – + +

Lab abnormalities

Elevated ALT/AST + + ±

AST/ALT > 2 + ± ±

Elevated PT/INR – ± +

Elevated ammonia – ± +

Low albumin – + +

Direct bilirubinemia – + ±

Indirect bilirubinemia + + ±

Urobilinogen + + +

Elevated blood urea nitrogen/creatinine – – ±

Radiologic findings

Ascites – + +

Fatty liver + – –

Cirrhosis – + +

Abbreviations: ALT, alanine aminotransferase; AST, aspartate aminotransferase; INR, International Normalized Ratio; PT, prothrombin time; +, typically present; –, typically absent; ±, variable.

48_Cydulka_Ch48.indd 276 16/03/17 12:25 PM

Source page 296

Diagnosis and Differential

Traditional liver function panels include a mix of markers of hepatocyte injury, usually include AST, ALT, and ALP, as well as indicators of hepatocyte catabolic activity (direct and indirect bilirubin).

Elevations of transaminases in the hundreds of units per liter suggest mild injury, or smoldering inflammation. Levels in the thousands suggest extensive acute hepatic necrosis. Acute viral hepatitis may cause the levels of ALT to rise several thousand units per liter. Less significant elevations, less than five times normal, are typical of alcoholic liver disease. ALT is a more specific marker of hepatocyte injury than AST. Patients with acute alcoholic hepatitis have AST and ALT levels that rise to several hundred

units per liter. A relative predominance of AST to ALT is expected in patients with alcoholic hepatitis.

ALP elevation is associated with biliary obstruction and cholestasis; elevations greater than four times normal strongly suggest cholestasis.

An elevated GGT in the setting of hepatitis suggests an alcoholic cause. It is also elevated by drugs such as phenobarbital and warfarin. It may rise in acute and chronic pancreatitis, acute myocardial infarction, uremia, COPD, rheumatoid arthritis, and diabetes mellitus.

Lactate dehydrogenase (LDH) is a nonspecific marker, which limits its utility.

Serum glucose level should immediately be checked in patients with altered mental status because severe hepatocellular injury can cause hypoglycemia. Hypoxia, sepsis, intoxication, intracranial lesions, or encephalopathy should also be considered in such a state.

Since nausea and vomiting may result in volume depletion and dehydration, serum electrolytes, BUN, and creatinine levels should be checked.

An elevated serum ammonia level is seen in patients with hepatic metabolic failure caused by acute and chronic liver diseases. Very high ammonia levels signify poor prognosis.

Prothrombin time serves as a true measure of liver function. Prolonged PT or elevated INR occurs in acute hepatitis and exacerbations of chronic compensated liver disease; however, it is a common complication of advancing cirrhosis and indicates significant liver dysfunction and a poor prognosis.

Albumin reflects the liver's synthetic function and may decrease in advancing cirrhosis or severe acute hepatitis. Low albumin suggests a poor short-term prognosis.

Serum bilirubin is somewhat insensitive for liver dysfunction. Hyperbilirubinemia in acute viral hepatitis varies in severity, and fractionation has no clinical value. However, the development of severe hyperbilirubinemia in primary biliary cirrhosis, alcoholic hepatitis, and acute liver failure suggests a poor prognosis.

Viral hepatitis serologies are often grouped into screening panels by hospital laboratories; however, these tests are rarely immediately available. Detection of immunoglobulin M anti-hepatitis A virus antibodies is standard for diagnosing acute infection with hepatitis A virus. Acute clinical illness in hepatitis B virus correlates with positive hepatitis B virus surface antigen (HBsAg). Hepatitis C virus is confirmed with positive anti-hepatitis C virus (anti-HCV) antibodies; however, this diagnosis

48_Cydulka_Ch48.indd 277 16/03/17 12:25 PM

Source page 297

may be masked by the 6- to 8-week delay between infection and antibody detection.

A CBC may be useful in detecting anemia, which suggests alcoholic hepatitis, decompensated cirrhosis, gastrointestinal bleeding, or a hemolytic process. Although white blood cell (WBC) count is not useful in diagnosis, a transient neutropenia followed by a relative lymphocytosis with atypical forms is seen with viral hepatitis.

The serum acetaminophen concentration is the basis for diagnosis and treatment in patients with the suspicion of toxic ingestion.

Differential diagnosis of acute hepatitis include viral hepatitis, drug- or toxin-induced hepatitis, alcoholic hepatitis, autoimmune hepatitis, cholecystitis and cholangitis, HELLP syndrome, hepatic, pancreatic, or biliary tumors, and metastatic liver diseases.

Emergency Department Care and Disposition

1. ■With the exception of acetaminophen toxicity (more completely discussed in Chapter 106), patients with acute hepatitis require supportive treatment with pain management, antiemetic medication, and fluid resuscitation.
2. ■Treat fluid and electrolyte imbalances secondary to poor oral intake, excessive diarrhea, or vomiting with intravenous crystalloids. Treat hypoglycemia with 1 ampule D50W IV. Administration of antiemetics may allow resumption of oral intake.
3. ■Admit high-risk patients, including the elderly and pregnant women, and patients with significant fluid and electrolyte imbalance or refractory vomiting. Other admission criteria include bilirubin ≥20 mg/dL, PT 50% above normal, hypoglycemia, low albumin, GI bleeding, encephalopathy, immunosupression, or suspected toxin-induced hepatitis.
4. ■Hospital admission is rarely required for patients with acute viral hepatitis. These patients can be managed as outpatients with the emphasis on rest, adequate oral intake, strict personal hygiene, and avoidance of hepatotoxins. Patients are recommended to return to the ED for poor oral intake, and worsening symptoms, particularly vomiting, jaundice, or abdominal pain. Follow-up visits should be arranged.
5. ■Management of alcoholic hepatitis is also supportive and can be managed as outpatients with emphasis on nutritional supplementations, including thiamine, magnesium, potassium, and folate, and adequate oral intake. All patients should be advised to avoid further alcohol inges-

tion and hepatotoxins. Patients and family members should be referred for detoxification or alcohol-dependency treatment. Prophylactic treatment for alcohol withdrawal is considered in patients who require hospital admission.

6. ■Patients with fulminant hepatic failure require critical care in the ED and admission to the intensive care unit. Aggressive support of circulation and respiration, monitoring and treatment of increased intracranial pressure if present, and correction of hypoglycemia and coagulopathy are warranted. Other treatments include oral lactulose, oral aminoglycoside antibiotics (neomycin, vancomycin), and diet modification for significant protein restriction. Early consultation with a hepatologist and liver transplant service is necessary.

48_Cydulka_Ch48.indd 278 16/03/17 12:25 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 294–301.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 49: Complications of General Surgical Procedures

Source pages 302–307 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 302–307

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Complications of General Surgical Procedures**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 49: Complications of General Surgical Procedures 283 tissue, poor perfusion, foreign bodies, and hematomas. [Source pages 302–307]

- Necrotizing fasciitis should be considered in a systemically ill patient with rapidly expanding infection and pain out of proportion to examination (see Chapter 90). [Source pages 302–307]

- Superficial or deep fascial wound dehiscence can occur due to diabetes, poor nutrition, chronic steroid use, and inadequate or improper closure of the wound. [Source pages 302–307]

- Operative exploration may be required to determine the extent of dehiscence. [Source pages 302–307]

- Vascular Complications Superficial thrombophlebitis manifests with erythema, warmth, and full- ness of the affected vein. [Source pages 302–307]

Clinical reasoning structure

- **Problem representation:** Complications of General Surgical Procedures; source pages 302–307.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Complications of General Surgical Procedures**.

- Source-page range: **302–307**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 302

CHAPTER 49: Complications of General Surgical Procedures 283

tissue, poor perfusion, foreign bodies, and hematomas. Necrotizing fasciitis should be considered in a systemically ill patient with rapidly expanding infection and pain out of proportion to examination (see Chapter 90).

Superficial or deep fascial wound dehiscence can occur due to diabetes, poor nutrition, chronic steroid use, and inadequate or improper closure of the wound. Operative exploration may be required to determine the extent of dehiscence.

Vascular Complications

Superficial thrombophlebitis manifests with erythema, warmth, and fullness of the affected vein. It usually occurs in the upper extremities after intravenous catheter insertion or in the lower extremities due to stasis in varicosities. DVT most commonly occurs in the lower extremities postoperatively (see Chapter 25).

Drug Therapy Complications

Numerous medications cause fever without associated concomitant infection. Many antibiotics prescribed perioperatively can cause antibiotic-induced diarrhea. PMC, the most serious diarrheal complication, is caused by Clostridium difficile toxin. Watery or even bloody diarrhea, fever, and crampy abdominal pain are the usual complaints.

■■ DIAGNOSIS AND DIFFERENTIAL

Postoperative patients with fever should have an evaluation focusing on the elements detailed above. Patients with suspected respiratory complications should have chest x-rays. Radiographs may demonstrate atelectasis, pneumonia, or pneumothorax. Additional imaging like CT or ultrasound may be indicated based on the operative procedure performed.

Patients with oliguria or anuria should be evaluated for signs of hypovolemia or urinary retention. Diagnosis of PMC is established by demonstrat-

ing C difficile cytotoxin in the stool. Nevertheless, in 27% of cases, the assay may be negative.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

Contact the surgeon who performed the procedure to discuss patients who present with postoperative complications. Patients who are toxic appearing, have underlying debilitating conditions or elderly require hospitalization.

1. ■Patients with mild atelectasis and no evidence of hypoxemia may be managed as outpatients with pain control and increased deep breathing.

2. ■Postoperative pneumonia may be polymicrobial. Admission and antibiotic therapy to cover nosocomial infections such as Pseudomonas and methicillin-resistant Staph aureus are usually recommended (see Chapter 30).

3. ■Nontoxic patients with UTI can be managed as outpatients with oral antibiotic therapy geared toward appropriate organisms. Consider gram-positive flora when instrumentation has occurred. Ill-appearing patients require admission.

4. ■Wound hematomas may require removal of some sutures and evacuation. Consultation with the surgeon before treatment is appropriate.

49_Cydulka_Ch49.indd 283 16/03/17 12:36 PM

Source page 303

Seromas can be confirmed and treated with needle aspiration. Admission may not be necessary for either of these processes.

5. ■Wound infections are often treated with oral antibiotics unless the patient shows signs of systemic toxicity or carries significant comorbidities. Perineal infections are often polymicrobial requiring parenteral antibiotics and admission. Necrotizing fasciitis needs immediate surgical debridement and broad-spectrum parenteral antibiotics (see Chapter 90).

6. ■Patients with superficial thrombophlebitis may be treated as outpatients with nonsteroidal anti-inflammatory drugs (NSAIDs), local heat application and elevation. Antibiotics may be indicated if surrounding cellulitis or lymphangitis are noted. Suppurative thrombophlebitis requires hospitalization and surgical excision.

7. ■Patients with suspected antibiotic-induced PMC will require fluid resuscitation and likely empiric therapy. Oral or intravenous metronidazole and oral vancomycin are treatments for this condition.

■■ SPECIFIC CONSIDERATIONS

Complications of Breast Surgery

Although overall rates of complications are low following breast surgery, wound infections, hematomas, seromas, pneumothorax, and necrosis of the skin flaps may be seen. Lymphedema of the ipsilateral arm may occur after

mastectomy.

Complications of Gastrointestinal Surgery

Stimulation of the splanchnic nerves during intraabdominal surgery may lead to dysmotility and a paralytic ileus. After gastrointestinal surgery, small bowel tone returns to normal within 24 hours and colonic function within 3 to 5 days.

Patients develop nausea, vomiting, constipation, and abdominal distention and pain. An adynamic ileus typically resolves after bowel rest, nasogastric suction and intravenous hydration. Prolonged ileus should prompt an investigation for nonneuronal causes like peritonitis, intra-abdominal abscesses, hemoperitoneum, pneumonia, sepsis, electrolyte imbalance, or medications. Abdominal imaging, complete blood cell count, basic metabolic panel, and urinalysis should be obtained. Occasionally, surgical intervention may be necessary for obstruction due to adhesions.

Intraabdominal abscesses are caused by preoperative contamination, intraoperative spillage of bowel contents or postoperative anastomotic leaks. Diagnosis is confirmed by computed tomography or ultrasonography. Antibiotic therapy as well as either percutaneous or surgical drainage will be required.

Pancreatitis occurs especially after direct manipulation of the pancreatic duct. The clinical spectrum extends from mild nausea and vomiting to severe abdominal pain and hemodynamic instability. Complications like pleural effusion and severe hemorrhage may occur. Serum amylase measurements are not specific and measurement of a lipase is more reliable.

Cholecystitis and biliary colic have been reported as postoperative complications. Elderly patients are more prone to develop acalculous cholecystitis. Characteristic lab findings of an obstructive process may be absent.

49_Cydulka_Ch49.indd 284 16/03/17 12:36 PM

Source page 304

Fistulas, internal or external, may result from either technical complications or direct bowel injury. Fistulas can lead to electrolyte abnormalities and require surgical consultation and possible hospitalization. Anastomotic leaks occur primarily after esophageal, gastric and colonic procedures and can cause devastating consequences as a result of infection. Esophageal leaks occur within 10 days of the operation and carry very high morbidity and mortality rates.

Complications of bariatric surgery remain common, although mortality after the procedures is low. In the weeks after surgery, patients are at risk for leaks and bleeding. Dumping syndrome is seen in gastric bypass proce-

dures due to the rapid influx of hyperosmolar chyme into the small intestine resulting in fluid sequestration and hypovolemia. Patients experience nausea, vomiting, epigastric discomfort, palpitations, dizziness, and sometimes syncope. Other complications include gastroesophageal reflux, vitamin and electrolyte deficiencies, ulcers, obstruction, gastric slippage, and band erosion.

Complications of laparoscopic procedures include problems related to pneumoperitoneum, traumatic injury from insertion of the needle and trocar, and retained stones after cholecystectomy.

Complications of transabdominal feeding tubes and percutaneous endoscopic gastrostomy tubes include infections, hemorrhage, peritonitis, aspiration, wound dehiscence, sepsis, and obstruction of the tube. Dislodged tubes should be replaced with the appropriately sized tube (same type if possible, or a temporary foley catheter).

Acute complications arising from stomas (ileostomy or colostomy) are usually due to technical errors of stoma placement. Later complications can be from the underlying disease, such as Crohn's disease or cancer. Ischemia, necrosis, skin maceration, bleeding, parastomal hernia, and prolapse may be seen.

The most common complications of colonoscopy are hemorrhage and perforation. Hemorrhage occurs typically due to polypectomy, biopsies, or mucosal lacerations or tearing. Perforation may be immediately apparent or symptoms may be delayed for several hours to days. Upright chest or abdominal radiographs may reveal free air but CT is most sensitive.

Rectal surgery complications include urinary retention (frequently after hemorrhoidectomy), constipation, prolapse, bleeding, and infections.

Tetanus has been known to occur in surgical wounds although, by far, this rare disease is more common after minor trauma.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 87, "Complications of General Surgical Procedures," by Edmond A. Hooker

49_Cydulka_Ch49.indd 285 16/03/17 12:36 PM

Source page 305

This page intentionally left blank

49_Cydulka_Ch49.indd 286 16/03/17 12:36 PM

References/provenance

- Source: *Tintinalli's Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.

- Source pages: 302–307.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 50: Acute Kidney Injury

Source pages 308–311 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 308–311

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Acute Kidney Injury**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 50: Acute Kidney Injury 289 A B Figure 50-1. [Source pages 308–311]

- Dilated inferior vena cava (arrows) with little respiratory variation as might be expected in volume overload. [Source pages 308–311]

- An almost fully collapsed inferior vena cava at inspiration (arrows) and expira- tion (arrowheads) as might be expected in prerenal acute kidney injury. [Source pages 308–311]

- Used with permission from Michael B, Stone, MD, RDMS. [Source pages 308–311]

- 50_Cydulka_Ch50.indd 289 16/03/17 12:37 PM 290 SECTION 7: Renal and Genitourinary Disorders loss of renal function occurs over 10 to 14 days with complete obstruction, and worsens with associated UTI. [Source pages 308–311]

Clinical reasoning structure

- **Problem representation:** Acute Kidney Injury; source pages 308–311.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Acute Kidney Injury**.

- Source-page range: **308–311**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 308

CHAPTER 50: Acute Kidney Injury 289

A

B

Figure 50-1. Ultrasound of the inferior vena cava. A. Dilated inferior vena cava (arrows) with little respiratory variation as might be expected in volume overload. B. An almost fully collapsed inferior vena cava at inspiration (arrows) and expiration (arrowheads) as might be expected in prerenal acute kidney injury. Used with permission from Michael B, Stone, MD, RDMS.

50_Cydulka_Ch50.indd 289 16/03/17 12:37 PM

Source page 309

290 SECTION 7: Renal and Genitourinary Disorders

loss of renal function occurs over 10 to 14 days with complete obstruction, and worsens with associated UTI.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

Emergency department goals in the initial care of patients with acute renal failure focus on treating the underlying cause with an emphasis on correcting fluid and electrolyte derangements. Efforts should be made to prevent further renal damage and provide supportive care while attempting to establish a diagnosis and facilitating further evaluation and treatment as appropriate. Urine output should be closely monitored and consideration should be given to placement of a Foley catheter. ECG should be performed to evaluate for hyperkalemia-associated changes. Appropriate labs including serum potassium, creatinine, and BUN should be obtained.

Prerenal Failure

1. Use isotonic fluids (normal saline or Lactated Ringer’s solution) for volume resuscitation.

2. If cardiac failure is causing prerenal azotemia, optimize cardiac output to improve renal perfusion. Reducing intravascular volume (i.e., with diuretics) may be appropriate. Bedside echocardiography as well as sonographic assessment of the IVC may help guide this decision.

Renal Failure (Intrinsic)

Adequate circulating volume must be restored first; hypovolemia potentiates and exacerbates all forms of renal failure. Ischemia or nephrotoxic agents are the most common causes of intrinsic renal failure. History, physical examination, and baseline laboratory tests should provide clues to the diagnosis. Nephrotoxic medications and intravenous contrast should be avoided.

1. Vasopressor support with norepinephrine or dopamine may be needed as a temporizing measure to support mean arterial pressure and support renal perfusion in the setting of shock.
2. Use caution with renally excreted drugs (digoxin, magnesium, sedatives, and narcotics) and intravenous contrast because therapeutic doses may accumulate to excess and cause serious side effects. Fluid restriction may be required. Restoration of adequate intravascular volume with crystalloid infusion may be useful in the prevention of radiocontrast nephropathy.

Postrenal Failure

Establish appropriate urinary drainage; the exact procedure depends on the level of obstruction.

1. Place a Foley catheter to relieve obstruction caused by prostatic hypertrophy. However, there is no data to support the practice of intermittent catheter clamping to prevent hypotension and hematuria; urine should be completely and rapidly drained.

50_Cydulka_Ch50.indd 290 16/03/17 12:37 PM

Source page 310

2. Percutaneous nephrostomy may be required for ureteral occlusion until definitive surgery to correct the obstruction can take place once the patient is stabilized.
3. For the acutely anuric patient, obstruction is the major consideration. If no urine is obtained on initial bladder catheterization and a bladder scan or bedside ultrasound reveals a full bladder, emergency urologic consultation may be necessary.
4. With chronic urinary retention, postobstructive diuresis may occur due to osmotic diuresis or tubular dysfunction. Patients may become suddenly hypovolemic and hypotensive. Urine output must be closely monitored, with appropriate fluid replacement as needed.

Dialysis

If treatment of the underlying cause fails to improve renal function, or in cases with severe hyperkalemia or hypervolemia, dialysis should be considered. It is prudent to consult with a nephrologist as soon as possible if

urgent dialysis is a consideration.

1. The nephrologist usually makes decisions about dialysis. Dialysis is often initiated when the blood urea nitrogen is greater than 100 mg/dL or serum creatinine is greater than 10 mg/dL.

2. Patients with complications of acute renal failure such as cardiac instability (due to metabolic acidosis and hyperkalemia), intractable volume overload, hyperkalemia, and uremia (i.e., encephalopathy, pericarditis, and bleeding diathesis) not easily corrected by other measures should be considered for emergency dialysis.

Disposition

Patients with new-onset renal failure usually require hospital admission, often to an intensive care unit. Consider transferring patients to another institution if nephrology consultation and dialysis facilities are not available.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 88, "Acute Kidney Injury," by Richard Sinert and Peter R. Peacock, Jr.

50_Cydulka_Ch50.indd 291 16/03/17 12:37 PM

Source page 311

Chapter Rhabdomyolysis

51 Annet Alenyo Ngabirano

Rhabdomyolysis is the destruction of skeletal muscle, caused by any mechanism that results in injury to myocytes and their membranes. Table 51-1 lists commonly recognized conditions associated with rhabdomyolysis. In general, the most common causes of rhabdomyolysis in adults are alcohol and drugs of abuse, followed by medications, muscle diseases, trauma, neuroleptic malignant syndrome, seizures, immobility, infection, strenuous physical activity, and heat-related illness. Multiple causes are present in more than half of patients. In children, rhabdomyolysis is less common and is thought to be more benign.

Table 51-1 Common Conditions Associated with Rhabdomyolysis in Adults

Trauma	Immunologic diseases	Ischemic injury
Crush injury involving muscle	Compartment syndrome	
Electrical or lightning injury	Dermatomyositis	Compression
Drugs of abuse	Polymyositis	Medications
Amphetamines (including ecstasy (3, 4-methylene-dioxymethamphetamine))	Bacterial infection Group A β-hemolytic streptococci	Antipsychotics Barbiturates Benzodiazepines Clofibrate Caffeine

Cocaine Legionella Colchicine

Ethanol Salmonella Corticosteroids

Heroin Shigella Diphenhydramine

Lysergic acid diethylamide Staphylococcus aureus Isoniazid

Methamphetamines Streptococcus pneumoniae Lithium

Opiates Viral infection Monoamine oxidase

Phencyclidine Coxsackievirus inhibitors

Environment and Cytomegalovirus Narcotics

excessive muscular Epstein–Barr virus Neuroleptic agents

activity Enterovirus Phenothiazines

Contact sports Hepatitis virus Propofol

Delirium tremens Herpes simplex virus Salicylates

Dystonia Human immunodeficiency Selective serotonin reuptake

Psychosis virus inhibitors

Seizures Influenza virus (A and B) Statins

Marathons, military basic Rotavirus Theophylline

training Mycoplasma Tricyclic antidepressants

Heat stroke Zidovudine

Some novel cancer

Genetic disorders chemotherapeutic agents

Glycolysis and glycogenoly-

sis disorders

Fatty acid oxidation

disorders

Mitochondrial and

respiratory chain

metabolism disorders

292

51_Cydulka_Ch51.indd 292 16/03/17 12:49 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.

- Content ID: SEG-TINTINALLI-EMM8-FULL.

- Source pages: 308–311.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 51: Rhabdomyolysis

Source pages 312–315 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 312–315

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Rhabdomyolysis**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 51: Rhabdomyolysis 293 ■■ CLINICAL FEATURES Symptoms are usually acute in onset and include myalgias, stiffness, weak- ness, malaise, low-grade fever, and dark (usually brown) urine. [Source pages 312–315]

- Nausea, vomiting, abdominal pain, and tachycardia can occur in severe rhabdomy- olysis. [Source pages 312–315]

- Muscle symptoms, however, may be present in only half of cases. [Source pages 312–315]

- Some patients may present with complications of rhabdomyolysis such as acute renal failure, metabolic derangements, disseminated intravascular coagulation, and mechanical complications (e.g., compartment syndrome or peripheral neuropathy) (Table 51-2). [Source pages 312–315]

- Acute rhabdomyolysis may be present without any of these signs or symptoms and with normal findings on physical examination. [Source pages 312–315]

Clinical reasoning structure

- **Problem representation:** Rhabdomyolysis; source pages 312–315.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Rhabdomyolysis**.

- Source-page range: **312–315**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 312

CHAPTER 51: Rhabdomyolysis 293

■■ CLINICAL FEATURES

Symptoms are usually acute in onset and include myalgias, stiffness, weakness, malaise, low-grade fever, and dark (usually brown) urine. Nausea, vomiting, abdominal pain, and tachycardia can occur in severe rhabdomyolysis. Muscle symptoms, however, may be present in only half of cases. Some patients may present with complications of rhabdomyolysis such as acute renal failure, metabolic derangements, disseminated intravascular coagulation, and mechanical complications (e.g., compartment syndrome or peripheral neuropathy) (Table 51-2).

Acute rhabdomyolysis may be present without any of these signs or symptoms and with normal findings on physical examination. For this reason, diagnosis is often made from a significant history, an elevated serum creatine kinase level or the presence of dark urine on routine laboratory testing.

■■ DIAGNOSIS AND DIFFERENTIAL

An elevated serum creatine kinase is the most sensitive and reliable indicator of muscle injury. Diagnosis requires a fivefold or greater increase above the upper threshold of normal serum creatine kinase level in the absence of cardiac or brain injury. Persistently elevated levels despite treatment suggest ongoing muscle necrosis.

Myoglobinuria develops once skeletal muscle injury is >100 gram.

Because myoglobin contains heme, qualitative tests such as the dipstick test do not differentiate among hemoglobin, myoglobin, and red blood cells. Therefore, suspect myoglobinuria when the urine dipstick test is positive for blood but no red blood cells are present on microscopic examination. Absence of an elevated serum myoglobin level or of myoglobinuria does not exclude the diagnosis as levels may return to normal within 1 to 6 hours after the onset of muscle necrosis.

Other laboratory studies are useful in identifying the common com-

plications of rhabdomyolysis and the underlying cause. Urinalysis should be obtained for all patients. Serum electrolyte, calcium, phosphorus, and uric acid levels help detect hyperkalemia, abnormal calcium and phosphorus levels, and hyperuricemia. Serum creatinine and BUN levels are needed to identify acute renal failure. Because disseminated

Table 51-2 Complications of Rhabdomyolysis

Acute renal failure

Metabolic derangements

Hypercalcemia (late)

Hyperkalemia

Hyperphosphatemia

Hyperuricemia

Hypocalcemia

Hypophosphatemia (late)

Disseminated intravascular coagulation

Mechanical complications

Compartment syndrome

Peripheral neuropathy

51_Cydulka_Ch51.indd 293 16/03/17 12:49 PM

Source page 313

intravascular coagulation is a potential complication, a baseline CBC should be obtained and a coagulopathic screen considered (e.g., prothrombin time, partial thromboplastin time, fibrin split products, and fibrinogen level). Other common laboratory findings in rhabdomyolysis include elevated levels of aldolase, lactate dehydrogenase, urea, creatine, and aminotransferases. Further laboratory testing should be guided by the medical history and clinical presentation.

Other causes of muscle pain and weakness besides rhabdomyolysis should be considered in the appropriate clinical setting. These include acute myopathies, periodic paralysis, polymyositis or dermatomyositis, and Guillain–Barré syndrome. Rhabdomyolysis associated with strenuous exercise, fasting, or repeat episodes of rhabdomyolysis suggests an inherited metabolic myopathy.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. Provide aggressive IV rehydration for 24 to 72 hours. One method of rehydration is rapid correction of the fluid deficit with IV crystalloids followed by infusion of 2.5 mL/kg/h, with the goal of maintaining a minimum urine output of 2 mL/kg/h. Another method titrates infusion rates to a goal of 200 to 300 mL of urine output per hour.

2. No prospective controlled studies have demonstrated benefit from alkalinization of the urine with sodium bicarbonate or forced diuresis with mannitol or loop diuretics. Bicarbonate is widely recommended but without an evidence base. If bicarbonate is given, maintain an isotonic solution and avoid metabolic alkalosis or hypokalemia. Mannitol may be harmful because it may cause osmotic diuresis in hypovolemic patients.

3. Place a urinary catheter in patients in critical condition and those with acute renal failure to monitor urine output. Institute cardiac monitoring as electrolyte and metabolic complications can cause dysrhythmias. For patients with heart disease, comorbid conditions, or preexisting renal disease or for elderly patients, hemodynamic monitoring may be necessary to avoid fluid overload. Serial measurements of urine pH, electrolytes, creatine kinase, calcium, phosphorus, BUN, and creatinine are needed.

4. Monitor electrolytes. Hypocalcemia observed early in rhabdomyolysis usually requires no treatment. Give calcium, if needed, to treat hyperkalemia-induced cardiotoxicity or profound signs and symptoms of hypocalcemia. If hypercalcemia is symptomatic, continue saline diuresis. Treat hyperphosphatemia with oral phosphate binders when serum levels are >7 mg/dL. Treat hypophosphatemia when the serum level is <1 mg/dL. Hyperkalemia, which is usually most severe in the first 12 to 36 hours after muscle injury, can be significant and prolonged. Traditional insulin and glucose therapy, although recommended, may not be as effective in rhabdomyolysis-induced hyperkalemia. The use of ion-exchange resins (e.g., sodium polystyrene sulfonate) is effective. Dialysis may be needed (see Chapter 17, Fluids and Electrolytes, Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed.).

5. Avoid prostaglandin inhibitors such as nonsteroidal anti-inflammatory drugs (NSAIDs) because of their vasoconstrictive effects on the kidney.

51_Cydulka_Ch51.indd 294 16/03/17 12:49 PM

Source page 314

6. Finally, treat the underlying cause.

7. The majority of healthy patients with exertional rhabdomyolysis and without comorbidities (i.e., heat stress, dehydration, trauma) can be treated with oral or IV rehydration, observed in the ED, and then released. Otherwise, patients should be admitted for IV hydration, diuresis, management of complications, and treatment of the underlying cause. Admit to a monitored bed to identify dysrhythmias if they occur.

Consult the nephrology service to evaluate the need for dialysis, especially for patients with hyperkalemia unresponsive to therapy.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 89, "Rhabdomyolysis," by Francis L. Counselman and Bruce M. Lo.

51_Cydulka_Ch51.indd 295 16/03/17 12:49 PM

Source page 315

C h apter Emergencies in Renal Failure

52 and Dialysis Patients

Jonathan A. Maisel

Patients with end-stage renal disease (ESRD) may sustain multiple complications of their disease process and treatment. Emergent dialysis is most commonly required for hyperkalemia, severe metabolic acidosis, and pulmonary edema resistant to alternative therapy. See the appropriate chapters for discussion of the management of hypertension, heart failure, bleeding disorders, and electrolyte disorders.

■■ CARDIOVASCULAR COMPLICATIONS

Elevated levels of troponin I and T are common, even in those who are asymptomatic, and are associated with increased long-term risk of ischemic heart disease. Myocardial infarction may be defined by a 20% or greater rise in the troponin value, with at least one value above the 99th percentile. Hypertension occurs in most patients starting dialysis. Management includes control of blood volume, followed by use of adrenergic-blocking drugs, angiotensin-converting enzyme inhibitors, or vasodilating agents. Congestive heart failure (CHF) may be caused by hypertension, coronary ischemia, and valvular disease, as well as uremic cardiomyopathy, fluid overload, and arteriovenous (AV) fistulas (high output failure). Treatment includes supplemental oxygen, bilevel positive airway pressure, nitrates, angiotensin converting enzyme inhibitors, and possibly furosemide 60 to 100 mg IV, as its pulmonary vasodilator effect may provide relief, even in oliguric patients. Preload can be further reduced by inducing diarrhea with sorbitol and with phlebotomy (minimum 150 mL). Blood should be collected in a transfusion bag so that it may be transfused back to the patient during subsequent dialysis. Hemodialysis (HD) is the definitive treatment. Pericarditis in ESRD patients is usually due to worsening uremia. Electrocardiographic (ECG) changes typical of acute pericarditis are not always seen. Pericardial friction rubs are louder than in most other forms of pericarditis, often palpable, and frequently persist after the metabolic abnormalities have been corrected. Uremic pericarditis is treated with intensive dialysis. Cardiac tamponade is

the most serious complication of uremic pericarditis. It presents with changes in mental status, hypotension, or dyspnea. An enlarged heart on chest x-ray may suggest the diagnosis, which can be confirmed with echocardiography. Hemodynamically significant pericardial effusions require pericardiocentesis under fluoroscopic or ultrasonographic guidance.

■■ NEUROLOGIC COMPLICATIONS

Uremic encephalopathy presents with altered mental status, cognitive defects, memory loss, slurred speech, and asterixis. The progressive neurologic symptoms of uremia are the most common indications for initiating HD. It should remain a diagnosis of exclusion until structural, vascular, infectious, toxic, and metabolic causes of neurologic dysfunction have been ruled out. Peripheral neuropathy, manifested by paresthesias, diminished deep tendon reflexes,

52_Cydulka_Ch52.indd 296 16/03/17 2:14 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 312–315.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 52: Emergencies in Renal Failure and Dialysis Patients

Source pages 316–319 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 316–319

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Emergencies in Renal Failure and Dialysis Patients**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 52: Emergencies in Renal Failure and Dialysis Patients 297 impaired vibration sense, muscle wasting, and weakness, occurs in 60% to 100% of patients with ESRD. [Source pages 316–319]

- Autonomic dysfunction, characterized by pos- tural dizziness, gastric fullness, bowel dysfunction, reduced sweating, reduced heart rate variability, and baroreceptor control impairment, is com- mon in ESRD patients. [Source pages 316–319]

- Stroke is seen in 6% of HD patients, with half of cases being hemorrhagic, typically subdural hematomas. [Source pages 316–319]

- Stroke may be caused by cerebrovascular disease, head trauma, bleeding dyscrasias, anticoagulation, excessive ultrafiltration, or hypertension. [Source pages 316–319]

- Stroke should be considered in any ESRD patient presenting with a change in mental status. [Source pages 316–319]

Clinical reasoning structure

- **Problem representation:** Emergencies in Renal Failure and Dialysis Patients; source pages 316–319.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Emergencies in Renal Failure and Dialysis Patients**.

- Source-page range: **316–319**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 316

CHAPTER 52: Emergencies in Renal Failure and Dialysis Patients 297

impaired vibration sense, muscle wasting, and weakness, occurs in 60% to 100% of patients with ESRD. Autonomic dysfunction, characterized by postural dizziness, gastric fullness, bowel dysfunction, reduced sweating, reduced heart rate variability, and baroreceptor control impairment, is common in ESRD patients. Stroke is seen in 6% of HD patients, with half of cases being hemorrhagic, typically subdural hematomas. Stroke may be caused by cerebrovascular disease, head trauma, bleeding dyscrasias, anticoagulation, excessive ultrafiltration, or hypertension. Stroke should be considered in any ESRD patient presenting with a change in mental status.

■■ HEMATOLOGIC COMPLICATIONS

Anemia is caused by decreased erythropoietin, blood loss from dialysis, frequent phlebotomy, and decreased red cell survival. Factitious anemia reflects changes in plasma volume related to dialysis. Abnormal hemostasis in ESRD is multifactorial in origin, resulting in an increased risk of gastrointestinal (GI) tract bleeding, subcapsular liver hematomas, subdural hematomas, and intraocular bleeding. Benefit may be seen following the infusion of desmopressin 0.3 μg/kg or cryoprecipitate. Immunologic compromise, caused by impaired leukocyte chemotaxis and phagocytosis, leads to high mortality rates from infection. Dialysis does not appear to improve immune system function.

■■ GASTROINTESTINAL COMPLICATIONS

Anorexia, nausea, and vomiting are common symptoms of uremia, and are used as an indication to initiate dialysis, and assess its efficacy. Chronic constipation is common, due to decreased fluid intake and the use of phosphate-binding gels.

■■ COMPLICATIONS OF HEMODIALYSIS

Hypotension is the most frequent complication of HD. Excessive ultrafiltration due to underestimation of the patient’s ideal blood volume (dry weight) is the

most common cause of intradialytic hypotension. Cardiac compensation for fluid loss may be compromised by diastolic dysfunction common in ESRD patients. Other causes of intradialytic hypotension include myocardial dysfunction from ischemia, hypoxia, arrhythmias, and pericardial tamponade, abnormalities of vascular tone secondary to sepsis or antihypertensive medications, and volume loss from inadequate oral intake, vomiting, diarrhea, GI bleeding, or blood tubing or filter leaks. Treatment consists of Trendelenberg positioning, oral salt solution, or infusion of normal saline solution. If these interventions fail, excessive ultrafiltration is unlikely, and further evaluation will be required. Dialysis disequilibrium, caused by cerebral edema following large solute clearances, is characterized by nausea, vomiting, and hypertension, which can progress to seizures, coma, and death. Treatment consists of terminating dialysis, and administering mannitol intravenously to increase serum osmolality.

■■ COMPLICATIONS OF VASCULAR ACCESS

Complications of vascular access account for more inpatient hospital days than any other complication of HD. Thrombosis or stenosis present with loss of the bruit and thrill over the access. These need to be treated within

52_Cydulka_Ch52.indd 297 16/03/17 2:14 PM

Source page 317

24 hours with angiographic clot removal, angioplasty, or direct injection of thrombolytic into the access. Vascular access infections often present with signs of systemic sepsis, including fever, hypotension, and an elevated white blood cell count. Classic signs of pain, erythema, swelling, and discharge are often missing. Obtain both peripheral and catheter blood cultures simultaneously. A fourfold higher colony count in the catheter culture suggests it is the source. Staphylococcus aureus is the most common infecting organism, followed by gram-negative bacteria. Patients usually require hospitalization along with treatment with vancomycin 15 mg/kg IV and an aminoglycoside, such as gentamycin 100 mg IV. Potential life-threatening hemorrhage from a vascular access may result from a ruptured aneurysm or anastomosis, or overanticoagulation. Bleeding can often be controlled with 5 to 10 minutes of pressure at the puncture site. If this fails, the addition of an adsorbable gelatin sponge soaked in reconstituted thrombin, or a prothrombotic gauze, followed by 10 minutes of direct pressure may be effective. Life-threatening hemorrhage may require placement of a tourniquet proximal to the access, and vascular surgery consultation. If the etiology is excessive anticoagulation, the effects of heparin can be reversed with protamine 0.01 mg per unit heparin dispensed during dialysis or 10 to 20 mg protamine if the heparin dose is unknown. If a newly inserted vascular

access continues to bleed, desmopressin 0.3 μg/kg IV can be given as an adjunct to direct pressure.

■■ COMPLICATIONS OF PERITONEAL DIALYSIS

Peritonitis is the most common complication of peritoneal dialysis (PD). Signs and symptoms are similar to those seen in other patients with peritonitis, and include fever, abdominal pain, and rebound tenderness. A cloudy effluent supports the diagnosis. Peritoneal fluid should be sent to the laboratory for cell count, Gram stain, culture, and sensitivity. With peritonitis, cell counts usually reveal >100 leukocytes/mm3, with >50% neutrophils. Gram stain is positive in only 10% to 40% of culture-proven peritonitis. Organisms isolated include Staphylococcus epidermidis, S. aureus, Streptococcus species, and gram-negative bacteria. Empiric therapy begins with a few rapid exchanges of dialysate to decrease the number of inflammatory cells within the peritoneum. The addition of heparin (500 to 1000 U/L dialysate) decreases fibrin clot formation. Empiric antibiotics, covering gram-positive organisms (e.g., cephalothin or vancomycin 500 mg/L dialysate) and gram-negative organisms (e.g., gentamycin 100 mg/L dialysate), are added to the dialysate. Inpatient versus outpatient treatment of PD-related peritonitis should be based on clinical presentation. Infections around the PD catheter are characterized by pain, erythema, swelling, and discharge. Causative organisms are S. aureus and Pseudomonas aeruginosa. Outpatient treatment consists of a first-generation cephalosporin or ciprofloxacin.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 90, "End-Stage Renal Disease," by Mathew Foley, Ninfa Mehta and Richard Sinert.

52_Cydulka_Ch52.indd 298 16/03/17 2:14 PM

Source page 318

C h a p ter Urinary Tract Infections

53 and Hematuria

David R. Lane

■■ URINARY TRACT INFECTIONS

Urinary tract infections (UTIs) are the most common bacterial infections treated in the outpatient setting. More than 50% of women experience one in their lifetimes, and approximately 12% of women have a UTI annually. UTIs are divided into two major categories: lower tract infections and upper tract infections. Lower UTIs include urethritis, typically caused by sexually transmitted diseases and differentiated by the presence of discharge, and cystitis, an acute bacterial infection of the urinary bladder. Pyelonephritis is

an infection of the upper urinary tract structures including the ureters and kidneys. Differentiation of upper and lower UTIs is based on history and physical exam.

Uncomplicated UTI occurs in young, healthy, nonpregnant women with structurally and functionally normal urinary tracts.

Complicated UTI occurs in patients who are older, healthy, nonpregnant women with normal urinary tracts: that is, all men, and women who have a structural or functional genitourinary abnormality or an underlying predisposing medical condition that increases the risk of infection and recurrence or reduces the effectiveness of antimicrobial therapy.

Asymptomatic bacteriuria (ABU) is the presence of significant bacteria in urine without signs or symptoms that are referable to a urinary tract infection; the usual cutoff is a single organism isolated in a quantity of at least 100,000 colony forming U/mL. Screening and treatment is not generally recommended, with the exception of women who are pregnant or men who are going to undergo a transurethral prostate resection.

More than 80% of UTIs are caused by Escherichia coli. Other causative organisms include Klebsiella, Proteus, Enterobacter, Pseudomonas, Chlamydia, and Staphylococcus saprophyticus.

Clinical Features

Typical symptoms of cystitis include frequency, urgency, hesitancy, and suprapubic pain. Pyelonephritis is characterized by the addition of flank pain or costovertebral angle tenderness, particularly in conjunction with fever, chills, or nausea and vomiting. If vaginal or urethral discharge is present, urethritis, vaginitis, cervicitis, or PID are more likely than UTI. As in many diseases, diabetics and the elderly have a predilection for presenting atypically, and weakness, malaise, generalized abdominal pain, or altered mental status may be the only signs or symptoms noted.

Diagnosis and Differential

Diagnosis of UTI can be inferred from history and physical, and confirmed by urine dipstick or urinalysis and culture. Clean-catch midstream collection of urine is as accurate as urine obtained by catheterization if the patient

53_Cydulka_Ch53.indd 299 01/05/17 11:38 am

Source page 319

follows instructions carefully. Catheterization should only be used in a patient that cannot void spontaneously, is too ill or immobilized, or is extremely obese. Note that 1% to 2% of patients develop a UTI after a single catheter insertion.

The two tests of interest on urine dipstick are nitrite reaction, a measure of bacteriuria, and leukocyte esterase, a measure of pyuria. Nitrite has a very high specificity (>90%), but a low sensitivity (50%); several organisms are not detected by this test (Enterococcus, Pseudomonas, and Acinetobacter). Leukocyte esterase sensitivity (77%) is affected by high levels of protein or glucose in the urine. In addition it may be falsely positive in vaginitis or cervicitis. If either nitrite or leukocyte esterase is positive, the sensitivity is 75% and specificity is 82%. If both are positive, the specificity is 98% to 100%. A positive urine dipstick nitrite or leukocyte test result supports the diagnosis of UTI; a negative test result does not exclude it.

Urinalysis also evaluates bacteriuria and pyuria, the urine white blood cell count. Bacteriuria is defined as any bacteria on a Gram-stained specimen of uncentrifuged urine, and has a sensitivity of 95% and specificity of over 60%. False-negative results may occur in patients with low-colony-count UTI or Chlamydia infections. False-positive results can occur with vaginal or fecal contamination. Pyuria is defined as greater than 2 to 5 white blood cells per high power field in women and 1 to 2 white blood cells per high power field in men. False-negative results may occur in symptomatic patients with dilute urine, incompletely treated UTI, an obstructed, infected kidney, or systemic leukopenia.

Urine culture should only be performed for the following patients: those with complicated UTI (including men and pregnant women), those with relapse or reinfection, children, and those who are septic. Finally, imaging (CT or ultrasound) should be limited to patients with severe or nonresponsive pyelonephritis, or if there is a significant concern for a renal stone.

No single test or combination of testing can effectively rule out UTI in women presenting to the ED with symptoms of cystitis, and if test results are equivocal, consider initiating empiric treatment.

The differential diagnosis for patients presenting with dysuria includes vaginitis and cervicitis in women and urethritis, prostatitis, and epididymitis in men. Intraabdominal pathology, bladder or urethral structural abnormalities or cancer, chemical irritation or allergies to hygienic products, and trauma are also considerations.

Emergency Department Care and Disposition

Appropriate antimicrobial treatment should rapidly improve symptoms in all UTIs and also minimize growing community antibiotic resistance. Consider using nitrofurantoin 100 mg extended release twice a day PO for 5 days as a first-line agent in uncomplicated UTI. Alternatives in uncomplicated UTI include TMP-SMX DS 160/800 mg twice a day for 3 days or

fosfomycin 3 g in a single dose. Avoid flouroquinolones in uncomplicated UTI to reduce antibiotic resistance.

For complicated UTI or pyelonephritis, ciprofloxacin 500 mg twice a day for 7 days is considered the first-line agent. Alternatives include levofloxacin 750 mg daily for 5 days or cefpodoxime 400 mg twice a day for 7 to 14 days. If local resistance is less than 20%, TMP-SMX DS

53_Cydulka_Ch53.indd 300 01/05/17 11:38 am

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 316–319.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 53: Urinary Tract Infections and Hematuria

Source pages 320–323 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 320–323

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Urinary Tract Infections and Hematuria**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 53: Urinary Tract Infections and Hematuria 301 160/800 mg twice a day for 7 to 14 days or amoxicillin-clavulanate 875/125 mg twice a day for 14 days are acceptable. [Source pages 320–323]

- For pregnant patients, ABU should be treated with 3 days of nitrofu- rantoin and UTI should be treated with 7 days of nitrofurantoin. [Source pages 320–323]

- Fluoro- quinolones and tetracyclines are teratogenic and should be avoided, and TMP-SMX should be used with caution in the first and third trimesters. [Source pages 320–323]

- If there is suspicion for concomitant infection with gonorrhea or Chla- mydia, antibiotic choice is more complex (see Chapter 87, "Sexually Trans- mitted Infections"). [Source pages 320–323]

- In addition to the antibiotic, prescribe a 2-day course of a bladder anal- gesic such as phenazopyridine 200 mg three times a day. [Source pages 320–323]

Clinical reasoning structure

- **Problem representation:** Urinary Tract Infections and Hematuria; source pages 320–323.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Urinary Tract Infections and Hematuria**.

- Source-page range: **320–323**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 320

CHAPTER 53: Urinary Tract Infections and Hematuria 301

160/800 mg twice a day for 7 to 14 days or amoxicillin-clavulanate 875/125 mg twice a day for 14 days are acceptable.

For pregnant patients, ABU should be treated with 3 days of nitrofurantoin and UTI should be treated with 7 days of nitrofurantoin. Fluoroquinolones and tetracyclines are teratogenic and should be avoided, and TMP-SMX should be used with caution in the first and third trimesters.

If there is suspicion for concomitant infection with gonorrhea or Chlamydia, antibiotic choice is more complex (see Chapter 87, "Sexually Transmitted Infections").

In addition to the antibiotic, prescribe a 2-day course of a bladder analgesic such as phenazopyridine 200 mg three times a day. Discharge instructions must include return precautions for uncontrolled pain, fever, vomiting, and also should encourage adequate hydration, completion of the entire antibiotic course, and outpatient follow-up with a primary care doctor.

Admit patients with pyelonephritis who exhibit intractable vomiting. Consider admission for pregnant females (20% will develop sepsis) and patients with inadequate outpatient follow-up. Parenteral antibiotic options for inpatient management of pyelonephritis include ceftriaxone 1 g IV daily, ciprofloxacin 400 mg IV every 12 hours, and piperacillin-tazobactam, 3.375 g IV every 6 hours.

■■ HEMATURIA

Gross hematuria, visible to the eye, suggests a lower urinary tract source; however, other pigments such as myoglobin can simulate hematuria.

Microscopic hematuria (>3 RBCs/HPF) suggests a renal source, though it can also be present in UTI.

Diagnosis and Differential

A urine dipstick is positive with approximately 5 to 20 red blood cells per milliliter of urine. A positive dipstick should be followed by microscopy. Urethral catheterization induces hematuria in 15% of patients. False-positives may also result from myoglobin, free hemoglobin due to hemolysis, and porphyrins. False-negative results may occur if the urine has a high specific gravity.

Any process that results in infection, inflammation, or injury to the kidneys, ureters, bladder, prostate, male genitalia, or urethra may result in hematuria. UTI is the most common cause of hematuria associated with urgency, dysuria, and nocturia, whereas painless hematuria is largely due to neoplastic, hyperplastic, or vascular causes. The most common causes of hematuria in patients <40 years are infections and inflammatory conditions, including UTI, prostatitis, urethritis, sexually transmitted diseases, epididymo-orchitis, nephrolithiasis, and tuberculosis. In older patients, consider renal and bladder cancer, as well as prostate tumors and benign prostatic hypertrophy in men. Consider strenuous exercise or poststreptococcal infection (in younger patients), as well as trauma, malignant hypertension, eroding abdominal aortic aneurysm, coagulopathy, foreign body, Henoch–Schonlein purpura, pulmonary-renal syndromes, sickle cell disease complications, and renal vein thrombosis. Hematuria in a patient taking oral anticoagulants should not be attributed to the anticoagulant

53_Cydulka_Ch53.indd 301 01/05/17 11:38 am

Source page 321

alone, as the incidence of underlying disease is up to 80% in patients referred to urology. Imaging in the ED for hematuria should be limited to patients with trauma, symptoms of nephrolithiasis, or high concern for abdominal aortic aneurysm.

Emergency Department Care and Disposition

Treatment of hematuria is directed at the cause. ED management consists of the minimization of complications and appropriate referral or admission for further evaluation. All patients with hematuria should be promptly followed up by either primary care or urology.

Admit patients with infection associated with an obstructive stone, intractable pain or vomiting, hemodynamic instability, newly diagnosed glomerulonephritis, significant anemia, renal insufficiency, bladder outlet obstruction, preeclampsia, or any other potentially life-threatening cause of hematuria.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 91, "Urinary Tract Infections and Hematuria," by Kim Askew.

53_Cydulka_Ch53.indd 302 01/05/17 11:38 am

Source page 322

C h a p ter Acute Urinary Retention

54 Casey Glass

■■ CLINICAL FEATURES

Urinary retention can be either acute or chronic. The most common cause of retention is outlet obstruction secondary to benign prostatic hyperplasia (BPH) in men, although medication use, acute neurologic dysfunction, urinary tract bleeding or calculi, and other anatomic obstruction are also common causes in both men and women. Acute syndromes typically present with rapid onset of lower abdominal pain occasionally radiating to the lower back. Patients typically complain of difficulty voiding but some may not volunteer this information. There is a 20% chance of a recurrence in the following 6 months after an episode of acute obstruction. Chronic obstruction usually presents with lower abdominal discomfort and the patient may note incomplete voiding or the need to void frequently. Overflow incontinence is often present.

The history should address previous episodes of obstruction, recent medication changes, and over-the-counter medicine use. Assess for any history of trauma or neurologic disability or symptoms of infection. It is critical to know if any recent urologic procedures or urinary catheterizations have been performed. The duration of symptoms is also important as it is associated with the development of postobstructive diuresis and renal dysfunction.

■■ DIAGNOSIS AND DIFFERENTIAL

Causes of urinary retention may include anatomic obstruction, neurologic dysfunction, medication side effect, trauma to the genitourinary tract, infection, and psychological stress. Fever may indicate infection as either the cause or as a result of urinary outlet obstruction. Tachycardia and hypotension may resolve after the obstruction is relieved. The physical examination should address the functional and anatomic assessment of the lower urinary tract. Palpate the abdomen for a suprapubic mass corresponding to the distended urinary bladder. The penis should be examined for stricture at the meatus or palpable abnormalities of the penile urethra. In men the prostate should be assessed for size, texture, and tenderness. The female lower urinary tract should be evaluated for bladder prolapse or stricture of the urethral meatus. A pelvic exam is generally indicated to screen for infection or

mass as a cause of obstruction. A comprehensive neurologic examination should be performed in all patients and perineal sensation and anal sphincter tone should be documented.

Bedside ultrasound can be very helpful both in distinguishing the degree of obstruction and in discriminating obstruction from the sensation of fullness associated with bladder spasm in conditions such as inflammatory or infectious cystitis. The patient should first be encouraged to attempt to void. After a voiding attempt the bladder is imaged with a low-frequency sector

54_Cydulka_Ch54.indd 303 16/03/17 2:29 PM

Source page 323

Figure 54-1. Transverse and sagittal views of the urinary bladder. The prostate is visualized as a medium echogenicity structure posterior and caudal to the bladder. Anterior-posterior, cradio-caudal, and transverse measurements of the bladder are obtained for calculation of the bladder volume. Reproduced with permission from Casey Glass, MD.

format probe in both the transverse and sagittal views (Figure 54-1). Many manufacturers have a calculation package available to estimate the retained urine volume. Residual volumes >50 to 150 cc are consistent with urinary retention; however, volumes in the setting of retention are typically greater than 300 cc.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

The goals of emergency department care are relieving the discomfort of retention, assessing for any secondary injury to the renal system, and treatment of the primary cause of retention.

1. Most patients with bladder outlet obstruction are in distress, and passage of a urethral catheter alleviates their pain and urinary retention. Copious intraurethral lubrication including a topical anesthetic (2% lidocaine jelly) should be used. A 16-French Coudé catheter is recommended if straight catheters fail. The catheter should be passed to its fullest extent to obtain free urine flow before inflating the balloon. A catheter should not be placed if there is suspicion of trauma to the urethra, either secondary to a traumatic event or recent instrumentation. In most cases a catheter should be left indwelling and connected to a leg drainage bag. For cases of postanesthesia-related retention it is appropriate to remove the catheter after bladder drainage and then attempt a trial of spontaneous voiding.

54_Cydulka_Ch54.indd 304 16/03/17 2:29 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 320–323.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 54: Acute Urinary Retention

Source pages 324–325 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 324–325

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Acute Urinary Retention**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- The patient with obstruction from hematuria represents a special case. [Source pages 324–325]

- A 3-port Foley catheter should be placed and the bladder irrigated until returning fluid is free of blood. [Source pages 324–325]

- These patients are likely to need admis- sion for continued irrigation as the catheter often becomes blocked with clot following placement. [Source pages 324–325]

- Failure to pass a urethral Foley catheter, or recent urologic procedure or instrumentation requires the involvement of a urologist for catheter placement and consultation should not be delayed. [Source pages 324–325]

- Urgent urologic con- sultation is also indicated for obstruction secondary to stricture, prosta- titis, or urethral trauma. [Source pages 324–325]

Clinical reasoning structure

- **Problem representation:** Acute Urinary Retention; source pages 324–325.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Acute Urinary Retention**.

- Source-page range: **324–325**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 324

CHAPTER 54: Acute Urinary Retention 305

2. The patient with obstruction from hematuria represents a special case. A 3-port Foley catheter should be placed and the bladder irrigated until returning fluid is free of blood. These patients are likely to need admission for continued irrigation as the catheter often becomes blocked with clot following placement.

3. Failure to pass a urethral Foley catheter, or recent urologic procedure or instrumentation requires the involvement of a urologist for catheter placement and consultation should not be delayed. Urgent urologic consultation is also indicated for obstruction secondary to stricture, prostatitis, or urethral trauma. Physicians familiar with the procedure may choose to place a suprapubic urinary catheter.

4. Send urine for routine analysis as well as culture. Assess electrolytes, blood urea nitrogen (BUN), and creatinine for postobstructive renal failure. Urine output should be monitored quantitatively.

5. Oxybutynin can be prescribed for control of bladder spasms. This anticholinergic medicine can itself cause a functional obstruction. Patients may also require pain medication for control of discomfort from bladder spasms. Treat male patients in whom BPH is the suspected etiology of their obstruction with an α-blocker.

6. Antibiotics are not indicated unless there is evidence of cystitis or other infection.

7. If urinary retention has been chronic, postobstructive diuresis may occur even in the presence of normal blood urea nitrogen and creatinine levels. In such patients, closely monitor urinary output, for 4 to 6 hours after catheterization. Hourly output of greater than 200 mL for more than 4 to 6 hours is an indication for admission and fluid replacement.

8. Address precipitating causes of retention. Discontinue offending medications. Infectious or neurologic causes must be completely evaluated; urgency of workup depends on patient acuity and comorbidities.

9. In all cases of urinary retention, urologic follow-up in 3 to 7 days is indicated for a complete genitourinary evaluation. Patients should usually expect to have the catheter removed at that visit.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 92, "Acute Urinary Retention," by David Hung-Tsang Yen and Chen-Hsen Lee.

54_Cydulka_Ch54.indd 305 16/03/17 2:29 PM

Source page 325

C h a p ter Male Genital Problems

55 Gavin R. Budhram

■■ TESTICULAR TORSION

Testicular torsion results from abnormal fixation of the testis within the tunica vaginalis, allowing the testis to twist. It exhibits a bimodal age distribution, peaking in the perinatal period and during puberty but may occur at any age. Although sometimes associated with trauma, torsion usually occurs in the absence of any preceding event.

Clinical Features

Because of the potential for infarction and infertility, testicular torsion must be the primary consideration in any male complaining of testicular pain. Pain usually occurs suddenly, is severe, and is felt in the lower abdominal quadrant, the inguinal canal, or the testis. The pain may be constant or intermittent but is not positional because torsion is primarily an ischemic event. Although symptom onset tends to occur after exertion, the testicle also may twist from unilateral cremasteric muscle contraction during sleep. Early in presentation, the affected testicle is firm, tender, elevated, and in a transverse lie compared to the contralateral testicle. The unilateral absence of the cremasteric reflex is a sensitive but nonspecific finding.

Diagnosis and Differential

Color-flow duplex ultrasound is the most commonly used confirmatory study, but sensitivity ranges from 69% to 90%. In addition, urinalysis is typically ordered, but pyuria does not rule out testicular torsion.

Torsion of the appendages is more common than testicular torsion but is not dangerous because the appendix testis and appendix epididymis have no known function. The diagnosis is supported by pain that is most intense near the head of the epididymis or testis, an isolated tender nodule, or the pathognomonic blue dot appearance of a cyanotic appendage with illumination through thin prepubertal scrotal skin. If normal intratesticular blood flow can be demonstrated with color Doppler, surgical exploration is not necessary because most appendages calcify or degenerate over 10 to

14 days and cause no harm. The differential for testicular torsion also includes epididymitis, inguinal hernia, hydrocele, and scrotal hematoma.

Emergency Department Care and Disposition

1. ■When the diagnosis of testicular torsion is obvious, immediate urologic consultation is indicated for exploration because imaging tests can be too time consuming. Testicular salvage rates are excellent with surgical detorsion within 6 hours of symptom onset, but decline rapidly thereafter.

2. ■The emergency physician can attempt manual detorsion. Most testes twist in a lateral to medial direction, so detorsion is performed in a

55_Cydulka_Ch55.indd 306 26/04/17 12:15 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 324–325.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 55: Male Genital Problems

Source pages 326–331 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 326–331

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Male Genital Problems**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 55: Male Genital Problems 307 medial to lateral direction, similar to the opening of a book. [Source pages 326–331]

- The endpoint for successful detorsion is pain relief; urologic referral is still indicated. [Source pages 326–331]

- ■Urology should be consulted early in the patient’s course even if confir- matory testing is planned. [Source pages 326–331]

- When the diagnosis of testicular torsion cannot be ruled out by diagnostic studies or examination, urologic consultation is still indicated if suspicion is high. [Source pages 326–331]

- ■■ EPIDIDYMITIS AND ORCHITIS Clinical Features Epididymitis is characterized by gradual onset of pain due to inflammation. [Source pages 326–331]

Clinical reasoning structure

- **Problem representation:** Male Genital Problems; source pages 326–331.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Male Genital Problems**.

- Source-page range: **326–331**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 326

CHAPTER 55: Male Genital Problems 307

medial to lateral direction, similar to the opening of a book. The endpoint for successful detorsion is pain relief; urologic referral is still indicated.

3. ■Urology should be consulted early in the patient's course even if confirmatory testing is planned. When the diagnosis of testicular torsion cannot be ruled out by diagnostic studies or examination, urologic consultation is still indicated if suspicion is high.

■■ EPIDIDYMITIS AND ORCHITIS

Clinical Features

Epididymitis is characterized by gradual onset of pain due to inflammation. Bacterial infection is the most common cause, with infecting agents dependent on the patient's age. In patients <35 years, epididymitis is due primarily to sexually transmitted diseases; culture or DNA probe for gonococcus and Chlamydia is indicated in males <35 years even in the absence of urethral discharge. Common urinary pathogens such as Escherichia coli and Klebsiella predominate in older men. Epididymitis causes lower abdominal, inguinal canal, scrotal, or testicular pain alone or in combination. Due to the inflammatory nature of the pain, patients with epididymitis may note transient pain relief when elevating the scrotal contents while recumbent (positive Prehn's sign).

Diagnosis and Differential

Initially, when tenderness is well localized to the epididymis, the clinical diagnosis is clear. However, progression of inflammation results in the physical examination finding of a single, large testicular mass (epididymo-orchitis), which is difficult to differentiate from testicular torsion or carcinoma. Testicular malignancy should be suspected in patients presenting with asymptomatic testicular mass, firmness, or induration. Ten percent of tumors present with pain due to hemorrhage within the tumor. Orchitis in isolation is rare; it usually occurs in conjunction with other systemic infections, such as mumps or other viral illnesses.

Emergency Department Care and Disposition

1. ■If the patient appears toxic, admission is indicated for intravenous antibiotics (e.g., ceftriaxone 1 to 2 g every 12 hours IV or trimethoprim/sulfamethoxazole 5 mg/kg IV of trimethoprim component every 6 hours).
2. ■Outpatient treatment is the norm in patients who do not appear toxic; urologic follow-up within 1 week is indicated. Age <35: Treat gonorrhea and Chlamydia with ceftriaxone 250 mg IM single dose plus doxycycline 100 mg PO twice daily for 10 days. Age >35: Treat gram-negative bacilli with levofloxacin 500 mg PO daily for 10 days or ofloxacin 300 mg PO twice daily for 10 days.
3. ■In addition, scrotal elevation, ice application, nonsteroidal anti-inflammatory drugs, opioids for analgesia, and stool softeners are indicated.
4. ■Orchitis is treated with disease-specific therapy, symptomatic support, and urologic follow-up.

55_Cydulka_Ch55.indd 307 16/03/17 2:34 PM

Source page 327

308 SECTION 7: Renal and Genitourinary Disorders

■■ ACUTE PROSTATITIS

Acute prostatitis is a bacterial inflammation of the prostate presenting with varying complaints of suprapubic or genital pain, back pain, perineal pain, voiding difficulties, frequency, dysuria, pain with ejaculation, and fever and chills. Patients at risk include those with anatomic or neurophysiologic lower urinary tract obstruction, acute epididymitis or urethritis, unprotected rectal intercourse, phimosis, and an indwelling urethral catheter. The causative organism is E. coli in most cases, with Pseudomonas, Klebsiella, Enterobacter, Serratia, or Staphylococcus causing the remainder. Physical examination usually reveals perineal, rectal, and prostate tenderness. The diagnosis is clinical as urinalysis and culture are often negative even after prostate massage. Treatment is ciprofloxacin 500 mg twice daily, or levofloxacin 500 mg PO daily, or ofloxacin 300 mg PO twice daily. All treatments should be for 2 to 4 weeks. Pain medicine may be required. Admission is not necessary unless the patient is septic, immunocompromised, has significant comorbidities, or has worsened on outpatient therapy. In this case, IV treatment with piperacillin-tazobactam 3.375 to 4.5 g or cefotaxime with an aminoglycoside may be initiated. Consider testing and treatment for gonorrhea and chlamydia in patients with risk factors for sexually transmitted disease.

■■ SCROTUM

Scrotal abscesses may be localized to the scrotal wall or may arise from

extensions of infections of intrascrotal contents (i.e., testis, epididymis, and bulbous urethra). A simple hair follicle scrotal wall abscess can be managed by incision and drainage; no antibiotics are required in immunocompetent patients. When a scrotal abscess is suspected to arise from an intrascrotal infection, ultrasound may demonstrate pathology in the testis and/or epididymis. Retrograde urethrography may be needed to identify a source in the urethra. Definitive care of any complex abscess calls for a urology consultation.

Fournier gangrene is a polymicrobial infection of the perineal subcutaneous tissues. Diabetic males are at highest risk, but any immunocompromised state can be associated with the disease. Prompt diagnosis is essential to prevent extensive tissue loss. Early surgical consultation is recommended for at-risk patients who present with scrotal, rectal, or genital pain. Treatment mainstays include aggressive fluid resuscitation with normal saline solution and broad-spectrum antibiotics to cover gram-positive, gram-negative, and anaerobic organisms: imipenem 1 g IV every 8 hours or meropenem 500 mg to 1 g IV every 8 hours plus vancomycin 1 g IV every 12 hours if methicillin-resistant Staphylococcus aureus is suspected. Care usually includes hyperbaric oxygen therapy (if readily available) and surgical debridement.

■■ PENIS

Balanoposthitis is inflammation of the glans (balanitis) and foreskin (posthitis). Upon foreskin retraction, the glans and prepuce appear purulent, excoriated, malodorous, and tender. Treatment consists of cleaning with mild soap, ensuring adequate dryness, applying antifungal creams (nystatin 100,000 U/g, four times daily or clotrimazole 1% cream bid), and using an

55_Cydulka_Ch55.indd 308 16/03/17 2:34 PM

Source page 328

Phimosis

Paraphimosis

Figure 55-1. Phimosis and paraphimosis.

oral azole (fluconazole 150 mg single dose, reevaluate at 7 days for repeat dosing). Urologic referral is needed for reassessment and possible circumcision in recurrent cases. An oral cephalosporin (e.g., cephalexin 500 mg, times daily) should be prescribed in cases of secondary bacterial infection. Balanoposthitis can be the sole presenting sign of diabetes.

Phimosis is the inability to retract the foreskin proximally (Fig. 55-1). Physiologic phimosis is common in boys but less than 10% of foreskins remain nonretractile at age 3 with nearly all resolving by adolescence. Pathologic phimosis occurs as a result of infection, poor hygiene, or previ-

ous injury with scarring. Hemostatic dilation of the preputial ostium relieves urinary retention until definitive dorsal slit or circumcision can be performed. Topical steroid therapy, such as hydrocortisone 1% cream for 4 to 6 weeks, along with daily preputial retraction, reduces the rate of required circumcision.

55_Cydulka_Ch55.indd 309 16/03/17 2:34 PM

Source page 329

310 SECTION 7: Renal and Genitourinary Disorders

Paraphimosis is the inability to reduce the proximal edematous foreskin distally over the glans (see Fig. 55-1). Paraphimosis is a true urologic emergency because the resulting glans edema and venous engorgement can progress to arterial compromise and gangrene. If surrounding tissue edema can be successfully compressed, as by wrapping the glans with 2 × 2-in. elastic bandages for 5 minutes, the foreskin may be reduced. Making several puncture wounds with a small (22- to 25-gauge) needle may help with expression of glans edema fluid. Local anesthetic block of the penis is helpful if patients cannot tolerate the discomfort associated with edema compression and removal. If arterial compromise is suspected or has occurred, local infiltration of the constricting band with 1% plain lidocaine followed by superficial vertical incision of the band will decompress the glans and allow foreskin reduction.

Penile entrapment injuries occur when various objects are wrapped around the penis potentially occluding vascular supply. Objects may be removed by a variety of techniques including compression, cutting, cooling the penis, and possibly even urologic surgical removal. Retrograde urethrogram and Doppler studies may be employed if indicated to assess urethral injury and distal penile arterial blood supply, respectively.

Penile fracture occurs when there is an acute tear of the penile tunica albuginea. The penis is acutely swollen, discolored, and tender in a patient with history of intercourse-associated trauma accompanied by a snapping sound and sudden loss of erection. Retrograde urethrography may be indicated for assurance of urethral integrity. Urologic consultation is indicated.

Peyronie disease produces progressive penile deformity, typically curvature with erections, that is painful and may result in erectile dysfunction or preclude successful vaginal intercourse. Examination shows a thickened plaque on the dorsal penile shaft. Assurance and urologic follow-up are indicated.

Priapism is a painful pathologic erection that may be associated with urinary retention. Infection and impotence are other complications. Corporal aspiration and instillation of an α-adrenergic antagonist (i.e., phenyleph-

rine) is the next step and may need to be performed by the emergency physician when urologic consultation is not available. Even when emergency physicians provide stabilizing care, urologic consultation is indicated in all cases.

■■ URETHRA

Urethritis is characterized by purulent urethral discharge. Diagnosis is clinical but is often confirmed with a first-void urine specimen. Causative agents are usually Chlamydia trachomatis or Neisseria gonorrhoeae, though Trichomonas, Ureaplasma urealyticum, and herpes simplex virus may also occasionally cause urethritis. Treatment is ceftriaxone 250 mg IM and azithromycin 1 g PO.

Urethral stricture is becoming more common due to the high incidence of sexually transmitted diseases and may result in urinary retention. A normal post-void residual volume should be less than 100 mL and can be estimated with bedside ultrasonography. If a patient's bladder cannot be cannulated, the differential diagnosis includes urethral stricture, voluntary external sphincter spasm, bladder-neck contracture, or benign prostatic

55_Cydulka_Ch55.indd 310 16/03/17 2:34 PM

References/provenance

- Source: *Tintinalli's Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 326–331.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 56: Urologic Stone Disease

Source pages 332–335 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 332–335

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Urologic Stone Disease**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 56: Urologic Stone Disease 313 the management if the stone fails to pass spontaneously. [Source pages 332–335]

- Noncontrast helical computed tomography (CT) has the overall highest sensitivity and specific- ity. [Source pages 332–335]

- Positive findings include changes in the ureteral caliber, suspicious calcifications, stranding of perinephric fat, and dilation of the collecting system. [Source pages 332–335]

- It can be performed quickly and can rule in or rule out other diagnoses. [Source pages 332–335]

- The disadvantage of CT is radiation exposure; therefore, many facilities have adopted a low-dose CT protocol. [Source pages 332–335]

Clinical reasoning structure

- **Problem representation:** Urologic Stone Disease; source pages 332–335.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Urologic Stone Disease**.

- Source-page range: **332–335**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 332

CHAPTER 56: Urologic Stone Disease 313

the management if the stone fails to pass spontaneously. Noncontrast helical computed tomography (CT) has the overall highest sensitivity and specificity. Positive findings include changes in the ureteral caliber, suspicious calcifications, stranding of perinephric fat, and dilation of the collecting system. It has a PPV of 96% and a NPV of 93% to 97%. It can be performed quickly and can rule in or rule out other diagnoses. The disadvantage of CT is radiation exposure; therefore, many facilities have adopted a low-dose CT protocol. It has similar sensitivities as regular dose radiation CT for stones >3 mm and individuals with a BMI >30 (Table 56-1).

The use of plain kidney-ureter-bladder film (KUB) is limited as only 90% of ureteral stones are radiopaque. A KUB cannot rule in or rule out a ureteral stone. The KUB is useful in following the progression of a stone, once visualized on CT, in the outpatient setting.

The IV pyelogram (IVP) is rarely used anymore in the ED. It provides information on renal function and anatomy and is used as an adjunct to CT.

Ultrasound (US), an anatomic rather than a functional test, is useful in patients for whom CT is not advised such as pregnant women and children. It detects hydronephrosis and larger stones but is not sensitive for midureteral or small, less than 5-mm stones. Sensitivity increases to 90% for stones >6 mm. A false-positive diagnosis of hydronephrosis may be related to a full bladder, renal cysts, and anatomic variation. A rapid bolus of crystalloid can result in a false-positive diagnosis of hydroureter. Bedside US by ED physicians may also decrease length of stay for selected patients.

Table 56-1 Ancillary Tests in Urologic Stone Disease

Test	Sensitivity (%)	Specificity (%)	LR+	LR–	Comments
Noncontrast CT	94–97	96–99	24–∞	0.02–0.04	Advantages: speed, no RCM, detects other diagnoses Disadvantages: radiation, no evaluation of renal function

IV urogram 64–90 94–100 15–∞ 0.11– Advantage: evaluates renal
0.15 function
Disadvantage: RCM (allergy,
nephrotoxicity, metformin
related acidosis)
Ultrasound 63–85 79–100 10–∞ 0.10– Advantages: safe in preg-
0.34 nancy, no RCM, no radia-
tion, no known side effects
Disadvantages: insensitive
in middle third of the ureter,
may miss smaller stones
(<5 mm)
Plain 29–58 69–74 1.9–2.0 0.58– Advantage: may be used to
abdominal 0.64 follow stones
radiograph Disadvantage: poor sensi-
tivity and specificity
Abbreviations: LR = likelihood ratio; RCM = radiocontrast media.

56_Cydulka_Ch56.indd 313 01/05/17 10:30 am

Source page 333

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

Medications

1. ■Analgesia: Nonsteroidal anti-inflammatory drugs administered parenterally are considered the analgesics of choice. The prostaglandins promote dilation of the ureter and thus aid in the alleviation of the source of the pain. Ketorolac 30 mg IVP is the recommended starting dose. Caution should be used in the elderly, those with bleeding tendencies, and those with renal impairment. Opioids, in titrated doses such as morphine 5 mg IV or hydromorphone 1 mg IV, may also be used to assist in pain control. Care must be used when utilizing these medications in senior, pediatric, and pregnant populations.
2. ■Intravenous crystalloid: It is recommended to correct any volume deficits from vomiting or decreased oral intake. Studies have not shown it to be effective in pain control or stone passage.
3. ■Antiemetics: Metoclopramide, 10 mg IV, the only antiemetic drug specifically studied for the treatment of renal colic, has been shown to be effective in providing relief from nausea and pain. Other antiemetics may also be used.
4. ■Medical expulsion therapy: The use of α-blockers results in a 2- to 6-day improvement in rate and time to expulsion and decreased pain. Options

include tamsulosin 0.4 mg daily up to 4 weeks, terazosin 5 to 10 mg daily, and doxazosin 4 mg daily. Steroids and calcium channel blockers are not recommended.

5. ■Antibiotics: If the urinalysis shows evidence for an infection, the choice of antibiotic is dictated by the local resistance patterns to gram-negative rods. Patients with an infected ureteral stone without significant obstruction, fever, or systemic illness may be discharged with a fluoroquinolone such as ciprofloxacin 500 mg twice a day for 10 to 14 days or levofloxacin 500 mg daily for 10 to 14 days, a third-generation cephalosporin such as cefpodoxime 200 mg twice a day for 10 to 14 days or other drugs with good local sensitivities. Close follow-up is essential. Patients with a more complicated clinical situation should be admitted. Intravenous antibiotic selections to consider include gentamicin or tobramycin 1 mg/kg/dose every 8 hours and ampicillin 1 to 2 g every 4 hours; piperacillin-tazobactam 3.375 g every 6 hours; cefepime 2 g every 8 hours; ticarcillin-clavulanic acid 3.1 g every 6 hours, or ciprofloxacin 400 mg every 12 hours.

Disposition

If the stone is passed in the ED, no treatment is necessary other than elective urologic follow-up. Patients with small unilateral stones less than 5 mm, no complicated infection, and pain controlled by oral analgesics may be discharged with a urine strainer, prescriptions for oral analgesics and medical expulsive therapy, and urologic or primary care physician follow-up within 7 days. The rate of stone passage depends on its size and location. A stone measuring 5–6 mm in diameter takes 7 to 30 days to pass. Instruct patients to return if they develop fever, persistent vomiting, or uncontrolled pain.

56_Cydulka_Ch56.indd 314 01/05/17 10:30 am

Source page 334

Table 56-2 Indications for Admission in Patients with Nephrolithiasis

Absolute Indications for Admission	Relative Indications for Admission
Intractable pain or vomiting	Fever
Urosepsis*	Solitary kidney or transplanted kidney without obstruction
Single or transplanted kidney with obstruction*	Obstructing stone with signs of urinary infection
Acute renal failure	Urinary extravasation
Hypercalcemic crisis	Significant medical comorbidities
Severe medical comorbidities/advanced age	Stone unlikely to pass—large stone above

the pelvic brim

*Indication for urgent decompression.

Urologic consultation on an emergent basis is needed in patients with a complete obstruction complicated by fever and/or urosepsis and in patients with a solitary or transplanted kidney. Discuss disposition with a urologist in patients with a stone larger than 5 mm, renal insufficiency, severe underlying disease, extravasation or complete obstruction, sloughed renal papillae, UTI, or failed outpatient management.

Hospitalization is indicated for patients who have a solitary or transplanted kidney with obstruction, uncontrolled severe pain, intractable emesis, acute renal failure, hypercalcemic crisis, severe medical comorbidities, and urosepsis. Consider admission for patients with a fever, solitary or transplanted kidney without obstruction, obstructing stone with infection, urinary extravasation, significant medical comorbitidies, or large proximal ureteral stone. (Table 56-2).

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 94, "Urologic Stone Disease," by David E. Manthey and Bret A. Nicks.

56_Cydulka_Ch56.indd 315 01/05/17 10:30 am

Source page 335

C hapter Complications of Urologic

57 Procedures and Devices

Steven Go

■■ COMPLICATIONS OF URINARY CATHETERS

Complications related to the use of urinary catheters are not infrequent and therefore the catheters should be used only when absolutely necessary. Silicone catheters are available for patients with latex allergies.

Catheter-Associated Urinary Tract Infection

Catheter-associated urinary tract infections (CA-UTIs) are a very common cause of nosocomial infections. Signs and symptoms of CA-UTI include fever, rigors, altered mental status, malaise, or lethargy, flank pain, costovertebral angle tenderness, acute hematuria, and pelvic discomfort. In those whose catheters have been removed, dysuria, urgent or frequent urination, or suprapubic pain or tenderness may indicate CA-UTI. Do not treat asymptomatic bacteriuria in a patient with a short-term catheterization, except in pregnancy or immediately posturologic procedure. CA-UTI in spinal cord injury patients may also present with increased spasticity, autonomic dysreflexia, and a sense of unease. Pyelonephritis is the most common complication of CA-UTI and should be suspected when fever

is present. Other related infections include prostatitis, epididymitis, and scrotal abscess.

Diagnosis of CA-UTI in the ED is made clinically with urine microscopy. Send urine for culture. Add blood cultures if the patient is septic or immunocompromised. An ultrasound may be useful to identify urinary obstruction. Remember that pyuria is universal for patients with long-term (>1 month) indwelling catheters and in the absence of symptoms, pyuria should not be used in the diagnosis of infection. Hematuria is a better indicator of infection. Remove the urinary catheter if clinically feasible or replace the catheter if it is >7 days old. In patients with mild symptoms, empirically treat with ciprofloxacin 500 mg twice a day, levofloxacin 500 mg once a day, or cefpodoxime 200 mg twice a day. Tailor specific antibiotic choice to local bacterial sensitivities. Seven days is the recommended duration of antimicrobial treatment for patients with CA-UTI who have prompt resolution of symptoms, and 10 to 14 days of treatment is recommended for those with a delayed response. In patients with catheter-associated pyelonephritis, admission is frequently required. Check urine cultures and blood cultures if there is concern for sepsis.

Nondeflation of Foley Retention Balloon

Nondeflation of the Foley retention balloon prevents the removal of the catheter. Cut the plastic catheter valve just proximal to the inflation port and insert a flexible guide wire to expand the channel and deflate the balloon. If this is not successful, insert a 22-G central venous catheter (CVC) over the guide wire. Once the tip of the catheter enters the balloon, remove the

57_Cydulka_Ch57.indd 316 16/03/17 2:47 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 332–335.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 57: Complications of Urologic Procedures and Devices

Source pages 336–339 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 336–339

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Complications of Urologic Procedures and Devices**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 57: Complications of Urologic Procedures and Devices 317 guide wire and deflate the balloon. [Source pages 336–339]

- If the inflated balloon persists, instill 10cc of mineral oil through the CVC and wait 15 minutes to dissolve the balloon (this step may be repeated). [Source pages 336–339]

- If these methods fail, consult a urolo- gist to remove the catheter. [Source pages 336–339]

- ■■ COMPLICATIONS OF PERCUTANEOUS NEPHROSTOMY Percutaneous nephrostomy is a urinary drainage procedure used for supra- vesical or ureteral obstruction. [Source pages 336–339]

- Check hemoglobin, hematocrit, and renal function studies. [Source pages 336–339]

Clinical reasoning structure

- **Problem representation:** Complications of Urologic Procedures and Devices; source pages 336–339.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Complications of Urologic Procedures and Devices**.

- Source-page range: **336–339**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 336

CHAPTER 57: Complications of Urologic Procedures and Devices 317

guide wire and deflate the balloon. If the inflated balloon persists, instill 10cc of mineral oil through the CVC and wait 15 minutes to dissolve the balloon (this step may be repeated). If these methods fail, consult a urologist to remove the catheter.

■■ COMPLICATIONS OF PERCUTANEOUS NEPHROSTOMY

Percutaneous nephrostomy is a urinary drainage procedure used for supravesical or ureteral obstruction. Bleeding may occur. Check hemoglobin, hematocrit, and renal function studies. Check platelet count and coagulation studies if a coagulopathy is suspected. Treat mild bleeding by irrigating the nephrostomy tube to clear the blood clots. If bleeding fails to resolve, a vascular injury may be present and emergent urology consult is indicated. Initiate fluid resuscitation and blood transfusion in severe cases.

Infectious complications from nephrostomy tubes include bacteriuria, pyelonephritis, renal abscess, and urosepsis. Culture the urine and wound drainage (if any) and administer antibiotics after consulting a urologist.

Catheter dislodgment, tube blockage, and residual stone fragments can occur. Obtain a CT scan and consult urology to treat these mechanical issues.

■■ LITHOTRIPSY

Common post-lithotripsy complications include abdominal and flank pain, gross hematuria, skin ecchymosis, and ureteral obstruction from stone fragments. However, ruptured or perforated organs, infections (pyelonephritis, urosepsis, psoas abscess), vascular injury, and pancreatitis may also occur.

The presence of hypotension, syncope, flank hematoma, drop in hemoglobin, and severe pain are all harbingers of serious complications (e.g., perinephric and renal hematomas). Administer supportive therapy with IV fluids, antiemetics, and analgesics. Check blood counts, creatinine, and urinalysis. Obtain cultures and administer antibiotics if indicated. If simple

obstruction is suspected, obtain a renal ultrasound. Obtain a CT scan if a more severe injury is suspected. Consult urology early in the process as surgical intervention may be indicated in these cases.

■■ COMPLICATIONS OF URETERAL STENTS

A range of infectious complications can occur from ureteral stents. Therefore, order a urinalysis and culture for urinary symptoms, incontinence, flank pain, or change in the baseline discomfort. Treat simple infections with outpatient antibiotics in consultation with a urologist, as stent removal is not always necessary. However, if pyelonephritis or urosepsis is suspected, obtain appropriate cultures, administer IV antibiotics, order imaging (abdominal series, ultrasound, and CT) to verify stent placement, and consult urology.

Late complications with long-term stent placement may be seen. New-onset abdominal pain, severe flank pain, fever, irritative bladder symptoms, or gross hematuria may be indicative of stent migration, obstruction, or infection. Obtain a urinalysis to evaluate for infection, imaging to evaluate stent position, and urologic consultation. However, new anemia without

57_Cydulka_Ch57.indd 317 16/03/17 2:47 PM

Source page 337

hematuria may be indicative of a retroperitoneal hematoma and is diagnosed with CT. Assume vascular fistulization from an eroding stent in the presence of severe gross hematuria, syncope, or hypotension. Resuscitate these patients, order lab studies, irrigate the bladder, and transfuse appropriately. Obtain urology consultation immediately.

■■ COMPLICATIONS OF ARTIFICIAL URINARY SPHINCTERS

The artificial sphincter is a device used for urinary incontinence from various causes. Skin flora cause early infections, while later infections are usually due to gram-negative pathogens. Never introduce a urinary drainage catheter through an artificial urinary sphincter. Obtain urinalysis, appropriate labs, cultures, and imaging, and administer IV antibiotics as indicated. Consult a urologist for further evaluation and management.

■■ COMPLICATIONS OF ERECTILE DYSFUNCTION DEVICES

Injections for erectile dysfunction can result in priapism, which requires emergent urologic consultation. The most common complication of vacuum devices is pain from improper use, but skin necrosis and infections can occur. Infection of implanted penile prostheses (both early and late), erosion of the prosthesis into the penile tissue, and prosthesis migration can occur. Administer broad-spectrum IV antibiotics if infection is present

and consult a urologist for device removal.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 95, "Complications of Urologic Procedures and Devices," by Elaine B. Josephson.

57_Cydulka_Ch57.indd 318 16/03/17 2:47 PM

Source page 338

SECTION

Gynecology and Obstetrics

8

Chapter Vaginal Bleeding and Pelvic Pain

58 in the Nonpregnant Patient

Joelle Borhart

■■ VAGINAL BLEEDING

Abnormal vaginal bleeding is a common complaint in females presenting to the ED. Determination of pregnancy status in patients of reproductive age is critical to formulate the appropriate differential diagnosis and to guide subsequent testing and decision making.

Clinical Features

All patients should be asked about the amount and duration of bleeding. Reproductive and sexual history, history of sexually transmitted infections, trauma, medications, and the possibility of foreign bodies should be elicited. Symptoms of a possible bleeding disorder (history of bruising, epistaxis, other abnormal bleeding), endocrine disorder, or liver disease should be noted. A complete abdominal and pelvic examination, including speculum and bimanual exam, should be performed on nonvirginal patients to look for structural or traumatic causes of bleeding. Skin or conjunctiva pallor, abnormal vital signs, or a report of dizziness, syncope, or weakness can indicate significant blood loss.

Diagnosis and Differential

In prepubertal girls, causes of vaginal bleeding include genital trauma and/or sexual abuse, vaginitis, tumors, and foreign bodies. Bleeding coupled with vaginal discharge raises concerns for retained foreign bodies. Up to 20% of adolescents with abnormal uterine bleeding may have a primary coagulation disorder such as von Willebrand disease. Anovulatory cycles are also common during the teenage years. In women of reproductive age and perimenopausal women, bleeding can arise from the uterus or cervix and is most commonly due to anovulation, pregnancy, exogenous hormone use, coagulopathy, uterine leiomyomas, cervical and endometrial polyps, pelvic infections, and thyroid dysfunction. In postmenopausal women, the

most common causes of vaginal bleeding are exogenous estrogens, atrophic vaginitis, endometrial lesions including cancer, and other tumors.

319

58_Cydulka_Ch58.indd 319 16/03/17 3:06 PM

Source page 339

320 SECTION 8: Gynecology and Obstetrics

The new term abnormal uterine bleeding (AUB) encompasses all causes of abnormal bleeding in nonpregnant women and divides etiologies of AUB into structural and nonstructural causes. The use of the term dysfunctional uterine bleeding is no longer recommended. AUB may be ovulatory or anovulatory. Anovulatory cycles are common at the extremes of reproductive age. Patients with anovulatory cycles may present with prolonged menses, irregular cycles, or intermenstrual bleeding. Usually the bleeding is painless and minimal, but severe bleeding can occur, resulting in anemia and iron depletion.

A pregnancy test must be obtained on all women of reproductive age to rule out pregnancy as a cause of bleeding. Other laboratory evaluation is guided by the history and physical examination. A CBC should be checked if signs of excessive bleeding or anemia are present. A prolonged PT or elevated INR may identify a coagulopathy. Obtain thyroid function tests in patients with symptoms and signs of thyroid dysfunction. Ultrasonography is an important imaging modality to determine uterine size, characteristics of the endometrium, and to detect structural abnormalities. Ultrasound may be deferred for outpatient evaluation in stable, nonpregnant patients as the results will rarely change ED management.

Emergency Department Care and Disposition

Most patients with vaginal bleeding are hemodynamically stable and need no acute intervention. Patients who are unstable with persistent bleeding require aggressive resuscitation: IV crystalloids, blood products, and gynecologic consultation for urgent D and C.

1. ■For unstable patients with severe bleeding high-dose IV estrogen is considered first-line treatment. Give conjugated estrogen (e.g., Premarin®) 25 mg IV every 2 to 6 hours until bleeding slows followed by an oral contraceptive. These patients should be admitted to a gynecologic service for further management.

2. ■In stable patients short-term hormonal therapy may be prescribed to temporize an acute bleeding episode. Choices include the following:

a. Oral contraceptive regimen: ethinyl estradiol 35 μg and norethindrone 1 mg (e.g., Ortho-Novum 1/35®) three tablets daily for 7 days. Alternatively, a taper may be given: four tablets for 2 days, three

tablets for 2 days, two tablets for 2 days, and one tablet for 3 days.

b. Medroxyprogesterone 20 mg three times per day for 7 days or 10 mg per day for 10 days.

c. Tranexamic acid (e.g., Lysteda®), an antifibrinolytic agent, 600 to 1300 mg every 8 hours for 3 days. Withdrawal bleeding may be heavy and typically occurs 3 to 10 days after the hormonal therapy has stopped. Obtain gynecologic consultation if administration is considered.

3. ■If there is any concern for malignancy, hormonal therapy is best deferred until the patient is evaluated by gynecology and a decision about a biopsy is made.

4. ■Nonsteroidal anti-inlammatory drugs (NSAIDs), such as naproxen 500 mg twice daily PO or ibuprofen 400 mg every 6 hours PO, may reduce bleeding.

5. ■Stable patients may be discharged home and instructed to follow-up with their gynecologic provider.

58_Cydulka_Ch58.indd 320 16/03/17 3:06 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 336–339.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 58: Vaginal Bleeding and Pelvic Pain in the Nonpregnant Patient

Source pages 340–343 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 340–343

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Vaginal Bleeding and Pelvic Pain in the Nonpregnant Patient**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 58: Vaginal Bleeding and Pelvic Pain in the Nonpregnant Patient 321 ■■ PELVIC PAIN Pelvic pain generally arises from gynecologic pathology but referred pain from extrapelvic conditions, such as inflammatory bowel disease, urinary tract infections or stones, diverticulitis, leaking abdominal aneurysm, or appendicitis, need to be considered. [Source pages 340–343]

- Pregnancy should be excluded in all women of reproductive age. [Source pages 340–343]

- Pelvic inflammatory disease is a common cause of pelvic pain and is discussed in Chapter 64. [Source pages 340–343]

- Clinical Features Pelvic pain may be acute or chronic, intermittent or continuous. [Source pages 340–343]

- Attention to the characteristics of the pain will aid in determining etiologies. [Source pages 340–343]

Clinical reasoning structure

- **Problem representation:** Vaginal Bleeding and Pelvic Pain in the Nonpregnant Patient; source pages 340–343.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Vaginal Bleeding and Pelvic Pain in the Nonpregnant Patient**.

- Source-page range: **340–343**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 340

CHAPTER 58: Vaginal Bleeding and Pelvic Pain in the Nonpregnant Patient 321

■■ PELVIC PAIN

Pelvic pain generally arises from gynecologic pathology but referred pain from extrapelvic conditions, such as inflammatory bowel disease, urinary tract infections or stones, diverticulitis, leaking abdominal aneurysm, or appendicitis, need to be considered. Pregnancy should be excluded in all women of reproductive age. Pelvic inflammatory disease is a common cause of pelvic pain and is discussed in Chapter 64.

Clinical Features

Pelvic pain may be acute or chronic, intermittent or continuous. Attention to the characteristics of the pain will aid in determining etiologies. Sudden onset of unilateral pain suggests an ovarian cyst, ovarian torsion, obstruction, or renal lithiasis. Gradual onset suggests an infectious process or slowly enlarging mass. Other attributes, such as the relationship of the pain with the menstrual cycle, aggravating and relieving factors, and associated urinary, GI, and systemic symptoms assist in developing the differential diagnosis.

A complete abdominal and pelvic examination, including speculum and bimanual exams should be performed. A pregnancy test should be done to rule out pregnancy in women of reproductive age. Other testings, such as urinalysis, CBC, and ultrasound, are guided by the history and physical examination.

Diagnosis and Differential

Ovarian Cysts

Symptomatic ovarian cysts typically cause sudden-onset unilateral pain. Cysts can be complicated by bleeding from the cyst wall (hemorrhagic cyst) or by cyst rupture. A ruptured cyst can causes abnormal vital signs and an acute abdomen. Cysts that are >8 cm, multiloculated, or solid are concerning for malignancy, dermoid cysts, or endometriomas (“chocolate

cysts”). In women who are postmenopausal, an ovarian mass is malignant until proven otherwise. All patients with cysts should follow-up with their gynecologic provider for further evaluation.

Ovarian Torsion

Ovarian torsion results in the sudden onset of severe adnexal pain from ischemia of the ovary. Torsion is a surgical emergency and rapid intervention is necessary to preserve ovarian function. A history of acute onset of severe unilateral pain may be obtained, though atypical presentations are common. Pain may develop after exertion. Risk factors for torsion include pregnancy (enlarged corpus luteum), large ovarian cysts or tumors, and chemical induction of ovulation (ovarian hyperstimulation syndrome). Ultrasound with Doppler flow imaging is the diagnostic procedure of choice but is not 100% sensitive. Imaging early in the process may show congestion from venous outflow obstruction with preserved arterial flow and images obtained during a transient period of detorsion may appear normal. Analgesia, gynecologic consultation, and preparation for surgery are warranted if the diagnosis is suspected.

58_Cydulka_Ch58.indd 321 16/03/17 3:06 PM

Source page 341

Endometriosis

Endometriosis results from endometrium-like tissue implanted outside of the uterus causing chronic inflammation. Symptoms include recurrent pelvic pain associated with menstrual cycles, dyspareunia, and infertility. The definitive diagnosis is usually not made in the ED. Treatment consists of analgesics and gynecologic referral.

Leiomyomas

Leiomyomas (uterine fibroids) are benign smooth muscle tumors, often multiple, seen most commonly in women in middle and later reproductive years. About 30% of women with leiomyomas will develop symptoms such as abnormal vaginal bleeding, dysmenorrhea, bloating, backache, urinary symptoms, and dyspareunia. Severe pain can result with torsion of a pedunculated fibroid, or ischemia and infarction of a fibroid. Bimanual examination may demonstrate a mass or an enlarged uterus. Pelvic ultrasound is confirmatory. Treatment consists of NSAIDs or other analgesics for pain, hormonal manipulation for excessive bleeding, and referral to a gynecologist for definitive therapy.

Emergency Department Care and Disposition

Most patients are ultimately discharged from the ED even though there may not be a specific diagnosis. It is common and appropriate to discharge

patients with a diagnosis of undifferentiated abdominal pain. Patients should receive detailed discharge instructions with strict return precautions outlined. Follow-up instructions should also be very specific. Reevaluation in 12 to 24 hours can be scheduled if any concern persists. Analgesics, such as NSAIDs, provide effective pain control for most outpatients, although some patients will require opioids, such as oxycodone/acetaminophen (5/325) one to two tablets every 4 to 6 hours PO for a few days.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 96, "Abnormal Uterine Bleeding," by Bophal Sarha Hang; Chapter 97, "Abdominal and Pelvic Pain in the Nonpregnant Female," by Melanie Heniff and Heather R.B. Fleming.

58_Cydulka_Ch58.indd 322 16/03/17 3:06 PM

Source page 342

C h apter Ectopic Pregnancy and
59 Emergencies in The First
20 Weeks of Pregnancy

Robert Jones

■■ ECTOPIC PREGNANCY

Ectopic pregnancy (EP) is the leading cause of maternal death in the first trimester. Major risk factors include history of pelvic inflammatory disease, surgical procedures on the fallopian tubes including tubal ligation, previous EP, diethylstilbestrol exposure, intrauterine device use, and assisted reproduction techniques. The most common extrauterine location is the fallopian tube. This diagnosis must be considered in every woman of childbearing age presenting with abdominal pain and/or vaginal bleeding.

Clinical Features

The classic triad of abdominal pain, vaginal bleeding, and amenorrhea used to describe EP may be present, but many cases occur with more subtle findings. Presenting signs and symptoms may be different in ruptured versus nonruptured EP. Location of the EP will also determine the clinical features. The vast majority of EPs implant in the ampullary portion of the fallopian tube but additional implantation locations include the cervix, abdomen, and cesarean scar. Ninety percent of women with EP complain of abdominal pain; 50% to 80% have vaginal bleeding; and 70% give a history of amenorrhea. The pain described may be sudden, lateralized, extreme, or relatively minor and diffuse. The presence of hemoperitoneum with diaphragmatic irritation may cause the pain to be referred to the shoulder or upper abdomen. Presenting vital signs may be entirely normal even with a ruptured EP. There is poor correlation with the volume of hemoperi-

toneum and vital signs in EP. Relative bradycardia, as a consequence of vagal stimulation, may be present even in cases with rupture and hemoperitoneum. Physical examination findings are highly variable. The abdominal examination may show signs of localizing or diffuse tenderness with or without peritoneal signs. The pelvic examination findings may be normal but more often show cervical motion tenderness, adnexal tenderness with or without a mass, and possibly an enlarged uterus. Vaginal bleeding, ranging from spotting to heavy, is often present. Fetal heart tones may be heard in cases of EP beyond 12 weeks of gestation.

Diagnosis and Differential

The definitive diagnosis of EP is made either by ultrasound (US) or by direct visualization during surgery. The diagnosis of pregnancy is central to the diagnosis of possible EP and needs to be confirmed first. Urine pregnancy testing (for urinary β-human chorionic gonadotropin [β-hCG]) is a qualitative screening test with a threshold for detection of >20 mIU/mL of β-hCG. Urine qualitative testing is 95% to 100% sensitive and specific as

59_Cydulka_Ch59.indd 323 16/03/17 3:16 PM

Source page 343

compared with serum testing. Dilute urine, particularly when β-hCG levels are <50 mIU/mL, may result in a false-negative result. Quantitative serum testing for the diagnosis of pregnancy is virtually 100% sensitive for detecting β-hCG levels >5 mIU/mL and should be performed when the diagnosis of EP is considered but urine results are negative.

The primary goal of US in suspected EP is to determine if an intrauterine pregnancy (IUP) is present, since US cannot rule out the presence of EP. The transabdominal examination is usually performed first due to its wider field of view; the transvaginal examination is performed if the transabdominal examination is not diagnostic. When US reveals an unequivocal IUP and no other abnormalities, EP is effectively excluded unless the patient is at high risk for heterotopic pregnancy. Actual visualization of an EP with US occurs in a minority of cases. Sonographic findings of an empty uterus without an adnexal mass or free fluid in a woman with a positive pregnancy test result are considered indeterminate. In such situations, the findings must be evaluated in context with the patient's quantitative β-hCG level. A high β-hCG level (>6000 mIU/mL for transabdominal US or >1500 mIU/mL for transvaginal US) with an empty uterus is suggestive of EP. If the β-hCG is low (<1500 mIU/mL transvaginal), then the pregnancy may indeed be intrauterine or ectopic but often too small to be

visualized on US or the patient may have already had a miscarriage. In this situation, repeat quantitative β-hCG testing in 2 days must be performed. β-hCG should increase at least 66% in that period; EP has a slower rate of increase. Since many EPs will have β-hCG levels <1500 mIU/mL, quantitative β-hCG levels should not be used to determine the need for US imaging.

Differential diagnosis in the patient presenting with abdominal pain, vaginal bleeding, and early pregnancy includes threatened, incomplete, or missed abortion, recent elective abortion, implantation bleeding, molar pregnancy, heterotopic pregnancy, or corpus luteum cyst.

Emergency Department Care and Disposition

Treatment of patients with suspected EP depends on the patient's vital signs, physical signs, and symptoms. Close communication with the obstetric-gynecologic consultant is essential.

1. For unstable patients, insert two large-bore intravenous lines and begin rapid infusion of crystalloid and/or packed red blood cells to maintain blood pressure.

2. Perform a bedside urine pregnancy test.

3. Notify the obstetric-gynecologic consultant immediately if the patient is unstable, even before laboratory and diagnostic tests are complete.

4. Draw blood for blood typing, and rhesus (Rh) factor determination (or cross-matching for the unstable patients), quantitative β-hCG determination (if indicated), and serum electrolyte determination, as required. Rh-negative women with EP should receive 50 μg of anti-Rho (D) immunoglobulin.

5. Reliable, stable patients with indeterminate US results and a β-hCG level below 1000 mIU/mL can be discharged with ectopic precautions and follow-up in 2 days for repeat β-hCG determination and obstetric-gynecologic reevaluation.

59_Cydulka_Ch59.indd 324 16/03/17 3:16 PM

References/provenance

- Source: *Tintinalli's Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 340–343.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 59: Ectopic Pregnancy and Emergencies

Source pages 344–347 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 344–347

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Ectopic Pregnancy and Emergencies**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 59: Ectopic Pregnancy and Emergencies 325 6. [Source pages 344–347]

- Definitive treatment, as determined by the obstetric-gynecologic consul- tant, may involve laparoscopy, dilation and curettage, or medical man- agement with methotrexate. [Source pages 344–347]

- Methotrexate therapy, even when used in properly selected patients with EP, has a treatment failure rate of up to 36%, so proper selection is essential. [Source pages 344–347]

- ■■ THREATENED ABORTION AND ABORTION According to the World Health Organization, between 20% and 40% of pregnancies will spontaneously abort. [Source pages 344–347]

- Approximately 75% of these sponta- neous abortions will occur before 8 weeks of gestation. [Source pages 344–347]

Clinical reasoning structure

- **Problem representation:** Ectopic Pregnancy and Emergencies; source pages 344–347.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Ectopic Pregnancy and Emergencies**.

- Source-page range: **344–347**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 344

CHAPTER 59: Ectopic Pregnancy and Emergencies 325

6. Definitive treatment, as determined by the obstetric-gynecologic consultant, may involve laparoscopy, dilation and curettage, or medical management with methotrexate. Methotrexate therapy, even when used in properly selected patients with EP, has a treatment failure rate of up to 36%, so proper selection is essential.

■■ THREATENED ABORTION AND ABORTION

According to the World Health Organization, between 20% and 40% of pregnancies will spontaneously abort. Approximately 75% of these spontaneous abortions will occur before 8 weeks of gestation.

Clinical Features

Vaginal bleeding in the first 20 weeks, with a closed cervical os, benign examination, and no passage of tissue, is termed threatened abortion. A dilated cervix increases the likelihood of abortion (inevitable abortion). Incomplete abortion is defined as partial passage of the conceptus and is more likely between 6 and 14 weeks of pregnancy. The patient may report passage of grayish white products of conception (POC), or the POC may be evident on pelvic examination. Complete abortion is passage of all fetal tissue before 20 weeks gestation. All recovered POC should be sent for pathologic examination. Missed abortion is fetal death at less than 20 weeks without passage of any fetal tissue for 4 weeks after fetal death. Septic abortion implies evidence of infection during any stage of abortion, with signs and symptoms of pelvic pain, fever, cervical motion or uterine tenderness, or purulent or foul-smelling drainage.

Diagnosis and Differential

Perform a pelvic examination and obtain a complete blood count (CBC), blood typing (and cross-matching for unstable patients) and rhesus (Rh) factor determination, urine pregnancy test, quantitative β-hCG, and uri-

nalysis. The differential diagnosis includes EP, implantation bleeding, and gestational trophoblastic disease (GTD). Implantation bleeding, which is usually scant and painless, is a diagnosis of exclusion. GTD is a proliferative disease of the trophoblast that includes complete hydatidiform mole, partial mole, trophoblastic tumor, and choriocarcinoma. Patients with GTD present with bleeding, an enlarged uterus out of proportion to menstrual age, and a significantly elevated β-hCG level. Also consider the possibility of a molar pregnancy in patients with hyperemesis gravidarum or pregnancy-induced hypertension before 24 weeks of gestation.

Emergency Department Care and Disposition

Treatment of patients with suspected threatened abortion and abortion depends on the patient's vital signs, physical signs, and symptoms. Close communication with the obstetric-gynecologic consultant is essential.

1. Patients who are symptomatic or demonstrate signs of hemodynamic instability should receive supplemental oxygen, be placed on a cardiac monitor, and have two large-bore intravenous (IV) lines established. The obstetric-gynecologic consultant is consulted emergently in the unstable patient.

59_Cydulka_Ch59.indd 325 16/03/17 3:16 PM

Source page 345

2. Initiate aggressive IV crystalloid and/or packed red blood cell infusion, when appropriate, to help correct hypovolemia.

3. Rh-negative women with threatened abortion or abortion should receive anti-Rho (D) immunoglobulin. There is no uniform agreement on dosing. Dosing recommendations range from 50 to 300 μg in these patients.

4. US imaging should be performed when the patient is stable.

5. Incomplete abortion or GTD requires dilation and curettage. The decision to proceed with medical treatment, such as misoprostol, or surgical treatment, such as dilation and curettage, should be made in conjunction with the consulting obstetrician. GTD patients must receive close follow-up until quantitative β-hCG has returned to zero. Failure of the β-hCG to return to normal may indicate choriocarcinoma.

6. Septic abortion requires gynecologic consultation and broad-spectrum antibiotics such as ampicillin/sulbactam 3 g IV or clindamycin 600 mg plus gentamicin 1 to 2 mg/kg IV.

7. Patients with threatened abortion or complete abortion may be discharged with close follow-up arranged. Discharge instructions include pelvic rest (no intercourse or tampons) and instructions to return for heavy bleeding, fever, or pain.

■■ NAUSEA AND VOMITING OF PREGNANCY

Nausea and vomiting of pregnancy generally are seen in the first 12 weeks and affect 60% to 80% of pregnant women. Cases can range from mild symptoms to hyperemesis gravidarum.

Clinical Features

Findings on physical examination are usually normal except for signs of volume depletion. The presence of abdominal pain in nausea and vomiting of pregnancy or hyperemesis gravidarum is highly unusual and should suggest another diagnosis.

Diagnosis and Differential

Diagnostic workup may include CBC, serum electrolytes, blood urea nitrogen, creatinine, and urinalysis. Differential diagnosis includes ruptured EP, cholelithiasis, cholecystitis, gastroenteritis, pancreatitis, appendicitis, hepatitis, peptic ulcer disease, pyelonephritis, fatty liver of pregnancy, and hemolysis with elevated liver enzymes and low platelets (known as HELLP syndrome).

Emergency Department Care and Disposition

1. Rehydration should begin with IV fluid such as 5% dextrose in normal saline or 5% dextrose in Lactated Ringer's solution. Failure to include dextrose may result in prolonged ketosis.

2. Frequently used antiemetics are metoclopramide 10 mg IV every 6 to 8 hours, promethazine 12 to 25 mg IV every 4 hours (pregnancy class C, but widely used), and ondansetron 4 to 8 mg IV every 4 to 8 hours.

3. If the patient improves in the emergency department (urine ketones clearing and tolerating oral liquids), she may be discharged with a prescription for an antiemetic. Doxylamine with pyridoxine 50 mg PO

59_Cydulka_Ch59.indd 326 16/03/17 3:16 PM

Source page 346

every evening may be added as maintenance therapy for nausea and vomiting.

4. Admission is indicated for intractable vomiting, persistent ketonuria, electrolyte abnormalities, or weight loss greater than 10% of prepregnancy weight. Systemic steroids, such as methylprednisolone, 16 mg PO or IV every 8 hours for 3 days and tapered over 2 weeks to the lowest effective dose can be initiated in consultation with an obstetrician.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 98, "Ectopic Pregnancy and Emergencies in the First 20 Weeks of Pregnancy," by Heather A. Heaton.

59_Cydulka_Ch59.indd 327 16/03/17 3:16 PM

Source page 347

C hapter Comorbid Diseases in Pregnancy

60 Abigail D. Hankin

Many medical conditions can present in the context of pregnancy, either as a preexisting condition or arising during pregnancy. This chapter focuses on conditions that require different management when encountered in the pregnant patient. Some disorders are covered in other chapters within this text, including hypertension (Chapters 26 and 61), HIV infection (Chapter 92), and cardiac rhythm disturbances (Chapter 2).

■■ DIABETES

Diabetics are at increased risk for complications of pregnancy and acute complications of diabetes. Many patients with gestational diabetes are managed with diet alone, though some patients require oral hypoglycemics such as metformin or glyburide. Insulin therapy is necessary for some patients with gestational diabetes and nearly all patients with type I or II diabetes. Insulin requirements increase as a pregnancy progresses, from 0.7 U/kg/d during the first trimester to 1.0 U/kg/d at term.

Pregnant patients are at increased risk of diabetic ketoacidosis (DKA), and DKA can occur more rapidly, and at lower glucose levels, than in nonpregnant patients. DKA should be considered for any pregnant diabetic patient who is ill appearing and/or has a blood glucose level ≥180 mg/dL. Treatment of DKA includes supplemental oxygen, left lateral decubitus positioning, and usual DKA care: isotonic fluid resuscitation to correct volume deficits, administration of continuous insulin, correction of electrolyte abnormalities, and treatment of the underlying cause (see Chapter 129).

Mild hypoglycemia (glucose <70 mg/dL and able to follow commands) is treated with a snack of milk and crackers. Severe hypoglycemia (glucose <70 mg/dL and unable to follow commands) should be treated with 50 mL of 50% dextrose in water IV, with 1 to 2 mg of glucagon given IM or subcutaneous if IV access is delayed. Subsequently, the patient should be given 5% dextrose solution IV at 50 to 100 mL/h.

■■ HYPERTHYROIDISM

Hyperthyroidism in pregnancy increases the risk of preeclampsia, congenital anomalies, and neonatal morbidity. Clinical features may be subtle and may mimic normal pregnancy. Treatment is with propylthiouracil (PTU). Patients on PTU are at risk for purpuric rash and agranulocytosis.

Thyroid storm presents with fever, volume depletion, or cardiac decompensation manifesting as high-output heart failure. Management is similar to nonpregnant patients (see Table 60-1).

■■ HYPERTENSION

Treatment is often considered for chronic hypertension, defined as blood pressure >140/90 mm Hg measured on two occasions before 20 weeks gestation or sustained beyond 12 weeks gestation. Chronic hypertension is

328

60_Cydulka_Ch60.indd 328 16/03/17 3:31 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 344–347.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 60: Comorbid Diseases in Pregnancy

Source pages 348–355 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 348–355

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Comorbid Diseases in Pregnancy**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 60: Comorbid Diseases in Pregnancy 329 Table 60-1 Principles of Treatment of Thyroid Storm during Pregnancy Principle Comment Inhibit thyroid hormone release Propylthiouracil (PTU) 600–1000 mg PO loading dose with thionamides (PTU is pre- followed by 200–250 mg PO every 4 h (first trimester) ferred over methimazole; also or blocks conversion of T4 to T3) Methimazole 40 mg PO loading dose followed by 25 mg PO every 4 h (second and third trimesters) Inhibit new thyroid hormone Lugol solution 8–10 drops every 6–8 h production (give at least 1 h or after above step) Potassium iodine 5 drops PO every 6 h or Iopanoic acid 1 g IV every 8 h Do not use radioactive iodine because the fetus will concentrate iodine-131 after the 10th to 12th week of gestation, resulting in congenital hypothyroidism Block peripheral thyroid Propranolol 1–2 mg IV every 10–15 min and start hormone effects Propranolol 40 mg PO every 6 h or Esmolol 500 μg/kg IV bolus, then 50 μg/kg/min maintenance Hold if evidence of heart failure is present Prevent conversion of T4 to T3 Hydrocortisone 100 mg IV every 8 h or Dexamethasone 2 mg IV every 6 h Supportive care Left lateral decubitus position Oxygen Cooling blankets IV fluids Acetaminophen 650 mg PO every 4 h typically treated with labetalol, nifedipine, or methyldopa while angio- tensin receptor blockers and angiotensin-converting enzyme inhibitors are avoided in pregnancy. [Source pages 348–355]

- Acute hypertensive crisis in pregnancy is treated with labetalol or hydralazine (see Table 60-2). [Source pages 348–355]

- While blood pressure is being managed, fetal well-being should be closely monitored, and clinicians should consider superimposed preeclampsia. [Source pages 348–355]

- ■■ DYSRHYTHMIAS Dysrhythmias may be precipitated by pregnancy and labor/delivery. [Source pages 348–355]

- Supra- ventricular tachycardias are treated with vagal maneuvers, β-blockers, adenosine, verapamil, and diltiazem at usual dosages. [Source pages 348–355]

Clinical reasoning structure

- **Problem representation:** Comorbid Diseases in Pregnancy; source pages 348–355.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Comorbid Diseases in Pregnancy**.

- Source-page range: **348–355**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 348

CHAPTER 60: Comorbid Diseases in Pregnancy 329

Table 60-1 Principles of Treatment of Thyroid Storm during Pregnancy

Principle Comment

Inhibit thyroid hormone release Propylthiouracil (PTU) 600–1000 mg PO loading dose

with thionamides (PTU is pre- followed by 200–250 mg PO every 4 h (first trimester)

ferred over methimazole; also or

blocks conversion of T4 to T3) Methimazole 40 mg PO loading dose followed by

25 mg PO every 4 h (second and third trimesters)

Inhibit new thyroid hormone Lugol solution 8–10 drops every 6–8 h

production (give at least 1 h or

after above step) Potassium iodine 5 drops PO every 6 h

or

Iopanoic acid 1 g IV every 8 h

Do not use radioactive iodine because the fetus

will concentrate iodine-131 after the 10th to

12th week of gestation, resulting in congenital

hypothyroidism

Block peripheral thyroid Propranolol 1–2 mg IV every 10–15 min and start

hormone effects Propranolol 40 mg PO every 6 h

or

Esmolol 500 μg/kg IV bolus, then 50 μg/kg/min

maintenance

Hold if evidence of heart failure is present

Prevent conversion of T4 to T3 Hydrocortisone 100 mg IV every 8 h

or

Dexamethasone 2 mg IV every 6 h

Supportive care Left lateral decubitus position

Oxygen

Cooling blankets

IV fluids

Acetaminophen 650 mg PO every 4 h

typically treated with labetalol, nifedipine, or methyldopa while angiotensin receptor blockers and angiotensin-converting enzyme inhibitors are avoided in pregnancy.

Acute hypertensive crisis in pregnancy is treated with labetalol or hydralazine (see Table 60-2).

While blood pressure is being managed, fetal well-being should be closely monitored, and clinicians should consider superimposed preeclampsia.

■■ DYSRHYTHMIAS

Dysrhythmias may be precipitated by pregnancy and labor/delivery. Supraventricular tachycardias are treated with vagal maneuvers, β-blockers, adenosine, verapamil, and diltiazem at usual dosages. Patients with atrial fibrillation who require anticoagulation should be managed with unfractionated or low-molecular-weight heparin (LMWH). Electrical cardioversion may be used to treat dysrhythmias when indicated and have not been shown to be harmful to the fetus. Amiodarone should only be used to treat resistant, life-threatening dysrhythmias, as its use has been linked to fetal neurotoxicity.

60_Cydulka_Ch60.indd 329 16/03/17 3:31 PM

Source page 349

Table 60-2 Treatment of Hypertension in Pregnancy

Urgent Potential

Adjunct Control Adverse

For Existing to Existing of Acute Effects

Agent Hypertension Treatment Hypertension (Maternal)

Hydralazine N/A 50–300 mg Loading dose of Delayed

daily in 2–4 5 mg IV or IM, hypotension

divided doses; maintenance

use with dose there after

methyldopa of 5–10 mg

or labetalol to every 20–40
prevent reflex min up to 300
tachycardia* mg; or constant
infusion of
0.5–10 mg/h
Hydrochlorothiazide N/A 12.5–50 mg N/A Volume
daily depletion and
electrolyte
disorders
Labetalol 200–2400 mg N/A Loading dose Headache
daily in 2–3 of 20 mg IV;
divided doses maintenance
dose of 20–80
mg up to 300
mg; or constant
infusion of 1–2
mg/min
Nifedipine 30–120 mg N/A 10–30 mg Headache,
daily as orally, repeated interference
slow-release after 45 min if with labor
preparation needed
Methyldopa 0.5–3.0 g daily N/A N/A Sedation,
in 2–3 divided elevated
doses liver function
tests,
depression
Risk of fetal bradycardia and neonatal thrombocytopenia.
*
Abbreviation: N/A, not applicable.

■■ THROMBOEMBOLISM

Deep venous thrombosis (DVT) and pulmonary embolism (PE) are the most common causes of maternal morbidity and mortality in industrialized nations. Factors associated with increased risk of thromboembolism include advanced maternal age, increasing parity, multiple gestations, operative delivery, bed rest, and obesity. Symptoms of DVT and PE may be mistaken for symptoms of normal pregnancy. Table 60-3 shows the various imaging modalities to consider for the diagnosis of PE in pregnancy. DVT and PE are treated with unfractionated or low-molecular-weight heparin at usual doses. Warfarin is contraindicated (see Chapter 25).

Source page 350

60_Cydulka_Ch60.indd 331

Table 60-3 Imaging Modalities for Diagnosis of Pulmonary Embolism in Pregnancy

Radiation Limitations Disadvantages Advantages Lactation

Chest radiograph Minimal Nonspecific and Results determine next imaging May identify another cause No change

nonsensitive study, requiring more time to of pulmonary symptoms

diagnosis

CT-PA High maternal breast Contrast allergy and renal Hyperdynamic state in pregnancy High sensitivity and No need to discard

radiation; lower fetal insufficiency can affect interpretation specificity; needed if breast milk

radiation than V/Q scan abnormal chest radiograph

V/Q scan Low breast radiation but Not useful if abnormal If negative or inconclusive and Negative perfusion study Discard breast milk

CHAPTER 60: Comorbid Diseases in Pregnancy

higher fetal radiation than chest radiograph, asthma, suspicion for PE remains, will effectively rules out PE for 12 h

CT-PA COPD, underlying need a CT-PA; limited availability

pulmonary disease and time for isotope preparation,

which delays diagnosis

MRI/MRV No radiation Gadolinium safety for Limited availability Can detect pelvic and iliac No need to discard

fetus is unknown; do not thrombosis breast milk

use in maternal renal

insufficiency

Abbreviations: COPD, chronic obstructive pulmonary disease; CT-PA, chest CT–pulmonary angiography; V/Q scan, ventilation-perfusion scan; MRV, magnetic resonance venography.

16/03/17 3:31 PM

331

Source page 351

332 SECTION 8: Gynecology and Obstetrics

■■ ASTHMA

Symptoms of acute asthma exacerbation include cough, wheezing, and dyspnea. Acute therapy includes maintenance of oxygen saturation >95%, administration of inhaled β2-agonists, systemic corticosteroids, and fetal monitoring. Aerosolized ipratropium may be added in severely obstructed patients. Terbutaline sulfate, 0.25 mg SC every 20 minutes, may also be used. Subcutaneous epinephrine should be avoided, if possible. Peak expiratory flow rate (PEFR) is not altered in pregnancy, with goal PEFR between 380 and 550 L/min. Patients who are discharged home should receive oral prednisone 40 to 60 mg/d for 5 to 10 days in addition to a rescue β-agonist.

■■ CYSTITIS AND PYELONEPHRITIS

Clinical features of cystitis and pyelonephritis are similar in pregnant and

nonpregnant women, as are the causative microbes. Cystitis may be treated with nitrofurantoin, 100 mg PO twice daily for 3 to 7 days, while trimethoprim-sulfamethoxazole should be avoided due to concerns of fetal kernicterus during the third trimester. Fluoroquinolones and tetracyclines should also be avoided in pregnancy. Pregnant patients with pyelonephritis are treated aggressively because of increased risk of preterm labor and sepsis, and should be admitted for IV hydration and a second- or third-generation cephalosporin. Many providers continue antibiotic suppression for the remainder of the pregnancy.

■■ SICKLE CELL DISEASE

Pregnant women with sickle cell disease are at higher risk of miscarriage, preterm labor, and vasoocclusive crises. Clinical features, evaluation, and treatment are similar to nonpregnant patients. Management of vasoocclusive crisis includes aggressive hydration, analgesia, identification and treatment of the underlying cause, and fetal monitoring. Opiates can be used, while NSAIDs should be avoided, particularly after 32 weeks gestation. Blood transfusion should be considered when conservative measures have failed, or with severe anemia, preeclampsia, hypoxemia, acute chest syndrome, or a new-onset neurological event.

■■ HEADACHES

Headaches can be a presenting symptom of a wide variety of benign and life-threatening disorders, including intracranial hemorrhage, central vein thrombosis, mass, infection, and preeclampsia/eclampsia. Warning symptoms of potentially life-threatening disease include acute onset, postpartum headaches, neurological deficits, fever, and papilledema or retinal hemorrhages. If indicated, CT scan of the brain should be performed with appropriate shielding of the uterus. Migraine headaches should be treated with acetaminophen, antiemetics and, if necessary, narcotics. Ergot alkaloids and triptans should not be used.

60_Cydulka_Ch60.indd 332 16/03/17 3:31 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 348–355.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 61: Emergencies After 20 Weeks

Source pages 356–359 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 356–359

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Emergencies After 20 Weeks**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- Hospitalize all patients with a sustained systolic blood pressure ≥140 mmHg or diastolic ≥90 mmHg plus any symptoms of preeclampsia. [Source pages 356–359]

- Definitive treatment requires delivery of the fetus. [Source pages 356–359]

- In patients with mild preeclampsia, outpatient management may be ap- propriate after consultation with an obstetrician. [Source pages 356–359]

- Arrangements must be made for frequent laboratory evaluation and close fetal surveillance. [Source pages 356–359]

- ■■ VAGINAL BLEEDING DURING THE SECOND HALF OF PREGNANCY Placental abruption, placenta previa, and preterm labor are the most com- mon causes of vaginal bleeding during the second half of pregnancy. [Source pages 356–359]

Clinical reasoning structure

- **Problem representation:** Emergencies After 20 Weeks; source pages 356–359.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Emergencies After 20 Weeks**.

- Source-page range: **356–359**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 356

CHAPTER 61: Emergencies After 20 Weeks 337

4. Hospitalize all patients with a sustained systolic blood pressure ≥140 mmHg or diastolic ≥90 mmHg plus any symptoms of preeclampsia.

5. Definitive treatment requires delivery of the fetus.

6. In patients with mild preeclampsia, outpatient management may be appropriate after consultation with an obstetrician. Arrangements must be made for frequent laboratory evaluation and close fetal surveillance.

■■ VAGINAL BLEEDING DURING THE SECOND HALF OF PREGNANCY

Placental abruption, placenta previa, and preterm labor are the most common causes of vaginal bleeding during the second half of pregnancy.

Clinical Features

Placental Abruption

Placental abruption, the premature separation of the placenta from the uterine wall, must be considered in all pregnant females near term who present with painful vaginal bleeding. Clinical features include vaginal bleeding, abdominal pain, uterine tenderness, hypertonic contractions, increased uterine tone, fetal distress, and, in severe cases, disseminated intravascular coagulation (DIC), and fetal and/or maternal death. Vaginal bleeding may be mild or severe, depending on whether the area of abruption communicates to the cervical os. Abruption of greater than 50% of the placenta usually results in fetal demise. Be aware of the possibility of abruption after trauma, even minor trauma.

Placenta Previa

Placenta previa is the implantation of the placenta over the cervical os. Clinical features include painless, bright red vaginal bleeding. The amount of bleeding is frequently large as opposed to normal “bloody show,” when a small amount of bright red blood and mucous are passed.

Diagnosis and Differential

Transabdominal ultrasound should be obtained prior to performing specu-

lum or digital pelvic examination to differentiate abruption placenta from placenta previa as it is contraindicated in previa. If transabdominal ultrasound is nondiagnostic, transvaginal ultrasound should be considered. The transvaginal ultrasound must be performed by someone experienced in transvaginal ultrasound, as positioning is extremely important. Ultrasound is very sensitive in detecting placenta previa but has limited sensitivity in diagnosing placental abruption.

Emergency Department Care and Disposition

1. Hemodynamic instability is managed with IV normal saline or packed red blood cells.
2. Obtain emergent obstetric consultation, complete blood count (CBC), type and cross, baseline coagulation studies, and electrolyte studies on all patients.
3. Obtain a DIC profile on patients with suspected placental abruption.

61_Cydulka_Ch61.indd 337 25/03/17 12:58 PM

Source page 357

4. Give Rh (D) immune globulin 300 μg IM to Rh-negative patients.
5. Patients with placental abruption or placenta previa may need emergent cesarean delivery.
6. Do not use tocolytics in patients with suspected abruption.

Premature Rupture of Membranes

Premature rupture of membranes (PROM) is rupture of membranes before the onset of labor. Clinical presentation includes a rush of fluid or continuous leakage of fluid from the vagina. Diagnosis is confirmed by finding a pool of fluid in the posterior fornix with pH greater than 7.0 and a ferning pattern on smear. Sterile speculum examination may be done; however, digital pelvic examination should be avoided because it increases the rate of infection and decreases the latent period from rupture until delivery. Tests for chlamydia, gonorrhea, bacterial vaginosis, and group B Streptococcus should be performed. Management of PROM depends on gestational age and maturity of the fetus, condition of the fetus, concern for infection, and presence of other complicating factors. Antibiotics and steroids may be appropriate. An obstetrics consultation should be obtained to assist with treatment and admission decisions.

■■ PRETERM LABOR

Clinical Features and Diagnosis

Preterm labor is defined as labor before 37 weeks' gestation. Clinical features include regular uterine contractions with effacement and dilatation of the cervix. Cervical fluid should be examined for possible PROM. Only after

premature rupture of membranes and placenta previa have been excluded should digital examination be performed; use sterile gloves. Obtain tests for chlamydia, gonorrhea, bacterial vaginosis, and group B Streptococcus. Perform an ultrasound for fetal age, weight, anatomy, and amniotic fluid level if possible but obstetric consultation should not be delayed awaiting results.

Emergency Department Care and Disposition

1. Monitor mother and fetus.
2. Consult an obstetrician for admission and decision making regarding tocolytics.
3. If tocolytics are initiated, administer glucocorticoids to the mother to hasten fetal lung maturity if the gestational age is less than 34 weeks. Dexamethasone 6 mg IM is commonly used.
4. Do not use tocolytics if placental abruption is suspected.
5. Gestational age younger than 34 weeks is associated with poorer outcomes; if possible, the patient should be transferred to a tertiary care center with a high-risk intensive care unit.

■■ POSTPARTUM HEMORRHAGE

The differential diagnosis of hemorrhage in the first postpartum day includes uterine atony, uterine rupture, laceration of the lower genital tract, retained placental tissue, uterine inversion, and coagulopathy. After the first 24 hours, retained products of conception, uterine polyps, or coagulopathy such as von

61_Cydulka_Ch61.indd 338 25/03/17 12:58 PM

Source page 358

Willebrand disease are more likely causes. An enlarged and “doughy” uterus suggests uterine atony; a vaginal mass suggests an inverted uterus. Bleeding despite good uterine tone and size may indicate retained products of conception or uterine rupture. The vagina and cervix must be inspected for lacerations. The first priority of ED management is stabilization of the patient with crystalloid IV fluids and/or packed red blood cells, if needed. CBC, clotting studies, and type and cross must be obtained. Uterine atony is treated with uterine massage and oxytocin 20 units in one liter of normal saline IV at 200 mL/h. Minor lacerations may be repaired in the ED. Extensive lacerations, retained products of conception, uterine inversion, or uterine rupture require emergency operative treatment by the obstetrician.

■■ POSTPARTUM ENDOMETRITIS

Postpartum endometritis is a polymicrobial infection with symptoms that usually begin several days after delivery. Clinical features include fever, lower abdominal pain, and foul-smelling lochia. Physical examination reveals uterine or cervical motion tenderness and discharge. CBC, urinaly-

sis, and cervical cultures should be obtained. Admission for antibiotic treatment is indicated for patients who appear ill, have had a cesarean section, or who have underlying comorbidities. Antibiotic regimens for outpatients may include clindamycin or doxycycline. Do not use doxycycline in patients who are breastfeeding. Inpatient regimens include clindamycin 450 to 900 mg IV every 8 hours plus gentamicin 1.5 mg/kg IV every 8 hours or cefoxitin 1 to 2 g IV every 6 hours.

■■ MASTITIS

Mastitis is cellulitis of the periglandular breast tissue. Clinical features include swelling, redness, and tender engorgement of the involved portion of the breast, with or without fever and chills. Milk stasis presents similarly, except it lacks erythema, fever, or chills. For cellulitis, initiate treatment with dicloxacillin 500 mg orally four times daily or cephalexin 500 mg orally four times daily. Clindamycin 300 mg PO every 6 hours may be used in patients with penicillin allergy or if concerns about MRSA exist. Oral analgesics may be needed. Patients should continue nursing on the affected breast. However, in cases of purulent discharge, the mother should pump and discard the milk rather than nurse. Differentiate mastitis from breast abscess using bedside ultrasound.

■■ AMNIOTIC FLUID EMBOLISM

Amniotic fluid embolism is a sudden, catastrophic illness with mortality rates of 60% to 80%. Clinical features include sudden cardiovascular collapse with hypoxemia, pulmonary edema, altered mental status, seizures, and DIC along with sudden onset of fetal distress. Death can occur rapidly. Treatment is supportive; treat hypoxia, hypotension, and hypoperfusion.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 100, "Maternal Emergencies After 20 Weeks of Pregnancy and in the Postpartum Period," by Janet Simmons Young.

61_Cydulka_Ch61.indd 339 25/03/17 12:58 PM

Source page 359

C h a p ter Emergency Delivery

62 Stacie Zelman

Precipitous delivery in an emergency setting can be a source of significant anxiety for an emergency physician. While emergency delivery is a relatively uncommon occurrence, careful preparation and education can help in avoiding serious complications, and result in a positive outcome for the mother and child.

■■ CLINICAL FEATURES

Any pregnant woman who is beyond 20 weeks' gestation and appears to be in active labor should be evaluated expeditiously. Initial evaluation

should include complete maternal vital signs and fetal heart monitoring. A persistently slow or fast fetal heart rate (less than 110 beats/min or greater than 160 beats/min) is an indicator of fetal distress. History includes frequency and time of onset of contractions, leakage of fluid, vaginal bleeding, estimated gestational age, and prenatal care. A focused physical examination should include an abdominal examination evaluating fundal height, abdominal or uterine tenderness, and fetal position. A bimanual or sterile speculum examination should be performed if no contraindications exist such as active vaginal bleeding. After the exam, place the patient in the left lateral decubitus position to prevent maternal hypotension.

False labor is characterized by irregular, brief contractions usually confined to the lower abdomen. These typically painless contractions, commonly called Braxton–Hicks contractions, are irregular in intensity and duration. True labor is characterized by painful, regular contractions of steadily increasing intensity and duration leading to progressive cervical dilatation. True labor typically begins in the fundal region and upper abdomen and radiates into the pelvis and lower back.

■■ DIAGNOSIS AND DIFFERENTIAL

Patients without vaginal bleeding should be assessed with sterile speculum and bimanual examinations to evaluate the progression of labor, cervical dilation, and rupture of membranes. Patients with active vaginal bleeding require initial evaluation with ultrasound to rule out placenta previa. Spontaneous rupture of membranes typically occurs with a gush of clear or blood-tinged fluid. If ruptured membranes are suspected, a sterile speculum examination should be performed and amniotic fluid obtained from the fornix or vaginal vault. Amniotic fluid is alkaline, will stain Nitrazine paper dark blue and will “fern” if dried on a slide. The presence of meconium in amniotic fluid should be noted. Avoid digital examinations in the preterm patient in whom prolongation of gestation is desired as even one examination increases the chance of infection.

62_Cydulka_Ch62.indd 340 16/03/17 5:32 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 356–359.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 62: Emergency Delivery

Source pages 360–363 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 360–363

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Emergency Delivery**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 62: Emergency Delivery 341 ■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION If the cervix is dilated in a woman experiencing active contractions, further transport, even short distances, may be hazardous. [Source pages 360–363]

- Preparations should be made for emergency delivery. [Source pages 360–363]

- Assess fetal position by physical examina- tion, and confirm by ultrasound, if possible. [Source pages 360–363]

- Place the patient in the dorsal lithotomy position. [Source pages 360–363]

- Control of the delivery of the neonate is the major challenge. [Source pages 360–363]

Clinical reasoning structure

- **Problem representation:** Emergency Delivery; source pages 360–363.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Emergency Delivery**.

- Source-page range: **360–363**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 360

CHAPTER 62: Emergency Delivery 341

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

If the cervix is dilated in a woman experiencing active contractions, further transport, even short distances, may be hazardous. Preparations should be made for emergency delivery. Assess fetal position by physical examination, and confirm by ultrasound, if possible. Place the patient in the dorsal lithotomy position. Notify an obstetrician, if one is available.

Emergency Delivery Procedure (Fig. 62-1)

1. Control of the delivery of the neonate is the major challenge.

a. As the infant's head emerges from the introitus, support the perineum with a sterile towel placed along the inferior portion of the perineum with one hand while supporting the fetal head with the other.

b. Exert mild counter-pressure to prevent the rapid expulsion of the fetal head, which may lead to third- or fourth-degree perineal tears.

c. As the infant's head presents, use the inferior hand to control the fetal chin while keeping the superior hand on the crown of the head, supporting the delivery.

d. Ask the mother to breathe through contractions rather than bearing down and attempting to push the baby out rapidly.

2. After delivery of the head, palpate the neck for the presence of a nuchal cord (may be present in up to 35% of deliveries).

a. If the cord is loose, reduce it over the infant's head; delivery may then proceed as usual.

b. If the cord is tightly wound, clamp it in the most accessible area using two clamps in close proximity and cut to allow delivery of the infant.

3. After delivery of the head, the head will restitute or turn to one side or the other.

a. As the head rotates, hands are placed on either side, providing gentle downward traction to deliver the anterior shoulder.

b. Guide the shoulders upward, delivering the posterior shoulder and allowing the remainder of the infant to be delivered.

4. Place the posterior (left) hand underneath the infant's axilla before delivering the rest of the body. Use the anterior hand to grasp the infant's ankles and ensure a firm grip.

5. Wrap the infant in a towel and stimulate him or her while drying.

6. Double clamp the umbilical cord and cut with sterile scissors.

7. Place the infant in a warm incubator, where postnatal care may be provided and calculate Apgar scores at 1 and 5 minutes after delivery. Scoring includes general color, tone, heart rate, respiratory effort, and reflexes.

8. If an episiotomy is necessary (e.g., with a breech presentation), it may be performed as follows:

a. Inject a solution of 5 to 10 mL of 1% lidocaine with a small-gauge needle into the posterior fourchette and perineum.

b. While protecting the infant's head, make a 2- to 3-cm cut with scissors to extend the vaginal opening at the midline or 45 degrees from midline.

c. Support the incision with manual pressure from below, taking care not to allow the incision to extend into the rectum (less likely with the mediolateral incision).

62_Cydulka_Ch62.indd 341 16/03/17 5:32 PM

Source page 361

A B

Occiput anterior

C D

E

F

FIGURE 62-1. Movements of normal delivery. Mechanism of labor and delivery for vertex presentations. A. Engagement, flexion, and descent. B. Internal rotation. C. Extension and delivery of the head. After delivery of the head, the neck is checked for encirclement by the umbilical cord. D. External rotation, bringing the thorax into the anteroposterior diameter of the pelvis. E. Delivery of the anterior shoulder. F. Delivery of the posterior shoulder. Note that after delivery, the head is supported and used to gently guide delivery of the shoulder. Traction should be minimized.

■■ CORD PROLAPSE

If bimanual examination shows a palpable, pulsating cord:

a. Do not remove the examining hand; use the hand to elevate the presenting fetal part to reduce compression of the cord.

b. Immediate obstetric assistance is necessary, as a cesarean section is

indicated.

c. Keep the examining hand in the vagina while the patient is transported and prepped for surgery to prevent further compression of the cord by the fetal head. Do not attempt to reduce cord.

62_Cydulka_Ch62.indd 342 16/03/17 5:32 PM

Source page 362

■■ SHOULDER DYSTOCIA

Initially recognized after the delivery of the fetal head, when routine downward traction is insufficient to deliver the anterior shoulder. The anterior shoulder is trapped behind the pubic symphysis.

a. After delivery of the infant's head, the head retracts tightly against the perineum ("Turtle sign").

b. Upon recognizing shoulder dystocia, suction the infant's nose and mouth and call for assistance to position the mother in the extreme lithotomy position, with legs sharply flexed up to the abdomen (McRoberts maneuver) and held by the mother or an assistant.

c. Next, an assistant should apply downward suprapubic pressure to disimpact the anterior shoulder from the pubic symphysis.

d. Do not apply fundal pressure because this will further force the shoulder against the pelvic rim.

e. A Woods Corkscrewmaneuver may be attempted—place fingertips behind the anterior shoulder and in back of the posterior shoulder of the infant, and rotate the shoulder girdle 180°.

■■ BREECH PRESENTATION

The primary concern with breech presentation is head entrapment.

a. Breech presentations may be classified as frank, complete, incomplete, or footling.

b. In any breech delivery, immediate obstetric consultation should be requested.

c. Frank and complete breech presentations:

■Serve as a dilating wedge nearly as well as the fetal head, and delivery

●

may proceed in an uncomplicated fashion.

■Main point is to allow the delivery to progress spontaneously. This lets

●

the presenting portion of the fetus to dilate the cervix maximally.

d. Consult obstetrical texts for a detailed description of maneuvers for breech delivery.

e. Footling and incomplete breech positions are not considered safe for vaginal delivery because of the possibility of cord prolapse or incom-

plete dilatation of the cervix.

■■ POSTPARTUM CARE

The placenta should be allowed to separate spontaneously and assisted with gentle traction. Aggressive traction on the cord risks uterine inversion, tearing of the cord, or disruption of the placenta, which can result in severe vaginal bleeding. After removal of the placenta, gently massage the uterus to promote contraction. Infuse oxytocin 10 to 40 U/1000 mL NS at a moderate rate to maintain uterine contraction. Oxytocin may also be given as 10 U IM. Episiotomy or laceration repair may be delayed until an experienced obstetrician is able to close the laceration and inspect the patient for fourth-degree (rectovaginal) tears.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 101, "Emergency Delivery," by Sarah Elisabeth Frasure.

62_Cydulka_Ch62.indd 343 16/03/17 5:32 PM

Source page 363

C h a p ter Vulvovaginitis

63 Robert R. Cooney

Causes of vulvovaginitis include infections, irritants, allergies, reaction to foreign bodies, and atrophy. The normal vaginal flora help maintain an acidic pH between 3.8 and 4.5, which decreases pathogen growth.

■■ BACTERIAL VAGINOSIS

Bacterial vaginosis (BV) is the most common cause of malodorous vaginal discharge. However, up to 50% of infected women are asymptomatic. BV occurs when vaginal lactobacilli are replaced by anaerobes such as Gardnerella vaginalis, Mycoplasma, and Ureaplasma.

Clinical Features

The most common symptom is malodorous or "fishy smelling" vaginal discharge. Vaginal irritation, excoriation, fissures, and edema are uncommon. A combination of history, vaginal examination, and point-of-care testing confirm the diagnosis.

Diagnosis and Differential

The diagnosis can be made if three of the following four criteria are present: (1) vaginal discharge, (2) vaginal pH greater than 4.5, (3) positive amine test (fishy odor when 10% KOH is added to the discharge), and (4) clue cells seen on saline wet preparation.

Emergency Department Care and Disposition

Treat with metronidazole 500 mg PO twice daily for 7 days. Clindamycin 2% intravaginal cream at night for 7 days or metronidazole 0.75% intravaginal gel daily for 5 days are alternatives. Treatment is not recommended

for male partners or asymptomatic women. To avoid a disulfram-like reaction, patients treated with metronidazole should refrain from alcohol use during treatment and for 24 hours after ending treatment.

All symptomatic pregnant women should be treated, and women at high risk of preterm labor should be considered for treatment. Preferred treatment in pregnancy is metronidazole 250 mg PO twice daily for 7 days.

■■ CANDIDA VAGINITIS

Candida albicans is the second most common cause of infectious vaginitis. Conditions that promote Candida vaginitis include systemic antibiotics, diabetes, pregnancy, hormone replacement therapy, and birth control pills. Incidence is decreased in postmenopausal patients. Candidiasis is not typically considered a sexually transmitted disease, though it can be transmitted sexually.

63_Cydulka_Ch63.indd 344 21/04/17 3:51 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 360–363.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 63: Vulvovaginitis

Source pages 364–367 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 364–367

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Vulvovaginitis**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 63: Vulvovaginitis 345 Clinical Features The most common symptom of Candida vaginitis is pruritus. [Source pages 364–367]

- Other symp- toms include vaginal discharge, external dysuria, and dyspareunia. [Source pages 364–367]

- Exam may reveal vulvar and vaginal edema, erythema, and a thick “cottage cheese” discharge. [Source pages 364–367]

- Diagnosis and Differential Diagnosis is confirmed if vaginal pH is 4 to 4.5 and budding yeast or pseu- dohyphae are present on microscopy. [Source pages 364–367]

- Ten percent KOH dissolves vaginal epithelial cells, leaving yeast buds and pseudohyphae intact and easier to see. [Source pages 364–367]

Clinical reasoning structure

- **Problem representation:** Vulvovaginitis; source pages 364–367.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Vulvovaginitis**.

- Source-page range: **364–367**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 364

CHAPTER 63: Vulvovaginitis 345

Clinical Features

The most common symptom of Candida vaginitis is pruritus. Other symptoms include vaginal discharge, external dysuria, and dyspareunia. Exam may reveal vulvar and vaginal edema, erythema, and a thick “cottage cheese” discharge.

Diagnosis and Differential

Diagnosis is confirmed if vaginal pH is 4 to 4.5 and budding yeast or pseudohyphae are present on microscopy. Ten percent KOH dissolves vaginal epithelial cells, leaving yeast buds and pseudohyphae intact and easier to see. Use of KOH increases the sensitivity to 80%, with a specificity approaching 100%.

Emergency Department Care and Disposition

Single-dose treatment with fluconazole, 150 mg PO, is as effective as topical treatments. Almost all topically applied azoles are equally efficacious and available over-the-counter. Treatment options include clotrimazole, butoconazole, miconazole, terconazole, and tioconazole in several strengths and formulations. Pregnant patients should avoid oral fluconazole and instead use intravaginal agents for 7 days. For nonpregnant patients with complicated candidiasis, fluconazole 150 mg PO is given on days 1 and 3.

■■ TRICHOMONAS VAGINITIS

Trichomoniasis is a common sexually transmitted disease caused by the protozoan Trichomonas vaginalis.

Clinical Features

Presenting symptoms include a frothy, malodorous vaginal discharge, pruritus, and vaginal irritation. However, up to 50% of women harboring the organism are asymptomatic.

Diagnosis and Differential

Saline wet prep shows motile, pear-shaped, flagellated trichomonads.

Microscopy should be performed immediately after obtaining the sample or the organisms may lose motility. The sensitivity of microscopy is 60% to 70%. The sensitivity of culture is 95% but results are not readily available in the ED. New point-of-care PCR-based tests are now available as well.

Emergency Department Care and Disposition

The treatment of choice for trichomoniasis is metronidazole 2 g single oral dose or tinidazole 2 g single oral dose. Metronidazole 500 mg twice daily for 7 days is recommended for patients who fail single-dose therapy. Metronidazole gel is much less efficacious and thus not recommended for use. Sexual partners should be treated to avoid retransmission of disease. Patients should also be counseled to abstain from sexual activity until treatment course is completed and they are asymptomatic.

63_Cydulka_Ch63.indd 345 16/03/17 5:36 PM

Source page 365

■■ CONTACT VULVOVAGINITIS

Common causes of contact vulvovaginitis include douches, soaps, bubble baths, deodorants, perfumes, feminine hygiene products, topical antibiotics, and tight undergarments. Patients complain of perineal burning, itching, and local swelling. Examination findings may include redness, edema, and ulceration. Vaginal pH changes may promote overgrowth of Candida, obscuring the primary problem.

Try to identify the precipitating agent and rule out infectious causes. Most cases resolve spontaneously when the precipitant is withdrawn. For more severe reactions, cool sitz baths, compresses with Burow's solution, and topical corticosteroids may help. Oral antihistamines are drying but may be helpful if a true allergy is identified. Concomitant Candida infections should be treated as previously discussed.

■■ VAGINAL FOREIGN BODIES

Patients with a vaginal foreign body present with chronic, foul-smelling or bloody discharge. In younger girls, common items include toilet paper, toys, and small household objects. Items seen in adults include a forgotten or irretrievable tampon, packets of illegal drugs, or items used for sexual stimulation. Removal of the object is usually curative.

■■ ATROPHIC VAGINITIS

After menopause, the lack of estrogen stimulation leads to vaginal mucosal atrophy with a subsequent loss of resistance to minor trauma or infection. Bleeding can occur. The vaginal pH also increases, and subsequent changes in the vaginal flora can predispose to bacterial infection with

purulent discharge. Treatment consists primarily of topical estrogen creams. Estrogen creams should not be prescribed in the emergency department for women with prior reproductive tract cancer or postmenopausal bleeding.

■■ BARTHOLIN GLAND CYST AND ABSCESS

Bartholin glands are located in the labia minora at the 4 and 8 o'clock position. Obstruction of the gland may result in cyst or abscess formation. Patients present with pain and swelling that develops over several days. Exam will reveal pain, induration, and fluctuance. Patients without an abscess may be treated with broad-spectrum antibiotics, analgesics, and sitz baths. In addition, patients with an abscess require an incision and drainage with the placement of a Word catheter and referral for definitive surgical management.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 102, "Vulvovaginitis," by Ciara J. Barclay-Buchanan and Melissa A. Barton.

63_Cydulka_Ch63.indd 346 16/03/17 5:36 PM

Source page 366

C hapter Pelvic Inflammatory Disease

64 Abigail D. Hankin

Pelvic inflammatory disease (PID) comprises a spectrum of infections of the female upper reproductive tract, including endometritis, salpingitis, tubo-ovarian abscess, and pelvic peritonitis. Neisseria gonorrhoeae and Chlamydiatrachomatis are frequently isolated pathogens; however, many other microorganisms have been associated with PID, including vaginal flora, gram-negative rods, streptococci, and mycoplasmas. Risk factors for PID include age (teenagers are at greatest risk), multiple sexual partners, a new sexual partner within the past 30 days, presence of other sexually transmitted diseases, and intrauterine device insertion within the prior 3 weeks. PID occurs less commonly in pregnancy, but is associated with increased maternal morbidity and preterm labor. Complications of PID include significantly increased risk for ectopic pregnancy, infertility, and chronic pain, even after only a single episode of PID.

■■ CLINICAL FEATURES

Clinical diagnosis of PID is complicated due to the wide variation in symptoms and clinical findings. Common presenting complaints include lower abdominal pain (seen in >90%), vaginal discharge (75%), vaginal bleeding, urinary discomfort, fever, nausea, and vomiting. Many women with PID may have nonspecific or very subtle symptoms. Physical exam may

show lower abdominal tenderness, cervical motion tenderness, or uterine/adnexal tenderness. Mucopurulent cervicitis is a common finding, and its absence should prompt consideration of alternative causes of the patient's symptoms.

■■ DIAGNOSIS AND DIFFERENTIAL

Given the clinical variability in the diagnosis of PID and the potential sequelae of delayed treatment, the United States Centers for Disease Control and Prevention recommends empiric treatment for PID based on the minimal criteria listed in Table 64-1. No single laboratory test is highly sensitive or specific for PID. Laboratory evaluation should include a pregnancy test, wet preparation of vaginal secretions, and DNA probe or culture for N. gonorroheae and Chlamydia trachomatis. Treatment, when indicated, should not be delayed pending test results.

The differential diagnosis includes gastroenteritis, diverticulitis, ectopic pregnancy, spontaneous or septic abortion, ovarian cyst, pyelonephritis, and renal colic.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. Treatment guidelines from the Centers for Disease Control and Prevention are outlined in Tables 64-2 and 64-3. Patients with mild to moderate symptoms may be treated with oral therapy as outpatients with appropriate oral antibiotics and adequate analgesia .

64_Cydulka_Ch64.indd 347 16/03/17 5:44 PM

Source page 367

Table 64-1 Diagnostic Criteria for Pelvic Inflammatory Disease

Group 1: Minimum criteria. Empiric treatment if no other cause to explain findings.

Uterine or adnexal tenderness

Cervical motion tenderness

Group 2: Additional criteria improving diagnostic specificity.

Oral temperature >101°F (38.3°C)

Abnormal cervical or vaginal mucopurulent secretions

Elevated erythrocyte sedimentation rate

Elevated C-reactive protein level

Laboratory evidence of cervical infection with Neisseria gonorrhoeae or Chlamydia trachomatis (i.e., culture or DNA probe techniques)

Group 3: Specific criteria for PID based on procedures that may be appropriate for some patients.

Laparoscopic confirmation

Transvaginal US (or MRI) showing thickened, fluid-filled tubes with or without free pelvic

fluid or tubo-ovarian complex

Endometrial biopsy results showing endometritis

Source: Reproduced with permission from Centers for Disease Control and Prevention; Workowski KA, Berman SM. Sexually transmitted diseases treatment guidelines, 2010. MMWR Recommend Rep. 2010;59(RR-12):12.

2. Suggested criteria for admission include toxic appearance, inability to tolerate oral medication, nausea and vomiting, inability to exclude alternative diagnoses or surgical emergency, failure to respond to outpatient management, pregnancy, immunosuppression, concern for noncompliance, and tubo-ovarian abscess.

3. Outpatients should be reevaluated in the emergency department or by their gynecologist within 72 hours of emergency department discharge.

4. Provide preventative counseling and test or refer for HIV testing. The patient and their sexual partner(s) must complete the full treatment course before resuming sexual activity to prevent reinfection.

Table 64-2 Parenteral Treatment Regimens for Pelvic Inflammatory Disease

Cefotetan, 2 g IV every 12 h, or cefoxitin, 2 g IV every 6 h

plus

Doxycycline, 100 mg PO or IV every 12 h*

or

Clindamycin, 900 mg IV every 8 h

plus

Gentamicin, 2 mg/kg IV or IM loading dose, followed by gentamicin, 1.5 mg/kg every 8 h maintenance dose†

Alternative Parenteral Regimen (limited data on effectiveness)

Ampicillin/sulbactam, 3 g IV every 6 h

plus

Doxycycline, 100 mg PO or IV every 12 h*

PO doxycycline has the same bioavailability as IV doxycycline and avoids painful infusion.

*

Gentamicin dosing may be 3 to 5 mg/kg every 24 h.

†

Source: Reproduced with permission from Centers for Disease Control and Prevention; Workowski KA, Berman SM. Sexually transmitted diseases treatment guidelines, 2010. MMWR Recomm Rep. 2010;59(RR-12):12.

64_Cydulka_Ch64.indd 348 16/03/17 5:44 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 364–367.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 64: Pelvic Inflammatory Disease

Source pages 368–369 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 368–369

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Pelvic Inflammatory Disease**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 64: Pelvic Inflammatory Disease 349 Table 64-3 Oral and Outpatient Treatment Regimens for Pelvic Inflammatory Disease Ceftriaxone, 250 mg IM once, or cefoxitin, 2 g IM once, and probenecid, 1 g PO once administered concurrently or Other parenteral third-generation cephalosporin (e.g., ceftizoxime or cefotaxime) plus Doxycycline, 100 mg PO twice a day for 14 days with or without Metronidazole, 500 mg PO twice a day for 14 days If parenteral cephalosporin therapy is not feasible and community prevalence of fluoroquinolone resistance is low: Levofloxacin, 500 mg PO, or ofloxacin, 400 mg twice daily every day for 14 days with or without Metronidazole, 500 mg PO twice a day for 14 days Note: Other parenteral third-generation cephalosporins can be substituted for ceftriaxone or cefoxitin. [Source pages 368–369]

- Since the Centers for Disease Control and Prevention guidelines were published in 2006, clinically significant resistance to the fluoroquinolones (6.7% of infections in heterosexual men, an 11-fold increase from 0.6% in 2001) has emerged in the United States. [Source pages 368–369]

- Fluoroquinolone antibiotics are no longer recommended to treat gonorrhea in the United States. [Source pages 368–369]

- Fluoroquinolones may be an alternative treatment option for disseminated gonococcal infection if antimicrobial susceptibility can be documented. [Source pages 368–369]

- ■■ FURTHER READING For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 8th edition, see Chapter 103, “Pelvic Inflamatory Disease,” by Suzanne M. [Source pages 368–369]

Clinical reasoning structure

- **Problem representation:** Pelvic Inflammatory Disease; source pages 368–369.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Pelvic Inflammatory Disease**.

- Source-page range: **368–369**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 368

CHAPTER 64: Pelvic Inflammatory Disease 349

Table 64-3 Oral and Outpatient Treatment Regimens for Pelvic Inflammatory

Disease

Ceftriaxone, 250 mg IM once, or cefoxitin, 2 g IM once, and probenecid, 1 g PO once

administered concurrently

or

Other parenteral third-generation cephalosporin (e.g., ceftizoxime or cefotaxime)

plus

Doxycycline, 100 mg PO twice a day for 14 days

with or without

Metronidazole, 500 mg PO twice a day for 14 days

If parenteral cephalosporin therapy is not feasible and community prevalence

of fluoroquinolone resistance is low:

Levofloxacin, 500 mg PO, or ofloxacin, 400 mg twice daily every day for 14 days

with or without

Metronidazole, 500 mg PO twice a day for 14 days

Note: Other parenteral third-generation cephalosporins can be substituted for ceftriaxone or cefoxitin.

Since the Centers for Disease Control and Prevention guidelines were published in 2006, clinically

significant resistance to the fluoroquinolones (6.7% of infections in heterosexual men, an 11-fold

increase from 0.6% in 2001) has emerged in the United States. Fluoroquinolone antibiotics are no longer

recommended to treat gonorrhea in the United States. Fluoroquinolones may be an alternative treatment

option for disseminated gonococcal infection if antimicrobial susceptibility can be documented.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study

Guide, 8th edition, see Chapter 103, “Pelvic Inflamatory Disease,” by Suzanne M.

Shepherd, Brian Weiss, and William H. Shoff.

64_Cydulka_Ch64.indd 349 16/03/17 5:44 PM

Source page 369

C hapter Complications of Gynecologic

65 Procedures

Robert R. Cooney

The most common reasons for emergency department visits during the postoperative period after gynecologic procedures are pain, fever, and vaginal bleeding. A focused but thorough evaluation should be performed, including sterile speculum and bimanual examination. Consultation with the gynecologist who performed the procedure is indicated.

■■ COMPLICATIONS OF ENDOSCOPIC PROCEDURES

Laparoscopy

The major complications associated with laparoscopy are thermal injury of the bowel, viscus perforation, hemorrhage, vascular injury, ureteral or bladder injuries, incisional hernia, and wound dehiscence. Bowel injury should be suspected if pain is greater than expected after laparoscopy. Thermal injury is easily missed due to delayed development of symptoms for several days to weeks postoperatively. Patients with the above typically present with bilateral lower abdominal pain, fever, elevated white blood cell count, and peritonitis. X-rays can show an ileus or free air under the diaphragm. Early gynecology consultation should be obtained.

Hysteroscopy

Complications of hysteroscopy are rare but include uterine perforation, postoperative bleeding, fluid overload from absorption of distention media, gas embolism, and infection. Bleeding may originate from the uterus after resection or the cervix due to lacerations or tears. Management includes packing of the vaginal vault and consultation with a gynecologist.

■■ OTHER COMPLICATIONS OF GYNECOLOGIC PROCEDURES

Vaginal Cuff Cellulitis

Cuff cellulitis, a common early complication after hysterectomy, is an infection of the contiguous retroperitoneal space immediately above the vaginal apex and the surrounding soft tissue. Patients typically present with fever, abdominal pain, pelvic pain, back pain, and purulent vaginal discharge. Cuff tenderness and induration plus purulent discharge are prominent during the gynecologic exam. Abscesses are rare, but will present as a fluctuant mass near the cuff approximately 10 to 14 days postoperatively.

Treat with broad-spectrum antibiotics. Recommended regimens include imipenem-cilastatin, gentamicin and clindamycin, or ciprofloxacin plus metronidazole. Admit for continuation of antibiotics and consideration of drainage by interventional radiology.

350

65_Cydulka_Ch65.indd 350 21/04/17 3:52 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 368–369.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 65: Complications of Gynecologic Procedures

Source pages 370–373 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 370–373

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Complications of Gynecologic Procedures**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 65: Complications of Gynecologic Procedures 351 Postoperative Wound Infection Patients with wound infections generally present with fever and increasing pain at the surgical site. [Source pages 370–373]

- Exam will reveal erythema, tenderness, induration, and possibly incisional drain- age. [Source pages 370–373]

- Treatment includes drainage and antibiotic treatment directed at methicillin-resistant Staphylococcus aureus and streptococci. [Source pages 370–373]

- Patients with invasive infections should be admitted. [Source pages 370–373]

- Ureteral Injury Ureteral injury can occur during abdominal hysterectomy, resulting from crushing, transecting, or ligating trauma. [Source pages 370–373]

Clinical reasoning structure

- **Problem representation:** Complications of Gynecologic Procedures; source pages 370–373.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Complications of Gynecologic Procedures**.

- Source-page range: **370–373**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 370

CHAPTER 65: Complications of Gynecologic Procedures 351

Postoperative Wound Infection

Patients with wound infections generally present with fever and increasing pain at the surgical site. Onset is typically within 2 weeks of surgery. Exam will reveal erythema, tenderness, induration, and possibly incisional drainage. Treatment includes drainage and antibiotic treatment directed at methicillin-resistant Staphylococcus aureus and streptococci. Patients with invasive infections should be admitted.

Ureteral Injury

Ureteral injury can occur during abdominal hysterectomy, resulting from crushing, transecting, or ligating trauma. These patients present soon after surgery with flank pain. They may also complain of fever, costovertebral angle tenderness, and hematuria. The workup includes a urinalysis and a CT scan with IV contrast or an intravenous pyelogram to evaluate for obstruction. These patients should be admitted for ureteral catheterization and possible repair, although delayed repair after percutaneous nephrostomy is also acceptable.

Vesicovaginal Fistula

Vesicovaginal fistulas can occur after abdominal hysterectomy. Patients typically present 10 to 14 days following surgery with a watery vaginal discharge. Placing a cotton tampon in the vagina and instilling methylene blue through a urinary catheter can confirm the diagnosis. The tampon should stain within 20 minutes. Management includes placement of a Foley catheter and prompt gynecologic consultation.

Postconization Bleeding

The most common complication associated with loop electrocautery, laser ablation, and cold-knife conization of the cervix is bleeding, which can be rapid and excessive. Delayed hemorrhage can occur 1 to 2 weeks postop-

eratively. Direct visualization of the bleeding site is required. Applying Monsel's solution, direct pressure for 5 minutes with a large cotton swab, or cauterization with silver nitrate is a reasonable first step. If unsuccessful, the bleeding site may be better visualized and treated in the OR.

Septic Pelvic Thrombophlebitis

Septic pelvic thrombophlebitis is a rare complication that may follow cesarean delivery or hysterectomy. The presenting complaint is typically abdominal pain and fever. CT and MRI aid in the diagnosis but do not exclude the disease if negative. Patients are admitted for anticoagulation and IV antibiotics.

Induced Abortion

Complications associated with induced abortion may be characterized by time of occurence. These include uterine perforation, cervical lacerations, retained products of conception, and postabortal endometritis (Table 65-1). Patients with retained products of conception usually present with excessive bleeding and abdominal pain. Pelvic examination reveals an enlarged and tender uterus with an open cervical os. A pelvic ultrasound should be

65_Cydulka_Ch65.indd 351 16/03/17 5:48 PM

Source page 371

Table 65-1 Complications Associated with Induced Abortion

Timing	Complication	Possible Etiologies
Immediate complications: within 24 h postprocedure	Bleeding, pain	Uterine perforation, cervical lacerations
Delayed complications: between 24 h and 4 weeks postprocedure	Bleeding	Retained products of conception, postabortive endometritis
Late complications: >4 weeks postprocedure	Amenorrhea, psychological problems, Rh isoimmunization	—

done to confirm the diagnosis. Treatment is dilatation and curettage or medical management with misoprostol. Endometritis can occur with or without retained products of conception and is treated with antibiotics. Women who are Rh negative require Rh0 immunoglobulin, 300 μg IM, after spontaneous or induced abortion.

Assisted Reproductive Technology

Complications related to ultrasound-guided aspiration of oocytes include ovarian hyperstimulation syndrome, pelvic infection, intraperitoneal bleeding, and adnexal torsion. Ovarian hyperstimulation syndrome can be a life-threatening complication of assisted reproduction. Mild cases present with abdominal distention, ovarian enlargement, and weight gain. In severe

cases, patients have rapid weight gain, tense ascites from third spacing of fluid into the abdomen, pleural effusions, hemodynamic instability, oliguria or electrolyte abnormalities. Renal impairment and increased coagulability may also be noted. Avoid bimanual pelvic exam to prevent rupturing the ovaries. Workup includes CBC, electrolytes, liver enzymes, coagulation studies, and type and crossmatch. Initiate IV volume replacement and consult with gynecology for admission.

Postembolization Syndrome

Postembolization syndrome consists of postprocedure pelvic pain, fever, and leukocytosis lasting up to 7 days caused by myometrial and fibroid ischemia and necrosis after uterine fibroid embolization. Exam may reveal vaginal discharge of fibroid expulsion. Evaluate patients for other causes of fever and provide pain control. Patients with inadequate pain control or those in whom an infection is present may require admission.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 105, "Complications of Gynecologic Procedures," by Nikki Waller.

65_Cydulka_Ch65.indd 352 16/03/17 5:48 PM

Source page 372

SECTION

Pediatrics

9

C h apter Fever and Serious Bacterial

66 Illness in Children

Todd P. Chang

■■ FEVER AND SERIOUS BACTERIAL ILLNESS (SBI)

Fever is the most common pediatric chief complaint presenting to an emergency department (ED) and accounts for 30% of outpatient visits. Infants and children are at relatively high risk for serious bacterial illness (SBI), which includes urinary tract infection (UTI), pneumonia, bacteremia or sepsis, and meningitis—in decreasing prevalence. Neonates are at the highest risk due to their immature immune response, while infants from 1 to 3 months of age gradually transition to the lower risk profile of older infants and children. The incidence of UTI is 5% overall in children 2 months to 2 years, with a prevalence of 3% to 8% in all febrile children visiting an ED. Widespread vaccination has dropped the incidence of occult bacteremia for children 3 to 36 months of age to 0.5% to 0.7%, with further decreases expected with the 13-valent pneumococcal conjugate vaccine. Meningitis risk decreases from about 1% in the first month of life to <0.1% later in

infancy and childhood.

Clinical Features

In the neonate or infant <3 months of age, the threshold for concerning fever is 38°C (100.4°F); in infants and children 3 to 36 months old, the threshold is 39°C (102.2°F). In general, higher temperatures are associated with a higher incidence of SBI. Immature development may make reliable examination of younger infants difficult. Persistent crying, inability to console, poor feeding, or temperature instability may be the only findings suggestive of an SBI. In the neonate, there may not be any findings other than fever.

Diagnosis and Differential

Infants <3 months

An ill-appearing infant should be managed aggressively for SBI, with full sepsis evaluation, parenteral antibiotics, and admission, but a well-appearing febrile infant less than 3 months of age presents a challenge because history

66_Cydulka_Ch66.indd 353 16/03/17 5:54 PM

Source page 373

and physical examination are rarely helpful in diagnosing or excluding SBI in this age group. Meningismus is rarely present; rales may not be appreciated without strong negative inspiratory forces; and bacteremia occurs even in the well-appearing infant. A history of cough, tachypnea, or hypoxemia (by pulse oximetry), however, should alert the examiner to a possible lower respiratory tract infection and consideration of the diagnosis of pneumonia.

All febrile infants 0 to 28 days of age should receive full SBI evaluations, admission, and empiric antibiotic treatment. Antibiotic coverage in this age group includes ampicillin 50 mg/kg IV for Listeria monocytogenes and either gentamicin 2.5 mg/kg IV or cefotaxime 50 mg/kg IV for other common organisms. Avoid using ceftriaxone in this age group. Sepsis testing includes complete blood count (CBC), blood culture, urinalysis and urine culture, and lumbar puncture for CSF indices and culture. Obtain a chest radiograph if any respiratory signs described above are present, and order stool studies if the infant has diarrhea.

Febrile infants 29 to 90 days old without a focal source may be stratified to low or high risk of SBI using one of three classic criteria: Rochester Criteria, Boston Criteria, or Philadelphia Protocol. To meet low-risk SBI status, the infant must be well-appearing with a normal urinalysis and normal White Blood Cell (WBC) count, between 5,000/

mm3 and 15,000/mm3 for the Rochester and Philadelphia, and 5,000/mm3 and 20,000/mm3 for Boston. A negative CSF (WBC < 10 per hpf in Boston and WBC < 8 per hpf in Philadelphia) is also required for Boston and Philadelphia, and each criteria set has other parameters to complete a low-risk status (Table 66-1). Obtain a chest radiograph for infants with a suggestion of lower respiratory tract disease. Although there is evidence to support the use of any of these three criteria, the overall incidence of bacteremia is dropping due to two factors: empiric antibiotic treatment of women testing positive for Group B streptococcus during the third trimester and the effects from herd immunity from Haemophilus and pneumococcal vaccines.

All ill-appearing infants should receive parenteral antibiotic therapy (Table 66-2) and be admitted to the hospital. Infants older than 28 days at low risk for SBI may be managed conservatively as inpatients with ceftriaxone 50 mg/kg IM/IV pending cultures; as inpatients without antibiotics; as outpatients with ceftriaxone; or as outpatients without antibiotics. The key deciding factor should be the physician’s comfort level and the ability for close follow-up, typically within 12 to 24 hours. If antibiotics are administered (inpatient or outpatient), obtain CSF and urine and blood cultures prior to administration of antibiotics.

Well-appearing febrile infants between the ages of 29 and 90 days with an identifiable viral source of infection (e.g., respiratory syncytial virus [RSV] or influenza) should have urinary tract infection (UTI) and bacteremia ruled out before being discharged from the ED. Chest radiographs should be obtained at the discretion of the clinician, but are not indicated for well-appearing infants over 29 days with RSV. Consider deferring lumbar puncture if a virus is identified as the cause of the symptoms.

Some evidence supports the use of serum biomarkers such as C-reactive protein and procalcitonin to differentiate SBI from viral illness.

66_Cydulka_Ch66.indd 354 16/03/17 5:54 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 370–373.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 66: Fever and Serious Bacterial Illness in Children

Source pages 374–381 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 374–381

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Fever and Serious Bacterial Illness in Children**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- 66_Cydulka_Ch66.indd 355 Table 66-1 Comparison of Low-Risk Rochester Criteria, Philadelphia Protocol, and Boston Criteria for Assessment of Fever in Well-Appearing Neonates and Infants* Low-Risk Criteria for Serious Bacterial Infection* Rochester Criteria Philadelphia Protocol Boston Criteria Fever T ≥ 38°C (100.4°F) T ≥ 38.2°C (100.8°F) T ≥ 38°C (100.4°F) Age ≤ 60 days 29–56 days 28–89 days CHAPTER 66: Fever and Serious Bacterial Illness in Children 355 Past medical history Term infant ≥ 37 weeks gestation No immunodeficiency syndrome No immunizations within 48 h No perinatal or postnatal antibiotics No recent antibiotics No treatment for jaundice No chronic illnesses or admissions Not hospitalized longer than mother Physical examination Well appearing Same Same Unremarkable examination Laboratory values Blood count WBC ≥ 5000, ≤ 15,000/mm3 WBC ≤ 15,000/mm3 WBC ≤ 20,000/mm3 Absolute band count ≤ 1500/mm3 Band-to-neutrophil ratio ≤ 0.2 Urinalysis WBC ≤ 10 per high-power field WBC ≤ 10 per high-power field WBC ≤ 10 per high-power field Stool WBC ≤ 5 per high-power field – – Lumbar puncture and None WBC ≤ 8 per high-power field WBC ≤ 10 per high-power field cerebrospinal fluid findings Negative Gram stain Chest radiograph None Negative Negative if obtained Comments Excluded lumbar puncture, so number of Sensitivity of low-risk criteria for SBI 5% of low-risk neonates and infants had SBI missed meningitis cases is unknown. [Source pages 374–381]

- UTIs 98%; specificity 44%; PPV 14%; NPV (8 bacteremia, 8 UTI, 10 bacterial missed in those with negative urinalysis. [Source pages 374–381]

- The 99.7% gastroenteritis); 96% sensitive to ceftriaxone least sensitive of the low-risk criteria. [Source pages 374–381]

- 16/03/17 5:54 PM Abbreviations: NPV, negative predictive value; PPV, positive predictive value; SBI, serious bacterial illness; T, temperature; UTI, urinary tract infection. [Source pages 374–381]

- Any single deviation from the criteria is interpreted as failure of low-risk criteria. [Source pages 374–381]

Clinical reasoning structure

- **Problem representation:** Fever and Serious Bacterial Illness in Children; source pages 374–381.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Fever and Serious Bacterial Illness in Children**.

- Source-page range: **374–381**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 374

66_Cydulka_Ch66.indd 355

Table 66-1 Comparison of Low-Risk Rochester Criteria, Philadelphia Protocol, and Boston Criteria for Assessment of Fever in Well-Appearing Neonates

and Infants*

Low-Risk Criteria for Serious

Bacterial Infection* Rochester Criteria Philadelphia Protocol Boston Criteria

Fever T ≥ 38°C (100.4°F) T ≥ 38.2°C (100.8°F) T ≥ 38°C (100.4°F)

Age ≤ 60 days 29–56 days 28–89 days

CHAPTER 66: Fever and Serious Bacterial Illness in Children 355

Past medical history Term infant ≥ 37 weeks gestation No immunodeficiency syndrome No immunizations within 48 h

No perinatal or postnatal antibiotics No recent antibiotics

No treatment for jaundice

No chronic illnesses or admissions

Not hospitalized longer than mother

Physical examination Well appearing Same Same

Unremarkable examination

Laboratory values

Blood count WBC ≥ 5000, ≤ 15,000/mm3 WBC ≤ 15,000/mm3 WBC ≤ 20,000/mm3

Absolute band count ≤ 1500/mm3 Band-to-neutrophil ratio ≤ 0.2

Urinalysis WBC ≤ 10 per high-power field WBC ≤ 10 per high-power field WBC ≤ 10 per high-power field

Stool WBC ≤ 5 per high-power field – –

Lumbar puncture and None WBC ≤ 8 per high-power field WBC ≤ 10 per high-power field

cerebrospinal fluid findings Negative Gram stain

Chest radiograph None Negative Negative if obtained

Comments Excluded lumbar puncture, so number of Sensitivity of low-risk criteria for SBI 5% of low-risk neonates and infants had SBI

missed meningitis cases is unknown. UTIs 98%; specificity 44%; PPV 14%; NPV (8 bacteremia, 8 UTI, 10 bacterial

missed in those with negative urinalysis. The 99.7% gastroenteritis); 96% sensitive to ceftriaxone

least sensitive of the low-risk criteria.

16/03/17 5:54 PM

Abbreviations: NPV, negative predictive value; PPV, positive predictive value; SBI, serious bacterial illness; T, temperature; UTI, urinary tract infection.

Any single deviation from the criteria is interpreted as failure of low-risk criteria.

*

Source page 375

356 SECTION 9: Pediatrics

Table 66-2 Suggested Guidelines for the Evaluation and Management of Neonates,

Infants, and Children with Fever Who Are Well Appearing, Have Had

All Relevant Immunizations, and Have No Clinical Source for Fever

Age Group Evaluation Treatment

Neonate, 0–28 days* CBC and blood culture Admit and treat with:

of age, ≥ 38°C and Parenteral antibiotic

(100.4°F) Urinalysis and urine culture therapy with ampicillin,

SBI incidence of ill and 50 mg/kg, and either

appearing: 13%–21%; CSF cell count, Gram stain, and culture cefotaxime, 50 mg/kg,

if not ill appearing: Chest x-ray is optional, if no or gentamicin, 2.5 mg/kg

< 5% respiratory symptoms

Stool culture if diarrhea is present

Abbreviations: CSF, cerebrospinal fluid; SBI, serious bacterial illness; UTI, urinary tract infection.

*

For preterm infants, count age by estimated postconception date and not by actual delivery date for the

first 90 days of life.

Infants 3 to 36 Months

Physical examination findings become more reliable with increasing age,

though meningeal signs remain unreliable up to 2 years old. Viral illnesses

account for most febrile illnesses in this age group. Diagnostic testing

should be guided by clinical findings, including ill appearance; however,

UTIs are still prevalent in this age group, even with no focal symptoms

besides fever. Females prior to toilet training, circumcised boys younger

than 6 months of age, and uncircumcised boys under 1 year of age are at

risk for occult UTI. Obtain both a urinalysis and urine culture (by catheter-

ization or suprapubic tap) if a source for the fever is not otherwise identi-

fied. Consider CBC and blood culture for bacteremia/sepsis and CSF studies for meningitis, although these are much less common.

Older Febrile Children

The risk for bacteremia in children older than 3 years is <0.2%. CBC and blood cultures are not routinely recommended in immunized older children with fever. Etiologies to consider in older febrile children include viral infections, streptococcal pharyngitis, pneumonia, and sinusitis. Testing is directed by clinical presentation.

Emergency Department Care and Disposition

For the management of pneumonia, see Chapter 71; for the management of UTI, see Chapter 75; and infections of the ears, nose, and throat are covered in Chapter 68. Fever is not typically harmful to children, although it makes children uncomfortable and may potentiate seizures. Counsel parents against "fever phobia." To reduce a fever:

1. Remove excessive clothing and blankets to increase heat loss through radiation.
2. Administer acetaminophen 15 mg/kg PO/PR every 4 hours (up to 5 times a day).
3. Consider ibuprofen 10 mg/kg PO in children older than 1 year of age; the dose can be repeated every 6 to 8 hours (maximum of 40 mg/kg in 24 hours), and can be given concurrently with acetaminophen.

66_Cydulka_Ch66.indd 356 16/03/17 5:54 PM

Source page 376

Patients who are called to return to the ED for evaluation of positive blood cultures require repeat evaluation. Patients with positive blood cultures without contaminants should be hospitalized and treated with parenteral antibiotics. Well-appearing afebrile children already on antibiotics should complete the course of therapy. If the patient is afebrile, clinically well, without a focus of infection, and not currently on antibiotics, repeat testing or treatment is not usually necessary. If the child with a positive blood culture remains febrile or continues to appear ill, a full sepsis workup (CBC, repeat blood culture, lumbar puncture, urinalysis, and urine culture) should be performed. The patient should be hospitalized and given parenteral antibiotics (Table 66-3).

Table 66-3 Suggested Guidelines for the Evaluation and Management of Neonates, Infants, and Children with Fever Who Are Well Appearing, Have Had All Relevant Immunizations, and Have No Clinical Source for Fever

Age Group	Evaluation	Treatment

Infants 57 days* to 6 Urinalysis and urine Discharge if negative
months*of age, ≥ 38°C culture alone Treat for UTI with cefixime, 8
(100.4°F) or mg/kg/dose daily, or cefpodox-
Non-UTI SBI incidence is For conservative ime, 5 mg/kg/dose twice a day,
estimated to be negligible; management, treat or cefdinir, 7 mg/kg/dose twice
UTI is 3%–8% infants 57–90 days daily for 10 days as outpatient
using Philadelphia Admit and treat with paren-
Protocol or Boston teral ceftriaxone if fails conser-
Criteria above vative criteria for discharge
Infants 57 days to 6 Urinalysis and urine Discharge if negative
months*of age ≥ 39°C culture alone Treat for UTI as above
(102.2°F) or If WBC ≥ 15,000/mm3,
SBI incidence is estimated Urinalysis and urine cul- consider treatment with
as < 1%; non-UTI SBI ture in addition to CBC ceftriaxone, 50 mg/kg IV/IM,
incidence is estimated to be and blood culture and follow-up in 24 h
negligible; UTI is 3%–8% If WBC ≥ 20,000/mm3, consider
chest x-ray and CSF testing†
Infants/children 6 to Consider Urinalysis and Discharge if negative
36 months of age urine culture, based on Treat for UTI as above as
Non-UTI SBI incidence is the following: outpatient
< 0.4% Girls 6–24 months
UTI in girls ≤ 8% Boys 6–12 months
UTI in boys (<12 months) Uncircumcised boys
≤ 2% 12–24 months
Uncircumcised boys (1–2
years) remains 2%

Abbreviations: CSF, cerebrospinal fluid; SBI, serious bacterial illness; UTI, urinary tract infection.

*
For preterm infants, count age by estimated postconception date and not by actual delivery date for the first 90 days of life.

†
Meningismus is difficult to discern in infants < 6 months of age, and especially in infants < 2 months of age. Therefore, we recommend routine CSF testing in infants < 2 months of age, but selective CSF testing in infants 2–6 months of age. There is no absolute cutoff point for prediction of meningitis with a peripheral WBC count.

66_Cydulka_Ch66.indd 357 16/03/17 5:54 PM

Source page 377

■■ SEPSIS

Sepsis (bacteremia with clinical evidence of systemic infection) can rapidly

progress to multiorgan failure and death. Risk factors include prematurity, immunocompromised state, recent invasive procedures, and indwelling foreign objects such as catheters.

Clinical Features

Clinical signs may be vague and subtle in the young infant, including lethargy, poor feeding, irritability, or hypotonia. Fever is common; however, very young infants may be hypothermic. Tachypnea and tachycardia are usually present as a result of fever but also may be secondary to hypoxia and metabolic acidosis. Sepsis can rapidly progress to shock, manifest as prolonged capillary refill, decreased peripheral pulses, altered mental status, and decreased urinary output. Hypotension is usually a late sign of septic shock in infants and children; sustained tachycardia for age is often the only sign available to the clinician of impending shock.

Diagnosis and Differential

Diagnosis is based on clinical findings and confirmed by positive blood culture results. Though international criteria for sepsis have been published, all infants who appear toxic should be considered septic. The laboratory evaluation of a child with presumed sepsis includes CBC, blood culture, complete metabolic panel, blood lactate, catheterized urinalysis with culture and sensitivities, chest radiograph, lumbar puncture, and stool studies in the presence of diarrhea. Infants are at particular risk for hypoglycemia with SBI and sepsis, and a glucose level should always be checked.

Emergency Department Care and Disposition

1. ABCs. Administer high-flow oxygen, institute cardiac monitoring, and secure IV or IO access immediately. Perform endotracheal intubation in the presence of respiratory failure or severe respiratory distress.
2. Treat shock with 20 mL/kg boluses of 0.9% normal saline solution; preferably 60 mL/kg within 15 minutes. For infants, push actual volumes using multiple saline flushes. Repeat boluses until vital signs, perfusion, mental status, and urine output improve, up to 100 mL/kg total volume.
3. Treat hypoglycemia with 5 mL/kg 10% dextrose.
4. Initiate antibiotic therapy promptly, as soon as IV access is achieved. Do not delay due to difficulty with procedures such as lumbar puncture. Empiric antibiotic choices include cefotaxime 50 mg/kg IV or piperacillin-tazobactam 100 mg/kg IV; because of ceftriaxone-resistant S. pneumoniae, add vancomycin 15 mg/kg IV for ill-appearing infants.
5. Treat volume-refractory shock with dopamine 5 to 20 μg/kg/min or norepinephrine 0.1 to 0.2 μg/kg/min.
6. Consider the presence of drug-resistant organisms, immune-incompetence, or infection with unusual or opportunistic organisms.

■■ MENINGITIS

Meningitis is usually a complication of a primary bacteremia and has a

peak incidence in children between birth and 2 years of age. Prematurity

and immature immunity put children at higher risk.

66_Cydulka_Ch66.indd 358 16/03/17 5:54 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 374–381.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 67: Common Neonatal Problems

Source pages 382–385 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 382–385

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Common Neonatal Problems**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 67: Common Neonatal Problems 363 in neonates may present with subtle signs and symptoms including eye deviation, tongue thrusting, eyelid fluttering, periods of apnea, pedaling movements or arching. [Source pages 382–385]

- All seizures in neonates, unlike some in older infants and young children, require an extensive workup including testing of serum calcium level and admission to the hospital. [Source pages 382–385]

- Seizures need to be differentiated from benign conditions such as a Moro reflex or benign neo- natal sleep myoclonus. [Source pages 382–385]

- Benign neonatal sleep myoclonus was first described in 1982 and consists of lightning-like rhythmic jerks of the extremities when the infant is sleeping, is stopped when the neonate is touched or awakened, occurs only during sleep, requires no workup, and is best handled with reassurance. [Source pages 382–385]

- ■■ FEVER AND SEPSIS Fever in the neonate (age 28 days or younger) is defined as the history of documented fever by a parent or presence of a rectal temperature of 38°C (100.4°F) or higher in the ED. [Source pages 382–385]

Clinical reasoning structure

- **Problem representation:** Common Neonatal Problems; source pages 382–385.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Common Neonatal Problems**.

- Source-page range: **382–385**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 382

CHAPTER 67: Common Neonatal Problems 363

in neonates may present with subtle signs and symptoms including eye deviation, tongue thrusting, eyelid fluttering, periods of apnea, pedaling movements or arching. All seizures in neonates, unlike some in older infants and young children, require an extensive workup including testing of serum calcium level and admission to the hospital. Seizures need to be differentiated from benign conditions such as a Moro reflex or benign neonatal sleep myoclonus. Benign neonatal sleep myoclonus was first described in 1982 and consists of lightning-like rhythmic jerks of the extremities when the infant is sleeping, is stopped when the neonate is touched or awakened, occurs only during sleep, requires no workup, and is best handled with reassurance.

■■ FEVER AND SEPSIS

Fever in the neonate (age 28 days or younger) is defined as the history of documented fever by a parent or presence of a rectal temperature of 38°C (100.4°F) or higher in the ED. Fever in the neonate must be taken seriously, and at this point in time the proper management includes a complete sepsis workup, administration of parenteral antibiotics (e.g., cefotaxime and ampicillin), and admission to the hospital. For further detailed discussion, see Chapter 66.

■■ ALTE/BRUE

An apparent life-threatening event (ALTE), recently renamed brief resolved unexplained event (BRUE) by the American Academy of Pediatrics, is defined as an event occurring in an infant younger than 12 months involving a brief, but now resolved episode of one or more of the following: (1) cyanosis or pallor; (2) irregular, decreased, or absent breathing; (3) hyper- or hypotonia; or (4) altered responsiveness. According to the definition of BRUE, these infants appear well on presentation to the ED and there is no

explanation of the qualifying event. The new guidelines emphasize categorization into lower risk and higher risk, with lower risk criteria being: (1) age > 60 days; (2) prematurity ≥ 32 weeks, and postconceptional age ≥ 45 weeks; (3) first BRUE; (4) duration of event < 1 minute; (5) no CPR by a healthcare provider; 6) no concerning historical or physical examination features. If not considered lower risk, they are categorized as higher risk. Recommendations for further evaluation of lower risk patients show limited benefit, but the strength of the evidence to support these recommendations is only moderate to poor. Evaluation for higher risk patients may include a complete blood count; electrolytes, calcium, phosphorous, magnesium, and ammonia levels; urine toxicological screen; chest radiograph; electrocardiogram; testing for respiratory syncytial virus or pertussis; and a sepsis workup, including blood, urine, and cerebrospinal fluid. A brain CT may be included in cases of suspected nonaccidental trauma. There is no relation between ALTE/BRUE and sudden infant death syndrome, which are now considered distinct entities.

■■ JAUNDICE

There are multiple causes of jaundice, and the likelihood of any specific cause is based on the age of onset. Jaundice that occurs within the first 24 hours of life tends to be serious in nature and usually is addressed while the

67_Cydulka_Ch67.indd 363 16/03/17 6:02 PM

Source page 383

patient is in the newborn nursery. Jaundice that develops during the second or third day of life is usually physiologic; if the neonate is gaining weight, feeding and stooling well, is not anemic, does not have an elevated direct (conjugated) bilirubin level greater than one-third the total, and does not have a total bilirubin level indicating the need for phototherapy (check www.bilitool.org) reassurance and close follow-up are appropriate. Jaundice that develops after the third day of life is generally serious. Causes include sepsis, UTI, congenital TORCH infections, hemolytic anemia, biliary atresia, breast milk jaundice, and hypothyroidism. Workup of these infants usually includes a complete sepsis evaluation, including a lumbar puncture, a peripheral blood smear, complete blood count, total and direct bilirubin levels, liver function tests, reticulocyte count, and a Coombs test. Empiric antibiotics are generally administered when sepsis is suspected.

■■ ORAL THRUSH AND DIAPER RASH

Intraoral lesions due to Candida are typically white and pasty and cover the tongue, lips, gingiva, and buccal mucosa. Treatment consists of topical application of oral nystatin suspension four times a day.

Two main types of diaper rash are common in neonates: contact derma-

titis and candidal diaper dermatitis. Contact dermatitis is macular, erythematous, and has sharply demarcated edges. Treatment consists of frequent diaper changes, air drying, and application of a barrier cream containing zinc oxide. Candidal dermatitis presents with erythematous plaques with a scalloped border and satellite lesions. Treatment consists of frequent diaper changes and application of nystatin cream at least four times a day.

■■ EYE COMPLAINTS

Red eyes in neonates run the spectrum from benign to sight threatening. A common benign finding is a corneal abrasion from uncoordinated hand movements and untrimmed fingernails in neonates. Fluorescein staining is diagnostic. Cloudy corneas suggest acute glaucoma. Prompt pediatric ophthalmologic consultation is indicated. Conjunctivitis in the neonatal period may be related to sexually transmitted infections. Gonococcal conjunctivitis typically occurs in the first few days of life and presents with copious discharge. In these cases, a sepsis workup, ocular irrigation, parenteral antibiotics, prompt ophthalmology consultation, and admission to the hospital are indicated. After the first week of life, chlamydial conjunctivitis is more common and may range from mild to severe. Topical drops are inadequate treatment for chlamydial conjunctivitis. Oral erythromycin is indicated.

■■ ABDOMINAL CATASTROPHES

Surgically correctable abdominal emergencies in neonates are uncommon, may present with nonspecific symptomatology, and, when suspected, require prompt consultation with an experienced pediatric surgeon. Common signs and symptoms include irritability and crying, poor feeding, vomiting, constipation, and abdominal distention. Bilious vomiting is suggestive of malrotation with midgut volvulus and requires

67_Cydulka_Ch67.indd 364 16/03/17 6:02 PM

Source page 384

emergent surgical consultation and radiologic evaluation (upper gastrointestinal series with small bowel follow through). Projectile vomiting following feeds suggests pyloric stenosis, which is evaluated with ultrasound. A groin mass may represent an incarcerated hernia; inguinal hernias are common among premature infants.

■■ INBORN ERRORS OF METABOLISM

Typical presentations of inborn errors of metabolism include lethargy or cardiorespiratory arrest. Most neonates will have inborn errors of metabolism diagnosed after successful resuscitation. Adding a properly obtained serum ammonia level to the usual laboratories obtained during resuscitation

may assist in facilitating the diagnosis of some inborn errors of metabolism.

■■ CONGENITAL ADRENAL HYPERPLASIA

These neonates present with virilization, ambiguous genitalia, and hyperpigmentation and may be in shock on presentation. Laboratory values include hyponatremia and hyperkalemia. In addition to usual resuscitation measures, hydrocortisone administered intravenously, intramuscularly, or intraosseously at a dose of 12.5 to 25 mg is indicated.

■■ CONGENITAL HEART DISEASE AND CYANOSIS

Many disorders can present with cyanosis, and differentiating among them can be a diagnostic challenge. However, symptom patterns may help differentiate various causes and assist in suggesting the correct diagnosis and course of action. Rapid, unlabored respirations and cyanosis that does not respond to oxygen therapy suggest cyanotic heart disease with right-to-left shunting. Irregular or shallow breathing and cyanosis suggest sepsis, CNS disease, or metabolic disorders. Labored breathing with grunting and retractions is suggestive of pulmonary disease such as pneumonia or bronchiolitis. All cyanotic neonates should be admitted to the hospital for monitoring, therapy, and further investigation.

■■ NONACCIDENTAL TRAUMA

A battered child may present with unexplained bruises at different ages, skull fractures, intracranial injuries identifiable on computed tomography of the head, extremity fractures, cigarette burns, retinal hemorrhages, unexplained irritability, lethargy, or coma.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 114, "Neonatal Emergencies and Common Neonatal Problems," by Quynh H. Doan and Niranjan Kissoon; Chapter 115, "Sudden Infant Death Syndrome and Apparent Life-Threatening Event," by Claudius Ilene and Joel S. Tieder.

67_Cydulka_Ch67.indd 365 16/03/17 6:02 PM

Source page 385

C h a p te r Common Infections of the Ears,
68 Nose, Neck, and Throat

Yu-Tsun Cheng

This chapter is limited to infections of the ears, nose, neck, and throat. Further information can be found in Chapter 69 "Upper Respiratory Emergencies—Stridor and Drooling," as well as Chapter 151 "Ear, Nose and Sinus Emergencies" and Chapter 153 "Neck and Upper Airway Disorders."

■■ ACUTE OTITIS MEDIA

Acute otitis media (AOM) accounts for 7.4% of all visits to emergency

departments in the United States. AOM is an infection of the middle ear space that commonly affects young children because of relative immaturity of the upper respiratory tract, especially the eustachian tube. The most common pathogens in the post-pneumococcal vaccine era are Streptococcus pneumoniae (49%) and nontypeable Haemophilus influenzae (29%).

Clinical Features

Peak age is 6 to 18 months. Symptoms include fever, poor feeding, irritability, vomiting, ear pulling, and earache. Signs include bulging, pus behind the tympanic membrane (Fig. 68-1), an immobile tympanic membrane (TM), loss of visualization of bony landmarks within the middle ear, and bullae on the TM (bullous myringitis). Mastoiditis is the most common suppurative complication of AOM. The primary symptoms of mastoiditis include fever, protrusion of the auricle, and tenderness over the mastoid area.

Diagnosis and Differential

Making an accurate diagnosis is the most important first step. The definition of AOM requires three equally important components: (a) acute onset (<48 hours) of signs and symptoms, (b) middle-ear effusion (see Fig. 68-1), and (c) signs and symptoms of middle ear inflammation. A red TM alone does not indicate the presence of an ear infection. Fever and prolonged crying can cause hyperemia of the TM alone. Pneumatic otoscopy can be a helpful diagnostic tool; however, a retracted drum for whatever reason will demonstrate decreased mobility. Other common causes of acute otalgia are a foreign body in the external ear canal or otitis externa.

Emergency Department Care and Disposition

1. Treatment of pain is essential for all children diagnosed with AOM. Topical analgesics such as benzocaine-antipyrene are recommended for routine use, unless there is a known perforation of the TM. Acetaminophen 15 mg/kg or ibuprofen 10 mg/kg can be used.
2. Consider the use of a wait-and-see prescription for the treatment of uncomplicated AOM. Parents are given a prescription, and if the child is not better or becomes worse within 48 to 72 hours, the parents are advised to fill the prescription. Contraindications to the use of a wait-and-see prescription are as follows: age <6 months, an immunocompromised

68_Cydulka_Ch68.indd 366 20/04/17 2:43 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 382–385.
- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 68: Common Infections of the Ears, Nose, Neck, and Throat

Source pages 386–391 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 386–391

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Common Infections of the Ears, Nose, Neck, and Throat**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 68: Common Infections of the Ears, Nose, Neck, and Throat 367 Figure 68-1. [Source pages 386–391]

- Acute otitis media in a 3-year-old child with an outward bulge of the tympanic membrane and an exudative process in the middle ear space. [Source pages 386–391]

- Shelagh Cofer, Department of Otolaryngology, Mayo Clinic. [Source pages 386–391]

- state, ill-appearance, recent use of antibiotics, or the diagnosis of another bacterial infection. [Source pages 386–391]

- If any of these conditions are met, the child should be prescribed an immediate antibiotic. [Source pages 386–391]

Clinical reasoning structure

- **Problem representation:** Common Infections of the Ears, Nose, Neck, and Throat; source pages 386–391.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Common Infections of the Ears, Nose, Neck, and Throat**.

- Source-page range: **386–391**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 386

CHAPTER 68: Common Infections of the Ears, Nose, Neck, and Throat 367

Figure 68-1. Acute otitis media in a 3-year-old child with an outward bulge of the tympanic membrane and an exudative process in the middle ear space. Used with permission from Dr. Shelagh Cofer, Department of Otolaryngology, Mayo Clinic.

state, ill-appearance, recent use of antibiotics, or the diagnosis of another bacterial infection. If any of these conditions are met, the child should be prescribed an immediate antibiotic.

3. Amoxicillin 40 to 45 mg/kg/dose PO given twice daily remains the first drug of choice for uncomplicated AOM.

4. Second-line antibiotics include amoxicillin/clavulanate 40 to 45 mg/kg/dose (based on the amoxicillin component) given twice daily. Cefpodoxime 5 mg/kg/dose twice daily, cefuroxime 15 mg/kg/dose twice daily, cefdinir 7 mg/kg/dose twice daily, and ceftriaxone 50 mg/kg/dose IM for one to three daily doses are alternatives. For patients allergic to the previously mentioned antibiotics, azithromycin 10 mg/kg/dose PO on the first day followed by 5 mg/kg/dose PO for four more days can be used.

5. Infants younger than 60 days with AOM are at risk of infection with group B Streptococcus, Staphylococcus aureus, and gram-negative bacilli and should undergo evaluation and treatment for presumed sepsis.

6. In uncomplicated AOM, symptoms resolve within 48 to 72 hours; however, the middle-ear effusion may persist as long as 8 to 12 weeks. Routine follow-up is not necessary unless the symptoms persist or worsen.

7. If mastoiditis is suspected, obtain a CT scan of the mastoid. If the diagnosis is confirmed, obtain consultation with an otorhinolaryngologist and start parenteral antibiotics.

68_Cydulka_Ch68.indd 367 20/04/17 2:43 pm

Source page 387

368 SECTION 9: Pediatrics

Uncomplicated AOM is treated as an outpatient, whereas mastoiditis

typically requires inpatient treatment.

■■ OTITIS EXTERNA

Otitis externa (OE) is an inflammatory process involving the auricle, external auditory canal (EAC), and surface of the TM. It is commonly caused by Pseudomonas aeruginosa and S. aureus, which often coexist.

Clinical Features

Peak seasons for OE are spring and summer, and the peak age is 7 to 12 years. Symptoms include earache, itching, and, less commonly, fever. Signs include erythema, edema of the EAC, white exudate on the EAC and TM, pain with motion of the tragus or auricle, and periauricular or cervical adenopathy.

Diagnosis and Differential

Diagnosis for OE is based on clinical signs and symptoms. A foreign body within the external canal should be excluded by carefully removing any debris that may be present.

Emergency Department Care and Disposition

1. Cleaning the ear canal with a small tuft of cotton attached to a wire applicator is the first step. Place a wick in the canal if significant edema obstructs the EAC.
2. Consider oral analgesics, such as ibuprofen at 10 mg/kg/dose every 6 hours.
3. Fluoroquinolone-otic drops are now considered the preferred agents over neomycin containing drops. Ciprofloxacin with hydrocortisone, 0.2% and 1% suspension (Cipro HC), 3 drops twice daily or ofloxacin 0.3% solution 5 to 10 drops twice daily can be used. Ofloxacin is used when TM rupture is found or suspected.
4. Oral antibiotics are indicated if auricular cellulitis is present.

Follow-up should be advised if improvement does not occur within 48 hours; otherwise routine follow-up is not necessary. Malignant OE is characterized by systemic symptoms and auricular cellulitis. This condition can result in serious complications and requires hospitalization with parenteral antibiotics.

■■ ACUTE BACTERIAL SINUSITIS

Sinusitis is an inflammation of the paranasal sinuses that may be secondary to infection or allergy, and may be acute, subacute, or chronic. Acute bacterial sinusitis is defined as an infection of the paranasal sinuses with complete resolution in <30 days. The major pathogens in childhood are Streptococcus pneumoniae, Moraxella catarrhalis, and nontypeable Haemophilus influenzae.

Clinical Features

Two major types of sinusitis may be differentiated on clinical grounds:

acute severe sinusitis and mild subacute sinusitis. Acute severe sinusitis is

68_Cydulka_Ch68.indd 368 20/04/17 2:43 pm

Source page 388

CHAPTER 68: Common Infections of the Ears, Nose, Neck, and Throat 369

associated with elevated temperature, headaches, and localized swelling and tenderness or erythema in the facial area corresponding to the sinuses. Such localized findings are seen most often in older adolescents. Mild subacute sinusitis is manifest in childhood as a protracted upper respiratory infection associated with purulent nasal discharge persisting in excess of two weeks. Fever is infrequent. Chronic sinusitis may be confused with allergies or upper respiratory infections.

Diagnosis and Differential

The diagnosis is made on clinical grounds without laboratory or radiographic studies. Transillumination of the maxillary or frontal sinuses is seldom helpful. Nasal congestion lasting 3 to 7 days often accompanies viral upper respiratory infections and should not be diagnosed as acute sinusitis, nor treated with antibiotics. Similarly, colored drainage from the nose as a solitary symptom does not suggest a diagnosis of sinusitis and should not be treated with antibiotics. Imaging studies are not needed to confirm a diagnosis of acute bacterial sinusitis in children <6 years of age with persistent symptoms.

Emergency Department Care and Disposition

Patients with mild symptoms suggestive of a viral infection can be observed for 7 to 10 days, with no antibiotics prescribed. Suspect acute bacterial sinusitis if symptoms persist or are severe: fever >39°C, purulent nasal drainage for >3 days and ill-appearance.

1. For children with mild to moderate sinusitis, treat with amoxicillin 40 to 45 mg/kg/dose PO twice daily for 10 to 14 days.
2. For children who present with severe symptoms, are in day care, or have recently been treated with antibiotics, prescribe amoxicillin/clavulanate 40 to 45 mg/kg/dose twice a day (based on the amoxicillin component), or oral second- and third-generation cephalosporins such as cefuroxime 15 mg/kg PO twice a day or cefpodoxime 5 mg/kg PO twice a day.

■■ STOMATITIS AND PHARYNGITIS

Herpangina, hand, foot, and mouth disease (HFMD), and herpes simplex gingivostomatitis are the primary infections that cause stomatitis in children and are all viral. The vast majority of pharyngitis is caused by viral infections; however, group A β-hemolytic Streptococcus (GABHS) and Neisseria gonorrhoeae are bacterial infections that require accurate diagnoses. The identification and treatment of GABHS pharyngitis is important to prevent the suppurative complications and the sequelae of acute rheumatic fever.

Clinical Features

Herpangina causes a vesicular enanthem of the tonsils and soft palate, affecting children 6 months to 10 years of age during late summer and early fall. The vesicles are painful and can be associated with fever and dysphagia. HFMD usually begins as macules which progress to vesicles of the palate, buccal mucosa, gingiva, and tongue. Similar lesions may present on the palms of hands, soles of feet, and buttocks. Herpes simplex gingivostomatitis often presents with abrupt onset of fever, irritability, and decreased oral

68_Cydulka_Ch68.indd 369 20/04/17 2:43 pm

Source page 389

intake with edematous and friable gingiva. Vesicular lesions often with ulcerations are seen in the anterior oral cavity.

Peak seasons for GABHS are late winter or early spring, the peak age is 5 to 15 years, and it is rare before the age of 2 years. Symptoms include sore throat, fever, headache, abdominal pain, enlarged anterior cervical nodes, palatal petechiae, and hypertrophy of the tonsils. With GABHS there is usually the absence of cough, coryza, laryngitis, stridor, conjunctivitis, and diarrhea. A scarlatinaform rash associated with pharyngitis may indicate GABHS and is commonly referred to as scarlet fever.

Epstein-Barr Virus (EBV) is a herpes virus and often presents much like streptococcal pharyngitis. Common symptoms are fever, sore throat, and malaise. Cervical adenopathy may be prominent and often is posterior. Hepatosplenomegaly and splenomegaly may be present. EBV should be suspected in the child with pharyngitis nonresponsive to antibiotics in the presence of a negative throat culture.

Gonococcal pharyngitis in children and nonsexually active adolescents should alert one to the possibility of sexual abuse. Gonococcal pharyngitis may be associated with infection elsewhere including proctitis, vaginitis, urethritis, or arthritis.

Diagnosis and Differential

The diagnoses of herpangina, HFMD, and herpes simplex gingivostomatitis are based on clinical findings. To diagnose GABHS, current guidelines recommend the use of Centor criteria to determine which patients require testing: (a) tonsillar exudates, (b) tender anterior cervical lymphadenopathy, (c) absence of cough, and (d) history of fever. With two or more criteria, testing should be performed with a rapid antigen detection test and/or culture. If the rapid antigen test is negative, a confirmatory throat culture is recommended.

Diagnosis of EBV is often clinical. A heterophile antibody (monospot)

can aid in the diagnosis. The monospot may be insensitive in children <2 years of age and is often negative in the first week of illness. If obtained, the white blood cell count may show a lymphocytosis with a preponderance of atypical lymphocytes. Diagnosis of gonococcal pharyngitis is made by culture on Thayer-Martin medium. Vaginal, cervical, urethral, and rectal cultures also should be obtained if gonococcal pharyngitis is suspected.

Emergency Department Care and Disposition

1. Treatment of herpangina, HFMD, and herpes simplex gingivostomatitis is primarily supportive. Systemic analgesics such as a combination of ibuprofen and acetaminophen should be considered. Parenteral hydration may be necessary if the child cannot tolerate oral fluids. Occasionally oral narcotics may be required.

2. Antibiotics for the treatment of GABHS pharyngitis should be reserved for patients with a positive rapid antigen test or culture. Antibiotic choices for GABHS include penicillin V (children 250 mg PO twice daily, adolescent/adult 500 mg PO twice daily) for 7 to 10 days; benzathine penicillin G 1.2 million units IM (600,000 units IM for patients weighing less than 27 kg) once; or amoxicillin 50 mg/kg once daily for 7 to 10 days.

68_Cydulka_Ch68.indd 370 20/04/17 2:43 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 386–391.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 69: Upper Respiratory Emergencies—Stridor and Drooling

Source pages 392–399 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 392–399

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Upper Respiratory Emergencies—Stridor and Drooling**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 69: Upper Respiratory Emergencies—Stridor and Drooling 373 5 mL, can be used. [Source pages 392–399]

- Children with stridor associated only with agitation do not need epinephrine. [Source pages 392–399]

- Although intubation should be performed when clinically indicated, aggressive treatment with epinephrine usually prevents intubation. [Source pages 392–399]

- When necessary, consider a smaller endotracheal tube than estimated by age to avoid trauma to the inflamed mucosa. [Source pages 392–399]

- Heliox (70% helium/30% oxygen mixture) has theoretical benefits for severe refractory croup given its decreased airway resistance, though studies have shown no definitive advantage and it is limited by the low fractional concentration of oxygen. [Source pages 392–399]

Clinical reasoning structure

- **Problem representation:** Upper Respiratory Emergencies—Stridor and Drooling; source pages 392–399.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Upper Respiratory Emergencies—Stridor and Drooling**.

- Source-page range: **392–399**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 392

CHAPTER 69: Upper Respiratory Emergencies—Stridor and Drooling 373

5 mL, can be used. Children with stridor associated only with agitation do not need epinephrine.

4. Although intubation should be performed when clinically indicated, aggressive treatment with epinephrine usually prevents intubation. When necessary, consider a smaller endotracheal tube than estimated by age to avoid trauma to the inflamed mucosa.

5. Heliox (70% helium/30% oxygen mixture) has theoretical benefits for severe refractory croup given its decreased airway resistance, though studies have shown no definitive advantage and it is limited by the low fractional concentration of oxygen.

6. Children with persistent stridor at rest, tachypnea, retractions, or hypoxia, or those who require more than two treatments of epinephrine should be admitted to the hospital.

7. Discharge criteria include the following: at least 3 hours since the last dose of epinephrine, nontoxic appearance, no clinical signs of dehydration, room air oxygen saturation greater than 90%, parental ability to recognize changes in the patient's condition, and no social concerns with access to telephone and a relatively short transit time to the hospital.

■■ EPIGLOTTITIS

Epiglottitis, or supraglottitis, is life threatening and can occur at any age. Historically caused by Haemophilus influenzae type b, vaccination has decreased the occurrence of epiglottitis. In immunized children, most cases are caused by Streptococcus and Staphylococcus species. Candida may be a cause in immunocompromised patients.

Clinical Features

Classically, there is abrupt onset of high fever, sore throat, and drooling. Symptoms may progress rapidly to stridor and respiratory distress. Cough

may be absent and the voice muffled. The patient is toxic in appearance and may assume a tripod or sniffing position to maintain the airway. The presentation in older children and adults can be subtler. The only complaint may be severe sore throat, with or without stridor. The diagnosis is suggested by severe sore throat, a normal-appearing oropharynx, and a striking tenderness with gentle movement of the hyoid.

Diagnosis and Differential

Radiographs are usually unnecessary to make the diagnosis in patients with a classic presentation. If the diagnosis is uncertain, then lateral neck films should be taken at the bedside in extension and during inspiration with a minimum of disturbance. If it is necessary for the patient to be moved to the radiology suite, a physician trained in airway management should be present at all times. The epiglottis is normally tall and thin, but in epiglottitis, it is very swollen and appears squat and fat like a thumbprint (called the "thumb sign") at the base of the hypopharynx (Fig. 69-1). False-negative radiographic evaluations do occur, and, if suspicion remains, gentle direct visualization of the epiglottis is necessary to exclude the diagnosis. Blood cultures are positive in up to 90% of patients, whereas cultures from the epiglottis are less sensitive.

69_Cydulka_Ch69.indd 373 20/04/17 2:43 pm

Source page 393

Figure 69-1. Lateral neck view of a child with epiglottitis. Courtesy of W. McAlister, MD, Washington University School of Medicine, St. Louis, MO.

Emergency Department Care and Disposition

1. Keep the patient seated and upright. Provide oxygen and administer nebulized racemic or L-epinephrine. Heliox also can be attempted.
2. In the event of total airway loss, attempt bag-valve-mask ventilation.
3. Alert a referral center or pediatric otolaryngologist or anesthesiologist to coordinate decisions regarding definitive airway management.
4. The most experienced individual should perform intubation as soon as the diagnosis is made. Use sedation, paralytics, and vagolytics as

69_Cydulka_Ch69.indd 374 20/04/17 2:43 pm

Source page 394

indicated. Multiple endotracheal tube sizes must be immediately available. For the patient who is able to maintain their airway, use of paralytics must be accompanied by the certainty that intubation will be successful or that a surgical airway can immediately be performed if

unsuccessful.

5. Steroids may be employed to decrease mucosal edema of the epiglottis. Use methylprednisolone 2 mg/kg IV or dexamethasone 0.15 to 0.6 mg/kg IV.

6. Administer antibiotics: cefuroxime 50 mg/kg/dose IV, cefotaxime 50 mg/kg/dose IV, or ceftriaxone 50 mg/kg/dose IV are appropriate empiric options. In regions with increased cephalosporin resistance, vancomycin 10 mg/kg/dose IV should be added.

■■ BACTERIAL TRACHEITIS

Bacterial tracheitis (membranous laryngotracheobronchitis or “bacterial croup”) is uncommon and can present as either a primary or secondary infection. The mean age of presentation is between 5 and 8 years of age, compared with younger ages as had been previously described. It is usually caused by Staphylococcus aureus, Streptococcus pneumoniae, or β-lactamase–producing gram-negative organisms (H. influenzae and Moraxella catarrhalis).

Clinical Features

Patients with bacterial tracheitis appear toxic and have more respiratory distress than do patients with croup. They commonly present with sudden worsening of an upper respiratory infection with fever, sore throat, stridor, and cough occasionally productive of thick sputum.

Diagnosis and Differential

If obtained, radiographs of the lateral neck and chest usually demonstrate subglottic narrowing of the trachea with irregular densities and ragged and indistinct borders. Bronchoscopy, however, is most diagnostic.

Emergency Department Care and Disposition

1. Manage patients’ airways as above for epiglottitis; patients frequently require intubation. Ideally, perform intubation and bronchoscopy in the operating room where cultures and Gram stain may be obtained to guide antibiotic therapy.

2. Administer empiric parenteral antibiotics: ampicillin/sulbactam 50 mg/kg/dose IV or ceftriaxone 50 mg/kg/dose IV plus clindamycin 10 mg/kg/dose IV. In areas with increasing S. aureus resistance, consider the addition of vancomycin 10 mg/kg/dose IV.

■■ AIRWAY FOREIGN BODY

Foreign body (FB) aspirations cause more than 3000 deaths each year and have a peak incidence between ages 1 and 3 years. In children younger than 6 months, the cause is usually secondary to a feeding by a well-meaning sibling. The most common FB aspirations fall into two groups: food and

Source page 395

toys. Commonly aspirated foods include peanuts, sunflower seeds, raisins, grapes, and hot dogs, but almost any object may be aspirated. Unlike small round metal objects, aspirated vegetable matter commonly causes intense pneumonitis and subsequent pneumonia and suppurative bronchitis. A FB aspiration should be suspected if there is a history of sudden onset of coughing or choking and should be considered in all children with unilateral wheezing.

Clinical Features

At presentation many patients are asymptomatic. There may or may not be a witnessed aspiration. The primary symptom is cough, which is classically abrupt in onset, and may be associated with gagging, choking, stridor, or cyanosis. Signs depend upon the location of the FB and the degree of obstruction: stridor and hoarseness with a FB in the laryngotracheal area; unilateral wheezing and decreased breath sounds with a bronchial FB. Symptoms, however, are unreliable in localizing the level of FB. Wheeze may be present in 30% of laryngotracheal FB aspirations and stridor in up to 10% of bronchial aspirations. Eighty to 90% of FBs are located in the bronchi. Patients with immediate onset of severe stridor and cardiac arrest usually have laryngotracheal aspirations.

Diagnosis and Differential

FB aspiration is easily confused with more common causes of respiratory diseases because patients may have fever, wheezing, or rales. While plain chest radiographs may be helpful to confirm a diagnosis, they can be normal in >50% of tracheal FB and 25% of bronchial FB. More than 75% of FB in children <3 years of age are radiolucent. In cases of complete obstruction, atelectasis may be found. In partial obstructions, a ball valve effect occurs, with air trapping caused by the FB leading to hyperinflation of the obstructed lung apparent during expiration. Thus, in a stable cooperative child, inspiratory and expiratory posteroanterior chest radiographs may be helpful. In a stable but noncooperative child, decubitus films may be used, but are less sensitive than fluoroscopy. FB aspiration is definitively diagnosed preoperatively in only one-third of cases; thus, if clinically suspected, bronchoscopy is indicated.

Upper esophageal FBs are usually radiopaque and can impinge on the posterior aspect of the trachea. Patients may present with stridor, and typically have dysphagia. Radiographically, flat FBs such as coins are usually oriented in the sagittal plane when located in the trachea and in the coronal

plane when in the esophagus.

Emergency Department Care and Disposition

1. If FB aspiration or airway obstruction is clearly present, perform basic life support (BLS) procedures to relieve airway obstruction (see Chapter 3).

2. If BLS maneuvers fail, perform direct laryngoscopy and extraction of the FB with Magill forceps. If the FB cannot be seen, orotracheal intubation with dislodgement of the FB distally may be lifesaving.

3. If an obstructive laryngotracheal FB cannot be removed or dislodged, an emergent surgical airway may be required.

69_Cydulka_Ch69.indd 376 20/04/17 2:43 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 392–399.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 70: Wheezing in Infants and Children

Source pages 400–405 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 400–405

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Wheezing in Infants and Children**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 70: Wheezing in Infants and Children 381 9. [Source pages 400–405]

- Admit children to the hospital who do not respond adequately to treat- ment (e.g., persistent hypoxemia or continuing to require albuterol therapy at least every 2 hours) or whose caretaker may not be able to provide necessary ongoing care. [Source pages 400–405]

- An extremely small number of children will require intubation and mechanical ventilation despite aggressive management. [Source pages 400–405]

- It is best to try to avoid this due to challenges in ventilatory management and the potential for barotrauma; additionally, laryngoscopy during intubation may precipitate laryngospasm. [Source pages 400–405]

- In this setting, noninvasive ventilation strategies such as bilevel positive airway pressure (BiPAP) may prevent the need for mechanical ventilation. [Source pages 400–405]

Clinical reasoning structure

- **Problem representation:** Wheezing in Infants and Children; source pages 400–405.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Wheezing in Infants and Children**.

- Source-page range: **400–405**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 400

CHAPTER 70: Wheezing in Infants and Children 381

9. Admit children to the hospital who do not respond adequately to treatment (e.g., persistent hypoxemia or continuing to require albuterol therapy at least every 2 hours) or whose caretaker may not be able to provide necessary ongoing care.

10. An extremely small number of children will require intubation and mechanical ventilation despite aggressive management. It is best to try to avoid this due to challenges in ventilatory management and the potential for barotrauma; additionally, laryngoscopy during intubation may precipitate laryngospasm. In this setting, noninvasive ventilation strategies such as bilevel positive airway pressure (BiPAP) may prevent the need for mechanical ventilation. Should a patient progress to this stage, the most experienced operator available should perform the procedure, and a carefully considered sequence of rapid-sequence intubation medications chosen. These often include premedication with atropine 0.02 mg/kg (minimum 0.5 mg, maximum 1 mg) and lidocaine 1.5 mg/kg and sedation with ketamine 2 mg/kg, followed by paralysis using succinylcholine 2 mg/kg or rocuronium 1 mg/kg to provide optimal intubating conditions. Permissive hypercapnia is a strategy that may minimize barotrauma.

11. Following initial therapy, most children will improve and may be observed off of albuterol therapy. The duration of observation will vary; in general, those who were sicker on arrival should be observed at least 2 to 3 hours post-albuterol before making a decision to discharge home. Admit children to the hospital who either do not respond adequately to initial treatment or don’t maintain that response (e.g., hypoxemia, work of breathing, significant tachypnea, and/or

failure to normalize aeration).

12. Discharge planning should include an “action plan” (available at http://www.nhlbi.nih.gov/health/public/lung/asthma/actionplan_text.htm), albuterol as MDI or nebulizer, oral corticosteroids, and follow-up with the primary care provider.

■■ BRONCHIOLITIS

Bronchiolitis is the most frequent lower respiratory infection in the first 2 years of life and is most commonly caused by respiratory syncytial virus (RSV). Infection causes acute airway inflammation and edema, small airway epithelial cell necrosis and sloughing, increased mucus production and mucus plugs, and bronchospasm, all of which can vary considerably between patients and during the course of the illness.

Clinical Features

Most patients have wheezing and rhinorrhea typical of an upper respiratory infection (URI) in addition to some combination of fever, tachypnea, cough, rales, use of accessory muscles, and nasal flaring. Apnea and dehydration are of great concern and may be more common in infants younger than 3 months or those born prematurely. Hypoxemia, cyanosis, altered mental status, and fatigue are ominous signs and may portend respiratory failure. Disposition decisions should be made with consideration of signs of severity as well as the typical time course for illness: severity increases over the first 3 to 5 days with total duration of illness 7 to 14 days.

70_Cydulka_Ch70.indd 381 20/04/17 2:44 pm

Source page 401

Diagnosis and Differential

Bronchiolitis, like asthma, typically has wheezing as its most prominent clinical sign. In contrast to asthma, however, most infants with bronchiolitis are unresponsive to both β-agonists and corticosteroids. Thus, it is important to differentiate the two conditions. In general, patients with bronchiolitis are <2 years old (peaking presentation is 6 to12 months of age), do not have a past history or family history of wheezing or atopic disease, have a URI, and present during a RSV or other respiratory viral pathogen outbreak. Diagnosis is clinical and does not require laboratory or radiologic studies.

Routine testing for RSV or other pathogens is expensive and is rarely helpful; it may be indicated only if the information is necessary for infection control measures for hospitalized patients. Routine performance of radiographs increases inappropriate antibiotic use and does not change time to recovery. Radiographs are indicated only if other diseases or foreign

body aspiration are suspected, or if the patient has severe disease.

Emergency Department Care and Disposition

1. Nasal suctioning and saline drops: Deep suctioning of the nasal passages after saline instillation may substantially decrease work of breathing, correct hypoxemia, and enable the patient to feed normally. Nasal vasoconstrictors are not indicated and have resulted in tachydysrhythmias. Mildly ill, well-hydrated patients need to be treated with suctioning only.

2. Provide oxygen to maintain saturations > 90%. The use of heated, humidified, high-flow nasal cannula (HFNC) therapy provides delivery of an air/oxygen blend, up to 12 L/min in infants. Its use approximates continuous positive airway pressure to improve ventilation in a minimally invasive manner and should be considered for all moderately to severely ill infants.

3. Nebulized α- and β-agonists should not be routinely used. However, use of a β2-agonist (albuterol) might be considered for a moderately ill infant, particularly if there is a personal or family history of asthma or atopic disease. Nebulized epinephrine 0.5 mL of 1:1000 in 2.5 mL saline is not routinely effective but may be reserved for those ill enough to require hospitalization and who had a suboptimal response to β2-agonists. If these medications are used, an objective measure (e.g., respiratory rate or bronchiolitis score) should be used to assess response; lack of response should lead to a discontinuation of that therapy.

4. As with epinephrine, consider nebulized hypertonic saline 3% or 5%, 3 to 5 mL by nebulizer to decrease mucus production and viscidity for hospitalized infants unresponsive to other interventions.

5. Provide isotonic IV fluids (normal saline) if necessary. Patients with bronchiolitis may not be able to feed normally, and when respiratory rates exceed 60 to 70 breaths/min there is increased risk of aspiration of feedings.

6. Corticosteroids should not be used routinely for patients with bronchiolitis. A large multicenter study found no benefit of dexamethasone over placebo, whereas a second study found that the combination of dexamethasone and epinephrine was beneficial. However, the benefit was

70_Cydulka_Ch70.indd 382 20/04/17 2:44 pm

Source page 402

marginal and the doses of dexamethasone were large. Potential adverse effects of these medications and lack of sufficient evidence for benefit preclude their routine use.

7. Ventilatory support: Noninvasive measures (CPAP or BiPAP) may improve oxygenation and ventilation, decrease work of breathing, and delay or obviate the need for endotracheal intubation. Additionally, application of CPAP may prevent further apnea in affected infants.
8. Decision for hospitalization: Caution should be the rule when making disposition decisions about very young infants in respiratory distress due to an illness without effective outpatient therapies. Careful and serial observations (including pulse oximetry) and reassessment are keys to determining disposition. Persistent hypoxia, significant tachypnea or work of breathing, and inability to feed or maintain hydration are all indications for hospitalization. Decision-makers must also consider where in the clinical course of illness the patient resides (severity increases over the first 3 to 5 days of illness), the ability of caretakers to manage the illness, and the availability of follow-up. In addition, patients with risk factors for apnea and/or severe disease with possible respiratory failure should be identified early, including (a) young developmental age (< 6 to12 weeks) or prematurity (< 37 weeks); (b) witnessed apnea; (c) hemodynamically significant congenital heart disease: on medication for CHF; moderate to severe pulmonary hypertension; cyanotic CHD; (d) chronic lung disease: bronchopulmonary dysplasia; congenital malformations; cystic fibrosis; and (e) immunocompromised state (Table 70–1).

Table 70-1 Dosages of Medications for Asthma Exacerbations

Medication	American Pediatric Dosages	Alternate Pediatric Dosages
Bronchodilators		
Salbutamol/ albuterol (aerosol via valved holding chamber: or nebulized)	●● MDI (90 μg/puff); delivered via valved holding chamber: ●● ≤ 1 year: 2 puffs/dose ●● 1–3 years: 4 puffs/dose ●● ≥ 4 years: 8 puffs/dose	●● MDI (100 μg/puff); delivered ●● ≤ 1 year: 2 puffs/dose ●● 1–3 years: 4 puffs/dose ●● 4–6 years: 6 puffs/dose ●● ≥ 7 years: 8 puffs/dose
	●● Nebulization (unit dose nebule or 5 mg/mL solution): 0.15–0.3 mg/kg/dose (minimum 2.5 mg)	●● Nebulization (unit dose nebule or 5 mg/mL solution): ●● < 10 kg: 1.25-mg nebule or 0.25 mL of 5-mg/mL solution* ●● 10–20 kg: 2.5-mg nebule or 0.5 mL of 5-mg/mL solution* ●● > 20 kg: 5-mg nebule or 1 mL of 5-mg/mL solution*
	●● IV:	●● IV:

●● Not available ●● 0.5–3 mg/kg/h continuous

infusion (maximum 15 mg/h)

●● PO: ●● PO:

●● Not recommended ●● Not recommended

(Continued)

70_Cydulka_Ch70.indd 383 20/04/17 2:44 pm

Source page 403

384 SECTION 9: Pediatrics

Table 70-1 Dosages of Medications for Asthma Exacerbations (Continued)

Medication American Pediatric Dosages Alternate Pediatric Dosages

Ipratropium ●● MDI (20 μg/puff); delivered ●● MDI (20 μg/puff); delivered via

bromide (aerosol or via valved holding chamber: valved holding chamber:

nebulized) ●● 4–8 puffs/dose PRN, alter- ●● 3 puffs/dose for all patients,

nated with salbutamol alternated with salbutamol

●● Nebulization: ●● Nebulization:

●● 250–500 μg for all patients, ●● 250 μg for all patients, mixed

mixed with salbutamol with salbutamol

Systemic corticosteroids

Prednisone (PO) 1–2 mg/kg/dose (maximum 2 mg/kg (maximum 60 mg) in ED,

60 mg) for 3–10 days then 2 mg/kg/d bid for 4 days

Methylprednisolone 2 mg/kg/dose load, then 1 mg/kg/dose (maximum 125 mg/

(IV, IM) 0.5–1 mg/kg/dose every 6 h dose), repeated every 6 h or

change to oral regimen

Dexamethasone 0.6 mg/kg/dose (maximum ●● Multiple dosing regimens:

(PO, IV, IM) 16 mg) for 1–2 days ●● 0.6 mg/kg/d for 1–2 days

●● 0.3 mg/kg/d for 3–5 days

Other medications

Magnesium sulfate 25–75 mg/kg/dose (maximum 50 mg/kg/dose × 1 (monitor

(IV) 2 g) × 1 blood pressure)

Ketamine (IV)† 1–2 mg/kg/dose × 1 2 mg/kg/dose × 1 (may alleviate

need for intubation, given as

induction agent for intubation)

*

Mixed in 3 mL of normal saline solution.

Consider only if standard therapies have failed in order to prevent intubation.

†

Abbreviation: MDI = metered-dose inhaler.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study

Guide, 8th ed., see Chapter 124, "Wheezing in Infants and Children," by Allan

Shefrin, Alia Busuttil and Roger Zemek.

70_Cydulka_Ch70.indd 384 20/04/17 2:44 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 400–405.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 71: Pneumonia in Infants and Children

Source pages 406–407 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 406–407

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Pneumonia in Infants and Children**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 71: Pneumonia in Infants and Children 387 Table 71-2 Empiric antibiotic therapy for Pneumonia Bacterial Outpatient Inpatient Age Pathogens Treatment Treatment Neonates Group B Streptococcus Initial outpatient Ampicillin + gentamycin Gram-negative enterics treatment not or cefotaxime Listeria monocytogenes indicated 1–3 months Streptococcus Initial outpatient Ampicillin or Ceftriaxone pneumonia treatment not or Cefotaxime or Chlamydia trachomatis indicated Vancomycin. [Source pages 406–407]

- Macrolide Haemophilus influenza should be utilized if Bortadella pertussis C. [Source pages 406–407]

- pneumonia Amoxicillin Ampicillin or Ceftriaxone H. [Source pages 406–407]

- influenzae type b +/– clavulanic or Cefotaxime or Nontypeable acid or cefurox- Vancomycin +/– H. [Source pages 406–407]

- aureus 5–18 years Mycoplasma pneumonia Azithromycin (if Ampicillin or Ceftriaxone S. [Source pages 406–407]

Clinical reasoning structure

- **Problem representation:** Pneumonia in Infants and Children; source pages 406–407.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Pneumonia in Infants and Children**.

- Source-page range: **406–407**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 406

CHAPTER 71: Pneumonia in Infants and Children 387

Table 71-2 Empiric antibiotic therapy for Pneumonia

Bacterial Outpatient Inpatient

Age Pathogens Treatment Treatment

Neonates Group B Streptococcus Initial outpatient Ampicillin + gentamycin

Gram-negative enterics treatment not or cefotaxime

Listeria monocytogenes indicated

1–3 months Streptococcus Initial outpatient Ampicillin or Ceftriaxone

pneumonia treatment not or Cefotaxime or

Chlamydia trachomatis indicated Vancomycin. Macrolide

Haemophilus influenza should be utilized if

Bortadella pertussis C. trachomatis or

Staphylococcus aureus B. pertussis is suspected

3 months–5 years S. pneumonia Amoxicillin Ampicillin or Ceftriaxone

H. influenzae type b +/– clavulanic or Cefotaxime or

Nontypeable acid or cefurox- Vancomycin +/–

H. influenza ime axetil macrolide

S. aureus

5–18 years Mycoplasma pneumonia Azithromycin (if Ampicillin or Ceftriaxone

S. pneumonia M. pneumoniae or Cefotaxime or

Chlamydophila suspected) or Vancomycin +/–

pneumonia Amoxicillin macrolide

H. influenzae type b +/– clavulanic

S. aureus acid or cefurox-

ime axetil

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study

Guide, 8th ed, see Chapter 125, “Pneumonia in Infants and Children,” by

Joseph E. Copeland.
71_Cydulka_Ch71.indd 387 20/04/17 2:44 pm

Source page 407

Chapter Pediatric Heart Disease

72 Garth D. Meckler

There are six common clinical presentations of pediatric heart disease: cyanosis, shock, congestive heart failure (CHF), pathologic murmur, hypertension, and syncope. Table 72-1 lists the most common lesions in each category. While cyanosis and shock typically appear in the first weeks of life and are often dramatic in their presentation, the symptoms of CHF may be subtle and include respiratory distress or feeding intolerance, which may be misdiagnosed as viral upper respiratory tract illness, especially in winter months. A high index of suspicion must therefore be maintained in order to make the correct diagnosis. This chapter focuses on conditions producing cardiovascular symptoms seen in the emergency department (ED) that require immediate recognition, therapeutic intervention, and prompt referral to a pediatric cardiologist.

The evaluation of an asymptomatic murmur is a nonemergent diagnostic workup that can be done on an outpatient basis. Innocent murmurs, often described as flow murmurs, are of low intensity, are brief, and occur during systole. In general, common pathologic murmurs in children are typically harsh, holosystolic, continuous, or diastolic in timing, and often radiate.

Table 72-1 Clinical Presentations of Congenital Heart Disease

Clinical Presentation	Causative Conditions in Neonates	Causative Conditions in Infants and Children
Cyanosis	Transposition of the great arteries, TOF, tricuspid atresia, truncus arteriosus, total anomalous pulmonary venous return	Eisenmenger’s complex
Cardiovascular shock	Critical AS, coarctation of the aorta, HLHS (infants)	Coarctation of the aorta
Congestive heart failure	Rare: PDA, HLHS	PDA, VSD, ASD, atrioventricular canal
Murmur	PDA, valvular defects (AS, PS)	VSD, ASD, PDA, outflow obstructions, valvular defects (AS, PS)
Syncope	-	AS, PS, Eisenmenger’s complex
Hypertension	-	Coarctation of the aorta
Dysrhythmias	-	ASD, Ebstein’s anomaly,

postsurgical complication

after repair of congenital

heart defect

Abbreviations: AS = aortic stenosis; ASD = atrial septal defect; HLHS = hypoplastic left heart

syndrome; PDA = patent ductus arteriosus; PS = pulmonic stenosis; TOF = tetralogy of Fallot;

VSD = ventricular septal defect.

388

72_Cydulka_Ch72.indd 388 01/05/17 10:33 am

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.

- Content ID: SEG-TINTINALLI-EMM8-FULL.

- Source pages: 406–407.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 72: Pediatric Heart Disease

Source pages 408–417 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 408–417

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Pediatric Heart Disease**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 72: Pediatric Heart Disease 389 They may be associated with abnormal pulses or symptoms such as syn- cope or CHF. [Source pages 408–417]

- The treatment of dysrhythmias is discussed in Chapter 3, pediatric hypertension is discussed in Chapter 26, and syncope is discussed in Chapter 78. [Source pages 408–417]

- Chest pain is usually of benign etiology in children, though may occasionally represent congenital (e.g., Anomalous Left Coronary Artery arising from Pulmonary Artery, ALCAPA) or acquired (e.g., Kawasaki disease, myocarditis, pericarditis, cardiomyopathy) heart dis- ease. [Source pages 408–417]

- Myocarditis and cardiomyopathy are covered in Chapter 24, chest pain and acute coronary syndrome in Chapters 17 and 18, and Kawasaki disease in Chapter 83. [Source pages 408–417]

- ■■ CYANOSIS AND SHOCK Cardiac causes of cyanosis and shock typically present in the first 2 weeks of life and present in the critically ill neonate. [Source pages 408–417]

Clinical reasoning structure

- **Problem representation:** Pediatric Heart Disease; source pages 408–417.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Pediatric Heart Disease**.

- Source-page range: **408–417**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 408

CHAPTER 72: Pediatric Heart Disease 389

They may be associated with abnormal pulses or symptoms such as syncope or CHF.

The treatment of dysrhythmias is discussed in Chapter 3, pediatric hypertension is discussed in Chapter 26, and syncope is discussed in Chapter 78. Chest pain is usually of benign etiology in children, though may occasionally represent congenital (e.g., Anomalous Left Coronary Artery arising from Pulmonary Artery, ALCAPA) or acquired (e.g., Kawasaki disease, myocarditis, pericarditis, cardiomyopathy) heart disease. Myocarditis and cardiomyopathy are covered in Chapter 24, chest pain and acute coronary syndrome in Chapters 17 and 18, and Kawasaki disease in Chapter 83.

■■ CYANOSIS AND SHOCK

Cardiac causes of cyanosis and shock typically present in the first 2 weeks of life and present in the critically ill neonate. The differential diagnosis, however, is broad at this age, and, in addition to congenital heart disease, the clinician should consider infection (sepsis, pneumonia), metabolic disease (see Chapter 79), and nonaccidental trauma. For the neonate presenting with cyanosis, the hyperoxia test helps differentiate respiratory disease from cyanotic congenital heart disease (although imperfectly). When placed on 100% oxygen, the infant with cyanotic congenital heart disease will fail to demonstrate an increase in Pao2 or pulse oximetry, while those with respiratory causes will often respond with an improvement in Pao2 or pulse oximetry.

Clinical Features

Acral cyanosis (blue discoloration of the distal extremities) can be normal in the neonate, but central cyanosis (including the mucus membranes of the mouth) is the cardinal feature of cyanotic congenital heart disease. Appre-

ciation of cyanosis in dark-skinned neonates may be difficult, and an accurate set of vital signs including preductal and postductal pulse oximetry (i.e., right upper extremity and either lower extremity) and four-extremity blood pressures are essential. Cyanosis associated with a heart murmur strongly suggests congenital heart disease, but the absence of a murmur does not exclude a structural heart lesion. The cyanotic infant may be tachypneic, as well, though the increased respiratory rate in cyanotic heart disease is often effortless and shallow unless associated with congestive heart failure, which is rare in the first week of life.

Shock with or without cyanosis, especially during the first 2 weeks of life, should alert the clinician to the possibility of ductal-dependent congenital heart disease in which systemic (shock) or pulmonary (cyanosis) blood flow depends on patency of the fetal ductus arteriosis. Shock in the neonate is recognized by inspection of the patient's skin for pallor (or, more often, an "ashen grey" appearance), mottling, cyanosis, and assessment of the mental status appropriate for age. Mental status changes include apathy, irritability, or frank lethargy. Tachycardia and tachypnea may be the initial signs of impending cardiovascular collapse. Distal pulses should be assessed for quality, amplitude, and duration, and a differential between preductal (right brachial) and postductal (femoral) pulses or blood pressure is classic for ductal-dependent lesions such as coarctation of the aorta.

72_Cydulka_Ch72.indd 389 01/05/17 10:33 am

Source page 409

Diagnosis and Differential

The workup for congenital heart disease begins with chest radiograph and electrocardiogram (ECG) with pediatric analysis. Chest radiographs are assessed for heart size, shape, and pulmonary blood flow. An abnormal right position of the aortic arch may be a clue to the diagnosis of congenital cardiac lesion. Increased pulmonary vascularity may be seen with significant left-to-right shunting or left-sided failure. Decreased pulmonary blood flow is seen with right-sided outflow lesions such as pulmonic stenosis. Cyanotic heart lesions often demonstrate right axis deviation and right ventricular hypertrophy on ECG while left outflow obstruction (e.g., coarctation of the aorta) may show left ventricular hypertrophy. Echocardiography is generally required to define the diagnosis.

The differential diagnosis for cyanosis or shock due to congenital heart disease typically includes cyanotic lesions: transposition of the great vessels, tetralogy of Fallot, and other forms of right ventricular outflow tract obstruction or abnormalities of right heart formation. Acyanotic lesions that

can present with shock include severe coarctation of the aorta, critical aortic stenosis, and hypoplastic left ventricle. It should be noted that cyanosis may accompany shock of any cause.

Transposition of the great vessels represents the most common cyanotic defect presenting in the first week of life. This entity is easily missed due to the absence of cardiomegaly or murmur, unless there is a coexistent ventricular septal defect (VSD). Symptoms (before shock) include central cyanosis, increased respiratory rate, and/or feeding difficulty. There is usually a loud and single S2. Chest radiographs may show an “egg on a string” shaped heart with a narrow mediastinum and increased pulmonary vascular markings. ECG may show right-axis deviation and right ventricular hypertrophy.

Tetralogy of Fallot is the most common cyanotic congenital heart disease overall, and can present with cyanosis later in infancy or childhood. Physical examination reveals a holosystolic murmur of VSD, a diamond-shape murmur of pulmonary stenosis, and cyanosis. Cyanotic spells in the toddler may be relieved by squatting. Chest radiograph may show a boot-shape heart with decreased pulmonary vascular markings or a right-sided aortic arch. The ECG often demonstrates right ventricular hypertrophy and right axis deviation.

Hypercyanotic episodes, or “tet spells,” may bring children with tetralogy of Fallot to the ED in dramatic fashion. Symptoms include paroxysmal dyspnea, labored respirations, increased cyanosis, and syncope. Episodes frequently follow exertion due to feeding, crying, or straining with stools and last from minutes to hours.

Left ventricular outflow obstruction syndromes may present with shock, with or without cyanosis. Several congenital lesions fall into this category, but in all these disorders, systemic blood flow is dependent on a large contribution of shunted blood through a patent ductus arteriosus. When the ductus closes, infants present with decreased or absent perfusion, pallor or an ashen appearance, hypotension, tachypnea, and severe lactic acidosis. Diminished lower extremity pulses and BP, particularly compared to right brachial pulse and BP, are classic for coarctation of the aorta.

72_Cydulka_Ch72.indd 390 01/05/17 10:33 am

Source page 410

Emergency Department Care and Disposition

1. Cyanosis and respiratory distress are first managed with high-flow oxygen, cardiac and oxygen monitoring, and a stable intravenous or

intraosseous line. Caveats: neonates tolerate low oxygen saturations well due to oxygen-avid fetal hemoglobin; oxygen is a potent pulmonary vasodilator and may lead to “pulmonary steal” of systemic blood flow, worsening systemic shock in ductal-dependent systemic blood flow such as coarctation of the aorta. Treatment with prostaglandins (see below) is critical in these cases.

2. For severe shock in infants suspected of having shunt-dependent lesions, give prostaglandin E1 (PGE1) in an attempt to reopen the ductus. Treatment begins with 0.05 μg/kg/min and should be tapered to the lowest effective dose; the rate may be increased to 0.2 μg/kg/min if there is no improvement. Side effects include fever, skin flushing, diarrhea, and periodic apnea.

3. Obtain immediate consultation with a pediatric cardiologist and, if the patient is in shock, a pediatric intensivist.

4. Management of hypercyanotic spells consists of positioning the patient in the knee-to-chest position and administration of morphine sulfate 0.2 mg/kg SC, IM, or IO. Resistant cases should prompt immediate consultation with a pediatric cardiologist for consideration of phenylephrine for hypotension or propranolol for tachycardia.

5. Consider and treat noncardiac causes of symptoms; administer a fluid challenge of 10 to 20 mL/kg of normal saline solution and empiric administration of antibiotics as indicated. Fluids should be administered more judiciously to neonates with congenital heart disease, typically using10 mL/kg boluses.

6. Epinephrine is the initial drug of choice for hypotension. An infusion is started at 0.05 to 0.5 μg/kg/min and titrated to the desired blood pressure.

By definition, these children are critically ill and require admission, usually to the neonatal or pediatric intensive care unit.

■■ CONGESTIVE HEART FAILURE

Clinical Features

Congestive heart failure from congenital or acquired heart disease typically presents after the neonatal period, usually in the second or third month of life (congenital) or later in childhood (acquired causes). The distinction between pneumonia and CHF in infants requires a high index of clinical suspicion and is often difficult. Pneumonia can cause a previously stable cardiac condition to decompensate; thus, both problems can present simultaneously. Presenting symptoms include poor feeding, diaphoresis, irritability or lethargy with feeding, weak cry, and, in severe cases, grunting, nasal flaring, and respiratory distress. Note that the early

tachypnea of CHF in infants is typically “effortless” and is the first manifestation of decompensation, followed by increased work of breathing and rales on examination.

72_Cydulka_Ch72.indd 391 01/05/17 10:33 am

Source page 411

Diagnosis and Differential

Cardiomegaly evident on chest radiograph is universally present except in constrictive pericarditis. A cardiothoracic index greater than 0.6 on the PA chest radiograph is abnormal. The primary radiographic signs of cardiomegaly on the lateral chest radiograph are an abnormal cardiothoracic index and lack of retrosternal air space due to the direct abutment of the heart against the sternum.

Once CHF is recognized, age-related categories simplify further differential diagnosis (Table 72-2). Congenital cardiac causes of CHF are best categorized by age of onset. Early-onset CHF is associated with ductal-dependent lesions such as coarctation of the aorta and may be abrupt in onset; persistent patent ductus arteriosis (PDA) may also present in the neonatal period with CHF. Rarely, sustained tachyarrhythmias may present with CHF in the neonatal period. By contrast, lesions that result in pulmonary overcirculation such as VSD or atrial septal defect (ASD) present with gradual development of failure in the second or third month of life. Onset of CHF after age 3 months usually signifies acquired heart disease such as cardiomyopathy or myocarditis. The exception is when pneumonia, endocarditis, or another complication causes a congenital lesion to decompensate.

Cardiomyopathy presents with respiratory distress and feeding difficulties and is easily confused with upper respiratory tract infection. A pathologic gallop (S3 and or S4) is key to recognition. Rales and organomegaly are often present, and cardiomegaly and pulmonary vascular congestion are noted on chest radiography.

Myocarditis is often preceded by a viral respiratory illness, presents with nonspecific symptoms, and is more common in school age children. Presenting symptoms include shortness of breath, vomiting, poor feeding, lethargy, and fever. Signs of poor perfusion, organomegaly, and

Table 72-2 Differential Diagnosis of Congestive Heart Failure Based on Age at Presentation

Cardiac Lesion	Chest Radiograph	Electrocardiogram
Tetralogy of Fallot	Boot-shaped heart, normal-sized heart, decreased ventricular hypertrophy	Right axis deviation, right

pulmonary vascular markings

Transposition of the Egg-shaped heart, narrow Right axis deviation, right

great arteries mediastinum, increased ventricular hypertrophy

pulmonary vascular marking

Total anomalous Snowman sign, significant Right axis deviation, right

pulmonary venous cardiomegaly, increased ventricular hypertrophy, right

return pulmonary vascular markings atrial enlargement

Tricuspid atresia Heart of normal to slightly Superior QRS axis with right atrial

increased size, decreased hypertrophy, left atrial hypertrophy,

pulmonary vascular markings left ventricular hypertrophy

Truncus arteriosus Cardiomegaly, increased Biventricular hypertrophy

pulmonary vascular markings

72_Cydulka_Ch72.indd 392 01/05/17 10:33 am

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 408–417.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 73: Vomiting and Diarrhea in Infants and Children

Source pages 418–419 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 418–419

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Vomiting and Diarrhea in Infants and Children**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 73: Vomiting and Diarrhea in Infants and Children 399 Vomiting 1. [Source pages 418–419]

- Treat dehydration, hypoglycemia, and electrolyte abnormalities as men- tioned earlier. [Source pages 418–419]

- Ondansetron 0.15 mg/kg/dose should be administered (or 2 mg for 8 to <15 kg; 4 mg for 15 to <30 kg; 8 mg if ≥30 kg) as an adjunct to oral rehydration therapy in children with persistent vomiting. [Source pages 418–419]

- Oral dosing is preferred as the main objective is to support the success of oral rehydra- tion; intravenous ondansetron administration results in greater QT pro- longation and its use should be avoided. [Source pages 418–419]

- Most children can be discharged if they are tolerating oral rehydration, have adequate urine output, and ongoing fluid losses have been mini- mized. [Source pages 418–419]

Clinical reasoning structure

- **Problem representation:** Vomiting and Diarrhea in Infants and Children; source pages 418–419.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Vomiting and Diarrhea in Infants and Children**.

- Source-page range: **418–419**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 418

CHAPTER 73: Vomiting and Diarrhea in Infants and Children 399

Vomiting

1. Treat dehydration, hypoglycemia, and electrolyte abnormalities as mentioned earlier.

2. Ondansetron 0.15 mg/kg/dose should be administered (or 2 mg for 8 to <15 kg; 4 mg for 15 to <30 kg; 8 mg if ≥30 kg) as an adjunct to oral rehydration therapy in children with persistent vomiting. Oral dosing is preferred as the main objective is to support the success of oral rehydration; intravenous ondansetron administration results in greater QT prolongation and its use should be avoided.

3. Most children can be discharged if they are tolerating oral rehydration, have adequate urine output, and ongoing fluid losses have been minimized. Continuation of a normal diet (including lactose-containing milk or formula) is recommended. Patients who cannot tolerate oral fluids, have significant ongoing losses, severe electrolyte abnormalities, or surgical abdominal processes, require admission to the hospital.

Diarrhea

1. Treat dehydration and hypoglycemia as above.

2. Children with mild diarrhea who are not dehydrated may continue routine feedings. Do not withhold feedings > 4 hours in a dehydrated child or for any length of time in a child who is not dehydrated. There is no need to dilute formula because more than 80% of children with acute diarrhea can tolerate full-strength milk safely.

3. Antidiarrheal and antimotility agents such as loperamide are not recommended in children and are contraindicated in young children.

4. Antibiotics are unnecessary for the vast majority of children with acute gastroenteritis and may be associated with increased risk for hemolytic uremic syndrome. See Table 73-2 for specific treatment recommenda-

tions by pathogen.

5. All infants and children who appear toxic or have high-risk social situations, significant dehydration, significant ongoing fluid losses, altered mental status, inability to drink, or laboratory evidence of hemolytic anemia, thrombocytopenia, azotemia, or a significant sodium abnormality should be admitted.

6. Children who respond to oral or intravenous hydration can be discharged. Instructions should be given to return to the emergency department or seek care with the primary care provider if the child becomes unable to tolerate oral hydration, develops bilious vomiting, becomes less alert, or exhibits signs of dehydration, such as no longer wetting diapers. Dietary recommendations include a diet high in complex carbohydrates, lean meats, vegetables, fruits, and yogurt. Fatty foods and foods high in simple sugars should be avoided. The BRAT diet is discouraged because it does not provide adequate energy sources.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th edition, see Chap. 128, "Vomiting, Diarrhea, and Dehydration in Infants and Children," by Stephen B. Freedman and Jennifer D. Thull-Freedman.

73_Cydulka_Ch73.indd 399 20/04/17 2:45 pm

Source page 419

C hap t e r Pediatric Abdominal

74 Emergencies

Janet Semple-Hess

■■ ABDOMINAL PAIN

The assessment of acute abdominal pain can be challenging given the preverbal state of young children, the varied number of diagnoses that present similarly, and increasing appreciation of risks associated with pediatric diagnostic imaging.

Clinical Features

Presenting signs and symptoms differ by age. The key gastrointestinal (GI) signs and symptoms include pain, vomiting, diarrhea, constipation, fever, jaundice, and masses. Abdominal pain in children younger than 2 years typically manifests as fussiness, irritability, lethargy, or grunting. Toddlers and school-age children often localize pain poorly and point to their umbilicus, or may present as refusal to ambulate. Pain may be peritoneal and exacerbated by motion, or spasmodic, and associated with restlessness. Abdominal pain may originate from non-GI sources, and associated symptoms may help localize extra-abdominal causes such as cough with pneumonia or sore throat in streptococcal pharyngitis.

Vomiting and diarrhea are common in children. These symptoms may be the result of a benign process or indicate the presence of a life-threatening condition. Bilious vomiting is almost always indicative of a serious process, especially in the neonate. Upper GI bleeding in children presents with hematemesis, which is often frightening to caretakers, but rarely serious in an otherwise healthy infant or child. Lower GI bleeding presents with melena or hematochezia, and the distinction between painless and painful rectal bleeding can help differentiate likely etiologies. Jaundice can be an ominous sign, and sepsis, congenital infections, hepatitis, anatomic problems, and enzyme deficiencies should be considered in the evaluation of these patients. Abdominal masses may be asymptomatic (e.g., Wilms tumor) or associated with painless vomiting (e.g., pyloric stenosis) or colicky abdominal pain (e.g., intussusception).

Diagnosis and Differential

Obtain a thorough history from parent and child (if possible), including the quality and location of pain, chronology of events, feedings, bowel habits, fever, weight changes, and other systemic signs and symptoms. Begin the physical examination with an assessment of the child's overall appearance, vital signs, and hydration status. Observation should precede auscultation and palpation. Extra-abdominal areas including the chest, pharynx, testes, scrotum, inguinal area, and neck should also be evaluated. Adolescent females who are sexually active with lower abdominal pain may require a bimanual exam. The likely etiologies of abdominal pain vary with age. Table 74-1 classifies emergent and nonemergent conditions by age group.

74_Cydulka_Ch74.indd 400 20/04/17 2:46 pm

References/provenance

- Source: *Tintinalli's Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 418–419.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 74: Pediatric Abdominal Emergencies

Source pages 420–429 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 420–429

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Pediatric Abdominal Emergencies**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 74: Pediatric Abdominal Emergencies 401 Table 74-1 Causes of Abdominal Pain by Age Group Age Emergent Nonemergent 0–3 months Necrotizing enterocolitis Constipation Volvulus Acute gastroenteritis Incarcerated hernia Colic Testicular torsion Nonaccidental trauma Hirschsprung’s enterocolitis 3 months to 3 years Intussusception Urinary tract infections Volvulus Constipation Testicular torsion Henoch–Schönlein purpura Appendicitis Acute gastroenteritis Vaso-occlusive crisis Tumor Pneumonia Urinary tract infections 3 years through Appendicitis Streptococcus pharyngitis ado lescence Diabetic ketoacidosis Inflammatory bowel disease Vaso-occlusive crisis Pregnancy Ectopic pregnancy Renal stones Ovarian torsion Peptic ulcer disease/gastritis Testicular torsion Ovarian cysts Cholecystitis Henoch–Schönlein purpura Pancreatitis Constipation Urinary tract infections Acute gastroenteritis Tumor Nonspecific viral syndromes Pneumonia Neonates and Young Infants (0 to 3 Months) Life-threatening abdominal conditions in young infants include necrotizing enterocolitis, malrotation with midgut volvulus, incarcerated hernias, and nonaccidental trauma. [Source pages 420–429]

- Other urgent conditions include pyloric stenosis and testicular torsion. [Source pages 420–429]

- Inconsolability, lethargy, and poor feeding may be the only indication of serious underlying disease. [Source pages 420–429]

- Bilious vomiting in an infant indicates intestinal obstruction and should be considered a surgical emer- gency until proven otherwise. [Source pages 420–429]

- Common non-life-threatening causes of abdominal pain in young infants include colic (see Chapter 67) milk- protein allergy or gastroesophageal reflux. [Source pages 420–429]

Clinical reasoning structure

- **Problem representation:** Pediatric Abdominal Emergencies; source pages 420–429.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Pediatric Abdominal Emergencies**.

- Source-page range: **420–429**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 420

CHAPTER 74: Pediatric Abdominal Emergencies 401

Table 74-1 Causes of Abdominal Pain by Age Group

Age Emergent Nonemergent

0–3 months Necrotizing enterocolitis Constipation

Volvulus Acute gastroenteritis

Incarcerated hernia Colic

Testicular torsion

Nonaccidental trauma

Hirschsprung’s enterocolitis

3 months to 3 years Intussusception Urinary tract infections

Volvulus Constipation

Testicular torsion Henoch–Schönlein purpura

Appendicitis Acute gastroenteritis

Vaso-occlusive crisis

Tumor

Pneumonia

Urinary tract infections

3 years through Appendicitis Streptococcus pharyngitis

ado lescence Diabetic ketoacidosis Inflammatory bowel disease

Vaso-occlusive crisis Pregnancy

Ectopic pregnancy Renal stones

Ovarian torsion Peptic ulcer disease/gastritis

Testicular torsion Ovarian cysts

Cholecystitis Henoch–Schönlein purpura

Pancreatitis Constipation

Urinary tract infections Acute gastroenteritis

Tumor Nonspecific viral syndromes

Pneumonia

Neonates and Young Infants (0 to 3 Months)

Life-threatening abdominal conditions in young infants include necrotizing enterocolitis, malrotation with midgut volvulus, incarcerated hernias, and nonaccidental trauma. Other urgent conditions include pyloric stenosis and testicular torsion. Inconsolability, lethargy, and poor feeding may be the only indication of serious underlying disease. Bilious vomiting in an infant indicates intestinal obstruction and should be considered a surgical emergency until proven otherwise. Common non-life-threatening causes of abdominal pain in young infants include colic (see Chapter 67) milk-protein allergy or gastroesophageal reflux. Fever requires thorough investigation for a source (see Chapter 66). Other causes of irritability in infants should be considered, including hair or thread tourniquets of the digits and genitalia, and corneal abrasions (see Chapter 67).

Helpful studies in this age group include abdominal two-view radiographs to identify obstruction, pneumatosis intestinalis, or free air; abdominal ultrasound to diagnose pyloric stenosis, testicular torsion, and hernias; and upper GI contrast studies to identify malrotation. Useful laboratory studies include serum electrolytes to identify abnormalities resulting from vomiting and dehydration and a CBC, coagulation panel, and type and screen if concern is for a surgical abdomen with perforation.

Malrotation of the intestine can present with life-threatening volvulus. Symptoms include bilious vomiting, abdominal distention, and obstipation,

74_Cydulka_Ch74.indd 401 20/04/17 2:46 pm

Source page 421

or, occasionally, streaks of blood in the stool. The vast majority of cases present within the first month of life. Patients are ill-appearing and may present in compensated or decompensated shock. A “bird’s beak” appearance on an upper GI series is suggestive of this diagnosis. Immediate surgical consultation and aggressive fluid resuscitation are critical to improve outcomes.

Pyloric stenosis usually presents with progressive nonbilious, projectile vomiting occurring just after feeding. Pyloric stenosis occurs most commonly in the third or fourth week of life, but may be seen anywhere from 1 to 12 weeks of age. A left upper quadrant pyloric mass, or “olive,” may be present, but is usually difficult to detect, and peristaltic waves may be

noted following a feeding trial in the ED. Ultrasound is the imaging modality of choice. Electrolytes may demonstrate a characteristic hypochloremic metabolic alkalosis, which must be corrected prior to definitive surgical care. While pyloric stenosis is not a surgical emergency, the resultant dehydration from persistent vomiting requires immediate medical treatment.

Necrotizing enterocolitis, although rare outside of the neonatal unit, should be considered in any infant younger than 3 weeks of age, especially if premature. Vomiting, poor feeding, and abdominal distention may be present, and abdominal films may show pneumatosis intestinalis, as well as portal vein gas. Immediate surgical/neonatology consultation, NPO, IVF hydration, and broad-spectrum antibiotics for intestinal flora should be instituted if suspected.

Hirschsprung's disease is a pathologic cause of constipation in this age group, as true constipation is unusual. Key history for Hirschprung's disease is delayed passage of meconium at birth (more than 24 hours). Symptoms include infrequent and often explosive bowel movements, poor feeding and growth, and progressive abdominal distension. Toxic megacolon may occur as a complication and if suspected, emergent surgical consultation is warranted.

Older Infants and Toddlers (3 Months to 3 Years)

The differential diagnosis of acute abdominal pain in this age group includes intussusception, gastroenteritis (see Chapter 73), constipation, urinary tract disease (see Chapter 75), and nonaccidental trauma. Though less common in this age group, acute appendicitis, diabetic ketoacidosis or other metabolic diseases, tumors such as Wilm's, neuroblastoma or hepatoblastoma and malrotation with midgut volvulus must be considered.

Imaging studies are guided by the differential diagnosis: radiographs can help rule out free air or obstruction, and confirm suspected constipation; ultrasound can help identify intussusception, abdominal or renal mass, or appendicitis, and air-contrast enema is both diagnostic and potentially therapeutic for intussusception. A CBC and electrolytes may be helpful in evaluating complications of vomiting and diarrhea, and urinalysis will identify pyelonephritis as a potential cause of abdominal pain in this age group. A bedside glucose is helpful in the lethargic child or in cases with persistent or prolonged vomiting.

Intussusception occurs when one portion of the bowel telescopes into another, which can result in a partial or complete obstruction, bowel-wall edema, and eventually bowel ischemia. The greatest incidence occurs between 6 to 18 months of age, and it is the most common cause of

Source page 422

A B

Figure 74-1. A and B. Ultrasound image of intussusception showing the classic target appearance of bowel-within-bowel. Reproduced with permission from Ma OJ, Mateer JR, Blaivas M: Emergency Ultrasound, 2nd ed. New York: McGraw-Hill; 2008.

intestinal obstruction in children under 2 years of age. The classic presentation of intermittent paroxysms of abdominal pain with pain-free intervals (or lethargy), vomiting (may be bilious), and “currant jelly stool” are not present in all patients, and bloody stools is a late sign. Stool testing may reveal occult blood in 70% of intussusception cases. Providers must have a high index of suspicion for intussusception in patients presenting with non-specific changes in mental status or who are ill-appearing without any apparent etiology. Ultrasound should be performed if the diagnosis is ambiguous, and may demonstrate the classic “target sign” (see Fig. 74-1). In more classic cases, air or barium enema should be performed, and can be both diagnostic and therapeutic. Radiologic reduction requires a readily available pediatric surgeon for irreducible cases or perforation during the procedure. Uncomplicated cases of intussusception successfully reduced by enema may be discharged after observation if tolerating oral fluids and has reliable follow-up, as there is a 10% recurrence risk.

Constipation, defined as infrequent, hard stools, is a common cause of abdominal pain in children and may be a sign of either a pathologic (e.g., Hirschsprung disease, cystic fibrosis, spinal cord abnormality, or infant botulism) or functional process. History is key to the diagnosis. For neonates, verify passage of meconium in the first 24 hours of life. A rectal examination is recommended to assess presence of stool, rectal tone, sensation, and size of the anal vault. A careful neurologic examination should be completed to assess for neuromuscular causes. A single upright abdominal radiograph may be helpful to visualize fecal retention or impaction, and to help rule out the concern for obstruction. Treatment in the ED with suppositories or enemas may be necessary, and outpatient maintenance therapy such as with polyethylene glycol for older children (older than 12 months) is essential to prevent recurrence. Admission is indicated for patients with impaction associated with vomiting, dehydration, and failure of outpatient treatment.

Children (3 to 15 Years Old)

Acute abdominal pain in children 3 to 15 years old includes a range of diagnoses, including appendicitis (Chapter 43), constipation, gastroenteritis,

74_Cydulka_Ch74.indd 403 20/04/17 2:46 pm

Source page 423

urinary tract infection, streptococcal pharyngitis (Chapter 68), pneumonia (Chapter 71), pancreatitis (Chapter 42), and functional abdominal pain. In adolescents, gall bladder disease (cholelithiasis, biliary colic, or cholecystitis), renal stones, ovarian pathology (cysts or torsion), and testicular torsion should be considered. Diabetes and diabetic ketoacidosis are more prevalent in this age group and commonly present as vague abdominal pain. The presence of a parent and allowing the child to sit on the parents' lap are helpful in examining younger children. Older children may not readily offer important history surrounding embarrassing topics such as constipation and genital pain. Adolescents should be interviewed alone to provide confidentiality and facilitate discussion of potentially important information surrounding sexual activity and other sensitive subjects.

Henoch–Schönlein purpura (HSP) is an idiopathic vasculitis of children between 2 and 11 years of age. HSP classically presents with palpable purpura, acute abdominal pain, and arthritis. A urinalysis should be obtained to identify hematuria, with subsequent renal function tests when blood is present, as up to 30% of children with HSP will develop renal involvement, sometimes months later. Abdominal pain in HSP is caused by bowel wall edema and GI vasculitis, and may occur before typical rash of HSP appears. Microscopic and even gross GI bleeding is not uncommon, though rarely life-threatening. Intussusception (ileo-ileo as compared to the classic ileo-colic in idiopathic intussception) occurs in 3.5%, and must be considered in patients with significant abdominal pain. Treatment of HSP is primarily supportive: joint pain responds well to nonsteroidal anti-inflammatory medications; severe abdominal pain may improve with corticosteroids. Consultation and follow-up with a pediatric rheumatologist or nephrologist may be necessary for more severe symptoms or renal involvement with hypertension.

■■ GASTROINTESTINAL BLEEDING

Upper GI (UGI) bleeding is distinguished from lower GI (LGI) bleeding anatomically at the ligament of Treitz. The presentation of GI bleeding in children varies, and includes bright red blood in small strands or clots in emesis or stool, vomiting of gross blood (hematemesis), black tarry stools (melena), or profuse bright red blood per rectum. Occult bleeding may present with pallor, fatigue, or anemia.

The first step in management is to evaluate the need for acute resuscitation, and distinguish the bleed as UGI or LGI. Confirmation through gastric

or fecal occult blood testing is useful to distinguish substances that grossly mimic blood. Foods such as beets, cranberries, and red food dyes can appear like blood in stool. A thorough history and physical examination is important to determine the location, amount, and etiology of the bleed. Examination for extra-GI sources is also important (e.g., epistaxis is a common cause of hematemesis in children). Ask about recent dental surgery or gingival bleeding. Post-tonsillectomy bleeding is an important and potentially serious cause of hematemesis. A common cause of hematemesis in the newborn is swallowed maternal blood (either post-delivery or from fissures in mother's nipples).

The differential diagnosis of UGI and LGI bleeding by age group is summarized in Table 74-2. In general, UGI bleeding in children without

74_Cydulka_Ch74.indd 404 20/04/17 2:46 pm

References/provenance

- Source: *Tintinalli's Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 420–429.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 76: Seizures and Status Epilepticus in Children

Source pages 430–431 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 430–431

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Seizures and Status Epilepticus in Children**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 76: Seizures and Status Epilepticus in Children 411 over 20 minutes; valproic acid 20 mg/kg IV/IO; and levetiracetam 20 to 30 mg/kg IV/IO. [Source pages 430–431]

- Fourth-line treatment strategies include continuous infusions in the ICU setting of propofol, midazolam, or pentobarbital, or induction of general inhaled anesthesia. [Source pages 430–431]

- Propofol dose is 0.5 to 2 mg/kg or 1.5 to 4 mg/kg/hr IV/IO; or midazolam continuous infusion: 0.05 to 0.4 mg/kg/hr IV/IO. [Source pages 430–431]

- Treat electrolyte abnormalities as follows: for hyponatremia < 120 mEq/L, 3% NaCl 4 to 6 mL/kg, for hypocalcemia, < 7 mg/dL of calcium and < 0.8 mmol/L of ionized calcium, 0.3 mL/kg of 10% calcium gluconate, over 10 minutes, and for hypomagnesemia < 1.5 mEq/L, 50 mg/kg of magnesium sulfate over 20 minutes. [Source pages 430–431]

- Children who present to the ED following a brief seizure and who regain a normal mental status without focal neurologic deficits are candidates for discharge and outpatient follow-up. [Source pages 430–431]

Clinical reasoning structure

- **Problem representation:** Seizures and Status Epilepticus in Children; source pages 430–431.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Seizures and Status Epilepticus in Children**.

- Source-page range: **430–431**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 430

CHAPTER 76: Seizures and Status Epilepticus in Children 411

over 20 minutes; valproic acid 20 mg/kg IV/IO; and levetiracetam 20 to 30 mg/kg IV/IO.

6. Fourth-line treatment strategies include continuous infusions in the ICU setting of propofol, midazolam, or pentobarbital, or induction of general inhaled anesthesia. Propofol dose is 0.5 to 2 mg/kg or 1.5 to 4 mg/kg/hr IV/IO; or midazolam continuous infusion: 0.05 to 0.4 mg/kg/hr IV/IO.

7. Treat electrolyte abnormalities as follows: for hyponatremia < 120 mEq/L, 3% NaCl 4 to 6 mL/kg, for hypocalcemia, < 7 mg/dL of calcium and < 0.8 mmol/L of ionized calcium, 0.3 mL/kg of 10% calcium gluconate, over 10 minutes, and for hypomagnesemia < 1.5 mEq/L, 50 mg/kg of magnesium sulfate over 20 minutes.

8. Children who present to the ED following a brief seizure and who regain a normal mental status without focal neurologic deficits are candidates for discharge and outpatient follow-up. Patients presenting to the ED with ongoing seizures meet the definition for status epilepticus and even when seizure termination is achieved in the ED, it is suggested that they be admitted for further observation. All infants with true seizures should be admitted to the hospital. Patients with refractory seizures, those with seizures from acute traumatic brain injury, CNS infection, or serious metabolic or electrolyte abnormalities typically require treatment in the ICU setting.

Febrile Seizure

Seizures in the setting of fever are common in children and are usually benign. Simple febrile seizures (SFS) occur between 6 months and 6 years of life, are brief (<15 minutes), and generalized. Patients with SFS require no specific ED evaluation or medication for their seizure, though evaluation

for a treatable source of fever such as a urinary tract infection, and antipyretics may be indicated. Children experiencing SFS have the same 1% risk of developing epilepsy as the general population. Complex febrile seizures (>15 minutes in duration, focal, or recurrent) carry a slightly higher risk for epilepsy, but also do not routinely require ED evaluation or treatment. Febrile status epilepticus is treated as above with the addition of neuroimaging, CSF analysis and culture, and potential antibiotics and acyclovir as indicated.

First Seizure

The overall risk of recurrence after a single afebrile seizure is about 40%. Emergent neuroimaging is not necessary in the ED in the patient with a normal neurologic exam, though outpatient MRI and EEG may be of benefit. Most neurologists do not recommend starting anticonvulsant medication after a first seizure.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 135, "Seizures in Infants and Children," by Maija Holsti.

76_Cydulka_Ch76.indd 411 20/04/17 2:47 pm

Source page 431

C ha p t er Altered Mental Status and
77 Headache in Children

Carlo Reyes

■■ HEADACHE

Pediatric headache accounts for 1% of emergency department visits. The vast majority of headaches in children have a benign etiology. Factors associated with serious or dangerous causes of headache include preschool age, occipital location, recent onset of headache, and inability of the child to describe the quality of the head pain.

Clinical Features

Headaches can be classified as primary or secondary. Primary headaches are associated with normal physical exam findings, are self-limited, and may be recurrent. Examples of primary headache include physiologic or functional (migraine, tension, cluster). Secondary headaches are usually anatomic (vascular malformation, tumor) or infectious (sinusitis, dental abscess), and have higher morbidity and mortality if left untreated. A careful history and physical examination should focus on identifying or excluding secondary headaches. Historical findings that suggest secondary headache include acute onset; associated fever or stiff neck; morning vomiting; behavioral changes; altered mental status; "worst ever" head-

ache; sleep disturbance; associated trauma or toxic exposure; or aggravation by coughing, Valsalva, or lying down. Physical findings suggestive of secondary headaches include blood pressure abnormalities, nuchal rigidity, head tilt, ptosis, retinal hemorrhage or papilledema, visual field defects, and neurologic findings (altered mental status, ataxia, and hemiparesis).

Diagnosis and Differential

Pertinent historical features of pediatric headache include severity and quality of pain, age of first occurrence, precipitants, rapidity of onset, location, duration, and associated symptoms. Physical examination should include a thorough general examination with careful attention to the neurologic examination, including cranial nerves, gait, strength, and mental status. The history and physical exam should guide the work up. For instance, CT is indicated to rule out suspected intracranial hemorrhage. Magnetic resonance imaging better visualizes the posterior fossa, the most common location of pediatric brain tumors, and either modality may be used for space-occupying lesions elsewhere. Importantly, neuroimaging in children is usually not indicated unless there is an abnormal neurologic exam, altered mental status, concurrent seizures, a recent “worst headache of life” complaint, or change in type of headache.

412

77_Cydulka_Ch77.indd 412 20/04/17 2:47 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 430–431.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 77: Altered Mental Status and Headache in Children

Source pages 432–437 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 432–437

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Altered Mental Status and Headache in Children**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 77: Altered Mental Status and Headache in Children 413 Emergency Department Care and Disposition 1. [Source pages 432–437]

- For secondary headaches, evaluate and treat underlying cause and pain. [Source pages 432–437]

- For primary headaches, narcotics are not recommended. [Source pages 432–437]

- Most pri- mary headaches respond to ibuprofen 10 mg/kg or acetaminophen 15 mg/kg. [Source pages 432–437]

- For migraines: prochlorperazine 0.15 mg/kg, metoclopramide 0.1 mg/kg, or ketorolac 0.5 mg/kg are widely used for abortive migraine therapy. [Source pages 432–437]

Clinical reasoning structure

- **Problem representation:** Altered Mental Status and Headache in Children; source pages 432–437.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Altered Mental Status and Headache in Children**.

- Source-page range: **432–437**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 432

CHAPTER 77: Altered Mental Status and Headache in Children 413

Emergency Department Care and Disposition

1. For secondary headaches, evaluate and treat underlying cause and pain.

2. For primary headaches, narcotics are not recommended. Most primary headaches respond to ibuprofen 10 mg/kg or acetaminophen 15 mg/kg.

3. For migraines: prochlorperazine 0.15 mg/kg, metoclopramide 0.1 mg/kg, or ketorolac 0.5 mg/kg are widely used for abortive migraine therapy. Metoclopramide and prochlorperazine are often given with diphenhydramine 1 mg/kg to prevent extrapyramidal side effects. “Migraine cocktails” that include ketorolac, prochlorperazine or metoclopramide, and diphenhydramine appear effective and safe. Sumatriptan 5 to 10 mg (in children weighing 20 to 39 kg) or 20 mg (> 40 kg children) nasal spray or 0.1 mg/kg subcutaneously is commonly used.

4. Cluster and tension headaches are managed much the same way as migraines. Sumatriptan as dosed above and high-flow oxygen (7 L/min non-rebreather mask) can be used for cluster headaches. Tension headaches usually respond to first-line oral therapy such as ibuprofen 10 mg/kg.

5. For headaches that disrupt activities of daily living or school performance, refer to their primary care provider or pediatric neurologist to consider prophylactic regimens.

6. In general, most patients may be discharged after relief of symptoms. Patients with life-threatening causes of headache require admission for definitive care. Patients with intractable pain also may need admission.

■■ ALTERED MENTAL STATUS

Altered mental status (AMS) in a child is defined as the failure to respond

to the external environment in a manner consistent with the child's developmental level. In treating children with AMS, aggressive resuscitation, stabilization, diagnosis, and treatment must occur simultaneously to prevent morbidity and death.

Clinical Features

The spectrum of AMS ranges from confusion to lethargy, stupor, and coma indicative of depression of the cerebral cortex or localized abnormalities of the reticular activating system. Important historical elements to consider include prodromal events (recent illnesses, exposures, trauma), risk factors (medications, social or family history, potential for abuse), and other associated symptoms. A head-to-toe exam should be performed to identify occult infection, trauma, toxicity, or metabolic disease. A child's mental status can be assessed by the Glasgow coma scale, or the simpler AVPU scale, where A means "alert," V means "responsive to verbal stimuli," P means "responsive to painful stimuli," and U means "unresponsive," corresponding to a GCS of 15, 13, 8, and 3, respectively.

Pathologic causes of altered mental status can be divided into supratentorial lesions, subtentorial lesions, and metabolic encephalopathy. Supratentorial lesions are due to compression of the brainstem, resulting

77_Cydulka_Ch77.indd 413 20/04/17 2:47 pm

Source page 433

in altered level of consciousness and focal motor abnormalities with a rostral-to-caudal progression of dysfunction, and fast component of nystagmus away from the stimulus during cold caloric testing. Subtentorial lesions produce rapid loss of consciousness, cranial nerve abnormalities, abnormal breathing patterns, asymmetric or fixed pupils, and no eye movement during cold caloric testing. Metabolic encephalopathy produces decreased level of consciousness before exhibiting motor signs, which are symmetrical when present. Pupillary reflexes are intact in metabolic encephalopathy except with profound anoxia, opiates, barbiturates, and anticholinergics.

Diagnosis and Differential

A thorough history and physical examination are paramount to determining the diagnosis. The familiar mnemonic AEIOU TIPS (alcohol, encephalopathy, insulin, opiates, uremia, trauma, infection, poisoning, and seizure) is helpful in organizing diagnostic possibilities (Table 77-1).

Table 77-1 AEIOU TIPS: A Mnemonic for Pediatric Altered Mental Status

A Alcohol. Ethanol. Isopropyl alcohol. Methanol. Concurrent hypoglycemia is

common.

Acid-base and metabolic. Hypotonic and hypertonic dehydration. Hepatic dysfunction, inborn errors of metabolism.

Arrhythmia/cardiogenic. Stokes–Adams, supraventricular tachycardia, aortic stenosis, heart block, pericardial tamponade, hypertensive encephalopathy.

E Encephalopathy. Reye's syndrome. Parainfectious encephalomyelitis. Autoimmune encephalitis.

Endocrinopathy. Addison's disease can present with AMS or psychosis. Thyrotoxicosis can present with ventricular dysrhythmias. Pheochromocytoma can present with hypertensive encephalopathy.

Electrolytes. Hypo-/hypernatremia and disorders of calcium, magnesium, and phosphorus can produce AMS.

I Insulin. AMS from hyperglycemia is rare in children, but diabetic ketoacidosis is the most common cause. Hypoglycemia can be the result of many disorders. Irritability, confusion, seizures, and coma can occur with blood glucose levels < 40 mg/dL.

Intussusception. AMS may be the initial presenting symptom.

O Opiates. Common household exposures are to Lomotil, Imodium, diphenoxylate, and dextromethorphan. Clonidine, an α-agonist, can also produce similar symptoms.

Oxygen. Disorders of airway, breathing, or circulation may adversely affect oxygen delivery to the brain; hypercapnia from primary lung disease or neurologic dysfunction also may result in altered mental status.

U Uremia. Encephalopathy occurs in over one-third of patients with chronic renal failure. Hemolytic-uremic syndrome can produce AMS in addition to abdominal pain.

Thrombocytopenic purpura and hemolytic anemia also can cause AMS. In children with chronic renal failure, neurologic dysfunction may develop secondary to stroke, hypertension, or metabolic derangements.

(Continued)

77_Cydulka_Ch77.indd 414 20/04/17 2:47 pm

Source page 434

Table 77-1 AEIOU TIPS: A Mnemonic for Pediatric Altered Mental Status (Continued)

T Trauma. Hypovolemia or hemorrhage from multisystem trauma may lead to insufficient cerebral perfusion and result in altered mental status. Consider concussion, hemorrhage or contusion, or epidural or subdural hematoma. Remember to look for signs of child abuse, particularly shaken baby syndrome with retinal hemorrhages.

Tumor. Primary, metastatic, or meningeal leukemic infiltration. Intracerebral tumors commonly produce focal neurologic signs, and posterior fossa tumors typically block the ventricular system and create signs and symptoms suggestive of hydrocephalus. Supratentorial and infratentorial tumors may present abruptly with altered mental status, fever, or meningismus after an intratumor hemorrhage.

Thermal. Hypo- or hyperthermia. Progressive hypothermia leads to insidious altered mental status. Temperatures > 41°C (105.8°F) result in headache, weakness, and dizziness followed by confusion, euphoria, combativeness, and altered mental status.

I Infection. Bacterial meningitis, encephalitis, and brain abscess are the most important causes of AMS in children, especially AMS with fever. Brain abscess is characterized by fever and headache before AMS changes. Presenting symptoms also include generalized or focal seizures. Any systemic infection associated with vasculitis or shock may lead to altered mental status secondary to cerebral hypoperfusion.

Intracerebral vascular disorders. Subarachnoid, intracerebral, or intraventricular hemorrhages can be seen with trauma, ruptured aneurysm, or arteriovenous malformations. Venous thrombosis can follow severe dehydration or pyogenic infection of the mastoid, orbit, middle ear, or sinuses. Arterial thrombosis is uncommon in children, except in those with homocystinuria. Intracerebral and intraventricular hemorrhages may follow birth asphyxia or trauma in neonates, but in older children, they may signify a congenital or acquired coagulopathy. Cerebral emboli from bacterial endocarditis may cause altered mental status. Acute confusional migraine may be associated with profound alterations in consciousness. Children with sickle cell anemia can develop cerebral thrombosis, status epilepticus, and coma.

P Psychogenic. Rare in children, characterized by decreased responsiveness with normal neurologic examination including oculovestibular reflexes. Psychogenic unresponsiveness may be a conversion reaction, an adjustment reaction, a panic state, or malingering.

Poisoning/ingestion. Drugs, toxins, or illicit substances can be ingested by accident, through neglect or abuse, or in a suicidal gesture.

S Seizure. Generalized motor seizures and absence status epilepticus are often associated with prolonged unresponsiveness in children. In a child with a history of seizures who presents with AMS, consider nonconvulsive status epilepticus. Seizures in a febrile child suggest intracranial infection. Shunt malfunction should be considered among patients with a ventriculoperitoneal shunt for hydrocephalus.

Abbreviation: AMS, altered mental status.

Diagnostic adjuncts largely depend on the clinical situation and must be

determined immediately after the primary and secondary survey. Rapid bedside glucose determination can immediately confirm either hypoglycemia, or hyperglycemia due to diabetic ketoacidosis. Toxicity due to poisoning or ingestion may require a reversal agent. If meningitis or encephalitis

77_Cydulka_Ch77.indd 415 20/04/17 2:47 pm

Source page 435

is suspected, lumbar puncture should be done immediately after initial resuscitation and stabilization. Neuroimaging may be indicated in the setting of trauma or suspicion of space-occupying lesion.

Emergency Department Care and Disposition

Treatment priorities should concentrate on stabilization and reversal of life-threatening conditions.

1. Ensure airway, breathing, and circulation. Immobilize the cervical spine if trauma is suspected and obtain appropriate imaging studies when the patient is stabilized.
2. Provide continuous pulse oximetry and supplemental oxygen as needed to correct hypoxia, including bag-valve-mask and intubation when appropriate. Consider capnometry for intubated patients.
3. Administer fluid resuscitation with 20 mL/kg fluid boluses of isotonic crystalloid for hypotension. Fluid boluses may be repeated up to 60 mL/kg. Intravenous pressors are indicated if hypotension persists.
4. Treat hypoglycemia with 10% dextrose 5 mL/kg in infants or 25% dextrose 2 mL/kg in older children.
5. Control core body temperature to minimize metabolic demands. Prevent hypothermia with warming lamps and treat hyperthermia when present.
6. Treat seizures with benzodiazepines (see Chapter 76).
7. For suspected opiate or clonidine overdose, administer naloxone 0.01 to 0.1 mg/kg IV every 2 minutes. For iatrogenic benzodiazepine overdose, consider flumazenil 0.01 mg/kg IV in the setting of known, isolated benzodiazepine ingestion.
8. Administer empiric antibiotics, ceftriaxone or cefotaxime 50 mg/kg/dose, and consider additional vancomycin 10 mg/kg/dose for suspected meningitis.
9. Patients with AMS will require admission and neurologic observation. Only those with a transient, rapidly reversible, and benign cause of AMS may be discharged from the emergency department after a period of observation with follow-up scheduled within 24 hours of discharge.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study

Guide, 8th ed., see Chapter 136, “Headache in Children,” by David C. Sheridan and Garth Meckler; Chapter 137 “Altered Mental Status in Children” by Sarah Mellion and Kathleen Adelgais.

77_Cydulka_Ch77.indd 416 20/04/17 2:47 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 432–437.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 78: Syncope and Sudden Death in Children and Adolescents

Source pages 438–439 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 438–439

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Syncope and Sudden Death in Children and Adolescents**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 78: Syncope and Sudden Death in Children and Adolescents 419 Table 78-3 Events Easily Mistaken for Cardiovascular Syncope Condition Distinguishing Characteristics Basilar migraine Headache, rarely loss of consciousness, other neurologic symptoms Seizure Loss of consciousness simultaneous with motor event, prolonged postictal phase Vertigo Rotation or spinning sensation, no loss of consciousness Hyperventilation Inciting event, paresthesias or carpopedal spasm, tachypnea Hysteria No loss of conciousness, indifference to event Hypoglycemia Confusion progressing to loss of consciousness, requires glucose administration to terminate chest x-ray, and echocardiography should be considered in this setting. [Source pages 438–439]

- Syncope associated with chest pain may require cardiac troponins as part of the evaluation to rule out ischemic heart disease (e.g., aberrant left coronary artery). [Source pages 438–439]

- Electrolytes, thyroid studies, chest x-ray, and urine pregnancy testing or drug screen should be considered in the appropriate clinical setting. [Source pages 438–439]

- ■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION 1. [Source pages 438–439]

- Strongly consider obtaining an electrocardiogram (ECG) in all patients. [Source pages 438–439]

Clinical reasoning structure

- **Problem representation:** Syncope and Sudden Death in Children and Adolescents; source pages 438–439.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Syncope and Sudden Death in Children and Adolescents**.

- Source-page range: **438–439**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 438

CHAPTER 78: Syncope and Sudden Death in Children and Adolescents 419

Table 78-3 Events Easily Mistaken for Cardiovascular Syncope

Condition Distinguishing Characteristics

Basilar migraine Headache, rarely loss of consciousness, other neurologic symptoms

Seizure Loss of consciousness simultaneous with motor event, prolonged postictal phase

Vertigo Rotation or spinning sensation, no loss of consciousness

Hyperventilation Inciting event, paresthesias or carpopedal spasm, tachypnea

Hysteria No loss of conciousness, indifference to event

Hypoglycemia Confusion progressing to loss of consciousness, requires glucose administration to terminate

chest x-ray, and echocardiography should be considered in this setting. Syncope associated with chest pain may require cardiac troponins as part of the evaluation to rule out ischemic heart disease (e.g., aberrant left coronary artery). Electrolytes, thyroid studies, chest x-ray, and urine pregnancy testing or drug screen should be considered in the appropriate clinical setting.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. Strongly consider obtaining an electrocardiogram (ECG) in all patients.
2. If vasovagal syncope is likely, patients may be discharged home with reassurance and advised to increase water and salt intake.
3. Consider an echocardiogram for patients with known or suspected cardiac disease. If an echocardiogram is not immediately available, the urgency for obtaining the study should be determined in consultation with a cardiologist.

4. If no clear cause is found, the child may be discharged to be further evaluated and followed by the primary care provider unless there are cardiac risk factors or exercise-induced symptoms for which referral to a cardiologist is warranted.
5. Patients with a normal ECG but a history suggesting a dysrhythmia are candidates for outpatient monitoring and cardiac workup.
6. Children with documented dysrhythmias should be admitted. All children admitted for syncope should undergo cardiac monitoring. Children who are survivors of sudden cardiac arrest should be admitted to a pediatric intensive care unit.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 127, "Syncope, Dysrhythmias, and ECG Interpretation," by Andrew C. Dixon.

78_Cydulka_Ch78.indd 419 20/04/17 2:48 pm

Source page 439

C hapter Hypoglycemia and Metabolic

79 Emergencies in Infants

and Children

Teresa J. Riech

■■ HYPOGLYCEMIA

Hypoglycemia in children may be due to inadequate oral intake, excess insulin, low levels of hyperglycemic hormones (e.g., cortisol or growth hormone), inborn errors of metabolism, or systemic infection. Prompt recognition and treatment of hypoglycemia are essential to avoid potentially severe and permanent neurologic injury, and bedside glucose testing should be considered in any neonate, infant, or child with altered mental status.

Clinical Features

Hypoglycemic children may manifest symptoms related to adrenergic hormone release including tachycardia, diaphoresis, tremors, anxiety, irritability, and tachypnea. Severe hypoglycemia may result in apnea or seizures, particularly in neonates and infants, who may not manifest typical signs and symptoms of older children and adults. Neonates and infants with hypoglycemia may also present with altered mental status and nonspecific symptoms such as poor feeding, an abnormal or high-pitched cry, temperature instability, and irritability or lethargy. Hypoglycemia often accompanies critical illness (sepsis) and the features of that illness may dominate the clinical picture, thereby masking the signs of hypoglycemia.

Diagnosis and Differential

Hypoglycemia is defined as a plasma glucose concentration less than

45 mg/dL in symptomatic children and less than 35 mg/dL in asymptomatic neonates. Bedside glucose testing is the most important diagnostic test in any neonate or infant who is critically ill or has altered mental status. Abnormal results should be confirmed with a venous sample, but treatment should not be delayed for confirmatory results. Urine testing for ketones is important, as ketonuria is associated with ketotic hypoglycemia, adrenal insufficiency, and other inborn errors of metabolism. Absent urine ketones are associated with hyperinsulinemic states, infants of a diabetic mother, as well as disorders of fatty acid oxidation and mitochondria.

Emergency Department Care and Disposition

1. For neonates administer 10% dextrose 5 mL/kg IV/IO/PO/NG. Treat infants with the same dose of 10% dextrose, or 25% dextrose 2 mL/kg IV/IO/PO/NG. Give older children 2 mL/kg of 25% dextrose IV/IO/PO/NG.
2. Administer maintenance dextrose for persistent hypoglycemia using 10% dextrose at 1.5 times maintenance.
3. When intravenous access is not immediately available, consider glucagon 0.03 mg/kg IM or subcutaneous (maximum 0.5 mg).

79_Cydulka_Ch79.indd 420 26/04/17 2:29 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 438–439.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 79: Hypoglycemia and Metabolic Emergencies in Infants and Children

Source pages 440–445 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 440–445

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Hypoglycemia and Metabolic Emergencies in Infants and Children**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 79: Hypoglycemia and Metabolic Emergencies in Infants and Children 421 4. [Source pages 440–445]

- If adrenal insufficiency is suspected, give hydrocortisone 25 mg IV/IM for neonates and infants, 50 mg for toddlers and school-age children, and 100 mg for everyone else. [Source pages 440–445]

- Steroids should be given early in patients with hypopituitarism and adrenal insufficiency. [Source pages 440–445]

- Consider empiric antibiotics for suspected sepsis. [Source pages 440–445]

- ■■ INBORN ERRORS OF METABOLISM Inborn errors of metabolism are challenging childhood disorders represent- ing a broad spectrum of diseases with nonspecific signs and symptoms. [Source pages 440–445]

Clinical reasoning structure

- **Problem representation:** Hypoglycemia and Metabolic Emergencies in Infants and Children; source pages 440–445.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Hypoglycemia and Metabolic Emergencies in Infants and Children**.

- Source-page range: **440–445**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 440

CHAPTER 79: Hypoglycemia and Metabolic Emergencies in Infants and Children 421

4. If adrenal insufficiency is suspected, give hydrocortisone 25 mg IV/IM for neonates and infants, 50 mg for toddlers and school-age children, and 100 mg for everyone else. Steroids should be given early in patients with hypopituitarism and adrenal insufficiency.

5. Consider empiric antibiotics for suspected sepsis.

■■ INBORN ERRORS OF METABOLISM

Inborn errors of metabolism are challenging childhood disorders representing a broad spectrum of diseases with nonspecific signs and symptoms. Delay in accurate diagnosis and treatment can lead to significant morbidity and mortality. Despite the myriad etiologies, the principles of initial emergency department (ED) diagnosis and management are relatively simple. The sudden acute deterioration of a healthy neonate should always prompt consideration of metabolic disease, and making a definitive diagnosis is less important than having a high index of suspicion and implementing supportive care.

Clinical Features

Vomiting, altered mental status, and poor feeding are the most common features of metabolic emergencies. Seizures may accompany some metabolic crises. Tachypnea due to metabolic acidosis and tachycardia from dehydration, as well as hypotension due to hypovolemia or adrenal insufficiency, may be noted. Rarely, some metabolic disorders may be associated with characteristic body or urine odors or other phenotypic stigmata (e.g., ambiguous genitalia and hyperpigmentation in congenital adrenal hyperplasia, as discussed separately below). Most metabolic toxins cross the placenta and are cleared by maternal enzymes; therefore, newborns are asymptomatic, and symptoms present after feeding begins.

Diagnosis and Differential

Screening laboratory tests for suspected inborn errors of metabolism

include a bedside glucose, urine for ketones, a blood gas analysis for metabolic acidosis, serum ammonia, basic metabolic panel, and lactate level. Figure 79-1 details the diagnostic evaluation recommended in the ED. Additional laboratory tests for definitive diagnosis that should be considered based on initial screening results include liver function tests, CBC, lactate dehydrogenase, creatine kinase, serum amino acids and acyl-carnitine profile, urine organic acids, and reducing substances. The differential diagnosis of shock in the neonate includes sepsis, congenital heart defects, and abdominal catastrophes.

Emergency Department Care and Disposition

Despite the diverse etiology and complexity of inborn errors of metabolism, ED resuscitation and stabilization of patients with these disorders are relatively simple. Neonates, infants, and children presenting in metabolic crisis, regardless of cause, show some combination of dehydration, metabolic acidosis, and encephalopathy, which must be immediately addressed. The goals of treatment are to improve circulatory status by restoring circulatory volume, provide energy substrate to halt catabolism, remove the

79_Cydulka_Ch79.indd 421 26/04/17 2:29 pm

Source page 441

Poor feeding Blood glucose/Calcium/

Vomiting Ammonia/pH/CO2

Urine ketones

Lethargy

Convulsion Crisis mimics sepsis

Coma (attain appropriate cultures)

Hypoglycemia—glycogen

storage disease

Hyperammonemia—

urea cycle defect

Blood glucose, calcium

Ammonia Hyperammonemia + acidosis

pH, CO2 Organic acid disorders

Urine ketones

Normal ammonia + acidosis

Aminoacidopathies

(e.g., maple syrup urine)

+ Acidosis/–Urinary ketones

Fatty acid oxidation defect

Figure 79-1. Approach to suspected metabolic disorders.

inciting metabolic substrate (formula or breast milk), and help eliminate toxic metabolites.

1. Attend to Airway, Breathing, Circulation, and Disability (ABCD). If apnea, hypoventilation, or hypoxemia necessitates endotracheal intubation, avoid prolonged use of paralytics because metabolic acidosis can be worsened by respiratory acidosis if insufficient ventilation is provided.
2. Restore circulation with 0.9% saline boluses of 20 mL/kg, reassessing after each bolus.
3. Make patients NPO, and correct hypoglycemia as above with D10 in neonates and infants or D25 in children.
4. Begin 10% dextrose infusion at twice maintenance rate.
5. Consider broad-spectrum antibiotics.
6. Specific metabolic treatments are guided by the underlying defect and should be determined in consultation with a metabolic specialist. Substrates to facilitate shunting of metabolic pathways or hemodialysis may be needed.
7. Reflexive use of bicarbonate should be avoided; however, in select cases of severe acidosis (pH < 7.0), treatment with 0.5 mEq/kg/h of sodium bicarbonate may be considered.

Patients with a newly diagnosed or suspected metabolic disorder and those who are dehydrated or otherwise decompensated require admission

79_Cydulka_Ch79.indd 422 26/04/17 2:29 pm

Source page 442

Table 79-1 Conditions Associated with Hypoglycemia in Infants and Children

Neonates	Infancy and Childhood
Infant of a diabetic mother	Idiopathic ketotic hypoglycemia/starvation
Congenital heart disease	Diabetes mellitus/endocrine disorder
Infection/sepsis	Infection/sepsis
Adrenal hemorrhage	Inborn errors of metabolism
Hypothermia	Hypothermia
Hypoglycemia-inducing drug use by mother	Drug induced (salicylates, etc.)
Hyperinsulinism	
Maternal eclampsia	Idiopathic
Fetal alcohol syndrome	
Hypopituitarism	

for dextrose and specific treatment. Transfer to a pediatric hospital may be required (Table 79-1).

■■ CONGENITAL ADRENAL HYPERPLASIA

Congenital adrenal hyperplasia results from deficiency in one of the enzymes involved in the production of cortisol, which leads to decreased cortisol levels sometimes accompanied by mineralocorticoid deficiency (“salt wasting syndrome”). Steroid precursors may be shunted to androgen production with virilization of females.

Clinical Features

Patients with salt-wasting adrenal hyperplasia typically present in the second to fifth week of life with nonspecific symptoms including shock, vomiting, lethargy, irritability, and poor feeding. On examination, females may have clitoromegaly and males may have a small penis or hypospadias. Hyperpigmentation may be noted on axillae and around the nipples.

Diagnosis and Differential

The most important laboratory tests include bedside glucose and electrolytes, as hyponatremia and hyperkalemia are often present and suggest the diagnosis. Serum potassium levels of 6 to 12 mEq/kg are not unusual but are rarely associated with ECG abnormalities. Definitive diagnosis depends on analysis of blood hormone levels. If possible, blood should be obtained for analysis before treatment with exogenous steroids; however, treatment should not be delayed in the critically ill neonate. The differential diagnosis includes sepsis, congenital heart disease, and other inborn errors of metabolism.

Emergency Department Care and Disposition

1. Administer 0.9% saline boluses of 20 mL/kg IV/IO.
2. Treat hypoglycemia as above.
3. Administer hydrocortisone IV/IO/IM: 25 mg for neonates, 50 mg for toddlers and school-age children, and 100 mg for adolescents.

79_Cydulka_Ch79.indd 423 26/04/17 2:29 pm

Source page 443

Table 79-2 Management of Hypoglycemia in the ED

Patient Age	Dextrose Bolus Dose	Dextrose Maintenance Dosage to Consider	Other Treatments to Consider
Neonate	D10 5 mL/kg PO/NG/IV/IO	6 mL/Kg/h D10 IM/IV	Glucagon, 0.03 mg/ kg IM/IV Hydrocortisone, 25 g PO/IM/IV/IO
Infant	D10 5 mL/kg PO/NG/IV/IO	6 mL/Kg/h D10 IM/IV	Glucagon, 0.03 mg/ kg IM/IV or Hydrocortisone, 25 g D25 2 mL/kg PO/IM/IV/IO

Child D25 2 mL/kg 6 mL/kg/h D10 for the first Glucagon, 0.03 mg/kg/
PO/NG/IV/IO 10 kg + 3 mL/kg/h for IM/IV,
11–20 kg + 1.5 mL/kg/h for Hydrocortisone, 50 g
each additional kg >20 kg PO/IM/IV/IO
Adolescent — 6 mL/kg/h D10 for the first Glucagon, 0.03 mg/
10 kg + 3 mL/kg/h for kg IM/IV
11–20 kg + 1.5 mL/kg/h for Hydrocortisone,100 g
each additional kg >20 kg PO/IM/IV/IO

Abbreviations: D10, 10% dextrose; D25, 25% dextrose; NG, (Via) nasogastric tube.

4. Treat hyperkalemia with calcium gluconate 100 mg/kg IV (1 mL/kg) and bicarbonate 1 mEq/kg IV. Insulin should be avoided as it may cause profound hypoglycemia.

Neonates with adrenal crisis from adrenal hyperplasia require hospitalization. Infants with shock or severe hyperkalemia should be admitted to the ICU with endocrine consultation. Those with a known diagnosis of adrenal hyperplasia who have normal vital signs and are able to tolerate oral intake may be discharged home after administration of hydrocortisone as above with instructions to triple their usual home dose of steroid until fever, vomiting, or diarrhea resolve and when next-day follow-up can be assured) (Table 79-2).

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 144, "Metabolic Emergencies in Infants and Children," by Garth Meckler, Nadeemuddin Qureshi, Mohammed Al-Mogbil, and Osama Y. Kentab.

79_Cydulka_Ch79.indd 424 26/04/17 2:29 pm

References/provenance

- Source: *Tintinalli's Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 440–445.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 80: Diabetes in Children

Source pages 446–447 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 446–447

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Diabetes in Children**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 80: Diabetes in Children 427 intensive care settings. [Source pages 446–447]

- Patients with cerebral edema require ICU admis- sion and possible intracranial pressure monitoring. [Source pages 446–447]

- Consultation with the patient’s primary care physician and a pediatric endocrinologist should be made early in the course of therapy. [Source pages 446–447]

- ■■ FURTHER READING For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 145, “Diabetes in Children,” by Adam Vella. [Source pages 446–447]

- 80_Cydulka_Ch80.indd 427 20/04/17 2:48 pm C h a p ter Fluid and Electrolyte Therapy 81 in Infants and Children Ron L. [Source pages 446–447]

Clinical reasoning structure

- **Problem representation:** Diabetes in Children; source pages 446–447.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Diabetes in Children**.

- Source-page range: **446–447**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 446

CHAPTER 80: Diabetes in Children 427

intensive care settings. Patients with cerebral edema require ICU admis-
sion and possible intracranial pressure monitoring. Consultation with
the patient’s primary care physician and a pediatric endocrinologist
should be made early in the course of therapy.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 145, “Diabetes in Children,” by Adam Vella.

80_Cydulka_Ch80.indd 427 20/04/17 2:48 pm

Source page 447

C h a p ter Fluid and Electrolyte Therapy

81 in Infants and Children

Ron L. Kaplan

The most common disorder of fluid balance in children requiring emer-
gency care is dehydration. Dehydration is the result of a negative fluid
balance that can result from decreased fluid intake, increased output (renal,
GI, or insensible losses from the skin or respiratory tract), or conditions
such as sepsis, burns, and diabetes.

■■ CLINICAL FEATURES

The presence of fluid and electrolyte disturbances is often suggested by a
thorough history. Children with chronic underlying disease are at particular
risk. The clinical appearance depends on the degree of dehydration, the rate
at which the fluid was lost, and the age of the patient.

Though the gold standard for assessing dehydration is comparison of
pre-illness weight with weight on presentation to the emergency depart-
ment (ED), a reliable and recent pre-illness weight is rarely available in
the ED. Physical examination can provide an estimation of the degree of
dehydration, which is typically classified as mild, moderate, or severe.
Clinical signs and symptoms of dehydration are listed in Table 81-1. An
important exception to the reliability of signs and symptoms to predict

degree of dehydration occurs in hypernatremic dehydration, when fluid

Table 81-1 Clinical Guidelines for Assessing Dehydration in Children

None to Mild Mild to Moderate

(< 3% Body (3% to 9% Body Severe (> 9% Body

Weight Loss) Weight Loss) Weight Loss)

Mental status Well, alert Fatigued, restless, Apathetic, lethargic,

irritable unconscious

Thirst Normal, slight Increased, eager to Very thirsty or too

increase, or drink lethargic to drink

refusing fluids

Heart rate Normal Normal to increased Tachycardia or

bradycardia in critically

ill children

Pulse quality Normal Normal to reduced Weak, thready

Eyes Normal Slightly sunken Sunken

Tears Present Decreased Absent

Mucous membranes Moist Dry Parched

Anterior fontanelle Normal Sunken Sunken

Capillary Refill Normal 3 to 4 s >4s

Extremities Warm Warm to cool Cold, mottled, cyanotic

Urine output Normal Decreased Minimal

Source: Used with permission from Stephen Freedman, MD and Jennifer Thull-Freedman.

428

81_Cydulka_Ch81.indd 428 20/04/17 2:50 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 446–447.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 81: Fluid and Electrolyte Therapy in Infants and Children

Source pages 448–451 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 448–451

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Fluid and Electrolyte Therapy in Infants and Children**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 81: Fluid and Electrolyte Therapy in Infants and Children 429 loss occurs primarily from the interstitial and intracellular spaces and clinical signs of intravascular volume depletion may be minimal. [Source pages 448–451]

- In this setting, however, the skin may have a characteristic doughy feel. [Source pages 448–451]

- ■■ DIAGNOSIS AND DIFFERENTIAL If available, the absolute and relative fluid deficit can be calculated from a pre-illness weight: 1 kg of weight loss is equivalent to 1 L of fluid deficit. [Source pages 448–451]

- In the absence of a reliable pre-illness comparison weight, the diagnosis of dehydration is based primarily on historical data and physical examination findings (Table 81-1). [Source pages 448–451]

- Routine laboratory testing is not required in mild to moderate dehydra- tion, but serum electrolytes and other studies may be indicated in the setting of severe dehydration, signs and symptoms of electrolyte abnormalities, or certain underlying medical conditions. [Source pages 448–451]

Clinical reasoning structure

- **Problem representation:** Fluid and Electrolyte Therapy in Infants and Children; source pages 448–451.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Fluid and Electrolyte Therapy in Infants and Children**.

- Source-page range: **448–451**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 448

CHAPTER 81: Fluid and Electrolyte Therapy in Infants and Children 429

loss occurs primarily from the interstitial and intracellular spaces and clinical signs of intravascular volume depletion may be minimal. In this setting, however, the skin may have a characteristic doughy feel.

■■ DIAGNOSIS AND DIFFERENTIAL

If available, the absolute and relative fluid deficit can be calculated from a pre-illness weight: 1 kg of weight loss is equivalent to 1 L of fluid deficit.

In the absence of a reliable pre-illness comparison weight, the diagnosis of dehydration is based primarily on historical data and physical examination findings (Table 81-1).

Routine laboratory testing is not required in mild to moderate dehydration, but serum electrolytes and other studies may be indicated in the setting of severe dehydration, signs and symptoms of electrolyte abnormalities, or certain underlying medical conditions. A rapid bedside glucose test should be done on any child presenting with altered sensorium. Infants are at particular risk of hypoglycemia.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

The management of fluid and electrolyte disturbances in infants and young children revolves around a few basic principles: (a) identification and treatment of shock; (b) administration of appropriate fluids to replace fluid deficits, ongoing losses, and maintenance fluid requirements; (c) identification and treatment of causes that have specific therapies (e.g., diabetic ketoacidosis, sepsis, and inborn errors of metabolism). The most common approaches to rehydration include oral rehydration therapy and parenteral therapy, though rehydration through nasogastric tube is also effective, simple, and well tolerated.

1. Treat hypovolemic shock with an initial bolus of 20 mL/kg of isotonic

crystalloid: normal saline (NS) or lactated Ringer (LR) solution IV/IO. Repeat boluses every 10 to 20 minutes until mental status, vital signs, and peripheral perfusion improve. Up to 60 mL/kg or more may be needed in the first hour; care must be taken to monitor for signs of volume overload or myocardial dysfunction.

2. Oral rehydration is as effective as intravenous therapy and is recommended by the World Health Organization (WHO) for mild to moderate dehydration. Administer frequent small sips of oral rehydration solution containing glucose and electrolytes (e.g., Rehydralyte®) by mouth or nasogastric tube. Give 50 mL/kg orally over 4 hours for mild dehydration and 100 mL/kg for moderate dehydration. Vomiting is not a contraindication to attempting oral rehydration. Ondansetron may be used to facilitate oral rehydration if nausea or vomiting persists. Give 0.1 mg/kg IV or 0.15 mg/kg PO or by oral disintegrating tablet with a maximum dose of 8 mg.

3. Treat patients who cannot tolerate oral or enteral rehydration with IV or IO administration of NS or LR. If shock is not present, administer 20 mL/kg over 1 hour and repeat as needed. After initial volume replacement, 5% dextrose in 0.9% saline may be started at one to two times maintenance while determining disposition.

81_Cydulka_Ch81.indd 429 20/04/17 2:50 pm

Source page 449

4. For maintenance treatment, use the following formula to determine fluid rate:

For the first 10 kg: 100 mL/kg/d or 4 mL/kg/h

For the second 10 kg: 50 mL/kg/d or 2 mL/kg/h

For each kg >20 kg: 20 mL/kg/d or 1 mL/kg/h

Isotonic fluids such as 5% dextrose in 0.9% saline should be used for maintenance treatment after the first week of life in order to avoid iatrogenic hyponatremia. 10% dextrose with no electrolytes should be used on day 1 of life and 5% dextrose in 0.45% saline can be used on days 2 to 7 of life.

5. Treat specific electrolyte disturbances, as listed in Table 81-2. Of note, overly rapid correction of hypo- or hypernatremia may lead to severe CNS sequelae. If neurologic symptoms such as seizures or altered mental status accompany hyponatremia, administer 2 to 3 mL/kg of 3% saline until symptoms resolve or the serum sodium is >120 mEq/L. Otherwise, correct the hyponatremia slowly over 48 hours to avoid CNS osmotic demyelination. Hypernatremia must be corrected slowly as well, in order

to prevent cerebral edema and central pontine myelinolysis.

Table 81-2 Electrolyte Disorders in Infants and Children and Initial Management

Electrolyte Disorder	Symptoms and Initial Treatment		
	Common Causes	Signs	Recommendations
Hyponatremia	Vomiting, diarrhea, excess free water intake hyporeflexia	Mental status changes, seizures, hyporeflexia	IV normal saline starting with a 20 mL/kg bolus For seizures: 4 mL/kg of 3% saline over 30 min Further correction slowly over 48 h
Hypernatremia	Vomiting, diarrhea, insensible losses, diabetes insipidus, renal disease	Diarrhea, mental status changes, ataxia, doughy skin, seizures, hyperreflexia	IV normal saline starting with a 20 mL/kg bolus Further correction to take place slowly over 48 h
Hypokalemia	Vomiting, DKA ileus	Muscle weakness, hyporeflexia	Generally tolerated well, replace orally over several days If severe: IV 0.2 to 0.3 mEq/kg/h of KCl
Hyperkalemia*	Cortical adrenal hyperplasia (neonates), renal failure. May be due to hemolysis of blood sample	ECG changes: peaked T waves, prolonged PR inter- val, widening of QRS	Insulin 0.1 unit/kg plus 25% glucose, 0.5 g/kg IV Calcium gluconate 10%, 1 mL/kg IV, no faster than 1 mL/min Albuterol, 0.5% solution, 2.5 mg via nebulization For other treatments, see Chapter 4.

(Continued)

Source page 450

Table 81-2 Electrolyte Disorders in Infants and Children and Initial Management (Continued)

Electrolyte Disorder	Symptoms and Initial Treatment		
	Common Causes	Signs	Recommendations
Hypocalcemia	Dietary or vitamin D deficiency, hypopar-	Vomiting, irritability, muscle weakness,	Calcium gluconate 10%, 1 mL/kg IV, no faster

athyroid and chronic tetany, seizures than 1 mL/min
renal failure
Hypercalcemia Malignancy, hyper- Fatigue, irritability, IV normal saline starting
vitaminosis D or A anorexia, vomiting, at 20 mL/kg
constipation Consider furosemide
1-2 mg/kg IV, maximum
dose 40 mg
Hypomagnesemia Diarrhea, short Muscle spasms, For seizures or
gut, diuretics, weakness, ataxia, dysrhythmia: IV
chemotherapy nystagmus, seizures magnesium sulfate
ECG changes: 1 mEq/kg slowly over 4 h
prolonged PR and Asymptomatic patients
QTc, torsades de can be treated with oral
pointes supplements
Hypermagnesemia Ingestion of Hypotension, respi- Remove exogenous
antacids or renal ratory failure, loss of source of magnesium
dysfunction deep tendon reflexes If severe: calcium
ECG changes: gluconate 10%
widening of QRS, 1 mL/kg IV no faster
PR, QTc than 1 mL/min
*
Mild hyperkalemia usually well tolerated in neonates.
Most children with mild to moderate dehydration can be managed as outpatients without any laboratory evaluation in the emergency department. Admission criteria include young infants with ongoing significant fluid losses; severe dehydration; significant electrolyte or metabolic derangements; persistent vomiting and failed attempts at oral rehydration; or an underlying diagnosis requiring ongoing inpatient treatment (e.g., DKA or inborn errors of metabolism).

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 129, "Fluid and Electrolyte Therapy in Infants and Children," by Chan Melissa and Enarson Paul.

81_Cydulka_Ch81.indd 431 20/04/17 2:50 pm

Source page 451

C hapter Musculoskeletal Disorders
82 in Children
Mark X. Cicero

■■ CHILDHOOD PATTERNS OF INJURY

The growth plate (physis) is the weakest point in children's long bones and

the frequent site of fractures. The ligaments and periosteum are stronger than the physis; therefore, they tolerate mechanical forces at the expense of physeal injury. The blood supply to the physis arises from the epiphysis, so separation of the physis from the epiphysis may result in growth arrest. The Salter–Harris classification is used to describe fractures involving the growth plate (Figs. 82-1 and 82-2).

Figure 82-1. Salter–Harris type II fracture of distal tibia. Lateral radiograph illustrating fracture extending through physeal growth plate and metaphysis with dorsal displacement of epiphysis requiring reduction. Used with permission from Wake Medical Center, Raleigh, NC.

82_Cydulka_Ch82.indd 432 20/04/17 2:50 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 448–451.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 82: Musculoskeletal Disorders in Children

Source pages 452–463 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 452–463

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Musculoskeletal Disorders in Children**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 82: Musculoskeletal Disorders in Children 433 Figure 82-2. [Source pages 452–463]

- An x-ray of a Salter–Harris type III fracture of the distal tibia, also known as Tillaux’s fracture. [Source pages 452–463]

- Note the intra-articular component extending through the growth plate and the medial epiphysis of the tibia. [Source pages 452–463]

- Reproduced with permission from Schafermeyer RW, Tenebein M, Macias CG: Strange and Schafermeyer’s Pediatric Emergency Medicine, 4th ed. [Source pages 452–463]

- Salter–Harris Type I Fracture In type I physeal fracture, the epiphysis separates from the metaphysis. [Source pages 452–463]

Clinical reasoning structure

- **Problem representation:** Musculoskeletal Disorders in Children; source pages 452–463.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Musculoskeletal Disorders in Children**.

- Source-page range: **452–463**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 452

CHAPTER 82: Musculoskeletal Disorders in Children 433

Figure 82-2. An x-ray of a Salter–Harris type III fracture of the distal tibia, also known as Tillaux's fracture. Note the intra-articular component extending through the growth plate and the medial epiphysis of the tibia. Reproduced with permission from Schafermeyer RW, Tenebein M, Macias CG: Strange and Schafermeyer's Pediatric Emergency Medicine, 4th ed. New York: McGraw-Hill, Inc; 2015.

Salter–Harris Type I Fracture

In type I physeal fracture, the epiphysis separates from the metaphysis. The reproductive cells of the physis stay with the epiphysis. There are no bony fragments, and these injuries have a low incidence of growth disturbance. Diagnosis is suspected clinically in children with point tenderness over a physis. On radiograph, there may be no abnormality; there may be an associated joint effusion; or there may be epiphyseal displacement from the metaphysis. In the absence of epiphyseal displacement, the diagnosis is clinical. Treatment consists of splint immobilization, ice, elevation, and referral to orthopedics. Type I fractures of the distal fibula are not associated with growth arrest and can be followed by a primary care physician (PCP) or by orthopedics after splinting.

82_Cydulka_Ch82.indd 433 20/04/17 2:50 pm

Source page 453

434 SECTION 9: Pediatrics

Salter–Harris Type II Fracture

A type II physeal fracture goes through the physis and out through the metaphysis. Growth is preserved because the physis remains with the epiphysis. Diagnosis is made by noting a metaphyseal triangular-shaped fragment without epiphyseal involvement on radiograph. Treatment is closed reduction (if necessary) with analgesia and sedation followed by splint or cast immobilization, and follow-up with orthopedics.

Salter–Harris Type III Fracture

The hallmark of type III physeal fracture is an intra-articular fracture of the epiphysis with the cleavage plane continuing along the physis. The prognosis for bone growth depends on the circulation to the epiphyseal bone fragment and is usually favorable. Diagnosis is made radiographically with an epiphyseal fragment without a metaphyseal fracture. Reduction of the unstable fragment with anatomic alignment of the articular surface is critical. Open reduction is sometimes required.

Salter–Harris Type IV Fracture

The fracture line of type IV physeal fractures begins at the articular surface and extends through the epiphysis, physis, and metaphysis. Diagnosis is made radiographically with both epiphyseal and metaphyseal fractures. Especially when there is displacement of the bony fragments, open reduction is usually required to reduce the risk of premature bone growth arrest.

Salter–Harris Type V Fracture

In type V physeal fracture, severe compressive forces essentially crush the physis. There is usually no epiphyseal displacement. The diagnosis is often difficult. An initial diagnosis of sprain or type I injury may prove incorrect when later growth arrest occurs. Radiographs may look normal or demonstrate focal narrowing of the epiphyseal plate. There is usually an associated joint effusion. Treatment consists of casting, nonweight bearing, and close orthopedic follow-up in anticipation of bone growth arrest.

Torus Fractures, Greenstick Fractures, and Plastic Deformities

Children’s long bones are more compliant than those of adults and tend to bow and bend under forces where an adult’s might fracture. Torus (buckle) fractures involve a bulging or buckling of the bony cortex, usually of the metaphysis. Patients have point tenderness over the fracture site and soft tissue swelling. Radiographs may be subtle but show cortical disruption. Torus fractures are not typically angulated, rotated, or displaced, so reduction is not necessary. Splinting in a position of function for 3 to 4 weeks is often preferred over casting. Orthopedic follow-up is usually recommended.

In greenstick fractures, the cortex and periosteum are disrupted on one side of the bone but intact on the other. If the degree of angulation is significant, treatment is closed reduction and immobilization.

Plastic deformities, also known as bowing or bending fractures, are seen in the forearm and lower leg in combination with a completed fracture in the companion bone. The diaphyseal cortex is deformed, but the periosteum

82_Cydulka_Ch82.indd 434 20/04/17 2:50 pm

Source page 454

is intact. Treatment requires prompt orthopedic consultation for closed reduction and realignment.

■■ FRACTURES ASSOCIATED WITH CHILD ABUSE

Certain injury patterns are consistently seen in abused children, particularly multiple fractures in various stages of healing. Please see Chapter 187 for details.

■■ SELECTED PEDIATRIC INJURIES

Clavicle Fracture

Clavicles are commonly fractured in children, and may occur in newborns during difficult deliveries, presenting in neonates with nonuse of the arm. If the fracture was not initially appreciated, parents may notice a bony callus at age 2 to 3 weeks. In older infants and children, the usual mechanism is a fall onto the outstretched arm or shoulder. Care of the patient with a clavicle fracture is directed toward pain control. Even displaced fractures usually heal well, although patients may have a residual bump at the fracture site. A simple sling is effective and less painful than other methods of clavicle immobilization. Newborns require no specific treatment. Orthopedic consultation in the emergency department (ED) is required for an open fracture (which also requires antibiotics), displacement of the medial clavicle, or a skin-tenting fracture fragment that has the potential to convert to an open fracture. Otherwise, routine follow-up with the PCP is usually adequate.

Supracondylar and Condylar Fractures

The most common elbow fracture in childhood is the supracondylar fracture of the distal humerus. The mechanism is commonly a fall onto the outstretched arm. Children complain of pain on passive elbow flexion and hold the forearm pronated. The close proximity of the brachial artery to the fracture predisposes the artery to injury. Subsequent arterial spasm or compression by casts may further compromise distal circulation. A forearm compartment syndrome (resulting in Volkmann ischemic contracture) may occur.

Radiographs show the injury, but the findings may be subtle. Type I fractures have no displacement or angulation, or may have a posterior fat pad as the only radiographic manifestation of a fracture. Confirmation of a fracture may be seen on an x-ray taken 2 to 4 weeks later, when a periosteal reaction is visible. Type II fractures are angulated, but the posterior cortex is intact, while type III fractures are completely displaced with no cortical contact.

In cases of neurovascular compromise, immediate fracture reduction is indicated. If an ischemic forearm compartment is suspected after

reduction, surgical decompression or arterial exploration may be indicated. Outpatient treatment is acceptable for type I fractures after appropriate immobilization. Such children need orthopedic reassessment within 2 to 7 days. Orthopedic consultation and admission is recommended for patients with type II or III fractures. Open reduction with operative pinning is usually required.

82_Cydulka_Ch82.indd 435 20/04/17 2:50 pm

Source page 455

Lateral and medial condylar fractures and intercondylar and transcondylar fractures carry risks of neurovascular compromise, especially to the ulnar nerve. These patients have soft tissue swelling and tenderness while maintaining the arm in flexion. Depending on the displacement visualized on x-ray, patients may require open reduction.

Radial Head Subluxation (“Nursemaid’s Elbow”)

Radial head subluxation is a very common injury seen most often in children 1 to 4 years of age. The typical history is that the child was lifted or pulled by the hand or wrist, though 50% have no such history and parents may report a fall or simply that their child refuses to use the arm. The arm is held in adduction, flexed at the elbow, with the forearm pronated. Gentle examination demonstrates no tenderness to direct palpation, but attempts to supinate the forearm or move the elbow cause pain. If the history and examination are strongly suggestive, radiographs are not needed. However, if the history is atypical or there is point tenderness or signs of trauma, radiographs should be obtained.

There are two maneuvers for reduction. The first, the supination/flexion technique, is performed by holding the patient’s elbow at 90° with one hand and then firmly supinating the wrist and simultaneously flexing the elbow so that the wrist is directed to the ipsilateral shoulder. There may be a “click” with reduction, and the child may transiently cry and resist. The second, the hyperpronation technique, is reported to be more successful. The hyperpronation technique is performed by holding the child’s elbow at 90° in one hand and then firmly pronating the wrist while extending the elbow. Usually the child will resume normal activity in 5 to 10 minutes if reduction is achieved. If the child is not better after a second reduction attempt, alternate diagnoses and radiographs should be considered. No specific therapy is needed after successful reduction. Parents should be reminded to avoid linear traction on the arm because there is a risk of recurrence.

Forearm Injuries

Torus Fractures

Torus fractures, also called buckle fractures, are among the most common pediatric bony injuries and may occur in the radius or ulna. Treatment consists of pain management with NSAIDs, application of a volar splint, and follow-up with the PCP or orthopedics in 1 to 3 weeks.

Fractures of the Radial and Ulnar Shafts

Metaphyseal Fractures

Any metaphyseal fracture with rotational deformity or more than 10 degrees of angulation in children above 8 years of age, or more than 15 to 20 degrees in younger children, requires consultation with an orthopedist to determine the need for reduction. Otherwise, immobilization in a splint with follow-up with orthopedics within 1 week is adequate treatment.

Diaphyseal Fractures

Most diaphyseal forearm fractures warrant urgent orthopedic consultation. Transverse fractures of one or both bones may remain unstable despite

82_Cydulka_Ch82.indd 436 20/04/17 2:50 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 452–463.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 83: Rashes in Children

Source pages 464–475 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 464–475

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Rashes in Children**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 83: Rashes in Children 445 tiny papules on an erythematous base with slightly raised edges. [Source pages 464–475]

- The eye- lids and chin are characteristically spared. [Source pages 464–475]

- As the illness progresses, and 1 to 2 days after the facial rash appears, a nonpruritic erythematous macular or maculopapu- lar rash appears on the trunk and limbs. [Source pages 464–475]

- This rash may last for 1 week and is not pruritic. [Source pages 464–475]

- As the rash fades, central clearing of the lesions occurs, leaving a lacy reticular appearance. [Source pages 464–475]

Clinical reasoning structure

- **Problem representation:** Rashes in Children; source pages 464–475.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Rashes in Children**.

- Source-page range: **464–475**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 464

CHAPTER 83: Rashes in Children 445

tiny papules on an erythematous base with slightly raised edges. The eyelids and chin are characteristically spared. Circumoral pallor is typical. The rash fades after 4 to 5 days. As the illness progresses, and 1 to 2 days after the facial rash appears, a nonpruritic erythematous macular or maculopapular rash appears on the trunk and limbs. This rash may last for 1 week and is not pruritic. As the rash fades, central clearing of the lesions occurs, leaving a lacy reticular appearance. Palms and soles are rarely affected. The exanthem may recur intermittently in the weeks after the onset of illness. Sun exposure or hot baths may exacerbate the rash. Associated symptoms include fever, malaise, headache, sore throat, cough, coryza, nausea, vomiting, diarrhea, and myalgias. Treatment is supportive.

Eczema Herpeticum

In children with existing eczema, this life-threatening, rare, viral infection can arise. The most frequent etiologic agent is herpes simplex virus. Bacterial superinfection with staphylococci or streptococci is presumed. Clinical manifestations of eczema herpeticum include fever and vesicular eruptions in areas of skin contemporaneously affected by eczematous lesions (Fig. 83-2). Treatment includes acyclovir 20 mg/kg/dose PO every

Figure 83-2. Typical appearance of eczema herpeticum. Used with permission from University of North Carolina Department of Dermatology.

83_Cydulka_Ch83.indd 445 20/04/17 2:51 pm

Source page 465

446 SECTION 9: Pediatrics

6 hours and either trimethoprim-sulfamethoxazole 5 mg/kg/dose twice daily or clindamycin 10 mg/kg/dose three times daily for 10 days. Inpatient admission is often necessary.

Varicella (Chicken Pox)

Due to immunizations, the incidence of varicella has declined dramatically. The etiologic agent is varicella-zoster virus, a herpes virus. It typically occurs in children younger than 10 years but may occur at all

ages. Varicella occurs most often in the late winter and early spring. Patients are highly contagious from the prodrome phase of the illness until all lesions are crusted over. The rash starts as faint red macules on the scalp or trunk. Within the first day, lesions begin to vesiculate and develop a red base, producing the characteristic appearance (Fig. 83-3). Over the next few days, groups of lesions develop, producing the classic appearance of crops of lesions in multiple stages of development. Over the next 1 to 2 weeks, lesions become dry and crusted. The rash typically spreads centrifugally (outward from the center). The palms and soles are usually spared. Low-grade fever, malaise, and headache are frequently seen but are typically mild. Treatment is symptomatic and includes

Figure 83-3. Typical rash of varicella (chicken pox). Used with permission from University of North Carolina Department of Dermatology.

83_Cydulka_Ch83.indd 446 20/04/17 2:52 pm

Source page 466

diphenhydramine 1.25 mg/kg/dose, every 6 hours as needed for itching and acetaminophen 15 mg/kg/dose, every 4 hours as needed for fever. Although not needed in healthy children, varicella-zoster immune globulin and acyclovir 20 mg/kg up to 800 mg PO five times daily may be needed for immunocompromised children.

Roseola Infantum (Exanthem Subitum)

Roseola is a common acute febrile illness in children ages 6 months to 3 years and thought to be caused by human herpesvirus 6. Roseola presents with an abrupt onset, high fever lasting 3 to 5 days. As the fever begins to resolve, blanching macular or maculopapular, rose or pink discrete lesions develop (Fig. 83-4). Areas typically affected include the neck, trunk, and buttocks but the face and proximal extremities may also be involved, though mucous membranes are spared. The rash lasts 1 to 2 days and rapidly fades. Associated symptoms are typically mild and may include irritability when the fever is highest, cough, coryza, anorexia, and abdominal discomfort. The treatment is symptomatic.

Figure 83-4. An infant with roseola infantum. Reproduced with permission from Knoop K, Stack L, Storrow A: Atlas of Emergency Medicine, 3rd ed. New York: McGraw-Hill Companies, Inc; 2010. Photo contributor: Raymond C. Baker, MD.

83_Cydulka_Ch83.indd 447 20/04/17 2:52 pm

Source page 467

■■ FUNGAL INFECTIONS

Tinea infections are common in infants and children and named for the body parts affected: tinea capitis (scalp), corporis (body), and pedis (foot) are common examples. Tinea infections typically manifest as scaly patches with pruritus of varying intensity. Successful treatment for all but tinea capitis and unguium is usually accomplished with topical creams including those available over-the-counter (clotrimazole, miconazole, tolnaftate) or by prescription (ketoconazole, oxiconazole, ciclopirox, terbinafine). Treatment is continued for 7 to 10 days after the resolution of lesions. Tinea capitis ranges from mild scalp scaliness with patchy alopecia to a painful, boggy mass known as a kerion. Tinea capitis is treated with oral griseofulvin ultramicrosize 15 mg/kg/dose once daily and selenium sulfide shampoo. Treatment of tinea capitis is usually for at least 8 weeks and close follow-up is important as treatment response and liver function tests need to be monitored. Tinea unguium is similarly treated with griseofulvin for longer periods of time.

■■ BACTERIAL INFECTIONS

Impetigo

Impetigo is a superficial skin infection, typically caused by group A β-hemolytic streptococci or Staphylococcus aureus. The lesions usually occur in small children, often in areas of insect bites or minor trauma. The lesions start as red macules and papules that form vesicles and pustules (Fig. 83-5). The formation of a golden crust results from rupture of the vesicles. The lesions may become confluent. With the exception of regional lymphadenopathy, fever and systemic signs are rare. Most commonly affected areas include the face, neck, and extremities. Diagnosis is

Figure 83-5. A young girl with crusting impetiginous lesions on her chin. Reproduced with permission from Knoop K, Stack L, Storrow A: Atlas of Emergency Medicine, 3rd ed. New York: McGraw-Hill Companies, Inc; 2010. Photo contributor: Michael J. Nowicki, MD.

83_Cydulka_Ch83.indd 448 20/04/17 2:52 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 464–475.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 84: Sickle Cell Anemia in Children

Source pages 476–481 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 476–481

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Sickle Cell Anemia in Children**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 84: Sickle Cell Anemia in Children 457 Clinical Features Consider ACS in all patients with SCA who present with complaints of chest pain, especially when associated with tachypnea, dyspnea, cough, or other symptoms of respiratory distress. [Source pages 476–481]

- Significant hypoxia and rapid dete- rioration to respiratory failure can occur. [Source pages 476–481]

- Diagnosis and Differential Chest radiographs should be obtained but may be normal during the first hours to days. [Source pages 476–481]

- The diagnosis is made in the setting of a new infiltrate on chest radiograph in the setting of chest pain and respiratory symptoms. [Source pages 476–481]

- Both pneumonia and acute chest syndrome typically cause leukocytosis. [Source pages 476–481]

Clinical reasoning structure

- **Problem representation:** Sickle Cell Anemia in Children; source pages 476–481.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Sickle Cell Anemia in Children**.

- Source-page range: **476–481**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 476

CHAPTER 84: Sickle Cell Anemia in Children 457

Clinical Features

Consider ACS in all patients with SCA who present with complaints of chest pain, especially when associated with tachypnea, dyspnea, cough, or other symptoms of respiratory distress. Significant hypoxia and rapid deterioration to respiratory failure can occur.

Diagnosis and Differential

Chest radiographs should be obtained but may be normal during the first hours to days. The diagnosis is made in the setting of a new infiltrate on chest radiograph in the setting of chest pain and respiratory symptoms. Both pneumonia and acute chest syndrome typically cause leukocytosis. Thrombocytopenia may accompany a severe crisis. Sputum and blood cultures are rarely positive. Pneumonia is most commonly due to atypical bacteria (chlamydia, mycoplasma), but can represent community acquired pathogens (Streptococcus pneumoniae) as well. Asthma, cardiac volume overload, and, in adulthood, pulmonary hypertension are also commonly associated with SCA.

Emergency Department Care and Disposition

Because deterioration can be rapid, close monitoring is required, typically in the ICU.

1. Initial stabilization: oxygen to maintain saturation >95%, hydration at 50% to 75% maintenance fluids, analgesia (see Pain Crisis).

2. Antibiotics: macrolide, such as azithromycin 10 mg/kg PO or IV, and third generation cephalosporin, such as ceftriaxone or cefotaxime 50 to 100 mg/kg daily IV.

3. Consider simple PRBC transfusion in children with symptomatic ACS who have dropped >1 g/dL from their baseline Hgb. However, raising hemoglobin above 10 g/dL is discouraged, so this recommendation typi-

cally applies to patients with a Hgb of <9 g/dL.

4. Consider exchange transfusion for patients with rapid progression of ACS, with saturation <90% despite oxygen, increasing respiratory distress, progressive pulmonary infiltrates, or a declining Hgb despite transfusion.

5. Admit all children with suspected ACS to the hospital.

■■ ACUTE CENTRAL NERVOUS SYSTEM EVENTS

Clinical Features

Consider acute stroke in any patient with SCA who presents with sudden onset headache or neurologic changes, including hemiparesis, seizures, speech defects, visual disturbances, transient ischemic attacks, vertigo, cranial nerve palsies, paresthesias, altered mental status or coma. Children with SCA are at significantly higher risk of acute ischemic stroke than unaffected children.

Diagnosis and Differential

Because of the challenges in diagnosing acute ischemic stroke (e.g., limited availability of MRI, CT negative in acute setting), providers may need to

84_Cydulka_Ch84.indd 457 20/04/17 2:53 pm

Source page 477

initiate treatment based exclusively on clinical suspicion. As SCA patients age, their risk of a hemorrhagic stroke increases as well, primarily due to aneurysm development. If clinical history is appropriate for a subarachnoid hemorrhage, workup should proceed as for a patient without SCA. Acute chest crisis, sudden severe anemia, and meningitis can cause neurologic symptoms, and appropriate tests should be sent to investigate these possibilities in the appropriate clinical setting.

Emergency Department Care and Disposition

Suspected CNS vasoocclusion necessitates immediate stabilization and careful monitoring. Urgent exchange transfusion is preferred to standard transfusion, and the Hgb should not be raised above 10 g/dL. Transfusion should be initiated with the goal of <30% HbS. As in any ischemic stroke, temperature, glucose, and oxygenation should be monitored and controlled. Limited data exists regarding the use of alteplase, and it currently has no role in the management of SCA-related ischemic stroke in children. Intracranial hemorrhage should be managed in conjunction with a neurosurgeon. All children with suspected stroke should be admitted to the pediatric intensive care unit.

■■ PRIAPISM

Clinical Features and Diagnosis

Priapism, a painful sustained erection >4 hours in the absence of sexual stimulation, occurs when sickled cells accumulate in the corpora cavernosa. It can affect any male with SCA regardless of age, and severe prolonged attacks can cause impotence. Typically, the patient has an edematous and tender penis, with difficulty urinating.

Emergency Department Care and Disposition

Patients with priapism should receive vigorous oral or IV hydration and appropriate analgesia. As in non-SCA priapism, needle aspiration of the corpora cavernosa and administration of a vasoconstrictor, such as 1:1,000,000 epinephrine solution, may be required. Transfusion is not recommended as immediate therapy for priapism. Management and admission decisions should be made promptly in consultation with urology and pediatric hematology.

■■ HEMATOLOGICAL CRISES

Acute Sequestration Crises

Clinical Features and Diagnosis

Intrasplenic trapping of red blood cells primarily affects children under 5 years of age, but can occur in older patients with combined forms of SCA (e.g., SC disease). Often preceded by a viral syndrome, splenic sequestration presents with sudden-onset left upper quadrant pain; pallor and lethargy; tender splenomegaly; and progresses to hypotension, shock, and death. A CBC shows profound anemia, with hemoglobin <6 g/dL, or >2 g/dL lower than the patient's baseline level. Minor episodes can occur with

84_Cydulka_Ch84.indd 458 20/04/17 2:53 pm

Source page 478

insidious onset of abdominal pain, slowly progressive splenomegaly, and a more gradual fall in hemoglobin level. Generally the hemoglobin level remains >6 g/dL.

Patients may have accompanying mild neutropenia or thrombocytopenia. Less commonly, sequestration can occur in the liver. Clinical features include an enlarged and tender liver with associated hyperbilirubinemia, severe anemia, and elevated reticulocyte count. Cardiovascular collapse is rare in this condition.

Emergency Department Care and Disposition

Hydration, PRBC transfusion, and admission are required for major episodes. Transfusion should be gradual (5 mL/kg), as return of the sequestered RBCs to the intravascular space can occur, causing hyperviscosity. Occasionally, children with minor episodes can be observed, and discharged with close follow-up with a hematologist.

■■ APLASTIC EPISODES

Potentially life-threatening aplastic episodes are precipitated primarily by viral infections, typically parvovirus B19, and present with gradual onset of pallor, dyspnea, and fatigue. The CBC shows an unusually low hemoglobin and reticulocyte count, with normal white blood cell and platelet counts. Recovery is spontaneous within 1 to 2 weeks, and patients can typically be temporized with PRBC transfusion in the ED with close outpatient follow-up in 7 to 10 days for repeat labs or inpatient setting.

■■ HEMOLYTIC CRISES

Bacterial and viral infections in children with SCA can precipitate rapid hemolysis, with sudden-onset of jaundice and pallor. A CBC shows hemoglobin level decreased from baseline, with markedly increased reticulocytosis. Specific therapy is rarely required, though transfusion may be helpful in symptomatic patients. Care should be directed toward treating the underlying infection. Close follow-up to monitor hemoglobin level and reticulocyte count should be arranged if discharged from the ED.

■■ INFECTIONS

Clinical Features

Poor splenic function renders children with SCA particularly susceptible to bacterial infections, particularly in early childhood. Currently, bacteremia rates are 0.3% to 1.1%.

Diagnosis and Differential

Risk factors for sepsis in children with SCA include temperature >40°C, WBC >30,000 or <5000 cells/mm3, platelet count <100,000/mm3, Hgb <5 g/dL, ill appearance, and history of pneumococcal sepsis.

84_Cydulka_Ch84.indd 459 20/04/17 2:53 pm

Source page 479

Emergency Department Care and Disposition

Children with a temperature above 38.5°C meeting any high-risk feature (see above) should have a CBC, blood culture, reticulocyte count, and, if clinically indicated, a urine culture sent and received a parenteral antibiotic with activity against Streptococcus pneumoniae and Haemophilus influenzae, such as ceftriaxone 50 mg/kg IV or IM, as soon as possible. Vancomycin should be added if the patient is at high risk for penicillin-resistant pneumococcal infection. Septic shock must be managed aggressively with early goal-directed therapy. Patients with any respiratory symptoms should also get a chest radiograph. Well-appearing patients who have reliable next-

day follow-up can be discharged following a parenteral dose of ceftriaxone pending culture results. Ill-appearing patients with temperatures >39.5°C should be admitted.

■■ VARIANTS OF SICKLE CELL DISEASE

Sickle cell trait is the carrier state of SCA (heterozygous). These patients are typically asymptomatic and experience sickling only in the presence of extreme hypoxia or high altitude. Baseline microscopic hematuria may exist. Sickle cell–hemoglobin C disease is a heterozygous condition characterized by mild to moderate anemia and fewer complications. However, splenomegaly can persist to adulthood, and these patients remain at risk for sequestration crises. Sickle cell β-thalassemia disease is a heterozygous condition with variable severity and symptoms.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 142, "Sickle Cell Disease in Children," by John Marshall.

84_Cydulka_Ch84.indd 460 20/04/17 2:53 pm

References/provenance

- Source: *Tintinalli's Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 476–481.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 85: Hematologic-Oncologic Emergencies in Children

Source pages 482–487 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 482–487

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Hematologic-Oncologic Emergencies in Children**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 85: Hematologic-Oncologic Emergencies in Children 463 5. [Source pages 482–487]

- Hyperleukocytosis (WBC >100,000/mm3) typically requires treatment if symptoms of stasis (e.g., stroke, visual change, dyspnea, priapism) occur, or if the WBC is >200,000/mm3 for AML and >300,000/mm3 for ALL in the absence of symptoms. [Source pages 482–487]

- Aggressive hydration with normal saline bolus of 20 mL/kg repeated as tolerated. [Source pages 482–487]

- If patient is symptomatic after hydration, arrange for leukapheresis. [Source pages 482–487]

- Avoid PRBC transfusions and diuretics if possible. [Source pages 482–487]

Clinical reasoning structure

- **Problem representation:** Hematologic-Oncologic Emergencies in Children; source pages 482–487.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Hematologic-Oncologic Emergencies in Children**.

- Source-page range: **482–487**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 482

CHAPTER 85: Hematologic-Oncologic Emergencies in Children 463

5. Hyperleukocytosis (WBC >100,000/mm3) typically requires treatment if symptoms of stasis (e.g., stroke, visual change, dyspnea, priapism) occur, or if the WBC is >200,000/mm3 for AML and >300,000/mm3 for ALL in the absence of symptoms. Treatment includes:

a. Aggressive hydration with normal saline bolus of 20 mL/kg repeated as tolerated.

b. If patient is symptomatic after hydration, arrange for leukapheresis.

c. Avoid PRBC transfusions and diuretics if possible.

d. Anticipate and initiate treatment for tumor lysis syndrome.

e. If asymptomatic with high levels, consider hydroxyurea which will reduce the WBC count to half in 24 to 48 hours.

■■ LYMPHOMA

Hodgkin’s lymphoma is a lymphoid neoplasm preferentially affecting adolescents. Most cases present in the cervical or supraclavicular lymph nodes causing nontender, nonerythematous, rubbery lymphadenopathy chains. Systemic symptoms (e.g., fever, night sweats, weight loss) occur in 39% to 50% of children and teens. A chest radiograph may demonstrate an anterior mediastinal mass. Non-Hodgkin’s lymphoma can originate in or outside of the lymphatic system, and occurs across ages, particularly in those with a history of immunosuppression. Because the tumor can occur in any organ, presenting signs and symptoms differ by location. A CBC, electrolytes and creatinine (looking for tumor lysis), and chest radiograph (looking for mediastinal mass) should be performed in the ED. ED care involves management of acute complications, avoidance of steroid therapy except in life-threatening situations, consultation with an oncologist, and recognition of the potential for intrathoracic airway compromise in patients with a mediastinal mass. Compression of the superior vena cava (SVC syndrome) or compression of both SVC and trachea (superior mediastinal syndrome) can occur, with respi-

ratory symptoms, upper body edema, headache, and altered mental status.

■■ CENTRAL NERVOUS SYSTEM TUMORS

Brain tumors are common pediatric malignancies, and typically present with nonspecific symptoms: headaches (especially early morning), irritability, emesis, and behavioral changes related to increased intracranial pressure. In infants, overt signs of increased pressure (e.g., bulging fontanel, sunsetting of eyes) can sometimes be appreciated. Vomiting, ataxia, cranial nerve palsies, or vague neurologic signs and symptoms can occur as well. CT scan or MRI are acceptable imaging studies in the ED. Spinal cord tumors can present with back pain with or without neurological signs and are evaluated by MRI. Seizures should be treated if present, and dexamethasone (1 mg/yr of age up to 10 mg) can be given to reduce vasogenic edema in patients not suspected to have leukemia or lymphoma. Further management should be determined by oncology and neurosurgery.

■■ EXTRACRANIAL SOLID TUMORS

Neuroblastoma is a primitive ganglion tumor that can arise along the sympathetic nervous system, most commonly in the adrenal gland, other abdominal location, chest, or neck. Patients can present with a painless mass,

85_Cydulka_Ch85.indd 463 20/04/17 2:53 pm

Source page 483

hepatomegaly, or symptoms of mass effect from compression of the bowel, bladder, lymphatics, spinal cord, trachea, or superior vena cava. Occasionally, retrobulbar metastasis can cause raccoon eyes and proptosis. Paraneoplastic manifestations can include hypertension, watery diarrhea, and opsoclonus-myoclonus syndrome (rapid, multidirectional eye movements and jerking of the extremities). CBC for evidence of bone marrow infiltration and chest radiograph for mediastinal mass should be obtained in the ED. Urine levels of homovanillic acid and vanillylmandelic acid are helpful diagnostically.

Wilms tumor, or nephroblastoma, primarily affects young children (<10 years of age) and typically presents with an abdominal mass with few symptoms other than those explained by local compression. Many patients present with lung metastasis or hypertension from increased renin production. These masses can rupture with trauma or vigorous palpation.

Germ cell tumors typically present as masses in the ovary or testicle, although extragonadal locations are common in younger children. They are particularly more common in boys with a history of an undescended testicle. Diagnosis is by ultrasound.

Retinoblastoma is a white-grey intraocular malignancy that typically presents in children less than 2 years of age with loss of the normal

red reflex (Fig. 85-1). Strabismus, decreased visual acuity, a fixed pupil, glaucoma, or an injected, painful eye are less common presentations.

Figure 85-1. Typical appearance of leukocoria in a patient with retinoblastoma. Reproduced with permission from Shah BR, Lucchesi M. Atlas of Pediatric Emergency Medicine. New York: McGraw-Hill Companies, Inc.; 2006.

85_Cydulka_Ch85.indd 464 20/04/17 2:53 pm

Source page 484

One-quarter of cases are bilateral, and diagnosis is made through ophthalmology consultation and CT scan or MRI.

Bone and tissue sarcomas include rhabdomyosarcoma (a painless tissue mass), osteosarcoma (a bony tumor around the metaphysis of the knee, proximal humerus, pelvis, or mandible), and Ewing's sarcoma (a bony tumor of the long bones and axial skeleton). Osteosarcoma and Ewing's sarcoma can present as a dull, aching pain, particularly at night or with activity, a tender mass that seems to appear following a minor trauma, or with systemic symptoms. Diagnosis of rhabdomyosarcoma is by ultrasound or CT scan, while osteosarcoma and Ewing's sarcoma often are seen on plain radiograph, with osteosarcoma demonstrating a mixed sclerotic/lytic "sunburst" appearance of periosteal reaction and Ewing's sarcoma having a destructive, "moth-eaten" appearance.

■■ HEMOPHILIA

Hemophilia is an inherited deficiency of factor 8 (hemophilia A) or factor 9 (hemophilia B) which can be mild (>5% factor activity), moderate (1% to 5% factor activity), or severe (<1% factor activity).

Clinical Features

Appropriate family history or bleeding/bruising out of proportion with injury, recurrent bleeding, or in bleeding into uncommon locations (GI, joints) should raise concern for a bleeding disorder. Common issues in hemophilia are hemarthroses (and resultant destructive arthropathy), muscle bleeds (occasionally causing compartment syndrome), and intracranial hemorrhage.

Diagnosis and Differential

Hemophilia causes an elevated PTT, with normal platelets, bleeding time, and PT. Quantitative factor levels are diagnostic. Other bleeding disorders include thrombocytopenias, vitamin K deficiency bleeding (typically 2 to 14 days after birth in neonates whose parents have refused prophylactic vitamin K administration at birth), and von Willebrand's disease (prolonged bleeding in the face of normal platelet numbers with or without factor 8 deficiency).

Emergency Department Care and Disposition

Treatment is with appropriate factor replacement: up to 100% correction with severe bleeds, and less with more minor bleeds.

■■ ANEMIA

Clinical Features

Anemia may be asymptomatic or accompanied by pallor and fatigue or heart failure.

Diagnosis and Differential

Iron deficiency anemia due primarily to excessive cow's milk intake typically affects children aged 6 months to 3 years of age. Children outside of this age range should be assessed for occult GI bleeding with stool guaiac testing.

85_Cydulka_Ch85.indd 465 20/04/17 2:53 pm

Source page 485

Laboratory studies demonstrate anemia with a low reticulocyte count and low mean corpuscular volume (MCV). Hemolytic anemia can be primary or secondary to an underlying infection or intrinsic disorder. Patients have isolated anemia with high MCV, spherocytes and schistocytes on peripheral smear, elevated indirect bilirubin, and a positive direct antibody test.

Emergency Department Care and Disposition

Most asymptomatic or minimally symptomatic patients with iron deficiency anemia can be safely discharged with careful follow-up, reduction of milk intake below 24 oz/d, and initiation of iron therapy (2 to 3 mg/kg/dose of elemental iron three times daily). Hemolytic anemia may require more careful observation or transfusion, based on the patient's clinical status and hemoglobin level. Steroids are indicated in the treatment of autoimmune hemolytic anemia. If transfusion is required, patients with very low hemoglobin levels (<5 g/dL) should receive 3 to 5 mL/kg of PRBCs over 2 to 3 hours to start, while patients with a pretransfusion Hb >5 g/dL can receive 10 mL/kg to start.

■■ IDIOPATHIC THROMBOCYTOPENIC PURPURA (ITP)

ITP is an autoimmune disorder of platelet destruction.

Clinical Features

While often an isolated condition in preschool-aged children, ITP can be a feature of rheumatologic disorders (e.g., lupus) or infections (e.g., HIV, hepatitis C), particularly in teens. Patients can present with petechiae or bleeding (nares or gingiva), and, in the younger child without a coexisting condition, ITP often follows a viral infection by days to weeks.

Diagnosis and Differential

Other autoimmune and infectious disorders (lupus, HIV, hepatitis C) should be considered, particularly in the teen. In ITP, other cell lines are not

affected and platelets are often below 20,000/mm3 with a large platelet volume. CBC and blood type should be sent. Disseminated intravascular coagulation and hemolytic-uremic syndrome should be considered in the ill patient with thrombocytopenia.

Emergency Department Care and Disposition

ITP in children carries an excellent prognosis, with 70% resolving spontaneously within 6 months, regardless of treatment. Children with minimal symptoms and platelets >20,000/mm3 are often followed without intervention. For children with significant bleeding or platelet counts <20,000/mm3, the following options exist:

1. Prednisone 1 to 2 mg/kg/d for 2 to 4 weeks. Steroids should only be started in conjunction with a hematologist once leukemia has been excluded.
2. Intravenous immunoglobulin (IVIG) 1 g/kg over 4 to 6 hours. Premedication with acetaminophen and diphenhydramine is recommended.
3. Anti-Rh (D) immunoglobulin (WinRho®) 75 μg/kg over 1 hour can be used in patients whose blood type is Rh+. Because these antibodies bind

85_Cydulka_Ch85.indd 466 20/04/17 2:53 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 482–487.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 86: Renal Emergencies in Infants and Children

Source pages 488–493 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 488–493

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Renal Emergencies in Infants and Children**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 86: Renal Emergencies in Infants and Children 469 such as acute tubular necrosis, hyaline casts may be seen on UA with a low to normal specific gravity. [Source pages 488–493]

- In glomerulonephritis, UA often shows hema- turia, casts, and proteinuria while isolated proteinuria is more indicative of nephrotic syndrome. [Source pages 488–493]

- Basic blood tests such as serum electrolytes (includ- ing sodium and potassium), BUN and creatinine, and a complete blood count (CBC) should be obtained in all cases of AKI to help identify the cause and guide management. [Source pages 488–493]

- Hyperkalemia and other findings may require emergency interventions. [Source pages 488–493]

- Additional blood tests may be indicated in specific scenarios. [Source pages 488–493]

Clinical reasoning structure

- **Problem representation:** Renal Emergencies in Infants and Children; source pages 488–493.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Renal Emergencies in Infants and Children**.

- Source-page range: **488–493**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 488

CHAPTER 86: Renal Emergencies in Infants and Children 469

such as acute tubular necrosis, hyaline casts may be seen on UA with a low to normal specific gravity. In glomerulonephritis, UA often shows hematuria, casts, and proteinuria while isolated proteinuria is more indicative of nephrotic syndrome. Basic blood tests such as serum electrolytes (including sodium and potassium), BUN and creatinine, and a complete blood count (CBC) should be obtained in all cases of AKI to help identify the cause and guide management. Hyperkalemia and other findings may require emergency interventions. Additional blood tests may be indicated in specific scenarios.

Emergency Department Care and Disposition

Treatment of AKI is based on the etiology. Goals of treatment include identifying the underlying cause and correcting fluid and electrolyte imbalances. Life-threatening complications such as severe hyperkalemia or hypertensive emergency should be addressed immediately. A pediatric nephrologist should be consulted for most cases of AKI as many cases require inpatient admission.

1. For prerenal AKI, resuscitate with crystalloid fluids, starting with 10 to 20 mL/kg of normal saline without added potassium. For prerenal AKI due to hemorrhagic shock, give crystalloid fluids until blood products are available then transfuse packed red blood cells in aliquots of 10 mL/kg.

2. For AKI due to intrinsic renal injury or disease, treatment depends on the clinical state of the patient and the etiology of the AKI. For example, it may be necessary to fluid restrict despite oliguria in order to achieve overall fluid balance in hypervolemic patients. It is crucial to monitor the patient’s weight and liquid intake and output. Consider discontinuing all nephrotoxic medications and treat hypertension with antihyperten-

sive agents.

3. For postrenal AKI, insert a Foley catheter to relieve the obstruction. Antihypertensive agents and/or diuretics may also be necessary to control significant hypertension.

■■ NEPHROTIC SYNDROME

Nephrotic syndrome is a chronic disease that affects glomerular capillary wall permeability causing urinary protein loss. The hallmarks of nephrotic syndrome are significant proteinuria, hypoalbuminemia, hyperlipidemia, and edema.

Clinical Features

Patients most commonly present with edema, which may involve the face, abdomen, scrotum or labia, or lower extremities. Because periorbital edema is nonspecific, patients are often misdiagnosed as having an allergic reaction. Patients may have foamy or bloody urine. Extreme hypoalbuminemia may cause pleural effusions, bowel wall edema, or ascites. Life-threatening complications of nephrotic syndrome include thromboembolic events (arterial and venous) and severe infections due to an immunocompromised state (due to loss of antibodies).

86_Cydulka_Ch86.indd 469 20/04/17 2:54 pm

Source page 489

Diagnosis and Differential

Nephrotic syndrome is classified as primary (only affecting the kidney) or secondary (multisystem disease with renal involvement). Primary nephrotic syndromes include minimal change disease, focal segmental sclerosis, membranous nephropathy, membranoproliferative nephritis, and proliferative nephritis (which may be diffuse, focal, or mesangial). Secondary forms of nephrotic syndrome include lupus, Henoch–Schönlein purpura, sickle cell anemia, and drug or toxin exposure (e.g., heavy metals). TORCH infections may cause congenital nephrotic syndrome.

There are four diagnostic criteria for nephrotic syndrome: (1) hypoproteinemia with albumin level <3 g/dL, (2) urine protein to creatinine ratio >2 on first morning void or 24-hour urine protein >50 mg/kg, (3) hypercholesterolemia (>200 mg/dL), and (4) generalized edema. Further testing may be required to distinguish primary from secondary nephrotic syndrome. Renal biopsy is not indicated during initial episodes of acute nephrotic syndrome.

Emergency Department Care and Disposition

In the emergency department (ED), the goal is to treat the acute symptoms. Managing the fluid status of the nephrotic syndrome patient can be chal-

lenging since patients may be intravascularly depleted but show signs of fluid overload with significant edema. Treatment principles are as follows:

1. Treat hypovolemic shock with 20 mL/kg normal saline.
2. Treat mild to moderate dehydration with small, frequent amounts of a low-salt oral solution.
3. Treat volume overload and edema with furosemide 1 to 2 mg/kg/dose. If the serum albumin is extremely low, diuretics may be less effective, so albumin 0.5 to 1 g/kg should be administered followed by a dose of furosemide.
4. Definitive treatment of nephrotic syndrome often includes oral corticosteroids; however, steroid therapy should be initiated in conjunction with a pediatric nephrologist.

Many patients with nephrotic syndrome can safely be discharged home on a low-salt diet (< 2 g/d) with close follow-up with their PCP or nephrologist. Indications for admission include severe edema, thrombotic complications, or signs and symptoms of systemic infection.

■■ ACUTE GLOMERULONEPHRITIS

Acute glomerulonephritis (GN) is the result of inflammation leading to glomerular injury and is characterized by hematuria and proteinuria. Glomerulonephritis may be immune-mediated, inherited, or post-infectious, and is classified as primary (isolated renal involvement) or secondary (resulting from a systemic disorder).

Clinical Features

Patients typically present with bloody, foamy (from proteinuria), or tea-colored urine along with oliguria, fatigue, or lethargy. Severe hypertension may result in headaches. In post-streptococcal GN, there may be a history

86_Cydulka_Ch86.indd 470 20/04/17 2:54 pm

Source page 490

of a sore throat, upper respiratory infection (URI), or skin infection preceding the onset of urinary symptoms by 1 to 2 weeks. In IgA nephropathy, children may report a concurrent URI or URI symptoms within the preceding week.

Diagnosis and Differential

Common causes of acute GN include post-infectious etiologies (post-streptococcal GN), Henoch–Schönlein purpura, systemic lupus erythematosus, and IgA nephropathy. Diagnosis of GN is made by examination of the urine. UA with microscopy demonstrates RBC casts, dysmorphic RBCs, and proteinuria. Other laboratory studies should include a CBC, electrolytes, BUN and creatinine, serum albumin, and urine culture. Serum

complement levels are low in more than 90% of patients with post-streptococcal GN. Streptococcal serologic tests such as antistreptolysin-O and streptozyme may also be helpful.

Emergency Department Care and Disposition

The treatment of glomerulonephritis is determined by the underlying cause. Treatment of post-streptococcal GN is primarily supportive as symptoms usually resolve within a few weeks. Depending on the severity of disease, immunosuppressive agents may be used to treat inflammation and ACE inhibitors may be required to treat hypertension, particularly in patients with proteinuria. Nephrology consultation is indicated for patients with new-onset GN. Patients with mild disease may be discharged home on a fluid-restricted low-sodium diet with close follow-up.

■■ HEMOLYTIC-UREMIC SYNDROME

Hemolytic-uremic syndrome (HUS) is characterized by acute renal failure, thrombocytopenia, and microangiopathic hemolytic anemia. Approximately 90% of cases are due to Shiga-toxin producing Escherichia coli O157:H7, found in unpasteurized milk, undercooked meats, and contaminated produce.

Clinical Features

The majority of patients with HUS present with nausea, vomiting, and bloody diarrhea. Within a week of symptom onset, oliguria, anemia, seizures, and encephalopathy may occur.

Diagnosis and Differential

Diagnosis of HUS is supported by labs. A CBC will demonstrate a hemoglobin level of 5 to 9 g/dL and platelets <150,000/mm3. Peripheral smear shows schistocytes, helmet cells, and burr cells. A stool specimen to test for Shiga toxin and E. coli O157:H7 should also be sent.

Emergency Department Care and Disposition

Treatment of HUS in the ED is primarily supportive. Hypovolemia should be treated with 10 to 20 mL/kg boluses of normal saline while monitoring for fluid overload in oliguric patients. Platelet transfusions are not

86_Cydulka_Ch86.indd 471 20/04/17 2:54 pm

Source page 491

recommended as they increase thrombotic complications. Antibiotics are contraindicated in pediatric diarrheal illnesses as they may increase the risk of HUS. Antiperistaltic agents (e.g., loperamide) are also contraindicated as they increase the risk for systemic complications of E. coli infection. All patients with HUS require hospitalization and up to

70% will require dialysis.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study

Guide, 8th ed., see Chapter 134, “Renal Emergencies in Children,” by Andrew

Dixon and Brandy Stauffer.

86_Cydulka_Ch86.indd 472 20/04/17 2:54 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.

- Content ID: SEG-TINTINALLI-EMM8-FULL.

- Source pages: 488–493.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 87: Sexually Transmitted Infections

Source pages 494–499 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 494–499

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Sexually Transmitted Infections**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 87: Sexually Transmitted Infections 475 2. [Source pages 494–499]

- Alternative treatment options include cefixime 400 mg PO single dose AND azithromycin 1 g PO single dose or doxycycline 100 mg PO twice daily for 7 days AND test-of-cure in 1 week. [Source pages 494–499]

- Disseminated gonorrhea is treated initially with parenteral ceftriaxone 1 g daily IM/IV AND azithromycin 1 g PO single dose. [Source pages 494–499]

- Duration of par- enteral therapy varies and should be determined in consultation with an infectious disease specialist. [Source pages 494–499]

- Recommend that patients avoid sexual contact for 7 days after complet- ing antibiotic treatment and symptoms have resolved. [Source pages 494–499]

Clinical reasoning structure

- **Problem representation:** Sexually Transmitted Infections; source pages 494–499.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Sexually Transmitted Infections**.

- Source-page range: **494–499**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 494

CHAPTER 87: Sexually Transmitted Infections 475

2. Alternative treatment options include cefixime 400 mg PO single dose AND azithromycin 1 g PO single dose or doxycycline 100 mg PO twice daily for 7 days AND test-of-cure in 1 week.

3. Disseminated gonorrhea is treated initially with parenteral ceftriaxone 1 g daily IM/IV AND azithromycin 1 g PO single dose. Duration of parenteral therapy varies and should be determined in consultation with an infectious disease specialist.

4. Recommend that patients avoid sexual contact for 7 days after completing antibiotic treatment and symptoms have resolved.

■■ TRICHOMONIASIS

Clinical Features

Patients with Trichomonas vaginalis parasite infections can remain asymptomatic or present with a range of symptoms up to severe inflammatory disease. In men this infection is often asymptomatic, but may present as urethritis. Symptoms in women may include vulvar irritation and a malodorous, thin watery discharge with associated burning, pruritus, dysuria, urinary frequency, dyspareunia, or lower abdominal pain.

Diagnosis and Differential

Microscopic visualization of the motile Trichomonas vaginalis parasites on a wet preparation of cervical smears or spun urine samples is diagnostic. Motility is only present for 10 to 20 minutes after collection thereby decreasing sensitivity. Culture is the most sensitive and specific test, but may take up to 7 days. DNA probes and monoclonal antibodies are used at some institutions.

Emergency Department Care and Disposition

1. Metronidazole 2 g PO, or tinidazole 2 g PO in a single dose is the treatment of choice.

2. An alternative treatment regimen is metronidazole 500 mg PO twice daily for 7 days.

3. Metronidazole (pregnancy category B) 2 g PO in a single dose is the drug of choice for treating symptomatic pregnant patients.

4. Patients should avoid intercourse for 1 week after the last dose of antibiotics.

■■ SYPHILIS

Clinical Features

Treponema pallidum, a spirochete, causes the three stages of syphilis infections: primary, secondary, and tertiary stages. Primary syphilis is characterized by a chancre (Fig. 87-1), a single painless ulcer with indurated borders that develops on the penis, vulva, or other areas of sexual contact. The primary chancre heals and disappears after 3 to 6 weeks, and the secondary stage occurs after several additional weeks. Rash and lymphadenopathy are the most common symptoms of secondary syphilis. The rash consists of nonpruritic dull red or pink papules that begin on the trunk and flexor surfaces of the extremities and subsequently spread to the palms and soles. Constitutional

87_Cydulka_Ch87.indd 475 20/04/17 2:54 pm

Source page 495

Figure 87-1. Syphilis chancre in a male. A painless ulcer caused by syphilis is seen on the distal penile shaft with a smaller erosion on the glans. The ulcer is quite firm on palpation. Reproduced with permission from Wolff K, Johnson RA: Fitzpatrick's Color Atlas and Synopsis of Clinical Dermatology, 7th ed. New York: McGraw-Hill Companies, Inc; 2013.

symptoms such as fever, malaise, headache, and sore throat are common before spontaneous resolution of this second stage. Tertiary or latent syphilis develops in about one-third of patients and can begin 3 to 20 years after the initial infection. Involvement of the nervous system (meningitis, dementia, tabes dorsalis), cardiovascular system (thoracic aneurysm), and widespread granulomatous lesions (gummata) are characteristic of tertiary syphilis.

Diagnosis and Differential

Treponema pallidum cannot be cultured in the laboratory and testing is a challenge because there is no single optimal diagnostic test. The sensitivity and specificity of tests for syphilis depend on the stage of the disease and type of test. Nontreponemal tests such as Venereal Disease Research Laboratory (VDRL) and rapid plasma regain (RPR) detect nonspecific antibodies that are indicative of infection but do not become detectable until 1 to 4 weeks after a chancre appears. Positive results for these screening tests must subsequently be confirmed with an immunoassay specific for T. pallidum antibodies. A presumptive diagnosis of syphilis is made if a positive result on a nontrepo-

nemal antibody test is supported by a positive confirmatory result on a trepo-
nemal antibody test. Direct visualization of organisms using darkfield
microscopy is diagnostic for primary, secondary, or early congenital syphilis.

Emergency Department Care and Disposition

1. For primary and secondary syphilis, treat patients with benzathine
penicillin G 2.4 million units IM as a single dose. Be aware of the

87_Cydulka_Ch87.indd 476 20/04/17 2:54 pm

Source page 496

Jarisch–Herxheimer reaction (fever, headache, and myalgias) that may
develop within the first 24 hours after treatment.

2. Alternative treatment options for primary, secondary, and early latent
syphilis include doxycycline 100 mg PO twice daily for 14 days, or
tetracycline 500 mg PO four times daily for 28 days.

3. Treat tertiary syphilis with benzathine penicillin G 2.4 million units IM
weekly for 3 weeks.

■■ HERPES SIMPLEX INFECTIONS

Clinical Features

Primary genital herpes simplex virus (HSV) infection often begins with a
viral prodrome lasting 2 to 24 hours that is characterized by localized or
regional pain, tingling, and burning. Constitutional symptoms such as head-
ache, fever, painful inguinal lymphadenopathy, anorexia, or malaise often
follow. Papules and vesicles on an erythematous base appear, with vesicular
erosion that occurs over a few hours to several days. In men, painful HSV
ulcers often appear on the shaft or glans of the penis (Fig. 87-2), while in

Figure 87-2. Genital herpes in a male. Classic vesicles are shown proximally on the penis; several formerly vesicular lesions have crusted over. Reproduced with permission from Goldsmith LA, Katz SI, Gilchrest BA, et al: Fitzpatrick's Dermatology in General Medicine, 8th ed. New York: McGraw-Hill Companies, Inc; 2012.

87_Cydulka_Ch87.indd 477 20/04/17 2:54 pm

Source page 497

women, the painful ulcers occur on the introitus, urethral meatus, labia, and
perineum. Viral shedding occurs for 10 to 12 days after the onset of the
rash, and lesions heal completely within 3 weeks.

Diagnosis and Differential

HSV infection is diagnosed based on a characteristic history and associated
physical findings. Viral cultures taken from vesicles or early ulcers are
more reliable for confirmation than the classically described Tzanck smear

to identify intranuclear inclusions.

Emergency Department Care and Disposition

1. Treat a patient's first episode of HSV with a 7- to 10-day course of acyclovir 400 mg PO three times daily, valacyclovir 1 g PO twice daily, or famciclovir 250 mg PO three times daily. Treatment hastens recovery but does not cure the infection.
2. For proctitis or oral infections, treat with acyclovir 400 mg five times daily for 7 to 10 days.
3. Patients with severe disease that require hospitalization should be treated with intravenous acyclovir 5 to 10 mg/kg body weight every 8 hours.
4. Recurrent episodes of genital herpes can be treated with a 5-day course of valacyclovir 1 g PO daily or acyclovir 400 mg PO three times daily. Antiviral therapy may reduce the severity and duration of recurrent episodes of HSV when initiated when symptoms begin.

■■ CHANCROID

Clinical Features

Caused by Haemophilus ducreyi, chancroid presents as a painful erythematous papule at the site of infection. This lesion erodes and becomes ulcerated and pustular over the next 2 days (Fig. 87-3). Up to 50% of patients have more than one lesion, with "kissing lesions" occurring from autoinoculation of adjacent areas of skin. Painful lymphadenopathy develops 1 to 2 weeks after primary infection, and if treatment is not initiated these lymph nodes become necrotic and infected buboes.

Diagnosis and Differential

Chancroid is typically diagnosed clinically, with care to consider the diagnoses of syphilis and HSV. Sometimes the organism may be cultured from a swab of an ulcer or bubo, but a special culture medium is required that is not found in many laboratories.

Emergency Department Care and Disposition

1. Treatment regimens include azithromycin 1 g PO as a single dose, ceftriaxone 250 mg IM as a single dose, erythromycin base 500 mg PO three times daily for 7 days, or ciprofloxacin 500 mg PO twice daily for 3 days. Symptoms often improve within 3 days, but larger ulcers may require 2 to 3 weeks to heal.
2. Incision and drainage of buboes can be considered for symptomatic relief.

87_Cydulka_Ch87.indd 478 20/04/17 2:54 pm

References/provenance

- Source: *Tintinalli's Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.

- Source pages: 494–499.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 88: Toxic Shock Syndromes

Source pages 500–503 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 500–503

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Toxic Shock Syndromes**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 88: Toxic Shock Syndromes 481 vomiting, diarrhea, or abdominal pain. [Source pages 500–503]

- The rash associated with TSS is described as a “painless sunburn” that typically fades within 3 days and is followed by full-thickness desquamation. [Source pages 500–503]

- Diagnosis and Differential TSS is a clinical diagnosis characterized by an acute febrile illness asso- ciated with erythroderma, hypotension, and multiorgan involvement. [Source pages 500–503]

- Laboratory evaluation reveals evidence of end-organ damage, and testing often includes CBC, comprehensive metabolic panel, coagulation stud- ies, creatine phosphokinase level, urinalysis, chest radiograph, ECG, blood gas, and cultures of blood and other potential sites of infection. [Source pages 500–503]

- Other infections to consider in the differential diagnosis of TSS include streptococcal toxic shock syndrome (STSS), myonecrosis due to Clos- tridium perfringens, TSS due to Clostridium sordellii, staphylococcal scalded skin syndrome, sepsis due to other bacterial organisms, Rocky Mountain spotted fever, leptospirosis, and meningococcemia. [Source pages 500–503]

Clinical reasoning structure

- **Problem representation:** Toxic Shock Syndromes; source pages 500–503.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Toxic Shock Syndromes**.

- Source-page range: **500–503**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 500

CHAPTER 88: Toxic Shock Syndromes 481

vomiting, diarrhea, or abdominal pain. The rash associated with TSS is described as a “painless sunburn” that typically fades within 3 days and is followed by full-thickness desquamation.

Diagnosis and Differential

TSS is a clinical diagnosis characterized by an acute febrile illness associated with erythroderma, hypotension, and multiorgan involvement. Laboratory evaluation reveals evidence of end-organ damage, and testing often includes CBC, comprehensive metabolic panel, coagulation studies, creatine phosphokinase level, urinalysis, chest radiograph, ECG, blood gas, and cultures of blood and other potential sites of infection. Other infections to consider in the differential diagnosis of TSS include streptococcal toxic shock syndrome (STSS), myonecrosis due to Clostridium perfringens, TSS due to Clostridium sordellii, staphylococcal scalded skin syndrome, sepsis due to other bacterial organisms, Rocky Mountain spotted fever, leptospirosis, and meningococcemia. Noninfectious illnesses such as Stevens–Johnson syndrome, toxic epidermal necrolysis, and Kawasaki’s disease may also be considered in the proper clinical circumstances.

Emergency Department Care and Disposition

TSS is mediated by pyrogenic exotoxins released by certain strains of S. aureus, and initial management includes aggressive resuscitation to treat shock and begin treating potential sources of infection. Abscesses should be drained, foreign bodies should be removed, surgical consultation should be considered for debridement of infected tissue, and appropriate broad-spectrum antibiotics are recommended.

1. Resuscitate patients with hypotension and shock using aggressive IV crystalloid intravenous fluids such as normal saline. Use vasopressors to

treat persistent hypotension.

2. Treat TSS with suspected methicillin-resistant Staphylococcus aureus (MRSA) with vancomycin 15 mg/kg IV every 12 hours (maximum dose 2.25 g) or linezolid 600 mg IV plus clindamycin 900 mg IV every 8 hours.

3. Patients with suspected methicillin-sensitive Staphylococcus aureus (MSSA) can be treated with nafcillin or oxacillin 2 g IV every 4 hours, plus clindamycin 900 mg IV every 8 hours.

4. Consider IV immunoglobulin therapy after consultation with infectious disease or critical caire specialists when no clinical improvement is seen after 6 hours of aggressive treatment.

■■ STREPTOCOCCAL TOXIC SHOCK SYNDROME

STSS is a clinical illness characterized by shock and multiorgan failure as a result of invasive infection due to group A Streptococcus (Streptococcus pyogenes) and other streptococcal pathogens. Risk factors for STSS include the presence of underlying medical conditions such as chronic heart, liver, and kidney disease, diabetes, alcoholism, drug abuse, surgical procedures, traumatic injuries, childbirth, varicella infection, and influenza.

88_Cydulka_Ch88.indd 481 20/04/17 2:55 pm

Source page 501

Clinical Features

Similar to TSS, fulminant STSS is often preceded by nonspecific symptoms such as fever, chills, sweats, malaise, arthralgias, cough, sore throat, rhinorrhea, anorexia, nausea, vomiting, abdominal pain, or diarrhea. A rash is seen in a minority of cases. Pain at a localized site, often an extremity or abdomen, that is out of proportion to physical findings is a common presentation of STSS due to the strong association of STSS and necrotizing soft tissue infections. Diagnostic criteria for fulminant STSS are listed in Table 88-2.

Diagnosis and Differential

STSS is a clinical diagnosis characterized by an acute febrile illness associated with hypotension and multiorgan involvement in the setting of a streptococcal infection. Laboratory evaluation is similar to TSS (see above) and findings identify evidence of end-organ damage. The differential diagnosis includes many of the same disease processes as TSS (see above).

Emergency Department Care and Disposition

1. Treat suspected STSS patients with aggressive isotonic fluid resuscitation to restore volume and tissue perfusion. Large volumes of IV crystalloid fluid (10 to 20 L/day) may be required. Use vasopressors or

inotropic agents for persistent hypotension.

Table 88-2 Case Definition for Streptococcal Toxic Shock Syndrome

Clinical criteria:

●● Hypotension: systolic blood pressure ≤90 mm Hg (adult) or < 5th percentile by age (children < 16 years of age)

●● Multiorgan involvement (≥ 2 organ systems):

●● Renal: serum Cr ≥ 2 mg/dL (≥ 177 μmol/L) for adults or ≥ 2 times the upper limit of normal for age. In patients with preexisting renal disease, > 2-fold elevation above baseline.

●● Coagulopathy: platelet count ≤ 100,000/mm3 (≤ 100 × 106/L) or disseminated intravascular coagulation, defined by prolonged clotting times, low fibrinogen level, and presence of fibrin degradation products

●● Hepatic: total bilirubin, ALT, or AST ≥ 2 times the upper limit of normal for the patient's age. In patients with preexisting liver disease, > 2-fold elevation above baseline.

●● Acute respiratory distress syndrome: acute onset of diffuse pulmonary infiltrates and hypoxemia in the absence of cardiac failure or by evidence of diffuse capillary leak manifested by acute onset of generalized edema, or pleural or peritoneal effusions with hypoalbuminemia.

●● Skin: generalized erythematous macular rash that may desquamate

●● Soft tissue necrosis, including necrotizing fasciitis or myositis, or gangrene

Laboratory criteria:

●● Isolation of group A Streptococcus

Case classification:

●● Probable: All clinical criteria met + absence of other identified etiology for illness + isolation of group A Streptococcus from a nonsterile site

●● Confirmed: All clinical criteria met + isolation of group A Streptococcus from a sterile site (e.g., blood, cerebrospinal fluid, synovial fluid, pleural fluid, or pericardial fluid)

Abbreviations: ALT, alanine aminotransferase; AST, aspartate aminotransferase; Cr, creatinine.

88_Cydulka_Ch88.indd 482 20/04/17 2:55 pm

Source page 502

2. Assess the possible need for airway management, as more than half of patients with STSS will develop ARDS and require intubation and mechanical ventilation.

3. Obtain blood and wound cultures prior to giving antibiotics when possible, as the majority of patients will have positive blood cultures to confirm a streptococcal pathogen.

4. Broad-spectrum antimicrobial therapy is recommended. Treat suspected STSS with piperacillin-tazobactam 4.5 g IV or meropenem 1 g IV

plus IV clindamycin 900 mg. Add vancomycin 15 mg/kg IV (maximum dose 2.25 g) when MRSA infection is suspected.

5. Administration of IV immunoglobulin remains controversial and consultation should be obtained prior to initiating this therapy.

6. Initiate an emergent surgical consultation for patients with a suspected necrotizing soft tissue infection. Patients with STSS may require debridement, fasciotomy, or amputation as the disease progresses.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 150, "Toxic Shock Syndromes," by Stephen Y. Liang

88_Cydulka_Ch88.indd 483 20/04/17 2:55 pm

Source page 503

C h a p ter Sepsis

89 John E. Gough

Sepsis is generally defined as a suspected or confirmed infection with evidence of systemic inflammation. Severe sepsis is sepsis with evidence of new organ dysfunction thought to be secondary to tissue hypoperfusion. Septic shock is present when cardiovascular failure occurs, reflected by persistent hypotension or the need for vasopressors despite adequate fluid resuscitation. The majority of sepsis cases are caused by gram-negative and gram-positive bacteria; however, sepsis is a heterogeneous clinical syndrome that can be caused by any class of microorganism including fungi, mycobacteria, viruses, rickettsiae, and protozoa. The most likely microorganisms that cause sepsis varies based on possible patient exposure to drug-resistant microorganisms (e.g., due to recent hospitalization or other health care–associated exposure) and the specific anatomic site of suspected infection. Pneumonia, intraabdominal infection, urinary tract infection, and skin or soft tissue infections are the most common infections that precipitate sepsis.

■■ CLINICAL FEATURES

The presence of sepsis may be apparent on the initial presentation of a critically ill patient. In some cases, however, sepsis can be challenging to identify early in a patient's evaluation. Abnormal vital signs such as hyperthermia or hypothermia, tachycardia, hypotension, and tachypnea may suggest sepsis as a diagnostic possibility in the right clinical circumstances. In particular, up to 40% of patients who present with undifferentiated hypotension are later diagnosed with septic shock when a source of infection is identified through diagnostic testing.

Traditionally sepsis has been categorized as a type of distributive shock associated with peripheral vasodilation, warm extremities, and a compensa-

tory increase in cardiac output. However, this does not accurately reflect the presentation of all patients with sepsis. The combination of intravascular volume depletion and septic cardiomyopathy may also manifest as “cold shock,” with impaired peripheral perfusion and cool extremities.

Many different abnormalities are possible with severe sepsis, reflecting different areas of end-organ damage. Severe sepsis is the leading cause of acute lung injury and acute respiratory distress syndrome (ARDS). Widespread inflammation secondary to sepsis commonly affects pulmonary function even in the absence of pneumonia. Refractory hypoxemia, decreased lung compliance, and a chest x-ray demonstrating bilateral pulmonary alveolar infiltrates are commonly found with ARDS. Renal manifestations of severe sepsis include acute renal failure with azotemia, oliguria, and anuria. The most frequent hepatic abnormality is cholestatic jaundice and is associated with elevation in transaminases, alkaline phosphatase, and/or bilirubin. The most frequent hematologic changes associated with sepsis are neutropenia or neutrophilia, thrombocytopenia, and disseminated intravascular coagulation (DIC). Hyperglycemia may develop,

89_Cydulka_Ch89.indd 484 20/04/17 2:56 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 500–503.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 89: Sepsis

Source pages 504–511 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 504–511

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Sepsis**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 89: Sepsis 485 and uncontrolled hyperglycemia is a significant risk for adverse outcome. [Source pages 504–511]

- Cutaneous lesions that occur as a result of sepsis can be divided into five categories: direct bacterial involvement of the skin and underlying soft tissues (cellulitis, erysipelas, and fasciitis); lesions from hematogenous seeding of the skin or the underlying tissue (petechiae, pustules, cellulitis, ecthyma gangrenosum); lesions resulting from hypotension and/or DIC (acrocyanosis and necrosis of peripheral tissues); lesions secondary to intravascular infections (microemboli and/or immune complex vasculitis); and lesions caused by toxins (toxic shock syndrome). [Source pages 504–511]

- ■■ DIAGNOSIS AND DIFFERENTIAL Consider the possibility of sepsis in patients who present with criteria for systemic inflammatory response syndrome (SIRS), as detailed in Table 89-1. [Source pages 504–511]

- Severe sepsis involves these criteria plus evidence of organ dysfunction such as hypotension, elevated serum lactate level, renal dysfunction, hypoxia, hyperbilirubinemia, thrombocytopenia, or coagulopathy. [Source pages 504–511]

- Septic shock involves refractory hypotension, and does not typically reverse with rapid volume replacement of at least 1.5 to 3.0 L of isotonic crystalloid (or 20 to 30 mL/kg in children). [Source pages 504–511]

Clinical reasoning structure

- **Problem representation:** Sepsis; source pages 504–511.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Sepsis**.

- Source-page range: **504–511**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 504

CHAPTER 89: Sepsis 485

and uncontrolled hyperglycemia is a significant risk for adverse outcome.

Cutaneous lesions that occur as a result of sepsis can be divided into five categories: direct bacterial involvement of the skin and underlying soft tissues (cellulitis, erysipelas, and fasciitis); lesions from hematogenous seeding of the skin or the underlying tissue (petechiae, pustules, cellulitis, ecthyma gangrenosum); lesions resulting from hypotension and/or DIC (acrocyanosis and necrosis of peripheral tissues); lesions secondary to intravascular infections (microemboli and/or immune complex vasculitis); and lesions caused by toxins (toxic shock syndrome).

■■ DIAGNOSIS AND DIFFERENTIAL

Consider the possibility of sepsis in patients who present with criteria for systemic inflammatory response syndrome (SIRS), as detailed in Table 89-1. Severe sepsis involves these criteria plus evidence of organ dysfunction such as hypotension, elevated serum lactate level, renal dysfunction, hypoxia, hyperbilirubinemia, thrombocytopenia, or coagulopathy. Septic shock involves refractory hypotension, and does not typically reverse with rapid volume replacement of at least 1.5 to 3.0 L of isotonic crystalloid (or 20 to 30 mL/kg in children). History and physical examination findings combined with initial laboratory and radiolographic investigations often identify a presumptive infection source for sepsis, which can guide empiric therapy. Focus particular attention to signs and symptoms of potential infections in the central nervous system, pulmonary system, intraabdominal structures, urinary tract, skin, and soft tissues.

Initial laboratory studies for patients with suspected sepsis often include a CBC with differential and platelet count, lactic acid level, serum electrolytes, liver function panel, renal function panel, arterial blood gas analysis, blood cultures, urinalysis, and urine culture. Consider type and cross-

matching blood for patients with acute anemia or refractory hypotension who may need emergent transfusion. A chest x-ray can assess for pneumonia as a potential source of infection. Additional radiographs, CT scans, ultrasound, or lumbar puncture may be utilized in specific patients based on clinical suspicion for potential infectious sources. Other laboratory tests for markers of sepsis may be considered, such as C-reactive protein and procalcitonin.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

Studies have shown that the cornerstones of the initial treatment and stabilization of severe sepsis are early recognition, early reversal of hemodynamic compromise, and early empiric antibiotics for infection control. The goals of resuscitation are to improve preload, tissue perfusion, and oxygen delivery.

1. For patients with hypoxia or respiratory distress, provide supplemental oxygen to keep oxygen saturation greater than 90%, and perform endotracheal intubation if necessary.
2. Initiate IV fluid resuscitation with an initial bolus of crystalloid IV fluid at 20 to 30 mL/kg.
3. Begin vasopressors for patients in septic shock with hypotension that does not respond to aggressive IV fluid bolus therapy. A central venous

89_Cydulka_Ch89.indd 485 20/04/17 2:56 pm

Source page 505

Table 89-1 Definition of Sepsis in Adults and Children

●● Systemic Inflammatory Response Syndrome (SIRS) criteria:

1. Fever (temperature >38.3°C) or hypothermia (temperature <36°C)
2. Pulse rate (>90 beats/min or >2 SDs above the normal value for age)
3. Tachypnea (respiratory rate >20 breaths/min)
4. Leukocytosis (WBC >12,000 cells/μL) or leukopenia (WBC <4000 cells/μL), or normal WBC with >10% immature forms

●● Sepsis: Infection (documented or suspected), and some of the following:

●● General Parameters: Fever (temperature >38.3°C)

●● Hypothermia (temperature <36°C)

●● Pulse rate (>90 beats/min or >2 SDs above the normal value for age)

●● Tachypnea

●● Altered mental status

●● Significant edema or positive fluid balance (>20 mL/kg during 24 h)

●● Hyperglycemia (plasma glucose >140 mg/dL or 7.7 mmol/L) in the absence of diabetes

●● Inflammatory Parameters:

●● Leukocytosis (WBC >12,000 cells/μL)

●● Leukopenia (WBC <4000 cells/μL)

●● Normal WBC with >10% immature forms

●● Plasma C-reactive protein (CRP) >2 SDs above the normal value

●● Plasma procalcitonin >2 SDs above the normal value

●● Hemodynamic Parameters:

●● Arterial hypotension (SBP <90 mm Hg, MAP <70 mm Hg, or an SBP decrease >40 mm Hg in adults or <2 SDs below normal for age)

●● Organ Dysfunction Parameters:

●● Arterial hypoxemia (PaO2/FIO2 <300)

●● Acute oliguria (urine output <0.5 mL/kg per hour for at least 2 h despite adequate fluid resuscitation)

●● Creatinine level increase >0.5 mg/dL

●● Coagulation abnormalities (INR >1.5 or aPTT >60 s)

●● Ileus (absent bowel sounds)

●● Thrombocytopenia (platelet count <100,000 cells/μL)

●● Hyperbilirubinemia (plasma total bilirubin >4 mg/dL)

●● Tissue Perfusion Parameters:

●● Hyperlactatemia (above upper limits of laboratory normal levels)

●● Decreased capillary refill or mottling

●● Severe Sepsis: Sepsis-induced tissue hypoperfusion or organ dysfunction (any of the following thought to be due to infection)

●● Sepsis-induced hypotension

●● Lactate level above upper limits of laboratory normal levels

●● Urine output <0.5 mL/kg/h for at least 2 h despite adequate fluid resuscitation

●● Acute lung injury with PaO2/FIO2 <250 in the absence of pneumonia as infectious source

●● Acute lung injury with PaO2/FIO2 <200 in the absence of pneumonia as infectious source

●● Creatinine level >2.0 mg/dL

●● Bilirubin level >2 mg/dL

●● Platelet count <100,000 cells/μL

●● Coagulopathy (INR >1.5)

Abbreviations: aPTT, activated partial thromboplastin time; FIO2, fraction of inspired oxygen; INR, international normalized ratio; MAP, mean arterial pressure; PaO2, arterial partial pressure of oxygen; SBP, systolic blood pressure; SD, standard deviation.

Source: Adapted with permission from Levy M M, Fink MP, Marshall J C, et al: 2001 SCCM/ESICM /ACCP/ATS/SIS International Sepsis Definitions Conference. Crit Care Med. April 2003; 31(4): 1250–1256.

89_Cydulka_Ch89.indd 486 20/04/17 2:56 pm

Source page 506

CHAPTER 89: Sepsis 487

line is preferred but large peripheral lines may be used temporarily to begin treatment. Start norepinephrine 0.5 to 30 µg/min with a goal mean arterial pressure of 65 mm Hg. Consider vasopressin 0.03 or 0.04 U/min as a second-line agent.

4. Consider repeating a serum lactate level after 1 to 2 hours as one way to assess restoration of adequate tissue oxygen perfusion.

5. Initiate appropriate broad-spectrum antibiotics based on the initial suspected source of sepsis (see Table 89-2). Empiric antibiotic therapy is

Table 89-2 Empiric Antibiotic Selection in Severe Sepsis and Septic Shock

Host	Likely Pathogens	Initial Antibiotic Selection
Adults (nonneutropenic) without obvious source of infection	Staphylococcus aureus, streptococci, gram-negative bacilli, others	Imipenem, 500 mg every 6 h to 1 g IV every 8 h or Meropenem, 1 g IV every 8 h or Doripenem, 500 mg IV every 8 h or Ertapenem*, 1 g IV every 24 h plus Vancomycin†, 15 mg/kg loading dose
Adults (nonneutropenic), suspected biliary source	Aerobic gram-negative bacilli, enterococci	Ampicillin/sulbactam, 3 g IV every 6 h or Piperacillin/tazobactam, 4.5 g IV every 6 h or Ticarcillin/clavulanate, 3.1 g IV every 4 h
Adults (nonneutropenic), suspected pneumonia	Streptococcus pneumoniae, methicillin-resistant S. aureus, gram-negative bacilli, Legionella	Ceftriaxone, 1–2 g IV every 12 h plus Azithromycin, 500 mg IV, then 250 mg IV every 24 h plus Levofloxacin, 750 mg IV every 24 h or moxifloxacin, 400 mg IV every 24 h plus Vancomycin†, 15 mg/kg loading dose

Adults (nonneutro- S. aureus Vancomycin†, 15 mg/kg loading dose
penic), suspected
illicit use of IV drugs
Adults with pete- Neisseria meningiti- Ceftriaxone, 2 g IV every 12 h
chial rash dis, RMSF or
Cefotaxime, 2 g IV every 4-6 h
Consider
Addition of doxycycline 100 mg IV every
12 h for possible RMSF
(continued)
89_Cydulka_Ch89.indd 487 20/04/17 2:56 pm

Source page 507

488 SECTION 10: Infectious and Immunologic Diseases
Table 89-2 Empiric Antibiotic Selection in Severe Sepsis and
Septic Shock (continued)
Host Likely Pathogens Initial Antibiotic Selection
Adults (nonneutro- Mixture of aerobic Imipenem, 500 mg IV every 6 h to 1 g
penic), suspected and anaerobic gram- IV every 8 h
intraabdominal negative bacilli or
source
Meropenem, 1 g IV every 8 h
or
Doripenem, 500 mg IV every 8 h
or
Ertapenem, 1 g IV every 24 h
or
Ampicillin/sulbactam, 3 g IV every 6 h
or
Piperacillin/tazobactam, 4.5 g IV every 6 h
Adults (nonneutro- Aerobic gram- Levofloxacin, 750 mg IV every 24 h
penic), suspected negative bacilli, or
urinary source enterococci
Moxifloxacin, 400 mg IV every 24 h
(hospitalized with
pyelonephritis) or
Piperacillin/tazobactam, 4.5 g IV every 6 h
or
Ceftriaxone, 1–2 g IV every 12–24 h
or
Ampicillin, 1–2 g IV every 4–6 h plus

gentamicin, 1.0–1.5 mg/kg every 8 h‡

Adults (nonneutro- Enterobacteriaceae, Piperacillin/tazobactam, 4.5 g IV every 6 h

penic), suspected Pseudomonas aeru- or

urinary source ginosa, entero-cocci,

Imipenem, 500 mg every 6 h to 1 g IV

(complicated rarely S. aureus

every 8 h

urinary tract

infection/urinary or

catheter) Meropenem, 1 g IV every 8 h

or

Doripenem, 500 mg IV every 8 h

or

Ampicillin, 1–2 g IV every 4–6 h plus

gentamicin, 1.0–1.5 mg/kg every 8 h‡

Neutropenic adults Aerobic gram-nega- Ceftazidime, 2 g IV every 8 h

tive bacilli, especially or

P. aeruginosa, S.

Cefepime, 2 g IV every 8 h

aureus

or

Imipenem, 500 mg IV every 6 h to 1 g

IV every 8 h

or

Meropenem, 1 g IV every 8 h

(continued)

89_Cydulka_Ch89.indd 488 20/04/17 2:56 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 504–511.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 90: Soft Tissue Infections

Source pages 512–517 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 512–517

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Soft Tissue Infections**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 90: Soft Tissue Infections 493 immunoglobulin remain controversial and may be considered in consul- tation with surgical colleagues. [Source pages 512–517]

- ■■ CELLULITIS Cellulitis is a subset of soft tissue infections characterized by a local soft tissue inflammatory response caused by bacterial invasion of the skin. [Source pages 512–517]

- This infection is more common in the elderly, immunocompromised patients, and patients with peripheral vascular disease. [Source pages 512–517]

- Clinical Features Cellulitis presents as localized tenderness, erythema, and induration of the skin. [Source pages 512–517]

- Lymphangitis and lymphadenitis can accompany cellulitis and may indicate a more severe infection. [Source pages 512–517]

Clinical reasoning structure

- **Problem representation:** Soft Tissue Infections; source pages 512–517.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Soft Tissue Infections**.

- Source-page range: **512–517**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 512

CHAPTER 90: Soft Tissue Infections 493

immunoglobulin remain controversial and may be considered in consultation with surgical colleagues.

■■ CELLULITIS

Cellulitis is a subset of soft tissue infections characterized by a local soft tissue inflammatory response caused by bacterial invasion of the skin. This infection is more common in the elderly, immunocompromised patients, and patients with peripheral vascular disease.

Clinical Features

Cellulitis presents as localized tenderness, erythema, and induration of the skin. Lymphangitis and lymphadenitis can accompany cellulitis and may indicate a more severe infection. Patients may present with associated fever and chills.

Diagnosis and Differential

History and physical examination are often sufficient to make the clinical diagnosis of cellulitis. Obtaining a white cell count or blood cultures rarely changes management of otherwise healthy patients with uncomplicated cellulitis. Cellulitis of the lower extremity may sometimes mimic or be complicated by deep venous thrombosis, and clinical uncertainty may prompt additional studies to evaluate for this possibility. In patients with signs of systemic toxicity, consider additional evaluation for an occult abscess using bedside ultrasound and/or cultures of blood and other suspected areas of infection.

Emergency Department Care and Disposition

1. Consider outpatient treatment of patients with uncomplicated cellulitis in which MRSA is not suspected using cephalexin 500 mg PO four times daily, dicloxacillin 500 mg PO four times daily, or clindamycin 300 PO mg four times daily.
2. When local epidemiology supports a high likelihood of MRSA, treat with antibiotics effective against MRSA (see section "Methicillin-Resistant Staphylococcus aureus" above).

3. Recommend follow-up within a few days to evaluate the cellulitis and response to therapy. Skin markers may be helpful to mark the extent of cellulitis for serial evaluations of treatment success.
4. Admit patients with systemic toxicity or evidence of bacteremia to the hospital for IV antibiotics. Consider risks of complications when making treatment decisions for patients with diabetes mellitus, alcoholism, or other immunosuppressive disorders.
5. Initiate IV antibiotics, such as clindamycin 600 to 900 mg IV every 8 hours, cefazolin 1 to 2 g IV every 8 hours, or nafcillin 1 to 2 g IV every 4 hours for patients admitted for inpatient care.

■■ ERYSIPELAS

Erysipelas is a superficial cellulitis with associated lymphatic involvement and is caused primarily by group A Streptococcus. Infection is often associated with an initial portal of entry or break in the skin.

90_Cydulka_Ch90.indd 493 20/04/17 2:56 pm

Source page 513

Clinical Features

Onset of this infection is often acute and characterized by sudden high fever, chills, malaise, and nausea. Over 1 to 2 days, a small area of erythema with a burning sensation develops. The erythema of erysipelas is sharply demarcated from surrounding skin and is tense and painful. Lymphangitis and lymphadenitis are common.

Diagnosis and Differential

The diagnosis is based primarily on physical findings and history. Although leukocytosis is common, white blood cell counts, cultures, ASO titers, and anti-DNase B titers are of limited use in an acute care setting. Differential diagnosis includes other forms of local cellulitis. Consider surgical consultation for patients with bullae, crepitus, pain out of proportion to examination, or rapidly progressing erythema with signs of systemic toxicity, as these signs and symptoms may suggest necrotizing infection.

Emergency Department Care and Disposition

Guidelines no longer differentiate the treatment of nonpurulent cellulitis from erysipelas. Patients with uncomplicated erysipelas can be treated with antibiotics as detailed above for cellulitis.

■■ CUTANEOUS ABSCESSES

Cutaneous skin abscesses are frequently the result of a breakdown in the cutaneous barrier, with a subsequent focal infection caused by resident bacterial flora. Incision and drainage is often the only necessary treatment

for simple skin abscesses.

Clinical Features and Diagnosis

Patients present with an area of swelling, tenderness, and overlying erythema. The area of swelling is often fluctuant. Cutaneous abscesses are usually well-localized infections, although they may cause systemic toxicity in immunosuppressed patients. Evaluate cutaneous abscesses for predisposing injury or foreign bodies. Needle aspiration or bedside ultrasound may be useful when it is unclear whether a patient has an abscess or cellulitis. There are several common abscess types with unique managements discussed below.

Emergency Department Care and Disposition

1. Cutaneous skin abscesses can often be managed with local incision and drainage. In healthy, immunocompetent patients, the routine use of antibiotics following abscess incision and drainage is typically not necessary unless significant surrounding cellulitis or signs of severe disease are present.

2. Bartholin gland abscess presents as unilateral painful swelling of the labia with a fluctuant 1 to 2 cm mass. Routine antimicrobial treatment is typically not necessary unless there is a suspicion of sexually transmitted disease. Treatment involves incision and drainage along the vaginal

90_Cydulka_Ch90.indd 494 20/04/17 2:56 pm

Source page 514

mucosal surface of the abscess, generally followed by the insertion of a Word catheter. The Word catheter can be left in place for up to 4 weeks. Sitz baths are recommended after 2 days. Follow-up with gynecology is recommended within 2 days in patients with severe symptoms and within 1 week in patients with mild symptoms.

3. Hidradenitis suppurativa is a recurrent chronic infection involving the apocrine sweat glands and commonly occurs in the axilla and groin. Acute abscesses can be treated with incision and drainage. Refer patients with recurrent disease to a surgeon for definitive treatment.

4. Infected epidermoid and pilar cysts are erythematous, tender, cutaneous nodules that are often fluctuant. Perform a simple incision and drainage and recommend wound rechecks in 2 to 3 days. The cyst contains a capsule that must be removed to prevent recurrence, and this capsule excision is typically done at a later follow-up visit.

5. Pilonidal abscess presents as a tender, swollen, and fluctuant mass along the superior gluteal fold. Treatment in the acute care setting

includes incision and drainage and the patient should be rechecked in 2 to 3 days. Surgical referral is usually necessary for definitive treatment.

6. Folliculitis is characterized by pruritic erythematous lesions that are usually <5 mm in diameter, with pustules sometimes present at the centers. These inflamed hair follicles are usually caused by bacterial invasion with S. aureus. The lesions usually resolve spontaneously with symptomatic care, but can be treated with warm compresses or topical antibiotics such as bacitracin. For painful or more extensive cases, oral antibiotics such as cephalexin, dicloxacillin, or azithromycin are recommended.

■■ SPOROTRICHOSIS

Sporotrichosis is caused by traumatic inoculation of the fungus Sporothrix schenckii, which is commonly found on plants and in the soil.

Clinical Features

After a 3-week incubation period, three types of infection may occur. The fixed cutaneous type of infection occurs at the site of inoculation and looks like a crusted ulcer or verrucous plaque. The local cutaneous type also remains at the site of inoculation but presents as a subcutaneous nodule or pustule with surrounding erythema. The lymphocutaneous type is the most common of the three and presents as a painless nodule at the site of inoculation with subcutaneous nodules that migrate along lymphatic channels.

Diagnosis and Differential

The diagnosis of sporotrichosis is based on the history and physical examination. Tissue biopsy cultures are often diagnostic but of limited use in the acute care setting. The differential diagnosis includes tuberculosis, tularemia, cat-scratch disease, leishmaniasis, nocardiosis, and staphylococcal lymphangitis.

90_Cydulka_Ch90.indd 495 20/04/17 2:56 pm

Source page 515

Emergency Department Care and Disposition

1. Itraconazole 100 to 200 mg/d PO for 3 to 6 months is highly effective when treating localized or systemic sporotrichosis.

2. Most cases of cutaneous sporotrichosis can be treated on an outpatient basis. Patients who have systemic symptoms or who are acutely ill with disseminated disease should be admitted for possible treatment with amphotericin B 0.5 mg/kg/d.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 152, “Soft Tissue Infections,” by Elizabeth W. Kelly and David Magilner.

90_Cydulka_Ch90.indd 496 20/04/17 2:56 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 512–517.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 91: Serious Viral Infections

Source pages 518–525 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 518–525

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Serious Viral Infections**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 91: Serious Viral Infections 499 Less common manifestations of HSV infection include herpetic whit- low on a finger or herpes gladiatorum elsewhere on the body in wrestlers and other athletes involved in close contact sports. [Source pages 518–525]

- Herpetic keratitis presents as a painful, red eye, and can be identified by characteristic den- dritic lesions on fluorescein slit lamp exam. [Source pages 518–525]

- Either HSV-1 or HSV-2 can cause genital lesions, and while genital HSV-2 is more common and has a higher relapse rate, the proportion of genital disease caused by HSV-1 is increasing. [Source pages 518–525]

- HSV infection within the central nervous system (CNS) is one of the most common viral causes of encephalitis in the United States. [Source pages 518–525]

- This occurs most commonly in patients between 20 and 50 years of age and HSV encephalitis carries a mortality rate >70% if left untreated. [Source pages 518–525]

Clinical reasoning structure

- **Problem representation:** Serious Viral Infections; source pages 518–525.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Serious Viral Infections**.

- Source-page range: **518–525**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 518

CHAPTER 91: Serious Viral Infections 499

Less common manifestations of HSV infection include herpetic whitlow on a finger or herpes gladiatorum elsewhere on the body in wrestlers and other athletes involved in close contact sports. Herpetic keratitis presents as a painful, red eye, and can be identified by characteristic dendritic lesions on fluorescein slit lamp exam. Either HSV-1 or HSV-2 can cause genital lesions, and while genital HSV-2 is more common and has a higher relapse rate, the proportion of genital disease caused by HSV-1 is increasing.

HSV infection within the central nervous system (CNS) is one of the most common viral causes of encephalitis in the United States. This occurs most commonly in patients between 20 and 50 years of age and HSV encephalitis carries a mortality rate >70% if left untreated. HSV encephalitis is associated with fever and neurologic symptoms such as focal motor or cranial nerve deficits, ataxia, seizures, altered mental status, or behavioral abnormalities. Meningitis can occur in up to 25% of females with primary HSV-2 infections, although this typically has a benign course. HSV infections in immunocompromised hosts can lead to widespread dissemination with multi-organ involvement.

Diagnosis and Differential

The diagnosis and initial treatment of suspected HSV infection is often based on clinical findings, as confirmatory testing may take days or weeks before results are available. Because mucocutaneous HSV infection is a lifelong condition, laboratory testing is often recommended for confirmation. Viral culture can be obtained from an unroofed vesicle, or polymerase chain reaction or direct fluorescent antibody testing can be performed on swabbed tissue. A Tzanck test is generally not useful.

Consider the possibility of HSV infection in patients with suspected infection of the CNS, particularly in patients with altered mental status,

cognitive deficits, psychiatric symptoms, or seizures (see Chapter 148, “Central Nervous System and Spinal Infections”). Temporal lobe lesions on CT scan or MRI are suggestive of HSV encephalitis and cerebrospinal fluid analysis may demonstrate a lymphocytic pleocytosis and/or erythrocytosis. PCR testing of spinal fluid is the testing modality of choice for HSV when suspected, and the viral DNA can be detected within the first 24 hours, with results remaining positive for a week or longer.

Emergency Department Care and Disposition

1. Treat severe systemic HSV infections, including suspected CNS infection or disseminated disease, with acyclovir 10 mg/kg IV every 8 hours. Adjust dosing for immunocompromised patients with severe mucocutaneous involvement to acyclovir 5 mg/kg IV every 8 hours based on ideal body weight.
2. Treat mild to moderate HSV infection in immunocompetent patients with oral acyclovir 400 mg three times a day, valacyclovir 1000 mg every 12 hours, or famciclovir 250 mg three times a day. Treatment is most effective when initiated within 48 to 72 hours of symptom onset and is typically continued for 7 to 10 days.
3. Treat recurrent HSV with or acyclovir 400 mg three times a day for 5 days, valacyclovir 2000 mg every 12 hours for 1 day (labialis) or

91_Cydulka_Ch91.indd 499 20/03/17 9:43 AM

Source page 519

500 mg every 12 hours for 3 days (genital), or famciclovir 1500 mg for one dose (orolabial) or 1000 mg for one dose (genital). Less severe outbreaks may be treated with topical 5% acyclovir ointment applied six times per day for 7 days.

■■ VARICELLA AND HERPES ZOSTER

Varicella zoster virus (VZV) is the causative organism for both varicella (chickenpox) and herpes zoster (shingles) infections. Herpes zoster is reactivation of the latent virus within a dorsal root ganglion that can occur with a waning immune response due to advancing age or immunosuppression. The presence of herpes zoster in an otherwise young and healthy person may be a sign of HIV infection or other immunosuppressive disorder.

Clinical Features

Primary varicella infection is a febrile illness with a superficial vesicular rash that appears in crops and is concentrated on the torso and face. Lesions are typically at varying stages of development, including papules, vesicles, and crusted lesions. Headache, malaise, and loss of appetite are often present. Most infections are minor and self-limited with lesions crusting over

and sloughing off after 1 to 2 weeks. Varicella is contagious until all of the lesions have crusted over.

Complications are more common in very young and very old patients and those with immunocompromise. Bacterial infections of skin lesions, often with group A streptococci, can cause serious illnesses including necrotizing fasciitis. CNS complications such as cerebellar ataxia, meningitis, and meningoencephalitis have been described. Infection of cerebral arteries resulting in vasculopathy and stroke may also occur. Pneumonitis can be severe and is more common in pregnant women.

Herpes zoster causes lesions similar to those of a primary varicella infection but that occur in a unilateral and dermatomal distribution on the skin. A prodrome of malaise, headache, and photophobia may occur with localized pain, itching, and parasthesias preceding the outbreak of the rash. Although any dermatomal level may be affected, involvement of the trigeminal or thoracic dermatomes are common. Ocular involvement with herpes zoster ophthalmicus may occur when the ophthalmic distribution of the trigeminal nerve is involved (see Chapter 149, “Ocular Emergencies”), and is associated with zoster lesions on the tip or side of the nose (Hutchinson’s sign). Consider ophthalmology consultation when these are present or a patient has facial lesions with associated eye symptoms.

Herpes zoster lesions typically last about 2 weeks but may persist for a month. Postherpetic neuralgia is pain that continues 30 days after the initial eruption and occurs more often with advancing age. Rash involving multiple dermatomes or crossing the midline may represent disseminated disease, which can occur in immunocompromised patients when the virus spreads to visceral organs to cause pneumonitis, hepatitis, or encephalitis.

Diagnosis and Differential

Most cases of varicella infection can be diagnosed with history and physical examination findings. Laboratory diagnosis with viral culture, antigen

91_Cydulka_Ch91.indd 500 20/03/17 9:43 AM

Source page 520

testing, or PCR testing of vesicle fluid may be appropriate in patients with atypical illness or severe disease. Although smallpox has been eradicated, it remains a potential threat as a biologic weapon, and the lesions could be confused with those of varicella. The lesions of smallpox are larger, distributed more on the extremities, and all lesions are at the same stage of development. Obtain a chest radiograph if pneumonitis is suspected. Consider advanced testing when CNS infection is suspected, which may include an MRI of the brain, lumbar puncture, and/or PCR testing.

Emergency Department Care and Disposition

1. Provide supportive care for healthy children with primary varicella infection. Oral acyclovir 20 mg/kg (maximum dose of 800 mg) five times a day for 5 days may be considered for high-risk patients such as children >12 years of age, adults, those with chronic skin or pulmonary disorders, those receiving long-term salicylate therapy, and immuno-compromised patients.
2. Antiviral agents for the treatment of herpes zoster will hasten lesion resolution, reduce new lesions, reduce viral shedding, and decrease acute pain. Use of antiviral therapy in immunocompromised patients may reduce the risk of disseminated disease. Ideally begin antiviral medication within 72 hours of the onset of rash, although treatment can be considered at >72 hours if new vesicles are developing. Treat with a 7- to 10-day course of oral acyclovir 800 mg five times per day, valacy-clovir 1 g three times daily or famciclovir 500 mg three times daily.
3. Provide appropriate analgesia for patients with herpes zoster, as lesions can be very painful.
4. Treat immunocompromised patients with acute varicella infections, severe herpes zoster outbreaks, visceral involvement, or disseminated disease with acyclovir 10 mg/kg IV every 8 hours.

■■ EPSTEIN–BARR VIRUS INFECTION

Epstein–Barr virus (EBV) is the causative agent of heterophile-positive infectious mononeucleosis. EBV infection is also associated with cancers such as B-cell lymphoma, Hodgkin’s disease, Burkitt’s lymphoma, and nasopharyngeal carcinoma.

Clinical Features

Transmission of EBV is via salivary secretions. Close contact is required for transmission, often from an asymptomatic individual. Clinical mani-festations depend on age and immune status with peaks of infection in early childhood and young adulthood. Infants and young children are often asymptomatic or have only mild pharyngitis. Teenagers and young adults can develop infectious mononucleosis, which presents with fever, lymphadenopathy, and pharyngitis with tonsillar exudates. Splenomegaly occurs in more than half of patients. Symptoms generally resolve over 2 to 3 weeks, and most patients recover uneventfully.

Epstein–Barr virus can affect nearly all organ systems. Neurologic com-plications such as encephalitis, meningitis, and Guillain–Barré syndrome have been described. Hepatitis, myocarditis, and hematologic disorders are

91_Cydulka_Ch91.indd 501 20/03/17 9:43 AM

Source page 521

also known complications. Rarely death results from splenic rupture, CNS complications, and airway obstruction.

Diagnosis and Differential

When infectious mononeucleosis is suspected based on history and physical exam, consider a CBC and a monospot test to provide confirmation, when indicated. Lymphocytosis with >50% lymphocytes and presence of atypical lymphocytes are suggestive of the diagnosis. The monospot test may be negative early in the course of disease, and repeat testing later in the course of illness may be helpful when the diagnosis remains uncertain.

Emergency Department Care and Disposition

1. Rest and analgesia are the mainstays of therapy for EBV infections.
2. Use of corticosteroids is associated with increased complications and is recommended only for patients with severe disease, such as upper airway obstruction, neurologic disease, or hemolytic anemia.
3. Advise patients to avoid all contact sports for a minimum of 3 weeks after illness onset to avoid splenic injury.

■■ MEASLES

Measles is caused by an RNA virus that is highly infectious and is communicable before symptoms begin. Measles outbreaks have occurred recently in communities where children are not appropriately vaccinated.

Clinical Features

Acute measles infection occurs with fever, malaise, cough, runny nose, and conjunctivitis. Small, white Koplik's spots appear on the buccal mucosa early in the illness, followed by a red maculopapular rash that typically begins on the head and spreads throughout the body.

Diagnosis and Differential

The diagnosis of measles is typically made based on clinical findings such as the pathognomonic Koplik's spots and characteristic rash. Confirmatory testing can be done by detection of immunoglobulin M antibodies.

Emergency Department Care and Disposition

1. Treatment for measles is supportive care, with particular attention to ensuring adequate nutrition, especially vitamin A intake.
2. Report suspected measles cases to the local health department. Coordinate with health department officials to identify and offer post-exposure prophylaxis to unvaccinated contacts when appropriate.
3. See http://www.cdc.gov/measles/hcp/ for current CDC guidelines for the diagnosis and treatment of measles.

■■ ARBOVIRAL INFECTIONS

Arboviral infections are spread by mosquito, tick, and fly bites with

increased incidence in warmer months due to the breeding patterns of these

arthropod vectors. West Nile virus and the viruses that cause La Cross

91_Cydulka_Ch91.indd 502 20/03/17 9:43 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.

- Content ID: SEG-TINTINALLI-EMM8-FULL.

- Source pages: 518–525.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 92: HIV Infection and AIDS

Source pages 526–533 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 526–533

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **HIV Infection and AIDS**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 92: HIV Infection and AIDS 507 The spectrum of disease caused by HIV infection varies, from those with asymptomatic infection to those seeking care from involvement of any organ system. [Source pages 526–533]

- In the assessment of patients with known HIV, knowledge of recent CD4 count and HIV viral load can help with management, as CD4 counts below 200 cells/mm3 and viral load greater than 50,000 copies/mL are asso- ciated with an increased risk of AIDS-defining illness. [Source pages 526–533]

- When these levels are unavailable, a total lymphocyte count of <1700 cells/mm3 is strongly predictive of a CD4 count of <200 cells/mm3. [Source pages 526–533]

- Febrile Illness Systemic symptoms such as fever, weight loss, and malaise are common in HIV-infected patients and account for the majority of HIV-related ED presentations. [Source pages 526–533]

- Laboratory investigation may include electrolytes, com- plete blood count, blood cultures, urinalysis and culture, liver function tests, and chest radiographs in the appropriate clinical settings. [Source pages 526–533]

Clinical reasoning structure

- **Problem representation:** HIV Infection and AIDS; source pages 526–533.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **HIV Infection and AIDS**.

- Source-page range: **526–533**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 526

CHAPTER 92: HIV Infection and AIDS 507

The spectrum of disease caused by HIV infection varies, from those with asymptomatic infection to those seeking care from involvement of any organ system. In the assessment of patients with known HIV, knowledge of recent CD4 count and HIV viral load can help with management, as CD4 counts below 200 cells/mm3 and viral load greater than 50,000 copies/mL are associated with an increased risk of AIDS-defining illness. When these levels are unavailable, a total lymphocyte count of <1700 cells/mm3 is strongly predictive of a CD4 count of <200 cells/mm3.

Febrile Illness

Systemic symptoms such as fever, weight loss, and malaise are common in HIV-infected patients and account for the majority of HIV-related ED presentations. Laboratory investigation may include electrolytes, complete blood count, blood cultures, urinalysis and culture, liver function tests, and chest radiographs in the appropriate clinical settings. Selected patients may benefit from specific serologic testing for syphilis, cryptococcosis, toxoplasmosis, cytomegalovirus (CMV), and coccidioidomycosis. Consider lumbar puncture after neuroimagingwhen CNS infection is suspected.

In HIV patients without specific signs or symptoms to localize the source of fever, common causes can vary by stage of disease. Patients with CD4 counts higher than 500 cells/mm3 generally have febrile illnesses from sources similar to those in nonimmunocompromised patients. Those with CD4 counts between 200 and 500 cells/mm3 are more likely to have bacterial respiratory infections. For patients with CD4 counts lower than 200 cells/mm3, infections without localizing symptoms or findings can include Pneumocystis jirovecii pneumonia (PCP), Mycobacterium avium complex (MAC), Mycobacterium tuberculosis (TB), CMV, drug fever, and

sinusitis. Other causes may include endocarditis (see Chapter 93), lymphoma, Histoplasma capsulatum, and Cryptococcus neoformans. Fever caused by the primary HIV infection itself tends to occur in the afternoon or evening and is generally responsive to antipyretics.

Immune reconstitution inflammatory syndrome mimics an autoimmune event, with fever, lymphadenitis, and systemic symptoms that begin weeks to months after starting antiretroviral treatment and often associated with tuberculosis therapy. This rarely requires cessation of medications, but suspected cases should be discussed with an infectious disease specialist.

CMV is a common cause of serious opportunistic viral disease in HIV-infected patients. The most important manifestation is retinitis, which can often be diagnosed via clinical evaluation. Disseminated disease commonly involves the gastrointestinal, pulmonary, and central nervous systems and a specific diagnosis may require a tissue biopsy with histologic evidence of viral inclusions and inflammation. Cultures, antigen-based tests, and nucleic acid testing can be used to confirm CMV infection.

Disseminated MAC occurs predominately in patients with CD4 counts below 100 cells/mm3 who are not on antiretroviral medications. Persistent fever and night sweats are typical, often with weight loss, diarrhea, malaise, and anorexia. Diagnosis is made with acid-fast stain of stool or other body fluids or culture. Treatment reduces bacteremia and improves symptoms but does not completely eradicate the infection.

92_Cydulka_Ch92.indd 507 20/03/17 11:32 AM

Source page 527

Pulmonary Complications

Pulmonary presentations are among the most common reasons for ED visits by HIV-infected patients. Bacterial pneumonia outnumbers opportunistic infections, and Streptococcus pneumoniae is the most common cause of pneumonia in HIV-infected patients in the United States and Western Europe. Productive cough, leukocytosis, and the presence of a focal infiltrate suggest bacterial pneumonia. Common HIV-related causes of pulmonary abnormalities include PCP, TB, CMV, C. neoformans, Histoplasma capsulatum, and neoplasms. Evaluation may include pulse oximetry, arterial blood gas analysis, sputum culture and Gram stain, acid-fast stain, blood cultures, and chest radiography.

Pneumocystis jirovecii pneumonia is the most frequent opportunistic infection, and the most common identifiable cause of death among AIDS patients in the United States. The symptoms of PCP are often insidious,

and include fever, nonproductive cough, and progressive shortness of breath. Chest radiographs typically demonstrate diffuse interstitial infiltrates, although negative radiographs are reported in up to 20% of patients. Elevated lactate dehydrogenase levels, although nonspecific, are often seen. Hypoxia at rest or with exertion and an increased alveolar-arterial oxygen gradient can identify patients at risk, and a presumptive diagnosis of PCP may be made in the setting of unexplained hypoxemia. Definitive diagnosis often requires bronchoscopy.

TB frequently occurs in patients with CD4 T-cell counts of 200 to 500 cells/mm3. Consider TB in HIV-infected patients with pulmonary symptoms, and initiate appropriate respiratory isolation. As HIV-infection progresses, many patients do not report classic manifestations of TB such as cough with hemoptysis, night sweats, prolonged fevers, weight loss, and anorexia. Classic upper lobe involvement and cavitary lesions are less common, particularly among late-stage AIDS patients. False-negative purified protein derivative TB test results are frequent among AIDS patients due to immunosuppression. Late-stage HIV is also associated with extrapulmonary tuberculosis, typically within lymph nodes, bone marrow, CNS, GI, and urogenital systems.

Consider disseminated fungal infections such as C. neoformans and Aspergillus fumigatus in severely immunosuppressed patients. Noninfectious causes of pulmonary complaints such as neoplasms and lymphocytic interstitial pneumonitis may also occur.

Neurologic Complications

Neurologic disease may be caused by a variety of opportunistic infections, neoplasms, the HIV virus itself, or even as a side effect of antiretroviral treatment. Headaches, seizures, altered mental status, meningismus, or focal neurologic deficits can be caused by AIDS dementia, Toxoplasma gondii, or C. neoformans. Consider neuroimaging and lumbar puncture for patients with CD4 <200 and neurologic deficits or headaches. Initial neuroimaging with a noncontrast head computed tomography (CT) scan is appropriate, while a contrast-enhanced CT scan or magnetic resonance imaging (MRI) may also be considered in appropriate clinical circumstances. When obtaining cerebrospinal fluid, measure an opening pressure and order a complete cell count, glucose, protein, Gram stain, India

92_Cydulka_Ch92.indd 508 20/03/17 11:32 AM

Source page 528

ink stain, bacterial, viral, and fungal cultures, toxoplasmosis and cryptococcus antigen, and coccidioidomycosis titer.

AIDS dementia complex is a progressive process heralded by gradual impairment of recent memory and other cognitive deficits caused by direct HIV infection. Less common CNS infections that may present with neurologic symptoms include bacterial meningitis, histoplasmosis, CMV, progressive multifocal leukoencephalopathy, herpes simplex virus, neurosyphilis, and TB. HIV patients may experience HIV neuropathy characterized by painful sensory symptoms of the feet.

Ophthalmologic Manifestations

CMV retinitis is the most frequent and serious ocular infection and the leading cause of blindness in AIDS patients. Early infection may be asymptomatic, but may progress to changes in visual acuity, visual field abnormalities, photophobia, scotoma, eye redness, or pain. Herpes zoster ophthalmicus is another diagnosis to consider and is recognized by the typical zoster rash in the distribution of the ophthalmic branch of cranial nerve V.

Gastrointestinal Complications

Diarrhea is the most frequent gastrointestinal complaint and is estimated to occur in 50% to 90% of AIDS patients. Common causes include bacteria, parasites, viruses, fungi, and side effects of antiviral therapy. In many cases, a causative organism is never found. Evaluate stool for leukocytes, ova, and parasites, and consider obtaining acid-fast staining and bacterial culture. If bacterial infection is suspected, treat empirically with ciprofloxacin to cover common enteric pathogens. Cryptosporidium and Isospora are common parasitic causes of diffuse watery diarrhea, and both can be identified by acid-fast staining of a stool specimen. In late-stage AIDS, the most common diarrheal pathogens are MAC and CMV, each requiring biopsy for diagnosis.

Oral candidiasis affects more than 80% of AIDS patients and can be identified by its classic appearance of white plaques easily scraped from an erythematous base. Esophageal involvement may occur with Candida, herpes simplex, and CMV. Complaints of odynophagia or dysphagia are usually indicative of esophagitis and may be extremely debilitating and may lead to malnutrition.

Hepatomegaly with elevated alkaline phosphatase occurs in approximately 50% of AIDS patients. Co-infection with hepatitis B and hepatitis C is common, especially among IV drug users.

Anorectal disease is common in AIDS patients. Proctitis is characterized by painful defecation, rectal discharge, and tenesmus. Common causative organisms include Neisseria gonorrhoeae, Chlamydia trachomatis, syphilis, and herpes simplex.

Cutaneous Manifestations

Generalized conditions such as xerosis, seborrheic eczema, and pruritus are

common. Kaposi sarcoma is a rash characterized by painless, raised, non-blanching brown-black or purple papules and nodules and is more common in homosexual men with HIV infection. Common sites are the face, chest, genitals, and oral cavity. Reactivation of varicella-zoster virus is more common in patients with HIV infection and AIDS than in the general population. Herpes simplex virus infections are common. HIV patients may

92_Cydulka_Ch92.indd 509 20/03/17 11:32 AM

Source page 529

develop bullous impetigo and Pseudomonas-associated chronic ulcerations. Bacterial skin infections, scabies, human papillomavirus, and hypersensitivity reactions to medications are also common.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. Maintain an awareness of possible undiagnosed HIV infection in all patient populations. ED-based studies have demonstrated that a substantial number of patients have unsuspected HIV infection that cannot be accurately predicted based on demographic factors.
2. Base the decision to admit a patient with AIDS on severity of illness, with attention to new presentation of fever of unknown origin, hypoxia worse than baseline or PaO2 below 60 mm Hg, suspected PCP, suspected TB, new CNS symptoms, intractable diarrhea, suspected CMV retinitis, herpes zoster ophthalmicus, or a patient unable to perform self-care.
3. Treatment regimens for common HIV-related infections are listed in Table 92-2. Suspected bacterial sepsis and focal bacterial infections should be treated with standard antibiotics (see Chapter 89).

Table 92-2 Treatment Recommendations for Common Human Immunodeficiency Virus-Related Infections

Organ System	Infection	Therapy
Systemic	Mycobacterium avium-intracellulare	Clarithromycin, 500 mg PO twice a day and Ethambutol, 15 mg/kg PO once a day and Rifabutin, 300 mg/kg PO once a day
	CMV infection	Ganciclovir, 5 mg/kg IV twice a day for 2 weeks, then 5 mg/kg/d or Foscarnet, 90 mg/kg every 12 h for 3 weeks, then 90 mg/kg once a day
Pulmonary	Pneumocystis jirovecii (Pneumocystis carinii)	Trimethoprim-sulfamethoxazole dose using 15–20 mg of trimethoprim component per

pneumonia kilogram per day PO or IV in divided doses

three times a day for 3 weeks

If partial pressure of arterial oxygen is

<70 mm Hg or alveolar-arterial gradient is

>35 mm Hg, then add

Prednisone, 40 mg twice a day for 5 days, then

40 mg once a day for 5 days, then 20 mg once

a day for 11 days

or

Pentamidine, 4 mg/kg/d IV or IM for 3 weeks

Mycobacterium Isoniazid, 5 mg/kg PO once a day

tuberculosis infection* and

Rifampin, 10 mg/kg PO once a day or rifabutin

5 mg/kg PO once a day

and

Pyrazinamide, 15–30 mg/kg PO once a day

and

Ethambutol, 15–20 mg/kg PO once a day

(Continued)

92_Cydulka_Ch92.indd 510 20/03/17 11:32 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 526–533.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 93: Infective Endocarditis

Source pages 534–537 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 534–537

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Infective Endocarditis**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 93: Infective Endocarditis 515 Table 93-2 Modified Duke Criteria for Infective Endocarditis Definite Infective Endocarditis Pathologic criteria Microorganisms demonstrated by culture or histologic examination of a vegetation or in a vegetation that has embolized, or in an intracardiac abscess or Pathologic lesions: vegetation or intracardiac abscess present, confirmed by histology showing active endocarditis Clinical Criteria, Using Specific Definitions Listed in Table 93-1 Two major criteria or One major and three minor criteria or Five minor criteria Possible infective endocarditis One major criterion and one minor criterion Three minor criteria Rejected Firm alternate diagnosis for manifestations of endocarditis or Resolution of manifestations of endocarditis with antibiotic therapy for 4 days or less or No pathologic evidence of infective endocarditis at surgery or autopsy after antibiotic therapy for 4 days Does not meet criteria for possible infective endocarditis elevations in inflammatory markers. [Source pages 534–537]

- Investigate for potential infections from other sources based on individual presenting complaints. [Source pages 534–537]

- ■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION 1. [Source pages 534–537]

- Focus initial interventions to stabilize any impairments of airway, breathing, and circulation. [Source pages 534–537]

- Patients with acute infective endocarditis may present with respiratory compromise and require emergent airway stabilization. [Source pages 534–537]

Clinical reasoning structure

- **Problem representation:** Infective Endocarditis; source pages 534–537.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Infective Endocarditis**.

- Source-page range: **534–537**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 534

CHAPTER 93: Infective Endocarditis 515

Table 93-2 Modified Duke Criteria for Infective Endocarditis

Definite Infective Endocarditis

Pathologic criteria

Microorganisms demonstrated by culture or histologic examination of a vegetation or in a vegetation that has embolized, or in an intracardiac abscess

or

Pathologic lesions: vegetation or intracardiac abscess present, confirmed by histology showing active endocarditis

Clinical Criteria, Using Specific Definitions Listed in Table 93-1

Two major criteria

or

One major and three minor criteria

or

Five minor criteria

Possible infective endocarditis

One major criterion and one minor criterion

Three minor criteria

Rejected

Firm alternate diagnosis for manifestations of endocarditis

or

Resolution of manifestations of endocarditis with antibiotic therapy for 4 days or less

or

No pathologic evidence of infective endocarditis at surgery or autopsy after antibiotic therapy for 4 days

Does not meet criteria for possible infective endocarditis

elevations in inflammatory markers. Investigate for potential infections

from other sources based on individual presenting complaints.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. Focus initial interventions to stabilize any impairments of airway, breathing, and circulation. Patients with acute infective endocarditis may present with respiratory compromise and require emergent airway stabilization.

2. Pulmonary edema may be due to left-sided valvular rupture and may benefit from afterload reduction along with usual care for acute heart failure.

3. Initiate antibiotics to patients with suspected endocarditis as soon as appropriate cultures are obtained. Table 93-3 lists empiric treatment regimens. Definitive therapy is based on culture and sensitivity results and typically requires 4 to 6 weeks of antibiotics.

4. Admit patients with suspected infective endocarditis to the hospital for definitive diagnosis and management. Successful treatment often utilizes a team approach with cardiology, infectious disease, and thoracic surgery. Patients with neurologic or other embolic phenomena may benefit from additional specialist consultation.

5. Guidelines for antibiotic prophylaxis were updated in 2007. Only patients in the highest risk group for infective endocarditis need antibiotic prophylaxis for dental procedures involving manipulation of gingival tissue or procedures on infected skin or soft tissue. These high-risk

93_Cydulka_Ch93.indd 515 20/03/17 11:51 AM

Source page 535

Table 93-3 Empiric Therapy of Suspected Bacterial Endocarditis*

Patient Characteristics	Recommended Agents, Initial Dose
Uncomplicated history	Ceftriaxone, 1–2 g IV or Nafcillin, 2 g IV or Oxacillin, 2 g IV or Vancomycin, 15 mg/kg plus Gentamicin, 1–3 mg/kg IV or Tobramycin, 1 mg/kg IV
Injection drug use, congenital heart disease, hospital-acquired, suspected methicillin-	Nafcillin, 2 g IV plus

resistant Staphylococcus aureus, or already Gentamicin, 1–3 mg/kg IV

on oral antibiotics plus

Vancomycin, 15 mg/kg IV

Prosthetic heart valve Vancomycin, 15 mg/kg IV

plus

Gentamicin, 1–3 mg/kg IV

plus

Rifampin, 300 mg PO

*

Based on American Heart Association, endorsed by the Infectious Disease Society of America, http://www.idsociety.org/Organ_System/, accessed April 1, 2014. Because of controversy in the literature regarding the optimal regimen for empiric treatment, antibiotic selection should be based on patient characteristics, local resistance patterns, and current authoritative recommendations.

patients include those with a prior history of infective endocarditis, a prosthetic heart valve, unrepaired or failed repair of a congenital heart defect, or a cardiac transplant recipient with valvular regurgitation due to a structurally abnormal valve. Full antibiotic prophylaxis recommendations may be found at http://circ.ahajournals.org/content/116/15/1736.full.pdf.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 155, "Infective Endocarditis," by Richard Rothman, Catherine A. Marco, and Samuel Yang.

93_Cydulka_Ch93.indd 516 20/03/17 11:51 AM

Source page 536

C h a p te r Tetanus and Rabies

94 Michael T. Fitch

■■ TETANUS

Tetanus is an acute and frequently fatal disease resulting from an infection with the organism Clostridium tetani. The disease is exotoxin mediated and can occur after any of a variety of tissue injuries, including clean and contaminated wounds, elective surgery, burns, puncture wounds, otitis media, dental infections, animal bites, abortion, and pregnancy. Due to successful vaccination campaigns, tetanus is a rare disease in developed countries but remains a significant problem elsewhere in the world.

Clinical Features

C. tetani is prevalent throughout the environment in soil, dust, skin surfaces, and animal and human feces. Spores are resistant to destruction, survive years on environmental surfaces, and are introduced into the body following tissue injury. Anaerobic tissue conditions lead to toxin formation

when the spores begin to germinate, and therefore crushed or devitalized tissue, a retained foreign body, or infection can favor growth of the toxin-producing form of C. tetani. The incubation period ranges from <24 hours to >1 month, and while most cases occur within 14 days some patients may present several months after injury. Many cases of tetatnus occur in patients where no specific injury is recognized, and injuries are often minor and occur indoors.

Clinically, tetanus is categorized into three forms: local, cephalic, and generalized. Local tetanus is uncommon and presents with persistent muscle contractions in proximity to the injury site. Cephalic tetanus is rare, occurs occasionally with otitis media or head injuries, and has localized involvement of the face and cranial nerves. Generalized tetanus is the most common form (80% of cases) and often presents with a descending pattern of symptoms that begin with pain and stiffness in the jaw. Progression of the symptoms leads to trismus (lockjaw) and development of the classically described facial expression, risus sardonicus. Violent spasms and tonic contractions of muscle groups are responsible for the symptoms of the disease including dysphagia, opisthotonos, flexing of the arms, fist clenching, rigidity of abdominal muscles, and extension of the lower extremities. Autonomic symptoms include fever, sweating, hypertension, and tachycardia. Neonatal tetanus is a form of generalized tetanus and can occur due to inadequate maternal immunization and poor umbilical cord care, with symptoms typically presenting by the second week of life and associated with an extremely high mortality rate.

Diagnosis and Differential

Tetanus is diagnosed clinically, and early symptoms may include trismus, neck stiffness or sore throat, and dysphagia. The tetanospasmin toxin interferes with central and peripheral nervous system inhibitory pathways leading

94_Cydulka_Ch94.indd 517 20/03/17 11:56 AM

Source page 537

to unopposed muscle contraction, seizures, and autonomic nervous system dysfunction. Prior immunization does not eliminate tetanus as a diagnostic possibility. There are no confirmatory laboratory tests. Other potential diagnoses with muscular contraction or other symptoms that may be similar to tetatnus include strychnine poisoning, dystonic reactions to phenothiazine, hypocalcemic tetany, rabies, peritonsillar abscess, peritonitis, and meningitis.

Emergency Department Care and Disposition

1. Patients with tetanus require hospitalization to manage frequent muscle

spasms that may last 3 to 4 weeks. Symptoms can be triggered by noise or touch and thus environmental stimuli should be minimized to prevent precipitation of convulsive spasms. The potential for laryngospasm or respiratory muscle spasms are best managed initially in an intensive care unit due to concern for progression to respiratory compromise.

2. Administer tetanus immune globulin 3000 to 6000 units IM in a single injection away from the site of tetanus toxoid administration. Give before wound debridement as additional exotoxin may be released during wound manipulation.

3. Administer tetanus toxoid 0.5 mL IM at presentation, and 6 weeks and 6 months after presentation.

4. Identify and debride any wound or devitalized tissue where suspected inoculation occurred to minimize further toxin production.

5. Antibiotics are of uncertain value in the treatment of tetanus. Metronidazole 500 mg IV every 6 hours is recommended when indicated.

6. Administer benzodiazepines to treat muscle spasms. Midazolam, 0.05 to 0.15 mg/kg/h IV may be given in appropriately monitored settings as a continuous drip when needed. Lorazepam 2 mg IV to effect may be used in small quantities.

7. Neuromuscular blockade may be required to control ventilation and muscular spasm and to prevent fractures and rhabdomyolysis. In such cases, intubated patients may receive a vecuronium bolus of 0.1 mg/kg IV followed by continuous infusion at 1 μg/kg/min. Appropriate sedation during neuromuscular blockade should be provided.

8. Labetalol 0.25 to 1 mg/min continuous IV infusion (0.3 to 1 mg/kg/h in children) can be used to treat manifestations of sympathetic hyperactivity. Magnesium sulfate 40 mg/kg loading dose followed by 2 g/h IV (1.5 g/h if ≤45 kg) has been suggested as an additional treatment for this condition to maintain blood levels of 2.0–4.0 mmol/L. Morphine sulfate 0.5 to 1 mg/kg/h is also useful and may provide sympathetic control without compromising cardiac output. Clonidine 0.3 mg every 8 hours PO or NG may be helpful in the management of cardiovascular instability.

9. Patients who recover from clinical tetanus should undergo active immunization as recovery from an active infection does not confer future immunity.

■■ RABIES

Infection with the rabies virus leads to a fatal encephalitis in humans and animals, and is most often transmitted by contact with infected animal saliva via a bite or scratch wound. While most emergency physicians never encounter patients suffering from acute rabies encephalitis, it is very

94_Cydulka_Ch94.indd 518 20/03/17 11:56 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 534–537.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 94: Tetanus and Rabies

Source pages 538–543 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 538–543

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Tetanus and Rabies**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 94: Tetanus and Rabies 519 common for patients to present for evaluation and treatment after animal bites or other potential exposures to the rabies virus. [Source pages 538–543]

- In the United States, dog bites, cat bites, and exposure to bats are common reasons that patients present for postexposure prophylaxis. [Source pages 538–543]

- Rabies cases in animals within the United States are most often found in raccoons, bats, skunks, foxes, cats, dogs, and cattle. [Source pages 538–543]

- Ferrets, rabbits, guinea pigs, and squirrels are generally not known to transmit rabies to humans and thus these kinds of exposures are considered very low risk for rabies transmission and providers should contact local public health authorities if there are concerns about how to manage these low-risk exposures. [Source pages 538–543]

- Clinical Features Typical incubation periods for rabies virus infections are from 20 to 90 days. [Source pages 538–543]

Clinical reasoning structure

- **Problem representation:** Tetanus and Rabies; source pages 538–543.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Tetanus and Rabies**.

- Source-page range: **538–543**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 538

CHAPTER 94: Tetanus and Rabies 519

common for patients to present for evaluation and treatment after animal bites or other potential exposures to the rabies virus. In the United States, dog bites, cat bites, and exposure to bats are common reasons that patients present for postexposure prophylaxis. Rabies cases in animals within the United States are most often found in raccoons, bats, skunks, foxes, cats, dogs, and cattle. Ferrets, rabbits, guinea pigs, and squirrels are generally not known to transmit rabies to humans and thus these kinds of exposures are considered very low risk for rabies transmission and providers should contact local public health authorities if there are concerns about how to manage these low-risk exposures.

Clinical Features

Typical incubation periods for rabies virus infections are from 20 to 90 days. However, incubation periods as short as 4 days or as long as 6 years have been reported, emphasizing the importance of postexposure prophylaxis for patients regardless of the amount of time that has elapsed since a suspected exposure. The natural history of clinical rabies in humans is presented in Table 94-1.

The viral prodrome associated with human rabies infection includes nonspecific symptoms such as malaise, lethargy, headache, fever, nausea, vomiting, anxiety, and pain at the site of a bite wound. Encephalitic rabies (80% of cases) is characterized by episodic hyperexcitability, disorientation, hallucinations, and bizarre behavior and is associated with autonomic dysfunction, hypersalivation, hyperthermia, tachycardia, and cardiac arrhythmias. Paralytic rabies (20% of cases) generally begins with

Table 94-1 Natural History of Clinical Rabies in Humans after Incubation Period

Clinical Stage	Event	Duration	Defining Usual Common Symptoms and Signs*
Prodrome	First symptom	2–10 days	●● Pain or paresthesia at site of bite

●● Malaise, lethargy

●● Headache

●● Fever

●● Nausea, vomiting, anorexia

●● Anxiety, agitation, depression

Acute neurologic First neurologic 2–7 days ●● Anxiety, agitation, depression

phase sign ●● Hyperventilation, hypoxia

●● Aphasia, incoordination

●● Paresis, paralysis

●● Hydrophobia, pharyngeal spasms

●● Confusion, delirium, hallucinations

●● Marked hyperactivity

Coma Onset of coma 0–14 days ●● Coma

●● Hypotension, hypoventilation, apnea

●● Pituitary dysfunction

●● Cardiac arrhythmia, cardiac arrest

Death or recovery Death or initia- Months ●● Pneumothorax

(extremely rare) tion of recovery (recovery) ●● Intravascular thrombosis

●● Secondary infections

*

Not every symptom or sign may be present in each case.

94_Cydulka_Ch94.indd 519 20/03/17 11:56 AM

Source page 539

progressively worsening extremity paresis and bilateral facial weakness and progresses to coma and organ failure.

Rabies infections lead to coma within 10 days of onset. The clinical course is characterized by complications such as pituitary dysfunction, seizures, respiratory dysfunction, cardiac dysrhythmias, autonomic dysfunction, renal failure, and secondary bacterial infections. Almost all rabies infections are fatal with only a few case reports of patient survival.

Diagnosis and Differential

During the incubation period for rabies infection, there are no definitive diagnostic tests to confirm infection. The diagnosis of rabies infection is therefore clinical, and may be considered for a patient with an unexplained and rapidly progressive encephalitis, especially if autonomic instability, dysphagia, hydrophobia or neurologic symptoms are present. Once symptoms are present, antigen and antibody testing of serum, cerebrospinal fluid (CSF), saliva, and/or tissue biopsies can detect evidence of rabies infection. CSF analysis often demonstrates pleocytosis with a mononuclear predomi-

nance. The final diagnosis of rabies is made by postmortem analysis of brain tissue.

Emergency Department Care and Disposition

For the rare patient who presents with an acute rabies infection and symptoms of clinical disease, there are no specific therapies that have demonstrated clinical benefit. Treatment of an active clinical rabies infection is supportive care with admission to a critical care environment where treatment of the associated complications and further diagnostic testing can be completed.

The more typical focus for emergency providers is to assess patients with possible exposure to the rabies virus, manage public health and animal control notification, and appropriately administer postexposure prophylaxis for rabies exposures.

1. Evaluate all patients for potential postexposure prophylaxis who have been bitten, scratched, or exposed to saliva from a wild animal or from a domestic animal that could be rabid.

2. Exposure to Bats: Provide postexposure prophylaxis to patients who have had direct contact with a bat unless the person can be certain that a bite, scratch, or mucus membrane exposure did not occur. Consider postexposure prophylaxis for persons who were in the same room as a bat and may have been unaware if a bite or direct contact occurred. When possible, capture and submit animals for testing to the public health department as negative testing means postexposure prophylaxis is not necessary.

3. Exposures from Healthy-Appearing Domestic Animals that Can Be Observed: Postexposure prophylaxis may not be immediately necessary if a healthy domestic animal is available for confinement and observation for 10 days after the incident to watch for signs of rabies infection. Consult with local animal control and/or public health officials for guidance in these circumstances.

4. Exposures from Stray Animals and Wild Animals: Begin postexposure prophylaxis for exposures from wild animals or stray animals with

94_Cydulka_Ch94.indd 520 20/03/17 11:56 AM

Source page 540

uncertain immunization histories that cannot be observed. Report exposures to the appropriate animal control and/or public health authorities.

5. Patients with wounds associated with potential rabies exposures should receive tetanus prophylaxis, wound cleansing with soap and water, irrigation with a dilute solution of povidone-iodine (1 mL povidone-iodine

in 9 mL of water or normal saline), antibiotics (if indicated), and rabies prophylaxis.

6. In the United States, postexposure prophylaxis consists of a regimen of one dose of human rabies immunoglobulin (HRIG) and four doses of rabies vaccine over a 14-day period, except for immunocompromised persons, who should receive a five-dose series of vaccine over a 28-day period. Postexposure prophylaxis recommendations are provided in Table 94-2.

Table 94-2 Rabies Postexposure Prophylaxis Schedule—United States, 2010

Immunization Status	Treatment Regimen*
Not previously immunized	Wound cleansing Cleanse all wounds with soap and water; irrigate wounds with 9:1 diluted solution of povidone-iodine (if available).
	HRIG Administer 20 IU/kg actual body weight. If anatomically feasible, infiltrate the full dose around the wound(s) and give any remaining volume IM at an anatomic site distant from vaccine administration; do not give HRIG in the same syringe as vaccine. HRIG may partially suppress active production of rabies virus antibody, so do not give more than the recommended dose.
	Vaccine HDCV or PCECV 1.0 mL (deltoid area†), one dose on days 0,‡ 3, 7, and 14.§
Previously immunized■	Wound cleansing Cleanse all wounds with soap and water; irrigate wounds with 9:1 povidone-iodine solution.
	HRIG HRIG should not be administered.
	Vaccine HDCV or PCECV 1.0 mL (deltoid area†), one dose on days 0‡ and 3.

*
These regimens are appropriate for all age groups, including children.

†
The deltoid area is the only acceptable site of vaccination for adults and older children. For younger children, the outer aspect of the thigh (anterolateral aspect) may be used. Vaccine should never be administered in the gluteal area.

‡
Day 0 is the day the first dose of vaccine is administered.

§

Day 28 vaccine dose no longer recommended by the Advisory Committee on Immunization Practices, unless the patient is immunocompromised. See http://www.cdc.gov/rabies/resources/acip_recommendations.html.

■

Any persons with a history of preexposure prophylaxis with HDCV or PCECV; prior postexposure prophylaxis with HDCV or PCECV; or previous immunization with any other type of rabies vaccine and a documented history of antibody response to the prior immunization.

Abbreviations: HDCV, human diploid cell vaccine; HRIG, human rabies immunoglobulin; PCECV, purified chick embryo cell culture vaccine.

94_Cydulka_Ch94.indd 521 20/03/17 11:56 AM

Source page 541

7. HRIG provides passive immunity and is administered once at the beginning of therapy up to the seventh day after the initial exposure. The dose is 20 IU/kg, with as much as possible infiltrated locally at the exposure site and the remainder administered intramuscularly in the deltoid or thigh away from the site where rabies vaccine is given.

8. Provide active immunization with rabies vaccine (human diploid cell vaccine or purified chick embryo cell culture vaccine) in the deltoid or lateral thigh given on days 0, 3, 7, and 14 for patients with normal immune systems. Patients with immunocompromise should receive a fifth dose of vaccine on day 28.

9. The Centers for Disease Control and Prevention (CDC) and state or county health departments can provide assistance in the management of postexposure prophylaxis for rabies. The most current information available from the CDC can be accessed at www.cdc.gov/rabies/.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 156, "Tetanus," by Joel L. Moll and Donna L. Carden; and Chapter 157, "Rabies," by David Weber.

94_Cydulka_Ch94.indd 522 20/03/17 11:56 AM

References/provenance

- Source: *Tintinalli's Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 538–543.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 95: Malaria

Source pages 544–547 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 544–547

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Malaria**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 95: Malaria 525 Table 95-1 Drug Options for Therapy of Uncomplicated Plasmodium falciparum Malaria Treatment of Choice Is an Artemisinin-Containing Combination Therapy (ACT) Drug Adult Dose Pediatric Dose Artemether-lumefantrine 4 tablets twice daily 5–15 kg: 1 tab initially, 1 tablet ●● (CoArtem®; each tablet for 3 days (the first two in 8 h, then 1 tablet every contains artemether doses should be about 12 h × 2 days 20 mg and lumefantrine 8 h apart) ●● 15–25 kg: 2 tablets initially, 120 mg) 2 tablets in 8 h, then 2 tablets every 12 h × 2 days ●● 25–35 kg: 3 tablets initially, 3 tablets in 8 h, then 3 tablets every 12 h × 2 days ●● >35 kg: follow adult dosing Artesunate-amodiaquine 2 tablets once daily ●● 5 to <9 kg: 1 tablet/day of arte- (where available; not for 3 d sunate (AS) 25 mg/amodiaquine available in the United (AQ) 67.5 mg States; each adult tablet ●● 9 to <18 kg: 1 tablet/day of AS contains artesunate 50 mg/AQ 135 mg 100 mg and amodia- ●● 18 to <36 kg: 1 tablet/day of AS quine hydrochloride salt 100 mg/AQ 270 mg 270 mg) ●● ≥36 kg: 2 tablets/day of AS 100 mg/AQ 270 mg (adult dose) Alternatives to ACT Drug Adult Dose Pediatric Dose Atovaquone-proguanil 4 tablets once daily for ●● 5–8 kg: 2 pediatric tablets × 3 days (Malarone®; each 3 days. [Source pages 544–547]

- Do not use for ●● 9–10 kg: 3 pediatric tablets × 3 days adult tablet contains treatment if atovaquone- ●● 11–20 kg: 1 adult tablet × 3 days atovaquone 250 mg and proguanil has been taken ●● 21–30 kg: 2 adult tablets × 3 days proguanil 100 mg; each as chemoprophylaxis ●● 31–40 kg: 3 adult tablets × 3 days pediatric tablet contains and the patient’s current ●● >41 kg: adult dose atovaquone 62.5 mg and illness is a suspected proguanil 25 mg) treatment failure. [Source pages 544–547]

- Quinine sulfate (plus dox- 650 mg PO every 8 h for 10 mg sulfate salt/kg, up to adult ycycline or clindamycin) 3–7 days dose PO every 8 h for 3–7 days Plus Doxycycline 100 mg PO every 12 h for 2.2 mg/kg (up to adult dose of 7 days 100 mg) PO every 12 h for 7 days Or in children under age 8 years Clindamycin 7 mg/kg PO every 8 h for 7 days 4. [Source pages 544–547]

- Chloroquine has no effect on the exoerythrocytic forms of P. [Source pages 544–547]

- Unless treated with prima- quine, relapse will occur. [Source pages 544–547]

Clinical reasoning structure

- **Problem representation:** Malaria; source pages 544–547.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Malaria**.

- Source-page range: **544–547**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 544

CHAPTER 95: Malaria 525

Table 95-1 Drug Options for Therapy of Uncomplicated Plasmodium falciparum

Malaria

Treatment of Choice Is an Artemisinin-Containing Combination Therapy (ACT)

Drug Adult Dose Pediatric Dose

Artemether-lumefantrine 4 tablets twice daily 5–15 kg: 1 tab initially, 1 tablet

●●

(CoArtem®; each tablet for 3 days (the first two in 8 h, then 1 tablet every

contains artemether doses should be about 12 h × 2 days

20 mg and lumefantrine 8 h apart) ●● 15–25 kg: 2 tablets initially,

120 mg) 2 tablets in 8 h, then 2 tablets

every 12 h × 2 days

●● 25–35 kg: 3 tablets initially,

3 tablets in 8 h, then 3 tablets

every 12 h × 2 days

●● >35 kg: follow adult dosing

Artesunate-amodiaquine 2 tablets once daily ●● 5 to <9 kg: 1 tablet/day of arte-

(where available; not for 3 d sunate (AS) 25 mg/amodiaquine

available in the United (AQ) 67.5 mg

States; each adult tablet ●● 9 to <18 kg: 1 tablet/day of AS

contains artesunate 50 mg/AQ 135 mg

100 mg and amodia- ●● 18 to <36 kg: 1 tablet/day of AS

quine hydrochloride salt 100 mg/AQ 270 mg

270 mg) ●● ≥36 kg: 2 tablets/day of AS
100 mg/AQ 270 mg (adult dose)
Alternatives to ACT
Drug Adult Dose Pediatric Dose
Atovaquone-proguanil 4 tablets once daily for ●● 5–8 kg: 2 pediatric tablets × 3 days
(Malarone®; each 3 days. Do not use for ●● 9–10 kg: 3 pediatric tablets × 3 days
adult tablet contains treatment if atovaquone- ●● 11–20 kg: 1 adult tablet × 3 days
atovaquone 250 mg and proguanil has been taken ●● 21–30 kg: 2 adult tablets × 3 days
proguanil 100 mg; each as chemoprophylaxis ●● 31–40 kg: 3 adult tablets × 3 days
pediatric tablet contains and the patient's current ●● >41 kg: adult dose
atovaquone 62.5 mg and illness is a suspected
proguanil 25 mg) treatment failure.
Quinine sulfate (plus dox- 650 mg PO every 8 h for 10 mg sulfate salt/kg, up to adult
ycycline or clindamycin) 3–7 days dose PO every 8 h for 3–7 days
Plus
Doxycycline 100 mg PO every 12 h for 2.2 mg/kg (up to adult dose of
7 days 100 mg) PO every 12 h for 7 days
Or in children under age
8 years
Clindamycin 7 mg/kg PO every 8 h for 7 days
4. Chloroquine has no effect on the exoerythrocytic forms of P. vivax and P. ovale, which remain dormant in the liver. Unless treated with primaquine, relapse will occur. This treatment can be given after full recovery from the initial illness. Treat adults with primaquine phosphate 30 mg of base per day for 14 days. Treat children with primaquine phosphate 0.5 mg/kg base for 14 days. Primaquine should be avoided in patients with glucose-6-phosphate dehydrogenase deficiency because of hemolysis.
95_Cydulka_Ch95.indd 525 20/03/17 12:10 PM

Source page 545

526 SECTION 10: Infectious and Immunologic Diseases
Table 95-2 Antimalarial Drug Options for Severe (Complicated) Malaria
Drug Adult Dose Pediatric Dose
Artesunate* (available from the CDC if quinidine fails to provide improvement; call 770-488-7788) if necessary. 2.4 mg/kg IV at 0, 8, and 24 h, then daily. 2.4 milligrams/kg IV at 0, 8, and 24 h, then daily. Artesunate can be given IM
Quinidine gluconate (plus doxycycline or clindamycin)† 6.25 mg base (= 10 mg salt)/kg IV load over 2 h (maximum, 600 mg), follow with 0.0125 mg 6.25 mg base (= 10 mg salt)/kg IV load over 2 h (maximum, 600 mg), follow with 0.0125 mg

base (= 0.02 mg salt)/kg/min base (= 0.02 mg salt)/kg/min

continuous infusion. continuous infusion.

Plus

Doxycycline 2.2 mg/kg IV (up to adult 2.2 mg/kg IV (up to adult

dose of 100 mg) every 12 h dose of 100 mg) every 12 h

for 7 days. for 7 days

Or in children under age

8 years

Clindamycin 10 mg base/kg loading dose

IV followed by 5 mg base/kg

IV every 8 h for 7 days.

Artesunate is considered the drug of choice by World Health Organization guidelines.

*

†

Quinine dihydrochloride is an alternative to quinidine gluconate (20 mg [salt]/kg infused IV over 2–4 hours, then 10 mg/kg every 8 hours, can be given IM if necessary, as 50 mg/mL solution).

Abbreviation: CDC = Centers for Disease Control and Prevention.

5. For severe or complicated malaria infections, antimalarial drug options are listed in Table 95-2. Continue the treatment by a parenteral route for at least 24 hours. Monitor blood glucose as parenteral quinidine and quinine can cause severe hypoglycemia. These agents are also myocardial depressants, and thus contraindicated in patients with heart disease. Cardiac monitoring is recommended during administration.

6. Provide aggressive supportive care to all hospitalized ill patients, including appropriate IV fluid replacement, correction of metabolic derangements, and advanced support (e.g., dialysis or mechanical ventilation), as necessary.

7. The CDC has a malaria hotline: 770-488-7788 M-F, 9 am to 5 pm EST, and 770-488-7100 after hours, weekends, and holidays. The CDC website can be accessed at http://www.cdc.gov/malaria/ to obtain resistance patterns in various countries and information on malaria prophylaxis and treatment. When resistance patters are uncertain, concern for chloroquine resistance for initial treatment should be considered.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 158, "Malaria," by Malcolm Molyneaux.

95_Cydulka_Ch95.indd 526 20/03/17 12:10 PM

Source page 546

Chapter Foodborne and

96 Waterborne Diseases

Benjamin Weston

Foodborne disease may occur after consumption of food contaminated with bacteria, viruses, parasites, chemicals, or biotoxins. Viruses such as norovirus, astrovirus, rotavirus, and enteric adenovirus are the most common sources, with norovirus causing over half of all cases. Bacterial disease is often more severe and includes nontyphoidal Salmonella, which is the most common cause for hospitalization and associated death in the United States. Other bacterial causes may include Clostridium perfringens, Campylobacter spp., Listeria monocytogenes, Shigella spp., Shiga toxin-producing Escherichia coli, and Staphylococcus aureus. Parasitic causes include Giardia lamblia, Toxoplasma gondii, Entamoeba histolytica, and Cryptosporidium. In addition, patients may experience symptoms of scombroid or ciguatera poisoning after eating some types of fish associated with these toxin-induced syndromes.

Waterborne diseases occur from ingestion or contact with contaminated water. Symptoms are most commonly gastrointestinal or dermatologic in nature. Common organisms include those associated with foodborne illness as well as Vibrio species, Aeromonas species, Pseudomonas aeruginosa, Yersinia species, hepatitis A, nontuberculous Mycobacterium, and other less common organisms. The most common pathogen responsible for recreational waterborne disease outbreaks is Cryptosporidium.

■■ CLINICAL FEATURES

Symptoms of both foodborne and ingested waterborne illness include nausea, vomiting, diarrhea, abdominal cramping, fever, dehydration, and malaise. A history of multiple family members or closely associated individuals with simultaneous symptoms is supportive of a suspected foodborne or ingested waterborne illness. Physical exam may reveal findings of dehydration, and some patients will have stool positive for frank or occult blood. Prolonged illness beyond two weeks suggests protozoan parasites. Shiga-toxin producing Escherichia coli (E. coli) infections may present with vomiting, abdominal cramping, bloody diarrhea, and mild fever, and may be complicated by hemolytic uremic syndrome, especially if antibiotic treatment is prescribed.

Patients with scombroid fish or ciguatera poisoning have symptoms similar to other foodborne illnesses and typically present one to six hours after ingestion. In addition to nausea and vomiting, patients with scombroid poisoning frequently have flushing and headache due to a histamine-mediated reaction. Those with ciguatera poisoning may have headaches, muscle aches, paresthesias, reversal of hot and cold sensation, or sensitivity to extreme temperatures, due to sodium channel-mediated nerve

depolarizations.

The skin manifestations of waterborne illness may include localized cellulitis, painful indurated plaques of Mycobacterium marinum, or

96_Cydulka_Ch96.indd 527 20/03/17 12:18 PM

Source page 547

necrotizing infections and hemorrhagic bulla associated with Vibrio vulnificus. Patients with Aeromonas hydrophila skin infections often have a history of trauma associated with freshwater exposure, and may have foul smelling wounds.

■■ DIAGNOSIS AND DIFFERENTIAL

Diagnostic testing is often not indicated as most of these illnesses are self-limited. For those patients who are more acutely ill, or if other significant diagnoses are being entertained, consider stool cultures with Gram stain, the neutrophil marker lactoferrin, electrolytes, and complete blood count. Fecal leukocyte testing is neither sensitive nor specific for invasive disease, and is a poor predictor of response to antibiotic treatment. Shiga toxin-producing E. coli and Vibrio cultures may require specific laboratory procedures (check local laboratory guidelines). Reserve ova and parasite testing for those patients with symptoms over two weeks, immunocompromise, community waterborne outbreaks, recent travel to endemic areas, or men who have sex with men. Clostridium difficile antigen testing may be indicated in those with prolonged symptoms, recent antibiotic use, significant comorbidities, or extremes of age. Likely pathogens to consider in the differential can often be identified based on clinical features of the presenting illness (Table 96-1).

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

Most cases are self-limited and improve with supportive treatment.

1. Initiate rehydration with oral fluids, if tolerated. Intravenous rehydration with normal saline will benefit patients who are significantly dehydrated or those with continued vomiting.
2. Antiemetics, such as metoclopramide 10 mg PO or IV, or ondansetron 4 PO or IV may facilitate oral rehydration. Antihistamines, such as diphenhydramine 25 mg PO or IV, may improve the symptoms of scombroid fish poisoning.
3. Loperamide 4 mg initially, then 2 mg after every unformed stool up to a maximum of 16 mg per day can be recommended for mild to moderate, nonbloody diarrhea in adults without fever.

4. Antibiotics are recommended for consideration in patients with moderate to severe traveler’s diarrhea, symptoms for more than 1 week, patients requiring hospitalization, and immunocompromised hosts. Empiric antibiotics may include ciprofloxacin 500 mg PO twice per day for 3 to 5 days or levofloxacin 500 mg PO once daily for 3 to 5 days. Antibiotics and antimotility agents are contraindicated in patients with suspected E. coli 0157:H7 as these can increase the risks of developing hemolytic–uremic syndrome. For specific pathogen treatment protocols, see Tables 159-3 to 159-6 in Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 8th edition, Chapter 159.

96_Cydulka_Ch96.indd 528 20/03/17 12:18 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 544–547.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 96: Foodborne and Waterborne Diseases

Source pages 548–551 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 548–551

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Foodborne and Waterborne Diseases**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 96: Foodborne and Waterborne Diseases 529 Table 96-1 Clinical Features of Foodborne Infections Clinical Presentation Foodborne Pathogens Gastroenteritis with vomiting as the Viral pathogens: Norovirus, Rotavirus, primary symptom and Astrovirus; preformed toxins: Staphylococcus aureus and Bacillus cereus Noninflammatory diarrhea Can be any enteric pathogen, but classically: (watery, nonbloody) ETEC Giardia Vibrio cholera Enteric viruses Cryptosporidium Cyclospora Inflammatory diarrhea Shigella (grossly bloody, fever) Campylobacter Salmonella EIEC Shiga toxin-producing Escherichia coli O157:H7 and non-O157:H7 Vibrio parahaemolyticus Yersinia Entamoeba Persistent diarrhea (>14 days) Parasites: Giardia Cyclospora Entamoeba Cryptosporidium Neurologic manifestations Botulism (Clostridium botulinum toxin) Scombroid fish poisoning Ciguatera fish poisoning Tetrodotoxin Toxic mushroom ingestion Paralytic shellfish poisoning Guillain–Barré syndrome Systemic illness Listeria monocytogenes Brucella Salmonella typhi Salmonella paratyphi Vibrio vulnificus Hepatitis A and E Abbreviations: EIEC, enteroinvasive E. [Source pages 548–551]

- Treat waterborne skin infections with tetanus vaccination and empiric antibiotics (see Table 96-2). [Source pages 548–551]

- Patients with necrotizing infections should have surgical consultation to consider operative debridement. [Source pages 548–551]

- Most patients can be treated as outpatients; admission is indicated in those appearing toxic, those with significant dehydration or uncontrolled vomiting, the immunocompromised, or those at the extremes of age with significant symptoms. [Source pages 548–551]

- 96_Cydulka_Ch96.indd 529 20/03/17 12:18 PM 530 SECTION 10: Infectious and Immunologic Diseases Table 96-2 Treatment of Waterborne Skin Infections Clinical Features of Pathogen Skin Infection Treatment Vibrio vulnificus Cellulitis with hemorrhagic Doxycycline 100 mg IV twice bullae, septicemia per day plus fourth-generation cephalosporin; necrotizing infec- tions require emergent surgical debridement Aeromonas species Cellulitis, necrotizing wound Mild infections: ciprofloxacin infections 500 mg PO twice per day; severe infections: ciprofloxacin 400 mg IV twice per day plus an IV antipseudomonal penicillin or fourth-generation cephalosporin; necrotizing infections require emergent surgical debridement Pseudomonas Hot-tub folliculitis, cellulitis Hot-tub folliculitis is usually aeruginosa in immunocompromised/ self-limited. [Source pages 548–551]

Clinical reasoning structure

- **Problem representation:** Foodborne and Waterborne Diseases; source pages 548–551.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Foodborne and Waterborne Diseases**.

- Source-page range: **548–551**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 548

CHAPTER 96: Foodborne and Waterborne Diseases 529

Table 96-1 Clinical Features of Foodborne Infections

Clinical Presentation Foodborne Pathogens

Gastroenteritis with vomiting as the Viral pathogens: Norovirus, Rotavirus,

primary symptom and Astrovirus; preformed toxins:

Staphylococcus aureus and Bacillus cereus

Noninflammatory diarrhea Can be any enteric pathogen, but classically:

(watery, nonbloody) ETEC

Giardia

Vibrio cholera

Enteric viruses

Cryptosporidium

Cyclospora

Inflammatory diarrhea Shigella

(grossly bloody, fever) Campylobacter

Salmonella

EIEC

Shiga toxin-producing Escherichia coli

O157:H7 and non-O157:H7

Vibrio parahaemolyticus

Yersinia

Entamoeba

Persistent diarrhea (>14 days) Parasites:

Giardia

Cyclospora

Entamoeba

Cryptosporidium

Neurologic manifestations Botulism (Clostridium botulinum toxin)

Scombroid fish poisoning

Ciguatera fish poisoning

Tetrodotoxin

Toxic mushroom ingestion

Paralytic shellfish poisoning

Guillain–Barré syndrome

Systemic illness Listeria monocytogenes

Brucella

Salmonella typhi

Salmonella paratyphi

Vibrio vulnificus

Hepatitis A and E

Abbreviations: EIEC, enteroinvasive E. coli; ETEC, enterotoxigenic E. coli.

5. Treat waterborne skin infections with tetanus vaccination and empiric antibiotics (see Table 96-2). Patients with necrotizing infections should have surgical consultation to consider operative debridement.

6. Most patients can be treated as outpatients; admission is indicated in those appearing toxic, those with significant dehydration or uncontrolled vomiting, the immunocompromised, or those at the extremes of age with significant symptoms.

96_Cydulka_Ch96.indd 529 20/03/17 12:18 PM

Source page 549

Table 96-2 Treatment of Waterborne Skin Infections

Pathogen	Clinical Features of Skin Infection	Treatment
Vibrio vulnificus	Cellulitis with hemorrhagic bullae, septicemia	Doxycycline 100 mg IV twice per day plus fourth-generation cephalosporin; necrotizing infections require emergent surgical debridement
Aeromonas species	Cellulitis, necrotizing wound	Mild infections: ciprofloxacin

infections 500 mg PO twice per day; severe
infections: ciprofloxacin 400
mg IV twice per day plus an IV
antipseudomonal penicillin or
fourth-generation cephalosporin;
necrotizing infections require
emergent surgical debridement

Pseudomonas aeruginosa | Hot-tub folliculitis, cellulitis in immunocompromised/ diabetics | Hot-tub folliculitis is usually self-limited. Severe infection: ciprofloxacin 400 mg IV twice per day plus an IV antipseudomonal penicillin or fourth-generation cephalosporin

Mycobacterium marinum | Granulomatous skin infections | Clarithromycin 500 mg PO twice per day or doxycycline 100 mg PO twice per day for 3 months; severe cases: combine with rifampin or ethambutol

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 159, "Food and Waterborne Diseases," by Lane M. Smith and Simon A. Mahler.

96_Cydulka_Ch96.indd 530 20/03/17 12:18 PM

Source page 550

Chapter Zoonotic Infections

97 David Gordon

Zoonotic infections are a class of specific diseases and infections that are naturally transmitted between vertebrate animals and humans. Transmission may occur via direct contact with an infected animal or animal product, by ingestion of contaminated water or food products, by inhalation, or through arthropod vectors—most commonly ticks. Diagnosis can be challenging because of the nonspecific nature of symptoms (e.g., fever, headache, and myalgias) which overlap with the presentation of many other infectious and autoimmune diseases. Consider zoonotic infections in patients presenting with viral-like illnesses in conjunction with occupational or geographical risk factors for zoonotic disease exposure and transmission. See Table 97-1 for a list of common systemic zoonotic

Table 97-1 Common Systemic Zoonotic Infections

Agent	Animal Reservoir	Physical Findings	Diagnostic Tests	Treatment

Aeromonas Fish, reptiles Nonspecific — See Chapter 159,
species fever, severe “Food and Waterborne
crepitant cel- Illnesses,” in Tintinalli’s
lulitis with sys- Emergency Medicine:
temic toxicity, A Comprehensive
gastroenteritis Study Guide, 8th ed.
Brucella canis Dogs Nonspecific Serologic Doxycycline plus
fever testing, blood
gentamicin or rifampin.
culture TMP-SMX plus
gentamicin in children
Capnocytophaga Dogs and Fever, septic Culture of bite Amoxicillin-clavulanate
cats shock, and wound or clindamycin.
meningitis from Pip-Tazo or a carbape-
infected bite nem plus clindamycin/
vancomycin for shock
Chlamydophila Birds Fever, flu-like Serologic Doxycycline.
psittaci illness, pneu- testing and Azithromycin and
monia, endocar- sputum levofloxacin are
ditis, sepsis culture alternatives
Coxiella burnetii Cattle, Fever, pneumo- Serologic Doxycycline, with the
sheep, goats. nia, hepatitis, testing, PCR possible alternative of
Occasionally meningitis, a fluoroquinolone or
cats endocarditis macrolide
Ehrlichia species Ticks Nonspecific Clinical diagno- Doxycycline
fever, sepsis, sis, serologic recommended
meningitis, testing, periph- for all patients
hepatitis eral blood (even children and
smear, immu- pregnancy). Rifampin
nocytologic is alternative
testing, PCR
(Continued)
531
97_Cydulka_Ch97.indd 531 20/03/17 2:03 PM

Source page 551

532 SECTION 10: Infectious and Immunologic Diseases
Table 97-1 Common Systemic Zoonotic Infections (Continued)
Animal Physical Diagnostic
Agent Reservoir Findings Tests Treatment
Leptospira Birds, dogs, Fever, Darkfield Penicillin G IV.

species rodents pneumonia, microscopic Ceftriaxone IV alterna-
conjunctivitis, examination tive. Mild disease: oral
lymphade- of body fluids, doxycycline or amoxicil-
nopathy serologic lin or azithromycin
testing
Francisella Rabbits, Fever, sepsis, Serologic IV aminoglyco-
tularensis cats, wild meningitis, testing (poses sides. Alternative:
animals, bit- pneumonia, hazards to Doxycycline or cipro-
ing insects hepatitis, rash laboratory floxacin
staff)
Rickettsia Ticks Fever, diar- Clinical diag- Doxycycline or chlor-
rickettsii rhea, or nosis, rise in amphenicol
typical presen- antibody titer
tation of Rocky between acute
Mountain and convales-
spotted fever cent serum,
skin biopsy
Salmonella Dogs, cats Fever, abdomi- Blood or stool Fluoroquinolones
enterica (rarely), rep- nal pain, sep- culture or third-generation
tiles (turtles) sis, cellulitis, cephalosporins
meningitis,
endocarditis,
septic arthritis
Streptococcus Fish, seafood Fever, cellulitis Wound β-Lactams except
iniae cellulitis culture, blood aztreonam.
culture Alternatives:
Azithromycin, clindamy-
cin, fluoroquinolones
Yersinia pestis Dogs, cats, Bubonic: fever, Blood culture, Doxycycline, fluoro-
rodents headache, culture of sus- quinolone, gentami-
buboes, or pected sites cin, streptomycin, or
pneumonic: chloramphenicol
cough, chills,
dyspnea, shock

Abbreviations: PCR, polymerase chain reaction; Pip-Tazo, piperacillin-tazobactam; TMP-SMX, trimethoprim-sulfamethoxazole.

infections and Table 97-2 for specific treatment recommendations for tick-borne diseases.

■■ ROCKY MOUNTAIN SPOTTED FEVER

Rocky Mountain spotted fever (RMSF) is a tick-borne disease caused by

Rickettsia rickettsii, a pleomorphic obligate intracellular organism. The primary vectors for transmission are the American dog tick (Dermacentor variabilis), Rocky Mountain wood tick (Dermacentor andersoni) and the brown dog tick (Rhipicephalus sanguineus). Most cases occur between April and September with peak occurrences in June and July. While

97_Cydulka_Ch97.indd 532 20/03/17 2:03 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 548–551.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 97: Zoonotic Infections

Source pages 552–559 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 552–559

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Zoonotic Infections**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- 97_Cydulka_Ch97.indd 533 Table 97-2 Tick-borne Zoonotic Infections and Specific Treatment Tick-borne Zoonotic Infection Specific Treatment Rocky Mountain spotted fever Doxycycline 100 mg PO or IV twice a day for 7 days, or for 2 days after temperature normalizes. [Source pages 552–559]

- Some recommendations exist for an initial loading dose of 200 mg. [Source pages 552–559]

- For children weighing <45 kg, the dose is 2.2 mg/kg twice daily. [Source pages 552–559]

- Although doxycycline is contraindicated for use in pregnancy, it may be warranted in life-threatening situations. [Source pages 552–559]

- Chloramphenicol is an alternative; however, it has multiple toxic effects and contraindications, and may be difficult to obtain. [Source pages 552–559]

Clinical reasoning structure

- **Problem representation:** Zoonotic Infections; source pages 552–559.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Zoonotic Infections**.

- Source-page range: **552–559**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 552

97_Cydulka_Ch97.indd 533

Table 97-2 Tick-borne Zoonotic Infections and Specific Treatment

Tick-borne Zoonotic Infection Specific Treatment

Rocky Mountain spotted fever Doxycycline 100 mg PO or IV twice a day for 7 days, or for 2 days after temperature normalizes. Some recommendations exist

for an initial loading dose of 200 mg. For children weighing <45 kg, the dose is 2.2 mg/kg twice daily. Although doxycycline

is contraindicated for use in pregnancy, it may be warranted in life-threatening situations. Chloramphenicol is an alternative;

however, it has multiple toxic effects and contraindications, and may be difficult to obtain. Dosing is 50 mg/kg/d divided into four

doses for 7 days.

Lyme disease Primary stage or mild secondary: 14–21 days of doxycycline (100 mg PO twice a day), amoxicillin (500 mg PO three times a day in

adults, 50 mg/kg/d divided three times a day in children), or cefuroxime (500 mg PO twice a day in adults, 30 mg/kg/d divided three

times a day in children). Macrolides possible but less effective.

Severe illness, CNS positive, or high-degree heart block: ceftriaxone 2 g IV for 14–30 days.

A single 200-mg oral dose of doxycycline given within 72 h of a high-risk deer tick bite is effective in preventing Lyme disease.

Tick-borne relapsing fever Doxycycline (100 mg PO/IV twice a day for 7–10 days). Alternative: erythromycin (500 mg PO/IV four times a day for 7–10 days).

Chloramphenicol is an alternate.

Colorado tick fever Treatment is supportive.

CHAPTER 97: Zoonotic Infections

Tularemia Adults: streptomycin, 1 g IM/IV twice a day, or gentamicin/tobramycin, 5 mg/kg IV divided every 8 h. Treat for 10 days.

Children: streptomycin, 15 mg/kg IM twice daily (should not exceed 2 g/d).

Mild disease: ciprofloxacin 750 mg PO twice a day or doxycycline 100 mg PO twice a day. Treat for 21 days.

Prophylaxis for lab exposures: doxycycline 100 mg PO twice a day or ciprofloxacin 500 mg PO twice a day. Treat for 14 days.

Babesiosis Atovaquone (750 mg PO every 12 h) plus azithromycin (500 mg PO on day 1, then 250–1000 mg daily). Treat for 10 days.

If relapse occurs, treat for the longer duration: 6 weeks or 2 weeks after negative blood smear.

Severe disease in adults: clindamycin (1200 mg IV twice a day or 600 mg PO three times a day) + quinine (650 mg PO three times a

day). Treat for 7–10 days.

Ehrlichiosis and anaplasmosis Doxycycline, 100 mg PO twice a day for 7–14 days. For children weighing <45 kg, the dose is 2.2 mg/kg twice a day.

20/03/17 2:03 PM

533

Source page 553

cases have been reported in the majority of the continental United States, more than 60% have originated from five states: North Carolina, Tennessee, Oklahoma, Missouri, and Arkansas.

Clinical Features

Early findings of RMSF include fever, headache, myalgias, and malaise. Patients may also present with lymphadenopathy, nausea, vomiting, and diarrhea. In some patients, particularly children, abdominal pain and edema may be present. Late in the disease course, complications may include meningitis, encephalitis, renal failure, respiratory failure, and myocarditis. The classic clinical triad for RMSF of fever, rash, and history of tick bite is uncommon as only about 50% of infected patients can recall a tick bite and the rash is absent in about 10% of cases. When the rash does occur, it typically develops 2 to 4 days after the onset of fever as a maculopapular eruption that may subsequently become petechial. Rash often appears on the ankles and wrists and then spreads centripetally to the trunk, usually sparing the face, and characteristically involves the palms and soles.

Diagnosis and Differential

The diagnosis of RMSF is largely clinical, with confirmation coming after treatment is initiated. In the absence of an alternative explanation, consider the possibility of RMSF in a febrile patient with a headache who lives in or recently traveled through an endemic region in the spring or summer seasons with possible exposure to ticks. Laboratory abnormalities are usually nonspecific, but RMSF is suggested by the combination of normal white and red cells, thrombocytopenia, elevated liver function tests, and hyponatremia. RMSF can be confirmed—typically outside of the emergency department setting—with a rise in antibody titer between acute and convalescent sera or via skin biopsy of the rash with immunofluorescence testing.

The differential diagnosis for patients with symptoms of RMSF includes viral illnesses, pneumonia, meningococcemia, ehrlichiosis, Kawasaki's disease, toxic shock syndrome, scarlet fever, and leptospirosis.

Emergency Department Care and Disposition

1. Recommended treatment for adults is doxycycline 100 mg PO or IV twice a day. Some recommend an initial loading dose of 200 mg. Treatment should continue for 7 days or until 3 days after resolution of fever.

2. Antibiotic therapy for children weighing <45 kg is doxycycline 2.2 mg/kg PO twice a day. Doxycycline has been used for short courses in chil-

dren without significant staining of the teeth, and is recommended as therapy by both the American Academy of Pediatrics and Centers for Disease Control and Prevention as the treatment of choice for rickettsial diseases in children of all ages.

■■ LYME DISEASE

Lyme disease remains the most common vector-borne zoonotic infection in the United States. The responsible organism is the spirochete Borrelia burgdorferi and the vector is the Ixodes deer tick (also known as the

97_Cydulka_Ch97.indd 534 20/03/17 2:03 PM

Source page 554

black-legged tick). In highly endemic areas, the transmission rate from a deer tick bite is about 3%, although there is minimal risk if the duration of attachment is <36 hours. Cases have been reported in all 48 continental states, but 95% of cases originate from 13 states in the northeast and northern midwest with peak disease transmission in May through August.

Clinical Features

Lyme disease is typically divided into three distinct stages although some patients may not progress through all stages, and stages may overlap or be separated by periods of remission. The first stage involves early localized disease and is characterized by the rash of erythema migrans which may be in conjunction with nonspecific constitutional and rheumatologic symptoms. The characteristic rash occurs within 2 to 30 days in 60% to 80% of cases, often begins at the site of the tick bite, and manifests as an erythematous plaque with central clearing. Other associated symptoms in the first stage of lyme disease include fatigue, headache, fever, neck stiffness, myalgias, arthralgias, and adenopathy.

The second stage, early disseminated disease, occurs within a few days to 6 months after initial infection. This stage is characterized by multiple secondary annular target-shaped skin lesions, fever, adenopathy, neuropathies, cardiac abnormalities, and arthritic problems. Approximately 15% of untreated patients develop neurologic complications such as cranial neuritis that may lead to the classically described bilateral facial nerve palsy. Other neurologic manifestations include headache, neck stiffness, difficulty with mentation, cerebellar ataxia, myelitis, encephalitis, motor or sensory radiculoneuritis, and mononeuritis multiplex. Oligoarticular arthritis can occur, usually affecting the large joints such as the knees. In 8% of patients, there are cardiac manifestations such as myopericarditis or varying degrees of atrioventricular (AV) block,

which sometimes can require a temporary pacemaker for stabilization.

The third stage, late disseminated disease, occurs months to years after initial infection. It can be characterized by chronic arthritis, myocarditis, subacute encephalopathy, axonal polyneuropathy, and leukoencephalopathy. Symptoms can persist for a decade or more.

Diagnosis and Differential

Initial diagnosis of Lyme disease is based primarily on clinical features. Confirmation may be obtained via polymerase chain reaction (PCR) testing, polyvalent fluorescence immunoassay, or western immunoblot testing. Differential diagnosis depends on clinical manifestation of the disease stage, and may include southern tick-associated rash illness (STARI), cellulitis, erythema multiforme, tinea corporis, viral or bacterial meningitis, encephalitis, rheumatic fever, septic arthritis, endocarditis, and other inflammatory or autoimmune conditions.

Emergency Department Care and Disposition

1. Treat primary stage or mild secondary Lyme disease with 14 to 21 days of doxycycline 100 mg PO twice a day in adults, 2.2 mg/kg twice daily for children <45 kg, amoxicillin 500 mg PO three times a day in adults,

97_Cydulka_Ch97.indd 535 20/03/17 2:03 PM

Source page 555

50 mg/kg/d divided three times a day in children, or cefuroxime 500 mg PO twice a day in adults, 30 mg/kg/d divided three times a day in children. Macrolides are less effective.

2. Treat severe illness, meningitis, radiculopathy, or high-degree heart block from lyme disease with ceftriaxone 2 g IV daily in adults, 50 to 75 mg/kg daily in children for 14 to 30 days.

3. Routine antibiotic prophylaxis for Lyme disease following a tick bite is not recommended. A single dose of doxycycline 200 mg in adults and 4 mg/kg in children at least 8 years old may be considered for a deer tick bite in areas highly endemic for Lyme disease when the suspected time of tick attachment is at least 36 hours and prophylaxis can be administered within 72 hours of tick removal.

■■ EHRLICHIOSIS

Ehrlichiosis is a group of zoonotic diseases caused by infection from the Ehrlichia genus. These are small, gram-negative pleomorphic coccobacilli that infect circulating leukocytes. The lone star tick (Amblyomma americanum) serves as the primary vector while the white-tailed deer is the major animal reservoir in the southeastern United States.

Clinical Features

Symptoms usually develop within 1 to 2 weeks after a tick bite. Clinical manifestations may include fever, headache, malaise, nausea, vomiting, diarrhea, abdominal pain, and arthralgias. Fever is present in a majority of cases (97%). A rash of variable appearance may be present but is not common. A minority of patients progress to serious complications that may include renal failure, respiratory failure, and encephalitis.

Diagnosis and Differential

Diagnosis initially must be made on clinical grounds. Laboratory clues may include leukocytopenia, thrombocytopenia, and elevation of hepatic enzymes. During the first week of illness, a peripheral blood smear may show colonies of bacteria in the white blood cells in 20% of infected patients. PCR offers a specific test for confirmation but is not sensitive enough to be used to exclude illness based only on a negative result. An increase in antibody titer between the acute and convalescent phases of the illness may be used for diagnosis. The gold standard for diagnosis is an immunofluorescence assay.

The differential diagnosis includes RMSF, Lyme disease, babesiosis, anaplasmosis, viral illnesses, cholecystitis/cholangitis, malaria, meningitis, and typhoid.

Emergency Department Care and Disposition

1. Doxycycline 100 mg PO twice a day in adults, 2.2 mg/kg twice daily for children <45 kg is first-line treatment for adults and children of all ages. Treatment should continue until 3 to 5 days after fever resolution or 10 to 14 days after resolution of CNS symptoms in severe disease.

2. Rifampin 300 mg twice a day for adults can be used for patients with contraindications to doxycycline.

97_Cydulka_Ch97.indd 536 20/03/17 2:03 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 552–559.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 98: World Travelers

Source pages 560–565 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 560–565

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **World Travelers**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 98: World Travelers 541 Table 98-2 Physical Findings in Selected Tropical Infections Physical Finding Likely Infection or Disease Rash Dengue fever, typhus, syphilis, gonorrhea, Ebola fever, brucellosis, chikungunya, HIV seroconversion, Zika virus Jaundice Hepatitis, malaria, yellow fever, leptospirosis, relapsing fever Lymphadenopathy Rickettsial infections, brucellosis, HIV, Lassa fever, leishmaniasis, Epstein–Barr virus, cytomegalovirus, toxoplasmosis, trypanosomiasis Hepatomegaly Amebiasis, malaria, typhoid, hepatitis, leptospirosis Splenomegaly Malaria, relapsing fever, trypanosomiasis, typhoid, brucellosis, kala-azar, typhus, dengue fever, schistosomiasis Eschar Typhus, borreliosis, Crimean–Congo hemorrhagic fever, anthrax Hemorrhage Lassa, Marburg, or Ebola viruses; Crimean–Congo hemorrhagic fever; meningococcemia, epidemic louse-borne typhus Abbreviation: HIV, human immunodeficiency virus. [Source pages 560–565]

- abnormal findings such as fever, skin rash, hepatosplenomegaly, lymph- adenopathy, and jaundice (Table 98-2). [Source pages 560–565]

- ■■ DIAGNOSIS AND DIFFERENTIAL Evaluate the results of laboratory studies to assess for abnormalities sug- gestive of specific disease processes. [Source pages 560–565]

- On a CBC with differential, look for lymphopenia (dengue, HIV, typhoid), eosinophilia (parasites, fungal dis- ease), or thrombocytopenia (malaria, dengue, acute HIV, typhoid). [Source pages 560–565]

- Con- sider a blood smear and dipstick antigen test for malaria, if available, for travelers with fever who have recently visited endemic locations. [Source pages 560–565]

Clinical reasoning structure

- **Problem representation:** World Travelers; source pages 560–565.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **World Travelers**.

- Source-page range: **560–565**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 560

CHAPTER 98: World Travelers 541

Table 98-2 Physical Findings in Selected Tropical Infections

Physical Finding Likely Infection or Disease

Rash Dengue fever, typhus, syphilis, gonorrhea, Ebola fever, brucellosis,

chikungunya, HIV seroconversion, Zika virus

Jaundice Hepatitis, malaria, yellow fever, leptospirosis, relapsing fever

Lymphadenopathy Rickettsial infections, brucellosis, HIV, Lassa fever, leishmaniasis,

Epstein–Barr virus, cytomegalovirus, toxoplasmosis, trypanosomiasis

Hepatomegaly Amebiasis, malaria, typhoid, hepatitis, leptospirosis

Splenomegaly Malaria, relapsing fever, trypanosomiasis, typhoid, brucellosis,

kala-azar, typhus, dengue fever, schistosomiasis

Eschar Typhus, borreliosis, Crimean–Congo hemorrhagic fever, anthrax

Hemorrhage Lassa, Marburg, or Ebola viruses; Crimean–Congo hemorrhagic

fever; meningococcemia, epidemic louse-borne typhus

Abbreviation: HIV, human immunodeficiency virus.

abnormal findings such as fever, skin rash, hepatosplenomegaly, lymph-

adenopathy, and jaundice (Table 98-2).

■■ DIAGNOSIS AND DIFFERENTIAL

Evaluate the results of laboratory studies to assess for abnormalities sug-

gestive of specific disease processes. On a CBC with differential, look for

lymphopenia (dengue, HIV, typhoid), eosinophilia (parasites, fungal dis-

ease), or thrombocytopenia (malaria, dengue, acute HIV, typhoid). Con-

sider a blood smear and dipstick antigen test for malaria, if available, for

travelers with fever who have recently visited endemic locations. Urinalysis

may show proteinuria and hematuria in cases of leptospirosis. Obtain liver

function tests if physical examination identifies jaundice. Consider testing for specific diseases suspected by symptoms and risk of exposure. Obtain a chest radiograph when respiratory symptoms are present, and consider a liver ultrasound if amoebic liver abscess is suspected.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. Dengue fever is commonly known as break-bone fever, and is spread by the day-biting Aedes aegypti mosquito and caused by one of four different dengue viruses. Incubation is 4 to 7 days. Symptoms are high fever, headache, nausea, vomiting, myalgias, and rash. Dengue hemorrhagic fever, while classically associated with dengue in Southeast Asia, continues to expand globally and is characterized by high fever, vascular permeability, bleeding, hepatomegaly, and shock. Make the diagnosis using PCR (1 to 8 days after symptom onset) or IgM ELISA (after 4 days of symptoms). Daily blood counts are recommended. Outpatient treatment can be recommended in mild cases, with oral hydration as tolerated and close follow-up for blood work. Avoid aspirin and NSAIDs. Inpatient treatment is recommended for a drop of hemoglobin or platelets, hemorrhagic symptoms, or abnormal vital signs.

2. Chikungunya is a common arbovirus infection in returning travelers. Since 2013 when local transmission was first recognized in the

98_Cydulka_Ch98.indd 541 20/03/17 2:10 PM

Source page 561

Americas, it has expanded to 45 countries or territories. This disease spreads by day-biting mosquitos (Aedes aegypti and Aedes albopictus), and presents very much like classic dengue fever with generalized arthralgia that progresses to chronic arthropathy in 5% to 30% of patients. Use PCR to diagnose 1 to 4 days after symptom onset or test for IgM levels after 5 days of symptoms. There is no preventative vaccine and no specific medication to treat chikungunya virus infection. Treatment is supportive and can include rest, fluids, and NSAIDs to relieve pain and fever. People infected with chikungunya should avoid further mosquito exposure during the first week of illness to reduce the risk of local transmission.

3. Zika virus is spread through the Aedes species mosquito. First discovered in 1947, outbreaks have been reported in Africa, Southeast Asia, and the Pacific Islands—including an outbreak in Brazil and the Americas starting in 2015. Symptoms are commonly fever, rash, joint pain, and conjunctivitis. The illness is usually mild and lasts for less than a week. Development of Guillain-Barré syndrome and congenital

microcephaly has been associated with Zika virus infection. Treatment is supportive as no specific antiviral treatment is available for Zika virus disease.

4. Typhoid fever, also known as enteric fever, is caused by Salmonella typhi and Salmonella paratyphi. Transmission is from contaminated food or water, after contact with the infected urine or feces of symptomatic individuals, or asymptomatic carriers. Incubation is 1 to 3 weeks, and typhoid fever commonly lasts more than 10 days in returning travelers. Initial symptoms include fever with headache, and this progresses to high fevers with chills, headache, cough, abdominal distention, myalgias, constipation, and prostration. A classic presentation is a relative bradycardia despite a high fever, but this is often absent. After several days, a pale red macular rash known as “rose spots” appears on the trunk. Complications may include small bowel ulceration, anemia, disseminated intravascular coagulopathy (DIC), pneumonia, meningitis, myocarditis, and renal failure. Diagnosis is made based on clinical findings and can be confirmed by stool culture. Treat with ceftriaxone 2 g IV or IM for 14 days, or ciprofloxacin 500 to 750 mg PO twice daily for 14 days. For severe typhoid fever complicated by delirium, coma, shock, or DIC, administer an initial dose of dexamethasone 3 mg/kg IV. Blood transfusions may be required in severe cases.

5. Brucellosis is caused by the bacteria Brucella, and is most commonly contracted by contact with cattle, goats, camels, dogs, pigs, or after ingestion of unpasteurized milk or cheese. Symptoms include fever, abdominal pain, back pain, fatigue, headache, joint pain, and loss of appetite. Relapsing fever or a chronic low-grade fever is characteristic. Patients often have lymphadenopathy, hepatomegaly, and splenomegaly, and may develop a septic arthritis. Make the diagnosis based on blood culture results or confirmatory serology testing. Consult infectious disease to align treatment with doxycycline, rifampicin, streptomycin or gentamicin for 2 weeks with appropriate follow-up.

6. Rickettsial spotted fevers are transmitted by the bite, body fluid, or feces of ixodid arthropod ticks. Mortality without treatment approaches 25%. Scrub typhus (Rickettsia orientalis) and African tick

98_Cydulka_Ch98.indd 542 20/03/17 2:10 PM

Source page 562

typhus (R. conorii) are the most common forms in travelers returning from Southeast Asia and Africa, respectively. Incubation period is 3 to 14 days. Symptoms may include fever, malaise, myalgias, severe

headache, rash, nausea, and vomiting followed by lymphadenopathy and splenomegaly. The skin lesion in scrub typhus starts as a papule at the bite site, which later becomes necrotic and forms a crusted black eschar. Make the diagnosis based on clinical findings and use serologic tests to confirm the diagnosis after empiric treatment with doxycycline 100 mg twice daily for 7 to 10 days.

7. Epidemic Louse-Borne Typhus is common in Mexico, Guatemala, Ethiopia, and the Himalayas, and is caused by R. prowazekii. This illness is distinct from the disease caused by S. typhi. Incubation period is 8 to 12 days, and patients may or may not be aware of the louse exposure. Symptoms include high fevers, severe headache, and a maculopapular rash between 4 and 7 days. Make the diagnosis based on clinical features and use serologic tests to confirm the diagnosis after empiric treatment with doxycycline 100 mg twice daily for 7 to 10 days.

8. Leptospirosis occurs after fresh water exposure to Leptospira interrogans or after exposure to infected dogs. Incubation period is 2 to 20 days. Symptoms include high fever, severe headache, chills, myalgias, hepatitis, and conjunctival injection. Confirmation of this diagnosis is by serology. Treat mild disease within 3 days of symptoms with amoxicillin 500 mg three times daily or doxycycline 100 mg twice daily. Treat severe cases with penicillin G 5 MU every 6 hours IV, or ceftriaxone 1 g IV or IM daily for 7 to 14 days.

9. Crimean–Congo hemorrhagic fever is a tick-borne viral disease that is rising in frequency in Africa, Asia, Eastern Europe, and the Middle East. Agricultural workers are at the greatest risk, but this infection can also be acquired from contact with the blood of infected individuals. Symptoms include sudden onset of fever, headache, myalgia, dizziness, and possibly mental confusion. The hemorrhagic period starts the third to fifth day of illness with epistaxis, hemoptysis, GI bleeding, vaginal bleeding, or hematuria. Patients may have laboratory abnormalities such as thrombocytopenia, elevated liver enzymes and creatinine, or prolonged prothrombin time and activated partial thromboplastin time. Make the diagnosis based on clinical findings and confirm with serology. Treatment is supportive and may require transfusions or respiratory support in severe illness. For moderate to severe illness, treat with ribavirin.

10. Yellow fever is caused by a flavivirus transmitted by a day-biting mosquito, and occurs along a broad equatorial belt in South and Central America and Africa. Symptoms range from a mild flu-like illness to

hemorrhagic fever with shock. While only 15% of patients progress to severe disease, these patients have an associated 20% to 50% mortality. After an incubation period of 3 to 6 days, typical early symptoms include fever, headache, myalgias, conjunctival injection, abdominal pain, prostration, facial flushing, and relative bradycardia. Symptoms later progress to jaundice, black emesis, and albuminuria. Severe illness may progress to shock, multiorgan failure, and bleeding. Treatment is supportive including transfusion as needed.

98_Cydulka_Ch98.indd 543 20/03/17 2:10 PM

Source page 563

11. Cysticercosis is a systemic parasitic infection caused by dissemination of the larval cysts of the pork tapeworm. Humans become infected by ingesting contaminated food or inadvertent contact with contaminated soil. Involvement of almost any tissue can occur. CNS infection is known as neurocysticercosis, and is a common infectious cause of seizures worldwide. Additional symptoms of neurocysticercosis may include headache, visual or mental status changes, stroke, meningoencephalitis, and obstructive hydrocephalus. Noncontrast CT shows calcifications of inactive disease, and may reveal hydrocephalus. Treat with praziquantel 17 mg/kg three times daily. Steroids are recommended for those with encephalitis, hydrocephalus, or vasculitis.

12. African trypanosomiasis (African sleeping sickness) is caused by the microscopic parasite Trypanosoma brucei, and is transmitted by the tsetse fly. After a bite, a localized inflammatory reaction occurs and is followed in 2 to 3 days by a painless chancre that increases in size for 2 to 3 weeks and then gradually regresses. Intermittent fevers then develop with malaise and rash. When CNS infection develops, behavioral and neurologic changes occur with a progression from encephalitis to coma and eventual death. Other complications can include hemolysis, anemia, pancarditis, and meningoencephalitis. Make the diagnosis with rapid evaluation of blood smears for the mobile parasite. The specific drug and treatment course will depend on the type of infection. Consult infectious disease expert for diagnosis and treatment with pentamidine or other agents.

13. Chagas disease (American trypanosomiasis) is caused by the protozoan T. cruzi and is endemic in regions of Latin America and is reported as far north as Texas. This disease is spread by the nocturnal reduviid “kissing bug” or “assassin” bug. Acute phase illness begins with unilateral periorbital edema or painful cutaneous edema at the site of skin

penetration (chagoma). The subsequent toxemic phase is characterized by parasitemia with lymphadenopathy and hepatosplenomegaly. The acute phase diagnosis is made by blood culture or examination of peripheral blood smears demonstrating motile parasites. In the chronic phase, serologic tests or tissue biopsy are useful to make the diagnosis. Consult with infectious disease specialists for recommendations to treat with nifurtimox, benznidazole, or other agents.

14. Visceral leishmaniasis is caused by the intracellular protozoan Leishmania that is transmitted by Lutzomyia or Phlebotomus sandflies. This disease may be found in patients who serve in the military, adventure travelers, field biologists, and emigrants from endemic zones. Visceral leishmaniasis presents with fever, weight loss, hepatosplenomegaly, pancytopenia, and hypergammaglobulinemia. Treat with pentavalent antimonial compounds, such as sodium stibogluconate (available through the CDC) or miltefosine.

15. Schistosomiasis larvae are released into fresh water by snails. Soon after exposure, patients develop a macular-papular pruritic dermatitis over the lower legs. Four to 8 weeks later, fever occurs with headache, cough, urticaria, diarrhea, hepatosplenomegaly, and hypereosinophilia. Worms mature in the venous blood and deposit eggs in the bladder, GI tract, brain, skin, and liver. Diagnosis is suspected from eosinophilia and microscopic identification of eggs in midday urines or stools. Treat

98_Cydulka_Ch98.indd 544 20/03/17 2:10 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 560–565.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 99: The Transplant Patient

Source pages 566–575 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 566–575

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **The Transplant Patient**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 99: The Transplant Patient 547 Table 99-1 Infections Stratified by Posttransplant Period (Continued) Period after Transplant/ Conditions Infection Comments <1 mo: HSCT-specific ●● Additional bacterial patho- Neutropenia and mucocuta- infections gens: Streptococcus viridans neous injury increase risk for and enterococci HSCT patients. [Source pages 566–575]

- Lungs, blood- ●● Viral infections include stream, and GI tract most respiratory syncytial virus commonly affected sites. [Source pages 566–575]

- and HSV 1–6 mo: in patients with ●● Polyomavirus BK infection, Activation of latent infec- Pneumocystis jirovecii nephropathy tions, relapse, residual, and pneumonia and antiviral ●● C. [Source pages 566–575]

- difficile colitis opportunistic infections (CMV, HBV) prophylaxis ●● HCV infection occur during this period. [Source pages 566–575]

- ●● Adenovirus infection, Viral pathogens and allograft influenza rejection cause the majority ●● Cryptococcus neoformans of febrile episodes during infection this period. [Source pages 566–575]

Clinical reasoning structure

- **Problem representation:** The Transplant Patient; source pages 566–575.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **The Transplant Patient**.

- Source-page range: **566–575**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 566

CHAPTER 99: The Transplant Patient 547

Table 99-1 Infections Stratified by Posttransplant Period (Continued)

Period after Transplant/ Conditions	Infection	Comments
<1 mo: HSCT-specific infections	●● Additional bacterial pathogens: Streptococcus viridans and enterococci ●● Viral infections include respiratory syncytial virus and HSV ●● Pneumocystis jirovecii pneumonia ●● Cryptococcus neoformans infection ●● Mycobacterium tuberculosis infection ●● Anastomotic complications ●● Infection with Listeria, Nocardia, Toxoplasma, Strongyloides, Leishmania,	Neutropenia and mucocutaneous injury increase risk for HSCT patients. Lungs, bloodstream, and GI tract most commonly affected sites.
1–6 mo: in patients with Pneumocystis jirovecii nephropathy and antiviral (CMV, HBV) prophylaxis	●● Polyomavirus BK infection, pneumonia and antiviral ●● C. difficile colitis ●● HCV infection ●● Adenovirus infection, influenza ●● Cryptococcus neoformans infection this period. Polyomavirus BK, ●● Mycobacterium tuberculosis adenovirus infections, and infection recurrent HCV are becoming ●● Anastomotic complications more common.	Activation of latent infections, relapse, residual, and opportunistic infections occur during this period. Viral pathogens and allograft rejection cause the majority of febrile episodes during
1–6 mo: in patients with- out prophylaxis	●● Pneumocystis ●● Infection with herpesviruses (HSV, varicella-zoster virus, CMV, Epstein–Barr virus) ●● HBV infection ●● Infection with Listeria, Nocardia, Toxoplasma, Strongyloides, Leishmania,	Discontinuation of prophylaxis at the end of this period may prompt active infection, especially CMV. Graft-versus-host disease and mucocutaneous injury increase risk for HSCT patients.

Trypanosoma cruzi

>6 mo: general ●● Community-acquired pneu- Community-acquired

monia and urinary tract organisms dominate during

infections this period. Transplant

●● Infection with Aspergillus, recipients have a persistently

atypical molds, Mucor increased risk of infection

species due to community-acquired

●● Infection with Nocardia, pathogens.

Rhodococcus species

>6 mo: late viral ●● CMV infection (colitis and In some patients, chronic

infections retinitis) viral infections may cause

●● Hepatitis (HBV, HCV) allograft injury (e.g., cir-

●● HSV encephalitis rhosis from HCV infection in

●● Community-acquired viral liver transplant recipients,

infections (severe acute bronchiolitis obliterans in

respiratory syndrome, West lung transplant recipients,

Nile) accelerated vasculopathy in

●● JC polyomavirus infection heart transplant recipients

(progressive multifocal leu- with CMV infection) or a

koencephalopathy) malignant condition such as

●● Skin cancer, lymphoma PTLD or skin or anogenital

(PTLD)

Source page 567

548 SECTION 10: Infectious and Immunologic Diseases

Clinical Features

The most common reason for transplant recipients to present to the ED is fever, although as many as half of transplant patients with serious infections can present without fever. When fever is present, the underlying cause in addition to a possible infection can be drug side effects, hypersensitivity reactions, rejection, or malignancy.

Diagnosis and Differential

Pneumonia and urinary tract infections are the most common infections found in this population. Blood counts, inflammatory markers, and tests of renal and liver function may be helpful to obtain in this group of complex patients. Maintain a low threshold for obtaining chest radiographs. Obtain appropriate cultures of urine, blood, and other bodily fluids based on patient symptoms, presentation, physical examination findings, and disease severity. Consider the possibility of central nervous system infections such as meningitis or encephalitis and perform lumbar puncture when appropriate. Liver transplant patients are especially susceptible to intraabdominal

infections during the first postoperative month. Lung transplant patients are prone to pneumonia. Cardiac transplant patients may develop mediastinitis during the first postoperative month.

Emergency Department Care and Disposition

1. When an infection is diagnosed in a patient with an organ transplant, consider consultation with the transplant team to help determine recommended treatment medications and duration.
2. For infections of skin and superficial wounds, a broad-spectrum antibiotic plus an agent specific to MRSA is recommended. Common treatment options include imipenem 500 mg IV every 6 hours, or meropenem 1 g IV every 8 hours, or piperacillin/tazobactam 3.375 IV every 6 hours plus vancomycin 1 g IV every 12 hours or linezolid 600 mg IV every 12 hours.
3. Pneumonia may be caused by a wide variety of organisms from common to atypical to opportunistic. Treatment options include imipenem 500 mg IV every 6 hours, meropenem 1 g IV every 8 hours, cefotaxime 1 to 2 g IV every 6 to 8 hours plus gentamicin 1 to 2 mg/kg IV every 8 hours, or piperacillin/tazobactam 3.375 g IV every 6 hours. Add MRSA-specific therapy, listed above, and fungal therapy, listed below, if suspected.
4. Intraabdominal infections may be due to enteric gram-negative aerobic, obligate anaerobic bacilli and facultative bacilli, or enteric gram-positive streptococci. Recommended coverage is to combine metronidazole 500 mg IV every 12 hours plus one of the following agents: imipenem 500 mg IV every 6 hours, meropenem 1 g IV every 8 hours, doripenem 500 mg IV every 8 hours, or piperacillin/tazobactam 3.375 g IV every 6 hours.
5. The initial treatment of suspected fungal disease is fluconazole 400 mg daily IV; amphotericin B 0.7 mg/kg/d IV has been a mainstay of treatment but has more toxicity than fluconazole. For oral or esophageal Candida, treat with fluconazole 200 mg PO on day 1 followed by 100 mg PO daily.

99_Cydulka_Ch99.indd 548 20/03/17 2:27 PM

Source page 568

6. Suspected CMV disease is treated with ganciclovir 5 mg/kg IV twice daily. For bone marrow transplant patients, add immunoglobulin.
7. Disseminated or ocular varicella and herpes simplex virus are typically treated with acyclovir 800 mg IV five times a day. Adjust dosing in the presence of renal insufficiency. Alternatives include valacyclovir

1000 mg every 8 hours or famciclovir 500 mg every 8 hours.

8. Treatment for Pneumocystis jirovecii pneumonia starts with prednisone 80 mg/d followed immediately by antimicrobial therapy. First choice is trimethoprim/sulfamethoxazole (TMP-SMX) 15 mg/kg/d IV divided every 8 hours while critically ill. Oral therapy is TMP-SMX double strength (DS) two tablets PO every 8 hours for 3 weeks of total therapy. Pentamidine 4 mg/kg/d IV or IM for 3 weeks, or clindamycin 600 mg IV plus primaquine 30 mg orally daily are reserved as alternative therapies if TMP-SMX is not tolerated.

9. Toxoplasmosis can be treated initially with pyrimethamine 200 mg PO initially then 50 to 75 mg PO daily plus sulfadiazine 1 to 4 g PO daily plus folinic acid 10 mg PO daily.

10. Urinary tract infections (see Chapter 53), invasive gastroenteritis (due to Salmonella, Campylobacter, and Listeria, see Chapter 37 and Chapter 96), and diverticulitis (see Chapter 44) can be treated with commonly recommended antimicrobial agents.

■■ COMPLICATIONS OF IMMUNOSUPPRESSIVE AGENTS

Therapeutic immunosuppression is associated with a number of adverse effects and complications. These adverse effects are typically gradual in onset and may include pancreatitis, bleeding, hypoglycemia or hyperglycemia, bradycardia or tachycardia, hyperkalemia, hypertension or hypotension, cardiotoxicity, pulmonary edema, seizures, thromboembolic events, and thrombocytopenia. Medication side effects such as fever or rigors may be similar to symptoms that would otherwise suggest an infection. A headache syndrome often indistinguishable from migraine is common in transplant recipients and usually develops within the first two months of immunosuppression, and may require an initial workup to exclude life-threatening conditions such as a mass lesion or infection. As the number of immunosuppressive drugs has increased dramatically, a complete listing of adverse effects is beyond the scope of this manual. Please refer to the parent textbook, referenced at the end of this chapter, or to web resources for a more complete listing of side effects of these medications.

An illness that prevents a transplant patient from successfully taking his or her immunosuppressive therapy may warrant hospital admission for IV therapy. Consider discussing new medications with a member of the patient's transplant team, as starting even simple medications can precipitate complications.

■■ CARDIAC TRANSPLANTATION

Heart transplantation results in a denervated heart that does not respond with centrally mediated tachycardia in response to stress or exercise, but

does respond appropriately to circulating catecholamines. Patients therefore may experience fatigue or shortness of breath at the onset of exercise

99_Cydulka_Ch99.indd 549 20/03/17 2:27 PM

Source page 569

before an appropriate increase in heart rate occurs over time to meet the metabolic demand. A donor heart is implanted with its sinus node intact to preserve normal atrioventricular conduction, and therefore a normal heart rate is typically 90 to 100 beats/min. However, electrocardiograms frequently will have two distinct P waves as the technique of cardiac transplantation also preserves the recipient's native sinus node at the superior cavoatrial junction. The sinus node of the donor heart is easily identified by its constant 1:1 relation to the QRS complex, whereas the native P wave appears independently through the donor heart rhythm.

Clinical Features

Because the heart is denervated, patients experiencing myocardial ischemia do not present with angina. Instead, heart transplant recipients may present with heart failure secondary to silent myocardial infarctions. Evaluate transplant recipients who have new-onset shortness of breath, chest fullness, or symptoms of congestive heart failure with an electrocardiogram and serial troponin levels to evaluate for the possible presence of myocardial ischemia or infarction.

Although most episodes of acute cardiac transplant rejection are asymptomatic, the most common symptoms that do occur are dysrhythmias and generalized fatigue. Consider the possibility of transplant rejection for patients who present with atrial or ventricular dysrhythmia or symptoms of acute heart failure. In children, rejection may present with low-grade fever, fussiness, and poor feeding.

Emergency Department Care and Disposition

1. Treat acute rejection with methylprednisolone 1 g IV after consultation with a representative from the transplant center. Treatment for rejection is typically begun after biopsy confirmation unless the patient is hemodynamically unstable.
2. Atropine has no effect on a denervated heart and is not recommended for bradydysrhythmias in heart transplant patients. Instead, administer isoproterenol 2 to 10 μg/min IV infusion.
3. Treat hypotension with inotropic agents such as dopamine 2 to 20 μg/kg/min IV or dobutamine 2 to 20 μg/kg/min IV.
4. Symptoms concerning for acute rejection or another illness requiring hospitalization should prompt transfer to a transplantation center, when

possible.

■■ LUNG TRANSPLANTATION

Clinical Features

A lung transplant patient suffering rejection may develop cough, chest tightness, fever, and hypoxemia. Infection, such as interstitial pneumonia, may present with a clinical picture similar to acute rejection. Acute rejection can manifest very quickly, with a severe and rapid decline in patient status and in some circumstances an isolated fever may be the only initial finding. Spirometry may show a >10% drop in forced expiratory volume in 1 second, and physical examination may reveal rales and adventitious sounds. Chest radiograph may demonstrate bilateral interstitial infiltrates or

99_Cydulka_Ch99.indd 550 20/03/17 2:27 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 566–575.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 100: General Management of the Poisoned Patient

Source pages 576–583 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 576–583

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **General Management of the Poisoned Patient**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 100: General Management of the Poisoned Patient 557 Table 100-2 Drug Concentrations That May Assist Patient Assessment or Management Acetaminophen Methanol Carbamazepine Methotrexate Carbon monoxide Paraquat Digoxin Phenobarbital Ethanol Phenytoin Ethylene glycol Salicylate Iron Theophylline Lithium Valproic acid Methemoglobin potentially distracting. [Source pages 576–583]

- Consider empiric testing for acetaminophen and aspirin in all potentially poisoned patients. [Source pages 576–583]

- Complementary testing includes blood glucose concentration, serum lactate/electrolytes, renal/liver func- tion, urine HCG, and electrocardiography. [Source pages 576–583]

- ■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION 1. [Source pages 576–583]

- Gross decontamination should occur prior to patient entry into the ED. [Source pages 576–583]

Clinical reasoning structure

- **Problem representation:** General Management of the Poisoned Patient; source pages 576–583.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **General Management of the Poisoned Patient**.

- Source-page range: **576–583**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 576

CHAPTER 100: General Management of the Poisoned Patient 557

Table 100-2 Drug Concentrations That May Assist Patient Assessment or Management

Acetaminophen	Methanol
Carbamazepine	Methotrexate
Carbon monoxide	Paraquat
Digoxin	Phenobarbital
Ethanol	Phenytoin
Ethylene glycol	Salicylate
Iron	Theophylline
Lithium	Valproic acid
Methemoglobin	

potentially distracting. Consider empiric testing for acetaminophen and aspirin in all potentially poisoned patients. Complementary testing includes blood glucose concentration, serum lactate/electrolytes, renal/liver function, urine HCG, and electrocardiography.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. Gross decontamination should occur prior to patient entry into the ED. Observe universal precautions and don personal protective gear. Ensure that the victim's clothing and jewelry are removed/bagged.

2. The primary goal in management is resuscitation. Focus on assessment and stabilization of the airway, breathing, and circulation. Supportive care is often the only required intervention in the poisoned patient. Early endotracheal intubation may be necessary given the anticipated clinical course of some toxidromes. Treat hypotension with IV crystalloid infusions (10 to 20 mL/kg in children) before initiating pressors. Treat ventricular dysrhythmias according to standard ACLS/PALS protocols. Potentially useful medications in the management of toxin-induced car-

diac arrest are listed in Table 100-3.

3. The proper and timely use of antidotes (Table 100-4) is paramount in the management for select poisoned patients. Rarely, however, does antidotal therapy trump standard resuscitation steps. Focus first on resuscitation measures as listed above.

4. Altered mental status and coma are common presentations of many intoxicants. Reasonable empiric treatments include supplemental oxygen, naloxone (0.2 to 2 mg IV/IO/IM), glucose (1 to 1.5 g/kg IV/IO), and thiamine (10 to 100 mg IV/IO) in the adult patient.

5. Seizures from toxins generally respond to benzodiazepines (first-line) or barbiturates (second-line). Isoniazid (INH) poisoning may precipitate seizures which are refractory to benzodiazepines or barbiturates. Treatment for INH poisoning requires pyridoxine. Hypoglycemia and hyponatremia should be excluded as the etiology for seizures. Phenytoin is not effective in toxin-induced seizures and is not recommended.

100_Cydulka_Ch100.indd 557 20/03/17 2:47 PM

Source page 577

Table 100-3 Potential Interventions in Toxin-Induced Cardiac Arrest

Toxin or Toxin/Drug Class	Intervention
Toxins with a specific antidote (examples)	Antidote
Digoxin	Digoxin Fab
Organophosphates	Atropine
Envenomation	Antivenom
Sodium channel blocker or wide-complex tachycardia	Sodium bicarbonate
Calcium channel blocker or β-blocker	High-dose insulin
Local anesthetic agents	IV lipid emulsion
Lipophilic cardiotoxins	
Other Therapies to Consider	
Cardiac pacing	
Intra-aortic balloon pump	
Extracorporeal membrane oxygenation	

Table 100-4 Common Antidotes: Initial Dosages and Indications

Antidote	Pediatric Dose	Adult Dose	Indication
Calcium chloride 10% 27.2 mg/mL elemental Ca	0.2–0.25 mL/kg IV	10 mL IV	Calcium channel antagonists
Calcium	0.6–0.8 mL/kg IV	10–30 mL IV	Hypermagnesemia

gluconate 10% Hypocalcemia

9 mg/mL elemental Ca

Cyanide Not typically used 1 ampule O2 cham- Cyanide

antidote kit ber of ventilation Hydrogen sulfide

Amyl nitrite bag 30 s on/30 s off (use only sodium

nitrite)

Sodium nitrite (3% 0.33 mL/kg IV 10 mL IV Cyanide

solution)

Sodium thiosulfate 1.65 mL/kg IV 50 mL IV Cyanide

(25% solution)

Dextrose 0.5 g/kg IV 1 g/kg IV Insulin

(glucose) Oral hypoglycemics

Digoxin Fab 1–2 vials IV 5–10 vials Digoxin and other

Acute toxicity cardioactive steroids

Flumazenil 0.01 mg/kg IV 0.2 mg IV Benzodiazepines

Glucagon 50–150 μg/kg IV 3–10 mg IV Calcium channel

blockers

β-Blockers

Hydroxocobalamin 70 mg/kg IV (maximum 5 g). Can be Cyanide

repeated up to 3 times. Administer with Nitroprusside

sodium thiosulfate.

(continued)

100_Cydulka_Ch100.indd 558 20/03/17 2:47 PM

Source page 578

CHAPTER 100: General Management of the Poisoned Patient 559

Table 100-4 Common Antidotes: Initial Dosages and Indications (continued)

Antidote Pediatric Dose Adult Dose Indication

IV lipid emulsion 1.5 mL/kg IV bolus 100-mL IV bolus Local anesthetic

20% over 1 min (may over 1 min, fol- toxicity

be repeated two lowed by 400 mL Rescue therapy for

times at 5-min IV over 20 min lipophilic cardiotoxins

intervals), followed

by 0.25 mL/kg per

minute

Methylene blue 1–2 mg/kg IV 1–2 mg/kg IV Oxidizing toxins

Neonates: 0.3– (e.g., nitrites, benzo-

1.0 mg/kg IV caine, sulfonamides)

Naloxone As much as As much as required Opioids

required Start: 0.1–0.4 mg IV Clonidine

Start: 0.01 mg IV

Pyridoxine Gram for gram if amount isoniazid Isoniazid
ingested is known Gyromitra esculenta
Hydrazine
70 mg/kg IV 5 g IV
(maximum 5 g)
Sodium 1–2 mEq/kg IV bolus followed by 2 Sodium channel
bicarbonate mEq/kg per h IV infusion blockers
Urinary alkalinization
Thiamine 5–10 mg IV 100 mg IV Wernicke's syndrome
Wet beriberi

6. Once stabilized, surface decontamination is the next priority in care. If not previously done, completely disrobe the patient. Dermal toxins must be removed from the skin by irrigation. Ocular exposure often requires pain control with topical agents. Copiously irrigate the eye with isotonic crystalloid. This may require several liters before restoration of physiologic pH. Ophthalmology consultation is recommended for all ocular alkali injuries.

7. Gastrointestinal decontamination is achieved via removal of the toxin from the stomach, binding toxin within the GI tract, or enhancing transit time through the gut. The particular method(s) utilized, if any, depends on the route, timing, amount, and nature of the toxin. Consider GI decontamination if (1) the exposure is expected to cause significant toxicity, (2) GI decontamination is expected to positively affect clinical outcome, and (3) the risk/benefit ratio favors GI decontamination. GI decontamination should only be instituted when an airway is patent/protected. Table 100-5 lists the indications, contraindications, and complications of several GI decontamination modalities.

a. Inducing emesis with syrup of ipecac is no longer recommended for routine use in the ED. Orogastric lavage (OG) is now rarely used in the management of poisoned patients due to potentially serious consequences and unproven benefit. Only consider OG lavage when the poison was ingested within an hour of presentation and the nature/dose of the poison is likely to cause clinical deterioration despite supportive care or antidotal therapy. This procedure requires advance-

100_Cydulka_Ch100.indd 559 20/03/17 2:47 PM

Source page 579

Table 100-5 Indications, Contraindications, and Complications of Gastrointestinal Decontamination Procedures

Orogastric Lavage

Indications Rarely indicated

Consider for recent (<1 h) ingestion of life-threatening amount of a toxin for which there is no effective treatment once absorbed

Contraindications Corrosive/hydrocarbon ingestion

Supportive care/antidote likely to lead to recovery

Unprotected airway

Unstable, requiring further resuscitation (hypotension, seizures)

Complications Aspiration pneumonia/hypoxia

Water intoxication

Hypothermia

Laryngospasm

Mechanical injury to gastrointestinal tract

Time consuming, resulting in delay instituting other definitive care

Activated Charcoal Adults 50 g orally, children 1 g/kg orally

Indications Ingestion within the previous hour of a toxic substance known to be adsorbed by activated charcoal, where the benefits of administration are judged to outweigh the risks

Contraindications Nontoxic ingestion

Toxin not adsorbed by activated charcoal

Recovery will occur without administration of activate charcoal

Unprotected airway

Corrosive ingestion

Possibility of upper gastrointestinal perforation

Complications Vomiting

Aspiration of the activated charcoal

Impaired absorption of orally administered antidotes

Whole-Bowel Polyethylene glycol 2 L/h in adults, children

Irrigation 25 mL/kg per hour (maximum 2 L/h)

Indications (potential) Iron ingestion >60 mg/kg with opacities on abdominal radiograph

Life-threatening ingestion of diltiazem or verapamil

Body packers or stuffers

Slow-release potassium ingestion

Lead ingestion (including paint flakes containing lead)

Symptomatic arsenic trioxide ingestion

Life-threatening ingestions of lithium

Contraindications Unprotected airway

Gastrointestinal perforation, obstruction or ileus, hemorrhage

Intractable vomiting

Cardiovascular instability

Complications Nausea, vomiting

Pulmonary aspiration

Time consuming; possible delay instituting other

definitive care

100_Cydulka_Ch100.indd 560 20/03/17 2:47 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 576–583.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 101: Anticholinergic Toxicity

Source pages 584–585 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 584–585

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Anticholinergic Toxicity**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 101: Anticholinergic Toxicity 565 encephalitis, Reye syndrome, head trauma, other intoxications, neuroleptic malignant syndrome, delirium tremens, acute psychiatric disorders, and sympathomimetic toxicity. [Source pages 584–585]

- ■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION Treatment is primarily supportive. [Source pages 584–585]

- The goal is to prevent life-threatening complications, which include status epilepticus, hyperthermia, cardiovas- cular collapse, and rhabdomyolysis. [Source pages 584–585]

- Place the patient on a cardiac monitor and secure intravenous or intraosseous access. [Source pages 584–585]

- Activated charcoal may decrease drug absorption, even beyond 1 hour of ingestion. [Source pages 584–585]

Clinical reasoning structure

- **Problem representation:** Anticholinergic Toxicity; source pages 584–585.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Anticholinergic Toxicity**.

- Source-page range: **584–585**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 584

CHAPTER 101: Anticholinergic Toxicity 565

encephalitis, Reye syndrome, head trauma, other intoxications, neuroleptic malignant syndrome, delirium tremens, acute psychiatric disorders, and sympathomimetic toxicity.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

Treatment is primarily supportive. The goal is to prevent life-threatening complications, which include status epilepticus, hyperthermia, cardiovascular collapse, and rhabdomyolysis.

1. Place the patient on a cardiac monitor and secure intravenous or intraosseous access.
2. Activated charcoal may decrease drug absorption, even beyond 1 hour of ingestion.
3. Monitor temperature. Treat hyperthermia using conventional methods.
4. Hypertension usually does not require intervention, but should be treated conventionally as necessary.
5. Standard antidysrhythmics are usually effective, but avoid class IA medications (e.g., procainamide). Treat dysrhythmias, widened QRS complexes, and hypotension from sodium blocking agents (e.g., cyclic antidepressants) with IV sodium bicarbonate 1 mEq/kg.
6. Treat agitation with benzodiazepines, such as lorazepam 2 to 4 mg IV or 0.1 mg/kg). Avoid use of phenothiazines.
7. Treat seizures with benzodiazepines, such as lorazepam 2 mg IV or 0.1 mg/kg.
8. Physostigmine treatment is controversial. It is indicated if conventional therapy fails to control seizures, severe agitation, unstable dysrhythmias, coma with respiratory depression, malignant hypertension, or hypotension. The initial dose is 0.5 to 2 mg IV (0.02 mg/kg in children, maximum dose 2 mg), slowly administered over 5 minutes. When effective, a significant decrease in agitation may be apparent within 15 to 20 minutes. Physostigmine may worsen cyclic antidepressant toxicity and lead to bradycardia and asystole. It is contraindicated in patients

with cardiovascular or peripheral vascular disease, bronchospasm, intestinal or bladder obstruction, cardiac conduction disturbances, and suspected concomitant sodium channel antagonist poisoning. Observe the patient for cholinergic excess.

9. If diagnosis of anticholinergic poisoning is unclear, then do not administer a diagnostic challenge of physostigmine in order to avoid the small but increased risk of adverse effects in patients without anticholinergic toxicity.

10. Patients with mild anticholinergic toxicity can be discharged after 6 hours of observation if their symptoms have resolved. Admit more symptomatic patients for 24 hours of observation. Patients who receive physostigmine usually require a 24-hour admission.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 202, "Anticholinergics," by Dan Quan and Frank Lovecchio.

101_Cydulka_Ch101.indd 565 20/03/17 2:58 PM

Source page 585

C h apter Psychopharmacologic Agents

102 Shan Yin

■■ CYCLIC ANTIDEPRESSANTS

Cyclic antidepressants inhibit reuptake of norepinephrine and serotonin, block sodium channels, and antagonize postsynaptic serotonin receptors. They can produce severe toxicity in overdose. Their use has declined as newer, safer agents have been developed.

Clinical Features

Toxicity may present with altered mental status, seizures, cardiac conduction or rhythm disturbances, hypotension, respiratory depression, and, in severe cases, coma. If serious toxicity is going to develop, it generally develops within 6 hours of the ingestion.

Diagnosis

The diagnosis is clinical. Characteristic ECG changes include sinus tachycardia; right axis deviation of the terminal 40 ms; PR, QRS, and QT interval prolongation. Right bundle-branch block, A-V blocks, and the Brugada pattern are less common. Amoxapine can notably cause toxicity without characteristic ECG changes.

Emergency Department Care and Disposition

Care is primarily supportive.

1. Obtain IV access, and initiate cardiac rhythm and ECG monitoring.

2. Consider using 1 g/kg of activated charcoal PO if no contraindications exist.

3. Hypotension is treated with isotonic crystalloids. Norepinephrine or epinephrine is indicated if hypotension persists.

4. Treat conduction disturbances and ventricular dysrhythmias with sodium bicarbonate as an IV bolus of 1 to 2 mEq/kg, repeated until the patient improves or until blood pH is 7.50 to 7.55. A continuous IV infusion (150 mEq added to 1 L of 5% dextrose in water) may be used at a rate of 2 to 3 mL/kg/h.

5. Treat torsades de pointes with 2 g of IV magnesium sulfate.

6. Control agitation with benzodiazepines. Avoid physostigmine.

7. Treat seizures with benzodiazepines. Phenobarbital, starting at 10 to 15 mg/kg IV, may be required for refractory seizures.

8. Patients who remain asymptomatic after 6 hours do not need admission for toxicologic reasons. Admit symptomatic patients to a monitored bed or intensive care unit (ICU).

566

102_Cydulka_Ch102.indd 566 20/03/17 3:02 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 584–585.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 102: Psychopharmacologic Agents

Source pages 586–593 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 586–593

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Psychopharmacologic Agents**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 102: Psychopharmacologic Agents 567 ■■ ATYPICAL ANTIDEPRESSANTS, SEROTONIN REUPTAKE INHIBITORS, AND SEROTONIN SYNDROME Newer antidepressants include trazodone, bupropion, mirtazapine, selective serotonin reuptake inhibitors, and serotonin/norepinephrine reuptake inhib- itors. [Source pages 586–593]

- They are safer than older agents but can still cause toxicity, including the serotonin syndrome. [Source pages 586–593]

- ■■ TRAZODONE Clinical Features Symptoms of toxicity include central nervous system depression, ataxia, dizziness, seizures, orthostatic hypotension, vomiting, and abdominal pain. [Source pages 586–593]

- ECG abnormalities include QT interval prolongation, sinus bradycardia and tachycardia, and torsades de pointes. [Source pages 586–593]

- Emergency Department Care and Disposition Supportive care is generally sufficient in isolated overdoses. [Source pages 586–593]

Clinical reasoning structure

- **Problem representation:** Psychopharmacologic Agents; source pages 586–593.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Psychopharmacologic Agents**.

- Source-page range: **586–593**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 586

CHAPTER 102: Psychopharmacologic Agents 567

■■ ATYPICAL ANTIDEPRESSANTS, SEROTONIN REUPTAKE INHIBITORS, AND SEROTONIN SYNDROME

Newer antidepressants include trazodone, bupropion, mirtazapine, selective serotonin reuptake inhibitors, and serotonin/norepinephrine reuptake inhibitors. They are safer than older agents but can still cause toxicity, including the serotonin syndrome.

■■ TRAZODONE

Clinical Features

Symptoms of toxicity include central nervous system depression, ataxia, dizziness, seizures, orthostatic hypotension, vomiting, and abdominal pain. ECG abnormalities include QT interval prolongation, sinus bradycardia and tachycardia, and torsades de pointes. Priapism can be seen at therapeutic doses.

Emergency Department Care and Disposition

Supportive care is generally sufficient in isolated overdoses.

1. Initiate cardiac rhythm monitoring and obtain a 12-lead ECG.
2. Consider single-dose activated charcoal if no contraindications exist.
3. Treat hypotension with isotonic IV fluids, followed by norepinephrine.
4. Treat torsades de pointes with IV magnesium sulfate or overdrive pacing.
5. Discharge patients who remain asymptomatic for at least 6 hours, with psychiatric evaluation as indicated. Admit those with neurologic and/or cardiac symptoms to a monitored bed.

■■ BUPROPION

Clinical Features

Toxicity manifests as agitation, dizziness, tremor, vomiting, drowsiness, and tachycardia. Seizures are more common than with other atypical antidepressants and can be the initial presenting sign. ECG changes include

sinus tachycardia, QRS interval widening, and QT interval prolongation.

Emergency Department Care and Disposition

Seizures should be anticipated.

1. Start a peripheral IV line and initiate cardiac rhythm monitoring.

2. Consider activated charcoal if no contraindications exist.

3. Treat seizures with benzodiazepines, followed by phenobarbital.

4. Administer IV sodium bicarbonate for QRS prolongation.

5. Consider IV lipid emulsion for refractory cardiovascular instability.

6. Observe asymptomatic patients for 8 hours. Monitor patients ingesting >450 mg of sustained-release bupropion for up to 24 hours. Admit those with seizures, persistent tachycardia, or lethargy.

102_Cydulka_Ch102.indd 567 20/03/17 3:02 PM

Source page 587

■■ MIRTAZAPINE

Clinical Features

Toxicity causes sedation, confusion, sinus tachycardia, and hypertension. Coma and respiratory depression are seen in severe cases or with co-ingestion of other sedatives.

Emergency Department Care and Disposition

1. Isolated overdoses can generally be managed with supportive care.

2. Consider single-dose activated charcoal if no contraindications exist.

3. Admit symptomatic patients to a monitored bed. Discharge asymptomatic patients after 6 hours.

■■ SELECTIVE-SEROTONIN REUPTAKE INHIBITORS

Clinical Features

Selective-serotonin reuptake inhibitors (SSRIs) include fluoxetine, fluvoxamine, paroxetine, sertraline, citalopram, and escitalopram. Signs and symptoms may include vomiting, sedation, tremor, sinus tachycardia, mydriasis, seizures, diarrhea, and hallucinations. Sinus bradycardia is more common with fluvoxamine than with other SSRIs. QRS and QT interval prolongation has been reported in citalopram and escitalopram ingestions.

Emergency Department Care and Disposition

Supportive care is generally sufficient.

1. Establish IV access and initiate cardiac monitoring.

2. Single-dose activated charcoal is appropriate for most ingestions if there are no contraindications.

3. Benzodiazepines are recommended for management of seizures.

4. Administer IV sodium bicarbonate for QRS prolongation.

5. Observe patients for at least 6 hours. Admit patients who are tachycardic,

have altered mental status, have conduction abnormalities, or have signs of serotonin syndrome.

■■ SEROTONIN-NOREPINEPHRINE REUPTAKE INHIBITORS

Clinical Features

Serotonin-norepinephrine reuptake inhibitors (SNRIs) include venlafaxine, duloxetine, levomilnacipran, and desvenlafaxine. Overdose may cause hypotension, diaphoresis, tremor, mydriasis, sedation, coma, and seizures. ECG changes include sinus tachycardia and QRS or QT interval widening. Rhabdomyolysis has been reported with venlafaxine overdoses with and without seizures.

Emergency Department Care and Disposition

There are no established guidelines for treating SNRI overdoses.

1. Initiate peripheral IV access and cardiac monitoring.
2. Consider single-dose activated charcoal if no contraindications exist. Consider whole-bowel irrigation for venlafaxine overdoses >4000 mg.

102_Cydulka_Ch102.indd 568 20/03/17 3:02 PM

Source page 588

3. Benzodiazepines are the anticonvulsants of choice.
4. Treat hypotension with fluids and a direct-acting vasopressor.
5. Administer IV sodium bicarbonate for QRS prolongation.
6. Administer IV fluid for rhabdomyolysis.
7. All patients require at least 6 hours of observation, or 24 hours for those ingesting extended-release preparations. Admit symptomatic patients to a monitored bed.

■■ SEROTONIN SYNDROME

Serotonin syndrome is a potentially life-threatening adverse reaction to serotoninergic medications. It can be produced by any drug or combination of drugs that increase central serotonin neurotransmission, most commonly antidepressants.

Clinical Features

Signs and symptoms are altered mental status, hyperthermia, seizures, and increased muscle tone, particularly myoclonus. Hyperthermia is the most common cause of death.

Diagnosis

Symptoms are nonspecific, and there are no confirmatory laboratory tests. Diagnostic criteria emphasize exposure to a serotoninergic drug and presence of myoclonus.

Emergency Department Care and Disposition

Treatment is supportive. Watch patients for rhabdomyolysis and metabolic

acidosis.

1. Endotracheal intubation and ventilatory support may be required in severe cases.
2. Use benzodiazepines to decrease discomfort and promote muscle relaxation.
3. The antiserotonergic agent cyproheptadine may be given at an initial dose of 4 to 12 mg PO, repeated with 2 mg doses at 2-hour intervals if no response.
4. Admit all patients until symptoms resolve. Severely ill patients require admission to an ICU. Discontinue serotonergic drugs.

■■ MONOAMINE OXIDASE INHIBITORS

Monoamine oxidase inhibitors (MAOIs) are used to treat refractory depression. They cause accumulation of neurotransmitters such as norepinephrine in presynaptic nerve terminals and increased systemic availability of dietary amines, such as tyramine. MAOIs can lead to fatal food and drug interactions and cause severe toxicity in overdose.

Clinical Features

Symptoms include headache, agitation, palpitations, chest pain, and tremor. Signs include sinus tachycardia, hyperreflexia, diaphoresis, fasciculations, mydriasis, hyperventilation, nystagmus, flushing, muscle rigidity, and

102_Cydulka_Ch102.indd 569 20/03/17 3:02 PM

Source page 589

hypertension. Signs and symptoms often are delayed after ingestion and the delay can be up to 24 hours. Coma, seizures, bradycardia, hypotension, hypoxia, and hyperthermia may develop. Death usually results from multiorgan failure.

Diagnosis and Differential

Diagnosis is made on clinical grounds. Laboratory tests can identify complications, including rhabdomyolysis, renal failure, hyperkalemia, metabolic acidosis, and disseminated intravascular coagulation. The differential diagnosis includes drugs and conditions that produce a hyperadrenergic state, altered mental status, and/or muscle rigidity.

Emergency Department Care and Disposition

Treatment consists of supportive care and management of complications.

1. Obtain IV access and initiate cardiac rhythm monitoring.
2. Consider activated charcoal if no contraindications exist.
3. Hypertension may be treated with phentolamine, 2.5 to 5 mg IV every 10 to 15 minutes, followed by an infusion. An alternative agent is nitroprusside. Avoid beta-blockers.

4. Nitroglycerin is indicated for anginal chest pain and signs of myocardial ischemia.
5. Treat hypotension with isotonic IV fluid boluses, followed by norepinephrine.
6. Treat ventricular dysrhythmias with lidocaine or procainamide.
7. Treat bradycardia with atropine, isoproterenol, dobutamine, and pacing.
8. Treat seizures with benzodiazepines. General anesthesia and muscle paralysis using vecuronium, with ongoing EEG monitoring, may be necessary.
9. Treat hyperthermia with benzodiazepines to reduce muscle rigidity plus cooling measures. Chemical paralysis with a nondepolarizing agent may be needed for severe rigidity. Dantrolene 0.5 to 2.5 mg/kg IV every 6 hours could be considered if other measures have failed.
10. Patients who have ingested >1 mg/kg require ICU admission. Those who have ingested less can be admitted to a monitored bed. Observe asymptomatic patients for 24 hours.

■■ ANTIPSYCHOTICS

Antipsychotics are used to treat psychosis, agitation, nausea, headaches, hiccups, and involuntary motor disorders. Their therapeutic action involves blockade of dopamine receptors.

Clinical Features

Central nervous system effects include lethargy, ataxia, dysarthria, confusion, and coma. Seizures are more common with loxapine and clozapine. Antimuscarinic toxicity may be seen. Cardiovascular manifestations include orthostatic hypotension; sinus tachycardia; PR, QRS, and QT interval prolongation; and ST and T wave abnormalities. Extrapyramidal symptoms such as dystonia, akathisia, and tardive dyskinesia can be seen during therapeutic use.

102_Cydulka_Ch102.indd 570 20/03/17 3:02 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 586–593.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 103: Sedatives and Hypnotics

Source pages 594–599 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 594–599

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Sedatives and Hypnotics**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 103: Sedatives and Hypnotics 575 blood urea nitrogen and creatinine levels, complete blood count, and cre- atinine phosphate kinase. [Source pages 594–599]

- Emergency Department Care and Disposition Treatment begins with airway management and supportive care. [Source pages 594–599]

- Once pul- monary and cardiovascular function have been adequately assessed and stabilized, enhancing elimination can be considered. [Source pages 594–599]

- Endotracheal intubation in the severely poisoned patients is commonly required and should be initiated early in the ED course. [Source pages 594–599]

- Due to the potential for myocardial depression, place two large-bore IVs and initiate fluid resuscitation with isotonic saline for hypotension. [Source pages 594–599]

Clinical reasoning structure

- **Problem representation:** Sedatives and Hypnotics; source pages 594–599.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Sedatives and Hypnotics**.

- Source-page range: **594–599**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 594

CHAPTER 103: Sedatives and Hypnotics 575

blood urea nitrogen and creatinine levels, complete blood count, and creatinine phosphate kinase.

Emergency Department Care and Disposition

Treatment begins with airway management and supportive care. Once pulmonary and cardiovascular function have been adequately assessed and stabilized, enhancing elimination can be considered.

1. Stabilize the airway. Endotracheal intubation in the severely poisoned patients is commonly required and should be initiated early in the ED course.
2. Due to the potential for myocardial depression, place two large-bore IVs and initiate fluid resuscitation with isotonic saline for hypotension.
3. Consider empiric treatment with naloxone and thiamine early in the management.
4. Dopamine or norepinephrine may be needed if fluid boluses fail to reverse hypotension. Hypothermia between 30°C (86°F) and 36°C (96.8°F) requires standard rewarming techniques.
5. Activated charcoal helps reduce absorption. In the awake, cooperative patient, 50 to 100 g orally (1 g/kg in children) should be administered. For sedated or unconscious patients, airway protection should precede the administration of activated charcoal. Multidose charcoal may reduce serum levels, but has not been shown to change clinical outcome.
6. Forced diuresis is not indicated due to risks of sodium and fluid overload and a lack of proven efficacy.
7. Urinary alkalinization is not considered first-line therapy. While it may enhance the clearance of phenobarbital and primidone, it is less effective than multidose charcoal alone, and has no role in the management of short-acting barbiturates.
8. Hemodialysis, hemoperfusion, and hemodiafiltration are reserved for

patients who deteriorate despite aggressive medical support, but are only effective with phenobarbital toxicity.

9. Disposition depends on the degree of intoxication: evidence of toxicity greater than 6 hours from time of arrival requires hospital admission. Obtain psychiatric consult for intentional overdose. Toxicology or poison center consultation is recommended to assist with management.

■■ BENZODIAZEPINES

Clinical Features

Benzodiazepine overdoses are common but carry a low mortality rate in isolation. There is, however, variation in the clinical outcome among agents due to differences in potency. Parenteral administration in the ED may produce significant complications. Benzodiazepines, when mixed with other sedative-hypnotics, may produce profound toxicity including respiratory depression, hypotension, and death.

The primary effect of benzodiazepines is central nervous system depression and is characterized by somnolence, dizziness, slurred speech, confusion, ataxia, incoordination, and general impairment in intellectual function. Neurologic effects in the elderly, very young, or malnourished may be prolonged or enhanced. Disinhibition, extrapyramidal reactions, and

103_Cydulka_Ch103.indd 575 20/03/17 3:10 PM

Source page 595

paradoxical excitation are uncommon but reported. Short-term, anterograde amnesia is a common, sometimes desirable, effect with the administration of certain benzodiazepines.

Chronic use of benzodiazepines is associated with physiologic addiction. Withdrawal from benzodiazepines may occur following abrupt cessation. Symptoms are more intense following withdrawal from short-acting agents. Clinical findings may mimic alcohol withdrawal and include anxiety, irritability, insomnia, nausea, vomiting, tremor, and sweating. Serious manifestations include hallucinations, psychosis, disorientation, and seizures. Treatment begins with reintroduction of a benzodiazepine with subsequent, gradual tapering.

Diagnosis and Differential

There is limited value in toxicological testing as serum levels do not correlate well with clinical findings. Qualitative urine screening is unreliable and a positive test does not prove causation of clinical signs as clinical features are nonspecific and may be seen with overdose of any sedative-hypnotic.

Emergency Department Care and Disposition

1. Priorities include assessment and stabilization of the airway, breathing,

and circulation (see the preceding section on barbiturates for guidance regarding initial management, resuscitation, and laboratory monitoring).

2. Do not induce emesis. Activated charcoal (1 g/kg in children or 50 to 100 g in adults) will adsorb benzodiazepines. Exercise caution in the sedated patient, and secure the airway before administration. Multidose charcoal is not indicated. Gastric lavage, forced dieresis, enhanced elimination, hemodialysis, and hemoperfusion are ineffective and unnecessary.

3. Flumazenil is a unique, selective antagonist of the central effects of benzodiazepines. Unlike naloxone, flumazenil should not be used empirically for the undifferentiated sedative-hypnotic toxidrome as it may precipitate seizures (Table 103-1). The ED applications of flumazenil are generally limited to the setting of respiratory depression following procedural sedation with benzodiazepines. The dose is 0.2 mg IV and titrated every minute to a maximum total dose of 3 mg (0.01 to 0.02 mg/kg in children). The half-life of flumazenil is approximately 1 hour and rebound sedation can occur in the setting of overdose with long-acting benzodiazepines.

4. In general, care is supportive. Admit patients with significant alterations of mental status, respiratory depression, or hypotension. Consultation with mental health specialists may be appropriate.

Table 103-1 Contraindications to Flumazenil

Overdose of unknown agents
Suspected or known physical dependence on benzodiazepines
Suspected cyclic antidepressant overdose
Coingestion of seizure-inducing agents
Known seizure disorder
Suspected increased intracranial pressure

103_Cydulka_Ch103.indd 576 20/03/17 3:10 PM

Source page 596

■■ NONBENZODIAZEPINE SEDATIVES

Clinical Features

The hallmark of all sedative-hypnotic medications, regardless of subclass, is sedation. Exposure to nonbenzodiazepine agents is common and coingestion with other sedatives may be synergistic and produce profound sedation. Three agents, ethchlorvynol, glutethimide, and methaqualone, have been removed from the markets in the United States and Canada.

Buspirone

Buspirone has a complex mechanism of action and rapid absorption. Side

effects include sedation, GI distress, vomiting, and dizziness. The symptoms with overdose are exaggerations of the side effects noted with therapeutic dosing. The drug is generally well tolerated in overdose, and treatment is primarily supportive. Due to effects on the serotoninergic system, serotonin syndrome has been reported. Seizures, hypotension, priapism, and dystonia are rare complications.

Carisoprodol and Meprobamate

Carisoprodol and its active metabolite, meprobamate, are marketed as central-acting muscle relaxants and anxiolytics, respectively. In overdoses, both may cause sedation, coma, and cardiopulmonary depression. Carisoprodol, but not meprobamate, can cause myoclonus, which may be a clue in an unknown overdose. Meprobamate has been associated with pharmacobezoars, and may be a cause of prolonged toxicity.

Chloral Hydrate

Chloral hydrate is the oldest sedative-hypnotic available today. At therapeutic doses, chloral hydrate produces mental status depression without loss of airway and respiratory reflexes. Vomiting and paradoxical excitation can occur in a small percentage of children. In overdose, coma and respiratory depression can occur. Chloral hydrate is a myocardial sensitizer and cardiac dysrhythmias, decreased cardiac contractility, and asystole have been reported. When combined with alcohol, a potent, “knock-down” cocktail known as a “Mickey Finn” is created. A withdrawal syndrome similar to ethanol has been described.

Chloral hydrate may produce a characteristic pear-like odor. Abdominal radiographs may aid in the diagnosis of pharmacobezoars, as chloral hydrate is radiopaque. Respiratory depression and coma are treated supportively. Treat ventricular dysrhythmias with IV β-blockers.

γ -Hydroxybutyrate

γ-Hydroxybutyrate (GHB) has been marketed as a muscle builder, fat burner, antidepressant, anxiolytic, hypnotic, and cholesterol-lowering medication. The only approved use in the United States is for the treatment of narcolepsy and is prescribed as its sodium salt (sodium oxybate). GHB is often used illicitly and has street names including “liquid ecstasy,” “Georgia home boy,” “G,” and “grievous bodily harm.” GHB has a narrow therapeutic window and may produce a range of toxicity from mild sedation to coma. Seizures, bradycardia, hypothermia, and cardiac depression

103_Cydulka_Ch103.indd 577 20/03/17 3:10 PM

Source page 597

may occur. Rapid sedation with abrupt recovery 6 to 12 hours later is a

common feature with GHB. Due to its amnestic effects and rapid sedation, GHB has been illicitly used for drug-facilitated sexual assault. Two compounds, 1,4 butanediol and γ-butyrolactone are GHB precursors and have been abused to produce similar effects. Treatment is primarily supportive with focus on airway management. Physostigmine and neostigmine should be avoided. Rapid awaking and self-extubation may be a clue to GHB intoxication. Toxicologic detection of GHB is difficulty owing to its very short half-life and rapid elimination. Withdrawal from GHB mimics alcohol withdrawal and may be severe, lasting from 3 days to 2 weeks.

Melatonin

The endogenous hormone melatonin is secreted by the pineal gland and is believed to help regulate the sleep–wake cycle. Melatonin can be purchased without prescription. Side effects following therapeutic dosing include headache, dizziness, fatigue, and irritability. Overdose data are limited, and signs and symptoms exaggerate the side effects from therapeutic doses.

Ramelteon

Ramelteon is a relatively new medication used to treat insomnia. It binds to melatonin receptors in the brain. Absorption following oral dosing is rapid. In overdose, sedation is common and treatment is supportive. Abuse and withdrawal have not been reported.

Tasimelteon

Tasimelteon is used to treat non-24-hour sleep–wake disorder. It peaks quickly following ingestion. There is no data on overdoses currently. Abuse and withdrawal have not been reported.

Zolpidem, Zaleplon, and Zopiclone

Zolpidem, zaleplon, and zopiclone have gained increased popularity for the treatment of insomnia. Though initially thought to produce little or no psychomotor impairment, addiction or withdrawal, experience has proved otherwise. Zaleplon appears to be less likely to cause withdrawal than zolpidem or zopiclone. Side effects in therapeutic doses include nausea and somnolence. Sedation would be the expected primary effect after overdose. Vivid dreams, sleep-walking, and driving have been reported with zolpidem. Fatalities following zolpidem overdoses have been reported, but are usually associated with mixed ingestions. Methemoglobinemia and hemolytic anemia have been reported after large zoplicone overdoses.

Emergency Department Care and Disposition

1. In general, management for the nonbenzodiazepines is supportive (see the preceding section on benzodiazepines for treatment priorities).

2. For ventricular dysrhythmias in the setting of chloral hydrate intoxication, IV β-blockers (e.g., propranolol 1 mg IV or 0.01 to 0.1 mg/kg IV

in children) are first-line agents.

3. Disposition is largely guided by the degree of symptoms. Have a low threshold for admission of any patient with altered mental status, abnormal vital signs, or dysrhythmias. Consult psychiatric services when appropriate.

103_Cydulka_Ch103.indd 578 20/03/17 3:10 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 594–599.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 104: Alcohols

Source pages 600–603 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 600–603

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Alcohols**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 104: Alcohols 581 Clinical Features Clinically, isopropanol intoxication is similar to that of ethanol but pro- duces a greater degree of intoxication than ethanol. [Source pages 600–603]

- Severe poisoning pres- ents as coma, respiratory depression, and hypotension. [Source pages 600–603]

- Hemorrhagic gastritis is common and causes nausea, vomiting, abdominal pain, and upper gastrointestinal (GI) bleeding. [Source pages 600–603]

- Diagnosis and Differential A glucose level should be measured in all patients with altered mental sta- tus. [Source pages 600–603]

- Laboratory studies may reveal an elevated osmolal gap, ketonemia, and ketonuria, without acidosis. [Source pages 600–603]

Clinical reasoning structure

- **Problem representation:** Alcohols; source pages 600–603.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Alcohols**.

- Source-page range: **600–603**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 600

CHAPTER 104: Alcohols 581

Clinical Features

Clinically, isopropanol intoxication is similar to that of ethanol but produces a greater degree of intoxication than ethanol. Severe poisoning presents as coma, respiratory depression, and hypotension. Hemorrhagic gastritis is common and causes nausea, vomiting, abdominal pain, and upper gastrointestinal (GI) bleeding.

Diagnosis and Differential

A glucose level should be measured in all patients with altered mental status. Laboratory studies may reveal an elevated osmolal gap, ketonemia, and ketonuria, without acidosis. In the setting of upper GI bleeding, coagulation studies, a complete blood count, and a type and screen should be obtained. When available, serum isopropanol and acetone levels confirm the diagnosis, but are not required for management.

Emergency Department Care and Disposition

1. Treat hypotension with aggressive infusion of IV crystalloids, although persistent hypotension may require vasopressors. Treat significant bleeding from hemorrhagic gastritis with transfusion of packed red blood cells and plasma as indicated.

2. Do not administer metabolic blockade with fomepizole or ethanol since acetone, the metabolite of isopropanol, is no more toxic than the parent compound.

3. Hemodialysis removes both isopropanol and acetone, and may be indicated for refractory hypotension due to isopropanol.

4. Patients with prolonged CNS depression require admission. Those who are asymptomatic after 6 to 8 hours of observation can be discharged or referred for psychiatric evaluation if indicated.

5. Charcoal does not bind alcohols, and is useful only if there is coingestion of an absorbable substance.

■■ METHANOL AND ETHYLENE GLYCOL

Methanol is commonly found in windshield washing fluid, solid fuel for

stoves, and solvents. Ethylene glycol is commonly found in antifreeze. Toxicity from these alcohols results from the formation of toxic metabolites, which produce a significant anion gap metabolic acidosis. Methanol leads to the formation of toxic formic acid, while ethylene glycol is metabolized into the toxic compounds glycolic acid, glyoxylic acid, and oxylic acid with subsequent formation of calcium oxylate crystals.

Clinical Features

The clinical features of methanol toxicity include metabolic acidosis, CNS depression, and visual changes (classically, a complaint of looking at a snowstorm). The onset of symptoms may be delayed with concurrent ethanol consumption. Funduscopic examination may reveal retinal edema or hyperemia of the optic disk.

Ethylene glycol poisoning often exhibits three distinct clinical phases after ingestion. First, within 12 hours, CNS effects predominate: the patient appears intoxicated without the odor of ethanol on the breath. Second,

104_Cydulka_Ch104.indd 581 20/03/17 3:24 PM

Source page 601

12 to 24 hours after ingestion, cardiopulmonary effects predominate: elevated heart rate, respiratory rate, and blood pressure are common. Congestive heart failure, respiratory distress syndrome, and circulatory collapse may develop. Third, 24 to 72 hours after ingestion, renal effects predominate which are characterized by flank pain, costovertebral angle tenderness, and acute tubular necrosis with acute renal failure. Hypocalcemia may result from precipitation of calcium oxalate into tissues leading to tetany and typical ECG changes.

Diagnosis and Differential

The diagnosis is based on clinical presentation and laboratory findings of an elevated osmol gap, and/or an anion gap metabolic acidosis (which may take hours to develop). Elevated levels of methanol or ethylene glycol can confirm toxicity. An elevated osmolal gap is present early, and may be useful when methanol or ethylene glycol testing is not immediately available. Basic laboratory investigations include a bedside glucose, electrolytes, renal function arterial blood gas, urinalysis, and methanol or ethylene glycol level. Ethylene glycol poisoning differs from methanol poisoning in that visual disturbances and fundoscopic abnormalities are absent and calcium oxalate crystals may be present in the urine.

The differential diagnosis includes other causes of an anion gap metabolic acidosis such as salicylate or isoniazid toxicity, diabetic ketoacidosis, alcoholic ketoacidosis, uremia, and lactic acidosis.

Emergency Department Care and Disposition

Treatment is based on metabolic blockade and removing toxic metabolites from the body. Both fomepizole and ethanol have a greater affinity for alcohol dehydrogenase than methanol and ethylene glycol. Indications for metabolic blockade are listed in Table 104-1.

1. Administer fomepizole 15 mg/kg IV load followed by 10 mg/kg every 12 hours for four additional doses. Subsequent doses should be at 15 mg/kg. Fomepizole is a potent inhibitor of alcohol dehydrogenase with greater affinity and fewer side effects than ethanol. If fomepizole is not available, use ethanol 800 mg/kg IV load, followed by a continuous

Table 104-1 Indications for Metabolic Blockade with Fomepizole or Ethanol

1. ■Elevated plasma levels: methanol >20 mg/dL (>6 mmol/L) or ethylene glycol > 20 mg/dL (>3 mmol/L)

2. If methanol or ethylene glycol level not available:

A. ■Documented or suspected significant methanol or ethylene glycol ingestion with ethanol level lower than approximately 100 mg/dL (22 mmol/L)

B. Coma or altered mental status in patient with unclear history and:

Unexplained serum osmolar gap of >10 mOsm/L

or

Unexplained metabolic acidosis and ethanol level of <100 mg/dL (<22 mmol/L)*

*

If serum ethanol level is >100 mg/dL (>22 mmol/L), patient will be protected from the formation of toxic metabolites by coingestion of ethanol, and specific metabolic blockade treatment can be delayed until toxic alcohol level is available. If the ethanol level is likely to fall to <100 mg/dL before the toxic alcohol results are back, then initiate metabolic blockade.

104_Cydulka_Ch104.indd 582 20/03/17 3:24 PM

Source page 602

infusion of 100 mg/kg/h in the average drinker and 150 mg/kg/h in the heavy drinker. Adjust the infusion accordingly to maintain a blood ethanol level at 100 to 150 mg/dL. If resources are limited, oral therapy with commercial 80 proof liquor can be initiated. A load of 3 to 4 oz with maintenance of 1 to 2 oz/h is a typical dose for a 70-kg patient.

2. Monitor serum glucose during treatment with ethanol as hypoglycemia may be induced, especially in children. Treat hypoglycemia with 1 mL/kg 50% dextrose in water in adults and 5 mL/kg 10% dextrose in water in children.

3. Dialysis eliminates both methanol and ethylene glycol and their toxic metabolites. Indications include metabolic acidosis or evidence of end organ impairment (e.g., blindness, renal failure). Fomepizole or ethanol

treatments do not alter the indications for dialysis; however, both fomepizole and ethanol are dialyzed and, therefore, increase the dosing interval of fomepizole to every 4 hours. Double the infusion rate of ethanol during dialysis and adjust accordingly to maintain the level at 100 to150 mg/dL.

4. Continue dialysis, fomepizole, or ethanol treatment until the methanol or ethylene glycol level is <20 mg/dL and the metabolic acidosis has resolved.

5. In methanol poisoning, administer folinic acid (leucovorin) 50 mg IV. In ethylene glycol poisoning, administer pyridoxine 100 mg IV and thiamine 100 mg IV.

6. Administer sodium bicarbonate 1 to 2 mEq/kg and titrated to maintain a normal pH in methanol toxicity to increase renal excretion of formic acid. Sodium bicarbonate may also be given to an ethylene glycol toxic patient with metabolic acidosis.

7. Treat documented and symptomatic hypocalcemia in ethylene glycol toxicity with calcium gluconate or calcium chloride.

8. Consult a medical toxicologist or regional poison control center (800 222-1222) to aid in the management of symptomatic methanol or ethylene glycol ingestion.

9. Patients with suspected ethylene glycol or methanol ingestion who are asymptomatic after 6 hours with no ethanol detected and no osmolar gap or metabolic acidosis may be safely discharged. In the setting of co-ingestion with ethanol, it is important to ensure no metabolic acidosis develops 6 hours after ethanol has been metabolized. Patients with significant signs and symptoms should be admitted to an intensive care unit.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 185, "Alcohols," by Jennifer P. Cohen and Dan Quan.

104_Cydulka_Ch104.indd 583 20/03/17 3:24 PM

Source page 603

C h ap t e r Drugs of Abuse

105 D. Adam Algren

■■ OPIOIDS

The term opioid refers to any drug that is active at the opioid receptor, while opiates refers to naturally occurring derivatives of the opium plant, such as morphine. Narcotic is a legal term and generically refers to any drug that causes sedation. Emergency physicians commonly utilize opioids as analgesics and must be familiar with the clinical presentation and treatment of

opioid toxicity.

Clinical Features

Opioid overdose produces a clinical toxidrome: miosis, and central nervous system (CNS) and respiratory depression. However, normal or large pupils may be seen with diphenoxylate, meperidine, pentazocine, and propoxyphene toxicities. Severe hypoxia or co-ingestants could also produce mydriasis. Severe opioid overdose, especially with heroin, may be associated with acute lung injury. Seizures have been reported with tramadol, propoxyphene, and meperidine overdoses. Meperidine, tramadol, and dextromethorphan have been associated with serotonin syndrome when combined with other serotonergic agents. Methadone may prolong the QT interval and result in torsade de pointes. Opioid withdrawal is manifest by abdominal pain, nausea, vomiting, diarrhea, dysphoria, piloerection, and lacrimation.

Diagnosis and Differential

The diagnosis is clinical. Qualitative immunoassay urine drug screens are commonly available but must be interpreted with caution as there are numerous agents that produce false-positive and false-negative results for opioids. Toxicity from clonidine, guanfacine, and imidazolines mimics opioid intoxication and produces varying degrees of miosis and CNS/respiratory depression. Because response to naloxone with these agents is less reliable, consider clonidine intoxication in patients who appear opioid poisoned but do not respond to naloxone. Other possible causes of CNS depression and miosis include pontine hemorrhage and toxicity from organophosphates/carbamates, antipsychotics, and sedative-hypnotics (i.e., GHB).

Emergency Department Care and Disposition

1. Naloxone is the primary treatment for opioid toxicity. Administer a small dose of 0.1 to 0.4 mg IV, SC, or IM initially for CNS and respiratory depression to avoid precipitating withdrawal. Larger doses of 2 mg could be used in apneic patients. Repeat doses may be administered every 3 minutes. The pediatric dose is 0.1 mg/kg. Given the relatively short duration of action of naloxone, monitor closely for recurrent toxicity.

105_Cydulka_Ch105.indd 584 20/03/17 3:30 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 600–603.
- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 105: Drugs of Abuse

Source pages 604–611 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 604–611

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Drugs of Abuse**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- In large overdoses, consider an infusion of naloxone: two-thirds of the dose required to initially “wake up” the patient per hour. [Source pages 604–611]

- Consider endotracheal intubation in patients who respond poorly to naloxone and those with acute lung injury from overdose. [Source pages 604–611]

- Patients with heroin overdose who are awake and asymptomatic 2 hours after the last naloxone dose can be discharged. [Source pages 604–611]

- Patients with immediate-acting opioid exposures require observation for at least 6 hours, or 6 hours following naloxone administration if given. [Source pages 604–611]

- Patients with exposure to long-acting opioids (sustained release mor- phine/oxycodone, methadone, and buprenorphine) require admission for prolonged observation. [Source pages 604–611]

Clinical reasoning structure

- **Problem representation:** Drugs of Abuse; source pages 604–611.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Drugs of Abuse**.

- Source-page range: **604–611**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 604

CHAPTER 105: Drugs of Abuse 585

2. In large overdoses, consider an infusion of naloxone: two-thirds of the dose required to initially “wake up” the patient per hour.

3. Consider endotracheal intubation in patients who respond poorly to naloxone and those with acute lung injury from overdose.

4. Patients with heroin overdose who are awake and asymptomatic 2 hours after the last naloxone dose can be discharged. Patients with immediate-acting opioid exposures require observation for at least 6 hours, or 6 hours following naloxone administration if given. Patients with exposure to long-acting opioids (sustained release morphine/oxycodone, methadone, and buprenorphine) require admission for prolonged observation.

5. Treatment of opioid withdrawal is supportive (i.e., fluids, antiemetics). Clonidine 0.1 to 0.3 mg PO may be used to treat withdrawal.

■■ COCAINE AND AMPHETAMINES

Cocaine and amphetamines are widely used drugs of abuse and produce similar clinical manifestations.

Clinical Features

Cocaine and amphetamines induce euphoria and produce complications secondary to the enhanced release and blockade of catecholamines at the synaptic receptors. Onset of effect via intranasal, inhalational (crack use), and intravenous use is rapid. Repeated drug administration leads to prolonged effects and increased toxicity. Symptoms of sympathomimetic overdose include mydriasis, hypertension, tachycardia, diaphoresis, agitation, rhabdomyolysis, acute kidney injury, and hyperthermia. Cardiovascular complications include dysrhythmias, myocardial ischemia, cardiomyopathy, and aortic/coronary artery dissection. Chest pain is common and often atypical. Patients are at risk of developing an acute coronary syndrome (STEMI and NSTEMI) related to vasospasm and

increased thrombogenesis. Cocaine dysrhythmias result from sodium and potassium blockade. Following cocaine use patients may develop supraventricular tachycardia (SVT), atrial fibrillation/flutter along with wide complex tachycardia/dysrhythmias similar to tricyclic antidepressants. The Brugada pattern and prolonged QT may also be seen with cocaine. Although amphetamines may result in SVT, other dysrhythmias are less commonly seen.

Neurologic complications of cocaine and amphetamines include seizures, spinal cord infarctions, cerebral vasculitis, and ischemic/hemorrhagic cerebrovascular accidents. Choreoathetoid movements and dystonic reactions may occur. A “washout” syndrome following heavy use may result in CNS depression that generally resolves within 24 hours. Amphetamine users may develop severe psychosis.

Cocaine users can develop pulmonary complications including pneumomediastinum, pneumothorax, bronchospasm, and pulmonary edema and hemorrhage. Cocaine abuse during pregnancy increases risk for spontaneous abortion, abruptio placentae, fetal prematurity, and intrauterine growth retardation. “Body stuffers” (hasty ingestion of drugs to avoid police) and “body packers” (ingestion of large amounts of tightly packed pure drug for

105_Cydulka_Ch105.indd 585 20/03/17 3:30 PM

Source page 605

importation) may be asymptomatic or demonstrate signs of severe cocaine toxicity if a bag ruptures.

Diagnosis and Differential

Diagnosis of cocaine, amphetamine, or stimulant intoxication is usually clinical. Urine drug screening for cocaine is reliable and can detect exposure within 72 hours. Urine screens for amphetamines are less specific and have high false-negative and false-positive results.

Additional laboratory evaluation for intoxicated patients includes a complete metabolic panel to assess acid/base status, kidney function, and a creatine kinase (CK) to assess for rhabdomyolysis. The evaluation of altered mental status or seizure may include a head CT to exclude intracranial hemorrhage. Consider ECG, chest radiograph, and cardiac enzymes in cocaine- or amphetamine-intoxicated patients presenting with chest pain.

The differential diagnosis includes drug intoxications (anticholinergic, PCP, serotonin syndrome), sedative-hypnotic withdrawal, stroke, CNS infections, hypoglycemia, hyponatremia, and thyrotoxicosis. Concomitant use of substances such as alcohol or opioids may significantly alter the

presentation.

Emergency Department Care and Disposition

1. Benzodiazepines are the mainstay of treatment for cardiovascular and CNS effects. Administer lorazepam 2 mg IV, or 0.1 mg/kg in children; or diazepam 5 mg for agitation, hypertension, and tachycardia, and titrate to effect. Judicious use of antipsychotic medications can be considered for those with refractory agitation and violent behavior.
2. Treat seizures with benzodiazepines as noted above. In addition, phenobarbital 15 to 20 mg/kg, propofol, and neuromuscular blockade with continuous EEG monitoring may be necessary for status epilepticus.
3. Treat cardiac ischemia or acute coronary syndrome with aspirin, nitrates, morphine, and benzodiazepines. Diltiazem 20 mg IV can be used in those with ischemic ST changes. β-blockers are contraindicated due to unopposed β-receptor stimulation. Although coronary angiography is preferred for persistent, concerning ST changes, fibrinolytic therapy may be used in those without contraindications.
4. Treat cocaine-induced wide complex tachydysrhythmia and QRS interval prolongation with sodium bicarbonate 1 to 2 mEq/kg titrated to a serum pH of 7.45 to 7.55. Diltiazem 20 mg IV can be used to treat SVT. Torsades de pointes can be treated with magnesium and/or overdrive pacing. Consider lipid emulsion therapy 1.5 mL/kg IV over 2 minutes then 0.25 mL/kg/min until improvement; maximum dose 10 mL/kg for those with severe, refractory cardiac toxicity.
5. Treat hypertension unresponsive to benzodiazepines with nitroprusside 0.3 μg/kg/min IV or phentolamine 2.5 to 5 mg IV.
6. Treat “body stuffers” with activated charcoal and supportive care as above. Treat asymptomatic “body packers” with activated charcoal and whole-bowel irrigation using polyethylene glycol. Symptomatic “body packers” with presumed rupture of ingested packets are treated for acute toxicity as above and require immediate surgical consultation

105_Cydulka_Ch105.indd 586 20/03/17 3:30 PM

Source page 606

for possible laparotomy. Endoscopy should be avoided due to risk of rupture.

7. Patient disposition depends on initial presentation, response to treatment, stimulant involved, and expected duration of effect. Amphetamines have a longer duration of effect than cocaine and may require longer periods of observation or hospital admission. Patients with mild toxicity that resolves completely during observation may be discharged. Admit

“body packers” and those patients with more severe toxicity or who remain symptomatic.

■■ HALLUCINOGENS

Clinical Features

Table 105-1 summarizes the classification, features, complications, and specific treatments of commonly abused hallucinogens.

Diagnosis and Differential

Diagnosis is primarily clinical. Routine drug screens will not detect LSD, psilocybin, or mescaline. Urine tests for phencyclidine (PCP) are unreliable. Some amphetamine screens will detect methylenedioxy-methamphetamine (MDMA). Urine tests for marijuana are unreliable indicators of acute use because patients may be positive for days to weeks after their last use. Check glucose, electrolytes, renal function, CK, and urinalysis to evaluate for hyponatremia, rhabdomyolysis, and acute kidney injury.

Exclude other causes of altered mental status, including traumatic injuries, hypoglycemia, and infection in patients with hyperthermia.

The differential diagnosis of hallucinogen intoxication includes alcohol and benzodiazepine withdrawal, hypoglycemia, anticholinergic poisoning, thyrotoxicosis, CNS infections, structural CNS lesions, and acute psychosis.

Emergency Department Care and Disposition

Most hallucinogen intoxications are managed by monitoring, providing a calm environment, and the use of benzodiazepines for agitation and sympathomimetic symptoms. Antipsychotic medications should be used with caution as these may lower seizure threshold. β-Blockers should not be used as noted above.

1. Treat agitation, hyperthermia, seizures, tachycardia, and hypertension with lorazepam 1 to 2 mg IV or PO, or 0.1 mg/kg in children; or diazepam 5 to10 mg IV or PO. Repeat dosing as needed. Consider propofol, intubation, and paralysis for severe, refractory toxicity.
2. Consider nitroprusside or phentolamine for severe hypertension refractory to benzodiazepines.
3. Treat rhabdomyolysis with aggressive isotonic IV fluid administration.
4. Most patients with hallucinogen intoxication can be safely discharged from the ED after a period of observation. Admit patients with persistent altered mental status or serious medical complications, such as severe hyperthermia, hypertension, seizures, and rhabdomyolysis.

105_Cydulka_Ch105.indd 587 20/03/17 3:30 PM

Source page 607

105_Cydulka_Ch105.indd 588

588

Table 105-1 Common Hallucinogens

Typical Hallucinogenic Duration of

SECTION 11: Toxicology and Pharmacology

Drug Dose Action Clinical Features Complications Specific Treatment

Lysergic acid diethylamide 20–80 μg 8–12 h ●● Mydriasis ●● Coma ●● Reassurance

●● Tachycardia ●● Hyperthermia ●● Benzodiazepines

●● Anxiety ●● Coagulopathy

●● Muscle tension ●● Persistent psychosis

●● Hallucinogen persisting

perception disorder

Psilocybin 5–100 mushrooms 4–6 h ●● Mydriasis ●● Seizures (rare) ●● Reassurance

4–6 mg of psilocybin ●● Tachycardia ●● Hyperthermia (rare) ●● Hydration

●● Muscle tension ●● Benzodiazepines

●● Nausea and vomiting

Mescaline 3–12 “buttons” 6–12 h ●● Mydriasis ●● Rare ●● Supportive

200–500 mg of mescaline ●● Abdominal pain ●● Benzodiazepines

●● Nausea/vomiting

●● Dizziness

●● Nystagmus

●● Ataxia

Methylenedioxymethamphetamine 50–200 mg 4–6 h ●● Mydriasis ●● Hyponatremia ●● Benzodiazepines

(“Ecstasy”) ●● Bruxism ●● Hypertension ●● Hydration

●● Jaw tension ●● Seizures ●● Active cooling

●● Ataxia ●● Hyperthermia ●● Dantrolene

●● Dry mouth ●● Dysrhythmias ●● Specific serotonin

●● Nausea ●● Rhabdomyolysis antagonists

(Continued)

20/03/17 3:30 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.

- Content ID: SEG-TINTINALLI-EMM8-FULL.

- Source pages: 604–611.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 106: Analgesics

Source pages 612–617 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 612–617

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Analgesics**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 106: Analgesics 593 ■■ ACETAMINOPHEN (APAP) Clinical Features Acute APAP toxicity can present in four classic stages outlined in Table 106-2. [Source pages 612–617]

- Massive APAP ingestions (4-hour APAP level >800 micro- grams/mL) may cause early coma or agitation and lactic acidosis, as opposed to delayed symptom onset seen in the majority of APAP overdoses. [Source pages 612–617]

- Diagnosis and Differential Toxic exposure to APAP is likely when a patient >6 years old ingests either: (a) >10 g or 200 mg/kg in a single ingestion or over a 24-hour period, or (b) >6 g or 150 mg/kg/d for two consecutive days. [Source pages 612–617]

- For children <6 years old, either: (a) >200 mg/kg in a single ingestion or over an 8-hour period, or (b) 150 mg/kg/day for two consecutive days would be considered toxic. [Source pages 612–617]

- Confirm toxicity with a serum APAP concentration and an exact time of ingestion. [Source pages 612–617]

Clinical reasoning structure

- **Problem representation:** Analgesics; source pages 612–617.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Analgesics**.

- Source-page range: **612–617**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 612

CHAPTER 106: Analgesics 593

■■ ACETAMINOPHEN (APAP)

Clinical Features

Acute APAP toxicity can present in four classic stages outlined in Table 106-2. Massive APAP ingestions (4-hour APAP level >800 micrograms/mL) may cause early coma or agitation and lactic acidosis, as opposed to delayed symptom onset seen in the majority of APAP overdoses.

Diagnosis and Differential

Toxic exposure to APAP is likely when a patient >6 years old ingests either: (a) >10 g or 200 mg/kg in a single ingestion or over a 24-hour period, or (b) >6 g or 150 mg/kg/d for two consecutive days. For children <6 years old, either: (a) >200 mg/kg in a single ingestion or over an 8-hour period, or (b) 150 mg/kg/day for two consecutive days would be considered toxic. Confirm toxicity with a serum APAP concentration and an exact time of ingestion. Plot the serum APAP level on the Rumack–Matthew nomogram (Fig. 106-1); this nomogram applies only to the setting of a single acute exposure during the window between 4 hours and 24 hours postingestion. Obtain additional laboratory studies including electrolytes, glucose, BUN, creatinine, transaminases, CBC, PT, ASA, urine toxicology screen, and ECG as clinically indicated (e.g., potential coingestion in the suicidal patient).

Emergency Department Care and Disposition

1. Early administration of activated charcoal 1 g/kg PO, NG, or OG if possible.

2. N-acetylcysteine (NAC) is the antidote for APAP poisoning. The treatment algorithm is depicted in Fig. 106-2 and dosing of NAC is outlined in Table 106-3.

Table 106-2 Clinical Stages of Acute Acetaminophen Toxicity

Stage 1 Stage 2 Stage 3 Stage 4

Timing First 24 h Days 2 to 3 Days 3 to 4 After day 5

Clinical Anorexia Improvement in Recurrence Clinical improve-

manifestations Nausea anorexia, nausea, of anorexia, ment and recov-

Vomiting and vomiting nausea, and ery (7–8 days)

Malaise Abdominal pain vomiting or

Hepatic Encephalopathy Deterioration

tenderness Anuria to multi-organ

Jaundice failure and

death

Laboratory Hypokalemia Elevated serum Hepatic failure Improvement and

abnormalities transaminases resolution

Elevated bilirubin Metabolic acidosis or

and prolonged Coagulopathy Continued

prothrombin time Renal failure deterioration

if severe Pancreatitis

106_Cydulka_Ch106.indd 593 26/04/17 4:52 pm

Source page 613

594 SECTION 11: Toxicology and Pharmacology

300 2000

200 1300

150 1000

900

800

700

100 600

90

80 500

70

60 400

Acetaminophen Plasma Concentration

50 300

40 250 Possible

Toxicity

30 200

μg/mL

μM/L

20

100

90 Toxicity

80

Unlikely

70

10 60

9

8 50

7

40

5

30

4

Recommend treatment

20 if level is above broken line

2

Take level

at least 10

4h

postingestion

4 8 12 16 20 24 26

Hours Postingestion

Figure 106-1. Rumack–Matthew nomogram.

3. If APAP is still detectable after the dosing regimens described in Table 106-3 are complete, continue NAC until APAP is undetectable in serum.

4. Treat abnormalities related to fulminant hepatic failure by correcting coagulopathy and acidosis, managing cerebral edema, and supporting multi-organ failure. Consider early transfer to a transplant center.

5. Patients with nontoxic APAP levels based on the Rumack–Matthew nomogram may be medically cleared from the ED if there is no evidence of other drug ingestion. Admit all patients receiving NAC therapy.

106_Cydulka_Ch106.indd 594 26/04/17 4:52 pm

Source page 614

APAP Ingestion

<4 h from Between Unknown

ingestion 4 and 24 h or >24 h

from ingestion from ingestion

Consider GI Consider GI

decontamination decontamination

for unknown

ingestion

Send >4 h

APAP level

Send APAP levels

Send LFTs

(AST, ALT, PT)

Level Level Give 1st dose AC

available not available

<8 h until >8 h

Plot on Give 1st APAP >10 µg/mL

nomogram dose AC or AST/ALT

(within 8 h) increased

Not toxic: Yes: No:

Toxic:

symptomatic Continue Supportive

AC Rx

Rx AC Rx

If

pH <7.3

PT >100

Cr >3.3

AMS

Refer to liver

transplant unit

Figure 106-2. Treatment guidelines for acetaminophen (APAP) ingestion. All times noted are postingestion. AC, acetylcysteine; ALT, alanine aminotransferase; AMS, altered mental status; AST, aspartate aminotransferase; Cr, creatinine; LFTs, liver function tests; PT, prothrombin time; Rx, treatment.

■■ NONSTEROIDAL ANTI-INFLAMMATORY DRUGS (NSAIDs)

Clinical Features

NSAID toxicity is more commonly associated with chronic therapeutic use as opposed to acute overdoses. The clinical features of acute NSAID toxicity are outlined in Table 106-4.

106_Cydulka_Ch106.indd 595 26/04/17 4:52 pm

Source page 615

Table 106-3 Acetylcysteine Dosing Regimens

IV Pediatric

Oral IV Adult (<40 kg)

Preparation Available as 10% Available as 20% Available as 20%

and 20% solutions solution solution
Dilute to 5% Dilute to 2%
solution for oral solution by mixing
administration 50 mL in 450 mL 5%
dextrose in water
Loading dose 140 mg/kg 150 mg/kg in 200 mL 150 mg/kg (7.5 mL/
5% dextrose in water kg) infused over
infused over 15–60 min 15 to 60 min
Maintenance dose 70 mg/kg every 4 h 50 mg/kg in 500 mL 50 mg/kg (2.5 mL/kg)
for 17 doses 5% dextrose in water infused over 4 h
infused over 4 h followed by
followed by 100 mg/kg (5 mL/kg)
100 mg/kg in 1000 mL infused over 16 h
5% dextrose in water
infused over 16 h
Duration of therapy 72 h 20 h 20 h
Comments Dilute with pow- Monitor for drug- Monitor for drug-
dered drink mix, related adverse effects related adverse
juice, or soda and anaphylactoid effects and anaphy-
Serve chilled reactions lactoid reactions
Drink through a 500 mL of the 2%
straw to reduce solution prepared
disagreeable smell as described above
is enough to treat a
33-kg child for the
full 20-h course

Diagnosis and Differential

The manifestations of NSAID toxicity are generally nonspecific. NSAID levels are not readily available or clinically useful in assessing toxicity. Laboratory evaluation should include electrolytes, glucose, renal and hepatic function tests, and an APAP level. Consideration of evaluation for co-ingestions should be made. Bedside glucose determination is indicated for altered mental status or seizures.

Table 106-4 NSAID Toxicity After an Acute Overdose

Initial symptoms within 4 h of ingestion Abdominal pain, nausea, vomiting
Central nervous system Headache, nystagmus, diplopia, altered mental status, coma, muscle twitching, and seizures
(mefenamic acid)
Cardiovascular Hypotension, shock, bradydysrhythmia
Electrolyte Hyperkalemia, hypocalcemia, hypomagnesemia

GI and hepatic Continued abdominal pain, nausea, vomiting,

hepatic injury, pancreatitis (rare)

Renal Acute kidney injury

106_Cydulka_Ch106.indd 596 26/04/17 4:52 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.

- Content ID: SEG-TINTINALLI-EMM8-FULL.

- Source pages: 612–617.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 107: Xanthines and Nicotine

Source pages 618–621 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 618–621

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Xanthines and Nicotine**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 107: Xanthines and Nicotine 599 caffeine toxicity includes serial measurement of the serum theophylline or caffeine concentration, serum electrolytes, and a 12-lead ECG. [Source pages 618–621]

- The differ- ential diagnosis includes other sympathomimetic overdose (e.g., amphet- amines, cocaine) as well as anticholinergic toxicity. [Source pages 618–621]

- Emergency Department Care and Disposition Treatment of methylxanthine poisoning consists of stabilization, gastric decontamination and elimination, treatment of life-threatening toxic effects, and, in severe cases, hemoperfusion or dialysis. [Source pages 618–621]

- Table 107-1 lists appropriate methods of GI decontamination and elimi- nation in methylxanthine toxicity. [Source pages 618–621]

- Place all patients on cardiac monitors and establish intravenous (IV) or intraosseous (IO) access. [Source pages 618–621]

Clinical reasoning structure

- **Problem representation:** Xanthines and Nicotine; source pages 618–621.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Xanthines and Nicotine**.

- Source-page range: **618–621**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 618

CHAPTER 107: Xanthines and Nicotine 599

caffeine toxicity includes serial measurement of the serum theophylline or caffeine concentration, serum electrolytes, and a 12-lead ECG. The differential diagnosis includes other sympathomimetic overdose (e.g., amphetamines, cocaine) as well as anticholinergic toxicity.

Emergency Department Care and Disposition

Treatment of methylxanthine poisoning consists of stabilization, gastric decontamination and elimination, treatment of life-threatening toxic effects, and, in severe cases, hemoperfusion or dialysis.

1. Table 107-1 lists appropriate methods of GI decontamination and elimination in methylxanthine toxicity.
2. Place all patients on cardiac monitors and establish intravenous (IV) or intraosseous (IO) access.
3. Treat nausea and vomiting with ondansetron 4 to 8 mg IV or PO (0.1 to 0.15 mg/kg). Consider ranitidine for gastric hypersecretion but avoid cimetidine, which can prolong the half-life of theophylline.
4. Treat seizures with benzodiazepines, barbiturates, or even propofol. Rapid escalation of anticonvulsant treatment is warranted when a benzodiazepine treatment fails. Lorazepam 0.1 mg/kg IV, or 1 to 2 mg empirically for adults may be used as initial treatment, repeated within 5 to 10 minutes if seizures do not cease. Phenobarbital 20 mg/kg or pentobarbital 5 mg/kg may be used in cases of benzodiazepine failure. Anticipate respiratory depression and the need for ventilator assistance when using phenobarbital after a benzodiazepine. Phenytoin is contraindicated in theophylline toxicity. Due to the common occurrence of severe seizures and status epileptics, use phenobarbital to prophylax against seizures in cases of acute toxicity with theophylline concentration of 80 μg/dL or chronic toxicity with serum concentrations of >40 μg/dL.
5. Administer IV isotonic crystalloid for hypotension. Consider cardiose-

lective beta blockers such as esmolol or metoprolol in patients with hypotension unresponsive to IV fluids or conventional vasopressors.

6. Treat cardiac dysrhythmias with cardioselective blockers such as metoprolol and esmolol. Consider a calcium channel blocker such as diltiazem for atrial fibrillation. Hypokalemia may be severe but is typically

Table 107-1 GI Decontamination for Methylxanthine Toxicity

GI Decontamination Technique	Indication*	Dosing
Activated charcoal (single dose)	Acute ingestion	<12 y old: 0.5–1 g/kg PO >12 y old: 25–100 g PO
Multidose activated charcoal	Acute ingestion	Normal activated charcoal loading dose, followed by: 0.25–0.5 g/kg PO every 2–4 h for 12 h (frequency and duration may vary)
Whole-bowel irrigation using iso-osmolar polyethylene glycol electrolyte solution	Acute ingestion of sustained-release preparations	9 mo–6 y: 25 mL/kg/h 6–12 y old: 1000 mL/h >12 y old: 1500–2000 mL/h Duration: 4–6 h or until clear rectal effluent

*

Consider contraindications; for details see http://www.clintox.org/positionstatements.cfm.

107_Cydulka_Ch107.indd 599 20/03/17 3:41 PM

Source page 619

unresponsive or poorly responsive to potassium supplementation because the force driving potassium intracellularly continues despite potassium supplementation. In cases of potassium supplementation, hyperkalemia may occur as massive quantities of potassium move extracellularly as the effect of theophylline or caffeine abates. Therefore, potassium supplementation is not recommended unless the patient experiences a dysrhythmia likely to be the result of hypokalemia or empirically if the serum potassium is <2.5 mmol/L

7. Consider hemodialysis or hemoperfusion in a symptomatic patient with a serum theophylline level >90 μg/mL after acute ingestion, or >40 μg/mL in the setting of chronic toxicity and in patients with life-threatening seizures or dysrhythmias.

8. Admit patients with seizures or ventricular dysrhythmias to an intensive care unit. Patients with mild symptoms and theophylline levels below 25 μg/mL usually do not require admission or specific treatment other than symptomatic care. Decrease the dose or discontinue theophylline or caffeine. Symptomatic patients using theophylline with serum concentrations >30 μg/mL may be given 1 to 2 doses of oral activated

charcoal, regardless of when the last ingestion of theophylline medication occurred, as activated charcoal works by “gut dialysis” to draw theophylline from the blood across the GI surface and into the charcoal sink, effectively lowering serum concentration.

■■ NICOTINE

Nicotine is present in tobacco products and smoking cessation medications, and is also used as an outdoor pesticide. Pediatric exposures to nicotine gum or discarded cigarette butts placed in a beverage such as a soda can are relatively common. Nicotine is rapidly absorbed through the lungs, mucous membranes, intestinal tract, and skin. Once absorbed, it binds to nicotinic receptors throughout the body including the central nervous system, autonomic system, and neuromuscular junction.

Clinical Features

Nicotine toxicity affects the GI, neurologic, cardiovascular, and respiratory systems. Nausea, vomiting, bradycardia, dysrhythmias, hypoventilation, coma, and seizures can occur. In severe poisoning, nicotine can result in paralysis and respiratory arrest. Table 107-2 lists the clinical effects of nicotinic receptor stimulation.

Diagnosis and Differential

Diagnosis of acute nicotine toxicity is largely based on history and physical examination. The differential diagnosis for nicotine toxicity is primarily cholinergic poisoning by other substances, such as organophosphate and carbamate pesticides, as well as cholinergic medication overdose.

Emergency Department Care and Disposition

1. Place patients on a cardiac monitor and establish vascular access.
2. In cases of dermal exposure to nicotine from a pesticide product, perform decontamination as for other HAZMAT substances. This is

107_Cydulka_Ch107.indd 600 20/03/17 3:41 PM

Source page 620

Table 107-2 Clinical Effects of Nicotine Toxicity

Signs and Symptoms of Nicotine Toxicity*	
Organ System	Immediate (<1 h) Delayed (>1 h)
GI	Hypersalivation Diarrhea
	Nausea
	Vomiting
Cardiovascular	Tachycardia Dysrhythmias
	Hypertension Bradycardia
	Hypotension
Neurologic	Tremor Hypotonia

Headache Seizure

Ataxia Coma

Respiratory Bronchorrhea Hypoventilation

Apnea

*

Onset of toxicity is varied and can be delayed for hours following dermal exposure.

typically done in a dedicated location, often external to the ED. GI

decontamination for nicotine exposure is not recommended, and

urine acidification is contraindicated.

3. Treat nausea and vomiting with ondansetron 4 to 8 mg IV or PO

(0.15 mg/kg).

4. Treat seizures or agitation with lorazepam 2 mg IV (0.1 mg/kg).

5. Administer isotonic crystalloid for hypotension.

6. Anticipate neuromuscular weakness or respiratory depression in severe

toxicity and be prepared for endotracheal intubation and mechanical

ventilation.

7. Patients who remain asymptomatic at least 3 hours after ingestion of

nicotine-containing products can be discharged. However, patients who

ingest intact transdermal patches should be monitored for at least

6 hours.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study

Guide, 8th ed., see Chapter 192, "Methylxanthines and Nicotine," by Chip

Gresham and Daniel E. Brooks.

107_Cydulka_Ch107.indd 601 20/03/17 3:41 PM

Source page 621

C h a p te r Cardiac Medications

108 Michael Levine

■■ DIGITALIS GLYCOSIDES

Digoxin is used to treat atrial fibrillation, especially with concurrent con-

gestive heart failure. Other cardiac glycosides are found in plants such as

foxglove, oleander, and lily of the valley.

Clinical Features

Toxicity can occur following acute ingestion or develop during chronic

therapy (Table 108-1). Acute toxicity typically presents with abrupt onset of

nausea and vomiting. Characteristic cardiac effects include bradydysrhyth-

mias and/or supraventricular tachycardia with atrioventricular block.

Severe toxicity can result in ventricular dysrhythmias. Chronic toxicity is

more common in the elderly and often occurs as a result of renal failure or

diuretic therapy. Neuropsychiatric symptoms are more common with

chronic toxicity, though cardiac effects are similar to those seen with acute toxicity.

Diagnosis and Differential

Hyperkalemia is often seen in acute poisoning, but may be absent in chronic toxicity. Serum digoxin levels are neither sensitive nor specific for toxicity. However, those patients with higher levels (>2 ng/mL) are more likely to experience toxicity. Almost any dysrhythmia, except for rapidly conducted atrial dysrhythmias, may be seen with toxicity; however, the

Table 108-1 Clinical Presentation of Digitalis Glycoside Toxicity

Acute toxicity

Clinical history Intentional or accidental ingestion

GI effects Nausea and vomiting

Central nervous system effects Headache, dizziness, confusion, coma

Cardiac effects Bradydysrhythmias or supraventricular tachydysrhythmias with atrioventricular block

Electrolyte abnormalities Hyperkalemia

Digoxin level Marked elevation (if obtained within 6 h)

Chronic toxicity

Clinical history Typically in elderly cardiac patients taking diuretics; may have renal insufficiency

GI effects Nausea, vomiting, diarrhea, abdominal pain

Central nervous system effects Fatigue, weakness, confusion, delirium, coma

Cardiac effects Almost any dysrhythmia, other than a rapidly conducted atrial fibrillation, can occur; ventricular dysrhythmias are common

Electrolyte abnormalities Normal, low, or high serum potassium, hypomagnesemia

Digoxin level Minimally elevated or within “therapeutic” range

602

108_Cydulka_Ch108.indd 602 22/04/17 4:38 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 618–621.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 108: Cardiac Medications

Source pages 622–633 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 622–633

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Cardiac Medications**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 108: Cardiac Medications 603 most common finding is premature ventricular beats. [Source pages 622–633]

- The differential diag- nosis includes sinus node disease or toxicity from calcium channel block- ers, β-blockers, class IA antidysrhythmics, clonidine, other cardiotoxic plants, or intrinsic conduction disease. [Source pages 622–633]

- Emergency Department Care and Disposition All patients require continuous cardiac monitoring, intravenous (IV) access, and frequent reevaluation (Table 108-2). [Source pages 622–633]

- Administer activated charcoal 1 g/kg in cases of acute toxicity in which the patient is awake and cooperative. [Source pages 622–633]

- Use atropine 0.5 to 1 mg (0.02 mg/kg, minimum dose 0.1 mg) IV to treat bradydysrhythmias. [Source pages 622–633]

Clinical reasoning structure

- **Problem representation:** Cardiac Medications; source pages 622–633.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Cardiac Medications**.

- Source-page range: **622–633**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 622

CHAPTER 108: Cardiac Medications 603

most common finding is premature ventricular beats. The differential diagnosis includes sinus node disease or toxicity from calcium channel blockers, β-blockers, class IA antidysrhythmics, clonidine, other cardiotoxic plants, or intrinsic conduction disease.

Emergency Department Care and Disposition

All patients require continuous cardiac monitoring, intravenous (IV) access, and frequent reevaluation (Table 108-2).

1. Administer activated charcoal 1 g/kg in cases of acute toxicity in which the patient is awake and cooperative.
2. Use atropine 0.5 to 1 mg (0.02 mg/kg, minimum dose 0.1 mg) IV to treat bradydysrhythmias.
3. Administer digoxin-specific Fab for ventricular dysrhythmias, hemodynamically significant bradydysrhythmias, and hyperkalemia greater than 5.5 mEq/L. Dosing of digoxin-specific Fab is calculated according to Table 108-3.
4. Hyperkalemia that is thought to be due to cardiac glycoside toxicity is best treated with Fab fragments. Hyperkalemia that is felt to be due to intrinsic renal disease may be treated with dextrose followed by insulin;

Table 108-2 Treatment of Digitalis Glycoside Poisoning

Asymptomatic patients

Obtain accurate history

Continuous cardiac monitoring

IV access

GI decontamination (for awake, cooperative patients within 1 h of ingestion): activated charcoal, 1 g/kg orally

Frequent reevaluation

Calculate digoxin-specific Fab antibody fragment dose in anticipation of potential need: may bring to drug bedside, depending on ready availability

Symptomatic patients

Obtain accurate history

IV access

Continuous cardiac monitoring

GI decontamination (for awake, cooperative patients within 1 h of ingestion): activated charcoal, 1 g/kg orally

Life-threatening dysrhythmias

Digoxin-specific Fab antibody fragments: IV infusion

Atropine (for symptomatic bradycardia): 0.5 to 1 mg IV (0.02 mg/kg, minimum dose 0.1 mg)

Pacer: external or transvenous

Abbreviation: Fab, antigen-binding fragment.

Table 108-3 Calculation of Digoxin-Specific Fab Antibody Fragment Dose*

Acute life-threatening ingestion with unknown digoxin level and unknown amount ingested: 10 vials intravenous digoxin-specific Fab

A simple and accurate variation using serum digoxin level

Number of vials = [serum digoxin level (ng/mL) × patient's weight (kg)]/100

*

The digoxin-specific Fab antibody fragments commercially available in the United States contain 38 or 40 mg per vial, depending on the manufacturer, but both bind approximately 0.5 mg of digoxin.

Abbreviation: Fab, antigen-binding fragment.

108_Cydulka_Ch108.indd 603 22/04/17 4:38 pm

Source page 623

other options are sodium bicarbonate, potassium-binding resin, or hemodialysis. Historically, IV calcium use has been discouraged because of reports of ventricular dysrhythmias. However, recent evidence suggests that IV calcium use is likely safe.

5. Admit patients with signs of mild toxicity to a monitored setting and manage those with significant toxicity in an intensive care unit.

Repeated digoxin levels following digoxin Fab are not accurate and should not be obtained.

■■ β-BLOCKERS

β-Blockers are used in the management of hypertension, acute coronary syndromes, dysrhythmias, congestive heart failure, thyrotoxicosis, social phobias, migraines, and glaucoma. In overdose, their negative inotropic and chronotropic effects result in progressive bradycardia and hypotension.

Clinical Features

Toxicity usually develops within 6 hours of ingestion of an immediate-release product. With sustained-release preparations toxicity is generally seen within 8 hours of ingestion. The cardiovascular system is the primary organ system affected; however, other noncardiac manifestations may occur (Table 108-4). Sotalol, unlike other β-blockers, is also a class III

antidysrhythmic, and thus may cause QT-interval prolongation and torsades de pointes. The onset of toxicity from sotalol may be delayed up to 12 hours postingestion.

Diagnosis and Differential

The diagnosis is made based on clinical findings. An ECG should be obtained in all cases. Laboratory studies are directed at identifying underlying medical conditions or complications. Specific drug levels are not commonly available and correlate poorly with clinical effects. Table 108-5 lists other agents that result in bradycardia and hypotension.

Table 108-4 Common Findings with β-Blocker Toxicity

Cardiac

Hypotension

Bradycardia

Conduction delays and blocks

Ventricular dysrhythmias (seen with sotalol)

Asystole

Decreased contractility

Central nervous system

Depressed mental status

Coma

Psychosis

Seizures

Pulmonary

Bronchospasm

Endocrine

Hypoglycemia

108_Cydulka_Ch108.indd 604 22/04/17 4:38 pm

Source page 624

Table 108-5 Toxicologic Causes of Bradycardia and Hypotension

Causes	Differentiating Features
β-Blockers	Relative or absolute hypoglycemia
Calcium channel blockers	Marked hyperglycemia
Naturally occurring cardiac glycosides (oleander, foxglove, lily of the valley, rhododendron, and toad-derived bufotoxin)	Ventricular ectopy Hyperkalemia May cross-react with digoxin immunoassay
Class IC antidysrhythmic drugs (propafenone)	Wide-complex bradycardia
Clonidine	Opioid-like manifestations: coma, miosis, decreased respirations

Digoxin (acute) Hyperkalemia

Elevated level on digoxin immunoassay

Organophosphates Muscarinic toxidrome

Emergency Department Care and Disposition

The goal of therapy is to restore cardiac output by increasing heart rate and improving myocardial contractility (Fig. 108-1). Establish continuous cardiac monitoring and IV access. IV fluids may be administered for hypotension.

Hypotension

Evaluation

(e.g., ECG, cardiac ultrasound, or pulmonary artery catheter)

Decreased Decreased Bradycardia

QRS >120 ms

contractility SVR

Sodium Glucagon Vasopressors Glucagon

bicarbonate High-dose insulin Adrenergic

Adrenergic agents agents

Calcium salts Cardiac pacing

Figure 108-1. Management strategies in β-blocker toxicity. Cardiac function is evaluated using ECG and cardiac ultrasound and/or a pulmonary artery catheter. For patients with a wide QRS interval, consider sodium bicarbonate therapy. For patients with impaired myocardial contractility, consider glucagon, norepinephrine, insulin and glucose, and/or calcium therapy. For patients with preserved cardiac contractility, administer IV fluids. For patients with decreased systemic vascular resistance, consider norepinephrine. For patients with bradycardia, consider atropine, glucagon, and/or cardiac pacing. (See the text for details.) IVF, IV fluids; SVR, systemic vascular resistance.

108_Cydulka_Ch108.indd 605 22/04/17 4:38 pm

Source page 625

1. Administer activated charcoal 1 g/kg within 1 to 2 hours of ingestion if no contraindications exist.

2. Glucagon has inotropic and chronotropic effects and is the agent of choice for the treatment of toxicity. Administer as an IV bolus of 3 to 10 mg (0.05 mg/kg). Follow with a continuous infusion of 1 to 5 mg/h. Nausea and vomiting are common side effects. Pretreatment with antiemetics is recommended.

3. Use vasopressors, such as norepinephrine 2 to 30 μg/kg/min or epinephrine 1 to 20 μg/kg/min, for refractory bradycardia and hypotension. Significantly higher doses may be necessary.

4. Calcium may be of limited benefit in cases of refractory hypotension as either calcium gluconate or calcium chloride: 10 mL of 10% (0.15 mL/kg) repeated three to six times as necessary. Although calcium chloride contains more elemental calcium than calcium gluconate, it is irritating to soft tissues and should be administered via central line.
5. Hyperinsulinemia-euglycemia (HIE) therapy may improve myocardial contractility. Bolus regular insulin IV 1 U/kg followed by a continuous infusion of 1 U/kg/h. Serum glucose should be supplemented as needed and monitored frequently to avoid hypoglycemia. Monitor serum potassium for severe hypokalemia.
6. Cardiac pacing may be attempted but it is often unsuccessful. Aggressive measures include extracorporeal circulation or intra-aortic balloon pump placement. Hemodialysis may be of benefit in cases involving acebutolol, atenolol, nadolol, or sotalol.
7. Isoproterenol, a pure β-agonist, may provide benefit.
8. Use magnesium sulfate, lidocaine, and overdrive pacing to treat sotalol-induced ventricular dysrhythmias.
9. Symptomatic patients should be admitted to an ICU. Asymptomatic patients who ingest can be medically cleared 6 hours after an ingestion of an immediate-release drug or 8 hours after an ingestion of a sustained release drug. Patients who ingest sotalol should be observed for 12 hours.

■■ CALCIUM CHANNEL BLOCKERS

Calcium channel blockers are used in the treatment of hypertension, vasospasm, and rate control of supraventricular tachydysrhythmias. Three widely used classes of calcium channel blockers are the phenylalkylamines (verapamil), benzothiazepines (diltiazem), and dihydropyridines (nifedipine, etc.).

Clinical Features

Toxicity usually develops within 6 hours of ingestion of an immediate-release product. With sustained-release preparations toxicity can be delayed 12 to 24 hours. Toxicity primarily affects the cardiovascular system causing sinus bradycardia, atrioventricular block, and hypotension. Verapamil and diltiazem have a proportionally greater effect on the myocardium than the dihydropyridines. Dihydropyridine overdose can result in reflex tachycardia. With severe toxicity, all classes of calcium channel blockers can cause bradycardia, depressed myocardial contractility, and vasodilatation. Hyperglycemia, lactic acidosis, and noncardiogenic pulmonary edema may occur. Central nervous system effects are due to hypoperfusion and other etiologies should be sought if the blood pressure is normal.

108_Cydulka_Ch108.indd 606 22/04/17 4:38 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 622–633.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 109: Anticonvulsants

Source pages 634–637 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 634–637

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Anticonvulsants**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 109: Anticonvulsants 615 include ataxia, nystagmus, miosis or mydriasis, ileus, bowel obstruction, hypertonicity, increased deep tendon reflexes, dystonic reactions, and an anticholinergic toxidrome. [Source pages 634–637]

- Severe toxicity may result in coma and respira- tory failure, and seizures can occur at high serum concentrations. [Source pages 634–637]

- Though rare, carbamazepine can widen the QRS interval and cause cardiac dys- rhythmias. [Source pages 634–637]

- Patients taking carbamazepine therapeutically will develop toxicity with lesser acute ingestions because a large quantity of carbamaze- pine is already present in the body. [Source pages 634–637]

- Diagnosis and Differential Serum carbamazepine concentrations do not correlate with severity of poison- ing, though concentrations >40 µg/mL may be associated with an increased risk of complications, and those more than 60 to 80 µg/mL may be fatal. [Source pages 634–637]

Clinical reasoning structure

- **Problem representation:** Anticonvulsants; source pages 634–637.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Anticonvulsants**.

- Source-page range: **634–637**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 634

CHAPTER 109: Anticonvulsants 615

include ataxia, nystagmus, miosis or mydriasis, ileus, bowel obstruction, hypertonicity, increased deep tendon reflexes, dystonic reactions, and an anticholinergic toxidrome. Severe toxicity may result in coma and respiratory failure, and seizures can occur at high serum concentrations. Though rare, carbamazepine can widen the QRS interval and cause cardiac dysrhythmias. Patients taking carbamazepine therapeutically will develop toxicity with lesser acute ingestions because a large quantity of carbamazepine is already present in the body.

Diagnosis and Differential

Serum carbamazepine concentrations do not correlate with severity of poisoning, though concentrations >40 µg/mL may be associated with an increased risk of complications, and those more than 60 to 80 µg/mL may be fatal. A false-positive tricyclic antidepressant result on urine drug screen can occur.

Emergency Department Care and Disposition

1. Consider activated charcoal in the awake patient within 1 hour of ingestion or in patients chronically taking carbamazepine therapeutically.
2. Consider hemodialysis or hemodiafiltration for life-threatening overdose.
3. Obtain an ECG to evaluate for QRS interval widening. Treat QRS interval widening with sodium bicarbonate.
4. Patients can be discharged if asymptomatic, with declining serum levels (below 15 µg/mL), and a normal ECG.

■■ VALPROATE

Clinical Features

In acute overdose, valproate causes central nervous system depression. Other findings include respiratory depression, hypotension, hypoglycemia, hypernatremia, hypophosphatemia, and anion gap metabolic acidosis that may persist for days. Liver toxicity, cerebral edema, hyperammonemia,

pancreatitis, and thrombocytopenia have been reported after acute overdose. Hepatic failure (microvesicular steatosis) occurs in about 1 in 20,000 patients on long-term therapy, with children <3 years of age on multiple antiepileptics at greatest risk.

Diagnosis and Differential

Obtain a serum valproate level. The therapeutic range is 50 to 100 µg/mL; adverse effects increase at concentrations >150 µg/mL and frank coma may occur with levels above 800 µg/mL. Check serum ammonia and bedside glucose in patients with altered level of consciousness. Consider liver function tests, electrolytes, and a CBC.

Emergency Department Care and Disposition

1. Consider multidose activated charcoal after ingestion of enteric-coated, delayed-release preparations and measure serial concentrations to account for delayed peak serum concentration. Consider whole-bowel irrigation in any enteric-coated or extended-release preparations.

109_Cydulka_Ch109.indd 615 20/03/17 3:57 PM

Source page 635

2. L-carnitine, 50 mg/kg/day, may hasten recovery in patients with acute intoxication and increase survival from hepatotoxicity.

3. Hemodialysis may be useful in toxicity associated with very high serum concentrations >800 µg/mL. Hemoperfusion and hemodiafiltration can be used to treat severe overdose.

4. All symptomatic patients, other than those with mild symptoms, require admission. Asymptomatic patients with declining serum levels can be discharged.

■■ SECOND-GENERATION ANTICONVULSANTS

As a group, the second-generation anticonvulsants possess little toxicity in acute overdose.

●

■Felbamate may cause aplastic anemia and hepatic failure and can crystallize in the kidney, leading to acute renal failure in large overdose.

●

■Gabapentin may cause drowsiness, ataxia, nausea, and vomiting that generally resolve in about 10 hours. Gabapentin is increasingly abused for its benzodiazepine-like effects.

●

■Lacosamide may cause dizziness, headache, nausea, and diplopia during therapeutic use.

●

■Lamotrigine has been associated with autoimmune reactions during therapeutic use and drowsiness, vomiting, ataxia, and dizziness in overdose. Seizures, coma, cardiac toxicity (QRS and QT-interval prolongation), and acute pancreatitis have been reported. Treatment includes sodium bicarbonate.

●

■Levetiracetam may cause lethargy, coma, and respiratory depression.

●

■Oxcarbazepine may cause hyponatremia and a drug rash during therapeutic use.

●

■Pregabalin has been reported to cause somnolence and dizziness during long-term therapeutic use. Overdose may cause depressed level of consciousness. Pregabalin is increasingly abused for its benzodiazepine-like effects.

●

■Rufinamide may cause headache, dizziness, fatigue and somnolence during long-term therapy.

●

■Tiagabine may cause the rapid neurologic toxicity, including lethargy, coma, seizures, myoclonus, muscular rigidity, and delirium.

●

■Topiramate may cause nephrolithiasis and glaucoma during therapeutic use. In overdose, somnolence, vertigo, agitation, mydriasis, and seizures have been reported. Topiramate can produce a metabolic acidosis, which can last up to 7 days due to the long half-life of the drug.

●

■Zonisamide may promote renal stone formation and cause a drug rash during therapeutic use.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 197, “Anticonvulsants” by Frank LoVecchio.

109_Cydulka_Ch109.indd 616 20/03/17 3:57 PM

Source page 636

C h a p ter Iron

110 O. John Ma

■■ CLINICAL FEATURES

Iron toxicity from an intentional or accidental ingestion is a common poisoning. Based on clinical findings, iron poisoning can be divided into five stages.

The first stage develops within the first few hours of ingestion. Direct

irritative effects of iron on the gastrointestinal (GI) tract produce abdominal pain, vomiting, and diarrhea. Vomiting is the clinical sign most consistently associated with acute iron toxicity. The absence of these symptoms within 6 hours of ingestion essentially excludes a diagnosis of significant iron toxicity.

During the second stage, which may continue for up to 24 hours following ingestion, the patient's GI symptoms may resolve, providing a false sense of security despite toxic amounts of iron absorption. While patients may be asymptomatic, they often appear ill and may have abnormal vital signs reflecting hypovolemia and metabolic acidosis.

The third stage may appear early or develop hours after the second stage as shock and a metabolic acidosis evolve. Iron-induced coagulopathy may cause bleeding and worsen hypovolemia. Hepatic dysfunction, cardiomyopathy, and renal failure may also develop.

The fourth stage develops 2 to 5 days after ingestion and is characterized by elevation of aminotransferase levels and possible progression to hepatic failure.

The fifth stage, which occurs 4 to 6 weeks after ingestion, reflects the corrosive effects of iron on the pyloric mucosa and may cause gastric outlet obstruction.

■■ DIAGNOSIS AND DIFFERENTIAL

The diagnosis of iron poisoning is based on the clinical picture and the history provided by the patient, significant others, caretakers, or EMS providers. When determining a patient's potential for toxicity, the total amount of elemental iron must be used in calculations. Table 110-1 reviews the predicted clinical effects based on the amount of iron ingested.

Laboratory evaluation includes serum electrolytes, renal studies, serum glucose, coagulation studies, complete blood count, hepatic enzymes, and a serum iron level. A single serum iron level does not reflect what iron levels have been previously, what direction they are going, or the degree of iron toxicity in tissues; a single low serum level does not exclude the diagnosis of iron poisoning since there are variable times to peak level following ingestion of different iron preparations. Serum iron levels have limited use in directing management since toxicity is primarily intracellular rather than in the blood. The total iron binding capacity (TIBC) becomes falsely elevated in the presence of elevated serum iron levels or deferoxamine, and is of no clinical value.

110_Cydulka_Ch110.indd 617 26/04/17 2:33 pm

Source page 637

618 SECTION 11: Toxicology and Pharmacology

Table 110-1 Predicted Toxicity of Iron Ingestion

Elemental Iron Dose*	Serum Iron Concentration†	Predicted Clinical Effects
<20 mg/kg	<300 μg/dL (<54 μM/L)	Nontoxic or mild GI symptoms
20–60 mg/kg	300–500 μg/dL (54–90 μM/L)	Expected significant GI symptoms and potential for systemic toxicity
>60 mg/kg	>500 μg/dL (>90 μM/L)	Moderate to severe systemic toxicity
	>1000 μg/dL (>180 μM/L)	Severe systemic toxicity and increased morbidity

*

Elemental iron dose by history.

Serum iron concentration obtained within 4 to 6 hours of ingestion.

†

Plain radiographs may reveal iron in the GI tract; however, many iron preparations are not radiopaque, so normal radiographs do not exclude iron ingestion.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

Patients who have remained asymptomatic for 6 hours after ingestion of iron, have not ingested a potentially toxic amount, and who have a normal physical examination do not require medical treatment for iron toxicity. Patients whose symptoms resolve after a short period of time, and who have normal vital signs, usually have mild toxicity and require only supportive care. This subset of patients still requires an observation period. Figure 110-1 is an algorithm for the clinical management of patients after an acute iron ingestion.

Patients who are symptomatic or demonstrate signs of hemodynamic instability after iron ingestion require aggressive management in the ED.

1. Place the patient on supplemental oxygen and a cardiac monitor, and establish two large-bore IV lines.

2. Administer vigorous intravenous (IV) crystalloid infusion to help correct hypovolemia and hypoperfusion.

3. Perform gastric lavage in patients who present within 60 minutes of ingestion. Activated charcoal is not recommended.

4. Whole-bowel irrigation with a polyethylene glycol solution is efficacious. Administration of 250 to 500 mL/h in children or 2 L/h in adults via nasogastric tube may clear the GI tract of iron pills before absorption occurs.

5. Administer antiemetics such as ondansetron 4 mg IV in adults (0.1 mg/

kg to a maximum dose of 4 mg in pediatric patients) or promethazine 25 mg IV in adults.

6. Correct coagulopathy with vitamin K1 5 to 10 mg SC and fresh frozen plasma 10 to 25 mL/kg in adults (10 mL/kg in pediatric patients). Order blood for type and screen or crossmatch as necessary.

7. Deferoxamine is a chelating agent that can remove iron from tissues and free iron from plasma. Deferoxamine is safe to administer to children and pregnant women. Deferoxamine therapy is indicated in patients

110_Cydulka_Ch110.indd 618 26/04/17 2:33 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 634–637.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 110: Iron

Source pages 638–641 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 638–641

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Iron**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 110: Iron 619 Iron Ingested Asymptomatic GI symptoms Systemic only toxicity Quantity ingested Stabilize patient Volume resuscitation <20 mg/kg >60 mg/kg Assess acid-base status Obtain serum Fe 20–60 mg/kg Obtain abdominal radiograph or unknown Consider GI decontamination Consider antiemetics Consider abdominal Volume resuscitation radiograph Assess acid-base status Consider GI Obtain abdominal radiograph decontamination Consider GI decontamination Consider antiemetics <500 µg/dL or >500 µg/dL Obtain 4 h Unavailable AND Metabolic acidosis postingestion Normal acid-base Symptoms persistent serum Fe status Asymptomatic Symptoms develop Asymptomatic at 6 h Obtain baseline urine postingestion Start IV deferoxamine No Discharge Clinically Urine stable? [Source pages 638–641]

- Yes Stop Yes No Continue deferoxamine and deferoxamine supportive care Figure 110-1. [Source pages 638–641]

- Algorithm for clinical management of patients following iron ingestion. [Source pages 638–641]

- with systemic toxicity, metabolic acidosis, worsening symptoms, or a serum iron level predictive of moderate to severe toxicity. [Source pages 638–641]

- Intravenous infusion is the preferred route of deferoxamine administration because IM absorption is unpredictable in the hypovolemic patient. [Source pages 638–641]

Clinical reasoning structure

- **Problem representation:** Iron; source pages 638–641.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Iron**.

- Source-page range: **638–641**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 638

CHAPTER 110: Iron 619

Iron Ingested

Asymptomatic GI symptoms Systemic

only toxicity

Quantity

ingested Stabilize patient

Volume resuscitation

<20 mg/kg >60 mg/kg

Assess acid-base status

Obtain serum Fe

20–60 mg/kg Obtain abdominal radiograph

or unknown Consider GI decontamination

Consider antiemetics

Consider abdominal

Volume resuscitation

radiograph

Assess acid-base status

Consider GI

Obtain abdominal radiograph

decontamination

Consider GI decontamination

Consider antiemetics

<500 µg/dL or >500 µg/dL

Obtain 4 h

Unavailable AND Metabolic acidosis

postingestion

Normal acid-base Symptoms persistent

serum Fe

status Asymptomatic Symptoms develop

Asymptomatic

at 6 h Obtain baseline urine

postingestion Start IV deferoxamine

No

Discharge Clinically Urine

stable? colored?

Yes

Stop Yes No Continue deferoxamine and

deferoxamine supportive care

Figure 110-1. Algorithm for clinical management of patients following iron ingestion.

with systemic toxicity, metabolic acidosis, worsening symptoms, or a serum iron level predictive of moderate to severe toxicity.

Intravenous infusion is the preferred route of deferoxamine administration because IM absorption is unpredictable in the hypovolemic patient. The recommended initial dose is 50 mg/kg in children to an adult dose of 1000 mg IV. Since hypotension is the rate-limiting factor for IV infusion, it is recommended to begin with a slow IV infusion at 5 mg/kg/h. The deferoxamine infusion rate can be increased to 15 mg/kg/h, as tolerated, within the first hour of treatment. The recommended total amount of deferoxamine is 360 mg/kg or 6 g during the first 24 hours. Initiate deferoxamine therapy without waiting for the serum iron level in any clinically ill patient with a known iron ingestion.

Evaluate the efficacy of deferoxamine treatment through serial urine samples. As ferrioxamine is excreted, urine changes to a classic vin rose appearance. Clinical recovery is the most important factor guiding the termination of deferoxamine therapy.

110_Cydulka_Ch110.indd 619 26/04/17 2:33 pm

Source page 639

8. Patients who remain asymptomatic after 6 hours of observation, have a normal physical examination, and have a reliable history of an insignificant ingestion may be considered for discharge. Patients initially symptomatic who become asymptomatic should be admitted for further evaluation since this may represent the second stage of iron toxicity.

Admit all patients who receive deferoxamine therapy to an intensive care setting. Assess all patients for suicide risk. Consider child abuse or

neglect in pediatric cases.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 198, "Iron," by Stephanie H. Hernandez and Lewis S. Nelson.

110_Cydulka_Ch110.indd 620 26/04/17 2:33 pm

Source page 640

C h a p ter Hydrocarbons and Volatile

111 Substances

Allyson A. Kreshak

Products containing hydrocarbons are found in many household and workplace settings and include fuels, lighter fluids, paint removers, pesticides, polishers, degreasers, and lubricants. Some volatile substances may be recreationally abused. Exposure may cause mild to severe toxicity and, rarely, sudden death.

■■ CLINICAL FEATURES

Toxicity depends on route of exposure, physical characteristics, chemical characteristics, and the presence of toxic additives (e.g., lead or pesticides). See Table 111-1 for clinical features.

Chemical pneumonitis is the most common pulmonary complication and is most likely to occur following aspiration of a hydrocarbon with low viscosity (ability to resist flow), high volatility (tendency for a liquid to become a gas), and low surface tension (cohesive force between molecules). Symptoms occur quickly and include cough, gagging, choking, and dyspnea. Physical examination may reveal tachypnea, wheezing, grunting, and an elevated temperature. Radiographic abnormalities do not always occur. If radiographic findings occur, they may lag behind the clinical picture by 4 to 24 hours, but most are apparent within 6 hours.

Cardiac toxicity manifests as potentially lethal dysrhythmias resulting from myocardial sensitization to circulating catecholamines ("sudden sniffing death syndrome"). Halogenated hydrocarbon solvents are most frequently implicated, but all classes of hydrocarbons have been associated with dysrhythmias.

Table 111-1 Clinical Manifestations of Hydrocarbon Exposure

System	Clinical Manifestations
Pulmonary	Tachypnea, grunting respirations, wheezing, retractions
Cardiac	Ventricular dysrhythmias (may occur after exposure to halogenated hydrocarbons and aromatic hydrocarbons)
Central nervous	Slurred speech, ataxia, lethargy, coma
Peripheral nervous	Numbness and paresthesias in the extremities

GI and hepatic Nausea, vomiting, abdominal pain, loss of appetite (mostly with halogenated hydrocarbons)

Renal and metabolic Muscle weakness or paralysis secondary to hypokalemia in patients who abuse toluene

Hematologic Lethargy (anemia), shortness of breath (anemia), neurologic depression/syncope (carbon monoxide from methylene chloride), cyanosis (methemoglobinemia from amine-containing hydrocarbons)

Dermal Local erythema, papules, vesicles, generalized scarlatiniform eruption, exfoliative dermatitis, “huffer’s rash,” cellulitis

111_Cydulka_Ch111.indd 621 20/03/17 4:46 PM

Source page 641

Central nervous system toxicity may present as intoxication, ranging from initial giddiness, agitation, and hallucinations to seizures, slurred speech, ataxia, and coma. Chronic exposure may cause recurrent headaches, cerebellar ataxia, and mood lability.

Gastrointestinal toxicity can include vomiting (which can lead to aspiration), abdominal pain, anorexia, and hepatic damage (particularly from halogenated hydrocarbons such as carbon tetrachloride, methylene chloride, trichloroethylene, and tetrachloroethylene).

Dermal toxicity includes contact dermatitis and blistering with progression to full-thickness burns. Injection of hydrocarbons can cause tissue necrosis. Burns can result after cutaneous contact with hot tar and asphalt.

Less common acute toxicities include hematologic disorders such as hemolysis, methemoglobinemia, carboxyhemoglobinemia (from methylene chloride), and renal dysfunction.

■■ DIAGNOSIS AND DIFFERENTIAL

Diagnosis is made by history and physical examination findings, bedside monitoring, laboratory tests, and chest radiograph. An abdominal radiograph may reveal ingestion of radiopaque substances (e.g., chlorinated hydrocarbons).

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. Secure the airway and maintain ventilation support in patients with respiratory insufficiency or neurologic depression. Administer oxygen to symptomatic patients and place them on a cardiac monitor (see Table 111-2). An EKG should be obtained.

2. Treat hypotension with intravenous crystalloid infusion. Avoid catecholamines except in cases of cardiac arrest. Treat tachydysrhyth-

mias with propranolol, esmolol, or lidocaine. Avoid class IA and III agents.

3. Follow standard hazardous material measures for decontamination of the patient. Initial decontamination should ideally be done at the scene and should include removal of the patient from the exposure (including clothing or dermal contact). Skin may be irrigated with soap and water. Activated charcoal is not indicated following isolated hydrocarbon ingestions. The use of gastric emptying is not beneficial, and this decontamination technique is not regularly employed following ingestion.

4. Meticulous wound care with potential surgical debridement is indicated for dermal exposures. Treat tar and asphalt injuries with immediate cooling and cold water, and application of ointment combined with surface-active agents (e.g., Polysorbate 80).

5. Prophylactic antibiotics are not generally indicated. Corticosteroid use is not indicated.

6. Admit symptomatic patients, those exposed to hydrocarbons capable of producing delayed toxicity (e.g., halogenated hydrocarbons), and those exposed to hydrocarbons with toxic additives (e.g., pesticides or organic metal compounds). Patients with severe respiratory distress may need

111_Cydulka_Ch111.indd 622 20/03/17 4:46 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 638–641.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 111: Hydrocarbons and Volatile Substances

Source pages 642–643 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 642–643

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Hydrocarbons and Volatile Substances**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 111: Hydrocarbons and Volatile Substances 623 Table 111-2 Management of Hydrocarbon Exposures Airway and Secure airway. [Source pages 642–643]

- breathing Antidotes: Administer oxygen for carboxyhemoglobinemia and methy- lene blue for methemoglobinemia. [Source pages 642–643]

- Ventilatory support: Provide positive end-expiratory pressure or continuous positive airway pressure as needed to achieve adequate oxygenation. [Source pages 642–643]

- Cardiac Circulation: Administer IV crystalloid fluid for initial volume resuscitation of hypotensive patients. [Source pages 642–643]

- Do not use catecholamines in cases of halogenated hydrocarbon exposure. [Source pages 642–643]

Clinical reasoning structure

- **Problem representation:** Hydrocarbons and Volatile Substances; source pages 642–643.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Hydrocarbons and Volatile Substances**.

- Source-page range: **642–643**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 642

CHAPTER 111: Hydrocarbons and Volatile Substances 623

Table 111-2 Management of Hydrocarbon Exposures

Airway and Secure airway.

breathing Antidotes: Administer oxygen for carboxyhemoglobinemia and methy-

lene blue for methemoglobinemia.

Provide supplemental oxygen if wheezing.

Administer inhaled β2-agonists if wheezing.

Ventilatory support: Provide positive end-expiratory pressure or

continuous positive airway pressure as needed to achieve adequate

oxygenation.

Cardiac Circulation: Administer IV crystalloid fluid for initial volume

resuscitation of hypotensive patients.

Do not use catecholamines in cases of halogenated hydrocarbon

exposure.

Consider propranolol, esmolol, or lidocaine for ventricular dysrhythmias

induced by halogenated hydrocarbon exposure.

Consult the poison control center, toxicologist, and other appropriate

specialists as needed.

Decontamination Dermal: Remove hydrocarbon-soaked clothes, decontaminate skin with

soap and water, and decontaminate eyes with saline irrigation.

GI: Not indicated.

Other Laboratory tests: Order CBC, basic metabolic panel, liver function tests

(serum transaminase, bilirubin, albumin levels), prothrombin time,

partial thromboplastin time, carboxyhemoglobin level, methemoglobin

level, and/or radiologic studies as indicated (see the text).

Correct electrolyte abnormalities.

Do not give steroids.

Administer blood products as needed.

intubation and mechanical ventilation with high levels of PEEP. Patients exposed to nonhalogenated aliphatic hydrocarbons that are asymptomatic and with a normal chest radiograph may be discharged with return precautions after 6 to 8 hours of observation.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 199, "Hydrocarbons and Volatile Substances," by C. William Heise and Frank LoVecchio.

111_Cydulka_Ch111.indd 623 20/03/17 4:46 PM

Source page 643

C h a p ter Caustics

112 Jennifer Cullen

Caustics are substances that can cause histological and functional damage on contact and include both alkalis (pH >7) and acids (pH <7). The most common alkali exposure is household bleach (sodium hypochlorite with hydroxide), which is usually benign except in intentional ingestions. The most common acid exposures are drain cleaners (sulfuric acid), automobile batteries, and masonry cleaners (hydrochloric acid).

■■ CLINICAL FEATURES

Common features of caustic ingestions include dysphagia, odynophagia, epigastric pain, and vomiting with gastrointestinal (GI) tract injuries. Dysphonia, stridor, and respiratory distress can be seen with laryngotracheal injury. Esophageal injuries are graded by direct endoscopic visualization: (1) edema and hyperemia; (2) ulcerations, blisters, and exudates (2a—noncircumferential; 2b—circumferential); (3) deep ulceration and necrosis. Intentional ingestions are associated with higher-grade injury that can lead to the development of strictures. Most ingestions with serious injury are symptomatic with stridor, drooling, or vomiting, although distal GI injury without oral or facial burns is possible. Disc battery ingestions are often asymptomatic, though batteries >15 mm in diameter can become lodged in the esophagus and cause pressure necrosis.

Caustic exposures to the cornea are particularly serious if they involve alkalis. Dermal exposures to caustics usually produce only local pain and irritation. However, alkali and sodium hydrofluoric acid burns can penetrate deeply. Alkali exposures lead to liquefactive necrosis, which is more severe than the coagulation necrosis caused by most acids. Hydrofluoric acid can cause systemic hypocalcemia, hypomagnesemia, and hyperkalemia with subsequent ventricular dysrhythmias.

■■ DIAGNOSIS AND DIFFERENTIAL

The diagnosis is clinical. Routine laboratory tests are recommended for

severely affected patients and include electrolytes, assessment of acid–base status, and monitoring for potential gastrointestinal bleeding. Monitor serum calcium and magnesium levels and perform an ECG in patients with hydrofluoric acid exposures, especially ingestions. Consider chest and/or abdominal radiographs in symptomatic caustic ingestions to assess for free air or to investigate for foreign body in cases of suspected disc battery ingestion. Noncontrast CT of the chest and abdomen may be useful if perforated viscus is suspected, especially after ingestion of strong acids. Early endoscopic evaluation (ideally <12 hours after ingestion) is indicated for intentional caustic ingestions, and unintentional cases presenting with stridor, oral burns, vomiting, drooling, or inability to tolerate oral intake.

624

112_Cydulka_Ch112.indd 624 20/03/17 4:56 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 642–643.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 112: Caustics

Source pages 644–645 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 644–645

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Caustics**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 112: Caustics 625 ■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION Focus treatment on decontamination, early anticipatory airway manage- ment, stabilization of hemodynamic status, and delineation of extent of injury. [Source pages 644–645]

- Remove contaminated clothing and irrigate exposed skin with copious amounts of water. [Source pages 644–645]

- Alkali burns may require local debridement and re- moval of devitalized tissue followed by additional irrigation. [Source pages 644–645]

- Perform aggressive ocular decontamination with normal saline for a minimum of 15 minutes with frequent monitoring of ocular pH until a pH of 7.5 to 8.0 is achieved. [Source pages 644–645]

- Gastric decontamination in the form of activated charcoal, ipecac, or gastric lavage is contraindicated. [Source pages 644–645]

Clinical reasoning structure

- **Problem representation:** Caustics; source pages 644–645.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Caustics**.

- Source-page range: **644–645**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 644

CHAPTER 112: Caustics 625

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

Focus treatment on decontamination, early anticipatory airway management, stabilization of hemodynamic status, and delineation of extent of injury.

Decontamination

1. Remove contaminated clothing and irrigate exposed skin with copious amounts of water. Alkali burns may require local debridement and removal of devitalized tissue followed by additional irrigation.

2. Perform aggressive ocular decontamination with normal saline for a minimum of 15 minutes with frequent monitoring of ocular pH until a pH of 7.5 to 8.0 is achieved.

3. Gastric decontamination in the form of activated charcoal, ipecac, or gastric lavage is contraindicated.

Supportive Care

1. Perform early oral intubation with direct visualization in symptomatic patients with stridor, significant drooling, or dysphonia. Blind nasotracheal intubation is contraindicated.

2. Obtain IV access and administer isotonic IV fluids for hypotension.

3. Obtain surgical consultation for suspected or confirmed peritonitis or free air.

Special Considerations

1. Treat hydrofluoric acid dermal exposures with topical calcium gluconate gel (3.5 g mixed with 150 mL water-soluble lubricant). Consider intradermal 5% calcium gluconate for large burns and calcium gluconate infusion into the radial artery over 4 hours or given as a Bier block (10 mL of 10% calcium gluconate in 40 mL saline or 5% dextrose) for refractory distal extremity burns.

2. Oral ingestions of hydrofluoric acid within an hour can be suctioned via nasogastric tube followed by instillation of up to 300 mL of 10% cal-

cium gluconate. High doses of IV calcium and magnesium may be needed to treat systemic deficiencies and dysrhythmias.

3. Disc batteries lodged in the esophagus require emergent endoscopic removal.

4. Laundry detergent pod ingestions in young children may produce serious toxicity including caustic injury, respiratory distress, and neurologic depression. Treat supportively with airway protection as needed.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 200, "Caustic Ingestions," by Nicole C. Bouchard and Wallace A. Carter.

112_Cydulka_Ch112.indd 625 20/03/17 4:56 PM

Source page 645

C h a p te r Pesticides

113 Charles W. O'Connell

Pesticides include insecticides, herbicides, and rodenticides. In addition to active toxic ingredients, many pesticides also contain "inert" products such as petroleum distillates, which may be toxic as well. Although the mainstay of treatment is supportive care, some antidotes are essential.

■■ INSECTICIDES

Clinical Features

Organophosphate insecticides include agents such as diazinon, acephate, malathion, parathion, and chlorpyrifos. Absorption occurs through ingestion, inhalation (e.g., nerve gas agents), and dermal routes. Toxicity is produced through binding and inhibition of acetylcholinesterase, causing excess accumulation of acetylcholine and stimulation of cholinergic receptors, of both the muscarinic and nicotinic receptor types. The muscarinic receptor agonism results in a cholinergic crisis known as "SLUDGE" and "DUMBELS" effects (Table 113-1).

Onset of symptoms, ranging from minutes to hours, varies based on amount and route of exposure. Most patients become symptomatic within 8 hours of acute exposure, though some fat-soluble agents (e.g., fenthion) can cause delayed symptoms with dermal exposure. Nicotinic receptor stimulation leads to fasciculations and muscle weakness, which is most pronounced in the respiratory muscles, which leads to worsening of pulmonary dysfunction caused by bronchorrhea from muscarinic effects. Nicotinic effects can also cause tachycardia and mydriasis, paradoxical to the

Table 113-1 SLUDGE, DUMBELS, and "Killer Bees" Mnemonics for the Cholinergic Effects of Cholinesterase Inhibition

S Salivation

L Lacrimation

U Urinary incontinence

D Defecation

G GI pain

E Emesis

D Defecation

U Urination

M Muscle weakness, miosis

B Bradycardia, bronchorrhea, bronchospasm

E Emesis

L Lacrimation

S Salivation

Killer Bees Bradycardia, bronchorrhea, bronchospasm

626

113_Cydulka_Ch113.indd 626 20/03/17 5:01 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.

- Content ID: SEG-TINTINALLI-EMM8-FULL.

- Source pages: 644–645.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 113: Pesticides

Source pages 646–651 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 646–651

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Pesticides**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 113: Pesticides 627 expected muscarinic cholinergic effects. [Source pages 646–651]

- Central nervous system (CNS) effects, which often predominate in children, include tremor, restlessness, confusion, seizures, and coma. [Source pages 646–651]

- A variety of subacute and chronic effects are associated with organo- phosphate insecticide poisoning. [Source pages 646–651]

- An intermediate syndrome, 1 to 5 days after acute poisoning, may present with paralysis or weakness of neck, facial, and respiratory muscles, which can result in respiratory arrest if not treated. [Source pages 646–651]

- Organophosphate-induced delayed neuropathy can occur 1 to 3 weeks after acute poisoning, resulting in a distal motor-sensory polyneu- ropathy with leg weakness and paralysis. [Source pages 646–651]

Clinical reasoning structure

- **Problem representation:** Pesticides; source pages 646–651.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Pesticides**.

- Source-page range: **646–651**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 646

CHAPTER 113: Pesticides 627

expected muscarinic cholinergic effects. Central nervous system (CNS) effects, which often predominate in children, include tremor, restlessness, confusion, seizures, and coma.

A variety of subacute and chronic effects are associated with organophosphate insecticide poisoning. An intermediate syndrome, 1 to 5 days after acute poisoning, may present with paralysis or weakness of neck, facial, and respiratory muscles, which can result in respiratory arrest if not treated. Organophosphate-induced delayed neuropathy can occur 1 to 3 weeks after acute poisoning, resulting in a distal motor-sensory polyneuropathy with leg weakness and paralysis.

Diagnosis and Differential

Organophosphate poisoning is typically a clinical diagnosis based on the toxidrome and history of exposure (Table 113-1); laboratory cholinesterase assay can aid in confirmation. An ECG may be useful to monitor for prolonged QT, which is associated with increased morbidity and mortality in organophosphate poisoning.

Emergency Department Care and Disposition

Treatment of organophosphate poisoning is listed in Table 113-2, and should not be delayed pending confirmatory tests.

1. In symptomatic patients, administer 100% oxygen and focus on airway management, with gentle suctioning of secretions. Nondepolarizing agents should be used for rapid sequence intubation. The use of succinylcholine for intubation, which is metabolized by plasma cholinesterase, may result in prolonged paralysis.

2. The key in treatment is large amounts of atropine titrated to attenuation of tracheobronchial secretions. Tachycardia and dilated pupils are not contraindications to additional atropine. Of note, atropine will only reverse the muscarinic effects, but not the nicotinic effects of excess

Table 113-2 Treatment for Organophosphate Poisoning

Decontamination Protective clothing must be worn to prevent secondary poisoning of health care workers

Handle and dispose of all clothes as hazardous waste. Wash patient with soap and water

Handle and dispose of water runoff as hazardous waste

Monitoring Cardiac monitor, pulse oximeter, 100% oxygen

Gastric lavage No proven benefit

Activated charcoal No proven benefit

Urinary alkalinization No proven benefit

Atropine 1 mg or more IV in an adult or 0.01 to 0.04 mg/kg (but never <0.1 mg) IV in children. Repeat every 5 min until tracheobronchial secretions attenuate

Pralidoxime 1 to 2 g for adults or 20 to 40 mg/kg (up to 1 g) in children, mixed with normal saline and infused IV over 5 to 10 min

Continuous infusion may be employed

Seizures Benzodiazepines

113_Cydulka_Ch113.indd 627 20/03/17 5:01 PM

Source page 647

628 SECTION 11: Toxicology and Pharmacology

Table 113-3 Nonorganophosphate Insecticide Poisoning

Agent	Example	Clinical Feature	Diagnosis	Treatment
Carbamates	Carbaryl, pirimicarb, propoxur, trimethacarb	Similar to organophosphates, but cholinesterase levels	Depressed cholinesterase levels	Atropine
Organochlorines	Chlordane, heptachlor, dieldrin, aldrin	Neurological: excitability, seizures	History, special lab cooling	Benzodiazepine, cooling
Pyrethroids	Pyrethrins	parasthesia, vomiting, and oropharyngeal irritation, allergic hypersensitivty	History	Bronchodilators and antihistamines, supportive
Neonicotinoids	Imidacloprid	Nausea, headache, sedation	History	Supportive
N,N-Dietyl-3-methylbenzamide	DEET	Seizures	History	Benzodiazepines

acetylcholine. Glycopyrrolate or high-dose diphenhydramine can be substituted for atropine if it is not available.

3. Minimal exposures require only 6 to 8 hours of observation. Recrudescence can occur due to reexposure to contaminated clothing, particularly

leather. Significant poisonings require intensive care monitoring.

4. Other commonly encountered pesticides and their treatment are listed in Table 113-3.

■■ HERBICIDES

Herbicides are agents used to kill weeds. In addition to intrinsic toxicity, they may also be packaged with surfactants or solvents with their own toxic effects. The most dangerous are bipyridyl herbicides, namely paraquat and diquat. Nonbipyridyl herbicides and their treatments are depicted in Table 113-4.

Table 113-4 Nonbipyridyl Herbicides

Agent	Example	Clinical Feature	Treatment
Chlorophenoxy phenoxyacetic acid	2,4-dichloro-	Mucous membrane irritation, pulmonary edema, muscle toxicity, hyperthermia	Urine alkalization Hemodialysis
Glyphosate		Mucous membrane irritation, erosions, multiorgan failure, respiratory distress	Observe asymptomatic patients for 6 h
Urea-substituted	Chlorimuron, diuron, ■uomenturon, isoproturon	Methemoglobinemia	Methylene blue
Organophosphates	Butiphos	SLUDGE	Same as pesticides
N,N-Dietyl-3-methylbenzamide	DEET	Seizures	Benzodiazepines

113_Cydulka_Ch113.indd 628 20/03/17 5:01 PM

Source page 648

Paraquat is especially toxic with lethality seen with oral doses of 10 to 20 mL in an adult and 4 to 5 mL in children of 20% solution. Both agents are toxic via inhalation, dermal exposure, or ingestion. Due to their caustic properties, ulceration of skin and mucous membranes and gastrointestinal corrosion can occur. Paraquat ingestions result in renal, cardiac, and hepatic failure along with progressive pulmonary fibrosis from oxidative damage. Due to the latter, treatment often includes restriction of supplemental oxygen except for severe hypoxia. Decontamination of skin is important to prevent continued absorption. Early after ingestion, GI decontamination with activated charcoal, fuller's earth, or bentonite may be helpful. Charcoal hemoperfusion may remove paraquat. Because of poor prognosis, especially with paraquat ingestions, admit patients with ingestion for further observation and treatment.

■■ RODENTICIDES

The most commonly used rodenticides are superwarfarins, including brodifacoum, difenacoum, and bromadiolone. Coagulopathy typically develops within 48 hours and lasts for weeks to months due to the long half-lives of these agents. Single small ingestions in children usually do not result in toxic effects. Acute intentional or repeated ingestions may present with delayed hemorrhage, including hematuria, gastrointestinal hemorrhage, or epistaxis. Screening for toxic effect can be performed with an INR obtained 24 to 48 hours after ingestion. If the INR is elevated (>2), then start oral vitamin K1, typically at doses of 20 mg/d in adults (1 to 5 mg for children), which can be continued for up to 10 months until the anticoagulant effect is cleared. Acute hemorrhage requires more aggressive therapy with volume replacement, fresh frozen plasma or four-factor prothrombin complex, IV vitamin K1, and supportive care. Nonanticoagulant rodenticides are described in Table 113-5.

Table 113-5 Nonanticoagulant Rodenticides

Agent	Toxicity	Clinical Effects	Treatment
Arsenic	Severe	Vomiting and diarrhea, cardiovascular collapse	Supportive care, BAL for acute poisoning, Succimer
Barium carbonate	Severe	Vomiting and diarrhea, dysrhythmia, respiratory failure, hyopkalemia, paralysis	Gastric lavage with sodium or magnesium sulfate; potassium replacement
White phosphorus	Severe	Burns, cardiovascular collapse	Lavage with potassium permangante
N-3-Pyridylmethyl-N’-p-nitrophenyl urea (Vacor)	Severe	GI symptoms, insulin de■cient hyperglycemia and DKA	Nicotinamide IV
Sodium ■uroacetate	Severe	Vomiting, respiratory depression, impaired consciousness, twitching, seizure	Supportive care
Strychnine	Severe	Awake seizure like activity	Benzodiazepines

(Continued)

113_Cydulka_Ch113.indd 629 20/03/17 5:01 PM

Source page 649

Table 113-5 Nonanticoagulant Rodenticides (Continued)

Agent	Toxicity	Clinical Effects	Treatment
Thallium	Severe	Early GI irritation, respiratory failure and dysrhythmias	Oral Prussian blue

Metal phosphides Severe Vomiting, shock, Intragastric alkalization

hypocalcemia

α-Napthyl-thiourea Moderate Pulmonary edema Supportive care

Cholecalciferol Moderate Hypercalcemia Saline, furosemide,

steroids

Bromethalin Low Tremors, focal seizures Benzodiazepines

Norbormide Low Vasoconstrictive tissue Supportive care

hypoxia

Red squill Low Vomiting, diarrhea, hyper- Supportive care

kalemia, heart block with

ventricular dysrhythmias

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study

Guide, 8th ed., see Chapter 201, "Pesticides," by Burillo-Putze Guillermo and

Xarau Santiago Nogue.

113_Cydulka_Ch113.indd 630 20/03/17 5:01 PM

References/provenance

- Source: *Tintinalli's Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 646–651.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 114: Metals and Metalloids

Source pages 652–657 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 652–657

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Metals and Metalloids**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 114: Metals and Metalloids 633 Table 114-3 Guidelines for Chelation Therapy in Arsenic-Poisoned Patients Chelator Dose Dimercaprol 3–5 mg/kg IM every 4 h for 2 days, followed by 3–5 mg/kg IM every 6–12 h until able to switch to succimer Succimer 10 mg/kg PO every 8 h for 5 days, followed by 10 mg/kg PO every 12 h pointes have been reported. [Source pages 652–657]

- Chronic poisoning causes peripheral neuropa- thy, malaise, and confusion. [Source pages 652–657]

- Skin findings include alopecia, hyperpigmen- tation, keratoses, and transverse white nail lines (Mees’ lines). [Source pages 652–657]

- Diagnosis and Differential An exposure history is most useful in identifying arsenic poisoning. [Source pages 652–657]

- How- ever, consider the diagnosis in patients with hypotension preceded by pro- found vomiting and diarrhea. [Source pages 652–657]

Clinical reasoning structure

- **Problem representation:** Metals and Metalloids; source pages 652–657.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Metals and Metalloids**.

- Source-page range: **652–657**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 652

CHAPTER 114: Metals and Metalloids 633

Table 114-3 Guidelines for Chelation Therapy in Arsenic-Poisoned Patients

Chelator Dose

Dimercaprol 3–5 mg/kg IM every 4 h for 2 days, followed by 3–5 mg/kg IM every 6–12 h until able to switch to succimer

Succimer 10 mg/kg PO every 8 h for 5 days, followed by 10 mg/kg PO every 12 h

pointes have been reported. Chronic poisoning causes peripheral neuropathy, malaise, and confusion. Skin findings include alopecia, hyperpigmentation, keratoses, and transverse white nail lines (Mees' lines).

Diagnosis and Differential

An exposure history is most useful in identifying arsenic poisoning. However, consider the diagnosis in patients with hypotension preceded by profound vomiting and diarrhea. The CBC may demonstrate leukocytosis with acute poisoning. An ECG may show QT-interval prolongation. The diagnosis is confirmed by documenting an elevated 24-hour urine arsenic level. Other diagnoses to consider include gastroenteritis, septic shock, encephalopathy, peripheral neuropathy, Addison's disease, and lead, thallium, or mercury poisoning.

Emergency Department Care and Disposition

1. Support the ABCs: endotracheal intubation may be required to protect the airway; treat hypotension with volume resuscitation and vasopressors, but avoid overresuscitation that may result in pulmonary or cerebral edema. Manage dysrhythmias according to ACLS/PALS protocols, but avoid agents that prolong the QT interval (class IA, IC, and III antidysrhythmics). Magnesium sulfate, isoproterenol, and/or overdrive pacing should be considered for treatment of torsades de pointes.
2. Decontaminate the gastrointestinal tract with whole-bowel irrigation using polyethylene glycol solution in those with radiopaque GI fragments.
3. Treat symptomatic patients empirically with chelation therapy

(Table 114-3) using dimercaprol IM. Succimer PO is preferred for clinically stable patients who can tolerate oral intake.

4. Hospitalize acutely poisoned patients, patients requiring dimercaprol, or those with suspected suicidal or homicidal intent. Discharge stable patients with subacute or chronic poisoning if follow-up is ensured.

■■ MERCURY POISONING

Mercury poisoning can result from exposure to elemental, inorganic, or organic mercury compounds.

Clinical Features

Elemental mercury exposure is most likely to occur after contact with a broken thermometer, light bulb or other mercury spills. Mercury is primarily absorbed via inhalation (especially with heating or vacuuming). Ingestions of elemental mercury are nontoxic in those with normal GI tracts. Vapor exposure results in cough, fever, dyspnea, vomiting, and headache.

114_Cydulka_Ch114.indd 633 20/03/17 5:07 PM

Source page 653

Table 114-4 Guidelines for Chelation Therapy in Mercury-Poisoned Patients

	Elemental and Inorganic Mercury	Organic Mercury
Severe acute poisoning	Dimercaprol, 5 mg/kg IM every 4 h for 2 days, followed by 2.5 mg/kg every 8 h for 5 days, then IM every 6 h for 2 days, followed by every 12 h for 14 days	Succimer, 10 mg/kg PO every 4 h for 2 days, followed by every 12 h for 14 days
	2.5 mg/kg IM every 12–24 h until clinical improvement occurs or until able to switch to succimer therapy	
Mild acute poisoning and chronic poisoning	Succimer, 10 mg/kg PO every 8 h for 5 days, then every 12 h for 14 days	No proven benefit for chelation therapy

Acute lung injury can progress to respiratory failure. Classic findings include tremor, rash, and hypertension. Inorganic mercury is used as a disinfectant and in manufacturing. Ingestion of inorganic mercury results in corrosive injury to the GI tract, with vomiting, diarrhea, abdominal pain, and GI bleeding early, followed by acute kidney injury. Organic mercury is found in some fungicides and pesticides and can be absorbed when ingested. Poisoning tends to occur with chronic exposures and results in profound central nervous system dysfunction.

Diagnosis and Differential

A history of exposure to mercury is key to diagnosis and is confirmed by an elevated 24-hour urine mercury level when toxicity is due to elemental or inorganic mercury; an elevated whole blood mercury level is necessary in cases of organic mercury exposure. The differential diagnosis is exten-

sive and includes causes of metal fume fever (elemental), encephalopathy or tremor (elemental/inorganic/organic), thyroid disease, metabolic encephalopathy, stroke, and tumor. Consider alternative causes of corrosive gastroenteritis (ingestion of iron, arsenic, phosphorus, acids, and alkalis) if mercury salt ingestion is suspected.

Emergency Department Care and Disposition

1. Support the ABCs with airway management and crystalloid infusion.
2. Consider gastric lavage in cases of inorganic mercury ingestion or activated charcoal in cases of organic mercury ingestion.
3. Begin chelation therapy prior to confirming the diagnosis (Table 114-4). Dimercaprol is preferred for elemental and inorganic mercury poisonings. Succimer is the agent of choice in organic mercury poisonings and in cases of mild or chronic elemental/inorganic mercury toxicity.
4. Admit patients with respiratory symptoms following elemental mercury vapor exposure, those with inorganic mercury ingestions, or patients requiring dimercaprol chelation. Ensure prompt environmental assessment in cases of elemental mercury vapor exposures to prevent further toxicity.

■■ POISONING WITH OTHER METALS

Less common toxic heavy metals include bismuth, cadmium, chromium, cobalt, copper, silver, thallium, and zinc. Unique manifestations and treatments of these exposures are outlined in Table 114-5.

114_Cydulka_Ch114.indd 634 20/03/17 5:07 PM

Source page 654

Table 114-5 Miscellaneous Metal Poisoning: Unique Manifestations and Treatments of Patients Poisoned by Less Common Metals

Metal	Poisoning Source	Acute Clinical Manifestations	Chronic Clinical Manifestations	Specific Treatment
Bismuth	Antidiarrheals (bismuth subsalicylate), impregnated surgical packing paste	Abdominal pain, encephalopathy (limited evidence)	Myoclonic kidney injury evidence)	Dimercaprol (limited evidence; not generally indicated)
Cadmium	Contaminated soil in cadmium-rich areas; alloys used in welding, solder-ing, jewelry, and lung cancer	Ingestion: hemorrhagic gastroenteritis (itai-itai or ouch-monitis, acute lung injury (questionable) Pneumonitis:	Ingestion: suc-cimer (limited evidence) Proteinuria, osteomalacia disease), generally indicated	

batteries chelation not
indicated
Chromium Corrosion inhibi- Skin irritation Mucous mem- Acetylcysteine
tors (e.g., heat- and ulceration, brane irritation, (animal studies
ing systems), contact dermati- perforation of suggest efficacy
pigment produc- tis; GI irritation, nasal septum, as chelator)
tion, leather renal and pulmo- chronic cough,
tanning, metal nary failure contact derma-
finishing, dietary titis, skin ulcers
supplements, (“chrome holes”),
prosthetic joints lung cancer
Cobalt “Hard metal Contact dermati- Hard metal lung Acetylcysteine
dust” (tungsten– tis, asthma disease (spec- (animal studies
cobalt mixture), trum ranging suggest efficacy
flexible mag- from alveolitis to as chelator)
nets, drying fibrosis), cardio-
agents, pros- myopathy, thyroid
thetic joints hyperplasia
Copper Leaching from Ingestion: Hepatotoxicity Dimercaprol for
copper pipes resembles (childhood cirrho- hepatic or hema-
and containers; iron poisoning; sis or idiopathic tologic toxicity
fungicide (cop- blue vomitus copper toxicosis) Succimer in mild
per sulfate); (copper salts), poisoning
welding (copper hepatotoxicity,
oxide) hemolysis, met-
hemoglobinemia
Inhalation: metal
fume fever (self-
limited fever,
chills, cough,
dyspnea)
(Continued)
114_Cydulka_Ch114.indd 635 20/03/17 5:07 PM

Source page 655

636 SECTION 11: Toxicology and Pharmacology
Table 114-5 Miscellaneous Metal Poisoning: Unique Manifestations and
Treatments of Patients Poisoned by Less Common Metals (Continued)
Poisoning Acute Clinical Chronic Clinical Specific
Metal Source Manifestations Manifestations Treatment
Silver Colloidal (metal- Mucosal irrita- Argyria (perma- Selenium

lic) silver used tion (silver oxide nent skin discol- (possible role)
for medicinal and nitrate) oration due to
purposes as silver deposition
oral solutions, and melanocyte
aerosols, and stimulation)
douches; cau-
terizing and
antiseptic agent
(silver nitrate);
jewelry, wire
Thallium Rodenticides Early: nausea, Sensorimotor Multidose acti-
(use prohibited vomiting, neuropathy, psy- vated charcoal
in the United abdominal pain, chosis, dermati- Prussian
States); con- tachycardia tis, hepatotoxicity blue, 125 mg/
taminated herbal Intermediate kg PO every
products; medi- (>24 h): painful 12 h (usually
cal radioisotope ascending neu- dissolved in
(miniscule dose); ropathy, cardiac 50 mL of 15%
most poison- dysrhythmias, mannitol)
ings related to altered mental
homicide status
Delayed
(2 weeks):
alopecia
Zinc Smelting, Ingestion: nau- Copper defi- Edetate calcium
electroplating, sea, vomiting, ciency, sidero- disodium
military smoke abdominal pain blastic anemia, Supportive care
bombs, zinc loz- (resembles iron neutropenia for metal fume
enges, welding/ poisoning) fever
galvanizing (zinc Inhalation:
oxide) mucosal irrita-
tion, metal fume
fever (zinc oxide)

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 203, “Metals and Metalloids,” by Heather Long and Lewis S. Nelson.

114_Cydulka_Ch114.indd 636 20/03/17 5:07 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.

- Content ID: SEG-TINTINALLI-EMM8-FULL.

- Source pages: 652–657.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 115: Industrial Toxins

Source pages 658–663 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 658–663

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Industrial Toxins**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 115: Industrial Toxins 639 produced, leading to irritation of the mucous membranes of the eyes and upper airway. [Source pages 658–663]

- With increasing exposure, the more distal airways are affected, leading to pulmonary edema and acute respiratory distress syndrome, which may be delayed up to 6 hours. [Source pages 658–663]

- Treat respiratory symptoms with humidified oxygen and bronchodilators. [Source pages 658–663]

- Nebulized sodium bicarbonate may be of benefit within 4 hours of exposure. [Source pages 658–663]

- Consider steroids, either intravenous or inhaled. [Source pages 658–663]

Clinical reasoning structure

- **Problem representation:** Industrial Toxins; source pages 658–663.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Industrial Toxins**.

- Source-page range: **658–663**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 658

CHAPTER 115: Industrial Toxins 639

produced, leading to irritation of the mucous membranes of the eyes and upper airway. With increasing exposure, the more distal airways are affected, leading to pulmonary edema and acute respiratory distress syndrome, which may be delayed up to 6 hours.

Emergency Department Care and Disposition

1. Treat respiratory symptoms with humidified oxygen and bronchodilators.
2. Nebulized sodium bicarbonate may be of benefit within 4 hours of exposure.
3. Consider steroids, either intravenous or inhaled. However, there is insufficient evidence of improved outcomes with steroid therapy.
4. Evaluate for dermal and corneal burns in patients with significant exposure.
5. Monitor patients with moderate symptoms for 24 hours.

■■ NITROGEN DIOXIDE

Clinical Features

Nitrogen dioxide and nitrogen oxides are typically encountered in silo gas and military ordinance, and are formed in any instance of air-based combustion. A triphasic clinical course is often observed, manifested initially as dyspnea and flu-like symptoms, followed by transient improvement, then increasing dyspnea 12 hours post-exposure due to pulmonary edema. Methemoglobinemia has also been reported. Diagnosis is typically made by clinical history.

Emergency Department Care and Disposition

1. Initial treatment is supportive, ensuring airway integrity and adequate oxygenation/ventilation.
2. Give supplemental humidified oxygen and bronchodilators if bronchospasm is present.
3. Because the respiratory effects of NO2 exposure are delayed, patients may present with minimal symptoms despite significant exposure, and early intubation should be considered.
4. Early corticosteroids may reduce the risk of delayed pulmonary edema.
5. Most patients require hospitalization for 24 to 48 hours.

■■ AMMONIA

Clinical Features

Ammonia is found in many household chemicals, fertilizers, and plastic production. It has a characteristic pungent odor and rapidly irritates the mucous membranes of the upper airway. Lower airway injury may occur with massive exposure. Diagnosis is typically made by clinical history.

Emergency Department Care and Disposition

1. Treat exposed patients supportively with humidified oxygen, bronchodilators, and anticholinergics.

2. Provide copious irrigation and corneal burn evaluation for patients with eye discomfort.

115_Cydulka_Ch115.indd 639 20/03/17 5:17 PM

Source page 659

640 SECTION 11: Toxicology and Pharmacology

■■ CYANIDE

Clinical Features

Cyanide is an extremely toxic chemical generated during the combustion of synthetic materials, wool, and silk. It is used in vermin extermination, precious metal reclamation, and chemical laboratories. It is also found naturally in seeds of the Prunus species. Toxicity is caused by inhibition of cytochrome oxidase, which subsequently disrupts oxidative phosphorylation and ATP production. Patients with significant exposure present with altered consciousness, hyperventilation, hypotension, and bradycardia. They will typically have normal oxygen saturation on co-oximetry. A smell of bitter almonds may be detected. Cherry-red skin is an uncommon finding.

Diagnosis and Differential

A severe, unexplained anion gap metabolic acidosis is characteristic of cyanide exposure and is due to lactic acid production from anaerobic metabolism. The PaO2 of venous blood will approach that of arterial blood as a result of impaired tissue oxygen consumption. An elevation of carboxyhemoglobin may also be seen in fire victims. Whole blood cyanide levels confirm exposure, but are not rapidly available. The diagnosis is made clinically and treatment initiated as soon as possible.

Emergency Department Care and Disposition

1. Treat cyanide poisoning with aggressive supportive care and antidotal therapy, outlined in Tables 115-2 and 115-3. Nitrites induce the formation of methemoglobin, which binds and removes cyanide to allow reactivation of oxidative phosphorylation. Sodium thiosulfate provides a sulfate group that is transferred to cyanide to create less toxic thiocyanate. Hydroxocobalamin directly binds cyanide, removing it from

cytochrome oxidase, and forming cyanocobalamin. Nitrite therapy is
not contraindicated in patients with hypotension from severe cyanide

Table 115-2 Treatment of Cyanide Poisoning in Adults

100% supplemental oxygen

IV crystalloids and vasopressors for hypotension

Sodium bicarbonate for acidemia

AND

Cyanide antidote kit Amyl nitrite inhaler; crack vial and inhale over 30 s.*

Sodium nitrite 3% solution: 10 mL (300 mg) IV given
over no less than 5 min.†

Sodium thiosulfate 25% solution: 50 mL (12.5 g) IV.
Repeat sodium thiosulfate once at half dose (25 mL) if
symptoms persist.

OR

Hydroxocobalamin 5 g IV over 15 min.
If needed, may repeat 5 g for a total of 10 g.

*
Not necessary if IV is in place.

†
Avoid nitrites in the presence of severe hypotension if diagnosis is unclear; consider sodium thiosulfate
or hydroxocobalamin.

115_Cydulka_Ch115.indd 640 20/03/17 5:17 PM

Source page 660

CHAPTER 115: Industrial Toxins 641

Table 115-3 Treatment of Cyanide Poisoning in Children

100% oxygen

IV crystalloids and vasopressors for hypotension

Sodium bicarbonate for acidemia

AND

Cyanide antidote kit Amyl nitrite inhaler; crack vial and hold in front of nose
for 15–30 s.*,†

Sodium nitrite 3% solution: adjusted according to
hemoglobin level, given IV over no less than 5 min‡
(monitor methemoglobin level <30%)

Hemoglobin (g/100 mL) Sodium Nitrite 3% Solution (mL/kg)

7 0.19

8 0.22

9 0.25

10 0.27

11 0.30

12 0.33

13 0.36

14 0.39

Sodium thiosulfate 25% solution: 1.65 mL/kg IV.

Repeat sodium thiosulfate once at half dose (0.825 mL/kg) if symptoms persist.

*

Not necessary if IV is in place.

†

Consider withholding nitrites if suspected concomitant carbon monoxide poisoning.

‡

Avoid nitrites in the presence of severe hypotension if diagnosis is unclear.

toxicity. However, nitrites are relatively contraindicated in hypotensive patients with an uncertain exposure history and in victims of smoke inhalation who may have concomitant carbon monoxide poisoning. The latter contraindication is due to reduced oxygen delivery from induced methemoglobinemia. In these instances, hydroxocobalamin is the ideal antidote; however, sodium thiosulfate may also be used while omitting the use of nitrites from the cyanide antidote kit.

2. Consider activated charcoal in patients with a history of cyanide ingestion who present within 1 hour, are alert, and have a patent airway.

3. Admit patients receiving antidotal therapy or who may have ingested a substance with the potential for delayed toxicity.

■■ HYDROGEN SULFIDE

Clinical Features

Hydrogen sulfide is a colorless, flammable gas found in petroleum processing, as a byproduct of organic matter decomposition, and can be made from household products. The mechanism of toxicity is the same as cyanide, except that the inhibition of oxidative phosphorylation quickly ceases when removed from exposure. A characteristic rotten egg's odor is frequently described. Mucous membrane irritation is common, and with significant

115_Cydulka_Ch115.indd 641 20/03/17 5:17 PM

Source page 661

642 SECTION 11: Toxicology and Pharmacology

exposure, rapid loss of consciousness, seizures, delayed pulmonary edema, corneal injury, and death may occur.

Emergency Department Care and Disposition

1. Removal from the source of hydrogen sulfide is the most important aspect of care, followed by aggressive supportive care with humidified oxygen and decontamination.

2. Treat patients with persistent mental status changes or cardiovascular

instability with sodium nitrite.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study

Guide, 8th ed., see Chapter 204, “Industrial Toxins,” by Chip Gresham and Frank

LoVecchio

115_Cydulka_Ch115.indd 642 20/03/17 5:17 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.

- Content ID: SEG-TINTINALLI-EMM8-FULL.

- Source pages: 658–663.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 116: Vitamins and Herbals

Source pages 664–665 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 664–665

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Vitamins and Herbals**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 116: Vitamins and Herbals 645 Table 116-2 Commonly Used Herbal Agents and Potential Adverse Effects (Continued) Agent General Use Adverse Effect Wormwood Dyspepsia Absinthism: restlessness, vertigo, tremor (Absinthe)† paresthesias, delirium Yohimbine† Aphrodisiac Hallucinations, weakness, hypertension, paralysis Drug interactions: clonidine, cyclic antidepressants * Considered generally safe; rare adverse effects. [Source pages 664–665]

- † Increased potential for toxicity, adverse effects more common. [Source pages 664–665]

- ■■ DIAGNOSIS AND DIFFERENTIAL Diagnosis is usually made clinically. [Source pages 664–665]

- A history of massive acute inges- tion or chronic supratherapeutic use should be sought. [Source pages 664–665]

- Laboratory studies that may be helpful include a complete blood count, basic metabolic panel, hepatic enzymes, coagulation studies, bleeding time, creatine phosphokinase, toxicology screen, and urine pregnancy test. [Source pages 664–665]

Clinical reasoning structure

- **Problem representation:** Vitamins and Herbals; source pages 664–665.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Vitamins and Herbals**.

- Source-page range: **664–665**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 664

CHAPTER 116: Vitamins and Herbals 645

Table 116-2 Commonly Used Herbal Agents and Potential Adverse Effects (Continued)

Agent	General Use	Adverse Effect
Wormwood (Absinthe)†	Dyspepsia	Absinthism: restlessness, vertigo, tremor paresthesias, delirium
Yohimbine†	Aphrodisiac	Hallucinations, weakness, hypertension, paralysis
		Drug interactions: clonidine, cyclic antidepressants

*

Considered generally safe; rare adverse effects.

†

Increased potential for toxicity, adverse effects more common.

■■ DIAGNOSIS AND DIFFERENTIAL

Diagnosis is usually made clinically. A history of massive acute ingestion or chronic supratherapeutic use should be sought. Laboratory studies that may be helpful include a complete blood count, basic metabolic panel, hepatic enzymes, coagulation studies, bleeding time, creatine phosphokinase, toxicology screen, and urine pregnancy test. An ECG may be indicated if signs of sympathomimetic stimulation are present.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

Basic supportive care and discontinuation of the vitamin or herbal preparation are usually all that are needed to treat mild toxicity.

1. Consider activated charcoal 1 g/kg PO for large vitamin A or vitamin D ingestions.
2. Treat hypercalcemia from vitamin A or D overdose with normal saline, loop diuretics, and corticosteroids (to reduce GI absorption). Refractory cases may require adjunctive treatments, such as bisphosphonates and

calcitonin.

3. Consider diagnostic and therapeutic lumbar puncture to treat idiopathic intracranial hypertension (pseudotumor cerebri) from hypervitaminosis A.

4. Administer diphenhydramine 25 to 50 mg IV (1 mg/kg in children) or PO to patients with “niacin flush” symptoms.

5. Consider N-acetylcysteine 140 mg/kg PO or IV for treating severe hepatotoxicity from herbal preparations such as pennyroyal oil.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 205, “Vitamins and Herbals,” by Rick Tovar.

116_Cydulka_Ch116.indd 645 20/03/17 5:31 PM

Source page 665

C h a p ter Dyshemoglobinemias

117 Chulathida Chomchai

Dyshemoglobinemias result from the alteration of the hemoglobin molecule, which prevents it from carrying oxygen. Carboxyhemoglobin is created following exposure to carbon monoxide, which is usually considered an environmental emergency and is discussed in Chapter 127. Table 117-1 lists common pharmaceuticals capable of causing the formation of methemoglobin.

■■ CLINICAL FEATURES

Methemoglobinemia presents with grayish-brown discoloration of the skin that is recognized as cyanosis. Children up to the age of 4 months lack the enzyme activity that normally reduces methemoglobin, thus making them susceptible to oxidant stress-induced methemoglobinemia. Three scenarios occur with some frequency: children with acute gastroenteritis and increased nitrate production from bacteria in the GI tract; children exposed to nitrates in water of agricultural areas where fertilizer runoff contaminates water sources; and overconsumption of nitrogenous vegetables such as spinach.

In drug-induced methemoglobinemia, the slate-gray to blue discoloration of the skin is apparent with levels of 10% to 15%. Symptoms occur in proportion to declining oxygen delivery. Headache, nausea, and fatigue

Table 117-1 Drugs Commonly Implicated in Patients with Methemoglobinemia

Oxidant Comments

Analgesics Commonly reported

Phenazopyridine Rarely used

Phenacetin

Antimicrobials Common

Antimalarials Hydroxylamine metabolite formation is inhibited by

Dapsone cimetidine

Local anesthetics Most commonly reported of the local anesthetics

Benzocaine Rare

Lidocaine Common in topical anesthetics

Prilocaine Rare

Dibucaine

Nitrates/nitrites Cyanide antidote kit used to enhance sexual encounters

Amyl nitrite Used to enhance sexual encounters

Isobutyl nitrite Cyanide antidote kit

Sodium nitrite Cold packs

Ammonium nitrae Excessive topical use

Silver nitrate Problem in infants, due to nitrate fertilizer runoff

Well water Rare

Nitroglycerin

Sulfonamides Uncommon

Sulfamethoxazole

646

117_Cydulka_Ch117.indd 646 20/03/17 5:38 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 664–665.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 117: Dyshemoglobinemias

Source pages 666–669 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 666–669

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Dyshemoglobinemias**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 117: Dyshemoglobinemias 647 occur at lower levels (20% to 30%). [Source pages 666–669]

- Levels above 50% can cause loss of consciousness, myocardial ischemia, dysrhythmias, seizures, and metabolic acidosis. [Source pages 666–669]

- In patients with cardiopulmo- nary disease in which there is impaired oxygen delivery, the symptoms will be manifested at lower methemoglobin levels. [Source pages 666–669]

- ■■ DIAGNOSIS AND DIFFERENTIAL The diagnosis of methemoglobinemia should be considered in patients presenting with cyanosis that does not improve with administration of oxy- gen. [Source pages 666–669]

- During venipuncture, blood may appear chocolate brown, a visible effect that is easily identified when the blood is placed on filter paper with a normal patient’s blood for comparison. [Source pages 666–669]

Clinical reasoning structure

- **Problem representation:** Dyshemoglobinemias; source pages 666–669.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Dyshemoglobinemias**.

- Source-page range: **666–669**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 666

CHAPTER 117: Dyshemoglobinemias 647

occur at lower levels (20% to 30%). Levels above 50% can cause loss of consciousness, myocardial ischemia, dysrhythmias, seizures, and metabolic acidosis. Levels above 70% may be lethal. In patients with cardiopulmonary disease in which there is impaired oxygen delivery, the symptoms will be manifested at lower methemoglobin levels.

■■ DIAGNOSIS AND DIFFERENTIAL

The diagnosis of methemoglobinemia should be considered in patients presenting with cyanosis that does not improve with administration of oxygen. During venipuncture, blood may appear chocolate brown, a visible effect that is easily identified when the blood is placed on filter paper with a normal patient's blood for comparison. Levels are measured by co-oximetry on an arterial blood gas analyzer, with either an arterial or venous sample. Standard pulse oximetry will give an erroneously high oxygen saturation level at approximately 85% and does not change despite administration of 100% oxygen. Pulse co-oximeters are available that can noninvasively measure both methemoglobin and carboxyhemoglobin.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

Methemoglobinemia should be treated initially with close monitoring and high concentrations of inspired oxygen (Table 117-2). Methemoglobinemia at levels above 25% and symptomatic patients with lower levels should be treated with methylene blue. The initial dose of methylene blue is 1 to 2 mg/kg as a 10% solution IV, given over 15 minutes. Clinical improvement should occur within 20 minutes, after which the dose may be repeated if improvement has not occurred. Failure to respond to a second dose is usually due to one of five causes (in order or likelihood of occurrence):

1. Glucose-6-phosphate dehydrogenase deficiency (G6PD): Consider transfusion of packed red blood cells for severely elevated methemoglobin levels in patients with suspected G6PD deficiency.

2. Dapsone: Compounds with long half-life, such as dapsone, can produce prolonged oxidative stress. Activated charcoal should be given. Treat dapsone-induced methemoglobinemia with a repeated dose of methylene blue; consider the addition of IV cimetidine to impede the metabolism of dapsone to its oxidant metabolite, hydroxylamine.

Table 117-2 Management of Methemoglobinemia

Assess airway, breathing, and circulation

Place an IV line

Administer oxygen

Attach the patient to a cardiac and pulse oximetry monitor

Obtain an ECG

Decontaminate the patient as needed

Administer methylene blue—if symptomatic or methemoglobin >25%

Consider: cimetidine for patients taking dapsone

117_Cydulka_Ch117.indd 647 20/03/17 5:38 PM

Source page 667

3. NADPH-methemoglobin reductase deficiency: Patients with congenital absence of this enzyme are not chronically cyanotic, nor do they have resting methemoglobin levels above normal. However, they lack the ability to convert methylene blue to its active metabolite. As with G6PD deficiency, consider packed red cell or exchange transfusions for severe cases, especially those with hemolysis.

4. Methylene blue-induced hemolysis: Paradoxically, methylene blue can be a source of oxidant stress. Methylene blue doses, therefore, should not exceed 7 mg/kg/d.

5. Sulfhemoglobinemia: This rare drug-induced dyshemoglobinemia can occur with sulfur-containing pharmaceuticals and phenacetin. Patients appear cyanotic at sulfhemoglobin levels of 5%, and pulse oximetry may read in the 70% to 80% range, but are rarely symptomatic. Treat sulfhemoglobinemia with supplement oxygen.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 207, "Dyshemoglobinemias," by Brenna M. Farmer and Lewis S. Nelson.

117_Cydulka_Ch117.indd 648 20/03/17 5:38 PM

Source page 668

SECTION

Environmental Injuries

12

C h a p ter Cold Injuries

118 Gerald (Wook) Beltran

■■ NONFREEZING COLD INJURIES

Trench foot is a direct soft tissue injury that results from prolonged exposure to nonfreezing cold and moisture. The foot is initially pale, mottled, pulseless, and anesthetic and does not improve quickly with rewarming. Several hours after rewarming, the foot becomes hyperemic and painful as perfusion returns after 2 to 3 days. Bullae and edema are late findings. Anesthesia may persist for weeks or even permanently. Hyperhidrosis and sensitivity to cold are late features and may last for months to years. Chilblains (pernio) are painful inflammatory lesions typically affecting the ears, hands, and feet caused by chronic exposure to intermittent damp, nonfreezing conditions. Localized edema, erythema, and cyanosis appear up to 12 hours after the exposure and are accompanied by pruritis and burning paresthesias. Tender blue nodules may form after rewarming and can persist for several days. Treatment of trench foot and chilblains includes drying, elevation, warming, and bandaging of the affected body part. With chilblains, add nifedipine 20 mg PO three times daily, pentoxifyline 400 mg PO three times daily, or limaprost 20 μg PO three times daily, as well as topical corticosteroids, such as 0.025% fluocinolone cream.

■■ FROSTBITE

Clinical Features

Freezing of tissue causes frostbite. Patients initially complain of stinging, burning, and numbness. Frostbite injuries are classified by the depth of injury and amount of tissue damage based on appearance after rewarming. First-degree frostbite (frostnip) is characterized by partial thickness skin freezing, erythema, edema, lack of blistering, and no tissue loss. Second-degree frostbite is characterized by deeper, full-thickness skin freezing and results in the formation of clear bullae. The patient complains of numbness, followed by aching and throbbing. Deep cold injury or third-degree frostbite involves freezing of the skin and subdermal plexus leading to hemorrhagic bullae and skin necrosis. Fourth-degree frostbite extends into muscle,

649

118_Cydulka_Ch118.indd 649 22/04/17 4:39 pm

Source page 669

650 SECTION 12: Environmental Injuries

tendon, and bone with mottled skin, nonblanching cyanosis, and eventual dry, black, mummified eschar formation. Early injuries are better classified as superficial or deep because it is difficult to initially evaluate the depth of

injury. Laboratory testing and imaging are not needed to diagnose frostbite.

Treatment

1. Provide rapid rewarming in circulating water at 37°C to 39°C (98.6°F to 102.2°F) for 20 to 30 minutes until tissue is pliable and erythematous.
2. Debridement of clear blisters and aspiration of hemorrhagic blisters are controversial. Consult with a surgeon for local preference.
3. Apply topical aloe vera every 6 hours.
4. Provide pain management, local wound care, and dressing. Splint and elevate the affected extremities. Patients may require parenteral opioids initially, followed by oral NSAIDs.
5. Provide tetanus immunoprophylaxis, if needed.
6. Patients with superficial, local frostbite may be discharged home with close follow-up arranged.
7. Patients with deeper injuries require admission for ongoing care.
8. The use of prophylactic bacitracin ointment, antibiotics, and silver sulfadiazine is controversial.

■■ HYPOTHERMIA

Hypothermia, a core body temperature of <35°C (<95°F), results from heat loss through conduction, convection, radiation, or evaporation.

Clinical Features

The features of hypothermia represent a continuum. Mild hypothermia (32°C to 35°C [90°F to 95°F]) can present with tachycardia, tachypnea, hypertension, shivering, and a normal level of consciousness. As core temperatures fall below 32°C (90°F), shivering ceases and heart rate and blood pressure decrease. As temperature decreases further, patients become confused, lethargic, and then comatose. Pupillary reflexes are lost. Respiratory rate decreases, gag and cough reflexes are diminished, and bronchorrhea occurs. Aspiration is common. Impaired renal concentration results in a cold diuresis and hemoconcentration. The cardiac effects of worsening hypothermia progress from sinus bradycardia, to atrial fibrillation with slow ventricular response, to ventricular fibrillation, to asystole. At temperatures <32°C (<90°F), the risk for dysrhythmias increases. As core body temperature falls below 28°C (below 82°F), the risk for cardiac arrest increases as the potential for malignant cardiac dysrhythmias become more pronounced.

Diagnosis and Differential

The diagnosis of hypothermia is based on core temperature and may not be initially obvious, especially in cases where a history of prolonged environmental exposure is missing. Low-reading thermometers are required to measure and monitor temperature. Along with the history and physical exam, laboratory investigation is directed at determining the underlying cause and

complications. Helpful laboratory studies may include glucose, complete

blood count (CBC), electrolytes, thyroid-stimulating hormone, random

118_Cydulka_Ch118.indd 650 22/04/17 4:39 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 666–669.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 118: Cold Injuries

Source pages 670–675 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 670–675

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Cold Injuries**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 118: Cold Injuries 651 cortisol level, clotting profile, lactic acid, lipase, creatinine, blood gas, and electrocardiogram (ECG). [Source pages 670–675]

- Acid-base disorders are common, but do not follow a predictable pattern. [Source pages 670–675]

- Intravascular thrombosis, embolism, and disseminated intravascular coagulation (DIC) may occur. [Source pages 670–675]

- Electrocardiographic changes include PR, QRS, and QT prolongations, T-wave inversion, and a slow positive deflection at the end of the QRS (Osborn J wave). [Source pages 670–675]

- In addition to environmen- tal exposure, causes of hypothermia include hypoglycemia, hypothyroidism, hypoadrenalism, hypopituitarism, central nervous system (CNS) dysfunction, drug intoxication, sepsis, and dermal disease. [Source pages 670–675]

Clinical reasoning structure

- **Problem representation:** Cold Injuries; source pages 670–675.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Cold Injuries**.

- Source-page range: **670–675**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 670

CHAPTER 118: Cold Injuries 651

cortisol level, clotting profile, lactic acid, lipase, creatinine, blood gas, and electrocardiogram (ECG). Acid-base disorders are common, but do not follow a predictable pattern. Intravascular thrombosis, embolism, and disseminated intravascular coagulation (DIC) may occur. Electrocardiographic changes include PR, QRS, and QT prolongations, T-wave inversion, and a slow positive deflection at the end of the QRS (Osborn J wave). In addition to environmental exposure, causes of hypothermia include hypoglycemia, hypothyroidism, hypoadrenalism, hypopituitarism, central nervous system (CNS) dysfunction, drug intoxication, sepsis, and dermal disease.

Emergency Department Care and Disposition

1. Place patient in a warm environment. Initiate continuous monitoring of vital signs, pulse oximetry, and core temperature (rectal, bladder, or esophageal thermometer). Indications for intubation are similar to those for normothermic patients. Initiate warmed intravenous fluids titrated to a rate based on the patient's volume status. Remove wet clothing, dry, and cover patients.
2. Handle patients gently to avoid precipitation of a lethal dysrhythmia.
3. Attempt to palpate a pulse and detect respirations for 30 to 45 seconds. If none is detected, initiate cardiopulmonary resuscitation (CPR). Alternatively, use bedside ultrasonography to help determine cardiovascular status and need for CPR.
4. Sinus bradycardia, atrial fibrillation, and atrial flutter usually require no therapy and will resolve with rewarming. Ventricular fibrillation is typically refractory to therapy until the patient is rewarmed, although the American Heart Association guidelines do recommend a single defibrillation attempt.
5. Rewarming techniques include passive rewarming, active external rewarming, and active core rewarming (Table 118-1). Choice of technique depends primarily on cardiovascular status. Temperature is a secondary

Table 118-1 Staging and Treatment of Accidental Hypothermia

Typical Core

Stage Clinical Symptoms Temperature Treatment

Mild Conscious, shivering 35–32°C Warm environment and clothing,

(HT I) warm sweet drinks, and active

movement (if possible)

HT I patients with significant trauma

or comorbidities or those suspected

of secondary hypothermia should

receive HT II treatment

Moderate Impaired conscious- <32–28°C Active external and minimally inva-

(HT II) ness* (may or may sive rewarming techniques (warm

not be shivering) environment; chemical, electrical, or

forced air heating packs or blankets;

warm parenteral fluids)

Cardiac and core temperature

monitoring

Minimal and cautious movements to

avoid arrhythmias

Full-body insulation, horizontal

position, and immobilization

(Continued)

118_Cydulka_Ch118.indd 651 22/04/17 4:39 pm

Source page 671

Table 118-1 Staging and Treatment of Accidental Hypothermia (Continued)

Typical Core

Stage Clinical Symptoms Temperature Treatment

Severe Unconscious,* vital <28°C HT II management plus:

(HT III) signs present Airway management as required

Preference to treat in an ECMO/CPB

center, if available, due to the high

risk of cardiac arrest

Consider ECMO/CPB in cases with

cardiac instability that is refractory

to medical management

Consider ECMO/CPB for comorbid

patients who are unlikely to tolerate

the low cardiac output associated

with HT III

HT IV Vital signs absent Cardiac arrest CPR and up to three doses of

is possible epinephrine and defibrillation (further

below 32°C; the dosing guided by clinical response)
risk increases Airway management
substantially Transport to ECMO/CPB†
below 28°C and Prevent further heat loss (insulation,
continues to warm environment, do not apply
increase with heat to head)
ongoing cooling Active external and minimally invasive
rewarming (see HT II) during transport
is recommended but controversial; do
not apply heat to head
Abbreviations: CPB, cardiopulmonary bypass; CPR, cardiopulmonary resuscitation; ECMO, extracorporeal membrane oxygenation; HT, hypothermia.

*
Consciousness may be impaired by comorbid illness (e.g., trauma, CNS pathology, toxic ingestion) independent of core temperature.

†
Transfering an HT IV patient to an ECMO/CPB center may reduce mortality by 40% to 90% (number needed to treat, approximately 2); if ECMO/CPB is not available within a few hours of transport, consider on-site rewarming with hot packs or forced air blankets, warm IV fluid, ± warm thoracic lavage, ± warm bladder lavage, and ± warm peritoneal lavage; do not apply heat to the head.

consideration. Patients with a stable cardiovascular status (including sinus bradycardia and atrial fibrillation) and temperature above 30°C (above 90°F) may be passively rewarmed. All patients with cardiovascular instability require rapid core rewarming; extracorporeal circuit rewarming (e.g., extracorporeal membrane oxygenation or cardiopulmonary bypass) is the technique of choice for these patients. Invasive rapid core rewarming in patients without cardiac instability is controversial.

6. Continue resuscitative efforts until the core temperature reaches 32°C (90°F).
7. Address and treat underlying causes (e.g., dextrose 50 mL IV for hypoglycemia; treat suspected hypothyroidism and hypoadrenalism with hormone replacement).
8. Admit all patients with symptomatic hypothermia or with hypothermia secondary to an underlying condition. Healthy patients with mild environmental hypothermia that resolves quickly may be discharged home if social circumstances allow (Fig. 118-1).

118_Cydulka_Ch118.indd 652 22/04/17 4:39 pm

Source page 672

Patient’s trunk feels cold on examination or core temperature <35°C?

Vital signs present

Yes No

Impaired Obvious signs of irreversible death

consciousness Valid DNR order

Conditions unsafe for rescuer

Avalanche burial >35 min

No Yes

and airway packed with snow

No to all Yes to any

Trauma Prehospital cardiac Start CPR; do not delay Consider

Comorbidities instability transport termination

Suspected secondary SBP <90 mm Hg Prevent further heat loss of CPR

hypothermia Ventricular arrhythmias

Airway management

Core temperature <28°C

and up to 3 doses of

medication/defibrillation

No to all Yes to any No to all Yes to any

Cardiac arrest from alternate

cause prior to cooling

Major trauma

Witnessed normothermic

arrest

Avalanche burial <35 min

Transport to Transport to

nearest ECMO No to all Yes to any

appropriate hospital

hospital

Consider tests to confirm Transport to

need for ECMO/CPB nearest

Core temperature <32°C appropriate

Serum potassium <12 mmol/L hospital or

manage as per

supervising MD

Yes to all No to any

HT I HT I, II, or III HT IV

Warm environment Active external and Do NOT terminate CPR

and clothing minimally invasive Transport to ECMO

Warm sweet drinks rewarming (insulation, Prevent further heat loss

Active movement hot packs, or forced air (insulation, warm environment,

blankets, warm IV fluid, do not apply heat to head)

allow shivering) Active external and minimally

Minimal and cautious invasive rewarming during

movements to avoid transport is recommended

arrhythmias but controversial; do not

No ROSC

Airway management apply heat to head

if required Consider

Source page 673

654 SECTION 12: Environmental Injuries

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study

Guide, 8th ed., see Chapter 208, “Cold Injuries,” by Michael T. Paddock;

Chapter 209, “Hypothermia,” by Doug Brown.

118_Cydulka_Ch118.indd 654 22/04/17 4:39 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 670–675.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 119: Heat Emergencies

Source pages 676–677 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 676–677

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Heat Emergencies**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 119: Heat Emergencies 657 Table 119-1 Summary of Cooling Techniques Cooling Method Advantages Disadvantages Recommendations Evaporative Provides effective Can cause shivering Strongly recommended cooling cooling Readily available Less effective in humid environments Practical Makes it difficult to Well tolerated maintain electrode positions Immersion cooling Provides effective Can cause shivering Recommended cooling Poorly tolerated Not compatible with resuscitation settings Ice packs on neck, Practical Cooling times longer Can be used as adjunct axillae, and groin than other modalities cooling method Can be added Poorly tolerated to other cooling methods Cardiopulmonary Provides fast and Invasive Recommended in bypass effective cooling Not readily available severe or resistant cases when available Setup is labor intensive Cooling blankets Easy to apply Have limited cooling Not recommended efficacy when other methods Impede use of other available cooling methods Cold water – Invasive Effectiveness and gastric, urinary Labor intensive safety not established bladder, rectal, or peritoneal lavage May lead to water intoxication Human experience is limited low-potency topical corticosteroids, or calamine lotion. [Source pages 676–677]

- Advise patients to wear light, loose fitting clothing. [Source pages 676–677]

- Treatment of heat syncope and heat cramps consists of rest and oral or IV rehydration. [Source pages 676–677]

- ■■ FURTHER READING For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 210, “Heat Emergencies,” by Frank LoVecchio. [Source pages 676–677]

- 119_Cydulka_Ch119.indd 657 20/03/17 5:53 PM C hapter Bites and Stings 120 Michael Levine ■■ WASPS, BEES, AND STINGING ANTS (HYMENOPTERA) Wasps, bees, and stinging ants are members of the order Hymenoptera. [Source pages 676–677]

Clinical reasoning structure

- **Problem representation:** Heat Emergencies; source pages 676–677.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Heat Emergencies**.

- Source-page range: **676–677**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 676

CHAPTER 119: Heat Emergencies 657

Table 119-1 Summary of Cooling Techniques

Cooling Method Advantages Disadvantages Recommendations

Evaporative Provides effective Can cause shivering Strongly recommended

cooling cooling

Readily available Less effective in humid

environments

Practical Makes it difficult to

Well tolerated maintain electrode

positions

Immersion cooling Provides effective Can cause shivering Recommended

cooling Poorly tolerated

Not compatible with

resuscitation settings

Ice packs on neck, Practical Cooling times longer Can be used as adjunct

axillae, and groin than other modalities cooling method

Can be added Poorly tolerated

to other cooling

methods

Cardiopulmonary Provides fast and Invasive Recommended in

bypass effective cooling Not readily available severe or resistant

cases when available

Setup is labor intensive

Cooling blankets Easy to apply Have limited cooling Not recommended

efficacy when other methods

Impede use of other available
cooling methods
Cold water – Invasive Effectiveness and
gastric, urinary Labor intensive safety not established
bladder, rectal, or
peritoneal lavage May lead to water
intoxication
Human experience is
limited

low-potency topical corticosteroids, or calamine lotion. Advise patients to wear light, loose fitting clothing. Treatment of heat syncope and heat cramps consists of rest and oral or IV rehydration.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 210, "Heat Emergencies," by Frank LoVecchio.

119_Cydulka_Ch119.indd 657 20/03/17 5:53 PM

Source page 677

C hapter Bites and Stings

120 Michael Levine

■■ WASPS, BEES, AND STINGING ANTS (HYMENOPTERA)

Wasps, bees, and stinging ants are members of the order Hymenoptera. Local and generalized reactions may occur in response to an encounter. Systemic toxicity may result from an allergic reaction or from massive envenomation as occurs classically with the so-called Africanized bees, which often attack in large numbers. Fire ant venom may cross-react in individuals sensitized to other Hymenoptera stings.

Clinical Features

Local reactions consist of pain, erythema, edema, and pruritus at the sting site. Severe local reactions may increase the likelihood of serious systemic reactions if the patient is reexposed. The local reaction to a fire ant sting consists of a sterile pustule that evolves over 6 to 24 hours, sometimes resulting in necrosis and scarring. Systemic reactions may include urticaria, angioedema, hypotension, and bronchospasm. In addition, those with massive envenomation may develop severe vomiting, diarrhea, rhabdomyolysis, myocardial injury, hepatic failure, and renal failure. Delayed reactions, which manifest 5 to 14 days post-envenomation, infrequently occur, but resemble serum sickness. Such illness is characterized by fever, malaise, headache, urticaria, lymphadenopathy, and polyarthritis.

Emergency Department Care and Disposition

1. In addition to fluid resuscitation, patients with anaphylactic or anaphy-

lactoid reactions should receive the following:

a. Epinephrine 1:1000 concentration (0.3 to 0.5 cc IM in adults or 0.01 mg/kg up to a maximal dose of 0.5 cc in pediatrics). Patients whose symptoms persist despite IM epinephrine or those with evidence of shock should receive IV epinephrine as a continuous infusion.

b. Endotracheal intubation should be performed if needed for airway edema.

c. Steroids, such as methylprednisolone 2 mg/kg, with a maximal single dose of 125 mg, or prednisone 1 mg/kg orally, with a maximal single dose of 60 mg.

d. H1 antagonists, such as diphenhydramine 1 mg/kg IV with a maximal single dose of 50 mg.

e. H2 receptor antagonists, such as famotidine 0.5 mg/kg IV with a maximal dose of 40 mg.

f. Bronchospasm should be treated with beta agonists, such as albuterol 2.5 to 5 mg via nebulization.

2. Patients with urticaria, but no other systemic manifestations, should be treated with H1 and H2 blockers and steroids, but without epinephrine or albuterol.

3. The stinger should be removed after the patient has been stabilized.

658

120_Cydulka_Ch120.indd 658 20/03/17 5:59 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 676–677.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 120: Bites and Stings

Source pages 678–685 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 678–685

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Bites and Stings**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- Patients with a single sting who have only minor symptoms can be dis- charged home after a period of observation. [Source pages 678–685]

- For patients with multiple stings, disposition is as follows: a. [Source pages 678–685]

- Admit if more than 100 stings, substantial comorbidities, extremes of age, or severe systemic manifestations b. [Source pages 678–685]

- Patients with <100 stings, and who remain asymptomatic without laboratory evidence of complications (e.g., no rhabdomyolysis and normal renal function) following an observation period of 6 hours can be discharged from the Emergency Department. [Source pages 678–685]

- Refer all patients with Hymenoptera reactions to an allergist for further evaluation, prescribe a premeasured epinephrine injector (EpiPen®), and advise them to carry allergy alert identification. [Source pages 678–685]

Clinical reasoning structure

- **Problem representation:** Bites and Stings; source pages 678–685.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Bites and Stings**.

- Source-page range: **678–685**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 678

CHAPTER 120: Bites and Stings 659

4. Patients with a single sting who have only minor symptoms can be discharged home after a period of observation.

5. For patients with multiple stings, disposition is as follows:

a. Admit if more than 100 stings, substantial comorbidities, extremes of age, or severe systemic manifestations

b. Patients with <100 stings, and who remain asymptomatic without laboratory evidence of complications (e.g., no rhabdomyolysis and normal renal function) following an observation period of 6 hours can be discharged from the Emergency Department.

6. Refer all patients with Hymenoptera reactions to an allergist for further evaluation, prescribe a premeasured epinephrine injector (EpiPen®), and advise them to carry allergy alert identification. Instruct patients to use epinephrine at the first sign of a systemic reaction.

■■ BROWN RECLUSE SPIDER (Loxosceles reclusa)

Clinical Features

The initial L. reclusa bite is painless. It evolves into a firm erythematous lesion that heals over several days to weeks. Occasionally, a severe reaction with immediate pain, hemorrhagic blister formation, and local blanching may occur. These lesions often become necrotic over the next 3 to 4 days and form significant eschars. Loxoscelism is a common systemic reaction following the bite of some South American Loxosceles species, but occurs infrequently following the bite of L. reclusa. Systemic symptoms, which are more common in children, typically occur 1 to 3 days post-envenomation, and are characterized by fever, chills, vomiting, arthralgias, myalgias, petechiae, and hemolysis. Severe cases progress to seizures, renal failure, disseminated intravascular coagulation, and death.

Diagnosis and Differential

Loxosceles species are distinguished by three sets of paired eyes; most

other spiders have eight eyes in two rows. A pigmented, violin-shaped pattern on the cephalothorax of the brown recluse is characteristic but unreliable. The diagnosis of L. reclusa envenomation is commonly clinical since the bite is rarely witnessed. Assays to confirm L. reclusa poisoning are not clinically available. Patients with significant envenomation may exhibit hemolysis, coagulopathy, or abnormal renal function. Local abscesses are frequently misdiagnosed as Loxosceles envenomation

Emergency Department Care and Disposition

1. Treatment of a brown recluse spider bite includes supportive measures, such as pain medication, tetanus prophylaxis, and antibiotics if infection is present. In the United States, antivenom is not commercially available and is usually not needed.

2. Most wounds heal without intervention. Dapsone, which was once advocated for treatment of these envenomations, has not been clearly proven to be beneficial. Due to the risk of significant adverse events and the lack of proven efficacy, its use in this setting is not recommended. Arrange serial wound evaluations for outpatients.

120_Cydulka_Ch120.indd 659 20/03/17 5:59 PM

Source page 679

3. Patients with systemic reactions and hemolysis should be hospitalized.

4. Surgery is reserved for lesions larger than 2 cm and is deferred for 2 to 3 weeks after the bite.

■■ HOBO SPIDER (TEGENARIA AGRESTIS)

The hobo spider, also known as the Northwestern brown spider, causes clinical signs and symptoms that are similar to those of the brown recluse spider bite. The skin site is initially painless before developing induration, erythema, blistering, and necrosis. Patients may experience headache, vomiting, and fatigue. There is no specific diagnostic test or therapeutic intervention for hobo spider bites. Surgical repair for severe ulcerative lesions is delayed until the necrotizing process is complete.

■■ BLACK WIDOW SPIDER (Latrodectus mactans)

Clinical Features

Black widow spider bites induce an immediate pinprick sensation that may quickly spread to the entire extremity. Erythema at the site appears within 1 hour and often forms a “target” lesion. Systemic manifestations may include diffuse myalgias, especially involving the larger muscle groups, including the back, abdomen, and trunk. Severe pain may wax and wane for several days. Other signs and symptoms include hypertension, tachycardia, muscle fasciculations, headache, nausea, vomiting, and diaphoresis. The

fasciculations and diaphoresis may be localized or diffuse. Serious, albeit rare, acute complications include myocarditis, priapism, and dyspnea.

Emergency Department Care and Disposition

1. Initial therapy includes local wound treatment and supportive care. Liberal dosing of analgesics and benzodiazepines will relieve pain and cramping.
2. Lacrodectus antivenom, derived from horse serum, is rapidly effective for severe envenomation even when the presentation is delayed. Fatal cases of anaphylaxis have been described with this therapy, and its use should be reserved for severe cases refractory to opioid analgesics and benzodiazepines. The package insert provides dosing instructions.
3. Patients receiving antivenom may be discharged after a short observation period if symptoms of envenomation resolve.

■■ TARANTULAS

When threatened, tarantulas may flick barbed hairs into their victim. Although North American tarantula hairs rarely penetrate human skin, they can embed deeply into the conjunctiva and cornea and cause an inflammatory response. Any patient complaining of ocular symptoms after exposure to a tarantula should undergo a thorough slit lamp examination to search for embedded hairs. Treatment includes topical steroids and consultation with an ophthalmologist for surgical removal of the hairs.

Tarantula bites may also occur. They are painful, and cause local erythema and edema. Provide local wound care and analgesia.

120_Cydulka_Ch120.indd 660 20/03/17 5:59 PM

Source page 680

■■ SCORPION (SCORPIONIDA)

Clinical Features

Although highly toxic species are found in the Caribbean, Asia, and Africa, the only North American scorpion that produces systemic toxicity is the bark scorpion (Centruroides sculpturatus). Venom from C. sculpturatus causes immediate pain and paresthesia. A positive “tap test” (exquisite local tenderness when the area is lightly tapped) may be seen. Systemic effects are relatively infrequent and occur mainly in children. Systemic manifestations include diffuse motor involvement and/or cranial nerve dysfunction, the latter of which may result in roving eye movements, diplopia, and difficulty swallowing. Systemic symptoms may last 24 to 48 hours without antivenom therapy.

Emergency Department Care and Disposition

1. Most patients require only oral analgesics and/or benzodiazepines.

2. Patients with systemic manifestations may require IV benzodiazepines and opiates.
3. While uncommon, patients with significant hypersalivation and stridor may warrant endotracheal intubation. Atropine has been used to help with secretions, although data supporting its use for this purpose are lacking.
4. A F(ab) equine antivenom is available and highly effective for those patients with high-grade, systemic toxicity.
5. Patients without systemic symptoms may be observed briefly and discharged with analgesics.

■■ SCABIES (SARCOPTES SCABIEI)

Clinical Features

Scabies bites are often concentrated in the web spaces between fingers and toes. Other common areas include the axilla and genital area, children's faces and scalps, and the female nipple. Transmission is typically by direct contact. The distinctive feature of scabies infestation is intense pruritus with "burrows." The female mite is easily scraped out with a blade edge. Associated vesicles, papules, crusts, and eczematization may obscure the diagnosis.

Emergency Department Care and Disposition

Advise patients to apply permethrin cream from the neck down; infants may require additional application to the scalp, temple, and forehead. The patient should bathe before application, apply the medication, and then bathe again in 12 hours. Reapplication is necessary only if mites are found 2 weeks after treatment, although the pruritus may last for several weeks after successful therapy. Ivermectin, 200 μg/kg PO, followed by a second dose in 10 days, is an alternative treatment.

■■ TICKS

The spectrum of tick borne illness includes Lyme disease, Rocky Mountain spotted fever, ehrlichiosis, babesiosis, Colorado tick fever, tularemia, and tickborne encephalitis. Certain species of ticks have a neurotoxin capable

120_Cydulka_Ch120.indd 661 20/03/17 5:59 PM

Source page 681

of inducing tick paralysis, a symmetric ascending flaccid paralysis nearly identical to Guillain–Barré syndrome. Indeed, a diagnosis of Guillain–Barré syndrome should not be considered until a thorough search rules out the presence of an engorged tick. The recommended method of tick removal involves grasping the tick with forceps near the point of attachment and pulling with steady, gentle traction. Since disease transmission is time dependent, prompt tick removal is essential.

■■ CHIGGERS (TROMBICULIDAE)

Clinical Features

Chiggers are tiny mite larvae that cause intense pruritus when they feed on host epidermal cells. They tend to attach to skin in areas of tight-fitting clothing such as near waistbands. Children who have been sitting on lawns are prone to chigger lesions in the genital area. Itchiness begins within a few hours, followed by a papule that enlarges to a nodule (“chigger bite”) over the next 1 to 2 days. Single bites may cause soft tissue edema, whereas infestation has been associated with fever and erythema multiforme. The diagnosis of chigger bites is based on typical skin lesions and intense pruritus in the context of known outdoor exposure.

Emergency Department Care and Disposition

Treatment is symptomatic with oral or topical antihistamines, although oral steroids may be required in more severe cases. Annihilation of the mites requires topical application of permethrin or other topical scabicides. The package insert provides techniques for proper use.

■■ FLEAS (SIPHONAPTERA)

Flea bites are frequently found in zigzag lines, especially on the legs and waist. They are intensely pruritic lesions with hemorrhagic puncta, surrounding erythema, and urticaria. Discomfort is relieved with calamine lotion, cool soaks, and oral or topical antihistamines. Severe irritation may require topical steroid creams. Patients who develop impetigo and other local infections should be treated with topical or oral antibiotics.

■■ LICE (ANOPLURA)

Body lice concentrate on the waist, shoulders, axillae, and neck. Their bites produce red spots that progress to papules and wheals. They are so intensely pruritic that linear scratch marks are suggestive of infestation. The white ova of head lice are adherent to the hair shaft and therefore can be distinguished from dandruff. Pubic lice are spread by sexual contact. They cause intense pruritus, and their small white eggs (nits) are visible on hair shafts. As with scabies, permethrin is the primary treatment for body lice infestation. Treatment of any hair-borne infestation requires application of pyrethrin with piperonyl butoxide after hair washing, with reapplication in 10 days. Wet combing hair with a fine-tooth comb will remove dead lice and nits. Clothing, bedding, and personal articles should be washed in hot (>52°C [125.6°F]) water to prevent reinfestation.

120_Cydulka_Ch120.indd 662 20/03/17 5:59 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 678–685.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 121: Trauma and Envenomation from Marine Fauna

Source pages 686–689 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 686–689

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Trauma and Envenomation from Marine Fauna**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 121: Trauma and Envenomation from Marine Fauna 667 hypertension, tachycardia, diaphoresis, and even pulmonary edema. [Source pages 686–689]

- The most innocuous Cnidaria are the anthozoans (anemones) that occasionally cause a mild local reaction. [Source pages 686–689]

- Porifera (the sponges) can produce a stinging, pruritic dermatitis. [Source pages 686–689]

- Spicules of silica or calcium carbonate can become embedded in the skin along with toxic secretions from the sponge. [Source pages 686–689]

- Echinodermata include sea urchins and sea stars. [Source pages 686–689]

Clinical reasoning structure

- **Problem representation:** Trauma and Envenomation from Marine Fauna; source pages 686–689.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Trauma and Envenomation from Marine Fauna**.

- Source-page range: **686–689**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 686

CHAPTER 121: Trauma and Envenomation from Marine Fauna 667

hypertension, tachycardia, diaphoresis, and even pulmonary edema. The most innocuous Cnidaria are the anthozoans (anemones) that occasionally cause a mild local reaction.

Porifera (the sponges) can produce a stinging, pruritic dermatitis. Spicules of silica or calcium carbonate can become embedded in the skin along with toxic secretions from the sponge. Echinodermata include sea urchins and sea stars. Sea urchin spines produce immediate pain with trauma; some contain venom that leads to erythema and swelling. Retained spines can lead to infection and granuloma formation. The crown-of-thorns sea star, Acanthaster planci, has sharp rigid spines that cause burning pain and local inflammation. Another class of echinoderms are sea cucumbers, which can cause mild contact dermatitis. Annelida include bristle and fire worms, which embed bristles in the skin, causing pain and erythema. Mollusca include gastropods and octopuses. Both the Indo-Pacific cone shell, Conus, and the blue-ringed octopus, Hapaloch-laena, can deliver paralytic venom that can quickly lead to respiratory paralysis.

Vertebrate envenomations are primarily due to stingrays (order Rajif-ormes) and spined venomous fish (scorpion fish, lionfish, catfish, and weeverfish). The stingray whip tail has venomous spines, which puncture or lacerate causing an intense painful local reaction. The spines of venom-ous fish have glands that force venom into the wound after puncture and cause local pain, erythema, and edema. Retention of a spine tip can lead to infection. In tropical Indo-Pacific waters, bites by venomous sea snakes, although rare, can cause neurotoxicity and myotoxicity, and even death from ascending paralysis.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. Copiously irrigate lacerations, punctures, and bite wounds; explore for

foreign matter and debride devitalized tissue. Soft tissue radiographs or ultrasound may help locate foreign bodies, which usually require removal especially if intraarticular. Leave lacerations open for delayed primary closure. Update tetanus, if needed.

2. Prophylactic antibiotic therapy is not indicated for routine minor wounds in healthy patients but may be considered in selected patients (Table 121-1). Antibiotic therapy for infected wounds is first directed toward likely pathogens and later by culture and sensitivity results. Treat Staphylococcus and Streptococcus species with a first-generation cephalosporin, such as cephalexin 500 mg (10 to 25 mg/kg in children) four times daily PO or cefazolin 1 to 2 g (25 mg/kg in children) every 8 hours IV; alternatively doxycycline 100 mg PO/IV twice daily for MRSA coverage may be used in adults. Addition of a third-generation cephalosporin, such as ceftriaxone 1 g (50 mg/kg in children) IV daily or cefotaxime 2 g (50 mg/kg in children) IV every 8 hours, or a fluoroquinolone, such as levofloxacin or ciprofloxacin 500 mg PO/IV daily will cover ocean-related infections from Vibrio (avoid use in children). A fluoroquinolone or third-generation cephalosporin, or trimethoprim-sulfamethoxazole double strength, 1 tablet (4 mg/kg trimethoprim component in children) PO twice each day, or imipenem, 500 mg (10 to 25 mg/kg

121_Cydulka_Ch121.indd 667 01/05/17 11:15 am

Source page 687

Table 121-1 Recommendations for Antibiotic Treatment of Marine-Associated Wounds

No Antibiotics Indicated	Prophylactic/Outpatient Antibiotics	Hospital Admission for IV Antibiotics
Healthy patient	Late wound care	Predisposing medical conditions
Prompt wound care	Large lacerations or injuries	Long delays before definitive wound care
No foreign body	Early or local inflammation	Deep wounds, significant trauma
No bone or joint involvement	Wounds with retained foreign bodies	
Small or superficial injuries	Progressive inflammatory change	
	Penetration of periosteum, joint space, or body cavity	
	Major injuries associated with envenomation	
	Systemic illness	

in children) IV every 6 hours, will cover fresh water infections from Aeromonas. Granulomas from Mycobacterium marinum require several months of treatment with clarithromycin or rifampin plus ethambutol.

3. Treatment of jellyfish stings is controversial and may be species dependent. Regardless, initial irrigation with seawater to remove nematocyst-laden tentacles may be beneficial. Hot water (111°F (43.3°C) to 114°F (45.6°C)) and topical lidocaine may lessen local venom effects. See Table 121-2 summarizing early treatment of all marine envenomations.

Table 121-2 Early Treatment of Marine Envenomations

Marine Organism	Detoxification	Further Treatment
Penetrating Envenomations		
Catfish, lionfish, scorpionfish, stingray	Hot water immersion*; lidocaine infiltration	Usual wound care.† Assess for stingray spines (x-ray or exploration). Observe for development of systemic symptoms.
Stonefish	Hot water immersion*; lidocaine infiltration	Usual wound care.† Administer stonefish antivenin if severe systemic reaction.
Sea snake	—	Pressure immobilization. Administer antivenom if severe systemic reaction. Observe 8 h for myotoxicity or neurotoxicity.
Australian blue-ringed octopus	—	Use pressure immobilization. Provide supportive care.

(Continued)

121_Cydulka_Ch121.indd 668 01/05/17 11:15 am

Source page 688

Table 121-2 Early Treatment of Marine Envenomations (Continued)

Marine Organism	Detoxification	Further Treatment
Cone snail	—	Use pressure immobilization. Provide supportive care.
Sea urchin	Hot water immersion*; topical lidocaine	Explore wound and remove any spines.
Fireworms	Topical 5% acetic acid (vinegar).	Remove bristles. Consider topical corticosteroids.

Nonpenetrating Envenomations

Fire coral, hydroids, anemones Irrigate with seawater Topical corticosteroid for itching.

or saline.

Portuguese man-of-war, blue Hot water immersion,* Topical corticosteroid for itching.

bottles, and non-box jellyfish sea water irrigation, Observe for development of

topical lidocaine. systemic symptoms.

Remove tentacles and Supportive care.

nematocysts.

Box jellyfish Hot water immersion.* Topical corticosteroid for itching.

Irrigate with saline or Observe for development of

seawater. systemic symptoms.

Topical 5% acetic acid Supportive care.

(vinegar). Administer Chironex antivenin.

Remove tentacles and

nematocysts.

Topical lidocaine.

Irukandji's syndrome Irrigate with saline or Parenteral opioids for pain.

seawater. Magnesium for cardiac arrest.

Hot water immersion.*

Remove tentacles and

nematocysts.

*

Hot water immersion = 111°F (43.3°C) to 114°F (45.6°C) water until pain relieved.

†

Usual wound care = irrigate, explore, debride, consider antibiotics and analgesi.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study

Guide, 8th ed., see Chapter 213, "Marine Trauma and Envenomation," by John J.

Devlin and Kevin Knoop.

121_Cydulka_Ch121.indd 669 01/05/17 11:15 am

Source page 689

C h a p te r High-Altitude Disorders

122 Shaun D. Carstairs

High-altitude disorders are due primarily to hypoxia; the rapidity and

height of ascent influence the risk of occurrence.

■■ ACUTE MOUNTAIN SICKNESS

Clinical Features

Acute mountain sickness (AMS) is usually seen in nonacclimated people

making a rapid ascent to higher than 2000 m (6560 ft) above sea level.

Symptoms resembling a hangover may develop within 6 hours after arrival

at altitude but may be delayed as long as one day. Typical symptoms include bifrontal headache along with a combination of GI disturbance, dizziness, fatigue, or sleep disturbance. Worsening headache, vomiting, oliguria, dyspnea, and weakness indicate progression of AMS. Physical examination findings in early AMS are limited. Postural hypotension and peripheral and facial edema may occur. Localized rales are noted in up to 20% of cases. Funduscopy shows tortuous and dilated veins; retinal hemorrhages are common at altitudes higher than 5000 m (16,500 ft). Resting SaO2 is typically normal for altitude and correlates poorly with the diagnosis of AMS.

Diagnosis and Differential

The differential diagnosis includes hypothermia, carbon monoxide poisoning, pulmonary or central nervous system infections, migraine, dehydration, and exhaustion. The diagnosis is based largely on history of rapid ascent and symptoms.

Emergency Department Care and Disposition

The goals of treatment are to prevent progression, abort the illness, and improve acclimatization.

1. Terminate further ascent until symptoms resolve. For mild AMS, symptomatic therapy includes an analgesic, such as acetaminophen or an NSAID, and an antiemetic, such as ondansetron disintegrating tablets, 4 to 8 mg every 4 to 6 hours PO. Mild AMS usually improves or resolves in 12 to 36 hours if ascent is stopped.

2. A decrease in altitude of 300 to 1000 m should provide prompt relief of symptoms. Immediate descent and treatment are indicated for patients with moderate AMS or if there is a change in the level of consciousness, ataxia, or pulmonary edema.

3. Low-flow oxygen also relieves symptoms.

4. Consider hyperbaric therapy for moderate AMS if descent is not possible.

5. Pharmacologic therapy for moderate AMS includes acetazolamide 125 to 250 mg in adults and 2.5 mg/kg in children, PO twice daily, until

122_Cydulka_Ch122.indd 670 21/03/17 9:26 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 686–689.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 122: High-Altitude Disorders

Source pages 690–695 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 690–695

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **High-Altitude Disorders**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 122: High-Altitude Disorders 671 symptoms resolve and dexamethasone 4 mg PO, IM, or IV every 6 hours with a taper over several days. [Source pages 690–695]

- Indications for acetazolamide are (a) history of altitude illness, (b) abrupt ascent higher than 3000 m (9840 ft), (c) AMS, and (d) symptomatic periodic breathing during sleep at high altitude. [Source pages 690–695]

- Acetazolamide pharmacologically produces an acclimatization response by inducing a bicarbonate diuresis and metabolic acidosis. [Source pages 690–695]

- Acetazolamide is effective for both prophylaxis and treatment. [Source pages 690–695]

- Acetazolamide should be avoided in sulfa-allergic patients. [Source pages 690–695]

Clinical reasoning structure

- **Problem representation:** High-Altitude Disorders; source pages 690–695.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **High-Altitude Disorders**.

- Source-page range: **690–695**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 690

CHAPTER 122: High-Altitude Disorders 671

symptoms resolve and dexamethasone 4 mg PO, IM, or IV every 6 hours with a taper over several days.

a. Indications for acetazolamide are (a) history of altitude illness, (b) abrupt ascent higher than 3000 m (9840 ft), (c) AMS, and (d) symptomatic periodic breathing during sleep at high altitude.

b. Acetazolamide pharmacologically produces an acclimatization response by inducing a bicarbonate diuresis and metabolic acidosis. Acetazolamide is effective for both prophylaxis and treatment.

c. Acetazolamide should be avoided in sulfa-allergic patients.

6. Patients who respond well to treatment may be discharged. Provide counseling on preventing future episodes: graded ascent, prophylaxis using acetazolamide, and avoidance of overexertion, alcohol, and respiratory depressants. Begin prophylaxis with acetazolamide a day before ascent and continue for at least two days after reaching high altitude.

■■ HIGH-ALTITUDE PULMONARY EDEMA

Risk factors for high-altitude pulmonary edema (HAPE) include rapid ascent, heavy exertion, cold, pulmonary hypertension, and use of a sleep medication. Children with acute respiratory infections may be more susceptible to HAPE. HAPE may be fatal if not recognized and treated early.

Clinical Features

HAPE usually begins on the second to fourth night at a new altitude and may progress quickly from dry cough and impaired exercise capacity to resting dyspnea, productive cough, severe weakness, and cyanosis. Physical examination findings include tachycardia, tachypnea, localized or generalized rales, and signs of pulmonary hypertension, such as a prominent P2 and right ventricular heave. Resting SaO2 is low for altitude and drops significantly with exertion. CXR abnormalities progress from interstitial to localized to generalized alveolar infiltrates. Right axis deviation

and a right ventricular strain pattern are seen on EKG with progressive disease.

Diagnosis and Differential

The differential diagnosis includes pneumonia, acute asthma, congestive heart failure, myocardial ischemia, and pulmonary embolism. Decreased exercise performance and dry cough are enough to suspect early HAPE. A key to diagnosis is response to treatment.

Emergency Department Care and Disposition

Early recognition of HAPE is essential to prevent progression. General measures include rest and keeping patients warm.

1. Initiate supplemental oxygen and titrate to SaO2 ≥ 90%.

2. Immediate descent is the treatment of choice. Exertion by the patient must be minimized during descent. Hyperbaric treatment may be used if descent is not an option. Patients with very mild cases of HAPE may be managed with bed rest and oxygen alone.

122_Cydulka_Ch122.indd 671 21/03/17 9:26 AM

Source page 691

3. Pharmacologic treatment is usually unnecessary if descent and oxygen are available. In such cases (field conditions), options include nifedipine 20 to 30 mg extended release PO every 12 hours, or tadalafil 10 mg PO twice daily. Tadalafil blunts hypoxic pulmonary vasoconstriction. Nifedipine or tadalafil may also be used for HAPE prophylaxis in persons with prior episodes. Inhaled albuterol 2 to 4 puffs every 4 to 6 hours may be used for both prophylaxis and treatment but is not well studied.

4. Patients may be discharged if clinical and radiographic improvements are noted and room air SaO2 remains > 90%.

■■ HIGH-ALTITUDE CEREBRAL EDEMA

Clinical Features

High-altitude cerebral edema (HACE) is defined as progressive neurologic deterioration with AMS or HAPE. Patients present with altered mental status, ataxia, stupor, and progress to coma if untreated. Focal neurologic signs such as third and sixth cranial nerve palsies may be present.

Diagnosis and Differential

The differential diagnosis includes stroke or transient ischemic attack, tumor, meningitis, encephalitis, or metabolic disturbance. Increased T2 signaling in the splenium of the corpus callosum is seen on MRI. Laboratory testing to rule out other diagnoses may be considered but should not delay treatment.

Emergency Department Care and Disposition

1. Initiate supplemental oxygen and titrate to SaO2 ≥ 90%. Comatose patients require intubation and ventilation.

2. Immediate descent is needed. Initiate hyperbaric therapy if descent is not possible.

3. Administer dexamethasone 8 mg PO, IM or IV, followed by 4 mg PO, IM or IV every 6 hours.

4. In intubated patients, monitor arterial blood gases, taking care to avoid lowering PaCO2 below 30 mmHg. Monitor intracranial pressures and cerebral blood velocities by transcranial Doppler US, if possible.

5. Patients remaining ataxic or confused after descent require admission.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 221, "High-Altitude Disorders," by Peter H. Hackett and Christopher B. Davis.

122_Cydulka_Ch122.indd 672 21/03/17 9:26 AM

Source page 692

C h a p ter Dysbarism and Complications

123 of Diving

Christian A. Tomaszewski

Dysbarism is commonly encountered in scuba divers and refers to complications associated with changes in environmental ambient pressure and with breathing compressed gases. These effects are governed by three gas laws: Boyle's law states that pressure and volume are inversely related; Henry's law states that, at equilibrium, the quantity of gas in solution is proportional to the partial pressure of that gas; Dalton's law states that total pressure exerted by a mixture of gases is the sum of the partial pressures of each gas.

■■ CLINICAL FEATURES

Barotrauma is the most common diving-related affliction and is caused by the direct mechanical effects of pressure, as gas-filled cavities in the body contract or expand with pressure changes. The most common form of barotrauma occurs during descent and is middle ear squeeze, or barotitis media. It is caused by inability to equalize pressure causing tympanic membrane bleeding or rupture and may result in conductive hearing loss. A forceful Valsalva during equalization can cause inner ear barotrauma with rupture of the round or oval window. Symptoms include tinnitus, sensorineural hearing loss, and vertigo. If the sinus ostia are occluded on descent, an impending squeeze can cause bleeding from the maxillary or frontal sinuses, resulting in pain and epistaxis.

Barotrauma during ascent is due to expansion of gas in body cavities. In

the middle ear, the pressure differential from asymmetrical expansion can cause alternobaric vertigo. Although rare, “reverse squeeze” may affect the ear or sinuses during ascent with rupture. Pulmonary overinflation and even burst lung can occur during rapid, panicked ascents if divers fail to exhale or if intrinsic pulmonary air trapping exists (e.g., COPD) resulting in pneumomediastinum, subcutaneous emphysema, or pneumothorax. The most serious consequence is cerebral arterial gas embolism (CAGE). Neurologic symptoms occur on ascent or immediately upon surfacing and include loss of consciousness, seizure, blindness, disorientation, hemiplegia, or other signs of stroke.

Divers using compressed air, caisson (tunnel) workers, and high-altitude pilots can all present with decompression sickness (DCS). In divers, this usually results from exceeding the dive table limits for depth and time. DCS can occur within minutes to hours of surfacing, rarely days later. Excessive bubble formation in tissue or circulation from saturated gas can cause both acute occlusive and delayed inflammatory effects. Type I DCS, “pain-only,” includes mottled skin and deep pain of the joints, usually the shoulder or knee, and is unaffected by movement. Type II, “serious,” DCS involves the central nervous system, typically the spine. Patients may initially complain of truncal constriction with ascending paralysis. Prolonged exposure at depth can lead to cardiopulmonary “chokes” with cough and hemoptysis or vestibular “staggers” with vertigo and hearing loss. Because DCS and

123_Cydulka_Ch123.indd 673 21/03/17 9:37 AM

Source page 693

CAGE can be difficult to distinguish, or present simultaneously, the term “decompression illness” is now typically used for both.

■■ DIAGNOSIS AND DIFFERENTIAL

Dive profile (depth, duration, and repetitiveness) and time of symptom onset are the most useful historical factors in distinguishing dysbarism from other disorders. During descent, the most common maladies are the squeezes. A fistula test, insufflation of the tympanic membrane on the affected side causing the eyes to deviate to the contralateral side, may help diagnose inner ear barotrauma. During ascent, barotrauma or alternobaric vertigo is most likely to occur. A chest x-ray may reveal pneumomediastinum, pneumothorax or subcutaneous air after pulmonary overinflation. If accompanied by early neurological symptoms, CAGE should be considered. Laboratory testing may reveal elevated hematocrit from hemoconcentration or elevated creatine phosphokinase from circulatory distribution of bubbles.

The differential diagnosis for DCS is broad. Musculoskeletal complaints could be due to joint strain or symptomatic herniated cervical disk. Chest pain may represent cardiac ischemia from overexertion. Immersion pulmonary edema from noncardiogenic causes can occur during strenuous dives, particularly in cold water. Seizures at depth can result from breathing enriched mixtures of oxygen exceeding 1.4 atmospheres absolute. Regardless, DCS should be suspected in patients with vague pain or neurological symptoms especially if safe diving limits, depth and time, were exceeded. If DCS is suspected, an early trial of pressure with hyperbaric oxygen usually results in some improvement.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. Decompression Illness (DCS/CAGE):

a. Administer 100% oxygen and IV fluids.

b. If CAGE is suspected, place the patient in the supine position; place in the left lateral decubitus position if vomiting occurs.

c. Rapidly arrange for recompression therapy with hyperbaric oxygen. Divers Alert Network (1-919-684-9111) may help provide chamber locations.

d. Lidocaine 1 mg/kg IV bolus followed by a continuous infusion at 1 mg/min may provide neuroprotection in CAGE.

2. Treat middle ear barotitis with decongestants and analgesics. Consider antibiotics if infection is suspected. Advise patients against diving until healing is completed. Inner ear barotrauma requires bed rest with the head upright until otolaryngologic evaluation for possible surgical exploration.

3. Pulmonary overinflation with ascent may require needle decompression or tube thoracostomy acutely if a pneumothorax develops.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 214, “Diving Disorders,” by Brain Snyder and Tom Neuman.

123_Cydulka_Ch123.indd 674 21/03/17 9:37 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 690–695.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 124: Near Drowning

Source pages 696–697 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 696–697

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Near Drowning**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- Patients who arrive in the emergency department in asystole or cardiac arrest after warm water submersion and are normothermic have a poor prognosis for recovery without significant neurologic handicaps. [Source pages 696–697]

- ■■ FURTHER READING For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 215, “Drowning,” by Stephen John Cico and Linda Quan. [Source pages 696–697]

- 124_Cydulka_Ch124.indd 677 26/04/17 2:34 pm C ha p ter Thermal and Chemical Burns 125 Sandra L. [Source pages 696–697]

- Werner ■■ THERMAL BURNS The majority of burn patients are treated and released from the ED. [Source pages 696–697]

- Of those hospitalized, more than 60% are admitted to one of the country’s 127 burn centers. [Source pages 696–697]

Clinical reasoning structure

- **Problem representation:** Near Drowning; source pages 696–697.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Near Drowning**.

- Source-page range: **696–697**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 696

CHAPTER 124: Near Drowning 677

5. Patients who arrive in the emergency department in asystole or cardiac arrest after warm water submersion and are normothermic have a poor prognosis for recovery without significant neurologic handicaps.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 215, "Drowning," by Stephen John Cico and Linda Quan.

124_Cydulka_Ch124.indd 677 26/04/17 2:34 pm

Source page 697

C ha p ter Thermal and Chemical Burns

125 Sandra L. Werner

■■ THERMAL BURNS

The majority of burn patients are treated and released from the ED. Of those hospitalized, more than 60% are admitted to one of the country's 127 burn centers. The risk of death from a major burn is associated with larger burn size, advanced age, concomitant inhalation injury, and female sex.

Clinical Features

Burns are categorized by their size and depth. Burn size is calculated as the percentage of body surface area (BSA) involved. The most common method to estimate this is the Rule of Nines (Fig. 125-1). A more accurate tool, especially in infants and children, is the Lund and Browder burn diagram (Fig. 125-2). For smaller burns, the patient's hand can be used to estimate the size of the burn. The area of the back of the patient's hand represents approximately 1% of BSA, and the number of "hands" represents the BSA burned.

9%

18% front

18% back

9%

1%

18%

Figure 125-1. Rule of Nines to estimate the percentage of burn.

678

125_Cydulka_Ch125.indd 678 21/03/17 9:50 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 696–697.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 125: Thermal and Chemical Burns

Source pages 698–705 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 698–705

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Thermal and Chemical Burns**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 125: Thermal and Chemical Burns 679 A A 2 2 2 2 112 13 112 112 13 112 1 2 12 2 12 112 112 B B B B C C C C 1 34 1 34 1 34 Relative percentages of areas affected by growth (age in years) 0 1 5 10 15 Adult A: half of head 9 12 8 12 6 12 5 12 4 12 3 12 B: half of thigh 2 34 3 14 4 4 14 4 12 4 34 C: half of leg 2 12 2 12 2 34 3 3 14 3 12 Second-degree and Third-degree = Total percent burned Figure 125-2. [Source pages 698–705]

- Lund and Browder diagram to estimate the percentage of a burn. [Source pages 698–705]

- Burn depth historically has been described in degrees: first, second, third, and fourth. [Source pages 698–705]

- A more clinically relevant classification scheme catego- rizes burns as superficial partial thickness, deep partial thickness, and full thickness. [Source pages 698–705]

- Table 125-1 summarizes the characteristics of each type of burn. [Source pages 698–705]

Clinical reasoning structure

- **Problem representation:** Thermal and Chemical Burns; source pages 698–705.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Thermal and Chemical Burns**.

- Source-page range: **698–705**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 698

CHAPTER 125: Thermal and Chemical Burns 679

A

A

2 2

2 2

112 13 112 112

13

112

1 2 12 2 12

112 112

B B

B B

C C

C C

1 34 1 34 1 34

Relative percentages of areas affected by growth (age in years)

0 1 5 10 15 Adult

A: half of head 9 12 8 12 6 12 5 12 4 12 3 12

B: half of thigh 2 34 3 14 4 4 14 4 12 4 34

C: half of leg 2 12 2 12 2 34 3 3 14 3 12

Second-degree and

Third-degree =

Total percent burned

Figure 125-2. Lund and Browder diagram to estimate the percentage of a burn.

Burn depth historically has been described in degrees: first, second, third, and fourth. A more clinically relevant classification scheme categorizes burns as superficial partial thickness, deep partial thickness, and full thickness. Table 125-1 summarizes the characteristics of each type of

burn.

Inhalation injury occurs most frequently in closed-space fires and in patients with decreased cognition (intoxication, overdose, head injury). Both the upper and lower airway can be injured by heat, particulate matter, and toxic gases. Thermal injury is usually limited to the upper airway, and can result in acute airway compromise. Particulate matter can reach

125_Cydulka_Ch125.indd 679 21/03/17 9:50 AM

Source page 699

Table 125-1 Burn Depth Features Classified by Degree of Burn

Burn Depth	Histology/Anatomy	Example	Healing
Superficial (first degree)	Epidermis	Sunburn	7 days
No blisters, painful			
Superficial partial- thickness (superficial second degree)	Epidermis and superficial dermis	Hot water scald	14–21 days, no scar
Blisters, very painful			
Deep partial- thickness (deep second degree)	Epidermis and deep dermis, sweat glands, grease, and hair follicles	Hot liquid, steam, flame	3–8 weeks, permanent scar
Blisters, very painful			
Full-thickness (third degree)	Entire epidermis and dermis charred, pale, leathery; no pain necessary	Flame	Months, severe scarring, skin grafts
Fourth degree	Entire epidermis and dermis, as well as bone, fat, and/or muscle	Flame	Months, multiple surgeries usually required

the terminal bronchioles and lead to bronchospasm and edema. Clinical indicators of inhalation injury include facial burns, singed nasal hair, soot in the upper airway, hoarseness, carbonaceous sputum, and wheezing. Carbon monoxide poisoning should be suspected in all patients with inhalation injuries. Hydrogen cyanide poisoning should be considered in fires involving nitrogen-containing polymer products such as wool, silk, polyurethane, and vinyl.

Diagnosis and Differential

The American Burn Association (ABA) classifies burns into major, moderate, and minor. Table 125-2 summarizes the ABA burn classifications.

Table 125-2 Burn Depth Features: American Burn Association Burn Classification

Burn

Classification Burn Characteristics Disposition

Major burn Partial-thickness >25% BSA, age 10–50 years Burn center

Partial-thickness >20% BSA, age <10 years or >50 years treatment

Full-thickness >10% BSA in anyone

Burns involving hands, face, feet, or perineum

Burns crossing major joints

Circumferential burns of an extremity

Burns complicated by inhalation injury

Electrical burns

Burns complicated by fracture or other trauma

Burns in high-risk patients

(continued)

125_Cydulka_Ch125.indd 680 21/03/17 9:50 AM

Source page 700

CHAPTER 125: Thermal and Chemical Burns 681

Table 125-2 Burn Depth Features: American Burn Association Burn Classification (continued)

Burn

Classification Burn Characteristics Disposition

Moderate burn Partial-thickness 15–25% BSA, age 10–50 years Hospitalization

Partial-thickness 10–20% BSA, age <10 years or >50 years

Full-thickness burns ≤10% BSA in anyone

No major burn characteristics present

Minor burn Partial-thickness <15% BSA, age 10–50 years Outpatient

Partial-thickness <10% BSA, age <10 years or >50 years treatment

Full-thickness <2% in anyone

No major burn characteristics present

Abbreviation: BSA, body surface area.

Emergency Department Care and Disposition

Management of patients with moderate to major burns is divided into three phases: prehospital care, ED resuscitation and stabilization, and transfer to a burn center. Prehospital burn care consists of stopping the burning process, establishing an airway, initiating fluid resuscitation, relieving pain and protecting the burn wound.

1. In the ED, reevaluate the airway, administer 100% O2, and intubate and ventilate the patient if there are signs of airway compromise, an airway burn, or respiratory distress. Obtain an ABG, carboxyhemoglobin level, and CXR. Monitor vital signs and oxygen saturation.

2. Establish 2 IV lines in unburned areas. Use a burn formula, such as the

Parkland/Baxter formula, to guide initial fluid resuscitation (Table 125-3). Ongoing fluid resuscitation is further guided by vital signs, cerebral and peripheral perfusion, and adequate urine output.

3. Evaluate and treat traumatic injuries using standard trauma resuscitation guidelines (Chapter 156, “Trauma in Adults”; Chapter 157, “Trauma in Children”; Chapter 158, “Trauma in the Elderly”).

Table 125-3 Parkland Formula for Fluid Resuscitation

Adults

LR 4 mL × weight (kg) × % BSA burned* over initial 24 h

Half over the first 8 h from the time of burn

Other half over the subsequent 16 h

Example: 70-kg adult with 40% second- and third-degree burns:

4 mL × 70 kg × 40 = 11,200 mL over 24 h

Children

LR 3 mL × weight (kg) × % BSA burned* over initial 24 h plus maintenance

Half over the first 8 h from the time of burn

Other half over the subsequent 16 h

*

Partial- and full-thickness burns only.

Abbreviations: BSA, body surface area; LR, lactated Ringer’s solution.

125_Cydulka_Ch125.indd 681 21/03/17 9:50 AM

Source page 701

4. After initiating resuscitative measures, address burn wound care. Apply cool compresses to small burns. Cover large burns with sterile dry sheets as saline soaked dressings may induce hypothermia. Administration of empiric antibiotics and application of topical antimicrobials during resuscitation are not recommended.

5. Administer intravenous opioid analgesia early and titrate to pain.

6. Treat inhalation injuries with humidified oxygen, endotracheal intubation and mechanical ventilation, bronchodilators, and pulmonary toilet. Hyperbaric oxygen therapy is used for severe carbon monoxide poisoning.

7. Circumferential burns of the neck, chest, or limbs may compromise breathing and circulation. Escharotomy may be required.

8. Update tetanus prophylaxis, if needed.

9. Hospitalize patients with moderate and major burns. The ABA’s criteria for referral to a burn unit are listed in Table 125-4.

10. Table 125-5 summarizes the care of minor burns. Patients with minor burns may be discharged after ED treatment, provided close follow-up

is available.

■■ CHEMICAL BURNS

More than 25,000 products are capable of producing chemical burns.

Chemical burn injuries represent a smaller percentage of burn injuries but

account for 30% of burn deaths.

Table 125-4 American Burn Association Burn Unit Referral Criteria

Full-thickness/third-degree burns in any age group

Electrical burns, including lightning injury

Chemical burns

Inhalation injury

Burn injury in patients with preexisting medical disorders that could complicate manage-

ment, prolong recovery, or affect mortality

Burn injury in any patients with concomitant trauma (such as fractures) in whom the burn

injury poses the greatest risk of morbidity or mortality

Burn injury in children in hospitals without qualified personnel or equipment to care for children

Burn injury in patients who will require special social, emotional, or long-term rehabilita-

tive intervention

Burn injury in children <10 years and adults >50 years of age

Table 125-5 ED Care of Minor Burns

Provide appropriate analgesics before burn care and for outpatient use

Cleanse burn with mild soap and water or dilute antiseptic solution

Debride wound as needed

Apply topical antimicrobial:

1% silver sulfadiazine cream (not on the face or in patients with a sulfa allergy)

Bacitracin ointment

Triple-antibiotic ointment (neomycin, polymyxin B, bacitracin zinc)

Consider use of synthetic occlusive dressings

Provide detailed burn care instructions with follow-up in 24–48 h

125_Cydulka_Ch125.indd 682 21/03/17 9:50 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.

- Content ID: SEG-TINTINALLI-EMM8-FULL.

- Source pages: 698–705.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 126: Electrical and Lightning Injuries

Source pages 706–711 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 706–711

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Electrical and Lightning Injuries**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 126: Electrical and Lightning Injuries 687 Table 126-1 Immediate and Delayed Complications of Electrical Injuries (Continued) Skeletal Vertebral compression fractures, long bone fractures, shoulder dislocations (anterior and posterior), scapular fractures Ophthalmologic Corneal burns, delayed cataracts, intraocular hemorrhage or thrombosis, uveitis, retinal detachment, orbital fracture Auditory Hearing loss, tinnitus, tympanic membrane perforation (rare), delayed mastoiditis or meningitis Oral burns Delayed labial artery hemorrhage, scarring and facial deformity, delayed speech development, impaired mandibular/dentition development Obstetric Spontaneous abortion, fetal death Abbreviation: GI, gastrointestinal. [Source pages 706–711]

- Diagnosis and Differential The diagnosis of electrical injury is usually based on the history of contact with an electrical source and the typical skin or oral lesions in children. [Source pages 706–711]

- However, patients with amnesia or any other type of altered mental status, with no skin injuries, must undergo an interview and a more detailed physical examination, especially for possible thunderstorm incidents. [Source pages 706–711]

- Lab- oratory and radiographic evaluation of high-voltage injures should follow standard trauma guidelines. [Source pages 706–711]

- Atrial or ventricular arrhythmias, bradyarr- thymias, prolonged QT intervals or ST-T wave abnormalities may be noted on the ECG. [Source pages 706–711]

Clinical reasoning structure

- **Problem representation:** Electrical and Lightning Injuries; source pages 706–711.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Electrical and Lightning Injuries**.

- Source-page range: **706–711**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 706

CHAPTER 126: Electrical and Lightning Injuries 687

Table 126-1 Immediate and Delayed Complications of Electrical

Injuries (Continued)

Skeletal Vertebral compression fractures, long bone fractures, shoulder dislocations (anterior and posterior), scapular fractures

Ophthalmologic Corneal burns, delayed cataracts, intraocular hemorrhage or thrombosis, uveitis, retinal detachment, orbital fracture

Auditory Hearing loss, tinnitus, tympanic membrane perforation (rare), delayed mastoiditis or meningitis

Oral burns Delayed labial artery hemorrhage, scarring and facial deformity, delayed speech development, impaired mandibular/dentition development

Obstetric Spontaneous abortion, fetal death

Abbreviation: GI, gastrointestinal.

Diagnosis and Differential

The diagnosis of electrical injury is usually based on the history of contact with an electrical source and the typical skin or oral lesions in children. However, patients with amnesia or any other type of altered mental status, with no skin injuries, must undergo an interview and a more detailed physical examination, especially for possible thunderstorm incidents. Laboratory and radiographic evaluation of high-voltage injures should follow standard trauma guidelines. Atrial or ventricular arrhythmias, bradyarrthymias, prolonged QT intervals or ST-T wave abnormalities may be noted on the ECG. An elevated serum CK, myoglobin or urine myoglobin suggests extensive muscle injury and rhabdomyolysis. Assessment and treatment of complications associated with electrical injuries are summarized in

Table 126-2.

Table 126-2 Assessment and Treatment of Complications of Electrical Injuries

Circulatory Order cardiac monitoring and ECG. Assess indications for admission for cardiac monitoring.

Renal Initiate fluid resuscitation and monitor urine output with a Foley catheter.

Nervous system Order head CT as needed for altered mental status; assess for spinal cord and peripheral nerve injury.

Skin Assess and treat cutaneous burns.*

Musculoskeletal Perform a careful assessment of spine, pelvis, long bones, and joints. Assess for compartment syndrome and need for fasciotomy.†

Vascular Spasm may occur leading to delayed thrombosis, aneurysm formation, or muscle damage.

Coagulation Treat coagulation disorders by eliminating the precipitating factor through early surgical debridement. If hemorrhage is present, replace coagulation factors.

Lungs Assess for inhalation injury, carbon monoxide, alveolar injury from blast.

Eyes Document complete eye examination. Delayed cataracts may develop.

(Continued)

126_Cydulka_Ch126.indd 687 21/03/17 9:55 AM

Source page 707

688 SECTION 12: Environmental Injuries

Table 126-2 Assessment and Treatment of Complications of Electrical Injuries (Continued)

Ears Assess for blast injury. Document hearing. Middle and inner ear disorders and hearing loss may occur.

GI Intra-abdominal injury may occur from current or blast.

Lips and oral cavity Watch for delayed bleeding.

Refer to Chapter 125, “Thermal and Chemical Burns.”

*

†

Refer to Chapter 51, “Rhabdomyolysis.”

The differential diagnosis includes other causes of arrhythmias, such as myocardial ischemia and neurologic dysfunction, such as stroke, closed head injury, and spinal cord injury.

Emergency Department Care and Disposition

1. Victims should be treated as blunt multiple trauma patients with attention to advanced life-support protocols and cervical and spine immobilization.
2. Assess and stabilize the airway. In facial burns due to electric flash, burn injury of the airway is not common, and the need to control the airway even less so. Provide high-flow oxygen and monitor with pulse oximetry.
3. Treat ventricular fibrillation, asystole, or ventricular tachycardia using standard ACLS protocols. Other dysrhythmias are usually transient and do not need immediate therapy.
4. Seizures occur immediately after the electric shock. A new seizure requires standard therapy and it is necessary to consider focal cerebral injury.
5. In compartment syndrome, consultation with a general surgeon, trauma surgeon, or burn surgeon must be requested in a priority manner.
6. Monitor for hyperkalemia, rhabdomyolysis, and the potential to develop acute kidney injury. Monitor urine output with a Foley catheter. Initiate crystalloid resuscitation in patients with large burns or extensively damaged muscle. The usual fluid replacement formulas used for burn patients often underestimate fluid requirements. In severe rhabdomyolysis or myoglobinuria, use aggressive fluid rehydration aiming for a urine output of 2 mL/kg/h and continuing until the serum creatine kinase level is less than five times normal or urine myoglobin measurements return to normal.
7. Burns need to be covered with dry sterile dressings. Prophylactic systemic antibiotics are usually not necessary unless there are large open contaminated wounds.
8. Administer tetanus prophylaxis, if not up to date.
9. Provide pain control with opioids.
10. Reduce and immobilize fractures and dislocations.
11. For pregnant patients, consult an obstetrician for fetal monitoring and admission.
12. Patients with more than a minor low-voltage injury should be admitted for further monitoring (Table 126-3).

126_Cydulka_Ch126.indd 688 21/03/17 9:55 AM

Source page 708

Table 126-3 Indications for Admission for Patients with Electrical Injuries

High voltage >600 V
Symptoms suggestive of systemic injury

Cardiovascular: chest pain, palpitations

Neurologic: loss of consciousness, confusion, weakness, headache, paresthesias

Respiratory: dyspnea

Gastrointestinal: abdominal pain, vomiting

Evidence of neurologic or vascular injury to a digit or extremity

Burns with evidence of subcutaneous tissue damage

Dysrhythmia or abnormal electrocardiogram

Suspected foul play, abuse, suicidal intent, or unreliable social situation

Pregnancy

Associated injuries requiring admission

Comorbid diseases (cardiac, renal, neurologic)

13. Children with isolated oral injuries or isolated hand wounds can usually be discharged home after consultation with an ENT specialist or plastic surgeon. Provide parents with instructions for controlling delayed labial artery bleeding. Arrange close follow-up with the consulting surgeon to assess scarring and stricture.

14. Asymptomatic patients who sustained a low-voltage injury (≤240 V) and have a normal ECG on presentation and normal physical exam may be discharged home.

■■ INJURIES DUE TO ELECTRONIC CONTROL DEVICES

Electronic control devices, such as the cattle prod, stun gun, and the TASER®, deliver high-voltage, low-amperage electrical pulses that induce involuntary muscle contraction, neuromuscular incapacitation, and/or pain. The likelihood of electrical injury is minimal. Injuries are usually limited to superficial punctures, and minor lacerations and cutaneous burns. The majority of deaths that have followed the use of these devices have occurred in persons who were extremely ill and agitated due to psychosis, stimulant drugs, or other conditions. Ill-appearing and agitated patients should be evaluated and treated in the same way as all patients who may have sustained blunt trauma or who have ingested unknown substances. Cardiac monitoring and other testing are not needed just because a TASER® has been used.

■■ LIGHTNING INJURIES

Lightning is a unidirectional extremely high-voltage current that causes substantially different injuries from those caused by high-voltage AC electricity. Lightning often travels over the skin, rather than through the body (flashover), which may explain why most people survive. However, a significant number may have permanent sequelae.

Clinical Features

Clinical events occur at multiple levels due to the electric effect, the high

temperature generated, and mechanical effects due to the shock wave. The

126_Cydulka_Ch126.indd 689 21/03/17 9:55 AM

Source page 709

690 SECTION 12: Environmental Injuries

Table 126-4 Complications Associated with Lightning Injuries

System Injury

Cardiovascular Dysrhythmias (asystole, ventricular fibrillation/tachycardia, premature ventricular contractions), electrocardiographic changes, myocardial infarction (unusual)

Neurologic Immediate or delayed, permanent or transient; loss of consciousness, confusion, amnesia, intracranial hemorrhage, hemiplegia, amnesia, respiratory center paralysis, cerebral edema, neuritis, seizures, parkinsonian syndromes, cerebral infarction, myelopathy, progressive muscular atrophy, progressive cerebellar syndrome, transient paralysis, paresthesias, myelopathy, autonomic dysfunction

Cutaneous Burns (first to third degree), scars, contractures

Ophthalmologic Cataracts (often delayed), corneal lesions, uveitis, iridocyclitis, vitreous hemorrhage, macular degeneration, optic atrophy, diplopia, chorioretinitis, retinal detachment, hyphema

Otologic Tympanic membrane rupture, temporary or permanent deafness, tinnitus, ataxia, vertigo, nystagmus

Renal Myoglobinuria, hemoglobinuria, renal failure (rare)

Obstetric Fetal death, placental abruption

Miscellaneous Secondary blunt trauma, compartment syndrome, disseminated intravascular coagulation

most common immediate cause of death after lightning strike is primary or secondary hypoxic cardiac arrest due to respiratory arrest. In the patient with spontaneous circulation, hypertension and tachycardia are caused by sympathetic activation. Patients may experience temporary loss of consciousness, confusion, and amnesia. Feathering or fern-shaped burns on the skin are pathognomonic of lightning but are transient and rarely seen. Tympanic membrane rupture may occur by blast effect. Deep tissue injuries, myoglobinuria, and renal failure are uncommon. The details of specific immediate and delayed systemic injuries and complications are summarized in Table 126-4.

Diagnosis and Differential

The diagnosis of lightning injury is based on history and should be considered in any critically ill patient found outside during or after a thunderstorm. While increasingly rare, indoor exposures to lightning may occur via indoor pools, use of hard-wired telephones, or in poorly constructed struc-

tures. Carefully assess patients for neurologic and cardiovascular complications, otologic and ophthalmologic injuries, and blunt trauma. Burns are uncommon. Laboratory and radiographic evaluation of lightning injures should follow standard trauma guidelines. Differential diagnosis includes stroke or intracranial hemorrhage, seizure disorder, and cerebral, spinal cord, or other neurologic trauma.

Emergency Department Care and Disposition

1. Victims should be treated as blunt multiple trauma patients with attention to advanced life-support protocols and cervical and spine immobilization.

126_Cydulka_Ch126.indd 690 21/03/17 9:55 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 706–711.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 127: Carbon Monoxide

Source pages 712–715 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 712–715

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Carbon Monoxide**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 127: Carbon Monoxide 693 Standard pulse oximetry is unreliable in the presence of increasing COHb as oxygen saturation readings will be artificially high or normal. [Source pages 712–715]

- Additional laboratory and imaging abnormalities seen in symptomatic patients may include elevated anion gap metabolic acidosis, elevated lactate, elevated creatine phosphokinase, elevated troponin, ECG changes consistent with ischemia, and bilateral globus pallidus lesions on brain imaging. [Source pages 712–715]

- The differential diagnosis is wide due to the nonspecific nature of the symptoms and includes flu-like illness, gastroenteritis, exposure to other toxins, and infectious causes of mental status changes. [Source pages 712–715]

- Cardiovascular compromise after poisoning may represent a concomitant myocardial infarction. [Source pages 712–715]

- ■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION Remove patients from the source of exposure and address airway, breath- ing, and circulation. [Source pages 712–715]

Clinical reasoning structure

- **Problem representation:** Carbon Monoxide; source pages 712–715.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Carbon Monoxide**.

- Source-page range: **712–715**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 712

CHAPTER 127: Carbon Monoxide 693

Standard pulse oximetry is unreliable in the presence of increasing COHb as oxygen saturation readings will be artificially high or normal. Additional laboratory and imaging abnormalities seen in symptomatic patients may include elevated anion gap metabolic acidosis, elevated lactate, elevated creatine phosphokinase, elevated troponin, ECG changes consistent with ischemia, and bilateral globus pallidus lesions on brain imaging.

The differential diagnosis is wide due to the nonspecific nature of the symptoms and includes flu-like illness, gastroenteritis, exposure to other toxins, and infectious causes of mental status changes. Cardiovascular compromise after poisoning may represent a concomitant myocardial infarction.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

Remove patients from the source of exposure and address airway, breathing, and circulation.

1. Begin treatment in all patients suspected of CO poisoning with the highest concentration of supplemental oxygen available (e.g., 100% oxygen via facemask with reservoir) and continue until the patient is asymptomatic. Provide continuous monitoring of vital signs, heart rate, and rhythm. Establish IV access.

2. Guidelines for hyperbaric oxygen therapy (HBO) in patients with severe poisoning are listed in Table 127-2. Indications for pediatric and adult HBO are similar. The threshold COHb for initiating HBO in pregnant patients is lower because of concerns for the fetus. Consult with a hyperbaric specialist. Patients must have a secure airway and stable hemodynamics before transport and treatment with HBO as access may be limited en route and in the hyperbaric chamber.

3. Guidelines for disposition are listed in Table 127-3. Ensure that the

home or work environment is no longer a source of carbon monoxide exposure.

Table 127-2 Commonly Utilized Indications for Referral for Hyperbaric Oxygen Treatment

Syncope

Confusion/altered mental status

Seizure

Coma

Focal neurologic deficit

Pregnancy with carboxyhemoglobin >15%

Carboxyhemoglobin >25% in any patient

Evidence of acute myocardial ischemia

127_Cydulka_Ch127.indd 693 21/03/17 10:01 AM

Source page 713

694 SECTION 12: Environmental Injuries

Table 127-3 Disposition Considerations

Symptom Severity	Disposition	Comments
Minimal or no symptoms	Home	Assess safety issues
Headache	Home after symptom resolution	Administer 100% oxygen in ED
Vomiting	Observe 4 h	
Elevated carbon monoxide level	Assess safety issues	
Ataxia, seizure, syncope, chest pain, focal neurologic deficit, dyspnea, ECG changes	Hospitalize	Administer 100% oxygen in ED Consult with Carbon monoxide level, comorbid hyperbaric specialist conditions, including pregnancy, age, and stability of the patient must be considered if considering transfer for hyperbaric oxygen

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 222, "Carbon Monoxide," by Gerald Maloney.

127_Cydulka_Ch127.indd 694 21/03/17 10:01 AM

Source page 714

Chapter Mushroom and Plant Poisoning

128 Chulathida Chomchai

■■ MUSHROOMS

Mushroom poisoning occurs in four population groups: foragers who purposefully harvest mushrooms or plants for food; teenagers and young adults who use mushrooms to get “high”; preschool-age children who accidentally ingest mushrooms while playing outdoors; and, rarely, victims of attempted homicide or suicide.

Clinical Features

Mushroom toxicity is divided into those with early onset, defined as within 2 hours after ingestion, and delayed onset, defined as 6 hours to 20 days after ingestion. It is important to determine if patients ingested only one type or multiple types of mushrooms and the time elapsed from ingestion to symptoms. Foragers may be able to provide a description of the mushroom. Clinical features of common mushroom poisonings are listed in Table 128-1.

Diagnosis and Differential

Most patients who develop GI symptoms within 2 hours of ingestion have a reassuring clinical course and do not develop major organ failure (Table 128-1). An exception to this is Amanita smithiana ingestion, which results in early GI symptoms and delayed renal failure. Mushrooms with potential liver, kidney, and CNS effects are often associated with onset of vomiting that is delayed for 6 or more hours after ingestion. Toxic species include species of genera Amanita, Galerina, Gyromitra, and Lepiota. Ingestions may be misdiagnosed as viral gastrointestinal illness or food poisoning if a history of mushroom ingestion is not pursued.

Emergency Department Care and Disposition

Consultation with a poison center is advised as regional differences in mushrooms types and toxicity exist.

1. Treatment regimens for mushroom poisonings are listed in Table 128-1.
2. Admit all patients with delayed onset of vomiting or diarrhea for ongoing monitoring of renal and liver function and fluid status for 48 hours.
3. Rapid progression to hepatic encephalopathy, hepatorenal syndrome, or coagulopathy are indications for liver transplantation. Consider transfer to a liver transplant center early in the course of mushroom ingestion.
4. Patients who ingest hallucinogenic mushrooms or mushrooms with muscarinic effects only may be discharged when symptoms subside.

128_Cydulka_Ch128.indd 695 21/03/17 10:21 AM

Source page 715

696 SECTION 12: Environmental Injuries

Table 128-1 Mushrooms: Symptoms, Toxicity, and Treatment

Symptoms	Mushrooms	Toxicity	Treatment
Gastrointestinal symptoms Onset <2 h	Chlorophyllum molybdites Omphalotus illudens	Nausea, vomiting, diarrhea (occasional bloody) Cantharellus cibarius	IV hydration Antiemetics
Onset 6–24 h (delayed renal failure)	Amanita smithiana Amanita phalloides, Amanita verna, Amanita virosa	Initial: nausea, vomiting, diarrhea Day 2: rise in AST, glucose, renal, hepatic, ALT and coagulation functions Day 3: hepatic failure with coagulopathy (delayed liver failure) renal failure	IV hydration; closely monitor electrolytes, and coagulation functions For Amanita and Lepiota: Acetylcysteine, load 140 mg/kg PO/NG Penicillin G 300,000 U/kg IV per day
Lepiota sp. Gyromitra esculenta		hemolysis (delayed onset seizures)	As above + headache, tremor, ataxia, seizures Silymarin 5 mg/kg IV over 1 h, then 20 mg/kg IV per day (available PO in the United States) For Gyromitra, treat seizures with both benzodiazepines AND pyridoxine 5 g IV
Muscarinic syndrome (SLUDGE)	Inocybe sp. Clitocybe sp.	Salivation, lacrimation, diarrhea, gastrointestinal distress, emesis CNS excitement	Supportive, atropine 0.01 mg/kg IV repeated as needed for severe secretions
Onset <30 min	Amanita muscaria Amanita pantherina	Intoxication, dizziness, ataxia, visual disturbances, seizures	Supportive, sedation with diazepam 2–5 mg IV as needed for adults
Amanita gemmata		tachycardia, hypertension, warm dry skin, dry mouth, mydriasis (anticholinergic effects)	
Hallucinations	Paneolus	Visual hallucinations	Supportive, sedation with

Onset <30 min Psilocybe sp. tions, ataxia or diazepam 2–5 mg IV for

adults

Gymnopillus

spectabilis

Disulfiram Coprinus atramen- Headache, flush- Supportive IV hydration

tarius ing, tachycardia,

hyperventilation,

palpitations

Onset 2–72 h Clitocybe clavipes

after mushroom,

and <30 min

after alcohol

Abbreviations: ALT, alanine aminotransferase; AST, aspartate aminotransferase; BUN, blood urea nitrogen;

CNS, central nervous system; IV, intravenous; PT, prothrombin time; PTT, partial thromboplastin time; SLUDGE

syndrome, salivation, lacrimation, urination, defecation, gastrointestinal hypermotility, and emesis.

128_Cydulka_Ch128.indd 696 21/03/17 10:21 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 712–715.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 128: Mushroom and Plant Poisoning

Source pages 716–721 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 716–721

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Mushroom and Plant Poisoning**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 128: Mushroom and Plant Poisoning 697 ■■ PLANTS Most patients with plant-related exposures and ingestions require no treatment and may be discharged after a short period of observation. [Source pages 716–721]

- Tables 128-2 and 128-3 describe the clinical symptoms and treatment regimens of common and severe poisonous plant ingestions. [Source pages 716–721]

- Table 128-2 Symptoms and Treatment of Severely Poisonous Plant Ingestions Plant Symptoms Treatment Castor bean (Ricinus Delayed gastroenteritis, delirium, Whole-bowel irrigation communis) seizures, coma, death Supportive care Coyotillo (Karwinskia Ascending paralysis Supportive care humboldtiana) Foxglove (Digitalis Nausea, vomiting, diarrhea, GI decontamination with purpurea) abdominal pain, confusion, activated charcoal cardiac dysrhythmias Monitoring of potassium level Antidysrhythmics Digoxin-specific Fab antibody for dysrhythmias Jequirity bean (Abrus Delayed gastroenteritis, delirium, Whole-bowel irrigation precatorius) seizures, coma, death Supportive care Oleander (Nerium Nausea, vomiting, diarrhea, GI decontamination with oleander) abdominal pain, confusion, activated charcoal cardiac dysrhythmias Monitoring of potassium level Antidysrhythmics Digoxin-specific Fab antibody for dysrhythmias Poison hemlock Tachycardia, tremors, diaphoresis, GI decontamination with (Conium maculatum) mydriasis, muscle weakness, activated charcoal seizures, neuromuscular blockade Supportive care Water hemlock Nausea, vomiting, abdominal GI decontamination (Cicuta maculata) pain, delirium, seizures, death Supportive care Yew (Taxus species) Common: nausea, vomiting, GI decontamination with acti- abdominal pain vated charcoal Rare: seizures, cardiac dysrhyth- Consider whole-bowel irrigation mias, coma Supportive care Table 128-3 Symptoms and Treatment of Common Poisonous Plant Ingestions or Exposures Plant Symptoms Treatment Ackee (Blighia sapida) Hypoglycemia Glucose Aloe (Aloe barbaden- Abdominal pain, diarrhea, red Supportive care sis) urine, nephritis Azalea (Rhododendron Usually minor symptoms GI decontamination with species) Severe intoxication: salivation, activated charcoal

- Brayer and Lynette Froula; and Chapter 220, “Poisonous Plants,” by Betty C. [Source pages 716–721]

Clinical reasoning structure

- **Problem representation:** Mushroom and Plant Poisoning; source pages 716–721.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Mushroom and Plant Poisoning**.

- Source-page range: **716–721**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 716

CHAPTER 128: Mushroom and Plant Poisoning 697

■■ PLANTS

Most patients with plant-related exposures and ingestions require no
treatment and may be discharged after a short period of observation.
Tables 128-2 and 128-3 describe the clinical symptoms and treatment
regimens of common and severe poisonous plant ingestions.

Table 128-2 Symptoms and Treatment of Severely Poisonous Plant Ingestions

Plant	Symptoms	Treatment
Castor bean (Ricinus communis)	Delayed gastroenteritis, delirium, seizures, coma, death	Whole-bowel irrigation Supportive care
Coyotillo (Karwinskia humboldtiana)	Ascending paralysis	Supportive care
Foxglove (Digitalis purpurea)	Nausea, vomiting, diarrhea, abdominal pain, confusion, cardiac dysrhythmias	GI decontamination with activated charcoal Monitoring of potassium level Antidysrhythmics Digoxin-specific Fab antibody for dysrhythmias
Jequirity bean (Abrus precatorius)	Delayed gastroenteritis, delirium, seizures, coma, death	Whole-bowel irrigation Supportive care
Oleander (Nerium	Nausea, vomiting, diarrhea, GI decontamination with	

oleander) abdominal pain, confusion, activated charcoal
cardiac dysrhythmias Monitoring of potassium level
Antidysrhythmics
Digoxin-specific Fab antibody
for dysrhythmias
Poison hemlock Tachycardia, tremors, diaphoresis, GI decontamination with
(Conium maculatum) mydriasis, muscle weakness, activated charcoal
seizures, neuromuscular blockade Supportive care
Water hemlock Nausea, vomiting, abdominal GI decontamination
(Cicuta maculata) pain, delirium, seizures, death Supportive care
Yew (Taxus species) Common: nausea, vomiting, GI decontamination with acti-
abdominal pain vated charcoal
Rare: seizures, cardiac dysrhyth- Consider whole-bowel irrigation
mias, coma Supportive care
Table 128-3 Symptoms and Treatment of Common Poisonous Plant Ingestions
or Exposures
Plant Symptoms Treatment
Ackee (Blighia sapida) Hypoglycemia Glucose
Aloe (Aloe barbaden- Abdominal pain, diarrhea, red Supportive care
sis) urine, nephritis
Azalea (Rhododendron Usually minor symptoms GI decontamination with
species) Severe intoxication: salivation, activated charcoal
lacrimation, bradycardia, Atropine for symptomatic
hypotension, progressive bradycardia
paralysis Fluids or vasopressors for
hypotension
(Continued)
128_Cydulka_Ch128.indd 697 21/03/17 10:21 AM

Source page 717

698 SECTION 12: Environmental Injuries
Table 128-3 Symptoms and Treatment of Common Poisonous Plant Ingestions
or Exposures (Continued)
Plant Symptoms Treatment
Cactus Pain and irritation from embed- Removal of spines
ded spines Rubber cement peel
Caladium species Usually minor symptoms Ingest cold milk or ice cream for
Severe intoxication: burning oral burning bradycardia
and irritation of oral mucosa, Fluids or vasopressors for
swelling, drooling, dysphagia, hypotension
respiratory compromise

Colchicum (autumn crocus, meadow saffron, glory lily)	Delayed and severe gastroen- teritis → severe multi-system organ failure	GI decontamination with activated charcoal Aggressive fluid resuscitation
Dumbcane (Dieffenbachia amoena)	Usually minor symptoms Severe intoxication: burning oral burning and irritation of oral mucosa, swelling, drooling, dysphagia, respiratory compromise	Ingest cold milk or ice cream for symptoms Analgesics Consider steroids if severe
Fava beans (Vicia faba)	In persons with glucose-6-phosphate dehydrogenase deficiency: GI upset, fever, headache, hemolytic anemia, hemoglobinuria, jaundice	Treatment varies depending on degree of hemolysis seen
Henbane (Hyoscyamus niger)	Anticholinergic symptoms: hallucinations, mydriasis, severe cases tachycardia, agitation, seizures, coma	Consider physostigmine in
Jimsonweed (Datura species)	Anticholinergic symptoms: hallucinations, mydriasis, tachycardia, agitation, seizures, coma	GI decontamination with acti- vated charcoal Consider whole-bowel irrigation Supportive care
Lily of the valley (Convallaria majalis)	Nausea, vomiting, diarrhea, abdominal pain, confusion, cardiac arrhythmias	GI decontamination with activated charcoal Monitoring of potassium level Antiarrhythmics Digoxin-specific Fab antibody for arrhythmias
Monkshood (Aconitum species)	Bradycardia, heart block, tor- sades de pointes, ventricular fibrillation	GI decontamination with activated charcoal Supportive care
Nettle (stinging nettle, bull nettle) (Urtica species)	Localized burning	Symptomatic care
Nightshade, common or woody (Solanum species)	Nausea, vomiting, diarrhea, abdominal pain; with larger doses: delirium, hallucinations, coma	Supportive care
Nightshade, deadly (Atropa belladonna)	Anticholinergic symptoms: hallucinations, mydriasis, tachycardia, agitation, seizures,	GI decontamination with activated charcoal Supportive care

coma

(Continued)

128_Cydulka_Ch128.indd 698 21/03/17 10:21 AM

Source page 718

CHAPTER 128: Mushroom and Plant Poisoning 699

Table 128-3 Symptoms and Treatment of Common Poisonous Plant Ingestions or Exposures (Continued)

Plant	Symptoms	Treatment
Peach, apricot, pear, crab apple, yam bean, large amounts are ingested: activated charcoal and hydrangea (pits or seeds)	Acute cyanide toxicity if GI decontamination with abdominal pain, lethargy	antidote therapy
Pepper (Capsicum species)	Irritation and pain on contact	Copious irrigation with water Milk or ice cream for oral irritation Analgesics
Philodendron species	Usually minor symptoms Severe intoxication: burning irritation and irritation of oral mucosa, swelling, drooling, dysphagia, respiratory compromise	Cold milk or ice cream for oral Analgesics Consider steroids
Pokeweed (Phytolacca americana)	Mucosal irritation, abdominal pain, nausea, vomiting, profuse diarrhea Severe intoxication: coma, death	GI decontamination with activated charcoal Supportive care
Potato, eggplant (raw) (Solanum species)	Nausea, vomiting, diarrhea, abdominal pain With larger doses: delirium, hallucinations, coma	Supportive care
Pothos (devil's ivy, Epipremnum species)	Usually minor symptoms Severe intoxication: burning irritation and irritation of oral mucosa, swelling, drooling, dysphagia, respiratory compromise	Cold milk or ice cream for oral Analgesics Consider steroids
Yellow sage (Lantana camara)	Dilated pupils, vomiting, diar- rhea, weakness, coma	GI decontamination with activated charcoal Fluids
Toxicodendron species (poison ivy, oak, and	Dermatitis	Skin protection Antipruritic and topical

sumac) therapies

Systemic steroids for facial,

genital, or widespread

involvement

Holly (Ilex species) Gastroenteritis GI decontamination with

Can be fatal if significant activated charcoal

ingestion Supportive care

Poinsettia (Euphorbia Occasional local irritation —

pulcherrima)

American mistletoe Gastroenteritis GI decontamination with

(Phoradendron activated charcoal

flavescens) Supportive care

Easter lily (Lilium Toxicity has not been reported No treatment necessary

longiflorum) in humans

128_Cydulka_Ch128.indd 699 21/03/17 10:21 AM

Source page 719

700 SECTION 12: Environmental Injuries

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study

Guide, 8th ed., see Chapter 219, "Mushroom Poisoning," by Anne F. Brayer and

Lynette Froula; and Chapter 220, "Poisonous Plants," by Betty C. Chen and

Lewis S. Nelson.

128_Cydulka_Ch128.indd 700 21/03/17 10:21 AM

References/provenance

- Source: *Tintinalli's Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 716–721.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 129: Diabetic Emergencies

Source pages 722–731 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 722–731

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Diabetic Emergencies**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 129: Diabetic Emergencies 703 Clinical Features Hyperglycemia causes an osmotic diuresis resulting in dehydration, hypo- tension, and tachycardia. [Source pages 722–731]

- Ketonemia causes an acidosis with myocardial depression, vasodilation, and compensatory Kussmaul respiration. [Source pages 722–731]

- Nausea, vomiting, and abdominal pain are common. [Source pages 722–731]

- The absence of fever does not exclude infection. [Source pages 722–731]

- Acetone, formed from oxidation of ketone bodies, causes the characteristic fruity odor of the patient’s breath. [Source pages 722–731]

Clinical reasoning structure

- **Problem representation:** Diabetic Emergencies; source pages 722–731.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Diabetic Emergencies**.

- Source-page range: **722–731**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 722

CHAPTER 129: Diabetic Emergencies 703

Clinical Features

Hyperglycemia causes an osmotic diuresis resulting in dehydration, hypotension, and tachycardia. Ketonemia causes an acidosis with myocardial depression, vasodilation, and compensatory Kussmaul respiration. Nausea, vomiting, and abdominal pain are common. The absence of fever does not exclude infection. Acetone, formed from oxidation of ketone bodies, causes the characteristic fruity odor of the patient's breath.

Diagnosis and Differential

Diagnosis of DKA is based on laboratory values of glucose >250 mg/dL, anion gap >10 mEq/L, bicarbonate <15 mEq/L, pH <7.3, and a moderate ketonemia/ketonuria. Beware of "euglycemic ketoacidosis" where blood glucose is only mildly elevated but the other criteria are present. This may occur if the patient has just taken insulin, has impaired gluconeogenesis (alcoholics), or takes a sodium-glucose cotransporter 2 inhibitor, such as canaglifozin.

An anion gap metabolic acidosis results from formation of ketone bodies. In DKA, the conversion of acetoacetate to β-hydroxybutyrate is favored which can result in low acetoacetate levels and high β-hydroxybutyrate levels. If the nitroprusside test is used to detect serum or urine ketones, it may be falsely low or negative as it only detects acetoacetate, not β-hydroxybutyrate.

Osmotic diuresis results in loss of sodium, chloride, calcium, phosphorus, and magnesium. Serum and urine glucose and ketones are elevated. Pseudohyponatremia is common: for each 100 mg/dL increase in blood glucose, the sodium decreases by 1.6 mEq/L, though some recommend this correction factor be 2.4, especially if glucose >400 mg/dL. Serum potassium may be low from osmotic diuresis and vomiting, normal, or high from acidosis. In acidosis, potassium is driven extracellularly. Note the acidotic patient with normal or low potassium has marked depletion of total body potassium.

Laboratory investigation includes serum pH, glucose, electrolytes, blood urea nitrogen, creatinine, phosphorus, magnesium, complete blood count, urinalysis (and pregnancy if indicated), electrocardiogram, and chest radiograph to assess the severity of DKA and search for the underlying cause. Determine pH using venous blood. The difference between venous and arterial pH (0.03) is not clinically significant, so the risk and pain of arterial puncture is unnecessary.

The differential diagnosis includes hypoglycemia, hyperosmolar hyperglycemic state, alcoholic or starvation ketoacidosis, renal failure, lactic acidosis, and ingestions such as salicylate, methanol, ethylene glycol, iron, or isoniazid.

Emergency Department Care and Disposition

1. The goal of treatment is to correct the volume deficit, acid–base imbalance, and electrolyte abnormalities, administer insulin, and treat the underlying cause (Fig. 129-1). See Table 129-1 for admission guidelines.

129_Cydulka_Ch129.indd 703 22/04/17 4:41 pm

Source page 723

Treatment Time Comments

0

If glucose >250 (12.8 mosm/L), urine +

Brief history/examination ketones, assume DKA, search for

Monitor, glucose, ECG, urine/serum ketones precipitant, infection, check ECG for

IV #1 NS 15–20 mL/kg/h for first hour hyperkalemia, infarction, foley catheter

#2 0.5 NS TKO as needed

Send electrolytes, CBC, phosphate,

calcium, magnesium, VBG, consider Begin flow sheet of vital signs, mental

blood/urine culture. status, BS, lytes, AG, venous pH, I/Os

Perform detailed history and exam

1h

If initial [K+] >5.2 initiate IV infusion of regular Initial electrolytes: check osmolarity,

insulin at 0.1–0.14 U/kg/h*. Repeat [K+] AG, BS, corrected [Na+], potassium

STAT in 2 h (fluid rate guide below) Initial [K+] determines further therapy

Adequate urine output is essential

If initial [K+] is >3.3 and <5.2 and urine output before initiating K+ therapy

add 20–30 mEq of K+ to each liter of fluid

and insulin drip as above Repeat glucose, electrolytes, AG

If AG >25 or glucose >800

] is <3.3 hold insulin drip and give

If initial [K+ (44 mosmol/L) or significant

K+ @20–30 mEq/h until [K+] is >3.3 then 2h comorbidity, consider ICU disposition

initiate insulin drip as above

If AG <25 and glucose <800

(44 mosmol/L) and no significant

comorbidity, consider floor or diabetic

After NS bolus: Generally for eunatremia or unit disposition

hypernatremia give 0.45 NS @250–500 mL/h

with K+ supplement as above. Rate of hydration is dependent on

For hyponatremia continue NS at 250–500 mL/h hemodynamics, hydration status,

urine output

If pH <6.9 may give 100 mmol NaHCO3 in Patients with pH > 6.9 do not require

400 mL of water with 20 eq KCL at 200 mL/h. NaHCO3

Repeat every 2 h until pH > 7.0. Check [K+]

every 2 h 3h Recheck glucose, lytes, AG, VBG,

mental status, I/Os, check results of

initial phosphate, magnesium, calcium.

Check electrolytes every 2 h initially

in ED. Check glucose hourly

Active -Adequate fluid infusion

Goals: -Insulin infusing If blood glucose does not decrease

-Maintain [K+] 3.3–5.2 by 10% (or 3 mmol/L/h) after 1 h of

-Lower glucose by 75 mg/dL/h insulin therapy, give 0.14 U/kg bolus

-Maintain adequate electrolytes then resume previous rate

(Ca, Mg, Phos)

When glucose approaches 200 (11 mmol/L) If blood glucose decreasing faster than

change IV to D5 0.5 NS with 20–40 mEq KCL/L 50–75 mg/dL/h, decrease insulin drip.

and/or decrease insulin rate to 0.02–0.05 U/kg/h 4–12 h Check glucose hourly

In young and new-onset diabetics

avoid excess free water, monitor

carefully

Source page 724

CHAPTER 129: Diabetic Emergencies 705

2. Administer bicarbonate when the pH ≤6.9. Then, the benefits of correcting the effects of acidosis (vasodilation, hyperkalemia, depression of cardiac, respiratory, and CNS systems) outweigh the risk of bicarbonate treatment (paradoxical CSF acidosis, hypokalemia, impaired oxyhemoglobin dissociation, rebound alkalosis, sodium overload).

3. Monitor glucose, anion gap, potassium, and bicarbonate hourly until recovery is established: glucose <200 mg/dL, bicarbonate ≥18, and

pH >7.3.

4. Cerebral edema is a complication of treatment. Young age and new-onset diabetes are risk factors. If there is any change in neurologic status early in therapy, begin treatment with mannitol 1 g/kg before obtaining the diagnostic CT scan. Avoiding rapid correction of sodium, glucose, and hypovolemia may reduce risk.

■■ HYPEROSMOLAR HYPERGLYCEMIC STATE

Hyperosmolar hyperglycemic state (HHS) occurs in type 2 diabetics. Causes are similar to DKA (Table 129-2). Insulin deficiency or resistance combined with physiologic stress results in an inflammatory state with elevated counterregulatory hormones causing hyperglycemia. The resulting osmotic diuresis causes hypovolemia and electrolyte losses.

Clinical Features

The typical patient is elderly with type 2 diabetes and presents with nonspecific complaints of weakness, dyspnea, chest or abdominal pain, or mental status changes, and has preexisting lung, heart, renal, or neurologic disease.

Physical examination reveals signs of dehydration with orthostasis, dry skin and mucous membranes, and altered mental status. Focal deficits and seizures may occur.

Diagnosis and Differential

Hyperosmolar hyperglycemic state is defined by laboratory values of glucose >600 mg/dL, calculated serum osmolality (formula: Osm = 2 [Na+] + glucose/18) >315 mOsm/kg, bicarbonate >15 mEq/L, pH >7.3, and negative to mildly positive ketones. Marked elevation in glucose and serum osmolality and the absence of a significant ketosis help distinguish HHS from DKA. Further laboratory investigation is similar to that for DKA discussed above.

Emergency Department Care and Disposition

Treatment consists of correcting the volume deficit, electrolyte imbalance, and hyperosmolality, and treating the underlying cause (Fig. 129-2). See Table 129-1 for admission guidelines.

129_Cydulka_Ch129.indd 705 22/04/17 4:41 pm

Source page 725

Treatment of Hyperosmolar Hyperglycemic State

IV Fluids Potassium

Insulin

Administer NS 1.0 L/h 0.1–0.14 U/kg/h IV insulin

infusion

If serum K+ is <3.3

mEq/L, give 40 mEq

K+ until K+ ≥3.3 mEq/L.

Evaluate corrected Goal is glucose ↓ by 50 mg/dL/h

serum Na and (2.8 mmol/L/h).

hydration If glucose ↓ >70 mg/dL/h

(>3.9 mmol/L/h), decrease

insulin to 0.05 U/kg/h. If serum K+ ≥5.0 mEq/L,

If glucose ↓ <50 mg/dL/h do not give K+ but check

Serum Na high

Serum Na low (<2.8 mmol/L), increase potassium every 1–2 h.

or normal

insulin to 0.14–0.20 U/kg/h.

1

2NS at NS at

4–14 mL/kg/h 4–14 mL/kg/h When serum glucose

≤300 mg/dL (≤16.6 mmol/L) If serum K+ ≥3.3 but

<5.0 mEq/L, provide K+

to maintain serum

K = at 4–5 mEq/L.*

Change to D5 12 NS and decrease

insulin to 0.05 U/kg/h to maintain

serum glucose between 200–250

mg/dL (11–13.8 mmol/L) mg/dL

until plasma osmolality is

≤315 mOsm/kg.

Check electrolytes, blood urea

nitrogen, creatinine, and

glucose every 2 h until stable.

After resolution of HHS, provide

SC insulin as needed.

Figure 129-2. Protocol for the management of severely ill adult patients with hyperosmolar hyperglycemic state (HHS). *Concentrations of K+≥20 mEq/L should be administered via central line. D5 1/2 NS, 5% dextrose in half normal saline; HHS, hyperosmolar hyperglycemic state; NS, normal saline.

■■ DIABETIC FOOT ULCERS

These are classified and managed as nonlimb-threatening, limb-threatening, or life-threatening (Table 129-3). Recommended antibiotic therapy is listed in Table 129-4.

129_Cydulka_Ch129.indd 706 22/04/17 4:41 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 722–731.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 131: Thyroid Disease Emergencies

Source pages 732–735 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 732–735

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Thyroid Disease Emergencies**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 131: Thyroid Disease Emergencies 713 Table 131-1 Symptoms and Signs of Thyrotoxicosis Affected System Symptoms Signs Constitutional Lethargy Diaphoresis Weakness Fever Heat intolerance Weight loss Neuropsychiatric Emotional lability Fine tremor Anxiety Muscle wasting Confusion Hyperreflexia Coma Periodic paralysis Psychosis Ophthalmologic Diplopia Lid lag Eye irritation Dry eyes Exophthalmos Ophthalmoplegia Conjunctival infection Endocrine: thyroid gland Neck fullness Thyroid enlargement Tenderness Bruit Cardiorespiratory Dyspnea Widened pulse pressure Palpitations Systolic hypertension Chest pain Sinus tachycardia Atrial fibrillation or flutter Congestive heart failure GI Diarrhea Hyperactive bowel sound Reproductive Oligomenorrhea Gynecomastia Decreased libido Telangiectasia Gynecologic Menorrhagia Sparse pubic hair Irregularity Hematologic Pale skin Anemia Leukocytosis Dermatologic Hair loss Pretibial myxedema* Warm, moist skin Palmar erythema Onycholysis * Pretibial myxedema may be present in 5% of patients with Graves’ disease. [Source pages 732–735]

- Diagnosis and Differential An elevated FT4 or FT3 level and a suppressed TSH level are diagnostic of thyrotoxicosis. [Source pages 732–735]

- Fever and tachy- cardia are cardinal features; laboratory tests cannot distinguish it from thyrotoxicosis. [Source pages 732–735]

- The differential diagnosis for thyroid storm includes sepsis, heat stroke, delirium tremens, neuroleptic malignant syndrome, serotonin syndrome, pheochromocytoma, and sympathomimetic drug overdose. [Source pages 732–735]

- Emergency Department Care and Disposition Treatment includes: 1. [Source pages 732–735]

Clinical reasoning structure

- **Problem representation:** Thyroid Disease Emergencies; source pages 732–735.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Thyroid Disease Emergencies**.

- Source-page range: **732–735**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 732

CHAPTER 131: Thyroid Disease Emergencies 713

Table 131-1 Symptoms and Signs of Thyrotoxicosis

Affected System Symptoms Signs

Constitutional Lethargy Diaphoresis

Weakness Fever

Heat intolerance Weight loss

Neuropsychiatric Emotional lability Fine tremor

Anxiety Muscle wasting

Confusion Hyperreflexia

Coma Periodic paralysis

Psychosis

Ophthalmologic Diplopia Lid lag

Eye irritation Dry eyes

Exophthalmos

Ophthalmoplegia

Conjunctival infection

Endocrine: thyroid gland Neck fullness Thyroid enlargement

Tenderness Bruit

Cardiorespiratory Dyspnea Widened pulse pressure

Palpitations Systolic hypertension

Chest pain Sinus tachycardia

Atrial fibrillation or flutter

Congestive heart failure

GI Diarrhea Hyperactive bowel sound

Reproductive Oligomenorrhea Gynecomastia

Decreased libido Telangiectasia

Gynecologic Menorrhagia Sparse pubic hair

Irregularity

Hematologic Pale skin Anemia

Leukocytosis

Dermatologic Hair loss Pretibial myxedema*

Warm, moist skin

Palmar erythema

Onycholysis

*

Pretibial myxedema may be present in 5% of patients with Graves' disease.

Diagnosis and Differential

An elevated FT4 or FT3 level and a suppressed TSH level are diagnostic of thyrotoxicosis. Thyroid storm is a clinical diagnosis. Fever and tachycardia are cardinal features; laboratory tests cannot distinguish it from thyrotoxicosis. The differential diagnosis for thyroid storm includes sepsis, heat stroke, delirium tremens, neuroleptic malignant syndrome, serotonin syndrome, pheochromocytoma, and sympathomimetic drug overdose.

Emergency Department Care and Disposition

Treatment includes:

1. Supportive care, including airway stabilization, supplemental oxygen, and cardiac monitoring. Cooling is indicated for hyperthermia. Administer acetaminophen 325 to 650 mg PO/PR every 4 to 6 hours. Correct

131_Cydulka_Ch131.indd 713 21/03/17 11:13 AM

Source page 733

dehydration with IV fluids. Provide nutritional support: glucose, multivitamins, thiamine, folate.

2. Inhibit thyroid hormone release with thioamides: PTU loading dose of 600 to 1000 mg PO and follow with 200 to 250 mg every 4 hours or methimazole 40 mg PO as loading dose and follow with 25 mg every 4 hours. PTU is preferred as it also blocks peripheral conversation of T4 to T3.

3. Inhibit new thyroid hormone production (at least 1 hour after step 3) with Lugol solution 8-10 drops PO every 6-8 h or potassium iodide 5 drops PO every 6 h or IV iopanoic acid 1 gram every 8 h for the first 24 h, then 500 mg twice a day or ipodate 0.5 - 3 g/day for 3 days or lithium carbonate 300mg PO every 6 h.

4. Provide peripheral β-adrenergic receptor blockade using propranolol IV in slow 1 to 2 mg boluses and repeat every 10 to 15 minutes or esmolol 500 mcg/kg IV bolus followed by 50 to 200 mcg until desired effect is achieved. Propranolol 20 -120 mg PO may be used in patients with less toxicity. Avoid propranolol and esmolol in patients with bronchospastic disease or severe congestive heart failure.

5. Prevent peripheral conversion of T4 to T3 using hydrocortisone 100 mg initially, followed by three times daily until stable or dexamethasone 2 mg every 6 h.

6. Evaluate and treat precipitating causes. Search for an infectious source in febrile patients and administer appropriate antibiotics for identified infections.

7. Admit patients to a monitored or ICU setting.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 228, "Hyperthyroidism" by Alzamani Mohammad Idrose; Chapter 229, "Hyperthyroidism," by Alzamani Mohammad Idrose.

131_Cydulka_Ch131.indd 714 21/03/17 11:13 AM

Source page 734

C h apter Adrenal Insufficiency

132 Michael P. Kefer

Adrenal insufficiency results when the physiologic demand for glucocorticoids and mineralocorticoids exceeds the supply from the adrenal cortex. The pituitary secretes adrenocorticotropin hormone (ACTH) and associated melanocyte stimulating hormone (MSH). ACTH stimulates the adrenal cortex to secrete cortisol. Cortisol has negative feedback on the pituitary to inhibit secretion of ACTH and MSH.

■■ CLINICAL FEATURES

Primary adrenal insufficiency is due to adrenal gland failure, resulting in low cortisol and aldosterone production. Clinical features include weakness, dehydration, hypotension, anorexia, nausea, vomiting, weight loss, and abdominal pain. Hyperpigmentation of skin and mucous membranes occurs as a result of uninhibited MSH secretion in conjunction with ACTH.

Secondary adrenal insufficiency results from inadequate secretion of ACTH with resultant cortisol deficiency. Aldosterone levels are not affected because of regulation by the renin-angiotensin axis and serum potassium levels. Therefore, hyperpigmentation and hyperkalemia do not occur.

Adrenal crisis is the acute, life-threatening form of adrenal insuffi-

ciency. Clinical features described above are severe and accompanied by shock and altered mental status.

■■ DIAGNOSIS AND DIFFERENTIAL

All patients with adrenal insufficiency have low plasma cortisol levels. The ACTH level is high in primary and low in secondary adrenal insufficiency. Cortisol and ACTH levels are usually not available in the emergency department, so the diagnosis of primary adrenal insufficiency is based on the presence of the clinical features and lab findings of hyponatremia, hyperkalemia, hypoglycemia, anemia, metabolic acidosis, and prerenal azotemia. Secondary adrenal insufficiency is similarly diagnosed, but hyperkalemia is not seen as aldosterone secretion is normal.

The most common cause of acute adrenal insufficiency is adrenal suppression from prolonged steroid use with either abrupt steroid withdrawal or exposure to increased physiologic stress such as injury, illness, or surgery. It may take up to 1 year for the hypothalamic-pituitary-adrenal axis to recover following prolonged suppression with steroid treatment. Tables 132-1 and 132-2 list causes of primary and secondary adrenal insufficiency, respectively. Consider the diagnosis of adrenal crisis in any patient with unexplained hypotension refractory to pressors, especially if one of the causes listed is known to exist.

132_Cydulka_Ch132.indd 715 21/03/17 11:47 AM

Source page 735

Table 132-1 Causes of Primary Adrenal Insufficiency

Primary Adrenal Insufficiency	Examples
Autoimmune	Isolated adrenal insufficiency or associated with polyglandular insufficiencies (polyglandular autoimmune syndrome type I or II)
Adrenal hemorrhage or thrombosis	Necrosis caused by meningococcal sepsis
Coagulation disorders	Overwhelming sepsis (Waterhouse–Friderichsen syndrome)
Drugs	Adrenal enzyme inhibitors (affect those with limited pituitary or adrenal reserve)
	Etomidate
	Aminoglutethimide (can be used by body builders)
	Mitotane (orphan drug used to treat adrenocortical carcinoma)

Ketoconazole

Infections Tuberculosis

Fungal, bacterial sepsis

Acquired immunodeficiency syndrome involving

adrenal glands

Infiltrative disorders Sarcoidosis

Hemochromatosis

Amyloidosis

Lymphoma

Metastatic cancer

Surgery Bilateral adrenalectomy

Bariatric surgery

Hereditary Adrenal hypoplasia

Congenital adrenal hyperplasia

Adrenoleukodystrophy

Familial glucocorticoid deficiency

Table 132-2 Causes of Secondary Adrenal Insufficiency

Secondary Adrenal Insufficiency

(hypothalamic-pituitary dysfunction) Examples

Sudden cessation of prolonged Chronic use of steroid inhibits CRH and

glucocorticoid therapy ACTH production

Pituitary necrosis or bleeding Postpartum pituitary necrosis (Sheehan’s

syndrome)

Exogenous glucocorticoid administration Causes decreased production of CRH at

hypothalamus and ACTH at pituitary

Brain tumors Pituitary tumor

Hypothalamic tumor

Local invasion (craniopharyngioma)

Pituitary irradiation Disrupts CRH and ACTH production capacity

Pituitary surgery in hypothalamic-pituitary axis

Head trauma involving the pituitary gland

(Continued)

132_Cydulka_Ch132.indd 716 21/03/17 11:47 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 732–735.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 132: Adrenal Insufficiency

Source pages 736–743 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 736–743

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Adrenal Insufficiency**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 132: Adrenal Insufficiency 717 Table 132-2 Causes of Secondary Adrenal Insufficiency (Continued) Secondary Adrenal Insufficiency (hypothalamic-pituitary dysfunction) Examples Infiltrative disorders of the pituitary Sarcoidosis or hypothalamus Hemosiderosis Hemochromatosis Histiocytosis X Metastatic cancer Lymphoma CNS infections involving hypothalamus Tuberculosis or pituitary Meningitis Fungus Human immunodeficiency virus Abbreviations: ACTH, adrenocorticotropic hormone; CRH, corticotropin-releasing hormone. [Source pages 736–743]

- ■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION 1. [Source pages 736–743]

- Five percent dextrose in normal saline is the fluid of choice to correct hypoglycemia, hyponatremia, and hypotension. [Source pages 736–743]

- Hydrocortisone 100 mg IV provides glucocorticoid and mineralcorti- coid activity. [Source pages 736–743]

- If an ACTH stimulation test is planned, use dexametha- sone 4 mg IV instead, which will not affect the test result. [Source pages 736–743]

Clinical reasoning structure

- **Problem representation:** Adrenal Insufficiency; source pages 736–743.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Adrenal Insufficiency**.

- Source-page range: **736–743**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 736

CHAPTER 132: Adrenal Insufficiency 717

Table 132-2 Causes of Secondary Adrenal Insufficiency (Continued)

Secondary Adrenal Insufficiency

(hypothalamic-pituitary dysfunction) Examples

Infiltrative disorders of the pituitary Sarcoidosis

or hypothalamus Hemosiderosis

Hemochromatosis

Histiocytosis X

Metastatic cancer

Lymphoma

CNS infections involving hypothalamus Tuberculosis

or pituitary Meningitis

Fungus

Human immunodeficiency virus

Abbreviations: ACTH, adrenocorticotropic hormone; CRH, corticotropin-releasing hormone.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. Five percent dextrose in normal saline is the fluid of choice to correct

hypoglycemia, hyponatremia, and hypotension.

2. Hydrocortisone 100 mg IV provides glucocorticoid and mineralcorti-

coid activity. If an ACTH stimulation test is planned, use dexametha-

sone 4 mg IV instead, which will not affect the test result.

3. Vasopressors are added when shock is refractory to the above.

4. Patients on chronic steroid therapy for adrenal insufficiency with minor

illness or injury should have their glucocorticoid dose tripled daily with

prompt follow-up.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study

Guide, 8th ed., see Chapter 230, “Adrenal Insufficiency,” by Alzamani Mohammad

Idrose.

132_Cydulka_Ch132.indd 717 21/03/17 11:47 AM

Source page 737

132_Cydulka_Ch132.indd 718 21/03/17 11:47 AM

Source page 738

SECTION

Hematologic and Oncologic

Emergencies 14

C h a p ter Evaluation of Anemia and

133 the Bleeding Patient

Rita K. Cydulka

Anemia may be chronic and unrelated to the chief complaint, or it may result from acute blood loss as seen in trauma, gastrointestinal bleeding, or other acute hemorrhage. Suspect underlying bleeding disorders in patients presenting with spontaneous bleeding from multiple sites, bleeding from nontraumatized sites, delayed bleeding several hours after injury, or bleeding into deep tissues or joints.

■■ CLINICAL FEATURES

The rate of the development of the anemia, the extent of the anemia, the age of the patient, and the ability of the cardiovascular system to compensate for the decreased oxygen-carrying capacity determine the severity of the patient's symptoms and clinical presentation. Patients may complain of weakness, fatigue, palpitations, orthostatic symptoms, and dyspnea with minimal exertion. Patients may have pale conjunctiva, skin, and nail beds. Tachycardia, hyperdynamic precordium, and systolic murmurs may be present. Tachypnea at rest and hypotension are late signs. Use of ethanol, prescription drugs, and recreational drugs may alter the patient's ability to compensate for the anemia.

Patients with bleeding may or may not have an obvious site of hemorrhage. A history of excessive or abnormal bleeding in the patient and other family members may indicate an underlying bleeding disorder. Historical data about liver disease and drug use, such as use of ethanol, aspirin, nonsteroidal anti-inflammatory drugs, warfarin, and antibiotics should be gathered. Mucocutaneous bleeding (including petechiae, ecchymoses, purpura, and epistaxis), gastrointestinal, genitourinary, or heavy menstrual bleeding are features associated with qualitative or quantitative platelet disorders. Patients with deficiencies of coagulation factors often present with delayed bleeding, hemarthroses, or bleeding into potential spaces between fascial planes and into the retroperitoneum. Patients with combi-

nation abnormalities of platelets and coagulation factors, such as disseminated intravascular coagulation, present with both mucocutaneous and potential space bleeding.

719

133_Cydulka_Ch133.indd 719 21/04/17 10:47 am

Source page 739

720 SECTION 14: Hematologic and Oncologic Emergencies

Acquired hemolytic anemia may be autoimmune or drug-induced. Microangiopathic syndromes include thrombotic thrombocytopenic purpura (TTP) and hemolytic-uremic syndrome (HUS). The classic pentad of TTP is CNS abnormalities, renal disease, fever, microangiopathic hemolytic anemia, and thrombocytopenia. HUS consists of acute nephropathy or renal failure, microangiopathic hemolytic anemia, and thrombocytopenia. Macrovascular hemolysis can be caused by prosthetic heart valves.

■■ DIAGNOSIS AND DIFFERENTIAL

A decreased red blood count, hemoglobin, and hematocrit are diagnostic for anemia. The initial evaluation of newly diagnosed anemia includes a complete blood count, review of RBC indices, reticulocyte count, stool hemoccult examination, urine pregnancy test in females, and examination of the peripheral blood smear. The mean corpuscular volume (MCV) and reticulocyte count can assist in classifying the anemia and can aid in differential diagnosis (Fig. 133-1).

Laboratory studies used to diagnose bleeding disorders can be divided into the following three categories: (a) those that test the initial formation of a platelet plug (primary hemostasis); (b) those that assess the formation of cross-linked fibrin (secondary hemostasis); and (c) those that test the fibrinolytic system, which is responsible for limiting the size of the fibrin clots formed (see Tables 133-1 and 133-2). Initial studies to evaluate the patient with suspected bleeding disorders include complete blood count with platelet count, prothrombin time, and partial thromboplastin time and international normalized ratio (INR). If there is a suspicion for a hemolytic anemia, further studies may be warranted (Table 133-3).

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. Type and crossmatch blood in patients with anemia and ongoing blood loss so that it is available for transfusion, if necessary.

2. Consider immediate transfusion of packed RBCs in symptomatic patients who are hemodynamically unstable or have evidence of tissue hypoxia.

3. Admit patients with anemia and ongoing blood loss for further evaluation and treatment. Admit patients with chronic anemia or newly diagnosed anemia with unclear etiology if they are hemodynamically unstable, hypoxic, acidotic, or demonstrate cardiac ischemia.

4. Consider hematology consultation to assist in evaluation of those patients with anemia of unclear etiology, anemic patients with concomitant abnormalities of platelets and white blood cell counts, and patients with suspected bleeding disorders.

133_Cydulka_Ch133.indd 720 21/04/17 10:47 am

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 736–743.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 133: Evaluation of Anemia and the Bleeding Patient

Source pages 744–745 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 744–745

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Evaluation of Anemia and the Bleeding Patient**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 133: Evaluation of Anemia and the Bleeding Patient 725 Table 133-3 Basic Tests and Findings in the Evaluation of Hemolytic Anemia Purpose Test Finding Confirm anemia/blood Hemoglobin Decreased loss Hematocrit Decreased Confirm compensatory Reticulocyte count Increased RBC production Confirm hemolysis Peripheral smear Schistocytes—intravascular hemolysis, RBCs fragmented by shear mechanism Spherocytes—extravascular hemolysis, RBC phagocytosis by macrophages Confirm hemolysis Lactate dehydrogenase Increased, released by RBCs Potassium Increased, released by RBCs Confirm hemolysis Haptoglobin Decreased, indicative of intravascular hemolysis Free haemoglobin Increased, indicative of intravascular hemolysis Hemoglobinuria Present Confirm hemoglobin Total bilirubin Increased breakdown Indirect (unconjugated) Increased (hepatic conjugation of bilirubin bilirubin overwhelmed) Urinary urobilinogen Increased Abbreviation: RBC, red blood cell. [Source pages 744–745]

- ■■ FURTHER READING For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 231, “Anemia,” by John C. [Source pages 744–745]

- Hemphill; Chapter 232, “Tests of Hemostasis,” by Stephen John Cico and Robin R. [Source pages 744–745]

- Hemphill; Chapter 237, “Acquired Hemolytic Anemia,” by Laurie Ann Dixon and Robin R. [Source pages 744–745]

- 133_Cydulka_Ch133.indd 725 21/04/17 10:47 am C hapter Acquired Bleeding Disorders 134 Alisheba Hurwitz Platelet abnormalities, drugs, systemic illnesses, and endogenous antico- agulants can cause acquired bleeding disorders. [Source pages 744–745]

Clinical reasoning structure

- **Problem representation:** Evaluation of Anemia and the Bleeding Patient; source pages 744–745.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Evaluation of Anemia and the Bleeding Patient**.

- Source-page range: **744–745**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 744

CHAPTER 133: Evaluation of Anemia and the Bleeding Patient 725

Table 133-3 Basic Tests and Findings in the Evaluation of Hemolytic Anemia

Purpose Test Finding

Confirm anemia/blood Hemoglobin Decreased

loss Hematocrit Decreased

Confirm compensatory Reticulocyte count Increased

RBC production

Confirm hemolysis Peripheral smear Schistocytes—intravascular

hemolysis, RBCs fragmented by shear

mechanism

Spherocytes—extravascular

hemolysis, RBC phagocytosis by

macrophages

Confirm hemolysis Lactate dehydrogenase Increased, released by RBCs

Potassium Increased, released by RBCs

Confirm hemolysis Haptoglobin Decreased, indicative of intravascular

hemolysis

Free haemoglobin Increased, indicative of intravascular

hemolysis

Hemoglobinuria Present

Confirm hemoglobin Total bilirubin Increased

breakdown Indirect (unconjugated) Increased (hepatic conjugation of

bilirubin bilirubin overwhelmed)

Urinary urobilinogen Increased

Abbreviation: RBC, red blood cell.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 231, "Anemia," by John C. Ray and Robin R. Hemphill; Chapter 232, "Tests of Hemostasis," by Stephen John Cico and Robin R. Hemphill; Chapter 237, "Acquired Hemolytic Anemia," by Laurie Ann Dixon and Robin R. Hemphill.

133_Cydulka_Ch133.indd 725 21/04/17 10:47 am

Source page 745

C hapter Acquired Bleeding Disorders

134 Alisheba Hurwitz

Platelet abnormalities, drugs, systemic illnesses, and endogenous anticoagulants can cause acquired bleeding disorders.

■■ ACQUIRED PLATELET DEFECTS

Clinical Features

Patients with platelet disorders commonly present with petechiae or mucosal bleeding. Splenomegaly may be noted in patients experiencing platelet sequestration.

Diagnosis and Differential

Acquired platelet abnormalities include quantitative and qualitative defects. Quantitative platelet disorders may be caused by decreased platelet production (marrow infiltration, aplastic anemia, drugs, viral infections, and chronic alcohol use), or increased platelet destruction (idiopathic thrombocytopenic purpura (ITP), thrombotic thrombocytopenic purpura (TTP), hemolytic uremic syndrome, disseminated intravascular coagulation (DIC), viral infections, drugs, and HELLP syndrome). Other causes include acute hemorrhage, hemodialysis, and splenic sequestration. Qualitative platelet disorders are commonly associated with renal disease, liver disease, DIC, drugs (aspirin, NSAIDs, clopidogrel), and myeloproliferative disorders.

Emergency Department Care and Disposition

1. Circulatory stabilization of the acutely bleeding patient is paramount, including volume resuscitation, packed red cell transfusion, and pressors as necessary.
2. Perform a focused history and physical; order appropriate laboratory studies. Most evaluations of bleeding patients will include a CBC with peripheral smear. Other studies to consider include PT/INR, PTT, basic metabolic panel, type and screen, and fibrinogen.
3. Consider platelet transfusion in patients with a platelet count <10,000/mm3 or active bleeding with platelets <50,000/mm3.
4. Consult with a hematologist, as subtleties in diagnosis and treatment

exist. For example, some conditions may be worsened by platelet transfusion (DIC and TTP).

■■ BLEEDING IN IMMUNE THROMBOCYTOPENIA

Emergency Department Care and Disposition

In general, patients with a platelet count >50,000/mm3 require no treatment. Patients with a platelet count <30,000/mm3 and patients with a platelet count of <50,000/mm3 with bleeding typically require treatment.

1. Treatment options include corticosteroids, such as prednisone 60 to 100 mg PO daily (onset of action 4 to 14 days).

134_Cydulka_Ch134.indd 726 21/03/17 11:53 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 744–745.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 134: Acquired Bleeding Disorders

Source pages 746–749 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 746–749

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Acquired Bleeding Disorders**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 134: Acquired Bleeding Disorders 727 2. [Source pages 746–749]

- Immunoglobulin 1 g/kg daily IV (onset of action 1 to 3 days) is usually reserved for patients with very low platelet counts and bleeding. [Source pages 746–749]

- If transfusing platelets, generally two to three times the usual dose is required due to circulating antibodies. [Source pages 746–749]

- ■■ ACQUIRED COAGULATION DEFECTS Coagulation disorders often present with significant hemorrhage and fre- quently involve platelet defects as well. [Source pages 746–749]

- Bleeding in Liver Disease Patients with liver disease have an increased risk of bleeding for multiple reasons including decreased synthesis of vitamin K-dependent coagulation factors (II, VII, IX, and X), thrombocytopenia, and increased fibrinolysis. [Source pages 746–749]

Clinical reasoning structure

- **Problem representation:** Acquired Bleeding Disorders; source pages 746–749.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Acquired Bleeding Disorders**.

- Source-page range: **746–749**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 746

CHAPTER 134: Acquired Bleeding Disorders 727

2. Immunoglobulin 1 g/kg daily IV (onset of action 1 to 3 days) is usually reserved for patients with very low platelet counts and bleeding.

3. If transfusing platelets, generally two to three times the usual dose is required due to circulating antibodies.

■■ ACQUIRED COAGULATION DEFECTS

Coagulation disorders often present with significant hemorrhage and frequently involve platelet defects as well.

Bleeding in Liver Disease

Patients with liver disease have an increased risk of bleeding for multiple reasons including decreased synthesis of vitamin K-dependent coagulation factors (II, VII, IX, and X), thrombocytopenia, and increased fibrinolysis.

Emergency Department Care and Disposition

1. Transfuse red cells as needed.

2. Transfuse platelets for thrombocytopenia <60,000/mm3.

3. Consider transfusion of fresh frozen plasma 15 mL/kg IV with caution. It may increase the risk of thrombotic complications, and it may increase portal hypertension thus worsening variceal bleeding.

Bleeding in Renal Disease

A variety of hemostatic defects are associated with renal disease including platelet dysfunction due to anemia, deficiency of coagulation factors, and thrombocytopenia.

Emergency Department Care and Disposition

1. Transfuse red cells as needed.

2. Desmopressin 0.3 µg/kg SC or IV improves platelet function rapidly.

3. Hemodialysis improves platelet function transiently for 1 to 2 days.

4. Transfuse platelets and cryoprecipitate for life-threatening bleeding only.

Disseminated Intravascular Coagulation

The clinical features of DIC are a result of simultaneous hemorrhage and thrombosis and vary according to the underlying illness. Bleeding typically predominates and can range from petechiae to diffuse hemorrhaging from GI/GU tracts, surgical wounds, and venipuncture sites.

Emergency Department Care and Disposition

1. Treat the underlying illness and provide hemodynamic support including pressors and packed red cells as necessary.
2. Transfuse platelets if counts <20,000/mm3, or <50,000/mm3 with active bleeding.
3. Transfuse fresh frozen plasma (15 mL/kg) for bleeding patient with PT or PTT that are >1.5 times the upper limit of normal, or with fibrinogen level <150 mg/dL.

134_Cydulka_Ch134.indd 727 21/03/17 11:53 AM

Source page 747

4. Transfuse fibrinogen concentrate to patients with hypofibrinogenemia and active bleeding, with the goal of raising fibrinogen levels to 100 to 150 mg/dL.
5. Administer vitamin K IV to bleeding patients with elevated PT.
6. Consider heparin use for patients with a clinical picture dominated by thrombosis.
7. Tranexamic acid should be given only in the setting of DIC due to trauma.

Bleeding Due to Circulating Anticoagulants

Patients with acquired factor VIII inhibitors (acquired hemophilia A) present with spontaneous bruising and hematomas. Consult with a hematologist for management guidance. Treatment options for acute bleeding episodes include recombinant factor VIIa and activated prothrombin complex concentrate.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 233, "Acquired Bleeding Disorders," by Robert W. Shaffer and Sally A. Santen

134_Cydulka_Ch134.indd 728 21/03/17 11:53 AM

Source page 748

C hap t e r Hemophilias and von Willebrand

135 Disease

Colin G. Kaide

■■ HEMOPHILIAS

The most common hemophilias are caused by genetic deficiencies of factor VIII (hemophilia A) or factor IX (hemophilia B).

Clinical Features

Bleeding complications depend on the severity of the disease. Patients with severe disease (factor VIII or factor IX activity level <1%) experience spontaneous bleeds and difficult to control bleeding after trauma. Patients with moderate disease (1% to 5% factor activity level) may bleed spontaneously but more commonly bleed after trauma. Patients with mild disease (5% to 40% factor activity level) usually only bleed after trauma. Easy bruising, recurrent hemarthrosis, and muscle hematomas are the most common clinical manifestations. Central nervous system (CNS) bleeding accounts for the most common cause of death. Retroperitoneal, gastrointestinal (GI), and soft tissue bleeding can occur. Mucocutaneous bleeding (dental bleeding, epistaxis, GI tract bleeding, lungs bleeding) can occur, although this is rare. Neck hematomas may obstruct the airway. Unless there is another underlying disease, most patients with hemophilia do not have problems with minor cuts or abrasions. Almost all cases of hemophilia occur in men with women being genetic carriers of the disease. The unlikely exception occurs when a hemophiliac man reproduces with a woman who is a carrier of the hemophilia gene; hemophiliac females can result. Owing to spontaneous mutations and lyonization of the X chromosome women may develop varying degrees of decreased factor levels.

A rare form of hemophilia occurs when a person develops a spontaneous antibody to their intrinsic factor VIII. It is often (40%) associated with other conditions such as autoimmune diseases, cancers, certain drugs, the postpartum period. It usually happens in the sixth or seventh decade and affects men and women equally. Widespread purpura and internal bleeding are common with hemarthrosis less common.

Diagnosis and Differential

Clinically, it is impossible to differentiate between hemophilias A and B. Laboratory testing in patients with hemophilia most often shows a normal prothrombin time (PT), prolonged partial thromboplastin time (PTT), and a normal bleeding time. However, if greater than 30% to 40% of factor activity is present, the PTT may be normal. Specific factor assays may be used to differentiate between the types of hemophilia. Ten percent to 25% of patients with hemophilia A and 1% to 2% of patients with hemophilia B will develop an inhibitor, which is an antibody against the deficient factor. The presence of an inhibitor makes treatment more difficult, requiring

729
135_Cydulka_Ch135.indd 729 21/03/17 12:01 PM

Source page 749

alternative, pre-activated factors such as recombinant factor VIIa or factor eight inhibitor bypassing activity (FEIBA).

Emergency Department Care and Disposition

The mainstay of therapy is early factor replacement. Replacement factor products are listed in Table 135-1.

1. Determine the type of hemophilia and the presence or absence of inhibitor. See Table 135-2 for factor replacement guidelines. Factor replacement may need to be instituted before definitive imaging after head trauma and other life-threatening injuries. If an inhibitor is present, use therapy as outlined in Table 135-3.

2. Determine the desired factor activity level. Factor activity level determines how much factor replacement is required. Calculate the amount of factor needed using the patient's weight and the desired increase in factor:

Factor VIII required = (Target factor – Base line factor) × Weight (kg) × 0.5

Factor IX required = (Target factor – Base line factor)× Weight (kg)

For severe hemophilia, assume 0% intrinsic activity. The half-life of factor VIII is approximately 8 to 12 hours. For hemophilia B, the half-life of factor IX is approximately 16 to 24 hours.

3. Treat patients with undiagnosed bleeding disorders with fresh frozen plasma (FFP). FFP contains 1 unit of factor VIII/mL. Specific factor assays should guide further therapy.

Table 135-1 Replacement Factor Products for Hemophilia Treatment

Hemophilia Type	Available Products* (Manufacturer/Distributor)
Hemophilia A	Recombinant Factor VIII Concentrates
	Advate® (Baxter)
	Helixate FS® (Bayer/CSL Behring)
	Kogenate FS® (Bayer)
	Recombinate® (Baxter)
	Xyntha® (Pfizer)
	Human Plasma-Derived Factor VIII Concentrates
	Hemofil M® (Baxter)
	Monoclate-P® (CSL Behring)
	Human Plasma-Derived Factor VIII Concentrates That Contain von Willebrand Factor
	Alphanate® (Grifols)

Humate-P® (CSL Behring GmbH)

Koate-DVI® (Grifols/Kedrion Biopharma)

Hemophilia B Recombinant Factor IX Concentrate

BeneFIX® (Pfizer)

Human Plasma-Derived Factor IX Concentrates

AlphaNine SD® (Grifols)

Mononine® (CSL Behring)

Commercial trade names provided for ease of specific identification.

*

135_Cydulka_Ch135.indd 730 21/03/17 12:01 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.

- Content ID: SEG-TINTINALLI-EMM8-FULL.

- Source pages: 746–749.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 135: Hemophilias and von Willebrand Disease

Source pages 750–753 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 750–753

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Hemophilias and von Willebrand Disease**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 135: Hemophilias and von Willebrand Disease 731 Table 135-2 Initial Factor Replacement Guidelines in Severe Hemophilia Desired Factor Hemophilia Hemophilia B Severity and Level to Control A Initial Initial Dose Site Bleeding Dose (IU/ kg) (IU/ kg) Comments Minor: skin – – – Abrasions and (deep laceration) superficial lacera- tions usually do not require factor replacement. [Source pages 750–753]

- Minor: early 0.2–0.4 IU/mL 10–20 20–30 Repeat dose every hemarthrosis, (20%–40%) 12–24 h for 1–3 mild muscle days until bleeding bleeding, mild episode is resolved. [Source pages 750–753]

- oral bleeding Typical duration of replacement is 1–3 days. [Source pages 750–753]

- Moderate: defi- 0.3–0.6 IU/mL 15–30 25–50 Orthopedic consult nite hemarthro- (30%–60%) may be required for sis, moderate splinting, physical muscle bleed- therapy, and follow- ing, moderate up. [Source pages 750–753]

- Typical duration oral bleeding of replacement is 3–5 days. [Source pages 750–753]

Clinical reasoning structure

- **Problem representation:** Hemophilias and von Willebrand Disease; source pages 750–753.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Hemophilias and von Willebrand Disease**.

- Source-page range: **750–753**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 750

CHAPTER 135: Hemophilias and von Willebrand Disease 731

Table 135-2 Initial Factor Replacement Guidelines in Severe Hemophilia

Desired Factor Hemophilia Hemophilia B

Severity and Level to Control A Initial Initial Dose

Site Bleeding Dose (IU/ kg) (IU/ kg) Comments

Minor: skin – – – Abrasions and

(deep laceration) superficial lacera-

tions usually do not

require factor

replacement. Treat

with pressure and

topical thrombin.

Minor: early 0.2–0.4 IU/mL 10–20 20–30 Repeat dose every

hemarthrosis, (20%–40%) 12–24 h for 1–3

mild muscle days until bleeding

bleeding, mild episode is resolved.

oral bleeding Typical duration of

replacement is 1–3

days.

Moderate: defi- 0.3–0.6 IU/mL 15–30 25–50 Orthopedic consult

nite hemarthro- (30%–60%) may be required for

sis, moderate splinting, physical

muscle bleed- therapy, and follow-

ing, moderate up. Typical duration

oral bleeding of replacement is

3–5 days.

Major: retro- 0.6–1.0 IU/mL 30–50 30–50 Repeat dose every

pharyngeal, GI, (60%–100%) 8–24 h until resolu-
intra-abdominal, tion of bleeding
intrathoracic, episode. May
retroperitoneal require replacement
for up to 10 days.
CNS 1.0 IU/mL (100%) 50 50–100 Treat before CT.
Early neurosurgical
consultation.

Table 135-3 Replacement Therapy for Hemophilia A and B in Patients with Inhibitors

Type of Product*	Initial Dose	Dosing Interval	Comments
Human plasma-derived activated prothrombin complex concentrate	50–100 units/kg 200 units/kg.	6–12 h	Total daily doses should not normally exceed
FEIBA NF® (Baxter)			
Recombinant activated factor VII	90 µg/kg	2h	Repeat until hemostasis achieved or therapy
NovoSeven RT® (Novo Nordisk)			judged ineffective.

*
Commercial trade names provided for ease of specific identification.

135_Cydulka_Ch135.indd 731 21/03/17 12:01 PM

Source page 751

4. Treat minor bleeding in patients with mild hemophilia A with desmopressin (DDAVP) 0.3 µg/kg IV over 15 to 30 minutes or DDAVP 150 µg single spray in each nostril (for children >5 years, DDAVP 150 µg single spray in one nostril). Very mild mucosal bleeding can also be treated with antifibrinolytic agents, such as e-aminocaproic acid (EACA) 75 to 100 mg/kg (up to 6 g) IV/PO every 6 hours or tranexamic acid 10 to 25 mg/kg IV every 6 hours.
5. Indications for admission include bleeding involving the head, neck, pharynx, retropharynx, or retroperitoneum, potential compartment syndrome, and inability to control pain, and treatment requiring multiple factor replacement.

■■ VON WILLEBRAND DISEASE

Clinical Features

von Willebrand disease (vWD) is a group of disorders caused by a defect or deficiency of von Willebrand factor (vWF). vWF is a cofactor for platelet adhesion and a carrier protein for factor VIII in the plasma. Type

1 vWD is most common; patients have 20% to 50% of normal vWF levels and usually manifest with skin and mucosal bleeding symptoms. Patients with type 2 vWF have abnormal and dysfunctional vWF. Patients with type 3 vWF have complete vWF deficiency and have clinical presentations similar to hemophilia. Patients with mild vWF are frequently unaware of the disease until bleeding occurs after a traumatic episode or surgical procedure.

Diagnosis and Differential

The PT and PTT are usually normal. The bleeding time is prolonged and vWF activity is low. Bleeding time, however, is rarely used anymone and has been suplanted by more specific platelet activation and aggregation studies. Variability in vWF levels can make it difficult to distinguish vWD from hemophilia A.

Emergency Department Care and Disposition

The treatment of vWD depends on the type of disease and the severity of bleeding. Indications for admission are similar to patients with hemophilia.

1. The mainstay of treatment for bleeding in type 1 vWD is desmopressin (DDAVP) 0.3 µg/kg (up to 20 µg) SC/IV every 12 to 24 hours for up to four doses or DDAVP 150 µg single spray in each nostril (for children >5 years, DDAVP 150 µg single spray in 1 nostril). If there is no response to DDAVP, administer factor VIII concentrate or cryoprecipitate as described for type 2 and 3 vWD.
2. Factor VIII concentrate containing vWF is used to treat bleeding in patients with type 2 or 3 disease (Table 135-1).
3. Cryoprecipitate 10 bags IV every 12 to 24 hours can be used for type 1 refractory to DDAVP or type 2 or 3 vWD. There is, however, a risk of viral transmission.
4. Platelet transfusions may benefit patients with type 3 vWD who do not respond to vWF-containing plasma products.

135_Cydulka_Ch135.indd 732 21/03/17 12:01 PM

Source page 752

5. Oral contraceptives may help increase vWF levels and limit menstrual bleeding in women with vWD and menorrhagia.
6. Patients who have sustained dental injuries or who require dental procedures may need e-aminocaproic acid (EACA) 75 to 100 mg/kg (up to 6 grams) PO every 6 hours or tranexamic acid mouthwash for 5 to 10 days.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study

Guide, 8th ed., see Chapter 235, “Hemophilias and von Willebrand’s Disease,” by Robin R. Hemphill.

135_Cydulka_Ch135.indd 733 21/03/17 12:01 PM

Source page 753

Chapter Sickle Cell Disease and Other

136 Hereditary Hemolytic Anemias

Colleen Fant

Inherited anemias stem from abnormal hemoglobin structure or production or from abnormal RBC structure or metabolism. Anemia occurs when destruction of red blood cells exceeds production. Most patients with dyshemoglobinemia are aware of their status.

■■ SICKLE CELL DISEASE

Clinical Features

Most patients with sickle cell disease (SCD) (homozygous (HbSS)) or sickle trait (heterozygous (HbAS)) present with symptoms related to vasoocclusive crisis, or with pain, weakness, or infections. Physical examination findings commonly include pallor, venous stasis changes, jaundice, hepatosplenomegaly, cardiac flow murmurs, cardiomegaly, and high-output congestive heart failure (CHF).

Painful vasooclusive crisis in the musculoskeletal system or, in some cases, organ-specific symptoms are the most common presentations to the ED. Crisis occurs when sickled RBCs mechanically obstruct blood flow, causing ischemia, organ damage, and infarction. Crisis-causing stresses include fever or infection (especially encapsulated organisms Haemophilus influenzae or Pneumococcus), cold exposure or high altitude, dehydration or overexertion, medication noncompliance, or drug use.

Acute symptomatic anemia results from splenic sequestration, bone marrow failure (aplastic crisis), or hemolysis due to infection and presents with fatigue, weakness, dyspnea, chronic heart failure, or shock.

Life or limb-threatening events seen in SCD patients include acute chest syndrome (vasoocclusive pulmonary insult), stroke, renal infarct, mesenteric infarcts, sepsis, avascular necrosis, osteomyelitis, pneumonia, or priapism.

Diagnosis and Differential

The degree of illness guides the evaluation of an acute crisis. Although workups should be individualized, the more common, vague complaints (pain, weakness, fever) must include a search for an underlying, treatable event. See Table 136-1.

Anemia due to splenic sequestration may present as splenomegaly with or without left-upper quadrant pain and elevated reticulocyte count. Anemia

due to bone marrow failure may be suggested by a depressed reticulocyte count. Leukocytosis or left shift with increased bands suggests infection. Assessment of electrolytes will allow evaluation of dehydration and renal function. Liver function tests and lipase may help evaluate abdominal pain. Perform a pregnancy test on women. Febrile SCD patients without localizing symptoms should have blood cultures, urinalysis, and chest radiographs performed.

136_Cydulka_Ch136.indd 734 21/03/17 12:06 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 750–753.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 136: Sickle Cell Disease and Other Hereditary Hemolytic Anemias

Source pages 754–759 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 754–759

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Sickle Cell Disease and Other Hereditary Hemolytic Anemias**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 136: Sickle Cell Disease and Other Hereditary Hemolytic Anemias 735 Table 136-1 Guidelines for the Assessment and Management of Acute Vasoocclusive Crisis History Duration and location of pain History of fever History of focal swelling or redness Precipitating factors for acute episode Medications taken for pain relief Physical examination Assess degree of pain Inspect sites of pain, looking for swelling, warmth, redness General: respiratory distress, pallor, hydration, jaundice, rash Vital signs: especially temperature, pulse oximetry Respiratory: lung sounds Heart: cardiomegaly and systolic murmur common with chronic anemia Abdomen: tenderness, organomegaly Ancillary tests If moderate to severe pain, focal pathology is present, or pain is atypical for acute episode Obtain CBC, leukocyte differential, reticulocyte count, urinalysis Chest radiograph, if signs of lower respiratory tract pathology Blood cultures and additional blood tests: as indicated by clinical condition General management Bed rest, provide warmth and a calm, relaxing atmosphere Distractions where appropriate: television, music, etc. [Source pages 754–759]

- Oral fluids: typically about 3 L/d IV fluids to correct dehydration or if reluctant to drink or vomiting is present Oxygen: not routinely required unless hypoxemia is present Encourage deep breathing, incentive spirometry Consider adjuvant therapy: anxiolytic, tinzaparin, antiemetic Pain management Use analgesics appropriate to degree of pain Acetaminophen for mild pain NSAID for mild to moderate pain (avoid if renal insufficiency is present) Opioids for moderate to severe pain, typical initial doses include: Morphine, 0.3 mg/kg PO or 0.1–0.15 mg/kg IV Hydromorphone, 0.06–0.08 mg/kg PO or 0.015–0.020 mg/kg IV Reassess response in 15–30 minutes, may repeat with one- fourth to one-half initial dose, consider patient’s known effective dose Disposition and Consider admission to the hospital if: follow-up Acute chest syndrome is suspected Sepsis, osteomyelitis, or other serious infection is suspected Hemoglobin is <5 g/dL White blood cell count is >30,000/mm3 Platelet count is <100,000/mm3 Consider discharge if: Pain is under control and patient can take oral fluids and medications Appropriate oral analgesics are available Patient is able to comply with home care instructions Patient has resources for follow-up Abbreviations: CBC, complete blood count; NSAID, nonsteroidal anti-inflammatory drug. [Source pages 754–759]

- 136_Cydulka_Ch136.indd 735 21/03/17 12:06 PM 736 SECTION 14: Hematologic and Oncologic Emergencies Patients presenting with symptoms of acute chest syndrome (chest pain, cough, fever, dyspnea) need immediate evaluation (Table 136-2). [Source pages 754–759]

- Chest radiographs are important to identify the presence of a new infiltrate. [Source pages 754–759]

- Assess oxygenation; type and cross for possible exchange transfusion. [Source pages 754–759]

Clinical reasoning structure

- **Problem representation:** Sickle Cell Disease and Other Hereditary Hemolytic Anemias; source pages 754–759.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Sickle Cell Disease and Other Hereditary Hemolytic Anemias**.

- Source-page range: **754–759**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 754

CHAPTER 136: Sickle Cell Disease and Other Hereditary Hemolytic Anemias 735

Table 136-1 Guidelines for the Assessment and Management of Acute

Vasoocclusive Crisis

History Duration and location of pain

History of fever

History of focal swelling or redness

Precipitating factors for acute episode

Medications taken for pain relief

Physical examination Assess degree of pain

Inspect sites of pain, looking for swelling, warmth, redness

General: respiratory distress, pallor, hydration, jaundice, rash

Vital signs: especially temperature, pulse oximetry

Respiratory: lung sounds

Heart: cardiomegaly and systolic murmur common with chronic

anemia

Abdomen: tenderness, organomegaly

Ancillary tests If moderate to severe pain, focal pathology is present, or pain is atypical for acute episode
Obtain CBC, leukocyte differential, reticulocyte count, urinalysis
Chest radiograph, if signs of lower respiratory tract pathology
Blood cultures and additional blood tests: as indicated by clinical condition
General management Bed rest, provide warmth and a calm, relaxing atmosphere
Distractions where appropriate: television, music, etc.
Oral fluids: typically about 3 L/d
IV fluids to correct dehydration or if reluctant to drink or vomiting is present
Oxygen: not routinely required unless hypoxemia is present
Encourage deep breathing, incentive spirometry
Consider adjuvant therapy: anxiolytic, tinzaparin, antiemetic
Pain management Use analgesics appropriate to degree of pain
Acetaminophen for mild pain
NSAID for mild to moderate pain (avoid if renal insufficiency is present)
Opioids for moderate to severe pain, typical initial doses include:
Morphine, 0.3 mg/kg PO or 0.1–0.15 mg/kg IV
Hydromorphone, 0.06–0.08 mg/kg PO or 0.015–0.020 mg/kg IV
Reassess response in 15–30 minutes, may repeat with one-fourth to one-half initial dose, consider patient's known effective dose
Disposition and follow-up Consider admission to the hospital if:
Acute chest syndrome is suspected
Sepsis, osteomyelitis, or other serious infection is suspected
Hemoglobin is <5 g/dL
White blood cell count is >30,000/mm3
Platelet count is <100,000/mm3
Consider discharge if:
Pain is under control and patient can take oral fluids and medications
Appropriate oral analgesics are available
Patient is able to comply with home care instructions
Patient has resources for follow-up

Abbreviations: CBC, complete blood count; NSAID, nonsteroidal anti-inflammatory drug.

136_Cydulka_Ch136.indd 735 21/03/17 12:06 PM

Source page 755

736 SECTION 14: Hematologic and Oncologic Emergencies

Patients presenting with symptoms of acute chest syndrome (chest pain, cough, fever, dyspnea) need immediate evaluation (Table 136-2). Chest radiographs are important to identify the presence of a new infiltrate. Assess oxygenation; type and cross for possible exchange transfusion. Radiographs of the skeleton are indicated for atypical focal bone pain. Advanced imaging for abdominal pain or for neurologic manifestations will help to assess these symptoms.

Table 136-2 Assessment and Treatment of Acute Chest Syndrome

History Major presenting symptom: dyspnea, fever, cough

Accompanying chest, rib, bone, or joint pain

Assess degree or severity of pain

Recent or previous sepsis, infection, pneumonia, or hospitalization

Prior history of acute chest syndrome, especially if required intubation and ventilatory support

Potentially infectious contacts

Current medications

Immunization history: especially pneumococcal and Haemophilus influenzae type b

Baseline hemoglobin level and arterial oxygenation saturation

Physical examination General: respiratory distress, pallor, hydration, jaundice, rash

Vital signs: especially temperature, pulse oximetry

Respiratory: chest wall, lung sounds

Heart: cardiomegaly and systolic murmur common with chronic anemia

Abdomen: tenderness, organomegaly

Ancillary tests Complete blood count, leukocyte differential, reticulocyte count, urinalysis

Cross-match sample: if red blood cell transfusion is contemplated

Arterial blood gas: if moderate to severe respiratory distress and/or hypoxemia on pulse oximetry

Chest radiography

Blood cultures

Additional blood tests: as indicated by clinical condition

Treatment Oxygen: adjust according to pulse oximetry

Oral hydration: preferable

IV hydration: use hypotonic fluids, use a rate and dose at approximately 1.5 times of maintenance (overaggressive IV fluids can worsen acute chest syndrome)

Analgesics: if needed, generally potent parenteral opioids are used; monitor for signs of respiratory suppression

Antibiotics: empiric antibiotics recommended to treat community
acquired pneumonia
Bronchodilators: nebulized β2-adrenergic agonists
Chest physiotherapy
Transfusion: use if severe acute anemia is present
Exchange Consider when
transfusion Severe acute chest syndrome on admission and past history of
requiring ventilatory support: useful to prevent intubation
Deterioration despite above management: useful to prevent
intensive care unit admission
Patient already intubated and on ventilatory support: useful to
shorten duration of ventilatory need
Suspected or confirmed fat or bone marrow embolism
136_Cydulka_Ch136.indd 736 21/03/17 12:06 PM

Source page 756

CHAPTER 136: Sickle Cell Disease and Other Hereditary Hemolytic Anemias 737

The differential diagnosis for musculoskeletal pain includes osteomyelitis, bony infarcts, cellulitis, and acute arthritides. For abdominal pain, the differential includes pancreatitis, hepatitis, cholecystitis, pelvic inflammatory disease, and pyelonephritis. Chest pain may be due to acute chest, pneumonia, or pulmonary embolus. Neurologic symptoms may be due to meningitis, cerebral infarct, ischemia, or hemorrhage.

Emergency Department Care and Disposition

Initial management for acute crisis in SCD patients is primarily supportive and includes pain management and an assessment for the underlying cause of the crisis (see Tables 136-1 and 136-2).

1. Encourage oral rehydration if dehydration is suspected. IV crystalloid may be used as an alternative, up to 1.5 times maintenance. Hypotonic fluids may reduce the tendency for sickling.
2. Administer opioid pain medications for severe pain. Individualized treatment plans are warranted for patients with frequent relapses.
3. Administer supplemental oxygen for hypoxia.
4. ECG and cardiac monitoring are appropriate for patients with cardiopulmonary symptoms.
5. For symptoms of acute infection, cultures should be obtained and broad-spectrum antibiotics administered. For patients with suspected acute chest, antibiotics should be targeted toward community-acquired pneumonia.
6. Consider exchange transfusion for acute crisis or complications in specific circumstances—aplastic crisis, cardiopulmonary decompensa-

tion, pregnancy, stroke, respiratory failure, general surgery, and priapism (requires urologic consultation). Exchange transfusion is typically reserved for patients with severe crises, that is, PaO2 <60 mm Hg.

7. Patients with priapism lasting >4 hours should additionally receive corporal aspiration and injection of dilute α-adrenergic agonist.

8. Admission criteria include pulmonary, neurologic, aplastic, or infectious crises; splenic sequestration; intractable pain; persistent nausea and vomiting; or an uncertain diagnosis.

9. Patients able to be discharged should receive oral analgesics, close follow-up, and instructions to return immediately for temperature above 38°C or worsening symptoms.

■■ VARIANTS OF SICKLE CELL DISEASE

Other genetic variants of hemoglobinopathies exist and vary in presentation, from asymptomatic to SCD-like and depend upon the specific abnormality and whether homozygous, heterozygous, or combined with sickle cell trait.

Thalassemias

Thalassemias are hereditary disorders caused by defective synthesis of globin chains, resulting in microcytic, hypochromic, hemolytic anemia. The degree of illness depends upon the type and number of genetic abnormalities.

136_Cydulka_Ch136.indd 737 21/03/17 12:06 PM

Source page 757

Patients with β-thalassemia minor have mild microcytic anemia and are generally asymptomatic. Patients with β-thalassemia major (Cooley's anemia) develop hepatosplenomegaly, jaundice, and bony changes. They are at increased risk for infection and may develop severe anemia requiring blood transfusions. Iron overload from transfusions can cause significant morbidity and mortality.

Glucose-6-Phosphate Dehydrogenase Deficiency

Glucose-6-phosphate dehydrogenase (G6PD) deficiency is the most common enzymopathy of RBCs. This causes hemoglobin precipitation, RBC removal by the spleen, and hemolysis. The amount of hemolysis depends on degree of enzyme abnormality. Most patients are asymptomatic until an exposure to an oxidative stress (e.g., medication, infection, fava beans) causes hemolysis. A serious complication of G6PD deficiency is severe neonatal jaundice that presents in the first week of life. Evaluation includes a complete blood count and reticulocyte count, bilirubin levels, serum aminotransferases, and lactate dehydrogenase. Treatment is determined by the

patient’s overall clinical condition and includes early treatment of infection and may include blood transfusion for severe anemia.

Hereditary Spherocytosis

Hereditary spherocytosis results from an erythrocyte membrane defect creating small inflexible RBCs that are unable to pass through the spleen resulting in an increased rate of destruction and a compensatory increase in RBC production. Complications include aplastic or megaloblastic crises, cholecystitis or cholelithiasis, splenomegaly, hemolytic crisis, and neonatal hemolysis with jaundice. Treatments include blood transfusions and splenectomy in severe cases.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 236, “Sickle Cell Disease and Hereditary Hemolytic Anemias,” by Jean Williams-Johnson and Eric Williams.

136_Cydulka_Ch136.indd 738 21/03/17 12:06 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 754–759.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 137: Transfusion Therapy

Source pages 760–765 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 760–765

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Transfusion Therapy**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 137: Transfusion Therapy 741 Table 137-1 Transfusion Reactions Reaction Signs and Type Symptoms Management Evaluation Acute intra- Fever, chills, Immediately stop transfu- Retype and repeat vascular low back sion. [Source pages 760–765]

- hemolytic pain, flushing, IV hydration to maintain Direct and indirect reaction dyspnea, diuresis; diuretics may be Coombs test. [Source pages 760–765]

- CBC, creatinine, shock, Cardiorespiratory support as prothrombin time, hemoglobinuria indicated. [Source pages 760–765]

- Haptoglobin, indirect bilirubin, lactate dehydrogenase, plasma free hemoglobin. [Source pages 760–765]

- Delayed Often have Usually presents days to Hemolytic workup as extravascular low-grade weeks after transfusion. [Source pages 760–765]

Clinical reasoning structure

- **Problem representation:** Transfusion Therapy; source pages 760–765.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Transfusion Therapy**.

- Source-page range: **760–765**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 760

CHAPTER 137: Transfusion Therapy 741

Table 137-1 Transfusion Reactions

Reaction Signs and

Type Symptoms Management Evaluation

Acute intra- Fever, chills, Immediately stop transfu- Retype and repeat

vascular low back sion. cross-match.

hemolytic pain, flushing, IV hydration to maintain Direct and indirect

reaction dyspnea, diuresis; diuretics may be Coombs test.

tachycardia, necessary. CBC, creatinine,

shock, Cardiorespiratory support as prothrombin time,

hemoglobinuria indicated. activated partial

thromboplastin time.

Haptoglobin, indirect

bilirubin, lactate

dehydrogenase,

plasma free

hemoglobin.

Urine for hemoglobin.

Delayed Often have Usually presents days to Hemolytic workup as

extravascular low-grade weeks after transfusion. above to investigate

hemolytic fever but may Rarely causes clinical the possibility

reaction be entirely instability. of intravascular

asymptomatic hemolysis.

Febrile Fever, chills Stop transfusion. Hemolytic workup as

nonhemolytic Initially manage as in intra- above because may

transfusion vascular hemolytic reaction not be able to initially

reaction (above) because one cannot distinguish febrile from

initially distinguish between hemolytic transfusion

the two. reactions.

Can treat fever and chills

with acetaminophen.

Usually mild but can be life-threatening in patients with tenuous cardiopulmonary status.

Consider infectious workup. Premedication with acetaminophen can mask this reaction.

Allergic reaction pruritus

Mild: urticaria, If mild, reaction can be resolve with

Severe: treated with diphenhydramine, dyspnea, bronchospasm, hypotension, tachycardia, shock

Stop transfusion. For mild symptoms that diphenhydramine, amine; if symptoms resolve, no further workup is necessary, although can restart transfusion. blood bank should be notified. For severe reaction, do hemolytic workup as above because initially may be indistinguishable from a hemolytic reaction.

137_Cydulka_Ch137.indd 741 21/03/17 12:12 PM

Source page 761

■■ FRESH FROZEN PLASMA

Fresh frozen plasma contains all coagulation factors and fibrinogen. FFP is used for replacement of multiple coagulation deficiencies (liver failure, warfarin-induced over-anticoagulation and massive transfusion) and to correct coagulation defects for which no factor is available. FFP is also administered prior to high-risk invasive procedures if PT/INR > 1.5X normal, aPTT > 1.5X top of normal range, or coagulation factor assay < 25% normal activity. FFP is not indicated to treat a prolonged INR in the absence of active bleeding.

Fresh frozen plasma needs to be thawed for 20 to 40 minutes before it can be used. Each 200 to 250 mL pack of FFP contains 1 unit/mL of each coagulation factor and 2 mL fibrinogen. In general, 1 unit of FFP will increase most coagulation factors by 3% to 5% in a 70-kg adult. FFP should be ABO-type compatible. Type AB is the universal donor for FFP, and in emergencies, universal donor FFP can be given minutes after thawing.

Starting dose is 15 mL/kg (or 4 units in a 70-kg adult). If rapid reversal of a vitamin K antagonist coagulopathy is needed, prothrombin complex concentrate or recombinant activated factor VII is faster and more reliable.

■■ CRYOPRECIPITATE

Cryoprecipitate is derived from FFP. One bag of cryoprecipitate contains 80 units factor VIII, 200 to 300 mL von Willebrand factor, 40 to 60 units fibrinogen, factor XIII, and variable amounts of fibronectin.

Cryoprecipitate is indicated for (a) active bleeding in patients with afibrinogenemia or hypofibrinogenemia (fibrinogen levels < 100 mL/dL) as a result of a pathological process (DIC, severe liver disease, abruptio placentae, amniotic fluid embolus, massive transfusion); (b) active bleeding in patients with von Willebrand disease when desmopressin (DDAVP) is not effective or in patients with von Willebrand disease type 2B, or if factor VIII concentrate containing von Willebrand factor is not available; and (c) hemophilia type A when virally inactivated factor VIII concentrates are not available. Cryoprecipitate can also be used as fibrin surgical adhesives in surgical patients.

Cryoprecipitate should be ABO-compatible. The volume of each unit is 20 to 50 mL. The usual dose is 1 unit/5 kg of body weight (10 to 14 units for an adult) and will raise fibrinogen concentration to 75 mL/dL.

■■ FIBRINOGEN CONCENTRATE

Fibrinogen concentrate is derived from pooled human plasma. It used to treat bleeding episodes in patients with congenital fibrinogen deficiency. Fibrinogen is dosed according to the patient’s fibrinogen baseline level, the target level (usually >150 mL/dL), volume of distribution, and body weight. If the baseline level is unknown, the initial dose is fibrinogen concentrate 70 mL/kg.

Prothrombin complex concentrate (PCC) is a blood-derived concentration of vitamin K-dependent clotting factors (factors II, VII, IX, and X). Some PCC formulations may also contain the anticoagulant proteins C, S, and antitrombin, as well as heparin. PCC administration does not necessitate ABO-compatibility testing.

137_Cydulka_Ch137.indd 742 21/03/17 12:12 PM

Source page 762

■■ COAGULATION FACTOR VIIA (RECOMBINANT)

Coagulation factor VIIa is primarily used for treatment of hemophilia A and B in patients who have developed inhibitor antibodies to factors VII or IX, respectively.

■■ SPECIFIC FACTOR REPLACEMENT THERAPY

Table 137-2 outlines therapy for congenital coagulation factor deficiencies.

■■ COMPLICATIONS OF TRANSFUSIONS

Acute complications include febrile nonhemolytic transfusion reaction (most common), acute hemolytic reaction, ABO incompatibility, allergic reaction and anaphylaxis, and transfusion related acute lung injury (TRALI).

Table 137-2 Replacement Therapy for Congenital Factor Deficiencies

Coagulation Factor	Approximate Incidence*	Replacement Therapy
Factor I (fibrinogen)	1 per million	Cryoprecipitate Fibrinogen concentrate
Factor II (prothrombin)	1 per 2 million	3- or 4-factor PCC for major bleeding
Factor V	1 per million	FFP
Factor VII	1 per 500,000	4-factor PCC for major bleeding Coagulation factor VIIa (recombinant)
Factor VIII†	1 per 5000–10,000 males	Recombinant factor VIII Desmopressin for mild hemophilia
Von Willebrand's disease‡	Up to 1 per 100 persons	Desmopressin or factor VIII concentrates (or cryoprecipitate if either unavailable)
Factor IX†	1 per 30,000 males	Recombinant factor IX 3- or 4-factor PCC
Factor X	1 per million	FFP for minor bleeding episodes 3- or 4-factor PCC for major bleeding
Factor XI‡	3 per 10,000 Ashkenazi Jews 1 per million in general population	FFP
Factor XII	25 per 1000	No bleeding manifestations, replacement not required
Factor XIII	1 per million	FFP or cryoprecipitate

Abbreviations: FFP, fresh frozen plasma; PCC, prothrombin complex concentrate.

*

Source: van Herrewegen F, Meijers JC, Peters M, van Ommen CH: Clinical practice: the bleeding child. Part II: disorders of secondary hemostasis and fibrinolysis. Eur J Pediatr. 2012;171:207.

†

See Chapter 235, "Hemophilias and von Willebrand's Disease," in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed.

‡

Factor XI levels correlate poorly with bleeding complications; many patients have low levels but no bleeding complications.

137_Cydulka_Ch137.indd 743 21/03/17 12:12 PM

Source page 763

744 SECTION 14: Hematologic and Oncologic Emergencies

Two important first steps in any confirmed or suspected transfusion reaction are to (1) immediately stop the transfusion, and (2) contact the blood bank that issued the transfusion product. (3) Table 137-1 summarizes the clinical presentation, evaluation, and management of acute transfusion reactions.

Delayed transfusion complications present days or weeks after the index transfusion and include delayed hemolytic transfusion reaction, transfusion associated graft versus host disease, posttransfusion purpura, and infectious disease transmission.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 238, "Transfusion Therapy," by Clinton J. Coil and Sally A. Santen.

137_Cydulka_Ch137.indd 744 21/03/17 12:12 PM

References/provenance

- Source: *Tintinalli's Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 760–765.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 138: Anticoagulants, Antiplatelet Agents, and Fibrinolytics

Source pages 766–771 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 766–771

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Anticoagulants, Antiplatelet Agents, and Fibrinolytics**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 138: Anticoagulants, Antiplatelet Agents, and Fibrinolytics 747 also be used to reverse rivaroxaban, but has not been shown to have a major benefit in dabigatran reversal. [Source pages 766–771]

- ■■ HEPARINS, POLYSACCHARIDES, AND HIRUDINS Parenteral agents include unfractionated heparin (UFH), low-molecular-weight heparin (LMWH) (e.g., enoxaparin, dalteparin), Xa inhibitors (fondaparinux), and direct thrombin inhibitors (e.g., hirudin, lepirudin, bivalirudin, argatroban). [Source pages 766–771]

- UFH and LMWH are used to treat and prevent DVT, as well as PE, unstable angina, and acute myocardial infarction. [Source pages 766–771]

- Dos- ing regimens for UFH and LMWH are weight based. [Source pages 766–771]

- UFH requires moni- toring of the activated partial thromboplastin time. [Source pages 766–771]

Clinical reasoning structure

- **Problem representation:** Anticoagulants, Antiplatelet Agents, and Fibrinolytics; source pages 766–771.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Anticoagulants, Antiplatelet Agents, and Fibrinolytics**.

- Source-page range: **766–771**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 766

CHAPTER 138: Anticoagulants, Antiplatelet Agents, and Fibrinolytics 747

also be used to reverse rivaroxaban, but has not been shown to have a major benefit in dabigatran reversal.

■■ HEPARINS, POLYSACCHARIDES, AND HIRUDINS

Parenteral agents include unfractionated heparin (UFH), low-molecular-weight heparin (LMWH) (e.g., enoxaparin, dalteparin), Xa inhibitors (fondaparinux), and direct thrombin inhibitors (e.g., hirudin, lepirudin, bivalirudin, argatroban). UFH and LMWH are used to treat and prevent DVT, as well as PE, unstable angina, and acute myocardial infarction. Dosing regimens for UFH and LMWH are weight based. UFH requires monitoring of the activated partial thromboplastin time. Therapeutic range is 1.5 to 2.5 “normal” value. Heparin and LMWH may be used during pregnancy. Enoxaparin 1 mg/kg SC every 12 hours may be used in outpatient management of DVT. The two major complications of heparin are bleeding and heparin-induced thrombocytopenia (HIT). Hirudin, lepirudin, and argatroban can be used for anticoagulation in patients with HIT.

Although protamine can reverse the anticoagulation effects of heparin, the adverse effects of protamine are significant. LMWH carries a lower bleeding risk than UFH, but a higher bleeding risk in patients with renal disease. LMWH may also cause pruritis, local skin reaction, or rarely, skin necrosis.

■■ ANTIPLATELET AGENTS

Oral agents include aspirin, clopidogrel, prasugrel, ticagrelor, dipyridamole, and cilostazol. Aspirin is an irreversible cyclooxygenase inhibitor; its effects last for the life of the platelet. Nonenteric-coated aspirin (162 to 325 mg) should be administered in the setting of ACS, although it is contraindicated during active GI hemorrhage, it is generally considered safe to give to closely monitored patients with guaiac positive stool. Nonsteroidal anti-

inflammatories have the potential to reduce the efficacy of aspirin. Clopidogrel, prasugrel, and ticagrelor inhibit platelet activation by rendering the fibrinogen receptor ineffective. An oral loading dose of clopidogrel 600 mg results in full antiplatelet effect by 2 hours, and sustained effects last up to 48 hours. Side effects of aspirin are mainly GI and dose related. Complications of clopidogrel include dyspepsia, rash, or diarrhea. Omeprazole reduces clopidogrel's efficacy.

GPIIb/IIIa parenteral medications alter the common final pathway receptor in platelet aggregation. These agents should be used in consultation with the interventional cardiologist. Patients receiving GPIIb/IIIa agents are at increased risk of bleeding complications during cardiac catheterization or coronary artery bypass surgery, but are not at increased risk of intracranial hemorrhage.

■■ FIBRINOLYTICS

Fibrinolytic agents include streptokinase, anistreplase, alteplase/tPA, reteplase, and tenecteplase. Although mechanisms of action vary, each agent eventually converts plasminogen to plasmin, which breaks down the

138_Cydulka_Ch138.indd 747 20/04/17 3:58 pm

Source page 767

Table 138-1 General Contraindications to Fibrinolytic Therapy

Absolute

●● Active or recent (<14 days) internal bleeding

●● Ischemic stroke within the past 2–6 months

●● Any prior hemorrhagic stroke

●● Intracranial or intraspinal surgery or trauma within the past 2 months

●● Intracranial or intraspinal neoplasm, aneurysm, or arteriovenous malformation

●● Known severe bleeding diathesis

●● Current anticoagulant treatment (e.g., warfarin with INR >1.7 or heparin with increased aPTT)

●● Current use of a direct thrombin inhibitor or direct factor Xa inhibitor with evidence of anticoagulant effect by laboratory tests

●● Platelet count <100,000/mm3 (<100 × 109/L)

●● Uncontrolled hypertension (i.e., blood pressure >185/110 mm Hg)

●● Suspected aortic dissection or pericarditis

●● Pregnancy

Relative*

●● Active peptic ulcer disease

●● Cardiopulmonary resuscitation for longer than 10 min

●● Hemorrhagic ophthalmic conditions

●● Puncture of noncompressible vessel within the past 10 days

●● Significant trauma or major surgery within the past 2 weeks to 2 months

●● Advanced renal or hepatic disease

Concurrent menses is not a contraindication.

*

Abbreviation: aPTT = activated partial thromboplastin time.

fibrin in a thrombus. Alteplase/tPA theoretically causes less systemic fibrinolysis, without the antigenic side effects of streptokinase and anistreplase that prevent retreatment within 6 months and preclude treatment within 12 months of a streptococcal infection. Strict adherence to established guidelines and informed consent is essential. Hemorrhagic complications, including intracranial hemorrhage, may occur. Streptokinase and anistreplase may cause hypotension or anaphylaxis. Contraindications to fibrinolytic therapy are listed in Table 138-1.

■■ COMPLICATIONS OF ANTITHROMBOTIC USE

Emergency treatment of bleeding complications of antithrombotic therapy is listed in Table 138-2.

Table 138-2 Emergency Treatment of Bleeding Complications of Antithrombotic Therapy

Agent Management

Heparins

Minor bleeding Immediate cessation of heparin administration.

Supratherapeutic aPTT not always present.

Anticoagulation effect lasts up to 3 h from last IV dose.

Observation with serial aPTT may be sufficient.

(Continued)

138_Cydulka_Ch138.indd 748 20/04/17 3:58 pm

Source page 768

Table 138-2 Emergency Treatment of Bleeding Complications of Antithrombotic Therapy (Continued)

Agent Management

Major bleeding Protamine 1 mg IV per 100 units of total amount of IV UFH administered within the past 3 h.

Protamine is given slowly IV over 1–3 min with a maximum of 50 mg over any 10-min period.

Protamine has an anaphylaxis risk.

Protamine does not completely reverse low-molecular-weight heparin.

Enoxaparin: Protamine 1 mg IV (maximum dose, 50 mg)

for every 1 mg of enoxaparin given in the previous 8 h.
If 8–12 h since last enoxaparin dose, give protamine 0.5 mg IV for every 1 mg of enoxaparin given.
Dalteparin and tinzaparin: Protamine 1 mg IV per every 100 units of dalteparin or tinzaparin given. If aPTT (measured 2–4 h after the protamine infusion) remains prolonged, give a second dose of protamine 0.5 mg IV per 100 units of dalteparin or tinzaparin.
Pentasaccharides
Fondaparinux Antithrombotic effect of fondaparinux is 24–30 h.
For life-threatening bleeding, anecdotal evidence suggests rFVIIa 90 µg/kg IV is effective.
Oral Direct Thrombin Inhibitors
Dabigatran Oral activated charcoal if recent or excessive ingestion.
Maintain urine output.
Consider hemodialysis or charcoal hemoperfusion.
For life-threatening bleeding, consider in descending order: aPCC, rFVIIa, or PCC.
For severe bleeding or urgent reversal, administer Idarucizumab 5 g IV. Dose may be repeated x 1.
Oral Factor Xa Inhibitors
Rivaroxaban For major bleeding, experimental evidence suggests PCC can reverse the antithrombotic effect of rivaroxaban.
Apixaban For major bleeding, administer andexanet alfa (when available), dose not established.
Oral Antiplatelet Agents
Aspirin Desmopressin 0.3–0.4 µg/kg IV over 30 min.
Platelet transfusion to increase count by 50,000/mm3 (typically requires one single donor apheresis-collected platelet concentrate or six units of random donor platelets).
Aspirin-induced platelet inhibition may last for 7 days, so repeat platelet transfusions are sometimes required.
Other antiplatelet agents: clopidogrel, prasugrel, ticagrelor
Platelet transfusion to increase count by 50,000/mm3 (typically requires one single donor apheresis-collected platelet concentrate or six units of random donor platelets).
Desmopressin 0.3–0.4 µg/kg IV over 30 min.
NSAID-induced platelet inhibition typically lasts <1 day.
Clopidogrel-, prasugrel-, or ticagrelor-induced platelet inhibition may last up to 5–7 days.
(Continued)

138_Cydulka_Ch138.indd 749 20/04/17 3:58 pm

Source page 769

Table 138-2 Emergency Treatment of Bleeding Complications of Antithrombotic Therapy (Continued)

Agent	Management
Fibrinolytics	
	Minor external bleeding Manual pressure.
	Significant internal bleeding Immediate cessation of fibrinolytic agent, antiplatelet agent, and/or heparin.
	Reversal of heparin with protamine as above.
	Typed and cross-matched blood ordered with verification of aPTT, CBC, thrombin clotting time, and fibrinogen level.
	Volume replacement with crystalloid and packed red blood cells as needed.
	Major bleeding or hemody- All measures listed for significant internal bleeding.
	namic compromise Fibrinogen concentrate 70 mg/kg IV, and recheck fibrinogen level; if fibrinogen level <100 mg/dL, repeat fibrinogen concentrate dose.
	If bleeding persists after fibrinogen concentrate or despite fibrinogen level >100 mg/dL, administer FFP two units IV.
	If bleeding continues after FFP, administer an antifibrinolytic such as aminocaproic acid 5 g IV over 60 min followed by 1 g/h continuous IV infusion for 8 h or until bleeding stops, or tranexamic acid 10 mg/kg IV every 6–8 h.
	Consider platelet transfusion.
	Intracranial hemorrhage All measures listed for significant internal and major bleeding with hemodynamic compromise.
	Immediate neurosurgery consultation.

Abbreviations: aPCC = activated prothrombin complex concentrate (not available in the United States); aPTT = activated partial thromboplastin time; FFP = fresh frozen plasma; NSAID = nonsteroidal anti-inflammatory drug; PCC = prothrombin complex concentrate; rFVIIa = recombinant activated factor VII; UFH = unfractionated heparin.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 239, "Thrombotics and Antithrombotics," by David E. Slattery and Charles V. Pollack Jr.

138_Cydulka_Ch138.indd 750 20/04/17 3:58 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.

- Content ID: SEG-TINTINALLI-EMM8-FULL.

- Source pages: 766–771.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 139: Emergency Complications of Malignancy

Source pages 772–779 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 772–779

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Emergency Complications of Malignancy**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 139: Emergency Complications of Malignancy 753 ■■ HYPERCALCEMIA OF MALIGNANCY Hypercalcemia is most commonly seen with breast cancer, lung cancer, and multiple myeloma. [Source pages 772–779]

- The symptoms are nonspecific and include polydipsia, polyuria, generalized weakness, lethargy, anorexia, nausea, constipation, abdominal pain, volume depletion, and altered mentation. [Source pages 772–779]

- Signs and symptoms are related to the rate of rise and may occur above 12 mg/dL (ionized >5.5 mg/dL). [Source pages 772–779]

- ECG may show shortened QT interval, ST depression, and atrioventricular blocks. [Source pages 772–779]

- Normal saline infusion to restore normovolemia followed by an infusion of 250 mL/h if the patient can tolerate is the mainstay of treatment. [Source pages 772–779]

Clinical reasoning structure

- **Problem representation:** Emergency Complications of Malignancy; source pages 772–779.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Emergency Complications of Malignancy**.

- Source-page range: **772–779**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 772

CHAPTER 139: Emergency Complications of Malignancy 753

■■ HYPERCALCEMIA OF MALIGNANCY

Hypercalcemia is most commonly seen with breast cancer, lung cancer, and multiple myeloma. The symptoms are nonspecific and include polydipsia, polyuria, generalized weakness, lethargy, anorexia, nausea, constipation, abdominal pain, volume depletion, and altered mentation.

Signs and symptoms are related to the rate of rise and may occur above 12 mg/dL (ionized >5.5 mg/dL). ECG may show shortened QT interval, ST depression, and atrioventricular blocks.

Normal saline infusion to restore normovolemia followed by an infusion of 250 mL/h if the patient can tolerate is the mainstay of treatment. Furosemide is not recommended unless needed to prevent volume overload in patients with cardiac or renal failure. Further treatment should be discussed with the patient's oncologist. Bisphosphonates such as zoledronic acid 4 mg IV over 15 minutes or pamidronate 60 to 90 mg IV over 4 to 24 hours can prevent bone resorption. Calcitonin 4 IU/kg SC or IM causes a more rapid decrease in calcium levels. Glucocorticoids may be helpful in lymphoma and multiple myeloma. Consider hemodialysis for patients with profound mental status changes and renal failure, or those who cannot tolerate a normal saline infusion (see Chapter 4, "Fluids, Electrolytes, and Acid Base Disorders").

■■ HYPONATREMIA DUE TO SYNDROME OF INAPPROPRIATE ANTI-DIURETIC HORMONE (SIADH)

Inappropriate anti-diuretic hormone secretion is most commonly associated with bronchogenic lung cancer, but can also occur from chemotherapy or medications. Symptoms include anorexia, nausea, headache, altered mentation, and seizures. Mild hyponatremia (>125 mEq/L) is usually asymptomatic.

SIADH should be suspected in patients with cancer who present with normovolemic hyponatremia. Laboratory abnormalities include serum osmolality <280 mOsm/L, urine osmolality >100 mOsm/L, and urine sodium >20 mEq/L. The differential diagnosis includes hypothyroidism, renal failure, cirrhosis, adrenal crisis, and hypovolemia/hypervolemia. Mild hyponatremia >125 mEq/L is treated with a water restriction of 500 mL/d and close follow-up. More severe hyponatremia is treated with furosemide 0.5 to 1 mg/kg PO with normal saline infusion to maintain normovolemia. Three percent hypertonic saline is reserved for severe hyponatremia <120 mEq/L with seizures or coma. An infusion of 25 to 100 mL/h should be titrated to a correction of 0.5 to 1 mEq/h with a maximum of 12 mEq/L/d (see Chapter 4, “Fluids, Electrolytes, and Acid Base Disorders”).

■■ ADRENAL CRISIS

Adrenal crisis is most commonly caused by acute physiologic stress in the face of exogenous steroid-induced adrenal suppression or malignant infiltration of adrenal tissue. Symptoms include weakness or nausea, and hypotension unresponsive to fluids. Laboratory abnormalities may include hypoglycemia, hyponatremia, and hyperkalemia. A serum cortisol level is ideally drawn before giving steroids. Treat acute adrenal insufficiency

139_Cydulka_Ch139.indd 753 21/03/17 2:53 PM

Source page 773

empirically with stress-dose hydrocortisone 100 to 150 mg IV, methylprednisolone 20 to 30 mg IV, or dexamethasone 4 mg IV, IV crystalloids, and supportive care (see Chapter 132, “Adrenal Insufficiency and Adrenal Crisis”).

■■ TUMOR LYSIS SYNDROME

Tumor lysis syndrome occurs when massive quantities of potassium, phosphate, and uric acid are released. It usually occurs 1 to 3 days after chemotherapy for acute leukemia or lymphoma. The resulting hyperuricemia, hyperkalemia, hyperphosphatemia, and hypocalcemia may cause uric acid precipitates and calcium phosphate deposits in the kidney with renal failure, life-threatening arrhythmias, tetany, and seizures. Laboratory evaluation should include 12-lead ECG, basic electrolyte levels, complete blood count, uric acid, and phosphorus.

Preventative measures reduce the incidence of tumor lysis syndrome. Hyperkalemia is the most immediate life threat. Treatment of hyperkalemia is with insulin, glucose, bicarbonate (if acidotic), and albuterol (see Chapter 4, “Fluids, Electrolytes, and Acid Base Disorders”). Avoid calcium unless ventricular arrhythmias or widened QRS complexes are seen, as it

may worsen calcium phosphate precipitation in the kidney. Aggressive infusion of isotonic fluids reverses volume depletion and helps to prevent renal deposition of uric acid and calcium phosphate crystals. Manage hyperphosphatemia with IV insulin and glucose. Phosphate binders have a limited effect. Consider hemodialysis for potassium levels above 6.0 mEq/L, uric acid levels above 10.0 mg/dL, phosphate levels above 10 mg/dL, creatinine levels above 10 mg/dL, symptomatic hypocalcemia, or volume overload. Patients with tumor lysis syndrome should be admitted to an intensive care unit.

■■ FEBRILE NEUTROPENIA

Febrile neutropenia is defined by temperatures above 38°C for an hour or a single temperature above 38.3°C with an absolute neutrophil count below 1000 cells/mm3. Neutrophil counts typically reach a nadir 10 to 15 days after chemotherapy and rebound 5 days later.

Febrile neutropenic patients often lack localizing signs and symptoms because of an attenuated immune response. Meticulous attention must be paid to all skin surfaces, mucosal areas, and vascular access sites in which the patient may have an occult infection. Digital rectal examination is often withheld until after initial antibiotic administration because of the fear of inducing bacteremia.

Laboratory evaluation includes complete blood count with differential, blood cultures obtained through all lumens of indwelling catheters as well as a peripheral site, urinalysis, urine culture, and CXR, electrolytes, renal, and liver function tests. Additional studies based on symptoms may include stool culture (diarrhea), sputum culture (cough), lumbar puncture (headache, stiff neck, altered mental status), wound culture (drainage), and CT or ultrasound (abdominal pain).

The decision for empiric antibiotics and admission should be made with the patient's oncologist. Empiric antibiotics (Table 139-2) are generally

139_Cydulka_Ch139.indd 754 21/03/17 2:53 PM

Source page 774

Table 139-2 Initial Empiric Antibiotic Therapy in Febrile Neutropenia

Circumstance	Drug and Adult Dosage	Comments
Outpatient and daily assessments by a medical provider for the initial 3 days.	Ciprofloxacin, 500 mg PO every 8 h Amoxicillin/clavulanate, 500 mg PO every 8 h	Useful for low-risk patients with
Monotherapy	Cefepime, 2 g IV every 8 h	Monotherapy with these broad-or spectrum agents appears to be as

Ceftazidime, 2 g IV every 8 h good as dual-drug therapy in most
or circumstances.
Imipenem/cilastatin, 1 g IV every 8 h
or
Meropenem, 1 g IV every 8 h
or
Piperacillin/tazobactam, 4.5 g IV
every 6 h
Dual therapy One of the monotherapy agents Potential advances include syner-
plus gistic effects against some gram-
Gentamicin, 1.7 mg/kg IV every 8 h negative bacteria and reduced
or emergence of drug resistance.
Tobramycin, 1.7 mg/kg IV every 8 h There is an increased risk for
or adverse effects, including
Amikacin, 5 mg/kg IV every 8 h nephrotoxicity, ototoxicity, and
hypokalemia.
Risk factors Vancomycin, 1 g IV every 12 h Vancomycin is not usually
for severe plus necessary for initial empiric
gram-positive Cefepime, 2 g IV every 8 h antibiotic therapy.
infection (see or Vancomycin may be incorporated
text) Ceftazidime, 2 g IV every 8 h into initial therapeutic regimens
or of high-risk patients in institutions
Imipenem/cilastatin, 1 g IV every 8 h with increased gram-positive
or infection rates.
Meropenem, 1 g IV every 8 h

indicated for an ANC <500/mm3. For neutrophil counts between 500 and 1000, the decision is based on the patient's presentation. Add vancomycin for severe mucositis, catheter site infection, recent use of fluoroquinolone prophylaxis, hypotension, residence in an institution with hospital-associated methicillin-resistant Staphylococcus aureus, or known colonization with resistant gram-positive organisms.

■■ HYPERVISCOSITY SYNDROME

Hyperviscosity syndrome refers to impaired blood flow due to abnormal elevations of paraproteins or blood cells. It is most common in patients with dysproteinemia (e.g., Waldenstrom's macroglobulimenia and multiple myeloma), acute leukemia, or polycythemia. Hematocrits above 60% and WBC counts above 100,000/mm3 may cause hyperviscosity syndromes.

Initial symptoms include fatigue, abdominal pain, headache, blurry vision, dyspnea, fever, or altered mental status. Thrombosis or bleeding

may occur. Physical exam findings may include retinal hemorrhages,

139_Cydulka_Ch139.indd 755 21/03/17 2:53 PM

Source page 775

exudates, and “sausage-linked” vessels. Elevated serum viscosity (>4) or rouleaux formation (red cells stacked like coins) support the diagnosis. Serum viscosity measurement will not identify hyperviscosity caused by polycythemia or leukemia.

Initial therapy consists of intravenous isotonic fluids and plasmapheresis or leukopheresis in consultation with a hematologist. A temporizing measure in patients with coma is 1000 mL phlebotomy with simultaneous infusion of 2 to 3 L normal saline.

■■ THROMBOEMBOLISM

Thromboembolism is the second leading cause of death in cancer patients. See Chapter 25 for discussion of the diagnosis and management of deep vein thrombosis and pulmonary embolism. Low-molecular-weight heparin (LMWH) is the recommended initial treatment for cancer patients with venous thromboembolism and is usually continued for the duration of treatment for the clot. There is insufficient experience with the new oral anticoagulants to recommend them.

■■ NAUSEA AND VOMITING

Chemotherapy commonly causes nausea and vomiting. Other causes of nausea and vomiting include radiation enteritis, bowel obstruction, infection or tumor infiltration, and increased intracranial pressure. Treatment consists of rehydration, administration of antiemetics, and correction of electrolyte derangements (Table 139-3).

Table 139-3 Antiemetic Agents for Chemotherapy-Induced Vomiting

Class and Agent	Adult Dose	Comments
Dopamine receptor antagonists		
Metoclopramide	10 mg IV or IM	Dose-related extrapyramidal side effects
Promethazine	25 mg IV or IM	IV use common but not approved by the FDA
Serotonin antagonists		
Dolasetron	0.35 mg/kg IV, max 12.5 mg over 5 min (all)	Constipation, headaches
Granisetron	10 μg/kg IV over 5 min	
Ondansetron	0.15 mg/kg IV, max 16mg over 15 min	
Corticosteroids		

Dexamethasone 20 mg IV Mechanism unknown, no

immunosuppression

Benzodiazepines

Lorazepam 1 to 2 mg IV Sedation, anxiolysis

Histamine receptor antagonists

Diphenhydramine 50 mg IV or IM Minor therapeutic effect

Abbreviation: FDA, Food and Drug Administration.

139_Cydulka_Ch139.indd 756 21/03/17 2:53 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.

- Content ID: SEG-TINTINALLI-EMM8-FULL.

- Source pages: 772–779.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 140: Headache

Source pages 780–785 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 780–785

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Headache**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 140: Headache 761 cerebellar hemorrhage when headache is associated with vestibular symp- toms as this may need surgical consultation. [Source pages 780–785]

- Headaches associated with brain tumors may be bilateral, unilateral, constant, or intermittent. [Source pages 780–785]

- These are classically described as worse upon awakening, worsened by Valsalva maneuver, positional, and associated with nausea and vomiting. [Source pages 780–785]

- Consider the possibility of primary or metastatic brain lesions in patients with known cancer diagnoses, seizures, or mental status changes. [Source pages 780–785]

- Risk factors for cerebral venous thrombosis include patients with hypercoagulable states due to oral contraceptive use, postpartum or periop- erative status, clotting factor deficiencies, or polycythemia. [Source pages 780–785]

Clinical reasoning structure

- **Problem representation:** Headache; source pages 780–785.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Headache**.

- Source-page range: **780–785**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 780

CHAPTER 140: Headache 761

cerebellar hemorrhage when headache is associated with vestibular symptoms as this may need surgical consultation.

Headaches associated with brain tumors may be bilateral, unilateral, constant, or intermittent. These are classically described as worse upon awakening, worsened by Valsalva maneuver, positional, and associated with nausea and vomiting. Consider the possibility of primary or metastatic brain lesions in patients with known cancer diagnoses, seizures, or mental status changes.

Risk factors for cerebral venous thrombosis include patients with hypercoagulable states due to oral contraceptive use, postpartum or perioperative status, clotting factor deficiencies, or polycythemia. Papilledema may be present, and neurological findings can wax and wane. Lumbar puncture may demonstrate an increased opening pressure, and MR venography is most useful for diagnosis.

Temporal arteritis (see Table 140-3) presents most commonly in patients over the age of 50 and its incidence increases with age. Associated symptoms may include fatigue, fever, jaw claudication, and vision changes. The involved temporal artery may be tender, nonpulsatile, or have a diminished pulse, but can also be normal. ESR and CRP can be helpful in establishing the diagnosis, and a temporal artery biopsy allows a definitive diagnosis.

Migraine headaches are the most common benign cause of headache in ED patients. These generally have a gradual onset, last 4 to 72 hours, and are typically unilateral, pulsating, and worsened by physical activity. Nausea and vomiting, photophobia, and phonophobia are frequently present, and the headache may be associated with or without an aura.

Idiopathic intracranial hypertension, also known as pseudotumor cerebri, is most common in obese women ages 20 to 44. This may present with headaches, transient vision disturbances, back pain, and pulsatile

tinnitus, and when left untreated can lead to permanent vision loss. The diagnosis is most often made by the presence of papilledema, a normal neurological examination, and elevated lumbar puncture opening pressure. Intracranial hypotension can occur after a procedure that violates the dura such as lumbar puncture or epidural anesthesia. This headache is typically worsened with upright posture and improved with lying down. Alterations in hearing or vision, nausea, and vomiting may occur.

Table 140-3 American College of Rheumatology Criteria for Diagnosis of Temporal Arteritis

Clinical Features Comments

Age at disease onset ≥50 years

New headache Onset or type

Temporal artery abnormality Tenderness to palpation of temporal arteries

Decreased pulsation of temporal arteries

Erythrocyte sedimentation rate ≥50 mm/h Westergren method

Abnormal artery biopsy (can be done after Vasculitis initiating steroids) Predominance of mononuclear cell infiltration or granulomatous inflammation

Multinucleated giant cells

140_Cydulka_Ch140.indd 761 21/03/17 12:34 PM

Source page 781

Cluster headaches are uncommon and present with severe pain that is unilateral and located in the orbital, supraorbital, or temporal area. These are frequently associated with lacrimation, nasal congestion, rhinorrhea, conjunctival injection, and pacing in the exam room. These symptoms tend to occur daily for weeks, with periods of remission that may last for for weeks to years.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. Assess patients with an initial history and physical examination. When indicated, utilize appropriate diagnostic testing and/or neuroimaging based on risk factors, patient history, and physical findings. Disease-specific treatment and specialty consultation are indicated in the appropriate clinical circumstances.

2. When temporal arteritis is suspected in patients without vision loss, begin empiric treatment with oral prednisone 60 mg daily. Consult with an ophthalmologist or other appropriate specialist to arrange a temporal artery biopsy to confirm the diagnosis and for appropriate follow-up and continued treatment.

3. Treat patients with migraine headache with medications for symptom

relief that may include dihydroergotamine (DHE), sumatriptan, and dopamine-antagonist antiemetics such as metoclopramide, chlorpromazine, or prochlorperazine (see Table 140-4). Consider dexamethasone 6 to 10 mg IV as an adjunctive therapy that may reduce migraine recurrence.

4. Treat patients with idiopathic intracranial hypertension with a starting dose of oral acetazolamide 250 to 500 mg twice a day and recommend weight loss for obese patients. During a diagnostic lumbar puncture, excess cerebrospinal fluid can be removed to reduce intracranial pressure to 15 to 20 cm H2O. Consult with a neurologist and/or ophthalmologist for ongoing evaluation and treatment.

5. Patients with intracranial hypotension due to lumbar puncture or epidural anesthesia often respond well to symptomatic treatment. Evidence of efficacy for intravenous fluids and caffeine is limited. The most effective treatment is an epidural blood patch, typically performed by an anesthesiologist.

6. Treat cluster headaches with high-flow oxygen delivered at 12 L/min for 15 minutes via a nonrebreathing facemask. Consider sumitriptan 6 mg subcutaneous injection for pain that is unresolved with oxygen therapy.

7. Consult with appropriate specialists based on the suspected underlying cause of headache, if symptom control is not adequately achieved in the ED, or if admission for further diagnostic testing or treatment may be needed.

8. When patients are discharged home, discuss indications for return and reassessment such as worsening of symptoms, failure to improve, or the development of new symptoms such as fever, neck stiffness, vision changes, or neurological abnormalities. Advise patients about expected side effects of medications prescribed or given in the ED. Recommend appropriate follow-up and outpatient management instructions based on the suspected diagnosis.

140_Cydulka_Ch140.indd 762 21/03/17 12:34 PM

Source page 782

140_Cydulka_Ch140.indd 763

Table 140-4 Treatment Options for Migraine Headache

Drug Dosing Contraindications Precautions and Pregnancy Category Notes

Ketorolac 30 mg IV or IM History of peptic ulcer disease Pregnancy Category B (especially in elderly) Avoid in third trimester

Prochlorperazine 5–10 mg IV or PR Pregnancy Category C Antiemetic

Drowsiness Concurrent: diphenhydramine

Dystonic reactions

Metoclopramide 10 mg IV Pregnancy Category B Antiemetic

Drowsiness Concurrent: diphenhydramine

Dystonic reactions

Droperidol 2.5 mg IV slow, or Pregnancy Category C Concurrent: diphenhydramine

2.5 mg IM QT interval prolongation and/or torsade de pointes

Chlorpromazine 7.5 mg IV Pregnancy not classified Antiemetic

Hypotension Pretreat with: normal saline bolus to

Drowsiness minimize hypotension

Dystonic reactions Concurrent: diphenhydramine

Magnesium sulfate 2 g IV over 30 min Pregnancy Category D but effective in pre-eclampsia Nonvalidated

and eclampsia

Methylprednisolone 125 mg IV or IM Rescue therapy Nonvalidated

CHAPTER 140: Headache

Dexamethasone 6–10 mg IV Rescue therapy Adjunctive therapy to reduce recurrence

Sumatriptan 6 mg SC Ischemic Heart Disease Pregnancy Category C

Uncontrolled hypertension

Basilar or hemiplegic migraine

Dihydroergotamine 1 mg IV over 3 min Pregnancy Pregnancy Category X Pretreat with antiemetic

(DHE) Uncontrolled hypertension Nausea

Ischemic Heart Disease Vomiting

21/03/17 12:34 PM

Recent sumatriptan use (within 24 h) Diarrhea

Basilar or hemiplegic migraine Abdominal pain

763

Valproate 500 mg IV Pregnancy Pregnancy Category X Nonvalidated

Source page 783

764 SECTION 15: Neurology

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study

Guide, 8th ed., see Chapter 165, “Headache” by Michael Harrigan and Ana C.G.

Felix.

140_Cydulka_Ch140.indd 764 21/03/17 12:34 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 780–785.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 141: Stroke Syndromes and Spontaneous Subarachnoid Hemorrhage

Source pages 786–795 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 786–795

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Stroke Syndromes and Spontaneous Subarachnoid Hemorrhage**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 141: Stroke Syndromes and Spontaneous Subarachnoid Hemorrhage 767 Table 141-2 Anterior and Posterior Circulation of the Brain Circulation Major Arteries Major Regions of Brain Supplied Anterior (internal Ophthalmic Optic nerve and retina carotid system) Anterior cerebral Frontal pole Anteromedial cerebral cortex Anterior corpus callosum Middle cerebral Frontoparietal lobe Anterotemporal lobe Posterior (vertebral Vertebral Brainstem system) Posteroinferior cerebellar Cerebellum Basilar Thalamus Posterior cerebral Auditory/vestibular structures Medial temporal lobe Visual occipital cortex ■■ DIAGNOSIS AND DIFFERENTIAL Patients with stroke may present variably and sometimes subtly. [Source pages 786–795]

- Assess for stroke risk factors based on age and comorbidities such as atrial fibrillation, hypertension, diabetes, smoking, coronary atherosclerosis, valvular replace- ment, and recent myocardial infarction. [Source pages 786–795]

- Risk factors for SAH include excessive alcohol consumption, polycystic kidney disease, family history of SAH, Marfan’s syndrome, and Ehlers–Danlos syndrome. [Source pages 786–795]

- Inquire about a history of drug use, chiropractic manipulation, or recent activities associ- ated with blood-pressure elevation. [Source pages 786–795]

- Consider alternative diagnoses that may mimic stroke (Table 141-3). [Source pages 786–795]

Clinical reasoning structure

- **Problem representation:** Stroke Syndromes and Spontaneous Subarachnoid Hemorrhage; source pages 786–795.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Stroke Syndromes and Spontaneous Subarachnoid Hemorrhage**.

- Source-page range: **786–795**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 786

CHAPTER 141: Stroke Syndromes and Spontaneous Subarachnoid Hemorrhage 767

Table 141-2 Anterior and Posterior Circulation of the Brain

Circulation Major Arteries Major Regions of Brain Supplied

Anterior (internal carotid system) Ophthalmic Anterior cerebral Middle cerebral Optic nerve and retina Frontal pole Anteromedial cerebral cortex Anterior corpus callosum Frontoparietal lobe Anterotemporal lobe

Posterior (vertebral system) Vertebral Posteroinferior cerebellar Basilar Posterior cerebral Medial temporal lobe Brainstem Cerebellum Thalamus Auditory/vestibular structures Visual occipital cortex

■■ DIAGNOSIS AND DIFFERENTIAL

Patients with stroke may present variably and sometimes subtly. Assess for stroke risk factors based on age and comorbidities such as atrial fibrillation, hypertension, diabetes, smoking, coronary atherosclerosis, valvular replacement, and recent myocardial infarction. Risk factors for SAH include excessive alcohol consumption, polycystic kidney disease, family history of SAH, Marfan's syndrome, and Ehlers–Danlos syndrome. Inquire about a history of drug use, chiropractic manipulation, or recent activities associated with blood-pressure elevation. Consider alternative diagnoses that may mimic stroke (Table 141-3). If ischemic stroke is the primary working diagnosis and thrombolytic therapy is being considered, it is essential to

obtain an accurate determination of the time the patient was last known to be at their neurological baseline and whether the patient may be a candidate for thrombolytic therapy (Tables 141-4 and 141-5).

Focus patient assessment on the neurological examination, with particular emphasis on detecting motor weakness, sensory deficits, and cerebellar dysfunction. Depending on clinical circumstances, investigate other findings such as meningismus, carotid bruits, signs of embolic disease, papilledema, or preretinal hemorrhage. Calculate a National Institutes of Health Stroke Scale (NIHSS) score if thrombolytic therapy may be considered. Appropriately expedite the history, physical examination, laboratory, and radiographic imaging.

An emergent noncontrasted CT scan of the brain (best interpreted by a neuroradiologist) is essential to determine whether hemorrhage or a stroke mimic is present. Most acute ischemic strokes will not be visualized on a CT scan in the early hours of a stroke.

The differential diagnosis for SAH is broad and includes intracranial hemorrhage, drug toxicity, ischemic stroke, meningitis, encephalitis, intracranial tumor, venous sinus thrombosis, and primary headache syndromes. Modern CT scanners are reported to be 98% sensitive to detect SAH within 6 to 12 hours of symptom onset, 91% to 93% at 24 hours, and 50% at 1 week. If SAH is suspected and the CT is negative, many clinicians recommend a lumbar puncture to assess for the presence of red blood cells or xanthochromia as evidence of subarachnoid blood that was not visualized on CT scan. However, some recent studies have challenged this position and suggest that

141_Cydulka_Ch141.indd 767 21/03/17 12:54 PM

Source page 787

Table 141-3 Stroke Mimics

Disorder	Distinguishing Clinical Features
Seizures/postictal paralysis (Todd's paralysis)	Transient paralysis following a seizure, which typically disappears quickly; can be confused with transient ischemic attack. Seizures can be secondary to a cerebrovascular accident.
Syncope	No persistent or associated neurologic symptoms.
Meningitis/encephalitis	Fever, immunocompromised state may be present, meningismus, detectable on lumbar puncture.
Complicated migraine	History of similar episodes, preceding aura, headache.
Brain neoplasm or abscess	Focal neurologic findings, signs of infection, detectable by imaging.

Epidural/subdural hematoma History of trauma, alcoholism, anticoagulant use, bleeding disorder; detectable by imaging.

Subarachnoid hemorrhage Sudden onset of severe headache.*

Hypoglycemia Can be detected by bedside glucose measurement, history of diabetes mellitus.

Hyponatremia History of diuretic use, neoplasm, excessive free water intake.

Hypertensive encephalopathy Gradual onset; global cerebral dysfunction, headache, delirium, hypertension, cerebral edema.

Hyperosmotic coma Extremely high glucose levels, history of diabetes mellitus.

Wernicke's encephalopathy History of alcoholism or malnutrition; triad of ataxia, ophthalmoplegia, and confusion.

Labyrinthitis Predominantly vestibular symptoms; patient should have no other focal findings; can be confused with cerebellar stroke.

Drug toxicity (lithium, phenytoin, carbamazepine) Can be detected by particular toxidromes and elevated blood levels. Phenytoin and carbamazepine toxicity may present with ataxia, vertigo, nausea, and abnormal reflexes.

Bell's palsy Neurologic deficit confined to isolated peripheral seventh nerve palsy; often associated with younger age.

Ménière's disease History of recurrent episodes dominated by vertigo symptoms, tinnitus, deafness.

Demyelinating disease (multiple sclerosis) Gradual onset. Patient may have a history of multiple episodes of neurologic findings in multifocal anatomic distributions.

Conversion disorder No cranial nerve findings, nonanatomic distribution of findings (e.g., midline sensory loss), inconsistent history or examination findings.

*

Although subarachnoid hemorrhage is a type of stroke, it has special considerations in terms of diagnosis and management. See Chapter 166, "Spontaneous Subarachnoid and Intracerebral Hemorrhage," in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed.

141_Cydulka_Ch141.indd 768 21/03/17 12:54 PM

Source page 788

Table 141-4 American Heart Association (AHA)/American Stroke Association (ASA) 2013 Inclusion/Exclusion Criteria for IV Recombinant Tissue

Plasminogen Activator (rtPA) in Acute Ischemic Stroke

Inclusion Criteria

Measurable diagnosis of acute ischemic stroke

Use of NIHSS score recommended. Major strokes (NIHSS score >22) are more likely to have poor outcomes.

Onset of symptoms <3 h prior to rtPA administration

Must be well established and is defined as the time of the witnessed onset of symptoms or the time the patient was last known at baseline.

Age ≥18 years

No clear upper age limit.

Exclusion Criteria

Significant head trauma or prior stroke in previous 3 months

Symptoms suggest subarachnoid hemorrhage

Arterial puncture at noncompressible site ≤7 days ago

History of previous intracranial hemorrhage

Intracranial neoplasm, arteriovenous malformation, or aneurysm*

Recent intracranial or intraspinal surgery*

Pretreatment systolic blood pressure >185 mm Hg or diastolic blood pressure >110 mm Hg despite therapy

Active internal bleeding*

Platelet count <100,000/mm3

If patient has no history of thrombocytopenia, rtPA may be given before this lab result is available; however, rtPA should be stopped if the platelet count is <100,000/mm3

Use of heparin within preceding 48 h and a prolonged activated partial thromboplastin time (aPTT) greater than upper limit of normal

International normalized ratio (INR) >1.7 or partial thromboplastin time (PTT) >15 s

Oral anticoagulant use in and of itself is not a contraindication to rtPA. If patient is not taking oral anticoagulant or heparin, rtPA may be given before this lab result is available; however, rtPA should be stopped if these lab tests come back elevated above normal limits.

Current use of direct thrombin inhibitors or direct factor Xa inhibitors with elevated sensitive laboratory tests (such as aPTT, INR, platelet count, and ecarin clotting time [ECT]; thrombin time [TT]; or appropriate factor Xa activity assays)*

Blood glucose level <50 mg/dL

(2.7 mmol/L)

Non–contrast-enhanced CT (NECT) Do not give rtPA if CT shows acute intracranial

demonstrates multilobar infarction hemorrhage or neoplasm.

(hypodensity >1/3 cerebral

hemisphere)

(Continued)

141_Cydulka_Ch141.indd 769 21/03/17 12:54 PM

Source page 789

770 SECTION 15: Neurology

Table 141-4 American Heart Association (AHA)/American Stroke Association

(ASA) 2013 Inclusion/Exclusion Criteria for IV Recombinant Tissue

Plasminogen Activator (rtPA) in Acute Ischemic Stroke (Continued)

Relative Exclusion Criteria

Only minor or rapidly improving stroke Some patients may have a lower NIHSS score but

symptoms (clearing spontaneously) have a potentially disabling condition (e.g., apha-

sia, hemianopia). Some studies have shown poor

outcomes for untreated minor strokes.

Pregnancy No randomized controlled trials have been pub-

lished regarding safety or efficacy of rtPA for

ischemic stroke in pregnancy. Case series have

reported mixed results.

Seizure at onset with postictal resid- rtPA can be given if the residual impairments are

ual neurologic impairments thought to be secondary to the stroke as opposed

to the seizure.

Major surgery or serious trauma

within preceding 14 days†

Previous GI or urinary tract hemor-

rhage within preceding 21 days†

Previous myocardial infarction within Rationale for this criterion was a statement indi-

preceding 3 months† cating that myocardial rupture can result if rtPA

is given within a few days of acute myocardial

infarction.

Former Exclusion Criteria from 2007 AHA/ASA Guidelines

Evidence of acute trauma (fracture)

Failure of the patient or responsible

party to understand the risks and

benefits of, and alternatives to,

the proposed treatment after a full

discussion

*

New exclusion criteria for 2013 AHA/ASA guidelines.

Changed from absolute contraindication from 2007 AHA/ASA guidelines.

†

Abbreviation: NIHSS, National Institutes of Health Stroke Scale.

a lumbar puncture may be unnecessary if a third-generation CT scan done within 6 hours of symptom onset is interpreted as negative by a neuroradiologist. A growing body of literature is exploring the combination of a negative noncontrast CT with CT angiography to exclude SAH or an associated aneurysm, with a 99.4% reported negative predictive value of this combination. However, potential downsides to this diagnostic strategy are the additional ionizing radiation exposure and possible detection of incidental cerebral aneurysms.

Other diagnostic tests may be useful in certain patients to exclude stroke mimics or concurrent conditions. Tests such as bedside glucose, complete blood count, ECG, pulse oximetry, electrolyte and coagulation studies, cardiac enzyme levels, toxicology screen, blood alcohol level, echocardiogram, carotid duplex scanning. MRI, MR angiogram, and CT angiogram may be of value when the suspicion for alternative disease entities is high.

141_Cydulka_Ch141.indd 770 21/03/17 12:54 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 786–795.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 142: Altered Mental Status and Coma

Source pages 796–801 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 796–801

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Altered Mental Status and Coma**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 142: Altered Mental Status and Coma 777 disorientation, inability to perform self-care tasks, and personality changes. [Source pages 796–801]

- Anxiety, depression, and speech difficulties may be observed. [Source pages 796–801]

- Patients with vascular dementia may be noted to have exaggerated or asym- metric deep tendon reflexes, gait abnormalities, or extremity weakness. [Source pages 796–801]

- Diagnosis and Differential While dementia typically develops slowly over time, physical examination may identify a precipitant or underlying cause of an acute or subacute change in a patient’s level of functioning. [Source pages 796–801]

- Focal neurologic signs suggest the possibility of vascular dementia or a mass lesion. [Source pages 796–801]

Clinical reasoning structure

- **Problem representation:** Altered Mental Status and Coma; source pages 796–801.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Altered Mental Status and Coma**.

- Source-page range: **796–801**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 796

CHAPTER 142: Altered Mental Status and Coma 777

disorientation, inability to perform self-care tasks, and personality changes. Anxiety, depression, and speech difficulties may be observed. Patients with vascular dementia may be noted to have exaggerated or asymmetric deep tendon reflexes, gait abnormalities, or extremity weakness.

Diagnosis and Differential

While dementia typically develops slowly over time, physical examination may identify a precipitant or underlying cause of an acute or subacute change in a patient's level of functioning. Focal neurologic signs suggest the possibility of vascular dementia or a mass lesion. Increased motor tone and other extrapyramidal signs may suggest Parkinson's disease. Consider normal pressure hydrocephalus if urinary incontinence and gait disturbance are noted. This is further suggested by excessively large ventricles on head CT. Diagnostic studies may include a complete blood count, basic metabolic profile, urinalysis, thyroid profile, serum vitamin B12 level, testing for syphilis, erythrocyte sedimentation rate, serum folate level, human immunodeficiency virus testing, and chest radiography. Consider head CT or MRI as well as lumbar puncture if the diagnosis is not readily apparent and intracranial processes or central nervous system infections are suspected causes. The differential diagnosis includes delirium, depression, and other treatable causes.

Emergency Department Care and Disposition

1. Initiate an appropriate workup to assess for medical causes of worsening mental function.

2. Use antipsychotic drugs to manage persistent psychosis or severely disruptive or dangerous behavior, keeping in mind potential adverse reactions.

3. Aim treatment of vascular dementia at addressing risk factors, such as

hypertension.

4. Many patients with newly diagnosed dementia will need inpatient admission for further evaluation and treatment. Discharge is appropriate for those patients who have longstanding and stable symptoms, consistent caregivers, and reliable follow-up for outpatient evaluation.

■■ COMA

Clinical Features

Coma is a state of reduced alertness and responsiveness from which the patient cannot be aroused. Severity can be quantified using the Glasgow Coma Scale (Table 142-3) and the FOUR (Full Outline of UnResponsiveness) score. Pupillary findings, cranial nerve testing, findings of hemiparesis, and response to stimulation can help the clinician to determine a probable general category: diffuse CNS dysfunction such as a toxic-metabolic coma or focal CNS dysfunction characteristic of structural coma. Toxic-metabolic coma is characterized by lack of focal physical examination findings. The pupils are typically small and reactive, but may be large in severe sedative poisoning as from barbiturates.

Coma from supratentorial lesions or masses may present with progressive hemiparesis or asymmetric muscle tone and reflexes. Coma without

142_Cydulka_Ch142.indd 777 21/03/17 2:25 PM

Source page 797

Table 142-3 Glasgow Coma Scale

Component	Score	Adult	Child <5 years	Child >5 years
Motor	6	Follows commands	Normal spontaneous movements	Follows commands
	5	Localizes pain	Localizes to supraocular pain (>9 months)	
	4	Withdraws to pain	Withdraws from nail bed pressure	
	3	Flexion	Flexion to supraocular pain	
	2	Extension	Extension to supraocular pain	
	1	None	None	
Verbal	5	Oriented	Age-appropriate Oriented speech/vocalizations	
	4	Confused speech	Less than usual ability; Confused irritable cry	
	3	Inappropriate words	Cries to pain	Inappropriate words

2 Incomprehensible Moans to pain Incomprehensible

1 None No response to pain

Eye opening 4 Spontaneous Spontaneous

3 To command To voice

2 To pain To pain

1 None None

lateralizing signs may result from decreased cerebral perfusion from increased intracranial pressure (ICP). In addition, reflex changes in blood pressure and heart rate may be observed, such as the Cushing reflex (hypertension and bradycardia). Coma from posterior fossa or infratentorial lesions may be abrupt in onset, with abnormal extensor posturing and loss of pupillary reflexes and extraocular movements. Brainstem compression with loss of brainstem reflexes may develop rapidly. Pontine hemorrhage, another infratentorial cause of coma, may present with pinpoint pupils.

Pseudocoma or psychogenic coma is a diagnostic challenge. History taking and observation of responses to stimulation reveal findings that differ from typical syndromes. Pupillary responses, extraocular movements, muscle tone, and reflexes are intact. Valuable tests include responses to manual eye opening (there should be little or no resistance in an unresponsive patient) and extraocular movements. If avoidance of gaze is consistently seen with the patient always looking away from the examiner, or if nystagmus is demonstrated with caloric vestibular testing, this is strong evidence for nonphysiologic or feigned unresponsiveness.

Diagnosis and Differential

History, examination, laboratory studies, and neuroimaging will often identify potential causes for a comatose state. Abrupt onset of coma suggests a catastrophic stroke or status epilepticus. Gradual onset suggests a metabolic process or progressive lesion such as a tumor or a slowly developing intracranial bleed. Examination may reveal signs of trauma or suggest other

142_Cydulka_Ch142.indd 778 21/03/17 2:25 PM

Source page 798

possibilities, such as toxidromes. Fine neurologic testing is not feasible with a comatose patient, but asymmetric findings on pupillary examination, assessment of corneal reflexes, and testing of oculovestibular reflexes may suggest focal CNS lesions. Extensor or flexor posturing suggests profound CNS dysfunction. A head CT should be obtained followed by lumbar puncture if the scan is nondiagnostic and a subarachnoid hemorrhage or infection is suspected. Basilar artery thrombosis may be a concern in a

comatose patient with a nondiagnostic head CT and an absence of alternative diagnoses; MRI or cerebral angiography may be considered to evaluate for this potential diagnosis. Patients who have had seizures and remain unresponsive may be having electrical seizures without motor activity (nonconvulsive status epilepticus), and an EEG should be performed if this is suspected. Consider toxic ingestions, infections, and nonaccidental trauma in comatose children. The differential diagnosis of coma includes primary CNS disorders (Table 142-4) and generalized disease processes that also affect the brain.

Table 142-4 Differential Diagnosis of Coma

Coma from causes affecting the brain diffusely

Encephalopathies

●● Hypoxic encephalopathy

●● Metabolic encephalopathy

●● Hypertensive encephalopathy

Hypoglycemia

Hyperosmolar state (e.g., hyperglycemia)

Electrolyte abnormalities (e.g., hypernatremia or hyponatremia, hypercalcemia)

Organ system failure

●● Hepatic encephalopathy

●● Uremia/renal failure

Endocrine (e.g., Addison’s disease, hypothyroidism, etc.)

Hypoxia

Carbon dioxide narcosis

Toxins

Drug reactions (e.g., neuroleptic malignant syndrome)

Environmental causes—hypothermia, hyperthermia

Deficiency state—Wernicke’s encephalopathy

Sepsis

Coma from primary CNS disease or trauma

Direct CNS trauma

●● Diffuse axonal injury

●● Subdural hematoma

●● Epidural hematoma

Vascular disease

●● Intraparenchymal hemorrhage (hemispheric, basal ganglia, brainstem, cerebellar)

Subarachnoid hemorrhage

Infarction

●● Hemispheric, brainstem

CNS infections

Neoplasms

Seizures

●● Nonconvulsive status epilepticus

●● Postictal state

142_Cydulka_Ch142.indd 779 21/03/17 2:25 PM

Source page 799

780 SECTION 15: Neurology

Emergency Department Care and Disposition

Treatment of coma involves supportive care and identification of the underlying cause.

1. Stabilize the airway, breathing, and circulation.

2. Identify and treat reversible causes, such as hypoglycemia and opioid toxicity. Consider empiric naloxone. Administer thiamine before glucose in hypoglycemic patients with a history of alcohol abuse or malnutrition.

3. Consider nonconvulsive status epilepticus in patients who have had a seizure and have not returned to baseline neurological function.

4. If increased ICP is suspected, elevate the head of the bed to 30°. Consider giving mannitol, 0.5 to 1.0 g/kg. Dexamethasone, 10 mg IV reduces vasogenic brain edema associated with a tumor. Hyperventilation may transiently lower ICP. Avoid lowering $PaCO_2$ below 35 mm Hg.

5. Discharge patients with readily reversible causes of coma if home care and follow-up care are adequate and a clear cause of the episode is found and reversed. Admit all other patients for further evaluation and management.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 168, "Altered Mental Status and Coma," by J. Stephen Huff.

142_Cydulka_Ch142.indd 780 21/03/17 2:25 PM

References/provenance

- Source: *Tintinalli's Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 796–801.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 143: Ataxia and Gait Disturbances

Source pages 802–803 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 802–803

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Ataxia and Gait Disturbances**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 143: Ataxia and Gait Disturbances 783 while standing still, cerebellar damage may cause motor ataxia, and sen- sory neuropathy may also be present. [Source pages 802–803]

- Confabulation characterizes the memory loss of Korsakoff’s syndrome. [Source pages 802–803]

- Consider vitamin B12 deficiency in patients with loss of position sense in the second toe and a positive Rom- berg test. [Source pages 802–803]

- A serum cyanocobalamin level and complete blood count are the initial steps in evaluation, although neurologic manifestations often precede macrocytic anemia. [Source pages 802–803]

- Neurosyphilis will cause similar symptoms of posterior column disease and can be screened for with VDRL or RPR tests. [Source pages 802–803]

Clinical reasoning structure

- **Problem representation:** Ataxia and Gait Disturbances; source pages 802–803.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Ataxia and Gait Disturbances**.

- Source-page range: **802–803**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 802

CHAPTER 143: Ataxia and Gait Disturbances 783

while standing still, cerebellar damage may cause motor ataxia, and sensory neuropathy may also be present. Confabulation characterizes the memory loss of Korsakoff's syndrome. Consider vitamin B12 deficiency in patients with loss of position sense in the second toe and a positive Romberg test. A serum cyanocobalamin level and complete blood count are the initial steps in evaluation, although neurologic manifestations often precede macrocytic anemia. Neurosyphilis will cause similar symptoms of posterior column disease and can be screened for with VDRL or RPR tests. Gait impairments can also present in elderly patients as manifestations of Parkinson's disease or normal pressure hydrocephalus.

Ingestions are a common cause of acute ataxia in children. Ataxia in a 2- to 4-year-old may follow recent immunizations, viral illnesses, or varicella. Cranial nerve abnormalities or motor dysfunction along with cerebellar abnormalities may be present with posterior fossa mass lesions. Chaotic eye movements (opsoclonus) and myoclonic jerks of the head and chest are most commonly associated with neuroblastomas of the chest or abdomen. In the absence of post-varicella ataxia or another identifiable cause, clinicians can consider neuroimaging, lumbar puncture, and consultation as clinically indicated.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. Administer thiamine 100 mg IV to alcoholics and other malnourished adults who might have Wernicke–Korsakoff syndrome.
2. Admit patients with an acute unexplained inability to walk for further evaluation and diagnostic testing.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 169, "Ataxia and Gait Disturbances," by J. Stephen

Huff.

143_Cydulka_Ch143.indd 783 21/03/17 2:32 PM

Source page 803

C h a p ter Acute Vertigo

144 Steven Go

■■ CLINICAL FEATURES

Vertigo is a perception of movement when none exists that results from a mismatch between the visual, vestibular, and proprioceptive sensory systems. Symptoms of vertigo are classically described by patients as a sensation that “the room is spinning,” but can also include atypical sensations of other types of movement. Vertigo is classified as peripheral or central (Table 144-1). Peripheral vertigo (involving the vestibular apparatus and eighth cranial nerve) usually has a sudden onset with intense symptoms that include nausea, vomiting, intolerance of head movement, and diaphoresis. Central vertigo (involving central structures such as the brainstem or cerebellum) can present either abruptly or gradually, but usually is characterized by less severe symptoms that are not well characterized by patients. Clinicians can work to discriminate between these two types of vertigo during a care encounter in the Emergency Department (ED) while recognizing that some overlap may exist between these two types of vertigo.

■■ DIAGNOSIS AND DIFFERENTIAL

The differential diagnosis for an episode of vertigo (Table 144-2) may be extensive, and providers should evaluate for the presence or absence of some

Table 144-1 Differentiating Peripheral from Central Causes of Acute Undifferentiated Vertigo

	Peripheral	Central
Onset	Sudden or insidious	Sudden
Severity of vertigo	Intense spinning	Ill defined, less intense
Prodromal dizziness	Occurs in up to 25%; often single episode	Occurs in up to 25%; recurrent episodes suggest transient ischemic attacks
Intolerant of head movements/ Dix–Hallpike maneuver	Yes	Sometimes
Associated nausea/diaphoresis	Frequent	Variable
Auditory symptoms	Points to peripheral causes	May be present
Proportionality of symptoms	Usually proportional	Often disproportionate
Headache or neck pain	Unusual	More likely
Nystagmus	Rotatory-vertical, horizontal	Vertical
CNS symptoms/signs	Absent	Usually present
Head impulse test	Abnormal	Usually normal

HINTS examination (combined Peripheral signs on all three Central signs on at least

horizontal head impulse test, bedside tests one of three bedside tests

nystagmus, and test of skew)

784

144_Cydulka_Ch144.indd 784 22/04/17 4:42 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 802–803.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 144: Acute Vertigo

Source pages 804–811 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 804–811

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Acute Vertigo**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 144: Acute Vertigo 785 Table 144-2 Causes of Acute Undifferentiated Vertigo Vestibular/otologic Benign paroxysmal positional vertigo Traumatic: following head injury Infection: labyrinthitis, vestibular neuronitis, Ramsay Hunt syndrome Systemic conditions Ménière’s syndrome with vestibular/otologic Neoplastic effects Vascular Otosclerosis Paget’s disease Toxic or drug-induced: aminoglycosides Neurologic Vertebrobasilar insufficiency or vertebral artery dissection Lateral Wallenberg’s syndrome Anterior inferior cerebellar artery syndrome Neoplastic: cerebellopontine angle tumors Cerebellar disorders: hemorrhage, degeneration Basal ganglion diseases Multiple sclerosis Infections: neurosyphilis, tuberculosis Epilepsy Migraine headaches Cerebrovascular disease General Hematologic: anemia, polycythemia, hyperviscosity syndrome Toxic: alcohol Chronic renal failure Metabolic: thyroid disease, hypoglycemia key features during the history and physical examination. [Source pages 804–811]

- Inquire about the speed of symptom onset, severity, duration, temporal pattern, head or neck trauma, chiropractic manipulation, and any associated symptoms such as headache, neck pain, or loss of consciousness. [Source pages 804–811]

- Assess risk factors for stroke such as age, hypertension, cardiovascular disease, and coagulopathy. [Source pages 804–811]

- Physical examination should include eye (e.g., nystagmus), ear, vestibular, and neuro- logical examinations, with particular attention to potential abnormalities of the cranial nerves or cerebellar function. [Source pages 804–811]

- Since focal deficits are not universal in central vertigo, further specialized neurological tests may be of value in some patients. [Source pages 804–811]

Clinical reasoning structure

- **Problem representation:** Acute Vertigo; source pages 804–811.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Acute Vertigo**.

- Source-page range: **804–811**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 804

CHAPTER 144: Acute Vertigo 785

Table 144-2 Causes of Acute Undifferentiated Vertigo

Vestibular/otologic Benign paroxysmal positional vertigo

Traumatic: following head injury

Infection: labyrinthitis, vestibular neuronitis, Ramsay Hunt syndrome

Systemic conditions Ménière’s syndrome

with vestibular/otologic Neoplastic

effects Vascular

Otosclerosis

Paget’s disease

Toxic or drug-induced: aminoglycosides

Neurologic Vertebrobasilar insufficiency or vertebral artery dissection

Lateral Wallenberg’s syndrome

Anterior inferior cerebellar artery syndrome

Neoplastic: cerebellopontine angle tumors

Cerebellar disorders: hemorrhage, degeneration

Basal ganglion diseases

Multiple sclerosis

Infections: neurosyphilis, tuberculosis

Epilepsy

Migraine headaches

Cerebrovascular disease

General Hematologic: anemia, polycythemia, hyperviscosity syndrome

Toxic: alcohol

Chronic renal failure

Metabolic: thyroid disease, hypoglycemia

key features during the history and physical examination. Inquire about the speed of symptom onset, severity, duration, temporal pattern, head or neck trauma, chiropractic manipulation, and any associated symptoms such as headache, neck pain, or loss of consciousness. Assess risk factors for stroke such as age, hypertension, cardiovascular disease, and coagulopathy. Physical examination should include eye (e.g., nystagmus), ear, vestibular, and neurological examinations, with particular attention to potential abnormalities of the cranial nerves or cerebellar function. Since focal deficits are not universal in central vertigo, further specialized neurological tests may be of value in some patients. HINTS testing consists of the Head Impulse test (evaluation of the vestibulo-ocular reflex), examination for Nystagmus direction that changes with gaze direction, and Test of Skew (vertical ocular misalignment during the cover-uncover eye test). The presence of one or more findings on the HINTS exam that are consistent with central vertigo has been reported to be sensitive and specific for stroke when completed by experts, although it is unclear whether this specific testing is useful for physicians without special training. If benign paroxysmal positional vertigo (BPPV) is suspected, a Dix–Hallpike maneuver may be useful to help make the diagnosis (sensitivity 75% to 82%). Physical examination maneuvers that involve head or neck twisting should not be used if cervical artery dissection, vertebrobasilar insufficiency, or spinal pathology are suspected.

Laboratory investigations are not typically indicated in the routine workup for patients with symptoms consistent with peripheral vertigo, unless there is strong suspicion of a specific cause where laboratory analysis would be helpful. Neuroimaging studies such as an emergent CT scan,

144_Cydulka_Ch144.indd 785 22/04/17 4:42 pm

Source page 805

CT angiogram, MRI, or MR angiogram are not routinely indicated when peripheral vertigo is suspected, but may be useful when specific causes of central vertigo are significant concerns. Consider neuroimaging in elderly patients with signs or symptoms concerning for central vertigo (such as cranial nerve or cerebellar findings), significant stroke risks, coagulopathy (e.g., taking anticoagulant medications), headache, head or neck trauma, or intractable or persistent (>72 hours) symptoms. Figure 144-1 illustrates an approach to patients with vertigo.

1. Investigate for syncope, seizures,

LOSS OF YES intracranial process

CONSCIOUSNESS 2. ECG, neurodiagnostic imaging,

elecrolytes

NO 3. Consider admission

NO

TRUE VERTIGO ILL-DEFINED DIZZINESS

y You

erl ng

Eld

1. Consider disequilibrium 1. Obtain orthostatic BP,

of aging lytes, etc.

YES 2. Obtain orthostatic BP, 2. Consider psychiatric

lytes, CBC, tilt test, ECG dizziness

3. Consider referral to

cardiologist

CNS YES

SYMPTOMS & DURATION OF SYMPTOMS/SIGNS

SIGNS) Co

in ns

0 m ta

–1 nt

(5

ief

Br CENTRAL

VERTIGO/MS

TIAs

1. Consider CT, MRI, MRA

NO 1. Check for carotid bruits 2. Consider admission and

2. Consider echocardiogram neurology or neurosurgical

3. Consider antiplatelet or referral

anticoagulant therapy

VERTIGO

PRECIPITATED BY YES

CONSIDER BPPV

SUDDEN HEAD

MOVEMENTS 1. Confirm via Dix-Hallpike position test

2. Consider modified Epley maneuver

NO 3. Rx: antiemetics or antivertigo drugs

HEARING YES YES RECENT URI YES CONSIDER

TINNITUS

LOSS OR OTITIS MEDIA BACTERIAL

LABYRINTHITIS

NO NO 1. CT or MRI

NO

2. Antibiotics

3. ENT consultation

CONSIDER ACOUSTIC CONSIDER MÉNIÈRE

OTHER CONDITIONS

NEUROMA DISEASE

eg, VESTIBULAR

NEURONITIS 1. CT/MRI 1. Rx: antihistamines/antiemetics,

2. Refer to neurosurgeon diuretics

1. Rx: antihistamines/

or ENT 2. Diet <1 g added salt per day

antiemetics

3. ENT referral

2. Reassurance

Figure 144-1. Approach to a patient with vertigo. BP, blood pressure; BPPV, benign paroxysmal positional vertigo; CBC, complete blood count; CNS, central nervous system; ENT, ear, nose, and throat; MRA, magnetic resonance angiography; MS, multiple sclerosis; Rx, treatment; TIAs, transient ischemic attacks; URI, upper respiratory infection.

144_Cydulka_Ch14

Source page 806

CHAPTER 144: Acute Vertigo 787

■■ CAUSES OF PERIPHERAL VERTIGO

Benign paroxysmal positional vertigo (BPPV) is believed to be caused by loose otoconia that most commonly enter the posterior semicircular canal and cause the inappropriate sensation of motion. Findings suggestive of BPPV are listed in Table 144-3. The Dix–Hallpike maneuver can support the diagnosis. To perform this physical examination maneuver, begin with the patient seated with the head turned 45° to the right. Rapidly lower the patient to a supine position, with the head hanging over the edge of the bed and the neck in 20° of extension. A positive test is when patients experience a short-lived rotatory nystagmus, with rapid eye beating toward the affected (dependent) ear that is associated with an acute episode of vertigo. After any nystagmus or symptoms resolve, slowly return the patient to the sitting position. Repeat the maneuver with the head turned 45° to the left. The side with the positive test serves as the starting point for the potentially curative Epley maneuver (see below).

Ménière's syndrome is characterized by recurrent bouts of vertigo associated with unilateral tinnitus and a sense of fullness and diminished hearing in the affected ear. Because the diagnosis requires multiple epi-

sodes of attacks with progressive hearing loss, Ménière's syndrome is not typically diagnosed on the first presentation of vertigo.

A perilymph fistula presents with sudden onset of vertigo during activities that can cause barotrauma such as flying, scuba diving, heavy lifting, and coughing. Infection can also cause a perilymph fistula, and the diagnosis can be confirmed by nystagmus elicited by pneumatic otoscopy.

Vestibular neuronitis is characterized by the sudden onset of severe vertigo sometimes associated with unilateral tinnitus and hearing loss. It is the second most common cause of peripheral vertigo, and is thought to be viral in nature. Symptoms last several days to weeks before they resolve spontaneously without recurrence.

Vestibular ganglionitis causes vertigo when a neurotrophic virus such as varicella zoster reactivates. The most famous variant is Ramsay Hunt syndrome (deafness, vertigo, and facial nerve palsy), which is associated with vesicles inside the external auditory canal.

Labyrinthitis, although commonly viral, can also be caused by bacterial infection from otitis media, meningitis, and mastoiditis. Patients with labyrinthitis present with a sudden onset of vertigo with hearing loss and middle ear findings.

Ototoxicity may induce hearing loss and vertigo. Common offenders causing peripheral ototoxicity include NSAIDs, salicylates, aminoglycosides,

Table 144-3 Supportive Findings in Benign Paroxysmal Positional Vertigo

Latency period of <30 sec between the provocative head position and onset of nystagmus.

The intensity of nystagmus increases to a peak before slowly resolving.

Duration of vertigo and nystagmus ranges from 5 to 40 sec.

If nystagmus is produced in one direction by placing the head down, then the nystagmus reverses direction when the head is returned to the sitting position.

Repeated head positioning causes both the vertigo and accompanying nystagmus to fatigue and subside.

Abnormal horizontal head impulse test indicating abnormal vestibulo-ocular reflex function.

HINTS testing not indicative of stroke.

Abbreviation: HINTS, horizontal head impulse test, nystagmus, and test of skew.

144_Cydulka_Ch144.indd 787 22/04/17 4:42 pm

Source page 807

loop diuretics, and cytotoxic agents. Anticonvulsants, tricyclic antidepressants, neuroleptics, hydrocarbons, alcohol, and phencyclidine may cause centrally mediated vertigo.

Tumors of the eighth cranial nerve and cerebellopontine angle, such as meningiomas, acoustic neuromas, and acoustic schwannomas, may pres-

ent with hearing loss and subsequent gradual onset of mild vertigo. Ataxia, ipsilateral facial weakness, loss of the corneal reflex, and cerebellar signs can also occur.

Vertigo may occur after closed head injury (e.g., basilar skull fracture) and tends to resolve over several weeks. In the acute setting, injuries with an associated intracranial hemorrhage should be excluded with neuroimaging. Postconcussive syndrome can also develop subacutely and may include an unsteady gait with dizziness.

■■ CAUSES OF CENTRAL VERTIGO

Cerebellar hemorrhage or infarction typically causes moderate vertigo symptoms and can be associated with headache, nausea, and vomiting. Cerebellar findings such as truncal ataxia, abnormal Romberg testing, and tandem gait abnormalities are often present.

Lateral medullary infarction of the brainstem (Wallenberg syndrome) causes vertigo and ipsilateral facial numbness, loss of the corneal reflex, Horner syndrome, dysphagia, and dysphonia. Contralateral loss of pain and temperature sensation in the extremities also occur.

Vertebrobasilar insufficiency may result in sudden vertigo due to brainstem transient ischemic attack that typically lasts from minutes to 24 hours. Diplopia, dysphagia, dysarthria, vision changes, and syncope may also be present. Unlike other causes of central vertigo, vertebrobasilar insufficiency may be induced by movement of the head that is caused by positional decreases in vertebral artery blood flow.

Vertebral artery dissection can be caused by mechanisms that induce a sudden rotation of the head such as a motor vehicle crash, chiropractic adjustments, or violent sneezing. Patients present with central vertigo, headache, neck pain, and a unilateral Horner’s syndrome.

Other potential causes of central vertigo include multiple sclerosis, neoplasms of the fourth ventricle, and vestibular migraine.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. For most causes of peripheral vertigo, pharmacologic therapies in the ED may include anticholinergics and antihistamines (Table 144-4). Diphenhydramine 25 to 50 mg IM, IV, or PO and meclizine 25 mg PO are often effective in providing symptomatic relief. Transdermal scopolamine 0.5 mg is often recommended, but may be less useful for acute treatment due to its prolonged onset of action (4 to 8 hours). It may be most useful when prescribed as a discharge medication.

2. Second-line pharmacologic agents to consider when other medications are inadequate include calcium channel blockers, neuroleptics (avoid in orthostatic patients), and antiemetics such as ondansetron 4 mg IV.

Benzodiazepines prevent the process of vestibular rehabilitation and

can be used sparingly for symptomatic relief.

144_Cydulka_Ch144.indd 788 22/04/17 4:42 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.

- Content ID: SEG-TINTINALLI-EMM8-FULL.

- Source pages: 804–811.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 145: Seizures and Status Epilepticus in Adults

Source pages 812–815 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 812–815

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Seizures and Status Epilepticus in Adults**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 145: Seizures and Status Epilepticus in Adults 793 school-aged children and often last only a few seconds. [Source pages 812–815]

- Patients suddenly lose consciousness without losing postural tone and appear confused, detached, or withdrawn. [Source pages 812–815]

- An absence seizure typically ends abruptly with a return to normal functioning. [Source pages 812–815]

- Partial seizures are due to electrical discharges that begin in a localized region of the cerebral cortex. [Source pages 812–815]

- These seizures may remain localized to one area of the brain or may later spread to other regions. [Source pages 812–815]

Clinical reasoning structure

- **Problem representation:** Seizures and Status Epilepticus in Adults; source pages 812–815.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Seizures and Status Epilepticus in Adults**.

- Source-page range: **812–815**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 812

CHAPTER 145: Seizures and Status Epilepticus in Adults 793

school-aged children and often last only a few seconds. Patients suddenly lose consciousness without losing postural tone and appear confused, detached, or withdrawn. An absence seizure typically ends abruptly with a return to normal functioning.

Partial seizures are due to electrical discharges that begin in a localized region of the cerebral cortex. These seizures may remain localized to one area of the brain or may later spread to other regions. Partial seizures are described as simple, in which consciousness is not affected, or complex, in which consciousness is altered. Complex partial seizures are often due to discharges in the temporal lobe (also termed temporal lobe seizures) and may include automatisms, visceral complaints, hallucinations, memory disturbances, distorted perception, and affective disorders.

Status epilepticus is defined as a single seizure lasting for longer than 5 minutes, or as two or more seizures that occur sequentially without an intervening recovery of consciousness. Nonconvulsive status epilepticus is characterized by altered mental status without perceptible muscular convulsive activity and is confirmed by electroencephalogram (EEG).

Eclampsia is a disorder found in pregnant women from 20 weeks gestation up to 8 weeks postpartum and is characterized by generalized seizures, hypertension, edema, and proteinuria.

■■ DIAGNOSIS AND DIFFERENTIAL

When a patient presents with seizure-like activity, obtain a detailed history and inquire about the presence of preceding aura, abrupt or gradual onset, progression of motor activity, incontinence, whether the activity was local or generalized, symmetry of symptoms, duration of the episode, and presence of postictal confusion or lethargy. When a patient has a previously diagnosed seizure disorder, inquire about their baseline sei-

zure pattern, common precipitants, and any recent changes in antiepileptic regimen. For those patients without a history of seizures, ask about recent or remote head injury. Consider the possibility of intracranial pathology if persistent, severe, or sudden-onset headaches are present. Consider eclampsia in patients with a current or recent pregnancy. Other factors that may predispose to seizures include metabolic or electrolyte abnormalities, hypoxia, systemic illness, cancer, sequelae of coagulopathy or anticoagulation, exposure to industrial or environmental toxins, drug ingestion or withdrawal, and alcohol use. Seizures are a common manifestation of central nervous system (CNS) disease in patients with the human immunodeficiency virus.

Evaluate each patient for findings of physical injuries that may have occurred during the seizure, perform a neurologic exam, and closely follow the level of consciousness. A transient focal deficit following a simple or complex focal seizure is referred to as Todd's paralysis and typically resolves within 48 hours. Initiate appropriate further testing for neurologic findings that are new or cannot be readily attributed to a benign cause.

Laboratory testing should be individualized and may not be indicated in all circumstances. In a patient with a known seizure disorder who has had a typical seizure, a glucose level and pertinent anticonvulsant

145_Cydulka_Ch145.indd 793 21/03/17 3:23 PM

Source page 813

levels may be appropriate. In an adult with a first seizure, additional studies are often used to assess for medical causes of seizure and may include serum glucose, basic metabolic panel, calcium, magnesium, a pregnancy test, and toxicology studies, as indicated. A noncontrast head CT is appropriate for a patient with a first seizure or a change in seizure pattern to identify a structural lesion or an acute intracranial process. Additional neuroradiographic imaging may be indicated based on CT scan results or clinical circumstances. MRI is typically not necessary on an emergent basis, but is often recommended as an aspect of non-emergent outpatient follow-up for patients with a first-time seizure. Lumbar puncture is indicated if CNS infection or subarachnoid hemorrhage is suspected.

The differential diagnosis of seizures includes syncope, pseudoseizures, hyperventilation syndrome, movement disorders, and migraines.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. During an active seizure, take measures to protect the patient from injury

or aspiration. IV anticonvulsants are often not needed during an uncomplicated seizure that resolves spontaneously.

2. In patients with a known seizure disorder whose anticonvulsant levels are low or who have missed doses of prescribed medications, supplemental doses may be appropriate.

3. Oral loading of phenytoin 20 mg/kg divided into three doses given every 2 to 4 hours is time-consuming and results in a delay in reaching therapeutic levels. Alternatively, phenytoin 20mg/kg IV may be administered at a rate of 25 mg/min. The loading dose of fosphenytoin is 20 phenytoin equivalents (PE)/kg at a maximum IV rate of 150 PE/min.

4. Administer IV magnesium sulfate to patients with suspected eclampsia. Consult an obstetrician early in the patient's care.

5. Patients in status epilepticus should have IV access, cardiac monitoring, and pulse oximetry. Endotracheal intubation may be required. If a paralytic agent is used, a short-acting drug is preferred so that any ongoing seizure activity can be monitored. Otherwise, EEG monitoring should be initiated. See Fig. 145-1 for anticonvulsant medication options. Lorazepam is considered the initial anticonvulsant of choice, followed by phenytoin, fosphenytoin, or levetiracetam. In refractory cases, consider IV midazolam, propofol, phenobarbital, pentobarbital, or ketamine. Early neurology consultation should be requested.

6. Patients with a first seizure who have a normal neurologic examination, no acute or chronic medical comorbidities, normal diagnostic testing including neuroimaging, and a normal mental status can be discharged from the ED with appropriate recommendations for outpatient follow-up. Initiation of antiepileptic medication, brain MRI, and other additional testing may be deferred to the outpatient setting.

7. Instruct discharged patients to take precautions that minimize the risks for injury from further seizures. Swimming, working with hazardous tools or machines, and working at heights should be prohibited. Patients should not drive vehicles until cleared by a neurologist or primary care physician, and driving privileges should conform to state laws.

145_Cydulka_Ch145.indd 794 21/03/17 3:23 PM

Source page 814

Supportive Care:

Active

IV access, monitors, maintain airway, oxygen,

Seizure

check point-of-care glucose and provide IV glucose if indicated

0–5 minutes

Patient Protection

IV Fosphenytoin: Consider

20 PE/kg at intubation

IV Lorazepam: 150 mg/min

_____

Established 2 mg up to

One of the IV Phenytoin:

Status 0.1 mg/kg

following 20 mg/kg at

Epilepticus or

50 mg/min

_____

Treatment IV Diazepam:

Goal: 5–10 min 10–20 mg IV Levetiracetam:

2000–4000 mg

_____

IV Midazolam: IV Propofol: Intubate patient

IV

Refractory Load with 0.2 1 mg/kg IV, Neuro ICU

Phenobarbital:

Status mg/kg or then 1–10 or 20 mg/kg at admission

Epilepticus then infusion mg/kg/h;

50–75

Treatment of 0.05–2 or Ketamine, Continuous

mg/min

Goal: <30 min mg/kg/h 5 mg/kg/h EEG monitoring

FIGURE 145-1. Guidelines for management of active seizures and status epilepticus. ICU = intensive care unit; PE = phenytoin equivalent.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 171, "Seizures," by Joshua G. Kornegay.

145_Cydulka_Ch145.indd 795 21/03/17 3:23 PM

Source page 815

C h a p t er Acute Peripheral Neurologic

146 Lesions

Nicholas E. Kman

A systematic approach to evaluating neurologic symptoms includes localizing the problem anatomically and distinguishing peripheral disorders

from those with a central etiology. Peripheral nerve disorders may affect sensory, motor, and autonomic functions (Table 146-1).

■■ GUILLAIN-BARRÉ SYNDROME

Clinical Features

Guillain-Barré syndrome (GBS) is an acute polyneuropathy characterized by immune-mediated peripheral nerve myelin sheath or axon destruction. It can be associated with a viral or febrile illness, Campylobacter jejuni infection, or vaccination.

Diagnosis and Differential

Although numerous variants exist, the typical presentation from GBS includes ascending symmetric weakness or paralysis and loss of deep tendon reflexes. Respiratory failure and significant autonomic fluctuations may occur. Cerebrospinal fluid (CSF) analysis typically shows high protein and a normal cell count (Table 146-2).

Table 146-1 Differentiating Central Nervous System from Peripheral Nervous System Disorders

	Central	Peripheral
History	Cognitive changes Sudden weakness Nausea, vomiting Headache	Weakness confined to one limb Weakness with associated pain Posture- or movement-dependent pain Weakness after prolonged period in one position
Physical Examination		
Reflexes	Brisk reflexes (hyperreflexia) Babinski's sign Hoffman's sign	Hypoactive reflexes Areflexia
Motor	Asymmetric weakness of ipsilateral upper and lower extremity Facial droop Slurred speech	Symmetric proximal weakness
Sensory	Asymmetric sensory loss in ipsilateral upper and lower extremity	Reproduction of symptoms with movement (compressive neuropathy) All sensory modalities involved
Coordination	Discoordination without weakness	Loss of proprioception

146_Cydulka_Ch146.indd 796 21/03/17 3:55 PM

References/provenance

- Source: *Tintinalli's Emergency Medicine Manual*, Eighth Edition.

- Content ID: SEG-TINTINALLI-EMM8-FULL.

- Source pages: 812–815.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 146: Acute Peripheral Neurologic Lesions

Source pages 816–819 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 816–819

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Acute Peripheral Neurologic Lesions**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 146: Acute Peripheral Neurologic Lesions 797 Table 146-2 Diagnostic Criteria for Classic Guillain-Barré Syndrome Required Progressive weakness of more than one limb Areflexia Suggestive Progression over days to weeks Recovery beginning 2–4 weeks after cessation of progression Relative symmetry of symptoms Mild sensory signs and symptoms Cranial nerve involvement (Bell’s palsy, dysphagia, dysarthria, ophthalmoplegia) Autonomic dysfunction (tachycardia, bradycardia, dysrhythmias, wide variations in blood pressure, postural hypotension, urinary retention, constipation, facial flushing, anhydrosis, hypersalivation) Absence of fever at onset Cytoalbuminologic dissociation of cerebrospinal fluid (high protein and low white cell count) Typical findings on electromyogram and nerve conduction studies Emergency Department Care and Disposition 1. [Source pages 816–819]

- Initial treatment includes supportive care which may include respiratory support as the disease progresses. [Source pages 816–819]

- Consultation with a neurologist for admission to a monitored setting is recommended. [Source pages 816–819]

- Administration of IV immunoglobulin and plasmapheresis may shorten the time to recovery. [Source pages 816–819]

- ■■ BELL’S PALSY Clinical Features Bell’s palsy causes seventh cranial nerve dysfunction. [Source pages 816–819]

Clinical reasoning structure

- **Problem representation:** Acute Peripheral Neurologic Lesions; source pages 816–819.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Acute Peripheral Neurologic Lesions**.

- Source-page range: **816–819**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 816

CHAPTER 146: Acute Peripheral Neurologic Lesions 797

Table 146-2 Diagnostic Criteria for Classic Guillain-Barré Syndrome

Required

Progressive weakness of more than one limb

Areflexia

Suggestive

Progression over days to weeks

Recovery beginning 2–4 weeks after cessation of progression

Relative symmetry of symptoms

Mild sensory signs and symptoms

Cranial nerve involvement (Bell’s palsy, dysphagia, dysarthria, ophthalmoplegia)

Autonomic dysfunction (tachycardia, bradycardia, dysrhythmias, wide variations in blood pressure, postural hypotension, urinary retention, constipation, facial flushing, anhydrosis, hypersalivation)

Absence of fever at onset

Cytoalbuminologic dissociation of cerebrospinal fluid (high protein and low white cell count)

Typical findings on electromyogram and nerve conduction studies

Emergency Department Care and Disposition

1. Initial treatment includes supportive care which may include respiratory support as the disease progresses.

2. Consultation with a neurologist for admission to a monitored setting is recommended.

3. Administration of IV immunoglobulin and plasmapheresis may shorten the time to recovery.

■■ BELL’S PALSY

Clinical Features

Bell's palsy causes seventh cranial nerve dysfunction. Patients may complain of facial weakness, articulation problems, difficulty keeping an eye closed, or inability to keep food in the mouth on one side.

Diagnosis and Differential

Physical examination findings demonstrate weakness on one side of the face, including the forehead, without other focal neurologic findings. The differential diagnosis includes stroke, Lyme disease, GBS, parotid tumors, middle ear lesions, cerebellopontine angle tumors, eighth cranial nerve lesions, HIV, and vascular disease. Inspect the ear for ulcerations caused by cranial herpes zoster activation, which is diagnostic for Ramsey–Hunt syndrome. Bell's Palsy is not likely if muscle strength is retained in the forehead on examination, and such a finding suggests a possible central neurologic lesion warranting radiographic imaging of the brain.

Emergency Department Care and Disposition

1. Systemic corticosteroids increase the frequency of complete recovery. Prescribe prednisone 1 mg/ kg per day PO for 7 days.

2. There is no proven benefit from administering antiviral medications unless evidence of Ramsey–Hunt syndrome is noted.

146_Cydulka_Ch146.indd 797 21/03/17 3:55 PM

Source page 817

3. Treat patients with Ramsey–Hunt syndrome with antiviral medications, such as famciclovir 500 mg PO three times a day for 7 days or valacyclovir 1 g PO three times a day for 7 days.

4. Advise patients to use an ophthalmic moisturizing ointment at night to prevent corneal drying from incomplete eye closure.

5. Arrange follow-up with a primary care physician or ENT specialist to ensure appropriate resolution.

■■ FOCAL MONONEUROPATHIES

Clinical Features

Focal mononeuropathies are most often due to focal nerve compression, although some systemic processes may also lead to mononeuropathy. Examples include carpal tunnel syndrome, resulting from compression of the median nerve at the wrist, and cubital tunnel syndrome, resulting from compression of the ulnar nerve at the elbow.

Diagnosis and Differential

Carpal tunnel syndrome causes pain, paresthesias, and numbness in the distribution of the medial nerve. The presence of Tinel's sign (light percussion over median nerve at wrist results in electric shock sensation shooting into hand) and use of Phalen's maneuver (holding wrists in flexion for 60

seconds worsens symptoms) may help confirm the diagnosis. Cubital tunnel syndrome causes tingling in the fifth and lateral fourth fingers that may progress to paralysis and wasting of the intrinsic hand muscles. Other common focal neuropathies include deep peroneal entrapment (causing foot drop and numbness between the first and second toes), meralgia paresthetica (entrapment of the lateral femoral cutaneous nerve causing numbness and pain of the anterolateral thigh), and mononeuritis multiplex (dysfunction of multiple peripheral nerves separated temporally and anatomically).

Emergency Department Care and Disposition

1. Conservative treatment is recommended, including anti-inflammatory medications.
2. Consider applying an appropriate splint for comfort. For carpal tunnel syndrome, splint the wrist in neutral position.
3. Direct additional treatment at the underlying cause of the neuropathy.

■■ PLEXOPATHIES

Plexopathies can be caused by trauma, surgery, neoplasm, or radiation therapy. Brachial plexopathy causes weakness in the arm or shoulder girdle followed by pain and paresthesias. Patients have weakness in various distributions of the brachial plexus. ED evaluation is directed at identifying acutely reversible causes, such as joint dislocation or traumatic injury, and referral for other causes, such as neoplasm. Lumbosacral plexopathy is less likely due to injury and can be caused by radiation, diabetic amyotrophy, aortic aneurysm, retroperitoneal hemorrhage, or

146_Cydulka_Ch146.indd 798 21/03/17 3:55 PM

Source page 818

arteriovenous malformations. Symptoms are weakness, decreased sensation, and possibly decreased reflexes in the areas innervated by the affected portions of the plexus. Plain radiographs, MRI, and abdominal CT may be useful in determining the etiology. Direct specific treatment at the underlying cause.

■■ NEUROMUSCULAR JUNCTION DISORDERS

Botulism is a toxin-mediated disorder of the neuromuscular junction that is caused by Clostridium botulinum toxin. Foodborne botulism typically comes from improperly preserved canned foods. In infantile botulism, organisms arise from ingested spores, often in honey, and produce a systemically absorbed toxin. Clinical features appear 6 to 48 hours after ingestion and may be preceded by nausea, vomiting, and diarrhea. Consider wound botulism in patients with a wound or a history of intravenous drug

use. Early complaints involve the eye or bulbar musculature and progress to symmetric descending weakness and potential respiratory insufficiency. Treatment includes respiratory support, trivalent botulinum antitoxin 10 mL IV, and admission. For infants, human botulism immunoglobulin decreases mechanical ventilation requirements and length of intensive care unit stays.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 172, "Acute Peripheral Neurologic Disorders," by Phillip Andrus and J. Michael Guthrie.

146_Cydulka_Ch146.indd 799 21/03/17 3:55 PM

Source page 819

C hapter Chronic Neurologic Disorders

147 Michael T. Fitch

Patients with chronic neurologic disorders may present to the emergency department with complications of their underlying medical conditions. Emergency management of these disorders often centers on evaluation of potential respiratory complications and an assessment for appropriate inpatient or outpatient management of acute illness or exacerbation of chronic disease.

■■ AMYOTROPHIC LATERAL SCLEROSIS

Clinical Features

Patients with amyotrophic lateral sclerosis (ALS) experience progressive muscle atrophy and weakness. Limb spasticity, hyperreflexia, and emotional lability are common symptoms of upper motor neuron demyelination, while lower motor neuron dysfunction can cause muscle weakness, atrophy, fasciculations, dysarthria, dysphagia, and difficulty with mastication. In the early phase of disease, symptoms may be asymmetric. Respiratory muscle weakness causes progressive respiratory difficulty that presents as dyspnea with exertion or at rest. Sensory and cognitive function is often spared, although up to 15% of patients may also develop dementia and Parkinsonism.

Diagnosis and Differential

The diagnosis early in the course of illness can be challenging, which contributes to a median time to diagnosis of 14 months. Clinicians may suspect ALS when signs are found of upper and lower motor neuron dysfunction without other central nervous system dysfunction. Other diagnoses to consider that may have ALS-like symptoms are systemic illnesses such as diabetes, thyroid and parathyroid dysfunction, vitamin B12 deficiency, heavy metal toxicity, vasculitis, and CNS tumors.

Emergency Department Care and Disposition

1. A new diagnosis of ALS is uncommon to make in the emergency department. Stable patients with suspicious symptoms should be referred for appropriate testing by a neurologist if an initial workup does not reveal an emergent cause of neurologic symptoms.

2. Patients with a known diagnosis of ALS may present for emergency care with complications such as respiratory failure, aspiration pneumonia, choking episodes, or traumatic injuries.

3. Optimize pulmonary function based on an individual patient's respiratory status and comorbidities using interventions such as nebulizer treatments, steroids, antibiotics, and intubation, as indicated. While a blood gas may not reliably predict impending respiratory failure, a forced vital capacity <25 mL/kg or 50% decreased from predicted

147_Cydulka_Ch147.indd 800 21/03/17 4:29 PM

References/provenance

- Source: *Tintinalli's Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 816–819.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 147: Chronic Neurologic Disorders

Source pages 820–825 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 820–825

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Chronic Neurologic Disorders**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 147: Chronic Neurologic Disorders 801 normal increases the likelihood of pneumonia and respiratory failure in these patients. [Source pages 820–825]

- Patients with pneumonia, inability to handle secretions, respiratory dis- tress, or need for ventilator support should be admitted to an appropriate level of inpatient care. [Source pages 820–825]

- ■■ MYASTHENIA GRAVIS Clinical Features Myasthenia gravis is an autoimmune disorder of the neuromuscular junc- tion that is characterized by muscle weakness and fatigue. [Source pages 820–825]

- Most patients have weakness of proximal extremity muscles, neck extensors, and facial or bulbar muscles. [Source pages 820–825]

- Ptosis and diplopia are common presenting symptoms, and symptoms typically worsen as the day progresses and improve with rest. [Source pages 820–825]

Clinical reasoning structure

- **Problem representation:** Chronic Neurologic Disorders; source pages 820–825.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Chronic Neurologic Disorders**.

- Source-page range: **820–825**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 820

CHAPTER 147: Chronic Neurologic Disorders 801

normal increases the likelihood of pneumonia and respiratory failure in these patients.

4. Patients with pneumonia, inability to handle secretions, respiratory distress, or need for ventilator support should be admitted to an appropriate level of inpatient care.

■■ MYASTHENIA GRAVIS

Clinical Features

Myasthenia gravis is an autoimmune disorder of the neuromuscular junction that is characterized by muscle weakness and fatigue. Most patients have weakness of proximal extremity muscles, neck extensors, and facial or bulbar muscles. Ptosis and diplopia are common presenting symptoms, and symptoms typically worsen as the day progresses and improve with rest. Severe respiratory muscle weakness causing respiratory failure is characteristic of a myasthenic crisis.

Diagnosis and Differential

The symptoms of myasthenia gravis can mimic those seen in many other disorders, including Lambert-Eaton syndrome, botulism, thyroid disorders, and stroke. The diagnosis can be established through administration of edrophonium (an acetylecholinesterase inhibitor), electromyography, and serum testing for acetylcholine receptor antibodies. Edrophonium or neostigmine is expected to improve muscle strength in objectively weak limb and ocular or pharyngeal muscles and can help make an initial diagnosis.

Emergency Department Care and Disposition

1. Myasthenia gravis can be treated with the acetylcholinesterase inhibitors pyridostigmine or neostigmine, thymectomy, immune suppression with steroids or azathioprine, and immune modulation with plasma

exchange or IV immunoglobulin.

2. Many drugs used in the emergency department have the potential to impact neuromuscular function. Check carefully for drug interactions when treating patients with myasthenia gravis (Table 147-1).

3. Respiratory failure can occur in patients with infection, recent surgery, or recent changes in immunosuppressant medications. When possible, avoid depolarizing or nondepolarizing paralytic agents if intubation is required.

4. Myasthenic crisis requiring emergent intervention may occur in undiagnosed patients with myasthenia gravis, or in patients with known disease who have acute exacerbations or inadequate treatment.

5. Administration of edrophonium 1 mg slow IV can help to distinguish a myasthenic crisis (inadequately treated myasthenia gravis) from a cholinergic crisis (often from overmedication). Resolution of muscle weakness within a few minutes suggests a myasthenia exacerbation that can be further treated with neostigmine 0.5 to 2 mg IM, IV, SC or in increments of 15 mg PO.

147_Cydulka_Ch147.indd 801 21/03/17 4:29 PM

Source page 821

Table 147-1 Drugs to Avoid in Myasthenia Gravis

Steroids	Adrenocorticotropic hormone,* methylprednisolone,* prednisone*
Anticonvulsants	Phenytoin, ethosuximide, trimethadione, paraldehyde, magnesium sulfate, barbiturates, lithium
Antimalarials	Chloroquine,* quinine*
IV fluids	Sodium lactate solution
Antibiotics	Aminoglycosides, fluoroquinolones,* neomycin,* streptomycin,* kanamycin,* gentamicin, tobramycin, dihydrostreptomycin,* amikacin, polymyxin A, polymyxin B, sulfonamides, viomycin, colistimethate,* lincomycin, clindamycin, tetracycline, oxytetracycline, rolitetracycline, macrolides, metronidazole
Psychotropics	Chlorpromazine,* lithium carbonate,* amitriptyline, droperidol, haloperidol, imipramine
Antirheumatics	d-Penicillamine, colchicine, chloroquine
Cardiovascular	Quinidine,* procainamide,* β-blockers (propranolol, oxprenlol, practolol, pindolol, sotalol), lidocaine, trimethaphan; magnesium; calcium channel blockers (verapamil)
Local anesthetics	Lidocaine,* procaine*
Analgesics	Narcotics (morphine, hydrmorphone, codeine, Pantopon, meperidine)
Endocrine	Thyroid replacement*

Eye drops Timolol,* echothiophate

Others Amantadine, diphenhydramine, emetine, diuretics, muscle relaxants, central nervous system depressants, respiratory depresants, sedatives, procaine,* phenothiazines

Neuromuscular blocking agents Tubocurarine, pancuronium, rocuronium, gallamine, dimthyl tubocurarine, succinylcholine, decamethonium

Note: See also discussion on eMedicine from WebMD by William D. Goldenberg, MD, available at: http://emedicine.medscape.com/article/793136-overview#a1.

Case reports implicate drugs in exacerbations of myasthenia gravis.

*

■■ MULTIPLE SCLEROSIS

Clinical Features

Multiple sclerosis (MS) is caused by multifocal areas of CNS demyelination and manifests with motor, sensory, visual, and/or cerebellar dysfunction. Types of MS include relapsing and remitting (90%), relapsing and progressive, and chronically progressive. Lhermitte sign is commonly experienced and is described as an electric shock sensation, vibration, or pain radiating down the back and often into the arms or legs from neck flexion. Physical examination may show decreased strength, increased tone, hyperreflexia, clonus, decrease in vibratory sense and joint proprioception, a positive Babinski reflex, and reduced pain and temperature sense. Optic neuritis with loss of central vision is a presenting symptom in 30% of cases and may cause an afferent papillary defect (Marcus–Gunn pupil). Bilateral internuclear ophthalmoplegia causes abnormal eye adduction and

147_Cydulka_Ch147.indd 802 21/03/17 4:29 PM

Source page 822

horizontal nystagmus, and its presence is strongly suggestive of MS. Cognitive and emotional problems are common as the disease progresses.

Diagnosis and Differential

A diagnosis of MS is suspected when a patient experiences two or more prolonged episodes of neurologic dysfunction that suggest white matter pathology in more than one location in the brain and/or spinal cord. MRI shows multiple discrete lesions in the supratentorial white matter, homogeneous borders around the ventricles, or infratentorial or spinal cord lesions. CSF protein and γ-globulin levels are often elevated. Other conditions that mimic symptoms of MS include systemic lupus erythematosus, Lyme disease, neurosyphilis, HIV disease, and Guillain-Barré syndrome.

Emergency Department Care and Disposition

1. Consult with a neurologist for assistance in managing a patient with MS

and suspected exacerbation. Refer patients with new symptoms suspicious for MS to a neurologist for further evaluation.

2. High-dose methylprednisolone therapy, 500 to 1000 mg per day, shortens the duration of exacerbations.

3. Consider testing for urinary tract infections in the appropriate clinical setting, especially for patients with known bladder dysfunction and postvoid residuals >100 mL.

4. Evaluate for potential sources of infection when fever is present. Fever reduction measures can help to minimize the muscle weakness that is associated with temperature elevations.

5. Consider inpatient admission for patients with significant exacerbations who may require IV steroids or antibiotics.

■■ LAMBERT-EATON MYASTHENIC SYNDROME

Lambert-Eaton myasthenic syndrome is an autoimmune disorder with fluctuating proximal limb muscle weakness and fatigue. It is classically described in older men with lung cancer. Strength improves with sustained or repeated exercise (Lambert sign). Patients complain of myalgias, stiffness, paresthesias, metallic tastes, and autonomic symptoms. Eye movements are unaffected. Electromyography is abnormal, and serum tests are specific for antibodies to voltage-gated calcium channels. Treatment is mostly supportive, but neuromuscular transmission can also be enhanced with 3,4-diaminopyridine. Immunosuppressive drugs may reduce symptom severity.

■■ PARKINSON'S DISEASE

Clinical Features

Parkinson's disease is a progressive extrapyramidal movement disorder with four classic signs: resting tremor, cogwheel rigidity, bradykinesia or akinesia, and impaired posture and equilibrium. Initially, many patients develop a unilateral resting upper extremity tremor, referred to as a "pill rolling" tremor, which improves with intentional movement.

147_Cydulka_Ch147.indd 803 21/03/17 4:29 PM

Source page 823

Diagnosis and Differential

The diagnosis of Parkinson's disease is clinical and based on the presence of the four classic signs. Parkinsonism can secondarily develop from exposure to illicit drugs, toxins, neuroleptic medications, hydrocephalus, head trauma, and other rare neurologic conditions. Drug-induced Parkinson's disease most commonly presents with akinesia. No radiographic imaging or laboratory study is pathognomonic for making the diagnosis.

Emergency Department Care and Disposition

Most patients with known Parkinson's disease are treated with medications that increase central dopamine, have anticholinergic properties, or act as central dopamine receptor agonists. Medication side effects may include psychiatric or sleep disturbances, anorexia, nausea, vomiting, cardiac dysrhythmias, orthostatic hypotension, dyskinesias, and dystonia. Patients being treated with medication for Parkinson's disease who present with new motor or psychiatric disturbances or decreased drug efficacy, a "drug holiday" for 1 week can be considered in consultation with a neurologist.

■■ POLIOMYELITIS AND POSTPOLIO SYNDROME

Clinical Features

Poliomyelitis is caused by an enterovirus infection that can lead to paralysis by motor neuron destruction and muscle denervation in some patients. Most acutely infected patients are asymptomatic or have a mild viral syndrome without paralysis. Major illness occurs in only 1% to 2% of infections and most commonly involves the spinal cord, resulting in asymmetric proximal limb weakness and flaccidity, absent tendon reflexes, and fasciculations. Maximal paralysis occurs within 5 days and is followed by muscle atrophy. Paralysis will resolve within 1 year in most patients. Other sequelae include autonomic dysfunction, speech and swallowing dysfunction, and encephalitis. Polio was eradicated from the United States in 1979 but is still endemic in Nigeria, Pakistan, Afghanistan, and India.

Postpolio syndrome is the recurrence of motor symptoms after a latent period of several decades in a patient who has a history of acute paralytic poliomyelitis. Symptoms may include muscle fatigue, joint pain, or weakness of new and previously affected muscle groups. Patients may have new bulbar, respiratory, or sleep difficulties.

Diagnosis and Differential

Consider an acute polio virus infection when an at-risk patient from an endemic area presents with an acute febrile illness, aseptic meningitis, and asymmetric flaccid paralysis with loss of deep tendon reflexes and preserved sensation. The diagnosis can be made by evaluating cerebrospinal fluid (CSF), which will demonstrate a pleocytosis and positive viral cultures for poliovirus, or from throat or rectal swab testing.

Other diseases with similar neurologic symptoms include Guillain-Barré syndrome, peripheral neuropathies associated with other disorders (e.g., mononucleosis, Lyme disease, or porphyria), abnormal electrolyte

147_Cydulka_Ch147.indd 804 21/03/17 4:29 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 820–825.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 148: Central Nervous System and Spinal Infections

Source pages 826–833 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 826–833

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Central Nervous System and Spinal Infections**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 148: Central Nervous System and Spinal Infections 807 symptoms as bacterial meningitis, such as fever, stiff neck, or headache. [Source pages 826–833]

- Additionally, patients diagnosed with encephalitis typically present with altered mental status, cognitive deficits, psychiatric symptoms, or seizures. [Source pages 826–833]

- Diagnosis and Differential A promptly performed lumbar puncture (LP) to obtain cerebrospinal fluid (CSF) is the appropriate diagnostic procedure for patients with suspected meningitis or encephalitis. [Source pages 826–833]

- Consider a CT scan of the brain before the procedure when altered mental status, new onset seizures, immunocompromise, focal neurologic signs, or papilledema are present. [Source pages 826–833]

- The goal for imaging in this circumstance is to identify possible contraindications to LP such as an occult mass or signs of brain shift or herniation (see Fig. [Source pages 826–833]

Clinical reasoning structure

- **Problem representation:** Central Nervous System and Spinal Infections; source pages 826–833.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Central Nervous System and Spinal Infections**.

- Source-page range: **826–833**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 826

CHAPTER 148: Central Nervous System and Spinal Infections 807

symptoms as bacterial meningitis, such as fever, stiff neck, or headache. Additionally, patients diagnosed with encephalitis typically present with altered mental status, cognitive deficits, psychiatric symptoms, or seizures.

Diagnosis and Differential

A promptly performed lumbar puncture (LP) to obtain cerebrospinal fluid (CSF) is the appropriate diagnostic procedure for patients with suspected meningitis or encephalitis. Consider a CT scan of the brain before the procedure when altered mental status, new onset seizures, immunocompromise, focal neurologic signs, or papilledema are present. The goal for imaging in this circumstance is to identify possible contraindications to LP such as an occult mass or signs of brain shift or herniation (see Fig. 148-1).

Send CSF for cell count and differential, protein and glucose levels, and a Gram's stain and bacterial culture. Consider additional CSF studies for immunocompromised patients or if a specific CNS infection is suspected based on the initial CSF testing results. HSV or enterovirus PCR, bacterial antigen testing, or specialized fungal testing can be considered as additional testing in appropriate clinical situations. Consider other laboratory studies such as a complete blood count (CBC), serum glucose and electrolytes, blood urea nitrogen (BUN), and creatinine to assess for other potential diagnoses. A blood culture may be helpful to identify bacterial pathogens, particularly when empiric antibiotics are given prior to lumbar puncture. See Table 148-1 for classically described findings in bacterial, viral, fungal, and neoplastic meningitis, although several studies have demonstrated that no single laboratory finding can accurately categorize the cause of CSF pleocytosis in all patients. Although not typically part of an

ED workup, characteristic findings on MRI in the medial temporal and inferior frontal lobes are suggestive of HSV infection.

Figure 148-1. Cranial imaging to evaluate potential contraindications for lumbar puncture should be focused on identifying signs of a focal space-occupying lesion, evidence of brain shift, and/or signs of severe diffuse brain swelling: (A) normal brain, (B) meningitis-associated cerebral infarct causing pronounced brain shift, and (C) diffuse brain swelling associated with severe infection. Initial lumbar puncture should not be done when CT findings of significant brain shift are found, and empiric therapy for meningitis should be continued in such patients. Reproduced with permission from Fitch MT, van de Beek D. Emergency diagnosis and treatment of adult meningitis. Lancet Infect Dis. 2007 Mar;7(3):191–200.

148_Cydulka_Ch148.indd 807 21/03/17 4:35 PM

Source page 827

148_Cydulka_Ch148.indd 808

808

SECTION 15: Neurology

Table 148-1 Cerebrospinal Fluid (CSF) Diagnostic Evaluation

Opening Pressure (<170 mm H2O)*	Cell Count (<5 WBC, Clear)	Color (Clear)	Gram Stain (Negative)	Protein (>40 mg/dL)	Glucose (<50 mg/dL)	Cytology (Negative)
Bacterial Elevated	Cloudy, turbid Positive (60%–80% before antibiotic, 7%–41% after antibiotic)		>1000–2000/mm3 WBC, neutrophilic predominance, >80% PMN	<40 mg/dL, CSF/ blood glucose ratio <0.3–0.4	>200 mg/dL Negative	
Viral Normal	Clear or bloody Negative		<300/mm3 WBC, lymphocytic predominance, <20% PMN	Normal <200 mg/dL	Negative	
Fungal Normal to elevated	Clear or cloudy Negative		<500/mm3	Normal to slightly low	>200 mg/dL Negative	
Neoplastic Normal	Clear or cloudy Negative		<300/mm3	Normal to slightly low	>200 mg/dL Positive	

Normal values and findings are in parentheses.

*

Abbreviation: PMN, polymorphonuclear lymphocyte.

21/03/17 4:35 PM

Source page 828

CHAPTER 148: Central Nervous System and Spinal Infections 809

The differential diagnosis for patients with findings concerning for meningitis or encephalitis may include subarachnoid hemorrhage, meningeal neoplasm, brain abscess, cerebral toxoplasmosis, and other systemic viral or bacterial infections.

Emergency Department Care and Disposition

1. When bacterial meningitis is a likely diagnosis, begin empiric antibiotic

therapy based on patient risk factors and most likely causative pathogens. Start antibiotics as soon as possible after a promptly performed LP, or immediately after blood cultures are drawn if neuroimaging or other diagnostic testing will be performed.

2. When administering empiric antibiotics for suspected bacterial meningitis, adults less than 50 years old should receive a third-generation cephalosporin such as ceftriaxone 2 g IV plus vancomycin 15 mg/kg IV. Patients at risk for Listeria monocytogenes infection (e.g., age 50 years or older, pregnant women, alcoholics, immunocompromised patients) should additionally receive ampicillin 2 g IV.

3. For patients with bacterial meningitis, adjunctive corticosteroids treatment started before or at the same time as the first dose of antibiotics may decrease mortality and neurologic sequelae for some subsets of patients. Consider dexamethasone 10 mg IV every 6 hours for 4 days in adults or 0.15 mg/kg IV every 6 hours for 4 days in children 3 months and older when it is initiated before or at the same time antibiotics are given.

4. Patients with suspected encephalitis should be started on acyclovir 10 mg/kg IV to cover for possible HSV or herpes zoster virus infections. Cytomegalovirus encephalitis can be treated with ganciclovir 5 mg/kg IV. Most other causes of viral encephalitis have no specific antimicrobial treatment that will impact patient outcomes, and recommended treatment in these circumstances is supportive care.

5. Patients with suspected bacterial meningitis or encephalitis should be admitted to the hospital for intravenous antimicrobial agents and further care.

■■ BRAIN ABSCESS

Clinical Features

A brain abscess is caused by a bacterial infection of the brain parenchyma and is composed of a central purulent cavity ringed by a layer of granulation tissue and an outer fibrous capsule. Presenting signs and symptoms are often nonspecific and may include headache, neck stiffness, fever, vomiting, confusion, or changes in mental status. Patient symptoms may reflect the origin of the infection (e.g., ear or sinus pain), and symptoms may be present for 1 to 8 weeks before the diagnosis is made.

Diagnosis and Differential

Neuroimaging is needed to make the diagnosis of a brain abscess. A CT scan of the head without contrast may identify abnormalities or evidence of surrounding edema, but a CT scan with contrast is more likely to accurately identify an abscess with thin rings of enhancement surrounding a

148_Cydulka_Ch148.indd 809 21/03/17 4:35 PM

Source page 829

low-density center. MRI is another sensitive test for making the diagnosis. Avoid lumbar puncture if a brain abscess or other mass lesion is suspected or identified. Routine laboratory studies are usually nonspecific. When possible, obtain blood cultures or cultures of other sites of infection to guide future management.

The differential diagnosis includes cerebrovascular disease, meningitis, brain neoplasm, subacute cerebral hemorrhage, and other focal brain infections, such as toxoplasmosis.

Emergency Department Care and Disposition

1. Early empiric antibiotic therapy is recommended based on the likely source of infection (see Table 148-2).
2. Consult with a neurosurgeon for admission to the hospital as many patients will require surgery for diagnosis, pathogen identification, and definitive treatment.

■■ EPIDURAL ABSCESS

Clinical Features

Spinal epidural abscess is a rare infection consisting of pyogenic material that accumulates in the epidural space. Up to half of these infections originate from hematogenous spread of bacteria from soft tissue, urine, or

Table 148-2 Guidelines for Empiric Treatment of Brain Abscess Based on Presumed Source

Presumed Source	Primary Empiric Therapy	Alternative Therapy
Otogenic	Cefotaxime 2 g IV every 4–6 h or ceftriaxone 2 g IV every 12 h PLUS metronidazole 500 mg IV every 8 h	Piperacillin/tazobactam 4.5 g IV every 6 h PLUS metronidazole 500 mg IV every 8 h
Odontogenic	Penicillin G 4 million units IV every 4 h PLUS metronidazole 500 mg IV every 6 h	Ceftriaxone 2 g IV every 12 h
Sinogenic	Cefotaxime 2 g IV every 6 h or ceftriaxone 2 g IV every 12 h PLUS metronidazole 500 mg IV every 8 h	No recommendation
Penetrating trauma	Cefotaxime 2 g IV every 6 h or ceftriaxone 2 g IV every 12 h PLUS metronidazole 500 mg IV every 8 h ± rifampin 10 mg/kg every 24 h	No recommendation
After neurosurgical procedure	Vancomycin loading dose 25–30 mg/kg IV loading dose or linezolid 600 mg IV every 12 h	Can substitute linezolid instead of

600 mg IV every 12 h PLUS ceftazi- vancomycin. Can substitute

dime 2 g IV every 8 h ± rifampin meropenem 2 g IV every 8 h OR

10 mg/kg every 24 h piperacillin/tazobactam 4.5 g IV

every 6 h OR cefepime 2 g IV

every 8 h for ceftazidime.

Unknown source Cefotaxime 2 g IV every 6 h PLUS No recommendation

metronidazole 500 mg IV every 6 h

Note: See also http://www.hopkins-abxguide.org; accessed June 18, 2014.

148_Cydulka_Ch148.indd 810 21/03/17 4:35 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 826–833.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 149: Eye Emergencies

Source pages 834–845 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 834–845

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Eye Emergencies**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 149: Eye Emergencies 815 Table 149-1 Selected Ophthalmic Medications Used in the Emergency Department (Continued) Drug Indication Dose Ofloxacin Conjunctivitis, corneal Conjunctivitis: 1 to 2 drops every abrasions if a contact lens 2–4 h for 2 days, then 1 to 2 drops wearer four times a day for 5 days. [Source pages 834–845]

- Corneal ulcer: 1 to 2 drops every 30 min while awake and 1 to 2 drops every 4–6 h after retiring for 2 days, then 1 to 2 drops every hour while awake for 5–7 days, then 1 to 2 drops four times a day for 2 days or until treatment completion Trifluridine Herpes simplex keratitis 1 drop every 2 h Ketorolac Allergic conjunctivitis, 1 drop four times/day for 3–4 days corneal abrasions, ultraviolet keratitis Prednisolone Allergic conjunctivitis, 2 drops four times a day acetate† uveitis * Agents that affect pupillary dilation or serve as a conjunctival decongestant should be avoided in patients with glaucoma. [Source pages 834–845]

- † Should be prescribed ONLY in conjunction with a specific recommendation from ophthalmologist. [Source pages 834–845]

- The lens should be discarded and not replaced until the infection has completely resolved. [Source pages 834–845]

- A severe purulent discharge with a hyperacute onset (within 12 to 24 hours) should prompt an emergent ophthalmology consultation for an aggressive workup for possible gonococcal conjunctivitis. [Source pages 834–845]

Clinical reasoning structure

- **Problem representation:** Eye Emergencies; source pages 834–845.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Eye Emergencies**.

- Source-page range: **834–845**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 834

CHAPTER 149: Eye Emergencies 815

Table 149-1 Selected Ophthalmic Medications Used in the Emergency

Department (Continued)

Drug Indication Dose

Ofloxacin Conjunctivitis, corneal Conjunctivitis: 1 to 2 drops every

abrasions if a contact lens 2–4 h for 2 days, then 1 to 2 drops

wearer four times a day for 5 days. Corneal

ulcer: 1 to 2 drops every 30 min

while awake and 1 to 2 drops every

4–6 h after retiring for 2 days, then

1 to 2 drops every hour while awake

for 5–7 days, then 1 to 2 drops

four times a day for 2 days or until

treatment completion

Trifluridine Herpes simplex keratitis 1 drop every 2 h

Ketorolac Allergic conjunctivitis, 1 drop four times/day for 3–4 days

corneal abrasions,

ultraviolet keratitis

Prednisolone Allergic conjunctivitis, 2 drops four times a day

acetate† uveitis

*

Agents that affect pupillary dilation or serve as a conjunctival decongestant should be avoided in

patients with glaucoma.

†

Should be prescribed ONLY in conjunction with a specific recommendation from ophthalmologist.

such as ciprofloxacin or tobramycin. The lens should be discarded and

not replaced until the infection has completely resolved.

A severe purulent discharge with a hyperacute onset (within 12 to

24 hours) should prompt an emergent ophthalmology consultation for an aggressive workup for possible gonococcal conjunctivitis. Neisseria gonorrhoeae infections may be confirmed by Gram stain (gram-negative intracellular diplococci). Emergency department (ED) care for N. gonorrhoeae infections include culture, parenteral ceftriaxone, and saline solution irrigation to remove the discharge.

If bacterial conjunctivitis is present in a neonate, suspect a sexually transmitted disease (STD) etiology. These are sight-threatening conditions which can be difficult to diagnose, and an emergent ophthalmology evaluation in the ED is warranted. If gonorrhea is suspected, an inpatient workup for disseminated disease is indicated. If chlamydia is suspected, then associated pneumonia must be ruled out prior to discharge. Herpes conjunctivitis is also a possibility. (See Herpes Simplex Virus.)

Viral Conjunctivitis

Viral conjunctivitis presents as watery discharge, chemosis, and conjunctival inflammation. It is often associated with viral respiratory symptoms and occasionally preauricular lymphadenopathy. Fluorescein staining should be done as in bacterial conjunctivitis, specifically in this case to rule out dendritic lesions. Treatment consists of cool compresses four times daily, naphazoline/pheniramine 0.025%/0.3% one drop three times daily, as needed, for conjunctival congestion or itching, artificial tears five or six times a day, and ophthalmology follow-up in 7 to 14 days. If a clear distinction between viral and bacterial etiologies cannot be made, consideration

149_Cydulka_Ch149.indd 815 21/03/17 4:45 PM

Source page 835

should be made to add topical antibiotics (Table 149-1) until reexamination by an ophthalmologist; however, routine use of antibiotics is discouraged. All the cases of viral conjunctivitis are extremely contagious and appropriate transmission precautions must be taken.

Allergic Conjunctivitis

Allergic conjunctivitis presents as pruritus, watery discharge, and chemosis with a history of allergies. There should be no lesions with fluorescein staining and no preauricular nodes. Conjunctival papillae are seen on slit lamp examination. Treatment consists of elimination of the inciting agent, cool compresses four times daily, artificial tears five or six times daily, and naphazoline/pheniramine 0.025%/0.3% one drop four times daily. Severe cases may require a mild topical steroid, but steroids should be only administered in consultation with an ophthalmologist.

Herpes Simplex Virus

Herpes simplex virus (HSV) infection may involve the eyelids, conjunctiva, or cornea. Corneal lesions stain brightly with fluorescein and can appear as a classic “dendrite” (a linear branching, epithelial defect with terminal bulbs) or as a “geographic ulcer” (an amoeba-like ulceration with dendrites at the edges) (Figure 149-1), but the presentation can be very variable and subtle. Corneal sensation can be decreased and should be checked. It is essential that HSV infection should not be confused with conjunctivitis, hence the necessity of a slit lamp fluorescein examination in these patients. If HSV infection is suspected, obtain an ophthalmology consult. If the outbreak involves only the eyelids, acyclovir 800 mg PO five times daily

Figure 149-1. Herpes simplex corneal dendrite in an infant seen with fluorescein staining. Used with permission from Allen R. Katz, Department of Ophthalmology, University of Nebraska Medical Center.

149_Cydulka_Ch149.indd 816 21/03/17 4:45 PM

Source page 836

for 7 to 10 days should be prescribed. If the conjunctiva is involved, prescribe trifluorothymidine 1% drops nine times daily. In addition, erythromycin ophthalmic 0.5% ointment twice daily and warm soaks three times daily to skin lesions can help prevent secondary bacterial infections. If corneal or deeper structure involvement is present, then antibiotic choice should be determined by the consulting ophthalmologist. Topical steroids are to be strictly avoided in all cases of HSV infection. Close follow-up from the ED and strong precautions are essential.

When conjunctivitis is present in the neonate, especially with mucocutaneous lesions, HSV also should be suspected, even in the absence of maternal infection. As with other causes of neonatal conjunctivitis, an emergent ophthalmology consult at the bedside is warranted. If HSV is diagnosed or strongly suspected, treat with acyclovir 20 mg/kg IV every 8 hours and topical antivirals (1% trifluridine, 0.1% iododeoxyuridine, or 3% vidarabine). Further workup for sepsis is indicated. Steroid drops should not be given.

Herpes Zoster Ophthalmicus

Shingles in a trigeminal distribution with ocular involvement is termed herpes zoster ophthalmicus (HZO). The presence or eventual development of HZO should be suspected in any patient whose shingles involve the tip of the nose (Hutchinson’s sign). Photophobia and pain secondary to iritis are often present. Slit lamp examination may show a “pseudodendrite,” a poorly staining mucus plaque without epithelial erosion. Acyclovir 800 mg PO five times a day, famciclovir 500 mg three times daily, or valacy-

clovir 1000 mg three times daily for 7 to 10 days should be prescribed if the skin lesions have been present less than 7 days. In addition, erythromycin 2% ointment and warm compresses should be applied to skin lesions. Ocular involvement requires erythromycin 0.5% ophthalmic ointment to the eye twice daily. For comfort, oral opioid analgesia, cycloplegic agents (cyclopentolate 1% one drop three times daily), and cool compresses are helpful. If iritis is present, prednisolone acetate 1% one drop every 1 to 6 hours is effective. However, because topical steroid use in patients with herpes simplex keratoconjunctivitis may be catastrophic, it is imperative that there should be no corneal lesions present on slit lamp examination before topical steroids are used. In severe cases, admission and acyclovir IV may be required. For this reason, all the cases of suspected HZO require ophthalmology consultation. All patients younger than 40 years with HZO should undergo an outpatient medical evaluation for a possible immunocompromised state.

Corneal Ulcer

A corneal ulcer is a serious infection of the corneal stroma caused by bacteria including Pseudomonas aeruginosa, viruses including herpes simplex and varicella zoster, and fungi. Immunocompromised patients are at risk for fungal or viral etiologies. It is commonly associated with trauma, especially in patients who use extended-wear contact lenses and those who wear lenses while sleeping. Ulcers may cause pain, redness, tearing, photophobia, and blurry vision. Slit lamp examination shows a staining corneal defect with a surrounding white hazy infiltrate, and associated iritis and

149_Cydulka_Ch149.indd 817 21/03/17 4:45 PM

Source page 837

sometimes a hypopyon. Topical ofloxacin 0.3% or ciprofloxacin 0.3% ophthalmic solution should be administered, one drop in the affected eye each hour. Antifungal or antiviral antibiotics may be given to immunocompromised patients in consultation with an ophthalmologist. Topical cycloplegics, such as cyclopentolate 1% one drop three times daily, aid in pain relief. Eye patching is strictly contraindicated because of the risk of worsening a potential Pseudomonas infection. Steroid drops should be avoided unless an ophthalmologist advises administration. It would be ideal for the patient to be seen by an ophthalmologist in the ED in order to culture the ulcer prior to starting antibiotics. However, if this is not possible, they should see the patient within 12 to 24 hours.

Iritis

Iritis is inflammation of the anterior uveal tract (iris and ciliary body) that

has many causes (Table 149-2). It presents with red eye, photophobia, and decreased vision. A hallmark physical examination sign is consensual pain (pain in the affected eye when light is shined in the nonaffected eye). Slit lamp examination reveals WBCs in the anterior chamber, usually with associated flare, and a hypopyon can eventually occur. Perform an exam with fluorescein dye to detect corneal abrasions, ulcers, or dendritic lesions, and measure intraocular pressure (IOP), which can be elevated. Once iritis is diagnosed, undertake an appropriate ED workup for a systemic etiology and treat them, if found. Symptomatic treatment with tropicamide (Table 149-1) is helpful. Ophthalmology consultation is appropriate with follow-up in 24 to 48 hours. Steroid drops can be of value, but should only be given if directed by the ophthalmologist.

Endophthalmitis

Endophthalmitis is an infection involving the deep structures of the globe. Patients present with pain and visual loss. It is often seen as a complication of globe violation either from trauma or ocular surgery. Hematologic spread is possible. Pathogens include Staphylococcus, Streptococcus, Haemophilus, and Bacillus species. Emergency ophthalmology consultation

Table 149-2 Differential Diagnosis of Iritis

Systemic diseases Malignancies

Juvenile rheumatoid arthritis Leukemia

Ankylosing spondylitis Lymphoma

Ulcerative colitis Malignant melanoma

Reiter’s syndrome Trauma/environmental

Behçet’s syndrome Corneal foreign body

Sarcoidosis Posttraumatic (blunt trauma)

Infectious Ultraviolet keratitis

Tuberculosis

Lyme disease

Herpes simplex

Toxoplasmosis

Varicella zoster

Syphilis

Adenovirus

149_Cydulka_Ch149.indd 818 21/03/17 4:45 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 834–845.
- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 150: Face and Jaw Emergencies

Source pages 846–851 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 846–851

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Face and Jaw Emergencies**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 150: Face and Jaw Emergencies 827 Table 150-2 Antibiotic Doses for Facial Infections Antibiotic Dosage Topical Antibiotics Mupirocin ointment 2%, apply to lesions three times per day Retapamulin ointment 1%, apply to lesions two times per day Oral Antibiotics Amoxicillin/clavulanate 875/125 mg two times per day Cephalexin 500 mg four times per day Clindamycin 300–450mg four times per day Dicloxacillin 500 mg four times per day Doxycycline 100 mg two times per day Metronidazole 500 mg every 8 hours Minocycline 100 mg two times per day Penicillin V 500 mg four times per day Trimethoprim-sulfamethoxazole 1–2 double-strength tablets two times per day Parenteral Antibiotics Ampicillin-sulbactam 1.5–3.0 g every 6 hours Clindamycin 600 mg every 8 hours Cefazolin 1 g every 8 hours Metronidazole 1 g loading dose, then 500 mg every 8 hours Nafcillin 1–2 g every 4 hours Penicillin G 2–3 million units every 6 hours Vancomycin 1 g every 12 hours for patients with signs of systemic illness, failed outpatient therapy, or significant comorbidities. [Source pages 846–851]

- Erysipelas Erysipelas is a superficial form of cellulitis, involving the epidermis, upper levels of the dermis, and the lymphatic system. [Source pages 846–851]

- aureus is a rare etiology but is associated with bullous disease, trauma, or the presence of a foreign body. [Source pages 846–851]

- Clinical features include a red, raised, puffy appearance with a sharply defined, palpable border. [Source pages 846–851]

- Most patients are treated with oral antibiotics, but hospitalization and parenteral antibiotics should be considered for patients who fail outpatient therapy, are immunocompro- mised, or exhibit evidence of systemic illness. [Source pages 846–851]

Clinical reasoning structure

- **Problem representation:** Face and Jaw Emergencies; source pages 846–851.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Face and Jaw Emergencies**.

- Source-page range: **846–851**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 846

CHAPTER 150: Face and Jaw Emergencies 827

Table 150-2 Antibiotic Doses for Facial Infections

Antibiotic Dosage

Topical Antibiotics

Mupirocin ointment 2%, apply to lesions three times per day

Retapamulin ointment 1%, apply to lesions two times per day

Oral Antibiotics

Amoxicillin/clavulanate 875/125 mg two times per day

Cephalexin 500 mg four times per day

Clindamycin 300–450mg four times per day

Dicloxacillin 500 mg four times per day

Doxycycline 100 mg two times per day

Metronidazole 500 mg every 8 hours

Minocycline 100 mg two times per day

Penicillin V 500 mg four times per day

Trimethoprim-sulfamethoxazole 1–2 double-strength tablets two times per day

Parenteral Antibiotics

Ampicillin-sulbactam 1.5–3.0 g every 6 hours

Clindamycin 600 mg every 8 hours

Cefazolin 1 g every 8 hours

Metronidazole 1 g loading dose, then 500 mg every 8 hours

Nafcillin 1–2 g every 4 hours

Penicillin G 2–3 million units every 6 hours

Vancomycin 1 g every 12 hours

for patients with signs of systemic illness, failed outpatient therapy, or sig-

nificant comorbidities.

Erysipelas

Erysipelas is a superficial form of cellulitis, involving the epidermis, upper levels of the dermis, and the lymphatic system. Most cases are caused by S. pyogenes; S. aureus is a rare etiology but is associated with bullous disease, trauma, or the presence of a foreign body. Clinical features include a red, raised, puffy appearance with a sharply defined, palpable border. The diagnosis is clinical. Most patients are treated with oral antibiotics, but hospitalization and parenteral antibiotics should be considered for patients who fail outpatient therapy, are immunocompromised, or exhibit evidence of systemic illness.

Impetigo

Impetigo is a superficial epidermal infection that can be divided into bullous and nonbullous presentations. Bullous impetigo is caused by S. aureus and nonbullous impetigo is caused by S. aureus and S. pyogenes. Clinical features of nonbullous impetigo include an erythematous rash with vesicles that break and form the characteristic amber crusts. Bullous impetigo presents as vesicles that enlarge to form bullae with clear yellow

150_Cydulka_Ch150.indd 827 21/03/17 5:01 PM

Source page 847

fluid. Topical therapy with mupirocin is appropriate for simple, nonbullous disease. Oral antibiotics are prescribed for more extensive or bullous lesions.

■■ SALIVARY GLAND DISORDERS

Viral Parotitis (Mumps)

Viral parotitis is an infection that can present with unilateral or bilateral swelling of the parotid glands. It is most commonly caused by a paramyxovirus. Clinical features may include a prodrome of fever, malaise, myalgias, and headache followed by parotid gland swelling. The gland is tense and painful, but lacks erythema and warmth. Pus cannot be expressed from Stensen's duct.

The diagnosis is clinical and treatment is supportive. Swelling may persist for 5 days. The patient is contagious for approximately 9 days after the onset of parotid swelling. Extrasalivary gland involvement includes orchitis in 20% to 30% of males and oophoritis in 5% of females. Other systemic complications include pancreatitis, aseptic meningitis, hearing loss, myocarditis, arthritis, hemolytic anemia, and thrombocytopenia. Consider hospitalization for patients with systemic complications.

Suppurative Parotitis

Suppurative parotitis is a serious bacterial infection that occurs in patients with diminished salivary flow. Retrograde transmission of bacteria leads to infection. Factors that lead to decreased salivary flow include recent anesthesia, dehydration, prematurity, advanced age, medications (e.g., diuretics, β-blockers, antihistamines, phenothiazines, and tricyclic antidepressants), and certain disorders (Sjögren's syndrome, diabetes, hypothyroidism, cystic fibrosis, and human immunodeficiency virus). Clinical features may include fever, trismus, erythema, and pain over the parotid gland. Pus may be expressed from Stensen's duct.

The diagnosis is clinical. Ultrasound or CT may be ordered if an abscess is suspected. Treatment should optimize salivary flow by using sialogogues such as lemon drops and stopping any medications that cause dry mouth. Oral antibiotics are appropriate for those tolerating oral intake and are without trismus. Hospitalization is appropriate for patients with signs of systemic illness, immunocompromise, inability to tolerate oral intake, or those that have failed outpatient therapy. Close follow-up should be arranged.

Sialolithiasis

Sialolithiasis is the development of stones in a stagnant salivary duct. Eighty percent of stones occur in the submandibular duct. Sialolithiasis is typically unilateral and presents with pain, swelling, and tenderness that may be exacerbated with eating. The diagnosis is clinical. A stone may be palpated within the duct and the gland is firm. Treatment includes analgesics, massage, sialogogues such as lemon drops, and antibiotics if a concurrent infection is suspected. Palpable stones may be milked from the duct. Persistent retained calculi may be removed by an otolaryngologist.

150_Cydulka_Ch150.indd 828 21/03/17 5:01 PM

Source page 848

Masticator Space Abscess

The masticator space consists of potential spaces bounded by the muscles of mastication. Infection is usually polymicrobial and is commonly associated with an odontogenic source. Clinical features include facial swelling, pain, erythema, and trismus. In advanced cases, signs of sepsis may be present. The diagnosis is made with contrast-enhanced CT scan. Because the masticator spaces ultimately communicate with tissue planes that extend into the mediastinum, early treatment is imperative. ED treatment includes stabilization, antibiotics, otolaryngology consult, and hospitalization.

■■ MANDIBLE DISORDERS

Temporomandibular Joint Dysfunction

The temporomandibular joint (TMJ) combines a hinge and a gliding action. Anatomic derangements or systemic disease can cause dysfunction of this joint. Clinical features include pain over the muscles of mastication or in the region of the TMJ and there may be a limited range of motion. The diagnosis is usually clinical. For patients with acute trauma, imaging with CT or panoramic x-ray may be warranted. Treatment for nontraumatic conditions consists of analgesics, soft diet, and referral to a dental specialist. An oral and maxillofacial surgeon manages fractures.

Trigeminal Neuralgia

Trigeminal neuralgia presents with facial pain in the distribution of the fifth cranial nerve. It is characterized by paroxysms of severe pain lasting seconds, with normal findings on physical exam. Carbamazepine, starting at 100 mg PO twice a day and then increasing dosage as needed, is an effective treatment.

Bell's Palsy

Bell's palsy is characterized by an acute onset of unilateral upper and lower facial paralysis. A detailed neurologic exam is useful in ruling out other conditions, such as stroke. Sparing of the forehead muscles on the affected side is suggestive of a central process. Examine the ear for presence of cranial herpes zoster. Treatment with steroids within the first 72 hours of symptom onset is thought to improve chances of a full recovery. The recommended dosing is 1 mg/kg of prednisone or equivalent, up to 60 mg/d, for 6 days followed by a 10-day taper. The use of antivirals is controversial and benefit modest at best. Ocular lubricants should be used to prevent dry eyes.

Dislocation of the Mandible

The mandible can be dislocated in an anterior, posterior, lateral, or superior position. Anterior dislocation is the most common. Patients with an acute jaw dislocation present with pain, difficulty swallowing, and malocclusion. In anterior dislocations, a history of extreme mouth opening is typical and there is difficulty with jaw movement. Other dislocations usually require significant trauma.

150_Cydulka_Ch150.indd 829 21/03/17 5:01 PM

Source page 849

The diagnosis of atraumatic anterior dislocations is clinical. Imaging with panoramic x-ray or CT is indicated for all other dislocations. Treatment of anterior dislocations without fracture is closed reduction and this is

made easier with analgesia, muscle relaxants, or procedural sedation. Reduction is most commonly done in a seated patient. The thumbs are padded with gauze, placed over the molars, and pressure is applied downward and backward. Patients with open or nonreducible dislocations, fractures, or nerve injury should be referred emergently to an oral and maxillofacial surgeon. After reduction, patients should be instructed to not open their mouth more than 2 cm for 2 weeks. Elective referral to an oral and maxillofacial surgeon is recommended.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 243, "Face and Jaw Emergencies," by Stephanie A. Lareau and Corey R. Heitz.

150_Cydulka_Ch150.indd 830 21/03/17 5:01 PM

References/provenance

- Source: *Tintinalli's Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 846–851.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 151: Ear, Nose, and Sinus Emergencies

Source pages 852–857 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 852–857

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Ear, Nose, and Sinus Emergencies**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 151: Ear, Nose, and Sinus Emergencies 833 treatment consists of pain control and warm compresses. [Source pages 852–857]

- Trauma to the Ear A hematoma can develop from any type of trauma to the ear. [Source pages 852–857]

- Improper treatment of ear hematomas can result in stimulation of the perichondrium and development of asymmetric cartilage formation. [Source pages 852–857]

- The resultant deformed auricle has been termed “cauliflower ear.” Immediate incision and drainage of the hematoma with a compressive dressing is necessary to prevent reac- cumulation of the hematoma. [Source pages 852–857]

- aureus should be given to immunocompromised patients. [Source pages 852–857]

Clinical reasoning structure

- **Problem representation:** Ear, Nose, and Sinus Emergencies; source pages 852–857.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Ear, Nose, and Sinus Emergencies**.

- Source-page range: **852–857**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 852

CHAPTER 151: Ear, Nose, and Sinus Emergencies 833

treatment consists of pain control and warm compresses. Antibiotics can be given for concomitant OM.

Trauma to the Ear

A hematoma can develop from any type of trauma to the ear. Improper treatment of ear hematomas can result in stimulation of the perichondrium and development of asymmetric cartilage formation. The resultant deformed auricle has been termed “cauliflower ear.” Immediate incision and drainage of the hematoma with a compressive dressing is necessary to prevent reaccumulation of the hematoma. Antibiotic coverage for P. aeruginosa and S. aureus should be given to immunocompromised patients.

Thermal injury to the auricle may be caused by excessive heat or cold. Superficial injury of either type is treated with cleaning, topical nonsulfa-containing antibiotic ointment, and a light dressing. Frostbite is treated with rapid rewarming by using saline soaked gauze at 38°C to 40°C. The rewarming process may be very painful and analgesics will be necessary. Any second- or third-degree burn requires immediate ENT or burn center consultation.

Foreign Bodies in the Ear

On examination, the foreign body is usually visualized and signs of infection or TM perforation should be sought. Live insects should be immobilized with 2% lidocaine solution distilled into the ear canal before removal. Foreign bodies may be removed with forceps and direct visualization or with the aid of a hooked probe or suction catheter. Irrigation is often useful for small objects; however, organic material may absorb water and swell. ENT consultation is required for cases of foreign body with TM perforation or if the object cannot be safely removed.

Cerumen Impaction

Cerumen impaction can cause symptoms of hearing loss, pressure, dizziness, tinnitus and otalgia. Impaction is often precipitated by the use of cotton swabs in the ear canal. Cerumen scoops/loops can be used to remove the impaction. If significant amounts of cerumen are in the canal then half-strength hydrogen peroxide, sodium bicarbonate, mineral oil, or carbamide peroxide otic can be instilled in the canal to soften the cerumen and help with removal. Additionally cerumen impactions can be removed with irrigation of the EAC using body-temperature fluid via an 18ga IV catheter or ear syringe making sure that the irrigation is directed to the EAC wall not the TM. Cerumen removal in the emergency department (ED) is often not required; asymptomatic or minimally symptomatic patients with cerumen impaction may be safely discharged home with a cerumen-softening otic and follow-up.

Tympanic Membrane Perforation

TM perforations can result from middle-ear infections, barotrauma, blunt/penetrating/acoustic trauma, and, rarely, lightning strikes. Acute pain and hearing loss are usually noted, with or without bloody otorrhea. Vertigo and tinnitus, when present, are usually transient. As most TM perforations heal spontaneously, antibiotics are not necessary unless there is persistent

151_Cydulka_Ch151.indd 833 21/03/17 5:58 PM

Source page 853

foreign material in the canal or middle ear. Patients with perforations from isolated blunt or noise trauma can be discharged with expedited specialty referral and should be instructed not to allow water to enter the ear canal.

Tinnitus

Tinnitus is the perception of sound without external stimuli. It may be constant, pulsatile, high or low pitched, hissing, clicking, or ringing in nature. Objective tinnitus can be heard by the examiner, whereas the more common subjective tinnitus cannot. Causes of tinnitus include vascular, mechanical, neurologic, Ménière's disease, and others. Common medications resulting in tinnitus include aspirin, nonsteroidal anti-inflammatory drugs, aminoglycosides, loop diuretics, and chemotherapeutics. If the patient's condition allows, potentially offending drugs should be stopped. Accurate diagnosis usually requires referral to an otolaryngologist.

Hearing Loss

Causes of sudden hearing loss are varied and may be idiopathic (most common), infectious, vascular or hematologic, metabolic, rheumatologic, or conductive. Other causes include Ménière's disease, Cogan's syndrome, acoustic neuroma, cochlear rupture, and ototoxic medications. Indicators of

poor prognosis include severe hearing loss on presentation and the presence of vertigo. If the cause is not readily determined by history and physical examination, otolaryngologic consultation is necessary.

■■ NASAL EMERGENCIES AND SINUSITIS

Epistaxis

Epistaxis is classified as anterior or posterior. Posterior epistaxis is suggested if an anterior source is not visualized, if bleeding occurs from both nares, or if blood is seen draining into the posterior pharynx after anterior sources have been controlled.

1. Perform a quick history to determine the duration and severity of the hemorrhage and the contributing factors (trauma, anticoagulant use, infection, bleeding diathesis, etc.).

2. Have the patient blow his or her nose to dislodge any clots. Instill 0.05% oxymetazoline 2 sprays/nostril or 0.25% phenylephrine 2 sprays/nostril.

3. Inspect for anterior bleeding using a good light source, a nasal speculum, and a suction catheter.

4. Apply direct external pressure to the cartilage just distal to the nasal bones for 15 minutes while leaning forward in the “sniffing” position. Reexamine the patient. Repeat once if necessary.

5. If bleeding continues, and an anterior source of bleeding is visualized, proceed to chemical cautery. If no source is identified, proceed to packing.

6. Chemical cautery with silver nitrate is the standard of care for ED treatment of anterior epistaxis. Insert cotton swabs or pledgets soaked in a 1:1 mixture of a 4% lidocaine and 0.05% oxymetazoline into the nasal cavity with bayonet forceps. After hemostasis is achieved, the mucosa is cauterized by firmly rolling the tip of a silver nitrate applicator over the

151_Cydulka_Ch151.indd 834 21/03/17 5:58 PM

Source page 854

area until it turns silvery-black. A small surrounding area also should be cauterized to control local arterioles. Overzealous use of cautery and bilateral septal cautery is discouraged because they may cause septal perforation and unintended local tissue necrosis.

7. Anterior nasal packing may be performed with thrombogenic foams and gels, commercial devices, or gauze. Dehydrated nasal sponges are available in several lengths to control anterior and posterior epistaxis. A film of antibiotic ointment is applied, then the sponge is rapidly inserted along the floor of the nasal cavity where it expands upon contact with

blood or secretions. Expansion can be hastened by rehydrating the sponge with sterile water from a catheter-tipped syringe. The longer sponges used to control posterior hemorrhages have been associated with some morbidity and should be used only for posterior epistaxis. Inflatable epistaxis tamponade balloons can also be used to control anterior and/or posterior hemorrhage, are easy to use, and are generally more comfortable than nasal sponges. Thrombogenic foams and gels are bioabsorbable, do not require removal, and are generally well tolerated. All nonabsorbable nasal packs should be removed in 2 to 3 days by an ENT physician. Tranexamic acid 5 mL (500 mg) applied to the nasal mucosa via a 15-cm cotton pledget can also be used to stop epistaxis. If packing or local cautery fails to control anterior bleeding, ENT consultation is necessary.

8. Posterior epistaxis may be treated with a dehydrated posterior sponge pack, as outlined above, or an inflatable balloon tamponade device. The balloon devices use independently inflatable anterior and posterior balloons to quickly control refractory epistaxis at these sites; the instructions for insertion are included in the balloon kit. To protect against potentially serious complications, all patients with posterior packs require ENT consultation for possible hospital admission. Posterior packs are removed 2 to 3 days after placement.

9. All patients with nasal packs should be started on antibiotic prophylaxis with amoxicillin/clavulanate 500/125 mg PO three times daily.

Complications of nasal packing include vasovagal syncope, dislodgment of the pack, recurrent bleeding, sinusitis, and toxic shock syndrome. Treatment of elevated blood pressure during an acute episode of epistaxis is generally not advised except in consultation with an otolaryngologist for cases of persistent epistaxis not controlled by the above measures.

Nasal Fractures

Nasal fracture is a clinical diagnosis suggested by the injury mechanism, swelling, tenderness, crepitance, gross deformity, and periorbital ecchymosis. Radiographic diagnosis usually is not necessary in the ED. Intermittent ice application, analgesics, and over-the-counter decongestants are the normal treatment. ENT follow-up within 6 to 10 days for reexamination and possible fracture reduction is prudent.

A fracture of the cribriform plate may violate the subarachnoid space and cause cerebrospinal fluid rhinorrhea. Symptoms may be delayed for several weeks. If a cribriform plate injury is suspected, CT and immediate neurosurgical consultation should be obtained.

151_Cydulka_Ch151.indd 835 21/03/17 5:58 PM

Source page 855

Nasal Septal Hematoma

A nasal septal hematoma should be suspected whenever there is trauma to the nose. A thorough evaluation of the nasal septum, with good lighting, should be performed. The septum should be visualized and palpated to assess for a hematoma. A nasal septal hematoma elevates the vascular perichondrium off of the cartilage thus disrupting the blood supply. It is important to remove the hematoma to return blood supply and prevent ischemic necrosis of the septum. If a nasal septal hematoma is identified then evacuation of the hematoma is necessary.

1. Place patient in “sniffing position” and examine the anterior nares with a nasal speculum under good lighting. Have suction, irrigation, and nasal packing materials ready.
2. Anesthetize the nasal mucosa by placing cotton pledgets soaked in 4% lidocaine into the affected nare. Infiltrative anesthesia of lidocaine without epinephrine can be administered if needed.
3. Visualize the hematoma and make a horizontal incision through the mucosa, avoid incising the cartilage.
4. Remove the hematoma.
5. Insert bilateral anterior nasal packing coated in antibiotic ointment.
6. Prescribe prophylactic antibiotics (amoxicillin/clavulanate 500/125 mg PO three times daily) if the nasal packing will be in for greater than 24 hours.
7. Patients should follow up with ENT.

Nasal Foreign Bodies

Nasal foreign bodies should be suspected in patients with unilateral nasal obstruction, foul rhinorrhea, or persistent unilateral epistaxis. After topical vasoconstriction with 0.05% oxymetazoline and possibly local anesthesia with 4% nebulized lidocaine, the foreign body should be removed under direct visualization. Tools for removal include forceps, suction catheters, hooked probes, and balloon-tipped catheters. Consult ENT if removal is unsuccessful.

Sinusitis and Rhinosinusitis

Sinusitis is inflammation of the mucosal lining of the paranasal sinuses (maxillary, frontal, ethmoid). Rhinosinusitis is sinusitis also involving the nasal cavity, almost always involves rhinitis, and is extremely common. It can be classified as acute, subacute, or chronic.

Symptoms include nasal congestion or blockage, facial pain or pressure,

hyposmia, nasal discharge, tooth pain, fever, and sinus pressure with head/body movement. There may be pain and tenderness with sinus percussion, mucosal swelling, facial swelling, and redness.

Complications include meningitis, cavernous sinus thrombosis, intracranial abscess and empyema, orbital cellulitis, and osteomyelitis. Patients with these deeper complications usually appear systemically ill or have focal neurologic findings.

The diagnosis of uncomplicated acute rhinosinusitis is clinical, and imaging is not necessary. CT scans are helpful in evaluating toxic patients and possible intracranial extension. Patients with chronic rhinosinusitis or

151_Cydulka_Ch151.indd 836 21/03/17 5:58 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 852–857.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 152: Oral and Dental Emergencies

Source pages 858–865 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 858–865

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Oral and Dental Emergencies**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 152: Oral and Dental Emergencies 839 be used with great caution in young infants due to its potential to depress the gag reflex. [Source pages 858–865]

- Adults and teens may experience pericoronitis (pain and local inflam- mation) with the eruption of the third molars (“wisdom teeth”). [Source pages 858–865]

- Prescribe penicillin VK 500 mg PO four times daily or clindamycin 300 mg PO four times daily, ibuprofen 400 to 800 mg PO three times daily (with or without hydrocodone 5 mg/acetaminophen 325 mg one to two tablets PO four times daily), and warm saline mouth rinses. [Source pages 858–865]

- Refer the patient to an oral surgeon or a general dentist for consideration of third molar extraction. [Source pages 858–865]

- Dental Caries and Pulpitis Dental caries are caused by bacteriogenic acid eroding through the enamel. [Source pages 858–865]

Clinical reasoning structure

- **Problem representation:** Oral and Dental Emergencies; source pages 858–865.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Oral and Dental Emergencies**.

- Source-page range: **858–865**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 858

CHAPTER 152: Oral and Dental Emergencies 839

be used with great caution in young infants due to its potential to depress the gag reflex.

Adults and teens may experience pericoronitis (pain and local inflammation) with the eruption of the third molars (“wisdom teeth”). Prescribe penicillin VK 500 mg PO four times daily or clindamycin 300 mg PO four times daily, ibuprofen 400 to 800 mg PO three times daily (with or without hydrocodone 5 mg/acetaminophen 325 mg one to two tablets PO four times daily), and warm saline mouth rinses. Refer the patient to an oral surgeon or a general dentist for consideration of third molar extraction.

Dental Caries and Pulpitis

Dental caries are caused by bacteriogenic acid eroding through the enamel. Examination may reveal a grossly decayed tooth, although occasionally there is no visible pathology—in these cases, localization may be accomplished by percussing individual teeth with a metallic object. If dental caries are not treated, pulpitis is the result. Reversible pulpitis is characterized by sudden, transient pain lasting seconds, often triggered by heat or cold. In contrast irreversible pulpitis pain lasts minutes to hours. Although antibiotics such as penicillin VK 500 mg PO four times daily or clindamycin 300 mg PO four times daily are commonly prescribed for pulpitis, their efficacy is controversial. Treat with ibuprofen 400 to 800 mg PO three times daily, hydrocodone 5 mg/acetaminophen 325 mg one to two tablets PO four times daily, and warm saline mouth rinses. A dental block may also be performed for short-term pain relief. Refer to a dentist for definitive management. If an abscess is present, antibiotics and incision and drainage should be considered.

Facial Space Infections

Odontogenic infections can spread readily to the facial spaces. Therefore,

it is imperative to exclude deep-space involvement whenever a dental infection is encountered. Ludwig's angina is a cellulitis involving both submandibular spaces and the sublingual space that can spread to the neck and mediastinum, causing airway compromise, overwhelming infection, and even death. If dental infections spread to the infraorbital space, a cavernous sinus thrombosis may result. This condition may present with limitation of lateral gaze, meningeal signs, sepsis, and coma. Administer intravenous antibiotics and obtain emergent surgical consultation for both conditions, with anticoagulation added for cavernous sinus thrombosis.

Postextraction Pain and Postextraction Alveolar Osteitis (Dry Socket)

Pain experienced within 24 to 48 hours after a tooth extraction responds well to icepacks, head elevation, and analgesics. Postextraction alveolar osteitis ("dry socket") occurs 48 to 72 hours postoperatively when the clot from the socket is displaced. It is associated with smoking, carbonated beverage consumption, and drinking with straws. It presents with severe pain along with foul odor and taste. Perform gentle saline or 0.12% chlorhexidine irrigation of the socket, followed by packing the socket with eugenol-impregnated gauze, which will relieve the pain. Prescribe penicillin VK 500 mg PO four times daily or clindamycin 300 mg PO four times daily in severe cases, along with analgesia, daily packing changes,

152_Cydulka_Ch152.indd 839 22/03/17 9:33 AM

Source page 859

gentle 0.12% chlorhexidine mouth rinses twice daily, and dental follow-up in 24 hours.

Postextraction Bleeding

Control bleeding after dental extraction by instructing the patient to bite on gauze positioned over the bleeding site for 20 minutes. If bleeding persists, pack the socket with Gelfoam®, Avitene®, or Surgicel®. Loosely approximating sutures can be used to hold these packing agents in place. Other emergency department (ED) treatments may include local injection of 1% lidocaine with epinephrine or careful silver nitrate application. Failure of these measures warrants a screening coagulation profile and consultation with an oral surgeon.

Periodontal Abscess

A periodontal abscess results from plaque and debris entrapped between the tooth and gingiva. For small abscesses, administer penicillin VK 500 mg PO four times daily or clindamycin 300 mg PO four times daily, analgesics, and short-term 0.12% chlorhexidine mouth rinses twice daily. Larger abscesses require incision and drainage. All patients need prompt dental referral.

Acute Necrotizing Ulcerative Gingivitis

Acute necrotizing ulcerative gingivitis (ANUG) presents with pain, ulcerated or “punched out” interdental papillae, gingival bleeding, foul taste, lymphadenopathy, and fever. The associated putrid breath gives this disorder its nickname of “trench mouth.” It occurs mainly in young patients with compromised resistance to infection due to HIV, stress, malnourishment, substance abuse, and various infections. Prescribe metronidazole 500 mg PO three times daily and 0.12% chlorhexidine mouth rinses twice daily, along with a protein-rich soft diet, multivitamins, and PO fluids. Symptomatic improvement is typically dramatic within 24 hours, and the patient should be referred for dental debridement, scaling, and workup of predisposing factors.

Peri-implantitis

Dental implants are being increasingly used to replace teeth. If they become infected, they present similarly to a periodontal abscess. Irrigate around the implant with 0.12% chlorhexidine solution and administer metronidazole 500 mg PO three times daily or amoxicillin 500 mg PO three times daily for 10 days. Prescribe analgesics and refer the patient to a dentist for definitive management.

■■ SOFT TISSUE LESIONS OF THE ORAL CAVITY

Aphthous Stomatitis

Aphthous stomatitis, or aphthous ulcer, presents with painful lesions, which are frequently multiple, involve the labial and buccal mucosa, and measure from 2 mm to several centimeters in diameter. Treat with 0.1% triamcinolone acetonide dental paste (Kenalog® in Orobase®) applied topically to lesions two to three times a day or 0.12% chlorhexidine applied topically twice a day. The lesions often heal within 7 to 10 days.

152_Cydulka_Ch152.indd 840 22/03/17 9:33 AM

Source page 860

Oral Cancer

Emergency physicians should be vigilant for oral cancers and their precursors because early diagnosis is associated with improved outcomes. Cancers can present early as leukoplakia (nonremovable white mucosal patches) and erythroplakia (red patch that cannot be classified as any other disease). The most common site for oral cancer to develop is the posterolateral border of the tongue. Symptoms and signs of oral cancer include pain, paresthesias, persistent ulcers, bleeding, lesion rigidity, induration, lymphadenopathy, and functional impairment. Refer all suspicious lesions or lesions that persist >14 days despite treatment for urgent follow-up with an oral surgeon for biopsy.

■■ OROFACIAL TRAUMA

Dental Fractures

The Ellis system is used to classify the anatomy of fractured teeth. Ellis class 1 fractures solely involve the enamel. These injuries may be smoothed with an emery board or referred to a dentist for cosmetic repair. Ellis class 2 fractures reveal the creamy yellow dentin underneath the white enamel. The patient complains of air and temperature sensitivities. To decrease pulpal contamination, dry the dentin and promptly cover the lesion with calcium hydroxide paste or glass ionomer cement. All patients should see a dentist within 24 hours. Ellis class 3 fractures are tooth-threatening fractures that involve the pulp and can be identified by a red blush in the exposed dentin or a visible drop of blood after wiping the tooth. Apply a thin layer of calcium hydroxide paste to cover the pulp, followed by an overlying layer of glass ionomer cement to completely cover the patch and dentin as well. Ideally, a dentist should evaluate the patient in the ED, but if the amount of exposed pulp is very small, the patched lesion can be evaluated by a dentist within 24 hours. Oral analgesics may be needed, but topical anesthetics are contraindicated. The use of prophylactic antibiotics is controversial.

Concussions, Luxations, and Avulsions

Concussion injuries involve posttraumatic tenderness to percussion with no mobility. Posttraumatic mobility without evidence of dislodgment is called subluxation, which has a higher incidence of future pulp necrosis. Prescribe nonsteroidal anti-inflammatory drugs (NSAIDs) and soft diet, and refer urgently to a dentist to rule out a more severe injury.

Extrusive luxation occurs when a tooth is partly avulsed out from the alveolar bone. Treatment involves gentle repositioning of the tooth to its original location (often with the aid of a dental block) and splinting with zinc oxide periodontal dressing. A dentist should evaluate these patients within 24 hours. When the tooth is laterally displaced with a fracture of the alveolar bone, the condition is called lateral luxation. Although manual relocation is possible, the treatment of such injuries is best done in consultation with a dentist. If the alveolar bone fracture is significant, splinting by a dentist in the ED is required. An intrusive luxation occurs when the tooth is forced below the gingiva and often has a poor outcome. For all luxations, prescribe a soft diet, meticulous oral hygiene, and 0.12% chlorhexidine mouth rinses twice daily, and refer to a dentist to be seen within 24 hours.

152_Cydulka_Ch152.indd 841 22/03/17 9:33 AM

Source page 861

Dental avulsion is a bona fide dental emergency in which a tooth has

been completely removed from the socket. Primary teeth in children are NOT replaced because of potential damage to the permanent teeth (Fig. 152-2).

Replant permanent teeth that have been avulsed for less than 3 hours immediately to attempt to save the periodontal ligament fibers. At the scene, handle an avulsed tooth by the crown only, rinse with water <10 seconds, and replant immediately. If replantation at the scene is not possible due to risk of aspiration, rinse the tooth and place in a nutrient solution, such as Hank's balanced salt solution (preserves cell viability for up to 4 to 6 hours), cold milk, saliva, or sterile saline. Transport the tooth immediately with the patient to the ED. An effective, but somewhat unsettling, way to safely transport the avulsed tooth is underneath the tongue of an alert, sober, cooperative patient or parent (if the patient is a child). Upon arrival in the ED, if the tooth has not already been replanted, gently rinse the tooth with Hank's solution or sterile saline and soak in either fluid. Remove the clot in the empty socket and gently irrigate with sterile normal saline. Examine the tooth to determine whether the apex at the root tip is open, then proceed with the appropriate tooth replantation procedure if indicated. (See Table 152-1 for specific recommendations for tooth replantation.)

Early consultation with a dentist is imperative, but do not delay replantation while awaiting the arrival of the specialist. After replantation, adults should receive doxycycline 100 mg PO BID for 7 days. Children <12 years old should receive penicillin VK 12.5 mg/kg/dose four times a day

Patient's right Patient's left

Age of permanent

tooth eruption

Age of primary

tooth eruption 7–8 years

8–12 months 8–9 years

9–11 months 11–12 years

16–22 months 10–11 years

13–19 months 10–12 years

6–7 years

25–33 months

12–13 years

11–13 years

6–7 years

23–31 months

11–12 years

14–18 months

10–12 years

17–23 months

9–10 years

10–16 months

7–8 years

6–10 months

6–7 years

Figure 152-2. Normal eruptive patterns of the primary and permanent dentition.

152_Cydulka_Ch152.indd 842 22/03/17 9:33 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.

- Content ID: SEG-TINTINALLI-EMM8-FULL.

- Source pages: 858–865.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 153: Neck and Upper Airway Disorders

Source pages 866–871 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 866–871

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Neck and Upper Airway Disorders**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 153: Neck and Upper Airway Disorders 847 ■■ RETROPHARYNGEAL ABSCESS Clinical Features Common symptoms include sore throat, neck pain, torticollis, and dyspha- gia. [Source pages 866–871]

- Additionally, patients may have a muffled voice, cervical adenopathy, and respiratory distress. [Source pages 866–871]

- Diagnosis and Differential A CT of the neck with IV contrast is the gold standard diagnostic exam and differentiates cellulitis from an abscess. [Source pages 866–871]

- Prepare for emergent airway management as needed. [Source pages 866–871]

- Administer IV fluids, clindamycin, or cefoxitin IV. [Source pages 866–871]

Clinical reasoning structure

- **Problem representation:** Neck and Upper Airway Disorders; source pages 866–871.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Neck and Upper Airway Disorders**.

- Source-page range: **866–871**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 866

CHAPTER 153: Neck and Upper Airway Disorders 847

■■ RETROPHARYNGEAL ABSCESS

Clinical Features

Common symptoms include sore throat, neck pain, torticollis, and dysphagia. Additionally, patients may have a muffled voice, cervical adenopathy, and respiratory distress. Stridor is more common in children.

Diagnosis and Differential

A CT of the neck with IV contrast is the gold standard diagnostic exam and differentiates cellulitis from an abscess.

Emergency Department Care and Disposition

1. Prepare for emergent airway management as needed.

2. Consult with otolaryngology early.

3. Administer IV fluids, clindamycin, or cefoxitin IV.

■■ ODONTOGENIC ABSCESS

Clinical Features

Most deep space neck infections are odontogenic, usually from the mandibular teeth. Patients may present with trismus, fever, neck mass, dysphagia, and dyspnea. Infection can range from diffuse cellulitis to abscesses in the labial or buccal gingiva. Infections can spread through many fascial layers into potential spaces in the face, neck, and mouth.

Diagnosis and Differential

Bedside ultrasound can detect superficial odontogenic abscesses, but if deep space infection is suspected, CT of the neck with IV contrast is recommended.

Emergency Department Care and Disposition

1. Perform surgical drainage of the abscess, either with I&D in the emergency department or in the operating room.

2. Prescribe Penicillin VK or amoxicillin for patients who can be discharged after drainage. Penicillin allergic patients may be treated with

clindamycin or levofloxacin.

3. In patients with deep space neck infections, administer IV ampicillin/sulbactam along with clindamycin and ciprofloxacin.

Complications

Ludwig's angina is an infection of the submental, sublingual, and submandibular spaces. Clinical exam reveals trismus and edema of the entire upper neck and floor of the mouth. Critically ill patients may have a necrotizing infection. These are associated with skin discoloration, crepitus of the subcutaneous tissue, and systemic signs of illness including fever, tachycardia, hypotension, and confusion. Immediate surgical consultation is required, as these patients need fasciotomy with wide local debridement and broad-spectrum IV antibiotics.

153_Cydulka_Ch153.indd 847 22/04/17 4:53 pm

Source page 867

■■ NECK AND UPPER AIRWAY MASSES

Clinical Features

Neck masses can be congenital, infectious, glandular, or neoplastic. In adults >40 years old, up to 80% of lateral neck masses persistent for >6 weeks are malignant.

Diagnosis and Differential

Patients with airway compromise or significant dysphagia or odynophagia should be evaluated using flexible nasopharyngolaryngoscopybefore CT scan. CT scan is often required for planning surgical intervention and will help to delineate the extent of the mass (Table 153-1).

Emergency Department Care and Disposition

1. Inflammatory lymph nodes should be treated with empiric antibiotics such as cephalexin 500 mg PO three to four times daily or amoxicillin 500 mg three to four times daily or clindamycin 300 mg three to four times daily.

2. Empiric therapy for sialoadenitis should cover staphylococcal infection with an antibiotic such as clindamycin.

■■ POSTTONSILLECTOMY BLEEDING

Clinical Features

Postoperative bleeding is a well-known complication of tonsillectomy and can cause death from airway obstruction or hemorrhagic shock. Incidence ranges from 1% to 8.8% with about half of patients requiring surgical intervention to control bleeding. Most significant hemorrhage occurs between postoperative days 5 and 10, but bleeding can be seen within 24 hours of surgery.

Emergency Department Care and Disposition

1. Keep the patient NPO and sitting upright.

2. Consult otolaryngology early.

3. Apply direct pressure to the bleeding tonsillar bed using a tonsillar pack or 4 × 4 gauze on a long clamp moistened with either thrombin or lidocaine and epinephrine.

Table 153-1 Neck Masses in Young and Older Adults

Young Adult Adult

Reactive lymphadenopathy Metastatic aerodigestive tract carcinoma

Mononucleosis Salivary gland infection or neoplasm

Lymphoma

Branchial cleft cyst Lymphoma

Thyroglossal duct cyst Thyroid disorder

Tuberculosis

153_Cydulka_Ch153.indd 848 22/04/17 4:53 pm

Source page 868

4. Alternatively, if the bleeding site is visualized, locally infiltrate with lidocaine and epinephrine then cauterize with silver nitrate.

5. Massive bleeding is rare but when it occurs the only means of protecting the airway is intubation. Plan for emergent cricothyrotomy prior to attempting intubation.

■■ COMPLICATIONS OF AIRWAY DEVICES

Tracheostomy Tubes and Cannulas

Early complications include bleeding, obstruction, dislodgement, and infection. These typically occur within the first postoperative week. Late complications like granulation, tracheal stenosis, and fistula formation occur after the first postoperative week. Emergency medicine providers must be proficient in replacing uncuffed with cuffed tracheostomy tubes for mechanical ventilation, replacement of tracheostomy tubes after accidental decannulation, correction of tube obstruction, and control of bleeding or infection at the tracheostomy site.

Tracheostomy Tube with Airway Obstruction

If the tracheostomy tube is patent, leave it in place. Consider mucus plugging of the trachea or mainstem bronchi distal to the tube and attempt to suction. Preoxygenation and placement of sterile saline into the trachea will aid in suctioning. You may have to remove the inner cannula and occasionally the entire tracheostomy tube if suctioning fails.

Tracheostomy Dislodgement

If a suction catheter cannot be passed through the tube and on x-ray the

tracheostomy tube is seen extrinsically compressing the trachea, the tube is likely dislodged. Remove the entire tracheostomy tube. Insert a nasopharyngoscope or flexible bronchoscope into the visible stoma to identify the true tracheal opening. If the opening cannot be identified obtain otolaryngology or general surgery consultation. If the patient cannot maintain his or her airway, you should orally intubate.

Tracheostomy Site Infection

Stomal skin infection, tracheitis, and bronchitis can be recurring problems. Treat polymicrobial infections in stable patients with amoxicillin-clavulanate 875 mg PO twice daily. Unstable or systemically ill patients should receive piperacillin-tazobactam 3.375 g IV plus vancomycin 1 g IV. Add a fluoroquinolone for Pseudomonas. Local wound infections can be treated with gauze soaked in 0.25% acetic acid.

Tracheostomy Site Bleeding

Sources of hemorrhage include granulation tissue in the stoma, trachea, or thyroid or erosion of the thyroid vessels, the tracheal wall, or the innominate artery. Packing with saline soaked gauze can control slow bleeding from the stoma. If this is ineffective remove the tube and examine the stoma and tracheal wall. Local bleeding can be controlled with silver nitrate. If bleeding is brisk, replace the tracheostomy tube with a cuffed endotracheal tube with the cuff below the bleeding site.

153_Cydulka_Ch153.indd 849 22/04/17 4:53 pm

Source page 869

Tracheoinnominate artery fistula is a rare but life-threatening complication of tracheostomy. This is typically seen during the first 3 weeks after tracheostomy and is due to vessel erosion from high (>25 mm Hg) cuff pressures or direct pressure from the tip of the tracheal cannula against the innominate artery. Some patients present with hemoptysis or a sentinel arterial bleed. Immediate orolaryngologic or thoracic surgery consultation is required.

If the patient presents with massive bleeding, the first maneuver is to hyperinflate the cuff to control brisk bleeding while planning operative intervention. If bleeding persists, slowly withdraw the tube while exerting anterior pressure on the anterior trachea. If these interventions fail to control bleeding, place an endotracheal tube from above the tracheoinnominate fistula, using direct visualization with a flexible intubating scope while an assistant removes the tracheostomy tube allowing the endotracheal tube to pass. Then use digital pressure of the innominate artery against the manubrium to control stomal hemorrhage.

Tracheal Stenosis

Weeks to months after decannulation, mucosal necrosis, and subsequent scarring can cause tracheal stenosis. Patients with stenosis have dyspnea, wheezing, stridor, and inability to clear secretions. Treatment in the emergency department includes humidified oxygen, nebulized racemic epinephrine, and steroids. Further treatment may include rigid bronchoscopy laser excision of scar bands, stenting, or tracheal reconstruction.

Changing a Tracheostomy Tube

If the tracheostomy is less than 7 days old and the situation is not emergent, tracheostomy tubes should be changed by a surgeon familiar with the procedure, as tracts are not yet mature and manipulation may easily create a false passage. Tracts can also easily collapse in patients with obesity or neck masses.

■■ LARYNGECTOMY PATIENTS

It is impossible to orally intubate patients after laryngectomy. Laryngectomy patients can be emergently intubated by placing an endobronchial tube into the tracheostoma. Do not advance the tube too far, as the carina can be only 4 to 6 cm from the tracheostoma.

Speech Devices

A Passy-Muir valve is a one-way valve that fits directly over the opening of an uncuffed tracheostomy tube, allowing for hands free speech. Because patients exhale around the tracheostomy tube, a Passy Muir valve should not be used with a cuffed tube. If a patient with a Passy-Muir valve develops respiratory distress or an inability to speak, your first intervention should be to remove the speech valve. Then check the tracheostomy tube for obstruction.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 246, “Neck and Upper Airway,” by Nicholas D. Hartman; Chapter 247 “Complications of Airway Devices,” by John P. Gaillard.

153_Cydulka_Ch153.indd 850 22/04/17 4:53 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 866–871.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 154: Dermatologic Emergencies

Source pages 872–875 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 872–875

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Dermatologic Emergencies**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 154: Dermatologic Emergencies 853 Figure 154-2. [Source pages 872–875]

- Systemic toxicity as well as oral, ocular, and anogenital mucosal lesions are common. [Source pages 872–875]

- TEN has a mortality rate of 25% to 35%, and poor prognostic indicators include advanced age, extensive disease, multiple medication use, leukopenia, azotemia, and thrombocytopenia. [Source pages 872–875]

- Diagnosis and Differential Initial diagnosis is based on the clinical features, while skin biopsy results can confirm the diagnosis. [Source pages 872–875]

- The differential diagnosis includes erythema multiforme, exfoliative dermatitis, primary blistering disorders, Kawasaki disease, and toxic infectious erythemas. [Source pages 872–875]

Clinical reasoning structure

- **Problem representation:** Dermatologic Emergencies; source pages 872–875.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Dermatologic Emergencies**.

- Source-page range: **872–875**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 872

CHAPTER 154: Dermatologic Emergencies 853

Figure 154-2. Toxic epidermal necrolysis.

to the skin. Systemic toxicity as well as oral, ocular, and anogenital mucosal lesions are common. TEN has a mortality rate of 25% to 35%, and poor prognostic indicators include advanced age, extensive disease, multiple medication use, leukopenia, azotemia, and thrombocytopenia.

Diagnosis and Differential

Initial diagnosis is based on the clinical features, while skin biopsy results can confirm the diagnosis. The differential diagnosis includes erythema multiforme, exfoliative dermatitis, primary blistering disorders, Kawasaki disease, and toxic infectious erythemas.

Emergency Department Care and Disposition

1. Admit patients with TEN to a critical care setting, preferably a burn unit with expertise managing appropriate cardiopulmonary, fluid, electrolyte, and sepsis resuscitation.

2. Emergent dermatologic consultation is recommended.

■■ MENINGOCOCCEMIA

Clinical Features

Meningococcemia is a serious and potentially fatal systemic infection caused by Neisseria meningitides. Clinical disease usually develops within 3 to 4 days after exposure and can present with a fulminant course. Possible presenting features can include headache, fever, altered mental status, nausea, vomiting, myalgias, arthralgia, and neck stiffness. Dermatologic manifestations of this infection include petechia, urticaria, hemorrhagic vesicles, and macules that evolve into palpable purpura with grey necrotic centers.

Diagnosis and Differential

Consider meningococcemia in ill-appearing patients with a petechial rash and associated symptoms of infection. The differential diagnosis includes

Source page 873

Rocky Mountain spotted fever, toxic shock syndrome, gonococcemia, bacterial endocarditis, vasculitis, viral and bacterial infections, and disseminated intravascular coagulation.

Emergency Department Care and Disposition

1. Treat meningococcemia in adult patients with an IV third-generation cephalosporin such as ceftriaxone or cefotaxime. Consider additional empiric antibiotic coverage for patients with suspected bacteremia when the pathogen is uncertain.
2. Admit patients with meningococcal infection for close monitoring, supportive care, and continued antibiotic treatment.

■■ PEMPHIGUS VULGARIS

Clinical Features

Pemphigus vulgaris is a generalized, mucocutaneous, autoimmune, blistering eruption with a grave prognosis. Primary lesions are clear, tense vesicles or bullae that vary in diameter, and are first noted on the head, trunk, and mucous membranes. Within 2 to 3 days, the bullae become turbid and flaccid then rupture, producing painful, denuded areas that are slow to heal and prone to secondary infection (Fig. 154-3).

Diagnosis and Differential

Pemphigus vulgaris is suspected by the appearance of lesions and confirmed by skin biopsy and immunofluorescence testing. The differential diagnosis includes bullous pemphigoid, TEN, EM major, dermatitis herpetiformis, and other blistering skin diseases. Bullous pemphigoid is a mucocutaneous blistering disease typically found in the elderly.

Figure 154-3. Scattered bullous lesions intermixed with erosions and painful inflammatory plaques in a patient with pemphigus vulgaris.

154_Cydulka_Ch154.indd 854 22/03/17 9:47 AM

Source page 874

Emergency Department Care and Disposition

1. Treat fluid and electrolyte disturbances aggressively.
2. Consult with a dermatologist for management of pemphigus vulgaris. Corticosteroids and immunosuppressives are the mainstays of therapy. Plasmaphoresis and IV immunoglobins may also be needed.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 141, "Rashes in Infants and Children," by Gary Bonfante and Amy Dunn; Chapter 249, "Generalized Skin Disorders," by Mark Sochor, Amit Pandit, and William J. Brady.

154_Cydulka_Ch154.indd 855 22/03/17 9:47 AM

Source page 875

C h a pter Other Dermatologic Disorders

155 Jason P. Stopyra

■■ HERPES ZOSTER

Clinical Features

Herpes zoster results from cutaneous activation of latent varicella zoster virus along a sensory nerve root dermatome. Pain or dysesthesia in an involved dermatome begins 3 to 5 days before lesions emerge. Erythematous papules develop first, progress to vesicular clusters, and these lesions crust after about a week. Herpes zoster of the ophthalmic branch of the trigeminal nerve, especially if accompanied by lesions on the nose, are concerning for possible eye involvement that can lead to keratitis or corneal ulceration (Fig. 155-1). A thorough eye exam should be performed (see Chapter 149, “Ocular Emergencies”). Generalized eruptions involving more than one dermatome may occur in immunocompromised patients.

Diagnosis and Differential

The differential diagnosis for these kinds of skin eruptions includes herpes simplex, impetigo, and contact dermatitis. The characteristic skin rash of herpes zoster presents in a unilateral distribution along a single sensory dermatome and is accompanied by localized pain at the site. A swab of the

Gr. Sm.

occipital occipital

N. N.

*

Post Cervical

cerv. cutaneous

Ophthalmic

Rami of the *Overlapping of greater

Maxillary

trigeminal N. auricular and facial

Mandibular nerves

Ramus of

Auricular

vagus

A B

FIGURE 155-1. A. Herpes zoster in trigeminal nerve distribution. Note lesion on the tip of the nose, which suggests nasociliary branch involvement. B. Dermatomes of the head and neck. cerv. = cervical; Gr. = greater; N. = nerve; Sm. = smaller. A. Reproduced with permission from Fleischer A Jr, Feldman S, McConnell C,

et al. Emergency Dermatology: A Rapid Treatment Guide. New York, NY: McGraw-

Hill; 2002. B. Reproduced with permission from Wolff K, Johnson R, Saavedra AP.

Fitzpatrick’s Color Atlas & Synopsis of Clinical Dermatology, 7th ed. New York:

McGraw-Hill; 2013.

856

155_Cydulka_Ch155.indd 856 22/03/17 10:31 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.

- Content ID: SEG-TINTINALLI-EMM8-FULL.

- Source pages: 872–875.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 155: Other Dermatologic Disorders

Source pages 876–885 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 876–885

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Other Dermatologic Disorders**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 155: Other Dermatologic Disorders 857 base of the vesicle can be sent for viral PCR to confirm the clinical diagno- sis when uncertainty exists. [Source pages 876–885]

- Antiviral medications started in the first 72 hours of illness shorten healing time, decrease formation of new lesions, and help prevent postherpetic neuralgia. [Source pages 876–885]

- Antiviral choices include acyclovir 800 mg PO five times per day for 7 to 10 days or valacyclovir 1000 mg PO three times a day for 7 days. [Source pages 876–885]

- Immunocompromised patients with severe disease can be treated with acyclovir, 10 mg/kg IV every 8 hours for 7 to 10 days. [Source pages 876–885]

- Aluminum acetate compressions three times daily and analgesics pro- vide symptomatic relief. [Source pages 876–885]

Clinical reasoning structure

- **Problem representation:** Other Dermatologic Disorders; source pages 876–885.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Other Dermatologic Disorders**.

- Source-page range: **876–885**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 876

CHAPTER 155: Other Dermatologic Disorders 857

base of the vesicle can be sent for viral PCR to confirm the clinical diagnosis when uncertainty exists.

Emergency Department Care and Disposition

1. Antiviral medications started in the first 72 hours of illness shorten healing time, decrease formation of new lesions, and help prevent postherpetic neuralgia. Antiviral choices include acyclovir 800 mg PO five times per day for 7 to 10 days or valacyclovir 1000 mg PO three times a day for 7 days. Immunocompromised patients with severe disease can be treated with acyclovir, 10 mg/kg IV every 8 hours for 7 to 10 days.
2. Aluminum acetate compressions three times daily and analgesics provide symptomatic relief.
3. Advise patients that herpes zoster is contagious to anyone who has not had chicken pox or the varicella zoster vaccine.
4. Consult with an ophthalmologist if eye involvement is suspected.

■■ HERPES SIMPLEX VIRUS INFECTIONS

Herpes simplex virus (HSV) lesions are painful grouped vesicles with an erythematous base. Primary disease may be preceded with or accompanied by constitutional symptoms. Tingling or burning precedes recurrent lesions. Oral lesions (“cold sores”) are usually caused by HSV1, but may also be caused by HSV2. The diagnosis can be confirmed with an HSV PCR test if necessary. Treatment (Table 155-1) is most effective when started within 24 hours of symptom onset. See Chapter 87, “Sexually Transmitted Infections,” for discussion and treatment of genital herpes.

■■ TINEA INFECTIONS

Clinical Features

Tinea refers to skin infections caused by dermatophytes (fungi). Tinea

capitis is characterized by patchy areas of alopecia with broken off hairs and scales at the periphery. Tinea barbae presents with severe inflammatory plaques and follicular pustules in the beard area. Interdigital scaling, maceration, plantar or palmar erythema or scaling, and pruritis are seen in

Table 155-1 Treatment of Herpes Simplex Virus (HSV) Gingivostomatitis (Herpes Labialis)

Condition	Acyclovir Treatment	Valacyclovir Treatment
First episode	400 mg PO three times a day for 7 days or 200 mg PO five times a day for 7 days	1 g PO twice a day for 7 days
Recurrent episode	400 mg PO three times a day for 5 days or 800 mg PO three a day for 2–5 days	500 mg PO twice a day for 3 days
Suppression	400 mg PO twice a day	1 g PO a day, may decrease to 500 mg PO a day if <10 outbreaks/year

155_Cydulka_Ch155.indd 857 22/03/17 10:31 AM

Source page 877

Tinea pedis (also known as athlete's foot) and Tinea manuum (hand). Onchomycosis may occur. Findings in Tinea cruris (commonly called jock itch) include erythema with a peripheral annular scaly edge that extends onto the thighs and buttocks but spares the penis and scrotum. Candida intertrigo involves the skinfolds. Tinea Corporus (trunk, neck, arms, and legs lesions) are typically circular, covered with scales and surrounded by a raised border.

Diagnosis and Differential

Identification of fungal elements on a potassium hydroxide preparation or with fungal culture may be completed if the diagnosis is uncertain. The differential diagnosis includes psoriasis, atopic, seborrheic, and chronic dermatitis (Table 155-2).

Table 155-2 Comparison Features of Common Papulosquamous Eruptions

Condition	Clinical Features	Location	Special Signs	Distinguishing Comments
Psoriasis	Erythematous, well-marginated papules and plaques with silvery scale	Trunk, extensor surfaces, scalp	Auspitz sign; nail pitting	Hereditary phenomenon, onset in early 20s
Seborrheic	Greasy, yellow	Midchest, supra-	Can overlap	Debilitated,

dermatitis scales pubic, scalp, with psoriasis,elderly, or
facial creases “sebopsoriasis”infants (cradle
cap)
Atopic Ill-defined vesicles Flexures > trunk Spares the Pruritus
dermatitis forming plaques nose “itch that
with scale; chronic rashes”; atopic
lesions lichenified individuals
Lichen 5 P’s: purple, Any skin, mucous Wickham Age
planus pruritic, polygonal, membranes, hair striae; Koebner 20–60 years old
planar papules follicles phenomenon
Pityriasis Lines of skin Trunk, in Herald patch Spring and fall,
rosea tension, collarette Christmas tree 1–2 weeks age 15–40 years
of scale pattern following before general old; viral exan-
skin lines eruption them, herpes
6 and 7
Tinea Sharply demarcated, Trunk, legs, arm, May need All ages; from
corporis erythematous, scaly neck KOH/culture to pets, soil, or
annular plaques; diagnose; sep- autoinoculation
may coalesce into tate branching from hands/
gyrate patterns hyphae on KOH feet; incubation
days or months
Pityriasis Versicolored—red, Central upper Spaghetti and Young adults,
(tinea) salmon, light chest and back meatballs on summer, hot
versicolor brown, dark brown, KOH; nonsep- humid environ-
hypopigmented; tate pseudo- ments
well-demarcated hyphae and
scaly patches budding yeast
(Continued)
155_Cydulka_Ch155.indd 858 22/03/17 10:31 AM

Source page 878

CHAPTER 155: Other Dermatologic Disorders 859
Table 155-2 Comparison Features of Common Papulosquamous
Eruptions (Continued)
Distinguishing
Condition Clinical Features Location Special Signs Comments
Secondary At 2–10 weeks, Palms, soles, Serology Great
syphilis macular erythema trunk masquerader—
on trunk, abdomen, can take any
inner extremities; form; can be
followed by papular confused with

or papulosquamous pityriasis rosea

lesions

Scabies Pruritic papules Finger webs, Scrapings Can be chronic

and burrows with wrists, axillae, show mites, “7-year itch”;

crusting areolae, umbi- feces, eggs intensely pru-

licus, abdomen, ritic, especially

waistband, at night

genitals

Abbreviation: KOH = potassium hydroxide.

Emergency Department Care and Disposition

1. Treat Tinea capitis with liquid microsize griseofulvin 20 to 25 mg/kg PO daily (maximum, 1 g/d) for 8 weeks, ultramicrosize griseofulvin (tablets only), 10 to 15 mg/kg/d (maximum, 750 mg/d) for 8 weeks, or terbinafine granules 125 mg PO daily for patients weighing <25 kg, 187.5 mg/d for patients weighing 25 to 35 kg, and 250 mg/d for patients weighing >35 kg for 6 weeks. Consider baseline liver function tests as oral antifungal treatment in patients with abnormal hepatic function is not recommended. Advise patients to wash hair with selenium sulfide 2.5% shampoo or ketoconazole 2% shampoo three times per week for the first 2 weeks of therapy.

2. Treat Tinea barbae with microsize griseofulvin, 500 mg PO daily or ultramicrosize griseofulvin, 375 mg PO daily, for 4 to 6 weeks. Tervinafine can also be used. Consider baseline liver function tests. Shaving or depilating the hair along with warm compresses is recommended.

3. Treat nonbullous tinea pedis and manuum, intertrigo, tinea corpora, and tinea cruris, with topical antifungal agents such as clotrimazole, miconazole, ketoconazole, or econazole twice daily for 4 weeks. Continue treatment 1 week after clearing has occurred. Antifungal powders used on a daily basis help prevent recurrences of tinea cruris.

4. Onchomycosis usually requires long-term oral treatment.

5. Follow-up with a primary care physician or dermatologist is recommended, especially if the lesions are not resolved in 4 to 6 weeks.

■■ ALLERGIC CONTACT DERMATITIS

Clinical Features

Contact dermatitis occurs after direct contact with an irritant or allergen. Reactions occasionally occur after exposure to aerosolized particles, such as burned poison ivy or oak. Detergents and soaps are common irritants, while nickel, plants, cosmetic preservatives, contact lens solutions, and skin

Source page 879

tape are common allergens. Physical findings include erythema, papules, vesicles, and bullae. Scaling and fissuring are seen with chronic contact dermatitis.

Emergency Department Care and Disposition

1. Begin treatment by identifying and removing the likely offending agent.
2. Aluminum acetate (Burow's solution) compresses two or three times per day may ease acute irritation.
3. Depending on severity, topical or oral corticosteroids and oral antihistamines can be used in medical management. Patients with extensive, severe allergic contact dermatitis may benefit from a short course of systemic corticosteroids.

■■ SUNBURN AND PHOTOSENSITIVITY

Clinical Features

Patients with sunburn have an inflammatory response to ultraviolet (UV) radiation and may present with minimal discomfort in mild cases or extreme pain with extensive blistering for more severe exposures. Tender and warm areas of erythema in sun-exposed areas are characteristic of sunburn. Sunburn reactions develop 2 to 6 hours after exposure and peak in 1 to 3 days. Exogenous photosensitivity disorders result from topical application or ingestion of an agent that increases the skin's sensitivity when exposed to UV light. Photosensitivity disorders may be phototoxic (occur quickly with characteristics similar to sunburn) or photoallergic (delayed onset characterized by eczema-like changes with vesiculation). Photosensitivity eruptions occur on sun-exposed skin and are associated with topically applied furocoumarins (lime juice, various fragrances, figs, celery, parsnips), and numerous medications, including sulfonamides, thiazides, furosemide, fluoroquinolones, and tetracyclines.

Emergency Department Care and Disposition

1. Sunburns are treated symptomatically with tepid baths, NSAIDs, and local wound care including topical antibiotics to blistered areas.
2. Initial management of exogenous photosensitivity is similar to the sunburn reaction, including the avoidance of the sun until the eruption has cleared. Identify and discontinue the causative agent, if possible.

■■ SECONDARY SYPHILIS

Clinical Features

Early secondary syphilis eruptions emerge 2 to 10 weeks after the appearance of the primary chancre (see Chapter 87, "Sexually Transmitted Infections"). This evanescent macular rash on the trunk and inner extremities may last only a few hours or days. Subsequent rash, often referred to as the

“great imitator” because of the varied clinical appearance, lasts 2 to 6 weeks and may present in a variety of forms. These eruptions are often papular lesions 2 to 5 mm in size, generalized, and reddish to copper in color. Involvement of the palms and soles is a helpful diagnostic finding. Consider risk factors and the possibility of secondary syphilis rash when

155_Cydulka_Ch155.indd 860 22/03/17 10:31 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 876–885.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 156: Trauma in Adults

Source pages 886–889 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 886–889

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Trauma in Adults**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 156: Trauma in Adults 867 ■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION 1. [Source pages 886–889]

- The ED management of trauma patients begins prior to the patient’s arrival. [Source pages 886–889]

- Emergency medical service providers inform the receiving ED about the mechanism of trauma, vital sign values, suspected inju- ries, and treatments provided. [Source pages 886–889]

- Airway patency is confirmed at the outset of the primary survey. [Source pages 886–889]

- In patients making inadequate respiratory efforts, perform a jaw thrust, and then insert an oral or nasal airway. [Source pages 886–889]

Clinical reasoning structure

- **Problem representation:** Trauma in Adults; source pages 886–889.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Trauma in Adults**.

- Source-page range: **886–889**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 886

CHAPTER 156: Trauma in Adults 867

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. The ED management of trauma patients begins prior to the patient's arrival. Emergency medical service providers inform the receiving ED about the mechanism of trauma, vital sign values, suspected injuries, and treatments provided.

2. Airway patency is confirmed at the outset of the primary survey. In patients making inadequate respiratory efforts, perform a jaw thrust, and then insert an oral or nasal airway. Avoid nasal airway placement in patients with suspected basilar skull fractures. Endotracheal intubation is indicated in comatose patients (GCS < 8) to protect the airway and prevent secondary brain injury from hypoxemia. Agitated trauma patients who need further diagnostic or therapeutic interventions or patients whose expected course necessitates immediate operative management are also candidates for intubation. Whenever possible, use a two-person spinal stabilization technique in which one caregiver provides in-line immobilization of the cervical spine while the other manages the airway. Trauma patients are often difficult to intubate due to associated facial trauma, cervical spine immobilization, or the presence of blood or vomitus. In virtually all trauma patients requiring urgent intubation, a rapid sequence intubation technique should be used. In cases of extensive facial trauma or when endotracheal intubation is not possible, cricothyrotomy or another advanced airway technique may be used to secure the airway.

3. Once the airway is secured, examine the neck and thorax to detect abnormalities such as a deviated trachea (tension pneumothorax), crepitus (pneumothorax), paradoxical movement of a chest wall segment (flail chest), sucking chest wound, fractured sternum, or the absence of breath sounds on either side of the chest (simple or tension pneumothorax, massive hemothorax, or right mainstem intubation). Treat tension pneumothorax immediately with needle decompression

followed by tube thoracostomy.

4. Rapidly assess the patient's hemodynamic status during the primary survey by noting the level of consciousness, skin color, and the presence and magnitude of peripheral pulses. Note the heart rate, blood pressure, and pulse pressure (systolic minus diastolic blood pressure). Hemorrhage of up to 30% of total blood volume may be associated with only mild tachycardia and a decrease in pulse pressure, but may quickly progress to shock if not recognized early. Establish two large-bore peripheral IV lines, and obtain blood samples for laboratory studies, particularly the blood type and screen. Establish a central line in patients who are unstable or in whom upper extremity peripheral veins are not easily cannulated. Current evidence indicates that early administration of tranexamic acid may reduce mortality.

5. Reassess hemodynamically unstable patients without an obvious indication for surgery after infusion of 2 L of warm crystalloid solution. If there is no marked improvement, consider the need to transfuse type O blood (O-negative for females of childbearing age). An extended focused assessment with sonography for trauma (eFAST) examination screens for causes of shock immediately after the primary survey. If a patient is hemodynamically stable, definitive imaging can be performed

156_Cydulka_Ch156.indd 867 22/03/17 10:32 AM

Source page 887

with a CT scan of the abdomen and pelvis with IV contrast. Patients with penetrating abdominal trauma who are in shock should undergo early operative intervention.

6. Patients who sustain major trauma may develop a bleeding diathesis, which results in defective clotting and platelet function. If patients require >10 units of packed red blood cells (PRBC), patients should receive PRBC in a 1:1 ratio with fresh frozen plasma. Both acidosis and hypothermia contribute to the coagulopathy and should be corrected as soon as possible.

7. After the primary survey and stabilization, perform an abbreviated neurologic examination, including an assessment of the patient's level of consciousness, GCS, pupillary size and reactivity, and motor function. A search for the cause of depressed level of consciousness includes the measurement of capillary blood glucose levels and the consideration of possible intoxicants, though one should begin with the assumption this is due to significant traumatic brain injury (TBI). In order to quickly identify potentially operative intracranial injuries

among patients with suspected traumatic brain injury and coma (GCS of 3 to 8), defer any procedures that do not correct a specific problem during the primary survey until after the CT scan of the head is performed. Intubated patients should undergo continuous capnography. GCS assessment can be insensitive in patients with normal or near-normal scores, and a GCS score of 15 does not exclude the presence of TBI.

8. Once the patient is hemodynamically stable and the airway is secured, log-roll the patient with one team member assigned to maintain in-line cervical stabilization. Palpate the spinous processes of the thoracic and lumbar spine for tenderness or deformity. Consider performing a rectal examination to assess for gross blood, displaced prostate, or determining rectal tone in the setting of a suspected spinal injury.

9. Certain conditions, such as injuries to the esophagus, diaphragm, and small bowel, often remain undiagnosed even with diligent examination, and further imaging and hospital observation for delayed presentation may be required. The most frequently missed injuries are orthopedic.

10. Expeditiously transport patients with hemodynamic instability and ongoing bleeding to the operating room or transfer them to another facility with appropriate surgical or critical care resources. Serial examinations are essential in patients without obvious indications for surgery identified on the initial assessment. These examinations may be performed in the inpatient or, in some cases, the ED observation unit settings.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 254, "Trauma in Adults," by Peter Cameron and Barry J. Knapp.

156_Cydulka_Ch156.indd 868 22/03/17 10:32 AM

Source page 888

C ha p ter Trauma in Children

157 Matthew Hansen

Trauma is the most common cause of death in children older than 1 year. Differences in anatomy and physiology mandate modifications to trauma evaluation and management in children.

■■ CLINICAL FEATURES

Head trauma is the most frequent pediatric injury resulting in death. Overall, motor vehicle crash is the most common mechanism, and it is the leading mechanism of traumatic death in children older than 1 year.

Airway: Airway management in children can be challenging. Anatomic differences include a large occiput, large tongue, and cephalad location of the larynx.

Breathing: Observe the rate, depth, pattern, and work of breathing as well as symmetry of rise and fall of the chest wall. Agitation or somnolence could be a result of hypoxia or hypercapnea, respectively. Children experience oxygen desaturation more quickly due to high oxygen demand and small lungs.

Circulation: Recognize early signs of circulatory shock including tachycardia, mental status changes, and color and perfusion abnormalities, because hypotension is typically a terminal event in children. Estimate normal systolic blood pressure in children 1 to 10 years of age using the following formula: 90 + (2 × age) mm Hg; hypotension can be estimated as systolic blood pressure less than 70 + (2 × age) mm Hg.

Disability: In younger children, the Modified Pediatric Glasgow Coma Scale (Table 157-1) should be used. This mirrors the Glasgow Coma Scale for eye opening and motor responses, but incorporates age-appropriate modifications for verbal responses. Perform a pupillary examination and basic assessment of tone and strength.

Exposure: Disrobe and expose the child to completely assess for injuries. However, the ratio of surface area to mass is greater in children, putting them at greater risk for hypothermia. Care should be taken to maintain normothermia.

■■ DIAGNOSIS AND DIFFERENTIAL

Head Injury

Infants and neonates are at the highest risk of significant intracranial injury. Mental status assessment should account for developmental stage and patient anxiety. Parietal and occipital skull fractures are frequently

Table 157-1 Modified Pediatric Glasgow Coma Scale

Coos or babbles = 5

Irritable cry = 4

Cries to pain = 3

Moans to pain = 2

No response = 1

157_Cydulka_Ch157.indd 869 22/03/17 10:49 AM

Source page 889

associated with intracranial bleeding. Noncontrast CT is the imaging modality of choice for intracranial injury in children. Scalp injuries, particularly in neonates, may result in significant blood loss and shock. Please

see Chapter 160 for a discussion on CT imaging in head trauma.

Spine Injuries

Young children with spinal cord injuries often do not have associated fractures since the ligaments are relatively elastic. “Clearing the cervical spine” in children is challenging as there is little evidence to guide practice. Multisystem trauma and head trauma are general indications for neck immobilization and cervical spine imaging. Please see Chapter 161 for a detailed discussion on the management of spine and spinal cord injuries. Due to the nature of spine injuries and the need to lower ionizing radiation, plain films of the spine remain a useful tool.

Chest Trauma

The chest radiograph is an essential tool in the evaluation of any child with chest trauma and CT imaging of the chest rarely changes management. Pulmonary contusions are the most common thoracic injury in children. The relatively compliant chest wall of the child means that serious injuries to intrathoracic structures can be present without significant external signs. Rib fractures are less common in children and generally require a significant mechanism of injury.

Abdominal and Genitourinary Trauma

Indications for CT include suspicious mechanism of injury, tenderness on exam, seatbelt sign, distention, and vomiting. A study of over 12,000 pediatric patients with blunt abdominal trauma recently found the following factors predicted very low risk for significant intraabdominal injury: (1) no evidence of abdominal wall trauma/seat belt sign nor Glasgow Coma Scale <14 with blunt abdominal trauma; (2) no abdominal tenderness on examination; and (3) no thoracic wall trauma, and no complaints of abdominal pain, decreased bowel sounds, or vomiting. Identification of a pelvic fracture, particularly an anterior ring fracture, should prompt investigation for associated urethral or bladder injury. The role of ultrasound for pediatric abdominal trauma, especially in stable patients, is not well established.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. Perform basic airway maneuvers such as jaw thrust and oropharyngeal suctioning and maintain a sniffing position to align the airway axes, often by placing a towel roll under the shoulders, which is especially helpful in younger children to align the airway.

2. Orotracheal intubation is indicated for definitive airway management. The following formula is used to estimate the endotracheal tube size: size = 4 + (age/4), subtract ½ size for cuffed tubes which are often preferred. Maintain in-line cervical spine stabilization at all times. Rapid sequence intubation is the safest method of intubating a trauma patient

with a full stomach. When possible, limit positive-pressure ventilation

before intubation to avoid gastric insufflation and vomiting.

157_Cydulka_Ch157.indd 870 22/03/17 10:49 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.

- Content ID: SEG-TINTINALLI-EMM8-FULL.

- Source pages: 886–889.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 157: Trauma in Children

Source pages 890–893 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 890–893

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Trauma in Children**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- Place two proximal IVs in critically ill patients. [Source pages 890–893]

- If central venous access is needed, the femoral vein is the most easily accessed site. [Source pages 890–893]

- Administer fluids in 20 mL/kg boluses of crystalloid; if there is no response to two boluses, then administer 10 mL/kg boluses of packed red blood cells. [Source pages 890–893]

- Permissive hypotension is not an evidence-based prac- tice in children and may be harmful. [Source pages 890–893]

- Consider using a massive transfu- sion protocol if large volume resuscitation is anticipated. [Source pages 890–893]

Clinical reasoning structure

- **Problem representation:** Trauma in Children; source pages 890–893.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Trauma in Children**.

- Source-page range: **890–893**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 890

CHAPTER 157: Trauma in Children 871

3. Vascular access can be challenging. Place two proximal IVs in critically ill patients. Obtain intraosseous access early, as needed. If central venous access is needed, the femoral vein is the most easily accessed site. Administer fluids in 20 mL/kg boluses of crystalloid; if there is no response to two boluses, then administer 10 mL/kg boluses of packed red blood cells. Permissive hypotension is not an evidence-based practice in children and may be harmful. Consider using a massive transfusion protocol if large volume resuscitation is anticipated.

4. For pain control, fentanyl 1 µg/kg or morphine 0.05 to 0.1 mg/kg are appropriate.

5. If a head injured patient has clinical signs of impending herniation, maintain the PaCO2 at 35 to 40 mm Hg, optimize blood pressure with IV fluids, and elevate the head of the bed 20° to 30°. See Table 157-2 for head injury management suggestions.

Table 157-2 Management of Serious Traumatic Brain Injury in Children

Considerations Primary Goals Comments

Cervical spine Maintain spinal precautions

Airway Maintain airway, intubate for GCS <8 or as needed for oxygenation and ventilation

Oxygenation and ventilation Oxygen saturation >90%; No prophylactic hyperventilation PCO2 35–40 mm Hg

Blood pressure SBP > 70 + (2 × age) No permissive hypotension

Cerebral perfusion pressure (CPP) CPP = 40–65 mm Hg

Intracranial pressure (ICP) monitoring ICP ≤20 mm Hg Consultation with neurosurgeon

GCS GCS before paralytics if possible Serial GCS to document changes

Sedation and pain Midazolam Consider paralytics once

management Fentanyl thorough neurologic examination
is completed
Neuroimaging Identify intracranial injury Transcranial Doppler may be
(noncontrast head CT and signs of increased useful in infants with open
and cervical spine CT intracranial pressure or fontanelles but requires an
when indicated) herniation experienced pediatric radiologist
Glucose Treat hypoglycemia and Maintain normal blood glucose
hyperglycemia
Increased ICP/impend- Elevate head of bed 3% normal saline 5 mL/kg bolus
ing herniation 30 degrees over 10 min followed by infusion
of 0.1 mL/kg/h to maintain
serum Na within 155–165 mEq/L
or
Mannitol 0.5–1 g/kg if
normotensive (response is not
dose-dependent)
Core temperature Maintain temperature Hypothermia in children not rec-
36–38°C ommended; avoid hyperthermia
(Continued)
157_Cydulka_Ch157.indd 871 22/03/17 10:49 AM

Source page 891

872 SECTION 18: Trauma
Table 157-2 Management of Serious Traumatic Brain Injury in Children (Continued)
Considerations Primary Goals Comments
Seizure prophylaxis Optional: consider for Phenytoin 20 mg/kg (or
children with witnessed fosphenytoin 20 PE/kg)
posttraumatic seizures and or
intracranial blood Levetiracetam 10–20 mg/kg
(maximum, 500 mg/dose) for first
week following severe traumatic
brain injury
Anemia Transfuse for hemoglobin
<7 g/dL
Neurosurgery/transfer ICP monitoring, evacuation of
intracranial blood, or cerebro-
spinal fluid shunt for refrac-
tory intracranial hypertension
Abbreviations: GCS, Glasgow Coma Scale; PCO2, partial pressure of carbon dioxide; PE, phenytoin
equivalent; SBP, systolic blood pressure.
6. Admit children with intracranial hemorrhage, spinal trauma, significant

chest trauma, abdominal trauma with internal organ injury, significant burns, or other concerning injuries. Indications for referral to a pediatric trauma center are listed in Table 157-3.

Table 157-3 Indications for Transfer to a Pediatric Trauma Center

Mechanism of injury	Ejection from motor vehicle
	Fall from a significant height
	Motor vehicle collision with prolonged extrication
	Motor vehicle collision with death of another vehicle occupant
Anatomic injury	Multiple severe trauma
	More than three long-bone fractures
	Spinal fractures or spinal cord injury
	Amputations
	Severe head or facial trauma
	Penetrating head, chest, or abdominal trauma

Source: Reproduced with permission from Harris BH, Barlow BA, Ballantine TV, et al: American Pediatric Surgical Association principles of pediatric trauma care. J Pediatr Surg. 1992;27:423. Copyright Elsevier.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 110, "Pediatric Trauma," by Camilo E. Gutiérrez.

157_Cydulka_Ch157.indd 872 22/03/17 10:49 AM

Source page 892

C h a p ter Trauma in the Elderly

158 O. John Ma

Emergency physicians need to be aware of the many unique mechanisms of injury and clinical features associated with geriatric trauma patients and apply special management principles when caring for them.

■■ CLINICAL FEATURES

Falls are the most common cause of fatal and non-fatal injury in people ≥65 years of age. Syncope, which has been implicated in many cases, may be secondary to dysrhythmias, venous pooling, autonomic derangement, hypoxia, anemia, or hypoglycemia. Motor vehicle crashes are the second most common cause of injury in the elderly and are the leading cause of death. Elderly pedestrians struck by a motor vehicle are much more likely to die than younger pedestrians. Intentional injuries and those caused by neglect should also be considered.

Evaluate the geriatric trauma patient as both a medical and a trauma patient. Since elderly patients may have a significant past medical history that impacts their trauma care, obtaining a precise history is vital. Family members, medical records, and the patient's primary physician may be helpful in gathering information regarding the traumatic event and the patient's previ-

ous level of function. Document medications, such as cardiac agents, diuretics, psychotropic agents, and anticoagulants. Investigating the cause of a fall may uncover serious underlying medical causes or prevent future trauma.

On physical examination, frequent monitoring of vital signs is essential. Avoid feeling reassured by “normal” vital signs. A normal tachycardic response to pain, hypovolemia, or anxiety may be absent or blunted in the elderly trauma patient. Medications such as β-blockers may mask tachycardia and delay appropriate resuscitation. Blood pressures are also misleading in the elderly patient. Because of the high incidence of underlying hypertension, consider using a higher cutoff for hypotension than in younger patients. In blunt trauma patients ≥65 years, mortality increases when systolic blood pressure dips below 110 mm Hg and heart rates exceed 90 beats per minute.

Pay special attention to anatomical variations that may make airway management more difficult. These include the presence of dentures, cervical arthritis, or temporomandibular joint arthritis. A thorough secondary survey is essential to uncover less serious injuries. These “minor” injuries may not be severe enough to cause problems during the initial resuscitation, but cumulatively may cause significant morbidity and mortality. Seemingly stable geriatric trauma patients can deteriorate rapidly and without warning.

■■ DIAGNOSIS AND DIFFERENTIAL

Head Injury

Never assume that alterations in mental status are due solely to any underlying dementia or senility when evaluating the elderly patient’s mental status. Elderly persons suffer a much lower incidence of epidural

158_Cydulka_Ch158.indd 873 22/03/17 10:58 AM

Source page 893

hematomas than the general population; however, there is a higher incidence of subdural and intraparenchymal hematomas in the elderly than in younger patients. The rate of intracranial hemorrhage approaches 7% to 14% in anticoagulated patients with blunt head injury who are experiencing no or minimal symptoms. Order non-contrast head CT for patients who take warfarin and have a mechanism of injury concerning for even a minor head injury.

Cervical Spine Injuries

The pattern of cervical spine injuries in the elderly is different than in younger patients, as there is an increased incidence of C1 and C2 fractures with the elderly. Place special emphasis on maintaining cervical immobilization until the cervical spine is properly assessed. Because of

the higher pre-test probability of injury, as well as the difficulties in interpreting plain radiographs in a patient with age-related degeneration, CT scan is the preferred initial modality for assessing the geriatric cervical spine.

Chest Trauma

Rib fractures are more common after blunt trauma due to osteoporotic changes. The pain associated with rib fractures, along with any decreased physiologic reserve, may predispose patients to respiratory complications. More severe thoracic injuries, such as hemopneumothorax, pulmonary contusion, flail chest, and cardiac contusion, can quickly lead to decompensation in elderly individuals whose baseline oxygenation status may already be diminished.

Abdominal Trauma

The abdominal examination in elderly patients is notoriously unreliable. The focused assessment with sonography for trauma (FAST) examination assists in evaluating for hemoperitoneum and the need for exploratory laparotomy in hemodynamically unstable patients. The FAST examination has largely replaced diagnostic peritoneal lavage. Use CT with contrast for patients who are hemodynamically stable. Ensure adequate hydration and baseline assessment of renal function prior to the contrast load for the CT scan. Some patients may be volume-depleted due to medications, such as diuretics. This hypovolemia coupled with contrast administration may exacerbate any underlying renal pathology.

Orthopedic Injuries

Hip fractures occur primarily in four areas: intertrochanteric, transcervical, subcapital, and subtrochanteric. Intertrochanteric fractures are the most common, followed by transcervical fractures. Be aware that pelvic and long bone fractures are not infrequently the sole etiology for hypovolemia in elderly patients. Coordinate timely orthopedic consultation, evaluation, and treatment with open reduction and internal fixation with the diagnosis and management of other injuries. Consider ordering CT of the pelvis in stable patients with pelvic tenderness after an injury if plain radiographs are negative.

158_Cydulka_Ch158.indd 874 22/03/17 10:58 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 890–893.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 158: Trauma in the Elderly

Source pages 894–897 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 894–897

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Trauma in the Elderly**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 158: Trauma in the Elderly 875 Long bone fractures of the femur, tibia, and humerus may produce a loss of mobility with a resulting decrease in the independent lifestyle of elderly patients. [Source pages 894–897]

- Early orthopedic consultation for intramedullary rodding of these fractures may result in increased early mobilization. [Source pages 894–897]

- The incidence of Colles fractures and humeral head and surgical neck fractures in elderly patients is increased by falls on the outstretched hand or elbow. [Source pages 894–897]

- Localized tenderness, swelling, and ecchymosis to the proximal humerus are characteristic of these injuries. [Source pages 894–897]

- ■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION As in all trauma patients, perform the primary and secondary surveys expeditiously. [Source pages 894–897]

Clinical reasoning structure

- **Problem representation:** Trauma in the Elderly; source pages 894–897.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Trauma in the Elderly**.

- Source-page range: **894–897**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 894

CHAPTER 158: Trauma in the Elderly 875

Long bone fractures of the femur, tibia, and humerus may produce a loss of mobility with a resulting decrease in the independent lifestyle of elderly patients. Early orthopedic consultation for intramedullary rodding of these fractures may result in increased early mobilization.

The incidence of Colles fractures and humeral head and surgical neck fractures in elderly patients is increased by falls on the outstretched hand or elbow. Localized tenderness, swelling, and ecchymosis to the proximal humerus are characteristic of these injuries. Arrange for early orthopedic referral.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

As in all trauma patients, perform the primary and secondary surveys expeditiously.

1. The main therapeutic goal is maintaining adequate oxygen delivery. Consider prompt tracheal intubation and use of mechanical ventilation in patients with more severe injuries, respiratory rates > 40 breaths/min, or when the PaO_2 is < 60 mm Hg or $PaCO_2$ > 50 mm Hg. While nonventilatory therapy helps prevent respiratory infections and is always desirable, early mechanical ventilation may avert the disastrous results associated with hypoxia.

2. Geriatric trauma patients can decompensate with over-resuscitation just as quickly as they can with inadequate resuscitation. During the initial resuscitative phase, administer crystalloid judiciously since elderly patients with diminished cardiac compliance are more susceptible to volume overload. Consider early and more liberal use of packed red blood cell transfusion. This practice early in the resuscitation enhances oxygen delivery and helps minimize tissue ischemia.

3. Base deficit and lactate levels are useful initial indicators of shock, and serial measurements can guide resuscitation progress.

4. Early invasive monitoring has been advocated to help physicians assess

the elderly's hemodynamic status. One study found that urgent invasive monitoring provides important hemodynamic information early, aids in identifying occult shock, limits hypoperfusion, helps prevent multiple organ failure, and improves survival. Given the complications associated with pulmonary-artery catheters, research on the assessment of shock by serum lactate and central or mixed venous oxygen saturation, and ongoing advances in less invasive hemodynamic monitoring and echocardiography, the optimal approach to recognizing and treating occult hypoperfusion is uncertain. A general strategy is to perform the initial imaging necessary to identify life-threatening injuries (e.g., CT of the head, spine, chest, abdomen, and pelvis) and then transport to the ICU for aggressive optimization of hemodynamics, after which nonessential imaging and interventions (e.g., extremity radiographs and suturing) can be performed.

5. Rapidly reverse anticoagulation in patients taking warfarin with intracranial bleeding on CT scan. Lacking sufficient evidence for or against, it is reasonable to reverse other forms of anticoagulation (e.g., aspirin, clopidogrel, heparin) in patients with diagnosed intracranial hemorrhage.

158_Cydulka_Ch158.indd 875 22/03/17 10:58 AM

Source page 895

6. Maintain a low threshold for admitting elderly patients with rib fractures for a period of observation until good pain control and pulmonary toilet is assured.

7. Admit elderly patients with polytrauma, significant chest wall injuries, abnormal vital signs, or evidence of overt or occult hypoperfusion to the ICU.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 255, "Trauma in the Elderly," by Ross J. Fleischman and O. John Ma.

158_Cydulka_Ch158.indd 876 22/03/17 10:58 AM

Source page 896

C h a p ter Trauma in Pregnancy

159 John Ashurst

Trauma is the leading cause of nonobstetric morbidity and mortality in pregnant women. Motor vehicle collisions followed by falls and domestic violence are the most common causes of trauma in pregnancy and fetal

survival is highly dependent on maternal stabilization.

■■ CLINICAL FEATURES

Physiologic changes of pregnancy make it difficult to determine the severity of injury. Heart rate increases 10 to 20 beats per minute in the second trimester while systolic and diastolic blood pressures drop 10 to 15 mm Hg. Blood volume can increase by 45%, but red cell mass increases to a lesser extent, leading to a physiologic anemia of pregnancy. It may be difficult to determine whether tachycardia, hypotension, or anemia is due to blood loss or normal physiologic changes. Due to the relative hypervolemic state, the patient may lose 30% to 35% of blood volume before manifesting signs of shock. Pulmonary changes in pregnancy include elevation of the diaphragm and a decrease in residual volume and function residual capacity. Tidal volume increases, resulting in hyperventilation with associated respiratory alkalosis. However, renal compensation causes the serum pH to remain unchanged. Gastric emptying is also delayed, which places the pregnant trauma patient at a higher risk of aspiration.

The anatomic changes in pregnancy affect the types of injuries that are typically seen in the mother. Splenic injury remains the most common cause of abdominal hemorrhage in the pregnant trauma patient. After the 12th week of gestation, the enlarging uterus emerges from the pelvis and by 20 weeks reaches the level of the umbilicus. Uterine blood flow increases, making severe maternal hemorrhage from uterine trauma more likely. The uterus also can compress the inferior vena cava when the patient is supine, leading to the “supine hypotension syndrome.” As pregnancy progresses, the small intestines are pushed cephalad, which causes an increased likelihood of injury in penetrating trauma to the upper abdomen. The bladder moves into the abdomen in the third trimester, thereby increasing its susceptibility to injury.

Abdominal trauma affects not only the mother but also the fetus. Fetal injuries are more likely to be seen in the third trimester and are often associated with pelvic fractures or penetrating trauma in the mother. Uterine rupture is rare but is associated with a very high fetal mortality rate. More common complications of trauma include uterine irritability, preterm labor, and placental abruption. Classically, the mother will demonstrate abdominal pain, vaginal bleeding, and uterine contractions. Fetal–maternal hemorrhage occurs in more than 30% of cases of significant trauma and may result in rhesus (Rh) isoimmunization of Rh-negative women.

877

159_Cydulka_Ch159.indd 877 22/03/17 11:03 AM

Source page 897

■■ DIAGNOSIS AND DIFFERENTIAL

Maternal stability and survival offer the best chance for fetal well-being, and no critical interventions or diagnostic procedures should be withheld out of concern for potential adverse effects to the fetus. The initial sequence of trauma resuscitation is unchanged. Special attention should be directed to the gravid abdomen, examining for evidence of injury, tenderness, or uterine contractions. If abdominal or pelvic trauma is suspected, perform a sterile pelvic examination to assess for genital trauma, vaginal bleeding, or ruptured amniotic membranes after pelvic ultrasound to determine placental location. Fluid with a pH of 7 in the vaginal canal suggests amniotic rupture, as does “ferning,” a branch-like pattern on drying of vaginal fluid on a microscope slide.

Initial laboratory studies include a complete blood count, serum chemistries, blood type, Rh status, and coagulation studies including fibrin split products and fibrinogen to determine the presence of disseminated intravascular coagulation. The Apt test or Kleihauer–Betke test should be obtained to assess for the presence of fetal hemoglobin in the maternal blood.

Obtain radiographs based on fundamental principles of trauma management. Adverse fetal effects from radiation are negligible from doses <5 rad, which is an exposure far greater than that received from most plain radiographs. Reducing the number of imaging CT cuts and shielding of the abdomen and pelvis when possible may decrease radiation exposure from medical imaging. Bedside ultrasonography is a highly sensitive, specific, and radiation-free alternative for imaging the abdomen. In addition to evaluating fetal heart rate, ultrasonography can assess gestational age, fetal activity or demise, placental location, and amniotic fluid volume. Diagnostic peritoneal lavage has largely been replaced by ultrasonography. If it is indicated, use the open supraumbilical technique.

Auscultate fetal heart tones to determine fetal viability and identify fetal distress early in the evaluation. A Doppler stethoscope or ultrasound facilitates this assessment. A normal fetal heart rate ranges between 120 and 160 beats per minute. Fetal bradycardia is most likely a result of hypoxia due to maternal hypotension, respiratory compromise, or placental abruption. Fetal tachycardia is most likely due to hypoxia or hypovolemia. In the setting of blunt abdominal trauma, external fetal monitoring is indicated for at least 4 to 6 hours for all patients beyond week 20 of gestation. Fetal tachycardia, lack of beat-to-beat or long-term variability, or late decelera-

tions on tocodynamometry are diagnostic of fetal distress and may be indications for emergent cesarean section if beyond the viable gestational age.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

As is the case of all trauma patients, initial priorities are the primary and secondary surveys directed at the pregnant trauma patient. Coordinate care with surgical and obstetric consultants.

1. Aggressive maternal resuscitation generally leads to the best possible fetal resuscitation.

2. Initiate supplemental oxygen and crystalloid infusions at 50% above that given to nonpregnant patients. For patients beyond week 20 of gestation who must remain supine, place a wedge under the right hip,

159_Cydulka_Ch159.indd 878 22/03/17 11:03 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 894–897.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 159: Trauma in Pregnancy

Source pages 898–899 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 898–899

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Trauma in Pregnancy**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 159: Trauma in Pregnancy 879 tilting the patient 30° to the left, thus reducing the likelihood of supine hypotension syndrome. [Source pages 898–899]

- Otherwise, keep the patient in a left lateral decubitus position. [Source pages 898–899]

- Avoid vasopressors if possible as they can have deleterious effects on uterine perfusion. [Source pages 898–899]

- Administer tetanus prophylaxis when indicated. [Source pages 898–899]

- Give Rho (D) immunoglobulin for all Rh-negative pregnant patients if indicated. [Source pages 898–899]

Clinical reasoning structure

- **Problem representation:** Trauma in Pregnancy; source pages 898–899.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Trauma in Pregnancy**.

- Source-page range: **898–899**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 898

CHAPTER 159: Trauma in Pregnancy 879

tilting the patient 30° to the left, thus reducing the likelihood of supine hypotension syndrome. Otherwise, keep the patient in a left lateral decubitus position.

3. Avoid vasopressors if possible as they can have deleterious effects on uterine perfusion.

4. Administer tetanus prophylaxis when indicated.

5. Give Rho (D) immunoglobulin for all Rh-negative pregnant patients if indicated.

6. Institute cardiotocodynamometry as soon as possible to monitor for fetal distress and uterine irritability.

7. Tocolytics have a variety of side effects, including fetal and maternal tachycardia. Administer only in consultation with an obstetrician.

8. Indications for emergent laparotomy in the pregnant patient remain the same as those in the nonpregnant patient.

9. The decision to admit or discharge a pregnant trauma patient is first based on the nature and severity of the presenting injuries. Women at 20 weeks of gestation require observation or admission for a minimum of 4 to 6 hours for external tocodynamometric monitoring.

10. Screen for potential intimate partner violence.

11. Instruct discharged patients to seek medical attention immediately if they develop abdominal pain or cramps, vaginal bleeding, leakage of fluid, or perception of decreased fetal activity.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 256, “Trauma in Pregnancy,” by Nicole M. Deiorio.

159_Cydulka_Ch159.indd 879 22/03/17 11:03 AM

Source page 899

C h a p ter Head Trauma

160 O. John Ma

■■ CLINICAL FEATURES

Traumatic brain injury (TBI) is the impairment in brain function after direct or indirect forces to the brain. The force of an object striking the head or a penetrating injury causes direct injury. Indirect injuries occur from acceleration/deceleration forces that result in the movement of the brain within the skull.

Traumatic brain injury can be classified as mild, moderate, and severe. Mild TBI includes patients with a Glasgow Coma Scale (GCS, see Table 160-1) score ≥14. Patients may be asymptomatic with only a history of head trauma, or may be confused and amnestic of the event. They may have experienced a brief loss of consciousness and complain

Table 160-1 Glasgow Coma Scale for All Age Groups

4 Years to Adult Child <4 Years Infant

Eye opening

4 Spontaneous Spontaneous Spontaneous

3 To speech To speech To speech

2 To pain To pain To pain

1 No response No response No response

Verbal response

5 Alert and oriented Oriented, social, Coos, babbles

speaks, interacts

4 Disoriented Confused speech, Irritable cry

conversation disoriented, consolable,

aware

3 Speaking but Inappropriate words, Cries to pain

nonsensical inconsolable, unaware

2 Moans or unintel- Incomprehensible, Moans to pain

ligible sounds agitated, restless,

unaware

1 No response No response No response

Motor response

6 Follows commands Normal, spontaneous Normal, spontane-

movements ous movements

5 Localizes pain Localizes pain Withdraws to touch

4 Moves or withdraws Withdraws to pain Withdraws to pain

to pain

3 Decorticate flexion Decorticate flexion Decorticate flexion

2 Decerebrate Decerebrate extension Decerebrate

extension extension

1 No response No response No response

3–15

Note: In intubated patients, the Glasgow Coma Scale verbal component is scored as a 1, and the total score is marked with a “T” (or tube) denoting intubation (e.g., 8T).

880

160_Cydulka_Ch160.indd 880 22/03/17 11:10 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 898–899.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 160: Head Trauma

Source pages 900–905 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 900–905

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Head Trauma**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 160: Head Trauma 881 of a diffuse headache, nausea, and vomiting. [Source pages 900–905]

- Patients at high risk in this subgroup include those with a skull fracture, large subgaleal swelling, focal neurologic findings, coagulopathy, age >60 years, or drug/alcohol intoxication. [Source pages 900–905]

- Moderate TBI includes patients with a GCS score of 9 to 13. [Source pages 900–905]

- Overall, 40% of these patients have an abnormality on CT scan and 8% require neurosurgical intervention. [Source pages 900–905]

- The mortality of severe TBI (GCS score <9) approaches 40%. [Source pages 900–905]

Clinical reasoning structure

- **Problem representation:** Head Trauma; source pages 900–905.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Head Trauma**.

- Source-page range: **900–905**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 900

CHAPTER 160: Head Trauma 881

of a diffuse headache, nausea, and vomiting. Patients at high risk in this subgroup include those with a skull fracture, large subgaleal swelling, focal neurologic findings, coagulopathy, age >60 years, or drug/alcohol intoxication.

Moderate TBI includes patients with a GCS score of 9 to 13. Overall, 40% of these patients have an abnormality on CT scan and 8% require neurosurgical intervention.

The mortality of severe TBI (GCS score <9) approaches 40%. The immediate clinical priority in these patients is to prevent secondary brain injury, identify other life-threatening injuries, and identify treatable neurosurgical conditions.

Prehospital medical personnel often provide critical parts of the history, including mechanism and time of injury, presence and length of unconsciousness, initial mental status, seizure activity, vomiting, verbalization, and movement of extremities. For an unresponsive patient, contact family and friends to gather key information including past medical history, medications (especially anticoagulants), and recent use of alcohol or drugs.

Perform a detailed neurologic examination that includes assessing the mental status and GCS, pupils for size, reactivity, and anisocoria, cranial nerve function, motor and sensory function, and any development of decorticate or decerebrate posturing.

Specific Injuries

Skull Fractures

Depressed skull fractures are classified as open or closed, depending on the integrity of the overlying scalp. Although basilar skull fractures can occur at any point in the base of the skull, the typical location is in the petrous portion of the temporal bone. Findings associated with a basilar skull fracture include hemotympanum, cerebrospinal fluid (CSF) otorrhea or rhinorrhea, periorbital ecchymosis (“raccoon eyes”), and retroauricular

ecchymosis (Battle's sign).

Cerebral Contusion and Intracerebral Hemorrhage

Common locations for contusions are the frontal poles, the subfrontal cortex, and the temporal lobes. Contusions may occur directly under the site of impact or on the contralateral side (contrecoup lesion). The contused area is usually hemorrhagic with surrounding edema, and occasionally associated with subarachnoid hemorrhage. Neurologic dysfunction may be profound and prolonged, with patients demonstrating mental confusion, obtundation, or coma. Focal neurologic deficits are usually present.

Traumatic Subarachnoid Hemorrhage

This condition results from the disruption of subarachnoid vessels and presents with blood in the CSF. Patients may complain of diffuse headache, nausea, or photophobia. Traumatic subarachnoid hemorrhage may be the most common CT abnormality in patients with moderate or severe TBI. Some cases may be missed if the CT scan is obtained less than 6 hours after injury.

160_Cydulka_Ch160.indd 881 22/03/17 11:10 AM

Source page 901

882 SECTION 18: Trauma

Epidural Hematoma

An epidural hematoma results from an acute collection of blood between the inner table of the skull and the dura mater. It is typically associated with a skull fracture that lacerates a meningeal artery, most commonly the middle meningeal artery. Underlying injury to the brain may not necessarily be severe. In the classic scenario, the patient experiences loss of consciousness after a head injury. The patient may present to the ED with clear mentation, signifying the "lucid interval," and then begin to develop mental status deterioration in the ED. A fixed and dilated pupil on the side of the lesion with contralateral hemiparesis is a classic late finding. The high-pressure arterial bleeding of an epidural hematoma can lead to herniation within hours of injury. An epidural hematoma appears biconvex on CT scan.

Subdural Hematoma

A subdural hematoma (SDH), which is a collection of venous blood between the dura matter and the arachnoid, results from tears of the bridging veins that extend from the subarachnoid space to the dural venous sinuses. A common mechanism is sudden acceleration–deceleration. Patients with brain atrophy, such as in alcoholics or the elderly, are more susceptible to a SDH. In acute SDH, patients present within 14 days of the injury, and most become symptomatic within 24 hours of injury. After 2 weeks, patients are defined as having a chronic SDH. Symptoms may range from a headache to

lethargy or coma. It is important to distinguish between acute and chronic SDHs by history, physical examination, and CT scan. An acute SDH appears as a hyperdense, crescent-shaped lesion that crosses suture lines.

Herniation

Diffusely or focally increased intracranial pressure (ICP) can result in herniation of the brain at several locations. Transtentorial (uncal) herniation occurs when a SDH or temporal lobe mass forces the ipsilateral uncus of the temporal lobe through the tentorial hiatus into the space between the cerebral peduncle and the tentorium. This results in compression of the oculomotor nerve and parasympathetic paralysis of the ipsilateral pupil, causing it to become fixed and dilated. When the cerebral peduncle is further compressed, it results in contralateral motor paralysis. The increased ICP and brainstem compression result in progressive deterioration in the level of consciousness. Occasionally, the contralateral cerebral peduncle is forced against the free edge of the tentorium on the opposite side, resulting in paralysis ipsilateral to the lesion—a false localizing sign. Central transtentorial herniation occurs with midline lesions in the frontal or occipital lobes, or in the vertex. Bilateral pinpoint pupils, bilateral Babinski signs, and increased muscle tone are found initially, which eventually develop into fixed midpoint pupils, prolonged hyperventilation, and decorticate posturing. Cerebellotonsillar herniation through the foramen magnum occurs much less frequently. Medullary compression causes flaccid paralysis, bradycardia, respiratory arrest, and sudden death.

Penetrating Injuries

Gunshot wounds and penetrating sharp objects can result in penetrating injury to the brain. The degree of neurologic injury will depend on the

160_Cydulka_Ch160.indd 882 22/03/17 11:10 AM

Source page 902

energy of the missile, whether the trajectory involves a single or multiple lobes or hemispheres of the brain, the amount of scatter of bone and metallic fragments, and whether a mass lesion is present.

■■ DIAGNOSIS AND DIFFERENTIAL

Tables 160-2 and 160-3 provide evidence-based indications for obtaining a CT scan of the head after injury.

Table 160-2 New Orleans Criteria and Canadian CT Head Rule Clinical Decision Rules

New Orleans Criteria—GCS 15*	Canadian CT Head Rule—GCS 13–15*
Headache	GCS <15 at 2 h
Vomiting	Suspected open or depressed skull fracture

Age >60 years Age ≥65 years

Intoxication More than one episode of vomiting

Persistent antegrade amnesia Retrograde amnesia >30 min

Evidence of trauma above the clavicles Dangerous mechanism (fall >3 ft or struck as pedestrian)

Seizure Any sign of basal skull fracture

Identification of patients who have an intracranial lesion on CT

100% sensitive, 5% specific 83% sensitive, 38% specific

Identification of patients who will need neurosurgical intervention

100% sensitive, 5% specific 100% sensitive, 37% specific

*

Presence of any one finding indicates need for CT scan.

Abbreviation: GCS, Glasgow Coma Scale.

Table 160-3 CT Scanning for Adults with Brain Injury (American College of Emergency Physicians Guidelines)

Adults with a Glasgow Coma Scale score of <15 at the time of evaluation should undergo CT imaging

Mild traumatic brain injury with or without loss of consciousness: if one or more of the following is present:

Glasgow Coma Scale score <15

Focal neurologic findings

Vomiting more than two times

Moderate to severe headache

Age >65 years

Physical signs of basilar skull fracture

Coagulopathy

Dangerous mechanism of injury (e.g., fall >4 ft)

Mild traumatic brain injury with loss of consciousness or amnesia: if one or more of the following is present:

Drug or alcohol intoxication

Physical evidence above the clavicles

Persistent amnesia

Posttraumatic seizures

160_Cydulka_Ch160.indd 883 22/03/17 11:10 AM

Source page 903

Approximately 8% of patients suffering a severe TBI will have an associated cervical spine fracture. Obtain imaging studies of the cervical spine on all trauma patients who present with altered mental status, neck pain, intoxication, neurologic deficit, severe distracting injury, or if the mecha-

nism of injury is deemed serious enough to potentially produce cervical spine injury.

Laboratory work should include type and crossmatching, complete blood count, basic metabolic panel, arterial blood gas analysis, directed toxicologic studies, and coagulation studies.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. Initiate standard protocols for evaluation and stabilization of trauma patients (see Chapter 156). Diligently assess for other significant injuries.

2. Administer oxygen, and secure cardiac monitoring and two IV lines. For patients with severe TBI, endotracheal intubation (via rapid sequence intubation) to protect the airway and prevent hypoxemia is the top priority. Provide cervical spine immobilization, and use an adequate sedation/induction agent when securing the airway.

3. Hypotension is associated with increased mortality rates. Restoration of an adequate blood pressure is vital to maintain cerebral perfusion. Resuscitation with IV crystalloid fluid to a mean arterial pressure (MAP) of 80 mm Hg is indicated; if aggressive fluid resuscitation is not effective, then add vasopressors to maintain a MAP of 80 mm Hg.

4. Obtain immediate neurosurgical consultation after a head CT scan demonstrating intracranial injury has been identified. Patients with new neurologic deficits from an acute epidural or SDH require emergent neurosurgical treatment.

5. All patients who demonstrate signs of increased ICP should have the head of their bed elevated to 30° (provided that the patient is not hypotensive), adequate sedation, and maintenance of adequate arterial oxygenation. If the patient is not hypotensive, consider administering mannitol 0.25 to 1.0 g/kg IV in repetitive boluses. If the patient is hypotensive or inadequately fluid resuscitated, consider administering hypertonic saline as an alternative to mannitol. The adult dose of 3% NaCl is 250 mL over 30 minutes.

6. Hyperventilation is not recommended as a prophylactic intervention to lower ICP because of its potential to cause cerebral ischemia. Reserve hyperventilation as a last resort for decreasing ICP; if used, implement it as a temporary measure and monitor the PCO2 closely to maintain a range of 30 to 35 mm Hg.

7. Patients with signs of impending brain herniation may need emergency decompression by trephination (“burr holes”) when all other methods to control the elevated ICP have failed. CT scan prior to attempting trephination is recommended to localize the lesion and direct the

decompression site.

8. Treat seizures immediately with benzodiazepines, such as lorazepam,

and fosphenytoin at a loading dose of 18 to 20 mg PE per kilogram IV.

9. Use of prophylactic anticonvulsants remains controversial, and its

administration should be in consultation with the neurosurgeon.

160_Cydulka_Ch160.indd 884 22/03/17 11:10 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 900–905.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 161: Spine Trauma

Source pages 906–911 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 906–911

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Spine Trauma**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 161: Spine Trauma 887 C1 C1 C1 C2 2 2 C3 C5,C6 - Deltoid Arm abduction 3 3 C4 4 4 Biceps Elbow flexion 5 C5 5 6 C6,C7 - Extensor Wrist extension C6 6 7 8 carpi radialis C7 7 T1 C7,C8 - Triceps Elbow extension C8 T1 2 T1 3 C8,T1 - Hand intrinsics Finger abduction 2 4 T2 Flexor digitorum profundus Hand grasp 3 5 T3 4 6 T4 7 T2–T7 - Chest muscles 5 T5 6 8 T6 7 9 T7 8 10 T8 T9 9 11 T10 10 12 T9–T12 - Abdominal muscles 11 L1 T11 2 T12 12 3 4 L1,L2,L3 - Iliopsoas Hip flexion L1 5 L1 2 L2 L2,L3,L4 - Quadriceps Knee extension 3 L3 L4,L5,S1,S2 Knee flexion 4 Hamstrings L4 L4,L5 Ankle dorsiflexion 5 Tibialis anterior L5 S1 L5,S1 - Extensor Great toe extension S2 hallucis longus S3 S1,S2 Ankle plantar flexion S4 Gastrocnemius S5 S2,S3,S4 - Bladder Voluntary rectal tone Co Anal sphincter Figure 161-1. [Source pages 906–911]

- The spinal cord level of injury can be delineated by physical examination, including a detailed neurologic examination. [Source pages 906–911]

- High-resolution CT is more sensitive and specific for cervical spine fractures than plain films, and is the modality of choice at most trauma centers for suspected cervical spine injuries. [Source pages 906–911]

- For plain radiography of the cervical spine, at least three views (lateral, odontoid, and anteroposterior) are necessary. [Source pages 906–911]

- Both CT and plain radiography can miss purely ligamentous injuries. [Source pages 906–911]

Clinical reasoning structure

- **Problem representation:** Spine Trauma; source pages 906–911.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Spine Trauma**.

- Source-page range: **906–911**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 906

CHAPTER 161: Spine Trauma 887

C1

C1 C1

C2

2 2

C3

C5,C6 - Deltoid Arm abduction 3 3

C4 4 4

Biceps Elbow flexion 5

C5 5 6

C6,C7 - Extensor Wrist extension C6 6 7

8

carpi radialis C7 7 T1

C7,C8 - Triceps Elbow extension C8 T1 2

T1 3

C8,T1 - Hand intrinsics Finger abduction 2 4

T2

Flexor digitorum profundus Hand grasp 3 5

T3

4 6

T4 7

T2–T7 - Chest muscles 5

T5

6 8

T6

7 9

T7

8 10

T8

T9 9 11

T10 10

12

T9–T12 - Abdominal muscles 11 L1

T11

2

T12 12 3

4

L1,L2,L3 - Iliopsoas Hip flexion L1 5

L1

2

L2

L2,L3,L4 - Quadriceps Knee extension 3

L3

L4,L5,S1,S2 Knee flexion 4

Hamstrings L4

L4,L5 Ankle dorsiflexion 5

Tibialis anterior L5

S1

L5,S1 - Extensor Great toe extension S2

hallucis longus S3

S1,S2 Ankle plantar flexion S4

Gastrocnemius S5

S2,S3,S4 - Bladder Voluntary rectal tone Co

Anal sphincter

Figure 161-1. Spinal cord level. The spinal cord level of injury can be delineated by physical examination, including a detailed neurologic examination.

High-resolution CT is more sensitive and specific for cervical spine fractures than plain films, and is the modality of choice at most trauma centers for suspected cervical spine injuries. For plain radiography of the cervical spine, at least three views (lateral, odontoid, and anteroposterior) are necessary.

Both CT and plain radiography can miss purely ligamentous injuries. Sensitivity is not high enough to rely on flexion-extension films. When ligamentous injury is suspected, reliable patients can be discharged with

Source page 907

888 SECTION 18: Trauma

C2

C3

C4

C7 C6 C5 C5 C6 C7

C8 T2 C8

T1 T3 T1

T4

T5

T6

T7

T8

T9

T10

T11

T12

L1

L5 L2 L3 L2 L5

S2 S2

S1 S1

L4 L3 S3 S3 L3 L4

S5 S6

S4 Coc

C2

C3

C5 C4 C4 C5

C6 C6

C7 T1 T2 T1 C7

C8 T3 C8

T4

T5

T6

T7

T8

T9

T10

T11

S1 S1

T12

L5 L3 L2 L1 L2 L3 L5

L4 L4

S3 S3

Figure 161-2. Dermatomes for sensory examination.

Alternatively, MRI provides the most sensitive and specific view of the ligaments and neural structures. The mechanisms, characteristics, and stability of common cervical spine fractures are summarized in Table 161-4.

Table 161-1 Four Major Incomplete Spinal Cord Syndromes

General

Syndrome Mechanisms Symptoms Prognosis*

Anterior cord Direct anterior cord Complete paralysis below Poor

compression the lesion with loss of pain

Flexion of cervical spine and temperature sensation

Thrombosis of anterior Preservation of propriocep-

spinal artery tion and vibratory function

Source page 908

CHAPTER 161: Spine Trauma 889

Table 161-2 NEXUS Criteria

Absence of midline cervical tenderness

Normal level of alertness and consciousness*

No evidence of intoxication

Absence of focal neurologic deficit

Absence of painful distracting injury†

*

Abnormal defined as Glasgow coma scale score <15; disorientation to person, place, time, or events; inability to remember three objects at 5 minutes; delayed or inappropriate response to external stimuli.

†

Any injury thought “to have the potential to impair the patient’s ability to appreciate other injuries.”

With regard to the thoracic and lumbar spine, plain radiography may still have a role in the mildly injured patient, but CT imaging is the standard at most trauma centers as it is more sensitive than plain radiography and CT is better at defining the extent and stability of spinal fractures.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

Treat blunt and penetrating injuries to the spine with identification and stabilization of identified injuries, and prevention of secondary injuries.

1. Airway and breathing should be assessed with a low threshold for endotracheal intubation for patients with cervical spine injury at C5 and above since the diaphragm is innervated by C3 through C5. Airway compromise can also occur due to soft tissue swelling associated with high cervical spine fractures.

2. Hypotension should be assumed to be from hypovolemic shock until proven otherwise. Hypotensive neurogenic shock should be considered if the patient has a probable spinal cord injury, does not have acute blood

loss, and is warm and relatively bradycardic.

3. Collars and long boards should be removed as soon as clinically appro-

priate. In patients with penetrating neck injuries who are neurologically

Table 161-3 Canadian Cervical Spine Rule for Radiography: Cervical Spine

Imaging Unnecessary in Patients Meeting These Three Criteria

Assessment Definitions

Assessment #1: High-risk factors include:

There are no high-risk factors that Age 65 years or older

mandate radiography. A dangerous mechanism of injury*

The presence of paresthesias in the extremities

Assessment #2: Low-risk factors include:

There are low-risk factors that allow a Simple rear-end motor vehicle crashes

safe assessment of range of motion. Patient able to sit up in the ED

Patient ambulatory at any time

Delayed onset of neck pain

Absence of midline cervical tenderness

Assessment #3: Can rotate neck 45 degrees to the left and to

The patient is able to actively rotate the right

his/her neck (regardless of pain).

*

Defined as fall from a height of >3 feet; an axial loading injury; high-speed motor vehicle crash, roll-

over, or ejection; motorized recreational vehicle or bicycle collision.

161_Cydulka_Ch161.indd 889 22/03/17 11:17 AM

Source page 909

890 SECTION 18: Trauma

Table 161-4 Cervical Spine Injuries

Flexion

Anterior subluxation (hyperflexion sprain) (stable)*

Bilateral interfacetal dislocation (unstable)

Simple wedge (compression) fracture (usually stable)

Spinous process avulsion (clay-shoveler’s fracture) (stable)

Flexion teardrop fracture (unstable)

Flexion-rotation

Unilateral interfacetal dislocation (stable)

Pillar fracture

Fracture of lateral mass (can be unstable)

Vertical compression

Jefferson burst fracture of atlas (potentially unstable)

Burst (bursting, dispersion, axial-loading) fracture (unstable)

Hyperextension

Hyperextension dislocation (unstable)

Avulsion fracture of anterior arch of atlas (stable)

Extension teardrop fracture (unstable)

Fracture of posterior arch of atlas (stable)

Laminar fracture (usually stable)

Traumatic spondylolisthesis (hangman’s fracture) (unstable)

Lateral flexion

Uncinate process fracture (usually stable)

Injuries caused by diverse or poorly understood mechanisms

Occipital condyle fractures (can be unstable)

Occipitoatlantal dissociation (highly unstable)

Dens fractures (type II and III are unstable)

*

Usual occurrence. Overall stability is dependent on integrity of the other ligamentous structures.

intact and fully conscious, spinal immobilization is no longer recommended. Log rolling and maintenance of inline immobilization should be used while removing the board.

4. Patients with cervical fractures will typically require admission. Spine precautions should be maintained and assessment and stabilization of other injuries should be performed.

5. Thoracic and lumbar fractures are high-energy fractures and patients are at risk for other injuries, such as injuries to the aorta, spinal cord, intraabdominal or thoracic organs. Most patients with acute fractures should be admitted to or managed by a spine surgeon. Patients with compression or wedge fractures with <40% vertebral height loss may, in consultation with a spine surgeon, be treated as an outpatient, but patients with burst fractures and chance fractures should be admitted. Imaging with CT should be considered.

6. Injuries to the sacral spine and nerve roots are unusual, and are usually found in association with pelvic fractures. Bladder and bowel dysfunction, radiculopathy and damage to the cauda equina can all result from sacral fractures. Isolated coccyx fractures usually require pain control and a rubber doughnut pillow.

7. High-dose steroids (methylprednisolone) are no longer routinely recommended for spinal cord injuries as a result of blunt trauma due to

161_Cydulka_Ch161.indd 890 22/03/17 11:17 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 906–911.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 162: Facial Injuries

Source pages 912–917 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 912–917

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Facial Injuries**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 162: Facial Injuries 893 Table 162-1 Important Clinical Issues in Facial Trauma (Continued) Eye Examine early before swelling of lids, or use retractors. [Source pages 912–917]

- Fat through eyelid wound indicates an orbital septum perforation. [Source pages 912–917]

- Widening of the distance between the medial canthi, or telecanthus, suggests serious nasoethmoidal-orbital complex trauma. [Source pages 912–917]

- Widening of the distance between the pupils, or hypertelorism, results from orbital dislocation and often is associated with blindness. [Source pages 912–917]

- Limited upward gaze occurs with entrapment of the inferior rectus or inferior oblique muscles, or damage to the oculomotor nerve. [Source pages 912–917]

Clinical reasoning structure

- **Problem representation:** Facial Injuries; source pages 912–917.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Facial Injuries**.

- Source-page range: **912–917**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 912

CHAPTER 162: Facial Injuries 893

Table 162-1 Important Clinical Issues in Facial Trauma (Continued)

Eye

Examine early before swelling of lids, or use retractors. Document visual acuity.

Fat through eyelid wound indicates an orbital septum perforation.

Widening of the distance between the medial canthi, or telecanthus, suggests serious nasoethmoidal-orbital complex trauma. Widening of the distance between the pupils, or hypertelorism, results from orbital dislocation and often is associated with blindness.

Examine extraocular muscle movements. Limited upward gaze occurs with entrapment of the inferior rectus or inferior oblique muscles, or damage to the oculomotor nerve.

Systematically examine the eye. Specifically, the pupil for teardrop sign pointing to globe rupture, the anterior chamber for hyphema, and swinging flashlight test for afferent papillary defect. Perform a fluoroscein test for corneal abrasions or ulcers.

Check intraocular pressure for evidence of orbital compartment syndrome only in absence of globe injury.

Nose

Crepitus over any facial sinus suggests sinus fracture.

Septal hematoma appears as blue, boggy swelling on nasal septum. Should be incised and drained to avoid a saddle nose deformity.

Ears

Auricular hematomas should be incised and drained to avoid a cauliflower deformity.

Cerebrospinal fluid leak, auditory canal lacerations, and hemotympanum suggest basilar skull fracture.

Oral

Jaw deviation due to mandible dislocation or condyle fracture. Malocclusion occurs in mandible, zygomatic, and Le Fort fractures.

Assess for missing or injured teeth.

Lacerations and mucosal ecchymosis suggest mandible fracture.

Place finger in external ear while the patient gently opens and closes jaw to detect condyle fractures.

Tongue blade test: Patient without fracture can bite down on a tongue blade enough to break blade twisted by examiner.

■■ DIAGNOSIS AND DIFFERENTIAL

Diagnosis of many maxillofacial injuries is made clinically and with radiographs. Plain films are helpful if CT is not available or to screen for injuries in low-risk patients. Facial CT is frequently required to make the definitive diagnosis and guide surgical management. Imaging recommendations based on suspected injury sites and pretest clinical suspicion are summarized in Table 162-2.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

During the primary survey of facial trauma patients, the airway must be secured and stabilized as clinically indicated. When endotracheal intubation is required, the orotracheal route is preferred over the nasotracheal route because of concern for nasocranial intubation, worsening of injury, and severe epistaxis. While rapid sequence intubation is the preferred method of airway management in trauma, always plan for a difficult airway in patients with facial trauma. To prevent the “can’t intubate/can’t oxygenate” failed airway, do not administer paralytics unless a patient can be bagged effectively or alternative airway devices or plans are in place.

162_Cydulka_Ch162.indd 893 22/03/17 11:25 AM

Source page 913

Table 162-2 Recommendations for Imaging Based on Level of Injury and Clinical Findings

Level	Significant Clinical Low Suspicion Findings	Additional Considerations
Frontal bone	Head CT Head CT (skull windows) Cervical spine CT with significant clinical findings	Facial CT with orbital involvement.
Midface	Waters’ view Face CT with coronal sections	Coronal face sections require coronal and axial cervical spine clearance for sections positioning.
Computer-generated, three-dimensional reconstructions with complex injuries	Head CT can replace Waters’ view	
Mandible	Panorex Mandible CT	Facial CT detects mandible fractures

Awake intubation with sedation and local airway anesthesia may allow the emergency physician to determine the feasibility of orotracheal intubation while still preserving a patient’s airway reflexes. When endotra-

cheal intubation appears impossible or is unsuccessful, perform emergent cricothyroidotomy to secure the airway. The laryngeal mask airway may be used as a temporizing measure, but it does not protect the airway from aspiration of stomach contents and may not be possible with injuries involving the pharynx.

Severe midfacial and mandibular injuries can result in substantial hemorrhage from the sphenopalatine and greater palatine branches of the external carotid artery. Posterior nasal epistaxis can be controlled with nasal tampons, dual balloon devices, or Foley catheter placement with layered gauze packing anteriorly, again being careful to avoid intracranial placement in severe midfacial fractures. Rarely, reduction of significantly displaced nasal fractures and Le Fort injuries is needed to stop arterial bleeding. If bleeding persists, immediate operative intervention may be required to ligate injured vessels. Alternatively, arterial embolization may be pursued to control bleeding from branches of the external carotid artery.

Management decisions will be dictated by the location and severity of the facial fractures, as well as concurrent injuries. Typically patients with sinus fractures should receive oral or intravenous antibiotics, such as first-generation cephalosporins, clindamycin, or amoxicillin-clavulanate.

Frontal sinus fractures are uncommon, and increase the immediate risk of traumatic brain injury, additional facial fractures, and cervical spine injury due to the amount of force needed to fracture the thick frontal bone. Because the dura is adherent to the posterior table of the frontal sinus, operative repair of through-and-through frontal sinus fractures is necessary to prevent pneumocephalus, cerebrospinal fluid (CSF) leak, and infection. Patients with depressed fractures also require admission for IV antibiotics and operative repair. Patients with isolated anterior table fractures of the frontal sinus may be discharged with appropriate follow-up with a facial surgeon.

Naso-orbito-ethmoid fractures result from significant trauma to the nasal bridge, and often have associated injury to the lacrimal duct, dural

162_Cydulka_Ch162.indd 894 22/03/17 11:25 AM

Source page 914

tears, and traumatic brain injury. Patients with these fractures require admission for specialty consultation with facial surgery and neurosurgery.

Orbital blowout fractures occur when a blunt object strikes the globe, transmitting force through the fluid-filled eye, and fracturing the medial or inferior orbital wall. Surgery may be required if these injuries result in extraocular muscle or oculomotor nerve entrapment, or significant enoph-

thalmos. A fracture involving the superior orbital fissure can damage the oculomotor and ophthalmic divisions of the trigeminal nerve (the “orbital fissure syndrome”), and can involve the optic nerve as well (the “orbital apex syndrome”). Patients with either of these syndromes require emergent ophthalmologic consultation. All other patients with isolated orbital fractures can be managed expediently as an outpatient with oral amoxicillin-clavulanate, decongestants, and instructions to avoid nose blowing until the defect has been repaired. Emergent ophthalmological consultation should be requested for ocular injury associated with orbital fractures. An ocular compartment syndrome may occur with a retrobulbar hematoma or malignant orbital emphysema which can cause an acute ischemic optic neuropathy.

Zygoma fractures occur in two major patterns: tripod fractures and isolated zygomatic arch fractures. Tripod fractures involve disruption of the infraorbital rim, the zygomaticofrontal suture, and the zygomaticotemporal junction. These fractures require admission for IV antibiotics and surgical repair. Patients with isolated fractures of the zygomatic arch may have elective outpatient repair.

Midfacial fractures are high-energy injuries and are often seen in victims of multisystem trauma. Patients frequently require endotracheal intubation for airway control. Oral packing is often required for hemorrhage control with fractures of the hard palate. Le Fort injury patterns are illustrated in Figure 162-1. Visual acuity should be tested, especially with Le Fort III and IV fractures, in which the incidence of blindness is around

III III

II

II

I

Figure 162-1. Le Fort injury patterns. Illustration of the fracture lines of Le Fort I (alveolar), Le Fort II (zygomatic maxillary complex), and Le Fort III (craniofacial dysfunction) fractures. Reproduced with permission from Tintinalli JE, Stapczynski JS, Ma OJ, et al: Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 8th ed. New York: McGraw-Hill; 2015.

162_Cydulka_Ch162.indd 895 22/03/17 11:25 AM

Source page 915

3% most commonly secondary to injury of the optic nerve. Both Le Fort II and III injuries can result in CSF leaks. Le Fort injuries require admission for management of significant associated injuries, IV antibiotics, and surgical repair.

Mandible fractures are often diagnosed in the setting of malocclusion and pain with attempted movement. Always look for multiple mandibular fractures, with one injury at the site of impact and a second subtle injury on the opposite side of the ring. A careful intraoral examination is important to exclude small breaks in the mucosa seen with open fractures, sublingual hematomas, and dental or alveolar ridge fractures. Patients with closed fractures may be given urgent outpatient follow-up, while open fractures require admission for IV antibiotics and operative repair. In the patient with a stable airway, a Barton bandage, an ace wrap over the top of the head and underneath the mandible, will stabilize the fracture and help relieve pain.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 259, "Trauma to the Face," by John Bailitz and Tarlan Hedayati.

162_Cydulka_Ch162.indd 896 22/03/17 11:25 AM

References/provenance

- Source: *Tintinalli's Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 912–917.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 163: Neck Injuries

Source pages 918–923 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 918–923

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Neck Injuries**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 163: Neck Injuries 899 ■■ DIAGNOSIS AND DIFFERENTIAL The zone classification summarizes structures placed at risk for injury in penetrating neck trauma (Fig. [Source pages 918–923]

- Zone I structures include the lung apices, thoracic vessels, distal trachea, esophagus, cervical spine, and ver- tebral and carotid arteries. [Source pages 918–923]

- Zone II structures include the mid-carotid and vertebral arteries, jugular veins, esophagus, cervical spine, larynx, and tra- chea. [Source pages 918–923]

- Zone III structures include the proximal carotid and vertebral arteries, oropharynx, and cervical spine. [Source pages 918–923]

- In penetrating neck trauma, a careful, structured physical exam is >95% sensitive for clinically significant vascular and aerodigestive injuries. [Source pages 918–923]

Clinical reasoning structure

- **Problem representation:** Neck Injuries; source pages 918–923.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Neck Injuries**.

- Source-page range: **918–923**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 918

CHAPTER 163: Neck Injuries 899

■■ DIAGNOSIS AND DIFFERENTIAL

The zone classification summarizes structures placed at risk for injury in penetrating neck trauma (Fig. 163-1). Zone I structures include the lung apices, thoracic vessels, distal trachea, esophagus, cervical spine, and vertebral and carotid arteries. Zone II structures include the mid-carotid and vertebral arteries, jugular veins, esophagus, cervical spine, larynx, and trachea. Zone III structures include the proximal carotid and vertebral arteries, oropharynx, and cervical spine.

In penetrating neck trauma, a careful, structured physical exam is >95% sensitive for clinically significant vascular and aerodigestive injuries. Particular attention should be paid to whether the platysma has been violated. If violation has occurred, the presence of deep structure injury is assumed until proven otherwise. The combination of physical examination with multidetector CT angiography (MDCTA) (100% sensitive and 97.5% specific for significant vascular or aerodigestive injury) has been recommended over mandatory operative exploration and selective management strategies for diagnosis in stable patients. Additional studies may be required if MDCTA detects a penetrating injury, but these are typically ordered in conjunction with the specialist who will be repairing the suspected injury. An algorithm for the diagnosis and management of penetrating neck injuries is shown in Fig. 163-2.

Safely managing blunt neck trauma victims requires a heightened level of suspicion and an aggressive approach to diagnosis because catastrophic injuries can present subtly. Therefore, MDCTA should be used liberally. If the MDCTA is negative, but significant suspicion of vascular injury remains, formal cerebral angiography may be required. Likewise, if a strong suspicion of blunt laryngotracheal or esophageal injury remains after

Angle of
mandible

Zone III

Cricoid

cartilage Zone II

Zone I

Figure 163-1. Zones of the neck.

163_Cydulka_Ch163.indd 899 22/03/17 11:34 AM

Source page 919

900 SECTION 18: Trauma

Penetrating

neck trauma

Stable Unstable

Injury violates platysma Operating room or

interventional

angiography

No Yes

Yes Repair injury

Observation Hard signs

of injury

No

No Soft signs

Yes

of injury

CTA or selective

CTA

testing

Suspicion for Suspicion for

Suspicion for

pharyngo- laryngotracheal

vascular injury

esophageal injury injury

Esophagram or CTA*

esophagoscopy* Color Doppler

*Combined esophagram Panendoscopy angiography

and esophagoscopy have *Angiography if CTA

100% sensitivity indeterminate

Negative Positive Negative Positive Negative Positive

IV antibiotics for 7-

Operative repair

10 days, nasogastric

unless small

or total parenteral
mucosal or
nutrition, operative
undisplaced
repair or
laryngeal
conservative
framework fracture
management

Figure 163-2. Penetrating neck trauma protocol.

163_Cydulka_Ch163.indd 900 22/03/17 11:34 AM

Source page 920

a negative MDCTA, laryngobronchoscopy, endoscopy, or swallowing studies may be necessary.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. Initiate standard ATLS protocol for the stabilization of trauma patients. Establish IV access, administer oxygen, and begin cardiac and respiratory monitoring.

2. Emergent airway control is indicated for patients with the clinical factors listed in Table 163-3. Have adjunct airway devices at hand and use paralytics with extreme caution, as these airways can be extremely difficult to secure. If oral intubation is not possible or is contraindicated, a surgical airway is indicated.

3. Penetrating neck trauma is associated with pneumothorax and hemothorax 20% of the time. Diagnose tension pneumothorax clinically and perform immediate needle thoracostomy. Otherwise, pneumothorax and hemothorax may be detected by bedside ultrasound or chest radiograph, and tube thoracostomy is indicated.

4. Apply direct pressure with hemostatic dressings to control active hemorrhage without occluding carotid arteries or the airway. Blind clamping of blood vessels is contraindicated due to a risk of subsequent neurovascular injury. Topical hemostatic agents may be used along with direct pressure. If direct pressure fails, inflating a Foley catheter in the wound tract may help control hemorrhage. Administration of tranexamic acid 1 g intravenously over 10 minutes then 1 g intravenously over 8 hours should be considered in exsanguinating patients with wounds less than 3 hours old. Uncontrolled hemorrhage despite these measures requires emergent surgery.

5. Immobilize the cervical spine in patients with altered level of con-

sciousness, blunt trauma, or penetrating trauma with neurologic deficits after careful examination for penetrating injury, bleeding, hematoma, or bruits/thrills. Cervical collars should not be placed at the expense of monitoring injuries or the ability to perform life-saving procedures.

6. All penetrating wounds that violate the platysma muscle mandate surgical consultation. Unstable patients with penetrating neck trauma require an emergent surgical or interventional radiologic procedure. Therefore, egress from the ED should not be delayed for unnecessary procedures or tests. Stable patients should undergo diagnostic evaluation for deep structure injuries (Fig. 163-2).

7. Penetrating wounds that do not violate the platysma require standard wound care and closure and the patient may be discharged if they are asymptomatic.

8. Patients with blunt neck trauma and hard signs of injury require emergent surgical consultation (Table 163-1). Patients with stable, symptomatic blunt neck trauma should be aggressively evaluated for injuries to deep structures utilizing MDCTA and other modalities. Surgical consultation and admission for these symptomatic patients are advised even if no injury is seen on the initial study.

9. The optimal management of asymptomatic blunt neck trauma patients remains unclear. Consider liberal use of MDCTA (and other adjunctive tests as necessary) and extended observation (4 to 6 hours) to detect

163_Cydulka_Ch163.indd 901 22/03/17 11:34 AM

Source page 921

delayed respiratory and neurological dysfunction. This is especially recommended for patients with risk factors for blunt cerebral vascular injury (Table 163-2).

10. Address the associated psychosocial issues of strangulation victims.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 260, “Trauma to the Neck” by Ashley S. Bean.

163_Cydulka_Ch163.indd 902 22/03/17 11:34 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 918–923.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 164: Cardiothoracic Injuries

Source pages 924–929 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 924–929

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Cardiothoracic Injuries**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 164: Cardiothoracic Injuries 905 should raise suspicion for liver or splenic injuries. [Source pages 924–929]

- The focus of diagnostic imaging is to exclude other injuries such as pneumothorax, pulmonary contusion, and intraabdominal injury. [Source pages 924–929]

- Even in the absence of coexisting injury, the pain of rib fractures may eventually lead to splinting, ventilatory compromise, and pneumo- nia. [Source pages 924–929]

- Consider intercostal nerve blocks and epidural anesthesia for pain control. [Source pages 924–929]

- Patients being discharged should generally receive nonsteroidal anti-inflammatory drugs (NSAIDs) and opioid analgesics. [Source pages 924–929]

Clinical reasoning structure

- **Problem representation:** Cardiothoracic Injuries; source pages 924–929.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Cardiothoracic Injuries**.

- Source-page range: **924–929**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 924

CHAPTER 164: Cardiothoracic Injuries 905

should raise suspicion for liver or splenic injuries. The focus of diagnostic imaging is to exclude other injuries such as pneumothorax, pulmonary contusion, and intraabdominal injury.

Even in the absence of coexisting injury, the pain of rib fractures may eventually lead to splinting, ventilatory compromise, and pneumonia. Consider intercostal nerve blocks and epidural anesthesia for pain control. Patients being discharged should generally receive nonsteroidal anti-inflammatory drugs (NSAIDs) and opioid analgesics. Remind them to breathe deeply or perform incentive spirometry exercises.

Admit patients with multiple fractures, medical comorbidities, or older age for a period of observation until they are stabilized on a regimen of pain control and pulmonary toilet. Do not attempt to stabilize the chest wall with tape or binding. Assess patients with a sternal fracture for cardiac injury by ECG, serial troponin measurements, and cardiac monitoring.

Assume patients with subcutaneous emphysema have a pneumothorax even if not seen on the initial chest radiograph. Supine chest radiograph is a relatively insensitive screening tool for pneumothorax and for hemothoraces of <200 mL. Up to 1000 mL may appear as only diffuse haziness. Lung collapse from intubation of a mainstem bronchus can have a similar appearance. If the patient can safely sit up, upright and expiratory views can increase sensitivity. Ultrasound has been shown to have good sensitivity for pneumothorax, with loss of the sliding pleura sign while a hemothorax will show fluid in the dependent portion of the chest. CT scan is highly sensitive for both of these conditions. If subclavian venous cannulation is attempted, it should be done on the side of the suspected injury so that an iatrogenic pneumothorax does not result in bilateral pneumothoraces.

Penetrating wounds should never be deeply probed. A small stab wound may develop into a delayed pneumothorax; repeat the ultrasound or chest

radiograph at 4 to 6 hours after the initial presentation.

■■ PNEUMOMEDIASTINUM AND TRACHEOBRONCHIAL INJURIES

Pneumomediastinum is most often caused by ruptured alveoli with dissection of air to the mediastinum and does not require treatment in the asymptomatic patient. Coughing, heavy breathing (such as seen in drug inhalation), or exertion can rupture alveoli and release air into the mediastinum. However, pneumomediastinum can also be the result of injuries to the trachea and large airways from high energy or major deceleration. Dyspnea, hemoptysis, subcutaneous emphysema in the neck, a crunching sound with the cardiac cycle, and a massive continued air leak through a chest tube suggest tracheobronchial injury.

■■ DIAPHRAGMATIC INJURIES

All penetrating injuries from the level of the nipples to the umbilicus have the potential to injure the diaphragm. Small lacerations can be asymptomatic and then progress to the rupture of abdominal contents into the chest weeks to months later. The diagnosis is obvious if imaging shows herniation of abdominal contents into the chest or coiling of a gastric tube within

164_Cydulka_Ch164.indd 905 22/03/17 11:40 AM

Source page 925

the chest. Subtle abnormalities may also be seen on chest radiograph, CT, or upper GI series with contrast. Laparotomy and laparoscopy remain the gold standards to exclude diaphragmatic injuries. All diaphragmatic lacerations require surgical repair.

■■ PENETRATING CARDIAC INJURIES

The right ventricle is the most commonly injured portion of the heart because of its large anterior exposure. Accumulation of blood in the pericardial space compresses the heart, which prevents diastolic filling. The diagnosis is typically made by bedside ultrasound. Pericardiocentesis should only be attempted for a patient in shock with confirmed cardiac tamponade since it is technically difficult and may result in laceration of a coronary artery or injury to the myocardium. Stable patients should have a pericardial window or thoracotomy performed in the operating room. Patients with penetrating chest injuries with signs of life in the field but who subsequently become pulseless may be candidates for ED thoracotomy. Relieving cardiac tamponade, controlling hilar bleeding, cross clamping the descending aorta, or repairing a myocardial laceration may be lifesaving. Stab wounds to the heart are more likely to be amenable to repair than injuries from gunshot wounds. Once the chest is opened, the heart is delivered

from the pericardium so that potential injuries can be visualized. Direct digital pressure, staples, sutures, or a Foley catheter with inflated balloon may be used to temporize bleeding lacerations on the way to the operating room for definitive repair.

■■ BLUNT INJURIES TO THE HEART

Blunt cardiac injury can lead to death from damage to cardiac structures, coronary artery injury and thrombosis, and contusion of the myocardium resulting in impaired contractility and arrhythmias. ECG changes consistent with ischemia suggest coronary artery dissection or thrombosis, which are evaluated and treated by cardiac catheterization and stenting. A direct blow to the chest such as when a young athlete is struck by a hard ball can induce ventricular fibrillation cardiac arrest even without myocardial injury (commotio cordis). Treat according to advanced cardiac life support (ACLS) algorithms because there is usually no structural damage to the heart.

A patient with cardiac injury may present with chest pain, tachycardia unexplained by hemorrhage, and arrhythmias. Bedside echocardiography by the emergency provider should be performed as a first screen for cardiac tamponade and grossly impaired contractility. Treat tamponade the same as tamponade from penetrating cardiac injury. Patients with hypotension not explained by another cause, arrhythmias, and impaired contractility should undergo further evaluation by formal echocardiography and cardiac enzymes, with transesophageal echocardiogram being three times more sensitive than transthoracic echocardiogram for blunt myocardial injury. Give antiarrhythmic and inotropic medications according to ACLS algorithms. Indications for admission include abnormalities on echocardiogram, ECG, or cardiac enzymes. Discharge patients with normal vital signs, normal initial ECG, no underlying cardiac disease, and age <55 years after 4 to 6 hours of normal cardiac monitoring.

164_Cydulka_Ch164.indd 906 22/03/17 11:40 AM

Source page 926

■■ TRAUMA TO THE GREAT VESSELS

Trauma to the major thoracic vessels is often lethal, with 90% of those sustaining blunt aortic injury dying at the scene. The most common site of blunt aortic injury is at the proximal descending aorta between the left subclavian artery and the ligamentum arteriosum. Injury to the subclavian and innominate arteries can be related to shoulder belts, fractures of the first and second ribs, and proximal clavicle. Half of patients present without external physical findings, so suspicion for this injury needs to be high with mechanisms involving high-speed deceleration.

Table 164-3 shows radiographic findings of thoracic aortic injury, although a normal chest radiograph does not exclude major vascular injury. All the patients with a mechanism highly concerning for great vessel injury should undergo CT angiogram with IV contrast. Conventional aortography is still used in some cases to assess injuries and guide operative planning. Transesophageal echocardiogram is useful for diagnosing aortic intimal lesions, but is contraindicated in airway compromise or suspected cervical spinal injury.

Indications for immediate operation for vascular injury are hemodynamic instability, radiographic evidence of a rapidly expanding hematoma, or large-volume chest tube output. Control hypertension in order to decrease shear stress on the vessel wall by titration of narcotic pain medications and sedatives. A short-acting β-blocker, such as esmolol, may be titrated to a systolic blood pressure of 100 to 120 mm Hg and a heart rate above 60 beats/min. If bradycardia prevents further dosing of a β-blocker, infuse an arterial dilator such as sodium nitroprusside. Do not use sodium nitroprusside without a β-blocker secondary to the reflex tachycardia that may develop.

Table 164-3 Radiographic Findings Suggestive of a Great Vessel Injury

Fractures

Sternum

Scapula

Multiple ribs

Clavicle in multisystem-injured patients

First and second rib

Mediastinal Clues

Obliteration of the aortic knob contour

Widening of the mediastinum

Depression of the left mainstem bronchus

Loss of paravertebral pleural stripe

Calcium layering at aortic knob

Abnormal general appearance of mediastinum

Deviation of nasogastric tube to the right

Lateral displacement of the trachea

Other Findings

Apical pleural hematoma (cap)

Massive left hemothorax

Obvious diaphragmatic injury

Source: Reproduced with permission from Mattox KL, Moore EE, Feliciano DV: Trauma, 7th ed. New York, NY: McGraw-Hill; 2013.

164_Cydulka_Ch164.indd 907 22/03/17 11:40 AM

Source page 927

908 SECTION 18: Trauma

■■ ESOPHAGEAL AND THORACIC DUCT INJURIES

Penetrating and occasionally blunt trauma may cause injury to the thoracic esophagus. If suspected, evaluate the patient by esophagram with water-soluble contrast, which is less likely to cause mediastinitis. A negative study with water-soluble contrast should be followed by the use of barium contrast, which has a higher sensitivity for injury. Flexible esophagoscopy is an alternative modality for assessing injury. Delayed diagnosis of esophageal injury has a high mortality if mediastinitis ensues. Injuries to the area of the left proximal subclavian vein may result in chylothorax, causing delayed pleural effusion with a mortality rate of approximately 50%. Initial treatment is chest tube insertion.

■■ PERICARDITIS

Patients may develop chest pain, fever, and a friction rub 2 to 4 weeks after cardiac trauma or surgery which may indicate pericarditis. ECG may show diffuse ST-segment elevation consistent with pericarditis. Pericardial and pleural effusions may be seen on echocardiography and chest radiograph, respectively. Treatment is with nonsteroidal anti-inflammatory medications.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 254, "Trauma in Adults," by Peter Cameron and Barry J. Knapp; Chapter 261, "Pulmonary Trauma," by David Jones, Anna Nelson, O. John Ma; Chapter 262, "Cardiac Trauma," by Christopher Ross and Theresa Schwab.

164_Cydulka_Ch164.indd 908 22/03/17 11:40 AM

References/provenance

- Source: *Tintinalli's Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 924–929.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 165: Abdominal Injuries

Source pages 930–933 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 930–933

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Abdominal Injuries**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 165: Abdominal Injuries 911 FAST is the rapid identification of free intraperitoneal fluid in the hypoten- sive patient with blunt abdominal trauma. [Source pages 930–933]

- Since the FAST examination can reliably detect small amounts of free intraperitoneal fluid and can estimate the rate of hemorrhage through serial examinations, it has essentially replaced diagnostic peritoneal lavage (DPL) for blunt abdominal trauma in the majority of North American trauma centers. [Source pages 930–933]

- Computed Tomography Abdominopelvic CT with IV contrast (Table 165-2) is the noninvasive gold standard study for the diagnosis of abdominal injury (unless the patient has an allergy to iodinated contrast). [Source pages 930–933]

- The addition of PO contrast is too time- consuming to be practical in trauma management. [Source pages 930–933]

- The major advantage of IV contrast CT over other diagnostic modalities is that the precise location(s) and grade of injury can be identified. [Source pages 930–933]

Clinical reasoning structure

- **Problem representation:** Abdominal Injuries; source pages 930–933.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Abdominal Injuries**.

- Source-page range: **930–933**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 930

CHAPTER 165: Abdominal Injuries 911

FAST is the rapid identification of free intraperitoneal fluid in the hypotensive patient with blunt abdominal trauma. Since the FAST examination can reliably detect small amounts of free intraperitoneal fluid and can estimate the rate of hemorrhage through serial examinations, it has essentially replaced diagnostic peritoneal lavage (DPL) for blunt abdominal trauma in the majority of North American trauma centers.

Computed Tomography

Abdominopelvic CT with IV contrast (Table 165-2) is the noninvasive gold standard study for the diagnosis of abdominal injury (unless the patient has an allergy to iodinated contrast). The addition of PO contrast is too time-consuming to be practical in trauma management. The major advantage of IV contrast CT over other diagnostic modalities is that the precise location(s) and grade of injury can be identified. CT can quantify and differentiate the amount and type of free fluid in the abdomen. Because CT can evaluate for retroperitoneal injuries, it is the ideal study for assessment of the duodenum and pancreas. The use of multiphasic CT (arterial, portal, and equilibrium phases) accurately identifies life-threatening mesenteric hemorrhage and transmural bowel injuries.

Diagnostic Peritoneal Lavage

The wide availability of CT and ED ultrasound has relegated DPL (Table 165-3) to a second-line screening test for evaluating abdominal

TABLE 165-2 Advantages and Disadvantages of CT in Abdominal Trauma

Advantages	Disadvantages
Ability to precisely locate intra-abdominal lesions	Expense
Ability to evaluate the retroperitoneum	Need to transport the trauma patient to the radiology suite
Ability to identify injuries that may be	Need for contrast materials

managed nonoperatively

Noninvasive Radiation exposure

TABLE 165-3 Advantages and Disadvantages of Diagnostic Peritoneal Lavage in Abdominal Trauma

Advantages	Disadvantages
Sensitivity	Invasive
Availability	Potential for iatrogenic injury
Relative speed with which it can be performed	Lack of specificity
Low complication rate	Inability to identify injuries that may be managed nonoperatively
Ability to detect early evidence of bowel perforation	Misapplication for evaluation of retroperitoneal injuries
No nephrotoxic contrast material needed	
No radiation exposure	

165_Cydulka_Ch165.indd 911 22/03/17 11:46 AM

Source page 931

trauma. For blunt trauma, indications for DPL include (a) patients who are too hemodynamically unstable to leave the ED for CT and (b) unexplained hypotension in patients with an equivocal physical examination. DPL is considered positive if more than 10 mL of gross blood is aspirated immediately, the red blood cell count is higher than 100,000 cells/mm3, the white blood cell count is higher than 500 cells/mm3, bile is present, or if vegetable matter is present.

The only absolute contraindication to DPL is when surgical management is clearly indicated, in which case the DPL would delay patient transport to the operating room. Relative contraindications include patients with advanced hepatic dysfunction, severe coagulopathies, previous abdominal surgeries, or a gravid uterus.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. Initiate standard protocols for evaluation and stabilization of trauma patients. See Chapter 156.
2. Administer oxygen as needed, attach cardiac monitoring, and secure two large-bore IV lines.
3. Administer IV crystalloid fluid to hypotensive abdominal trauma patients. Transfuse with O-negative or type-specific packed red blood cells as indicated.
4. Order laboratory work for abdominal trauma patients based on the mechanism of injury (blunt vs penetrating); labs may include type and

crossmatching, complete blood count, electrolytes, lactate level, directed toxicologic studies, coagulation studies, hepatic enzymes, and lipase.

5. Table 165-4 lists the indications for exploratory laparotomy. When a patient presents to the ED with an obvious high-velocity gunshot wound to the abdomen, do not delay transport of the patient to the operating room by performing a FAST examination unless there is a suspicion for

TABLE 165-4 Indications for Laparotomy

Blunt	Penetrating
Absolute	Anterior abdominal injury with hypotension
Injury to abdomen, back, and flank with hypotension	
Abdominal wall disruption	Abdominal tenderness
Peritonitis	GI evisceration
Free air under diaphragm on chest radiograph	High suspicion for transabdominal trajectory after gunshot wound
Positive FAST or DPL in hemodynamically unstable patient	CT-diagnosed injury requiring surgery (i.e., ureter or pancreas)
CT-diagnosed injury requiring surgery (i.e., pancreatic transection, duodenal rupture, diaphragm injury)	
Relative	Positive FAST or DPL in hemodynamically stable patient
Positive local wound exploration after stab wound	
Solid visceral injury in stable patient	
Hemoperitoneum on CT without clear source	

Abbreviations: DPL, diagnostic peritoneal lavage; FAST, focused assessment with sonography for trauma.

165_Cydulka_Ch165.indd 912 22/03/17 11:46 AM

Source page 932

cardiac injury. If organ evisceration is present, cover the wound with a moist, sterile dressing before surgery.

6. For an equivocal stab wound to the abdomen, obtain surgical consultation for local wound exploration. If the local wound exploration demonstrates no violation of the anterior fascia, the patient can be discharged home.

7. For the hemodynamically stable, blunt trauma patient with a positive FAST examination, further evaluation with CT may be warranted before admission to the surgical service.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study

Guide, 8th ed., see Chapter 263, “Abdominal Trauma,” by L. Keith French, Stephanie Gordy, and O. John Ma.

165_Cydulka_Ch165.indd 913 22/03/17 11:46 AM

Source page 933

C h apter Penetrating Trauma to The Flank

166 and Buttocks

Sum Ambur

Challenges in evaluating penetrating trauma to the flank and buttocks include recognizing peritoneal and retroperitoneal injuries and determining which patients need immediate surgery and which can be managed more conservatively. Trajectories and resultant severity and pattern of injury can vary widely with both stab and gunshot wounds. Hemodynamically unstable patients need immediate operative intervention. Mechanism and time of injury, weapon characteristics, and determining the bullet path or stab wound depth may also assist in determining the initial diagnostic approach if the patient is otherwise stable.

■■ PENETRATING FLANK TRAUMA

Clinical Features

Presentation may vary from stable vital signs with an innocuous-appearing wound to hemodynamic shock and peritonitis. Gross blood on rectal examination suggests bowel injury. Blood at the urethral meatus or hematuria suggests genitourinary injury.

Diagnosis and Differential

CT imaging is the diagnostic modality of choice for hemodynamically stable patients. Protocols vary by institution, but may include PO or rectal contrast if there is suspicion for hollow viscous injury. Contrast-enhanced CT can also often help determine stab wound depth. Bedside ultrasound (eFAST exam) can be used to assess for intraabdominal free fluid and thus help predict the need for surgical intervention.

Emergency Department Care and Disposition

1. ■Follow standard trauma resuscitation protocols. Patients who require emergent exploratory laparotomy include those who are hemodynamically unstable, display peritonitis, and have sustained gunshot wounds to the flank.

2. ■If there is concern that the peritoneum is violated or the patient displays signs of peritonitis, administer broad-spectrum antibiotics, such as pipercillin/tazobactam 3.375 g IV.

3. ■Many patients with stab wounds can be managed conservatively. High-risk patients (stab wounds with penetration beyond deep fascia) require surgical consultation and admission. Low-risk patients (stab wounds

superficial to deep fascia) may be discharged if serial examinations are unremarkable and the patient remains stable throughout an observation period.

914

166_Cydulka_Ch166.indd 914 22/03/17 11:50 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 930–933.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 166: Penetrating Trauma to The Flank and Buttocks

Source pages 934–935 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 934–935

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Penetrating Trauma to The Flank and Buttocks**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 166: Penetrating Trauma to The Flank and Buttocks 915 ■■ PENETRATING BUTTOCK TRAUMA Clinical Features Gunshot wounds are much more likely to require laparotomy than stab wounds. [Source pages 934–935]

- Gunshot wounds above the level of the greater trochanter and gross hematuria predict the need for surgery. [Source pages 934–935]

- Rectal examination to assess for gross blood, evaluation of lower extremity pulses, and neurologic examination to assess for sciatic and femoral nerve injury should be performed. [Source pages 934–935]

- Diagnosis and Differential Hemodynamically stable patients should undergo CT with oral, IV, and rectal contrast (to avoid missed colon and rectal injuries). [Source pages 934–935]

- Cystourethro- gram should be performed on patients with findings of hematuria or wounds near the genitourinary tract. [Source pages 934–935]

Clinical reasoning structure

- **Problem representation:** Penetrating Trauma to The Flank and Buttocks; source pages 934–935.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Penetrating Trauma to The Flank and Buttocks**.

- Source-page range: **934–935**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 934

CHAPTER 166: Penetrating Trauma to The Flank and Buttocks 915

■■ PENETRATING BUTTOCK TRAUMA

Clinical Features

Gunshot wounds are much more likely to require laparotomy than stab wounds. Gunshot wounds above the level of the greater trochanter and gross hematuria predict the need for surgery. Rectal examination to assess for gross blood, evaluation of lower extremity pulses, and neurologic examination to assess for sciatic and femoral nerve injury should be performed.

Diagnosis and Differential

Hemodynamically stable patients should undergo CT with oral, IV, and rectal contrast (to avoid missed colon and rectal injuries). Cystourethrogram should be performed on patients with findings of hematuria or wounds near the genitourinary tract. CT angiography or traditional angiography and venography may be indicated if a pelvic hematoma is found on CT. Bedside ultrasound (eFAST exam) can also be used to assess for intraabdominal free fluid and thus help predict the need for surgical intervention.

Emergency Department Care and Disposition

1. ■Follow standard trauma resuscitation protocols. Patients who are hemodynamically unstable, display peritonitis, or are likely to have intrapelvic or transabdominal bullet paths require exploratory laparotomy.
2. ■If the patient displays signs of peritonitis or there is high suspicion for peritoneal violation, administer antibiotics, such as pipercillin/tazobactam 3.375 g IV.
3. ■Interventional angiography may be required to treat extensive intrapelvic bleeding.
4. ■Wound exploration is of limited value. Only very superficial stab wounds

may be managed and discharged from the ED. Most patients with penetrating trauma to the buttocks require admission and observation due to the risk of occult injuries.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 264, "Trauma to the Flank and Buttocks," by Alicia Devine.

166_Cydulka_Ch166.indd 915 22/03/17 11:50 AM

Source page 935

C h apter Genitourinary Injuries

167 Thomas Dalton

Genitourinary (GU) injuries frequently occur in the setting of polytrauma, so a thorough evaluation is necessary to avoid missing significant injuries.

■■ CLINICAL FEATURES

Injuries should be suspected with any blunt or penetrating trauma near the GU tract, including any rapid deceleration, which can cause major vascular or parenchymal injury even without specific signs or symptoms. Hematuria of any amount raises the index of suspicion for GU injury, and difficulty with urination can be due to bladder or urethral injury or associated concomitant spinal cord injury. Flank contusions or hematomas, evidence of lower rib fractures, or penetrating flank injuries raise concern for renal injury. Lower abdominal pain, tenderness, ecchymosis, or evidence of a pelvic fracture as well as perineal or scrotal edema are consistent with possible bladder injury. Vaginal bleeding, a high-riding prostate, a perineal hematoma, and/or blood at the urethral meatus are concerning for urethral disruption.

■■ DIAGNOSIS AND DIFFERENTIAL

There is no direct relationship between the degree of hematuria and the severity of renal injury. There is some evidence that microscopic hematuria in patients with a blood pressure <90 mm Hg or any gross hematuria is associated with a more significant renal injury. In children where renal trauma is being considered, isolated microscopic hematuria with <50 red blood cells per high-powered field makes a significant renal injury less likely. An IV contrast-enhanced abdominal/pelvic CT scan is the imaging "gold standard" for the stable trauma patient with a suspected kidney injury. A 10-minute delayed image is needed to ascertain whether there is any urine extravasation but can be omitted if the kidney is normal and there are no fluid collections. A retrograde cystogram (plain film or CT) is the "gold standard" for demonstrating bladder injury, and a retrograde urethrogram is indicated for demonstrating urethral injuries. Color Doppler ultra-

sonography is the preferred imaging technique for investigating closed scrotal and testicular injuries. A focused assessment with sonography in trauma (FAST) exam can detect intra-abdominal fluid collections but cannot reliably evaluate renal, bladder, or ureteral injuries Table 167-1.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

Take a standardized approach to all multiple trauma patients to identify and treat life-threatening injuries (primary survey) and then perform a thorough secondary survey, including a GU examination, to diagnose all injuries. Obtain appropriate diagnostic imaging and laboratory testing as indicated by the initial history and examination.

167_Cydulka_Ch167.indd 916 22/03/17 11:56 AM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 934–935.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 167: Genitourinary Injuries

Source pages 936–939 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 936–939

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Genitourinary Injuries**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 167: Genitourinary Injuries 917 Table 167-1 Imaging for Genitourinary Trauma Injury Imaging Comments Multisystem trauma or Abdominal-pelvic IV Include pelvis to view entire GU tract suspected renal paren- contrast CT scan Delayed films needed to identify chymal or vascular injury urinary extravasation Any visceral injury FAST Identifies free fluid, but does not resulting in free specify type of visceral injury and intraperitoneal fluid does not identify renal vascular injury Renal artery injury Renal angiography Details vascular injuries Ureteral injury Abdominal-pelvic IV Delayed films needed to identify contrast CT scan extravasation; obtain IV pyelogram or retrograde pyelogram if still suspicious with negative CT Bladder injury Retrograde cystogram Can use plain radiographs or CT scan Urethral injury Retrograde urethrogram Discuss sequencing with radiologist, because if performed prior to abdominal-pelvic contrast CT scan, can interfere with diagnosis Scrotal/testicular injury Color Doppler US Contrast-enhanced US or MRI if suspicion is high and initial US is negative ■■ MANAGEMENT OF SPECIFIC INJURIES Kidney Kidney injuries include contusions, hematomas, lacerations, and com- pletely shattered kidneys with or without vascular injuries. [Source pages 936–939]

- Eighty percent of patients with kidney injury have additional visceral or skeletal injuries that complicate their management. [Source pages 936–939]

- Most renal injuries are handled nonop- eratively, but indications for operative treatment include life-threatening bleeding from the kidney; expanding, pulsatile, or non-contained hema- toma (thought to be from an avulsion injury); renal avulsion injury; and extravasation from the renal pelvis or from a ureteral injury. [Source pages 936–939]

- There are little data to support specific treatment recommendations for patients with iso- lated renal trauma. [Source pages 936–939]

- Patients with microscopic hematuria and no indication for imaging can be discharged home with instructions for no strenuous activity and follow-up in 1 to 2 weeks for repeat urinalysis. [Source pages 936–939]

Clinical reasoning structure

- **Problem representation:** Genitourinary Injuries; source pages 936–939.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Genitourinary Injuries**.

- Source-page range: **936–939**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 936

CHAPTER 167: Genitourinary Injuries 917

Table 167-1 Imaging for Genitourinary Trauma

Injury Imaging Comments

Multisystem trauma or Abdominal-pelvic IV Include pelvis to view entire GU tract

suspected renal paren- contrast CT scan Delayed films needed to identify

chymal or vascular injury urinary extravasation

Any visceral injury FAST Identifies free fluid, but does not

resulting in free specify type of visceral injury and

intraperitoneal fluid does not identify renal vascular

injury

Renal artery injury Renal angiography Details vascular injuries

Ureteral injury Abdominal-pelvic IV Delayed films needed to identify

contrast CT scan extravasation; obtain IV pyelogram

or retrograde pyelogram if still

suspicious with negative CT

Bladder injury Retrograde cystogram Can use plain radiographs or CT scan

Urethral injury Retrograde urethrogram Discuss sequencing with radiologist,

because if performed prior to

abdominal-pelvic contrast CT scan,

can interfere with diagnosis

Scrotal/testicular injury Color Doppler US Contrast-enhanced US or MRI if

suspicion is high and initial US is

negative

■■ MANAGEMENT OF SPECIFIC INJURIES

Kidney

Kidney injuries include contusions, hematomas, lacerations, and completely shattered kidneys with or without vascular injuries. Eighty percent of patients with kidney injury have additional visceral or skeletal injuries that complicate their management. Most renal injuries are handled nonoperatively, but indications for operative treatment include life-threatening bleeding from the kidney; expanding, pulsatile, or non-contained hematoma (thought to be from an avulsion injury); renal avulsion injury; and extravasation from the renal pelvis or from a ureteral injury. There are little data to support specific treatment recommendations for patients with isolated renal trauma. Patients with microscopic hematuria and no indication for imaging can be discharged home with instructions for no strenuous activity and follow-up in 1 to 2 weeks for repeat urinalysis. Those with a contusion (normal imaging and microscopic hematuria) can be discharged as above. Those with a higher-grade injury and/or gross hematuria should be admitted for observation (to include repeat hematocrit and urinalysis), hydration, and rest until gross hematuria clears, or general improvement ensues.

Ureter

Ureteral injuries are almost always due to iatrogenic complications of instrumentation or penetrating trauma. Notably the absence of hematuria does not exclude an injury. In a stable patient a delayed CT scan of the

167_Cydulka_Ch167.indd 917 22/03/17 11:56 AM

Source page 937

abdomen and pelvis with IV contrast can identify ureteral injuries. If the CT scan is non-diagnostic and there is a high concern for an injury then an IV pyelogram or retrograde pyelogram is indicated. Treatment is operative, including stenting in some cases.

Bladder

Bladder injuries occur in about 2% of blunt abdominal trauma patients and 80% are associated with pelvic fractures. Gross hematuria is present in about 95% of patients with significant injury and warrants a retrograde cystogram. Bladder injuries can also be present in pelvic fractures with only the presence of microscopic hematuria but the degree of microscopic hematuria warranting a cystogram is unclear. Extraperitoneal rupture is most common and can usually be treated by bladder catheter drainage alone. Intraperitoneal rupture always requires surgical exploration and repair. A retrograde cystogram can be performed by infusing ~350 ml of contrast material to distend the bladder. Passive bladder filling is not sensi-

tive enough to exclude a bladder rupture. Sonographic diagnosis of a bladder injury is not accurate.

Urethra

Posterior urethral injuries (membranous and prostatic urethra) are typically related to major blunt force trauma and are associated with pelvic fractures. Treatment is via suprapubic bladder drainage followed by surgical repair in several weeks. Because a urinary catheter can disrupt a partial posterior urethral injury, one should not be placed if there is suspicion of injury without first obtaining a retrograde urethrogram. Anterior urethral injuries usually occur due to direct trauma such as from a straddle injury or a direct blow to the bulbar or penile urethra. The absence of hematuria does not rule-out a urethral injury. If there is concern for an injury, then avoid placing a foley catheter and obtain a retrograde urethrogram by injecting 20 to 30 mL of contrast into the urethra and obtain a radiograph. If a foley catheter has already been placed, then a 16-gauge angiocatheter can be used to inject contrast between the catheter and urethra. Treatment is supportive, which may include a urinary catheter. Penetrating trauma to the anterior urethra generally requires operative repair.

Testicles and Scrotum

Evaluate blunt testicular trauma with an ultrasound examination. If testicular rupture is present, exploration and repair is indicated. If the testicle is intact, conservative treatment with ice, elevation, scrotal support, and pain medication is appropriate. Hematomas and hematoceles are managed on a case-by-case basis. Penetrating testicular trauma warrants surgical exploration and repair. Scrotal lacerations can be directly repaired and scrotal avulsions require surgical repair with the testicle covered in the remaining scrotum.

Penis

Simple contusions are managed conservatively with cold packs, rest, and pain medications. Simple lacerations involving skin only can be directly

167_Cydulka_Ch167.indd 918 22/03/17 11:56 AM

Source page 938

repaired, but deeper lacerations and/or penetrating injuries require operative exploration and repair. Amputation requires microsurgical reimplantation if the amputated segment is viable. Penile fractures require exploration and repair.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study

Guide, 8th ed., see Chapter 265, “Genitourinary Trauma,” by Matthew C. Gratton and French L. Keith.

167_Cydulka_Ch167.indd 919 22/03/17 11:56 AM

Source page 939

C h a p ter Trauma to The Extremities

168 Amy M. Stubbs

Isolated penetrating trauma to the extremities that is associated with vascular injury has a nearly 10% incidence of mortality or limb loss. Early identification of injuries requiring imaging and/or surgical intervention has reduced the rates of limb loss and disability.

■■ CLINICAL FEATURES

As with all traumatic injuries, a thorough primary survey with attention to life-threatening injuries should be accomplished prior to a detailed extremity exam. Direct pressure or a tourniquet should be applied to actively bleeding injuries. Once an extremity injury is identified, a meticulous vascular and neuromuscular exam should then be performed.

Note pulses distal to the injury, capillary refill, and the color and temperature of the limb. Use a Doppler to detect a signal in the absence of a pulse. Any “hard” signs of arterial injury should prompt immediate surgical consultation and intervention. Soft signs of arterial injury should also be noted and require further evaluation, typically imaging and observation (Table 168-1).

Document the size and shape of each wound, as well as any bony deformities or soft tissue defects. Evaluate the surrounding area for pain with palpation or range of motion. Carefully evaluate joints in the proximity of the wound for the possibility of an open joint. Perform detailed strength and sensory exams on the affected limb to check for peripheral nerve injury. Consult the appropriate surgical specialist for signs of injury to an artery, nerve, joint, or bone, or suspicion of compartment syndrome (see Fig. 168-1).

Table 168-1 Clinical Manifestations of Extremity Vascular Trauma

Hard signs

Absent or diminished distal pulses

Obvious arterial bleeding

Large expanding or pulsatile hematoma

Audible bruit

Palpable thrill

Distal ischemia (pain, pallor, paralysis, paresthesias, coolness)

Soft signs

Small, stable hematoma

Injury to anatomically related nerve

Unexplained hypotension

History of hemorrhage

Proximity of injury to major vascular structures

Complex fracture

920

168_Cydulka_Ch168.indd 920 21/04/17 3:57 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.

- Content ID: SEG-TINTINALLI-EMM8-FULL.

- Source pages: 936–939.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 168: Trauma to The Extremities

Source pages 940–943 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 940–943

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Trauma to The Extremities**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 168: Trauma to The Extremities 921 Penetrating Extremity Trauma Medical history and physical examination No signs of vascular injury (no hard or soft signs) Hard signs Soft signs ABI CT angiography <0.9 >0.9 Immediate Positive Negative or Observation vascular (occlusion or minimal surgery consult extravasation) (no occlusive arterial injury) Observation Figure 168-1. [Source pages 940–943]

- ■■ DIAGNOSIS AND DIFFERENTIAL Obtain ankle-brachial indexes (ABIs) on the affected and unaffected limb using a Doppler and manual blood pressure cuff. [Source pages 940–943]

- Though they have vari- able sensitivity and specificity for arterial injury and do not reliably detect nonocclusive injuries such as intimal flaps or pseudoaneurysms, a result of <0.9 is concerning for arterial injury. [Source pages 940–943]

- Soft signs of vascular injury or an abnormal ABI requires further imaging. [Source pages 940–943]

- At minimum, AP and lateral films of the affected limb are necessary to evaluate for bone or joint injuries. [Source pages 940–943]

Clinical reasoning structure

- **Problem representation:** Trauma to The Extremities; source pages 940–943.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Trauma to The Extremities**.

- Source-page range: **940–943**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 940

CHAPTER 168: Trauma to The Extremities 921

Penetrating

Extremity

Trauma

Medical history and physical examination

No signs of

vascular injury

(no hard or

soft signs)

Hard signs Soft signs

ABI

CT angiography <0.9 >0.9

Immediate Positive Negative or Observation

vascular (occlusion or minimal

surgery consult extravasation) (no occlusive

arterial injury)

Observation

Figure 168-1. Algorithm for penetrating extremity trauma. ABI, ankle-brachial index.

■■ DIAGNOSIS AND DIFFERENTIAL

Obtain ankle-brachial indexes (ABIs) on the affected and unaffected limb using a Doppler and manual blood pressure cuff. Though they have variable sensitivity and specificity for arterial injury and do not reliably detect nonocclusive injuries such as intimal flaps or pseudoaneurysms, a result of <0.9 is concerning for arterial injury. Soft signs of vascular injury or an abnormal ABI requires further imaging.

At minimum, AP and lateral films of the affected limb are necessary to evaluate for bone or joint injuries. Retained foreign bodies or embolized bullet fragments may also be seen. Image the joint above and below the

injury site.

168_Cydulka_Ch168.indd 921 22/03/17 12:00 PM

Source page 941

CT angiography has largely supplanted conventional angiography in the evaluation of vascular injuries as it is widely available, rapid, relatively noninvasive, and provides high-quality images that may also be useful in evaluating concurrent injuries. Ultrasound is neither as sensitive nor as specific as angiography and cannot reliably rule out a vascular injury. Laboratory testing is not required, though a CBC, type and screen, or creatinine may be indicated in certain settings, such as when there is profuse bleeding or concern exists for renal insufficiency when angiography is a consideration.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. ■Immediate operative intervention is typically indicated when hard signs of vascular injury are identified. For certain injuries, imaging, in consultation with the vascular surgeon, may be appropriate prior to surgical repair.
2. ■In the absence of arterial injury, 24-hour admission for observation and serial exams may be reasonable.
3. ■Control bleeding with direct pressure, pressure dressings, or a tourniquet. Vessels should not be clamped or ligated to avoid injury to adjacent nerves.
4. ■Bone or joint capsule injuries should be evaluated by an orthopedic surgeon as patients are at risk for infection, posttraumatic arthritis, and loss of function. Fractures due to penetrating injuries should be treated as open fractures, requiring surgical debridement and admission for intravenous antibiotics, such as a cephalosporin plus gentamycin.
5. ■Update tetanus and irrigate wounds with, at minimum, 500 to 1000 cc of saline or tap water at high pressure (15 to 20 pounds per square inch). Grossly contaminated or older wounds may require gentle scrubbing and/or debridement.
6. ■Closure of the wound depends on the timing of presentation and amount of contamination. Repair low-risk wounds and arrange follow-up in 24 to 48 hours. Wounds with tissue destruction, gross contamination, or retained foreign bodies should be repaired via delayed primary closure after 72 to 96 hours. If there is no fracture, antibiotics are generally not indicated for low risk injuries, but should be considered for high-risk cases such as hand injuries, gross contamination, or immunocompromised patients.

7. ■The decision to remove foreign bodies such as bullet fragments depends on the size, location, and composition of the object. As aggressive exploration may cause further tissue damage and increase infection risk; the risk/benefit ratio should be carefully considered.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 266, "Trauma to the Extremities," by James Heilman.

168_Cydulka_Ch168.indd 922 22/03/17 12:00 PM

Source page 942

SECTION

Injuries to the Bones, Joints, and Soft Tissue

19

C h a p ter 169 Initial Evaluation and Management of Orthopedic Injuries

Gregory M. Johnston

Orthopedic injuries have implications beyond localized pain and swelling at the site of injury. The clinician must methodically evaluate the patient to prevent missing an occult or concomitant injury. Prompt recognition and appropriate treatment are needed to prevent prolonged pain, temporary or permanent disability, or even death.

■■ CLINICAL FEATURES

Key to any emergency department encounter is obtaining a concise history from the patient and, when this is not feasible, from pre-hospital personnel or family members, and performing a meticulous physical examination. For instance, a patient may present with a deformed lower extremity, but a history of a fall from height will cue the clinician to consider the possibility of a concomitant spinal injury, or the spontaneous occurrence of a fracture in a patient with a history of malignancy may indicate a pathologic fracture. The clinician should conduct systematic palpation, since it is not uncommon for the pain of a fracture or dislocation to be referred to another area. Other essential aspects of the physical examination include inspection for swelling, discoloration, and deformity, documentation of both active and passive range of motion proximal and distal to the reported injury, and assessment of neurovascular status to include sensorimotor testing of peripheral nerve function. Vascular status should be assessed as soon as possible, since rapid correction of circulatory compromise can prevent ischemia, tissue loss, or amputation. Diagnostic imaging should not be used as a substitute for physical examination, since it is possible for an occult fracture to escape detection

on initial radiographic imaging. In addition, some fractures may only be detected on specialized radiographic views. If a patient has significant tenderness to palpation, pain with weight bearing, or with passive range of motion, then the possibility of an occult fracture should be considered. A negative plain film report does not exclude a significant injury. In such instances the fracture may only be detected by more advanced imaging modalities such as CT, bone scan, or MRI.

923

169_Cydulka_Ch169.indd 923 22/03/17 12:14 PM

Source page 943

924 SECTION 19: Injuries to the Bones, Joints, and Soft Tissue

■■ DIAGNOSIS AND DIFFERENTIAL

Consider obtaining studies of joints above and below the suspected fracture site, since additional injuries may be present. Pediatric patients present a special challenge due to the presence of ossifying growth centers. If a fracture is suspected, it is often helpful to obtain comparison films of the unaffected side. While many injuries may be treated in the emergency department and then routinely referred for outpatient follow-up, more significant injuries mandate immediate discussion with the orthopedic surgeon. Digital imaging is becoming more common, so it is often possible for the consulting orthopedist to conduct a contemporaneous review of the diagnostic studies when consulted by the emergency physician. This is not always feasible in rural medical facilities or military environments; therefore, overreliance on technology should not supplant the fundamental skill of describing radiographs. The following is a recommended way to verbally describe the fracture to the consultant in order to best communicate the findings of the radiograph:

●

■Open (overlying skin disrupted) versus closed (overlying skin intact): Open injuries frequently require urgent operative intervention, and mandate thorough irrigation and prophylactic antibiotics to prevent the morbid sequelae of osteomyelitis and sepsis. Tetanus prophylaxis, administration of first-generation cephalosporin, i.e., Cefazolin, and the inclusion of an aminoglycoside for contaminated wounds, i.e., Gentamicin, are recommended.

●

■Location of the fracture: Orthopedic conventions include midshaft, junction of the proximal and middle thirds, junction of middle and distal thirds, anatomic boney reference points, e.g., supracondylar or intertrochanteric, and intra-articular. The later has operative implications due to

the association of traumatic arthritis in cases not operatively managed.

●

■Orientation of the fracture: (see Fig. 169-1).

●

■Displacement and separation: Amount (in millimeters or percentage) and direction the distal fragment is detached and deviated from the proximal fragment. A separated fragment is maintained in anatomic alignment, but is offset from normal anatomic position.

●

■Shortening: reduction (in millimeters or centimeters) in bone length due to impaction or overriding fragments. Frequently requires surgical intervention.

●

■Angulation: Degree and direction of the angle formed by the distal fragment.

●

■Rotational deformity: Degree distal fragment is twisted on axis relative to the proximal fragment. Found on physical examination, and is not typically discernable by radiographs.

●

■Concomitant dislocation or subluxation: Complete loss of proper joint alignment (dislocation) compared to partial loss of joint surface contact (subluxation). Frequently requires surgical intervention.

●

■Salter-Harris fractures: Injuries to the epiphyseal plate of a growing child (see Fig. 169-2 and Table 169-1). Types I and V may not be noted on radiographs, and should be suspected clinically, immobilized and referred for follow-up. Unlike type I, type V injuries are associated with an axial loading force.

169_Cydulka_Ch169.indd 924 22/03/17 12:14 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 940–943.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 169: Initial Evaluation and Management of Orthopedic Injuries

Source pages 944–949 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 944–949

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Initial Evaluation and Management of Orthopedic Injuries**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 169: Initial Evaluation and Management of Orthopedic Injuries 925 A B C D E F G Figure 169-1. [Source pages 944–949]

- 169_Cydulka_Ch169.indd 925 22/03/17 12:14 PM 926 SECTION 19: Injuries to the Bones, Joints, and Soft Tissue I II III IV V Figure 169-2. [Source pages 944–949]

- Epiphyseal plate fractures based on the classification of Salter and Harris. [Source pages 944–949]

- ■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION 1. [Source pages 944–949]

- Control pain and swelling: Swelling intensifies pain and when un- checked can result in neurovascular compromise. [Source pages 944–949]

Clinical reasoning structure

- **Problem representation:** Initial Evaluation and Management of Orthopedic Injuries; source pages 944–949.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Initial Evaluation and Management of Orthopedic Injuries**.

- Source-page range: **944–949**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 944

CHAPTER 169: Initial Evaluation and Management of Orthopedic Injuries 925

A B C

D E

F G

Figure 169-1. Fracture Line Orientation. A. Transverse. B. Oblique. C. Spiral.

D. Comminuted. E. Segmental. F. Torus. G. Greenstick.

169_Cydulka_Ch169.indd 925 22/03/17 12:14 PM

Source page 945

926 SECTION 19: Injuries to the Bones, Joints, and Soft Tissue

I II III

IV V

Figure 169-2. Epiphyseal plate fractures based on the classification of Salter

and Harris.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. Control pain and swelling: Swelling intensifies pain and when un-

checked can result in neurovascular compromise. Administer narcotic

analgesics. Avoid use of oral analgesics in individuals with possible

operative lesions.

2. Remove constricting items such as jewelry, watches, and rings as soon

as possible.

3. Apply cold packs to reduce swelling and also to reduce pain.

4. Elevate the affected extremity.

Table 169-1 Description of Salter-Harris Fractures

Salter Type What Is Broken Off

I The entire epiphysis

II The entire epiphysis along with a portion of the metaphysis

III A portion of the epiphysis

IV A portion of the epiphysis along with a portion of the metaphysis

V Compression injury of the epiphyseal plate (nothing is “broken off”)

169_Cydulka_Ch169.indd 926 22/03/17 12:14 PM

Source page 946

5. Withhold oral intake: Pending complete evaluation and treatment. A patient who does not require an operation may still require procedural sedation.

6. Reduce fracture deformity and dislocations: Application of gradual and steady longitudinal traction is sufficient to reduce most fractures. Reduction will decrease swelling and pain, relieve tension on neurovascular structures adjacent to the deformity, reduce the risk of converting a closed fracture to an open one, and restore circulation to a pulseless extremity. Techniques for dislocation reduction vary based upon the site of injury. The best results are achieved when the patient is relaxed and sedated.

7. Recognize injuries that require orthopedic consultation: Many orthopedic injuries can be treated in the ED and discharged to home with outpatient follow-up. Some injuries require inpatient management due to co-morbid health conditions or issues related to pain control or ambulatory dysfunction, such as hip fractures. Other injuries, such as open fractures, compartment syndrome, irreducible dislocations, circulatory compromise, and injuries needing surgical intervention, require direct communication and urgent consultation in the ED with the on-call orthopedic surgeon.

8. Splinting and immobilization: Reduces pain, facilitates healing, and helps to prevent re-injury of the affected extremity (see Table 169-2).

Table 169-2 Immobilization Devices and Uses

Immobilization Technique Clinical Application

Shoulder immobilizer Clavicle fracture

Acromioclavicular separation

Shoulder dislocation (postreduction)

Humeral neck fracture

Sling A variety of upper-extremity injuries, in conjunction with other immobilization techniques; may be used alone for nondisplaced or clinically suspected fracture of the radial head

Long-arm gutter Elbow fracture other than nondisplaced radial head fracture

Reduced elbow dislocation

Sugar-tong Wrist or forearm fracture

Short-arm gutter Metacarpal or proximal phalanx fracture. (Ulnar gutter for

fourth or fifth ray; radial gutter for second [index] or third
[middle] ray)
Thumb spica Scaphoid fracture (proven or suspected)
Thumb metacarpal or proximal thumb phalanx fracture
Knee immobilizer Fracture or reduced subluxation of patella
Knee dislocation, postreduction (temporary)
Tibial plateau fracture
Knee ligament injury
Suspected meniscal tear (provided the knee can be fully
extended)
(Continued)
169_Cydulka_Ch169.indd 927 22/03/17 12:14 PM

Source page 947

928 SECTION 19: Injuries to the Bones, Joints, and Soft Tissue
Table 169-2 Immobilization Devices and Uses (Continued)
Immobilization Technique Clinical Application
Posterior ankle mold Ankle dislocation or fracture-dislocation
(consider above-the-knee Unstable ankle fracture (high distal fibular fracture or medial
extension and/or adjunctive and/or posterior malleolar fracture)
use of ankle sugar-tong for Widened medial mortise (indicates disruption of stabilizing
unstable ankle injuries) medial structures)
Metatarsal fracture (alternative immobilization dressings
may be used)
Ankle stirrup Simple ankle sprain
Stable lateral malleolus fracture (below the superior border
of the talus) without other ankle involvement (no medial
swelling or tenderness, posterior malleolus intact)
Hard-soled shoe Toe fracture
Some metatarsal fractures
Short-leg walking boot Some toe or foot contusions or fractures where
weightbearing is allowed
■■ FURTHER READING
For Further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study
Guide, 8th ed., see Chapter 267, “Initial Evaluation and Management of Ortho-
pedic Injuries,” by Jeffrey S. Menkes.
169_Cydulka_Ch169.indd 928 22/03/17 12:14 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.

- Source pages: 944–949.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 170: Hand and Wrist Injuries

Source pages 950–953 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 950–953

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Hand and Wrist Injuries**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 170: Hand and Wrist Injuries 931 Thumb MCP ulnar collateral ligament rupture (gamekeeper’s or skier’s thumb) results from forced radial abduction at the MCP joint, with the rupture occurring at the insertion on the proximal phalanx. [Source pages 950–953]

- The joint capsule and volar plate are usually involved. [Source pages 950–953]

- A complete tear is diagnosed when abduction stress on the proximal phalanx causes more than 30° to 35° of radial angulation relative to the metacarpal. [Source pages 950–953]

- Distal phalanx fractures most commonly involve the tuft. [Source pages 950–953]

- These are associated with nail bed laceration. [Source pages 950–953]

Clinical reasoning structure

- **Problem representation:** Hand and Wrist Injuries; source pages 950–953.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Hand and Wrist Injuries**.

- Source-page range: **950–953**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 950

CHAPTER 170: Hand and Wrist Injuries 931

Thumb MCP ulnar collateral ligament rupture (gamekeeper's or skier's thumb) results from forced radial abduction at the MCP joint, with the rupture occurring at the insertion on the proximal phalanx. The joint capsule and volar plate are usually involved. A complete tear is diagnosed when abduction stress on the proximal phalanx causes more than 30° to 35° of radial angulation relative to the metacarpal. Place in a thumb spica splint. Complete tears mandate surgical referral.

Distal phalanx fractures most commonly involve the tuft. These are associated with nail bed laceration. Place a volar or hairpin splint to the DIP joint. Avulsion fractures of the base often represent associated tendon involvement.

Proximal and middle phalanx fractures of the base and neck that are nondisplaced and stable can be treated with buddy taping. Transverse or spiral midshaft fractures or intraarticular fractures often require surgical fixation. Place a gutter splint with the MCP joint flexed at 70°, the IP joints in extension.

Metacarpal (MC) fractures most commonly involve the fourth or fifth MC neck (boxer's fracture). Skin lacerations over the fracture should raise concern for a "fight bite." Angulation more than 20° in the fourth MC, 40° in the fifth MC, or 15° in the second or third MC should be reduced. Place an ulnar gutter splint for fractures of the fourth or fifth MC and a radial gutter splint for fractures of the second or third MC with the wrist extended at 20° and the MCP joint flexed at 70°. Thumb MC fractures usually involve the base with intraarticular involvement (Bennett and Rolando fractures). Place in a thumb spica splint.

Compartment syndrome of the hand may result from crush injury or extravasation of IV fluids or radiocontrast media. The patient will complain of pain that is out of proportion to exam findings. On examination, the hand, at a resting position, is extended at the MCP joint and slightly

flexed at the PIP joint. There is tense edema and pain with passive stretch of the involved compartment. This is an orthopedic emergency and the diagnosis is made clinically due to the difficulty in obtaining accurate compartment pressures.

High-pressure injection injury occurs when substances in a high pressure device (>2000 psi), such as grease, paint, or hydraulic fluid, are injected into the hand. Oil-based paint causes a severe tissue reaction that can result in ischemia and eventual need for amputation. Obtain hand and forearm radiographs searching for radiopaque substances and subcutaneous air. While it appears benign, this injury requires emergent surgical evaluation.

■■ WRIST INJURIES

Scapholunate dissociation presents with wrist tenderness and swelling at the scapholunate joint. The PA radiograph demonstrates a space between the scaphoid and lunate that is >3 mm. Treat with a radial gutter splint and prompt referral.

Perilunate and lunate dislocations are best noted on lateral wrist radiograph. In both injuries, the normal alignment of the radius-lunate-capitate (the “3 C’s” sign) is lost. With a perilunate dislocation, the lunate remains aligned with the radius, but the carpals are dislocated, usually dorsal to the lunate. With a lunate dislocation, the lunate dislocates volar to

170_Cydulka_Ch170.indd 931 22/03/17 12:27 PM

Source page 951

Table 170-3 Summary of Carpal Bone Fractures and ED Management

Carpal Bone of Injury	Mechanism	Initial ED Examination	ED Management
Scaphoid	Fall on outstretched hand	Snuffbox tenderness. Pain with radial deviation and flexion with radial deviation	Short arm, thumb spica, in dorsiflexion
Triquetrum	Avulsion fracture—twisting of hand against resistance or hyperextension	Tenderness at the dorsum of the wrist, distal to the ulnar styloid	Short arm, sugar-tong splint
Body	fracture—direct trauma		
Lunate	Fall on outstretched hand	Tenderness at middorsum of the wrist, ulnar and distal to Lister tubercle	Short arm, thumb spica splint

Trapezium Direct blow to thumb; Painful thumb movement Short arm, thumb
force to wrist while and weak pinch strength spica splint
dorsiflexed and Snuffbox tenderness
radially deviated
Pisiform Fall directed on the Tender pisiform, promi- Short arm, volar splint
hypothenar eminence nent at the base of the in 30° of flexion and
hypothenar eminence ulnar deviation
Hamate Interrupted swing of Tenderness at the hook Short arm, volar wrist
a golf club, bat, or of the hamate, just splint with fourth and
racquet distal and radial to the fifth metacarpal joints
pisiform in flexion
Capitate Forceful dorsiflexion Tenderness over the Short arm, volar wrist
of the hand with capitate just proximal to splint
radial impact the third metacarpal
Trapezoid Axial load onto the Tenderness over the Short arm, thumb
index metacarpal radial aspect of the base spica splint
of the index metacarpal

the radius, but the remainder of the carpus aligns with the radius. Lunate dislocation on PA radiograph has a triangular shape, the “piece of pie” sign, and on lateral view, the “spilled teacup” sign. Emergent consult for closed reduction or surgical repair is indicated.

Carpal bone fractures are managed as summarized in Table 170-3.

The scaphoid is the most common carpal bone fractured. Fracture of the scaphoid, lunate, or capitate can cause avascular necrosis of the bone. Scaphoid and lunate fractures are often not detected on plain radiographs, so ED diagnosis and treatment should be based on clinical findings alone.

Colles, Smith, and Barton fractures involve the distal radius at the metaphysis (Table 170-4). Most of these fractures can be treated with closed reduction and a sugar-tong splint.

Radial styloid fracture can produce carpal instability with scapholunate dissociation as major carpal ligaments insert here. Splint the wrist in mild flexion and ulnar deviation.

170_Cydulka_Ch170.indd 932 22/03/17 12:27 PM

Source page 952

Table 170-4 Radiographic Appearance of Distal Radius Fractures

Colles’ fracture
Dorsal angulation of the plane of the distal radius
Distal radius fragment is displaced proximally and dorsally
Radial displacement of the carpus

Ulnar styloid may be fractured

Smith’s fracture

Volar angulation of the plane of the distal radius

Distal radius fragment is displaced proximally and volarly

Radial displacement of the carpus

The fracture line extends obliquely from the dorsal surface to the volar surface 1 to 2 cm proximal to the articular surface

Barton’s fracture

Volar and proximal displacement of a large fragment of radial articular surface

Volar displacement of the carpus

Radial styloid may be fractured

Ulnar styloid fracture may result in radioulnar joint instability. Place an ulnar gutter splint with the wrist in neutral position and slight ulnar deviation.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 268, “Injuries to the Hand and Digits,” by Moira Davenport and Peter Tang; Chapter 269, “Wrist Injuries,” by Robert Escarza, Maurice F. Loeffel ,III, and Dennis T. Uehara.

170_Cydulka_Ch170.indd 933 22/03/17 12:27 PM

Source page 953

C h a p ter Forearm and Elbow Injuries

171 Sandra L. Najarian

■■ SOFT TISSUE INJURIES

Biceps and Triceps Tendon Ruptures

Clinical Features

Patients with proximal long-head biceps tendon ruptures typically describe a “snap” or “pop” and complain of pain in the anterior shoulder. Examination reveals tenderness, swelling, and crepitus over the bicipital groove in the anterior shoulder. A mid-arm “ball” (the distally retracted biceps) appears when the elbow is flexed. Elbow flexion strength is maintained due to the preserved action of the brachialis and supinators. This is in contrast to distal biceps tendon rupture where elbow flexion and supination is weak. Examination of distal biceps rupture reveals swelling, ecchymosis, tenderness, and inability to palpate the tendon in the antecubital fossa. With the patient seated, the elbow flexed 60 to 80 degrees, and forearm resting on the patient’s lap, the examiner squeezes the muscle belly of the biceps causing the forearm to supinate (biceps squeeze test). If no supination is noted, then this is a positive test indicating a distal biceps tendon rupture. Triceps tendon ruptures are rare, and the majority occur distally. Patients present

with pain, swelling, and tenderness proximal to the olecranon; a sulcus with a proximal mass (the proximally retracted triceps tendon) may be palpable. Forearm extension is weak. A modified Thompson test can be used to assess triceps function. With the arm supported, elbow flexed at 90 degrees, and forearm hanging in a relaxed position, squeezing the triceps muscle should produce extension of the forearm unless a complete tear is present.

Diagnosis and Differential

Diagnosis is clinical. Obtain radiographs to exclude an associated avulsion fracture.

Emergency Department Care and Disposition

Treatment includes sling, ice, analgesics, and referral to an orthopedic surgeon for definitive management. Complete tendon tears in young active individuals often require surgical repair.

Overuse Syndromes

Clinical Features

Lateral epicondylitis or "tennis elbow" is more common than medial epicondylitis or "golfer’s elbow." Lateral epicondylitis affects the forearm and wrist extensors, and medial epicondylitis affects the forearm and wrist flexors. Patients with medial epicondylitis may develop an ulnar neuropathy. Both syndromes result from repetitive activity involving these muscle groups.

171_Cydulka_Ch171.indd 934 22/03/17 12:47 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 950–953.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 171: Forearm and Elbow Injuries

Source pages 954–961 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 954–961

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Forearm and Elbow Injuries**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 171: Forearm and Elbow Injuries 935 Diagnosis and Differential Diagnosis is clinical. [Source pages 954–961]

- Radiographs may help in ruling out an associated avulsion fracture of the lateral or medial epicondyle. [Source pages 954–961]

- Emergency Department Care and Disposition Management is conservative including rest, ice, anti-inflammatory medica- tions, and bracing. [Source pages 954–961]

- Occupational therapy can be useful in treating these syndromes. [Source pages 954–961]

- ■■ ELBOW DISLOCATIONS Clinical Features The majority of elbow dislocations are posterolateral and often occur as a result of a fall on an outstretched hand. [Source pages 954–961]

Clinical reasoning structure

- **Problem representation:** Forearm and Elbow Injuries; source pages 954–961.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Forearm and Elbow Injuries**.

- Source-page range: **954–961**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 954

CHAPTER 171: Forearm and Elbow Injuries 935

Diagnosis and Differential

Diagnosis is clinical. Radiographs may help in ruling out an associated avulsion fracture of the lateral or medial epicondyle.

Emergency Department Care and Disposition

Management is conservative including rest, ice, anti-inflammatory medications, and bracing. Occupational therapy can be useful in treating these syndromes. Surgery is reserved for refractory cases.

■■ ELBOW DISLOCATIONS

Clinical Features

The majority of elbow dislocations are posterolateral and often occur as a result of a fall on an outstretched hand. On examination, the patient holds the elbow in 45 degrees of flexion. Significant swelling of the elbow often obscures the olecranon, which is directed posteriorly. Neurovascular assessment is essential (Table 171-1). An open dislocation, absence of radial pulse before reduction, and presence of systemic injuries are all factors associated with arterial injury.

Diagnosis and Differential

Radiographs confirm the diagnosis. The lateral view reveals both the ulna and radius displaced posterior. The AP view reveals either medial or lateral displacement of the ulna and radius with maintenance of their normal relationship to each other. The presence of associated fractures, especially to the radial head and coronoid process, can render the elbow joint unstable and complicate treatment.

Emergency Department Care and Disposition

The goals of treatment are reduction with procedural sedation and recognition of neurovascular complications, associated fractures, and postreduction instability. Intra-articular lidocaine may provide analgesia for closed reduction.

1. Several methods can be used for reduction. With the first two-person method (Fig. 171-1), start by placing the patient supine, and apply

gentle longitudinal traction on the wrist and forearm while an assistant applies countertraction on the arm. Correct any medial or lateral

Table 171-1 Sensory and Motor Function Testing of the Radial, Median, and Ulnar Nerves

Radial	Median	Ulnar
Test for sensory function	Dorsum of the thumb and index web space	Two-point discrimination over the tip of the index finger
Test for motor function	Extend both wrist and fingers against resistance	“OK” sign with thumb and index finger; abduction against resistance
	Abduct index finger over the tip of the little finger	Abduct thumb (recurrent branch)

171_Cydulka_Ch171.indd 935 22/03/17 12:47 PM

Source page 955

A

B

Figure 171-1. Traction and flexion method of reducing a posterior elbow dislocation. A. Side-to-side manipulation is used to correct medial or lateral displacement. B. The elbow is then flexed while maintaining longitudinal traction.

displacement with the other hand. Then, apply downward pressure on the proximal forearm to disengage the coronoid process from the olecranon. A palpable clunk indicates a successful reduction. With the second two-person method (Fig. 171-2), start with the patient supine with the arm adducted across the torso and elbow slightly flexed. The patient may also be positioned prone with the affected arm abducted and the elbow slightly flexed. The assistant applies longitudinal traction

171_Cydulka_Ch171.indd 936 22/03/17 12:47 PM

Source page 956

A

B

Figure 171-2. Olecranon manipulation method of reducing a posterior elbow dislocation with the patient positioned prone (A) or supine (B).

on the wrist and forearm, while the operator holds the elbow with both thumbs behind the olecranon. Apply firm pressure with thumbs to push the olecranon over the trochlea and back into position. The single-person method (Fig. 171-3) starts with the patient prone and the affected arm on the stretcher with the elbow flexed over the edge of the

stretcher. Place a pillow or folded blanket under the humerus and hang a 5-lb weight to the wrist. The elbow should spontaneously reduce over several minutes. Gentle pressure to the olecranon can aid the reduction.

171_Cydulka_Ch171.indd 937 22/03/17 12:47 PM

Source page 957

A B

Figure 171-3. A. Hanging arm method of reducing a posterior elbow dislocation. B. Gentle manipulation can be applied to the olecranon, if necessary.

2. After the reduction, range the elbow fully to assess stability.

3. Assess neurovascular status after reduction and over a period of observation.

4. Obtain post reduction films.

5. Immobilize the elbow in a long arm posterior splint with the elbow in slightly less than 90 degrees of flexion and the forearm in mild pronation. Arrange for close orthopedic follow-up.

6. Immediate orthopedic consultation is necessary for patients with instability, neurovascular compromise, or open dislocations. Admission may be necessary to monitor neurovascular status.

■■ ELBOW FRACTURES

Clinical Features

Radial head fractures present with pain, swelling, and tenderness over the lateral elbow, and inability to fully extend the elbow. Supracondylar and intercondylar fractures present with significant swelling, tenderness, and limited range of motion at the elbow. Supracondylar fractures may resemble a posterior elbow dislocation. Olecranon fractures present with pain, swelling, and crepitus over the posterior elbow. Epicondyle, condyle, trochlear, capitellum, and coronoid fractures are uncommon and can occur in association with other elbow injuries such as dislocations. An Essex–Lopresti lesion, a radial head fracture with associated distal radioulnar joint dissociation, presents with wrist and forearm pain in addition to lateral elbow pain. A neurovascular assessment is essential in all elbow fractures. Potential complications of supracondylar fractures are numerous (Table 171-2). A decreased or absent radial pulse is common in children and often secondary to brachial artery injury, even if the hand appears warm and perfused. Signs of Volkmann ischemic contracture

171_Cydulka_Ch171.indd 938 22/03/17 12:47 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.

- Content ID: SEG-TINTINALLI-EMM8-FULL.

- Source pages: 954–961.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 172: Shoulder and Humerus Injuries

Source pages 962–969 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 962–969

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Shoulder and Humerus Injuries**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 172: Shoulder and Humerus Injuries 943 fractures that are open, have neurovascular compromise, or have persistent skin tenting. [Source pages 962–969]

- Clavicle fractures that are severely comminuted or displaced may benefit from operative intervention so consider early referral to ortho- pedics in those instances. [Source pages 962–969]

- Presence of a scapula fracture mandates investi- gation for associated intrathoracic injuries. [Source pages 962–969]

- Displaced glenoid articular fractures, angulated glenoid neck fractures, and certain acromial and cora- coid fractures may require surgical intervention. [Source pages 962–969]

- ■■ ACROMIOCLAVICULAR JOINT INJURIES Clinical Features Acromioclavicular joint injuries range from mild sprain to complete disrup- tion of all ligaments that attach the scapula and clavicle. [Source pages 962–969]

Clinical reasoning structure

- **Problem representation:** Shoulder and Humerus Injuries; source pages 962–969.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Shoulder and Humerus Injuries**.

- Source-page range: **962–969**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 962

CHAPTER 172: Shoulder and Humerus Injuries 943

fractures that are open, have neurovascular compromise, or have persistent skin tenting. Clavicle fractures that are severely comminuted or displaced may benefit from operative intervention so consider early referral to orthopedics in those instances. Presence of a scapula fracture mandates investigation for associated intrathoracic injuries. Displaced glenoid articular fractures, angulated glenoid neck fractures, and certain acromial and coracoid fractures may require surgical intervention.

■■ ACROMIOCLAVICULAR JOINT INJURIES

Clinical Features

Acromioclavicular joint injuries range from mild sprain to complete disruption of all ligaments that attach the scapula and clavicle. AC joint injuries occur from a direct force to the joint with the arm adducted or from a fall on the outstretched hand with an indirect force transmitted to the joint. Classification of these injuries and their physical findings are described in Table 172-1.

Diagnosis and Differential

Diagnosis is clinical. Acromioclavicular radiographs can help determine the severity of the injury and identify any associated fractures.

Table 172-1 Classification and Physical Findings in Acromioclavicular Joint Injuries

Type Injury Mechanism Radiograph/Exam

I Sprained Radiograph: Normal

acromioclavicular Exam: Tenderness over

ligaments acromioclavicular joint

II Acromioclavicular Radiograph: Slight widening

ligaments of acromioclavicular joint;

ruptured; clavicle elevated 25%–50%

coracoclavicular above acromion; may be
ligaments sprained slight widening of the
coracoclavicular interspace
Exam: Tenderness and
mild step-off deformity of
acromioclavicular joint
III Acromioclavicular Clavicle elevated
ligaments 100% above acromion;
ruptured; coracoclavicular interspace
coracoclavicular widened 25%–100%
ligaments Exam: Distal end of clavicle
ruptured; deltoid prominent; shoulder droops
trapezius muscles
detached
(Continued)
172_Cydulka_Ch172.indd 943 22/03/17 12:57 PM

Source page 963

944 SECTION 19: Injuries to the Bones, Joints, and Soft Tissue
Table 172-1 Classification and Physical Findings in Acromioclavicular Joint
Injuries (Continued)
Type Injury Mechanism Radiograph/Exam
IV Rupture of all sup- Radiograph: May appear
porting structures; similar to type II and III;
clavicle displaced axillary radiograph required
posteriorly in to visualize posterior
or through the dislocation
trapezius Exam: Possible posterior
displacement of clavicle
V Rupture of all Radiograph:
supporting Acromioclavicular joint
structures (more dislocated; generally
severe form of 200%–300% disparity
type III injury) of coracoclavicular inter-
space compared to normal
shoulder
Exam: More pain; gross
deformity of clavicle
VI Acromioclavicular Radiograph:
ligaments dis- Acromioclavicular joint
rupted; coracocla- dislocated; clavicle

vicular ligaments displaced inferiorly
may be disrupted; Exam: Severe swelling;
deltoid and multiple associated injuries
trapezius muscles
disrupted

Emergency Department Care and Disposition

Treatment for type I and II injuries includes sling immobilization, rest, ice, and analgesics. Early range-of-motion exercises are recommended at 7 to 14 days post injury. Treatment for type III is controversial, but the trend favors conservative management with sling immobilization rather than operative management. Treatment for type IV through VI is operative and warrants orthopedic consultation.

■■ GLENOHUMERAL JOINT DISLOCATION

Clinical Features

Anterior shoulder dislocations, the most common type of shoulder dislocations, are generally the result of an abduction, extension, and external rotation force on the arm. Patients present with pain, holding their arm in slight abduction and external rotation. The affected shoulder loses its normal contour. Patients resist any adduction or internal rotation. The humeral head is sometimes palpable anteriorly. Decreased pinprick sensation over the skin of the lateral deltoid is a sign of axillary nerve injury. Posterior shoulder dislocations often result from an indirect, abduction and internal

172_Cydulka_Ch172.indd 944 22/03/17 12:57 PM

Source page 964

rotational force or from a direct blow to the anterior shoulder. Patients present with pain, anterior flattening of the normal shoulder contour, and prominence of the posterior shoulder. Patients are unable to externally rotate or abduct the affected arm. Inferior shoulder dislocations (luxatio erecta) result from a hyper abduction force. Patients present with their arm completely abducted, elbow flexed, and hand resting on or behind their head. The humeral head is palpable on the lateral chest wall. Superior shoulder dislocations are very rare.

Diagnosis and Differential

Anteroposterior and scapular “Y” view radiographs confirm the type of dislocation and identify any associated fractures. The presence of minor fractures, such as Hill–Sachs lesions (humeral head defects) or Bankart lesions (glenoid labral defects), does not change ED management. Consider omitting prereduction radiographs in patients, who have a history of recurrent shoulder dislocation, presenting with signs and symptoms of a

recurrence in the absence of trauma.

Emergency Department Care and Disposition

Reduction techniques include traction, leverage, and scapular manipulation. Be familiar with at least two or three techniques. The most common techniques are detailed below. Procedural sedation is recommended; however, an intraarticular injection of 10 to 20 mL of 1% lidocaine can facilitate reduction and may obviate the need for sedation.

1. Modified Hippocratic technique. This method uses traction-countertraction. Place the patient supine with their arm abducted and the elbow flexed 90°. A sheet is placed across the thorax of the patient and tied around the waist of the assistant. Another sheet is placed around the patient's flexed elbow and the clinician's waist. Gradually apply traction to the patient's proximal forearm while the assistant provides countertraction. Gentle internal and external rotation may help the reduction.

2. Snowbird technique. This is another version of the traction-countertraction technique where the patient is sitting and the elbow is flexed to 90°. A belt or strap is placed around the proximal forearm with the loop of the belt hanging. The clinician can step on the belt loop to provide traction using downward pressure while an assistant uses a sheet around the patient's torso to provide countertraction. Gentle external rotation of the arm can aid the reduction.

3. Stimson technique. Place the patient prone with the affected arm hanging over the side of the stretcher. Attach at 10-lb weight to the wrist. Inject intraarticular lidocaine. Reduction generally occurs in 20 to 30 minutes after the muscles have completely relaxed.

4. Scapular manipulation technique. Scapular manipulation accomplishes reduction by repositioning the glenoid fossa rather than the humeral head. The first step is to apply traction to the patient's arm held in 90° of forward flexion. This can be accomplished in the prone position or in a seated position with an assistant applying traction. Position the arm in slight external rotation. Push the scapular tip as far medially

172_Cydulka_Ch172.indd 945 22/03/17 12:57 PM

Source page 965

as possible while stabilizing the superior aspect of the scapula with the other hand. A small amount of dorsal displacement of the scapula tip is recommended.

5. External rotation (Kocher's) technique. Place the patient supine with the arm adducted to the side. With the patient's elbow flexed

to 90°, slowly and gently externally rotate the arm. No traction is applied. Reduction is subtle and usually occurs before reaching the coronal plane.

6. Milch technique. With the patient supine, slowly abduct and externally rotate the arm to the overhead position. Apply gentle traction with the elbow fully extended. If reduction is not achieved, attempt to manipulate the humeral head into the glenoid fossa with the free hand.

7. Cunningham technique. With the patient seated and the clinician sitting to the side of the patient, have the patient rest their hand of the affected extremity on the clinician's shoulder. The affected arm should be fully adducted with the elbow flexed. The clinician then gently massages the trapezius and deltoid, which helps relax the patient. It is important for the patient to remain seated in an upright position with their scapula and shoulders retracted as much as possible. Continue to massage the biceps at mid-humeral level as the patient elevates and shrugs or retracts their shoulders. Once the patient is relaxed, the humerus will reduce back into place.

Once reduced, assess neurovascular status and provide sling immobilization in order to maintain the shoulder in adduction and internal rotation. Post reduction radiographs are useful for confirmation and documentation of successful reduction. Urgent orthopedic follow-up is necessary. Complications include recurrence, rotator cuff tears, humeral head defects, glenoid labral defects, and neurovascular injuries. Early operative repair may decrease the incidence of recurrence.

■■ HUMERUS FRACTURES

Clinical Features

Patients with proximal humeral fractures have pain, swelling, tenderness, ecchymosis, and crepitus about the shoulder. Range of motion is severely limited; patients hold their arm closely against the chest wall. Patients with humeral shaft fractures present with pain, swelling, localized tenderness, limited mobility, and crepitus on palpation. Shortening of the arm can be seen in displaced fractures. A careful neurovascular exam is essential. Injuries to the axillary nerve and artery are common in proximal humerus fractures. The radial nerve is most frequently injured in humeral shaft fractures and manifests as weak wrist extension, wrist drop, or altered sensation in the dorsal web space of the thumb index web space.

Diagnosis and Differential

Radiographs confirm the diagnosis. The Neer classification system divides the proximal humerus into four parts (articular surface of the humeral head, greater tubercle, lesser tubercle, and diaphysis of the humerus) and is used

to guide treatment.

172_Cydulka_Ch172.indd 946 22/03/17 12:57 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 962–969.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 173: Pelvis, Hip, and Femur Injuries

Source pages 970–975 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 970–975

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Pelvis, Hip, and Femur Injuries**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 173: Pelvis, Hip, and Femur Injuries 951 Pelvis Fracture Patient is hemodynamically stable or responded to initial resuscitation Unstable patient to whom a pelvic binding device has been applied CT scan Results of FAST exam performed during primary survey Negative Positive Angiography* Laparotomy *If unavailable or significant delay, then consider extraperitoneal packing in OR while waiting. [Source pages 970–975]

- Suggested algorithm for pelvic fracture treatment. [Source pages 970–975]

- FAST, focused abdominal sonogram for trauma; OR, operating room. [Source pages 970–975]

- After excluding other sources of hemorrhage, treatment for ongoing hemodynamic instability in patients with pelvic fractures includes angi- ography with embolization and external fixation. [Source pages 970–975]

- An algorithm for treating severe pelvic fractures is listed in Figure 173-1. [Source pages 970–975]

Clinical reasoning structure

- **Problem representation:** Pelvis, Hip, and Femur Injuries; source pages 970–975.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Pelvis, Hip, and Femur Injuries**.

- Source-page range: **970–975**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 970

CHAPTER 173: Pelvis, Hip, and Femur Injuries 951

Pelvis Fracture Patient is

hemodynamically

stable or responded

to initial resuscitation

Unstable patient

to whom a pelvic

binding device has

been applied CT scan

Results of FAST

exam performed

during primary

survey

Negative Positive

Angiography* Laparotomy

*If unavailable or significant delay, then consider

extraperitoneal packing in OR while waiting.

Figure 173-1. Suggested algorithm for pelvic fracture treatment. FAST, focused

abdominal sonogram for trauma; OR, operating room.

4. After excluding other sources of hemorrhage, treatment for ongoing

hemodynamic instability in patients with pelvic fractures includes angi-

ography with embolization and external fixation.

5. An algorithm for treating severe pelvic fractures is listed in Figure 173-1.

6. With the exception of lateral compression type I and AP compression

type I injuries, all other pelvic ring fractures require open reduction and

internal fixation (ORIF).

7. Acetabular fractures require orthopedic consultation and hospital admis-

sion. Nondisplaced fractures may be treated with analgesia and bed rest.

Displaced fractures are treated with early ORIF.

8. The treatment and disposition of single bone fractures are listed in Table 173-1.

■■ HIP FRACTURES

Clinical Features

The vast majority of hip fractures occur in elderly patients with osteoporosis or other bony pathology who present after a fall (Table 173-2). The affected leg is classically shortened and externally rotated. Patients with hip fractures may complain of pain at the site of injury or in the groin and knee. After performing a primary survey and stabilizing the patient, examine the patient for pain, shortening, rotation, deformities, pelvic or sacral tenderness, and neurovascular status. If no significant abnormalities are found, carefully evaluate range of motion. A history of fall or significant trauma should prompt the examiner to evaluate for other injuries.

173_Cydulka_Ch173.indd 951 22/04/17 4:44 pm

Source page 971

Table 173-2 Proximal Femur Fractures: Demographics and Clinical Features

Fracture	Demographics	Mechanism	Clinical Findings	Concomitant Injuries
Femoral head	Isolated fracture rare; seen in 6–16% of hip trauma; dashboard knee most common	Usually result of high-energy and exterior dislocations	Limb shortened and internally rotated (posterior dislocation)	Closed head injury; intrathoracic and/or intraabdominal injuries; pelvic fracture, knee injuries
Femoral neck	Common in older patients with osteoporosis; younger patients or stress fractures in young may be shortened and externally rotated	Low-impact falls or torsion	Ranges from pain with weight-bearing to inability to ambulate; limb shortened and externally rotated	Ipsilateral femoral shaft fracture
Greater trochanteric	Uncommon; older patients or adolescents	Direct trauma (older patients); avulsion due to	Ambulatory; — pain with palpation or	—

contraction of abduction gluteus medius (young patients)

Lesser trochanteric > adults (adolescents); rare in younger patients

Uncommon; avulsion due to forceful contraction of iliopsoas with flexion or rotation

Usually ambulatory; pain

avulsion of pathologic bone (older adults)

Intertrochanteric

Common in older patients with osteoporosis; concomitant traumatic injuries

Falls; high-energy trauma; shortened thigh; concomitant traumatic injuries

Severe pain; swelling; ecchymosis; shortened thigh; concomitant traumatic injuries

Anemia from blood loss into limb

Subtrochanteric

Similar to intertrochanteric; 15% of hip fractures pathologic

Falls; high-energy trauma; abducted, and externally rotated

Severe pain; Vascular injuries, anemia/hypovolemic shock from fracture itself or other traumatic injuries

Diagnosis and Differential

Radiographic evaluation of the hip includes AP and lateral views. Other radiographic views that may be helpful include an AP pelvis, which allows a comparison of both sides, and Judet views. Radiographs of the femur and knee may also be indicated. Significant pain with weight-bearing or inability

173_Cydulka_Ch173.indd 952 22/04/17 4:44 pm

Source page 972

Table 173-3 Proximal Femur Fractures: Treatment Issues

Fracture	ED Management	Disposition and Follow-Up	Complications
Femoral head	Immediate orthopedic consultation; emergent closed reduction of dislocation; ORIF if closed reduction is unsuccessful	Admission to trauma service	AVN; posttraumatic arthritis; sciatic nerve injury; heterotopic ossification

Femoral neck Orthopedic consul- Admission to AVN; infection; DVT
tation; ranges from orthopedic service and/or pulmonary
nonoperative to embolus
total hip arthroplasty
Greater trochanteric Analgesics; Orthopedic follow- Nonunion rare
protected weight- up 1–2 weeks;
bearing possible ORIF if dis-
placement >1 cm
Lesser trochanteric Analgesics; weight- Orthopedic or PCP Nonunion rare
bearing as toler- follow-up in
ated; evaluate for 1–2 weeks; admit
possible pathologic or urgent follow-up
fracture for pathologic
fracture
Intertrochanteric Orthopedic Admit for eventual DVT and/or
consultation ORIF; may need pulmonary embolism;
preoperative testing infection
and clearance by
PCP or hospitalist
Subtrochanteric Orthopedic con- Admit for ORIF DVT and/or
sultation; consider pulmonary embolism;
Hare® or Sager® infection; malunion
splint (shortened limb);
nonunion

Abbreviations: AVN, avascular necrosis; DVT, deep venous thrombosis; ORIF, open reduction and internal fixation; PCP, primary care physician.

to bear weight in patients with normal radiographs should raise suspicion for occult fracture, especially at the femoral neck or acetabulum. MRI is very sensitive (nearly 100%) for identifying occult hip fractures and may identify other sources of pain. CT may be useful in identifying fractures not seen on radiographs, but it is not as sensitive as MRI. The differential diagnosis includes pelvic fracture, hip dislocation, femur fracture, sprains, and strains.

Emergency Department Care and Disposition

1. The treatment of hip fractures is listed in Table 173-3.
2. If clinical suspicion of an occult fracture is high, obtain an MRI scan in the ED. Alternatively, arrange follow-up MRI within 24 to 48 hours and have the patient remain non-weight-bearing.
3. Traction devices may be used for immobilization of subtrochanteric fractures; however, they are contraindicated in femoral neck fractures.

Source page 973

954 SECTION 19: Injuries to the Bones, Joints, and Soft Tissue

Upward pull

on femur

Downward pressure

on pelvis

A

Downward pressure on pelvis

External and internal rotation

and upward pull on femur

Following reduction

B

Figure 173-2. A and B. Allis maneuver for reduction of posterior hip dislocation.

■■ HIP DISLOCATIONS

Hip dislocations may be anterior or posterior, and commonly result from a high-speed motor vehicle crash. Ninety percent of hip dislocations are posterior, and they may be associated with acetabular fractures. On examination, the extremity is shortened, internally rotated, and adducted. With anterior dislocations the extremity is held in abduction and external rotation. Assess the patient's neurovascular status. Obtain radiographs of the hip and pelvis to evaluate for hip dislocation. Further assessment of the acetabulum and femur may be done with Judet views or CT.

Hip dislocations are true orthopedic emergencies and should be reduced within 6 hours because delays in reduction are associated with a higher incidence of avascular necrosis. One of the most common methods for hip reduction is described in Figure 173-2. After reduction, gently test range of motion and repeat the neurovascular exam. Order post-reduction radiographs or CT to confirm reduction and evaluate for injuries not apparent on initial radiographs. Coordinate post-reduction care with the orthopedist.

173_Cydulka_Ch173.indd 954 22/04/17 4:44 pm

References/provenance

- Source: *Tintinalli's Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 970–975.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 174: Knee and Leg Injuries

Source pages 976–981 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 976–981

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Knee and Leg Injuries**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 174: Knee and Leg Injuries 957 Fall or blunt-trauma mechanism Yes No Age <12 or age >50 years No knee radiography Yes No Knee radiography Able to walk four weight-bearing steps in ED No Yes Knee radiography No knee radiography Figure 174-1. [Source pages 976–981]

- Reproduced with permis- sion from Seaberg DC, Yealy DM, Lukens T, et al: Multicenter comparison of two clinical decision rules for the use of radiography in acute, high-risk knee injuries, Ann Emerg Med. [Source pages 976–981]

- Table 174-1 Ottawa Knee Rules: Radiograph If One Criterion Is Met Patient age >55 years (rules have been validated for children 2–16 years of age) Tenderness at the head of the fibula Isolated tenderness of the patella Inability to flex knee to 90 degrees Inability to transfer weight for four steps both immediately after the injury and in the ED Table 174-2 Mechanism of Knee Injury and Treatment Fracture Mechanism Treatment Patella Direct blow (i.e., fall, Nondisplaced fracture with intact extensor motor vehicle crash) or mechanism: knee immobilizer, rest, ice, forceful contraction of analgesia. [Source pages 976–981]

- Follow-up for serial radiographs quadriceps muscle Displaced >3 mm, articular incongruity >2 mm, or with disruption of extensor mechanism: above treatment plus early referral for ORIF Severely comminuted fracture: surgical debridement of small fragments and suturing of quadriceps and patellar tendons Open fracture: irrigation and antistaphylococcal antibiotics in the ED; debridement and irrigation in the operating room Femoral Fall with axial load with Incomplete or nondisplaced fractures in any age condyles valgus/varus/rotational group or stable impacted fractures in the elderly: forces, or a blow to the long leg splinting and orthopedic referral distal femur Displaced fractures or fractures with any degree of joint incongruity: splinting and orthopedic consult for ORIF (Continued) 174_Cydulka_Ch174.indd 957 22/04/17 4:44 pm 958 SECTION 19: Injuries to the Bones, Joints, and Soft Tissue Table 174-2 Mechanism of Knee Injury and Treatment (Continued) Fracture Mechanism Treatment Tibial spines Force directed against Incomplete or nondisplaced fractures: immobi- and tuberosity flexed proximal tibia in lization in full extension (knee immobilizer) and an anterior or posterior orthopedic referral in 2–7 days direction (i.e., motor Complete or displaced fracture: early vehicle crash, sporting orthopedic referral, often requires ORIF injury) Tibial Sudden force to flexed Incomplete or small avulsion fracture: tuberosity knee with quadriceps immobilization contracted Complete avulsion: ORIF Tibial plateau Valgus or varus forces Nondisplaced, lateral fracture: knee immobi- combined with axial lizer with non-weight-bearing and orthopedic load that drives the referral in 2–7 days femoral condyle into the Depression of articular surface: early tibia (i.e., fall, leg hit by

- ■■ DISLOCATIONS Clinical Features Patella dislocation results in pain and deformity of the knee. [Source pages 976–981]

Clinical reasoning structure

- **Problem representation:** Knee and Leg Injuries; source pages 976–981.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Knee and Leg Injuries**.

- Source-page range: **976–981**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 976

CHAPTER 174: Knee and Leg Injuries 957

Fall or blunt-trauma mechanism

Yes No

Age <12 or age >50 years No knee radiography

Yes No

Knee radiography Able to walk four

weight-bearing steps

in ED

No Yes

Knee radiography No knee radiography

Figure 174-1. Pittsburgh Knee Rules for radiography. Reproduced with permis-

sion from Seaberg DC, Yealy DM, Lukens T, et al: Multicenter comparison of two

clinical decision rules for the use of radiography in acute, high-risk knee injuries,

Ann Emerg Med. 1998;Jul;32(1):8-13.

Table 174-1 Ottawa Knee Rules: Radiograph If One Criterion Is Met

Patient age >55 years (rules have been validated for children 2–16 years of age)

Tenderness at the head of the fibula

Isolated tenderness of the patella

Inability to flex knee to 90 degrees

Inability to transfer weight for four steps both immediately after the injury and in the ED

Table 174-2 Mechanism of Knee Injury and Treatment

Fracture	Mechanism	Treatment
Patella	Direct blow (i.e., fall, motor vehicle crash) or forceful contraction of quadriceps muscle	Nondisplaced fracture with intact extensor mechanism: knee immobilizer, rest, ice, analgesia. Follow-up for serial radiographs
		Displaced >3 mm, articular incongruity >2 mm, or with disruption of extensor mechanism: above treatment plus early referral for ORIF
		Severely comminuted fracture: surgical debridement of small fragments and suturing of quadriceps and patellar tendons
		Open fracture: irrigation and antistaphylococcal antibiotics in the ED; debridement and irrigation in the operating room
Femoral condyles	Fall with axial load with valgus/varus/rotational forces, or a blow to the distal femur	Incomplete or nondisplaced fractures in any age group or stable impacted fractures in the elderly: long leg splinting and orthopedic referral
		Displaced fractures or fractures with any degree of joint incongruity: splinting and orthopedic consult for ORIF

(Continued)

Source page 977

Table 174-2 Mechanism of Knee Injury and Treatment (Continued)

Fracture	Mechanism	Treatment
Tibial spines and tuberosity	Force directed against flexed proximal tibia in full extension (knee immobilizer) and an anterior or posterior direction (i.e., motor vehicle crash, sporting injury)	Incomplete or nondisplaced fractures: immobilization in full extension (knee immobilizer) and orthopedic referral in 2–7 days
		Complete or displaced fracture: early orthopedic referral, often requires ORIF
Tibial tuberosity	Sudden force to flexed knee with quadriceps contracted	Incomplete or small avulsion fracture: immobilization
		Complete avulsion: ORIF
Tibial plateau	Valgus or varus forces combined with axial load that drives the femoral condyle into the	Nondisplaced, lateral fracture: knee immobilizer with non-weight-bearing and orthopedic referral in 2–7 days
		Depression of articular surface: early

tibia (i.e., fall, leg hit by orthopedic consult for ORIF
car bumper)

Abbreviation: ORIF, open reduction internal fixation.

■■ DISLOCATIONS

Clinical Features

Patella dislocation results in pain and deformity of the knee. Tearing of the medial knee joint capsule can occur. Knee dislocation results in significant ligamentous and capsular disruption. Spontaneous reduction occurs in 50% of patients. Multidirectional instability of the knee should raise the suspicion for a spontaneously reduced knee dislocation. Knee dislocations have a high incidence of associated injures, such as popliteal artery injury and peroneal nerve injury.

Diagnosis and Differential

Radiographs may help exclude associated fractures. Some authors recommend arteriography for all patients with confirmed knee dislocations. Others advocate for Doppler pressure indices (ankle-brachial index) and serial vascular exams if distal pulses are present, and there are no other signs of vascular injury (including ischemia, hemorrhage, or an expanding hematoma).

Emergency Department Care and Disposition

Procedural sedation is recommended to facilitate otherwise painful reductions. For patellar dislocation, flexing the hip and hyperextending the knee help slide the patella back into place. Knee immobilization and orthopedic or primary care physician follow-up are necessary. For knee dislocation, early reduction is essential along with documentation of pre-and post-reduction neurovascular status. Splinting the affected knee in 20 degrees of flexion is essential. Immediate orthopedic and vascular surgery consultation is warranted for all knee dislocations, and admission is mandatory for observation of neurovascular status.

174_Cydulka_Ch174.indd 958 22/04/17 4:44 pm

Source page 978

■■ TENDON, LIGAMENTOUS, and MENISCAL INJURIES

Clinical Features

Patients with quadriceps or patellar tendon rupture have pain and swelling about the knee and will not be able to extend the knee against resistance. A palpable defect is present above or below the knee depending on which tendon is involved. Patients with Achilles tendon rupture have severe pain in the calf and are unable to perform toe walk, run, or climb stairs. A palpable defect is often present in the Achilles tendon 2 to 6 cm above the

calcaneus. A positive Thompson test is diagnostic; with the patient lying prone and knee flexed at 90 degrees, the foot fails to plantar flex when the calf is squeezed. Most ligamentous injuries present with hemarthroses, though serious ligamentous injuries may present with little pain and no hemarthrosis due to complete disruption of the capsule. Patients with anterior cruciate ligament (ACL) tears often describe a “pop” and significant swelling over the next several hours after injury. Lachman’s test is the most sensitive test for ACL injuries. The anterior drawer and pivot shift test are also useful for diagnosis. Posterior cruciate ligament (PCL) injuries may result in a positive posterior drawer test; the composite history and exam findings, however, are more accurate for diagnosis of PCL injury. Medial and lateral collateral ligament injuries are diagnosed with abduction (valgus) and adduction (varus) stress testing in 30 degrees flexion. Laxity greater than 1 cm without a firm end point compared with the other knee is diagnostic for a complete torn medial or lateral collateral ligament. Stress testing should be repeated in extension. If laxity is present in extension, then this indicates injury to the cruciate ligament and posterior or posterolateral capsule. Peroneal nerve injury may occur with lateral injuries. Symptoms of meniscal injury include painful locking of the knee, a popping, clicking or snapping sensation, or a sense of instability with activity or joint swelling after activity. McMurray’s test and other tests for meniscal injury are not sensitive. Ligamentous injuries may be present as well. A patient with a ‘‘locked knee’’ from a torn meniscus will not be able to actively or passively range the affected knee and will have significant pain.

Diagnosis and Differential

The diagnosis is largely clinical. A high-riding patella may be seen on the lateral radiograph of the knee with patellar tendon rupture. Ultrasound or MRI can help identify Achilles tendon rupture when the diagnosis is not clear. Radiographs are usually normal or show a joint effusion in ligamentous or meniscal injuries. An avulsion fracture at the site of the lateral capsular ligament on the lateral tibial condyle (Segond fracture) is associated with ACL rupture. Outpatient MRI or arthroscopy provides definitive diagnosis.

Emergency Department Care and Disposition

Treatment of patellar or quadriceps tendon rupture includes knee immobilization and orthopedic consultation for surgical repair, usually within the first 7 to 10 days after the injury. Treatment of Achilles tendon rupture includes splinting in plantar flexion, non-weight-bearing, and referral to

174_Cydulka_Ch174.indd 959 22/04/17 4:44 pm

Source page 979

960 SECTION 19: Injuries to the Bones, Joints, and Soft Tissue

orthopedics for possible surgical repair. Treatment for ligamentous and meniscal injuries includes knee immobilization, ice, elevation, analgesics, and orthopedic referral. Treatment for a ‘‘locked knee’’ includes analgesia and closed reduction. Procedural sedation may be necessary. Place the patient supine with the knee flexed 90 degrees over the side of the bed. Apply longitudinal traction to the knee, and at the same time, apply internal and external rotation to the knee to unlock the joint. Consult orthopedics if the reduction is not successful.

■■ OVERUSE INJURIES

Patellar tendonitis or “jumper’s knee” presents with pain over the patellar tendon worsened by running up hills, jumping, or standing from a seated position. Treatment includes heat, nonsteroidal anti-inflammatory drugs (NSAIDs), and quadriceps-strengthening exercises. Steroid injections should be avoided. Shin splints and stress fractures can present with pain over the anterior leg. Patients typically describe a change or sudden increase in their training pattern. Patients present with activity-induced pain relieved by rest, which can progress to constant pain. Radiographs are often normal. Discontinuation of the activity for several weeks is the treatment for both shin splints and stress fractures. If a stress fracture is suspected, an outpatient bone scan or MRI can confirm the diagnosis.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 274, “Knee Injuries” by Rachel R. Bengtzen, Jeffrey N. Glaspy and Mark T. Steele; Chapter 275, “Leg Injuries” by Paul R. Heller.

174_Cydulka_Ch174.indd 960 22/04/17 4:44 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 976–981.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 175: Ankle and Foot Injuries

Source pages 982–985 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 982–985

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Ankle and Foot Injuries**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 175: Ankle and Foot Injuries 963 Table 175-1 Associated and Occult Injuries of the Ankle Injury Clinical Suspicion Confirmatory Test Important to identify in the ED Maisonneuve fracture Examine proximal fibula and Fibula radiograph shaft, tenderness to palpation; proximal fracture and syndes- mosis tear indicate unstable fracture Peroneal tendon dislocation Palpable anterior tendon Clinical examination dislocation or subluxation Usually identified in follow-up of ankle sprains Osteochondral injuries Diffuse ankle swelling, passive Ankle mortise view/CT plantarflexion Syndesmosis tear Significant ankle pain, positive Widened mortise with squeeze test weight bearing Anterior calcaneal process Tenderness more inferoante- Lateral ankle fracture rior than a typical ankle sprain radiograph/CT Lateral talar process Tenderness just distal to the Ankle mortise view/CT fracture tip of fibula Os trigonum Tenderness anterior to Achilles Lateral ankle radiograph tendon ■■ FOOT INJURIES Clinical Features The foot is divided into the hindfoot, midfoot, and forefoot. [Source pages 982–985]

- The Chopart joint separates the hindfoot from the midfoot. [Source pages 982–985]

- The Lisfranc joint separates the midfoot from the forefoot. [Source pages 982–985]

- Diagnosis and Differential Pay special attention on physical examination to the base of the fifth metatarsal and the base of the second metatarsal. [Source pages 982–985]

- If plain radiographs are nondiagnostic, CT is indicated for suspected Lisfranc joint injuries. [Source pages 982–985]

Clinical reasoning structure

- **Problem representation:** Ankle and Foot Injuries; source pages 982–985.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Ankle and Foot Injuries**.

- Source-page range: **982–985**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 982

CHAPTER 175: Ankle and Foot Injuries 963

Table 175-1 Associated and Occult Injuries of the Ankle

Injury Clinical Suspicion Confirmatory Test

Important to identify in the ED

Maisonneuve fracture Examine proximal fibula and Fibula radiograph

shaft, tenderness to palpation;

proximal fracture and syndes-

mosis tear indicate unstable

fracture

Peroneal tendon dislocation Palpable anterior tendon Clinical examination

dislocation or subluxation

Usually identified in follow-up

of ankle sprains

Osteochondral injuries Diffuse ankle swelling, passive Ankle mortise view/CT

plantarflexion

Syndesmosis tear Significant ankle pain, positive Widened mortise with

squeeze test weight bearing

Anterior calcaneal process Tenderness more inferoante- Lateral ankle

fracture rior than a typical ankle sprain radiograph/CT

Lateral talar process Tenderness just distal to the Ankle mortise view/CT

fracture tip of fibula

Os trigonum Tenderness anterior to Achilles Lateral ankle radiograph

tendon

■■ FOOT INJURIES

Clinical Features

The foot is divided into the hindfoot, midfoot, and forefoot. The Chopart

joint separates the hindfoot from the midfoot. The Lisfranc joint separates

the midfoot from the forefoot.

Diagnosis and Differential

Pay special attention on physical examination to the base of the fifth metatarsal and the base of the second metatarsal. If plain radiographs are nondiagnostic, CT is indicated for suspected Lisfranc joint injuries.

Hindfoot Injuries

Calcaneal injuries require a large force, and associated injuries are common. Measure Boehler's angle on a lateral radiograph view (formed by the intersection of a line connecting the posterior tuberosity and apex of the posterior facet, and a line from the posterior facet to the apex of the anterior facet) if concerned for a calcaneal compression fracture. An angle <20° is suggestive of a fracture. Treat with a posterior splint, elevation, analgesics, and orthopedic consultation. Manage small avulsion fractures of the talus with posterior splinting and timely orthopedic follow-up. Subtalar dislocations, as well as major fractures of the talar neck or body, require immediate orthopedic consultation.

Midfoot Injuries

Injuries of the tarsometatarsal joint combined with pain with midfoot torsion are suspicious for a Lisfranc injury. Lisfranc joint injuries are often

175_Cydulka_Ch175.indd 963 22/03/17 2:18 PM

Source page 983

175_Cydulka_Ch175.indd 964

Table 175-2 Summary of Emergent Care of Bony Foot Injuries

Fracture or Injury Type	ED Imaging	ED Care*	Orthopedic Referral (immediate: within 24 h; early: within 2 wk)	Weight-Bearing Status	Home Care/ Special Care and Management Considerations
Calcaneal, intra- and extra-articular	Plain films, Boehler angle; CT for subtle findings	Posterior splint	Intra-articular: immediate; extra-articular: early	NWBS; RICE	Possible surgery
Talus fracture	CT	Posterior splint	Major: immediate; minor: early	NWBS; RICE	Possible surgery Risk of avascular necrosis
Lisfranc	CT	Splint	Displaced: ortho consult in ED; nondisplaced: early	NWBS; RICE	Possible surgery Risk of compartment syndrome; arthritis
Navicular fracture displaced: immediate	Plain films or CT	Splint	Nondisplaced: early;	NWBS; RICE	Possible surgery Risk of avascular necrosis; non-union
Cuboid fracture	Plain films or CT	Splint	Early	NWBS; RICE	Comminuted: possible surgery
Cuneiform fracture	Plain films or CT	Splint	Early	NWBS; RICE	Medial: possible surgery
Jones	Plain films; CT for athletes	Splint	Early	NWBS; RICE	Athletes: possible surgery

Metatarsal fracture Plain films Posterior splint Within a week for a cast NWBS; RICE Surgery not likely

Stress fracture Clinical Cessation of

causative activity

Phalange fracture Plain films Buddy taping Hard-soled shoe and

weight bearing as

tolerated

Open fractures of Consider antibiotics, Td Pain control Ortho consult in ED

any kind

*

All patients with fractures should receive adequate analgesia, and splints should be well padded.

22/03/17 2:18 PM

Abbreviations: NWBS, nonweightbearing status; ortho, orthopedist; RICE, rest, ice, compression, elevation; Td, Tetanus booster.

Source page 984

associated with a fracture, most frequently noted at the base of the second metatarsal. A gap of >1 mm between the bases of the first and second metatarsal bones on plain film is considered unstable. If standard radiographs are nondiagnostic, complete workup may require CT imaging. Isolated navicular, cuboid, and cuneiform injuries are rare and treated conservatively.

Forefoot

Treat nondisplaced metatarsal shaft fractures with a posterior splint or orthopedic shoe. Fractures with greater than 3 to 4 mm displacement on plain films, however, require surgical reduction. Treat pseudo Jones fractures (nondisplaced avulsion fractures of the tuberosity of the fifth metatarsal) with a walking cast. Manage true Jones fractures (metaphyseal-diaphyseal junction fracture of the fifth metatarsal) with a nonweightbearing cast and timely orthopedic follow-up for potential surgery. Nondisplaced phalangeal fractures require buddy taping and a stiff-sole (orthopedic) shoe. Treat displaced phalangeal fractures and dislocations with a digital block, reduction by manual traction, and buddy taping. Recommended imaging and care for foot injuries can be found in Table 175-2.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 276, "Ankle Injuries," by Daniel A. Handel and Sarah Andrus Gaines; and Chapter 277, "Foot Injuries," by Sarah Andrus Gaines and Daniel A. Handel.

Source page 985

C h a p ter Compartment Syndrome

176 Sandra L. Najarian

Elevated pressures within a confined muscle compartment can lead to functional and circulatory impairment of that limb. The most common compartments affected are in the lower leg and forearm. This syndrome develops as a result of external compressive forces on a limb or from any mechanism that increases the compartmental size and pressure (Table 176-1).

■■ CLINICAL FEATURES

Severe and difficult-to-control pain, pain out of proportion to examination, and pain with passive stretch of the limb are the hallmark symptoms of this disease. Nerve dysfunction often accompanies the pain and causes burning or dysesthesias in the sensory distribution of the nerve. Motor function can be impaired as well. On exam, the compartment is often swollen, firm, and tender to palpation. Since tissue pressure does not typically exceed arterial pressure, the affected limb often has normal distal pulse, temperature, and color. Symptoms can begin within a few hours after the injury or up to 48 hours after the inciting event.

■■ DIAGNOSIS AND DIFFERENTIAL

Diagnosis is largely clinical. Direct measurement of the compartment is necessary when the diagnosis is in question or in patients who are obtunded or sedated. Several commercial devices are available to measure compartment pressures. Normal compartment pressure is < 10 mm Hg. The exact pressure elevation at which cell death occurs is unclear. Pressures between 30 to 50 mm Hg are felt to be detrimental if left untreated for several hours. The diastolic blood pressure minus the measured tissue pressure, or “delta pressure,” better predicts the potential for irreversible muscle damage. A delta pressure of 30 mm Hg is most commonly used to diagnose acute

Table 176-1 Causes of Compartment Syndrome

Orthopedic	Tibial fractures
	Forearm fractures
Vascular	Ischemic-reperfusion injury
	Hemorrhage
Iatrogenic	Vascular puncture in anticoagulated patients
	IV/intra-arterial drug injection
	Constrictive casts
Soft tissue injury	Prolonged limb compression
	Crush injury
	Burns
Hematologic	Hemophilia

Adverse effects of anticoagulants (warfarin)

966

176_Cydulka_Ch176.indd 966 22/03/17 2:56 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.

- Content ID: SEG-TINTINALLI-EMM8-FULL.

- Source pages: 982–985.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 176: Compartment Syndrome

Source pages 986–989 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 986–989

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Compartment Syndrome**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 176: Compartment Syndrome 967 compartment syndrome. [Source pages 986–989]

- Hypotensive patients do not tolerate elevated com- partment pressures as well as normotensive patients. [Source pages 986–989]

- The differential diag- nosis for compartment syndrome includes other causes of pain, such as fracture, hematoma, or infection, and other causes of neurologic or vascular compromise. [Source pages 986–989]

- ■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION Once the diagnosis is confirmed, surgical fasciotomy is necessary. [Source pages 986–989]

- Admit all patients to the operating room or appropriate inpatient service for obser- vation and serial examinations. [Source pages 986–989]

Clinical reasoning structure

- **Problem representation:** Compartment Syndrome; source pages 986–989.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Compartment Syndrome**.

- Source-page range: **986–989**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 986

CHAPTER 176: Compartment Syndrome 967

compartment syndrome. Hypotensive patients do not tolerate elevated compartment pressures as well as normotensive patients. The differential diagnosis for compartment syndrome includes other causes of pain, such as fracture, hematoma, or infection, and other causes of neurologic or vascular compromise.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

Once the diagnosis is confirmed, surgical fasciotomy is necessary. Admit all patients to the operating room or appropriate inpatient service for observation and serial examinations.

1. While definitive management is being arranged, administer supplemental oxygen, correct hypotension, remove constrictive casts or dressings, and place the affected limb at the level of the heart.
2. Reverse anticoagulation in anticoagulated patients, and replace factor levels in hemophiliacs.
3. Functional outcomes are favorable when diagnosis and treatment of compartment syndrome occurs within 6 hours of its onset. Fasciotomy may be futile if tissue pressures have been elevated to 24 to 48 hours as permanent dysfunction may already be present.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 278, “Compartment Syndrome” by Paul R. Haller.

176_Cydulka_Ch176.indd 967 22/03/17 2:56 PM

Source page 987

This page intentionally left blank

176_Cydulka_Ch176.indd 968 22/03/17 2:56 PM

Source page 988

SECTION

Nontraumatic Musculoskeletal

Disorders 20

C h a p ter Neck and Back Pain

177 Amy M. Stubbs

Neck and back pain are major causes of disability in the United States and are commonly seen in the ED. While the majority of cases have nonspecific or benign etiologies, serious underlying pathology may exist. Clinicians must be vigilant in obtaining a thorough history, focusing on risk factors for pathology, and physical examination that includes a detailed neurologic examination.

■■ CLINICAL FEATURES

Neck and back pain have myriad causes, but can often be classified into two groups: uncomplicated musculoskeletal pain and those with pain attributable to radiculopathy (spinal nerve root compression) or myelopathy (spinal cord compression) (see Tables 177-1, 177-2, and 177-3). Thoracolumbar pain may also be categorized by symptom duration: acute (<6 weeks), subacute (6 to 12 weeks), or chronic (>12 weeks).

The history should include the onset, circumstances, location, radiation, duration, and exacerbating/alleviating factors of the pain. Historical risk factors or symptoms concerning for serious pathology should be sought (see Table 177-4).

Examination should include evaluation of mobility and range of motion. The patient should be exposed, observing and palpating the neck and back for point tenderness, deformity, lymphadenopathy, or signs of infection. A thorough neurologic examination, including assessment of specific nerve roots' strength, sensation, and reflexes, is essential (see Tables 177-2 and 177-3). Specific examination maneuvers should be performed as indicated for the complaint (see Table 177-5).

A rectal examination evaluating for tone, sensation, and masses is requisite if concern exists for spinal cord compression or other serious pathology.

■■ DIAGNOSIS AND DIFFERENTIAL

The differentials for neck and back pain are broad but can generally be guided by the history and physical examination. The majority of patients with neck and back pain will not require emergent imaging or diagnostic testing.

969

177_Cydulka_Ch177.indd 969 22/03/17 2:51 PM

Source page 989

970 SECTION 20: Nontraumatic Musculoskeletal Disorders

Table 177-1 Differentiating Cervical Radiculopathy from Uncomplicated Musculoskeletal Neck Pain

Factors Favoring Cervical Radiculopathy Factors Favoring Uncomplicated or Myelopathy* Musculoskeletal Neck Pain*

Pain from the neck radiates down the arm in dermatome pattern.

Tenderness of involved muscles; examiner may find a focal point of tenderness.

Sensory changes along dermatome distribution.

Atrophy or thinning of shoulder muscles may occur after rotator cuff injury.

Pushing down on top of head, with neck in extension (chin up) and head leaning toward on the symptomatic side elicits pain, typically toward or down the arm (positive Spurling's sign); 90% specific, 45% sensitive.

Pain increases with shoulder abduction (increased pain could derive from rotator cuff-related pain; pain may decrease with this maneuver).

Pain may worsen with Valsalva, which increases intrathecal pressure.

Repetitive movement of arm or shoulder at work or play; may be new activity.

Flex neck forward until chin meets chest or pain stops movement.

History of recent injury or recent event of an electric shock sensation radiating down spine into both arms is a positive result (Lhermitte's sign).

Occasionally, paresthesias occur.

Depressed reflexes or, uncommonly, increased reflexes (see also Table 177-2).

Pain is accompanied by "stiffness" of involved muscle group.

Patient in either pain category is not expected to have all, or majority, of listed signs.

*

Table 177-2 Signs and Symptoms of Cervical Radiculopathy

Disk Space	Cervical Root	Sensory Pain Complaint	Motor Abnormality	Altered Weakness	Reflex
C1–C2	C2	Neck, scalp	Scalp		
C4–C5	C5	Neck, shoulder, upper arm	Shoulder deltoid, biceps	Infraspinatus, biceps	Reduced biceps reflex
C5–C6	C6	Neck, shoulder, upper medial, index finger, scapular area, lateral forearm	Thumb and proximal forearm, reflex	Deltoid, biceps, wrist extensors	Reduced brachioradialis
C6–C7	C7	Neck, posterior arm, dorsum forearm	Middle finger thumb, index finger	Triceps, pronator teres	Reduced triceps reflex
		proximal forearm, chest, medial third of scapula, middle finger			

C7–T1 C8 Neck, posterior Ring and little Triceps, flexor Reduced

arm, ulnar side of fingers carpi ulnaris, triceps reflex

forearm, medial hand intrinsics

inferior scapular

border, medial

hand, ring, and

little fingers

177_Cydulka_Ch177.indd 970 22/03/17 2:51 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.

- Content ID: SEG-TINTINALLI-EMM8-FULL.

- Source pages: 986–989.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 177: Neck and Back Pain

Source pages 990–995 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 990–995

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Neck and Back Pain**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 177: Neck and Back Pain 971 Table 177-3 Symptoms and Signs of Lumbar Radiculopathies Disk Nerve Altered Space Root Pain Complaint Sensory Change Motor Weakness Reflex L2–3 L3 Medial thigh, knee Medial thigh, knee Hip flexors None L3–4 L4 Medial lower leg Medial lower leg Quadriceps Knee jerk L4–5 L5 Anterior tibia, Medial foot Extensor hallicus Biceps great toe longus femoris L5–S1 S1 Calf, little toe Lateral foot Foot plantar flexors Achilles Table 177-4 Risk Factors for Serious Causes of Neck and Back Pain Risk Factors Concern Historical Risk Factors Pain >6 weeks Tumor, infection Age <18, >50 Congenital anomaly, tumor, infection Major trauma Fracture Minor trauma in elderly or rheumatologic disease Fracture History of cancer Tumor Fever and rigors Infection Weight loss Tumor, infection Injection drug use Infection Immunocompromised Infection Night pain Tumor, infection Unremitting pain, even when supine Tumor, infection Incontinence Epidural compression Saddle anesthesia Epidural compression Severe/progressive neurologic deficit Epidural compression Anticoagulants and coagulopathy Epidural compression Physical Risk Factors Fever Infection Patient writhing in pain Infection, vascular cause Unexpected anal sphincter laxity Epidural compression Perianal/perineal sensory loss Epidural compression Major motor weakness/gait disturbance Nerve root or epidural compression Positive straight leg raise test Herniated disk Table 177-5 Diagnostic Maneuvers for Neck and Back Pain Exam Finding/Maneuver Significance Spurling’s sign: Downward pressure with head in exten- Cervical radiculopathy sion and leaning toward symptomatic sign produces pain Abduction relief sign: Relief placing hand of affected Disc protrusion/radiculopathy arm on top of head Lhermitte’s sig

- Plain films of the spine have low sensitivity but may be considered as an initial imaging step if clinical suspicions include tumor, fracture, or infection. [Source pages 990–995]

- For patients with pain and neurologic deficits, MRI is the definitive test. [Source pages 990–995]

- CT scan is helpful for identification of disorders of the bony skeleton, but its sensitivity for nerve root or spinal cord disorders is poor. [Source pages 990–995]

- CT myelography can serve as an alternative to MRI, if the latter is contraindicated or unattainable. [Source pages 990–995]

Clinical reasoning structure

- **Problem representation:** Neck and Back Pain; source pages 990–995.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Neck and Back Pain**.

- Source-page range: **990–995**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 990

CHAPTER 177: Neck and Back Pain 971

Table 177-3 Symptoms and Signs of Lumbar Radiculopathies

Disk Nerve Altered

Space Root Pain Complaint Sensory Change Motor Weakness Reflex

L2–3 L3 Medial thigh, knee Medial thigh, knee Hip flexors None

L3–4 L4 Medial lower leg Medial lower leg Quadriceps Knee jerk

L4–5 L5 Anterior tibia, Medial foot Extensor hallicus Biceps

great toe longus femoris

L5–S1 S1 Calf, little toe Lateral foot Foot plantar flexors Achilles

Table 177-4 Risk Factors for Serious Causes of Neck and Back Pain

Risk Factors Concern

Historical Risk Factors

Pain >6 weeks Tumor, infection

Age <18, >50 Congenital anomaly, tumor, infection

Major trauma Fracture

Minor trauma in elderly or rheumatologic disease Fracture

History of cancer Tumor

Fever and rigors Infection

Weight loss Tumor, infection

Injection drug use Infection

Immunocompromised Infection

Night pain Tumor, infection

Unremitting pain, even when supine Tumor, infection

Incontinence Epidural compression

Saddle anesthesia Epidural compression

Severe/progressive neurologic deficit Epidural compression

Anticoagulants and coagulopathy Epidural compression

Physical Risk Factors

Fever Infection

Patient writhing in pain Infection, vascular cause

Unexpected anal sphincter laxity Epidural compression

Perianal/perineal sensory loss Epidural compression

Major motor weakness/gait disturbance Nerve root or epidural compression

Positive straight leg raise test Herniated disk

Table 177-5 Diagnostic Maneuvers for Neck and Back Pain

Exam Finding/Maneuver Significance

Spurling's sign: Downward pressure with head in exten- Cervical radiculopathy
sion and leaning toward symptomatic sign produces pain

Abduction relief sign: Relief placing hand of affected Disc protrusion/radiculopathy
arm on top of head

Lhermitte's sign: Forward neck flexion causes electric Cord compression
shock like pain in spine/arms

Hoffman's sign: Flexion of thumb and index finger when Upper motor neuron lesion
flicking tip of middle finger with hand in relaxed position

Straight leg test: Hip flexion to ~70 with knee in L4/L5 or L5/S1 disc herniation
extension produces pain in affected leg below knee

Crossed leg test: Hip flexion to ~70 with knee in extension L4/L5 or L5/S1 disc herniation
in opposing leg produces pain in affected leg below knee

177_Cydulka_Ch177.indd 971 22/03/17 2:51 PM

Source page 991

Imaging is usually not indicated in cases of acute, nontraumatic neck or back pain without concerning historical features or exam findings. Plain films of the spine have low sensitivity but may be considered as an initial imaging step if clinical suspicions include tumor, fracture, or infection. For patients with pain and neurologic deficits, MRI is the definitive test. CT scan is helpful for identification of disorders of the bony skeleton, but its sensitivity for nerve root or spinal cord disorders is poor. CT myelography can serve as an alternative to MRI, if the latter is contraindicated or unattainable.

Plain films or CT scan of the cervical spine should be considered in patients with chronic neck pain not previously imaged or those with a history or trauma, surgery, malignancy, or rheumatologic disease. Flexion-extension films or MRI may be considered if instability or ligamentous

injury is suspected. If radiographs are normal, or show only degenerative disease, and the patient has a benign examination, no further imaging is required. For patients with back pain, in whom plain thoracic and/or lumbar radiographs are indicated (as discussed above), anteroposterior and lateral views suffice. Careful attention should be paid to end plate erosions, disc space narrowing, or lytic lesions, which are worrisome for infection or neoplasm. Advanced imaging is necessitated by the presence of neurological deficits or abnormal plain radiographs.

Laboratory testing will not be useful in the majority of patients with neck and back pain. If serious pathology is suspected, a complete blood count, erythrocyte sedimentation rate (ESR), and urinalysis should be ordered. ESR has a sensitivity of 90% to 98% for infectious causes of back pain. Urinalysis is useful to rule out urinary or renal pathology as a pain source. Postvoid residual (PVR) assessment, with ultrasound or catheterization, should be performed on patients with complaints or exam findings concerning for epidural compression. Urinary retention is the most common finding in cauda equina syndrome, and PVR volume exceeding 500 mL, alone or in combination with other concerning findings, is an important predictor of MRI-proven cauda equina compression.

Mechanical disorders, such as strain caused by trauma, are often characterized by delayed pain and paraspinous pain and stiffness. See Chapters 161 and 163 for further discussion of traumatic neck and back injuries.

Cervical disc herniation, spondylosis, or stenosis can lead to radiculopathy or myelopathy. Signs and symptoms of radiculopathy include pain and weakness in a dermatomal distribution. Lower extremity hyperreflexia, a positive Babinski sign, and loss of sphincter tone are suspicious for myelopathy; any of these findings necessitate MRI. Metastatic cancer may cause a radiculopathy or myelopathy and should be included in the differential for chronic pain. Osteomyelitis, epidural abscess, hematoma, and transverse myelitits may cause neck pain with neurologic deficits, although these disorders are more commonly observed in the thoracic or lumbar spine.

Neck pain can be caused by a variety of other disorders such as myofascial pain syndrome, lymphadenopathy, temporal arteritis, ischemic heart disease, and neurodegenerative disorders. The distribution of symptoms in these disorders will typically not be dermatomal, and other historical features will usually distinguish them.

The majority of patients with back pain have nonspecific back pain; they have no radiculopathy or myelopathy and no specific etiology is found.

177_Cydulka_Ch177.indd 972 22/03/17 2:51 PM

Source page 992

These patients typically have benign examinations and movement often exacerbates their pain. Diagnostic evaluation is negative.

As with neck pain, disc herniation and degenerative changes of the thoracic or lumbar spine can be a cause of acute or chronic back pain. More than 95% of disc herniations are at the L4–L5 or L5–S1 nerve roots, compression of which can cause sciatica. These patients may have leg pain, neurologic deficits localized to a unilateral nerve root, and a positive straight-leg test. Spinal nerve root compression can occur at any level in the thoracic or lumbar spine, with associated radicular symptoms. Epidural compression syndromes, which include spinal cord compression, cauda equina syndrome, and conus medullaris syndrome, may cause pain, neurologic deficits, or autonomic dysfunction at any cord level. The thoracic spine is the most common site of compression due to neoplasm; however, if suspected, the entire spinal cord should be imaged, as 10% of patients will have additional metastatic lesions.

Infectious causes of back pain, including osteomyelitis, discitis, and epidural abscess, are often missed on initial presentation. A high level of suspicion should be maintained in patients with risk factors such as immunocompromise, recent back surgery, retained hardware, or intravenous drug use. Plain films may show bony destruction in patients with osteomyelitis, but x-rays are often initially normal in cases of infectious back pain. MRI is typically required for diagnosis. The ESR will be elevated in the majority of cases.

Other causes of back pain may originate from the spine itself or from nonspinal causes. Spinal stenosis can cause low back and leg pain that mimics claudication, worsening with walking and improving with rest. Ankylosing spondylitis, an autoimmune arthritis, causes chronic back pain and is identifiable by a characteristic squaring of vertebral bodies on radiography. Transverse myelitis, an inflammatory disorder of the spinal cord, may mimic a compression syndrome. Ruptured abdominal aneurysm, pyelonephritis, pancreatitis, or renal infarction should be considered in patients with nonspecific back pain accompanied by concerning history or exam findings.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. Patients with neck and back pain that have progressive neurologic deficits, myelopathy, or intractable pain should be imaged as indicated (typically MRI) and admitted to the appropriate service for further

management. Dexamethasone 10 mg IV should be given prior to imaging for suspected epidural compression.

2. Suspected or proven spinal infections require emergent consultation with a spinal surgeon. Empiric broad-spectrum antibiotics, such as vancomycin 1 g IV and piperacillin-tazobactam 3.375 g IV, should be given unless explicit instructions to withhold are given by the consultant.

3. Patients with a stable or mild radiculopathy may be managed conservatively with pain medication, routine activity as tolerated, and strict return precautions for worsening symptoms. Outpatient MRI and neurosurgical follow-up should be considered for patients who have failed conservative treatment.

177_Cydulka_Ch177.indd 973 22/03/17 2:51 PM

Source page 993

974 SECTION 20: Nontraumatic Musculoskeletal Disorders

4. Pain management may include NSAIDs, acetaminophen, or muscle relaxants used alone or in combination; all have been shown to be effective. A short course of oral opioids may be prescribed for patients with moderate to severe pain, but long-term use is discouraged. The evidence for manipulation, soft-collars, corticosteroids, or other alternative treatments is limited and should generally not be prescribed from the ED. Chronic pain often requires a multidisciplinary approach.

5. The majority of patients with neck or back pain will have a benign course and improve with time. Patients may be prescribed pain medicine as appropriate and activity as tolerated. Return precautions and instruction for follow-up should be provided.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 279, "Neck and Back Pain," by David Della-Giustina, Jeffrey S. Dubin, and William Frohna.

177_Cydulka_Ch177.indd 974 22/03/17 2:51 PM

References/provenance

- Source: *Tintinalli's Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 990–995.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 178: Shoulder Pain

Source pages 996–999 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 996–999

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Shoulder Pain**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 178: Shoulder Pain 977 primarily in patients older than 40 years and are associated with muscular weakness (especially with abduction and external rotation) and cuff ten- derness. [Source pages 996–999]

- Ninety percent will be chronic tears with a history of minimal or no trauma; in severe disease, muscular atrophy may be present. [Source pages 996–999]

- Acute tears may occur in patients of any age and result from significant force produc- ing a tearing sensation with immediate pain and disability. [Source pages 996–999]

- In patients between the ages of 30 and 50 years, abnormal calcifications on radio- graph in the clinical setting of a painful shoulder with rotator cuff tender- ness and often crepitus suggest the diagnosis of calcific tendonitis. [Source pages 996–999]

- Osteoarthritis is characteristically associated with degenerative disease in other joints (primary) or previous fracture or other underlying disorder (secondary). [Source pages 996–999]

Clinical reasoning structure

- **Problem representation:** Shoulder Pain; source pages 996–999.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Shoulder Pain**.

- Source-page range: **996–999**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 996

CHAPTER 178: Shoulder Pain 977

primarily in patients older than 40 years and are associated with muscular weakness (especially with abduction and external rotation) and cuff tenderness. Ninety percent will be chronic tears with a history of minimal or no trauma; in severe disease, muscular atrophy may be present. Acute tears may occur in patients of any age and result from significant force producing a tearing sensation with immediate pain and disability. In patients between the ages of 30 and 50 years, abnormal calcifications on radiograph in the clinical setting of a painful shoulder with rotator cuff tenderness and often crepitus suggest the diagnosis of calcific tendonitis. Osteoarthritis is characteristically associated with degenerative disease in other joints (primary) or previous fracture or other underlying disorder (secondary). The hallmark of adhesive capsulitis is significantly painful and limited range of motion often, but not always, associated with a period of immobilization. Radiographs should be obtained to exclude posterior glenohumeral dislocation. Biceps tendon pathology can also manifest as shoulder pain. The proximal long head of the biceps inserts on the superior aspect of the shoulder. Tendinopathy, subluxation, dislocation, and partial or complete tears can produce pain. Biceps tendinopathy can arise due to inflammation (tendonitis) or collagen tears around the tendon (tendinosis) and may be acute or chronic. Diagnosis is made by palpating the proximal tendon along the bicipital groove or resisted forearm supination. Reproduction of pain by these maneuvers suggests biceps pathology. Tendinitis and subluxation can be managed with sling immobilization, analgesics, anti-inflammatory agents, and frequent ice application, followed by early mobilization.

Other causes of shoulder pain that should be considered are a number of extrinsic conditions. Pancoast tumor may compress the brachial plexus and manifest as shoulder pain. Degenerative disease of the cervical spine,

brachial plexus disorders, and suprascapular nerve compression are neurologic processes that should be sought in patient evaluation. Vascular pathology, notably axillary artery thrombosis, also may cause shoulder pain. Acute cardiac, aortic, pulmonary, and abdominal pathology may cause pain referred to the shoulder, and the clinician must remain alert to this possibility.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. For intrinsic disease, the primary goals of emergency department care are to reduce pain and inflammation and prevent progression of disease. For most conditions, this translates to relative rest of the joint assisted by use of a sling as full immobilization is not suggested. Treatment also includes nonsteroidal anti-inflammatory drugs, opioid analgesics as needed, and the application of cold packs. Range-of-motion exercises should be encouraged as soon as pain allows to prevent loss of flexibility and to maintain strength.

2. Joint space injection with glucocorticoids such as triamcinolone 20 to 40 mg with or without a local anesthetic such as lidocaine should be used judiciously in view of the potential deleterious effects on soft tissues, tendon rupture with direct injection, and a recommended limitation of three injections into a single area. For all intrinsic disorders, follow-up with a primary care physician with expertise in joint disease

178_Cydulka_Ch178.indd 977 22/03/17 3:22 PM

Source page 997

or orthopedic referral is suggested within 7 to 14 days. Physical therapy referral for stretching and strengthening also may be valuable.

3. In extrinsic disease, the treatment and referral pattern will depend on the diagnosis. Neurologic problems will require analgesia and anti-inflammatory medications and may require neurology or neurosurgical follow-up. Vascular causes of shoulder pain must be evaluated carefully and, with axillary artery thrombosis, immediate consultation made to initiate thrombolysis. Treatment of other extrinsic conditions depends on the specific diagnosis.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 280, “Shoulder Pain,” by David Della-Giustina and David Hile.

178_Cydulka_Ch178.indd 978 22/03/17 3:22 PM

Source page 998

C h apter Hip and Knee Pain

179 Augusta Czysz

Hip and knee pain are common complaints, especially among athletes and the obese due to increased forces on their joints. Knee pain is usually due to local pathology, whereas hip pathology commonly causes referred pain in the buttocks and lower extremity and may be due to extraarticular pathology. Each pain has a broad differential diagnosis, but a focused history and physical examination will often lead to the diagnosis (Table 179-1).

■■ PSOAS ABSCESS

Abscess of the psoas muscle usually occurs through hematogenous spread and may present with abdominal pain radiating to the hip or flank, fever, and limp, or other constitutional symptoms. Eighty percent of the time, it is caused by Staphylococcus aureus. The diagnosis is made by CT. Treatment includes organism-specific antibiotics and surgical drainage.

■■ REGIONAL NERVE ENTRAPMENT SYNDROMES

Meralgia paresthetica, a compressive inflammation of the lateral femoral cutaneous nerve, causes pain in the hip, thigh, or groin, along with burning or tingling paresthesias, and hypersensitivity to light touch. Address the source of nerve irritation (e.g., obesity, pregnancy, tight pants, belt) and provide nonsteroidal anti-inflammatory drugs (NSAIDs) to treat in the ED. Obturator nerve entrapment usually occurs after pelvic fractures, with masses, or in athletes with a fascial band at the distal obturator canal, which causes pain in the groin and down the inner thigh. Surgery may be needed for pain relief. Ilioinguinal nerve entrapment is associated with pregnancy or hypertrophy of the abdominal wall musculature and causes groin pain. Piriformis syndrome, irritation of the sciatic nerve from the piriformis muscle, manifests as pain in the buttocks and hamstring muscles that worsens with sitting, climbing stairs, or squatting. ED treatment is conservative for all of these nerve entrapment syndromes (Table 179-2).

Table 179-1 Suggested Clues for the Differential Diagnosis of Hip and Knee Pain

Determine the location of the pain to narrow down the potential diagnosis.

Determine the activities that bring on the pain.

Complaints that the joint “gives out” or “buckles” generally are due to pain and reflex muscle inhibition rather than an acute neurologic emergency. This complaint may also represent patellar subluxation or ligamentous injury and joint instability.

Poor conditioning or quadriceps weakness generally causes anterior knee pain of the patellofemoral syndrome; therapy should address this weakness.

Locking of the knee suggests a meniscal injury, which may be chronic.

A popping sensation or sound at the onset of pain is reliable for a ligamentous injury.

A recurrent knee effusion after activity suggests a meniscal injury.

Pain at the joint line of the knee (palpable indentation between distal femur and proximal tibia) suggests a meniscal injury.

979

179_Cydulka_Ch179.indd 979 22/03/17 3:26 PM

Source page 999

980 SECTION 20: Nontraumatic Musculoskeletal Disorders

Table 179-2 Selected Syndromes by Location

Diagnosis Category	Diagnosis	Pain Location
Nerve entrapment	Meralgia paresthetica	Anterolateral thigh pain or paresthesias
	Obturator nerve entrapment	Groin and inner thigh pain
	Ilioinguinal nerve entrapment	Groin pain
	Piriformis syndrome (sciatic nerve compression by piriformis muscle)	Buttocks and hamstrings pain
Hip bursitis	Trochanteric bursitis	Hip pain when lying on side or with hip abduction and adduction
	Ischiogluteal bursitis	Ischial pain
	Iliopectineal and iliopsoas bursitis	Anterior pelvis and groin, hip extension
Knee bursitis	Pes anserine bursitis	Anterior medial knee pain
	Prepatellar bursitis	Pain anterior to patella
Hip overuse syndromes	External snapping hip syndrome (coxa saltans)	Posterior lateral hip pain
	Fascia lata syndrome	Lateral thigh pain
Knee overuse syndromes (runner’s knee)	Patellofemoral syndrome	Anterior knee pain, worse with prolonged knee flexion
	Medial plica syndrome	Anterior medial knee pain, knee snapping during repeated flexion/extension
	Iliotibial band syndrome or snapping knee syndrome	Pain over lateral epicondyles, or snapping when iliotibial band passes over femoral condyle
	Popliteus tendinitis	Posterior lateral knee pain, worse on downhill exercise
	Patellar tendinitis (jumper’s knee)	Inferior patellar or proximal patellar tendon pain
	Quadriceps tendinitis	Proximal patellar pain
	Popliteal (Baker) cyst	Posterior knee pain

■■ BURSAL SYNDROMES OF THE HIP AND KNEE

Inflamed or infected hip and knee bursae may cause localized pain. Inflammatory changes may be due to repetitive minor trauma, rheumatologic disorders such as psoriatic arthritis, rheumatoid arthritis or ankylosing spondylitis and crystalline disease including gout and pseudogout. Infection may be difficult to distinguish clinically from more benign disorders; analysis of bursal aspirate will help. Treatment is directed at the underlying cause. NSAIDs, rest, heat, and time are the basis of treatment for inflammatory conditions. Steroid injections into readily accessible bursae may be useful if infection has been excluded. Avoid injecting tendons with steroids. Treat infections with antibiotics (see Chapter 180, “Acute Disorders of the Joints and Bursae,” for specific recommendations). Admit immunocompromised patients with suspected infections for IV antibiotics and orthopedic surgery consultation (Table 179-2).

179_Cydulka_Ch179.indd 980 22/03/17 3:26 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 996–999.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 179: Hip and Knee Pain

Source pages 1000–1003 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 1000–1003

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Hip and Knee Pain**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 179: Hip and Knee Pain 981 ■■ MYOFASCIAL SYNDROMES/OVERUSE SYNDROMES Repetitive microtrauma that outpaces the body’s ability to heal results in overuse syndromes. [Source pages 1000–1003]

- Treatment generally consists of NSAIDs, heat, and rest, followed by gradual resumption of activities, physical therapy, and strengthening where appropriate (Table 179-2). [Source pages 1000–1003]

- ■■ BONE/ARTICULAR DERANGEMENTS Osteonecrosis (also called aseptic necrosis, ischemic necrosis, or avascular necrosis) is bone infarction caused by a lack of blood supply. [Source pages 1000–1003]

- It may be an idiopathic or a primary disorder, secondary to a systemic condition, or due to trauma. [Source pages 1000–1003]

- Conditions associated with osteonecrosis of the femoral head include femoral neck fracture, hip dislocation, occult or minor trauma, sickle cell disease, collagen vascular diseases, alcohol abuse, renal trans- plant, systemic lupus erythematosus, dysbarism, chronic pancreatitis, exog- enous steroid administration, Cushing disease, decompression sickness, Gaucher disease, and renal osteodystrophy. [Source pages 1000–1003]

Clinical reasoning structure

- **Problem representation:** Hip and Knee Pain; source pages 1000–1003.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Hip and Knee Pain**.

- Source-page range: **1000–1003**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 1000

CHAPTER 179: Hip and Knee Pain 981

■■ MYOFASCIAL SYNDROMES/OVERUSE SYNDROMES

Repetitive microtrauma that outpaces the body's ability to heal results in overuse syndromes. Treatment generally consists of NSAIDs, heat, and rest, followed by gradual resumption of activities, physical therapy, and strengthening where appropriate (Table 179-2).

■■ BONE/ARTICULAR DERANGEMENTS

Osteonecrosis (also called aseptic necrosis, ischemic necrosis, or avascular necrosis) is bone infarction caused by a lack of blood supply. It may be an idiopathic or a primary disorder, secondary to a systemic condition, or due to trauma. Conditions associated with osteonecrosis of the femoral head include femoral neck fracture, hip dislocation, occult or minor trauma, sickle cell disease, collagen vascular diseases, alcohol abuse, renal transplant, systemic lupus erythematosus, dysbarism, chronic pancreatitis, exogenous steroid administration, Cushing disease, decompression sickness, Gaucher disease, and renal osteodystrophy. Osteonecrosis of the hip may cause pain anywhere from the buttock to the knee. X-ray, CT, or MRI may aid in diagnosis. Joint replacement may be needed.

Osteomyelitis is an infection of the bone that results in bony destruction commonly caused by Staphylococcus aureus. Patients have local pain and may have associated warmth, swelling, and erythema. Imaging may be normal. MRI is the preferred imaging study but bone biopsy is required for confirmation. The acutely ill patient should receive high-dose, broad-spectrum, parenteral antibiotics based on the patient's risk factors and most likely organisms (Table 179-3).

Table 179-3 Risk Factors, Likely Infecting Organism, and Recommended Initial Empiric Antibiotic Therapy for Osteomyelitis

Recommended Initial

Risk Factor Likely Infecting Organism Empiric Antibiotic Therapy*
Elderly, hematog- Staphylococcus aureus, including Vancomycin plus piperacillin-
enous spread MRSA, gram-negative bacteria tazobactam, or imipenem
Sickle cell disease Salmonella, gram-negative Ciprofloxacin; consider
bacteria (S. aureus becoming vancomycin
more common)
Diabetes mellitus, or Polymicrobial: S. aureus, Vancomycin plus piperacillin-
vascular insufficiency Streptococcus agalactiae, and tazobactam, or imipenem
Streptococcus pyogenes plus
coliforms and anaerobes
Injection drug user S. aureus, including MRSA, and Vancomycin
Pseudomonas
Developing nations Mycobacterium tuberculosis See Chapter 31, “Tuberculosis”
Newborn S. aureus including MRSA, Vancomycin plus ceftazidime
gram-negative bacteria, group B
Streptococcus
Children S. aureus including MRSA Vancomycin plus ceftazidime
(Continued)
179_Cydulka_Ch179.indd 981 22/03/17 3:26 PM

Source page 1001

Table 179-3 Risk Factors, Likely Infecting Organism, and Recommended Initial
Empiric Antibiotic Therapy for Osteomyelitis (Continued)
Recommended Initial
Risk Factor Likely Infecting Organism Empiric Antibiotic Therapy*
Postoperative with S. aureus and coagulase- Vancomycin
or without retained negative staphylococci
orthopedic hardware
Human bite Streptococci or anaerobic Piperacillin-tazobactam or
bacteria imipenem
Animal bite Pasteurella multocida, Eikenella Cefuroxime if known P. multo-
corrodens cida, piperacillin-tazobactam
or imipenem
Abbreviation: MRSA, methicillin-resistant Staphylococcus aureus.
All patients require bone biopsy and debridement of infected/dead bone.
*
Osteitis pubis occurs following pregnancy, in athletes due to overuse of
the adductors and gracilis muscles, and after bladder and prostate surgery.
It causes pain in the region of the pubis and generally resolves over a period
of months with rest and NSAIDs. Myositis ossificans (also known as het-

erotopic calcification) is the deposition of bone in abnormal sites after direct trauma. Pain and a palpable mass will be present. Pain or physical obstruction may limit motion in the affected muscle or joint. Surgery may be required.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 281, "Hip and Knee Pain," by Kelly P. O'Keefe and Tracy G. Sanson.

179_Cydulka_Ch179.indd 982 22/03/17 3:26 PM

Source page 1002

C h apter Acute Disorders of The Joints

180 and Bursae

Andrew D. Perron

Acute disorders of the joints and bursae are common emergency conditions that involve a wide spectrum of ages, acuities, and etiologies. Mismanagement of certain pathologic entities can lead to significant morbidity for the patient.

■■ CLINICAL FEATURES

Multiple pathways can cause disruption of the normal joint milieu leading to acute joint complaints. These pathways include degeneration of articular cartilage with osteoarthritis, deposition of immune complexes as in rheumatoid arthritis, crystal-induced inflammation in gout and pseudogout, seronegative spondyloarthropathies such as ankylosing spondylitis and Reiter's syndrome, and the bacterial or viral invasion of septic arthritis. These pathologic events invariably lead to pain, the most common complaint of patients with a joint problem. Important historical factors to elicit include a determination of previous joint or bursal disease, presence of constitutional symptoms and whether the pain is acute, chronic, or acute on chronic. Determining the number and distribution of joints affected can help narrow the differential diagnosis (Table 180-1). A migratory pattern of

Table 180-1 Differential Diagnosis of Arthritis by Number of Affected Joints

Number of Joints	Differential Considerations for Typical Presentations
1 = Monoarthritis	85% of nongonococcal septic arthritis*
	Crystal-induced (gout, pseudogout)
	Gonococcal septic arthritis
	Trauma-induced arthritis
	Osteoarthritis (acute)
	Lyme disease
	Avascular necrosis
	Tumor

2–3 = Oligoarthritis† 15% of nongonococcal septic arthritis, more common with

Staphylococcus aureus and Streptococcus pneumonia

Lyme disease

Reactive arthritis (Reiter’s syndrome)

Gonococcal arthritis

Rheumatic fever

>3 = Polyarthritis† Rheumatoid arthritis

Systemic lupus erythematosus

Viral arthritis

Osteoarthritis (chronic)

Serum sickness

Serum sickness-like reactions

*

Involvement of more than one joint does not rule out septic arthritis.

†

The distinction between oligoarthritis and polyarthritis varies in the literature with a cut point of either three or four joints.

983

180_Cydulka_Ch180.indd 983 22/03/17 3:31 PM

Source page 1003

984 SECTION 20: Nontraumatic Musculoskeletal Disorders

joint pain can be seen with systemic lupus erythematosus and in many infectious etiologies such as gonococcal arthritis, acute rheumatic fever, Lyme disease, and viral arthritis.

On physical examination, arthritis should be distinguished from more focal periarticular inflammatory processes such as cellulitis, bursitis, and tendonitis. True arthritis produces joint pain exacerbated by active and passive motions. Skin erythema and warmth overlying a joint should be noted, although it is nonspecific regarding etiology of joint pathology.

■■ DIAGNOSIS AND DIFFERENTIAL

With the exception of recent joint surgery or cellulitis overlying a prosthetic knee or hip, history, physical examination, and routine blood tests do not distinguish acute septic arthritis from other forms of arthritis. Clinicians who suspect septic arthritis based on the patient’s presentation should perform arthrocentesis. Synovial fluid should be sent for culture, Gram stain, cell count, and crystal evaluation (Table 180-2). Except in pediatric septic arthritis, where the erythrocyte sedimentation rate has been shown to have 90% sensitivity, the serum white blood cell count and erythrocyte sedimentation rate lack the sensitivity and specificity to be reliable discriminators

in disorders of the joints and bursae. Adults with risk factors for sexually transmitted disease and migratory symptoms or tenosynovitis should be evaluated for gonococcal arthritis.

Radiographs should be obtained when the differential diagnosis includes trauma, tumor, osteomyelitis, ankylosing spondylitis, or avascular necrosis. More sophisticated modalities such as computed tomography, magnetic resonance imaging, and radioisotope scanning are used in isolated cases.

Table 180-2 Examination of Synovial Fluid

Normal	Noninflammatory	Inflammatory	Septic
Clarity	Transparent	Transparent	Cloudy	Cloudy
Color	Clear	Yellow	Yellow	Yellow
WBC/μL	<200	<200 to 2000	200 to 50 000	>50 000
PMNs (%)*	<25	<25	>50%	>50%
Culture	Negative	Negative	Negative	>50% positive†
Crystals	None	None	Multiple or none	None
Associated conditions	Osteoarthritis, trauma, rheumatic fever	Gout, pseudogout, RA, Lyme disease, SLE	Septic arthritis

*

The WBC and PMNs are affected by a number of factors, including disease progression, affecting organism, and host immune status. The joint aspirate WBC and %PMNs should be considered part of a continuum for each disease, particularly septic arthritis, and should be correlated with other clinical information.

†

Gonococcal arthritis is frequently culture negative.

Abbreviations: RA, rheumatoid arthritis, SLE, systemic lupus erythematosus, PMNs, polymorphonuclear neutrophils.

180_Cydulka_Ch180.indd 984 22/03/17 3:31 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 1000–1003.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 180: Acute Disorders of The Joints and Bursae

Source pages 1004–1007 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 1004–1007

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Acute Disorders of The Joints and Bursae**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 180: Acute Disorders of The Joints and Bursae 985 ■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION Septic arthritis is a condition that can rapidly lead to irreversible joint destruction if inadequately treated. [Source pages 1004–1007]

- It typically presents as a monoarticular arthritis and may be associated with fever, chills, or malaise, although absence of these symptoms does not exclude the diagnosis. [Source pages 1004–1007]

- The synovial fluid may help confirm the diagnosis, although there is significant overlap in fluid findings between a number of pathological entities that can affect a joint. [Source pages 1004–1007]

- Therapy requires admission for parenteral antibiotics and repeated needle aspiration, arthroscopy, or open surgical drainage. [Source pages 1004–1007]

- Specific patient demographics can help guide empiric antibiotic therapy in septic arthritis (Table 180-3). [Source pages 1004–1007]

Clinical reasoning structure

- **Problem representation:** Acute Disorders of The Joints and Bursae; source pages 1004–1007.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Acute Disorders of The Joints and Bursae**.

- Source-page range: **1004–1007**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 1004

CHAPTER 180: Acute Disorders of The Joints and Bursae 985

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

Septic arthritis is a condition that can rapidly lead to irreversible joint destruction if inadequately treated. It typically presents as a monoarticular arthritis and may be associated with fever, chills, or malaise, although absence of these symptoms does not exclude the diagnosis. The synovial fluid may help confirm the diagnosis, although there is significant overlap in fluid findings between a number of pathological entities that can affect a joint. Therapy requires admission for parenteral antibiotics and repeated needle aspiration, arthroscopy, or open surgical drainage. Specific patient demographics can help guide empiric antibiotic therapy in septic arthritis (Table 180-3). For most of these conditions, arthrocentesis will guide diagnosis, therapy, and disposition. As a general rule, skin overlying the intended joint should be free of cellulitis. Skin is cleansed with povidone-iodine which should be allowed to air-dry. The intended site should then be cleaned with an alcohol wipe. Skin can be anesthetized with either lidocaine or with ethyl chloride spray. In general, a large-bore needle, 18-G or 19-G, should be used for larger joints such as shoulders and knees while it may be necessary to use a smaller needle for digits.

Traumatic hemarthrosis is associated with intraarticular fracture or ligamentous injury. Aspiration of large effusions may decrease pain and increase range of motion. Treatment is supportive. Spontaneous hemarthrosis may be associated with coagulopathies requiring specific clotting factor replacement. It is usually not recommended to aspirate spontaneous hemarthroses.

Table 180-3 Commonly Encountered Organisms in Septic Arthritis in Adolescents and Adults*

Patient/Condition Expected Organisms Antibiotic Considerations

Young healthy adults, Staphylococcus, N. gon- Vancomycin, 15 mg/kg IV load, if	
or patients with risk orrhoeae, Streptococcus, Gram stain reveals gram-positive	
factors for Neisseria gram-negative bacteria organisms in clusters. Ceftriaxone,	
gonorrhoeae 1 g IV, or imipenem, 500 mg IV,	
should be used/added if either gram-	
negative organisms are present or no	
organisms are present on Gram stain	
and N. gonorrhoeae is suspected (also	
culture urethra, cervix, or anal canal	
as indicated).	
Adults with comorbid Staphylococcus, gram- Vancomycin, 15 mg/kg IV load, plus	
disease (rheumatoid negative bacilli cefepime, 2 g IV, or imipenem, 500 mg IV.	
arthritis, human Meropenem 1 g IV may be used as an	
immunodeficiency alternative agent.	
virus, cancer) or	
injection drug users	
Sickle cell patients Salmonella (increasingly Vancomycin, 15 mg/kg IV load, plus	
Staphylococcus) ciprofloxacin, 400 mg IV. Imipenem,	
500 mg IV, may be used as an	
alternative agent.	

*

Recommendations differ from the 2006 British Society of Rheumatology treatment guidelines due to the rising incidence of methicillin-resistant Staphylococcus aureus septic arthritis.

180_Cydulka_Ch180.indd 985 22/03/17 3:31 PM

Source page 1005

Crystal-induced synovitis generally affects middle-age to elderly patients. Gout involves deposition of uric acid crystals, typically affecting the great toe, tarsal joints, and knee, whereas pseudogout involves deposition of calcium pyrophosphate crystals, typically affecting the knee, wrist, ankle, and elbow. Pain with gout usually evolves over hours, whereas the pain associated with pseudogout occurs over a day or more. Either condition may be precipitated by trauma, surgery, significant illness, dietary or alcohol indiscretions, or certain medications. The synovial fluid is inflammatory with negative birefringent needle-shaped crystals in gout, and weakly positive birefringent rhomboid crystals in pseudogout. Treatment is with nonsteroidal anti-inflammatory drugs such as ibuprophen 400 to 800 mg three times daily for 3 to 5 days and opioid analgesics. Although not routinely necessary, colchicine 0.6 mg/h orally until resolution of symptoms or intolerable gastrointestinal side effects may be used as a

complementary therapy.

Osteoarthritis is a chronic, symmetric, polyarticular destruction of joints distinguished by a lack of constitutional symptoms. Patients may present with acute monoarticular exacerbations with small, noninflammatory synovial fluid collections and characteristic joint space narrowing on radiographs. Treatment involves rest and analgesics.

Lyme arthritis is a monoarticular or symmetric oligoarticular arthritis, especially of the large joints, with brief exacerbations followed by complete remission occurring weeks to years after the primary infection. Synovial fluid is inflammatory, typically with negative cultures. Treatment with appropriate antibiotics such as doxycycline, erythromycin, or amoxicillin for 3 to 4 weeks is effective.

Gonococcal arthritis is an immune-mediated infectious arthritis that typically affects adolescents and young adults. Fever, chills, and a migratory tenosynovitis or arthralgias typically precede mono- or oligoarthritis. Vesiculopustular lesions on the distal extremities are characteristic. Synovial fluid is usually inflammatory and often culture negative; cultures of the pharynx, urethra, cervix, and rectum increase the culture yield. The patient should be admitted for pain control and parenteral antibiotic therapy. Orthopedic consultation is advised.

Reiter's syndrome is a seronegative reactive spondyloarthropathy characterized by acute asymmetric oligoarthritis, especially of the lower extremities, preceded 2 to 6 weeks prior by an infectious illness such as urethritis caused by Ureaplasma or Chlamydia, or enteric infection caused by Salmonella or Shigella. The classic triad of arthritis, conjunctivitis, and urethritis is not required for the diagnosis. Synovial fluid is inflammatory. Treatment is symptomatic, and antibiotics have not been found to be useful.

Ankylosing spondylitis is a seronegative spondyloarthropathy that primarily affects the spine and pelvis and may be associated with morning stiffness and constitutional symptoms such as fatigue and weakness. Hereditary predilection, with HLA-B27 antigen or absence of rheumatoid factor, is significant. Radiographic findings include sacroiliitis and squaring of the vertebral bodies, known as bamboo spine. Treatment is symptomatic.

Rheumatoid arthritis is a chronic, symmetric, polyarticular joint disease, with sparing of the distal interphalangeal joints and associated with

180_Cydulka_Ch180.indd 986 22/03/17 3:31 PM

Source page 1006

morning stiffness, depression, fatigue, and generalized myalgias. Pericarditis, myocarditis, pleural effusion, pneumonitis, and mononeuritis multiplex may occur. Synovial fluid is inflammatory. Treatment of an acute exacerbation involves immobilization, nonsteroidal anti-inflammatory drugs, and, occasionally, corticosteroids. Antimalarials, immunosuppressants, disease-modifying antirheumatic drugs, and cytotoxic agents are used for long-term therapy.

Bursitis refers to an inflammatory process involving any of the more than 150 bursae throughout the human body and may be caused by infection, trauma, rheumatologic diseases, or crystal deposition. Certain repetitive activities may also precipitate bursitis: prepatellar bursitis, known as “carpet layer’s knee” or olecranon bursitis, known as “student’s elbow.” A suspicion for septic bursitis, especially in olecranon bursitis, necessitates aspiration of bursal fluid. Septic bursal fluid characteristically is purulent in appearance, with more than 1500 leukocytes/mm3 and positive culture. Treatment principles include drainage, rest, compressive dressing, analgesics, and antibiotics for septic bursitis. Septic bursitis generally responds well to oral antibiotics, with emphasis on coverage of Staphylococcus, including methicillin-resistant Staphylococcus aureus and Streptococcus species.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 284, “Joints and Bursae,” by John H. Burton and Timothy J. Fortuna.

180_Cydulka_Ch180.indd 987 22/03/17 3:31 PM

Source page 1007

Chapter Emergencies in Systemic

181 Rheumatic Diseases

Nicholas Genes

Patients who present to the ED with systemic rheumatic disease manifestations are often at risk for pain and discomfort, organ damage, infection, adverse drug events, and significant morbidity or mortality, especially if their disease presentation is not properly recognized and managed. A new diagnosis of rheumatic disease is usually not possible in an ED visit, but may be completed in the outpatient or inpatient setting, depending on the severity of symptoms. Given the various emergent manifestations, this chapter approaches systemic rheumatic diseases by pertinent organ systems affected (Table 181-1).

Table 181-1 Common Features and Complications of Systemic Rheumatic Diseases

Common and Characteristic

Disorder Clinical Features Complications

Antiphospholipid Multiple and recurrent venous and ARDS, pulmonary
syndrome arterial thromboses, recurrent abortions. embolism, ischemic
Secondary form is associated with sys- complications of vascular
temic lupus erythematosus, rheumatoid occlusion, bleeding,
arthritis, systemic sclerosis, and Sjögren's severe anemia, vision
syndrome. Thrombophlebitis and deep loss, catastrophic
vein thrombosis, thrombocytopenia, hemo- antiphospholipid
lytic anemia, microangiopathic hemolytic syndrome.
anemia, livedo reticularis, stroke, tran-
sient ischemic attack, eye vascular com-
plications. Coronary, renal, mesenteric,
and cerebral vascular occlusion.

Ankylosing Chronic inflammatory disease of the axial Acute spinal cord or nerve
spondylitis skeleton, with progressive stiffness of compression, subluxation
the spine. Young adults (peak at 20 and of the atlantoaxial joint,
30 years old). Back pain (improves with aortic regurgitation.
exercise), buttock, hip, or shoulder pain,
systemic complaints (fever, malaise,
fatigue, weight loss, myalgias), uveitis,
and restrictive pulmonary failure due to
costovertebral rigidity, ILD, renal impair-
ment, fracture of the ankylosed spine,
asymptomatic ileal and colonic mucosal
ulcerations. Secondary amyloidosis.

Adult Still's Inflammatory disorder (similar to systemic ARDS, arrhythmias, heart
disease onset juvenile rheumatoid arthritis). failure, fulminant hepatic
Systemic complaints (fever, malaise, fatigue, failure, red cell aplasia,
weight loss, myalgia), arthritis, myalgia, disseminated intravas-
evanescent rash, pharyngitis, lymphadenop- cular coagulation, micro-
athy, splenomegaly, anemia, thrombocytope- angiopathic hemolytic
nia. Pericarditis, myocarditis, pleurisy. anemia.

(Continued)

988

181_Cydulka_Ch181.indd 988 22/03/17 3:37 PM

References/provenance

- Source: *Tintinalli's Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 1004–1007.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 181: Emergencies in Systemic Rheumatic Diseases

Source pages 1008–1015 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 1008–1015

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Emergencies in Systemic Rheumatic Diseases**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 181: Emergencies in Systemic Rheumatic Diseases 989 Table 181-1 Common Features and Complications of Systemic Rheumatic Diseases (Continued) Common and Characteristic Disorder Clinical Features Complications Behçet’s disease Chronic, relapsing, inflammatory disease. [Source pages 1008–1015]

- Hypopyon, retinal vasculi- Systemic vasculitis involving arteries and tis, optic neuritis, eye vas- veins of all sizes (carotid, pulmonary, aortic, cular complication. [Source pages 1008–1015]

- Dural and inferior extremity vessels are most sinus thrombosis, aseptic commonly involved, with aneurysm, dis- meningitis and encephali- section, rupture, or thrombosis). [Source pages 1008–1015]

- Superior complaints (fever, malaise, fatigue, weight and inferior vena cava loss, myalgia), recurrent painful skin and syndrome. [Source pages 1008–1015]

- Abdominal mucosal lesions; asymmetric, nondeforming aorta or pulmonary artery arthritis of the medium and large joints; emergencies. [Source pages 1008–1015]

Clinical reasoning structure

- **Problem representation:** Emergencies in Systemic Rheumatic Diseases; source pages 1008–1015.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Emergencies in Systemic Rheumatic Diseases**.

- Source-page range: **1008–1015**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 1008

CHAPTER 181: Emergencies in Systemic Rheumatic Diseases 989

Table 181-1 Common Features and Complications of Systemic Rheumatic Diseases (Continued)

Common and Characteristic Disorder Clinical Features	Complications
Behçet's disease Chronic, relapsing, inflammatory disease. Hypopyon, retinal vasculi-	
Systemic vasculitis involving arteries and tis, optic neuritis, eye vas-	
veins of all sizes (carotid, pulmonary, aortic, cular complication. Dural	
and inferior extremity vessels are most sinus thrombosis, aseptic	
commonly involved, with aneurysm, dis- meningitis and encephali-	
section, rupture, or thrombosis). Systemic tis. Arrhythmias. Superior	
complaints (fever, malaise, fatigue, weight and inferior vena cava	
loss, myalgia), recurrent painful skin and syndrome. Abdominal	
mucosal lesions; asymmetric, nondeforming aorta or pulmonary artery	
arthritis of the medium and large joints; emergencies. Bowel	
thrombophlebitis and deep vein thrombosis; perforation.	
ocular complications. Neuropsychiatric	
manifestations. Pericarditis, myocarditis.	
Churg–Strauss Vasculitis with a multisystemic involve-	Heart failure, acute myo-
syndrome ment. Systemic complaints (fever, malaise,	cardial infarction, acute
fatigue, weight loss, myalgia), allergic rhi-	and constrictive pericar-
nitis, nasal obstruction, recurrent sinusitis,	ditis, GI bleeding, bowel
asthma, and peripheral blood eosinophilia.	perforation.
Systemic hypertension, pericarditis,	
abdominal pain, peripheral symmetric	
neuropathy; skin lesions and rash.	
Dermatomyositis/ Idiopathic inflammatory myopathies.	ARDS. Respiratory

polymyositis Muscle weakness, myalgia, and muscle failure and arrest due to
tenderness. Elevated serum creatine diaphragmatic or chest
kinase. Systemic complaints (fever, wall muscle weakness,
malaise, fatigue, weight loss, myalgia), alveolar hemorrhage,
Raynaud phenomenon, nonerosive inflam- and ILD. Heart failure,
matory polyarthritis, esophageal dysfunc- arrhythmias, and
tion, ILD, aspiration lung infections. conduction disturbances.
Giant cell arte- Chronic vasculitis of large- and medium- Ischemic optic neuropa-
ritis (temporal sized vessels. Elderly (mean age at thy, eye vessel occlusion.
arteritis) diagnosis: 70 years old). Associated with Aortitis (especially the
polymyalgia rheumatica in 50% of cases. thoracic tract) and aortic
Localized headache of new onset, tender- emergencies. Stroke.
ness of the temporal artery, and biopsy
revealing a necrotizing arteritis. Temporal
artery may be normal on clinical examina-
tion. Gradual onset, systemic complaints,
jaw or tongue claudication, eye complaints
and visual loss. Aortic regurgitation and aor-
tic arch syndrome. Neurologic complications
due to carotid and vertebrobasilar vasculitis.
Henoch– Systemic vasculitis associated with Respiratory failure and
Schönlein immunoglobulin A deposition, generally alveolar hemorrhage.
purpura in children. Frequently, acute presentation Seizures, intracranial
follows an upper respiratory infection. bleeding, GI hemorrhage,
Palpable purpura (in patients with neither bowel ischemia or perfo-
thrombocytopenia nor coagulopathy), ration, acute pancreatitis,
arthritis/arthralgia, abdominal pain, and intussusception (children).
renal impairment (adult), ILD. Acute scrotum.

Source page 1009

990 SECTION 20: Nontraumatic Musculoskeletal Disorders

Table 181-1 Common Features and Complications of Systemic Rheumatic Diseases (Continued)

Disorder	Common and Characteristic Clinical Features	Complications
Microscopic polyangiitis	Small-vessel systemic vasculitis, characterized by rapidly progressive glomerulonephritis and pulmonary involvement. Lung complications differentiate microscopic polyangiitis from polyarteritis nodosa. Systemic complaints (fever,	Rapidly progressive glomerulonephritis, severe lung hemorrhage, GI bleeding.

malaise, fatigue, weight loss, myalgia),
arthralgias, skin lesions, hemoptysis,
abdominal pain, renal impairment, sys-
temic hypertension.

Polyarteritis Systemic necrotizing vasculitis of Acute scrotum, ischemic
nodosa the medium-sized muscular arteries. and hemorrhagic stroke,
Systemic complaints (fever, malaise, acute coronary syndrome,
fatigue, weight loss, myalgia), arthral- heart failure, peripheral
gias, skin lesions, abdominal pain, renal artery ischemia, mesen-
impairment, systemic hypertension, teric ischemia and bowel
peripheral mononeuropathy typically with perforation, GI bleeding,
both motor and sensory deficits, eye acute pancreatitis, malig-
complications, leukocytosis, and normo- nant hypertension.
chromic anemia.

Relapsing Immune-mediated condition. Ears (vio- Airway obstruction. Acute
polychondritis laceous and erythematous auricula), renal failure, aortitis and
nose (saddle nose deformity), and other aortic emergencies, heart
cartilaginous structures inflammation block, ACS, scleritis,
(especially joints and respiratory tract). peripheral ulcerative kera-
One-third of cases associated with titis, and acute scrotum.
other systemic rheumatic diseases.
Sternoclavicular, costochondral, and
manubriosternal arthritis, upper airway
involvement, aortic or mitral valvular
regurgitation, pericarditis, renal impair-
ment, peripheral neuropathies, ocular
complications.

Rheumatoid Chronic, systemic, inflammatory disorder. Airway obstruction,
arthritis Symmetric and potentially destructive obliterative bronchiolitis,
arthritis. Systemic symptoms (fever, acute respiratory failure.
malaise, fatigue, weight loss, myalgia), ACS, heart failure, tho-
skin lesions, splenomegaly. Cervical racic aorta dissection,
spine involvement, pleuritis, ILD, arrhythmias and conduc-
pericarditis, myocarditis, and aortitis. tion disturbances, sublux-
Cricoarytenoid arthritis with potential for ation of the atlantoaxial
airway obstruction, ocular involvement. joints, bowel ischemia
Peripheral artery disease, Sjögren’s syn- and perforation. Septic
drome, vasculitis, and renal impairment. arthritis. Scleritis.
Abdominal pain. Anemia, leukopenia,
thrombocytosis, and Felty’s syndrome.

Increased risk of lymphoproliferative
diseases, particularly non-Hodgkin's
lymphoma.
(Continued)
181_Cydulka_Ch181.indd 990 22/03/17 3:37 PM

Source page 1010

CHAPTER 181: Emergencies in Systemic Rheumatic Diseases 991

Table 181-1 Common Features and Complications of Systemic Rheumatic Diseases (Continued)

Disorder	Common and Characteristic Clinical Features	Complications
Systemic lupus erythematosus	Systemic autoimmune disease, characterized by relapses and remissions, and affecting virtually every organ. Systemic complaints (fever, malaise, fatigue, weight loss, myalgia), symmetric polyarticular arthritis (small joints of the hands, wrists, and knees), butterfly rash, and/or nasal ulcers, Raynaud's phenomenon. Neuropsychiatric manifestations, pleurisy, lupus pneumonitis, shrinking or vanishing lung syndrome, pulmonary hypertension. Libman–Sacks endocarditis, pericarditis, myocarditis, endocarditis. GI nonspecific complaints. Renal impairment, mild anemia, and thrombocytopenia. Antiphospholipid syndrome. Ocular complications.	Airway obstruction, and ARDS, respiratory failure and arrest, alveolar hemorrhage, ACS, cardiac tamponade, heart failure, arrhythmias, pulmonary embolism, stroke, acute renal failure, Guillain–Barré like syndrome, seizures. Bowel ischemia and perforation, GI bleeding. acute pancreatitis. Hemolytic anemia, thrombotic microangiopathic
Sjögren's syndrome	Autoimmune disease. May be primary; secondary form is mostly associated with rheumatoid arthritis, systemic lupus erythematosus, polymyositis, or dermatomyositis. Xerophthalmia and xerostomia, systemic symptoms, arthralgia, skin lesions, Raynaud's phenomenon. ILD, acute pancreatitis. pulmonary hypertension, pericarditis, neuropsychiatric manifestations, peripheral neuropathy, hepatic abnormalities,	Hypokalemic respiratory arrest. Heart block, pulmonary embolism, ischemic stroke, transverse myelitis, optic neuritis, renal tubular acidosis,

renal impairment, increased risk of non-
Hodgkin’s lymphoma.
Systemic Inappropriate and excessive accumula- Scleroderma renal crisis.
sclerosis tion of collagen and matrix in a variety Respiratory failure, ARDS,
(scleroderma) of tissue; widespread vascular lesions aspiration pneumonitis,
with endothelial dysfunction, vascular pulmonary hypertension,
spasm, thickening of the vascular wall alveolar hemorrhage,
and narrowing of the vascular lumen. heart failure, arrhythmias,
Systemic complaints (fever, malaise, and conduction
fatigue, weight loss, myalgia), skin disturbances.
lesions (fingers, hands, and face),
carpal tunnel syndrome, Raynaud’s
phenomenon. ILD, renal impairment,
GI dysmotility, gastroesophageal reflux
(aspiration pneumonitis), chronic esoph-
agitis and stricture formation. Vascular
ectasia in the stomach (“watermelon
stomach”).
(Continued)
181_Cydulka_Ch181.indd 991 22/03/17 3:37 PM

Source page 1011

992 SECTION 20: Nontraumatic Musculoskeletal Disorders
Table 181-1 Common Features and Complications of Systemic Rheumatic
Diseases (Continued)
Common and Characteristic
Disorder Clinical Features Complications
Takayasu’s Chronic vasculitis, young women, pre- ACS, bowel ischemia and
arteritis dominantly Asians. Systemic complaints perforation, GI bleeding,
(fever, malaise, fatigue, weight loss, myal- stroke.
gia), arthralgias, skin lesions, abdominal
pain and diarrhea. Aorta and its primary
branches, and pulmonary artery involve-
ment. Neurologic manifestations, syncope,
subclavian steal syndrome, extremity
ischemia. Renovascular hypertension.
Normochromic normocytic anemia.
Wegener’s Multiple organ system vasculitis and Airway obstruction, sub-
(granulomatosis necrotizing granulomas. Respiratory tract glottic stenosis, bronchi-
with polyangiitis) manifestations in approximately 100% olitis obliterans organizing
of cases, with nose, oral cavity, upper pneumonia, and alveolar

trachea, external and middle ear, and hemorrhage. ACS, arrhyth-
orbit inflammations. Upper airway and mias. Rapidly progressive
pulmonary manifestations. Constitutional glomerulonephritis.
symptoms, arthralgias, glomerulonephritis
and small vessel vasculitis (scleritis and
episcleritis, palpable purpura or cutane-
ous nodules, peripheral neuropathy, deaf-
ness). Systemic hypertension. Pericarditis,
myocarditis. Renal impairment. Anemia,
leukocytosis, and thrombocytosis.

Abbreviations: ACS, acute coronary syndrome; ARDS, adult respiratory distress syndrome; ILD, interstitial lung disease.

■■ CLINICAL FEATURES

Airway Emergencies

Cricoarytenoid joint arthritis, seen in rheumatoid arthritis (RA), systemic lupus erythematosis (SLE), and relapsing polychondritis (RP), can present with throat pain, a fullness in the throat, voice changes, and tenderness over cartilaginous structures. More severe cases lead to dyspnea and stridor, leading to acute upper airway obstruction. Evaluate using CT imaging or fiber optic laryngoscopy; initial treatment consists of high-dose steroids, such as methylprednisolone 250 mg IV.

Endotracheal intubation is often difficult in patients with systemic rheumatic disease; thus, consider “dual setup” preparation for cricothyrotomy. Moreover, neck hyperextension should be avoided in patients with advanced rheumatoid arthritis or ankylosing spondylitis due to the risk of cervical fractures, subluxation and dislocation, and spinal cord compression.

Pulmonary Emergencies

Systemic sclerosis, RA, SLE, Wegener’s granulomatosis, and polymyositis can cause chronic pulmonary symptoms through processes such as

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 1008–1015.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 182: Nontraumatic Disorders of The Hand

Source pages 1016–1017 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 1016–1017

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Nontraumatic Disorders of The Hand**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 182: Nontraumatic Disorders of The Hand 997 Table 182-2 Initial Antibiotic Coverage for Common Hand Infections (Continued) Infection Initial Antimicrobial Agent(s) Likely Organisms Comments Animal bites If no visible signs of infection: S. [Source pages 1016–1017]

- aureus, strep- All animal bite (including amoxicillin/clavulanate, tococci, Eikenella wounds should human) 875/125 mg PO twice daily corrodens (human), receive prophylactic for 5 days. [Source pages 1016–1017]

- For signs of infection: multocida (cat), Ampicillin-sulbactam, 1.5 g anaerobes, and IV every 6 h, or cefoxitin, 2 g gram-negative IV every 8 h, or piperacillin- bacteria tazobactam, 3.375 g every 6 h. [Source pages 1016–1017]

- For penicillin allergy, use clindamycin plus moxifloxacin or TMP-SMX and metronidazole. [Source pages 1016–1017]

- Herpetic Acyclovir, 400 mg PO three Herpes simplex No surgical drainage whitlow times daily for 10 days. [Source pages 1016–1017]

Clinical reasoning structure

- **Problem representation:** Nontraumatic Disorders of The Hand; source pages 1016–1017.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Nontraumatic Disorders of The Hand**.

- Source-page range: **1016–1017**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 1016

CHAPTER 182: Nontraumatic Disorders of The Hand 997

Table 182-2 Initial Antibiotic Coverage for Common Hand Infections (Continued)

Infection Initial Antimicrobial Agent(s) Likely Organisms Comments

Animal bites If no visible signs of infection: S. aureus, strep- All animal bite
(including amoxicillin/clavulanate, tococci, Eikenella wounds should
human) 875/125 mg PO twice daily corrodens (human), receive prophylactic
for 5 days. Pasteurella oral antibiotics.
For signs of infection: multocida (cat),
Ampicillin-sulbactam, 1.5 g anaerobes, and
IV every 6 h, or cefoxitin, 2 g gram-negative
IV every 8 h, or piperacillin- bacteria
tazobactam, 3.375 g every
6 h. For penicillin allergy, use
clindamycin plus moxifloxacin
or TMP-SMX and metronidazole.

Herpetic Acyclovir, 400 mg PO three Herpes simplex No surgical drainage
whitlow times daily for 10 days. is indicated.

*

While many sources recommend 7–10 days of therapy, the Infectious Disease Society of America recommends 5 days of therapy if symptoms resolve, continue therapy if symptoms persist.

Abbreviations: MRSA, methicillin-resistant Staphylococcus aureus; TMP-SMX, trimethoprim-sulfamethoxazole.

■■ NONINFECTIOUS CONDITIONS

Tendonitis and tenosynovitis are usually due to overuse. Examination reveals tenderness over the tendon and pain with activation or passive stretch. Treat with immobilization and nonsteroidal anti-inflammatory drugs (NSAIDs).

Trigger finger is a tenosynovitis of the flexor sheath. Inflammation or scarring results in impingement and snap release of the tendon as the finger is extended from a flexed position. Steroid injection may be effective.

Definitive treatment is surgery.

DeQuervain tenosynovitis involves the extensor pollicis brevis and abductor pollicis tendons. Pain occurs at the radial aspect of the wrist and radiates into the forearm. The Finkelstein test is diagnostic: the patient grasps the thumb in the fist and deviates the hand ulnarly, reproducing the pain. Treat with a thumb spica splint, NSAIDs, and referral.

Carpal tunnel syndrome results from compression of the median nerve by the transverse carpal ligament. The cause is usually edema from overuse, pregnancy, or congestive heart failure. Pain in the median nerve distribution tends to be worse at night. The pain may be reproduced by tapping over the nerve at the wrist (Tinel's sign) or by holding the wrist flexed maximally for >1 minute (Phalen's test). Treat with a wrist splint, NSAIDs, and referral.

Dupuytren contracture results from fibrous changes in the subcutaneous tissues of the palm which may lead to tethering and joint contractures. Refer to a hand surgeon.

Ganglion cyst is a collection of fluid from synovial tissue that herniates from a joint capsule or tendon sheath. Treat with NSAIDs and referral.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 283, "Nontraumatic Disorders of the Hand," by Carl A. Germann.

182_Cydulka_Ch182.indd 997 22/03/17 3:43 PM

Source page 1017

C hapter Soft Tissue Problems of The Foot

183 Gavin R. Budhram

■■ CORNS AND CALLUSES

Calluses represent a dermatologic reaction to focal pressure. They are protective and should not be treated unless they are painful. Ongoing pressure may cause calluses to develop into corns. Corns have a central hyperkaratotic core that is often painful. Hard corns can resemble warts, but can be differentiated with incision; warts will bleed, corns will not. The differential diagnosis includes syphilis, psoriasis, lichen planus, rosacea, arsenic poisoning, basal cell nevus syndrome, and malignancy. Treatment for corns often includes paring with a scalpel to include removal of central keratin plug, but topical salicylic acid treatments may be more effective.

■■ PLANTAR WARTS

Plantar warts are common, contagious, and caused by the human papillomavirus. The diagnosis is clinical and the differential includes corns and undiagnosed melanoma. Topical treatment with 15% to 20% salicylic acid is most effective. Nonhealing lesions should be referred to a dermatologist

or podiatrist.

■■ ONYCHOCRYPTOSIS (INGROWN TOENAIL)

Onychocryptosis is characterized by increased inflammation or infection of the lateral or medial aspects of the toenail. This occurs when the nail plate penetrates the nail sulcus and subcutaneous tissue (usually in the great toe). Patients with underlying diabetes, arterial insufficiency, cellulitis, ulceration, or necrosis are at risk for amputation if treatment is delayed. Treatment depends on the type of inflammation. If the toenail is uninfected, sufficient results will often be obtained with elevation of the nail with a wisp of cotton between the nail plate and the skin, daily foot soaks, and avoidance of pressure on the area. A second option (requiring digital block) is to remove a spicule of the nail and debride the nail groove (Fig. 183-1). If granulation tissue or infection is present, partial removal of the nail is indicated. If the toenail is infected, perform digital block and cut one-fourth or less of the nail, including beneath the cuticle, with a longitudinal incision (Fig. 183-2). A nonadherent bulky dressing should be placed and the wound should be checked in 24 to 48 hours.

■■ BURSITIS

Calcaneal bursitis causes pain over the posterior heel in contrast to Achilles tendinopathy that causes pain and tenderness 2 to 6 cm superior to the posterior calcaneus. Pathologic bursae of the foot are categorized as follows: (1) noninflammatory, (2) inflammatory, (3) suppurative, and (4) calcified. Noninflammatory bursae become painful as a result of

183_Cydulka_Ch183.indd 998 22/03/17 3:56 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 1016–1017.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 183: Soft Tissue Problems of The Foot

Source pages 1018–1023 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 1018–1023

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Soft Tissue Problems of The Foot**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 183: Soft Tissue Problems of The Foot 999 Figure 183-1. [Source pages 1018–1023]

- This method is used for small nail fold swellings without infection. [Source pages 1018–1023]

- After antiseptic skin preparation and digital nerve block, an oblique portion of the affected nail is trimmed about one-third to two- thirds of the way back to the posterior nail fold. [Source pages 1018–1023]

- Use scissors to cut the nail; use forceps to grasp and remove the nail fragment. [Source pages 1018–1023]

- This method is used for onychocryptosis in the setting of significant granulation tissue or infection. [Source pages 1018–1023]

Clinical reasoning structure

- **Problem representation:** Soft Tissue Problems of The Foot; source pages 1018–1023.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Soft Tissue Problems of The Foot**.

- Source-page range: **1018–1023**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 1018

CHAPTER 183: Soft Tissue Problems of The Foot 999

Figure 183-1. Partial toenail removal. This method is used for small nail fold swellings without infection. After antiseptic skin preparation and digital nerve block, an oblique portion of the affected nail is trimmed about one-third to two-thirds of the way back to the posterior nail fold. Use scissors to cut the nail; use forceps to grasp and remove the nail fragment.

Figure 183-2. Partial toenail removal (infection present). This method is used for onychocryptosis in the setting of significant granulation tissue or infection.

direct pressure and resolve with simple measures including comfortable footwear, rest, ice, and nonsteroidal anti-inflammatory drugs (NSAIDs).

Inflammatory bursitis results from gout, syphilis, or rheumatoid arthritis. Suppurative bursitis results from spread of pyogenic organisms (often Staphylococcus aureus) from adjacent wounds. Complications include hygroma, calcified bursae, fistula, and ulcer formation. Treatment for septic bursitis is discussed in Chapter 180, "Acute Disorders of the Joints and Bursae."

■■ PLANTAR FASCIITIS

The plantar fascia is connective tissue anchoring the plantar skin to the bone. Plantar fasciitis is the most common cause of heel pain due to over-use. Patients have pain on the plantar surface of the foot that is worse when initiating walking and point tenderness over the anterior–medial calcaneus

183_Cydulka_Ch183.indd 999 22/03/17 3:56 PM

Source page 1019

1000 SECTION 20: Nontraumatic Musculoskeletal Disorders

that is worsened by dorsiflexion of the toes. The presence of heel spurs on radiography is of no diagnostic value because many patients without plantar fasciitis have this finding. Treatment includes rest, ice, and NSAIDs. Most of the cases are self-limited and resolve spontaneously within 12 months. Plantar stretch exercises, including using the hands to dorsiflex the ankle and toes are helpful in the acute phase. Glucocorticoid injections

are not indicated in the ED. Patients with severe cases may require a short-leg walking cast and should be referred to a podiatrist or orthopedist.

■■ NERVE ENTRAPMENT SYNDROMES

Tarsal Tunnel Syndromes

Tarsal tunnel syndrome involves heel and foot pain due to compression of the posterior tibial nerve as it courses inferior to the medial malleolus. Causes include running, restrictive footwear, edema of pregnancy, post-traumatic fibrosis, ganglion cysts, osteophytes, and tumors. Symptoms include numbness or burning of the sole and may be limited to the heel. Pain is worse at night and after running or standing, in contrast to plantar fasciitis which is worse upon morning standing. Typically, pain is located at the medial malleolus, the heel, the sole, and the distal calf.

The differential diagnosis includes plantar fasciitis and Achilles tendonitis. Percussion inferior to the medial malleolus yields pain radiating to the medial or lateral plantar surface (Tinel sign), and eversion and dorsiflexion worsen symptoms. Treatment includes NSAIDs, rest, and possible orthopedic referral.

Deep Peroneal Nerve Entrapment

Entrapment of the deep peroneal nerve occurs most frequently where it courses beneath the extensor retinaculum at the anterior aspect of the ankle. Recurrent ankle sprains, soft tissue masses, and restrictive footwear represent the most common causes. Symptoms include dorsal and medial foot pain as well as sensory hypoesthesia at the first web space. Prolonged entrapment may cause loss of the ability to hyperextend the toes.

Pain and tenderness can be elicited by plantar flexion during inversion of the foot. Treatment includes NSAIDs, rest, and possible orthopedic referral.

■■ GANGLIONS

A ganglion is a benign synovial cyst attached to a joint capsule or tendon sheath near the anterolateral ankle. Typically a firm, nontender, cystic lesion is found on examination. The diagnosis is clinical, but MRI or ultrasound can be used if in doubt. Treatment includes aspiration and injection of glucocorticoids, but most require surgical excision.

■■ TENDON LESIONS

Tenosynovitis and Tendinitis

Tenosynovitis and tendonitis are usually due to overuse and present with pain over the involved tendon. The flexor hallucis longus, posterior tibialis, and Achilles tendon are most commonly involved. Treatment includes ice, rest, and NSAIDs.

CHAPTER 183: Soft Tissue Problems of The Foot 1001

Tendon Lacerations

Tendon lacerations in the foot are complex, and orthopedic consultation is needed. After repair, extensor tendons are immobilized in dorsiflexion and flexor tendons in equinus.

Tendon Ruptures

Spontaneous rupture of the Achilles tendon is common, but rupture of the other tendons of the foot can occur. Age and chronic corticosteroid and fluoroquinolone use are risk factors for spontaneous rupture. Achilles tendon rupture presents with pain and a palpable defect in the area of the tendon. Patients have an inability to stand on tiptoes and an absence of plantar flexion with squeezing of the calf (Thompson test). Treatment is generally surgical in younger patients and conservative (casting in equinus) in the elderly. Anterior tibialis tendon rupture results in a palpable defect and mild foot drop. In most cases surgery is not necessary. Posterior tibialis tendon rupture is usually chronic and presents with a flattened arch and swelling over the medial ankle. Examination may show weakness on inversion, a palpable defect, and inability to stand on tiptoes. Flexor hallucis longus rupture presents with loss of plantar flexion of the great toe. The need for surgical repair depends on the patient's occupation and lifestyle. Disruption of the peroneal retinaculum occurs after a direct blow during dorsiflexion and causes localized pain behind the lateral malleolus; there is clicking during walking, as the tendon is subluxed. Treatment is surgical.

■■ PLANTAR INTERDIGITAL NEUROMA (MORTON NEUROMA)

Neuromas are thought to occur from entrapment of the plantar digital nerve proximal to its bifurcation due to tight-fitting shoes; the third interspace is most commonly affected. Patients often present with burning, cramping, or aching over the affected metatarsal head, and numbness in the toe. Pain is easily reproduced upon palpation of the area and at times a mass is felt. Diagnosis is clinical, but ultrasound or MRI may be helpful. Conservative treatment includes wide shoes and glucocorticoid injections. Local glucocorticoid injections may be curative. Surgical neurolysis is occasionally required.

■■ COMPARTMENT SYNDROMES OF THE FOOT

The foot has nine compartments. Compartment syndromes in the foot are associated with high-energy crush injuries. Other causes include bleeding disorders and postischemic swelling after arterial injury, foot and ankle fractures (especially calcaneal and Lisfranc's fracture/dislocation), burns,

and chronic overuse. Pain out of proportion to injury is one of the early findings. Additional symptoms include pain that is worsened on active and passive movement, paresthesias, and neurovascular deficits. Absent pulse and complete anesthesia are late findings.

At-risk patients must have compartment pressures checked. Any difference of less than 30 mm Hg between the Stryker STIC Device (Stryker,

183_Cydulka_Ch183.indd 1001 22/03/17 3:56 PM

Source page 1021

Kalamazoo, MI) and diastolic blood pressure is considered positive.

Prompt consideration of emergent fasciotomy is indicated.

■■ MALIGNANT MELANOMA

Melanoma of the foot, which accounts for 15% of all cutaneous melanomas, may present as atypical nonpigmented or pigmented lesions; the nail may be included. Vigilance is key as these lesions often mimic more benign conditions. The differential diagnosis includes fungal infections, plantar warts, and foot ulcers. Because prognosis is directly related to early diagnosis, all atypical or nonhealing lesions should be referred for biopsy.

■■ FURTHER READING

For further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 285, “Soft Tissue Problems of the Foot,” by Mitchell C. Sokolosky.

183_Cydulka_Ch183.indd 1002 22/03/17 3:56 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 1018–1023.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 184: Clinical Features of Behavioral Disorders

Source pages 1024–1027 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 1024–1027

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Clinical Features of Behavioral Disorders**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 184: Clinical Features of Behavioral Disorders 1005 than manic episodes. [Source pages 1024–1027]

- Manic individuals experience an elated mood that can quickly deteriorate to irritability and hostility. [Source pages 1024–1027]

- They appear energetic and expansive, with a decreased need for sleep, poor impulse control, racing thoughts, auditory hallucinations, grandiose ideas and pressured speech. [Source pages 1024–1027]

- Complications may include suicide, substance abuse, and marital and occupational disruptions. [Source pages 1024–1027]

- Bipolar illness is treated with lithium and valproate. [Source pages 1024–1027]

Clinical reasoning structure

- **Problem representation:** Clinical Features of Behavioral Disorders; source pages 1024–1027.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Clinical Features of Behavioral Disorders**.

- Source-page range: **1024–1027**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 1024

CHAPTER 184: Clinical Features of Behavioral Disorders 1005

than manic episodes. Manic individuals experience an elated mood that can quickly deteriorate to irritability and hostility. They appear energetic and expansive, with a decreased need for sleep, poor impulse control, racing thoughts, auditory hallucinations, grandiose ideas and pressured speech. Complications may include suicide, substance abuse, and marital and occupational disruptions. Bipolar illness is treated with lithium and valproate.

Anxiety Disorders

Panic disorder consists of recurrent episodes of severe anxiety and sudden, extreme autonomic symptoms. It is a diagnosis of exclusion as these symptoms can also occur in life-threatening cardiovascular and pulmonary disorders. The diagnosis of generalized anxiety disorder can be made when a patient experiences persistent worry or tension without discrete panic attacks for at least 6 months. Phobias consist of symptoms of anxiety, recognized as excessive by the person, prompted by the exposure to, or the anticipated exposure to, a specific stimulus. Posttraumatic stress disorder is an anxiety reaction to a severe, psychosocial stressor, typically perceived as life threatening. The individual experiences repetitive, intrusive memories of the event. Nightmares, feelings of guilt and depression, and substance abuse are common. Individuals with obsessive-compulsive disorder experience intrusive thoughts or images that create anxiety (obsessions). To control these thoughts and anxiety, the individual engages in repetitive behaviors or rituals (compulsions). Anxiety disorders may be treated with benzodiazepines.

■■ PERSONALITY DISORDERS

Individuals with a personality disorder exhibit a lifelong pattern of maladaptive behavior that is not limited to periods of illness. Personality disorders include paranoid, schizoid, schizotypal, antisocial, borderline, histrionic, narcissistic, avoidant, dependent, and obsessive-compulsive.

Personality disorders are difficult to treat.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. Many of the patients with psychiatric illnesses presenting to the ED do not need immediate treatment. Delirium is one exception where identification and treatment of the cause of delirium in the ED is essential.

2. The most common treatment in the ED is for the agitation associated with psychiatric illness (Fig. 184-1). Many patients have multiple psychiatric, substance use and medical diagnoses making treatment in the ED more complicated.

3. Disposition depends on the patient's ability to function, risk of suicide and homicidal intent.

184_Cydulka_Ch184.indd 1005 22/04/17 4:45 pm

Source page 1025

Schizophrenia

Bipolar Illness

Depression

Psychiatric

Drug intoxication/

withdrawal

Medical

Delirium

Dementia

Hyperthyroidism

Head Trauma

Temporal Lobe Epilepsy

Figure 184-1. Diagram of frequent medical, substance use and psychiatric illness in the ED.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 286, "Mental Health Disorders: ED Evaluation and Disposition," by Leslie S. Zun.

184_Cydulka_Ch184.indd 1006 22/04/17 4:45 pm

Source page 1026

C h a p ter Emergency Assessment and

185 Stabilization of Behavioral

Disorders

Leslie S. Zun

■■ INCIDENCE

In the last two decades, mental health-related visits to emergency departments (EDs) in the United States increased by 38% to 23.6 per 1000 population.

■■ CLINICAL FEATURES

Psychiatric patients present to EDs with various complaints including agitation, depression, suicide ideation, mania, or other abnormal behaviors. Some patients have overt psychiatric presentations while some may present with less obvious presentations such as panic disorders, depression, or suicidal ideation.

■■ DIAGNOSIS AND DIFFERENTIAL

The role of the emergency physician is to determine whether the psychiatric presentation is due to a medical or psychiatric etiology in order to decide on an appropriate patient disposition. This differentiation is made more difficult because approximately 50% of these patients have concomitant medical illnesses, most frequently hypertension, diabetes, asthma, and substance use disorder. Some psychiatric patients need treatment for agitation prior to evaluation in order to obtain vital signs, pulse oxygenation, and glucose assessment. The evaluation process includes a detailed medical and psychiatric history, medication history, social history, and a physical examination, including a thorough neurologic and mental status evaluation. The “red flags” for medical illness include initial presentation over the age of 45, abnormal vital signs, focal neurologic deficit, exposure to drugs or toxins, abnormal physical exam findings, and cognitive deficit. Some of the elements of the mental status examination are routinely assessed as part of the history and physical exam: appearance, behavior and attitude, and mood and affect. A number of items need to be specifically tested such as disorders of thought inquiring about suicidal and homicidal ideation, insight and judgment about their illness, disorders of perception such as hallucinations, and cognitive impairment using the Mini Mental State Exam or the clock drawing test. Factors associated with high risk of suicide include male gender, unmarried or recent loss of relationship, family history of suicide, substance abuse, history of depression or psychosis, hopelessness, frequent and pervasive thoughts of suicide, previous attempts, availability of a lethal method, and poor social support. Laboratory testing and imaging of psychiatric patients should be clinically driven rather than rote. Patients with new onset of psychiatric symptoms and patients with chronic mental illness and a new psychiatric presentation need such testing. The testing may include

185_Cydulka_Ch185.indd 1007 22/03/17 4:46 PM

Source page 1027

1008 SECTION 21: Psychosocial Disorders

Protocol for Treatment of Agitation

Based on response to
interventions, medication
is now required

Agitation associated Agitation due to
with delirium Intoxication

Agitation associated Undifferentiated
with psychosis in agitation or
patient with known complex
psychiatric disorder presentation

ETOH ETOH
CNS
or BZN or BZN CNS
Depressant
withdrawal not withdrawal is Stimulant
(e.g., ETOH)
suspected suspected

Identify and correct any 1. Oral Avoid BZN if 1. Oral 2nd-generation No Psychosis
underlying medical Benzodlazepines possible Antipsychotics Evident
condition lorazepam 1–2 mg risperidone 2 mg‡ Same as agitation
chlordiazepoxide olanzapine 5–10 mg‡ due to withdrawal
50 mg 1. Oral
Avoid BZN 1st-generation
diazepam 5–10 mg
Antipsychotics 2. Oral 1st-generation Psychosis
1. Oral 2nd-generation haloperidol Antipsychotics Evident
Antipsychotics 2. Parenteral 2–10 mg haloperidol 2–10 mg Same as for
risperidone 2 mg Benzodiazepines with BZN primary psychiatric
olanzapine 5–10 mg lorazepam 1–2 mg disorder
2. Parenteral
IM or IV 3. Parenteral
1st-generation
Antipsychotics 2nd-generation
2. Oral 1st-generation
haloperidol Antipsychotics
Antipsychotics
2–10 mg IM olanzapine 10 mg IM‡
haloperidol (low dose)#
ziprasidone

10–20 mg IM‡

3. Parenteral

2nd-generation 4. Parenteral

Antipsychotics 1st-generation

olanzapine 10 mg IM Antipsychotics

ziprasidone 10–20 mg IM haloperidol 2–10 mg

IM with BZN

4. Parenteral

1st-generation ‡ If an antipsychotic

Antipsychotics alone does not work

haloperidol (low dose)# sufficiently, add

IM or IV (with caution)† lorazepam 1–2 mg

(oral or parenteral).

There is strong

evidence that doses

above 3 mg (per day)

in patients with delirium

are associated with

significant risk of EPS,

so patients receiving

> 3 mg/day should be

assessed carefully

for EPS.

† See FDA guidelines.

Figure 185-1. Suggested algorithm for the ED management of agitated patients.

Reproduced with permission from Wilson MP, Pepper D, Currier GW, e

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.

- Content ID: SEG-TINTINALLI-EMM8-FULL.

- Source pages: 1024–1027.

- Processing timestamp: 2026-08-15T15:32:40Z.

- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 185: Emergency Assessment and Stabilization of Behavioral Disorders

Source pages 1028–1029 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 1028–1029

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Emergency Assessment and Stabilization of Behavioral Disorders**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 185: Emergency Assessment and Stabilization of Behavioral Disorders 1009 complete blood count, serum electrolytes, liver function tests, urinalysis, urine drug screen, alcohol levels, and head CT scan. [Source pages 1028–1029]

- Elderly patients may need thyroid function testing, ECG, and CXR. [Source pages 1028–1029]

- Medical illnesses that may masquerade as psychiatric illness include hypoglycemia, hyperthyroidism, and delirium. [Source pages 1028–1029]

- Delirium has a high rate of mortality and is frequently missed in the ED. [Source pages 1028–1029]

- It has a fluctuating course with change in consciousness and alertness as well as confusion. [Source pages 1028–1029]

Clinical reasoning structure

- **Problem representation:** Emergency Assessment and Stabilization of Behavioral Disorders; source pages 1028–1029.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Emergency Assessment and Stabilization of Behavioral Disorders**.

- Source-page range: **1028–1029**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 1028

CHAPTER 185: Emergency Assessment and Stabilization of Behavioral Disorders 1009

complete blood count, serum electrolytes, liver function tests, urinalysis, urine drug screen, alcohol levels, and head CT scan. Elderly patients may need thyroid function testing, ECG, and CXR. Medical illnesses that may masquerade as psychiatric illness include hypoglycemia, hyperthyroidism, and delirium. Delirium has a high rate of mortality and is frequently missed in the ED. It has a fluctuating course with change in consciousness and alertness as well as confusion. Tests of delirium such as the Confusion Assessment Method may be useful to identify these patients.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

1. The first priority is safety and stabilization, including attention to airway, breathing, and circulation.

2. All psychiatric patients need to disrobe and be placed in a gown. The medical and psychiatric evaluations should proceed quickly to determine the need for treatment and admission.

3. Agitation treatment begins with verbal de-escalation. Agitation may lead to violence and require medications and/or restraints to avoid self-injury and the harm to others.

4. The choice of the treatment of agitation depends on the level of agitation and underlying causation, if known.

5. Severely agitated patients associated with a psychiatric illness may need restraints and intramuscular haloperidol and lorazepam or an atypical antipsychotic given IM (Fig. 185-1). The goal of treatment is to calm without over-sedation and to remove restraints as soon as possible. Treat moderate and mild agitation with oral medications, if possible. Treat severely agitated patients with alcohol withdrawal or stimulant intoxication with a benzodiazepine.

6. Patients judged to be at high risk of harming themselves or others or those who are unable to care for themselves require admission to a psy-

chiatric facility.

7. Patients with moderate suicide potential need a psychiatric consultation. Patients with low risk for suicide may go home.

8. All psychiatric patients need close psychiatric follow-up and can benefit from a referral to community mental health resources. Ensure that discharged patients have someone safe with whom to stay.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 286, "Mental Health Disorders: ED Evaluation and Disposition," by Leslie S. Zun.

185_Cydulka_Ch185.indd 1009 22/03/17 4:46 PM

Source page 1029

C h ap t er Panic and Conversion Disorders

186 Kimberly Nordstrom

■■ PANIC DISORDER

Panic disorder manifests with a wide variety of symptoms that mimic other major medical problems.

Clinical Features

Panic disorder is defined as recurrent panic attacks followed by 1 month or more of persistent worry of future episodes or maladaptive changes in behavior related to the attack. Episodes begin unexpectedly; severity peaks within 10 minutes and symptoms last for up to 1 hour. Symptoms of panic attacks include palpitations, sweating, shortness of breath, trembling or shaking, choking sensation, chest pain or discomfort, nausea or abdominal distress, dizziness or light-headedness, paresthesias, chills or hot flashes, fear of losing control, fear of dying, and derealization or depersonalization.

Diagnosis and Differential

Panic disorder is a diagnosis of exclusion because its symptoms and signs mimic those of many potentially life-threatening disorders. A thorough history and physical examination and, when indicated, other tests help rule out these life-threatening disorders. The differential diagnosis of panic attacks is listed in Table 186-1.

Table 186-1 Medical Differential Diagnosis of Panic Attacks

Cardiovascular	Neurologic	Drug withdrawal
Angina	Migraine headache	Alcohol
Myocardial infarction	Ménière disease	Barbiturates
Mitral valve prolapse	Complex partial seizures	Benzodiazepines
Congestive heart failure	Transient ischemic attacks	Opiates
Tachyarrhythmias: Drug induced	β-Antagonists	
premature atrial contrac-	Caffeine	Psychiatric

tions, supraventricular Cocaine Posttraumatic stress

tachycardia Sympathomimetics disorder

Pulmonary Theophylline Depressive disorders

Hyperventilation Thyroid preparations Other anxiety disorders

Asthma Selective serotonin Psychosocial

Pulmonary embolus reuptake inhibitors Partner violence

Endocrine Cannabis Sexual abuse or assault

Hyperthyroidism Corticosteroids Other situational

Hypoglycemia β-Agonists stressors

Hyponatremia Triptans

Pheochromocytoma Nicotine

Carcinoid syndrome Yohimbine

Cushing’s syndrome Hallucinogens

Anticholinergics

1010

186_Cydulka_Ch186.indd 1010 22/03/17 4:51 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 1028–1029.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 186: Panic and Conversion Disorders

Source pages 1030–1033 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 1030–1033

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Panic and Conversion Disorders**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 186: Panic and Conversion Disorders 1011 Emergency Department Care and Disposition 1. [Source pages 1030–1033]

- After excluding life-threatening causes of symptoms, educate and re- assure the patient that panic disorder is an illness that can be treated effectively. [Source pages 1030–1033]

- Benzodiazepines, such as alprazolam 0.25 to 1 mg PO or lorazepam 0.5 to 1 mg PO/IV, are used to control acute symptoms. [Source pages 1030–1033]

- Antidepressants, such as selective serotonin reuptake inhibitors (SSRI) and serotonin- norepinephrine reuptake inhibitors (SNRI), are preferred for mainte- nance therapy. [Source pages 1030–1033]

- Ideally, follow up should be in place if sending the patient out with a benzodiazepine or SSRI/SNRI prescription. [Source pages 1030–1033]

Clinical reasoning structure

- **Problem representation:** Panic and Conversion Disorders; source pages 1030–1033.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Panic and Conversion Disorders**.

- Source-page range: **1030–1033**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 1030

CHAPTER 186: Panic and Conversion Disorders 1011

Emergency Department Care and Disposition

1. After excluding life-threatening causes of symptoms, educate and reassure the patient that panic disorder is an illness that can be treated effectively.

2. Benzodiazepines, such as alprazolam 0.25 to 1 mg PO or lorazepam 0.5 to 1 mg PO/IV, are used to control acute symptoms. Antidepressants, such as selective serotonin reuptake inhibitors (SSRI) and serotonin-norepinephrine reuptake inhibitors (SNRI), are preferred for maintenance therapy. Ideally, follow up should be in place if sending the patient out with a benzodiazepine or SSRI/SNRI prescription. Caution should be used in prescribing benzodiazepines to certain populations, such as drug abusers, geriatric patients, and those with respiratory disorders.

3. Ask the patient about suicidal thoughts, as the patient may require psychiatric consultation and hospitalization.

4. Most patients can be discharged. Refer to outpatient psychiatry for continuation or initiation of pharmacotherapy.

■■ CONVERSION DISORDER

Clinical Features

Patients with conversion disorder develop voluntary motor or sensory function deficits. The symptoms are not consciously produced by the patient and are usually in response to a stressor or conflict. The symptoms or deficits are not explained by a known organic etiology or culturally sanctioned response pattern. Organic disease may be concurrently present.

Diagnosis and Differential

Other medical or mental illnesses must be excluded before the diagnosis of conversion disorder can be made. The differential diagnosis is broad and includes stroke, multiple sclerosis, polymyositis, infectious disorders, as

well as drug ingestions or poisonings. The examination techniques listed in Table 186-2 may help test for true neurologic deficits.

Table 186-2 Testing Techniques for Conversion Disorder

Function	Technique
Sensation	
Yes/no test	Patient closes eyes and responds yes or no to touch stimulus. No response in numb area favors conversion disorder.
Bowlus and Currier test	Patient extends and then crosses the arms, with thumbs pointed down and palms facing together. Fingers (but not thumbs) are then interlocked, and then the hands are rotated inward toward chest. Sharp stimuli are applied to each finger in turn and the patient is asked to indicate normal or abnormal sensation in each digit. Patients with conversion disorder make mistakes and are inconsistent with responses.
Strength test	Patient closes eyes. Test “strength” by touching finger to be moved. True lack of sensation would not allow patient to ascertain finger to be moved.

(Continued)

Source page 1031

Table 186-2 Testing Techniques for Conversion Disorder (Continued)

Function	Technique
Pain	
Gray test	With abdominal pain due to psychological factors, the patient will close eyes during palpation. In pain of organic basis, the patient is more likely to watch the examiner’s hand to anticipate pain.
Motor	
Drop test	When a patient with paralysis of nonorganic etiology lifts a thumb, the affected limb will drop more slowly or fall with exaggerated speed as compared with the unaffected limb. In addition, an extremity dropped from above the face will miss it.
Thigh adductor test	Examiner places hands against both inner thighs of the patient who is told to adduct the “good” leg against resistance. With pseudoparalysis, the adductor muscles of “bad” leg will also adduct.
Hoover test	Examiner’s hands cup both heels of the patient who is asked to elevate the “good” leg. With pseudoparalysis the “bad” leg will push downward. When the patient is asked to lift the “bad” leg, if there is no downward pressure in the “good” leg, the patient is not trying.
Sternocleidomastoid test	Contraction of normal sternocleidomastoid muscle causes face

to rotate away from side of the contracted muscle. Patient with conversion hemiplegia cannot turn head to the weak side.

Coma

Corneal reflex Corneal reflexes remain intact in an awake patient.

Bell phenomenon Eyes divert upward when lids are opened, whereas eyes remain in neutral position in true coma.

Lid closing In true coma, lids when opened close rapidly initially and then more slowly as lids descend. Awake patients will have lids stay open, snap shut, or flutter.

Seizures

Corneal reflex Usually intact in pseudoseizure.

Abdominal muscu- lature with pseudoseizure. Palpation of abdominal musculature reveals lack of contractions

Blindness

Opticokinetic drum Rotating drum with alternating black and white stripes or piece of tape with alternating black and white sections pulled laterally in front of a patient's open eyes will produce nystagmus in a patient with intact vision.

Emergency Department Care and Disposition

1. Direct confrontation is not helpful and can be harmful to the patient. After excluding an organic cause for symptoms, gently reassure the patient that a serious medical illness has not been identified.

2. Suggest to the patient that symptoms often resolve spontaneously.

3. Consult with a neurologist or psychiatrist if symptoms do not resolve and preclude discharge. Otherwise, refer for outpatient psychiatric treatment as repetitive reassurance or other forms of therapy may be beneficial.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th edition, see Chapter 289, "Mood and Anxiety Disorders," by Tracy M. DeSelm.

186_Cydulka_Ch186.indd 1012 22/03/17 4:51 PM

Source page 1032

SECTION

Abuse and Assault

22

C ha p te r Child and Elderly Abuse

187 Jonathan Glauser

■■ CHILD ABUSE

Child maltreatment includes physical abuse, sexual abuse, emotional abuse, neglect, parental substance abuse, and Munchausen syndrome by proxy.

Clinical Features

Child neglect in early infancy results in the syndrome of failure to thrive. These children are frequently brought to the ED because of other medical problems, such as skin infections, severe diaper dermatitis, or acute gastroenteritis. Infants may have little subcutaneous tissue, protruding ribs, or occipital alopecia from lying on their back all day. They are wide eyed, wary, and difficult to console. They may have increased muscle tone in their lower extremities. Weight gain in the hospital is thought to be diagnostic of failure to thrive. Children over the age of 2 with environmental neglect are termed psychosocial dwarfs. They exhibit short stature, bizarre and voracious appetites, and are hyperactive with unintelligible or delayed speech.

Physical abuse is suggested by a history that is inconsistent with the nature of the injuries. The history of the event given by the caretaker may keep changing, or may be different from that given by the child.

The following findings suggest physical abuse: bruises over multiple areas, bites with an intercanine diameter >3 cm, lacerations of the frenulum or oral mucosa from force-feeding, burns of an entire hand or foot, or burns of the buttocks or genitalia from toilet training punishment, cigarette burns, spiral fractures caused by twisting of long bones, metaphyseal chip fractures, periosteal elevation from new bone formation at sites of previous microfractures, multiple fractures at different stages of healing, fractures at unusual sites such as lateral clavicle, ribs, sternum, vomiting, irritability, seizures, change in mental status, apnea, or retinal hemorrhages from intracranial hemorrhage (shaken baby syndrome). Vomiting, abdominal pain, and tenderness with diminished bowel sounds or abdominal distention may be due to a duodenal hematoma, as evidenced by a “double-bubble” sign on abdominal x-ray films.

Munchausen syndrome by proxy is a synonym for medical child abuse. A parent fabricates illness in a child in order to secure prolonged contact with health care providers. Complaints may be numerous and agents such as ipecac

187_Cydulka_Ch187.indd 1013 22/03/17 5:08 PM

Source page 1033

or warfarin may have been given to precipitate these complaints. Parents typically encourage more diagnostic tests, and are happy if they are positive.

Sexual abuse is suggested with complaints referable to the anogenital area, such as bleeding, discharge, or the presence of a sexually transmitted disease. Clefts or concavities in the hymen typically present in the 6 o’clock position. Victims of child abuse may be overly compliant with painful medical procedures, overly protective of the abusing parent, or overly affectionate to medical staff.

Diagnosis and Differential

Any serious injury in a child under the age of 5 should be viewed with suspicion. Parents and caregivers may appear to be under the influence of drugs or alcohol and refuse diagnostic studies. Victims of neglect may appear dirty, improperly clothed, and may be unimmunized.

A skeletal survey of the long bones will help detect evidence of physical abuse. See Chapter 157, “Trauma in Children,” for further workup of traumatic injuries.

Inspect the genital area carefully for injury. Speculum examination in the preadolescent is not needed unless perforating vaginal trauma is suspected. Absence of physical findings does not rule out abuse.

Laboratory testing for sexual abuse should include cultures of the throat, vagina, and rectum for gonorrhea and chlamydia. Rapid antigen assays are not reliable forensic evidence in prepubescent children. Perform testing for syphilis if clinical concern exists. Test for HIV if clinically concerned and appropriate counseling is available. Syphilis, gonorrhea, and C trachomatis infections are considered presumptive evidence for sexual abuse, as is HIV, if not acquired perinatally or by transfusion.

Rarely, conditions such as leukemia, aplastic anemia, and osteogenesis imperfecta can mimic physical abuse.

Emergency Department Care and Disposition

Address all medical issues and injuries. Infants suspected of suffering from failure to thrive and children with suspected Munchausen syndrome by proxy should be admitted to the hospital. Involve social services during the ED visit. Ensure a safe environment for each child. Every state requires reporting of suspected cases of child abuse. The law protects physicians from legal retaliation from parents.

■■ ELDERLY ABUSE

Elderly abuse is an act or omission resulting in harm to the health or welfare of an elderly person.

Clinical Features

Physical abuse is the most easily recognized form of elderly abuse, although chemical restraint such as intentional overmedication may be subtle. Caregiver neglect, defined as failure of a caregiver to provide basic care, goods and services such as food, clothing, hygiene, medical care, and shelter, accounts for the majority of cases of elder abuse. Financial abuse is the second commonest form of abuse, and occurs when family members take control of or steal assets, checks, or pensions for personal gain.

187_Cydulka_Ch187.indd 1014 22/03/17 5:08 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 1030–1033.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 187: Child and Elderly Abuse

Source pages 1034–1035 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 1034–1035

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Child and Elderly Abuse**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 187: Child and Elderly Abuse 1015 Emotional abuse entails inflicting anguish, emotional pain, or distress. [Source pages 1034–1035]

- Verbal threats, social isolation, and harassment can contribute to depression and other mental health problems. [Source pages 1034–1035]

- Self-neglect includes those behaviors of an elderly person that threaten his or her own safety: failure to provide adequate food, medical care, hygiene, clothing, or shelter. [Source pages 1034–1035]

- The spectrum of sexual abuse ranges from unwanted touching or indecent exposure to rape itself. [Source pages 1034–1035]

- Caretakers may give a conflicting report of an injury or illness. [Source pages 1034–1035]

Clinical reasoning structure

- **Problem representation:** Child and Elderly Abuse; source pages 1034–1035.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Child and Elderly Abuse**.

- Source-page range: **1034–1035**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 1034

CHAPTER 187: Child and Elderly Abuse 1015

Emotional abuse entails inflicting anguish, emotional pain, or distress. Verbal threats, social isolation, and harassment can contribute to depression and other mental health problems. Self-neglect includes those behaviors of an elderly person that threaten his or her own safety: failure to provide adequate food, medical care, hygiene, clothing, or shelter. The spectrum of sexual abuse ranges from unwanted touching or indecent exposure to rape itself. Caretakers may give a conflicting report of an injury or illness. The patient may be fearful of his or her companion. The caretaker may seem indifferent or angry toward the patient, or may be overly concerned with costs of treatment needed by the patient.

Diagnosis and Differential

Risk factors for elder abuse may be associated with caregivers, perpetrators, or with the elders. Patient characteristics include (a) cognitive impairment, (b) female sex, (c) physical dependency, (d) alcohol abuse, (e) developmental disability, (f) special medical or psychiatric needs, (g) lack of social support, and (h) limited experience managing finances.

Risk factors for or characteristics of perpetrators of abuse include (a) history of violence within or outside of the family, (b) excessive dependence on the elder for financial support, (c) history of mental illness or substance abuse.

Patients should be interviewed in private. Screening questions for elder abuse are available. Areas of concern include whether anyone has touched or hurt them, forced them to do things, taken something of theirs without asking, threatened them, or made them feel afraid.

The following findings on physical examination are suggestive of abuse: bruising or trauma, poor general appearance and inappropriate hygiene or soiled clothing, malnutrition and dehydration, contusions and lacerations to normally protected areas of the body—inner thighs, mastoid, palms, soles, buttocks, unusual burns or multiple burns in different stages of healing, rope or restraint marks on ankles or wrists, spiral fractures of long bones,

midshaft ulnar (nightstick) fractures from attempts to shield blows, multiple deep/uncared for ulcers.

Emergency Department Care and Disposition

Elder abuse is widely underreported and underrecognized. Treatment entails three key components:

1. Address medical and psychosocial needs.
2. Ensure patient safety.
3. Adhere to local reporting requirements.

Medical problems and injuries may be best managed with hospital admission. If neglect is unintentional, educate the caregiver. All 50 states have reporting requirements for elder abuse and neglect. Requirements for reporting within one's practice area are available at www.nceaaoa.gov.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 148, "Child Abuse and Neglect," by Margaret Colbourne and Michelle S. Clarke; Chapter 295, "Abuse of the Elderly and Impaired," by Jonathan Glauser and Frederic M. Hustey.

187_Cydulka_Ch187.indd 1015 22/03/17 5:08 PM

Source page 1035

C ha p ter Sexual Assault and Intimate

188 Partner Violence and Abuse

Mary Hancock

Not all individuals who are sexually assaulted sustain an injury. Lack of injury does not mean that an assault did not occur. Often the perpetrator is known to the assault survivor.

Intimate partner violence and abuse is defined as a pattern of assaultive behavior that may include physical injury, sexual assault, psychological abuse, stalking, deprivation, intimidation, and threats. Intimate partner violence and abuse occurs in every race, ethnicity, culture, geographic region, and religious affiliation and occurs in gay, lesbian, and heterosexual relationships.

■■ CLINICAL FEATURES

Elements of the sexual assault history are listed in Table 188-1. The history for intimate partner violence and abuse can be more difficult to obtain. Risk factors for intimate partner violence and abuse include female sex, age between 18 and 24 years, low socioeconomic status,

Table 188-1 Assault History

Who?

Did the assault survivor know the assailant?

Was it a single assailant or multiple assailants?

Can the survivor recall any identifying features of the assailant (height, build, age, race, tattoos, scars, birthmarks, etc.)? (Document in the medical records.)

What happened?

Was the patient physically assaulted?

With what (e.g., gun, bat, or fist) and to what part of the body?

Was there actual or attempted vaginal, anal, or oral penetration?

Did ejaculation occur? If so, where?

Was a foreign object used?

Was a condom used?

When?

When did the assault occur?

(Emergency contraception is most effective when started within 72 hours of the assault.)

Where?

Where did the assault occur?

(Corroborating evidence may be found based on the location of the assault.)

Suspicion of drug-facilitated rape?

Was there a period of amnesia?

Is there a history of being out drinking and then suddenly feeling very intoxicated?

Is there a history of waking up naked or with genital soreness?

Douche, shower, or change of clothing?

Did the patient douche, shower, or change clothing after the assault? (Performing any of these activities prior to seeking medical attention may decrease the probability of sperm or acid phosphatase recovery, as well as recovery of other bits of trace evidence.)

1016

188_Cydulka_Ch188.indd 1016 22/03/17 5:14 PM

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 1034–1035.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 188: Sexual Assault and Intimate Partner Violence and Abuse

Source pages 1036–1041 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 1036–1041

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Sexual Assault and Intimate Partner Violence and Abuse**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 188: Sexual Assault and Intimate Partner Violence and Abuse 1017 separated relationship status, and residence in rental housing. [Source pages 1036–1041]

- Injuries inconsistent with the patient’s history, multiple injuries in various stages of healing, delay in the time of injury occurrence and presentation, a visit for vague complaints without evidence of injury, or suicide attempts should trigger suspicions of intimate partner violence and abuse. [Source pages 1036–1041]

- Patients may complain initially of chronic pain syndromes, gyne- cologic or psychiatric difficulties, and alcohol and substance abuse. [Source pages 1036–1041]

- The patient also may appear frightened when the partner is present or the partner may be hostile, defensive, aggressive, or overly solicitous. [Source pages 1036–1041]

- Recent and remote abuse, including dates, locations, details of abuse, and witnesses, should be documented. [Source pages 1036–1041]

Clinical reasoning structure

- **Problem representation:** Sexual Assault and Intimate Partner Violence and Abuse; source pages 1036–1041.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Sexual Assault and Intimate Partner Violence and Abuse**.

- Source-page range: **1036–1041**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 1036

CHAPTER 188: Sexual Assault and Intimate Partner Violence and Abuse 1017

separated relationship status, and residence in rental housing. Injuries inconsistent with the patient's history, multiple injuries in various stages of healing, delay in the time of injury occurrence and presentation, a visit for vague complaints without evidence of injury, or suicide attempts should trigger suspicions of intimate partner violence and abuse. Patients may complain initially of chronic pain syndromes, gynecologic or psychiatric difficulties, and alcohol and substance abuse. The patient also may appear frightened when the partner is present or the partner may be hostile, defensive, aggressive, or overly solicitous. Recent and remote abuse, including dates, locations, details of abuse, and witnesses, should be documented. Patients need to be asked about any suicidal or homicidal ideation and plans and get appropriate, immediate evaluation.

Physical Examination

Perform a general medical examination including vital signs, appearance and demeanor. Focus head-to-toe inspection on defensive injury areas, such as the extremities, and potential areas of injury, such as the oral cavity, breasts, thighs, and buttocks. Record all the signs of trauma, new and old, in detail using a body map. Speculum examination should note any trauma, discharge, abrasions, including on the cervix. If anal penetration is reported, the rectum should be examined for abrasions and lacerations. Use of anoscopy in male patients increases detection of trauma.

Characteristic injuries of intimate partner violence and abuse include fingernail scratches, bite marks, cigarette burns, rope burns, and forearm bruising or nightstick fractures, suggesting a defensive posture. Abdominal injuries are common in the pregnant intimate partner violence and abuse

patient (Table 188-2).

Evidence Examination

Evidence collection in sexual assault is performed only within the first 72 hours after the assault. Informed consent is required. Most hospitals have a prepackaged rape kit. Chain of custody of the evidence must be maintained and the kit should never be left unattended. Not every part of the forensic evidence kit needs to be used every time. Tailor the evidence collection to the specifics of the assault, if known. If >72 hours have elapsed or the patient declines an evidentiary examination, perform a history and physical examination, document injuries, and provide prophylaxis for pregnancy and sexually transmitted infections. Label evidence clearly with the patient's name, type and source of evidence, date and time, and name of the examiner collecting the evidence. If there is high suspicion of drug-facilitated rape, a urine sample can be sent to a laboratory for toxicologic testing.

■■ DIAGNOSIS AND DIFFERENTIAL

Sexual assault is a legal determination, not a medical diagnosis. The legal definition contains three elements: carnal knowledge, nonconsent, and compulsion or fear of harm. Because of the legal considerations, careful documentation and evidence collection are important.

188_Cydulka_Ch188.indd 1017 22/03/17 5:14 PM

Source page 1037

Table 188-2 Signs Suggestive of Intimate Partner Violence

Findings	Comments
Injuries characteristic of violence	Fingernail scratches, broken fingernails, bite marks, dental injuries, cigarette burns, bruises suggesting strangulation or restraint, and rope burns or ligature marks may be seen.
Injuries suggesting a defensive posture	Forearm bruises or fractures may be sustained when individuals try to fend off blows to the face or chest.
Injuries during pregnancy	Up to 45% of women report abuse or assault during pregnancy. Preterm labor, placental abruption, direct fetal injury, and stillbirth can occur.
Central pattern of injury	Injuries to the head, neck, face, and thorax and abdominal injuries in pregnant women may suggest violence.
Extent or type of injury inconsistent with the patient's explanation	Multiple injuries at different anatomic sites inconsistent with the described mechanism of injury. The most common explanation of injury is a "fall." Embarrassment, evasiveness, or lack of concern with the injuries may be noted.

Multiple injuries in various These may be reported as “accidents” or “clumsiness.”
stages of healing
Delay between the time of Victims may wait several days before seeking medical care
injury and the presentation for injuries.
for treatment Victims may seek care for minor or resolving injuries.
Visits for vague or minor Frequent ED visits for a variety of injuries or illnesses
complaints without evidence including chronic pelvic pain and other chronic pain
of physiologic abnormality syndromes.
Suicide attempts Women who attempt or commit suicide often have a history
of intimate partner violence.

Many experts recommend routine screening for intimate partner violence and abuse for all adolescent and adult women who present to the ED and for mothers of children. Providers educated about the dynamics of intimate partner violence and abuse should conduct screening in a safe and private environment. Document all the findings, interventions, referrals, and required reporting.

■■ EMERGENCY DEPARTMENT CARE AND DISPOSITION

Address life-threatening injuries and the psychological needs. Treating critical injuries are the initial priority of the ED physician. A social worker or trained advocate should counsel the patient in the ED. Ensure the safety of the patient and any children involved while they are in the ED. Be familiar with reporting laws in your state.

■■ SEXUAL ASSAULT

Emergency Contraception

1. Obtain a pregnancy test.
2. Offer emergency contraception. Levonorgestrel only, 1.5 mg PO in a single dose (available OTC) OR ulipristal acetate 30 mg PO single

188_Cydulka_Ch188.indd 1018 22/03/17 5:14 PM

Source page 1038

dose (requires prescription) OR commonly prescribed contraceptive pills, ie. Ovral 2 pills once, repeat 2 pills in 12 hours (each dose contains levonorgestrel 0.5 mg and ethinyl estradiol 100 mcg).

STD Prophylaxis

Recommended regimens for infection prophylaxis include a single dose of ceftriaxone 250 mg IM OR single dose of cefixime 400 mg PO PLUS a single dose of metronidazole 2 g PO PLUS a single dose of azithromycin 1 g PO, OR doxycycline 100 mg PO twice a day for 7 days.

STD Treatment

1. Gonorrhea: Ceftriaxone 250 mg IM single dose or cefixime 400 mg PO

single dose.

2. Chlamydia: Azithromycin 1 g PO as single dose OR doxycycline 100 mg PO twice per day for 7 days (do not use doxycycline during pregnancy).

3. Trichomonas and bacterial vaginosis: metronidazole 2 g PO in single dose (do not use during first trimester of pregnancy).

4. Syphillis: Penicillin G benzathine 2.4 million IU IM. Use erythromycin 500 mg PO four times per day for 15 days if penicillin allergic.

Hepatitis Prophylaxis

Administer vaccine at the time of initial exam if patient has not been previously vaccinated. Follow-up doses of vaccine should be administered 1 to 2 months and 4 to 6 months after the first dose

HIV: Prophylaxis and Counseling

Rates of human immunodeficiency virus (HIV) seroconversion are low but occur from sexual assault or sexual abuse. Circumstances should guide the decision to administer postexposure prophylaxis. An assailant known to be infected with HIV, high viral load exposure, vaginal trauma, and ejaculate on membranes increase the risk for HIV seroconversion. Post-HIV exposure recommendations are posted on the Centers for Disease Control and Prevention (CDC) website (www.cdc.gov). Additional assistance may be obtained by calling the 24-hour National HIV/AIDS Post Exposure Hotline at 1-888-448-4911. Routine prophylaxis is not recommended, and counseling and follow-up should be provided.

■■ INTIMATE PARTNER VIOLENCE AND ABUSE

1. An attempt to leave an abusive relationship is often the most dangerous time.

2. Assess for potentially lethal situations. These include increasing frequency or severity of violence, the threat or use of weapons, obsession with the patient, taking hostages, stalking, homicidal or suicidal threats, and substance abuse by the assailant, especially with crack cocaine or amphetamines.

3. Hospital admission is an option in high-risk situations if a safe location cannot be established before discharge.

188_Cydulka_Ch188.indd 1019 22/03/17 5:14 PM

Source page 1039

Table 188-3 Hotlines for Patients

National Domestic Violence Hotline: 24 hours; links caller to help in her (or his) area—emergency shelter, domestic violence shelters, legal advocacy and assistance programs, social services

800-799-SAFE (7233)

800-787-3224 (TTY)

Rape, Abuse, and Incest National Network: 24 hours; automatically 800-656-HOPE (4673) transfers caller to nearest rape crisis center anywhere in the nation http://www.rainn.org

Follow-up Care

Follow-up care often requires the coordinated efforts of physicians, law enforcement, survival advocates, and counselors. Some communities and hospitals use the Sexual Assault Nurse Examiner (SANE) programs; others have rape crisis centers, and 24-hours safe rooms. National hotlines are available (Table 188-3). Know the resources available in your community.

■■ FURTHER READING

For further reading in Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 293, "Female and Male Sexual Assault," by Lisa Moreno-Walton; Chapter 294, "Intimate Partner Violence and Abuse," by Mary Hancock.

188_Cydulka_Ch188.indd 1020 22/03/17 5:14 PM

References/provenance

- Source: *Tintinalli's Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 1036–1041.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.

Chapter 189: Palliative Care

Source pages 1042–1089 | Content ID SEG-TINTINALLI-EMM8-FULL | Medical review required

Source: Tintinalli’s Emergency Medicine Manual, Eighth Edition

Source pages: 1042–1089

Content ID: SEG-TINTINALLI-EMM8-FULL

Version ID: SEG-TINTINALLI-EMM8-FULL-v1.0

Review status: MEDICAL_REVIEW_REQUIRED

Learning objectives

1. Identify the source-described scope of **Palliative Care**.

2. Organize the source-described findings, evaluation elements, and management concepts by source page.

3. Distinguish directly source-supported statements from items requiring medical-review confirmation.

Structured medical explanation

This chapter is a source-grounded educational transformation of the cited pages. The following source-derived observations are preserved without adding external recommendations:

- CHAPTER 189: Palliative Care 1023 Table 189-2 Key Communication Phrases Determining Decision-Making Capacity Will you describe your current condition? [Source pages 1042–1089]

- Tell me about the treatment options we have just discussed. [Source pages 1042–1089]

- Quality of Life What symptoms bother you the most? [Source pages 1042–1089]

- Prognosis Has anyone talked to you about what to expect? [Source pages 1042–1089]

- Do you have any sense of how much time is left? [Source pages 1042–1089]

Clinical reasoning structure

- **Problem representation:** Palliative Care; source pages 1042–1089.

- **Source-described inputs:** Review the findings and terminology preserved in the source excerpt below.

- **Decision relationships:** [UNCLEAR — page-level medical review required to validate the complete diagnostic and management sequence.]

- **Disposition/management logic:** [UNCLEAR — do not infer beyond the supplied source pages.]

RED FLAG

[UNCLEAR — medical reviewer must identify and validate source-supported red flags for this chapter.]

HIGH YIELD

- Chapter scope: **Palliative Care**.

- Source-page range: **1042–1089**.

- Review all source tables, figures, and algorithms within this range.

EXAM POINTS

[UNCLEAR — source-supported examination points require medical review.]

CLINICAL PEARLS

[UNCLEAR — source-supported clinical pearls require medical review.]

COMMON MISTAKES

[UNCLEAR — source-supported common mistakes require medical review.]

MANAGEMENT STEPS

[UNCLEAR — source-supported management steps require medical review; no treatment instruction is inferred here.]

Tables and algorithms

[UNCLEAR — review cited source pages for tables, figures, and algorithms before reconstruction.]

Source excerpt

Source page 1042

CHAPTER 189: Palliative Care 1023

Table 189-2 Key Communication Phrases

Determining Decision-Making Capacity

Will you describe your current condition?

Tell me about the treatment options we have just discussed.

Explain to me why you feel that way.

Quality of Life

What symptoms bother you the most? What concerns you the most?

Prognosis

Has anyone talked to you about what to expect?

Do you have any sense of how much time is left? Is this something you would like to talk about?

Talking with Surrogate Decision Makers

These decisions are very hard; if [the patient] was sitting with us today, what do you think [he/she] would say?

Can you tell me why you feel that way?

It is not a question of whether we will care for your [loved one], but how we will care for them.

Then we will do everything possible to keep your [loved one] comfortable, but we won't be providing ineffective and burdensome therapies such as CPR or intubation.

Discussing Palliative Care or Hospice Referral

To meet the goals we've discussed, I've asked the palliative care team to visit with you; they are experts in treating the symptoms you are experiencing. They can help your family deal with the changes brought on by your illness.

Breaking Bad News—Death Pronouncement

I wish there is more we could have done, your loved one has died; I'm very sorry for your loss. This has to be really difficult for you. Is there anyone I can call to be with you now?

9. In the case of imminent patient death, notify family members and the primary physician, and try to reach agreement to offer comfort measures only. Continuous opiate infusions plus intermittent or infused midazolam may be initiated and titrated to patient comfort. Delirium therapy can be directed to underlying cause (pain, fever) or, if not known or

reversible, treated with IV haloperidol. Atropine ophthalmic drops, 2 drops sublingual every 6 hours, can relieve the sound of terminal secretions, which have not been shown to be distressing to the patient, but can be very upsetting to the family. Place patient in a relatively calm quiet bed in the ED, allow all family at bedside, offer chaplain support and attempt to maintain dignity and regard for patient's dying process.

■■ SYMPTOM MANAGEMENT

Pain

Introduction

Any validated pain scale can be used to assess pain and track response to treatment. Remember to evaluate as you would for any pain complaint: location, character, duration, quality, alleviating/exacerbating factors. Asking what pain level is tolerable to them and their baseline pain score can be helpful. Do not judge malignant pain based on outward appearance or vital signs. To begin to treat the pain, first find out if the patient is opiate naïve or opiate tolerant. If patient is opiate naïve, the starting dose of morphine is

189_Cydulka_Ch189.indd 1023 22/03/17 5:27 PM

Source page 1043

0.1 mg/kg IV. For opiate tolerant patients, consider starting with hydromorphone 1 or 2 mg IV. Opiates reach maximum therapeutic levels and have peak effects/side effects at 6 to 10 minutes. Therefore, IV pain medications can be safely re-dosed every 15 minutes until relief is reached. The biggest obstacle to aggressive pain management with opiates is the fear of respiratory depression. Respiratory depression is not a sudden occurrence, but instead is part of a progression that starts with sedation, somnolence, and finally respiratory depression. A patient can be safely dosed and re-dosed until the symptoms are palliated, as long as level of consciousness is monitored.

Dyspnea

While identifying the cause of dyspnea and treating the underlying pathology, treatment should also be offered to palliate the symptoms. Opioids are extremely beneficial in treating the agitation and anxiety provoked by dyspnea. For an opioid-naïve patient, start with a dose of morphine 0.05 mg/kg IV (half the starting dose used for pain), and monitor for sedation and hypoventilation. Use a goal of maintaining a respiratory rate of at least 10 to 12 breaths per minute

Nausea/Vomiting

The underlying cause of nausea can help identify the class of antiemetic drugs most likely to be therapeutic. For chemotherapy-induced nausea, high doses of serotonin 5-hydroxytryptamine-3 antagonists such as ondan-

setron 8 mg IVP or corticosteroids such as dexamethasone 10 mg IVP are helpful. Steroids can improve symptoms caused by increased intracranial pressure and bowel obstruction from cancer. For refractory nausea, try a dopamine antagonist such as haloperidol 0.5 to 4 mg IVP. Metoclopramide 10 mg in 100 mL NS IV is excellent for the symptoms of diabetic gastroparesis or compression of the stomach due to tumor or ascites, or small bowel obstruction, as it increases motility.

Agitation/Delirium

Patients with terminal illness may become agitated, with or without delirium. The causes of this agitation are multifactorial, including pain, anxiety, terminal restlessness, breathlessness, fecal impaction, and mental anguish. The indicated class of medications varies depending on the situation but includes antipsychotics such as haloperidol 1 to 5 mg IV, anxiolytics such as midazolam 1 to 4 mg IV, and opiates such as morphine 2 to 10 mg IV.There is no evidence that these palliative interventions hasten death.

Constipation

Constipation is a universal side effect for patients on opiates. These patients need a concurrent 2-drug bowel regimen, including an osmotic agent, such as lactulose 30 mL every 24 hours or propylene glycol 17 g powder in 8 oz of water every 24 hours, along with a stimulant laxative, such as senna 2 tabs every evening or bisacodyl 5 to 15 mg × 1 dose.

■■ FURTHER READING

For Further reading in Tintinalli’s Emergency Medicine: A Comprehensive Study Guide, 8th ed., see Chapter 299, “Palliative Care,” by Robert J. Zalenski and Erin Zimny.

189_Cydulka_Ch189.indd 1024 22/03/17 5:27 PM

Source page 1044

Index

Note: Page numbers followed by f indicate figures, page numbers followed by t indicate tables.

A for pericoronitis, 839
AAAs. See abdominal aortic aneurysms for pulpitis, 839
(AAAs) for stomatitis, 370
ABCD scoring system, 773–774, 773t for thyrotoxicosis, 713–714
abciximab, 134t, 137 toxicity, 278, 593–595, 593t, 595f, 596t
abdominal aortic aneurysms (AAA), for varicella, 447
181–182, 182f acetazolamide
abdominal emergencies, in neonates, for acute mountain sickness, 670
364–365 for glaucoma, 823
abdominal injuries, 909–913 for headache, 762

diaphragmatic, 910 acetic acid, 683t
hollow visceral, 909 for otitis externa, 831
retroperitoneal, 909–910 acid
solid organ, 909 See also nucleic acid amplification
abdominal pain burns, 683t–684t
acute, 217–221, 218f, 220t folinic, 583
in children, 400–404, 401t hydrofluoric, 684t
abdominal trauma, 870, 874, 877 tranexamic, 320, 728, 835, 901
abduction relief sign, 971t valproic, 441, 557t
aberrant drug-related behaviors, acid-base disorders, 53–59
79–80, 79t clinical features, 53
ABIs. See ankle-brachial indexes (ABIs) defining nature of, 54, 54f–55f
abnormal uterine bleeding (AUB), 320 diagnosis and differential, 53–54
abortion metabolic acidosis, 56–58, 56t, 57t
induced, 351–352, 352t metabolic alkalosis, 58
and threatened abortion, 325–326 respiratory acidosis, 58–59
abscess respiratory alkalosis, 59
anorectal, 268, 269f acid-fast staining, for TB, 205–206
brain, 809–810, 810t acidosis
cutaneous, 494–495 metabolic, 56–58, 56t, 57t
epidural, 810–811 respiratory, 58–59
masticator space, 829 acquired bleeding disorders
periodontal, 840 coagulation defects, 727–728
peritonsillar, 377–378, 845–846 disseminated intravascular
psoas, 979 coagulation, 727–728
retropharyngeal, 377, 847 due to circulating anticoagulants, 728
absence seizures, 792 in immune thrombocytopenia,
abuse 726–727
child, 1013–1014 liver disease, 727
elderly, 1014–1015 platelet defects, 726
intimate partner, 1016–1017, 1018t, renal disease, 727
1019–1020 acquired coagulation defects, 727–728
sexual, 1014 acquired platelet defects, 726
accelerated idioventricular rhythms acromioclavicular joint injuries,
(AIVRs), 14, 14f 943–944, 943t–944t
ACE inhibitors. See angiotensin-converting ACS. See acute coronary syndromes
enzyme (ACE) inhibitors (ACS)
acetaminophen, 557t activated charcoal, 560t, 561, 561t–562t
for acute mountain sickness, 670 for acetaminophen toxicity, 593
for acute otitis media, 366 for anticholinergic toxicity, 565

for acute pain control, 67 for antidepressants, 566

for fever, 356 for antihypertensive toxicity, 611

for headache, 413 for aspirin toxicity, 592

for orofacial pain, 838 for barbiturate overdose, 575

1025

190_Cydulka_Index.indd 1025 22/04/17 4:55 pm

Source page 1045

1026 Index

activated charcoal (Continued) Bell's palsy, 797–798

for β-blocker toxicity, 606 CNS, 796t

for benzodiazepine overdose, 576 focal mononeuropathies, 798

for calcium channel blocker Guillain-Barré syndrome, 796–797,

toxicity, 608 797t

for digitalis glycoside toxicity, 603 peripheral nervous system, 796t

for herbicide poisoning, 629 plexopathies, 798–799

for MAOI toxicity, 570 acute prostatitis, 308

for mirtazapine toxicity, 568 acute pulmonary edema, 177–178, 177t

for phentoin toxicity, 614 acute renal failure, 177t, 178, 287–291,

for SNRI toxicity, 568 288t

for SSRI toxicity, 568 acute respiratory distress syndrome

for trazodone, 567 (ARDS), 484

for vitamin toxicity, 645 acute rheumatic fever (ARF), 440–441

for xanthine toxicity, 600 acute septic arthritis, 439

acute abdominal pain, 217–221, 218f, 220t acute sequestration crises, 458–459

in children, 400–404 acute urinary retention, 303–305, 304t

laboratory studies for, 235t acute vertigo, 784–791, 784t, 785t

neuromuscular junction disorders, 799 acyclovir

acute allergic reactions, 64–66 for eczema herpeticum, 445–446

acute angle closure glaucoma, 823–825, for encephalitis, 809

823t for herpes simplex, 478, 499–500

acute aortic dissection, 177t for herpes zoster, 857

acute appendicitis, 252–255 for HSV, 816

acute bacterial sinusitis, 368–369 for HZO, 817

acute chest syndrome (ACS), 456–457, for posttransplant infections, 549

736t for varicella, 447, 501

acute coronary syndromes (ACS), 128t, adalimumab, for Crohn's disease, 230

131–137, 132t, 134t–135t, 177t, adenocoticotropin hormone (ACTH), 715

178 adenosine

evaluation process for, 142t for dysrhythmias, 34

low-probability, 141–145 for paroxysmal supraventricular

acute gastroenteritis, 395, 396t–397t, 398 tachycardia, 22
acute glomerulonephritis, 288t, 470–471 for pediatric pulmonary resuscitation,
acute heart failure, 149–151, 151f 33t
acute hepatitis, 275–278, 276t adhesive capsulitis, 977
acute infectious diarrhea, 226–228, 227t adhesive tape, 92, 94f–95f
acute interstitial nephritis, 288t admission, for TB, 207
acute ischemia, 143t adrenal crisis, 753–754
acute ischemic stroke, 177t, 179 adrenal insufficiency, 715–717, 716t–717t
acute kidney injury (AKI), 287–291, adult epiglottitis, 846
288t, 289f, 468–469, 468t adults
acute lymphoblastic leukemia (ALL), seizures in, 792–795
461–463 trauma in, 865–868, 866t
acute mastoiditis, 832 adult Still’s disease, 988t
acute mountain sickness (AMS), advanced airway support, 1–9
670–671 basic airway management, 1–2
acute myelogenous leukemia (AML), 461 extraglottic devices, 2–3
acute myocardial infarction (AMI), 125, impending/actual cardiac arrest, 1
126, 127t, 132t intubation, 4
evaluation process for, 142t mask ventilation, 2, 3f
acute necrotizing ulcerative gingivitis noninvasive positive pressure
(ANUG), 840 ventilation, 2
acute-on-chronic heart failure, 149 orotracheal intubation, 4–7, 5f
acute otitis media (AOM), 366–368, 367f rapid airway assessment, 1
acute pain management, 67–68 RSI induction, 4
acute pericarditis, 129, 164–165 surgical airway, 7–8, 8f, 9f
acute perioperative hypertension, 177t adverse drug reactions, 994
acute peripheral neurological lesions, Aeromonas hydrophila, 528
796–799 Aeromonas species, 530t, 531t
190_Cydulka_Index.indd 1026 22/04/17 4:55 pm

References/provenance

- Source: *Tintinalli’s Emergency Medicine Manual*, Eighth Edition.
- Content ID: SEG-TINTINALLI-EMM8-FULL.
- Source pages: 1042–1089.
- Processing timestamp: 2026-08-15T15:32:40Z.
- Medical review: required before release.

Evidence ledger

Every statement in this record is either a source-page pointer, a conservative transformation of the displayed excerpt, or explicitly marked `[UNCLEAR]` for medical review.